

PRINCIPLES OF CONTINUUM MECHANICS

Worked notes on Chapters 1 to 12

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Every result is derived from the algebra of Chapter 1, with no step omitted. Section and equation numbers match the book. All chapter problems are solved in full, and each appears in the table of contents and in the PDF bookmarks. Suspected misprints in the printed text are recorded where they occur and collected in **book_questions.pdf**.

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Chapter 1

Mathematical Preliminaries

A working knowledge of vector and tensor algebra and analysis is a prerequisite for continuum mechanics. This chapter assembles the results we will need. The style is deliberately slow: we begin each topic with the naive question that forces the next idea, then make it precise and prove it. Throughout, $\mathcal{V}_n$ is an n dimensional real vector space, $\mathbb{E}^n$ the same carrying an inner product, and $\mathbb{E}^3$ the three dimensional case; $\{\underline{e}_i\}$ is a fixed orthonormal basis, indices run over $\{1, 2, 3\}$ unless stated otherwise, and a repeated index in a term is summed. The symbol δ_{ij} is Kronecker's, ε_{ijk} the permutation symbol, and Lin the space of second order tensors. Appendix [A](#) maps every numbered book equation to the place it is produced here.

1.1 Vector and tensor theory

1.1.1 Vector spaces

Displacements in space, velocities, and forces share exactly two features: any two of them may be added tip to tail, and any one may be scaled by a real number. Nothing else about “arrows” is used in the algebra that follows, so instead of drawing pictures we simply write down those two operations and the rules they obey. The reward is leverage: a theorem

proved from the rules holds at once for displacements, forces, and later for tensors and fields. The list of rules is the definition of a vector space.

DEFINITION 1.1 (Vector space). A real vector space is a set $\mathcal{V}$ with maps $+$: $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ and $\cdot$: $\mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V}$ such that, for all $\underline{a}, \underline{b}, \underline{c} \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{R}$,

- (a) $\alpha \underline{a} + \beta \underline{b} \in \mathcal{V}$,
- (b) $(\underline{a} + \underline{b}) + \underline{c} = \underline{a} + (\underline{b} + \underline{c})$,
- (c) $\underline{a} + \underline{b} = \underline{b} + \underline{a}$,
- (d) $\exists \underline{0} : \underline{a} + \underline{0} = \underline{a}$,
- (e) $\exists (-\underline{a}) : \underline{a} + (-\underline{a}) = \underline{0}$,
- (f) $(\alpha\beta)\underline{a} = \alpha(\beta\underline{a})$,
- (g) $(\alpha + \beta)\underline{a} = \alpha\underline{a} + \beta\underline{a}$,
- (h) $\alpha(\underline{a} + \underline{b}) = \alpha\underline{a} + \alpha\underline{b}$,
- (i) $1\underline{a} = \underline{a}$,
- (j) $0\underline{a} = \underline{0}$.

Vectors are set in bold; real numbers (scalars) in light face.

We want to pin any vector down by a finite list of numbers. For that we need a set of reference vectors that neither wastes effort duplicating each other nor leaves gaps. “No duplication” is linear independence; “no gaps” together with independence is a basis.

DEFINITION 1.2 (Linear dependence). $\{\underline{v}_1, \dots, \underline{v}_p\} \subset \mathcal{V}$ is linearly independent iff $\alpha_i \underline{v}_i = \underline{0} \Rightarrow \alpha_1 = \dots = \alpha_p = 0$, and linearly dependent otherwise.

DEFINITION 1.3 (Dimension, basis). $\mathcal{V}$ is finite dimensional of dimension n , written $\mathcal{V}_n$, iff every linearly independent set has at most n elements and some such set has exactly n . Any linearly independent set of n elements is a basis.

THEOREM 1.4 (Coordinates exist and are unique). *Let $\{\underline{u}_1, \dots, \underline{u}_n\}$ be a basis of $\mathcal{V}_n$. Then every $\underline{v} \in \mathcal{V}_n$ has a representation $\underline{v} = \alpha_i \underline{u}_i$ with unique coefficients α_i .*

Proof. Existence. $\{\underline{u}_1, \dots, \underline{u}_n, \underline{v}\}$ has $n + 1 > n$ elements, hence is dependent: $\lambda \underline{v} + \lambda_i \underline{u}_i = \underline{0}$ with the coefficients not all zero. If $\lambda = 0$ then $\lambda_i \underline{u}_i = \underline{0}$ forces every $\lambda_i = 0$, a contradiction; so $\lambda \neq 0$ and $\underline{v} = (-\lambda_i/\lambda) \underline{u}_i$. *Uniqueness.* $\alpha_i \underline{u}_i = \beta_i \underline{u}_i \Rightarrow (\alpha_i - \beta_i) \underline{u}_i = \underline{0} \Rightarrow \alpha_i = \beta_i$ by independence. ■

The axioms can add and scale, but they cannot yet say how *long* a vector is or what *angle* two of them make: those words never appear in (a)–(j). To measure we add one operation that eats two vectors and returns a number, is symmetric, is linear in each slot, and is strictly positive on nonzero vectors. From it flow length, angle, orthogonality, and all of Euclidean geometry.

DEFINITION 1.5 (Inner product, norm). $\mathbb{E}^n$ is $\mathcal{V}_n$ with a map $\cdot : \mathbb{E}^n \times \mathbb{E}^n \rightarrow \mathbb{R}$ such that, for all $\underline{u}, \underline{v}, \underline{w}$ and $\alpha \in \mathbb{R}$,

$$\begin{aligned} \text{(a)} \quad \underline{u} \cdot \underline{v} &= \underline{v} \cdot \underline{u}, & \text{(b)} \quad \underline{u} \cdot (\underline{v} + \underline{w}) &= \underline{u} \cdot \underline{v} + \underline{u} \cdot \underline{w}, & \text{(c)} \quad (\alpha \underline{u}) \cdot \underline{v} &= \alpha(\underline{u} \cdot \underline{v}), \\ \text{(d)} \quad \underline{u} \cdot \underline{u} &\geq 0, & \underline{u} \cdot \underline{u} = 0 &\iff \underline{u} = \underline{0}. \end{aligned}$$

The norm is $|\underline{u}| = (\underline{u} \cdot \underline{u})^{1/2}$; $\underline{u}$ is a unit vector iff $|\underline{u}| = 1$; $\underline{u} \perp \underline{v}$ iff $\underline{u} \cdot \underline{v} = 0$.

Before we may define the angle by $\cos \theta = \underline{u} \cdot \underline{v} / (|\underline{u}| |\underline{v}|)$, the right side must lie in $[-1, 1]$, or the arccosine is meaningless. That bound is not a new assumption: positivity already forces it, because a particular squared length can never be negative.

THEOREM 1.6 (Cauchy–Schwarz). $|\underline{u} \cdot \underline{v}| \leq |\underline{u}| |\underline{v}|$ for all $\underline{u}, \underline{v} \in \mathbb{E}^n$, with equality iff $\underline{u}, \underline{v}$ are parallel.

Proof. If $\underline{v} = \underline{0}$ both sides vanish. Otherwise positivity (d) applied to $\underline{u} - \lambda \underline{v}$ gives, for every $\lambda \in \mathbb{R}$,

$$0 \leq |\underline{u} - \lambda \underline{v}|^2 = |\underline{u}|^2 - 2\lambda \underline{u} \cdot \underline{v} + \lambda^2 |\underline{v}|^2,$$

using bilinearity and symmetry. The right side is least at $\lambda = \underline{u} \cdot \underline{v} / |\underline{v}|^2$; substituting, $0 \leq |\underline{u}|^2 - (\underline{u} \cdot \underline{v})^2 / |\underline{v}|^2$, i.e. $(\underline{u} \cdot \underline{v})^2 \leq |\underline{u}|^2 |\underline{v}|^2$. Equality forces $\underline{u} - \lambda \underline{v} = \underline{0}$ by (d), i.e. $\underline{u} \parallel \underline{v}$. ■

COROLLARY 1.7 (Triangle inequality). $|\underline{u} + \underline{v}| \leq |\underline{u}| + |\underline{v}|$.

Proof. $|\underline{u} + \underline{v}|^2 = |\underline{u}|^2 + 2\underline{u} \cdot \underline{v} + |\underline{v}|^2 \leq |\underline{u}|^2 + 2|\underline{u}| |\underline{v}| + |\underline{v}|^2 = (|\underline{u}| + |\underline{v}|)^2$ by Theorem 1.6; take square roots. ■

DEFINITION 1.8 (Angle). For $\underline{u}, \underline{v} \neq \underline{0}$, $\theta = \arccos(\underline{u} \cdot \underline{v} / (|\underline{u}||\underline{v}|)) \in [0, \pi]$, well defined by Theorem 1.6.

A general basis reproduces vectors but scrambles the metric: components are not projections, and $\underline{u} \cdot \underline{v}$ is not $u_i v_i$. A basis of mutually orthogonal unit vectors removes every cross term, so a component becomes a projection and the inner product becomes the plain sum of products. This is why all computation below uses an orthonormal basis.

DEFINITION 1.9 (Orthonormal basis). $\{\underline{e}_1, \dots, \underline{e}_n\}$ is orthonormal iff $\underline{e}_i \cdot \underline{e}_j = \delta_{ij}$.

PROPOSITION 1.10. *Every orthonormal set is linearly independent.*

Proof. $\alpha_i \underline{e}_i = \underline{0} \Rightarrow 0 = (\alpha_i \underline{e}_i) \cdot \underline{e}_j = \alpha_i \delta_{ij} = \alpha_j$. ■

PROPOSITION 1.11 (Components and the dot product). *Relative to an orthonormal basis $\{\underline{e}_i\}$, every $\underline{v}$ satisfies $\underline{v} = v_i \underline{e}_i$ with $v_i = \underline{v} \cdot \underline{e}_i$, and for $\underline{u} = u_i \underline{e}_i$, $\underline{v} = v_i \underline{e}_i$, $\underline{u} \cdot \underline{v} = u_i v_i$.*

Proof. By Theorem 1.4, $\underline{v} = v_i \underline{e}_i$ uniquely; then $\underline{v} \cdot \underline{e}_j = v_i \delta_{ij} = v_j$, and $\underline{u} \cdot \underline{v} = u_i v_j \delta_{ij} = u_i v_i$. ■

REMARK 1.12 (the summation convention, stated once). In $v_i \underline{e}_i$ the index i occurs twice and is summed; it is a dummy and may be relabelled, $v_i \underline{e}_i = v_j \underline{e}_j$. An index occurring once per term is free and must match across every term, as j in $\underline{v} \cdot \underline{e}_j = v_j$. A free index must never share a label with a dummy in the same term, and a symbol appearing three or more times in a term is undefined. Intuition: the dummy index is a hidden $\sum$; the free index is the “address” of the component the equation is really about.

REMARK 1.13 (projection, and why $\underline{0}$ is orthogonal to all). $\underline{v} = (v \cdot \underline{e}_k) \underline{e}_k$ exhibits $v_i = \underline{v} \cdot \underline{e}_i$ as the orthogonal projection of $\underline{v}$ on $\underline{e}_i$: geometrically, drop a perpendicular from the tip of $\underline{v}$ onto the $\underline{e}_i$ axis. Axiom (d) with $\underline{u} = \underline{0}$ gives $\underline{0} \cdot \underline{v} = 0$ for all $\underline{v}$: the zero vector, having no direction, is perpendicular to everything.

PROPOSITION 1.14 (Gram–Schmidt). *Every basis $\{\underline{u}_1, \dots, \underline{u}_n\}$ yields an orthonormal basis $\{\underline{e}_i\}$ with*

$$\text{span}\{\underline{e}_1, \dots, \underline{e}_k\} = \text{span}\{\underline{u}_1, \dots, \underline{u}_k\}$$

for each k ; hence there are infinitely many orthonormal bases when $n \geq 1$.

Proof. Recursively $\underline{w}_1 = \underline{u}_1$ and $\underline{w}_k = \underline{u}_k - \sum_{i < k} (\underline{u}_k \cdot \underline{e}_i) \underline{e}_i$, $\underline{e}_k = \underline{w}_k / |\underline{w}_k|$. Independence of the $\underline{u}_i$ gives $\underline{w}_k \neq \underline{0}$, and for $j < k$, $\underline{w}_k \cdot \underline{e}_j = \underline{u}_k \cdot \underline{e}_j - \underline{u}_k \cdot \underline{e}_j = 0$. Infinitude: for any orthogonal Q (Definition 1.47), $\{Q\underline{e}_i\}$ is again orthonormal, and such Q form a continuum. ■

Many physical quantities are linear scalar functions of a vector: the work of a fixed force along a variable displacement, a single component, a directional derivative. Any such function is fixed by its values on the basis, and reassembling those values yields one vector whose dot product reproduces the function. Isolating this fact now pays off later, when the increment of a field becomes its gradient.

THEOREM 1.15 (Riesz representation). *$f : \mathbb{E}^n \rightarrow \mathbb{R}$ is linear iff there is a unique $\underline{a} \in \mathbb{E}^n$ with $f(\underline{v}) = \underline{a} \cdot \underline{v}$ for all $\underline{v}$; explicitly $\underline{a} = f(\underline{e}_i) \underline{e}_i$.*

Proof. $(\Leftarrow)$ $\underline{v} \mapsto \underline{a} \cdot \underline{v}$ is linear by (b),(c). $(\Rightarrow)$ By linearity and Proposition 1.11, $f(\underline{v}) = f(v_i \underline{e}_i) = v_i f(\underline{e}_i) = a_i v_i = \underline{a} \cdot \underline{v}$ with $a_i := f(\underline{e}_i)$. Uniqueness: $\underline{a} \cdot \underline{v} = \underline{a}' \cdot \underline{v} \forall \underline{v} \Rightarrow (\underline{a} - \underline{a}') \cdot \underline{v} = 0$; take $\underline{v} = \underline{a} - \underline{a}'$ and use (d). ■

EXAMPLE 1.16 (the Riesz construction in action). Suppose $f : \mathbb{E}^3 \rightarrow \mathbb{R}$ is linear and we know its values on the standard basis: $f(\underline{e}_1) = 3$, $f(\underline{e}_2) = -1$, $f(\underline{e}_3) = 2$. Set $a_i := f(\underline{e}_i)$, so $\underline{a} = 3\underline{e}_1 - \underline{e}_2 + 2\underline{e}_3$. Then for any $\underline{v} = v_1 \underline{e}_1 + v_2 \underline{e}_2 + v_3 \underline{e}_3$,

$$f(\underline{v}) = f(v_i \underline{e}_i) = v_i f(\underline{e}_i) = 3v_1 - v_2 + 2v_3 = \underline{a} \cdot \underline{v}.$$

The abstract function f is now the concrete operation “dot with $(3, -1, 2)$ ”. Every linear scalar function on $\mathbb{E}^3$ hides a single vector; finding that vector is the entire content of the Riesz theorem.

REMARK 1.17 (why Riesz is the engine of continuum calculus). The pattern “linear functional $\Rightarrow$ dot product with a fixed vector” will reappear every time we define a derivative. The gradient of a scalar field (§1.2.2) is the Riesz representer of the linear part of the increment. For a vector field the increment is linear and vector valued, so its representer is a *tensor* (§1.2.5). Recognising this template once makes every later gradient obvious.

1.1.2 Change-of-basis formulas

A vector is a geometric object, but its components $v_i = \underline{v} \cdot \underline{e}_i$ are read off a chosen basis. Turn the basis and the same $\underline{v}$ shows different numbers. Mechanics must track that change exactly, so that physical statements can be made basis independent. The bookkeeping is a single matrix of dot products between the old and new axes, and it turns out to be orthogonal.

DEFINITION 1.18 (Direction cosines). For two orthonormal bases $\{\underline{e}_i\}$, $\{\underline{e}'_i\}$, set $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$.

THEOREM 1.19 (Transformation law and orthogonality). *With $v_i = \underline{v} \cdot \underline{e}_i$, $v'_i = \underline{v} \cdot \underline{e}'_i$,*

$$v'_j = a_{ij}v_i, \quad v_i = a_{ij}v'_j, \quad a_{ik}a_{jk} = \delta_{ij} = a_{ki}a_{kj}.$$

Proof. Expanding each old axis in the new basis, $\underline{e}_i = \underline{e}_i \cdot \underline{e}'_j \underline{e}'_j = a_{ij} \underline{e}'_j$, so $\underline{v}'_j \underline{e}'_j = \underline{v} = v_i \underline{e}_i = v_i a_{ij} \underline{e}'_j$, whence $\underline{v}'_j = a_{ij} v_i$ by independence of $\{\underline{e}'_j\}$; swapping the two bases gives the inverse. Orthogonality: $a_{ik}a_{jk} = \underline{e}_i \cdot \underline{e}'_k \underline{e}_j \cdot \underline{e}'_k = \underline{e}_i \cdot (\underline{e}_j \cdot \underline{e}'_k \underline{e}'_k) = \underline{e}_i \cdot \underline{e}_j = \delta_{ij}$, and $a_{ki}a_{kj} = \delta_{ij}$ likewise. ■

REMARK 1.20 (what “orthogonal matrix” means here). $a_{ik}a_{jk} = \delta_{ij}$ says the rows of (a_{ij}) are orthonormal; it is the algebraic echo of the geometric fact that both frames are orthonormal. Nothing is rotated in space, only the labels; the vector itself is untouched. This is the passive view of a change of basis, to be contrasted later with the active view, where an orthogonal *tensor* genuinely moves vectors (§1.1.5).

1.1.3 Vector products in $\mathbb{E}^3$

Two vectors span a parallelogram. Its *area*, and the *normal* direction to its plane, are geometric data the inner product cannot supply. In exactly three dimensions a plane has a unique normal line, so “area times normal” can be packed into one vector, the cross product; feeding a third vector then measures the signed *volume* of the spanned parallelepiped, the box product. These are the tools behind area, volume, orientation, determinants, and the axial vector of a rotation.

DEFINITION 1.21 (Cross and box products). $\times : \mathbb{E}^3 \times \mathbb{E}^3 \rightarrow \mathbb{E}^3$ satisfies, for all $\underline{u}, \underline{v}, \underline{w}$ and α, β ,

$$(a) \ \underline{u} \times \underline{v} = -\underline{v} \times \underline{u}, \quad (b) \ |\underline{u} \times \underline{v}| = |\underline{u}||\underline{v}| \sin \theta,$$

$$(c) \ (\alpha \underline{u} + \beta \underline{v}) \times \underline{w} = \alpha \underline{u} \times \underline{w} + \beta \underline{v} \times \underline{w},$$

$$(d) \ [\underline{u}, \underline{v}, \underline{w}] := \underline{u} \cdot (\underline{v} \times \underline{w}) = \underline{v} \cdot (\underline{w} \times \underline{u}) = \underline{w} \cdot (\underline{u} \times \underline{v}).$$

PROPOSITION 1.22. $\underline{u} \times \underline{u} = \underline{0}$; $\underline{u} \times \underline{v} \perp \underline{u}, \underline{v}$; and $[\cdot, \cdot, \cdot]$ is *trilinear and alternating* (it changes sign under any transposition of its arguments).

Proof. (a) gives $\underline{u} \times \underline{u} = -\underline{u} \times \underline{u} = \underline{0}$; then (d) gives $\underline{u} \cdot (\underline{u} \times \underline{v}) = [\underline{u}, \underline{u}, \underline{v}] = 0$ and $\underline{v} \cdot (\underline{u} \times \underline{v}) = [\underline{v}, \underline{u}, \underline{v}] = 0$. Trilinearity in slot one is (c) via (d); the other slots follow from cyclicity in (d). Alternation: (a) gives $[\underline{u}, \underline{v}, \underline{w}] = -\underline{u} \cdot \underline{w} \times \underline{v} = -[\underline{u}, \underline{w}, \underline{v}]$, and cyclicity extends this to every transposition. ■

DEFINITION 1.23 (Right-handed basis; permutation symbol). An orthonormal basis of $\mathbb{E}^3$ is right handed iff $\underline{e}_1 \times \underline{e}_2 = \underline{e}_3$ and cyclically. Then

$$\varepsilon_{ijk} := [\underline{e}_i, \underline{e}_j, \underline{e}_k] = \begin{cases} +1, & (i, j, k) \text{ even perm. of } (1, 2, 3), \\ -1, & (i, j, k) \text{ odd perm.}, \\ 0, & \text{repeated index.} \end{cases}$$

PROPOSITION 1.24 (Components of the products). *In a right-handed orthonormal basis, $\underline{e}_i \times \underline{e}_j = \varepsilon_{ijk} \underline{e}_k$, $(\underline{a} \times \underline{b})_k = \varepsilon_{ijk} a_i b_j$, and $[\underline{a}, \underline{b}, \underline{c}] = \varepsilon_{ijk} a_i b_j c_k$.*

Proof. $\underline{e}_i \times \underline{e}_j = (\underline{e}_i \times \underline{e}_j) \cdot \underline{e}_k \underline{e}_k = \varepsilon_{ijk} \underline{e}_k$; then $\underline{a} \times \underline{b} = a_i b_j \varepsilon_{ijk} \underline{e}_k$ and $[\underline{a}, \underline{b}, \underline{c}] = (\underline{a} \times \underline{b}) \cdot \underline{c} = \varepsilon_{ijk} a_i b_j c_k$. ■

LEMMA 1.25 (Alternating trilinear forms are one dimensional). *If $\omega : \mathbb{E}^3 \times \mathbb{E}^3 \times \mathbb{E}^3 \rightarrow \mathbb{R}$ is trilinear and alternating, then $\omega(\underline{a}, \underline{b}, \underline{c}) = \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3) [\underline{a}, \underline{b}, \underline{c}]$.*

Proof.

$$\begin{aligned} \omega(\underline{a}, \underline{b}, \underline{c}) &= a_i b_j c_k \omega(\underline{e}_i, \underline{e}_j, \underline{e}_k) \\ &= a_i b_j c_k \varepsilon_{ijk} \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3) \\ &= \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3) [\underline{a}, \underline{b}, \underline{c}], \end{aligned}$$

using $\omega(\underline{e}_i, \underline{e}_j, \underline{e}_k) = \varepsilon_{ijk} \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3)$ from alternation. ■

REMARK 1.26 (the workhorse). Lemma 1.25 is the one fact that makes determinants and invariants painless: *any* quantity linear in three vectors and sign reversing under a swap is a fixed multiple of the box product. Each time we invoke it, a page of index juggling collapses to a line. Intuitively, in three dimensions there is essentially one way to measure oriented volume, and everything alternating and trilinear must be proportional to it.

DEFINITION 1.27 (Determinant). For $B = (b_{ij}) \in \mathbb{R}^{3 \times 3}$, $\det B := \varepsilon_{ijk} b_{i1} b_{j2} b_{k3}$.

PROPOSITION 1.28. $\varepsilon_{mnp} \det B = \varepsilon_{ijk} b_{im} b_{jn} b_{kp}$, hence $\varepsilon_{ijk} \varepsilon_{mnp} = \det(\delta_{\bullet\bullet})$, the 3×3 determinant of the Kronecker array with rows i, j, k and columns m, n, p .

Proof. $(\underline{x}, \underline{y}, \underline{z}) \mapsto \varepsilon_{ijk} x_i y_j z_k$ is trilinear and alternating in its three columns, so by Lemma 1.25 equals $[\underline{x}, \underline{y}, \underline{z}]$ times its value 1 at $(\underline{e}_1, \underline{e}_2, \underline{e}_3)$. Applied to columns $(b_{\cdot m}, b_{\cdot n}, b_{\cdot p})$ this gives $\varepsilon_{ijk} b_{im} b_{jn} b_{kp} = \varepsilon_{mnp} \det B$. Putting $b_{ij} = \delta_{ij}$ (so $\det B = 1$) yields the second identity. ■

THEOREM 1.29 (ε - δ identity). $\varepsilon_{ijk}\varepsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$, $\varepsilon_{ijk}\varepsilon_{mjk} = 2\delta_{im}$, $\varepsilon_{ijk}\varepsilon_{ijk} = 6$.

Proof. Expand the determinant of Proposition 1.28, set $p = k$, sum ($\delta_{kk} = 3$): $\varepsilon_{ijk}\varepsilon_{mnk} = \delta_{im}(3\delta_{jn} - \delta_{jn}) - \delta_{in}(3\delta_{jm} - \delta_{jm}) + (\delta_{in}\delta_{jm} - \delta_{im}\delta_{jn}) = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$. Set $n = j$, sum: $\varepsilon_{ijk}\varepsilon_{mjk} = 3\delta_{im} - \delta_{im} = 2\delta_{im}$. Set $m = i$, sum: $\varepsilon_{ijk}\varepsilon_{ijk} = 2\delta_{ii} = 6$. ■

COROLLARY 1.30 (Determinant, contracted).

$$\det B = \frac{1}{6} \varepsilon_{ijk}\varepsilon_{mnp} b_{im}b_{jn}b_{kp}.$$

Proof. Contract $\varepsilon_{ijk}b_{im}b_{jn}b_{kp} = \varepsilon_{mnp} \det B$ with ε_{mnp} and use $\varepsilon_{mnp}\varepsilon_{mnp} = 6$. ■

1.1.4 Tensors

Riesz handled linear rules returning a *number*. Physics abounds in linear rules returning a *vector*: stress sends a surface normal to a traction, the inertia tensor sends angular velocity to angular momentum, the deformation gradient sends a material fibre to its deformed image. The object encoding such a rule is a second order tensor. Its simplest instance, $\underline{a} \otimes \underline{b}$, is the rule “project onto $\underline{b}$, then point along $\underline{a}$ ”.

DEFINITION 1.31 (Tensor). A (second order) tensor is a linear map $\underline{A} : \mathbb{E}^3 \rightarrow \mathbb{E}^3$. Equality, scalar multiple and sum are defined by $\underline{A} = \underline{B} \iff \underline{A}\underline{v} = \underline{B}\underline{v} \forall \underline{v}$, $(\alpha \underline{A})\underline{v} = \alpha(\underline{A}\underline{v})$, $(\underline{A} + \underline{B})\underline{v} = \underline{A}\underline{v} + \underline{B}\underline{v}$. The set Lin is a vector space; $\underline{Q}\underline{v} = \underline{0}$ and $\underline{I}\underline{v} = \underline{v}$ for all $\underline{v}$.

DEFINITION 1.32 (Tensor product). For $\underline{a}, \underline{b} \in \mathbb{E}^3$, $(\underline{a} \otimes \underline{b})\underline{v} := (\underline{b} \cdot \underline{v}) \underline{a}$.

THEOREM 1.33 (Component representation). $\{\underline{e}_i \otimes \underline{e}_j\}$ is a basis of Lin , and $\underline{A} = A_{ij}\underline{e}_i \otimes \underline{e}_j$ with $A_{ij} = \underline{e}_i \cdot \underline{A}\underline{e}_j$; hence $\dim \text{Lin} = 9$.

Proof. Spanning. For $\underline{v} = v_i \underline{e}_i$, $\underline{A}\underline{v} = v_i \underline{A}\underline{e}_i = v_i \underline{e}_j \cdot \underline{A}\underline{e}_i \underline{e}_j = A_{ji}(\underline{e}_i \cdot \underline{v}) \underline{e}_j = (A_{ji} \underline{e}_j \otimes \underline{e}_i)\underline{v}$, so $\underline{A} = A_{ij}\underline{e}_i \otimes \underline{e}_j$ with $A_{ij} = \underline{e}_i \cdot \underline{A}\underline{e}_j$. *Independence.* $\alpha_{ij}\underline{e}_i \otimes \underline{e}_j = \underline{Q}$ applied to $\underline{e}_k$ gives $\alpha_{ik}\underline{e}_i = \underline{0}$, so $\alpha_{ik} = 0$. ■

EXAMPLE 1.34 (reading off components of tensor products). (i) *The tensor product $\underline{a} \otimes \underline{b}$.* By Definition 1.32, $(\underline{a} \otimes \underline{b})\underline{e}_j = (\underline{b} \cdot \underline{e}_j)\underline{a} = b_j\underline{a}$. Dotting with $\underline{e}_i$,

$$(\underline{a} \otimes \underline{b})_{ij} = \underline{e}_i \cdot (\underline{a} \otimes \underline{b})\underline{e}_j = \underline{e}_i \cdot b_j\underline{a} = b_j\underline{e}_i \cdot \underline{a} = a_ib_j.$$

Thus the component matrix is the outer product of the column (a_i) with the row (b_j) :

$$((\underline{a} \otimes \underline{b})_{ij}) = \begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{pmatrix}.$$

Such a matrix has rank ≤ 1 : every row is a multiple of (b_1, b_2, b_3) . Consequently $\det(\underline{a} \otimes \underline{b}) = 0$ unless $\underline{a}$ or $\underline{b}$ is $\underline{0}$.

(ii) *The identity tensor.* $\underline{I}\underline{e}_j = \underline{e}_j$, so $I_{ij} = \underline{e}_i \cdot \underline{e}_j = \delta_{ij}$. In the basis expansion, $\underline{I} = \delta_{ij}\underline{e}_i \otimes \underline{e}_j = \underline{e}_1 \otimes \underline{e}_1 + \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3 = \underline{e}_i \otimes \underline{e}_i$, with matrix $(I_{ij}) = \text{diag}(1, 1, 1)$.

(iii) *The zero tensor.* $\underline{O}\underline{v} = \underline{0}$ for all $\underline{v}$, so $O_{ij} = \underline{e}_i \cdot \underline{0} = 0$; every entry of the matrix is zero.

REMARK 1.35 (outer versus inner product). Two operations eat a pair of vectors and return very different objects. The *inner product* $\underline{a} \cdot \underline{b} = a_ib_i$ sums all matching pairs and returns one number: it measures overlap and forgets direction. The *outer product* (tensor product) stores every pair, returning nine numbers arranged in the matrix (a_ib_j) : it remembers both directions. In $\mathbb{E}^3$ one may think of $\underline{a} \otimes \underline{b}$ as the instruction “measure how much $\underline{v}$ aligns with $\underline{b}$ (via $\underline{b} \cdot \underline{v}$), then lay that amount along $\underline{a}$ ”. The trace connects the two: $\text{tr}(\underline{a} \otimes \underline{b}) = a_ib_j\delta_{ij} = a_ib_i = \underline{a} \cdot \underline{b}$, collapsing the nine entries back to one.

EXAMPLE 1.36 (a diagonal tensor: the reflection). Suppose a linear map $\underline{A}$ is specified by its action on the basis: $\underline{A}\underline{e}_1 = -\underline{e}_1$, $\underline{A}\underline{e}_2 = \underline{e}_2$, $\underline{A}\underline{e}_3 = \underline{e}_3$. We compute every component $A_{ij} = \underline{e}_i \cdot \underline{A}\underline{e}_j$ in turn.

Column $j = 1$: $\underline{A}\underline{e}_1 = -\underline{e}_1$, so $A_{11} = \underline{e}_1 \cdot -\underline{e}_1 = -1$, $A_{21} = \underline{e}_2 \cdot -\underline{e}_1 = 0$, $A_{31} = \underline{e}_3 \cdot -\underline{e}_1 = 0$.

Column $j = 2$: $\underline{A}\underline{e}_2 = \underline{e}_2$, so $A_{12} = 0$, $A_{22} = 1$, $A_{32} = 0$.

Column $j = 3$: $\underline{A}\underline{e}_3 = \underline{e}_3$, so $A_{13} = 0$, $A_{23} = 0$, $A_{33} = 1$.

Hence

$$(\underline{A}_{ij}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \text{diag}(-1, 1, 1),$$

and the tensor product form is $\underline{A} = -\underline{e}_1 \otimes \underline{e}_1 + \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3$. For any $\underline{v} = v_i \underline{e}_i$, $\underline{A}\underline{v} = -v_1 \underline{e}_1 + v_2 \underline{e}_2 + v_3 \underline{e}_3$: the component along $\underline{e}_1$ is negated while those along $\underline{e}_2, \underline{e}_3$ are unchanged. This is precisely the *mirror reflection* in the plane perpendicular to $\underline{e}_1$.

REMARK 1.37 (why the diagonal form matters). A diagonal tensor $\underline{A} = \lambda_i \underline{e}_i \otimes \underline{e}_i$ (no sum) acts by scaling each basis direction independently: $\underline{A}\underline{v} = \lambda_1 v_1 \underline{e}_1 + \lambda_2 v_2 \underline{e}_2 + \lambda_3 v_3 \underline{e}_3$. The λ_i are the eigenvalues and the $\underline{e}_i$ the eigenvectors. In continuum mechanics, when $\underline{A}$ is the (symmetric) stretch tensor, the eigenvectors are the principal directions and $|\lambda_i|$ the principal stretches. The spectral theorem (Chapter 3) guarantees that every symmetric tensor can be written in this form for a suitable orthonormal basis.

LEMMA 1.38. $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$.

Proof. $[\underline{A}(\underline{a} \otimes \underline{b})]\underline{v} = \underline{A}[(\underline{b} \cdot \underline{v})\underline{a}] = (\underline{b} \cdot \underline{v})\underline{A}\underline{a} = [(\underline{A}\underline{a}) \otimes \underline{b}]\underline{v}$. ■

THEOREM 1.39 (Transformation law for tensor components). *With $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, $A'_{ij} = a_{ki} a_{lj} A_{kl}$.*

Proof. $\underline{A}\underline{e}'_n = (A_{kl} \underline{e}_k \otimes \underline{e}_l) \underline{e}'_n = A_{kl} a_{ln} \underline{e}_k$, so $A'_{jn} = \underline{e}'_j \cdot \underline{A}\underline{e}'_n = A_{kl} a_{ln} \underline{e}'_j \cdot \underline{e}_k = A_{kl} a_{kj} a_{ln}$. ■

REMARK 1.40 (order = number of cosine factors). Scalars, vectors and tensors transform with the direction cosine matrix zero, one and two times: $\varphi' = \varphi$, $v'_i = a_{ki} v_k$, $A'_{ij} = a_{ki} a_{lj} A_{kl}$. The count of a 's is the tensor's *order*, and is often taken as the definition of a Cartesian tensor. To illustrate: the trace $\text{tr } \underline{A} = A_{ii}$ carries zero free indices, hence zero cosines and transforms as a scalar ($\text{tr } \underline{A}' = A'_{ii} = a_{ki} a_{li} A_{kl} = \delta_{kl} A_{kl} = A_{kk} = \text{tr } \underline{A}$). The traction vector $t_i = A_{ij} n_j$ carries one free index,

hence one cosine factor ($t'_i = a_{ki}t_k$), and transforms as a vector. The tensor itself carries two. Invariants (§1.1.5) exist precisely because even combinations of cosines can pair up into Kronecker deltas, cancelling all basis dependence.

1.1.5 Algebraic operations with second-order tensors

Transpose and symmetry. In $\underline{u} \cdot \underline{A}\underline{v}$ one often needs to let $\underline{A}$ act on the other factor. The tensor legalising this, $\underline{u} \cdot \underline{A}\underline{v} = \underline{A}^t \underline{u} \cdot \underline{v}$, is the transpose. Whether a tensor equals or negates its transpose splits Lin into the two parts that rule mechanics: strain is symmetric, spin is skew.

DEFINITION 1.41 (Transpose). $\underline{A}^t$ is defined by $\underline{u} \cdot \underline{A}\underline{v} = \underline{v} \cdot \underline{A}^t \underline{u}$ for all $\underline{u}, \underline{v}$.

PROPOSITION 1.42. $(\underline{A}^t)_{ij} = A_{ji}$, $(\underline{A}^t)^t = \underline{A}$, $(\underline{A}\underline{B})^t = \underline{B}^t \underline{A}^t$.

Proof. $(\underline{A}^t)_{ij} = \underline{e}_i \cdot \underline{A}^t \underline{e}_j = \underline{e}_j \cdot \underline{A} \underline{e}_i = A_{ji}$. The rest follow from Definition 1.41 and uniqueness: $\underline{u} \cdot (\underline{A}\underline{B})\underline{v} = \underline{A}^t \underline{u} \cdot \underline{B}\underline{v} = \underline{B}^t \underline{A}^t \underline{u} \cdot \underline{v}$. ■

DEFINITION 1.43 (Symmetric, skew). $\underline{A}$ is symmetric iff $\underline{A}^t = \underline{A}$, skew iff $\underline{A}^t = -\underline{A}$.

PROPOSITION 1.44 (Axial vector of a skew tensor). *For skew $\underline{A}$ there is a unique $\underline{w}$ with $\underline{A}\underline{v} = \underline{w} \times \underline{v}$ for all $\underline{v}$; $A_{ij} = -\varepsilon_{ijk} w_k$, $w_k = -\frac{1}{2} \varepsilon_{kij} A_{ij}$.*

Proof. Skewness gives $A_{ii} = 0$ and three free entries; the general skew array is $A_{ij} = -\varepsilon_{ijk} w_k$. Then $(\underline{A}\underline{v})_i = -\varepsilon_{ijk} w_k v_j = \varepsilon_{ikj} w_k v_j = (\underline{w} \times \underline{v})_i$. Inversion by Theorem 1.29: $-\varepsilon_{ijm} A_{ij} = \varepsilon_{ijm} \varepsilon_{ijk} w_k = 2w_m$. ■

Product and inverse.

DEFINITION 1.45 (Product, inverse). $(\underline{A}\underline{B})\underline{v} := \underline{A}(\underline{B}\underline{v})$. $\underline{A}$ is invertible iff $\underline{A}\underline{u} = \underline{v}$ has a unique solution $\underline{u} = \underline{A}^{-1}\underline{v}$ for every $\underline{v}$.

PROPOSITION 1.46. $(\underline{A}\underline{B})_{ij} = A_{ik} B_{kj}$; in general $\underline{A}\underline{B} \neq \underline{B}\underline{A}$; and if $\underline{A}$ is invertible, $\underline{A}^{-1}\underline{A} = \underline{A}\underline{A}^{-1} = \underline{I}$.

Proof. $(\underline{A}\underline{B})\underline{e}_j = \underline{A}(B_{kj}\underline{e}_k) = B_{kj}A_{ik}\underline{e}_i$, so $(\underline{A}\underline{B})_{ij} = A_{ik}B_{kj}$; and $(\underline{A}^{-1}\underline{A})\underline{u} = \underline{A}^{-1}\underline{v} = \underline{u}$, $(\underline{A}\underline{A}^{-1})\underline{v} = \underline{A}\underline{u} = \underline{v}$. ■

Orthogonal tensors. Rigid motions neither stretch nor shear a body: they preserve every length and angle, hence every inner product. The tensors doing this are the orthogonal ones, and they are exactly those whose transpose is their inverse. Those with $\det = +1$ are rotations; $\det = -1$ adds a reflection.

DEFINITION 1.47 (Orthogonal). $\underline{A}$ is orthogonal iff $\underline{A}\underline{u} \cdot \underline{A}\underline{v} = \underline{u} \cdot \underline{v}$ for all $\underline{u}, \underline{v}$.

PROPOSITION 1.48. $\underline{A}$ orthogonal $\iff \underline{A}^t \underline{A} = \underline{I} \iff \underline{A} \underline{A}^t = \underline{I}$; then $\underline{A}^{-1} = \underline{A}^t$ and $\det \underline{A} = \pm 1$.

Proof. $\underline{A}\underline{u} \cdot \underline{A}\underline{v} = \underline{u} \cdot \underline{A}^t \underline{A}\underline{v}$, so orthogonality is $\underline{u} \cdot (\underline{A}^t \underline{A} - \underline{I})\underline{v} = 0 \forall \underline{u}, \underline{v}$; take $\underline{u} = (\underline{A}^t \underline{A} - \underline{I})\underline{v}$ to get $\underline{A}^t \underline{A} = \underline{I}$, then $\underline{A}^{-1} = \underline{A}^t$ and $\underline{A} \underline{A}^t = \underline{I}$. Finally $(\det \underline{A})^2 = \det \underline{A}^t \det \underline{A} = \det \underline{I} = 1$ (Theorem 1.52). ■

Principal invariants. The nine components of a tensor change with the basis; its physics does not. So we seek scalar combinations surviving every change of basis. Building them from the box product, itself geometric, makes invariance automatic. Three exist, and they are the coefficients of the characteristic polynomial (Problem 23), hence carry the eigenvalue data used throughout deformation analysis.

DEFINITION 1.49 (Invariants). For all $\underline{a}, \underline{b}, \underline{c}$,

$$\begin{aligned} I_1(\underline{A})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{A}\underline{a}, \underline{b}, \underline{c}] + [\underline{a}, \underline{A}\underline{b}, \underline{c}] + [\underline{a}, \underline{b}, \underline{A}\underline{c}], \\ I_2(\underline{A})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{c}] + [\underline{A}\underline{a}, \underline{b}, \underline{A}\underline{c}] + [\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}], \\ I_3(\underline{A})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}]. \end{aligned}$$

PROPOSITION 1.50 (Well-posedness). I_1, I_2, I_3 do not depend on $\underline{a}, \underline{b}, \underline{c}$.

Proof. Each right side is trilinear and alternating in $(\underline{a}, \underline{b}, \underline{c})$; by Lemma 1.25 it equals a fixed scalar times $[\underline{a}, \underline{b}, \underline{c}]$. ■

THEOREM 1.51 (Trace). I_1 is linear, $I_1(\underline{A}) = \text{tr } \underline{A} := A_{ii}$, $\text{tr}(\underline{e}_i \otimes \underline{e}_j) = \delta_{ij}$, $\text{tr}(\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{v}$.

Proof. Definition 1.49₁ is linear in $\underline{A}$. With $[\underline{e}_1, \underline{e}_2, \underline{e}_3] = 1$,

$$\text{tr}(\underline{e}_i \otimes \underline{e}_j) = \delta_{j1}\varepsilon_{i23} + \delta_{j2}\varepsilon_{1i3} + \delta_{j3}\varepsilon_{12i} = \delta_{ij},$$

so $\text{tr } \underline{A} = A_{ij}\delta_{ij} = A_{ii}$ and $\text{tr}(\underline{u} \otimes \underline{v}) = u_i v_j \delta_{ij} = \underline{u} \cdot \underline{v}$. ■

THEOREM 1.52 (Determinant). $I_3(\underline{A}) = \det(A_{ij})$; $\underline{A}$ invertible $\iff \det \underline{A} \neq 0$; $\det(\underline{A}\underline{B}) = \det \underline{A} \det \underline{B}$; $\det \underline{A}^t = \det \underline{A}$.

Proof. $I_3(\underline{A}) = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{A}\underline{e}_3] = \varepsilon_{ijk} A_{i1} A_{j2} A_{k3} = \det(A_{ij})$ by Proposition 1.24 and Definition 1.27. Multiplicativity and transpose invariance follow from Proposition 1.28 applied to $(A_{ik} B_{kj})$ and to (A_{ji}) . Invertibility: $\det \underline{A} \neq 0 \iff$ the columns $\underline{A}\underline{e}_i$ are independent $\iff \underline{A}$ is a bijection. ■

REMARK 1.53 (determinant = oriented volume ratio). When $[\underline{a}, \underline{b}, \underline{c}] > 0$ it is the volume of the parallelepiped on $\underline{a}, \underline{b}, \underline{c}$, so Definition 1.49₃ reads $\det \underline{A} = [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}]/[\underline{a}, \underline{b}, \underline{c}]$: $\det \underline{A}$ is the factor by which $\underline{A}$ scales oriented volume. Hence $\det > 0$ means orientation preserving, and later $\det$ of the deformation gradient is the local volume ratio, with $\det = 1$ signalling incompressibility.

DEFINITION 1.54 (Cofactor). $\underline{A}^*$ is defined by $\underline{A}^*(\underline{a} \times \underline{b}) = \underline{A}\underline{a} \times \underline{A}\underline{b}$.

PROPOSITION 1.55. $I_2(\underline{A}) = \text{tr } \underline{A}^*$; I_2 is quadratic and I_3 cubic in $\underline{A}$.

Proof. Applying the trace formula to $\underline{A}^*$ and $\underline{A}^*(\underline{e}_2 \times \underline{e}_3) = \underline{A}\underline{e}_2 \times \underline{A}\underline{e}_3$ etc. (see Problem 22a for the full computation) gives $\text{tr } \underline{A}^* = I_2(\underline{A})$. Degrees are read off Definition 1.49_{2,3}. ■

REMARK 1.56 (the cofactor maps area elements). If $\underline{a} \times \underline{b}$ is the area vector of a parallelogram, then $\underline{A}^*(\underline{a} \times \underline{b}) = \underline{A}\underline{a} \times \underline{A}\underline{b}$ is the area vector of the deformed parallelogram. Thus the determinant tells how $\underline{A}$ scales *volumes* (Remark 1.53), while the cofactor tells how $\underline{A}$ scales *oriented areas*. In the notation of continuum mechanics, the deformation gradient

$\underline{\underline{F}}$ satisfies $n \, da = \underline{\underline{F}}^*(N \, dA)$ (Nanson's formula): the cofactor converts a reference area element into its current image. Because area is one dimension lower than volume, the cofactor is quadratic in $\underline{\underline{A}}$ while the determinant is cubic; this is exactly what the degrees in Proposition 1.55 express. The identity $\underline{\underline{A}}^t \underline{\underline{A}}^* = (\det \underline{\underline{A}}) \underline{\underline{I}}$ (Problem 22c) ties the two together.

Inner product of tensors. Lin is nine dimensional; to do analysis on it, e.g. to call a stress increment small, it needs a length. The natural one pairs components entrywise and is generated intrinsically by trace and transpose.

DEFINITION 1.57 (Tensor inner product). $\underline{\underline{A}} \cdot \underline{\underline{B}} := \text{tr}(\underline{\underline{A}} \underline{\underline{B}}^t)$.

THEOREM 1.58. $\underline{\underline{A}} \cdot \underline{\underline{B}} = A_{ij} B_{ij} = \underline{\underline{B}} \cdot \underline{\underline{A}}$; $|\underline{\underline{A}}| = (A_{ij} A_{ij})^{1/2}$ makes Lin isometric to $\mathbb{E}^9$, so $|\underline{\underline{A}} \cdot \underline{\underline{B}}| \leq |\underline{\underline{A}}| |\underline{\underline{B}}|$ and $|\underline{\underline{A}} + \underline{\underline{B}}| \leq |\underline{\underline{A}}| + |\underline{\underline{B}}|$; and $\underline{\underline{A}} \cdot (\underline{\underline{u}} \otimes \underline{\underline{v}}) = \underline{\underline{u}} \cdot \underline{\underline{A}} \underline{\underline{v}}$.

Proof. $(\underline{\underline{A}} \underline{\underline{B}}^t)_{ik} = A_{ij} B_{kj}$, so $\text{tr}(\underline{\underline{A}} \underline{\underline{B}}^t) = A_{ij} B_{ij}$, symmetric. This is the Euclidean inner product of nine components, so the inequalities transfer from Theorem 1.6 and Corollary 1.7. Finally, by Lemma 1.38, $\underline{\underline{A}} \cdot (\underline{\underline{u}} \otimes \underline{\underline{v}}) = \text{tr}((\underline{\underline{A}} \underline{\underline{v}}) \otimes \underline{\underline{u}}) = \underline{\underline{A}} \underline{\underline{v}} \cdot \underline{\underline{u}} = \underline{\underline{u}} \cdot \underline{\underline{A}} \underline{\underline{v}}$. ■

DEFINITION 1.59 (Two orthogonal splittings). $\underline{\underline{A}} = \text{sym } \underline{\underline{A}} + \text{skw } \underline{\underline{A}}$ with $\text{sym } \underline{\underline{A}} = \frac{1}{2}(\underline{\underline{A}} + \underline{\underline{A}}^t)$, $\text{skw } \underline{\underline{A}} = \frac{1}{2}(\underline{\underline{A}} - \underline{\underline{A}}^t)$; and $\underline{\underline{A}} = \text{Sph } \underline{\underline{A}} + \text{Dev } \underline{\underline{A}}$ with $\text{Sph } \underline{\underline{A}} = \frac{1}{3}(\text{tr } \underline{\underline{A}}) \underline{\underline{I}}$, $\text{Dev } \underline{\underline{A}} = \underline{\underline{A}} - \frac{1}{3}(\text{tr } \underline{\underline{A}}) \underline{\underline{I}}$.

PROPOSITION 1.60. *Each summand is of the named type, the splittings are unique, $\text{tr}(\text{Dev } \underline{\underline{A}}) = 0$, and $\text{sym } \underline{\underline{A}} \cdot \text{skw } \underline{\underline{B}} = 0$, $\text{Sph } \underline{\underline{A}} \cdot \text{Dev } \underline{\underline{B}} = 0$ for all $\underline{\underline{A}}, \underline{\underline{B}}$. (Details in Problems 14–15.)*

Proof. $\frac{1}{2}(\underline{\underline{A}} + \underline{\underline{A}}^t)$ is symmetric, $\frac{1}{2}(\underline{\underline{A}} - \underline{\underline{A}}^t)$ skew, and their sum is $\underline{\underline{A}}$; $\text{tr}(\underline{\underline{A}} - \frac{1}{3}(\text{tr } \underline{\underline{A}}) \underline{\underline{I}}) = \text{tr } \underline{\underline{A}} - \text{tr } \underline{\underline{A}} = 0$. For symmetric $\underline{\underline{S}}$ and skew $\underline{\underline{W}}$, $\underline{\underline{S}} \cdot \underline{\underline{W}} = S_{ij} W_{ij} = S_{ji} (-W_{ji}) = -\underline{\underline{S}} \cdot \underline{\underline{W}} = 0$; and $\text{Sph } \underline{\underline{A}} \cdot \text{Dev } \underline{\underline{B}} = \frac{1}{3}(\text{tr } \underline{\underline{A}}) \text{tr}(\text{Dev } \underline{\underline{B}}) = 0$. Uniqueness is proved in Problems 14a, 15a. ■

1.2 Scalar, vector and tensor fields; gradient, divergence and curl; integral formulae

1.2.1 Euclidean point spaces

The algebra above treats *arrows*: they add, scale, and share a zero. Mechanics needs instead the *places* occupied by particles, and places behave differently. Two places cannot be added: “the doorway plus the window” names nothing. There is no zero place until someone plants a flag and calls it the origin. What *is* intrinsic is the displacement from one place to another, and a displacement does not move when the flag moves. So positions are not vectors, but their differences are. We therefore work with a set of points whose subtraction lands in a vector space.

DEFINITION 1.61 (Euclidean point space). A Euclidean point space is a set $\mathbb{E}^3$ of points $\mathbf{x}, \mathbf{y}, \dots$ with a subtraction $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{y} - \mathbf{x}$ into a three dimensional inner product space $\mathbb{E}^3$, the translation space, such that

(P1) (transport) for every $\mathbf{x}$ and $\underline{v} \in \mathbb{E}^3$ there is a unique $\mathbf{y}$ with $\mathbf{y} - \mathbf{x} = \underline{v}$, written $\mathbf{y} = \mathbf{x} + \underline{v}$;

(P2) (Chasles) $(\mathbf{y} - \mathbf{x}) + (\mathbf{z} - \mathbf{y}) = \mathbf{z} - \mathbf{x}$.

REMARK 1.62 (what each axiom buys). (P1) says the translation space acts freely and transitively on points: every displacement carries each point to a unique new one, and every pair of points is joined by a unique displacement. This is precisely what lets us later compare a field’s value at $\mathbf{x}$ with its values at neighbours $\mathbf{x} + s\underline{u}$, the comparison a derivative is built from. Setting $\mathbf{z} = \mathbf{x}$ in (P2), with $\mathbf{x} - \mathbf{x} = \underline{0}$ (the unique solution of $\mathbf{y} - \mathbf{x} = \underline{0}$ being $\mathbf{y} = \mathbf{x}$), gives $\mathbf{x} - \mathbf{y} = -(\mathbf{y} - \mathbf{x})$: reversing a displacement negates its arrow.

The definition is origin free, which is clean but useless for computation. Naming a reference place, the origin, is planting the flag; every

point then acquires one arrow with three components. The cost is dependence on the flag, measured precisely below.

PROPOSITION 1.63 (Position vector, Cartesian coordinates). *Fix an origin $\mathbf{O}$. Then (a) every point $\mathbf{x}$ determines a unique position vector $\underline{x} := \mathbf{x} - \mathbf{O}$, and $\mathbf{x} \mapsto \underline{x}$ is a bijection; (b) $\underline{x} = x_i \underline{e}_i$ with $x_i = \underline{x} \cdot \underline{e}_i$; (c) under $\mathbf{O} \rightarrow \mathbf{O}'$, $\underline{x}' = \underline{x} - \underline{c}$ with $\underline{c} = \mathbf{O}' - \mathbf{O}$, and $x'_i = x_i - c_i$.*

Proof. (a) $\underline{x} \in \mathbb{E}^3$ is well defined. Injective: $\mathbf{x}_1 - \mathbf{O} = \mathbf{x}_2 - \mathbf{O}$ gives, by (P2), $\underline{0} = (\mathbf{x}_1 - \mathbf{O}) + (\mathbf{O} - \mathbf{x}_2) = \mathbf{x}_1 - \mathbf{x}_2$. Surjective: $\mathbf{x} = \mathbf{O} + \underline{v}$ has $\underline{x} = \underline{v}$. (b) $\underline{x} \cdot \underline{e}_i = x_j \delta_{ji} = x_i$. (c) By (P2), $\mathbf{x} - \mathbf{O}' = (\mathbf{x} - \mathbf{O}) - \underline{c}$; dot with $\underline{e}_i$. ■

REMARK 1.64 (the deliberate abuse $\underline{x}(= \underline{x} - \underline{o})$). The book writes the position vector as a bare $\underline{x}$, so the same letter $\underline{x}$ names both the point and its position vector. Part (a) makes this legitimate *after* an origin is fixed, since the correspondence is then a bijection; part (c) is the warning label, that any formula with a bare $\underline{x}$ silently carries the chosen origin. Good continuum theory keeps its final physical statements independent of that choice.

DEFINITION 1.65 (Fields). A scalar field is $\varphi : \mathbb{E}^3 \rightarrow \mathbb{R}$, a vector field $\underline{v} : \mathbb{E}^3 \rightarrow \mathbb{E}^3$, a tensor field $\underline{A} : \mathbb{E}^3 \rightarrow \text{Lin}$; their argument is the position $\underline{x}$.

1.2.2 Gradients

In one variable a differentiable f satisfies $f(x_0 + h) = f(x_0) + f'(x_0)h +$ (small): near x_0 the graph is a straight line, and $f'(x_0)$ its slope. The single idea of differential calculus is that a smooth function is, up close, linear. We now transplant it to a field, where the increment depends on the *vector* $\underline{h} = \underline{x} - \underline{x}_0$, so the linear part is a linear scalar function of $\underline{h}$. Two things must be made precise: what “small” means for the remainder, and how to replace the abstract linear part by one computable vector.

DEFINITION 1.66 (Landau symbols). Let r and g be defined for ε near 0, with $g(\varepsilon) > 0$ for $\varepsilon \neq 0$. Then

$$\begin{aligned} r(\varepsilon) = O(g(\varepsilon)) &\iff \exists C > 0, \delta > 0 : |r(\varepsilon)| \leq C g(\varepsilon) \text{ for } 0 < \varepsilon < \delta, \\ r(\varepsilon) = o(g(\varepsilon)) &\iff \forall \eta > 0 \exists \delta > 0 : |r(\varepsilon)| \leq \eta g(\varepsilon) \text{ for } 0 < \varepsilon < \delta. \end{aligned}$$

Equivalently, $r = O(g)$ says r/g stays bounded as $\varepsilon \rightarrow 0$ and $r = o(g)$ says $r/g \rightarrow 0$. In words, big-oh bounds the ratio by *some* constant, little-oh forces it below *every* constant. The unqualified $o(\varepsilon)$ of the classical usage is the case $g(\varepsilon) = \varepsilon$, and $o(1)$ is the case $g \equiv 1$, that is, a quantity tending to zero.

The equality sign in $r = O(g)$ is an abbreviation for membership in a class of functions and is not symmetric: it may be read from left to right only. Thus $\varepsilon^3 = O(\varepsilon^2)$ is true and $O(\varepsilon^2) = \varepsilon^3$ is meaningless.

LEMMA 1.67 (comparison of orders). *Let $p > q > 0$. Then, as $\varepsilon \rightarrow 0^+$,*

$$(i) \quad r = o(g) \implies r = O(g), \quad (ii) \quad r = O(\varepsilon^p) \implies r = o(\varepsilon^q).$$

Proof. (i) Apply the little-oh condition with the single choice $\eta = 1$: it supplies $\delta > 0$ with $|r| \leq g$ on $(0, \delta)$, which is the big-oh condition with $C = 1$.

(ii) By hypothesis there are $C > 0$ and $\delta_0 > 0$ with $|r(\varepsilon)| \leq C\varepsilon^p$ for $0 < \varepsilon < \delta_0$. Let $\eta > 0$ be given and put

$$\delta = \min \left\{ \delta_0, (\eta/C)^{1/(p-q)} \right\},$$

which is a positive number because $p - q > 0$. For $0 < \varepsilon < \delta$,

$$\frac{|r(\varepsilon)|}{\varepsilon^q} \leq \frac{C\varepsilon^p}{\varepsilon^q} = C\varepsilon^{p-q} < C\delta^{p-q} \leq C \cdot \frac{\eta}{C} = \eta,$$

so $|r(\varepsilon)| \leq \eta\varepsilon^q$. As η was arbitrary, $r = o(\varepsilon^q)$. ■

The content of (ii) is that a higher power is not merely comparable to a lower one but negligible beside it, and it is the step that converts a volume estimate into a statement about areas in Section 6.5.2: there $p = 3$ and $q = 2$, and $O(R^3) = o(R^2)$ is exactly (ii).

REMARK 1.68 (why “faster than linear” removes ambiguity). If a field had two linear parts f_0, g_0 in (1.96), subtracting gives $f_0(\underline{h}) - g_0(\underline{h}) = -o(|\underline{h}|)$; putting $\underline{h} = s\underline{u}$ ($|\underline{u}| = 1$) and using linearity, $s[f_0(\underline{u}) - g_0(\underline{u})] = -o(s)$, so $f_0(\underline{u}) - g_0(\underline{u}) = -o(s)/s \rightarrow 0$. The left side is independent of s , hence 0 for every unit $\underline{u}$, so $f_0 = g_0$. Demanding the remainder die *faster* than $\underline{h}$ is exactly what pins the linear part down.

DEFINITION 1.69 (Differentiable scalar field). φ is differentiable at $\underline{x}_0$ iff there is a linear $f_0 : \mathbb{E}^3 \rightarrow \mathbb{R}$ with

$$\varphi(\underline{x}) = \varphi(\underline{x}_0) + f_0(\underline{x} - \underline{x}_0) + o(|\underline{x} - \underline{x}_0|) \quad (\underline{x} \rightarrow \underline{x}_0). \quad (1.96)$$

THEOREM 1.70 (Existence, uniqueness, naming of the gradient). *If φ is differentiable at $\underline{x}_0$, there is a unique $\text{grad } \varphi|_{\underline{x}_0} \in \mathbb{E}^3$ with $f_0(\underline{h}) = \text{grad } \varphi|_{\underline{x}_0} \cdot \underline{h}$, hence*

$$\varphi(\underline{x}) = \varphi(\underline{x}_0) + \text{grad } \varphi|_{\underline{x}_0} \cdot (\underline{x} - \underline{x}_0) + o(|\underline{x} - \underline{x}_0|). \quad (1.99)$$

Proof. f_0 is a linear functional on $\mathbb{E}^3$; by Riesz (Theorem 1.15) there is a unique $\underline{c}$ with $f_0(\underline{h}) = \underline{c} \cdot \underline{h}$. Explicitly, with $c_i := f_0(\underline{e}_i)$ and $\underline{h} = h_i \underline{e}_i$, $f_0(\underline{h}) = h_i f_0(\underline{e}_i) = h_i c_i = \underline{c} \cdot \underline{h}$. Name $\underline{c} =: \text{grad } \varphi|_{\underline{x}_0}$ and substitute into (1.96). ■

REMARK 1.71 (the gradient is a slope, and it points uphill). $\text{grad } \varphi|_{\underline{x}_0}$ is one vector attached to the place $\underline{x}_0$. By (1.99) the first order change under a step $\underline{h}$ is $\text{grad } \varphi \cdot \underline{h}$, largest when $\underline{h} \parallel \text{grad } \varphi$ (Cauchy–Schwarz). So the gradient points in the direction of steepest increase and its length is that steepest rate; its level sets are the directions of no first order change, orthogonal to $\text{grad } \varphi$.

REMARK 1.72 (the double construction reused throughout). The argument has a fixed three step shape: increment $\rightarrow$ linear part $\rightarrow$ Riesz representer. For a scalar field the representer is a vector, $\text{grad } \varphi$. Applied to a *vector* field the linear part is a linear map and Riesz upgrades to “a linear map is a tensor”, producing $\text{grad } \underline{v} \in \text{Lin}$ (§1.2.5); applied to a flux it produces the divergence. Spotting this template once explains every later derivative at a glance.

1.2.3 Directional derivatives

The differentiability limit (1.96) lets $\underline{x} \rightarrow \underline{x}_0$ from all directions at once, a limit in three variables. The cheapest way to probe it is to fix a direction and move only along it: set $\underline{v} = \underline{x} - \underline{x}_0$, let $s = |\underline{v}|$ so that $\underline{v} = s\underline{u}$ with $|\underline{u}| = 1$, and watch $f(s) = \varphi(\underline{x}_0 + s\underline{u})$ as an ordinary function of one real variable. The coefficient of s in its expansion is the rate of change of φ along $\underline{u}$. Let us compute it and see that the gradient already contains the answer for every direction.

DEFINITION 1.73 (Directional derivative). For a unit vector $\underline{u}$ and fixed $\underline{x}_0$, with $f(s) = \varphi(\underline{x}_0 + s\underline{u})$, the derivative of φ at $\underline{x}_0$ along $\underline{u}$ is $f'(0)$.

THEOREM 1.74 (The gradient gives every directional derivative). *If φ is differentiable at $\underline{x}_0$, then for every unit $\underline{u}$*

$$\left(\frac{df}{ds} \right) \Big|_{s=0} = \underline{u} \cdot \text{grad } \varphi \Big|_{\underline{x}_0}. \quad (1.102)$$

Proof. With $\underline{h} = s\underline{u}$, $|\underline{h}| = s$ (as $|\underline{u}| = 1$, $s \geq 0$), (1.99) gives

$$f(s) = \varphi(\underline{x}_0) + \text{grad } \varphi \Big|_{\underline{x}_0} \cdot (s\underline{u}) + o(s) = \varphi(\underline{x}_0) + s(\underline{u} \cdot \text{grad } \varphi \Big|_{\underline{x}_0}) + o(s). \quad (1.100)$$

Subtract $f(0)$, divide by $s \neq 0$: $[f(s) - f(0)]/s = \underline{u} \cdot \text{grad } \varphi \Big|_{\underline{x}_0} + o(s)/s$ (1.101); let $s \rightarrow 0$. ■

Choosing the special directions $\underline{e}_i$, which change only one coordinate, turns directional derivatives into ordinary partial derivatives and hands us the components of the gradient.

COROLLARY 1.75 (Partials as gradient components). *With $\hat{\varphi}(x_1, x_2, x_3) := \varphi(x_i \underline{e}_i)$,*

$$\underline{e}_i \cdot \text{grad } \varphi \Big|_{\underline{x}_0} = \left(\frac{\partial \hat{\varphi}}{\partial x_i} \right) \Big|_{(x_1^0, x_2^0, x_3^0)} =: \hat{\varphi}_{,i} \Big|_{\underline{x}_0}. \quad (1.103\text{--}1.106)$$

Proof. Take $\underline{u} = \underline{e}_i$ in Theorem 1.74; $\underline{x}_0 + s\underline{e}_i$ increments only the i th coordinate, so $\underline{e}_i \cdot \text{grad } \varphi \Big|_{\underline{x}_0} = \lim_{s \rightarrow 0} \frac{1}{s} [\hat{\varphi}(\dots, x_i^0 + s, \dots) - \hat{\varphi}(\dots, x_i^0, \dots)] = \partial \hat{\varphi} / \partial x_i$. As $\underline{x}_0$ is arbitrary, $\underline{e}_i \cdot \text{grad } \varphi = \hat{\varphi}_{,i}$. ■

THEOREM 1.76 (Cartesian gradient). $\text{grad } \varphi = \hat{\varphi}_{,i} \underline{e}_i = \varphi_{,i} \underline{e}_i$. (1.107–1.108)

Proof. By orthonormal expansion (Proposition 1.11), $\underline{w} = (\underline{w} \cdot \underline{e}_i) \underline{e}_i$; take $\underline{w} = \text{grad } \varphi$ and use $\text{grad } \varphi \cdot \underline{e}_i = \hat{\varphi}_{,i}$. Dropping the caret gives (1.108). ■

REMARK 1.77 (caret and comma). $\varphi(\underline{x})$ eats a point; $\hat{\varphi}(x_1, x_2, x_3)$ eats three real numbers. Different domains make them different functions, yet they take equal values when $\underline{x} = x_i \underline{e}_i$. The comma abbreviates a Cartesian partial, $\varphi_{,i} = \partial\varphi/\partial x_i$, from here on; dropping the caret is the same controlled abuse as writing a bare $\underline{x}$ for the position (Remark 1.64).

1.2.4 Parameterized paths and the chain rule

Directional derivatives run along straight lines of fixed direction. A particle's trajectory, or a coordinate line in curvilinear coordinates, bends: its direction $\underline{x}'(u)$ varies along the way. To differentiate a field along such a curve we compose φ with the path and use the one variable derivative. The resulting rule, in its coordinate free form $d\varphi = \text{grad } \varphi \cdot d\underline{x}$, is powerful enough to generate $\text{grad } \varphi$ in any coordinate system.

THEOREM 1.78 (Chain rule). *For a smooth simple curve $\underline{x}(u)$ and $\psi(u) = \varphi(\underline{x}(u))$,*

$$\psi'(u_0) = \text{grad } \varphi|_{\underline{x}(u_0)} \cdot \underline{x}'(u_0). \quad (1.112)$$

Proof. With $\underline{x}_0 = \underline{x}(u_0)$, $\underline{x}_1 = \underline{x}(u_1)$, (1.99) gives

$$\psi(u_1) = \psi(u_0) + \text{grad } \varphi|_{\underline{x}_0} \cdot (\underline{x}_1 - \underline{x}_0) + o(|\underline{x}_1 - \underline{x}_0|). \quad (1.109)$$

Smoothness of the path gives

$$\underline{x}_1 - \underline{x}_0 = \underline{x}'(u_0)(u_1 - u_0) + o(u_1 - u_0), \quad (1.110)$$

so $|\underline{x}_1 - \underline{x}_0| = O(u_1 - u_0)$ and $o(|\underline{x}_1 - \underline{x}_0|) = o(u_1 - u_0)$. Insert (1.110) into (1.109), subtract $\psi(u_0)$, divide by $u_1 - u_0$:

$$\frac{\psi(u_1) - \psi(u_0)}{u_1 - u_0} = \text{grad } \varphi|_{\underline{x}_0} \cdot \underline{x}'(u_0) + \frac{o(u_1 - u_0)}{u_1 - u_0}, \quad (1.111)$$

and let $u_1 \rightarrow u_0$. ■

COROLLARY 1.79 (Coordinate free differential). *With $d\tilde{x} = \tilde{x}'(u) du$,*

$$\frac{d\varphi}{du} = \text{grad } \varphi \cdot \frac{d\tilde{x}}{du}, \quad d\varphi = \text{grad } \varphi \cdot d\tilde{x}. \quad (1.113\text{--}1.114)$$

This holds for every $d\tilde{x}$ and uses no coordinates.

REMARK 1.80 (the power of (1.114)). Equation (1.114), $d\varphi = \text{grad } \varphi \cdot d\tilde{x}$, is coordinate free: it never mentions a basis. Yet it can *generate* the gradient formula in *any* coordinate system. The recipe is always the same: write $d\varphi$ from the conventional chain rule, express the coordinate increments as dot products with $d\tilde{x}$, read off $\text{grad } \varphi$ by matching with (1.114). The next two examples execute this recipe in Cartesian and cylindrical coordinates.

EXAMPLE 1.81 (recovering the Cartesian formula). *Step 1 (conventional chain rule).* With $\hat{\varphi}(x_1, x_2, x_3) = \varphi(x_i \underline{e}_i)$ and $\{\underline{e}_i\}$ fixed,

$$d\hat{\varphi} = \frac{\partial \hat{\varphi}}{\partial x_1} dx_1 + \frac{\partial \hat{\varphi}}{\partial x_2} dx_2 + \frac{\partial \hat{\varphi}}{\partial x_3} dx_3 = \hat{\varphi}_{,i} dx_i.$$

Step 2 (express dx_i via $d\tilde{x}$). Since the basis is fixed, $\tilde{x} = x_i \underline{e}_i$ gives $d\tilde{x} = dx_i \underline{e}_i$. Dotting with $\underline{e}_i$ and using orthonormality, $dx_i = \underline{e}_i \cdot d\tilde{x}$. Insert:

$$d\hat{\varphi} = \hat{\varphi}_{,i} (\underline{e}_i \cdot d\tilde{x}) = (\hat{\varphi}_{,i} \underline{e}_i) \cdot d\tilde{x}.$$

Step 3 (match with (1.114)). Setting this equal to $\text{grad } \varphi \cdot d\tilde{x}$ gives $(\text{grad } \varphi - \hat{\varphi}_{,i} \underline{e}_i) \cdot d\tilde{x} = 0$ for all $d\tilde{x}$. Choose $d\tilde{x} = \text{grad } \varphi - \hat{\varphi}_{,i} \underline{e}_i$ itself; then its squared norm $|\text{grad } \varphi - \hat{\varphi}_{,i} \underline{e}_i|^2 = 0$, so

$$\text{grad } \varphi = \hat{\varphi}_{,i} \underline{e}_i. \quad (1.115)$$

This is identical to (1.108), confirming that both routes give the same formula. The Cartesian gradient has no extra factors because $\{\underline{e}_i\}$ is fixed and orthonormal; when the basis varies with position (as in polar coordinates), extra terms appear.

EXAMPLE 1.82 (cylindrical polar coordinates). With $\underline{k} = \underline{e}_3$, define the rotating frame

$$\underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2, \quad \underline{e}_\theta(\theta) = \underline{k} \times \underline{e}_r = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2, \quad (1.116)$$

orthonormal for each θ , with

$$\underline{e}'_r(\theta) = \underline{e}_\theta(\theta), \quad \underline{e}'_\theta(\theta) = -\underline{e}_r(\theta). \quad (\dagger)$$

A point has $\underline{x} = r \underline{e}_r(\theta) + z \underline{k}$ (1.117). Then, using $d\underline{e}_r = \underline{e}'_r(\theta)d\theta = \underline{e}_\theta d\theta$,

$$d\underline{x} = dr \underline{e}_r + r d\underline{e}_r + dz \underline{k} = dr \underline{e}_r + r d\theta \underline{e}_\theta + dz \underline{k}, \quad (1.119)$$

and dotting with the orthonormal $\{\underline{e}_r, \underline{e}_\theta, \underline{k}\}$,

$$dr = \underline{e}_r \cdot d\underline{x}, \quad r d\theta = \underline{e}_\theta \cdot d\underline{x}, \quad dz = \underline{k} \cdot d\underline{x}. \quad (1.120)$$

With $\varphi = \tilde{\varphi}(r, \theta, z)$, $d\tilde{\varphi} = \tilde{\varphi}_{,r}dr + \tilde{\varphi}_{,\theta}d\theta + \tilde{\varphi}_{,z}dz$; substituting (1.120),

$$d\tilde{\varphi} = (\tilde{\varphi}_{,r}\underline{e}_r + \frac{1}{r}\tilde{\varphi}_{,\theta}\underline{e}_\theta + \tilde{\varphi}_{,z}\underline{k}) \cdot d\underline{x}, \quad (1.121)$$

so by (1.114), dropping the tilde,

$$\text{grad } \varphi = \varphi_{,r} \underline{e}_r + \frac{1}{r} \varphi_{,\theta} \underline{e}_\theta + \varphi_{,z} \underline{k}. \quad (1.122)$$

The extra $1/r$, absent from the Cartesian (1.108), arises solely because $\{\underline{e}_r, \underline{e}_\theta, \underline{k}\}$ varies with position while $\{\underline{e}_i\}$ is fixed.

EXAMPLE 1.83 (gradient of the radius $r = |\underline{x}|$). *Index route.* Start from $r^2 = \underline{x} \cdot \underline{x} = x_j x_j$. Note that $x_{j,i} = \partial x_j / \partial x_i = \delta_{ji}$: the Cartesian coordinate x_j depends only on itself, so its derivative with respect to x_i is 1 when $j = i$ and 0 otherwise. Differentiate $r^2 = x_j x_j$ using the product rule:

$$2r r_{,i} = (x_j x_j)_{,i} = x_{j,i} x_j + x_j x_{j,i} = \delta_{ji} x_j + x_j \delta_{ji} = x_i + x_i = 2x_i, \quad (1.123)$$

where the Kronecker delta δ_{ji} collapsed the sum over j to its $j = i$ term: $\delta_{ji} x_j = x_i$. Divide both sides by $2r$: $r_{,i} = x_i / r$. Since $\text{grad } r = r_{,i} \underline{e}_i$ (by 1.108),

$$\text{grad } r = \frac{x_i}{r} \underline{e}_i = \frac{1}{r} \underline{x}. \quad (1.124)$$

Coordinate free route. The differential product rule for the inner product reads $d(\underline{a} \cdot \underline{b}) = d\underline{a} \cdot \underline{b} + \underline{a} \cdot d\underline{b}$ (proved by $d(a_i b_i) = da_i b_i + a_i db_i$). Apply it to $r^2 = \underline{x} \cdot \underline{x}$:

$$2r \, dr = d(\underline{x} \cdot \underline{x}) = d\underline{x} \cdot \underline{x} + \underline{x} \cdot d\underline{x} = 2 \underline{x} \cdot d\underline{x}. \quad (1.125)$$

Cancel 2 and divide by r : $dr = r^{-1} \underline{x} \cdot d\underline{x}$. By (1.114), $dr = \text{grad } r \cdot d\underline{x}$; matching the two gives $\text{grad } r = r^{-1} \underline{x}$. Both routes arrive at the same answer, but the coordinate free route never wrote a single index.

REMARK 1.84 ($\text{grad } r$ points radially outward). Since $\underline{x}/r = \underline{x}/|\underline{x}|$ is the unit vector in the radial direction, $\text{grad } r$ is the unit radial vector: $|\text{grad } r| = 1$, pointing from the origin toward the point $\underline{x}$. The level sets of r are concentric spheres, and $\text{grad } r$ is indeed perpendicular to them, consistent with the general fact that the gradient is orthogonal to the level sets of the field (Remark 1.71).

1.2.5 Gradients of vector fields

Repeat the linearisation of §1.2.2 on a vector field. Now the increment $\underline{v}(\underline{x}) - \underline{v}(\underline{x}_0)$ is a *vector*, so its linear part is a linear map $\mathbb{E}^3 \rightarrow \mathbb{E}^3$, that is, a tensor. This is the double construction one rung up: the fixed object representing the linear part is $\text{grad } \underline{v} \in \text{Lin}$.

DEFINITION 1.85 (Differentiable vector field). $\underline{v}$ is differentiable at $\underline{x}_0$ iff there is a tensor $\underline{L}_0$ with

$$\underline{v}(\underline{x}) = \underline{v}(\underline{x}_0) + \underline{L}_0(\underline{x} - \underline{x}_0) + \underline{o}(\underline{x} - \underline{x}_0), \quad |\underline{o}(\underline{x} - \underline{x}_0)| = o(|\underline{x} - \underline{x}_0|), \quad (1.126\text{--}1.127)$$

and one writes $\underline{L}_0 = \text{grad } \underline{v}|_{\underline{x}_0}$.

THEOREM 1.86 (Cartesian representation). $\text{grad } \underline{v} = v_{i,j} \underline{e}_i \otimes \underline{e}_j$, *i.e.* $(\text{grad } \underline{v})_{ij} = v_{i,j}$. (1.132)

Proof. With $\underline{x} - \underline{x}_0 = s\underline{u}$, $|\underline{u}| = 1$, $\underline{L}_0 \underline{u} = \lim_{s \rightarrow 0} \frac{1}{s} [\underline{v}(\underline{x}_0 + s\underline{u}) - \underline{v}(\underline{x}_0)]$ (1.129). Take $\underline{u} = \underline{e}_j$ and $\underline{v} = \hat{v}_i \underline{e}_i$ with fixed $\underline{e}_i$: $\underline{L}_0 \underline{e}_j = \underline{e}_i \lim_{s \rightarrow 0} \frac{1}{s} [\hat{v}_i(\dots, x_j^0 + s, \dots) - \hat{v}_i(\dots)] = \hat{v}_{i,j} \underline{e}_i$ (1.130–1.131). Hence $L_{ij} = \underline{e}_i \cdot \underline{L}_0 \underline{e}_j = v_{i,j}$. ■

COROLLARY 1.87 (Chain rule for vector fields). $d\underline{v} = (\text{grad } \underline{v}) d\underline{x}$.
(1.133)

Proof. $d\hat{v}_i = \hat{v}_{i,j} dx_j = \hat{v}_{i,j} \underline{e}_j \cdot d\underline{x}$, so $d\underline{v} = d\hat{v}_i \underline{e}_i = \hat{v}_{i,j} \underline{e}_i (\underline{e}_j \cdot d\underline{x}) = (\hat{v}_{i,j} \underline{e}_i \otimes \underline{e}_j) d\underline{x} = (\text{grad } \underline{v}) d\underline{x}$ (1.134). $\blacksquare$

EXAMPLE 1.88 ($\underline{u} = \mu \underline{x} + \nu(\underline{x} \otimes \underline{x}) \underline{a}$). The field $\underline{u}(\underline{x}) = \mu \underline{x} + \nu(\underline{x} \otimes \underline{x}) \underline{a}$ arises, for instance, in quadratic displacement fields. We compute its gradient by two routes.

Index route. Write $u_i = \mu x_i + \nu x_i x_j a_j$ (since $(\underline{x} \otimes \underline{x}) \underline{a} = \underline{x} \cdot \underline{a} \underline{x}$, its i th component is $x_i(\underline{x} \cdot \underline{a})$). Differentiate with respect to x_k :

$$\begin{aligned} u_{i,k} &= \mu \frac{\partial x_i}{\partial x_k} + \nu \left(\frac{\partial x_i}{\partial x_k} x_j a_j + x_i \frac{\partial x_j}{\partial x_k} a_j \right) \\ &= \mu \delta_{ik} + \nu (\delta_{ik} x_j a_j + x_i \delta_{jk} a_j) \\ &= \mu \delta_{ik} + \nu (\underline{x} \cdot \underline{a}) \delta_{ik} + \nu x_i a_k \\ &= [\mu + \nu(\underline{x} \cdot \underline{a})] \delta_{ik} + \nu x_i a_k. \end{aligned} \quad (1.136)$$

Recognise the first term as $[\mu + \nu(\underline{x} \cdot \underline{a})] I_{ik}$ and the second as $(\underline{x} \otimes \underline{a})_{ik} = x_i a_k$, so

$$\text{grad } \underline{u} = [\mu + \nu(\underline{x} \cdot \underline{a})] \underline{I} + \nu \underline{x} \otimes \underline{a}. \quad (1.137)$$

Differential route. The product rule $d(\underline{b} \otimes \underline{c}) = d\underline{b} \otimes \underline{c} + \underline{b} \otimes d\underline{c}$ (proved by $d(b_i c_j) = db_i c_j + b_i dc_j$) gives $d(\underline{x} \otimes \underline{x}) = d\underline{x} \otimes \underline{x} + \underline{x} \otimes d\underline{x}$, so

$$\begin{aligned} d\underline{u} &= \mu d\underline{x} + \nu(d\underline{x} \otimes \underline{x} + \underline{x} \otimes d\underline{x}) \underline{a} \\ &= \mu d\underline{x} + \nu(\underline{x} \cdot \underline{a}) d\underline{x} + \nu \underline{x} (\underline{a} \cdot d\underline{x}) \\ &= \{[\mu + \nu(\underline{x} \cdot \underline{a})] \underline{I} + \nu \underline{x} \otimes \underline{a}\} d\underline{x}. \end{aligned} \quad (1.138)$$

In line two we used $(d\underline{x} \otimes \underline{x}) \underline{a} = (\underline{x} \cdot \underline{a}) d\underline{x}$ and $(\underline{x} \otimes d\underline{x}) \underline{a} = (d\underline{x} \cdot \underline{a}) \underline{x} = \underline{x} (\underline{a} \cdot d\underline{x})$. Comparing with $d\underline{u} = (\text{grad } \underline{u}) d\underline{x}$ confirms (1.137).

EXAMPLE 1.89 (affine field $\underline{u} = \underline{x} + (\underline{b} \cdot \underline{x}) \underline{a}$). Here $\underline{a}, \underline{b}$ are fixed vectors. Write $\underline{u}(\underline{x}) = \underline{x} + \underline{a}(\underline{b} \cdot \underline{x})$; its differential is

$$d\underline{u} = d\underline{x} + \underline{a}(\underline{b} \cdot d\underline{x}) = d\underline{x} + \underline{a} \otimes \underline{b} d\underline{x} = (\underline{I} + \underline{a} \otimes \underline{b}) d\underline{x},$$

where we recognised $\underline{a}(\underline{b} \cdot d\underline{x}) = (\underline{a} \otimes \underline{b})d\underline{x}$ by Definition 1.32. Thus

$$\text{grad } \underline{u} = \underline{I} + \underline{a} \otimes \underline{b}. \quad (1.141)$$

Notice that $\text{grad } \underline{u}$ is *uniform*: it does not depend on $\underline{x}$. This is no accident. Write $\underline{u}(\underline{x}) = (\underline{I} + \underline{a} \otimes \underline{b})\underline{x} =: \underline{A}\underline{x}$; then for any $\underline{x}_0$,

$$\underline{u}(\underline{x}) = \underline{A}\underline{x} = \underline{A}\underline{x}_0 + \underline{A}(\underline{x} - \underline{x}_0) = \underline{u}(\underline{x}_0) + \underline{A}(\underline{x} - \underline{x}_0),$$

which is already in the form (1.127) with $\underline{L}_0 = \underline{A}$ and *zero remainder*. Whenever the field is affine (linear in $\underline{x}$), the gradient is constant and the approximation (1.127) is exact everywhere.

EXAMPLE 1.90 (polar gradient of a vector field). For $\underline{u} = u_r \underline{e}_r + u_\theta \underline{e}_\theta + u_z \underline{k}$ with components functions of (r, θ) , using $(\dagger)$ and the product rule,

$$\begin{aligned} d\underline{u} &= [u_{r,r}dr + (u_{r,\theta} - u_\theta)d\theta]\underline{e}_r \\ &\quad + [u_{\theta,r}dr + (u_{\theta,\theta} + u_r)d\theta]\underline{e}_\theta \\ &\quad + [u_{z,r}dr + u_{z,\theta}d\theta]\underline{k}, \end{aligned}$$

where the $-u_\theta$ and $+u_r$ come from $d\underline{e}_r = \underline{e}_\theta d\theta$, $d\underline{e}_\theta = -\underline{e}_r d\theta$. Inserting $dr = \underline{e}_r \cdot d\underline{x}$, $d\theta = \frac{1}{r}\underline{e}_\theta \cdot d\underline{x}$ and $(\underline{e}_a \cdot d\underline{x})\underline{e}_b = (\underline{e}_b \otimes \underline{e}_a)d\underline{x}$,

$$\begin{aligned} \text{grad } \underline{u} &= u_{r,r} \underline{e}_r \otimes \underline{e}_r + \frac{1}{r}(u_{r,\theta} - u_\theta) \underline{e}_r \otimes \underline{e}_\theta + u_{\theta,r} \underline{e}_\theta \otimes \underline{e}_r \\ &\quad + \frac{1}{r}(u_{\theta,\theta} + u_r) \underline{e}_\theta \otimes \underline{e}_\theta + u_{z,r} \underline{k} \otimes \underline{e}_r + \frac{1}{r}u_{z,\theta} \underline{k} \otimes \underline{e}_\theta. \end{aligned} \quad (1.145)$$

1.2.6 Divergence and curl

The tensor $\text{grad } \underline{v}$ carries nine numbers. Two coordinate free combinations carry the physics: its trace measures the local rate of volume change, the divergence, and its skew part measures local rotation, the curl. Defining each through the trace and box product machinery keeps them intrinsic, after which the component formulas drop out.

DEFINITION 1.91 (Divergence of a vector field). $\text{div } \underline{v} := \text{tr}(\text{grad } \underline{v})$.

PROPOSITION 1.92. $\text{div } \underline{v} = v_{i,i}$.

Proof. $\operatorname{div} \underline{v} = \operatorname{tr}(v_{i,j} \underline{e}_i \otimes \underline{e}_j) = v_{i,j} \delta_{ij} = v_{i,i}$ (1.147), by $\operatorname{tr}(\underline{e}_i \otimes \underline{e}_j) = \delta_{ij}$. ■

DEFINITION 1.93 (Divergence of a tensor field). $\operatorname{div} \underline{A}$ is the vector field with $\underline{c} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{c})$ for every fixed $\underline{c}$.

PROPOSITION 1.94. $\operatorname{div} \underline{A} = A_{ij,j} \underline{e}_i$. (1.150)

Proof. Take $\underline{c} = \underline{e}_i$: $\underline{A}^t \underline{e}_i = A_{ij} \underline{e}_j$, so $\underline{e}_i \cdot \operatorname{div} \underline{A} = \operatorname{div}(A_{ij} \underline{e}_j) = A_{ij,j}$ (i fixed, j summed). ■

PROPOSITION 1.95 (Product rule). $\underline{u} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{u}) - \operatorname{tr}[\underline{A}^t (\operatorname{grad} \underline{u})]$. (1.152)

Proof. The scalar product rule $u_i A_{ij,j} = (A_{ij} u_i)_{,j} - A_{ij} u_{i,j}$ (1.151) reads, term by term, $\underline{u} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{u}) - \operatorname{tr}[\underline{A}^t (\operatorname{grad} \underline{u})]$, since $(\underline{A}^t \underline{u})_j = A_{ij} u_i$ and $\operatorname{tr}[\underline{A}^t \operatorname{grad} \underline{u}] = A_{ij} u_{i,j}$. ■

DEFINITION 1.96 (Curl of a vector field). $\operatorname{curl} \underline{v}$ is the vector with $\underline{c} \cdot \operatorname{curl} \underline{v} = \operatorname{div}(\underline{v} \times \underline{c})$ for every fixed $\underline{c}$.

PROPOSITION 1.97. $\operatorname{curl} \underline{v} = \varepsilon_{kij} v_{i,k} \underline{e}_j$. (1.165)

Proof. Take $\underline{c} = \underline{e}_j$: $\underline{v} \times \underline{e}_j = v_i \varepsilon_{ijk} \underline{e}_k$, so $\underline{e}_j \cdot \operatorname{curl} \underline{v} = \operatorname{div}(v_i \varepsilon_{ijk} \underline{e}_k) = \varepsilon_{ijk} v_{i,k} = \varepsilon_{kij} v_{i,k}$ (1.164), using $\varepsilon_{ijk} = \varepsilon_{kij}$. ■

REMARK 1.98 (curl of a tensor field). $\operatorname{curl} \underline{A}$ is defined by $(\operatorname{curl} \underline{A}) \underline{c} = \operatorname{curl}(\underline{A}^t \underline{c})$. Taking $\underline{c} = \underline{e}_m$, $\underline{A}^t \underline{e}_m = A_{mi} \underline{e}_i$ has i th component A_{mi} , so $\operatorname{curl}(\underline{A}^t \underline{e}_m) = \varepsilon_{kij} A_{mi,k} \underline{e}_j$; matching $(\operatorname{curl} \underline{A}) \underline{e}_m = (\operatorname{curl} \underline{A})_{jm} \underline{e}_j$ gives $\operatorname{curl} \underline{A} = \varepsilon_{kij} A_{mi,k} \underline{e}_j \otimes \underline{e}_m$, equivalently $\varepsilon_{ilm} A_{jl,i} \underline{e}_m \otimes \underline{e}_j$ (Problem 28).

EXAMPLE 1.99 (divergence of a tensor field in polar coordinates). For $\underline{A} = A_{rr} \underline{e}_r \otimes \underline{e}_r + A_{r\theta} \underline{e}_r \otimes \underline{e}_\theta + \cdots$ with components functions of (r, θ) , decompose $\operatorname{div} \underline{A}$ on $\{\underline{e}_r, \underline{e}_\theta, \underline{k}\}$ and use (1.152). From (1.145) applied to the frame fields,

$$\operatorname{grad} \underline{e}_r = \frac{1}{r} \underline{e}_\theta \otimes \underline{e}_\theta, \quad \operatorname{grad} \underline{e}_\theta = -\frac{1}{r} \underline{e}_r \otimes \underline{e}_\theta, \quad (1.156)$$

and the scalar divergence of $\underline{w} = w_r \underline{e}_r + w_\theta \underline{e}_\theta + w_z \underline{k}$ is $\operatorname{div} \underline{w} = w_{r,r} + \frac{1}{r}(w_{\theta,\theta} + w_r)$ (1.158). Applying (1.152) with $\underline{A}^t \underline{e}_r = A_{rr} \underline{e}_r + A_{r\theta} \underline{e}_\theta + A_{rz} \underline{k}$ and the like, and $\operatorname{tr}[\underline{A}^t \operatorname{grad} \underline{e}_r] = \frac{1}{r} A_{\theta\theta}$, $\operatorname{tr}[\underline{A}^t \operatorname{grad} \underline{e}_\theta] = -\frac{1}{r} A_{r\theta}$,

$$\begin{aligned}\underline{e}_r \cdot \operatorname{div} \underline{A} &= A_{rr,r} + \frac{1}{r}(A_{rr} - A_{\theta\theta} + A_{r\theta,\theta}), \\ \underline{e}_\theta \cdot \operatorname{div} \underline{A} &= A_{\theta r,r} + \frac{1}{r}(A_{\theta r} + A_{r\theta} + A_{\theta\theta,\theta}), \\ \underline{k} \cdot \operatorname{div} \underline{A} &= A_{zr,r} + \frac{1}{r}(A_{zr} + A_{z\theta,\theta}).\end{aligned}\tag{1.162}$$

1.2.7 Integral formulae

Balance laws equate the rate of change of a quantity inside a region to fluxes through its boundary. Passing between the two needs a bridge from volume integrals of derivatives to surface integrals of values. That bridge is the divergence theorem; every conservation law later rests on it.

THEOREM 1.100 (Divergence theorem). *For a C^1 vector field $\underline{v}$ on a bounded region R with piecewise smooth orientable boundary ∂R and exterior unit normal $\underline{n}$,*

$$\int_R \operatorname{div} \underline{v} \, dv = \int_{\partial R} \underline{v} \cdot \underline{n} \, da.\tag{1.167}$$

The proof is classical and is taken as given. Three consequences, used constantly later, are immediate.

COROLLARY 1.101 (Component, tensor and scalar forms).

$$\int_R v_{i,i} \, dv = \int_{\partial R} v_i n_i \, da;$$

$$\int_R \operatorname{div} \underline{A} \, dv = \int_{\partial R} \underline{A} \underline{n} \, da; \quad \int_R \operatorname{grad} \varphi \, dv = \int_{\partial R} \varphi \underline{n} \, da.$$

Proof. The first is (1.167) in components. For the tensor form, fix $\underline{c}$ and apply (1.167) to $\underline{A}^t \underline{c}$: $\underline{c} \cdot \int_R \operatorname{div} \underline{A} \, dv = \int_{\partial R} \underline{A}^t \underline{c} \cdot \underline{n} \, da = \underline{c} \cdot \int_{\partial R} \underline{A} \underline{n} \, da$, true for all $\underline{c}$. The scalar form is the tensor form with $\underline{A} = \varphi \underline{I}$, since $\operatorname{div}(\varphi \underline{I}) = \operatorname{grad} \varphi$ and $\varphi \underline{I} \underline{n} = \varphi \underline{n}$. ■

THEOREM 1.102 (Stokes, for reference). *For a differentiable $\underline{v}$ on a piecewise smooth surface Ω bounded by a piecewise smooth closed curve $\partial\Omega$,*

$$\int_{\Omega} \text{curl } \underline{v} \cdot \underline{n} \, da = \int_{\partial\Omega} \underline{v} \cdot d\underline{x}, \quad (1.168)$$

with $\underline{n}$ oriented so that $\underline{\nu} = \underline{t} \times \underline{n}$ is the outward normal to $\partial\Omega$ in the tangent plane. (Stated for completeness; not used in the sequel.)

1.3 Problems

All twenty eight problems of the chapter are solved below, in the book's order, with every algebraic step shown. Two identities recur:

$$\underline{a} \times (\underline{b} \times \underline{c}) = (\underline{a} \cdot \underline{c})\underline{b} - (\underline{a} \cdot \underline{b})\underline{c}, \quad \varepsilon_{ijk}\varepsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}. \quad (\star)$$

Problems 1–23 use only the algebra of §1.1; Problems 24–28 use the field calculus of §1.2.

PROBLEM 1 (Cauchy–Schwarz, triangle, Pythagoras). Prove $|\underline{a} \cdot \underline{b}| \leq |\underline{a}||\underline{b}|$, the triangle inequality, and the Pythagorean theorem for orthogonal $\underline{a}, \underline{b}$.

Solution. Cauchy–Schwarz. If $\underline{a} = \underline{0}$ then $\underline{a} \cdot \underline{b} = 0$ and the inequality reads $0 \leq 0$; assume $\underline{a} \neq \underline{0}$. Following the hint, form the vector

$$\underline{w} = (\underline{a} \cdot \underline{a})\underline{b} - (\underline{a} \cdot \underline{b})\underline{a}.$$

Positivity of the norm gives $0 \leq |\underline{w}|^2 = \underline{w} \cdot \underline{w}$. Expand the dot product by bilinearity, keeping every one of the four cross terms (write $\alpha = \underline{a} \cdot \underline{a}$, $\beta = \underline{a} \cdot \underline{b}$ for brevity):

$$\begin{aligned} |\underline{w}|^2 &= (\alpha\underline{b} - \beta\underline{a}) \cdot (\alpha\underline{b} - \beta\underline{a}) \\ &= \alpha^2 \underline{b} \cdot \underline{b} - \alpha\beta \underline{b} \cdot \underline{a} - \beta\alpha \underline{a} \cdot \underline{b} + \beta^2 \underline{a} \cdot \underline{a} \\ &= \alpha^2 \underline{b} \cdot \underline{b} - 2\alpha\beta^2 + \beta^2\alpha \quad (\text{using } \underline{b} \cdot \underline{a} = \underline{a} \cdot \underline{b} = \beta, \underline{a} \cdot \underline{a} = \alpha) \\ &= \alpha[\alpha \underline{b} \cdot \underline{b} - \beta^2] = (\underline{a} \cdot \underline{a})[(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2]. \end{aligned}$$

Since $\underline{a} \cdot \underline{a} = |\underline{a}|^2 > 0$, dividing the inequality $|\underline{w}|^2 \geq 0$ by it leaves the bracket nonnegative:

$$(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2 \geq 0 \implies (\underline{a} \cdot \underline{b})^2 \leq |\underline{a}|^2 |\underline{b}|^2.$$

Taking the (nonnegative) square root, $|\underline{a} \cdot \underline{b}| \leq |\underline{a}| |\underline{b}|$. Equality requires $\underline{w} = \underline{0}$, i.e. $\underline{a}, \underline{b}$ parallel.

Triangle inequality. Expand the squared norm and insert Cauchy–Schwarz at the marked step, $\underline{a} \cdot \underline{b} \leq |\underline{a} \cdot \underline{b}| \leq |\underline{a}| |\underline{b}|$:

$$\begin{aligned} |\underline{a} + \underline{b}|^2 &= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) = |\underline{a}|^2 + 2 \underline{a} \cdot \underline{b} + |\underline{b}|^2 \\ &\leq |\underline{a}|^2 + 2|\underline{a}| |\underline{b}| + |\underline{b}|^2 = (|\underline{a}| + |\underline{b}|)^2. \end{aligned}$$

Both sides are nonnegative, so $|\underline{a} + \underline{b}| \leq |\underline{a}| + |\underline{b}|$.

Pythagoras. If $\underline{a} \perp \underline{b}$, i.e. $\underline{a} \cdot \underline{b} = 0$, the middle term above drops out and $|\underline{a} + \underline{b}|^2 = |\underline{a}|^2 + |\underline{b}|^2$. $\square$

REMARK 1.103 (what the hint vector is doing). The vector $\underline{w} = (\underline{a} \cdot \underline{a})\underline{b} - (\underline{a} \cdot \underline{b})\underline{a}$ is (a positive multiple of) the part of $\underline{b}$ *perpendicular* to $\underline{a}$: subtracting the projection $(\underline{a} \cdot \underline{b} / \underline{a} \cdot \underline{a})\underline{a}$ from $\underline{b}$ and scaling by $\underline{a} \cdot \underline{a}$ gives exactly $\underline{w}$. Cauchy–Schwarz then says nothing more than “a squared length is never negative”: the inequality is tight precisely when that perpendicular part is zero, i.e. when $\underline{b}$ is parallel to $\underline{a}$. This is why Cauchy–Schwarz underlies the very definition of angle (Definition 1.8): $\cos \theta = \underline{a} \cdot \underline{b} / (|\underline{a}| |\underline{b}|)$ lands in $[-1, 1]$ exactly because $|\underline{w}|^2 \geq 0$.

PROBLEM 2 (Is the norm linear?). Is $f(\underline{v}) = |\underline{v}|$ linear?

Solution. No. Linearity requires both $f(\underline{u} + \underline{v}) = f(\underline{u}) + f(\underline{v})$ and $f(\alpha \underline{v}) = \alpha f(\underline{v})$; each fails.

Additivity. Take $\underline{v} = -\underline{u}$ with $\underline{u} \neq \underline{0}$:

$$f(\underline{u} + \underline{v}) = f(\underline{0}) = 0, \quad f(\underline{u}) + f(\underline{v}) = |\underline{u}| + |-\underline{u}| = 2|\underline{u}| > 0.$$

Homogeneity. For $\alpha < 0$,

$$f(\alpha \underline{v}) = |\alpha \underline{v}| = |\alpha| |\underline{v}| = -\alpha |\underline{v}| \neq \alpha |\underline{v}| = \alpha f(\underline{v})$$

whenever $\underline{v} \neq \underline{0}$.

What the norm does satisfy is $f(\alpha \underline{v}) = |\alpha|f(\underline{v})$ and $f(\underline{u} + \underline{v}) \leq f(\underline{u}) + f(\underline{v})$, the triangle inequality of Problem 1. Those two make it a norm, not a linear function. $\square$

PROBLEM 3 (A constant map is not a tensor). For a fixed $\underline{m}$, is there a tensor $\underline{A}$ with $\underline{A}\underline{v} = \underline{m}$ for all $\underline{v}$?

Solution. No, unless $\underline{m} = \underline{0}$. A linear map fixes the origin:

$$\underline{A}\underline{0} = \underline{A}(0 \cdot \underline{0}) = 0 \cdot \underline{A}\underline{0} = \underline{0},$$

whereas the requirement gives $\underline{A}\underline{0} = \underline{m}$. So $\underline{m} = \underline{0}$.

Second route, without using $\underline{0}$. Homogeneity with $\alpha = 2$ gives

$$\underline{m} = \underline{A}(2\underline{v}) = 2 \underline{A}\underline{v} = 2\underline{m} \implies \underline{m} = \underline{0}.$$

In the problem $\underline{m} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2) \neq \underline{0}$, so no such tensor exists. Contrast Problem 4, where $\underline{A}\underline{v} = (\underline{v} \cdot \underline{n})\underline{m}$ is linear: the difference is the factor $\underline{v} \cdot \underline{n}$, which vanishes with $\underline{v}$. $\square$

PROBLEM 4 (A rank-one tensor). With $\underline{m} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2)$, $\underline{n} = \frac{\sqrt{2}}{2}(\underline{e}_3 - \underline{e}_1)$, find $\underline{A}$ with $\underline{A}\underline{v} = (\underline{v} \cdot \underline{n})\underline{m}$.

Solution. The map $\underline{v} \mapsto (\underline{v} \cdot \underline{n})\underline{m}$ is linear in $\underline{v}$ (the inner product is linear in its second argument) and returns a vector, so it is a tensor. By Definition 1.32, $(\underline{m} \otimes \underline{n})\underline{v} = (\underline{n} \cdot \underline{v})\underline{m}$. Hence $\underline{A} = \underline{m} \otimes \underline{n}$.

Components. $A_{ij} = m_i n_j$ (from Example 1.34). The components of $\underline{m}$ are $m_1 = \frac{\sqrt{2}}{2}$, $m_2 = \frac{\sqrt{2}}{2}$, $m_3 = 0$; those of $\underline{n}$ are $n_1 = -\frac{\sqrt{2}}{2}$, $n_2 = 0$, $n_3 = \frac{\sqrt{2}}{2}$. Each entry $A_{ij} = m_i n_j$:

$$\begin{aligned} A_{11} &= \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}, & A_{12} &= \frac{\sqrt{2}}{2} \cdot 0 = 0, & A_{13} &= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{1}{2}, \\ A_{21} &= -\frac{1}{2}, & A_{22} &= 0, & A_{23} &= \frac{1}{2}, & A_{31} &= 0, & A_{32} &= 0, & A_{33} &= 0. \end{aligned}$$

Hence

$$(A_{ij}) = \frac{1}{2} \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check. $\underline{A}$ has rank 1 (all rows are multiples of $(-1, 0, 1)$), confirming it is a rank one tensor: the image of $\underline{A}$ is the line spanned by $\underline{m}$. $\square$

PROBLEM 5 ($\varepsilon_{ijk}A_{jk} = 0 \iff \underline{A} = \underline{A}^t$).

Solution. Write $c_i := \varepsilon_{ijk}A_{jk}$ (a vector, since i is free while j, k are summed). We must show $c_i = 0$ for all i exactly when $A_{jk} = A_{kj}$.

($\Leftarrow$) *Symmetric* $\Rightarrow c_i = 0$. Suppose $A_{jk} = A_{kj}$. In $c_i = \varepsilon_{ijk}A_{jk}$ rename the two summed indices $j \leftrightarrow k$ (a dummy relabelling changes nothing):

$$c_i = \varepsilon_{ijk}A_{jk} = \varepsilon_{ikj}A_{kj}.$$

Now use symmetry $A_{kj} = A_{jk}$ and the antisymmetry of the permutation symbol under swapping its last two slots, $\varepsilon_{ikj} = -\varepsilon_{ijk}$:

$$c_i = \varepsilon_{ikj}A_{kj} = -\varepsilon_{ijk}A_{jk} = -c_i.$$

Thus $c_i = -c_i$, i.e. $2c_i = 0$, so $c_i = 0$ for every i .

($\Rightarrow$) $c_i = 0 \Rightarrow$ *symmetric*. Split $\underline{A}$ into its symmetric and skew parts, $A_{jk} = S_{jk} + W_{jk}$ with $S_{jk} = \frac{1}{2}(A_{jk} + A_{kj})$, $W_{jk} = \frac{1}{2}(A_{jk} - A_{kj})$. By the direction just proved, $\varepsilon_{ijk}S_{jk} = 0$, so

$$0 = c_i = \varepsilon_{ijk}A_{jk} = \varepsilon_{ijk}S_{jk} + \varepsilon_{ijk}W_{jk} = \varepsilon_{ijk}W_{jk}.$$

Every skew array is $W_{jk} = -\varepsilon_{jkl}w_l$ for a unique axial vector $\underline{w}$ (this is the general solution of $W_{jk} = -W_{kj}$; see Proposition 1.44). Substitute and contract:

$$\begin{aligned} \varepsilon_{ijk}W_{jk} &= \varepsilon_{ijk}(-\varepsilon_{jkl}w_l) = -\varepsilon_{ijk}\varepsilon_{jkl}w_l \\ &= -\varepsilon_{jki}\varepsilon_{jkl}w_l = -2\delta_{il}w_l = -2w_i, \end{aligned}$$

where we wrote $\varepsilon_{ijk} = \varepsilon_{jki}$ (cyclic shift) and used the contracted identity $\varepsilon_{jki}\varepsilon_{jkl} = 2\delta_{il}$. Hence $0 = c_i = -2w_i$ gives $w_i = 0$ for all i , so $\underline{w} = \underline{0}$, so $W_{jk} = 0$, so $A_{jk} = S_{jk} = A_{kj}$: $\underline{A}$ is symmetric. $\square$

REMARK 1.104 (what the operation $A_{jk} \mapsto \varepsilon_{ijk}A_{jk}$ extracts). The computation shows $\varepsilon_{ijk}A_{jk} = -2w_i$, twice the (negated) axial vector of the skew part of $\underline{A}$. So contracting a tensor with the permutation symbol

on two indices is a machine that *throws away the symmetric part and reads off the skew part as a vector*. This is exactly the device behind writing the vorticity of a velocity field, or the moment of a force, as a single axial vector; it works only in three dimensions, where a skew tensor (three independent entries) matches a vector (three components).

PROBLEM 6 (Skew/symmetric and the tensor inner product). With converses: (a) $\underline{A}^t = -\underline{A} \Rightarrow \underline{A} \cdot (\underline{a} \otimes \underline{a}) = 0$; (b) $\underline{T} \cdot \underline{S} = 0 \ \forall \underline{S} \Rightarrow \underline{T} = \underline{O}$; (c) $\underline{T} \cdot \underline{S} = 0 \ \forall \underline{S} = \underline{S}^t \Rightarrow \underline{T}$ skew; (d) $\underline{T} \cdot \underline{S} = 0 \ \forall \underline{S} = -\underline{S}^t \Rightarrow \underline{T}$ symmetric.

Solution. Preliminary. Recall: $\underline{A} \cdot \underline{B} = \text{tr}(\underline{A}\underline{B}^t) = A_{ij}B_{ij}$ (sum of all nine component products), and $\underline{A} \cdot (\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{A}\underline{v}$.

Lemma (the key orthogonality). Let $\underline{S} = \underline{S}^t$ be symmetric and $\underline{W} = -\underline{W}^t$ skew. Then $\underline{S} \cdot \underline{W} = S_{ij}W_{ij}$. Relabelling $i \leftrightarrow j$: $S_{ij}W_{ij} = S_{ji}W_{ji} = S_{ij}(-W_{ij}) = -S_{ij}W_{ij}$, using $S_{ji} = S_{ij}$ and $W_{ji} = -W_{ij}$. Thus $2(\underline{S} \cdot \underline{W}) = 0$, i.e. $\underline{S} \cdot \underline{W} = 0$: a symmetric tensor is orthogonal to every skew tensor.

(a) $\underline{A} \cdot (\underline{a} \otimes \underline{a}) = \underline{a} \cdot \underline{A}\underline{a}$. Since $\underline{A}^t = -\underline{A}$, $\underline{a} \cdot \underline{A}\underline{a} = \underline{A}^t \underline{a} \cdot \underline{a} = -\underline{A}\underline{a} \cdot \underline{a} = -\underline{a} \cdot \underline{A}\underline{a}$, hence $\underline{a} \cdot \underline{A}\underline{a} = 0$. *Converse:* if $\underline{a} \cdot \underline{A}\underline{a} = 0$ for all $\underline{a}$, put $\underline{a} = \underline{e}_i + \underline{e}_j$ ($i \neq j$): $(\underline{e}_i + \underline{e}_j) \cdot \underline{A}(\underline{e}_i + \underline{e}_j) = A_{ii} + A_{ij} + A_{ji} + A_{jj} = 0$. With $\underline{a} = \underline{e}_i$: $A_{ii} = 0$; with $\underline{a} = \underline{e}_j$: $A_{jj} = 0$. Hence $A_{ij} + A_{ji} = 0$ for all $i \neq j$, together with $A_{ii} = 0$: $\underline{A}$ is skew.

(b) $\underline{T} \cdot \underline{S} = 0$ for all $\underline{S}$: choose $\underline{S} = \underline{T}$, getting $|\underline{T}|^2 = T_{ij}T_{ij} = 0$, so every $T_{ij} = 0$, i.e. $\underline{T} = \underline{O}$.

(c) $\underline{T} \cdot \underline{S} = 0$ for all symmetric $\underline{S}$. Decompose $\underline{T} = \underline{T}_s + \underline{T}_w$ (symmetric + skew). Choose $\underline{S} = \underline{T}_s$: by the Lemma, $\underline{T}_w \cdot \underline{T}_s = 0$, so $\underline{T} \cdot \underline{T}_s = \underline{T}_s \cdot \underline{T}_s = |\underline{T}_s|^2 = 0$, hence $\underline{T}_s = \underline{O}$ and $\underline{T} = \underline{T}_w$ is skew.

(d) $\underline{T} \cdot \underline{S} = 0$ for all skew $\underline{S}$. Choose $\underline{S} = \underline{T}_w$: by the Lemma, $\underline{T}_s \cdot \underline{T}_w = 0$, so $\underline{T} \cdot \underline{T}_w = |\underline{T}_w|^2 = 0$, hence $\underline{T}_w = \underline{O}$ and $\underline{T} = \underline{T}_s$ is symmetric. $\square$

PROBLEM 7 (Is $\underline{A}(\underline{u} \times \underline{v}) = \underline{A}\underline{u} \times \underline{v} + \underline{u} \times \underline{A}\underline{v}$?).

Solution. Write $D(\underline{A}) := \underline{A}(\underline{u} \times \underline{v}) - \underline{A}\underline{u} \times \underline{v} - \underline{u} \times \underline{A}\underline{v}$. The question is when $D(\underline{A}) = \underline{0}$ for all $\underline{u}, \underline{v}$. Note D is linear in $\underline{A}$.

No in general. Take $\underline{A} = \underline{I}$:

$$D(\underline{I}) = \underline{u} \times \underline{v} - \underline{u} \times \underline{v} - \underline{u} \times \underline{v} = -\underline{u} \times \underline{v} \neq \underline{0}.$$

Skew tensors do satisfy it. Let $\underline{W}x = \underline{\omega} \times x$, Problem 16. The Jacobi identity is

$$\underline{\omega} \times (\underline{u} \times \underline{v}) + \underline{u} \times (\underline{v} \times \underline{\omega}) + \underline{v} \times (\underline{\omega} \times \underline{u}) = \underline{0}.$$

Use $\underline{v} \times (\underline{\omega} \times \underline{u}) = -(\underline{\omega} \times \underline{u}) \times \underline{v}$ and $\underline{u} \times (\underline{v} \times \underline{\omega}) = -\underline{u} \times (\underline{\omega} \times \underline{v})$:

$$\underline{\omega} \times (\underline{u} \times \underline{v}) = (\underline{\omega} \times \underline{u}) \times \underline{v} + \underline{u} \times (\underline{\omega} \times \underline{v}),$$

that is $D(\underline{W}) = \underline{0}$.

Only skew tensors do. Suppose $D(\underline{A}) = \underline{0}$. Split $\underline{A} = \underline{S} + \underline{W}$ by Problem 14; $D(\underline{W}) = \underline{0}$, so $D(\underline{S}) = \underline{0}$ by linearity. Take the principal basis of $\underline{S}$, $\underline{S}\underline{e}_i = \lambda_i \underline{e}_i$, and put $\underline{u} = \underline{e}_i$, $\underline{v} = \underline{e}_j$ with ijk cyclic, so $\underline{e}_i \times \underline{e}_j = \underline{e}_k$:

$$\lambda_k \underline{e}_k = \lambda_i \underline{e}_i \times \underline{e}_j + \underline{e}_i \times \lambda_j \underline{e}_j = (\lambda_i + \lambda_j) \underline{e}_k \implies \lambda_k = \lambda_i + \lambda_j.$$

The three cyclic triples give

$$\lambda_3 = \lambda_1 + \lambda_2, \quad \lambda_1 = \lambda_2 + \lambda_3, \quad \lambda_2 = \lambda_3 + \lambda_1.$$

Adding, $\sum \lambda_i = 2 \sum \lambda_i$, so $\sum \lambda_i = 0$; then the first equation gives $\lambda_3 = -\lambda_3$, hence $\lambda_3 = 0$, and likewise $\lambda_1 = \lambda_2 = 0$. So $\underline{S} = \underline{0}$ and $\underline{A} = \underline{W}$ is skew. $\square$

REMARK 1.105 (what the answer says). $D(\underline{A}) = \underline{0}$ is the Leibniz rule for $\underline{A}$ acting on the product $\underline{u} \times \underline{v}$. The result is therefore: *a tensor differentiates the cross product exactly when it is skew*, i.e. when it is an infinitesimal rotation. This is the statement that $\mathfrak{so}(3)$ is the Lie algebra of the rotation group, and it recurs in Chapter 4, where the spin $\underline{W}$ is skew for precisely this reason.

PROBLEM 8 (Reciprocal bases). With $\underline{e}_i = a_{ij} \underline{e}'_j$, $\underline{e}'_i = a'_{ij} \underline{e}_j$, show (a) $a_{ik} a'_{kj} = \delta_{ij}$; (b) $(a') = (a)^t$.

Solution. (a) Substitute the second relation into the first:

$$\underline{e}_i = a_{ij}\underline{e}'_j = a_{ij}(a'_{jk}\underline{e}_k) = a_{ij}a'_{jk}\underline{e}_k.$$

But also $\underline{e}_i = \delta_{ik}\underline{e}_k$. The $\{\underline{e}_k\}$ are independent, so the coefficients match:

$$a_{ij}a'_{jk} = \delta_{ik}.$$

(b) Dot $\underline{e}_i = a_{ij}\underline{e}'_j$ with $\underline{e}'_k$ and use $\underline{e}'_j \cdot \underline{e}'_k = \delta_{jk}$:

$$\underline{e}_i \cdot \underline{e}'_k = a_{ij}\delta_{jk} = a_{ik}.$$

Dot $\underline{e}'_i = a'_{ij}\underline{e}_j$ with $\underline{e}_k$ and use $\underline{e}_j \cdot \underline{e}_k = \delta_{jk}$:

$$\underline{e}'_i \cdot \underline{e}_k = a'_{ij}\delta_{jk} = a'_{ik}.$$

The two left sides are the same number, since the dot product is symmetric:

$$a'_{ik} = \underline{e}'_i \cdot \underline{e}_k = \underline{e}_k \cdot \underline{e}'_i = a_{ki}, \quad \text{i.e.} \quad (a') = (a)^t.$$

Part (a) then reads $a_{ij}a_{kj} = \delta_{ik}$: the matrix (a) is orthogonal, $(a)(a)^t = (a)^t(a) = (\delta)$. This is the fact used in Problem 9 and again in Problem 3 of Chapter 2. $\square$

PROBLEM 9 (Invariance of trace and dot product; a computation). (a) $A_{ii} = A'_{ii}$; (b) deduce invariance of $\underline{u} \cdot \underline{v}$; (c) compute A'_{ij} for the given data.

Solution. (a) With $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, so that $\underline{e}'_j = a_{kj}\underline{e}_k$, the components transform as

$$A'_{ij} = \underline{e}'_i \cdot A\underline{e}'_j = a_{ki}\underline{e}_k \cdot A a_{lj}\underline{e}_l = a_{ki}a_{lj}\underline{e}_k \cdot A\underline{e}_l = a_{ki}a_{lj}A_{kl}.$$

Orthonormality of the primed basis fixes (a) :

$$\delta_{ij} = \underline{e}'_i \cdot \underline{e}'_j = a_{ki}a_{lj}\underline{e}_k \cdot \underline{e}_l = a_{ki}a_{lj}\delta_{kl} = a_{ki}a_{kj},$$

so (a) is orthogonal. Set $j = i$ and sum:

$$A'_{ii} = a_{ki}a_{li}A_{kl} = \delta_{kl}A_{kl} = A_{kk}.$$

(b) Vectors carry one factor, $u'_i = \underline{e}'_i \cdot \underline{u} = a_{ki}u_k$. Hence

$$u'_i v'_i = a_{ki}a_{li} u_k v_l = \delta_{kl} u_k v_l = u_k v_k.$$

Alternatively, apply (a) to $\underline{A} = \underline{u} \otimes \underline{v}$, whose trace is $\underline{u} \cdot \underline{v}$ by Theorem 1.51.

(c) The data are

$$(A_{ij}) = \begin{pmatrix} 1 & 5 & -5 \\ 5 & 0 & 0 \\ -5 & 0 & 1 \end{pmatrix}, \quad \underline{e}'_1 = \frac{1}{\sqrt{5}}(-1, 0, 2), \quad \underline{e}'_2 = \underline{e}_2 = (0, 1, 0).$$

Fix $\underline{e}'_3$ by right-handedness, $\underline{e}'_3 = \underline{e}'_1 \times \underline{e}'_2$. The cross product $(-1, 0, 2) \times (0, 1, 0)$ has components

$$(0 \cdot 0 - 2 \cdot 1, \quad 2 \cdot 0 - (-1) \cdot 0, \quad (-1) \cdot 1 - 0 \cdot 0) = (-2, 0, -1),$$

so $\underline{e}'_3 = \frac{1}{\sqrt{5}}(-2, 0, -1)$. The change-of-basis matrix $\underline{Q} = (a_{ij})$, $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, carries the components of $\underline{e}'_j$ in its j th column:

$$\underline{Q} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 & 0 & -2 \\ 0 & \sqrt{5} & 0 \\ 2 & 0 & -1 \end{pmatrix}.$$

The law $A'_{ij} = a_{ki}a_{lj}A_{kl}$ is $\underline{A}' = \underline{Q}^t \underline{A} \underline{Q}$, which we read as $A'_{ij} = \underline{e}'_i \cdot \underline{A} \underline{e}'_j$. Compute the three columns $\underline{A} \underline{e}'_j$ (matrix times vector); for example

$$\underline{A} \underline{e}'_1 = \frac{1}{\sqrt{5}} \underline{A} \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1(-1) + 5(0) - 5(2) \\ 5(-1) + 0 + 0 \\ -5(-1) + 0 + 1(2) \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} -11 \\ -5 \\ 7 \end{pmatrix},$$

and likewise $\underline{A} \underline{e}'_2 = (5, 0, 0)^t$, $\underline{A} \underline{e}'_3 = \frac{1}{\sqrt{5}}(3, -10, 9)^t$. Now $A'_{ij} = \underline{e}'_i \cdot \underline{A} \underline{e}'_j$ entry by entry:

$$A'_{11} = \frac{1}{\sqrt{5}}(-1, 0, 2) \cdot \frac{1}{\sqrt{5}}(-11, -5, 7) = \frac{11+0+14}{5} = 5,$$

$$A'_{12} = \frac{1}{\sqrt{5}}(-1, 0, 2) \cdot (5, 0, 0) = -\frac{5}{\sqrt{5}} = -\sqrt{5},$$

$$A'_{13} = \frac{1}{\sqrt{5}}(-1, 0, 2) \cdot \frac{1}{\sqrt{5}}(3, -10, 9) = \frac{-3+0+18}{5} = 3,$$

$$A'_{22} = (0, 1, 0) \cdot (5, 0, 0) = 0,$$

$$A'_{23} = (0, 1, 0) \cdot \frac{1}{\sqrt{5}}(3, -10, 9) = -\frac{10}{\sqrt{5}} = -2\sqrt{5},$$

$$A'_{33} = \frac{1}{\sqrt{5}}(-2, 0, -1) \cdot \frac{1}{\sqrt{5}}(3, -10, 9) = \frac{-6+0-9}{5} = -3,$$

with $A'_{21} = A'_{12}$, $A'_{31} = A'_{13}$, $A'_{32} = A'_{23}$ since $\underline{A}'$ is symmetric ($\underline{A}$ is). Therefore

$$\underline{A}' = \begin{pmatrix} 5 & -\sqrt{5} & 3 \\ -\sqrt{5} & 0 & -2\sqrt{5} \\ 3 & -2\sqrt{5} & -3 \end{pmatrix}.$$

Check: $\text{tr } \underline{A}' = 5 + 0 - 3 = 2 = 1 + 0 + 1 = \text{tr } \underline{A}$, confirming the invariance proved in part (a). $\square$

PROBLEM 10 (Projection and reflection tensors). (a) Consider a plane P with unit normal $\underline{n}$. Let $\underline{v} \in \mathbb{E}^3$ be arbitrary and $\underline{v}_p$ its projection onto P . Verify $\underline{v} = \underline{v}_p + (\underline{v} \cdot \underline{n})\underline{n}$ and find $\mathbb{P}_{(\underline{n})}$ in $\underline{v}_p = \mathbb{P}_{(\underline{n})}\underline{v}$. (b) A tensor $\underline{A}$ is a *projection* iff $\underline{A}^t = \underline{A}$ and $\underline{A}^2 = \underline{A}$. Show $\mathbb{P}_{(\underline{n})}$ is a projection, and that $\underline{Q}$, $\underline{I}$, $\underline{n} \otimes \underline{n}$ are projections. (c) Let $\mathbb{R}_{(\underline{n})}$ be the *reflection* tensor for P . Find $\mathbb{R}_{(\underline{n})}$ and its components relative to $\{\underline{e}_i \otimes \underline{e}_j\}$ for (i) $\underline{n} = \underline{e}_1$, (ii) $\underline{n} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2)$.

Solution. The normal is a unit vector and the plane through the origin with that normal consists of the vectors orthogonal to it, so throughout

$$\underline{n} \cdot \underline{n} = 1, \quad P = \{\underline{w} \in \mathbb{E}^3 : \underline{w} \cdot \underline{n} = 0\} = \{\underline{n}\}^\perp, \quad \mathcal{N} = \text{span}\{\underline{n}\}, \quad (\star_1)$$

P being two dimensional and $\mathcal{N}$ one dimensional.

(a) The shadow of $\underline{v}$ on the direction $\underline{n}$ has signed length $\underline{v} \cdot \underline{n}$, so the part of $\underline{v}$ along the normal, and what is left after removing it, are

$$\underline{v}_n := (\underline{v} \cdot \underline{n})\underline{n}, \quad \underline{v}_p := \underline{v} - (\underline{v} \cdot \underline{n})\underline{n}. \quad (\star_2)$$

That $\underline{v}_p$ lies in P is not an assumption but a computation; dot it with $\underline{n}$ and use bilinearity together with $\underline{n} \cdot \underline{n} = 1$:

$$\underline{v}_p \cdot \underline{n} = [\underline{v} - (\underline{v} \cdot \underline{n})\underline{n}] \cdot \underline{n} = \underline{v} \cdot \underline{n} - (\underline{v} \cdot \underline{n})\underline{n} \cdot \underline{n} \stackrel{(\star_1)}{=} \underline{v} \cdot \underline{n} - \underline{v} \cdot \underline{n} = 0,$$

hence $\underline{v}_p \in \{\underline{n}\}^\perp = P$, and by construction $\underline{v} = \underline{v}_p + \underline{v}_n$, which is the required identity.

The splitting is unique. Suppose $\underline{v} = \underline{w}_p + \alpha\underline{n}$ with $\underline{w}_p \in P$ and $\alpha \in \mathbb{R}$. Subtracting the two expressions for $\underline{v}$,

$$\underline{v}_p - \underline{w}_p = (\alpha - \underline{v} \cdot \underline{n})\underline{n},$$

whose left side lies in P and whose right side lies in $\mathcal{N}$. A vector of $P \cap \mathcal{N}$ satisfies both $\underline{w} = \lambda \underline{n}$ and $\underline{w} \cdot \underline{n} = 0$, so $\lambda = \lambda \underline{n} \cdot \underline{n} = \underline{w} \cdot \underline{n} = 0$ and $P \cap \mathcal{N} = \{0\}$. Therefore $\underline{w}_p = \underline{v}_p$ and $\alpha = \underline{v} \cdot \underline{n}$, and $\mathbb{E}^3 = P \oplus \mathcal{N}$.

Before writing $\underline{v}_p = \mathbb{P}_{(\underline{n})}\underline{v}$ we must know the map $\underline{v} \mapsto \underline{v}_p$ is linear, since only linear maps are tensors:

$$\begin{aligned} (\alpha \underline{u} + \beta \underline{v})_p &= \alpha \underline{u} + \beta \underline{v} - ((\alpha \underline{u} + \beta \underline{v}) \cdot \underline{n}) \underline{n} \\ &= \alpha \underline{u} + \beta \underline{v} - (\alpha \underline{u} \cdot \underline{n} + \beta \underline{v} \cdot \underline{n}) \underline{n} \\ &= \alpha [\underline{u} - (\underline{u} \cdot \underline{n}) \underline{n}] + \beta [\underline{v} - (\underline{v} \cdot \underline{n}) \underline{n}] = \alpha \underline{u}_p + \beta \underline{v}_p. \end{aligned}$$

Now recognise the subtracted term as a tensor acting on $\underline{v}$. By the definition of the tensor product, $(\underline{n} \otimes \underline{n})\underline{v} = (\underline{n} \cdot \underline{v})\underline{n} = (\underline{v} \cdot \underline{n})\underline{n}$, so $(\star_2)$ becomes $\underline{v}_p = \underline{v} - (\underline{n} \otimes \underline{n})\underline{v} = [\underline{I} - \underline{n} \otimes \underline{n}]\underline{v}$ for every $\underline{v}$, and equality of tensors is equality of their action. Hence

$$\mathbb{P}_{(\underline{n})} = \underline{I} - \underline{n} \otimes \underline{n}, \quad (\mathbb{P}_{(\underline{n})})_{ij} = \delta_{ij} - n_i n_j, \quad (\star_3)$$

the components following from $\underline{I} = \delta_{ij} \underline{e}_i \otimes \underline{e}_j$ and $(\underline{n} \otimes \underline{n})_{ij} = n_i n_j$.

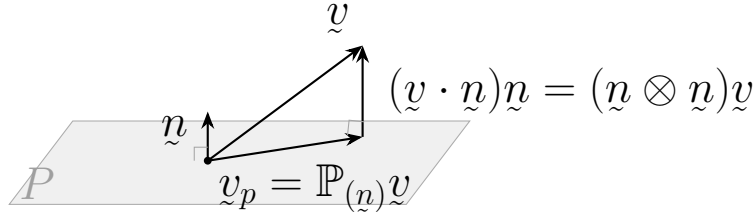


FIGURE P10.1. $\underline{v} = \underline{v}_p + (\underline{v} \cdot \underline{n})\underline{n}$ with $\underline{v}_p \cdot \underline{n} = 0$; $\mathbb{P}_{(\underline{n})}$ returns the leg in P , $\underline{n} \otimes \underline{n}$ the leg along $\underline{n}$, and $\mathbb{P}_{(\underline{n})} + \underline{n} \otimes \underline{n} = \underline{I}$.

(b) Everything rests on one property of the dyad, obtained by applying it twice to an arbitrary $\underline{v}$ and unfolding the definition at each step:

$$(\underline{n} \otimes \underline{n})^2 \underline{v} = (\underline{n} \otimes \underline{n})[(\underline{n} \cdot \underline{v})\underline{n}] = (\underline{n} \cdot \underline{v})(\underline{n} \cdot \underline{n})\underline{n} \stackrel{(\star_1)}{=} (\underline{n} \cdot \underline{v})\underline{n} = (\underline{n} \otimes \underline{n})\underline{v}, \quad (\star_4)$$

true for every $\underline{v}$, so $(\underline{n} \otimes \underline{n})^2 = \underline{n} \otimes \underline{n}$. Note where the hypothesis enters: the middle step used $\underline{n} \cdot \underline{n} = 1$, and for a normal of length other than one the dyad would not be idempotent. Its symmetry follows from the

definition of the transpose, since for all $\underline{u}, \underline{v}$

$$\underline{u} \cdot (\underline{n} \otimes \underline{n}) \underline{v} = \underline{u} \cdot (\underline{n} \cdot \underline{v}) \underline{n} = (\underline{n} \cdot \underline{v}) (\underline{u} \cdot \underline{n}) = \underline{v} \cdot (\underline{n} \cdot \underline{u}) \underline{n} = \underline{v} \cdot (\underline{n} \otimes \underline{n}) \underline{u},$$

which says $(\underline{n} \otimes \underline{n})^t = \underline{n} \otimes \underline{n}$. Transposition being linear, and $\underline{I}^t = \underline{I}$,

$$\mathbb{P}_{(\underline{n})}^t = \underline{I}^t - (\underline{n} \otimes \underline{n})^t = \underline{I} - \underline{n} \otimes \underline{n} = \mathbb{P}_{(\underline{n})},$$

and expanding the square, the two equal cross terms combining into one,

$$\mathbb{P}_{(\underline{n})}^2 = \underline{I} - \underline{n} \otimes \underline{n} - \underline{n} \otimes \underline{n} + (\underline{n} \otimes \underline{n})^2 \stackrel{(\star_4)}{=} \underline{I} - \underline{n} \otimes \underline{n} = \mathbb{P}_{(\underline{n})}.$$

The same two facts in components, which is the form used in calculations; recall $(\underline{AB})_{ij} = A_{ik} B_{kj}$, and that a Kronecker delta collapses the sum it is contracted with, $\delta_{ik} n_k = n_i$ and $\delta_{ik} \delta_{kj} = \delta_{ij}$:

$$\begin{aligned} [(\underline{n} \otimes \underline{n})^2]_{ij} &= (n_i n_k)(n_k n_j) = n_i (n_k n_k) n_j = n_i n_j, \\ [\mathbb{P}_{(\underline{n})}^2]_{ij} &= (\delta_{ik} - n_i n_k)(\delta_{kj} - n_k n_j) \\ &= \delta_{ik} \delta_{kj} - \delta_{ik} n_k n_j - n_i n_k \delta_{kj} + n_i n_k n_k n_j \\ &= \delta_{ij} - n_i n_j - n_i n_j + n_i n_j \\ &= \delta_{ij} - n_i n_j = (\mathbb{P}_{(\underline{n})})_{ij}, \\ (\mathbb{P}_{(\underline{n})})_{ji} &= \delta_{ji} - n_j n_i = \delta_{ij} - n_i n_j = (\mathbb{P}_{(\underline{n})})_{ij}. \end{aligned}$$

The remaining three are immediate from the definitions. For the zero tensor, $\underline{u} \cdot \underline{O} \underline{v} = \underline{u} \cdot \underline{0} = 0 = \underline{v} \cdot \underline{0} = \underline{v} \cdot \underline{O} \underline{u}$ gives $\underline{O}^t = \underline{O}$, and $\underline{O}^2 \underline{v} = \underline{O}(\underline{O} \underline{v}) = \underline{O} \underline{0} = \underline{0} = \underline{O} \underline{v}$ gives $\underline{O}^2 = \underline{O}$; it projects onto the trivial subspace $\{\underline{0}\}$. For the identity, $\underline{u} \cdot \underline{I} \underline{v} = \underline{u} \cdot \underline{v} = \underline{v} \cdot \underline{u} = \underline{v} \cdot \underline{I} \underline{u}$ gives $\underline{I}^t = \underline{I}$, and $\underline{I}^2 \underline{v} = \underline{I}(\underline{I} \underline{v}) = \underline{I} \underline{v}$ gives $\underline{I}^2 = \underline{I}$; it projects onto all of $\mathbb{E}^3$. For $\underline{n} \otimes \underline{n}$, symmetry and idempotency are exactly what was proved above, and it projects onto the normal line, since $(\underline{n} \otimes \underline{n}) \underline{v} = \underline{v}_n$ by $(\star_2)$.

The two projections $\mathbb{P}_{(\underline{n})}$ and $\underline{n} \otimes \underline{n}$ are not independent of each other: they are *complementary*, and this is the algebraic face of $\mathbb{E}^3 = P \oplus \mathcal{N}$. Directly from $(\star_3)$ and $(\star_4)$,

$$\begin{aligned} \mathbb{P}_{(\underline{n})} + \underline{n} \otimes \underline{n} &= \underline{I}, \\ \mathbb{P}_{(\underline{n})}(\underline{n} \otimes \underline{n}) &= (\underline{I} - \underline{n} \otimes \underline{n})(\underline{n} \otimes \underline{n}) = \underline{n} \otimes \underline{n} - (\underline{n} \otimes \underline{n})^2 \\ &\stackrel{(\star_4)}{=} \underline{O} = (\underline{n} \otimes \underline{n}) \mathbb{P}_{(\underline{n})}, \end{aligned}$$

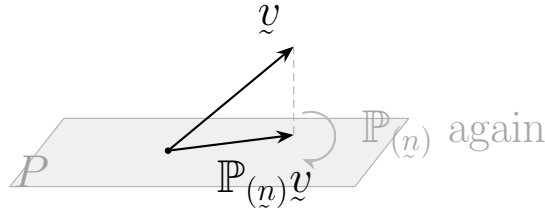


FIGURE P10.2. $\mathbb{P}_{(n)}^2 = \mathbb{P}_{(n)}$: the second application meets a vector already in P , since

$$(\mathbb{P}_{(n)}v) \cdot n = 0.$$

so each annihilates the range of the other. Their traces count the dimensions of what they keep; using linearity of the trace with $\text{tr } \underline{I} = 3$ and $\text{tr}(\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{v}$,

$$\text{tr } \mathbb{P}_{(n)} = \text{tr } \underline{I} - \text{tr}(\underline{n} \otimes \underline{n}) = 3 - \underline{n} \cdot \underline{n} = 3 - 1 = 2 = \dim P,$$

$$\text{tr}(\underline{n} \otimes \underline{n}) = \underline{n} \cdot \underline{n} = 1 = \dim \mathcal{N}.$$

Finally $\mathbb{P}_{(n)}$ is singular, since it kills the nonzero vector $\underline{n}$:

$$\mathbb{P}_{(n)}\underline{n} = \underline{n} - (\underline{n} \otimes \underline{n})\underline{n} = \underline{n} - (\underline{n} \cdot \underline{n})\underline{n} = \underline{n} - \underline{n} = \underline{0},$$

$$\underline{n} \neq \underline{0} \implies \det \mathbb{P}_{(n)} = 0,$$

so no projection other than $\underline{I}$ is invertible.

(c) The mirror image must leave the part lying in the plane alone and send the normal part to its negative. In the notation of $(\star_2)$ this says $\mathbb{R}_{(n)}\underline{v} = \underline{v}_p - \underline{v}_n$, and subtracting the normal part once to reach the plane and once more to pass through it produces the factor 2:

$$\mathbb{R}_{(n)}\underline{v} = \underline{v}_p - \underline{v}_n = [\underline{v} - (\underline{v} \cdot \underline{n})\underline{n}] - (\underline{v} \cdot \underline{n})\underline{n} = \underline{v} - 2(\underline{v} \cdot \underline{n})\underline{n} = [\underline{I} - 2\underline{n} \otimes \underline{n}]\underline{v}, \quad (1.1)$$

valid for every $\underline{v}$, whence

$$\mathbb{R}_{(n)} = \underline{I} - 2\underline{n} \otimes \underline{n} = \mathbb{P}_{(n)} - \underline{n} \otimes \underline{n}, \quad (\mathbb{R}_{(n)})_{ij} = \delta_{ij} - 2n_i n_j. \quad (\star_5)$$

Three properties can be established before any component is computed, and they will serve as checks on the two matrices below. Symmetry is inherited from $\underline{I}$ and $\underline{n} \otimes \underline{n}$, and the square collapses because the dyad is idempotent:

$$\mathbb{R}_{(n)}^t = \underline{I}^t - 2(\underline{n} \otimes \underline{n})^t = \underline{I} - 2\underline{n} \otimes \underline{n} = \mathbb{R}_{(n)},$$

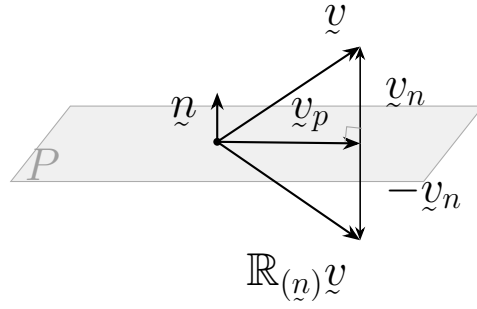


FIGURE P10.3. $\mathbb{R}_{(n)}v = v_p - v_n$; the normal part is removed twice, giving the factor 2 in

$$\mathbb{R}_{(n)} = \underline{I} - 2 \underline{n} \otimes \underline{n}.$$

$\mathbb{R}_{(n)}^2 = \underline{I} - 2 \underline{n} \otimes \underline{n} - 2 \underline{n} \otimes \underline{n} + 4(\underline{n} \otimes \underline{n})^2 = \underline{I} - 4 \underline{n} \otimes \underline{n} + 4(\underline{n} \otimes \underline{n})^2 \stackrel{(\star_4)}{=} \underline{I}$,
and the same in components,

$$\begin{aligned} [\mathbb{R}_{(n)}^2]_{ij} &= (\delta_{ik} - 2n_i n_k)(\delta_{kj} - 2n_k n_j) \\ &= \delta_{ij} - 2n_i n_j - 2n_i n_j + 4n_i(n_k n_k)n_j = \delta_{ij}, \end{aligned}$$

Thus $\mathbb{R}_{(n)}$ is its own inverse, $\mathbb{R}_{(n)}^{-1} = \mathbb{R}_{(n)}$: reflecting twice restores the original, as it must. Combining this with symmetry gives $\mathbb{R}_{(n)}^t \mathbb{R}_{(n)} = \mathbb{R}_{(n)}^2 = \underline{I}$, so $\mathbb{R}_{(n)}$ is orthogonal and therefore preserves every inner product, hence every length and angle:

$$\mathbb{R}_{(n)}u \cdot \mathbb{R}_{(n)}v = u \cdot \mathbb{R}_{(n)}^t \mathbb{R}_{(n)}v = u \cdot \underline{I}v = u \cdot v.$$

For the determinant, write $\mathbb{R}_{(n)} = \underline{I} + \underline{a} \otimes \underline{b}$ with $\underline{a} = -2\underline{n}$ and $\underline{b} = \underline{n}$; Problem 21(f) states $\det(\underline{I} + \underline{a} \otimes \underline{b}) = 1 + \underline{a} \cdot \underline{b}$, so

$$\det \mathbb{R}_{(n)} = 1 + (-2\underline{n}) \cdot \underline{n} = 1 - 2 \underline{n} \cdot \underline{n} = 1 - 2 = -1,$$

the negative sign saying that a reflection reverses orientation, in contrast to a rotation. The eigenvalues follow from the geometry: the normal is reversed and the plane is fixed,

$$\begin{aligned} \mathbb{R}_{(n)}\underline{n} &= \underline{n} - 2(\underline{n} \cdot \underline{n})\underline{n} = \underline{n} - 2\underline{n} = -\underline{n}, \\ \mathbb{R}_{(n)}\underline{w} &= \underline{w} - 2(\underline{w} \cdot \underline{n})\underline{n} = \underline{w} \quad (\underline{w} \in P), \end{aligned}$$

so $\lambda = -1$ once and $\lambda = +1$ twice, since $\dim P = 2$. This is confirmed by the characteristic polynomial: with

$$I_1 = \text{tr } \mathbb{R}_{(n)} = \text{tr } \underline{I} - 2 \text{tr}(\underline{n} \otimes \underline{n}) = 3 - 2 \underline{n} \cdot \underline{n} = 1,$$

$$I_2 = \frac{1}{2} [I_1^2 - \text{tr}(\mathbb{R}_{(\underline{n})}^2)] = \frac{1}{2} [1 - \text{tr} \underline{I}] = \frac{1}{2} [1 - 3] = -1, \quad I_3 = \det \mathbb{R}_{(\underline{n})} = -1,$$

Problem 23 gives

$$\det(\mathbb{R}_{(\underline{n})} - \lambda \underline{I}) = -\lambda^3 + I_1 \lambda^2 - I_2 \lambda + I_3 = -\lambda^3 + \lambda^2 + \lambda - 1 = -(\lambda - 1)^2(\lambda + 1),$$

whose roots are $+1, +1, -1$ as expected.

(i) $\underline{n} = \underline{e}_1$. Here $n_i = \underline{e}_1 \cdot \underline{e}_i = \delta_{i1}$, so $n_1 = 1$ and $n_2 = n_3 = 0$, and the outer product has entries $n_i n_j = \delta_{i1} \delta_{j1}$, which vanish unless $i = j = 1$:

$$(n_i n_j) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Substituting into $(\star_5)$ and subtracting entry by entry,

$$(R_{ij}) = \delta_{ij} - 2n_i n_j = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Reassembling the tensor from its components, $\mathbb{R}_{(\underline{e}_1)} = R_{ij} \underline{e}_i \otimes \underline{e}_j$, the only surviving terms being the diagonal ones,

$$\mathbb{R}_{(\underline{e}_1)} = -\underline{e}_1 \otimes \underline{e}_1 + \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3.$$

Its trace is $R_{11} + R_{22} + R_{33} = -1 + 1 + 1 = 1$ and, the matrix being diagonal, its determinant is the product of the diagonal entries, $(-1)(1)(1) = -1$, matching the general results.

(ii) $\underline{n} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2)$. First confirm the hypothesis $\underline{n} \cdot \underline{n} = 1$:

$$|\underline{n}|^2 = \left(\frac{\sqrt{2}}{2}\right)^2 (1^2 + 1^2 + 0^2) = \frac{1}{2} \cdot 2 = 1.$$

The components are $n_1 = n_2 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$ and $n_3 = 0$, so the nonzero products are

$$n_1 n_1 = n_1 n_2 = n_2 n_1 = n_2 n_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2},$$

every product involving the index 3 vanishing because $n_3 = 0$. Hence

$$(n_i n_j) = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad 2(n_i n_j) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and $(\star_5)$ gives, subtracting entry by entry,

$$(R_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Reassembling $\mathbb{R}_{(\underline{n})} = R_{ij} \underline{e}_i \otimes \underline{e}_j$ with the three nonzero entries $R_{12} = R_{21} = -1$ and $R_{33} = 1$,

$$\mathbb{R}_{(\underline{n})} = -\underline{e}_1 \otimes \underline{e}_2 - \underline{e}_2 \otimes \underline{e}_1 + \underline{e}_3 \otimes \underline{e}_3.$$

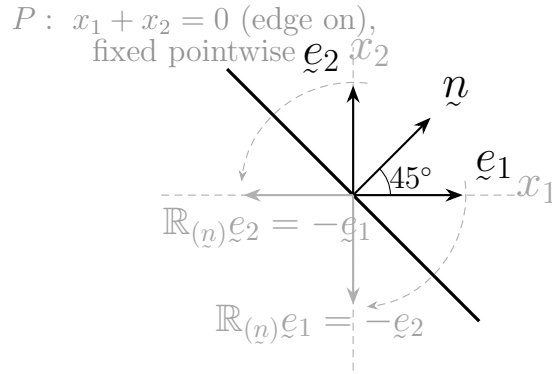


FIGURE P10.4. Case (ii): the mirror $x_1 + x_2 = 0$ seen edge on, normal at 45° . The axes are swapped and negated, which is why (R_{ij}) has zero diagonal and -1 off it.

It remains to verify that this matrix does what the general theory predicted. Compute the action on each basis vector from $\mathbb{R}_{(\underline{n})}\underline{e}_j = R_{ij}\underline{e}_i$, that is by reading the j th column and multiplying it against $\{\underline{e}_i\}$:

$$\mathbb{R}_{(\underline{n})}\underline{e}_1 = R_{11}\underline{e}_1 + R_{21}\underline{e}_2 + R_{31}\underline{e}_3 = 0 \underline{e}_1 + (-1)\underline{e}_2 + 0 \underline{e}_3 = -\underline{e}_2,$$

$$\mathbb{R}_{(\underline{n})}\underline{e}_2 = R_{12}\underline{e}_1 + R_{22}\underline{e}_2 + R_{32}\underline{e}_3 = (-1)\underline{e}_1 + 0 \underline{e}_2 + 0 \underline{e}_3 = -\underline{e}_1,$$

$$\mathbb{R}_{(\underline{n})}\underline{e}_3 = R_{13}\underline{e}_1 + R_{23}\underline{e}_2 + R_{33}\underline{e}_3 = 0 \underline{e}_1 + 0 \underline{e}_2 + 1 \underline{e}_3 = \underline{e}_3.$$

The normal is reversed. Using linearity and the three images just computed,

$$\mathbb{R}_{(\underline{n})}\underline{n} = \frac{\sqrt{2}}{2}(\mathbb{R}_{(\underline{n})}\underline{e}_1 + \mathbb{R}_{(\underline{n})}\underline{e}_2) = \frac{\sqrt{2}}{2}(-\underline{e}_2 - \underline{e}_1) = -\frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2) = -\underline{n}.$$

The plane is fixed pointwise. A basis of P is $\{\underline{e}_1 - \underline{e}_2, \underline{e}_3\}$; both really lie in P , since

$$(\underline{e}_1 - \underline{e}_2) \cdot \underline{n} = \frac{\sqrt{2}}{2}(1 - 1) = 0, \quad \underline{e}_3 \cdot \underline{n} = \frac{\sqrt{2}}{2}(0 + 0) = 0,$$

and both are fixed,

$$\mathbb{R}_{(\underline{n})}(\underline{e}_1 - \underline{e}_2) = \mathbb{R}_{(\underline{n})}\underline{e}_1 - \mathbb{R}_{(\underline{n})}\underline{e}_2 = -\underline{e}_2 - (-\underline{e}_1) = \underline{e}_1 - \underline{e}_2, \quad \mathbb{R}_{(\underline{n})}\underline{e}_3 = \underline{e}_3.$$

So the eigenvalues really are -1 once and $+1$ twice. *The involution, checked on the matrix.* Multiplying R by itself and writing out all nine entries,

$$\begin{aligned} R^2 &= \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} (0)(0) + (-1)(-1) + 0 & (0)(-1) + (-1)(0) + 0 & 0 \\ (-1)(0) + (0)(-1) + 0 & (-1)(-1) + (0)(0) + 0 & 0 \\ 0 & 0 & (1)(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \end{aligned}$$

so $R^2 = I$, and since $R^t = R$ this also confirms $R^t R = I$ entry by entry.

Trace and determinant. Summing the diagonal, $\text{tr } R = R_{11} + R_{22} + R_{33} = 0 + 0 + 1 = 1$; expanding the determinant along the third row, whose only nonzero entry is $R_{33} = 1$,

$$\det R = 1 \cdot \det \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = 1 \cdot [(0)(0) - (-1)(-1)] = -1.$$

Both agree with $I_1 = 1$ and $I_3 = -1$ found from $(\star_5)$, and with the spectrum $\{-1, +1, +1\}$. $\square$

PROBLEM 11 (Triple products; two induced tensors). (a) Prove $\underline{a} \times (\underline{b} \times \underline{c}) = (\underline{a} \cdot \underline{c})\underline{b} - (\underline{a} \cdot \underline{b})\underline{c}$ and $\underline{v} = (\underline{v} \cdot \underline{e})\underline{e} + \underline{e} \times (\underline{v} \times \underline{e})$ ($\underline{e}$ unit); (b) tensor for $f(\underline{v}) = \underline{e} \times (\underline{v} \times \underline{e})$; (c) tensor for $f(\underline{v}) = \underline{v} + \underline{a} \times \underline{v}$.

Solution. **(a)** Write the inner cross product first, then the outer one:

$$(\underline{b} \times \underline{c})_k = \varepsilon_{klm} b_l c_m, \quad [\underline{a} \times (\underline{b} \times \underline{c})]_i = \varepsilon_{ijk} a_j (\underline{b} \times \underline{c})_k = \varepsilon_{ijk} \varepsilon_{klm} a_j b_l c_m.$$

Contract the two symbols. Theorem 1.29 contracts on the last index, and $\varepsilon_{ijk} = \varepsilon_{kij}$ by cyclic shift, so

$$\varepsilon_{ijk}\varepsilon_{klm} = \varepsilon_{kij}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}.$$

Substituting and letting the deltas collapse the sums,

$$[\underline{a} \times (\underline{b} \times \underline{c})]_i = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})a_j b_l c_m = a_j c_j b_i - a_j b_j c_i = (\underline{a} \cdot \underline{c})b_i - (\underline{a} \cdot \underline{b})c_i.$$

Now put $\underline{a} = \underline{c} = \underline{e}$ with $|\underline{e}| = 1$, so $\underline{e} \cdot \underline{e} = 1$:

$$\underline{e} \times (\underline{v} \times \underline{e}) = (\underline{e} \cdot \underline{e})\underline{v} - (\underline{e} \cdot \underline{v})\underline{e} = \underline{v} - (\underline{v} \cdot \underline{e})\underline{e},$$

which rearranges to $\underline{v} = (\underline{v} \cdot \underline{e})\underline{e} + \underline{e} \times (\underline{v} \times \underline{e})$.

(b) From (a), $f(\underline{v}) = \underline{v} - (\underline{e} \cdot \underline{v})\underline{e}$. Both terms are linear in $\underline{v}$, and $(\underline{e} \cdot \underline{v})\underline{e} = (\underline{e} \otimes \underline{e})\underline{v}$, so

$$f(\underline{v}) = (\underline{I} - \underline{e} \otimes \underline{e})\underline{v}, \quad \underline{A} = \underline{I} - \underline{e} \otimes \underline{e} = \mathbb{P}_{(\underline{e})},$$

the projection of Problem 10 onto the plane perpendicular to $\underline{e}$. Check: $f(\underline{e}) = \underline{e} - \underline{e} = \underline{0}$ and $f(\underline{w}) = \underline{w}$ for $\underline{w} \perp \underline{e}$.

(c) $f(\underline{v}) = \underline{v} + \underline{a} \times \underline{v}$. In components, $(\underline{a} \times \underline{v})_i = \varepsilon_{ijk}a_j v_k$, so the second term is $\underline{\Omega}\underline{v}$ with $\Omega_{ik} = \varepsilon_{ijk}a_j = -\varepsilon_{ikj}a_j$, skew by Problem 16. Hence

$$\underline{A} = \underline{I} + \underline{\Omega}, \quad A_{ij} = \delta_{ij} - \varepsilon_{ijk}a_k.$$

Linearity of f is then visible, and $\det \underline{A} = 1 + |\underline{a}|^2 \neq 0$ by Problem 21(g), so $\underline{A}$ is invertible for every $\underline{a}$. $\square$

PROBLEM 12 (Two-point tensors). For orthonormal bases $\{\underline{e}_i\}$ of $\mathbb{E}^3$ and $\{\hat{\underline{E}}_A\}$ of $\hat{\mathbb{E}}^3$, show a linear $\underline{A} : \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ is $\underline{A} = A_{iB}\underline{e}_i \otimes \hat{\underline{E}}_B$ with $A_{iB} = \underline{e}_i \cdot \underline{A}\hat{\underline{E}}_B$.

Solution. *Step 1.* $\underline{A}\hat{\underline{E}}_B \in \mathbb{E}^3$, so resolve it on the spatial basis and name the coefficients:

$$\underline{A}\hat{\underline{E}}_B = (\underline{e}_i \cdot \underline{A}\hat{\underline{E}}_B)\underline{e}_i =: A_{iB}\underline{e}_i.$$

The dot product is legal, both arguments being in $\mathbb{E}^3$.

Step 2. For $\underline{V} = V_B \hat{\underline{E}}_B \in \hat{\mathbb{E}}^3$, by linearity,

$$\underline{A}\underline{V} = V_B \underline{A}\hat{\underline{E}}_B = A_{iB} V_B \underline{e}_i.$$

Step 3. The mixed dyad acts by $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$:

$$(\underline{e}_i \otimes \hat{\underline{E}}_B)\underline{V} = (\hat{\underline{E}}_B \cdot \underline{V})\underline{e}_i = V_B \underline{e}_i,$$

legal since $\hat{\underline{E}}_B, \underline{V} \in \hat{\mathbb{E}}^3$ while $\underline{e}_i \in \mathbb{E}^3$. Hence $(A_{iB}\underline{e}_i \otimes \hat{\underline{E}}_B)\underline{V} = A_{iB}V_B\underline{e}_i = \underline{A}\underline{V}$ for all $\underline{V}$, and

$$\underline{A} = A_{iB} \underline{e}_i \otimes \hat{\underline{E}}_B, \quad A_{iB} = \underline{e}_i \cdot \underline{A}\hat{\underline{E}}_B.$$

Step 4, consistency. Feed $\hat{\underline{E}}_C$ to the dyadic form, then dot with $\underline{e}_j$:

$$(A_{iB}\underline{e}_i \otimes \hat{\underline{E}}_B)\hat{\underline{E}}_C = A_{iB}\delta_{BC}\underline{e}_i = A_{iC}\underline{e}_i, \quad \underline{e}_j \cdot \underline{A}\hat{\underline{E}}_C = A_{iC}\delta_{ji} = A_{jC}.$$

□

REMARK 1.106 (which dots are legal).

$$\begin{aligned} \underline{e}_i \cdot \underline{A}\hat{\underline{E}}_B & \text{ defined, both in } \mathbb{E}^3, \\ \hat{\underline{E}}_B \cdot \underline{V} & \text{ defined, both in } \hat{\mathbb{E}}^3, \\ \underline{e}_i \cdot \hat{\underline{E}}_B & \text{ not defined,} \\ \underline{e}_i \otimes \hat{\underline{E}}_B & \text{ defined, a map } \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3. \end{aligned}$$

Also $\underline{A}^t = A_{iB}\hat{\underline{E}}_B \otimes \underline{e}_i$ maps $\mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, so $\underline{A} + \underline{A}^t$ is meaningless: a two-point tensor is neither symmetric nor skew. The object built here is $\underline{F} = F_{iA}\underline{e}_i \otimes \hat{\underline{E}}_A$ of (2.39).

PROBLEM 13 (Riesz representation for tensors). Show a linear $g : \text{Lin} \rightarrow \mathbb{R}$ is $g(\underline{V}) = \underline{A} \cdot \underline{V}$ for a unique $\underline{A}$.

Solution. Existence. Define the nine numbers

$$A_{ij} := g(\underline{e}_i \otimes \underline{e}_j), \quad \underline{A} := A_{ij} \underline{e}_i \otimes \underline{e}_j.$$

Resolve an arbitrary $\underline{V}$ on the same basis, $\underline{V} = V_{ij}\underline{e}_i \otimes \underline{e}_j$ with $V_{ij} = \underline{e}_i \cdot \underline{V} \underline{e}_j$. Since g is linear and the V_{ij} are scalars,

$$g(\underline{V}) = g(V_{ij}\underline{e}_i \otimes \underline{e}_j) = V_{ij} g(\underline{e}_i \otimes \underline{e}_j) = V_{ij} A_{ij} = \underline{A} \cdot \underline{V},$$

the last step being the definition $\underline{A} \cdot \underline{V} = A_{ij}V_{ij}$ of the tensor inner product.

Uniqueness. If $\underline{A} \cdot \underline{V} = \underline{A}' \cdot \underline{V}$ for every $\underline{V}$, then $(\underline{A} - \underline{A}') \cdot \underline{V} = 0$ for every $\underline{V}$. Choose $\underline{V} = \underline{A} - \underline{A}'$:

$$0 = (\underline{A} - \underline{A}') \cdot (\underline{A} - \underline{A}') = |\underline{A} - \underline{A}'|^2 \implies \underline{A} = \underline{A}'.$$

This is the tensor version of Problem 13's vector counterpart, and the proof is the same one twice: linearity gives existence through a basis, positive definiteness of the inner product gives uniqueness. $\square$

PROBLEM 14 (Symmetric-skew decomposition). $\underline{A} = \underline{S} + \underline{W}$, $\underline{S} = \frac{1}{2}(\underline{A} + \underline{A}^t)$, $\underline{W} = \frac{1}{2}(\underline{A} - \underline{A}^t)$. (a) uniqueness; (b) sym, skw are orthogonal subspaces, $\mathbb{E}^9 = \text{sym} \oplus \text{skw}$.

Solution. Existence. With $\underline{S} = \frac{1}{2}(\underline{A} + \underline{A}^t)$ and $\underline{W} = \frac{1}{2}(\underline{A} - \underline{A}^t)$,

$$\underline{S} + \underline{W} = \frac{1}{2}(\underline{A} + \underline{A}^t) + \frac{1}{2}(\underline{A} - \underline{A}^t) = \frac{1}{2}(\underline{A} + \underline{A}^t + \underline{A} - \underline{A}^t) = \frac{1}{2}(2\underline{A}) = \underline{A},$$

and, by $(\underline{A} + \underline{B})^t = \underline{A}^t + \underline{B}^t$ and $(\underline{A}^t)^t = \underline{A}$,

$$\begin{aligned} \underline{S}^t &= \frac{1}{2}(\underline{A}^t + (\underline{A}^t)^t) = \frac{1}{2}(\underline{A}^t + \underline{A}) = \underline{S}, \\ \underline{W}^t &= \frac{1}{2}(\underline{A}^t - \underline{A}) = -\frac{1}{2}(\underline{A} - \underline{A}^t) = -\underline{W}. \end{aligned}$$

(a) Uniqueness. Let $\underline{A} = \underline{S}' + \underline{W}'$ with $(\underline{S}')^t = \underline{S}'$, $(\underline{W}')^t = -\underline{W}'$. Transpose:

$$\underline{A}^t = (\underline{S}' + \underline{W}')^t = (\underline{S}')^t + (\underline{W}')^t = \underline{S}' - \underline{W}'.$$

Now add and subtract this from $\underline{A} = \underline{S}' + \underline{W}'$:

$$\begin{aligned} \underline{A} + \underline{A}^t &= 2\underline{S}' \implies \underline{S}' = \frac{1}{2}(\underline{A} + \underline{A}^t) = \underline{S}, \\ \underline{A} - \underline{A}^t &= 2\underline{W}' \implies \underline{W}' = \frac{1}{2}(\underline{A} - \underline{A}^t) = \underline{W}. \end{aligned}$$

(b) Orthogonal subspaces. Each is a subspace: $(\alpha \underline{S}_1 + \beta \underline{S}_2)^t = \alpha \underline{S}_1 + \beta \underline{S}_2$, and the same with a minus sign for skw. For orthogonality, with $\underline{S} \cdot \underline{W} = S_{ij} W_{ij}$,

$$\underline{S} \cdot \underline{W} = S_{ij} W_{ij} \stackrel{(1)}{=} S_{ji} W_{ij} \stackrel{(2)}{=} S_{ij} W_{ji} \stackrel{(3)}{=} -S_{ij} W_{ij} = -\underline{S} \cdot \underline{W},$$

with (1) $S_{ij} = S_{ji}$, (2) a relabelling, (3) $W_{ji} = -W_{ij}$. Hence $\underline{S} \cdot \underline{W} = 0$.

Dimensions. $\underline{S}$ is fixed by $S_{11}, S_{22}, S_{33}, S_{12}, S_{13}, S_{23}$, so $\dim \text{sym} = 6$. For $\underline{W}$, $W_{ii} = -W_{ii}$ gives a zero diagonal, leaving W_{12}, W_{13}, W_{23} , so $\dim \text{skw} = 3$. Their sum is 9, and they meet only in $\underline{O}$, since $\underline{A} = \underline{A}^t = -\underline{A}$ forces $2\underline{A} = \underline{O}$. Therefore

$$\mathbb{E}^9 = \text{sym} \oplus \text{skw}.$$

□

REMARK 1.107 (three uses). (i) $\underline{a} \cdot \underline{A}\underline{a} = \underline{a} \cdot \underline{S}\underline{a}$ by Problem 6(a), so positive definiteness (3.13) sees only $\underline{S}$. (ii) Chapter 4 splits $\underline{L} = \text{grad } \underline{v} = \underline{D} + \underline{W}$ into stretching and spin; part (a) is what makes them unique. (iii) By Problem 16 every skew $\underline{W}$ is $\underline{W}\underline{u} = \underline{\omega} \times \underline{u}$, which is why $\dim \text{skw} = 3$.

Expanding $\underline{A} \cdot \underline{A} = (\underline{S} + \underline{W}) \cdot (\underline{S} + \underline{W})$ and killing the cross terms by (b) gives $|\underline{A}|^2 = |\underline{S}|^2 + |\underline{W}|^2$: the splitting is an orthogonal projection, not merely a decomposition.

PROBLEM 15 (Spherical–deviatoric decomposition). $\underline{T} = \underline{A} + \alpha \underline{I}$, $\text{tr } \underline{A} = 0$, $3\alpha = \text{tr } \underline{T}$. (a) uniqueness; (b) Sph, Dev orthogonal, $\mathbb{E}^9 = \text{Sph} \oplus \text{Dev}$.

Solution. Existence. Put $\alpha = \frac{1}{3} \text{tr } \underline{T}$ and $\underline{A} = \underline{T} - \alpha \underline{I}$. Then $\underline{T} = \underline{A} + \alpha \underline{I}$ and, using $\text{tr } \underline{I} = \delta_{ii} = 3$,

$$\text{tr } \underline{A} = \text{tr } \underline{T} - \alpha \text{tr } \underline{I} = \text{tr } \underline{T} - \frac{1}{3}(\text{tr } \underline{T})(3) = 0.$$

(a) Uniqueness. Let $\underline{T} = \underline{B} + \beta \underline{I}$ with $\text{tr } \underline{B} = 0$. Take the trace:

$$\text{tr } \underline{T} = \text{tr } \underline{B} + \beta \text{tr } \underline{I} = 0 + 3\beta \implies \beta = \frac{1}{3} \text{tr } \underline{T} = \alpha,$$

and then $\underline{B} = \underline{T} - \beta \underline{I} = \underline{T} - \alpha \underline{I} = \underline{A}$.

(b) Orthogonal subspaces. Both are subspaces: Sph is the span of $\underline{\underline{I}}$, and tr is linear so Dev = $\{\underline{\underline{D}} : \text{tr } \underline{\underline{D}} = 0\}$ is its kernel. For $\lambda \underline{\underline{I}} \in \text{Sph}$ and $\underline{\underline{D}} \in \text{Dev}$, using $\underline{\underline{I}} \cdot \underline{\underline{D}} = \delta_{ij} D_{ij} = D_{ii}$,

$$(\lambda \underline{\underline{I}}) \cdot \underline{\underline{D}} = \lambda \text{tr } \underline{\underline{D}} = \lambda \cdot 0 = 0.$$

Dimensions: $\dim \text{Sph} = 1$; $\text{tr} : \mathbb{E}^9 \rightarrow \mathbb{R}$ is onto, so its kernel has $\dim \text{Dev} = 9 - 1 = 8$. Since $1 + 8 = 9$ and $\text{Sph} \cap \text{Dev} = \{\underline{\underline{O}}\}$, because $\lambda \underline{\underline{I}} \in \text{Dev}$ forces $3\lambda = 0$,

$$\mathbb{E}^9 = \text{Sph} \oplus \text{Dev}.$$

□

REMARK 1.108 (the two splittings are independent). Problems 14 and 15 split $\mathbb{E}^9$ two different ways, $\text{sym} \oplus \text{skw}$ and $\text{Sph} \oplus \text{Dev}$, and the two are compatible rather than competing: $\text{Sph} \subset \text{sym}$, so a single tensor decomposes as

$$\underline{\underline{T}} = \underbrace{\alpha \underline{\underline{I}}}_1 + \underbrace{(\text{sym } \underline{\underline{T}} - \alpha \underline{\underline{I}})}_5 + \underbrace{\text{skw } \underline{\underline{T}}}_3, \quad \alpha = \frac{1}{3} \text{tr } \underline{\underline{T}},$$

with dimensions $1 + 5 + 3 = 9$. In Chapter 3 the middle piece is what carries distortion at constant volume and the first is what carries volume change: there $\text{tr}(\ln \underline{\underline{U}}) = \ln J$ exactly, so the spherical part of the Hencky strain is $\frac{1}{3}(\ln J) \underline{\underline{I}}$.

PROBLEM 16 (Axial vector of a skew tensor). $f(\underline{\underline{v}}) = \underline{\underline{\omega}} \times \underline{\underline{v}} = \underline{\underline{\Omega}} \underline{\underline{v}}$. (a) $\Omega_{ij} = \varepsilon_{jik} \omega_k$, so $\underline{\underline{\Omega}} \in \text{skw}$; (b) $\omega_k = \frac{1}{2} \Omega_{ij} \varepsilon_{jik}$.

Solution. **(a)** Write both sides of $\underline{\underline{\Omega}} \underline{\underline{v}} = \underline{\underline{\omega}} \times \underline{\underline{v}}$ in components:

$$(\underline{\underline{\Omega}} \underline{\underline{v}})_i = \Omega_{ij} v_j, \quad (\underline{\underline{\omega}} \times \underline{\underline{v}})_i = \varepsilon_{ikj} \omega_k v_j,$$

the second by Proposition 1.24 with the summation indices named so that v carries j in both. Equating for every v ,

$$\Omega_{ij} = \varepsilon_{ikj} \omega_k.$$

Now $\varepsilon_{ikj} = -\varepsilon_{ijk} = \varepsilon_{jik}$: one swap gives a minus, a second swap gives it back. So $\Omega_{ij} = \varepsilon_{jik}\omega_k$, and interchanging i, j ,

$$\Omega_{ji} = \varepsilon_{ijk}\omega_k = -\varepsilon_{jik}\omega_k = -\Omega_{ij},$$

i.e. $\Omega \in \text{skw}$.

(b) Substitute $\Omega_{ij} = \varepsilon_{jil}\omega_l$ and contract:

$$\frac{1}{2} \Omega_{ij} \varepsilon_{jik} = \frac{1}{2} \varepsilon_{jil} \varepsilon_{jik} \omega_l = \frac{1}{2} (2\delta_{lk}) \omega_l = \omega_k,$$

the middle step by Theorem 1.29: $\varepsilon_{jil} \varepsilon_{jik} = \varepsilon_{ilj} \varepsilon_{ikj} = 2\delta_{lk}$ after a cyclic shift brings the repeated pair to the front.

So $\omega \mapsto \Omega$ and $\Omega \mapsto \omega$ invert one another: skw and $\mathbb{E}^3$ are in one-to-one correspondence, which is why $\dim \text{skw} = 3$ in Problem 14. $\square$

PROBLEM 17 (Identities for the tensor product). Verify (a) $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$; (b) $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$; (c) $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$; (d) $\underline{A} = (\underline{A}\underline{e}_i) \otimes \underline{e}_i = \underline{e}_i \otimes (\underline{A}^t \underline{e}_i)$.

Solution. Throughout, the definition is $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$: the second slot dots, the first survives.

(a) Test against arbitrary $\underline{u}, \underline{v}$ and use the symmetry of the dot product:

$$\underline{u} \cdot (\underline{a} \otimes \underline{b})\underline{v} = (\underline{b} \cdot \underline{v})(\underline{a} \cdot \underline{u}) = (\underline{a} \cdot \underline{u})(\underline{b} \cdot \underline{v}) = \underline{v} \cdot (\underline{b} \otimes \underline{a})\underline{u}.$$

By the definition of the transpose this says $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$.

(b) Apply both sides to arbitrary $\underline{v}$ and use linearity of $\underline{A}$:

$$[\underline{A}(\underline{a} \otimes \underline{b})]\underline{v} = \underline{A}[(\underline{b} \cdot \underline{v})\underline{a}] = (\underline{b} \cdot \underline{v}) \underline{A}\underline{a} = [(\underline{A}\underline{a}) \otimes \underline{b}]\underline{v}.$$

(c) Same test, now moving $\underline{A}$ across the dot product:

$$[(\underline{a} \otimes \underline{b})\underline{A}]\underline{v} = (\underline{b} \cdot \underline{A}\underline{v}) \underline{a} = (\underline{A}^t \underline{b} \cdot \underline{v}) \underline{a} = [\underline{a} \otimes (\underline{A}^t \underline{b})]\underline{v}.$$

So post-multiplication touches only the second slot, pre-multiplication only the first.

(d) With $\underline{A} = A_{ij}\underline{e}_i \otimes \underline{e}_j$ one has $\underline{A}\underline{e}_i = A_{ki}\underline{e}_k$ and $\underline{A}^t\underline{e}_i = A_{ik}\underline{e}_k$. Hence, summing on i ,

$$(\underline{A}\underline{e}_i) \otimes \underline{e}_i = A_{ki}\underline{e}_k \otimes \underline{e}_i = \underline{A}, \quad \underline{e}_i \otimes (\underline{A}^t\underline{e}_i) = A_{ik}\underline{e}_i \otimes \underline{e}_k = \underline{A},$$

the second after renaming the two dummy indices. Both are (3.11) with $\underline{A}$ inserted on one side or the other. $\square$

PROBLEM 18 (Transpose of the inverse). Show $(\underline{A}^t)^{-1} = (\underline{A}^{-1})^t =: \underline{A}^{-t}$.

Solution. Transpose the two defining relations of $\underline{A}^{-1}$, using $(\underline{B}\underline{C})^t = \underline{C}^t\underline{B}^t$ from Problem 20 and $\underline{I}^t = \underline{I}$:

$$\underline{A}\underline{A}^{-1} = \underline{I} \implies (\underline{A}^{-1})^t \underline{A}^t = \underline{I}, \quad \underline{A}^{-1}\underline{A} = \underline{I} \implies \underline{A}^t(\underline{A}^{-1})^t = \underline{I}.$$

So $(\underline{A}^{-1})^t$ is a two-sided inverse of $\underline{A}^t$, and inverses are unique: if $\underline{P}\underline{A}^t = \underline{I} = \underline{A}^t\underline{Q}$ then $\underline{P} = \underline{P}(\underline{A}^t\underline{Q}) = (\underline{P}\underline{A}^t)\underline{Q} = \underline{Q}$. Hence $(\underline{A}^t)^{-1} = (\underline{A}^{-1})^t$, and the common value may be written $\underline{A}^{-t}$ without ambiguity. $\square$

PROBLEM 19 (Inverse of a product). $(\underline{B}\underline{A})^{-1} = \underline{A}^{-1}\underline{B}^{-1}$.

Solution. Multiply both ways, cancelling the inner pair first:

$$(\underline{A}^{-1}\underline{B}^{-1})(\underline{B}\underline{A}) = \underline{A}^{-1}(\underline{B}^{-1}\underline{B})\underline{A} = \underline{A}^{-1}\underline{I}\underline{A} = \underline{A}^{-1}\underline{A} = \underline{I},$$

$$(\underline{B}\underline{A})(\underline{A}^{-1}\underline{B}^{-1}) = \underline{B}(\underline{A}\underline{A}^{-1})\underline{B}^{-1} = \underline{B}\underline{I}\underline{B}^{-1} = \underline{B}\underline{B}^{-1} = \underline{I}.$$

By uniqueness of inverses, $(\underline{B}\underline{A})^{-1} = \underline{A}^{-1}\underline{B}^{-1}$. The order reverses, as it must: the last map applied is the first to be undone. $\square$

PROBLEM 20 (Transpose of a product). $(\underline{B}\underline{A})^t = \underline{A}^t\underline{B}^t$.

Solution. Move the two factors across the dot product one at a time, each by the definition $\underline{u} \cdot \underline{C}\underline{v} = \underline{C}^t\underline{u} \cdot \underline{v}$:

$$\underline{u} \cdot (\underline{B}\underline{A})\underline{v} = \underline{u} \cdot \underline{B}(\underline{A}\underline{v}) = \underline{B}^t\underline{u} \cdot \underline{A}\underline{v} = \underline{A}^t\underline{B}^t\underline{u} \cdot \underline{v} = \underline{u} \cdot (\underline{A}^t\underline{B}^t)\underline{v}.$$

Comparing the first and last for all $\underline{u}, \underline{v}$, and using that a tensor is determined by its bilinear form, $(\underline{B}\underline{A})^t = \underline{A}^t\underline{B}^t$. Again the order reverses. $\square$

PROBLEM 21 (Determinant identities via box products). Verify (a) $\det(\alpha \underline{A}) = \alpha^3 \det \underline{A}$; (b) $\det(\underline{A}\underline{B}) = \det \underline{A} \det \underline{B}$; (c) $\det \underline{I} = 1$; (d) $\det(\underline{A}^{-1}) = (\det \underline{A})^{-1}$; (e) $\det(\underline{a} \otimes \underline{b}) = 0$; (f) $\det(\underline{I} + \underline{a} \otimes \underline{b}) = 1 + \underline{a} \cdot \underline{b}$; (g) $\det(\underline{A} + \underline{a} \otimes \underline{b}) = \det \underline{A} (1 + \underline{a} \cdot \underline{A}^{-t} \underline{b})$.

Solution. The master identity is $\det \underline{A} [\underline{a}, \underline{b}, \underline{c}] = [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}]$ for all $\underline{a}, \underline{b}, \underline{c}$ (Definition 1.49₃). Every part below follows from it.

(a)

$$\begin{aligned} \det(\alpha \underline{A}) [\underline{a}, \underline{b}, \underline{c}] &= [(\alpha \underline{A})\underline{a}, (\alpha \underline{A})\underline{b}, (\alpha \underline{A})\underline{c}] \\ &= \alpha^3 [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}] = \alpha^3 \det \underline{A} [\underline{a}, \underline{b}, \underline{c}]; \end{aligned}$$

cancel $[\underline{a}, \underline{b}, \underline{c}] \neq 0$.

(b)

$$\begin{aligned} [(\underline{A}\underline{B})\underline{a}, (\underline{A}\underline{B})\underline{b}, (\underline{A}\underline{B})\underline{c}] &= [\underline{A}(\underline{B}\underline{a}), \underline{A}(\underline{B}\underline{b}), \underline{A}(\underline{B}\underline{c})] \\ &= \det \underline{A} [\underline{B}\underline{a}, \underline{B}\underline{b}, \underline{B}\underline{c}] \\ &= \det \underline{A} \det \underline{B} [\underline{a}, \underline{b}, \underline{c}]. \end{aligned}$$

(c) $\det \underline{I} [\underline{a}, \underline{b}, \underline{c}] = [\underline{I}\underline{a}, \underline{I}\underline{b}, \underline{I}\underline{c}] = [\underline{a}, \underline{b}, \underline{c}]$, so $\det \underline{I} = 1$.

(d) From (b) and (c), $\det \underline{A} \det(\underline{A}^{-1}) = \det(\underline{A}\underline{A}^{-1}) = \det \underline{I} = 1$.

(e) $(\underline{a} \otimes \underline{b})\underline{u} = (\underline{b} \cdot \underline{u})\underline{a}$: each image is a scalar multiple of $\underline{a}$. Hence $[(\underline{a} \otimes \underline{b})\underline{e}_1, (\underline{a} \otimes \underline{b})\underline{e}_2, (\underline{a} \otimes \underline{b})\underline{e}_3] = [b_1\underline{a}, b_2\underline{a}, b_3\underline{a}]$. A box product with two parallel entries vanishes (alternation), so $\det(\underline{a} \otimes \underline{b}) = 0$.

(f) Set $\underline{T} = \underline{I} + \underline{a} \otimes \underline{b}$. Then $\underline{T}\underline{e}_i = \underline{e}_i + b_i\underline{a}$, and

$$\det \underline{T} = [(\underline{e}_1 + b_1\underline{a}), (\underline{e}_2 + b_2\underline{a}), (\underline{e}_3 + b_3\underline{a})].$$

Expand multilinearly: the box product is trilinear, so there are $2^3 = 8$ terms, classified by how many of the three factors contribute $b_i\underline{a}$:

$$\begin{aligned} &\underbrace{[\underline{e}_1, \underline{e}_2, \underline{e}_3]}_{=1} + b_1[\underline{a}, \underline{e}_2, \underline{e}_3] + b_2[\underline{e}_1, \underline{a}, \underline{e}_3] + b_3[\underline{e}_1, \underline{e}_2, \underline{a}] \\ &\quad + b_1b_2\underbrace{[\underline{a}, \underline{a}, \underline{e}_3]}_{=0} + b_1b_3\underbrace{[\underline{a}, \underline{e}_2, \underline{a}]}_{=0} + b_2b_3\underbrace{[\underline{e}_1, \underline{a}, \underline{a}]}_{=0} + b_1b_2b_3\underbrace{[\underline{a}, \underline{a}, \underline{a}]}_{=0}. \end{aligned}$$

Every term with $\underline{a}$ appearing twice or thrice vanishes by alternation. The surviving terms are: $1 + b_1a_1 + b_2a_2 + b_3a_3 = 1 + b_i a_i = 1 + \underline{a} \cdot \underline{b}$.

(g) For invertible $\underline{A}$, factor: $\underline{A} + \underline{a} \otimes \underline{b} = \underline{A}(\underline{I} + \underline{A}^{-1}(\underline{a} \otimes \underline{b})) = \underline{A}(\underline{I} + (\underline{A}^{-1}\underline{a}) \otimes \underline{b})$, where Lemma 1.38 gives $\underline{A}^{-1}(\underline{a} \otimes \underline{b}) = (\underline{A}^{-1}\underline{a}) \otimes \underline{b}$. Apply (b), then (f):

$$\det(\underline{A} + \underline{a} \otimes \underline{b}) = \det \underline{A} \det(\underline{I} + (\underline{A}^{-1}\underline{a}) \otimes \underline{b}) = \det \underline{A} (1 + \underline{A}^{-1}\underline{a} \cdot \underline{b}).$$

Finally, $\underline{A}^{-1}\underline{a} \cdot \underline{b} = \underline{a} \cdot (\underline{A}^{-1})^t \underline{b} = \underline{a} \cdot \underline{A}^{-t} \underline{b}$, using the definition of the transpose. $\square$

PROBLEM 22 (Cofactor identities). Let $\underline{A}^*$ satisfy $\underline{A}^*(\underline{a} \times \underline{b}) = \underline{A}\underline{a} \times \underline{A}\underline{b}$. Show: (a) $I_2 = \text{tr } \underline{A}^*$; (b) $I_2 = \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)]$; (c) $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$, hence $\underline{A}^* = (\det \underline{A}) \underline{A}^{-t}$; (d) recover $\det \underline{A} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{lmn} A_{il} A_{jm} A_{kn}$.

Solution. Step 0: is $\underline{A}^$ even well defined?* Definition 1.54 prescribes the value of $\underline{A}^*$ only on vectors that are presented to us in the form $\underline{a} \times \underline{b}$. Before the symbol may be used at all, two questions must be answered. Does the prescription reach every vector, so that at most one tensor can satisfy it? And does a tensor satisfying it exist, given that the same vector can be written as a cross product in infinitely many ways, so that the prescription might contradict itself?

Uniqueness. Every vector is a cross product. If $\underline{v} = \underline{0}$, then $\underline{v} = \underline{a} \times \underline{a}$ for any $\underline{a}$, since $\underline{a} \times \underline{a} = \underline{0}$. If $\underline{v} \neq \underline{0}$, set $\underline{u}_3 = \underline{v}/|\underline{v}|$, a unit vector, and complete it to a right-handed orthonormal triad $\{\underline{u}_1, \underline{u}_2, \underline{u}_3\}$, so that $\underline{u}_1 \times \underline{u}_2 = \underline{u}_3$. Then

$$\underline{u}_1 \times (|\underline{v}| \underline{u}_2) = |\underline{v}| (\underline{u}_1 \times \underline{u}_2) = |\underline{v}| \underline{u}_3 = \underline{v}.$$

So the defining relation does prescribe $\underline{A}^* \underline{v}$ for every $\underline{v}$, and two tensors obeying it take the same value on every vector, hence are the same tensor.

Existence. We produce $\underline{A}^*$ explicitly and verify the definition. Put

$$A_{mi}^* := \frac{1}{2} \varepsilon_{mjk} \varepsilon_{ipq} A_{jp} A_{kq}, \quad \underline{A}^* := A_{mi}^* \underline{e}_m \otimes \underline{e}_i. \quad (*)$$

This is a tensor by construction, being a linear combination of dyads. Now compute its action on $\underline{a} \times \underline{b}$, whose components are $(\underline{a} \times \underline{b})_i =$

$\varepsilon_{irs}a_rb_s$ by Proposition 1.24:

$$\begin{aligned} [\tilde{A}^*(\underline{a} \times \underline{b})]_m &= A_{mi}^* (\underline{a} \times \underline{b})_i \\ &= \frac{1}{2} \varepsilon_{mjk} \varepsilon_{ipq} A_{jp} A_{kq} \varepsilon_{irs} a_r b_s \\ &= \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} (\varepsilon_{ipq} \varepsilon_{irs}) a_r b_s. \end{aligned}$$

The bracket is contracted on its *first* index, whereas Theorem 1.29 contracts on the last; the two are the same statement because ε is unchanged by a cyclic shift of its indices, $\varepsilon_{ipq} = \varepsilon_{pqi}$ and $\varepsilon_{irs} = \varepsilon_{rsi}$, so

$$\varepsilon_{ipq} \varepsilon_{irs} = \varepsilon_{pqi} \varepsilon_{rsi} = \delta_{pr} \delta_{qs} - \delta_{ps} \delta_{qr}.$$

Substituting,

$$\begin{aligned} [\tilde{A}^*(\underline{a} \times \underline{b})]_m &= \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} (\delta_{pr} \delta_{qs} - \delta_{ps} \delta_{qr}) a_r b_s \\ &= \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_p b_q - \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_q b_p, \end{aligned}$$

the deltas having collapsed the sums by forcing $r = p$, $s = q$ in the first term and $s = p$, $r = q$ in the second. The two terms are in fact equal. In the second, first interchange the *names* of the summation indices p and q , which changes nothing since both are summed over 1, 2, 3:

$$-\frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_q b_p = -\frac{1}{2} \varepsilon_{mjk} A_{jq} A_{kp} a_p b_q;$$

then interchange the names of the summation indices j and k :

$$-\frac{1}{2} \varepsilon_{mjk} A_{jq} A_{kp} a_p b_q = -\frac{1}{2} \varepsilon_{mkj} A_{kq} A_{jp} a_p b_q = +\frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_p b_q,$$

the last equality using $\varepsilon_{mkj} = -\varepsilon_{mjk}$. Hence the two halves add rather than cancel, and

$$[\tilde{A}^*(\underline{a} \times \underline{b})]_m = \varepsilon_{mjk} (A_{jp} a_p) (A_{kq} b_q) = \varepsilon_{mjk} (\underline{Aa})_j (\underline{Ab})_k = (\underline{Aa} \times \underline{Ab})_m.$$

So $(*)_1$ does satisfy Definition 1.54, and by the uniqueness just proved it is *the* cofactor. Both the object and its components are now in hand.

(a) $I_2(\underline{A}) = I_1(\underline{A}^*)$. By Theorem 1.51, $I_1(\underline{A}^*) = \text{tr } \underline{A}^* = A_{11}^* + A_{22}^* + A_{33}^*$ with $A_{ii}^* = \underline{e}_i \cdot \underline{A}^* \underline{e}_i$ (no sum). For a right-handed orthonormal basis, $\underline{e}_1 = \underline{e}_2 \times \underline{e}_3$, $\underline{e}_2 = \underline{e}_3 \times \underline{e}_1$, $\underline{e}_3 = \underline{e}_1 \times \underline{e}_2$, so the definition of $\underline{A}^*$ evaluates

on each basis vector separately. The three computations, written out in full rather than left to a cyclic instruction:

$$\begin{aligned} \underline{A}^* \underline{e}_1 &= \underline{A}^*(\underline{e}_2 \times \underline{e}_3) = \underline{A}\underline{e}_2 \times \underline{A}\underline{e}_3, & \underline{A}_{11}^* &= \underline{e}_1 \cdot \underline{A}\underline{e}_2 \times \underline{A}\underline{e}_3 = [\underline{e}_1, \underline{A}\underline{e}_2, \underline{A}\underline{e}_3], \\ \underline{A}^* \underline{e}_2 &= \underline{A}^*(\underline{e}_3 \times \underline{e}_1) = \underline{A}\underline{e}_3 \times \underline{A}\underline{e}_1, & \underline{A}_{22}^* &= \underline{e}_2 \cdot \underline{A}\underline{e}_3 \times \underline{A}\underline{e}_1 = [\underline{e}_2, \underline{A}\underline{e}_3, \underline{A}\underline{e}_1], \\ \underline{A}^* \underline{e}_3 &= \underline{A}^*(\underline{e}_1 \times \underline{e}_2) = \underline{A}\underline{e}_1 \times \underline{A}\underline{e}_2, & \underline{A}_{33}^* &= \underline{e}_3 \cdot \underline{A}\underline{e}_1 \times \underline{A}\underline{e}_2 = [\underline{e}_3, \underline{A}\underline{e}_1, \underline{A}\underline{e}_2]. \end{aligned}$$

The box product is unchanged by a cyclic permutation of its three slots, $[\underline{u}, \underline{v}, \underline{w}] = [\underline{w}, \underline{u}, \underline{v}]$, because a cyclic permutation is two transpositions and each transposition reverses the sign. Applying it to the second and third lines so that all three entries are ordered as in Definition 1.49₂:

$$[\underline{e}_2, \underline{A}\underline{e}_3, \underline{A}\underline{e}_1] = [\underline{A}\underline{e}_1, \underline{e}_2, \underline{A}\underline{e}_3], \quad [\underline{e}_3, \underline{A}\underline{e}_1, \underline{A}\underline{e}_2] = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{e}_3].$$

Adding the three diagonal entries,

$$\text{tr } \underline{A}^* = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{e}_3] + [\underline{A}\underline{e}_1, \underline{e}_2, \underline{A}\underline{e}_3] + [\underline{e}_1, \underline{A}\underline{e}_2, \underline{A}\underline{e}_3].$$

The right side is exactly Definition 1.49₂ evaluated at $(\underline{a}, \underline{b}, \underline{c}) = (\underline{e}_1, \underline{e}_2, \underline{e}_3)$, for which $[\underline{e}_1, \underline{e}_2, \underline{e}_3] = 1$. Hence $\text{tr } \underline{A}^* = I_2(\underline{A})$, which is the assertion.

(b) $I_2(\underline{A}) = \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)]$. By (a) it suffices to contract $(*_1)$ on $m = i$:

$$I_2(\underline{A}) = \text{tr } \underline{A}^* = \underline{A}_{ii}^* = \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ipq} \underline{A}_{jp} \underline{A}_{kq}.$$

Again shifting cyclically to expose the contraction pattern of Theorem 1.29, $\varepsilon_{ijk} \varepsilon_{ipq} = \varepsilon_{jki} \varepsilon_{pqi} = \delta_{jp} \delta_{kq} - \delta_{jq} \delta_{kp}$, so

$$I_2(\underline{A}) = \frac{1}{2}(\delta_{jp} \delta_{kq} \underline{A}_{jp} \underline{A}_{kq} - \delta_{jq} \delta_{kp} \underline{A}_{jp} \underline{A}_{kq}).$$

Take the two pieces one at a time. In the first, δ_{jp} forces $p = j$ and δ_{kq} forces $q = k$, leaving

$$\delta_{jp} \delta_{kq} \underline{A}_{jp} \underline{A}_{kq} = \underline{A}_{jj} \underline{A}_{kk} = (\text{tr } \underline{A})(\text{tr } \underline{A}) = (\text{tr } \underline{A})^2.$$

In the second, δ_{jq} forces $q = j$ and δ_{kp} forces $p = k$, leaving

$$\delta_{jq} \delta_{kp} \underline{A}_{jp} \underline{A}_{kq} = \underline{A}_{jk} \underline{A}_{kj}.$$

This last expression is a trace: the components of $\underline{\underline{A}}^2 = \underline{\underline{A}}\underline{\underline{A}}$ are $(\underline{\underline{A}}^2)_{jl} = A_{jk}A_{kl}$, so setting $l = j$ and summing gives $(\underline{\underline{A}}^2)_{jj} = A_{jk}A_{kj} = \text{tr}(\underline{\underline{A}}^2)$. Therefore

$$I_2(\underline{\underline{A}}) = \frac{1}{2}[(\text{tr } \underline{\underline{A}})^2 - \text{tr}(\underline{\underline{A}}^2)].$$

(c) $\underline{\underline{A}}^t \underline{\underline{A}}^* = (\det \underline{\underline{A}}) \underline{\underline{I}}$. Two independent proofs are given, the first free of components.

Without coordinates. Let $\underline{\underline{a}}, \underline{\underline{b}}, \underline{\underline{c}}$ be arbitrary and set $\underline{\underline{v}} = \underline{\underline{a}} \times \underline{\underline{b}}$. Using the definition of the transpose, $\underline{\underline{c}} \cdot \underline{\underline{A}}^t \underline{\underline{w}} = \underline{\underline{A}} \underline{\underline{c}} \cdot \underline{\underline{w}}$, and then the definition of the cofactor,

$$\begin{aligned} \underline{\underline{c}} \cdot \underline{\underline{A}}^t \underline{\underline{A}}^* \underline{\underline{v}} &= \underline{\underline{A}} \underline{\underline{c}} \cdot \underline{\underline{A}}^*(\underline{\underline{a}} \times \underline{\underline{b}}) \\ &= \underline{\underline{A}} \underline{\underline{c}} \cdot \underline{\underline{A}} \underline{\underline{a}} \times \underline{\underline{A}} \underline{\underline{b}} \\ &= [\underline{\underline{A}} \underline{\underline{c}}, \underline{\underline{A}} \underline{\underline{a}}, \underline{\underline{A}} \underline{\underline{b}}] \\ &= [\underline{\underline{A}} \underline{\underline{a}}, \underline{\underline{A}} \underline{\underline{b}}, \underline{\underline{A}} \underline{\underline{c}}], \end{aligned}$$

the last step being the cyclic shift again. By Definition 1.49₃ together with Theorem 1.52, which identify I_3 with $\det$,

$$[\underline{\underline{A}} \underline{\underline{a}}, \underline{\underline{A}} \underline{\underline{b}}, \underline{\underline{A}} \underline{\underline{c}}] = (\det \underline{\underline{A}}) [\underline{\underline{a}}, \underline{\underline{b}}, \underline{\underline{c}}] = (\det \underline{\underline{A}}) \underline{\underline{c}} \cdot \underline{\underline{a}} \times \underline{\underline{b}} = (\det \underline{\underline{A}}) \underline{\underline{c}} \cdot \underline{\underline{v}},$$

where the middle equality is one more cyclic shift, $[\underline{\underline{a}}, \underline{\underline{b}}, \underline{\underline{c}}] = [\underline{\underline{c}}, \underline{\underline{a}}, \underline{\underline{b}}] = \underline{\underline{c}} \cdot \underline{\underline{a}} \times \underline{\underline{b}}$. Comparing the two displays,

$$\underline{\underline{c}} \cdot [\underline{\underline{A}}^t \underline{\underline{A}}^* - (\det \underline{\underline{A}}) \underline{\underline{I}}] \underline{\underline{v}} = 0 \quad \text{for every } \underline{\underline{c}}.$$

A vector orthogonal to every vector is the zero vector, so $[\underline{\underline{A}}^t \underline{\underline{A}}^* - (\det \underline{\underline{A}}) \underline{\underline{I}}] \underline{\underline{v}} = \underline{\underline{0}}$ for every $\underline{\underline{v}}$ of the form $\underline{\underline{a}} \times \underline{\underline{b}}$, that is, by Step 0, for every $\underline{\underline{v}}$. A tensor sending every vector to $\underline{\underline{0}}$ is $\underline{\underline{O}}$, so $\underline{\underline{A}}^t \underline{\underline{A}}^* = (\det \underline{\underline{A}}) \underline{\underline{I}}$.

In components. Start from $(\underline{\underline{A}}^t)_{ik} = A_{ki}$ and $(*)_1$:

$$(\underline{\underline{A}}^t \underline{\underline{A}}^*)_{ij} = (\underline{\underline{A}}^t)_{ik} A_{kj}^* = A_{ki} A_{kj}^* = \frac{1}{2} \varepsilon_{kmn} \varepsilon_{j pq} A_{ki} A_{mp} A_{nq}.$$

The triple sum $\varepsilon_{kmn} A_{ki} A_{mp} A_{nq}$ is precisely Proposition 1.28 read with the matrix (A_{ij}) and columns i, p, q :

$$\varepsilon_{kmn} A_{ki} A_{mp} A_{nq} = \varepsilon_{ipq} \det \underline{\underline{A}}.$$

(For completeness, here is why. The left side changes sign whenever two of i, p, q are interchanged: swapping i and p gives $\varepsilon_{kmn}A_{kp}A_{mi}A_{nq}$, and renaming the dummies $k \leftrightarrow m$ turns this into $\varepsilon_{mkn}A_{mp}A_{ki}A_{nq} = -\varepsilon_{kmn}A_{ki}A_{mp}A_{nq}$; the other two swaps are identical in form. Being trilinear and alternating in the columns, it equals a fixed multiple of ε_{ipq} by Lemma 1.25, and evaluating at $(i, p, q) = (1, 2, 3)$ identifies that multiple as $\varepsilon_{kmn}A_{k1}A_{m2}A_{n3} = [\underline{A}e_1, \underline{A}e_2, \underline{A}e_3] = \det \underline{A}$.) Substituting and using $\varepsilon_{j pq}\varepsilon_{ipq} = 2\delta_{ij}$ from Theorem 1.29,

$$(\underline{A}^t \underline{A}^*)_{ij} = \frac{1}{2} (\det \underline{A}) \varepsilon_{j pq} \varepsilon_{ipq} = \frac{1}{2} (\det \underline{A}) 2\delta_{ij} = (\det \underline{A}) \delta_{ij},$$

which is the component statement of $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$.

The consequence. Suppose now $\underline{A}$ is invertible, equivalently $\det \underline{A} \neq 0$ by Theorem 1.52. Then $\underline{A}^t$ is invertible too, and multiplying $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$ on the left by $(\underline{A}^t)^{-1}$ gives

$$\underline{A}^* = (\det \underline{A}) (\underline{A}^t)^{-1} = (\det \underline{A}) \underline{A}^{-t},$$

the last equality being $(\underline{A}^t)^{-1} = (\underline{A}^{-1})^t$, proved in Problem 18, which is what licenses the single symbol $\underline{A}^{-t}$.

(d) *The contracted determinant formula.* Take the trace of (c). On the right, $\text{tr}[(\det \underline{A}) \underline{I}] = (\det \underline{A}) \delta_{ii} = 3 \det \underline{A}$, since $\delta_{ii} = 1 + 1 + 1 = 3$. On the left, setting $j = i$ in the component computation just performed,

$$\text{tr}(\underline{A}^t \underline{A}^*) = (\underline{A}^t \underline{A}^*)_{ii} = A_{ki} A_{ki}^* = \frac{1}{2} \varepsilon_{kmn} \varepsilon_{ipq} A_{ki} A_{mp} A_{nq}.$$

Equating,

$$3 \det \underline{A} = \frac{1}{2} \varepsilon_{kmn} \varepsilon_{ipq} A_{ki} A_{mp} A_{nq}, \quad \text{so} \quad \det \underline{A} = \frac{1}{6} \varepsilon_{kmn} \varepsilon_{ipq} A_{ki} A_{mp} A_{nq}.$$

Renaming the summation indices, $k \rightarrow i, m \rightarrow j, n \rightarrow k$ in the first ε and $i \rightarrow l, p \rightarrow m, q \rightarrow n$ in the second, so that the three factors read A_{il}, A_{jm}, A_{kn} ,

$$\det \underline{A} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{lmn} A_{il} A_{jm} A_{kn},$$

in agreement with Corollary 1.30, which reached the same formula by contracting Proposition 1.28 directly. The two routes are genuinely different: there the $\frac{1}{6}$ came from $\varepsilon_{mnp} \varepsilon_{mnp} = 6$, here from $\frac{1}{2} \cdot 3$, the $\frac{1}{2}$ being built into the cofactor and the 3 being $\text{tr } \underline{I}$.

Checks. Take $\underline{A} = \underline{I}$ first. Then $\underline{I}\underline{a} \times \underline{I}\underline{b} = \underline{a} \times \underline{b}$, so $\underline{I}^* = \underline{I}$ and $\text{tr } \underline{I}^* = 3$; part (b) gives $\frac{1}{2}[3^2 - 3] = \frac{1}{2} \cdot 6 = 3$, and part (c) gives $\underline{I}^t \underline{I}^* = \underline{I} = (\det \underline{I})\underline{I}$ since $\det \underline{I} = 1$. Part (d) gives $\frac{1}{6}\varepsilon_{ijk}\varepsilon_{lmn}\delta_{il}\delta_{jm}\delta_{kn} = \frac{1}{6}\varepsilon_{ijk}\varepsilon_{ijk} = \frac{1}{6} \cdot 6 = 1$.

Take next $\underline{A} = \alpha \underline{e}_1 \otimes \underline{e}_1 + \beta \underline{e}_2 \otimes \underline{e}_2 + \gamma \underline{e}_3 \otimes \underline{e}_3$, so $\underline{A}\underline{e}_1 = \alpha \underline{e}_1$, $\underline{A}\underline{e}_2 = \beta \underline{e}_2$, $\underline{A}\underline{e}_3 = \gamma \underline{e}_3$. From the definition,

$$\begin{aligned}\underline{A}^* \underline{e}_1 &= \underline{A}\underline{e}_2 \times \underline{A}\underline{e}_3 = \beta\gamma (\underline{e}_2 \times \underline{e}_3) = \beta\gamma \underline{e}_1, \\ \underline{A}^* \underline{e}_2 &= \underline{A}\underline{e}_3 \times \underline{A}\underline{e}_1 = \gamma\alpha (\underline{e}_3 \times \underline{e}_1) = \gamma\alpha \underline{e}_2, \\ \underline{A}^* \underline{e}_3 &= \underline{A}\underline{e}_1 \times \underline{A}\underline{e}_2 = \alpha\beta (\underline{e}_1 \times \underline{e}_2) = \alpha\beta \underline{e}_3,\end{aligned}$$

so $\text{tr } \underline{A}^* = \beta\gamma + \gamma\alpha + \alpha\beta$. Part (b) predicts $\frac{1}{2}[(\alpha + \beta + \gamma)^2 - (\alpha^2 + \beta^2 + \gamma^2)]$, and expanding the square, $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$, the squares cancel and the $\frac{1}{2}$ removes the 2, leaving $\alpha\beta + \beta\gamma + \gamma\alpha$, which agrees. Part (c) predicts $\underline{A}^t \underline{A}^* = (\alpha\beta\gamma)\underline{I}$, and indeed $\underline{A}^t = \underline{A}$ here while $\underline{A}\underline{A}^* \underline{e}_1 = \beta\gamma \underline{A}\underline{e}_1 = \alpha\beta\gamma \underline{e}_1$, and likewise $\underline{A}\underline{A}^* \underline{e}_2 = \alpha\beta\gamma \underline{e}_2$ and $\underline{A}\underline{A}^* \underline{e}_3 = \alpha\beta\gamma \underline{e}_3$, while $\det \underline{A} = \alpha\beta\gamma$. $\square$

REMARK 1.109 (what the cofactor is for). Part (c) is not a curiosity; it is the identity that makes surface deformation computable. If $\underline{a} \times \underline{b}$ is the vector area of the parallelogram on $\underline{a}$ and $\underline{b}$, then Definition 1.54 says that $\underline{A}^*$ carries vector areas exactly as $\underline{A}$ carries vectors, and (c) converts that geometric statement into the working formula $\underline{A}^* = (\det \underline{A})\underline{A}^{-t}$. Applied in Chapter 2 to the deformation gradient $\underline{F}$ with $J = \det \underline{F}$, the same two lines become $\underline{F}^* = J\underline{F}^{-t}$ and the Piola–Nanson formula $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$. Part (b) is the identity that makes I_2 computable from $\underline{A}$ alone, without ever forming $\underline{A}^*$, and it is what turns the characteristic polynomial of Problem 23 into something one can evaluate.

PROBLEM 23 (Characteristic polynomial). Show $\det(\underline{A} - \lambda \underline{I}) = -\lambda^3 + I_1 \lambda^2 - I_2 \lambda + I_3$.

Solution. By Definition 1.49₃ applied to the tensor $\underline{A} - \lambda \underline{I}$,

$$\begin{aligned}\det(\underline{A} - \lambda \underline{I}) [\underline{a}, \underline{b}, \underline{c}] &= [(\underline{A} - \lambda \underline{I})\underline{a}, (\underline{A} - \lambda \underline{I})\underline{b}, (\underline{A} - \lambda \underline{I})\underline{c}] \\ &= [\underline{A}\underline{a} - \lambda \underline{a}, \underline{A}\underline{b} - \lambda \underline{b}, \underline{A}\underline{c} - \lambda \underline{c}].\end{aligned}$$

Expand by linearity in each slot separately. Each slot contributes either $\underline{A}(\cdot)$ or $-\lambda(\cdot)$, so there are $2^3 = 8$ terms. Group them by how many slots took the $-\lambda$:

$$0 \text{ slots : } [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}] = I_3[\underline{a}, \underline{b}, \underline{c}],$$

$$1 \text{ slot : } -\lambda \left[[\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}] + [\underline{A}\underline{a}, \underline{b}, \underline{A}\underline{c}] + [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{c}] \right] = -\lambda I_2[\underline{a}, \underline{b}, \underline{c}],$$

$$2 \text{ slots : } \lambda^2 \left[[\underline{A}\underline{a}, \underline{b}, \underline{c}] + [\underline{a}, \underline{A}\underline{b}, \underline{c}] + [\underline{a}, \underline{b}, \underline{A}\underline{c}] \right] = \lambda^2 I_1[\underline{a}, \underline{b}, \underline{c}],$$

$$3 \text{ slots : } (-\lambda)^3 [\underline{a}, \underline{b}, \underline{c}] = -\lambda^3 [\underline{a}, \underline{b}, \underline{c}],$$

the three groupings being exactly Definition 1.49_{3, 2, 1} in turn. Adding and cancelling the non-zero $[\underline{a}, \underline{b}, \underline{c}]$,

$$\det(\underline{A} - \lambda \underline{I}) = -\lambda^3 + I_1(\underline{A})\lambda^2 - I_2(\underline{A})\lambda + I_3(\underline{A}).$$

The alternating signs are simply the $(-\lambda)^n$ of the grouping, and this is why the invariants enter (3.2) in that order: I_k is the coefficient produced by putting $\underline{A}$ into exactly k of the three slots. $\square$

PROBLEM 24 (Field product rules). For scalar fields φ, ψ and vector fields $\underline{u}, \underline{v}$, show (a) $\text{grad}(\varphi + \psi) = \text{grad } \varphi + \text{grad } \psi$; (b) $\text{div}(\underline{u} + \underline{v}) = \text{div } \underline{u} + \text{div } \underline{v}$; (c) $\text{div}(\varphi \underline{v}) = \text{grad } \varphi \cdot \underline{v} + \varphi \text{div } \underline{v}$; (d) $\text{curl}(\text{grad } \varphi) = \underline{0}$; (e) $\text{div}(\text{curl } \underline{v}) = 0$; (f) $\text{grad}(\varphi \underline{u}) = \underline{u} \otimes \text{grad } \varphi + \varphi \text{grad } \underline{u}$; (g) $\text{div}(\underline{u} \otimes \underline{v}) = (\text{grad } \underline{u})\underline{v} + (\text{div } \underline{v})\underline{u}$.

Solution. All follow from the Cartesian representations and the ordinary product and equality-of-mixed-partial rules; write everything in components.

(a) $[\text{grad}(\varphi + \psi)]_i = (\varphi + \psi)_{,i} = \varphi_{,i} + \psi_{,i}$. (b) $\text{div}(\underline{u} + \underline{v}) = (u_i + v_i)_{,i} = u_{i,i} + v_{i,i}$. (c) $\text{div}(\varphi \underline{v}) = (\varphi v_i)_{,i} = \varphi_{,i} v_i + \varphi v_{i,i} = \text{grad } \varphi \cdot \underline{v} + \varphi \text{div } \underline{v}$. (d) $[\text{curl}(\text{grad } \varphi)]_j = \varepsilon_{kij}(\varphi_{,i})_{,k} = \varepsilon_{kij}\varphi_{,ik}$; since $\varphi_{,ik} = \varphi_{,ki}$ is symmetric in (i, k) while ε_{kij} is antisymmetric under $i \leftrightarrow k$, the contraction vanishes. (e) $\text{div}(\text{curl } \underline{v}) = (\varepsilon_{kij}v_{i,k})_{,j} = \varepsilon_{kij}v_{i,kj}$; $v_{i,kj}$ is symmetric in (k, j) and ε_{kij} antisymmetric under $k \leftrightarrow j$, so the sum is 0. (f) $[\text{grad}(\varphi \underline{u})]_{ij} = (\varphi u_i)_{,j} = u_i \varphi_{,j} + \varphi u_{i,j} = (\underline{u} \otimes \text{grad } \varphi)_{ij} + \varphi (\text{grad } \underline{u})_{ij}$. (g) $[\text{div}(\underline{u} \otimes \underline{v})]_i =$

$(u_i v_j)_{,j} = u_{i,j} v_j + u_i v_{j,j} = [(\text{grad } \underline{u}) \underline{v}]_i + (\text{div } \underline{v}) u_i$. (In (d),(e) twice continuous differentiability is assumed, so mixed partials commute.)

Parts (d) and (e) are the same statement twice: a symmetric array contracted with an antisymmetric one vanishes, since $A_{pq} S_{pq} = -A_{qp} S_{pq} = -A_{pq} S_{qp} = -A_{pq} S_{pq}$ after relabelling, so the quantity equals minus itself. The same argument proves the Piola identity in Problem 2 of Chapter 2. $\square$

PROBLEM 25 (Gradient in spherical coordinates). With $\underline{x} = r \underline{e}_\rho(\theta, \varphi)$, $\underline{e}_\rho = \cos \varphi \underline{e}_r(\theta) + \sin \varphi \underline{e}_3$, $\underline{e}_\varphi = \underline{e}_\rho \times \underline{e}_\theta$, find the components of $\text{grad } f$ in the variable basis $\{\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi\}$.

Solution. Step 1: build the frame from the Cartesian basis. Nothing is assumed about spherical coordinates; everything is constructed. The angle θ is the azimuth, measured in the equatorial plane from $\underline{e}_1$, and φ is the elevation above that plane, so φ is latitude and not the colatitude used in much of the physics literature. Start with the unit vector pointing along the projection of $\underline{x}$ onto the equatorial plane,

$$\underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2,$$

and differentiate it with respect to θ , the $\underline{e}_i$ being fixed:

$$\begin{aligned} \underline{e}_\theta(\theta) &:= \underline{e}_r'(\theta) = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2, \\ \underline{e}_\theta'(\theta) &= -\cos \theta \underline{e}_1 - \sin \theta \underline{e}_2 = -\underline{e}_r(\theta). \end{aligned}$$

Both are unit vectors, since $\cos^2 \theta + \sin^2 \theta = 1$ in each case, and they are orthogonal:

$$\underline{e}_r \cdot \underline{e}_\theta = \cos \theta (-\sin \theta) + \sin \theta (\cos \theta) = 0.$$

Also $\underline{e}_r \cdot \underline{e}_3 = 0$ and $\underline{e}_\theta \cdot \underline{e}_3 = 0$, both vectors having no $\underline{e}_3$ component. Tilting $\underline{e}_r$ up out of the equatorial plane by the angle φ produces the radial direction given in the statement,

$$\underline{e}_\rho(\theta, \varphi) = \cos \varphi \underline{e}_r(\theta) + \sin \varphi \underline{e}_3 = \cos \varphi \cos \theta \underline{e}_1 + \cos \varphi \sin \theta \underline{e}_2 + \sin \varphi \underline{e}_3,$$

so that $\underline{x} = r\underline{e}_\rho$ has Cartesian coordinates $x_1 = r \cos \varphi \cos \theta$, $x_2 = r \cos \varphi \sin \theta$, $x_3 = r \sin \varphi$, and $|\underline{x}| = r$ as the notation intends.

Step 2: the third leg of the frame, computed rather than quoted. The statement defines $\underline{e}_\varphi = \underline{e}_\rho \times \underline{e}_\theta$. Using $(\underline{u} \times \underline{v})_1 = u_2 v_3 - u_3 v_2$ and its two companions from Proposition 1.24, with

$$\underline{e}_\rho = (\cos \varphi \cos \theta, \cos \varphi \sin \theta, \sin \varphi), \quad \underline{e}_\theta = (-\sin \theta, \cos \theta, 0),$$

one obtains

$$\begin{aligned} (\underline{e}_\rho \times \underline{e}_\theta)_1 &= (\cos \varphi \sin \theta)(0) - (\sin \varphi)(\cos \theta) \\ &= -\sin \varphi \cos \theta, \\ (\underline{e}_\rho \times \underline{e}_\theta)_2 &= (\sin \varphi)(-\sin \theta) - (\cos \varphi \cos \theta)(0) \\ &= -\sin \varphi \sin \theta, \\ (\underline{e}_\rho \times \underline{e}_\theta)_3 &= (\cos \varphi \cos \theta)(\cos \theta) - (\cos \varphi \sin \theta)(-\sin \theta) \\ &= \cos \varphi (\cos^2 \theta + \sin^2 \theta) = \cos \varphi. \end{aligned}$$

Collecting, and recognising $\cos \theta \underline{e}_1 + \sin \theta \underline{e}_2 = \underline{e}_r$,

$$\underline{e}_\varphi = -\sin \varphi \cos \theta \underline{e}_1 - \sin \varphi \sin \theta \underline{e}_2 + \cos \varphi \underline{e}_3 = -\sin \varphi \underline{e}_r(\theta) + \cos \varphi \underline{e}_3.$$

Step 3: the frame is orthonormal and right-handed. Each of the six conditions is verified, using $\underline{e}_r \cdot \underline{e}_r = 1$, $\underline{e}_\theta \cdot \underline{e}_\theta = 1$, $\underline{e}_r \cdot \underline{e}_\theta = 0$, $\underline{e}_r \cdot \underline{e}_3 = \underline{e}_\theta \cdot \underline{e}_3 = 0$, $\underline{e}_3 \cdot \underline{e}_3 = 1$:

$$\begin{aligned} \underline{e}_\rho \cdot \underline{e}_\rho &= \cos^2 \varphi \underline{e}_r \cdot \underline{e}_r + 2 \cos \varphi \sin \varphi \underline{e}_r \cdot \underline{e}_3 + \sin^2 \varphi \underline{e}_3 \cdot \underline{e}_3 \\ &= \cos^2 \varphi + \sin^2 \varphi = 1, \\ \underline{e}_\varphi \cdot \underline{e}_\varphi &= \sin^2 \varphi \underline{e}_r \cdot \underline{e}_r - 2 \sin \varphi \cos \varphi \underline{e}_r \cdot \underline{e}_3 + \cos^2 \varphi \underline{e}_3 \cdot \underline{e}_3 \\ &= \sin^2 \varphi + \cos^2 \varphi = 1, \\ \underline{e}_\theta \cdot \underline{e}_\theta &= 1, \\ \underline{e}_\rho \cdot \underline{e}_\theta &= \cos \varphi \underline{e}_r \cdot \underline{e}_\theta + \sin \varphi \underline{e}_3 \cdot \underline{e}_\theta = 0 + 0 = 0, \\ \underline{e}_\varphi \cdot \underline{e}_\theta &= -\sin \varphi \underline{e}_r \cdot \underline{e}_\theta + \cos \varphi \underline{e}_3 \cdot \underline{e}_\theta = 0 + 0 = 0, \\ \underline{e}_\rho \cdot \underline{e}_\varphi &= (\cos \varphi)(-\sin \varphi) \underline{e}_r \cdot \underline{e}_r + (\sin \varphi)(\cos \varphi) \underline{e}_3 \cdot \underline{e}_3 \\ &= -\cos \varphi \sin \varphi + \sin \varphi \cos \varphi = 0. \end{aligned}$$

For the handedness, use $\underline{u} \times (\underline{v} \times \underline{w}) = \underline{u} \cdot \underline{w} \underline{v} - \underline{u} \cdot \underline{v} \underline{w}$ from Problem 11(a) with $\underline{u} = \underline{w} = \underline{e}_\theta$ and $\underline{v} = \underline{e}_\rho$:

$$\underline{e}_\theta \times \underline{e}_\varphi = \underline{e}_\theta \times (\underline{e}_\rho \times \underline{e}_\theta) = \underline{e}_\theta \cdot \underline{e}_\theta \underline{e}_\rho - \underline{e}_\theta \cdot \underline{e}_\rho \underline{e}_\theta = 1 \cdot \underline{e}_\rho - 0 \cdot \underline{e}_\theta = \underline{e}_\rho,$$

so $[\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi] = \underline{e}_\rho \cdot \underline{e}_\theta \times \underline{e}_\varphi = \underline{e}_\rho \cdot \underline{e}_\rho = 1 > 0$. The triad is a right-handed orthonormal basis at every point where it is defined.

Step 4: how the frame turns. Differentiate $\underline{e}_\rho$ with respect to each angle, remembering that $\underline{e}_3$ is constant and $\underline{e}'_r = \underline{e}_\theta$:

$$\frac{\partial \underline{e}_\rho}{\partial \theta} = \cos \varphi \underline{e}'_r(\theta) + 0 = \cos \varphi \underline{e}_\theta, \quad \frac{\partial \underline{e}_\rho}{\partial \varphi} = -\sin \varphi \underline{e}_r + \cos \varphi \underline{e}_3 = \underline{e}_\varphi.$$

The second is exactly the vector computed in Step 2, which is why the problem defines $\underline{e}_\varphi$ by a cross product: the cross product and the φ derivative deliver the same direction, so the frame that is natural geometrically is also the frame that is natural for differentiation. Hence, treating r, θ, φ as independent,

$$d\underline{e}_\rho = \frac{\partial \underline{e}_\rho}{\partial \theta} d\theta + \frac{\partial \underline{e}_\rho}{\partial \varphi} d\varphi = \cos \varphi d\theta \underline{e}_\theta + d\varphi \underline{e}_\varphi.$$

Step 5: the line element. Differentiate $\underline{x} = r\underline{e}_\rho(\theta, \varphi)$ by the product rule:

$$d\underline{x} = dr \underline{e}_\rho + r d\underline{e}_\rho = \underbrace{1}_{h_r} dr \underline{e}_\rho + \underbrace{r \cos \varphi}_{h_\theta} d\theta \underline{e}_\theta + \underbrace{r}_{h_\varphi} d\varphi \underline{e}_\varphi. \quad (*_2)$$

The three coefficients $h_r = 1$, $h_\theta = r \cos \varphi$, $h_\varphi = r$ are the only content of the problem. They are lengths per unit coordinate: increasing θ by $d\theta$ slides the point along the circle of latitude, whose radius is not r but $r \cos \varphi$ (Figure 1.1), so the distance travelled is $r \cos \varphi d\theta$; increasing φ by $d\varphi$ slides it along a meridian of radius r , a distance $r d\varphi$; increasing r moves it radially a distance dr , whence $h_r = 1$.

REMARK 1.110 (the three scale factors, read off the picture). Each coefficient in $(*_2)$ is the distance moved per unit coordinate:

$$\begin{array}{lll} dr \text{ at fixed } \theta, \varphi : & \text{outward along } \underline{e}_\rho, \text{ distance } dr, & h_r = 1, \\ d\theta \text{ at fixed } r, \varphi : & \text{along the parallel, radius } \varrho = r \cos \varphi, & h_\theta = r \cos \varphi, \\ d\varphi \text{ at fixed } r, \theta : & \text{along the meridian, radius } r, & h_\varphi = r. \end{array}$$

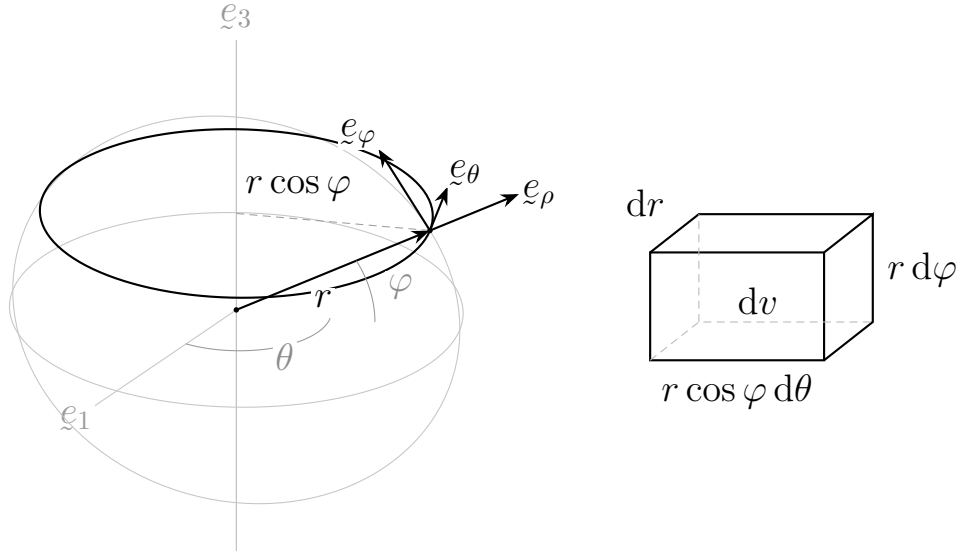


Figure 1.1: Left: the spherical frame at $\underline{x} = r\underline{e}_\rho$. The heavy circle is the parallel of latitude φ , and its radius is the dashed distance $r \cos \varphi$ from the polar axis, not r ; this single fact is the whole reason $h_\theta = r \cos \varphi$. The triad $\{\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi\}$ points outward, east and north, and $\underline{e}_\theta$ appears short only because it is foreshortened, pointing towards the reader. Right: the box built on the three coordinate increments, with edge lengths dr , $r \cos \varphi d\theta$ and $r d\varphi$; each edge is a coordinate increment multiplied by its scale factor.

The middle one is the trap: the parallel has radius $\varrho = r \cos \varphi$, the distance to the *axis*, not r . Two checks. At the equator $\cos \varphi = 1$ and $h_\theta = r$. At a pole $\cos \varphi \rightarrow 0$ and $h_\theta \rightarrow 0$, the degeneracy that excludes the poles below.

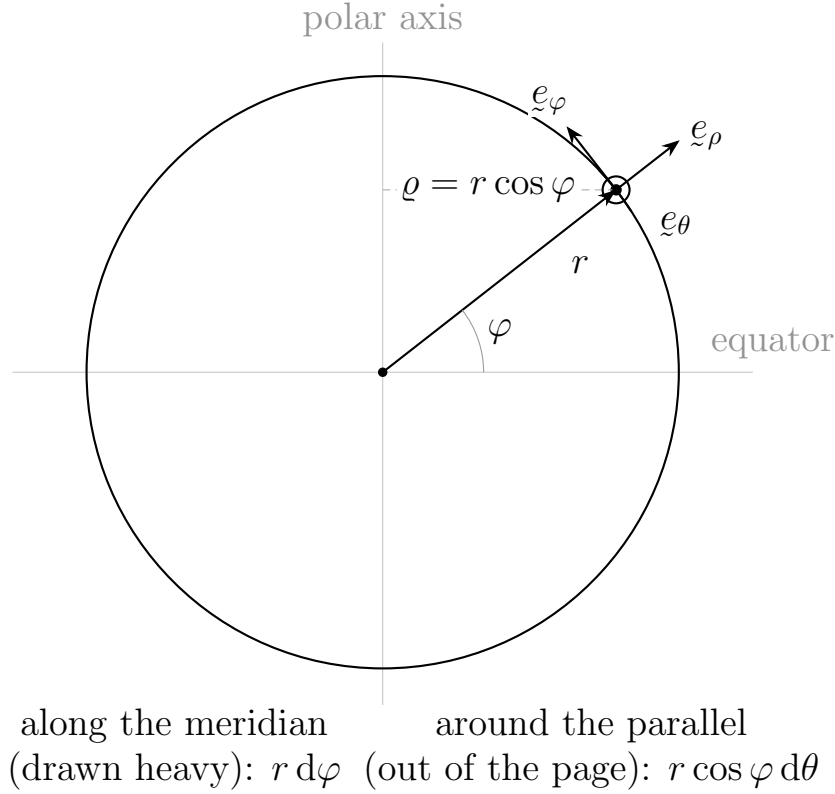


Figure 1.2: A meridional section, showing why $h_\theta = r \cos \varphi$ while $h_\varphi = r$. Since φ is measured from the equator, the point is at distance r from the origin but only $\varrho = r \cos \varphi$ from the polar axis. A change of longitude sweeps it around that smaller circle, a change of latitude along a meridian of radius r . The azimuthal direction $\underline{e}_\theta$ comes out of the page, marked by the circled dot.

Step 6: invert the line element. Because $\{\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi\}$ is orthonormal, the three coefficients in $(*_2)$ are recovered by dotting with the corresponding leg, every cross term vanishing:

$$\begin{aligned}\underline{e}_\rho \cdot d\tilde{x} &= dr \underline{e}_\rho \cdot \underline{e}_\rho + r \cos \varphi d\theta \underline{e}_\rho \cdot \underline{e}_\theta + r d\varphi \underline{e}_\rho \cdot \underline{e}_\varphi = dr, \\ \underline{e}_\theta \cdot d\tilde{x} &= dr \underline{e}_\theta \cdot \underline{e}_\rho + r \cos \varphi d\theta \underline{e}_\theta \cdot \underline{e}_\theta + r d\varphi \underline{e}_\theta \cdot \underline{e}_\varphi = r \cos \varphi d\theta, \\ \underline{e}_\varphi \cdot d\tilde{x} &= dr \underline{e}_\varphi \cdot \underline{e}_\rho + r \cos \varphi d\theta \underline{e}_\varphi \cdot \underline{e}_\theta + r d\varphi \underline{e}_\varphi \cdot \underline{e}_\varphi = r d\varphi.\end{aligned}$$

Solving for the coordinate increments, which requires $r \neq 0$ and $\cos \varphi \neq 0$,

$$dr = \underline{e}_\rho \cdot d\mathbf{x}, \quad d\theta = \frac{1}{r \cos \varphi} \underline{e}_\theta \cdot d\mathbf{x}, \quad d\varphi = \frac{1}{r} \underline{e}_\varphi \cdot d\mathbf{x}.$$

Step 7: the chain rule and the identification. Write $f = \hat{f}(r, \theta, \varphi)$ for the composite that assigns to the coordinates the value of the field at the corresponding point. The ordinary chain rule in three variables gives

$$df = f_{,r} dr + f_{,\theta} d\theta + f_{,\varphi} d\varphi, \quad f_{,r} := \frac{\partial \hat{f}}{\partial r}, \text{ etc.}$$

Substituting the three expressions just obtained and collecting the common factor $d\mathbf{x}$ on the right of every dot product,

$$\begin{aligned} df &= f_{,r} \underline{e}_\rho \cdot d\mathbf{x} + f_{,\theta} \frac{1}{r \cos \varphi} \underline{e}_\theta \cdot d\mathbf{x} + f_{,\varphi} \frac{1}{r} \underline{e}_\varphi \cdot d\mathbf{x} \\ &= \left(f_{,r} \underline{e}_\rho + \frac{1}{r \cos \varphi} f_{,\theta} \underline{e}_\theta + \frac{1}{r} f_{,\varphi} \underline{e}_\varphi \right) \cdot d\mathbf{x}. \end{aligned}$$

Compare with (1.114), $df = \text{grad } f \cdot d\mathbf{x}$. Subtracting, $\underline{w} \cdot d\mathbf{x} = 0$ where $\underline{w}$ is the difference of $\text{grad } f$ and the bracket. As $(dr, d\theta, d\varphi)$ ranges over $\mathbb{R}^3$ the vector $d\mathbf{x}$ in $(*_2)$ ranges over all of $\mathbb{E}^3$, precisely because the three scale factors are non-zero and the triad is a basis; so $\underline{w}$ is orthogonal to every vector and therefore $\underline{w} = \underline{0}$. Hence

$$\text{grad } f = f_{,r} \underline{e}_\rho + \frac{1}{r \cos \varphi} f_{,\theta} \underline{e}_\theta + \frac{1}{r} f_{,\varphi} \underline{e}_\varphi \quad (*_3)$$

and the components sought are $(\text{grad } f)_\rho = f_{,r}$, $(\text{grad } f)_\theta = f_{,\theta}/(r \cos \varphi)$, $(\text{grad } f)_\varphi = f_{,\varphi}/r$.

Where it fails. The derivation used $r \neq 0$ and $\cos \varphi \neq 0$. The first excludes the origin, where no radial direction exists; the second excludes the two poles $\varphi = \pm\pi/2$, where the circle of latitude degenerates to a point, $\underline{e}_\theta$ is undefined, and longitude carries no information. These are failures of the coordinate system, not of the field: $\text{grad } f$ itself is perfectly well defined there, as (1.114) makes plain, since it never mentions coordinates.

Checks. Three independent ones, each computed both ways.

Take $f = r = |\underline{x}|$. Then $f_{,r} = 1$ and $f_{,\theta} = f_{,\varphi} = 0$, so $(*_3)$ gives $\text{grad } r = \underline{e}_\rho$. Independently, from $r^2 = \underline{x} \cdot \underline{x}$ one has $2r \, dr = 2\underline{x} \cdot d\underline{x}$, hence $dr = (\underline{x}/r) \cdot d\underline{x}$ and $\text{grad } r = \underline{x}/r = \underline{e}_\rho$, in agreement.

Take $f = x_3 = r \sin \varphi$. Then $f_{,r} = \sin \varphi$, $f_{,\theta} = 0$, $f_{,\varphi} = r \cos \varphi$, so

$$\text{grad } f = \sin \varphi \, \underline{e}_\rho + \frac{1}{r} r \cos \varphi \, \underline{e}_\varphi = \sin \varphi \, \underline{e}_\rho + \cos \varphi \, \underline{e}_\varphi.$$

Substituting the Cartesian forms of $\underline{e}_\rho$ and $\underline{e}_\varphi$,

$$\begin{aligned} \text{grad } f &= \sin \varphi (\cos \varphi \, \underline{e}_r + \sin \varphi \, \underline{e}_3) + \cos \varphi (-\sin \varphi \, \underline{e}_r + \cos \varphi \, \underline{e}_3) \\ &= (\sin \varphi \cos \varphi - \cos \varphi \sin \varphi) \, \underline{e}_r + (\sin^2 \varphi + \cos^2 \varphi) \, \underline{e}_3 = \underline{e}_3, \end{aligned}$$

which is right, since $\text{grad } x_3 = \underline{e}_3$ directly from $dx_3 = \underline{e}_3 \cdot d\underline{x}$.

Take $f = \theta$. Then $(*_3)$ gives $\text{grad } \theta = \underline{e}_\theta / (r \cos \varphi)$. This is the familiar polar result $\text{grad } \theta = \underline{e}_\theta / \varrho$ with $\varrho = r \cos \varphi$ the distance from the polar axis, which is exactly what the dashed segment in Figure 1.1 measures, and it correctly blows up on the axis, where θ is discontinuous. $\square$

REMARK 1.111 (the scale-factor recipe, and the colatitude convention). Nothing in Steps 5 to 7 used the sphere. Whenever coordinates (q_1, q_2, q_3) produce a line element of the orthogonal form $d\underline{x} = \sum_k h_k dq_k \underline{\tilde{f}}_k$ with $\{\underline{\tilde{f}}_k\}$ orthonormal, the same three steps give

$$\text{grad } f = \sum_k \frac{1}{h_k} \frac{\partial f}{\partial q_k} \underline{\tilde{f}}_k,$$

the h_k being called scale factors. Cartesian coordinates have $h_1 = h_2 = h_3 = 1$ and the formula returns $f_{,i} \underline{e}_i$; the polar coordinates of §1.2.5 have $h_r = 1$, $h_\theta = r$, $h_z = 1$ and it returns $f_{,r} \underline{e}_r + r^{-1} f_{,\theta} \underline{e}_\theta + f_{,z} \underline{e}_3$. The division by h_k is not a convention but a units statement: $\partial f / \partial q_k$ is a change per unit *coordinate*, while a gradient component must be a change per unit *length*, and h_k is the number of lengths per coordinate.

One warning on conventions. Most physics texts measure the polar angle ϑ down from the north pole, the colatitude, and obtain $h_\vartheta = r$,

$h_\theta = r \sin \vartheta$. Here φ is measured up from the equator, so $\varphi = \pi/2 - \vartheta$ and $\sin \vartheta = \cos \varphi$, which is why $\cos \varphi$ appears in $(*_3)$ where the familiar formula shows a sine. The two agree; only the angle being named has moved.

PROBLEM 26 (Laplace's equation between two spheres). For $f = f(r)$, $r = |\underline{x}|$, solve $\operatorname{div}(\operatorname{grad} f) = 0$ on $a \leq r \leq b$ with $f(a) = f_a$, $f(b) = f_b$.

Solution. By Problem 25 with $f_\theta = f_\varphi = 0$, $\operatorname{grad} f = f'(r)\underline{e}_\rho$. Put $\underline{v} = \operatorname{grad} f = f'(r)\underline{e}_\rho$; then, using $d\underline{e}_\rho$ as above,

$$d\underline{v} = f''(r) dr \underline{e}_\rho + f'(r) (\cos \varphi \underline{e}_\theta d\theta + \underline{e}_\varphi d\varphi),$$

and inserting $dr = \underline{e}_\rho \cdot d\underline{x}$, $d\theta = \frac{1}{r \cos \varphi} \underline{e}_\theta \cdot d\underline{x}$, $d\varphi = \frac{1}{r} \underline{e}_\varphi \cdot d\underline{x}$ and $(\underline{e}_a \cdot d\underline{x})\underline{e}_b = (\underline{e}_b \otimes \underline{e}_a)d\underline{x}$,

$$\operatorname{grad} \underline{v} = f'' \underline{e}_\rho \otimes \underline{e}_\rho + \frac{f'}{r} \underline{e}_\theta \otimes \underline{e}_\theta + \frac{f'}{r} \underline{e}_\varphi \otimes \underline{e}_\varphi, \quad \operatorname{div} \underline{v} = \operatorname{tr}(\operatorname{grad} \underline{v}) = f'' + \frac{2f'}{r}.$$

So Laplace's equation is $f'' + 2f'/r = 0$, i.e. $(r^2 f')' = 0$, giving $r^2 f' = \text{const}$, $f' = \alpha/r^2$, $f(r) = \beta - \alpha/r$. Rewriting the constants through the boundary conditions,

$$f(r) = f_a + (f_b - f_a) \frac{\frac{1}{a} - \frac{1}{r}}{\frac{1}{a} - \frac{1}{b}}$$

which indeed gives $f(a) = f_a$, $f(b) = f_b$. □

PROBLEM 27 (A curl identity and the curl of a polar field). Establish $\underline{u} \cdot \operatorname{curl} \underline{v} = \operatorname{div}(\underline{v} \times \underline{u}) + \underline{v} \cdot \operatorname{curl} \underline{u}$ and derive the curl of the field (1.145)'s underlying $\underline{u} = u_r \underline{e}_r + u_\theta \underline{e}_\theta + u_z \underline{k}$.

Solution. Identity. Write $(\operatorname{curl} \underline{v})_i = \varepsilon_{ijk} v_{k,j}$. Then

$$\operatorname{div}(\underline{v} \times \underline{u}) = \varepsilon_{ijk} (v_j u_{k,i})_{,i} = \varepsilon_{ijk} v_{j,i} u_k + \varepsilon_{ijk} v_j u_{k,i}.$$

In the first term $\varepsilon_{ijk} v_{j,i} = \varepsilon_{kij} v_{j,i} = (\operatorname{curl} \underline{v})_k$, so it is $\underline{u} \cdot \operatorname{curl} \underline{v}$; in the second $\varepsilon_{ijk} u_{k,i} = -\varepsilon_{jik} u_{k,i} = -(\operatorname{curl} \underline{u})_j$, so it is $-\underline{v} \cdot \operatorname{curl} \underline{u}$. Hence $\operatorname{div}(\underline{v} \times \underline{u}) = \underline{u} \cdot \operatorname{curl} \underline{v} - \underline{v} \cdot \operatorname{curl} \underline{u}$, which rearranges to the stated identity.

Curl of the polar field. For any field, $\text{curl } \underline{u}$ is the axial vector of $\text{grad } \underline{u} - (\text{grad } \underline{u})^t$ (that is, $(\text{grad } \underline{u} - (\text{grad } \underline{u})^t)\underline{a} = (\text{curl } \underline{u}) \times \underline{a}$). From (1.145),

$$\begin{aligned} \text{grad } \underline{u} - (\text{grad } \underline{u})^t &= c(\underline{e}_r \otimes \underline{e}_\theta - \underline{e}_\theta \otimes \underline{e}_r) + u_{z,r}(\underline{k} \otimes \underline{e}_r - \underline{e}_r \otimes \underline{k}) \\ &\quad + \frac{1}{r}u_{z,\theta}(\underline{k} \otimes \underline{e}_\theta - \underline{e}_\theta \otimes \underline{k}), \end{aligned}$$

with $c = \frac{1}{r}(u_{r,\theta} - u_\theta) - u_{\theta,r}$. The axial vector of $\underline{e}_a \otimes \underline{e}_b - \underline{e}_b \otimes \underline{e}_a$ is $\underline{e}_b \times \underline{e}_a$, so

$$\begin{aligned} \text{curl } \underline{u} &= c(\underline{e}_\theta \times \underline{e}_r) + u_{z,r}(\underline{e}_r \times \underline{k}) + \frac{1}{r}u_{z,\theta}(\underline{e}_\theta \times \underline{k}) \\ &= \frac{1}{r}u_{z,\theta}\underline{e}_r - u_{z,r}\underline{e}_\theta + \left[u_{\theta,r} + \frac{1}{r}(u_\theta - u_{r,\theta})\right]\underline{k}, \end{aligned}$$

using $\underline{e}_\theta \times \underline{e}_r = -\underline{k}$, $\underline{e}_r \times \underline{k} = -\underline{e}_\theta$, $\underline{e}_\theta \times \underline{k} = \underline{e}_r$. □

PROBLEM 28 (Cartesian formula for the curl of a tensor). Establish $\text{curl } \underline{A} = \varepsilon_{ilm}A_{jl,i}\underline{e}_m \otimes \underline{e}_j$.

Solution. By definition $(\text{curl } \underline{A})\underline{c} = \text{curl}(\underline{A}^t\underline{c})$ for fixed $\underline{c}$. Take $\underline{c} = \underline{e}_j$; then $\underline{A}^t\underline{e}_j = A_{jl}\underline{e}_l$ is a vector field with l th component A_{jl} , and with $(\text{curl } \underline{w})_m = \varepsilon_{miq}w_{q,i}$,

$$\text{curl}(\underline{A}^t\underline{e}_j) = \varepsilon_{miq}A_{jq,i}\underline{e}_m = \varepsilon_{ilm}A_{jl,i}\underline{e}_m$$

(renaming $q \rightarrow l$ and using $\varepsilon_{miq} = \varepsilon_{qmi}$, i.e. writing the antisymmetric symbol in the cyclic order ilm). Writing $\text{curl } \underline{A} = C_{mj}\underline{e}_m \otimes \underline{e}_j$ and matching $(\text{curl } \underline{A})\underline{e}_j = C_{mj}\underline{e}_m$ gives $C_{mj} = \varepsilon_{ilm}A_{jl,i}$, i.e. $\text{curl } \underline{A} = \varepsilon_{ilm}A_{jl,i}\underline{e}_m \otimes \underline{e}_j$. □

Chapter 2

Bodies, Configurations, and Motions

This chapter builds the *kinematics* of continuum mechanics: the language for describing where the material of a body is, how fast it moves, and how it deforms, all before any mention of force. The single new idea is that we carry *two* pictures of the body at once, a fixed reference copy and the moving current one, and translate between them. Everything computational rests on the vector and tensor algebra of Chapter 1; we cite it freely and prove each new relation from it.

2.1 Position, velocity and acceleration

A *body* B is a collection of *material points*, labelled p . In the older literature these are sometimes called *particles*, to convey the notion of a small indivisible object; that name has some heuristic value but is not really what is intended. A material point is better regarded as a *primitive* notion: not a tiny lump of matter, but an abstract label for “a bit of the material”. The body B itself is not a region of space; we never see B directly, only the region of space it fills at a given instant. In the book’s figures B is drawn with a *dashed* outline for exactly this reason (Figure 2.1): it is a set of labels, not a place.

REMARK 2.1 (Figures 2.1–2.3: body versus configuration). Figure 2.1

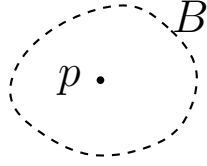


Figure 2.1: A material body.

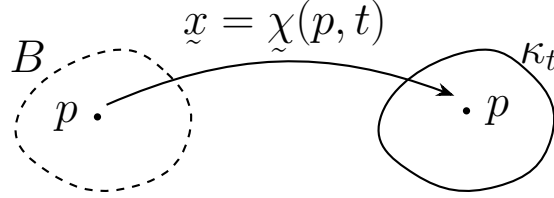


Figure 2.2: Configuration of the body at time t .

shows the abstract body B (a dashed blob with a marked point p): the reservoir of material points. Figures 2.2 and 2.3 each show B on the left mapped by an arrow to a *solid* blob on the right, the difference being only which placement of the body the arrow produces: the *current* configuration κ_t at time t in Figure 2.2, and a fixed *reference* configuration κ in Figure 2.3. Solid outlines mark genuine regions of Euclidean space; the dashed B is not one. The whole chapter is the art of never confusing the label p with either of its placements $\underline{x}$ or $\underline{X}$.

DEFINITION 2.2 (Configuration and motion). At time t the body occupies a *configuration* κ_t , a region of the Euclidean point space $\mathbb{E}^3$, and $p \in B$ sits at the position $\underline{x} \in \kappa_t$. Distinct material points occupy distinct positions at a fixed time, so there is an invertible *motion* $\underline{\chi}(p, t)$ with

$$\underline{x} = \underline{\chi}(p, t), \quad p = \underline{\chi}^{-1}(\underline{x}, t). \quad (2.1)$$

At fixed t , $\underline{\chi}$ tells us where each material point is, and $\underline{\chi}^{-1}$ tells us

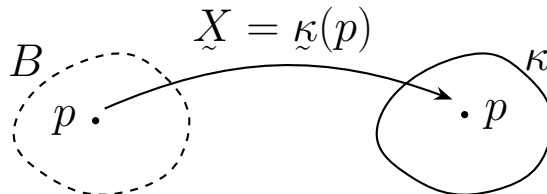


Figure 2.3: A reference configuration for the body.

which material point sits at a given place. Freezing p and letting t run traces the path of one material point through space; its time derivatives are its velocity and acceleration.

DEFINITION 2.3 (Velocity, acceleration).

$$\underline{v}(p, t) = \frac{\partial}{\partial t} \underline{\chi}(p, t), \quad \underline{a}(p, t) = \frac{\partial^2}{\partial t^2} \underline{\chi}(p, t). \quad (2.2)$$

The derivatives are taken at fixed p : they follow the *same* material point.

Working with the ethereal B is awkward, because a “label” has no coordinates. The remedy is to name each material point by the fixed place it occupies in one chosen, frozen configuration, the *reference* configuration κ .

DEFINITION 2.4 (Reference configuration). Fix a configuration κ and let $\underline{X} \in \kappa$ be positions in one-to-one correspondence with material points through an invertible placement map $\underline{\kappa}$,

$$\underline{X} = \underline{\kappa}(p), \quad p = \underline{\kappa}^{-1}(\underline{X}). \quad (2.3)$$

A common, though not obligatory, choice is the configuration actually occupied at some instant t_0 , $\kappa = \kappa_{t_0}$, for which

$$\underline{\kappa}(p) = \underline{\chi}(p, t_0). \quad (2.4)$$

Composing the placement’s inverse with the motion expresses the current position directly in terms of the reference position (Figure 2.4):

$$\underline{x} = \underline{\chi}_\kappa(\underline{X}, t), \quad \underline{\chi}_\kappa(\underline{X}, t) = \underline{\chi}(\underline{\kappa}^{-1}(\underline{X}), t). \quad (2.5\text{--}2.6)$$

The subscript records the dependence on the chosen κ . Since each p corresponds to exactly one $\underline{x} \in \kappa_t$ and one $\underline{X} \in \kappa$, the map $\underline{\chi}_\kappa(\cdot, t)$ is invertible at each t ,

$$\underline{X} = \underline{\chi}_\kappa^{-1}(\underline{x}, t). \quad (2.7)$$

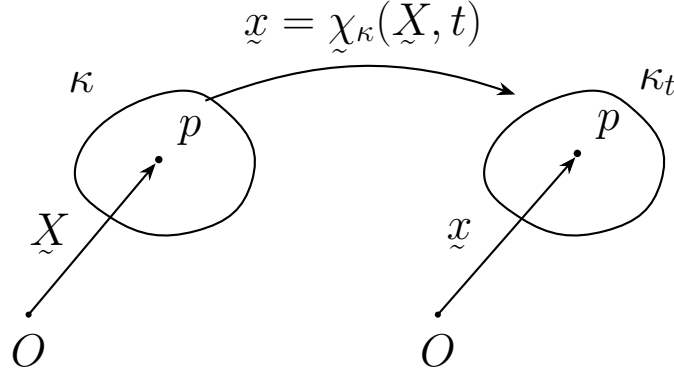


Figure 2.4: Mapping of reference configuration to current configuration.

REMARK 2.5 (Figure 2.4: the deformation map). Figure 2.4 draws two *solid* blobs, κ (reference, left) and κ_t (current, right), with the same material point p appearing in each, at $\underline{X}$ and at $\underline{x}$. The connecting arrow is χ_κ : it is the object a deformation actually is, a rule sending each reference place to its current place. Because both blobs are real regions of space, this map is the one we can differentiate, and its gradient will be the deformation gradient of §2.4.

Because B and κ are in fixed one-to-one correspondence, using (p, t) as independent variables is the same as using $(\underline{X}, t)$. The velocity therefore has a *referential* form (a function of $\underline{X}, t$) and a *spatial* form (a function of $\underline{x}, t$),

$$\hat{v}(\underline{X}, t) = \underline{v}(\kappa^{-1}(\underline{X}), t) = \frac{\partial}{\partial t} \chi_\kappa(\underline{X}, t), \quad \tilde{v}(\underline{x}, t) = \underline{v}(\chi_\kappa^{-1}(\underline{x}, t), t), \quad (2.8-2.9)$$

the partial in (2.8) being taken at fixed $\underline{X}$, i.e. at fixed p . The two forms are related through (2.5) and (2.7),

$$\hat{v}(\underline{X}, t) = \tilde{v}(\chi_\kappa(\underline{X}, t), t), \quad \tilde{v}(\underline{x}, t) = \hat{v}(\chi_\kappa^{-1}(\underline{x}, t), t). \quad (2.10)$$

2.2 Material, referential and spatial descriptions

Any field f carried by the material, temperature, density, velocity, may be written as a function of the material point, of the current position, or

of the reference position:

$$f = \bar{f}(p, t) = \tilde{f}(\underline{x}, t) = \hat{f}(\underline{X}, t), \quad (2.11)$$

where

$$\tilde{f}(\underline{x}, t) = \bar{f}(\chi^{-1}(\underline{x}, t), t), \quad \hat{f}(\underline{X}, t) = \bar{f}(\kappa^{-1}(\underline{X}), t), \quad (2.12)$$

and, relating the last two directly,

$$\hat{f}(\underline{X}, t) = \tilde{f}(\chi_{\kappa}(\underline{X}, t), t), \quad \tilde{f}(\underline{x}, t) = \hat{f}(\chi_{\kappa}^{-1}(\underline{x}, t), t). \quad (2.13)$$

These are the *material*, *spatial*, and *referential* descriptions of f .

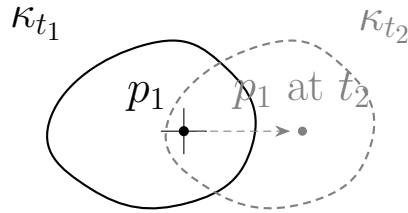
REMARK 2.6 (same value, three different functions: why the decorations matter). This is the single most important notational point of the chapter, and it is the exact analogue of the caret in Chapter 1, where $\hat{\varphi}(x_1, x_2, x_3)$ and $\varphi(\underline{x})$ shared their values but had different domains ($\mathbb{R}^3$ versus a point space) and so were different functions. Here the *same number* f , the temperature of a given bit of matter at a given instant, is produced by *three different machines*:

$$\begin{aligned} \bar{f} &\text{ eats a material label } p, \\ \tilde{f} &\text{ eats a current place } \underline{x}, \\ \hat{f} &\text{ eats a reference place } \underline{X}. \end{aligned}$$

They agree in value only when p , $\underline{x}$, $\underline{X}$ describe the same material point at the same time, i.e. when $\underline{x} = \chi_{\kappa}(\underline{X}, t)$. Their *derivatives* are genuinely different, because “hold the variable fixed” means hold p , hold $\underline{x}$, or hold $\underline{X}$, and these are three different physical instructions. The bar, tilde and caret are the honest bookkeeping of which machine is meant; the book (and we) will soon drop the decorations “to minimize visual distraction”, exactly as the caret was dropped in Chapter 1, on the understanding that the reader remembers which function a symbol denotes.

In the *spatial* description $\underline{x}$ and t are the independent variables. Fixing $\underline{x}$ fixes a *place*, not a material point, so as time passes different

material points stream through that fixed $\underline{x}$ (Figure 2.5). The set of fixed positions $\underline{x}$ is a *control volume*, a window in space through which material flows.



$\underline{x}$ stays put; at t_2 a *different* point p_2 sits there

Figure 2.5: Different material points occupy position $\underline{x}$ at different times.

REMARK 2.7 (Figure 2.5: the control-volume viewpoint). Figure 2.5 shows one fixed position $\underline{x}$ occupied by different material points at different times: the Eulerian, or spatial, standpoint of an observer nailed to space (a thermometer on a wall) rather than riding with the matter (a thermometer floating down a river). The distinction between “rate of change at a fixed place” and “rate of change following the matter” is precisely what the material derivative below measures.

Take the differential of f in each description. Using the coordinate free chain rule of Chapter 1, $df = \text{grad } f \cdot d\underline{x}$ (there written $d\varphi = \text{grad } \varphi \cdot d\underline{x}$), and allowing time to vary,

$$df = \tilde{f}' dt + \text{grad } \tilde{f} \cdot d\underline{x}, \quad (2.14)$$

$$df = (\hat{f})^\bullet dt + \text{Grad } \hat{f} \cdot d\underline{X}. \quad (2.15)$$

Here $\tilde{f}' = \partial \tilde{f} / \partial t$ is the *spatial time derivative* (at fixed $\underline{x}$) and grad is the gradient in $\underline{x}$; while $(\hat{f})^\bullet = \partial \hat{f} / \partial t$ is the *referential* (equivalently *material*) time derivative (at fixed $\underline{X}$, i.e. fixed p) and Grad is the gradient in $\underline{X}$. Since (2.14) and (2.15) are two expressions for the *same* increment df , dropping decorations,

$$\dot{f} dt + \text{Grad } f \cdot d\underline{X} = f' dt + \text{grad } f \cdot d\underline{x}. \quad (2.16)$$

We call $\dot{f}$ the *material derivative*: the rate of change of f following a fixed material point.

THEOREM 2.8 (Material derivative in the spatial description). *For a scalar field, with $\underline{v}$ the (spatial) material velocity,*

$$\dot{f} = f' + \text{grad } f \cdot \underline{v}. \quad (2.17)$$

For a vector field $\underline{u}$,

$$\dot{\underline{u}} = \underline{u}' + (\text{grad } \underline{u})\underline{v}, \quad (2.20)$$

and in particular the material acceleration is

$$\underline{a} = \dot{\underline{v}} = \underline{v}' + \underline{L}\underline{v}, \quad \underline{L} = \text{grad } \underline{v}. \quad (2.21)$$

Proof. Follow one material point: hold p , hence $\underline{X}$, fixed, so $d\underline{X} = \underline{0}$ in (2.16). Moving with the point, $d\underline{x} = d\underline{\chi}_\kappa(\underline{X}, t) = \underline{v} dt$ (the position changes at the velocity). Substituting into (2.16),

$$\dot{f} dt = f' dt + \text{grad } f \cdot \underline{v} dt = (f' + \text{grad } f \cdot \underline{v})dt,$$

and cancelling dt gives (2.17). For a vector field the identical argument applied componentwise, or directly from the vector chain rule $d\underline{u} = \underline{u}' dt + (\text{grad } \underline{u})d\underline{x}$ of Chapter 1, gives

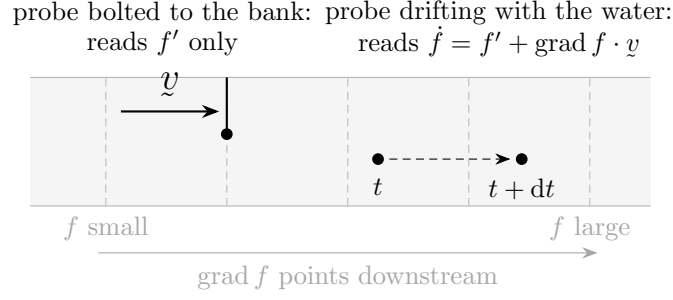
$$\dot{\underline{u}} dt + (\text{Grad } \underline{u})d\underline{X} = d\underline{u} = \underline{u}' dt + (\text{grad } \underline{u})d\underline{x}, \quad (2.18\text{--}2.19)$$

and setting $d\underline{X} = \underline{0}$, $d\underline{x} = \underline{v} dt$ yields (2.20). Taking $\underline{u} = \underline{v}$ gives the acceleration (2.21), with $\underline{L} = \text{grad } \underline{v}$ the spatial velocity gradient. ■

REMARK 2.9 (the convective term, intuitively). The material derivative splits into two clearly physical pieces:

$$\dot{f} = \underbrace{f'}_{\text{local: change at a fixed place}} + \underbrace{\text{grad } f \cdot \underline{v}}_{\text{convective: change from being carried to new places}}.$$

Float down a river whose temperature is steady in time ($f' = 0$) but rises downstream ($\text{grad } f \neq 0$): you still warm up, at the rate $\text{grad } f \cdot \underline{v}$, purely because the current carries you into warmer water. The acceleration (2.21) shows the same effect is why steady flows ($\underline{v}' = 0$) can still accelerate fluid particles, through the nonlinear term $\underline{L}\underline{v} = (\text{grad } \underline{v})\underline{v}$; this convective nonlinearity is the source of most of the difficulty in fluid and solid dynamics.



SUPPLEMENTARY FIGURE A. The two time derivatives of a field. A probe bolted to the bank samples one *place* and records only the explicit time change f' . A probe drifting with the water samples one *material point*; even in a perfectly steady field ($f' = 0$) it registers a change, because the current carries it into regions of different f . That extra reading is the convective term $\text{grad } f \cdot v$, and the sum is the material derivative $\dot{f}$ of (2.17).

2.3 Cartesian coordinates

To compute, we resolve everything on orthonormal bases. The current position uses a fixed basis $\{\underline{e}_i\}$ of the spatial translation space $\mathbb{E}^3$, the reference position a second fixed basis $\{\hat{\underline{E}}_A\}$ of the referential translation space $\hat{\mathbb{E}}^3$ (Figure 2.6):

$$\underline{x} = x_i \underline{e}_i \quad (i \in \{1, 2, 3\}), \quad \underline{X} = X_A \hat{\underline{E}}_A \quad (A \in \{1, 2, 3\}). \quad (2.22-2.23)$$

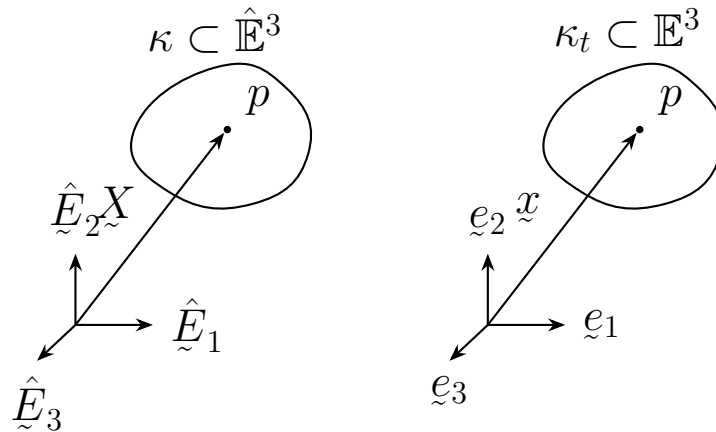


Figure 2.6: Position vectors in κ and κ_t .

REMARK 2.10 (Figure 2.6, and why two bases). Figure 2.6 draws the reference blob $\kappa \subset \hat{\mathbb{E}}^3$ with its own frame $\{\hat{\underline{E}}_A\}$ and the current blob

$\kappa_t \subset \mathbb{E}^3$ with a separate frame $\{\underline{e}_i\}$. Keeping the two frames *conceptually distinct*, even though both are copies of ordinary three-space, is the whole trick of the chapter: a quantity carrying one index of each kind (a lower-case i from $\{\underline{e}_i\}$ and an upper-case A from $\{\hat{E}_A\}$) is a *two-point tensor*, an object living with one foot in each configuration. The deformation gradient is the first and most important example. This is why the book uses capital letters X_A , $\hat{E}_A$ for reference quantities and lower-case x_i , $\underline{e}_i$ for current ones: the alphabet itself tracks which space you are in.

Regarding χ_κ as a vector in $\mathbb{E}^3$ and using that the $\hat{E}_A$ are fixed,

$$x_i \underline{e}_i = \chi_\kappa(X_A \hat{E}_A, t) = \chi_i(X_A, t) \underline{e}_i, \quad x_i = \chi_i(X_A, t), \quad (2.24)$$

shorthand for $x_i = \chi_i(X_1, X_2, X_3, t)$. The velocity, in the referential description, is obtained by differencing at fixed $\underline{X}$ and passing to the limit:

$$\begin{aligned} \underline{v} &= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha} [\chi_\kappa(\underline{X}, t + \alpha) - \chi_\kappa(\underline{X}, t)] \\ &= \underline{e}_i \lim_{\alpha \rightarrow 0} \frac{1}{\alpha} [\chi_i(X_A, t + \alpha) - \chi_i(X_A, t)] = \dot{\chi}_i \underline{e}_i, \end{aligned} \quad (2.25-2.26)$$

so $\hat{v}_i = \dot{\chi}_i = \partial \chi_i / \partial t$; likewise $\hat{a}_i = \ddot{\chi}_i = \partial^2 \chi_i / \partial t^2$ (2.27). In the spatial description $\underline{v} = \tilde{v}_i(x_k, t) \underline{e}_i$, $\underline{a} = \tilde{a}_i(x_k, t) \underline{e}_i$ (2.28). Resolving the material derivative (2.21), $\underline{a} = \underline{v}' + (\text{grad } \underline{v}) \underline{v}$, on $\{\underline{e}_i\}$ gives the component identity

$$\tilde{a}_i = \tilde{v}'_i + \tilde{v}_{i,j} \tilde{v}_j, \quad (2.29)$$

and applying (2.20) to the position field $\underline{x}$ itself, with $\underline{x}' = \underline{0}$ and $x_{i,j} = \delta_{ij}$, recovers the sanity check $\dot{x}_i = x_{i,j} \tilde{v}_j = \delta_{ij} \tilde{v}_j = \tilde{v}_i$: the material derivative of position is velocity, as it must be.

REMARK 2.11 (why $\underline{x}' = \underline{0}$, and why this is not circular). This step confuses almost everyone on first reading, because $\underline{x}$ is manifestly moving: the material point does change its position with time. The resolution is that the prime is *not* a material derivative.

Take the field $\underline{u}$ in (2.18) to be *position itself*. Its three descriptions are

$$\bar{\underline{u}}(p, t) = \chi(p, t), \quad \hat{\underline{u}}(\underline{X}, t) = \chi_\kappa(\underline{X}, t), \quad \tilde{\underline{u}}(\underline{x}, t) = \underline{x}.$$

The first two genuinely depend on t : hold the material point fixed and its location changes. The third does not, and the reason is purely definitional. In the spatial description the independent variable *is* $\underline{x}$. Asking for the value of the position field at the place $\underline{x}$ is asking “what position is the position $\underline{x}$?”, and the answer is $\underline{x}$, at every instant. The function $\tilde{\underline{u}}$ is the identity map of $\mathbb{E}^3$, and the identity map carries no t in it at all. Hence

$$\underline{u}' = \frac{\partial}{\partial t} \Big|_{\underline{x} \text{ fixed}} \tilde{\underline{u}}(\underline{x}, t) = \frac{\partial \underline{x}}{\partial t} \Big|_{\underline{x} \text{ fixed}} = \underline{0}, \quad \text{componentwise } \tilde{u}'_i = \frac{\partial x_i}{\partial t} = 0.$$

Nothing is being asserted about the motion here. What is being asserted is that a photograph of space, labelled by its own coordinates, does not change when the clock advances. The matter flowing through the point changes; the point does not.

The second ingredient is $\text{grad } \underline{u} = \underline{I}$, for the same reason: from $\tilde{u}_i = x_i$ we get $u_{i,j} = \partial x_i / \partial x_j = \delta_{ij}$, so $\text{grad } \underline{u} = \delta_{ij} \underline{e}_i \otimes \underline{e}_j = \underline{I}$. Substituting both into (2.20),

$$\dot{\underline{u}} = \underline{u}' + (\text{grad } \underline{u})\underline{v} = \underline{0} + \underline{I}\underline{v} = \underline{v},$$

that is $\dot{x}_i = \tilde{v}_i$. Read backwards this is a consistency check on the whole apparatus rather than a new result: (2.17) and (2.20) were constructed precisely so that $\dot{\underline{f}} = \underline{f}' + \text{grad } \underline{f} \cdot \underline{v}$ would reproduce the rate seen by a material point, and the material point’s rate of change of position is, by (2.2)₁, its velocity. Had the formula returned anything other than $\underline{v}$, the definition of the material derivative would have been wrong.

The whole content of the identity is therefore the split

$$\underbrace{\dot{x}_i}_{\text{the point moves}} = \underbrace{x'_i}_{=0: \text{ the place does not move}} + \underbrace{x_{i,j}\tilde{v}_j}_{= \tilde{v}_i: \text{ convection}}.$$

All of the motion sits in the convective term, none of it in the spatial time derivative. This is the cleanest possible illustration of the difference

between $'$ and $\dot{}$: for the position field the two are as far apart as they can be, $\underline{0}$ against $\underline{v}$.

EXAMPLE 2.12 (converting a motion between descriptions). Consider the motion (linear in $\underline{X}$, hence atypical, but transparent)

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} e^t & 0 & 0 \\ 0 & 1 & t \\ 0 & -t & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}, \quad \text{i.e.} \quad \begin{aligned} x_1 &= e^t X_1, \\ x_2 &= X_2 + tX_3, \\ x_3 &= -tX_2 + X_3. \end{aligned} \quad (2.30)$$

Referential velocity and acceleration. Differentiate at fixed $\underline{X}$:

$$\begin{aligned} \hat{v}_1 &= \dot{x}_1 = e^t X_1, & \hat{v}_2 &= X_3, & \hat{v}_3 &= -X_2; \\ \hat{a}_1 &= e^t X_1, & \hat{a}_2 &= 0, & \hat{a}_3 &= 0. \end{aligned}$$

Inversion. To get the spatial forms we invert (2.30). Immediately $X_1 = e^{-t}x_1$. The lower 2×2 block $\begin{pmatrix} 1 & t \\ -t & 1 \end{pmatrix}$ has determinant $1 + t^2$ and inverse $\frac{1}{1+t^2} \begin{pmatrix} 1 & -t \\ t & 1 \end{pmatrix}$, so

$$X_1 = e^{-t}x_1, \quad X_2 = \frac{x_2 - tx_3}{1 + t^2}, \quad X_3 = \frac{tx_2 + x_3}{1 + t^2}. \quad (2.31)$$

Spatial velocity. Substitute (2.31) into $\hat{v}_i$:

$$\begin{aligned} \tilde{v}_1 &= e^t X_1 = e^t e^{-t} x_1 = x_1, \\ \tilde{v}_2 &= X_3 = \frac{tx_2 + x_3}{1 + t^2}, & \tilde{v}_3 &= -X_2 = \frac{tx_3 - x_2}{1 + t^2}. \end{aligned}$$

Spatial acceleration, directly. Substitute (2.31) into $\hat{a}_i$: $\tilde{a}_1 = e^t X_1 = x_1$, $\tilde{a}_2 = 0$, $\tilde{a}_3 = 0$, i.e. $\underline{a} = x_1 \underline{e}_1$.

Spatial acceleration, via (2.29) (a check on the material derivative). The spatial velocity gradient is

$$(\tilde{v}_{i,j}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{t}{1+t^2} & \frac{1}{1+t^2} \\ 0 & -\frac{1}{1+t^2} & \frac{t}{1+t^2} \end{pmatrix}, \quad (2.33)$$

and the spatial time derivatives $\tilde{v}'_i = \partial \tilde{v}_i / \partial t$ (at fixed $\underline{x}$) are $\tilde{v}'_1 = 0$ and, by the quotient rule,

$$\tilde{v}'_2 = \frac{(1 - t^2)x_2 - 2tx_3}{(1 + t^2)^2}, \quad \tilde{v}'_3 = \frac{(1 - t^2)x_3 + 2tx_2}{(1 + t^2)^2}. \quad (2.32)$$

Then $\tilde{a}_2 = \tilde{v}'_2 + \tilde{v}_{2,j}\tilde{v}_j = \tilde{v}'_2 + \tilde{v}_{2,2}\tilde{v}_2 + \tilde{v}_{2,3}\tilde{v}_3$; with $\tilde{v}_{2,2} = t/(1 + t^2)$, $\tilde{v}_{2,3} = 1/(1 + t^2)$,

$$\tilde{v}_{2,2}\tilde{v}_2 + \tilde{v}_{2,3}\tilde{v}_3 = \frac{t(tx_2 + x_3) + (tx_3 - x_2)}{(1 + t^2)^2} = \frac{(t^2 - 1)x_2 + 2tx_3}{(1 + t^2)^2} = -\tilde{v}'_2,$$

so $\tilde{a}_2 = 0$, matching the direct computation; likewise $\tilde{a}_1 = x_1$, $\tilde{a}_3 = 0$. The two routes agree, which is the point of the example.

REMARK 2.13 (a correction to the printed $\tilde{v}'_2$). The material derivative check forces $\tilde{v}'_2 = [(1 - t^2)x_2 - 2tx_3]/(1 + t^2)^2$: only this value makes $\tilde{a}_2 = \tilde{v}'_2 + \tilde{v}_{2,j}\tilde{v}_j$ vanish, as it must since $\hat{a}_2 = 0$. (The text's printed coefficient of x_2 in $\tilde{v}'_2$ is a misprint; the denominator is squared and the numerator is $1 - t^2$, not $1 - 2t^2$.)

2.4 The deformation gradient

Assume the motion $\chi_\kappa(\underline{X}, t)$ is differentiable in $\underline{X}$. Holding t fixed, its linear part is a tensor, the *deformation gradient*:

$$d\underline{x} = \underline{F} d\underline{X}, \quad \underline{F} = \text{Grad } \chi_\kappa. \quad (2.34)$$

Allowing t to vary as well,

$$d\underline{x} = \underline{F} d\underline{X} + \underline{v} dt. \quad (2.35)$$

THEOREM 2.14 (The two gradients of a field are related by $\underline{F}$). *For any field u ,*

$$\text{Grad } u = (\text{grad } u)\underline{F}. \quad (2.37)$$

Proof. Write the increment $d\mathbf{u}$ referentially and spatially, as in (2.19), and substitute (2.35) and the material derivative $\dot{\mathbf{u}} = \mathbf{u}' + (\text{grad } \mathbf{u})\mathbf{v}$:

$$\dot{\mathbf{u}} dt + (\text{Grad } \mathbf{u})d\mathbf{X} = [\mathbf{u}' + (\text{grad } \mathbf{u})\mathbf{v}] dt + (\text{grad } \mathbf{u})\mathbf{F} d\mathbf{X}. \quad (2.36)$$

The dt terms cancel by (2.20), leaving $(\text{Grad } \mathbf{u})d\mathbf{X} = (\text{grad } \mathbf{u})\mathbf{F} d\mathbf{X}$ for every $d\mathbf{X}$; hence $\text{Grad } \mathbf{u} = (\text{grad } \mathbf{u})\mathbf{F}$. $\blacksquare$

To read off components, differentiate $\mathbf{x} = \chi_i \mathbf{e}_i$ at fixed t using $d\chi_i = \chi_{i,A} dX_A = \chi_{i,A} \hat{\mathbf{E}}_A \cdot d\mathbf{X}$:

$$d\mathbf{x} = \chi_{i,A} \mathbf{e}_i \hat{\mathbf{E}}_A \cdot d\mathbf{X} = (\chi_{i,A} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A) d\mathbf{X}, \quad (2.38)$$

so comparison with (2.34) gives

$$\mathbf{F} = F_{iA} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A, \quad F_{iA} = \chi_{i,A}. \quad (2.39)$$

The mixed basis $\{\mathbf{e}_i \otimes \hat{\mathbf{E}}_A\}$ shows $\mathbf{F}$ is a *two-point tensor*, a linear map $\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ (exactly the object of Problem 12 of Chapter 1). Its action on a reference basis vector,

$$\mathbf{F} \hat{\mathbf{E}}_B = (F_{iA} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A) \hat{\mathbf{E}}_B = F_{iA} (\hat{\mathbf{E}}_A \cdot \hat{\mathbf{E}}_B) \mathbf{e}_i = F_{iA} \delta_{AB} \mathbf{e}_i = F_{iB} \mathbf{e}_i, \quad (2.40)$$

gives the component extraction $F_{iA} = \mathbf{e}_i \cdot \mathbf{F} \hat{\mathbf{E}}_A$ (2.41).

LEMMA 2.15 (product of two dyads). *For all vectors, $(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d}) = (\mathbf{b} \cdot \mathbf{c}) \mathbf{a} \otimes \mathbf{d}$.*

Proof. Act on an arbitrary $\mathbf{v}$ and unfold using $(\mathbf{c} \otimes \mathbf{d})\mathbf{v} = (\mathbf{d} \cdot \mathbf{v})\mathbf{c}$ and $(\mathbf{a} \otimes \mathbf{b})\mathbf{c} = (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$:

$$\begin{aligned} [(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d})]\mathbf{v} &= (\mathbf{a} \otimes \mathbf{b})[(\mathbf{d} \cdot \mathbf{v})\mathbf{c}] = (\mathbf{d} \cdot \mathbf{v})(\mathbf{b} \cdot \mathbf{c})\mathbf{a} \\ &= (\mathbf{b} \cdot \mathbf{c})(\mathbf{d} \cdot \mathbf{v})\mathbf{a} = (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \otimes \mathbf{d})\mathbf{v}, \end{aligned} \quad (2.43-2.44)$$

true for all $\mathbf{v}$, so the tensors are equal. $\blacksquare$

Applying Lemma 2.15 to (2.37) with $\text{grad } \mathbf{u} = \tilde{u}_{i,k} \mathbf{e}_i \otimes \mathbf{e}_k$ and $\mathbf{F} = F_{jA} \mathbf{e}_j \otimes \hat{\mathbf{E}}_A$,

$$\begin{aligned} \text{Grad } \mathbf{u} &= (\tilde{u}_{i,k} \mathbf{e}_i \otimes \mathbf{e}_k)(F_{jA} \mathbf{e}_j \otimes \hat{\mathbf{E}}_A) = \tilde{u}_{i,k} F_{jA} (\mathbf{e}_k \cdot \mathbf{e}_j) \mathbf{e}_i \otimes \hat{\mathbf{E}}_A \\ &= \tilde{u}_{i,k} x_{j,A} \delta_{kj} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A = \tilde{u}_{i,j} x_{j,A} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A = \hat{u}_{i,A} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A, \end{aligned} \quad (2.42)$$

the last step being the ordinary chain rule $\hat{u}_{i,A} = \tilde{u}_{i,j} x_{j,A}$. So $\text{Grad } \underline{u}$ is a two-point tensor too.

Material curves. Let $\underline{X}(S)$, S arclength, parametrise a curve $C \subset \kappa$ of fixed material points; at time t^* it is convected to $\underline{x}_*(S) = \chi_\kappa(\underline{X}(S), t^*)$ (Figure 2.7). The relative position of two of its points is obtained by integrating the tangent,

$$\begin{aligned} \underline{x}_*(S_1) - \underline{x}_*(S_0) &= \underline{e}_i \int_{S_0}^{S_1} x'_{*i}(S) \, dS \\ &= \underline{e}_i \int_{S_0}^{S_1} F_{iA}(X_B(S), t^*) X'_A(S) \, dS, \end{aligned} \quad (2.45-2.46)$$

using $x'_{*i} = \chi_{i,A} X'_A = F_{iA} X'_A$: knowing $\underline{F}$ lets us reconstruct relative positions of connected material points.

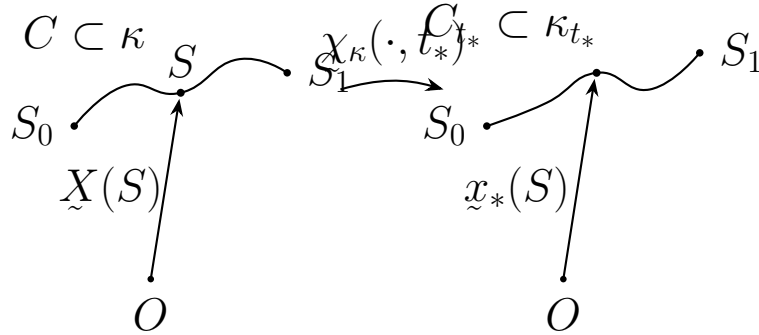


Figure 2.7: A material curve and its image in κ_{t^*} .

REMARK 2.16 (Figure 2.7). Figure 2.7 shows a curve C of fixed material points in κ and its deformed image C_{t^*} in κ_{t^*} ; the arclength S labels the material points and rides along with them. The tangent transforms by $\underline{F}$, $\underline{x}'_* = \underline{F} \underline{X}'$, the seed of the stretch formula $\underline{\mu} \underline{m} = \underline{F} \underline{M}$ of §2.6.

Volume. Invertibility of χ_κ at each t makes $\underline{F}$ invertible (inverse function theorem), and invertibility of a tensor is $\det \underline{F} \neq 0$, where, from Chapter 1,

$$\det \underline{F} = \frac{[\underline{F}\underline{a}, \underline{F}\underline{b}, \underline{F}\underline{c}]}{[\underline{a}, \underline{b}, \underline{c}]} \quad (2.47)$$

for any independent $\{\underline{a}, \underline{b}, \underline{c}\}$ (well defined for the two-point $\underline{F}$ because numerator and denominator each use vectors of a single space).

REMARK 2.17 (invertibility of $\underline{F}$ is only *local* information). The implication runs one way only. Global invertibility of the motion $\underline{\chi}_\kappa(\cdot, t)$ gives, by the inverse function theorem, an inverse $\underline{F}^{-1}$ at each $\underline{X}$. But the existence of $\underline{F}^{-1}$ at a *particular* $\underline{X}$ does *not* imply that $\underline{\chi}_\kappa$ is invertible at all $\underline{X}$: it implies only that $\underline{\chi}_\kappa$ is invertible in some *neighbourhood* of the point considered. Nothing local can prevent a motion from folding a distant part of the body onto the piece near $\underline{X}$; $\det \underline{F} \neq 0$ everywhere forbids local self-penetration, not global. This is the same phenomenon as $x \mapsto e^{ix}$ having nonzero derivative everywhere while being far from injective, and it is why global injectivity of the motion is imposed as a separate physical assumption in Definition 2.2 rather than deduced from $\det \underline{F} > 0$.

Three material curves through $\underline{X}$ carry tangents $d\underline{X}, d\underline{Y}, d\underline{Z}$ spanning a reference parallelepiped of volume $dV = [d\underline{X}, d\underline{Y}, d\underline{Z}]$ (Figure 2.8); their images $d\underline{x} = \underline{F}d\underline{X}$ etc. span $d\underline{v} = [d\underline{x}, d\underline{y}, d\underline{z}]$. Taking $\{\underline{a}, \underline{b}, \underline{c}\} = \{d\underline{X}, d\underline{Y}, d\underline{Z}\}$ in (2.47),

$$\begin{aligned} d\underline{v} &= [\underline{F}d\underline{X}, \underline{F}d\underline{Y}, \underline{F}d\underline{Z}] = \det \underline{F} [d\underline{X}, d\underline{Y}, d\underline{Z}] \\ &= J dV, \quad J = \det \underline{F} > 0. \end{aligned} \tag{2.48–2.50}$$

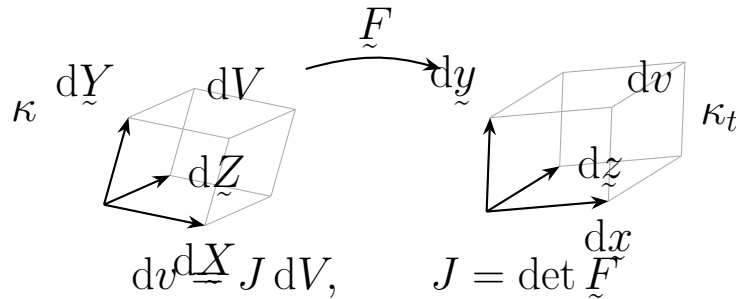


Figure 2.8: Three material curves intersecting at a material point.

REMARK 2.18 (Figure 2.8, and J as a volume ratio). Figure 2.8 shows the reference parallelepiped (left, κ) and its image (right, κ_t): $J =$

$\det \underline{\underline{F}}$ is the factor by which local volume is scaled, exactly the “oriented volume ratio” reading of the determinant from Chapter 1. If the reference configuration is *occupiable* (could be physically occupied), then $J > 0$; if not, one only knows $J = |\det \underline{\underline{F}}|$. Incompressibility is the constraint $J \equiv 1$.

Area (the Piola–Nanson formula). Let $d\underline{\underline{X}}, d\underline{\underline{Y}}$ span the tangent plane of a material surface at $\underline{\underline{X}}$, of area $dA = |d\underline{\underline{X}} \times d\underline{\underline{Y}}|$, with unit normal $\underline{\underline{N}}$ defined by

$$\underline{\underline{N}} dA = d\underline{\underline{X}} \times d\underline{\underline{Y}} \quad (2.51)$$

(Figure 2.9). The deformed edges $d\underline{\underline{x}} = \underline{\underline{F}}d\underline{\underline{X}}, d\underline{\underline{y}} = \underline{\underline{F}}d\underline{\underline{Y}}$ span a parallelogram of area da with unit normal $\underline{\underline{n}}, \underline{\underline{n}} da = d\underline{\underline{x}} \times d\underline{\underline{y}}$. Using the cofactor $\underline{\underline{F}}^*$, defined (as in Chapter 1) by $\underline{\underline{F}}^*(\underline{\underline{a}} \times \underline{\underline{b}}) = \underline{\underline{F}}\underline{\underline{a}} \times \underline{\underline{F}}\underline{\underline{b}}$,

$$\underline{\underline{n}} da = \underline{\underline{F}}d\underline{\underline{X}} \times \underline{\underline{F}}d\underline{\underline{Y}} = \underline{\underline{F}}^*(d\underline{\underline{X}} \times d\underline{\underline{Y}}) = \underline{\underline{F}}^*\underline{\underline{N}} dA, \quad (2.52\text{--}2.54)$$

the *Piola–Nanson formula*, in components $n_i da = F_{iB}^* N_B dA$. By Problem 22(c) of Chapter 1, $\underline{\underline{A}}^t \underline{\underline{A}}^* = (\det \underline{\underline{A}}) \underline{\underline{I}}$, valid also for the two-point $\underline{\underline{F}}$, so

$$\underline{\underline{F}}^* = J \underline{\underline{F}}^{-t}. \quad (2.55)$$

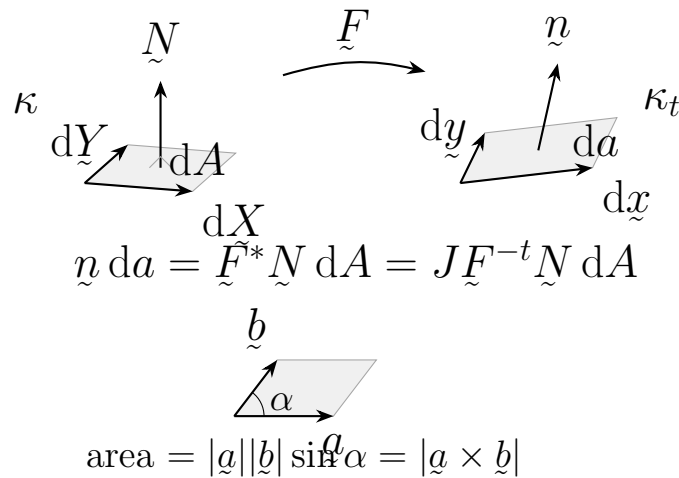


Figure 2.9: Deformation of oriented surface measure.

REMARK 2.19 (Figure 2.9: area is not a passive rider). Figure 2.9 shows the reference patch with edges $d\tilde{X}, d\tilde{Y}$, area dA , normal $\tilde{N}$, and its deformed image with $d\tilde{x}, d\tilde{y}$, area da , normal $\tilde{n}$; the small inset recalls $\text{area} = |\underline{a}||\underline{b}|\sin\alpha = |\underline{a} \times \underline{b}|$ from Chapter 1. The point of $\tilde{n} da = J\tilde{F}^{-t}\tilde{N} dA$ is that the normal does *not* transform by $\tilde{F}$ (that would be the tangent's rule): it transforms by the cofactor $\tilde{F}^* = J\tilde{F}^{-t}$, because a normal is defined by a cross product, and cross products go with cofactors. This single formula converts every surface traction and flux between reference and current descriptions in the balance laws.

2.5 Mass and density

Consider a sub-body $S \subset B$, occupying a part $P = \kappa(S) \subset \kappa$ of the reference configuration and a part $P_t = \chi_\kappa(P, t) \subset \kappa_t$ of the current one. Mass is assumed absolutely continuous in volume (more volume, more mass) and strictly positive on regions of positive volume, so there are positive *density* fields $\rho(\tilde{x}, t)$ and $\rho_\kappa(\tilde{X}, t)$ with

$$m(S, t) = \int_{P_t} \rho(\tilde{x}, t) dv = \int_P \rho_\kappa(\tilde{X}, t) dV. \quad (2.56)$$

(Conservation of mass, treated later, will make m independent of t .)

Change variables in the first integral by $\tilde{x} = \chi_\kappa(\tilde{X}, t)$, $dv = J dV$ from (2.50):

$$\int_{P_t} \rho(\tilde{x}, t) dv = \int_P \rho(\chi_\kappa(\tilde{X}, t), t) J(\tilde{X}, t) dV, \quad (2.57)$$

so equality of the two forms in (2.56) becomes

$$\int_P [\rho_\kappa(\tilde{X}, t) - \rho(\chi_\kappa(\tilde{X}, t), t) J(\tilde{X}, t)] dV = 0 \quad \text{for all } P \subset \kappa. \quad (2.58)$$

It is *sufficient* for this that the bracket vanish pointwise,

$$\rho_\kappa = \rho J \quad \text{at every } \tilde{X} \in \kappa. \quad (2.59)$$

That it is also *necessary*, when the integrand is continuous, is the content of the localization theorem.

THEOREM 2.20 (Localization). *Let $\varphi(\tilde{X})$ be continuous and suppose*

$$\int_P \varphi(\tilde{X}) \, dV = 0 \quad \text{for all } P \subset \kappa. \quad (2.60)$$

Then

$$\varphi(\tilde{X}) = 0 \quad \text{at every } \tilde{X} \in \kappa. \quad (2.61)$$

(The converse statement is obvious: if φ vanishes identically its integral over every P is zero.)

Proof. Fix $\tilde{X}_0 \in \kappa$ and take $P = R_\varepsilon$, the ball of radius ε and volume V_ε centred at $\tilde{X}_0$ and contained in κ . Consider the average discrepancy

$$I_\varepsilon = \left| \frac{1}{V_\varepsilon} \int_{R_\varepsilon} [\varphi(\tilde{X}_0) - \varphi(\tilde{X})] \, dV \right|. \quad (2.62)$$

By the triangle inequality for integrals, then bounding the integrand by its maximum over the (compact) ball,

$$\begin{aligned} I_\varepsilon &\leq \frac{1}{V_\varepsilon} \int_{R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})| \, dV \\ &\leq \frac{1}{V_\varepsilon} \int_{R_\varepsilon} \max_{\tilde{X} \in R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})| \, dV \\ &= \max_{\tilde{X} \in R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})|, \end{aligned} \quad (2.63)$$

the last step because the maximum is a constant and $\int_{R_\varepsilon} dV = V_\varepsilon$. A continuous function on a compact set attains its maximum, and by continuity of φ that maximum tends to 0 as the ball shrinks:

$$0 \leq I_\varepsilon \leq \max_{\tilde{X} \in R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})| \longrightarrow 0 \quad (\varepsilon \rightarrow 0). \quad (2.64)$$

But the hypothesis (2.60) makes $\int_{R_\varepsilon} \varphi(\tilde{X}) \, dV = 0$, so

$$\varphi(\tilde{X}_0) = \frac{1}{V_\varepsilon} \int_{R_\varepsilon} \varphi(\tilde{X}_0) \, dV = \frac{1}{V_\varepsilon} \int_{R_\varepsilon} [\varphi(\tilde{X}_0) - \varphi(\tilde{X})] \, dV \xrightarrow{\varepsilon \rightarrow 0} 0, \quad (2.65)$$

i.e. $|\varphi(\tilde{X}_0)| = \lim_{\varepsilon \rightarrow 0} I_\varepsilon = 0$. Since $\tilde{X}_0$ was arbitrary, $\varphi \equiv 0$. ■

Applying the theorem to the continuous bracket in (2.58) yields the pointwise mass relation (2.59), $\rho_\kappa = \rho J$. Continuity of that bracket is guaranteed by the assumed differentiability of χ_κ (which makes $\tilde{F}$, and hence $J = \det \tilde{F}$, continuous) together with continuity of ρ .

REMARK 2.21 (what localization really says). The theorem is the bridge from *integral* (global) balance laws to *pointwise* (local) field equations, and it is used again and again: conservation of mass, momentum, and energy are all first stated as “the integral over every sub-body vanishes”, and localization converts each into a differential equation holding at every point. Intuitively, if an average over *every* shrinking ball around $\underline{X}_0$ is forced to be zero, the value at $\underline{X}_0$ cannot be anything but zero, since a continuous function is nearly constant on a small enough ball and its average there nearly equals its central value. The relation $\rho_\kappa = \rho J$ itself is just conservation of mass in disguise: the same matter in a smaller current volume ($J < 1$) must be denser, $\rho = \rho_\kappa / J$.

2.6 Material curves and the shifter

Return to the material curve $C \subset \kappa$ and its convected image $C_{t_*} = \chi_\kappa(C, t_*) \subset \kappa_{t_*}$ (Figure 2.10). The material point at arclength station S on C occupies $\underline{x}_*(S) = \chi_\kappa(\underline{X}(S), t_*)$. Differentiating in S at fixed t_* , by the chain rule and (2.34),

$$\underline{x}'_*(S) = \tilde{F}(\underline{X}(S), t_*) \underline{M}(S), \quad \underline{M}(S) = \underline{X}'(S), \quad (2.66)$$

$\underline{M}$ being a *unit* tangent to C (because S is arclength, $|\underline{X}'| = 1$). The image $\underline{x}'_*$ is tangent to C_{t_*} but need not be a unit vector; writing $\underline{x}'_* = \mu \underline{m}$ with $|\underline{m}| = 1$ and $\mu = |\underline{x}'_*|$ the local *stretch*,

$$\mu \underline{m} = \tilde{F} \underline{M}. \quad (2.67)$$

REMARK 2.22 (Figure 2.10: what $\tilde{F}$ does to a fibre). Figure 2.10 shows the curve $C \subset \kappa$ with station S , position $\underline{X}(S)$ and unit tangent $\underline{M}$, beside its image $C_{t_*} \subset \kappa_{t_*}$ with $\underline{x}_*(S) = \chi_\kappa(\underline{X}(S), t_*)$. Equation (2.67)

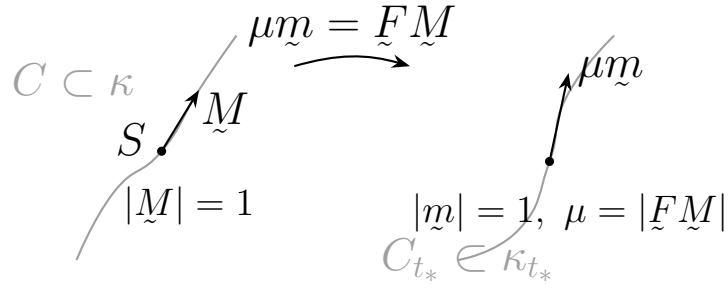


Figure 2.10: Orientation of material curves at point p with arclength S in κ .

is the sentence “ $\tilde{F}$ takes the unit material fibre $\tilde{M}$ and returns the deformed fibre, whose length μ is the stretch and whose direction $\tilde{m}$ is the new orientation”. Every deformation measure in §2.7 is extracted from this one statement by taking lengths and angles of $\tilde{F}\tilde{M}$.

Interpreting (2.67) for an *undeformed* body raises a genuine conceptual difficulty. We stipulated $\tilde{X} \in \hat{\mathbb{E}}^3$ but $\kappa_t \subset \mathbb{E}^3$ for every t , including t_0 ; so if we want to say “at t_0 the body is where its reference copy is”, we are comparing vectors from two different spaces, which is meaningless. The repair is a fixed two-point tensor that carries one space onto the other.

DEFINITION 2.23 (The shifter). Introduce fixed orthonormal vectors $\tilde{E}_A \in \mathbb{E}^3$ and the uniform two-point tensor

$$\underline{1} = \tilde{E}_A \otimes \hat{E}_A. \quad (2.68)$$

If $\{\hat{E}_A\}$ and $\{\tilde{E}_A\}$ have the same handedness, $\underline{1}$ rotates the former onto the latter and is a two-point rotation; otherwise it is merely orthogonal. Resolving $\tilde{E}_A \in \mathbb{E}^3$ on $\{\tilde{e}_i\}$ by the orthonormal expansion of Chapter 1,

$$\tilde{E}_A = \delta_{iA} \tilde{e}_i, \quad \delta_{iA} = \tilde{e}_i \cdot \tilde{E}_A, \quad (2.69)$$

(the dot product is legitimate: both factors lie in $\mathbb{E}^3$), giving the standard two-point decomposition

$$\underline{1} = \delta_{iA} \tilde{e}_i \otimes \hat{E}_A. \quad (2.70)$$

The δ_{iA} are Kronecker deltas if and only if $\{\underline{E}_A\} = \{\underline{e}_i\}$; otherwise they are the cosines of the fixed angles between the two sets.

PROPOSITION 2.24 (Algebra of the shifter). *With $\underline{I}$ and $\hat{\underline{I}}$ the identities of $\mathbb{E}^3$ and $\hat{\mathbb{E}}^3$,*

$$\underline{1} = \underline{1}\hat{\underline{I}} = \underline{I}\underline{1}, \quad \underline{1}^t \underline{1} = \hat{\underline{I}}, \quad \underline{1}\underline{1}^t = \underline{I}, \quad (2.71-2.73)$$

equivalently, in components,

$$\delta_{iA}\delta_{iB} = \delta_{AB}, \quad \delta_{iA}\delta_{jA} = \delta_{ij}. \quad (2.74)$$

Proof. For (2.71): $\underline{E}_A = \underline{I}\underline{E}_A$ and $\hat{\underline{E}}_A = \hat{\underline{I}}\hat{\underline{E}}_A$, so by the rules $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$ and $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$ (Problems 17(b),(c) of Chapter 1),

$$\begin{aligned} \underline{I}\underline{1} &= \underline{I}(\underline{E}_A \otimes \hat{\underline{E}}_A) = (\underline{I}\underline{E}_A) \otimes \hat{\underline{E}}_A = \underline{1}, \\ \underline{1}\hat{\underline{I}} &= \underline{E}_A \otimes (\hat{\underline{I}}^t \hat{\underline{E}}_A) = \underline{E}_A \otimes \hat{\underline{E}}_A = \underline{1}. \end{aligned}$$

For (2.72), transpose (2.68) by Problem 17(a) of Chapter 1, which gives $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$, and then multiply using Lemma 2.15:

$$\begin{aligned} \underline{1}^t \underline{1} &= (\hat{\underline{E}}_A \otimes \underline{E}_A)(\underline{E}_B \otimes \hat{\underline{E}}_B) = (\underline{E}_A \cdot \underline{E}_B) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B \\ &= \delta_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B = \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A = \hat{\underline{I}}. \end{aligned} \quad (2.72)$$

Identically, $\underline{1}\underline{1}^t = (\hat{\underline{E}}_A \cdot \hat{\underline{E}}_B)\underline{E}_A \otimes \underline{E}_B = \underline{E}_A \otimes \underline{E}_A = \underline{I}$. In components, (2.72) reads $\delta_{iA}\delta_{iB} = \delta_{AB}$ and (2.73) reads $\delta_{iA}\delta_{jA} = \delta_{ij}$, which is (2.74). ■

Acting with the shifter on (2.23) produces a *spatial* copy of the reference position,

$$\underline{1}X = X_A \underline{1}\hat{\underline{E}}_A = X_A \underline{E}_A \in \mathbb{E}^3, \quad (2.75)$$

and the statement $\kappa = \kappa_{t_0}$ is then properly written

$$\chi_\kappa(X, t_0) = \underline{1}X \quad (2.76)$$

(taking the origins of Figure 2.6 to coincide, without loss of generality).

This makes the *displacement* well defined:

$$\underline{u}(X, t) = \chi_\kappa(X, t) - \underline{1}X \in \mathbb{E}^3, \quad \hat{u}_i(X_B, t) = \chi_i(X_B, t) - \delta_{iA}X_A. \quad (2.77-2.78)$$

REMARK 2.25 (why the shifter is not pedantry). The definition of displacement found in most of the literature omits $\underline{1}$ and writes $\underline{u} = \underline{\chi}_\kappa - \underline{X}$. That expression subtracts a vector of $\mathbb{E}^3$ from a vector of $\hat{\mathbb{E}}^3$: an impermissible operation between different vector spaces, and a documented source of confusion in using displacement to characterise deformation. The shifter is the “two-point identity tensor”: when $\underline{E}_A$ and $\hat{\underline{E}}_A$ are aligned it simply *parallel transports* a vector from one copy of space to the other and changes nothing else, so its components δ_{iA} then really are Kronecker deltas and every classical formula is recovered. Its only job is to make the bookkeeping honest, exactly as the caret did for functions (Remark 2.6). The shifter and the displacement are put to substantive use later, in the book’s Chapters 7 and 12.

PROPOSITION 2.26 (Undeformed state). *If (2.76) holds then*

$$\underline{F}(\underline{X}, t_0) = \underline{1}, \quad F_{iA} = \delta_{iA}, \quad (2.79)$$

and consequently $\mu = 1$ for every material curve:

$$\underline{\mu} \underline{m} = \underline{1} \underline{M}, \quad \mu = \sqrt{\underline{1} \underline{M} \cdot \underline{1} \underline{M}} = \sqrt{\underline{M} \cdot \underline{M}} = 1. \quad (2.80\text{--}2.81)$$

Moreover, in the general case (2.67) forces $\mu > 0$.

Proof. (2.79): $\underline{F} = \text{Grad } \underline{\chi}_\kappa$ and $\underline{\chi}_\kappa(\underline{X}, t_0) = \underline{1} \underline{X}$ is *linear* in $\underline{X}$ with the uniform (position independent) coefficient $\underline{1}$, so its gradient is $\underline{1}$ itself; in components $F_{iA} = \partial(\delta_{jB} X_B) / \partial X_A \big|_{i=j} = \delta_{iA}$. Substituting into (2.67) gives (2.80), and since the shifter is orthogonal, $\underline{1} \underline{M} \cdot \underline{1} \underline{M} = \underline{M} \cdot \underline{1}^t \underline{1} \underline{M} = \underline{M} \cdot \hat{\underline{I}} \underline{M} = \underline{M} \cdot \underline{M} = 1$ by (2.72), so $\mu = 1$: an undeformed body stretches no fibre. Finally $\mu > 0$ always, for $\mu = 0$ would give $\underline{F} \underline{M} = \underline{0}$ with $\underline{M} \neq \underline{0}$ (as $|\underline{M}| = 1$), contradicting invertibility of $\underline{F}$. ■

2.7 The Cauchy–Green deformation tensors and the Lagrange–Euler strain tensors

The stretch is the positive square root of μ^2 , and (2.67) turns that into a quadratic form in $\underline{M}$. Using only the definitions of the transpose and

of the product of two tensors (Chapter 1),

$$\mu^2 = |\mu \underline{m}|^2 = \underline{F} \underline{M} \cdot \underline{F} \underline{M} = \underline{M} \cdot \underline{F}^t(\underline{F} \underline{M}) = \underline{M} \cdot (\underline{F}^t \underline{F}) \underline{M}, \quad (2.82)$$

so that

$$0 < \mu^2 = \underline{M} \cdot \underline{C} \underline{M}, \quad \underline{C} = \underline{F}^t \underline{F} \quad (2.83)$$

defines the *right Cauchy–Green deformation tensor*.

PROPOSITION 2.27 (Properties of $\underline{C}$). $\underline{C}$ is a referential tensor ($\hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^3$), symmetric, and positive definite; its components are

$$\begin{aligned} \underline{C} &= C_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B, & C_{AB} &= F_{iA} F_{iB} = C_{BA}, \\ \mu^2 &= C_{AB} M_A M_B. \end{aligned} \quad (2.86\text{--}2.88)$$

Proof. Domain. $\underline{F} : \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, hence $\underline{F}^t : \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, hence $\underline{C} = \underline{F}^t \underline{F}$ maps $\hat{\mathbb{E}}^3$ into itself. *Positive definiteness.* For any $\underline{a} \neq \underline{0}$ in $\hat{\mathbb{E}}^3$, moving $\underline{F}^t$ across the dot product by the definition of the transpose,

$$\underline{a} \cdot \underline{C} \underline{a} = \underline{a} \cdot \underline{F}^t \underline{F} \underline{a} = \underline{F} \underline{a} \cdot \underline{F} \underline{a} = |\underline{F} \underline{a}|^2 > 0, \quad (2.84)$$

strictly, because $\underline{F}$ is invertible ($\det \underline{F} \neq 0$) so $\underline{F} \underline{a} \neq \underline{0}$. *Symmetry*, using $(\underline{B} \underline{A})^t = \underline{A}^t \underline{B}^t$ (Problem 20 of Chapter 1) and $(\underline{F}^t)^t = \underline{F}$:

$$\underline{C}^t = (\underline{F}^t \underline{F})^t = \underline{F}^t (\underline{F}^t)^t = \underline{F}^t \underline{F} = \underline{C}. \quad (2.85)$$

Components. With $\underline{F} = F_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A$ and $\underline{F}^t = F_{iA} \hat{\underline{E}}_A \otimes \underline{e}_i$ (Problem 17(a) of Chapter 1), multiply by Lemma 2.15:

$$\begin{aligned} \underline{C} &= (F_{iA} \hat{\underline{E}}_A \otimes \underline{e}_i)(F_{jB} \underline{e}_j \otimes \hat{\underline{E}}_B) = F_{iA} F_{jB} (\underline{e}_i \cdot \underline{e}_j) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B \\ &= F_{iA} F_{jB} \delta_{ij} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B = C_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B, & C_{AB} &= F_{iA} F_{iB}, \end{aligned} \quad (2.86\text{--}2.87)$$

manifestly symmetric since $F_{iA} F_{iB} = F_{iB} F_{iA}$. Finally

$$\mu^2 = \underline{M} \cdot (C_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B) \underline{M} = \underline{M} \cdot C_{AB} M_B \hat{\underline{E}}_A = C_{AB} M_A M_B, \quad (2.88)$$

so (C_{AB}) is a positive definite matrix. ■

Inverting (2.67) gives the companion, spatial, measure. Applying F^{-1} to $\mu \underline{m} = \underline{F} M$,

$$\begin{aligned} \mu \underline{F}^{-1} \underline{m} &= \underline{F}^{-1}(\underline{F} M) = (\underline{F}^{-1} \underline{F}) M = \hat{I} M = M, \\ \text{i.e. } \mu^{-1} M &= \underline{F}^{-1} \underline{m}, \end{aligned} \quad (2.89\text{--}2.90)$$

and taking the squared length of the left side (recall $|M| = 1$),

$$\begin{aligned} \mu^{-2} &= |\mu^{-1} M|^2 = \underline{F}^{-1} \underline{m} \cdot \underline{F}^{-1} \underline{m} \\ &= \underline{m} \cdot (\underline{F}^{-1})^t \underline{F}^{-1} \underline{m} = \underline{m} \cdot \underline{F}^{-t} \underline{F}^{-1} \underline{m} = \underline{m} \cdot \underline{B}^{-1} \underline{m}, \end{aligned} \quad (2.91)$$

where

$$\underline{B} = \underline{F} \underline{F}^t \quad (2.92)$$

is the *left Cauchy–Green deformation tensor*; indeed $\underline{B}^{-1} = (\underline{F} \underline{F}^t)^{-1} = \underline{F}^{-t} \underline{F}^{-1}$ by Problems 18–19 of Chapter 1.

PROPOSITION 2.28 (Properties of $\underline{B}$). *$\underline{B}$ is a spatial tensor ($\mathbb{E}^3 \rightarrow \mathbb{E}^3$), symmetric, with*

$$\underline{B} = B_{ij} \underline{e}_i \otimes \underline{e}_j, \quad B_{ij} = F_{iA} F_{jA} = B_{ji}, \quad (2.93\text{--}2.94)$$

and $\underline{B}^{-1}$ is positive definite.

Proof. $\underline{F}^t : \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$ then $\underline{F} : \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, so $\underline{B}$ maps $\mathbb{E}^3$ to itself; symmetry is $\underline{B}^t = (\underline{F} \underline{F}^t)^t = \underline{F} \underline{F}^t = \underline{B}$. Components, again by Lemma 2.15:

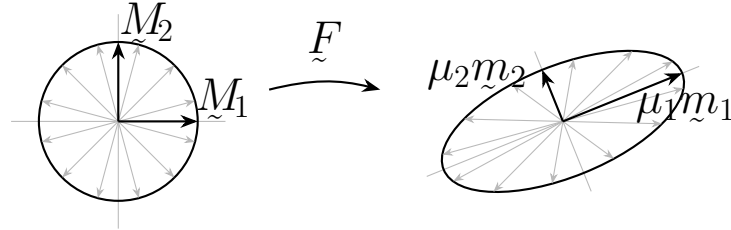
$$\begin{aligned} \underline{B} &= (F_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A)(F_{jB} \hat{\underline{E}}_B \otimes \underline{e}_j) = F_{iA} F_{jB} (\hat{\underline{E}}_A \cdot \hat{\underline{E}}_B) \underline{e}_i \otimes \underline{e}_j \\ &= F_{iA} F_{jB} \delta_{AB} \underline{e}_i \otimes \underline{e}_j = B_{ij} \underline{e}_i \otimes \underline{e}_j, \quad B_{ij} = F_{iA} F_{jA}. \end{aligned} \quad (2.93\text{--}2.94)$$

Positive definiteness of $\underline{B}^{-1}$: for $\underline{a} \neq \underline{0}$ in $\mathbb{E}^3$,

$$\underline{a} \cdot \underline{B}^{-1} \underline{a} = \underline{a} \cdot \underline{F}^{-t} \underline{F}^{-1} \underline{a} = \underline{F}^{-1} \underline{a} \cdot \underline{F}^{-1} \underline{a} = |\underline{F}^{-1} \underline{a}|^2 > 0, \quad (2.95)$$

since $\underline{F}^{-1}$ is invertible (with inverse $\underline{F}$). In Chapter 3 this is shown equivalent to positive definiteness of $\underline{B}$ itself. ■

REMARK 2.29 ($\underline{\mathcal{C}}$ and $\underline{\mathcal{B}}$: which space you measure in). $\underline{F}$ has one foot in each configuration, so it cannot be symmetric and carries rotation as well as stretch. Pairing $\underline{F}$ with its transpose cancels the rotation and lands the result wholly in one space: $\underline{\mathcal{C}} = \underline{F}^t \underline{F}$ lives in the *reference* space and is fed reference fibres $\underline{M}$, while $\underline{\mathcal{B}} = \underline{F} \underline{F}^t$ lives in the *current* space and is fed current fibres $\underline{m}$. Concretely, (2.83) says “give me the undeformed direction and I return the square of the stretch”, whereas (2.91) says “give me the deformed direction and I return the inverse square of the stretch”. If $\underline{F}$ is a pure rotation, $\underline{F}^t \underline{F} = \hat{\underline{I}}$ and no fibre stretches, which is exactly why these products, and not $\underline{F}$, are the measures of *distortion*.



unit directions $|\underline{M}| = 1$ in κ images $\underline{F}\underline{M}$: an ellipse

SUPPLEMENTARY FIGURE B. What $\underline{\mathcal{C}} = \underline{F}^t \underline{F}$ measures. Feed $\underline{F}$ every unit direction $\underline{M}$ at a material point: the images $\underline{F}\underline{M}$ sweep out an *ellipse* (an ellipsoid in three dimensions), because $\mu^2 = \underline{M} \cdot \underline{\mathcal{C}} \underline{M}$ is a quadratic form in $\underline{M}$. The stretch μ is the length of the image, so it depends on direction; its largest and smallest values occur along the principal axes of the ellipse, which are the eigenvectors of $\underline{\mathcal{C}}$, the corresponding μ^2 being its eigenvalues. A general fibre is both stretched *and* rotated, and the change in angle between two fibres is the shear recorded by the off-diagonal $\underline{C}_{AB}$. A rigid motion sends the circle to a circle of equal radius, i.e. $\underline{\mathcal{C}} = \hat{\underline{I}}$.

REMARK 2.30 (reading Supplementary Figure B: why $\underline{F}$ splits). The picture already contains the polar decomposition theorem of Chapter 3. Turning the circle into the ellipse can be done in two stages: first stretch along a fixed pair of orthogonal directions of κ , then rotate the stretched figure into place. Symbolically $\underline{F} = \underline{R}\underline{U}$ with $\underline{U}$ symmetric positive definite (the stretch) and $\underline{R}$ orthogonal (the rotation). Squaring kills the rotation, exactly as the picture suggests:

$$\underline{\mathcal{C}} = \underline{F}^t \underline{F} = (\underline{R}\underline{U})^t (\underline{R}\underline{U}) = \underline{U}^t \underline{R}^t \underline{R} \underline{U} = \underline{U} \hat{\underline{I}} \underline{U} = \underline{U}^2,$$

using $(\underline{B}\underline{A})^t = \underline{A}^t \underline{B}^t$, $\underline{R}^t \underline{R} = \underline{\hat{I}}$ and $\underline{U}^t = \underline{U}$. So $\underline{C}$ records the ellipse's shape and size but nothing about its final orientation, which is the precise sense in which it measures distortion alone; $\mu^2 = \underline{M} \cdot \underline{C} \underline{M} = \underline{M} \cdot \underline{U}^2 \underline{M} = |\underline{U} \underline{M}|^2$ says the stretch is done entirely by $\underline{U}$. Symmetrically $\underline{B} = \underline{F} \underline{F}^t = \underline{R} \underline{U}^2 \underline{R}^t = \underline{V}^2$ with $\underline{V} = \underline{R} \underline{U} \underline{R}^t$, the same stretch seen from the current configuration.

DEFINITION 2.31 (Lagrange and Euler strains). The deviation of the stretch from its undeformed value 1 is measured by

$$\frac{1}{2}(\mu^2 - 1) = \frac{1}{2}(\underline{M} \cdot \underline{C} \underline{M} - 1) = \underline{M} \cdot \underline{E} \underline{M}, \quad \underline{E} = \frac{1}{2}(\underline{C} - \underline{\hat{I}}), \quad (2.96-2.97)$$

the *Lagrange strain tensor*, and by

$$\frac{1}{2}(1 - \mu^{-2}) = \underline{m} \cdot \underline{e} \underline{m}, \quad \underline{e} = \frac{1}{2}(\underline{I} - \underline{B}^{-1}), \quad (2.98-2.99)$$

the *Euler strain tensor*. Both are symmetric.

Verification. Since $|\underline{M}| = 1$, $\underline{M} \cdot \underline{\hat{I}} \underline{M} = \underline{M} \cdot \underline{M} = 1$, so $\underline{M} \cdot \underline{E} \underline{M} = \frac{1}{2}(\underline{M} \cdot \underline{C} \underline{M} - \underline{M} \cdot \underline{M}) = \frac{1}{2}(\mu^2 - 1)$, which is (2.96); the Euler case is identical using (2.91) and $|\underline{m}| = 1$. ■

REMARK 2.32 (why the factor $\frac{1}{2}$, and why strain at all). For $\mu \simeq 1$ write $\mu = 1 + \varepsilon$ with ε small; then $\mu^2 - 1 = (1 + \varepsilon)^2 - 1 = 2\varepsilon + \varepsilon^2 \simeq 2(\mu - 1)$. The factor $\frac{1}{2}$ is chosen precisely so that $\underline{M} \cdot \underline{E} \underline{M} \simeq \mu - 1$, the *extension* (change in length per unit original length), recovering the engineer's strain in the small-deformation limit; likewise $\underline{m} \cdot \underline{e} \underline{m} \simeq \mu - 1$. The reason for introducing strains at all, rather than working with $\underline{C}$ and $\underline{B}$, is that they *vanish* in the undeformed state: by (2.79), $\underline{F} = \underline{1}$ gives $\underline{C} = \underline{1}^t \underline{1} = \underline{\hat{I}}$ and $\underline{B} = \underline{1} \underline{1}^t = \underline{I}$ by (2.72)–(2.73), hence $\underline{E} = \underline{\hat{Q}}$ and $\underline{e} = \underline{Q}$. (Henceforth we drop the caret on the referential zero tensor and let context decide.) The names used here are not universal, and there are infinitely many equivalent strain measures; see Ogden's book for a survey.

REMARK 2.33 (symmetry is meaningless for two-point tensors). $\underline{E}$ and $\underline{e}$ are symmetric, but note carefully that symmetry and skewness are

defined only for tensors mapping a vector space *to itself*: $\tilde{A}^t = \pm \tilde{A}$ compares $\tilde{A}$ with an object of the same type. For a two-point tensor such as $\tilde{F}$, the transpose $\tilde{F}^t$ maps $\mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$ while $\tilde{F}$ maps $\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, so the equation $\tilde{F}^t = \pm \tilde{F}$ cannot even be written. Two-point tensors are therefore *neither* symmetric nor skew, which is another reason the deformation measures are built as $\tilde{F}^t \tilde{F}$ and $\tilde{F} \tilde{F}^t$.

Arclength of a deformed curve. On C , $d\tilde{X}(S) = \tilde{M} dS$ with $dS = |d\tilde{X}|$ (arclength parametrisation), so the deformed arclength measure is

$$ds = |d\tilde{x}| = \sqrt{\tilde{F} d\tilde{X} \cdot \tilde{F} d\tilde{X}} = \sqrt{\tilde{M} \cdot \tilde{C} \tilde{M}} dS = \mu(S, t_*) dS, \quad (2.100)$$

that is $ds/dS = \mu(S, t_*)$ (2.101), and the total length of the deformed curve, and its change, are

$$l(C_{t_*}) = \int_{S_0}^{S_1} \mu(S, t_*) dS, \quad l(C_{t_*}) - l(C) = \int_{S_0}^{S_1} [\mu(S, t_*) - 1] dS. \quad (2.102-2.103)$$

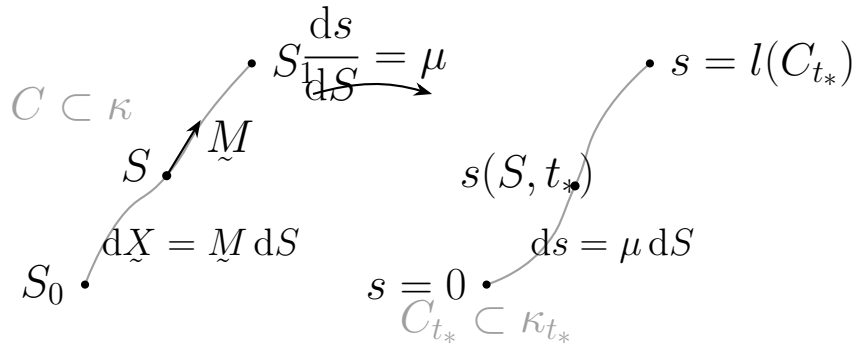


Figure 2.11: Arclength of a material point on material curves in κ and κ_{t_*} .

REMARK 2.34 (Figure 2.11). Figure 2.11 shows $C \subset \kappa$ carrying the stations S_0, S, S_1 , the unit tangent $\tilde{M}$ and the element $d\tilde{X} = \tilde{M} dS$, beside its image C_{t_*} carrying the deformed arclength $s(S, t_*)$. The content of (2.101) is that the stretch μ is literally the local ratio of deformed to undeformed arclength, so integrating it along the fibre gives the new length: a one-dimensional version of $dv = J dV$.

Angles between material curves. Let two material curves cross at $\tilde{X}$ with unit tangents $\tilde{M}_1, \tilde{M}_2$ and angle Θ ($\cos \Theta = \tilde{M}_1 \cdot \tilde{M}_2$); after deformation they cross at $\tilde{x}$ with unit tangents $\tilde{m}_1, \tilde{m}_2$ and angle θ ($\cos \theta = \tilde{m}_1 \cdot \tilde{m}_2$), where

$$\mu_1 \tilde{m}_1 = \tilde{F} \tilde{M}_1, \quad \mu_2 \tilde{m}_2 = \tilde{F} \tilde{M}_2 \quad (2.104)$$

(Figure 2.12). Dotting the two and moving $\tilde{F}^t$ across,

$$\mu_1 \mu_2 \cos \theta = \tilde{F} \tilde{M}_1 \cdot \tilde{F} \tilde{M}_2 = \tilde{M}_1 \cdot \tilde{C} \tilde{M}_2, \quad (2.105)$$

and dividing by $\mu_1 \mu_2 = \sqrt{\tilde{M}_1 \cdot \tilde{C} \tilde{M}_1} \sqrt{\tilde{M}_2 \cdot \tilde{C} \tilde{M}_2}$ from (2.83),

$$\cos \theta = \frac{\tilde{M}_1 \cdot \tilde{C} \tilde{M}_2}{\sqrt{(\tilde{M}_1 \cdot \tilde{C} \tilde{M}_1)(\tilde{M}_2 \cdot \tilde{C} \tilde{M}_2)}}, \quad (2.106)$$

in which $\tilde{M}_1 \cdot \tilde{C} \tilde{M}_2 = \tilde{M}_2 \cdot \tilde{C} \tilde{M}_1$ because $\tilde{M}_1 \cdot \tilde{C} \tilde{M}_2 = \tilde{C}^t \tilde{M}_1 \cdot \tilde{M}_2 = \tilde{M}_2 \cdot \tilde{C} \tilde{M}_1$ by the symmetry of $\tilde{C}$. Running the same argument backwards with $\mu_1^{-1} \tilde{M}_1 = \tilde{F}^{-1} \tilde{m}_1$, $\mu_2^{-1} \tilde{M}_2 = \tilde{F}^{-1} \tilde{m}_2$ from (2.90) gives the undeformed angle in terms of current data,

$$\cos \Theta = \frac{\tilde{m}_1 \cdot \tilde{B}^{-1} \tilde{m}_2}{\sqrt{(\tilde{m}_1 \cdot \tilde{B}^{-1} \tilde{m}_1)(\tilde{m}_2 \cdot \tilde{B}^{-1} \tilde{m}_2)}}. \quad (2.107-2.108)$$

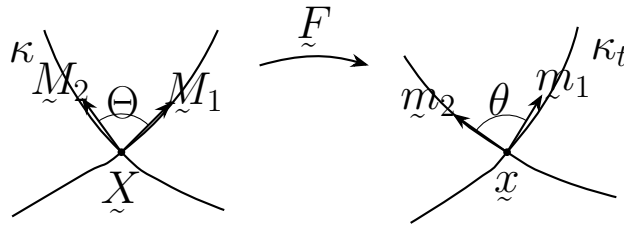


Figure 2.12: Intersecting material curves in κ and κ_{t*} .

REMARK 2.35 (Figure 2.12, and why $\tilde{C}$ deserves the name “distortion”). Figure 2.12 shows two material curves meeting at $\tilde{X}$ at angle Θ in κ , and their images meeting at $\tilde{x}$ at angle θ in κ_t . Formulas (2.106) and (2.108) say that $\tilde{C}$ (or $\tilde{B}$) determines *every* stretch and *every* angle change at the point, and these are the entire local geometric content of

a deformation: hence the Cauchy–Green tensors are said to characterise the *state of distortion* at p . Note the shear reading of the off-diagonal entries: taking $\tilde{M}_1 = \hat{\tilde{E}}_1$, $\tilde{M}_2 = \hat{\tilde{E}}_2$ (initially orthogonal, $\Theta = \pi/2$), (2.106) collapses to $\cos \theta = C_{12}/\sqrt{C_{11}C_{22}}$, so a nonzero C_{12} is the loss of perpendicularity, while the diagonal entries $C_{AA} = \mu^2$ (no sum) are squared stretches along the axes.

2.8 Problems

The three problems of §2.8 are solved below in full index notation, using only Chapter 1 algebra together with (2.39), $F_{iA} = \chi_{i,A}$. Recall the referential and spatial permutation symbols $\varepsilon_{ABC} = [\hat{\tilde{E}}_A, \hat{\tilde{E}}_B, \hat{\tilde{E}}_C]$ and $\varepsilon_{ijk} = [\underline{e}_i, \underline{e}_j, \underline{e}_k]$, and the contracted identities $\varepsilon_{ABC}\varepsilon_{DBC} = 2\delta_{AD}$, $\varepsilon_{ijk}\varepsilon_{ijk} = 6$.

LEMMA 2.36 (symmetric contracted with antisymmetric vanishes). *If $S_{BC} = S_{CB}$ and $A_{BC} = -A_{CB}$, then $A_{BC}S_{BC} = 0$, the sum running over both indices.*

Proof. Use the antisymmetry of A , then relabel the two dummy indices, then use the symmetry of S :

$$A_{BC}S_{BC} \stackrel{(1)}{=} -A_{CB}S_{BC} \stackrel{(2)}{=} -A_{BC}S_{CB} \stackrel{(3)}{=} -A_{BC}S_{BC}.$$

Step (1) is $A_{BC} = -A_{CB}$; step (2) renames the summation indices; step (3) is $S_{CB} = S_{BC}$. Hence $2A_{BC}S_{BC} = 0$. ■

The lemma is used twice below, at 2(a) and at 2(b).

Problem 1: components of the cofactor of $\tilde{F}$. *Show that $F_{iA}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC}$.*

Solution. The cofactor is defined by $\tilde{F}^*(\underline{a} \times \underline{b}) = \tilde{F}\underline{a} \times \tilde{F}\underline{b}$ for all $\underline{a}, \underline{b} \in \hat{\mathbb{E}}^3$. Take $\underline{a} = \hat{\tilde{E}}_B$, $\underline{b} = \hat{\tilde{E}}_C$, with

$$\hat{\tilde{E}}_B \times \hat{\tilde{E}}_C = \varepsilon_{BCA}\hat{\tilde{E}}_A$$

by Proposition 1.24. The left side becomes

$$\tilde{F}^*(\varepsilon_{BCA}\hat{\tilde{E}}_A) = \varepsilon_{BCA}\tilde{F}^*\hat{\tilde{E}}_A = \varepsilon_{BCA}F_{iA}^*\varepsilon_i,$$

using $\tilde{F}^*\hat{\tilde{E}}_A = F_{iA}^*\varepsilon_i$ (the analogue of (2.40)). The right side, expanding each image on $\{\varepsilon_i\}$ and using $\varepsilon_j \times \varepsilon_k = \varepsilon_{jki}\varepsilon_i$, is

$$\tilde{F}\hat{\tilde{E}}_B \times \tilde{F}\hat{\tilde{E}}_C = (F_{jB}\varepsilon_j) \times (F_{kC}\varepsilon_k) = F_{jB}F_{kC}\varepsilon_{jki}\varepsilon_i.$$

Equating components on ε_i and writing $\varepsilon_{jki} = \varepsilon_{ijk}$ (cyclic),

$$\varepsilon_{BCA}F_{iA}^* = \varepsilon_{ijk}F_{jB}F_{kC}.$$

Contract both sides with ε_{BCD} (sum over B, C). On the left, $\varepsilon_{BCA}\varepsilon_{BCD} = 2\delta_{AD}$, giving $2\delta_{AD}F_{iA}^* = 2F_{iD}^*$; on the right, $\varepsilon_{BCD} = \varepsilon_{DBC}$ (cyclic). Hence

$$F_{iD}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{DBC}F_{jB}F_{kC},$$

which upon renaming $D \rightarrow A$ is the required $F_{iA}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC}$.

□

The identity $\varepsilon_{BCA}\varepsilon_{BCD} = 2\delta_{AD}$ is Theorem 1.29. Directly: for $A \neq D$ every term repeats an index and dies; for $A = D = 3$ the survivors are $\varepsilon_{123}^2 + \varepsilon_{213}^2 = 2$. The factor 2 is why $\frac{1}{2}$ appears, each unordered pair $\{B, C\}$ being counted twice.

Check. For the simple shear of Chapter 3, $F_{iA} = \delta_{iA} + \gamma\delta_{i1}\delta_{A2}$, whose matrix is $\begin{pmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Since $\det \tilde{F} = 1$ and $\tilde{F}^* = J\tilde{F}^{-t} = \tilde{F}^{-t}$ by (2.55), the answer must be $\begin{pmatrix} 1 & 0 & 0 \\ -\gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Test the formula on the entry $A = 1, i = 2$:

$$F_{21}^* = \frac{1}{2}\varepsilon_{2jk}\varepsilon_{1BC}F_{jB}F_{kC}.$$

The only non-zero ε_{2jk} are $\varepsilon_{231} = 1$ and $\varepsilon_{213} = -1$; the only non-zero ε_{1BC} are $\varepsilon_{123} = 1$ and $\varepsilon_{132} = -1$. So four terms survive:

$$\begin{aligned} 2F_{21}^* &= \varepsilon_{231}\varepsilon_{123}F_{32}F_{13} + \varepsilon_{231}\varepsilon_{132}F_{33}F_{12} \\ &\quad + \varepsilon_{213}\varepsilon_{123}F_{12}F_{33} + \varepsilon_{213}\varepsilon_{132}F_{13}F_{32} \\ &= (+1)(+1)(0)(0) + (+1)(-1)(1)(\gamma) \\ &\quad + (-1)(+1)(\gamma)(1) + (-1)(-1)(0)(0) \\ &= -\gamma - \gamma = -2\gamma, \end{aligned}$$

so $F_{21}^* = -\gamma$, as required.

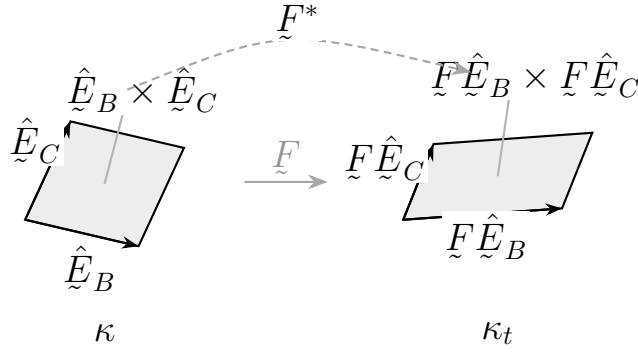


Figure 2.13: Problem 1. $\tilde{F}$ carries the edge vectors of a material parallelogram, $\tilde{F}^*$ carries its area vector. The problem converts the implicit definition into components, $F_{iA}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC}$; (2.55) then converts those into $J\tilde{F}^{-t}$.

Problem 2(a): F_{iA}^* as a referential divergence. Show $F_{iA}^* = G_{iAB,B}$, where $G_{iAB} = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}\chi_j\chi_{k,C}$.

Solution. Differentiate G_{iAB} with respect to X_B and sum, using the product rule (the ε 's are constants):

$$G_{iAB,B} = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}(\chi_{j,B}\chi_{k,C} + \chi_j\chi_{k,CB}).$$

Second term: for fixed A, k put $S_{BC} = \chi_{k,CB}$ and $A_{BC} = \varepsilon_{ABC}$. Mixed partials commute, so $S_{BC} = S_{CB}$; ε changes sign under a swap, so $A_{BC} = -A_{CB}$. By Lemma 2.36,

$$\varepsilon_{ABC}\chi_{k,CB} = 0.$$

First term, by $F_{jB} = \chi_{j,B}$ and Problem 1,

$$\frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC} = F_{iA}^*.$$

Hence $G_{iAB,B} = F_{iA}^*$. □

Problem 2(b): the Piola identity. Show $G_{iAB} = -G_{iBA}$ and deduce $F_{iA,A}^* = 0$.

Solution. Swapping the free indices A, B in the definition and using only the antisymmetry $\varepsilon_{BAC} = -\varepsilon_{ABC}$,

$$G_{iBA} = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{BAC}\chi_j\chi_{k,C} = -\frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}\chi_j\chi_{k,C} = -G_{iAB}.$$

Now take the referential divergence of Problem 2(a):

$$F_{iA,A}^* = (G_{iAB,B})_{,A} = G_{iAB,BA}.$$

Apply Lemma 2.36 again on the pair (A, B) : $(\cdot)_{,BA}$ is symmetric, G_{iAB} antisymmetric. Hence

$$F_{iA,A}^* = 0, \quad \text{i.e.} \quad \text{Div } \tilde{F}^* = \mathfrak{O},$$

the *Piola identity*. □

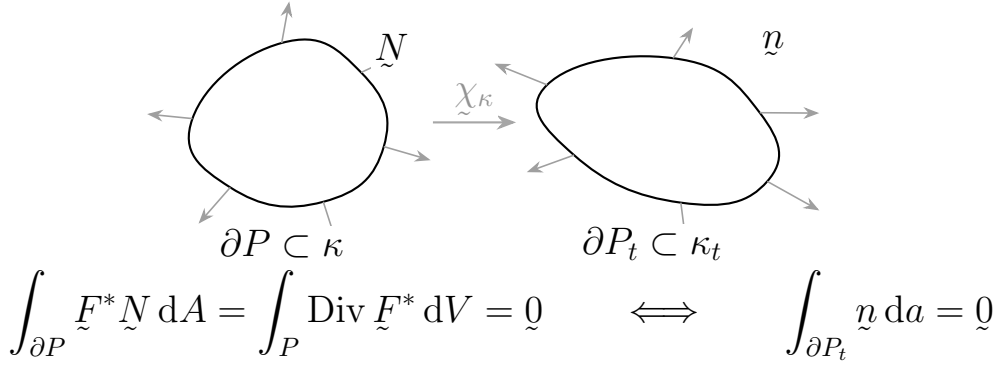


Figure 2.14: The Piola identity. A closed material surface stays closed, so its normals integrate to zero in κ_t as they do in κ . By Piola–Nanson (2.54) that is the divergence theorem applied to $\tilde{F}^*$, and it holds for every P and every motion only if $\text{Div } \tilde{F}^* = \mathfrak{O}$.

REMARK 2.37 (what the Piola identity buys). $\text{Div } \tilde{F}^* = \mathfrak{O}$ says the co-factor field is automatically divergence free, no matter what motion is prescribed. Combined with Piola–Nanson (2.54), it is exactly the statement that the deformed images of closed material surfaces stay closed: applying the divergence theorem, $\int_{\partial P} \tilde{F}^* \tilde{N} \, dA = \int_P \text{Div } \tilde{F}^* \, dV = \mathfrak{O}$, i.e. $\int_{\partial P_t} \tilde{n} \, da = \mathfrak{O}$, the elementary fact that the outward normals of any closed surface integrate to zero. It is the kinematic identity that later converts the Cauchy stress balance into its Piola form.

Problem 2(c): $\det \tilde{F}$ as a divergence. Show $\det \tilde{F} = \frac{1}{3} F_{iA} F_{iA}^*$, and hence that there is a field $\tilde{H}(X, t) = H_A \hat{E}_A$ with $\det \tilde{F} = H_{A,A}$.

Solution. The cofactor identity $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$ of Problem 22(c) of Chapter 1 holds verbatim for the two-point $\underline{F}$, where it reads, in components,

$$F_{iA} F_{iB}^* = (\det \underline{F}) \delta_{AB}.$$

Setting $B = A$ and summing, with $\delta_{AA} = 3$,

$$F_{iA} F_{iA}^* = 3 \det \underline{F} \quad \implies \quad \det \underline{F} = \frac{1}{3} F_{iA} F_{iA}^*.$$

Now substitute $F_{iA} = \chi_{i,A}$ and use the Piola identity of part (b) to move the derivative outside:

$$\det \underline{F} = \frac{1}{3} \chi_{i,A} F_{iA}^* = \frac{1}{3} \left[(\chi_i F_{iA}^*)_{,A} - \chi_i \underbrace{F_{iA,A}^*}_{=0} \right] = \left(\frac{1}{3} \chi_i F_{iA}^* \right)_{,A}.$$

Hence $\det \underline{F} = H_{A,A}$ with

$$H_A = \frac{1}{3} \chi_i F_{iA}^*,$$

i.e. $\underline{H} = \frac{1}{3} \underline{F}^{*t} \underline{\chi}_\kappa$. □

REMARK 2.38 (why one wants J as a divergence). Writing $J = \det \underline{F} = \text{Div } \underline{H}$ turns the volume integral of J into a surface integral by the divergence theorem, $\int_P J \, dV = \int_{\partial P} \underline{H} \cdot \underline{N} \, dA$. Since $\int_P J \, dV = \int_{P_t} dv$ is the deformed volume, this expresses the volume enclosed by a deformed surface purely in terms of data *on* that surface, the familiar “volume from a surface integral” trick, and it is the kinematic origin of several variational identities in nonlinear elasticity.

Problem 2(d): $\det \underline{F}^t = \det \underline{F}$. *Solution.* Substituting Problem 1 into the result of 2(c),

$$\det \underline{F} = \frac{1}{3} F_{iA} \cdot \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ABC} F_{jB} F_{kC} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{ABC} F_{iA} F_{jB} F_{kC}, \quad (*)$$

the two-point version of the contracted determinant formula of Chapter 1. The transpose has components $(\underline{F}^t)_{Ai} = F_{iA}$, and $\underline{F}^t$ maps

$\mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, so the same formula with the roles of the two index families exchanged gives

$$\begin{aligned}\det \underline{F}^t &= \frac{1}{6} \varepsilon_{ABC} \varepsilon_{ijk} (\underline{F}^t)_{Ai} (\underline{F}^t)_{Bj} (\underline{F}^t)_{Ck} \\ &= \frac{1}{6} \varepsilon_{ABC} \varepsilon_{ijk} F_{iA} F_{jB} F_{kC} = \det \underline{F},\end{aligned}$$

since the two scalar expressions differ only in the order in which the same factors are written. $\square$

Problem 3: change of basis for a two-point tensor. *With $\{\hat{E}_A\}, \{\hat{E}'_A\}$ orthonormal in $\hat{\mathbb{E}}^3$ and $\{\underline{e}_i\}, \{\underline{e}'_i\}$ orthonormal in $\mathbb{E}^3$, and $\underline{F} = F_{iA} \underline{e}_i \otimes \hat{E}_A = F'_{iA} \underline{e}'_i \otimes \hat{E}'_A$, find F'_{iA} .*

Solution. Introduce the two independent sets of direction cosines, one for each space,

$$a_{ij} = \underline{e}_i \cdot \underline{e}'_j \quad (\text{spatial}), \quad \hat{a}_{AB} = \hat{E}_A \cdot \hat{E}'_B \quad (\text{referential}),$$

so that, by the orthonormal expansion of Chapter 1,

$$\underline{e}'_i = (\underline{e}_j \cdot \underline{e}'_i) \underline{e}_j = a_{ji} \underline{e}_j, \quad \hat{E}'_A = (\hat{E}_B \cdot \hat{E}'_A) \hat{E}_B = \hat{a}_{BA} \hat{E}_B.$$

Extract the primed components with (2.41), $F'_{iA} = \underline{e}'_i \cdot \underline{F} \hat{E}'_A$. First act with $\underline{F}$ on the primed referential basis vector, using linearity and $\underline{F} \hat{E}_B = F_{jB} \underline{e}_j$ from (2.40):

$$\underline{F} \hat{E}'_A = \hat{a}_{BA} \underline{F} \hat{E}_B = \hat{a}_{BA} F_{jB} \underline{e}_j.$$

Then dot with $\underline{e}'_i = a_{ki} \underline{e}_k$ and use $\underline{e}_k \cdot \underline{e}_j = \delta_{kj}$:

$$F'_{iA} = a_{ki} \underline{e}_k \cdot \hat{a}_{BA} F_{jB} \underline{e}_j = a_{ki} \hat{a}_{BA} F_{jB} \delta_{kj} = a_{ji} \hat{a}_{BA} F_{jB}.$$

Hence

$$F'_{iA} = a_{ji} \hat{a}_{BA} F_{jB}$$

$\square$

In matrix form. With $(a)_{ij} = a_{ij}$ and $(\hat{a})_{AB} = \hat{a}_{AB}$, the index pattern $a_{ji}(\cdot)_{jB} \hat{a}_{BA}$ is

$$(F') = (a)^t (F) (\hat{a}).$$

Both (a) and $(\hat{a})$ are orthogonal, by the computation of Problem 9(a) of Chapter 1 applied in each space separately:

$$a_{ki}a_{kj} = \delta_{ij}, \quad \hat{a}_{CA}\hat{a}_{CB} = \delta_{AB}.$$

Check 1: J is unchanged. Taking determinants,

$$\det(F') = \det(a)^t \det(F) \det(\hat{a}) = (\pm 1) \det(F) (\pm 1),$$

and both signs are $+1$ when the two changes of basis preserve handedness. So J is basis independent, as (2.50) requires.

Check 2: $\tilde{C}$ transforms with one matrix only. From $C_{AB} = F_{iA}F_{iB}$,

$$\begin{aligned} C'_{AB} &= F'_{iA}F'_{iB} = (a_{ji}\hat{a}_{CA}F_{jC})(a_{ki}\hat{a}_{DB}F_{kD}) \\ &= \hat{a}_{CA}\hat{a}_{DB}\delta_{jk}F_{jC}F_{kD} = \hat{a}_{CA}\hat{a}_{DB}C_{CD}, \end{aligned}$$

using $a_{ji}a_{ki} = \delta_{jk}$. The spatial cosines have cancelled: $\tilde{C}$ is a referential tensor and cannot feel a change of spatial basis. The same computation on $B_{ij} = F_{iA}F_{jA}$ cancels the referential cosines instead.

REMARK 2.39 (the transformation law identifies the tensor's type). Compare with Chapter 1: a vector picks up one factor of the cosine matrix, $v'_i = a_{ki}v_k$; an ordinary (one-point) tensor picks up two factors of the *same* matrix, $A'_{ij} = a_{ki}a_{lj}A_{kl}$. A two-point tensor also picks up two factors, but one from *each* space, a_{ji} for the lower-case index and $\hat{a}_{BA}$ for the upper-case one. The two rotations are independent: one may re-label the reference frame without touching the current frame, and $\tilde{F}$ responds only in its capital index. This is the precise sense in which $\tilde{F}$ “has one foot in each configuration”, and it is why the alphabet convention of Remark 2.10 is more than cosmetic.

Chapter 3

Further Preliminaries, with Applications to the Analysis of Deformation

Orientation. Chapter 2 produced the deformation gradient $\underline{F}$ and, from it, the Cauchy–Green tensors $\underline{C} = \underline{F}^t \underline{F}$ and $\underline{B} = \underline{F} \underline{F}^t$. It stopped there, because the next question cannot be answered with the algebra of Chapter 1 alone. The question is this. A deformation both *stretches* material and *rotates* it. The stretch is physical: it costs energy, it is what a material resists. The rotation is not: carrying a rigid body across the room changes $\underline{F}$ but does nothing to the material. If the two are mixed together inside a single $\underline{F}$, no constitutive law can be written. So we need a theorem that separates them, cleanly and uniquely.

That theorem is the polar decomposition $\underline{F} = \underline{R} \underline{U} = \underline{V} \underline{R}$ of §3.4. Getting to it requires three pieces of linear algebra, and the chapter assembles them in order:

- §3.1, the spectral theorem: a symmetric tensor is a set of three stretches along three orthogonal axes, and nothing more.
- §3.2, positive definiteness: the criterion that says the three numbers are positive, so that square roots exist.

- §3.3, the square root theorem: $\underline{U} = \sqrt{\underline{C}}$ exists and is unique, which is what makes $\underline{R} = \underline{F}\underline{U}^{-1}$ well defined.

§3.5 then computes everything explicitly for four deformations, §3.6 isolates the class in which $\underline{F}$ is uniform, and §3.7 shows that rigid motions are exactly the case $\underline{U} = \hat{\underline{I}}$. Two ideas that the chapter uses constantly but states only in passing, the meaning of a *material* vector and the difference between a principal direction and a material direction, are given their own treatment in Remarks 3.26 and 3.27.

3.1 Principal values and vectors: spectral representation of a symmetric tensor

Why we want this before anything else

A tensor is a machine that eats a vector and returns a vector, and in general it does two things at once: it changes the length of what it eats and it turns it. Most vectors come out pointing somewhere new. But it can happen that for some special input direction the output points the same way, and only the length has changed. Such a direction is a direction the tensor does not rotate, and along it the tensor is nothing but multiplication by a number. If we could find enough such directions to build a basis, the tensor would be completely described by that handful of numbers: it would be three independent stretches along three fixed axes, with no turning at all.

That is what the spectral theorem delivers, and it delivers it whenever the tensor is symmetric. The relevance to Chapter 2 is immediate: $\underline{C}$ and $\underline{B}$ were proved symmetric there, at (2.85), so the theorem applies to them, and the numbers it produces will turn out to be the squares of the stretches of (2.83).

DEFINITION 3.1 (Principal value, principal vector). A unit vector $\underline{u}$ is a *principal vector* of a tensor $\underline{A}$ if there is a scalar λ , a *principal value*,

such that

$$\underline{\underline{A}}\underline{\underline{u}} = \lambda\underline{\underline{u}}, \quad \text{or} \quad (\underline{\underline{A}} - \lambda\underline{\underline{I}})\underline{\underline{u}} = \underline{\underline{0}}. \quad (3.1)$$

For definiteness vectors are taken in $\mathbb{E}^3$ and tensors as linear maps of $\mathbb{E}^3$ to itself. Everything below holds verbatim in $\hat{\mathbb{E}}^3$ with hats attached, and this is used without comment when $\underline{\underline{C}}$ and $\underline{\underline{U}}$ are treated in §3.4.

REMARK 3.2 (why the second form is the useful one). The two forms in (3.1) say the same thing, but the second is the one that can be attacked. Read as a statement about the tensor $\underline{\underline{A}} - \lambda\underline{\underline{I}}$, it says that this tensor kills a non-zero vector, hence is not invertible, hence has zero determinant by Theorem 1.52 of Chapter 1. The unknown λ has thereby been isolated in a scalar equation, and the unknown $\underline{\underline{u}}$ has been eliminated. This is the whole trick.

Because $\underline{\underline{u}}$ is a unit vector it is not $\underline{\underline{0}}$, so $\underline{\underline{A}} - \lambda\underline{\underline{I}}$ is singular and, by Problem 23 of Chapter 1,

$$0 = \det(\underline{\underline{A}} - \lambda\underline{\underline{I}}) = -\lambda^3 + I_1(\underline{\underline{A}})\lambda^2 - I_2(\underline{\underline{A}})\lambda + I_3(\underline{\underline{A}}), \quad (3.2)$$

where, by Problem 22 of Chapter 1,

$$I_1(\underline{\underline{A}}) = \text{tr } \underline{\underline{A}}, \quad I_2(\underline{\underline{A}}) = \frac{1}{2}[(\text{tr } \underline{\underline{A}})^2 - \text{tr}(\underline{\underline{A}}^2)], \quad I_3(\underline{\underline{A}}) = \det \underline{\underline{A}} \quad (3.3)$$

are the *principal invariants* of $\underline{\underline{A}}$. Equation (3.2) is the *characteristic equation*.

REMARK 3.3 (invariant means basis free, and why that matters here). The three numbers (3.3) were constructed in Chapter 1 out of box products alone, with no basis anywhere in the definition, so they do not change when the basis changes. Consequently the polynomial (3.2) does not change, and therefore neither do its roots. This is not a technicality: it is the statement that the principal values are properties of the *tensor*, not of the arithmetic we use to compute them. If they depended on the basis they could not be used in a constitutive law, and the whole chapter would collapse.

Equation (3.2) is a real cubic. Its complex roots occur in conjugate pairs, so it has either three real roots or exactly one; in particular *every tensor has at least one real principal value*. For symmetric tensors the situation is better still.

THEOREM 3.4 (Reality). *If $\underline{\underline{A}} = \underline{\underline{A}}^t$ then every root of (3.2) is real.*

Proof. Suppose a root λ and the associated $\underline{\underline{u}}$ are complex,

$$\lambda = a + ib, \quad \underline{\underline{u}} = u_j \underline{\underline{e}}_j \quad \text{with} \quad u_j = a_j + ib_j, \quad (3.4)$$

where $i^2 = -1$ and a, b, a_j, b_j are real. Here the basis $\{\underline{\underline{e}}_j\}$ is still real, and the inner product is extended to complex components *bilinearly*, $\underline{\underline{u}} \cdot \underline{\underline{v}} = u_j v_j$, without conjugation; this is the only reading under which the algebra of Chapter 1 continues to hold unchanged, and it is what makes the computation below work.

The components A_{ij} are real, so conjugating (3.1) term by term, and using that the conjugate of a product is the product of the conjugates,

$$\underline{\underline{A}}\underline{\underline{u}} = \bar{\lambda}\underline{\underline{u}}, \quad (3.5)$$

where

$$\bar{\lambda} = a - ib, \quad \underline{\underline{u}} = \bar{u}_j \underline{\underline{e}}_j \quad \text{with} \quad \bar{u}_j = a_j - ib_j. \quad (3.6)$$

Dot (3.1) with $\underline{\underline{u}}$ and (3.5) with $\underline{\underline{u}}$:

$$\underline{\underline{u}} \cdot \underline{\underline{A}}\underline{\underline{u}} = \lambda \underline{\underline{u}} \cdot \underline{\underline{u}}, \quad \underline{\underline{u}} \cdot \underline{\underline{A}}\underline{\underline{u}} = \bar{\lambda} \underline{\underline{u}} \cdot \underline{\underline{u}}. \quad (3.7)$$

Now use the symmetry of $\underline{\underline{A}}$ on the left-hand side of the second:

$$\underline{\underline{u}} \cdot \underline{\underline{A}}\underline{\underline{u}} = \underline{\underline{A}}^t \underline{\underline{u}} \cdot \underline{\underline{u}} = \underline{\underline{A}}\underline{\underline{u}} \cdot \underline{\underline{u}} = \underline{\underline{u}} \cdot \underline{\underline{A}}\underline{\underline{u}},$$

so the two left-hand sides in (3.7) are equal. Subtracting,

$$(\lambda - \bar{\lambda}) \underline{\underline{u}} \cdot \underline{\underline{u}} = 0, \quad (3.8)$$

in which, by the bilinear reading of the dot product,

$$\underline{\underline{u}} \cdot \underline{\underline{u}} = \bar{u}_j u_j = (a_j - ib_j)(a_j + ib_j) = a_j a_j + b_j b_j,$$

the cross terms ia_jb_j and $-ib_ja_j$ cancelling. This is a sum of squares of real numbers, hence strictly positive unless every a_j and every b_j vanishes, that is unless $\underline{u} = \underline{0}$; and $\underline{u} = \underline{0}$ is excluded by hypothesis. Therefore $\lambda - \bar{\lambda} = 0$, i.e. $2ib = 0$, i.e. $b = 0$ and λ is real. ■

THEOREM 3.5 (Orthogonality). *If $\underline{A} = \underline{A}^t$ and $\underline{A}\underline{u}_1 = \lambda_1\underline{u}_1$, $\underline{A}\underline{u}_2 = \lambda_2\underline{u}_2$ with $\lambda_1 \neq \lambda_2$, then $\underline{u}_1 \cdot \underline{u}_2 = 0$.*

Proof. Move $\underline{A}$ across the dot product twice:

$$\lambda_1\underline{u}_1 \cdot \underline{u}_2 = \underline{A}\underline{u}_1 \cdot \underline{u}_2 = \underline{u}_1 \cdot \underline{A}^t\underline{u}_2 = \underline{u}_1 \cdot \underline{A}\underline{u}_2 = \lambda_2\underline{u}_1 \cdot \underline{u}_2, \quad (3.9-3.10)$$

the third equality being the symmetry of $\underline{A}$. Hence $(\lambda_1 - \lambda_2)\underline{u}_1 \cdot \underline{u}_2 = 0$, and since $\lambda_1 \neq \lambda_2$ the second factor must vanish. ■

So if the three roots of (3.2) are distinct, the three principal vectors are mutually orthogonal and, being unit vectors, form an orthonormal basis. The book records that this remains true when roots repeat and refers the argument to its Appendix D. Since the whole chapter rests on it, here is a complete proof that needs nothing beyond Chapter 1 and does not care about multiplicities at all.

LEMMA 3.6 (The orthogonal complement of a principal direction is invariant). *Let $\underline{A} = \underline{A}^t$ and $\underline{A}\underline{u} = \lambda\underline{u}$ with $|\underline{u}| = 1$. Put $P = \{\underline{u}\}^\perp$, the plane through the origin perpendicular to $\underline{u}$. Then $\underline{A}\underline{w} \in P$ for every $\underline{w} \in P$.*

Proof. Let $\underline{w} \cdot \underline{u} = 0$. Then

$$\underline{u} \cdot \underline{A}\underline{w} = \underline{A}^t\underline{u} \cdot \underline{w} = \underline{A}\underline{u} \cdot \underline{w} = \lambda\underline{u} \cdot \underline{w} = \lambda \cdot 0 = 0,$$

so $\underline{A}\underline{w}$ is perpendicular to $\underline{u}$, that is $\underline{A}\underline{w} \in P$. ■

THEOREM 3.7 (Spectral theorem). *Every symmetric $\underline{A}$ possesses an orthonormal basis $\{\underline{u}_1, \underline{u}_2, \underline{u}_3\}$ of principal vectors, with real principal values $\lambda_1, \lambda_2, \lambda_3$.*

Proof. The cubic (3.2) has at least one real root λ_1 , and Theorem 3.4 says every root is real in any case. Since $\det(\underline{A} - \lambda_1 \underline{I}) = 0$, the tensor $\underline{A} - \lambda_1 \underline{I}$ is not invertible, so by Theorem 1.52 of Chapter 1 it is not injective, and there is a non-zero $\underline{p}$ with $(\underline{A} - \lambda_1 \underline{I})\underline{p} = \underline{0}$. Normalise, $\underline{u}_1 = \underline{p}/|\underline{p}|$.

By Lemma 3.6 the plane $P_1 = \{\underline{u}_1\}^\perp$ is carried into itself by $\underline{A}$, so the restriction $\underline{A}|_{P_1}$ is a linear map of the two-dimensional space P_1 to itself, and it is symmetric there because $\underline{w} \cdot \underline{A}\underline{s} = \underline{A}\underline{w} \cdot \underline{s}$ holds for all vectors, hence in particular for those lying in P_1 . Choose any orthonormal pair $\{\underline{s}_1, \underline{s}_2\}$ spanning P_1 ; the matrix of $\underline{A}|_{P_1}$ is $\begin{pmatrix} \alpha & \beta \\ \beta & \delta \end{pmatrix}$ with real entries, whose characteristic equation $\mu^2 - (\alpha + \delta)\mu + (\alpha\delta - \beta^2) = 0$ has discriminant

$$(\alpha + \delta)^2 - 4(\alpha\delta - \beta^2) = (\alpha - \delta)^2 + 4\beta^2 \geq 0,$$

a sum of squares. So there is a real root λ_2 and a unit $\underline{u}_2 \in P_1$ with $\underline{A}\underline{u}_2 = \lambda_2 \underline{u}_2$, and by construction $\underline{u}_1 \cdot \underline{u}_2 = 0$.

Finally let $\underline{u}_3 = \underline{u}_1 \times \underline{u}_2$, a unit vector orthogonal to both. It lies in P_1 , and it is orthogonal to $\underline{u}_2$ inside P_1 ; applying Lemma 3.6 once more, this time inside P_1 with $\underline{u}_2$ in the role of $\underline{u}$, gives $\underline{A}\underline{u}_3 \perp \underline{u}_2$, while $\underline{A}\underline{u}_3 \in P_1$ means $\underline{A}\underline{u}_3 \perp \underline{u}_1$. A vector of $\mathbb{E}^3$ orthogonal to both $\underline{u}_1$ and $\underline{u}_2$ is a multiple of $\underline{u}_3$, so $\underline{A}\underline{u}_3 = \lambda_3 \underline{u}_3$ for some real λ_3 . The triad is orthonormal by construction. ■

REMARK 3.8 (what the proof buys that the determinant argument does not). Nothing above counted multiplicities. If $\lambda_1 = \lambda_2$, the argument still produces two orthogonal principal vectors, because the second was obtained by restricting to a plane rather than by looking for a second root. This is exactly the case that a naive treatment mishandles, and it is not exotic: it occurs in §3.5.2 below whenever $f' = f/R$, and in every isotropic dilatation, where all three principal values coincide and *every* direction is principal.

Having an orthonormal basis of principal vectors, we can write the

tensor down. First,

$$\underline{I} = \underline{u}_i \otimes \underline{u}_i, \quad (3.11)$$

which is checked by feeding it an arbitrary $\underline{v}$: expanding $\underline{v} = (\underline{v} \cdot \underline{u}_i)\underline{u}_i$ on the orthonormal basis and using $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$,

$$(\underline{u}_i \otimes \underline{u}_i)\underline{v} = (\underline{u}_i \cdot \underline{v})\underline{u}_i = \underline{v} \quad \text{for every } \underline{v}, \quad \text{so} \quad \underline{u}_i \otimes \underline{u}_i = \underline{I}.$$

Then, using Problem 17(b) of Chapter 1, $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$,

$$\underline{A} = \underline{A}\underline{I} = \underline{A}(\underline{u}_i \otimes \underline{u}_i) = (\underline{A}\underline{u}_i) \otimes \underline{u}_i = \sum_{i=1}^3 \lambda_i \underline{u}_i \otimes \underline{u}_i, \quad (3.12)$$

the *spectral representation*. The explicit summation sign is written because the repeated index i now also sits on λ_i , and the summation convention would be ambiguous.

The matrix of $\underline{A}$ in the basis $\{\underline{u}_i \otimes \underline{u}_j\}$ is $(A_{ij}) = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, since $A_{ij} = \underline{u}_i \cdot \underline{A}\underline{u}_j = \lambda_j \underline{u}_i \cdot \underline{u}_j = \lambda_j \delta_{ij}$ (no sum). A symmetric tensor can therefore be diagonalised by resolving it on its own principal vectors.

REMARK 3.9 (reading (3.12) as a machine). Equation (3.12) says that $\underline{A}$ works like this: take the input $\underline{v}$, measure its three components $\underline{u}_i \cdot \underline{v}$ along the principal axes, multiply the i th by λ_i , and reassemble. Explicitly,

$$\underline{A}\underline{v} = \sum_i \lambda_i (\underline{u}_i \cdot \underline{v}) \underline{u}_i.$$

There is no rotation anywhere in that description. This is the precise content of the sentence “a symmetric tensor is pure stretch”, which is the sentence §3.4 will need. A dyad $\underline{u}_i \otimes \underline{u}_i$ is the projection $\mathbb{P}_{(\underline{u}_i)}$ of Problem 10 of Chapter 1 onto the i th axis, so (3.12) reads: *decompose, scale each piece, add up*.

REMARK 3.10 (powers and functions of a symmetric tensor). From (3.12) and the dyad product rule $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c}) \underline{a} \otimes \underline{d}$,

$$\begin{aligned} \underline{A}^2 &= \left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i \right) \left(\sum_j \lambda_j \underline{u}_j \otimes \underline{u}_j \right) \\ &= \sum_{i,j} \lambda_i \lambda_j \delta_{ij} \underline{u}_i \otimes \underline{u}_j = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i, \end{aligned}$$

the Kronecker delta arising from $\underline{u}_i \cdot \underline{u}_j = \delta_{ij}$ and collapsing the double sum to a single one. By induction $\underline{A}^n = \sum_i \lambda_i^n \underline{u}_i \otimes \underline{u}_i$, and if all $\lambda_i \neq 0$ the same formula with $n = -1$ gives the inverse, as one verifies at once by multiplying. *Every* function of $\underline{A}$ is obtained by applying that function to the three numbers and leaving the axes alone. This single observation is the entire content of §3.3.

REMARK 3.11 (what the three invariants *are*: lengths, areas, volume). Chapter 1 produced I_1, I_2, I_3 by a formal recipe, three ways of inserting $\underline{A}$ into a box product, and the pattern looked arbitrary. For a symmetric tensor the spectral representation makes the pattern obvious, and it is worth seeing, because it is the reason these three and no others appear in every constitutive law.

Let $\underline{A}$ be symmetric with principal values $\lambda_i > 0$ and principal axes $\underline{u}_i$, and apply it to the unit cube built on those axes. By (3.12) the cube goes to a rectangular box with edges λ_i along the same axes. Then, by Problem 7 of §3.8,

$$\begin{aligned} I_1 &= \lambda_1 + \lambda_2 + \lambda_3 && \text{sum of the three } \textit{edge} \text{ lengths,} \\ I_2 &= \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 && \text{sum of the three } \textit{face} \text{ areas,} \\ I_3 &= \lambda_1\lambda_2\lambda_3 && \text{the } \textit{volume}. \end{aligned}$$

So the three invariants measure how $\underline{A}$ scales length, area and volume, in that order. Nothing else of that kind exists in three dimensions, because a box has only edges, faces and an interior.

This also explains Problem 22(a) of Chapter 1, $I_2(\underline{A}) = \text{tr } \underline{A}^*$, which there looked like an identity with no content. The cofactor is the tensor that carries *areas*, Remark 1.55; its trace collects the three face areas; and that is exactly I_2 . The same three levels appear again in Chapter 2 as the three deformation measures: $\underline{F}$ carries line elements, (2.34); $\underline{F}^*$ carries area elements, (2.54); and $J = \det \underline{F}$ carries volume elements, (2.50).

For the deformation tensors themselves the reading is immediate.

With $\underline{C} = \underline{U}^2$ and λ_i the principal stretches,

$$\begin{aligned} I_1(\underline{C}) &= \lambda_1^2 + \lambda_2^2 + \lambda_3^2, & I_2(\underline{C}) &= \sum_{i < j} \lambda_i^2 \lambda_j^2, \\ I_3(\underline{C}) &= (\lambda_1 \lambda_2 \lambda_3)^2 = J^2, \end{aligned}$$

so $I_1(\underline{C})$ is the sum of squared *length* stretches, $I_2(\underline{C})$ the sum of squared *area* stretches, and $I_3(\underline{C})$ the squared *volume* stretch. An isotropic elastic material stores energy as a function of these three numbers and nothing else, which is why they are worth naming.

3.2 Positive definiteness and the extremal property of principal values

The square root of §3.3 will need the λ_i to be positive. The criterion is the following, and Chapter 2 has already verified it for $\underline{C}$ at (2.84).

DEFINITION 3.12 (Positive definite). A tensor $\underline{A}$ is *positive definite* if

$$\underline{a} \cdot \underline{A} \underline{a} > 0 \quad \text{for all } \underline{a} \neq \underline{0}. \quad (3.13)$$

Two observations before proceeding, both from Chapter 1. By Problem 14 any $\underline{A}$ splits as $\underline{A} = \underline{S} + \underline{W}$ with $\underline{S} = \text{sym } \underline{A}$ symmetric and $\underline{W} = \text{skw } \underline{A}$ skew, and by Problem 6(a) $\underline{a} \cdot \underline{W} \underline{a} = 0$ for skew $\underline{W}$; hence

$$\underline{a} \cdot \underline{A} \underline{a} = \underline{a} \cdot \underline{S} \underline{a} + \underline{a} \cdot \underline{W} \underline{a} = \underline{a} \cdot \underline{S} \underline{a}.$$

So (3.13) sees only the symmetric part, and a skew tensor can never be positive definite. We therefore assume $\underline{A}$ symmetric with no loss.

THEOREM 3.13 (Positive definite $\iff$ positive principal values). *For symmetric $\underline{A}$ with principal values λ_i ,*

$$\underline{A} \text{ positive definite} \iff \lambda_i > 0, \quad i = 1, 2, 3.$$

Moreover $\det \underline{A} = \lambda_1 \lambda_2 \lambda_3 > 0$.

Proof. ($\Rightarrow$) Put $\underline{a} = \underline{u}$, a principal vector with value λ . Then $\underline{u} \neq \underline{0}$ and

$$0 < \underline{u} \cdot \underline{A}\underline{u} = \underline{u} \cdot \lambda \underline{u} = \lambda \underline{u} \cdot \underline{u} = \lambda, \quad (3.14)$$

using $|\underline{u}| = 1$. Applying this to each principal vector,

$$\lambda_i > 0; \quad i = 1, 2, 3. \quad (3.15)$$

($\Leftarrow$) Conversely, if every $\lambda_i > 0$ then, by the spectral representation and $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$,

$$\underline{a} \cdot \underline{A}\underline{a} = \underline{a} \cdot \sum_i \lambda_i (\underline{u}_i \cdot \underline{a}) \underline{u}_i = \sum_i \lambda_i (\underline{a} \cdot \underline{u}_i)^2 > 0 \quad \text{for all } \underline{a} \neq \underline{0}, \quad (3.16)$$

the strict inequality holding because the λ_i are positive and, $\{\underline{u}_i\}$ being a basis, at least one component $\underline{a} \cdot \underline{u}_i$ is non-zero when $\underline{a} \neq \underline{0}$; were all three zero, $\underline{a}$ would be orthogonal to a basis and hence $\underline{0}$.

For the determinant, use the box-product definition with the principal triad itself as the test triad, which is legitimate because $\{\underline{u}_i\}$ is a basis and therefore linearly independent, so $[\underline{u}_1, \underline{u}_2, \underline{u}_3] \neq 0$. Feed the triad to $\underline{A}$, one member at a time:

$$\underline{A}\underline{u}_1 = \lambda_1 \underline{u}_1, \quad \underline{A}\underline{u}_2 = \lambda_2 \underline{u}_2, \quad \underline{A}\underline{u}_3 = \lambda_3 \underline{u}_3,$$

by the very definition of a principal vector. Now pull the three scalars out of the box product one slot at a time, using that it is linear in each slot *separately*: first the λ_1 from the first slot, leaving the other two untouched, then the λ_2 from the second, then the λ_3 from the third,

$$\begin{aligned} \det \underline{A} &= \frac{[\underline{A}\underline{u}_1, \underline{A}\underline{u}_2, \underline{A}\underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \frac{[\lambda_1 \underline{u}_1, \lambda_2 \underline{u}_2, \lambda_3 \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} \\ &= \frac{\lambda_1 [\underline{u}_1, \lambda_2 \underline{u}_2, \lambda_3 \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \frac{\lambda_1 \lambda_2 \lambda_3 [\underline{u}_1, \underline{u}_2, \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} \\ &= \lambda_1 \lambda_2 \lambda_3 > 0. \end{aligned} \quad (3.17)$$

The last step is a cancellation of one and the same non-zero number, $[\underline{u}_1, \underline{u}_2, \underline{u}_3]$, which appears identically above and below. For an orthonormal triad that number is $+1$ or -1 according to handedness, but

its value never enters: what matters is only that it is the *same* in numerator and denominator, so the answer cannot depend on which of the two orientations the triad happens to have. ■

The extremal property

The remaining result of the section says that the principal values bracket the diagonal entries in *any* orthonormal basis. It is proved by optimisation, and the optimisation is worth following because the same argument, constrained stationarity, recurs throughout mechanics.

Fix a symmetric positive definite $\underline{A}$ and define

$$f(\underline{u}) = \underline{u} \cdot \underline{A}\underline{u} \quad \text{with} \quad |\underline{u}| = 1. \quad (3.18)$$

f is continuous and positive on the unit sphere $S = \{\underline{v} : |\underline{v}| = 1\}$, which is closed and bounded, hence compact; a continuous function on a compact set attains its extrema. So there exist $\underline{u}_1, \underline{u}_2 \in S$ with

$$\begin{aligned} 0 < \lambda_1 &= \min_{\underline{u} \in S} f(\underline{u}), & 0 < \lambda_2 &= \max_{\underline{u} \in S} f(\underline{u}), \\ \lambda_1 &= f(\underline{u}_1), & \lambda_2 &= f(\underline{u}_2). \end{aligned} \quad (3.19)$$

REMARK 3.14 (a warning about the labels). In this subsection λ_1 and λ_2 denote the smallest and largest principal values, whereas everywhere else in the book the subscript is a plain enumeration index. In three dimensions the extremal pair is the smallest and the largest, so the middle principal value is not named here at all. Keeping the two conventions apart is the only difficulty in (3.26) below.

At an extremum f is stationary. Differentiating (3.18) with the product rule and using symmetry,

$$0 = df(\underline{u}) = d\underline{u} \cdot \underline{A}\underline{u} + \underline{u} \cdot \underline{A}(d\underline{u}) = d\underline{u} \cdot \underline{A}\underline{u} + \underline{A}^t \underline{u} \cdot d\underline{u} = 2 \underline{A}\underline{u} \cdot d\underline{u}, \quad (3.20)$$

so we seek $\underline{u}$ such that

$$\underline{A}\underline{u} \cdot d\underline{u} = 0. \quad (3.21)$$

The increments $d\underline{u}$ are not arbitrary: differentiating the constraint $\underline{u} \cdot \underline{u} = 1$ gives $2\underline{u} \cdot d\underline{u} = 0$, so $d\underline{u}$ ranges over the plane $\{\underline{u}\}^\perp$ and nothing more.

Thus (3.21) says that $\underline{A}\underline{u}$ is orthogonal to every vector of $\{\underline{u}\}^\perp$. By Problem 10 of Chapter 1 the orthogonal complement of $\{\underline{u}\}^\perp$ is the line spanned by $\underline{u}$, so $\underline{A}\underline{u}$ must be a multiple of $\underline{u}$:

$$\underline{A}\underline{u} = \mu\underline{u} \quad (3.22)$$

for some scalar μ . Dotting with $\underline{u}$ identifies the multiplier as $\mu = \underline{u} \cdot \underline{A}\underline{u} = f(\underline{u})$. Applying this at the minimiser and at the maximiser,

$$\underline{A}\underline{u}_1 = \lambda_1\underline{u}_1, \quad (3.23)$$

$$\underline{A}\underline{u}_2 = \lambda_2\underline{u}_2, \quad (3.24)$$

so λ_1 and λ_2 are principal values of $\underline{A}$: the extremes of the quadratic form on the unit sphere are attained *at* principal directions, and *equal* the corresponding principal values. By construction,

$$\lambda_1 = f(\underline{u}_1) \leq f(\underline{u}) \leq f(\underline{u}_2) = \lambda_2 \quad \text{for every } \underline{u} \in S. \quad (3.25)$$

Finally take $\underline{u} = \underline{e}_1$, an element of an arbitrary orthonormal basis. Then $f(\underline{e}_1) = \underline{e}_1 \cdot \underline{A}\underline{e}_1 = A_{11}$, and similarly for $\underline{e}_2, \underline{e}_3$, so (3.25) gives the *extremal property*

$$\lambda_1 \leq \min\{A_{11}, A_{22}, A_{33}\} \leq \max\{A_{11}, A_{22}, A_{33}\} \leq \lambda_2. \quad (3.26)$$

REMARK 3.15 (what (3.26) is good for). Two things. First, it is a free bound: having computed the matrix of $\underline{C}$ in any convenient basis, one reads off three diagonal entries and knows immediately that the largest principal stretch squared is at least the largest of them and the smallest is at most the smallest. In the simple shear of §3.5.1, (C_{AB}) has diagonal $1, 1 + \gamma^2, 1$, so $\lambda_{\max}^2 \geq 1 + \gamma^2$ and $\lambda_{\min}^2 \leq 1$ without solving anything. Second, it explains why the principal values are called principal: they are the extreme values of the stretch over all directions, so the material's worst-stretched fibre lies along a principal axis. That is the fact a failure criterion needs.

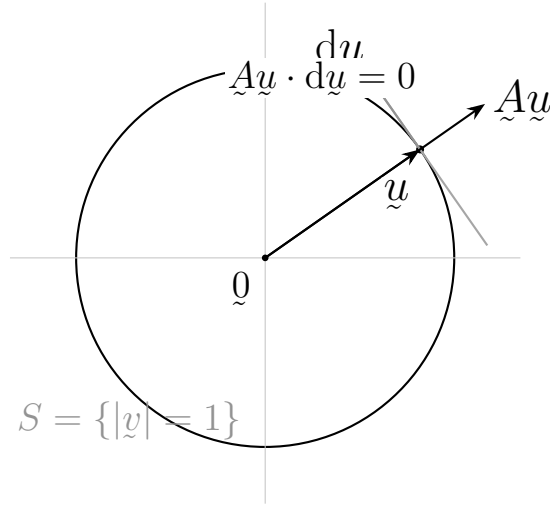


Figure 3.1: The unit sphere S . The constraint $|u| = 1$ confines the admissible increments du to the tangent plane at u , so stationarity of $f(u) = u \cdot Au$ requires Au to be orthogonal to that whole plane, hence parallel to u itself. This is (3.22): an extremum of the quadratic form is a principal direction.

3.3 Square root theorem

Remark 3.10 already contains the answer: a symmetric positive definite tensor is three positive numbers attached to three orthogonal axes, and the square root is obtained by taking the square roots of the three numbers. What has to be proved is that this is the *only* possible answer, because it is uniqueness, not existence, that makes $R = FU^{-1}$ a definition rather than a choice.

THEOREM 3.16 (Square root). *Let A be symmetric and positive definite. Then there is exactly one symmetric positive definite U with $A = U^2$. We write $U = \sqrt{A}$.*

Proof. Existence. Write A in spectral form, which Theorem 3.7 permits and Theorem 3.13 makes positive:

$$A = \sum_i \lambda_i u_i \otimes u_i, \quad \lambda_i > 0, \quad (3.27)$$

and define

$$U = \sum_i \sqrt{\lambda_i} u_i \otimes u_i. \quad (3.28)$$

This is symmetric, because each dyad $\underline{u}_i \otimes \underline{u}_i$ is symmetric by Problem 17(a) of Chapter 1, $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$, and a sum of symmetric tensors is symmetric. It is positive definite, because its principal values are the positive numbers $\sqrt{\lambda_i}$ and Theorem 3.13 applies. And squaring it,

$$\begin{aligned} \underline{U}^2 &= \underline{U} \left(\sum_i \sqrt{\lambda_i} \underline{u}_i \otimes \underline{u}_i \right) = \sum_i \sqrt{\lambda_i} (\underline{U} \underline{u}_i) \otimes \underline{u}_i \\ &= \sum_i \sqrt{\lambda_i} (\sqrt{\lambda_i} \underline{u}_i) \otimes \underline{u}_i = \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i = \underline{A}, \end{aligned} \quad (3.29)$$

where the second equality is Problem 17(b), $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$, applied term by term, and the third uses $\underline{U}\underline{u}_i = \sqrt{\lambda_i}\underline{u}_i$, which is read off (3.28) by feeding it $\underline{u}_i$ and using orthonormality. So this $\underline{U}$ has the stated properties.

A preparatory fact. Suppose $\underline{U}$ is symmetric positive definite with $\underline{U}^2 \underline{u} = \lambda \underline{u}$ for some unit $\underline{u}$ and $\lambda > 0$. Then, since $\underline{U}$ commutes with $\underline{I}$,

$$\begin{aligned} \underline{0} &= (\underline{U}^2 - \lambda \underline{I})\underline{u} = (\underline{U} + \sqrt{\lambda} \underline{I})(\underline{U} - \sqrt{\lambda} \underline{I})\underline{u} = (\underline{U} + \sqrt{\lambda} \underline{I})\underline{v}, \\ \text{where } \underline{v} &:= (\underline{U} - \sqrt{\lambda} \underline{I})\underline{u}, \end{aligned} \quad (3.30)$$

the factorisation being legitimate because $(\underline{U} + \sqrt{\lambda} \underline{I})(\underline{U} - \sqrt{\lambda} \underline{I}) = \underline{U}^2 - \sqrt{\lambda} \underline{U} + \sqrt{\lambda} \underline{U} - \lambda \underline{I} = \underline{U}^2 - \lambda \underline{I}$. Hence

$$\underline{U}\underline{v} = -\sqrt{\lambda} \underline{v}. \quad (3.31)$$

If $\underline{v} \neq \underline{0}$ this exhibits $-\sqrt{\lambda} < 0$ as a principal value of $\underline{U}$, contradicting Theorem 3.13. Therefore $\underline{v} = \underline{0}$, that is

$$\underline{U}\underline{u} = \sqrt{\lambda} \underline{u}.$$

In words: *a symmetric positive definite square root of $\underline{A}$ is forced to act on each principal vector of $\underline{A}$ by the positive square root of the corresponding principal value.* It has no freedom at all.

Uniqueness. Let $\tilde{\underline{U}}$ be symmetric positive definite with $\tilde{\underline{U}}^2 = \underline{A}$. Applying the preparatory fact to $\tilde{\underline{U}}$ with $\underline{u} = \underline{u}_i$ and $\lambda = \lambda_i$ gives $\tilde{\underline{U}}\underline{u}_i = \sqrt{\lambda_i} \underline{u}_i = \underline{U}\underline{u}_i$ for each i . Two linear maps agreeing on a basis agree everywhere: for arbitrary $\underline{w} = w_i \underline{u}_i$,

$$\tilde{\underline{U}}\underline{w} = w_i \tilde{\underline{U}}\underline{u}_i = w_i \underline{U}\underline{u}_i = \underline{U}\underline{w},$$

so $\tilde{U} = U$. ■

REMARK 3.17 (why positive definiteness is not decoration). Drop it and uniqueness fails immediately. The tensor $\tilde{I}$ has, among symmetric square roots, not only $\tilde{I}$ but also $\tilde{I} - 2 \underline{e}_1 \otimes \underline{e}_1 = \text{diag}(-1, 1, 1)$, whose square is $\text{diag}(1, 1, 1) = \tilde{I}$. There are eight such sign patterns. Positive definiteness is precisely the condition that picks one. In the polar decomposition this matters physically: a fibre must stretch by a positive factor, and the sign ambiguity is exactly the ambiguity between a stretch and a stretch followed by a reflection.

REMARK 3.18 (what §§3.1–3.3 really buy: every strain measure at once). Chapter 2 remarked, just after (2.99), that “there are infinitely many equivalent ways to define strain tensors” and sent the reader to Ogden. It did not say what makes them definable, and the answer is the two sections just completed. Once $\tilde{U}$ exists and is diagonal on a fixed orthonormal triad, *any* function of a single real variable can be applied to a deformation:

$$g(\tilde{U}) := \sum_i g(\lambda_i) \underline{u}_i \otimes \underline{u}_i.$$

This is well defined precisely because Theorem 3.7 supplies the triad and Theorem 3.16 makes $\tilde{U}$ unique, so there is nothing to choose.

A strain measure is such a g subject to two demands: it must vanish when nothing happens, $g(1) = 0$, so that rigid motions are strain free by (3.122); and it must be increasing near $\lambda = 1$ with $g'(1) = 1$, so that it reduces to the elongation for small stretches. The standard family satisfying both is the Seth–Hill family

$$\tilde{E}^{(m)} = \frac{1}{m} (\tilde{U}^m - \hat{I}) \quad (m \neq 0), \quad \tilde{E}^{(0)} = \ln \tilde{U} = \sum_i \ln \lambda_i \underline{u}_i \otimes \underline{u}_i,$$

which contains everything the literature uses:

$m = 2 :$	$\frac{1}{2}(\tilde{U}^2 - \hat{I}) = \frac{1}{2}(\tilde{C} - \hat{I}) = \tilde{E}$	Lagrange, (2.97),
$m = 1 :$	$\tilde{U} - \hat{I}$	Biot,
$m = -2 :$	$\frac{1}{2}(\hat{I} - \tilde{C}^{-1})$	referential Euler,
$m = 0 :$	$\ln \tilde{U}$	Hencky.

They all agree to first order, and that is why linear elasticity cannot tell them apart. Put $\underline{U} = \underline{\hat{I}} + \underline{\hat{H}}$ with $\underline{\hat{H}}$ small. Then $\underline{U}^m = \underline{\hat{I}} + m\underline{\hat{H}} + O(|\underline{\hat{H}}|^2)$ by the binomial expansion applied to each principal value, so

$$\underline{E}^{(m)} = \frac{1}{m}(m\underline{\hat{H}} + O(|\underline{\hat{H}}|^2)) = \underline{\hat{H}} + O(|\underline{\hat{H}}|^2) \quad \text{for every } m,$$

and $\ln(1 + h) = h + O(h^2)$ gives the same for $m = 0$. Every member of the family is the same tensor $\underline{\hat{H}}$ to first order. It is only in finite deformation that the choice matters, and then it matters a great deal.

Why $m = 0$ is the distinguished member. Two properties single out the logarithmic strain, and neither holds for any other m .

First, its trace is *exactly* the logarithmic volume change, with no approximation:

$$\text{tr}(\ln \underline{U}) = \sum_i \ln \lambda_i = \ln(\lambda_1 \lambda_2 \lambda_3) = \ln \det \underline{U} = \ln J,$$

using (3.17) and $\det \underline{U} = \det \underline{F}$ from (3.35) with $\det \underline{R} = 1$. For comparison $\text{tr} \underline{E} = \frac{1}{2}(I_1(\underline{C}) - 3) = \frac{1}{2} \sum (\lambda_i^2 - 1)$, which is not any simple function of J . So Hencky strain, and only Hencky strain, splits cleanly into a volumetric part $\frac{1}{3}(\ln J)\underline{\hat{I}}$ and a traceless distortional part.

Second, coaxial stretches *add*. If $\underline{U}_2$ and $\underline{U}_1$ share principal axes, then $\ln(\underline{U}_2 \underline{U}_1) = \ln \underline{U}_1 + \ln \underline{U}_2$, because on each axis the principal values multiply and their logarithms add. Stretching by 2 and then again by 2 gives Hencky strain $2 \ln 2$, exactly twice the strain of a single doubling, which is the property one expects of a strain and which $\underline{E}$ conspicuously lacks: there $\frac{1}{2}(4 - 1) = 1.5$ against $\frac{1}{2}(16 - 1) = 7.5$, five times as much rather than twice. The additivity fails as soon as the axes rotate, which is why the logarithmic strain is easy in one dimension and delicate in three.

3.4 Polar decomposition theorem

THEOREM 3.19 (Polar decomposition). *Let $\underline{F}$ have $\det \underline{F} > 0$, regarded as a two-point tensor from $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$. Then there exist unique*

symmetric positive definite $\underline{U}$ and $\underline{V}$, and a unique rotation $\underline{R}$, such that

$$\underline{F} = \underline{R}\underline{U} = \underline{V}\underline{R}. \quad (3.32)$$

In components,

$$F_{iA} = R_{iB}U_{BA} = V_{ij}R_{jA}. \quad (3.33)$$

The construction runs as follows, and each ingredient has already been prepared.

Step 1: build $\underline{U}$. By (2.85) and (2.84), $\underline{C} = \underline{F}^t \underline{F}$ is symmetric and positive definite, and by (2.86) it maps $\hat{\mathbb{E}}^3$ to itself, so it is referential. Theorem 3.16, applied in $\hat{\mathbb{E}}^3$, produces a unique symmetric positive definite $\underline{U}$ with

$$\underline{U}^2 = \underline{C}.$$

It is invertible, its principal values being positive, and by §3.3 it too is referential. Hence $\underline{R} := \underline{F}\underline{U}^{-1}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$, a two-point tensor, and $\underline{V}$, defined below, will map $\mathbb{E}^3$ to itself.

Step 2: the determinant bookkeeping. For a linearly independent $\{\underline{a}, \underline{b}, \underline{c}\}$ in $\hat{\mathbb{E}}^3$, using linearity of the box product and $\underline{U}\underline{a}, \underline{U}\underline{b}, \underline{U}\underline{c} \in \hat{\mathbb{E}}^3$,

$$\begin{aligned} (\det \underline{F})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{R}(\underline{U}\underline{a}), \underline{R}(\underline{U}\underline{b}), \underline{R}(\underline{U}\underline{c})] \\ &= (\det \underline{R})[\underline{U}\underline{a}, \underline{U}\underline{b}, \underline{U}\underline{c}] \\ &= (\det \underline{R})(\det \underline{U})[\underline{a}, \underline{b}, \underline{c}], \end{aligned} \quad (3.34)$$

so that, cancelling the non-zero box product,

$$\det \underline{F} = (\det \underline{R})(\det \underline{U}). \quad (3.35)$$

Since $\det \underline{U} = \lambda_1 \lambda_2 \lambda_3 > 0$ by (3.17) and $\det \underline{F} > 0$ by hypothesis,

$$\det \underline{R} > 0. \quad (3.36)$$

The book disposes of the other factorisation with “in the same way we easily establish”. Written out, it is the same three lines with the factors interchanged. Using (3.32)₂, and noting that $\underline{R}\underline{a}, \underline{R}\underline{b}, \underline{R}\underline{c}$ all lie in $\mathbb{E}^3$,

which is what makes the middle box product legitimate,

$$\begin{aligned} (\det \underline{\tilde{F}})[\underline{\tilde{a}}, \underline{\tilde{b}}, \underline{\tilde{c}}] &= [\underline{\tilde{V}}(\underline{\tilde{R}}\underline{\tilde{a}}), \underline{\tilde{V}}(\underline{\tilde{R}}\underline{\tilde{b}}), \underline{\tilde{V}}(\underline{\tilde{R}}\underline{\tilde{c}})] \\ &= (\det \underline{\tilde{V}})[\underline{\tilde{R}}\underline{\tilde{a}}, \underline{\tilde{R}}\underline{\tilde{b}}, \underline{\tilde{R}}\underline{\tilde{c}}] \\ &= (\det \underline{\tilde{V}})(\det \underline{\tilde{R}})[\underline{\tilde{a}}, \underline{\tilde{b}}, \underline{\tilde{c}}], \end{aligned}$$

so $\det \underline{\tilde{F}} = (\det \underline{\tilde{V}})(\det \underline{\tilde{R}})$. Comparing with (3.35) and cancelling the non-zero $\det \underline{\tilde{R}}$ gives the by-product $\det \underline{\tilde{V}} = \det \underline{\tilde{U}}$, which is not obvious from the definitions and will be confirmed at (3.42), where $\underline{\tilde{U}}$ and $\underline{\tilde{V}}$ are found to have the same principal values. Since $\det \underline{\tilde{F}} > 0$ and $\det \underline{\tilde{R}} > 0$, also $\det \underline{\tilde{V}} > 0$.

REMARK 3.20 (an extension smuggled in). Problem 21(b) of Chapter 1 proved $\det(\underline{A}\underline{B}) = \det \underline{A} \det \underline{B}$ for tensors of a single space. Equation (3.34) silently extends it to the case where the first factor is two-point and the second referential. The extension is legitimate for exactly the reason given after (2.47): at each step the three vectors inside a box product belong to a single space, so every box product written is defined. Nothing else is needed.

Step 3: $\underline{\tilde{R}}$ is orthogonal. Using $(\underline{A}\underline{B})^t = \underline{B}^t \underline{A}^t$ from Problem 20, $\underline{U}^{-t} = \underline{U}^{-1}$ from the symmetry of $\underline{U}$ (its inverse is symmetric because its spectral form has the same axes, see Remark 3.10), and $\underline{U}\underline{U} = \underline{C}$,

$$\underline{\tilde{R}}^t \underline{\tilde{R}} = \underline{U}^{-t} \underline{\tilde{F}}^t \underline{\tilde{F}} \underline{U}^{-1} = \underline{U}^{-1} \underline{C} \underline{U}^{-1} = \underline{U}^{-1} \underline{U} \underline{U} \underline{U}^{-1} = \underline{\hat{I}}. \quad (3.37)$$

So $\underline{\tilde{R}}$ is orthogonal, with $\underline{\tilde{R}}^{-1} = \underline{\tilde{R}}^t : \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$.

Step 4: rotate the principal frame. Write the spectral forms of $\underline{U}$ and, by squaring, of $\underline{C}$:

$$\underline{U} = \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i, \quad \underline{C} = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i, \quad (3.38)$$

with $\lambda_i > 0$ the *principal stretches* and $\underline{u}_i \in \hat{\mathbb{E}}^3$ orthonormal. Set

$$\underline{v}_i = \underline{\tilde{R}} \underline{u}_i \in \mathbb{E}^3. \quad (3.39)$$

These are orthonormal:

$$\underline{v}_i \cdot \underline{v}_j = \underline{R}\underline{u}_i \cdot \underline{R}\underline{u}_j = \underline{u}_i \cdot \underline{R}^t \underline{R}\underline{u}_j = \underline{u}_i \cdot \hat{I} \underline{u}_j = \underline{u}_i \cdot \underline{u}_j = \delta_{ij}. \quad (3.40)$$

Consequently, by (3.11) applied in $\mathbb{E}^3$ and then Problems 17(b),(c) of Chapter 1 in the combined form $(\underline{R}\underline{a}) \otimes (\underline{R}\underline{b}) = \underline{R}(\underline{a} \otimes \underline{b})\underline{R}^t$,

$$\underline{I} = \underline{v}_i \otimes \underline{v}_i = (\underline{R}\underline{u}_i) \otimes (\underline{R}\underline{u}_i) = \underline{R}(\underline{u}_i \otimes \underline{u}_i)\underline{R}^t = \underline{R}\hat{I}\underline{R}^t = \underline{R}\underline{R}^t. \quad (3.41)$$

The intermediate identity deserves its own line, since it is used again in §3.5. For any $\underline{w} \in \mathbb{E}^3$,

$$\begin{aligned} [(\underline{R}\underline{a}) \otimes (\underline{R}\underline{b})]\underline{w} &= (\underline{R}\underline{b} \cdot \underline{w}) \underline{R}\underline{a} = (\underline{b} \cdot \underline{R}^t \underline{w}) \underline{R}\underline{a} \\ &= \underline{R}[(\underline{a} \otimes \underline{b})\underline{R}^t \underline{w}] = [\underline{R}(\underline{a} \otimes \underline{b})\underline{R}^t]\underline{w}. \end{aligned}$$

Step 5: build $\underline{V}$. Take (3.32)₂ as the definition of $\underline{V}$, i.e. $\underline{V} := \underline{F}\underline{R}^t$; then $\underline{V} = \underline{V}\underline{I} = \underline{V}\underline{R}\underline{R}^t = \underline{F}\underline{R}^t$ is consistent, and

$$\begin{aligned} \underline{V} &= \underline{R}\underline{U}\underline{R}^t = \underline{R}\left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i\right)\underline{R}^t \\ &= \sum_i \lambda_i (\underline{R}\underline{u}_i) \otimes (\underline{R}\underline{u}_i) = \sum_i \lambda_i \underline{v}_i \otimes \underline{v}_i. \end{aligned} \quad (3.42)$$

This is manifestly symmetric and positive definite, and it exhibits the spectral decomposition $\underline{B} = \underline{V}^2 = \sum_i \lambda_i^2 \underline{v}_i \otimes \underline{v}_i$ of the left Cauchy–Green tensor. Note that $\underline{U}$ and $\underline{V}$ have the *same* principal values and differently placed axes. Also

$$\underline{F} = \underline{R}\underline{U} = \underline{R}\left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i\right) = \sum_i \lambda_i (\underline{R}\underline{u}_i) \otimes \underline{u}_i = \sum_i \lambda_i \underline{v}_i \otimes \underline{u}_i. \quad (3.43)$$

Step 6: $\det \underline{R} = 1$. Take the principal triad $\{\underline{u}_i\}$ of $\underline{U}$ as the test triad in the two-point determinant (2.47), which is allowed because it is a basis of $\hat{\mathbb{E}}^3$ and hence independent. By the definition (3.39) of $\underline{v}_i$,

$$(\det \underline{R})[\underline{u}_1, \underline{u}_2, \underline{u}_3] = [\underline{R}\underline{u}_1, \underline{R}\underline{u}_2, \underline{R}\underline{u}_3] = [\underline{v}_1, \underline{v}_2, \underline{v}_3].$$

Both triads are orthonormal, $\{\underline{u}_i\}$ by Theorem 3.7 and $\{\underline{v}_i\}$ by (3.40), so by Lemma 3.28

$$[\underline{u}_1, \underline{u}_2, \underline{u}_3] = \pm 1, \quad [\underline{v}_1, \underline{v}_2, \underline{v}_3] = \pm 1,$$

each sign being $+$ or $-$ according to the handedness of its triad. Substituting, $\det \underline{R} = (\pm 1)/(\pm 1)$, so $\det \underline{R} = \pm 1$ and the value is $+1$ exactly when the two triads have the *same* handedness. But (3.36) has already established $\det \underline{R} > 0$, which rules out -1 . Therefore

$$\det \underline{R} = +1,$$

and as a by-product the two triads must have the same handedness: $\underline{R}$ cannot turn a right-handed frame into a left-handed one. Thus $\underline{R}$ is a two-point rotation and Theorem 3.19 is proved apart from uniqueness.

Step 7: uniqueness. Suppose $\underline{F} = \underline{R}\underline{U} = \underline{R}'\underline{U}'$ with $\underline{U}, \underline{U}'$ symmetric positive definite and $\underline{R}, \underline{R}'$ rotations. Then

$$\underline{C} = \underline{F}^t \underline{F} = (\underline{R}'\underline{U}')^t (\underline{R}'\underline{U}') = \underline{U}'^t \underline{R}'^t \underline{R}' \underline{U}' = \underline{U}' \hat{I} \underline{U}' = \underline{U}'^2,$$

so $\underline{U}'$ is a symmetric positive definite square root of $\underline{C}$; by Theorem 3.16 it equals $\underline{U}$. Then $\underline{R}' = \underline{F}\underline{U}'^{-1} = \underline{F}\underline{U}^{-1} = \underline{R}$. Finally $\underline{V}' = \underline{F}\underline{R}'^t = \underline{F}\underline{R}^t = \underline{V}$. The three factors are therefore uniquely determined by $\underline{F}$, and only the positive definiteness of $\underline{U}$ was used to get there. $\square$

REMARK 3.21 (what the theorem says in words). $\underline{F} = \underline{R}\underline{U}$ says: *first* stretch by λ_i along the three referential axes $\underline{u}_i$, *then* rotate the result rigidly. $\underline{F} = \underline{V}\underline{R}$ says: *first* rotate, *then* stretch by the same three factors along the rotated axes $\underline{v}_i = \underline{R}\underline{u}_i$. The order can be exchanged provided the axes are carried along, and the stretches themselves are the same numbers in both readings. The physically important half is $\underline{U}$, or equivalently $\underline{C} = \underline{U}^2$: it is the part that a material feels. The rotation $\underline{R}$ costs nothing, which is why constitutive laws are written in terms of $\underline{C}$ and never in terms of $\underline{F}$ alone.

REMARK 3.22 (why it is called *polar*). The name is never explained in the text and is not decoration. Every non-zero complex number has exactly one representation

$$z = r e^{i\vartheta}, \quad r > 0,$$

as a positive magnitude times a unit-modulus factor: the polar form. Theorem 3.19 is the same statement for tensors, item by item:

$$\begin{aligned}
\text{the object:} \quad & z \neq 0 \quad \longleftrightarrow \quad \underline{\underline{F}}, \det \underline{\underline{F}} > 0 \\
\text{the magnitude:} \quad & r > 0 \quad \longleftrightarrow \quad \underline{\underline{U}} \text{ sym. pos. def.} \\
\text{the phase:} \quad & |e^{i\vartheta}| = 1 \quad \longleftrightarrow \quad \underline{\underline{R}}^t \underline{\underline{R}} = \hat{\underline{\underline{I}}} \\
\text{how it is got:} \quad & r = \sqrt{\bar{z}z} \quad \longleftrightarrow \quad \underline{\underline{U}} = \sqrt{\underline{\underline{F}}^t \underline{\underline{F}}} \\
\text{rigidity:} \quad & |z| = 1 \quad \longleftrightarrow \quad \underline{\underline{U}} = \hat{\underline{\underline{I}}}, \quad (3.122)
\end{aligned}$$

Conjugation $z \mapsto \bar{z}$ corresponds to transposition $\underline{\underline{F}} \mapsto \underline{\underline{F}}^t$, and $\bar{z}z = |z|^2$ to $\underline{\underline{F}}^t \underline{\underline{F}} = \underline{\underline{C}} = \underline{\underline{U}}^2$. Even the proof runs parallel: one forms $\bar{z}z$ to eliminate the phase, takes the positive square root, and divides it out. That is exactly Steps 1 and 3 of §3.4.

The analogy also shows where tensors are harder, and the difference is worth stating because it is the source of most of the difficulty in finite elasticity. Complex multiplication commutes, so $re^{i\vartheta} = e^{i\vartheta}r$ and there is only *one* polar form. Tensor multiplication does not, so $\underline{\underline{R}}\underline{\underline{U}} \neq \underline{\underline{U}}\underline{\underline{R}}$ in general and there are *two*, $\underline{\underline{F}} = \underline{\underline{R}}\underline{\underline{U}}$ and $\underline{\underline{F}} = \underline{\underline{V}}\underline{\underline{R}}$, with $\underline{\underline{V}} = \underline{\underline{R}}\underline{\underline{U}}\underline{\underline{R}}^t \neq \underline{\underline{U}}$ unless $\underline{\underline{R}}$ happens to fix the principal axes. Nothing in the complex case prepares one for the fact that “stretch then rotate” and “rotate then stretch” require different stretch tensors. Note also that $\underline{\underline{U}}$ and $\underline{\underline{V}}$ are not independent: by (3.42) they have the same principal values, only differently placed, so there is exactly one set of three stretches and two ways to say where they act.

Principal stretches are stretches: the link back to (2.67)

Nothing so far has said why the numbers λ_i deserve the name *stretch*. The following computation supplies the reason and, at the same time, forces us to be careful about what a material vector is.

Fix an instant t_* and a material point with referential position $\underline{\underline{X}}$. The principal vectors $\underline{\underline{u}}_i$ of $\underline{\underline{U}}(\underline{\underline{X}}, t_*)$ are three directions in $\hat{\mathbb{E}}^3$. Since $\hat{\mathbb{E}}^3$ is where material directions also live, we may select a material curve

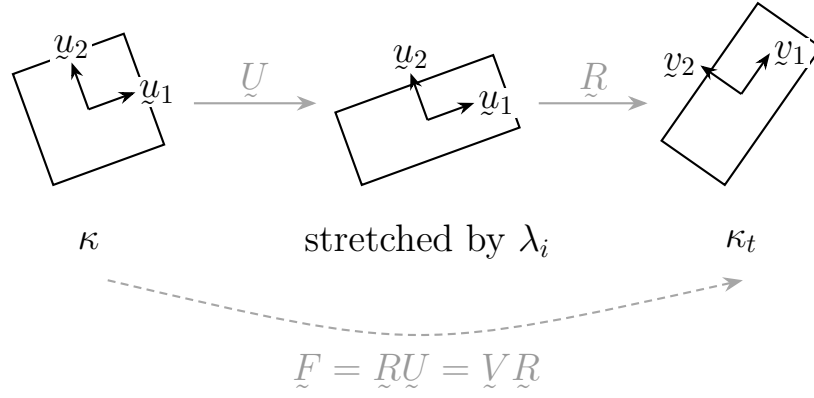


Figure A: The polar decomposition. Reading $\tilde{F} = \tilde{R}\tilde{U}$, the referential principal axes $\underline{u}_i$ are stretched by λ_i and the stretched configuration is then rotated rigidly onto κ_t . Reading $\tilde{F} = \tilde{V}\tilde{R}$ reverses the order and stretches along the rotated axes $\underline{v}_i = \tilde{R}\underline{u}_i$ instead, by the same factors. The two routes reach the same κ_t , and by Step 7 the factors are unique.

through $\tilde{X}$ whose unit tangent $\tilde{M} = \tilde{X}'(S)$ happens to equal a given $\underline{u} \in \{\underline{u}_i\}$ at that point and at that instant. For this curve, (2.67) reads $\mu\tilde{m} = \tilde{F}\tilde{M}$, so using (3.32)₁, (3.38)₁ and (3.39),

$$\begin{aligned}
 \mu\tilde{m} &= \tilde{F}\tilde{M} = \tilde{F}\underline{u} = \tilde{R}(\tilde{U}\underline{u}) \\
 &= \tilde{R}(\lambda\underline{u}) \\
 &= \lambda\tilde{R}\underline{u} \\
 &= \lambda\underline{v},
 \end{aligned} \tag{3.44}$$

where λ is the principal value belonging to $\underline{u}$ and $\underline{v} \in \{\underline{v}_i\}$. Since $|\underline{v}| = 1$ and $\mu > 0$, comparing lengths gives $\mu = \lambda$ and comparing directions gives $\tilde{m} = \underline{v}$. So: *a material fibre momentarily aligned with the i th principal direction of $\tilde{U}$ stretches by exactly λ_i , and ends up aligned with $\underline{v}_i$.* That is the justification for the name.

The same conclusion from the other factorisation:

$$\mu\tilde{m} = \tilde{F}\tilde{M} = \tilde{F}\underline{u} = \tilde{V}(\tilde{R}\underline{u}) = \tilde{V}\underline{v} = \lambda\underline{v}, \tag{3.45}$$

using (3.42) and $\tilde{V}\underline{v}_i = \lambda_i\underline{v}_i$. Line (3.44) reads the deformation as *stretch then rotate*, line (3.45) as *rotate then stretch*.

Computing $\underline{U}$ and $\underline{R}$ without eigenvectors

Theorem 3.19 proves that $\underline{U}$ exists but constructs it through (3.28), which needs the principal triad. Finding that triad means solving a cubic and then three linear systems, and for a $\underline{U}$ that varies from point to point, as in §§3.5.2–3.5.4, that is unusable. Problem 9 of §3.8 gives a closed form in two dimensions. Here is the three-dimensional one. It is not in the book, and it makes every example of §3.5 a finite computation.

THEOREM 3.23 (Closed form for the right stretch tensor). *Let $\underline{C} = \underline{F}^t \underline{F}$ have principal invariants I_1, I_2, I_3 , and let $i_1 = \text{tr } \underline{U}$, $i_2, i_3 = \det \underline{U}$ be those of $\underline{U} = \sqrt{\underline{C}}$. Then*

$$\underline{U} = \frac{1}{i_1 i_2 - i_3} \left[-\underline{C}^2 + (i_1^2 - i_2) \underline{C} + i_1 i_3 \hat{I} \right], \quad (*_1)$$

where $i_3 = \sqrt{I_3}$, $i_2 = \frac{1}{2}(i_1^2 - I_1)$, and i_1 is the largest real root of

$$i_1^4 - 2I_1 i_1^2 - 8\sqrt{I_3} i_1 + (I_1^2 - 4I_2) = 0. \quad (*_2)$$

The rotation then follows from $\underline{R} = \underline{F} \underline{U}^{-1}$, with $\underline{U}^{-1}$ supplied by Cayley–Hamilton as $\underline{U}^{-1} = (\underline{C} - i_1 \underline{U} + i_2 \hat{I})/i_3$.

Proof. The invariant relations. Since $\underline{U}$ and $\underline{C} = \underline{U}^2$ share principal axes with values λ_i and λ_i^2 , Problem 7 of §3.8 gives

$$I_1 = \sum_i \lambda_i^2, \quad I_2 = \sum_{i < j} \lambda_i^2 \lambda_j^2, \quad I_3 = (\lambda_1 \lambda_2 \lambda_3)^2,$$

while $i_1 = \sum \lambda_i$, $i_2 = \sum_{i < j} \lambda_i \lambda_j$, $i_3 = \lambda_1 \lambda_2 \lambda_3$. Squaring i_1 and subtracting twice i_2 ,

$$i_1^2 - 2i_2 = \left(\sum_i \lambda_i \right)^2 - 2 \sum_{i < j} \lambda_i \lambda_j = \sum_i \lambda_i^2 = I_1,$$

the cross terms cancelling exactly; squaring i_2 and subtracting $2i_1 i_3$ gives I_2 by the identical computation one level up; and $i_3^2 = I_3$ is immediate. Hence

$$i_3 = \sqrt{I_3} > 0, \quad i_2 = \frac{1}{2}(i_1^2 - I_1), \quad i_2^2 = I_2 + 2i_1 i_3,$$

the positive root being forced by $\lambda_i > 0$. Substituting the middle relation into the last,

$$\frac{1}{4}(i_1^2 - I_1)^2 = I_2 + 2i_1\sqrt{I_3},$$

and clearing the fraction gives $(*_2)$. Of its real roots, i_1 is the sum of three positive numbers and is the largest.

The formula. Apply Cayley–Hamilton, Problem 7(b), to $\underline{U}$:

$$\underline{U}^3 - i_1\underline{U}^2 + i_2\underline{U} - i_3\hat{I} = \hat{O}.$$

Multiply by $\underline{U}$ and use $\underline{U}^2 = \underline{C}$ repeatedly to eliminate all odd powers except one:

$$\underline{U}^4 - i_1\underline{U}^3 + i_2\underline{U}^2 - i_3\underline{U} = \hat{O}, \quad \text{i.e.} \quad \underline{C}^2 - i_1\underline{U}\underline{C} + i_2\underline{C} - i_3\underline{U} = \hat{O},$$

since $\underline{U}^4 = \underline{C}^2$ and $\underline{U}^3 = \underline{U}\underline{C}$. From the first display, $\underline{U}\underline{C} = \underline{U}^3 = i_1\underline{C} - i_2\underline{U} + i_3\hat{I}$; substituting,

$$\underline{C}^2 - i_1(i_1\underline{C} - i_2\underline{U} + i_3\hat{I}) + i_2\underline{C} - i_3\underline{U} = \hat{O},$$

that is

$$(i_1i_2 - i_3)\underline{U} = -\underline{C}^2 + (i_1^2 - i_2)\underline{C} + i_1i_3\hat{I},$$

which is $(*_1)$. The coefficient never vanishes: expanded, $i_1i_2 - i_3 = (\lambda_1 + \lambda_2)(\lambda_2 + \lambda_3)(\lambda_3 + \lambda_1)$, a product of three positive numbers.

The inverse. Divide the Cayley–Hamilton identity by $i_3 \neq 0$ and rearrange, using $\underline{U}^2 = \underline{C}$ again: $\underline{U}^{-1} = (\underline{U}^2 - i_1\underline{U} + i_2\hat{I})/i_3 = (\underline{C} - i_1\underline{U} + i_2\hat{I})/i_3$. ■

REMARK 3.24 (the same statement one dimension down). Run the proof in two dimensions and it collapses to Problem 9. There Cayley–Hamilton reads $\underline{U}^2 - I\underline{U} + J\hat{I} = \hat{O}$ with $I = i_1$, $J = i_2$, and $\underline{U}^2 = \underline{C}$ turns it directly into $I\underline{U} = \underline{C} + J\hat{I}$ with no need to multiply by $\underline{U}$ at all; the quartic $(*_2)$ degenerates to $I^2 = \text{tr } \underline{C} + 2J$ and $J = \sqrt{\det \underline{C}}$. The extra work in three dimensions is entirely the price of the third invariant.

Two checks on $(*_1)$. Put $\underline{C} = \hat{I}$: then $I_1 = 3$, $I_2 = 3$, $I_3 = 1$, and $(*_2)$ becomes $i_1^4 - 6i_1^2 - 8i_1 + 3 = 0$, whose largest root is $i_1 = 3$; then $i_2 = \frac{1}{2}(9 - 3) = 3$, $i_3 = 1$, and $(*_1)$ gives $\underline{U} = \frac{1}{9-1}[-\hat{I} + 6\hat{I} + 3\hat{I}] = \frac{8}{8}\hat{I} =$

$\hat{\underline{I}}$, as it must. Put $\underline{C} = \text{diag}(\lambda_1^2, \lambda_2^2, \lambda_3^2)$: every tensor in $(*_1)$ is diagonal, and the k th entry of the right side is $[-\lambda_k^4 + (i_1^2 - i_2)\lambda_k^2 + i_1 i_3] / (i_1 i_2 - i_3)$, which equals λ_k precisely because λ_k^4 can be reduced by the characteristic equation of $\underline{U}$, $\lambda_k^3 = i_1 \lambda_k^2 - i_2 \lambda_k + i_3$, multiplied by λ_k .

Material vectors

The word “material” has been used repeatedly since Chapter 2 and never defined sharply. It is the hinge of the argument just given, so it is worth pinning down completely.

DEFINITION 3.25 (Material vector). A vector field $\underline{M}$ attached to the body is *material* if, expressed as a function of the referential position and time, it does not depend on time:

$$\underline{\hat{M}} = \underline{\hat{M}}(\underline{X}), \quad \text{equivalently} \quad \underline{\dot{M}} = \left. \frac{\partial \underline{\hat{M}}}{\partial t} \right|_{\underline{X}} = \underline{0}.$$

REMARK 3.26 (what a material vector really is, with examples). The definition looks like a triviality and is not. Here is the content.

A material vector is a *label glued to matter*. It is decided once and for all in the reference configuration and then never touched again. It does not move, because κ does not move; it is $\underline{X}$ -dependent data, like a serial number painted on a fibre. What moves is its *image* $\underline{F}\underline{M}$, and that is a different object living in a different space.

The test is mechanical: write the field as a function of $(\underline{X}, t)$ and look for a t . If there is none, the field is material.

EXAMPLES OF MATERIAL VECTORS.

(i) *The tangent to a material curve.* Let $\mathcal{C} \subset \kappa$ be a curve consisting of a fixed set of material points, with arclength parametrisation $\underline{X}(S)$, and let $\underline{M}(S) = \underline{X}'(S)$ as in (2.66). Since $\mathcal{C}$ is a fixed subset of the fixed configuration κ , the function $\underline{X}(S)$ carries no t , so neither does $\underline{M}$. Material.

(ii) *The normal to a material surface.* If $d\underline{X}$ and $d\underline{Y}$ span the tangent plane of a material surface at $\underline{X}$, then $\underline{N} dA = d\underline{X} \times d\underline{Y}$ of (2.51) involves only referential data. Material.

(iii) *The referential basis* $\{\hat{\underline{E}}_A\}$ *and the position* $\underline{X}$. Both are fixed by hypothesis. Material, trivially.

(iv) *A fibre direction in a fibre-reinforced solid.* The direction in which the reinforcing fibres were laid up is decided when the material is manufactured, that is, in κ . Material. This is the example that makes the concept indispensable: an anisotropic constitutive law must be written in terms of such a vector, and it must be a *referential* one, or the law would change when the body is merely carried across the room.

EXAMPLES THAT ARE *not* MATERIAL VECTORS.

(v) *The current tangent* $\underline{m}$. By (2.67), $\underline{m} = \underline{F}\underline{M}/\mu$ with $\underline{F} = \underline{F}(\underline{X}, t)$; the t is right there. Not material. It also lives in $\mathbb{E}^3$ rather than $\hat{\mathbb{E}}^3$, so it could not be material in the sense of Definition 3.25 even in principle.

(vi) *The current normal* $\underline{n}$. By Piola–Nanson (2.54), $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$, again t -dependent. Not material.

(vii) *The principal directions* $\underline{u}_i$ *of* $\underline{U}$. These are referential, so they at least live in the right space, but they are properties of $\underline{U}(\underline{X}, t)$ and therefore inherit its time dependence. Not material. This is the case the book flags immediately after (3.44), and it is genuinely subtle, because $\underline{u}_i$ passes the “lives in $\hat{\mathbb{E}}^3$ ” test and fails only the “no t ” test. Remark 3.27 computes $\dot{\underline{u}}_1 \neq \underline{0}$ explicitly.

The algebraic consequence of being material is worth stating once. If $\underline{M}$ is material then, from (2.67),

$$\left. \frac{\partial}{\partial t} \right|_{\underline{X}} (\mu \underline{m}) = \left. \frac{\partial}{\partial t} \right|_{\underline{X}} (\underline{F} \underline{M}) = \dot{\underline{F}} \underline{M},$$

with no term $\underline{F} \dot{\underline{M}}$, because $\dot{\underline{M}} = \underline{0}$. Every rate formula for material fibres in Chapter 4 rests on the disappearance of that second term. If $\underline{M}$ were allowed to depend on time the formulas of Chapter 4 would all acquire an extra term and would describe nothing physical, since one would be comparing a different fibre at each instant.

REMARK 3.27 (the principal directions are not material: the algebra). Take the simple shear of §3.5.1 below and let the amount of shear grow

with time, $\gamma = \gamma(t)$, which is what happens in any actual shearing experiment. By (3.62) and (3.64) the first principal direction of $\underline{U}$ is

$$\underline{u}_1 = \cos A \hat{\underline{E}}_1 + \sin A \hat{\underline{E}}_2, \quad \tan A = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2}).$$

Differentiate at fixed $\underline{X}$, remembering that $\hat{\underline{E}}_A$ is constant:

$$\dot{\underline{u}}_1 = \dot{A}(-\sin A \hat{\underline{E}}_1 + \cos A \hat{\underline{E}}_2) = \dot{A} \underline{u}_2, \quad \dot{A} = \frac{dA}{d\gamma} \dot{\gamma}.$$

From $\tan A = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2})$,

$$\sec^2 A \frac{dA}{d\gamma} = \frac{1}{2} \left(1 + \frac{\gamma}{\sqrt{4 + \gamma^2}} \right) = \frac{\sqrt{4 + \gamma^2} + \gamma}{2\sqrt{4 + \gamma^2}} = \frac{\tan A}{\sqrt{4 + \gamma^2}},$$

so $dA/d\gamma = \sin A \cos A / \sqrt{4 + \gamma^2} > 0$ for $A \in (0, \pi/2)$. Hence $\dot{\underline{u}}_1 = \dot{A} \underline{u}_2 \neq \underline{0}$ whenever $\dot{\gamma} \neq 0$: the principal axes rotate through the material as the shear proceeds. At $\gamma = 0$ the rate is $dA/d\gamma = (\frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}})/2 = \frac{1}{4}$, so $A \approx \pi/4 - \dots$ turns at a quarter of the rate of γ .

Contrast this with $\underline{M} = \hat{\underline{E}}_1$, which is material. Its image under the deformation is, from (3.51),

$$\underline{F} \hat{\underline{E}}_1 = \underline{\hat{e}}_1 + \gamma \underline{\hat{e}}_2 (\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_1) = \underline{\hat{e}}_1 + \gamma \underline{\hat{e}}_2 \cdot \underline{0} = \underline{\hat{e}}_1, \quad \mu = |\underline{\hat{e}}_1| = 1.$$

So the X_1 fibres neither stretch nor turn, at any γ : they are the fibres lying in the shearing direction. Meanwhile $\underline{M} = \hat{\underline{E}}_2$, also material, has

$$\underline{F} \hat{\underline{E}}_2 = \underline{\hat{e}}_2 + \gamma \underline{\hat{e}}_1, \quad \mu = \sqrt{1 + \gamma^2}, \quad \underline{m} = \frac{\gamma \underline{\hat{e}}_1 + \underline{\hat{e}}_2}{\sqrt{1 + \gamma^2}},$$

so those fibres stretch and tilt. Two material fibres that were perpendicular are no longer so: by (2.106),

$$\begin{aligned} \cos \theta &= \frac{\hat{\underline{E}}_1 \cdot \underline{C} \hat{\underline{E}}_2}{\sqrt{(\hat{\underline{E}}_1 \cdot \underline{C} \hat{\underline{E}}_1)(\hat{\underline{E}}_2 \cdot \underline{C} \hat{\underline{E}}_2)}} = \frac{C_{12}}{\sqrt{C_{11}C_{22}}} \\ &= \frac{\gamma}{\sqrt{1 \cdot (1 + \gamma^2)}} = \frac{\gamma}{\sqrt{1 + \gamma^2}}, \end{aligned}$$

using the matrix (3.55); the same number follows directly from $\underline{e}_1 \cdot \underline{m}$ above, which is the point of (2.106).

The moral: *material* directions are the ones you may follow through the motion and still be talking about the same matter; *principal* directions are the ones along which, at one instant, the stretching is extremal. They coincide instant by instant, never as fields.

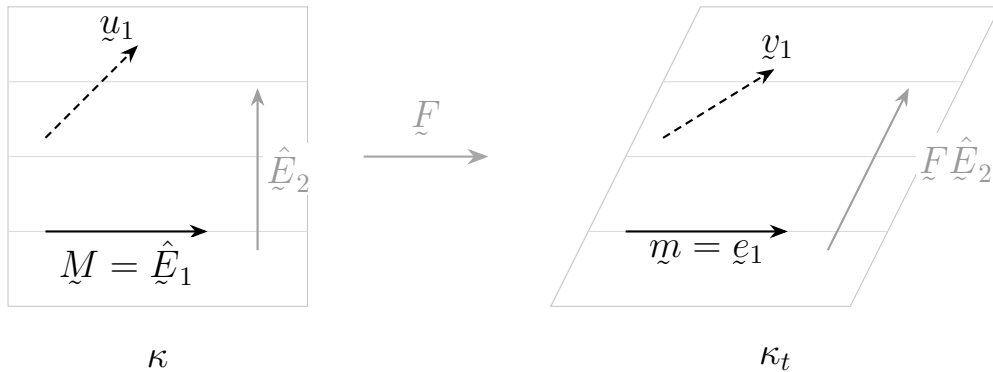


Figure B: Material versus principal directions in simple shear. The solid arrow is the material fibre $\hat{E}_1$: it is decided in κ and its image is $\underline{e}_1$, unstretched and unturned for every γ . The grey arrow is the material fibre $\hat{E}_2$, whose image tilts and lengthens by $\sqrt{1 + \gamma^2}$. The dashed arrow is the principal direction $\underline{u}_1$: it is *not* attached to any fixed matter, and as γ grows it sweeps across the fibres at the rate computed in Remark 3.27.

3.5 Examples

Four deformations are now carried through completely. The first is homogeneous and is the standard test case for every idea in the chapter; the other three are cylindrical and are the standard test cases for elasticity. In each, the recipe is the same and is worth memorising: *compute $d\underline{x}$ in the natural coordinates, express the coordinate increments as dot products with $d\underline{X}$, read off $\tilde{F}$ by comparison with $d\underline{x} = \tilde{F}d\underline{X}$, then form $\underline{C}$, $\underline{B}$, $\underline{U}$, $\underline{V}$, $\underline{R}$.*

The four rules used on every page below

Everything in §3.5 is built from four facts, all from Chapter 1. They are collected here so that the computations can quote them by name, and each is then still carried out in full at its point of use.

(R1) A dyad acting on a vector. $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$. The *second* slot does the dotting; the first slot is what comes out. So on an orthonormal basis $(\underline{a} \otimes \hat{\underline{E}}_B)\hat{\underline{E}}_A = \delta_{AB}\underline{a}$: a dyad picks out one basis vector and ignores the others.

(R2) A dyad times a dyad. $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c})\underline{a} \otimes \underline{d}$. The two *inner* slots dot; the two outer slots survive. So a product of two three-term sums has nine terms, and orthonormality kills six of them.

(R3) The box product. $[\underline{a}, \underline{b}, \underline{c}] = \underline{a} \cdot \underline{b} \times \underline{c} = \varepsilon_{ijk}a_ib_jc_k$. It is *linear in each slot separately*, so scalars come out one at a time; it is *alternating*, so it vanishes whenever two slots are equal and changes sign when two are swapped; and a cyclic permutation, being two swaps, leaves it unchanged.

(R4) The determinant. For a two-point $\underline{F}$, equation (2.47) gives

$$\det \underline{F} = \frac{[\underline{F}\underline{a}, \underline{F}\underline{b}, \underline{F}\underline{c}]}{[\underline{a}, \underline{b}, \underline{c}]}$$

for any independent $\underline{a}, \underline{b}, \underline{c}$ in the referential space. Both box products are then formed from vectors of a single space, which is what makes the ratio meaningful.

LEMMA 3.28 (Box product of an orthonormal triad). *If $\{\underline{a}, \underline{b}, \underline{c}\}$ is orthonormal, then $[\underline{a}, \underline{b}, \underline{c}] = +1$ when the triad is right handed and -1 when it is left handed.*

Proof. Right handed means, by definition, $\underline{b} \times \underline{c} = \underline{a}$. Then by (R3)

$$[\underline{a}, \underline{b}, \underline{c}] = \underline{a} \cdot \underline{b} \times \underline{c} = \underline{a} \cdot \underline{a} = |\underline{a}|^2 = 1.$$

Left handed means $\underline{b} \times \underline{c} = -\underline{a}$, and the same line gives -1 . ■

In particular $[\underline{e}_1, \underline{e}_2, \underline{e}_3] = 1$ and $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = 1$, both bases being right handed by assumption. The cylindrical frames need one line of checking, and it is done once here.

LEMMA 3.29 (The polar frames are right-handed and orthonormal).
With $\underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2$ and $\underline{e}_\theta(\theta) = \underline{e}'_r(\theta) = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2$,

$$\underline{e}_r \times \underline{e}_\theta = \underline{e}_3, \quad [\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = 1,$$

and likewise $\hat{E}_R \times \hat{E}_\Theta = \hat{E}_3$, $[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3] = 1$.

Proof. Orthonormality was checked in Problem 25 of Chapter 1; $|\underline{e}_r|^2 = \cos^2 \theta + \sin^2 \theta = 1$, similarly for $\underline{e}_\theta$, and $\underline{e}_r \cdot \underline{e}_\theta = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0$. For the cross product, expand bilinearly, keeping all four terms:

$$\begin{aligned} \underline{e}_r \times \underline{e}_\theta &= (\cos \theta \underline{e}_1 + \sin \theta \underline{e}_2) \times (-\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2) \\ &= -\cos \theta \sin \theta (\underline{e}_1 \times \underline{e}_1) + \cos^2 \theta (\underline{e}_1 \times \underline{e}_2) \\ &\quad - \sin^2 \theta (\underline{e}_2 \times \underline{e}_1) + \sin \theta \cos \theta (\underline{e}_2 \times \underline{e}_2). \end{aligned}$$

The first and last vanish because $\underline{v} \times \underline{v} = \underline{0}$. In the remaining two, $\underline{e}_1 \times \underline{e}_2 = \underline{e}_3$ and $\underline{e}_2 \times \underline{e}_1 = -\underline{e}_3$, so they contribute $\cos^2 \theta \underline{e}_3 + \sin^2 \theta \underline{e}_3 = \underline{e}_3$. Hence, by (R3) with a cyclic shift,

$$[\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = \underline{e}_3 \cdot \underline{e}_r \times \underline{e}_\theta = \underline{e}_3 \cdot \underline{e}_3 = 1.$$

The referential computation is the same with hats and Θ . ■

3.5.1 Simple shear

What this example is for. It is the simplest deformation that is neither a pure stretch nor a pure rotation, so it is the smallest case in which the polar decomposition has anything to say. Everything in §§3.1–3.4 gets used here at least once: the characteristic equation at (3.58), the extremal property implicitly at (3.60), the square root at $\underline{U} = \sqrt{\underline{C}}$, and the uniqueness of $\underline{R}$ at (3.70). It is also homogeneous, so §3.6 applies to it, and isochoric, so the constraint (3.57) cuts the work in half. The one

surprise it contains is Remark 3.33: the material rotates *backwards*, and seeing why is the point of the whole calculation.

Consider the deformation

$$\underline{x} = \underline{1}\underline{X} + \underline{u}, \quad (3.46)$$

where $\underline{1}$ is the shifter of (2.68) from $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$ and the displacement $\underline{u}$ of (2.77) is

$$\underline{u} = \gamma(\underline{E}_2 \cdot \underline{1}\underline{X})\underline{e}_1 = \gamma(\underline{1}^t \underline{E}_2 \cdot \underline{X})\underline{e}_1 = \gamma(\hat{\underline{E}}_2 \cdot \underline{X})\underline{e}_1, \quad (3.47)$$

in which γ , the *amount of shear*, is a dimensionless constant. The middle step moves $\underline{1}$ across the dot product by the definition of the transpose, and the last uses $\underline{1}^t \underline{E}_2 = \underline{1}^t \underline{1} \hat{\underline{E}}_2 = \hat{\underline{I}} \hat{\underline{E}}_2 = \hat{\underline{E}}_2$, from (2.72).

The Cartesian coordinates after deformation are $x_i = \underline{e}_i \cdot \underline{x}$, so

$$x_1 = \underline{e}_1 \cdot \underline{1}\underline{X} + \gamma \underline{E}_2 \cdot \underline{1}\underline{X}, \quad x_2 = \underline{e}_2 \cdot \underline{1}\underline{X}, \quad x_3 = \underline{e}_3 \cdot \underline{1}\underline{X}. \quad (3.48)$$

Using $\underline{1}\underline{X} = X_A \underline{E}_A$, which is (2.75), and choosing $\{\underline{E}_A\} = \{\underline{e}_i\}$ so that $\underline{e}_i \cdot \underline{E}_A = \delta_{iA}$,

$$x_1 = X_1 + \gamma X_2, \quad x_2 = X_2, \quad x_3 = X_3. \quad (3.49)$$

REMARK 3.30 (reading (3.49)). Planes $X_2 = \text{const}$ slide rigidly in the $\underline{e}_1$ direction, by an amount proportional to their height X_2 . Nothing moves vertically and nothing moves out of plane. A vertical line $X_1 = \text{const}$ tilts to a line of slope $1/\gamma$; the angle it makes with the vertical has tangent γ , which is where the name “amount of shear” comes from. Note carefully that γ is a *tangent of an angle*, not the angle itself; the two agree only to first order.

Differentiating (3.46) and (3.47) at fixed t ,

$$\begin{aligned} \underline{F} d\underline{X} &= d\underline{x} = \underline{1} d\underline{X} + \gamma(\hat{\underline{E}}_2 \cdot d\underline{X})\underline{e}_1 \\ &= \underline{1} d\underline{X} + \gamma(\underline{e}_1 \otimes \hat{\underline{E}}_2) d\underline{X} = (\underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2) d\underline{X}, \end{aligned} \quad (3.50)$$

the middle step being the definition of the dyad, $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$, read backwards. Since $d\underline{X}$ is arbitrary,

$$\underline{F} = \underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2. \quad (3.51)$$

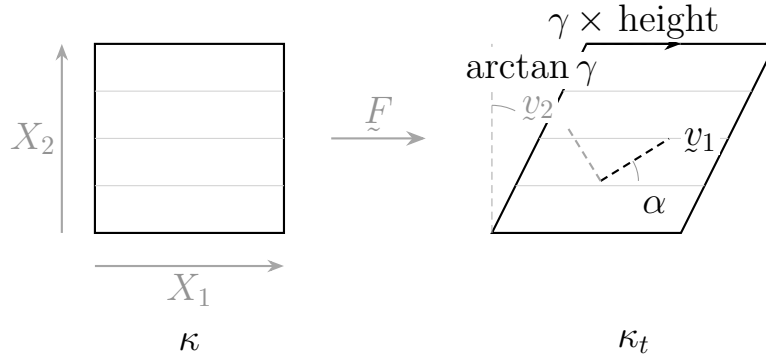


Figure 3.2: Simple shear. Horizontal material planes slide by an amount proportional to their height; the amount of shear γ is the tangent of the tilt angle of an initially vertical line. The dashed directions are the principal axes $\underline{v}_1, \underline{v}_2$ of $\underline{V}$ in κ_t , inclined at the angle α of (3.69); as $\gamma \rightarrow 0$ they approach $\pm 45^\circ$.

The Jacobian. Rule (R4) requires the images of three independent referential vectors, so take the basis and compute all three, using (R1) on the dyad in (3.51) and $\underline{\hat{E}}_A = \underline{E}_A = \underline{e}_A$:

$$\begin{aligned} F \underline{\hat{E}}_1 &= \underline{\hat{E}}_1 + \gamma (\underline{\hat{E}}_2 \cdot \underline{\hat{E}}_1) \underline{e}_1 = \underline{e}_1 + \gamma \cdot 0 \cdot \underline{e}_1 = \underline{e}_1, \\ F \underline{\hat{E}}_2 &= \underline{\hat{E}}_2 + \gamma (\underline{\hat{E}}_2 \cdot \underline{\hat{E}}_2) \underline{e}_1 = \underline{e}_2 + \gamma \cdot 1 \cdot \underline{e}_1 = \underline{e}_2 + \gamma \underline{e}_1, \\ F \underline{\hat{E}}_3 &= \underline{\hat{E}}_3 + \gamma (\underline{\hat{E}}_2 \cdot \underline{\hat{E}}_3) \underline{e}_1 = \underline{e}_3 + \gamma \cdot 0 \cdot \underline{e}_1 = \underline{e}_3. \end{aligned}$$

Only the second is moved, and it is moved by γ in the $\underline{e}_1$ direction: that is the whole of (3.49) read at the level of fibres. Now form the ratio (R4). The denominator is $[\underline{\hat{E}}_1, \underline{\hat{E}}_2, \underline{\hat{E}}_3] = 1$ by Lemma 3.28, the referential basis being right-handed orthonormal. The numerator is expanded by linearity of the box product in its second slot, (R3):

$$\begin{aligned} J = \det \underline{\hat{F}} &= \frac{[F \underline{\hat{E}}_1, F \underline{\hat{E}}_2, F \underline{\hat{E}}_3]}{[\underline{\hat{E}}_1, \underline{\hat{E}}_2, \underline{\hat{E}}_3]} = \frac{[\underline{e}_1, \underline{e}_2 + \gamma \underline{e}_1, \underline{e}_3]}{1} \\ &= [\underline{e}_1, \underline{e}_2, \underline{e}_3] + \gamma [\underline{e}_1, \underline{e}_1, \underline{e}_3] \\ &= 1 + \gamma \cdot 0 = 1. \end{aligned} \tag{3.52}$$

The first box product is 1 by Lemma 3.28; the second vanishes because the box product is *alternating* and its first two slots are the same vector, (R3). If one prefers to see it through the cross product instead of through

alternation, the same number comes out:

$$\begin{aligned} [\underline{e}_1, \underline{e}_2 + \gamma \underline{e}_1, \underline{e}_3] &= \underline{e}_1 \cdot (\underline{e}_2 + \gamma \underline{e}_1) \times \underline{e}_3 = \underline{e}_1 \cdot \underline{e}_2 \times \underline{e}_3 + \gamma \underline{e}_1 \cdot \underline{e}_1 \times \underline{e}_3 \\ &= \underline{e}_1 \cdot \underline{e}_1 + \gamma \underline{e}_1 \cdot -\underline{e}_2 = 1 - \gamma \cdot 0 = 1, \end{aligned}$$

using $\underline{e}_2 \times \underline{e}_3 = \underline{e}_1$ and $\underline{e}_1 \times \underline{e}_3 = -(\underline{e}_3 \times \underline{e}_1) = -\underline{e}_2$.

The geometry behind the algebra is worth one sentence: the sheared block of Figure 3.2 has the same base and the same height as the reference block, and a parallelepiped's volume is base times height regardless of how far the top has slid. That is what $\gamma[\underline{e}_1, \underline{e}_1, \underline{e}_3] = 0$ is saying. By (2.50) there is therefore no change of volume, and by (2.59) the densities agree, $\rho_\kappa = \rho$. Deformations with $J = 1$ are called *isochoric*.

The Cauchy–Green tensors. From (3.51) and Problem 17(a) of Chapter 1, $\underline{F}^t = \underline{1}^t + \gamma \hat{\underline{E}}_2 \otimes \underline{e}_1$, so

$$\begin{aligned} \underline{C} &= (\underline{1}^t + \gamma \hat{\underline{E}}_2 \otimes \underline{e}_1)(\underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2) \\ &= \underline{1}^t \underline{1} + \gamma \underline{1}^t (\underline{e}_1 \otimes \hat{\underline{E}}_2) + \gamma (\hat{\underline{E}}_2 \otimes \underline{e}_1) \underline{1} + \gamma^2 (\hat{\underline{E}}_2 \otimes \underline{e}_1)(\underline{e}_1 \otimes \hat{\underline{E}}_2) \\ &= \hat{\underline{I}} + \gamma (\underline{1}^t \underline{e}_1) \otimes \hat{\underline{E}}_2 + \gamma \hat{\underline{E}}_2 \otimes (\underline{1}^t \underline{e}_1) + \gamma^2 (\underline{e}_1 \cdot \underline{e}_1) \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 \\ &= \hat{\underline{I}} + \gamma (\hat{\underline{E}}_1 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_1) + \gamma^2 \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 \\ &= \hat{\underline{E}}_1 \otimes \hat{\underline{E}}_1 + \gamma (\hat{\underline{E}}_1 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_1) + (1 + \gamma^2) \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3, \end{aligned} \tag{3.53}$$

Each line deserves its justification, since nothing here is meant to be taken on trust.

Line 2 is the distributive law: the product of two two-term sums has four terms, and they are written in the order (first)(first), (first)(second), (second)(first), (second)(second).

Line 3 treats the four terms one at a time. The first is $\underline{1}^t \underline{1} = \hat{\underline{I}}$ by (2.72). The second uses Problem 17(b) of Chapter 1, $\underline{A}(a \otimes b) = (\underline{A}a) \otimes b$, with $\underline{A} = \underline{1}^t$: only the *first* slot of the dyad is touched, giving $(\underline{1}^t \underline{e}_1) \otimes \hat{\underline{E}}_2$. The third uses Problem 17(c), $(a \otimes b)\underline{A} = a \otimes (\underline{A}^t b)$, with $\underline{A} = \underline{1}$ so that $\underline{A}^t = \underline{1}^t$: now only the *second* slot is touched, giving $\hat{\underline{E}}_2 \otimes (\underline{1}^t \underline{e}_1)$. The fourth is rule (R2), the dyad product: the two inner slots, $\underline{e}_1$ and $\underline{e}_1$, dot to give $\underline{e}_1 \cdot \underline{e}_1 = 1$, and the two outer slots, both $\hat{\underline{E}}_2$, survive.

Line 4 evaluates $\underline{1}^t \underline{e}_1$. From (2.68), $\underline{1} = \underline{E}_A \otimes \underline{E}_A$, so by Problem

17(a) $\underline{\underline{1}}^t = \hat{\underline{\underline{E}}}_A \otimes \underline{\underline{E}}_A$, and by (R1)

$$\underline{\underline{1}}^t \underline{\underline{e}}_1 = (\underline{\underline{E}}_A \cdot \underline{\underline{e}}_1) \hat{\underline{\underline{E}}}_A = \delta_{A1} \hat{\underline{\underline{E}}}_A = \hat{\underline{\underline{E}}}_1,$$

using $\underline{\underline{E}}_A = \underline{\underline{e}}_A$ from (3.49). Substituting it into both the second and third terms produces the symmetric pair.

Line 5 writes out the identity, $\hat{\underline{\underline{I}}} = \hat{\underline{\underline{E}}}_1 \otimes \hat{\underline{\underline{E}}}_1 + \hat{\underline{\underline{E}}}_2 \otimes \hat{\underline{\underline{E}}}_2 + \hat{\underline{\underline{E}}}_3 \otimes \hat{\underline{\underline{E}}}_3$ by (3.11), and adds its $\hat{\underline{\underline{E}}}_2 \otimes \hat{\underline{\underline{E}}}_2$ term to the $\gamma^2 \hat{\underline{\underline{E}}}_2 \otimes \hat{\underline{\underline{E}}}_2$ already present, giving the coefficient $1 + \gamma^2$. It is a rearrangement, not a new computation, and its only purpose is to display the matrix (3.55) entry by entry.

The book writes “similarly, the left Cauchy–Green tensor is” and gives the answer. The computation is not quite the mirror image of the one above, because the shifter now appears the other way round, so here it is in full:

$$\begin{aligned} \underline{\underline{B}} &= \underline{\underline{F}} \underline{\underline{F}}^t = (\underline{\underline{1}} + \gamma \underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2)(\underline{\underline{1}}^t + \gamma \hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1) \\ &= \underline{\underline{1}} \underline{\underline{1}}^t + \gamma \underline{\underline{1}}(\hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1) + \gamma (\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2) \underline{\underline{1}}^t \\ &\quad + \gamma^2 (\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2)(\hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1). \end{aligned}$$

Take the four terms one at a time. The first is $\underline{\underline{I}}$, by (2.73). The second, by Problem 17(b), is $\gamma(\underline{\underline{1}} \hat{\underline{\underline{E}}}_2) \otimes \underline{\underline{e}}_1 = \gamma \underline{\underline{E}}_2 \otimes \underline{\underline{e}}_1 = \gamma \underline{\underline{e}}_2 \otimes \underline{\underline{e}}_1$, using $\underline{\underline{1}} \hat{\underline{\underline{E}}}_2 = \underline{\underline{E}}_2$ from (2.68) and the choice $\{\underline{\underline{E}}_A\} = \{\underline{\underline{e}}_i\}$ made at (3.49). The third, by Problem 17(c) with $\underline{\underline{A}} = \underline{\underline{1}}^t$ so that $\underline{\underline{A}}^t = \underline{\underline{1}}$, is $\gamma \underline{\underline{e}}_1 \otimes (\underline{\underline{1}} \hat{\underline{\underline{E}}}_2) = \gamma \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_2$. The fourth, by the dyad product rule, is $\gamma^2 (\hat{\underline{\underline{E}}}_2 \cdot \hat{\underline{\underline{E}}}_2) \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1 = \gamma^2 \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$. Adding, and expanding $\underline{\underline{I}} = \underline{\underline{e}}_i \otimes \underline{\underline{e}}_i$ so as to collect the $\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$ terms,

$$\underline{\underline{B}} = (1 + \gamma^2) \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1 + \underline{\underline{e}}_2 \otimes \underline{\underline{e}}_2 + \underline{\underline{e}}_3 \otimes \underline{\underline{e}}_3 + \gamma (\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_2 + \underline{\underline{e}}_2 \otimes \underline{\underline{e}}_1). \quad (3.54)$$

REMARK 3.31 ($\underline{\underline{C}}$ and $\underline{\underline{B}}$ are not the same tensor with new labels). Compare (3.53) and (3.54): the coefficient $1 + \gamma^2$ sits on $\hat{\underline{\underline{E}}}_2 \otimes \hat{\underline{\underline{E}}}_2$ in $\underline{\underline{C}}$ and on $\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$ in $\underline{\underline{B}}$. The index has moved from 2 to 1. That is not a slip; it is the rotation $\underline{\underline{R}}$ making itself felt. $\underline{\underline{C}}$ measures stretching as recorded on referential fibres, and the fibre that stretches most is $\hat{\underline{\underline{E}}}_2$ -ish; $\underline{\underline{B}}$ records it on the current fibres, and after the shear the most-stretched current direction is $\underline{\underline{e}}_1$ -ish. Anyone who writes $\underline{\underline{B}}$ by putting hats on $\underline{\underline{C}}$ will get this wrong.

In matrix form, with $C_{AB} = \hat{\underline{E}}_A \cdot \underline{C} \hat{\underline{E}}_B$ and $B_{ij} = \underline{e}_i \cdot \underline{B} \underline{e}_j$,

$$(C_{AB}) = \begin{pmatrix} 1 & \gamma & 0 \\ \gamma & 1 + \gamma^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (3.55)$$

$$(B_{ij}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3.56)$$

Principal stretches. First, one principal vector is free. Apply $\underline{C}$ of (3.53) to $\hat{\underline{E}}_3$ with (R1), taking the four dyads in turn:

$$\begin{aligned} \underline{C} \hat{\underline{E}}_3 &= (\hat{\underline{E}}_1 \cdot \hat{\underline{E}}_3) \hat{\underline{E}}_1 + \gamma (\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_3) \hat{\underline{E}}_1 + \gamma (\hat{\underline{E}}_1 \cdot \hat{\underline{E}}_3) \hat{\underline{E}}_2 \\ &\quad + (1 + \gamma^2) (\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_3) \hat{\underline{E}}_2 + (\hat{\underline{E}}_3 \cdot \hat{\underline{E}}_3) \hat{\underline{E}}_3 \\ &= 0 + 0 + 0 + 0 + \hat{\underline{E}}_3 = \hat{\underline{E}}_3, \end{aligned}$$

so $\hat{\underline{E}}_3$ is a principal vector of $\underline{C}$ with principal value 1, that is $\lambda_3^2 = 1$ and hence $\lambda_3 = 1$, the positive root being the one that is a stretch.

The other two are constrained before any computation, because the deformation is isochoric. Here every step is spelled out: (3.35) with $\det \underline{R} = 1$ from Step 6 turns $\det \underline{F}$ into $\det \underline{U}$; then the box product definition is applied with the principal triad of $\underline{U}$ as test triad; then $\underline{U} \underline{u}_i = \lambda_i \underline{u}_i$; then the scalars come out one slot at a time by (R3); then the common box product cancels:

$$\begin{aligned} 1 = \det \underline{F} &= \det(\underline{R} \underline{U}) = (\det \underline{R})(\det \underline{U}) = 1 \cdot \det \underline{U} \\ &= \frac{[\underline{U} \underline{u}_1, \underline{U} \underline{u}_2, \underline{U} \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \frac{[\lambda_1 \underline{u}_1, \lambda_2 \underline{u}_2, \lambda_3 \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} \\ &= \frac{\lambda_1 \lambda_2 \lambda_3 [\underline{u}_1, \underline{u}_2, \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \lambda_1 \lambda_2 \lambda_3. \end{aligned} \quad (3.57)$$

This is (3.17) again, applied to $\underline{U}$ instead of to a general symmetric tensor, and it says what it should: the volume ratio is the product of the three stretches, one per direction. With $\lambda_3 = 1$ it gives $\lambda_2 = \lambda_1^{-1}$: in simple shear the two in-plane stretches are reciprocal, so whatever is

stretched in one principal direction is compressed by the same factor in the other.

Now solve $\det(\mathcal{C} - \lambda^2 \hat{I}) = 0$. From (3.55), expanding along the third row,

$$\begin{aligned} 0 &= \begin{vmatrix} 1 - \lambda^2 & \gamma & 0 \\ \gamma & 1 + \gamma^2 - \lambda^2 & 0 \\ 0 & 0 & 1 - \lambda^2 \end{vmatrix} \\ &= (1 - \lambda^2) [(1 - \lambda^2)(1 + \gamma^2 - \lambda^2) - \gamma^2] \\ &= (1 - \lambda^2) [(1 - \lambda^2)^2 - \gamma^2 \lambda^2], \end{aligned} \quad (3.58)$$

the last step because $(1 - \lambda^2)(1 + \gamma^2 - \lambda^2) - \gamma^2 = (1 - \lambda^2)(1 - \lambda^2) + \gamma^2(1 - \lambda^2) - \gamma^2 = (1 - \lambda^2)^2 - \gamma^2 \lambda^2$. One root is $\lambda_3^2 = 1$ as noted. The remaining factor gives $1 - \lambda^2 = \pm \gamma \lambda$, i.e. $\lambda^2 \mp \gamma \lambda - 1 = 0$, whence $\lambda = \frac{1}{2}(\pm \gamma \pm \sqrt{\gamma^2 + 4})$; squaring the admissible (positive) roots,

$$4\lambda_{1,2}^2 = 4 + 2\gamma^2 \pm 2\gamma\sqrt{4 + \gamma^2}. \quad (3.59)$$

Identifying λ_1 with the $+$ root,

$$\lambda_1 = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2}), \quad \lambda_2 = \lambda_1^{-1}, \quad \lambda_3 = 1. \quad (3.60)$$

That $\lambda_2 = \lambda_1^{-1}$ is consistent with (3.57) can be checked directly:

$$\frac{1}{2}(\sqrt{4 + \gamma^2} - \gamma) \cdot \frac{1}{2}(\sqrt{4 + \gamma^2} + \gamma) = \frac{1}{4}(4 + \gamma^2 - \gamma^2) = 1.$$

The spectral representation of $\mathcal{C}$ is therefore

$$\mathcal{C} = \lambda_1^2 \underline{u}_1 \otimes \underline{u}_1 + \lambda_1^{-2} \underline{u}_2 \otimes \underline{u}_2 + \hat{E}_3 \otimes \hat{E}_3, \quad (3.61)$$

with $\underline{u}_1, \underline{u}_2$ orthonormal and perpendicular to $\hat{E}_3$:

$$\underline{u}_1 = \cos A \hat{E}_1 + \sin A \hat{E}_2, \quad \underline{u}_2 = -\sin A \hat{E}_1 + \cos A \hat{E}_2. \quad (3.62)$$

To find A , dot $\mathcal{C}\underline{u}_1 = \lambda_1^2 \underline{u}_1$ with $\hat{E}_1$ and move $\mathcal{C}$ across using its symmetry:

$$\underline{u}_1 \cdot \mathcal{C} \hat{E}_1 = \hat{E}_1 \cdot \mathcal{C} \underline{u}_1 = \lambda_1^2 \hat{E}_1 \cdot \underline{u}_1. \quad (3.63)$$

From (3.53), $\mathcal{C}\hat{E}_1 = \hat{E}_1 + \gamma\hat{E}_2$, so the left side is $\cos A + \gamma \sin A$, while the right side is $\lambda_1^2 \cos A$. Hence $\gamma \sin A = (\lambda_1^2 - 1) \cos A$ and

$$\tan A = \frac{1}{\gamma}(\lambda_1^2 - 1) = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2}) = \lambda_1, \quad (3.64)$$

the middle equality following from $\lambda_1^2 = \frac{1}{4}(\gamma + \sqrt{4 + \gamma^2})^2 = \frac{1}{4}(2\gamma^2 + 4 + 2\gamma\sqrt{4 + \gamma^2}) = \frac{1}{2}(\gamma^2 + 2 + \gamma\sqrt{4 + \gamma^2})$, so that $\lambda_1^2 - 1 = \frac{1}{2}\gamma(\gamma + \sqrt{4 + \gamma^2}) = \gamma\lambda_1$.

As $\gamma \rightarrow 0$, $\tan A \rightarrow 1$ and $A \rightarrow \pi/4$: for small shears the principal axes sit at approximately $\pm 45^\circ$ to the shearing direction. This is the familiar statement that a small shear is a small extension along one diagonal and an equal contraction along the other.

REMARK 3.32 (why the second scalar equation is redundant). Dotting $\mathcal{C}\underline{u}_1 = \lambda_1^2 \underline{u}_1$ with $\hat{E}_2$ instead of $\hat{E}_1$ produces no new information, and the reason is general. Suppose $\det \mathcal{A} = 0$, equivalently $\mathcal{A}\underline{u} = \underline{0}$ for some $\underline{u} \neq \underline{0}$. That statement is equivalent to $\underline{a} \cdot \mathcal{A}\underline{u} = 0$ for arbitrary $\underline{a}$, since a vector orthogonal to everything is $\underline{0}$. Expanding $\underline{a}$ on an orthonormal basis,

$$a_1 \underline{e}_1 \cdot \mathcal{A}\underline{u} + a_2 \underline{e}_2 \cdot \mathcal{A}\underline{u} + a_3 \underline{e}_3 \cdot \mathcal{A}\underline{u} = 0 \quad (3.65)$$

for arbitrary a_1, a_2, a_3 . Here $\mathcal{A} = \mathcal{C} - \lambda_1^2 \hat{I}$, whose determinant vanishes by construction, so the three scalar equations $\underline{e}_i \cdot \mathcal{A}\underline{u} = 0$ are linearly dependent; solving two of them forces the third. This is the reason an eigenvector is determined only up to scale and one uses normalisation to fix it.

The left tensor. From (3.54), $\mathcal{B}\underline{e}_3 = \underline{e}_3$, so

$$\mathcal{B} = \lambda_1^2 \underline{v}_1 \otimes \underline{v}_1 + \lambda_1^{-2} \underline{v}_2 \otimes \underline{v}_2 + \underline{e}_3 \otimes \underline{e}_3, \quad (3.66)$$

with

$$\underline{v}_1 = \cos \alpha \underline{e}_1 + \sin \alpha \underline{e}_2, \quad \underline{v}_2 = -\sin \alpha \underline{e}_1 + \cos \alpha \underline{e}_2 \quad (3.67)$$

for some angle α . Dot $\mathcal{B}\underline{v}_1 = \lambda_1^2 \underline{v}_1$ with $\underline{e}_2$. From (3.54), $\mathcal{B}\underline{v}_1 = \cos \alpha \mathcal{B}\underline{e}_1 + \sin \alpha \mathcal{B}\underline{e}_2$ with $\mathcal{B}\underline{e}_1 = (1 + \gamma^2)\underline{e}_1 + \gamma\underline{e}_2$ and $\mathcal{B}\underline{e}_2 = \gamma\underline{e}_1 + \underline{e}_2$;

therefore $\underline{e}_2 \cdot \underline{B} \underline{v}_1 = \gamma \cos \alpha + \sin \alpha$, while $\lambda_1^2 \underline{e}_2 \cdot \underline{v}_1 = \lambda_1^2 \sin \alpha$. Dividing by $\sin \alpha$,

$$1 + \gamma \cot \alpha = \lambda_1^2. \quad (3.68)$$

Hence $\cot \alpha = (\lambda_1^2 - 1)/\gamma = \lambda_1$ by the same identity used in (3.64), so

$$\tan \alpha = \frac{2}{\gamma + \sqrt{4 + \gamma^2}} = \cot A; \quad \text{hence} \quad A = \frac{\pi}{2} - \alpha. \quad (3.69)$$

Accordingly $\sin A = \cos \alpha$ and $\cos A = \sin \alpha$, and $\underline{v}_1, \underline{v}_2$ are determined.

The rotation. By (3.43), $\underline{R} = \sum_i \underline{v}_i \otimes \underline{u}_i$. Substituting (3.62) and (3.67) with $\cos A = \sin \alpha$, $\sin A = \cos \alpha$ and expanding both dyads term by term,

$$\begin{aligned} \underline{v}_1 \otimes \underline{u}_1 &= (\cos \alpha \underline{e}_1 + \sin \alpha \underline{e}_2) \otimes (\sin \alpha \hat{\underline{E}}_1 + \cos \alpha \hat{\underline{E}}_2) \\ &= \cos \alpha \sin \alpha \underline{e}_1 \otimes \hat{\underline{E}}_1 + \cos^2 \alpha \underline{e}_1 \otimes \hat{\underline{E}}_2 \\ &\quad + \sin^2 \alpha \underline{e}_2 \otimes \hat{\underline{E}}_1 + \sin \alpha \cos \alpha \underline{e}_2 \otimes \hat{\underline{E}}_2, \\ \underline{v}_2 \otimes \underline{u}_2 &= (-\sin \alpha \underline{e}_1 + \cos \alpha \underline{e}_2) \otimes (-\cos \alpha \hat{\underline{E}}_1 + \sin \alpha \hat{\underline{E}}_2) \\ &= \sin \alpha \cos \alpha \underline{e}_1 \otimes \hat{\underline{E}}_1 - \sin^2 \alpha \underline{e}_1 \otimes \hat{\underline{E}}_2 \\ &\quad - \cos^2 \alpha \underline{e}_2 \otimes \hat{\underline{E}}_1 + \cos \alpha \sin \alpha \underline{e}_2 \otimes \hat{\underline{E}}_2. \end{aligned}$$

Adding, and using $2 \sin \alpha \cos \alpha = \sin 2\alpha$ and $\cos^2 \alpha - \sin^2 \alpha = \cos 2\alpha$,

$$\underline{R} = \underline{e}_3 \otimes \hat{\underline{E}}_3 + \sin 2\alpha (\underline{e}_1 \otimes \hat{\underline{E}}_1 + \underline{e}_2 \otimes \hat{\underline{E}}_2) + \cos 2\alpha (\underline{e}_1 \otimes \hat{\underline{E}}_2 - \underline{e}_2 \otimes \hat{\underline{E}}_1). \quad (3.70)$$

As a check, $\underline{R}$ should carry the right-handed orthonormal triad $\{\hat{\underline{E}}_A\}$ to a right-handed orthonormal triad $\{\underline{w}_A\}$. Reading the columns of (3.70), that is picking out the terms whose second slot is $\hat{\underline{E}}_A$,

$$\begin{aligned} \underline{w}_1 &= \underline{R} \hat{\underline{E}}_1 = \sin 2\alpha \underline{e}_1 - \cos 2\alpha \underline{e}_2, \\ \underline{w}_2 &= \underline{R} \hat{\underline{E}}_2 = \cos 2\alpha \underline{e}_1 + \sin 2\alpha \underline{e}_2, \\ \underline{w}_3 &= \underline{R} \hat{\underline{E}}_3 = \underline{e}_3. \end{aligned} \quad (3.71)$$

Each has unit length, $\sin^2 2\alpha + \cos^2 2\alpha = 1$; they are mutually orthogonal, $\underline{w}_1 \cdot \underline{w}_2 = \sin 2\alpha \cos 2\alpha - \cos 2\alpha \sin 2\alpha = 0$; and $\underline{w}_1 \times \underline{w}_2 = (\sin^2 2\alpha + \cos^2 2\alpha) \underline{e}_1 \times \underline{e}_2 = \underline{e}_3 = \underline{w}_3$, so the triad is right handed.

REMARK 3.33 (the material turns backwards). The matrix of (3.70) in the plane is $\begin{pmatrix} \sin 2\alpha & \cos 2\alpha \\ -\cos 2\alpha & \sin 2\alpha \end{pmatrix}$, which is the rotation through the angle β with $\cos \beta = \sin 2\alpha$ and $\sin \beta = -\cos 2\alpha$, i.e. $\beta = 2\alpha - \pi/2$. As $\gamma \rightarrow 0$ we found $\alpha \rightarrow \pi/4$, so $\beta \rightarrow 0$ as it must. Expanding, $\tan \alpha = 2/(\gamma + \sqrt{4 + \gamma^2})$ gives $\alpha \approx \pi/4 - \gamma/4$, hence

$$\beta \approx -\frac{\gamma}{2}.$$

The polar rotation in simple shear is *clockwise* by half the amount of shear, even though the shear pushes the top of the block to the right. The check is independent: the skew part of $\underline{F} - \underline{I}$ is $\frac{\gamma}{2}(\underline{e}_1 \otimes \underline{e}_2 - \underline{e}_2 \otimes \underline{e}_1)$, which sends $\underline{e}_1 \mapsto -\frac{\gamma}{2}\underline{e}_2$, a clockwise turn of $\gamma/2$. Problem 9 below recovers the same β a third way, from $\tan \beta = -\gamma/2$ exactly.

3.5.2 Radial deformation of a right circular cylinder

What this example is for. Two new things. First, it is the first deformation in the chapter that is *not* homogeneous: $\underline{F}$ changes from point to point, so $\underline{C}$, $\underline{U}$, $\underline{R}$ and $\underline{J}$ are all fields and the shortcuts of §3.6 are unavailable. Second, it is written in curvilinear coordinates, in which the basis vectors themselves depend on position; handling that correctly is the entire content of (3.74) and is the skill the next two examples reuse without comment. The compensation is that the geometry does the diagonalisation for us, so no characteristic equation is ever solved: see Remark 3.35.

Let κ and κ_t be hollow right circular cylinders. Describe position in κ_t by the polar parametrisation (1.117),

$$\underline{x} = r \underline{e}_r(\theta) + z \underline{e}_3, \quad \underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2, \quad (3.72)$$

and position in κ by

$$\underline{X} = R \hat{\underline{E}}_R(\Theta) + Z \hat{\underline{E}}_3, \quad \hat{\underline{E}}_R(\Theta) = \cos \Theta \hat{\underline{E}}_1 + \sin \Theta \hat{\underline{E}}_2. \quad (3.73)$$

Suppose circles $R = \text{const}$ map to circles $r = \text{const}$ and that the azimuthal and axial coordinates are unchanged, $\theta = \Theta$ and $z = Z$; then $r = f(R)$ for some function f , and f determines the deformation completely.

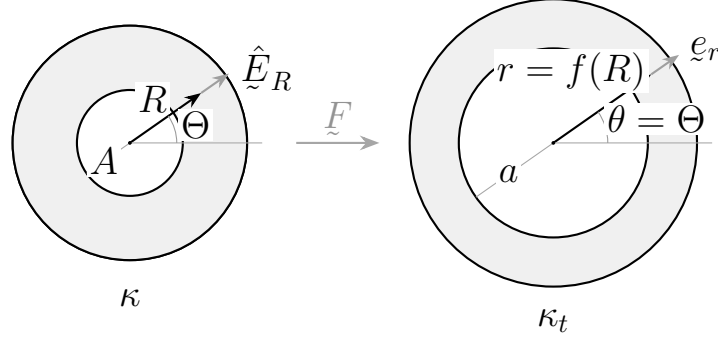


Figure 3.3: Annular cross-section of a right circular cylinder in κ and κ_t . Each material circle stays a circle and each radius stays a radius; only the radial coordinate changes, by $r = f(R)$. The inner radius A goes to $a = f(A)$.

What we are about to do, and why

The goal is $\underline{\underline{F}}$, because everything else in the chapter is built from it: $\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}}$ gives the stretches, $\underline{\underline{U}} = \sqrt{\underline{\underline{C}}}$ and $\underline{\underline{R}} = \underline{\underline{F}} \underline{\underline{U}}^{-1}$ split stretch from rotation, and $J = \det \underline{\underline{F}}$ gives the volume change. But $\underline{\underline{F}}$ is defined by $d\underline{\underline{x}} = \underline{\underline{F}} d\underline{\underline{X}}$, (2.34), so the only way to find it is to compute $d\underline{\underline{x}}$ and $d\underline{\underline{X}}$ and compare. The obstacle is that the natural description here is in polar coordinates, in which the basis vectors themselves move, and the whole of (3.74) is the bookkeeping that handles that. There are four steps and it is worth naming them before doing them:

Step A. Differentiate the position $\underline{\underline{x}} = r \underline{\underline{e}}_r(\theta) + z \underline{\underline{e}}_3$, being careful that $\underline{\underline{e}}_r$ depends on θ .

Step B. Do the same for $\underline{\underline{X}}$, and invert the result to express the three referential coordinate increments $dR, d\Theta, dZ$ as dot products with $d\underline{\underline{X}}$. This is the step that puts $d\underline{\underline{X}}$ into evidence so that $\underline{\underline{F}}$ can be read off.

Step C. Use the deformation itself, $r = f(R)$, $\theta = \Theta$, $z = Z$, to replace $dr, d\theta, dz$ by $dR, d\Theta, dZ$.

Step D. Compare with $d\tilde{x} = \tilde{F}d\tilde{X}$ and strip off the arbitrary $d\tilde{X}$.

Step A. From (3.72), $\tilde{x} = r \underline{e}_r(\theta) + z \underline{e}_3$, in which r, θ, z all vary and $\underline{e}_3$ is a constant vector. Differentiating with the product rule,

$$d\tilde{x} = dr \underline{e}_r(\theta) + r d[\underline{e}_r(\theta)] + dz \underline{e}_3,$$

and $\underline{e}_r$ depends on position only through θ , so the chain rule gives $d[\underline{e}_r(\theta)] = \underline{e}'_r(\theta) d\theta = \underline{e}_\theta(\theta) d\theta$, by the definition $\underline{e}_\theta = \underline{e}'_r$ of Lemma 3.29. Hence

$$d\tilde{x} = dr \underline{e}_r(\theta) + r d\theta \underline{e}_\theta(\theta) + dz \underline{e}_3.$$

Note the factor r in the middle term: it is there because moving the angle by $d\theta$ moves the point by the *arc length* $r d\theta$, not by $d\theta$. Forgetting it is the commonest error in this computation.

Step B. Exactly the same differentiation applied to (3.73), $\tilde{X} = R \hat{\underline{E}}_R(\Theta) + Z \hat{\underline{E}}_3$, gives

$$d\tilde{X} = dR \hat{\underline{E}}_R(\Theta) + R d\Theta \hat{\underline{E}}_\Theta(\Theta) + dZ \hat{\underline{E}}_3.$$

Now invert this. The frame $\{\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta, \hat{\underline{E}}_3\}$ is orthonormal by Lemma 3.29, so dotting with each member in turn picks out one coefficient and kills the other two:

$$\begin{aligned} \hat{\underline{E}}_R \cdot d\tilde{X} &= dR (\hat{\underline{E}}_R \cdot \hat{\underline{E}}_R) + R d\Theta (\hat{\underline{E}}_R \cdot \hat{\underline{E}}_\Theta) + dZ (\hat{\underline{E}}_R \cdot \hat{\underline{E}}_3) \\ &= dR \cdot 1 + R d\Theta \cdot 0 + dZ \cdot 0 = dR, \end{aligned}$$

$$\hat{\underline{E}}_\Theta \cdot d\tilde{X} = dR \cdot 0 + R d\Theta \cdot 1 + dZ \cdot 0 = R d\Theta,$$

$$\hat{\underline{E}}_3 \cdot d\tilde{X} = dR \cdot 0 + R d\Theta \cdot 0 + dZ \cdot 1 = dZ.$$

These three are $(*_2)$ of Problem 25 of Chapter 1 in cylindrical form. The middle one is the one to watch: what the dot product returns is $R d\Theta$, an arc length, not $d\Theta$ itself. So whenever $d\Theta$ is needed alone one must divide by R , and that division is where the f/R in (3.75) comes from.

Step C. Differentiate the three defining relations of the deformation, $r = f(R), \theta = \Theta, z = Z$:

$$dr = f'(R) dR, \quad d\theta = d\Theta, \quad dz = dZ.$$

Substitute these into Step A, and use $r = f(R)$ in the coefficient of the middle term:

$$d\tilde{x} = f'(R) dR \underline{e}_r(\theta) + f(R) d\Theta \underline{e}_\theta(\theta) + dZ \underline{e}_3.$$

Now replace each of dR , $d\Theta$, dZ by its expression from Step B. The first and third go straight in. The second needs the division just flagged: $d\Theta = \frac{1}{R}(R d\Theta) = \frac{1}{R} \hat{E}_\Theta \cdot d\tilde{X}$, so the coefficient $f(R)$ becomes f/R . Assembling,

$$\begin{aligned} \underline{F} d\tilde{X} &= d\tilde{x} = dr \underline{e}_r(\theta) + r d\theta \underline{e}_\theta(\theta) + dz \underline{e}_3 \\ &= f'(R) dR \underline{e}_r(\theta) + f(R) d\Theta \underline{e}_\theta(\theta) + dZ \underline{e}_3 \\ &= f'(R) \underline{e}_r(\theta) [\hat{E}_R(\Theta) \cdot d\tilde{X}] + (f/R) \underline{e}_\theta(\theta) [\hat{E}_\Theta(\Theta) \cdot d\tilde{X}] \\ &\quad + \underline{e}_3 [\hat{E}_3 \cdot d\tilde{X}]. \end{aligned} \tag{3.74}$$

Step D. Every term on the last line of (3.74) has the shape $\underline{a}(\underline{b} \cdot d\tilde{X})$, which by rule (R1) read backwards is $(\underline{a} \otimes \underline{b}) d\tilde{X}$. Collecting the three,

$$\underline{F} d\tilde{X} = \left[f' \underline{e}_r \otimes \hat{E}_R + (f/R) \underline{e}_\theta \otimes \hat{E}_\Theta + \underline{e}_3 \otimes \hat{E}_3 \right] d\tilde{X}.$$

This holds for *every* $d\tilde{X}$. Two tensors agreeing on every vector are the same tensor: if $\underline{A} d\tilde{X} = \underline{B} d\tilde{X}$ for all $d\tilde{X}$ then $(\underline{A} - \underline{B}) d\tilde{X} = \underline{0}$ for all $d\tilde{X}$, so $\underline{A} - \underline{B} = \underline{0}$. Therefore the brackets may be equated, and using $\Theta = \theta$, which is the second of the defining relations and lets the two angle labels be written the same way,

$$\underline{F} = f'(R) \underline{e}_r(\theta) \otimes \hat{E}_R(\theta) + (f/R) \underline{e}_\theta(\theta) \otimes \hat{E}_\Theta(\theta) + \underline{e}_3 \otimes \hat{E}_3. \tag{3.75}$$

REMARK 3.34 (three warnings about (3.75)). *The two frames are not the same objects.* $\underline{e}_r(\theta)$ is spatial and $\hat{E}_R(\Theta)$ referential; they carry the same angle here only because this particular deformation happens to leave the azimuth alone. In §3.5.3 they will not, and the two arguments must then be kept apart.

$\underline{F}$ is not constant. The coefficients $f'(R)$ and f/R depend on R , and the frame vectors depend on Θ . So this deformation is *not* homogeneous in the sense of §3.6, and every quantity derived from $\underline{F}$ below is a field.

The middle coefficient is a ratio of two arc lengths. The azimuthal stretch is f/R , the deformed arc $f \, d\Theta$ divided by the reference arc $R \, d\Theta$. It equals 1 only where $f(R) = R$; a purely radial motion that leaves radii radial still stretches the material circumferentially, and that is what a pressurised tube feels.

REMARK 3.35 ((3.75) is already a polar decomposition). Look at (3.75) against (3.43), $\tilde{F} = \sum_i \lambda_i \tilde{v}_i \otimes \tilde{u}_i$. The referential triad $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ is orthonormal and so is the spatial triad $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$, and $\tilde{F}$ is written as a sum of three dyads pairing them. So the principal stretches and both principal frames can be read off without solving any characteristic equation. This is why the cylindrical examples are the ones every textbook uses: the geometry has done the diagonalisation in advance.

The Jacobian, computed in full

Rule (R4) needs the images of three independent referential vectors. Take the frame $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$, which is orthonormal, and apply (3.75) to each member. Each application uses (R1) three times, once per dyad, and the inner products are Kronecker deltas because the frame is orthonormal:

$$\begin{aligned} \tilde{F} \hat{E}_R &= f'(\hat{E}_R \cdot \hat{E}_R) \underline{e}_r + \frac{f}{R}(\hat{E}_\Theta \cdot \hat{E}_R) \underline{e}_\theta + (\hat{E}_3 \cdot \hat{E}_R) \underline{e}_3 \\ &= f' \cdot 1 \cdot \underline{e}_r + \frac{f}{R} \cdot 0 \cdot \underline{e}_\theta + 0 \cdot \underline{e}_3 = f'(R) \underline{e}_r(\theta), \\ \tilde{F} \hat{E}_\Theta &= f'(\hat{E}_R \cdot \hat{E}_\Theta) \underline{e}_r + \frac{f}{R}(\hat{E}_\Theta \cdot \hat{E}_\Theta) \underline{e}_\theta + (\hat{E}_3 \cdot \hat{E}_\Theta) \underline{e}_3 \\ &= f' \cdot 0 \cdot \underline{e}_r + \frac{f}{R} \cdot 1 \cdot \underline{e}_\theta + 0 \cdot \underline{e}_3 = \frac{f}{R} \underline{e}_\theta(\theta), \\ \tilde{F} \hat{E}_3 &= f' \cdot 0 \cdot \underline{e}_r + \frac{f}{R} \cdot 0 \cdot \underline{e}_\theta + 1 \cdot \underline{e}_3 = \underline{e}_3. \end{aligned}$$

So the three referential frame vectors go to the three spatial frame vectors, each merely rescaled: radially by f' , azimuthally by f/R , axially by 1. Nothing is tilted, which is the content of Remark 3.35.

Both box products in (R4) are now box products of orthonormal right-handed triads, hence equal to 1 by Lemma 3.28 and Lemma 3.29:

$$[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3] = 1, \quad [\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = 1.$$

Assuming κ occupiable, so that $J > 0$ by (2.50), and extracting the three scalars one slot at a time by the linearity of the box product (R3),

$$\begin{aligned}
0 < J = \det \tilde{F} &= \frac{[\tilde{F}\hat{E}_R, \tilde{F}\hat{E}_\Theta, \tilde{F}\hat{E}_3]}{[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3]} = \frac{[f'e_r, \frac{f}{R}e_\theta, e_3]}{1} \\
&= f' [e_r, \frac{f}{R}e_\theta, e_3] = f' \cdot \frac{f}{R} [e_r, e_\theta, e_3] \\
&= f' \cdot \frac{f}{R} \cdot 1 = \frac{ff'}{R}.
\end{aligned} \tag{3.76}$$

The middle two lines are the point of “linear in each slot separately”: the factor f' is pulled out of the first slot with the other two left alone, then f/R out of the second, and only then is the surviving box product of pure unit vectors evaluated.

REMARK 3.36 (reading (3.76) without any algebra). The answer can be checked against the picture. Take the small curvilinear brick between radii R and $R + dR$, azimuths Θ and $\Theta + d\Theta$, heights Z and $Z + dZ$. Its volume is

$$dV = dR \cdot R d\Theta \cdot dZ,$$

the middle factor being an arc length, radius times angle. After the deformation the radial edge has length $dr = f'dR$, the azimuthal edge has length $r d\theta = f d\Theta$, and the axial edge is unchanged, so

$$dv = f'dR \cdot f d\Theta \cdot dZ.$$

Dividing,

$$J = \frac{dv}{dV} = \frac{f'dR \cdot f d\Theta \cdot dZ}{dR \cdot R d\Theta \cdot dZ} = \frac{ff'}{R},$$

which is (3.76) again. The R in the denominator is not an accident of the algebra: it is there because the reference arc length is $R d\Theta$ while the deformed one is $f d\Theta$, so the azimuthal stretch is f/R , a ratio of two arc lengths at two different radii.

Since $f > 0$ and $R > 0$, $J > 0$ implies $f'(R) > 0$: f is strictly increasing, so $R_2 > R_1$ gives $f(R_2) > f(R_1)$. Concentric circles go to

concentric circles with the ordering preserved, outer to outer and inner to inner.

If the deformation is isochoric then $J = 1$, i.e. $ff' = R$, i.e. $(\frac{1}{2}f^2)' = R$. Integrating and writing $f(R) = r$,

$$r^2 - a^2 = R^2 - A^2, \quad a = f(A), \quad (3.77)$$

A being a positive constant, since evaluating at $R = A$ gives $a^2 - a^2 = A^2 - A^2$, which fixes the constant of integration.

The book calls the area statement “an immediate consequence”. Written out: (3.77) says $r^2 - a^2 = R^2 - A^2$, and multiplying by π ,

$$\underbrace{\pi r^2 - \pi a^2}_{\text{area of the deformed annulus}} = \underbrace{\pi R^2 - \pi A^2}_{\text{area of the reference annulus}}.$$

So the annulus $[A, R]$ has exactly the area of its image $[a, r]$. That this had to happen follows from $J = 1$ and $z = Z$: a slab of the cylinder of axial thickness ΔZ over the annulus has volume (area) $\times \Delta Z$ before and (area) $\times \Delta Z$ after, since the thickness is unchanged; equality of the volumes therefore forces equality of the areas.

The transpose of (3.75), by Problem 17(a), $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$ applied to each of the three dyads, is

$$\underline{F}^t = f'(R) \hat{\underline{E}}_R(\theta) \otimes \underline{e}_r(\theta) + (f/R) \hat{\underline{E}}_\Theta(\theta) \otimes \underline{e}_\theta(\theta) + \hat{\underline{E}}_3 \otimes \underline{e}_3. \quad (3.78)$$

Now form $\underline{C} = \underline{F}^t \underline{F}$. The product of the two three-term sums has nine terms, each of the shape $(\hat{\underline{E}}_A \otimes \underline{e}_i)(\underline{e}_j \otimes \hat{\underline{E}}_B) = (\underline{e}_i \cdot \underline{e}_j) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B$ by the dyad rule $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c}) \underline{a} \otimes \underline{d}$. Writing $\lambda_1 = f'$, $\lambda_2 = f/R$, $\lambda_3 = 1$ and $(\hat{\underline{E}}_A) = (\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta, \hat{\underline{E}}_3)$, $(\underline{e}_i) = (\underline{e}_r, \underline{e}_\theta, \underline{e}_3)$, the nine terms are $\lambda_A \lambda_B (\underline{e}_A \cdot \underline{e}_B) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B$, and the inner product is δ_{AB} : the six off-diagonal terms vanish, for instance

$$(f' \hat{\underline{E}}_R \otimes \underline{e}_r)((f/R) \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta) = f'(f/R) (\underline{e}_r \cdot \underline{e}_\theta) \hat{\underline{E}}_R \otimes \hat{\underline{E}}_\Theta = 0,$$

and the three diagonal ones survive with their coefficients squared, for instance

$$(f' \hat{\underline{E}}_R \otimes \underline{e}_r)(f' \underline{e}_r \otimes \hat{\underline{E}}_R) = (f')^2 (\underline{e}_r \cdot \underline{e}_r) \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R = (f')^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R.$$

Multiplying in the other order for $\underline{B} = \underline{F}\underline{F}^t$ produces terms $(\underline{e}_i \otimes \hat{\underline{E}}_A)(\hat{\underline{E}}_B \otimes \underline{e}_j) = (\hat{\underline{E}}_A \cdot \hat{\underline{E}}_B) \underline{e}_i \otimes \underline{e}_j$, and now it is the *referential* triad whose orthonormality kills the cross terms. What survives is

$$\begin{aligned}\underline{C} &= (f')^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + (f/R)^2 \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3, \\ \underline{B} &= (f')^2 \underline{e}_r \otimes \underline{e}_r + (f/R)^2 \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3.\end{aligned}\tag{3.79}$$

These are spectral representations, in the sense of (3.12): each is a sum of three dyads built on an orthonormal triad, so the coefficients are the principal values and the triad members are the principal vectors. The principal stretches are therefore

$$\lambda_1 = f'(R), \quad \lambda_2 = f/R, \quad \lambda_3 = 1,$$

all three positive, since $f > 0$ and $R > 0$ give $\lambda_2 > 0$ and (3.76) gave $f' > 0$. The book says the stretch tensors “follow immediately”; the reason is Theorem 3.16. Since $\underline{C}$ is symmetric and positive definite with the spectral form above, the *unique* symmetric positive definite square root is obtained by replacing each principal value by its positive square root and leaving the axes untouched, which is (3.28):

$$\begin{aligned}\underline{U} &= f' \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + (f/R) \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3, \\ \underline{V} &= f' \underline{e}_r \otimes \underline{e}_r + (f/R) \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3.\end{aligned}\tag{3.80}$$

Finally the rotation. The inverse of a tensor in spectral form is obtained by inverting the principal values, Remark 3.10, so

$$\underline{U}^{-1} = \frac{1}{f'} \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + \frac{R}{f} \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,$$

which one verifies at once: $\underline{U}\underline{U}^{-1}$ has nine terms, the six off-diagonal ones vanishing on orthonormality and the three diagonal ones giving $f'(1/f')\hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + (f/R)(R/f)\hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3 = \hat{\underline{I}}$. Multiplying (3.75) on the right by it, again nine terms of which six vanish,

$$\begin{aligned}\underline{F}\underline{U}^{-1} &= f' \cdot \frac{1}{f'} (\hat{\underline{E}}_R \cdot \hat{\underline{E}}_R) \underline{e}_r \otimes \hat{\underline{E}}_R + \frac{f}{R} \cdot \frac{R}{f} (\hat{\underline{E}}_\Theta \cdot \hat{\underline{E}}_\Theta) \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta \\ &\quad + 1 \cdot 1 \cdot \underline{e}_3 \otimes \hat{\underline{E}}_3 \\ &= \underline{e}_r \otimes \hat{\underline{E}}_R + \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta + \underline{e}_3 \otimes \hat{\underline{E}}_3.\end{aligned}$$

The left factorisation gives the same thing: with $\underline{V}^{-1} = (1/f')\underline{e}_r \otimes \underline{e}_r + (R/f)\underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3$, multiplying (3.75) on the *left* produces the identical three surviving terms, now with the spatial triad supplying the Kronecker deltas. Hence

$$\underline{R} = \underline{F}\underline{U}^{-1} = \underline{V}^{-1}\underline{F} = \underline{e}_r(\theta) \otimes \hat{\underline{E}}_R(\theta) + \underline{e}_\theta(\theta) \otimes \hat{\underline{E}}_\Theta(\theta) + \underline{e}_3 \otimes \hat{\underline{E}}_3. \quad (3.81)$$

By Lemma 3.39 of §3.5.3, which applies verbatim, this $\underline{R}$ is a two-point rotation; indeed it is exactly the tensor called $\underline{Q}$ there.

Why bother interpreting $\underline{R}$ at all? Because (3.81) pairs a spatial vector with a referential one in each dyad, and such an object cannot be compared with $\underline{I}$: they live in different spaces. To say the sentence “there is no rotation here” one must first move everything into one space, and the shifter of (2.68) is the only tool for that. This is exactly the manoeuvre of §2.6, and (3.82)–(3.83) are its payoff.

So introduce the shifted vectors

$$\underline{E}_R(\Theta) = \underline{1}\hat{\underline{E}}_R(\Theta), \quad \underline{E}_\Theta(\Theta) = \underline{1}\hat{\underline{E}}_\Theta(\Theta), \quad \underline{E}_3 = \underline{1}\hat{\underline{E}}_3,$$

all three now in $\mathbb{E}^3$. Each satisfies $\underline{1}^t \underline{E}_R = \underline{1}^t \underline{1} \hat{\underline{E}}_R = \hat{\underline{I}} \hat{\underline{E}}_R = \hat{\underline{E}}_R$ by (2.72), and likewise for the other two. Problem 17(c) of Chapter 1 says $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$: post-multiplying a dyad by a tensor acts on the *second* slot, through the transpose. Apply it to each of the three dyads with $\underline{A} = \underline{1}$, so that $\underline{A}^t = \underline{1}^t$:

$$\begin{aligned} (\underline{e}_r \otimes \underline{E}_R) \underline{1} &= \underline{e}_r \otimes (\underline{1}^t \underline{E}_R) = \underline{e}_r \otimes \hat{\underline{E}}_R, \\ (\underline{e}_\theta \otimes \underline{E}_\Theta) \underline{1} &= \underline{e}_\theta \otimes (\underline{1}^t \underline{E}_\Theta) = \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta, \\ (\underline{e}_3 \otimes \underline{E}_3) \underline{1} &= \underline{e}_3 \otimes (\underline{1}^t \underline{E}_3) = \underline{e}_3 \otimes \hat{\underline{E}}_3. \end{aligned}$$

Adding the three lines reproduces (3.81) exactly. Reading the addition backwards, that is factoring the common $\underline{1}$ out on the right,

$$\underline{R} = [\underline{e}_r(\theta) \otimes \underline{E}_R(\theta) + \underline{e}_\theta(\theta) \otimes \underline{E}_\Theta(\theta) + \underline{e}_3 \otimes \underline{E}_3] \underline{1}. \quad (3.82)$$

The choices $\underline{E}_R(\theta) = \underline{e}_r(\theta)$, $\underline{E}_\Theta(\theta) = \underline{e}_\theta(\theta)$, $\underline{E}_3 = \underline{e}_3$ reduce the bracketed expression to $\underline{e}_r \otimes \underline{e}_r + \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3 = \underline{I}$, the spatial identity, by (3.11) applied to the orthonormal spatial triad. Therefore

$$\underline{R} = \underline{I} \underline{1} = \underline{1}. \quad (3.83)$$

REMARK 3.37 (what (3.83) says). There is no rotation in this deformation. Every material point moves radially outward or inward, and the local principal axes are the radial, azimuthal and axial directions both before and after. The apparatus of shifters is needed here only to say the sentence: in the one-space reading of Chapter 2 the statement is simply $\underline{R} = \underline{I}$. Note also that when $f' = f/R$, i.e. $f = cR$, two principal stretches coincide, every direction in the $(\underline{e}_r, \underline{e}_\theta)$ plane is principal, and $\underline{U} = c(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta) + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3$: this is the repeated-root case anticipated in Remark 3.8, and the theory handles it without modification.

3.5.3 Azimuthal shear

What this example is for. §3.5.2 stretched the cylinder without distorting it; this one distorts it without stretching it. It is the first deformation in which a rotation is genuinely present and has to be extracted, and the first in which the amount of shear varies from point to point. The pay-off is Remark 3.45: once this is done, the next example, torsion, is the same algebra with two letters changed.

Now the radial and axial coordinates are unchanged and the azimuth is shifted by an amount depending on R :

$$r = R, \quad \theta = \Theta + \phi(R), \quad z = Z. \quad (3.84)$$

Differentiating these three relations, and remembering that ϕ depends on R alone,

$$dr = dR, \quad d\theta = d\Theta + \phi'(R) dR, \quad dz = dZ.$$

Substitute into the polar line element $d\tilde{x} = dr \underline{e}_r(\theta) + r d\theta \underline{e}_\theta(\theta) + dz \underline{e}_3$, using $r = R$ in the middle term:

$$\begin{aligned} d\tilde{x} &= dR \underline{e}_r(\theta) + R[d\Theta + \phi'(R)dR] \underline{e}_\theta(\theta) + dZ \underline{e}_3 \\ &= dR \underline{e}_r(\theta) + R\phi'(R) dR \underline{e}_\theta(\theta) + R d\Theta \underline{e}_\theta(\theta) + dZ \underline{e}_3, \end{aligned}$$

and collect the two terms carrying dR :

$$\underline{F}d\tilde{X} = [\underline{e}_r(\theta) + R\phi'(R)\underline{e}_\theta(\theta)]dR + R \underline{e}_\theta(\theta) d\Theta + \underline{e}_3 dZ. \quad (3.85)$$

REMARK 3.38 (where the two terms in the bracket come from). This bracket is the whole of the deformation, and it is worth saying in words what each term is. Move a material point outward by dR at fixed Θ and Z . Two things happen at once. Its radius grows by $dr = dR$, which is the $\underline{e}_r$ term and is no deformation at all, merely the fact that the point has moved outward. But it also lands on a cylinder that has been twisted through a *different* angle, since ϕ depends on R ; the extra azimuthal displacement is $r \Delta\theta = R\phi'(R)dR$, which is the $\underline{e}_\theta$ term. So the coefficient $\gamma = R\phi'$ is not ϕ' : the factor R is there because an angle becomes a length only after multiplication by the radius. That single factor is what makes γ vary with R even when ϕ' is constant.

Substituting $dR = \hat{E}_R \cdot d\tilde{X}$, $Rd\Theta = \hat{E}_\Theta \cdot d\tilde{X}$ and $dZ = \hat{E}_3 \cdot d\tilde{X}$, which is the referential line element of (3.74) read backwards, (3.85) becomes

$$\begin{aligned} \underline{F} d\tilde{X} &= [\underline{e}_r + R\phi' \underline{e}_\theta] (\hat{E}_R \cdot d\tilde{X}) + \underline{e}_\theta (\hat{E}_\Theta \cdot d\tilde{X}) + \underline{e}_3 (\hat{E}_3 \cdot d\tilde{X}) \\ &= \left[\underline{e}_r \otimes \hat{E}_R + \underline{e}_\theta \otimes \hat{E}_\Theta + \underline{e}_3 \otimes \hat{E}_3 + R\phi' \underline{e}_\theta \otimes \hat{E}_R \right] d\tilde{X}, \end{aligned}$$

the R in $Rd\Theta$ having cancelled the R in the coefficient $R\underline{e}_\theta$ of $d\Theta$. Since $d\tilde{X}$ is arbitrary,

$$\underline{F} = \underline{Q} + R\phi'(R) \underline{e}_\theta(\theta) \otimes \hat{E}_R(\Theta), \quad (3.86)$$

where

$$\underline{Q} = \underline{e}_r(\theta) \otimes \hat{E}_R(\Theta) + \underline{e}_\theta(\theta) \otimes \hat{E}_\Theta(\Theta) + \underline{e}_3 \otimes \hat{E}_3. \quad (3.87)$$

LEMMA 3.39 ($\underline{Q}$ of (3.87) is a two-point rotation). $\underline{Q}^t \underline{Q} = \underline{I}$, $\underline{Q} \underline{Q}^t = \underline{I}$, $\det \underline{Q} = 1$, and $\underline{Q} \hat{E}_R = \underline{e}_r$, $\underline{Q} \hat{E}_\Theta = \underline{e}_\theta$, $\underline{Q} \hat{E}_3 = \underline{e}_3$.

Proof. The action on the basis is immediate from $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$ and orthonormality of $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$; for instance

$$\underline{Q} \hat{E}_R = (\hat{E}_R \cdot \hat{E}_R) \underline{e}_r + (\hat{E}_\Theta \cdot \hat{E}_R) \underline{e}_\theta + (\hat{E}_3 \cdot \hat{E}_R) \underline{e}_3 = \underline{e}_r + 0 + 0 = \underline{e}_r.$$

By Problem 17(a), $\underline{Q}^t = \hat{E}_R \otimes \underline{e}_r + \hat{E}_\Theta \otimes \underline{e}_\theta + \hat{E}_3 \otimes \underline{e}_3$. Multiplying out $\underline{Q}^t \underline{Q}$ produces nine terms, each of the form $(\hat{E}_A \otimes \underline{e}_i)(\underline{e}_j \otimes \hat{E}_B) =$

$(\underline{e}_i \cdot \underline{e}_j) \hat{E}_A \otimes \hat{E}_B$ by the dyad product rule; the six with $i \neq j$ vanish because the spatial triad is orthonormal, and the three with $i = j$ survive with coefficient 1:

$$\underset{\sim}{Q}^t \underset{\sim}{Q} = \hat{E}_R \otimes \hat{E}_R + \hat{E}_\Theta \otimes \hat{E}_\Theta + \hat{E}_3 \otimes \hat{E}_3 = \hat{I}$$

by (3.11). The same computation in the other order gives $\underset{\sim}{Q} \underset{\sim}{Q}^t = \underline{e}_r \otimes \underline{e}_r + \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3 = \underline{I}$. Finally the determinant. Use rule (R4) with the referential frame as test triad; its images were just computed. The denominator is $[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3] = 1$ and the numerator is $[\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = 1$, both by Lemma 3.28 together with Lemma 3.29, which established that each of the two frames is orthonormal and right handed. Hence

$$\det \underset{\sim}{Q} = \frac{[\underset{\sim}{Q} \hat{E}_R, \underset{\sim}{Q} \hat{E}_\Theta, \underset{\sim}{Q} \hat{E}_3]}{[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3]} = \frac{[\underline{e}_r, \underline{e}_\theta, \underline{e}_3]}{1} = \frac{1}{1} = 1.$$

No scalars had to be extracted here, because $\underset{\sim}{Q}$ maps unit vectors to unit vectors: it is pure rotation with no stretching, and the determinant of a rotation is 1 precisely because a rotation moves a unit cube to a unit cube. ■

Since $\underline{e}_\theta(\theta) = \underset{\sim}{Q} \hat{E}_\Theta(\Theta)$, factoring $\underset{\sim}{Q}$ out on the left of (3.86) with Problem 17(b), $\underset{\sim}{A}(\underset{\sim}{a} \otimes \underset{\sim}{b}) = (\underset{\sim}{A}\underset{\sim}{a}) \otimes \underset{\sim}{b}$,

$$\underset{\sim}{F} = \underset{\sim}{Q} \hat{G}, \quad \hat{G} = \hat{I} + R\phi'(R) \hat{E}_\Theta(\Theta) \otimes \hat{E}_R(\Theta), \quad (3.88)$$

which is checked forwards: $\underset{\sim}{Q} \hat{G} = \underset{\sim}{Q} \hat{I} + R\phi' \underset{\sim}{Q}(\hat{E}_\Theta \otimes \hat{E}_R) = \underset{\sim}{Q} + R\phi'(\underset{\sim}{Q} \hat{E}_\Theta) \otimes \hat{E}_R = \underset{\sim}{Q} + R\phi' \underline{e}_\theta \otimes \hat{E}_R$. Since $\hat{E}_R(\Theta) = \underset{\sim}{Q}^t \underline{e}_r(\theta)$, factoring on the right with Problem 17(c), $(\underset{\sim}{a} \otimes \underset{\sim}{b}) \underset{\sim}{A} = \underset{\sim}{a} \otimes (\underset{\sim}{A}^t \underset{\sim}{b})$,

$$\underset{\sim}{F} = \underset{\sim}{G} \underset{\sim}{Q}, \quad \underset{\sim}{G} = \underline{I} + R\phi'(R) \underline{e}_\theta(\theta) \otimes \underline{e}_r(\theta), \quad (3.89)$$

checked the same way: $\underset{\sim}{G} \underset{\sim}{Q} = \underset{\sim}{Q} + R\phi'(\underline{e}_\theta \otimes \underline{e}_r) \underset{\sim}{Q} = \underset{\sim}{Q} + R\phi' \underline{e}_\theta \otimes (\underset{\sim}{Q}^t \underline{e}_r) = \underset{\sim}{Q} + R\phi' \underline{e}_\theta \otimes \hat{E}_R$.

The deformation is isochoric: four independent proofs

The book disposes of this in one line, “it is straightforward to show, using either representation”. Here is the whole of it, four ways, because each uses a different piece of Chapter 1 and the agreement is a check on all four. Write $\gamma = R\phi'(R)$ throughout.

(i) *From (3.88).* By the multiplicativity established at (3.34)–(3.35), valid here because the first factor is two-point and the second referential,

$$\det \underline{\underline{F}} = (\det \underline{\underline{Q}})(\det \hat{\underline{\underline{G}}}) = 1 \cdot \det \hat{\underline{\underline{G}}}.$$

By Problem 21(f) of Chapter 1, $\det(\hat{\underline{\underline{I}}} + \underline{\underline{a}} \otimes \underline{\underline{b}}) = 1 + \underline{\underline{a}} \cdot \underline{\underline{b}}$, so

$$\det \hat{\underline{\underline{G}}} = 1 + \gamma \hat{\underline{\underline{E}}}_\Theta \cdot \hat{\underline{\underline{E}}}_R = 1 + \gamma \cdot 0 = 1,$$

the inner product vanishing because $\hat{\underline{\underline{E}}}_\Theta \perp \hat{\underline{\underline{E}}}_R$. Hence $\det \underline{\underline{F}} = 1$. Note where the hypothesis enters: it is the *orthogonality of the shear pair* that kills the determinant, not the smallness of γ . A shear of any amount whatever preserves volume.

(ii) *From (3.89).* Symmetrically, $\det \underline{\underline{F}} = (\det \underline{\underline{G}})(\det \underline{\underline{Q}})$ with $\det \underline{\underline{G}} = 1 + \gamma \underline{\underline{e}}_\theta \cdot \underline{\underline{e}}_r = 1 + 0 = 1$, so again $\det \underline{\underline{F}} = 1$.

(iii) *Directly from (3.86), without factoring at all.* Problem 21(g) of Chapter 1 gives $\det(\underline{\underline{A}} + \underline{\underline{a}} \otimes \underline{\underline{b}}) = (\det \underline{\underline{A}})(1 + \underline{\underline{a}} \cdot \underline{\underline{A}}^{-t}\underline{\underline{b}})$ for invertible $\underline{\underline{A}}$. Take $\underline{\underline{A}} = \underline{\underline{Q}}$, $\underline{\underline{a}} = \gamma \underline{\underline{e}}_\theta$, $\underline{\underline{b}} = \hat{\underline{\underline{E}}}_R$. By Lemma 3.39, $\underline{\underline{Q}}^{-1} = \underline{\underline{Q}}^t$, hence $\underline{\underline{Q}}^{-t} = (\underline{\underline{Q}}^{-1})^t = (\underline{\underline{Q}}^t)^t = \underline{\underline{Q}}$, so

$$\det \underline{\underline{F}} = (\det \underline{\underline{Q}})(1 + \gamma \underline{\underline{e}}_\theta \cdot \underline{\underline{Q}} \hat{\underline{\underline{E}}}_R) = 1 \cdot (1 + \gamma \underline{\underline{e}}_\theta \cdot \underline{\underline{e}}_r) = 1.$$

(iv) *From $\underline{\underline{C}}$, with no reference to $\underline{\underline{F}}$ at all.* By Problem 21(b), $\det \underline{\underline{C}} = \det(\underline{\underline{F}}^t \underline{\underline{F}}) = (\det \underline{\underline{F}}^t)(\det \underline{\underline{F}}) = (\det \underline{\underline{F}})^2$. From (3.92) below, the matrix of $\underline{\underline{C}}$ in the ordered orthonormal basis $(\hat{\underline{\underline{E}}}_R, \hat{\underline{\underline{E}}}_\Theta, \hat{\underline{\underline{E}}}_3)$ is

$$(C_{AB}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

whose determinant, expanded along the third row, is $1 \cdot [(1 + \gamma^2)(1) - \gamma \cdot \gamma] = 1 + \gamma^2 - \gamma^2 = 1$. Hence $(\det \tilde{F})^2 = 1$ and $\det \tilde{F} = \pm 1$; the positive sign is selected by the occupiability of κ , (2.50).

All four give $\det \tilde{F} = 1$: by (2.50) no material volume changes, and by (2.59) the densities coincide, $\rho_\kappa = \rho$.

The image of a radial line

Take a radial material curve in κ , that is the set $\Theta = \Theta_0$, $Z = Z_0$ with R varying, whose unit tangent is $\tilde{M} = \hat{\tilde{E}}_R(\Theta_0)$. Its stretch and current tangent follow from (2.67) and (3.86), using $\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_R = 1$ and $\tilde{Q} \hat{\tilde{E}}_R = \underline{e}_r$:

$$\mu \tilde{m} = \tilde{F} \hat{\tilde{E}}_R(\Theta) = \underline{e}_r(\theta) + R\phi'(R)\underline{e}_\theta(\theta); \quad \mu = \sqrt{1 + (R\phi')^2}, \quad (3.90)$$

the length following from $|\underline{e}_r + \gamma \underline{e}_\theta|^2 = \underline{e}_r \cdot \underline{e}_r + 2\gamma \underline{e}_r \cdot \underline{e}_\theta + \gamma^2 \underline{e}_\theta \cdot \underline{e}_\theta = 1 + \gamma^2$.

REMARK 3.40 (a plane spiral, not a helix). The book calls the image a *spiral* and leaves it there. It is worth being explicit, because the next section produces a curve with a different name and the two are easy to confuse.

Parametrise the radial material line by R and push it forward through (3.84):

$$\tilde{x}(R) = R \underline{e}_r(\Theta_0 + \phi(R)) + Z_0 \underline{e}_3.$$

The axial coordinate is the constant Z_0 , so the image lies entirely in the single plane $z = Z_0$: it is a *plane* curve. Its polar equation is obtained by eliminating R through $r = R$:

$$\theta = \Theta_0 + \phi(r),$$

a curve winding around the origin with radius increasing monotonically, which is precisely what the word spiral means. Two familiar cases: $\phi(R) = k(R - A)$ gives $r = A + (\theta - \Theta_0)/k$, an Archimedean spiral; $\phi(R) = c \ln(R/A)$ gives $r = A e^{(\theta - \Theta_0)/c}$, a logarithmic spiral.

The angle ψ that the deformed curve makes with the radial direction is read off (3.90): the tangent is proportional to $\underline{e}_r + \gamma \underline{e}_\theta$, so

$$\tan \psi = \frac{\gamma |\underline{e}_\theta|}{|\underline{e}_r|} = \gamma = R\phi'(R).$$

So γ is the tangent of the angle by which the shear tilts a radius away from being radial, exactly as in (3.49) it was the tangent of the angle by which the shear tilts a vertical line. The logarithmic spiral is the case $\gamma = c$ constant, the only azimuthal shear in which the tilt is the same at every radius.

REMARK 3.41 (three curves that get confused: spiral, cylindrical helix, spherical helix). The chapter produces two curves and names them differently, and a third name is often attached to neither correctly. They are genuinely different objects.

Plane spiral (§3.5.3, above). Lies in one plane. Radius varies, axial coordinate constant. Curvature varies along it; it is not a curve of constant curvature.

Cylindrical, or circular, helix (§3.5.4, (3.98)). Lies on a cylinder $r = \text{const}$. Radius constant, axial coordinate varies linearly with the azimuth. It has *constant* curvature and *constant* torsion, and it is the only curve in three-space with both constant, which is why it is the standard example. Remark 3.44 computes both.

Spherical helix. A curve lying on a *sphere* and crossing all the meridians at one fixed angle, the loxodrome or rhumb line. Neither $\tilde{F}$ of this chapter produces one, and nothing in §3.5 is a spherical helix. The word *spherical* occurs in Chapter 3 only in §3.6.1(c), and there it qualifies a *surface*, not a curve: a spherical material *surface* is carried to an ellipsoidal surface, (3.112). A sphere going to an ellipsoid and a curve winding on a sphere are unrelated statements, and the only thing they share is the adjective. The discriminating question is always the same: *on what surface does the curve lie, and which of r, θ, z is held fixed?* Plane spiral: z fixed. Cylindrical helix: r fixed. Spherical helix: $|\underline{x}|$ fixed. Problem 5(b) below asks for the stretch of a curve with $\underline{M} = \frac{\sqrt{2}}{2}(\hat{E}_\Theta + \hat{E}_3)$, which has R fixed and is therefore a cylindrical helix, not a spiral and not a spherical anything.

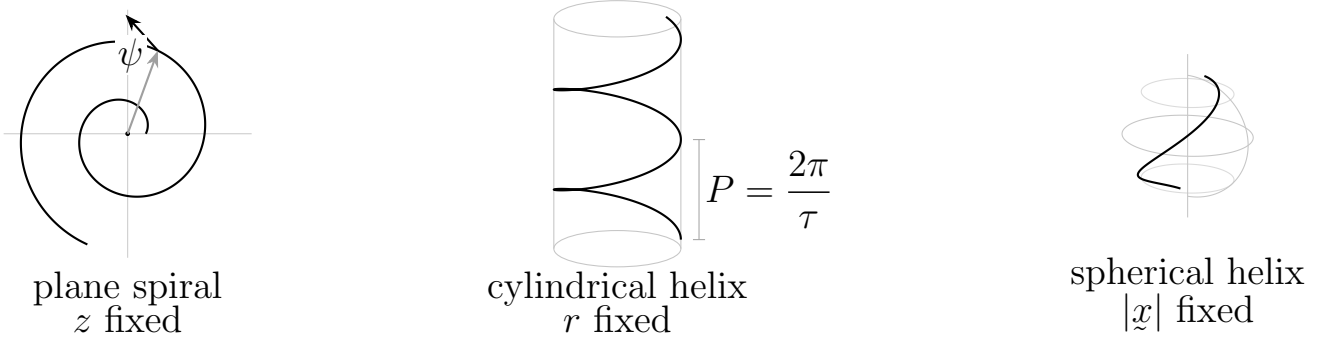


Figure C: The three curves of Remark 3.41, distinguished by which coordinate is held fixed. Left, the image of a radius under azimuthal shear: a plane curve, z constant, meeting every radius at the angle ψ with $\tan \psi = \gamma$. Centre, the image of a generator under torsion: a curve on the cylinder $r = R_0$, of constant curvature and torsion, rising by the pitch $P = 2\pi/\tau$ per turn, the same pitch on every coaxial cylinder. Right, a spherical helix, or loxodrome: a curve on a sphere crossing every meridian at one fixed angle. No deformation in this chapter produces one; it is drawn only so that the name is attached to the right object.

The Cauchy–Green tensors collapse because $\underline{\underline{Q}}$ is orthogonal:

$$\underline{\underline{C}} = \underline{\underline{\hat{G}}}^t \underline{\underline{Q}}^t \underline{\underline{Q}} \underline{\underline{\hat{G}}} = \underline{\underline{\hat{G}}}^t \underline{\underline{\hat{I}}} \underline{\underline{\hat{G}}} = \underline{\underline{\hat{G}}}^t \underline{\underline{\hat{G}}}, \quad \underline{\underline{B}} = \underline{\underline{G}} \underline{\underline{Q}} \underline{\underline{Q}}^t \underline{\underline{G}}^t = \underline{\underline{G}} \underline{\underline{I}} \underline{\underline{G}}^t = \underline{\underline{G}} \underline{\underline{G}}^t. \quad (3.91)$$

Writing $\gamma(R) = R\phi'(R)$ and expanding,

$$\begin{aligned} \underline{\underline{C}} &= (\underline{\underline{\hat{I}}} + \gamma \underline{\underline{\hat{E}}}_R \otimes \underline{\underline{\hat{E}}}_\Theta)(\underline{\underline{\hat{I}}} + \gamma \underline{\underline{\hat{E}}}_\Theta \otimes \underline{\underline{\hat{E}}}_R) \\ &= \underline{\underline{\hat{I}}} + \gamma [\underline{\underline{\hat{E}}}_R(\Theta) \otimes \underline{\underline{\hat{E}}}_\Theta(\Theta) + \underline{\underline{\hat{E}}}_\Theta(\Theta) \otimes \underline{\underline{\hat{E}}}_R(\Theta)] \\ &\quad + \gamma^2(R) \underline{\underline{\hat{E}}}_R(\Theta) \otimes \underline{\underline{\hat{E}}}_R(\Theta), \end{aligned} \quad (3.92)$$

the last term arising from $(\underline{\underline{\hat{E}}}_R \otimes \underline{\underline{\hat{E}}}_\Theta)(\underline{\underline{\hat{E}}}_\Theta \otimes \underline{\underline{\hat{E}}}_R) = (\underline{\underline{\hat{E}}}_\Theta \cdot \underline{\underline{\hat{E}}}_\Theta) \underline{\underline{\hat{E}}}_R \otimes \underline{\underline{\hat{E}}}_R = \underline{\underline{\hat{E}}}_R \otimes \underline{\underline{\hat{E}}}_R$, and likewise

$$\underline{\underline{B}} = \underline{\underline{I}} + \gamma(R) [\underline{\underline{e}}_r(\theta) \otimes \underline{\underline{e}}_\theta(\theta) + \underline{\underline{e}}_\theta(\theta) \otimes \underline{\underline{e}}_r(\theta)] + \gamma^2(R) \underline{\underline{e}}_\theta(\theta) \otimes \underline{\underline{e}}_\theta(\theta). \quad (3.93)$$

Why (3.92) and (3.53) are the same formula

The book states that “on replacing $\underline{\underline{\hat{E}}}_1$ by $\underline{\underline{\hat{E}}}_\Theta(\Theta)$, $\underline{\underline{\hat{E}}}_2$ by $\underline{\underline{\hat{E}}}_R(\Theta)$ and $\underline{\underline{\hat{E}}}_3$ by $-\underline{\underline{\hat{E}}}_3$, $\underline{\underline{C}}$ is seen to have exactly the same form as its counterpart

(3.53)". The claim is correct, but what a *replacement* of basis vectors is allowed to mean needs saying, and the substitution is short enough to do in full.

What the replacement is. Define

$$K := \hat{E}_\Theta \otimes \hat{E}_1 + \hat{E}_R \otimes \hat{E}_2 - \hat{E}_3 \otimes \hat{E}_3,$$

so that $K\hat{E}_1 = \hat{E}_\Theta$, $K\hat{E}_2 = \hat{E}_R$, $K\hat{E}_3 = -\hat{E}_3$. This is a referential tensor, and by exactly the computation of Lemma 3.39 it is orthogonal, since it pairs the orthonormal triad $\{\hat{E}_A\}$ with the triad $\{\hat{E}_\Theta, \hat{E}_R, -\hat{E}_3\}$. That second triad is orthonormal because $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ is, and it is *right handed* because

$$\hat{E}_\Theta \times \hat{E}_R = -(\hat{E}_R \times \hat{E}_\Theta) = -\hat{E}_3,$$

which is the third member. Hence $\det K = +1$: the replacement is a *rotation of the referential frame*, not merely a relabelling.

The substitution, term by term. Apply the replacement to (3.53). The identity goes to itself:

$$\begin{aligned} \hat{I} &= \hat{E}_1 \otimes \hat{E}_1 + \hat{E}_2 \otimes \hat{E}_2 + \hat{E}_3 \otimes \hat{E}_3 \\ &\longmapsto \hat{E}_\Theta \otimes \hat{E}_\Theta + \hat{E}_R \otimes \hat{E}_R + (-\hat{E}_3) \otimes (-\hat{E}_3) \\ &= \hat{E}_R \otimes \hat{E}_R + \hat{E}_\Theta \otimes \hat{E}_\Theta + \hat{E}_3 \otimes \hat{E}_3 = \hat{I}, \end{aligned}$$

the two minus signs multiplying to +1 and (3.11) closing the sum because $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ is orthonormal. The shear term goes to

$$\gamma(\hat{E}_1 \otimes \hat{E}_2 + \hat{E}_2 \otimes \hat{E}_1) \longmapsto \gamma(\hat{E}_\Theta \otimes \hat{E}_R + \hat{E}_R \otimes \hat{E}_\Theta),$$

and the square term to

$$\gamma^2 \hat{E}_2 \otimes \hat{E}_2 \longmapsto \gamma^2 \hat{E}_R \otimes \hat{E}_R.$$

Adding the three lines,

$$\hat{I} + \gamma(\hat{E}_R \otimes \hat{E}_\Theta + \hat{E}_\Theta \otimes \hat{E}_R) + \gamma^2 \hat{E}_R \otimes \hat{E}_R,$$

which is (3.92) exactly. Nothing is left over and nothing is missing, so the two equations are the same tensor written in two frames. In symbols the statement is $\underline{C}_{\text{azim}} = \underline{K} \underline{C}_{\text{simple}} \underline{K}^t$ with γ replaced by $\gamma(R)$.

The same works for $\underline{B}$. Replacing $\underline{e}_1$ by $\underline{e}_\theta(\theta)$, $\underline{e}_2$ by $\underline{e}_r(\theta)$ and $\underline{e}_3$ by $-\underline{e}_3$ in (3.54),

$$\begin{aligned} (1 + \gamma^2) \underline{e}_1 \otimes \underline{e}_1 &\longmapsto (1 + \gamma^2) \underline{e}_\theta \otimes \underline{e}_\theta, \\ \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3 &\longmapsto \underline{e}_r \otimes \underline{e}_r + \underline{e}_3 \otimes \underline{e}_3, \\ \gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1) &\longmapsto \gamma(\underline{e}_\theta \otimes \underline{e}_r + \underline{e}_r \otimes \underline{e}_\theta), \end{aligned}$$

and adding, with $\underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_r \otimes \underline{e}_r + \underline{e}_3 \otimes \underline{e}_3 = \underline{I}$, gives $\underline{I} + \gamma(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r) + \gamma^2 \underline{e}_\theta \otimes \underline{e}_\theta$, which is (3.93). The book omits the third replacement here; it is invisible in $\underline{B}$ for the same reason it was invisible in $\underline{C}$, since $\underline{e}_3$ enters only through $\underline{e}_3 \otimes \underline{e}_3$, but it is needed for the same reason, to keep the frame right handed.

REMARK 3.42 (so is anything wrong? two caveats the book does not state). Nothing is wrong. But two things are true that the bare word “replacement” hides, and both matter later.

(i) *The sign on $\hat{\underline{E}}_3$ is invisible in $\underline{C}$ and indispensable elsewhere.* As the substitution above shows, $\hat{\underline{E}}_3$ enters (3.53) only inside $\hat{\underline{I}}$, as $\hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3$, which is insensitive to the sign. So one could drop the minus sign and still land on (3.92). One may not drop it, because without it $\det \underline{K} = -1$ and $\underline{K}$ is a reflection; conclusions about $\underline{C}$ would survive, but conclusions about the *rotation* $\underline{R}$, about the handedness of the principal triad, and about the sign of $\det \underline{F}$ would not. The minus sign is doing work that only becomes visible one equation later.

(ii) *The order of the shear pair is reversed, and this flips the sign of the rotation angle.* In (3.53) the shear direction is $\hat{\underline{E}}_1$ and the shear normal is $\hat{\underline{E}}_2$; the replacement sends these to $\hat{\underline{E}}_\theta$ and $\hat{\underline{E}}_r$ respectively. But the natural ordering of the cylindrical frame is $(\hat{\underline{E}}_r, \hat{\underline{E}}_\theta, \hat{\underline{E}}_3)$, radial first. So the pair that carries the shear appears in the *opposite* order from the one in §3.5.1, and the polar rotation comes out with the opposite sign: Remark 3.33 found $\tan \beta = -\gamma/2$ for simple shear, whereas the computation below gives $\tan \hat{\beta} = +\gamma/2$ for azimuthal shear. There is

no contradiction, and the physical check confirms it: with $\phi' > 0$ the outer cylinders lead, a radius is dragged towards increasing θ , and the material turns counterclockwise, so a positive angle is right.

(iii) *The replacement is different at every material point.* Since $\hat{\underline{E}}_R$ and $\hat{\underline{E}}_\Theta$ depend on Θ , the tensor $\underline{K}$ above is $\underline{K}(\Theta)$, and $\gamma = \gamma(R)$ depends on R . So the transfer of results from §3.5.1 is valid *pointwise* and only pointwise. The principal *stretches* depend on R alone, but the principal *directions* depend on Θ as well, and $\underline{U}$, $\underline{V}$, $\underline{R}$ are fields, not constants, as they were in the homogeneous case.

The polar factors, explicitly

The book says only that $\underline{U}$ and $\underline{V}$ “are obtained as they were in Section 3.5.1 with the indicated replacements”, and that the rotations of (3.94) “are computed in the same way”. Since Problem 9 of §3.8 gives a closed form that needs no eigenvectors, here they are.

Work in the plane spanned by $(\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta)$, the third direction being inert since $\underline{C}\hat{\underline{E}}_3 = \hat{\underline{E}}_3$. From (3.92),

$$(C_{AB}) = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix}, \quad (\hat{G}_{AB}) = \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix},$$

in the ordered basis $(\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta)$; the matrix of $\hat{\underline{G}}$ has its γ in the (Θ, R) slot because $\hat{\underline{G}} = \hat{\underline{I}} + \gamma \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_R$. Then

$$J = \sqrt{\det \underline{C}} = \sqrt{(1 + \gamma^2) - \gamma^2} = 1, \\ I = \sqrt{\text{tr } \underline{C} + 2J} = \sqrt{2 + \gamma^2 + 2} = \sqrt{4 + \gamma^2},$$

and Problem 9 gives $\underline{U} = (\underline{C} + J\hat{\underline{I}})/I$:

$$\underline{U} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta) \right. \\ \left. + \gamma(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_R) + \gamma^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R \right] + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,$$

with matrix $\frac{1}{I} \begin{pmatrix} 2+\gamma^2 & \gamma \\ \gamma & 2 \end{pmatrix}$ in the plane. The check $\underline{U}^2 = \underline{C}$ is the computation of Remark 3.51 with the two indices interchanged. The principal

stretches are the roots of $\lambda^2 - I\lambda + J = 0$, that is

$$\lambda_{1,2} = \frac{1}{2} \left(\sqrt{4 + \gamma^2} \pm \gamma \right), \quad \lambda_3 = 1, \quad \gamma = R\phi'(R),$$

which vary from point to point, unlike §3.5.1.

For the rotation, invert $\underline{U}$ by the same identity: multiplying $\underline{U}^2 - I\underline{U} + J\hat{I} = \hat{Q}$ by $\underline{U}^{-1}$ gives $\underline{U}^{-1} = (I\hat{I} - \underline{U})/J = I\hat{I} - \underline{U}$, so in the plane

$$(U_{AB}^{-1}) = \frac{1}{I} \begin{pmatrix} (4 + \gamma^2) - (2 + \gamma^2) & -\gamma \\ -\gamma & (4 + \gamma^2) - 2 \end{pmatrix} = \frac{1}{I} \begin{pmatrix} 2 & -\gamma \\ -\gamma & 2 + \gamma^2 \end{pmatrix},$$

and therefore

$$\begin{aligned} (R_{\hat{G}})_{AB} &= (\hat{G}_{AC})(U_{CB}^{-1}) = \frac{1}{I} \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix} \begin{pmatrix} 2 & -\gamma \\ -\gamma & 2 + \gamma^2 \end{pmatrix} \\ &= \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 & -\gamma \\ \gamma & 2 \end{pmatrix}, \end{aligned}$$

the (2, 1) entry being $2\gamma - \gamma = \gamma$ and the (2, 2) entry $-\gamma^2 + 2 + \gamma^2 = 2$. This is orthogonal, each column having length $\sqrt{4 + \gamma^2}/I = 1$ and the columns being orthogonal since $-2\gamma + 2\gamma = 0$, with determinant $(4 + \gamma^2)/I^2 = 1$. So $\underline{R}_{\hat{G}}$ is the rotation of the $(\hat{E}_R, \hat{E}_\Theta)$ plane through the angle $\hat{\beta}$ with

$$\cos \hat{\beta} = \frac{2}{\sqrt{4 + \gamma^2}}, \quad \sin \hat{\beta} = \frac{\gamma}{\sqrt{4 + \gamma^2}}, \quad \tan \hat{\beta} = \frac{\gamma}{2} = \frac{R\phi'(R)}{2},$$

positive, that is counterclockwise, as Remark 3.42(ii) predicted.

The spatial side is identical with hats removed. From (3.93), $(B_{ij}) = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix}$ in the ordered basis $(\underline{e}_r, \underline{e}_\theta)$, with the same I and J since $\underline{B} = \underline{R}\underline{C}\underline{R}^t$ has the same invariants, so

$$\underline{V} = \frac{2(\underline{e}_r \otimes \underline{e}_r + \underline{e}_\theta \otimes \underline{e}_\theta) + \gamma(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r) + \gamma^2 \underline{e}_\theta \otimes \underline{e}_\theta}{\sqrt{4 + \gamma^2}} + \underline{e}_3 \otimes \underline{e}_3,$$

and $\underline{R}_G = \underline{V}^{-1}\underline{G}$ has the same matrix $\frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 & -\gamma \\ \gamma & 2 \end{pmatrix}$ in $(\underline{e}_r, \underline{e}_\theta)$. Notice which dyad carries the γ^2 : it is $\hat{E}_R \otimes \hat{E}_R$ in $\underline{U}$ and $\underline{e}_\theta \otimes \underline{e}_\theta$ in $\underline{V}$, the index

having moved from the shear normal to the shear direction, exactly as Remark 3.31 says it must.

Finally, from (3.91) the right polar factorisation of $\hat{G}$ and the left polar factorisation of G are

$$\hat{G} = \underline{R}_{\hat{G}} \underline{U}, \quad G = \underline{V} \underline{R}_G, \quad (3.94)$$

with $\underline{R}_{\hat{G}}$ referential and $\underline{R}_G$ spatial. Substituting into (3.88) and (3.89),

$$\underline{F} = \underline{Q} \hat{G} = \underline{Q} \underline{R}_{\hat{G}} \underline{U}, \quad \underline{F} = \underline{G} \underline{Q} = \underline{V} \underline{R}_G \underline{Q}.$$

In the first, $\underline{U}$ is symmetric and positive definite and $\underline{Q} \underline{R}_{\hat{G}}$ is a rotation, being a product of rotations by Problem 2 of §3.8; in the second, $\underline{V}$ is symmetric positive definite and $\underline{R}_G \underline{Q}$ is a rotation for the same reason. Both are therefore polar decompositions of $\underline{F}$, and by the uniqueness proved in Step 7 of §3.4 their rotation factors must be the one and only $\underline{R}$:

$$\underline{R} = \underline{Q} \underline{R}_{\hat{G}} = \underline{R}_G \underline{Q}. \quad (3.95)$$

The two expressions for $\underline{R}$ are equal as a consequence, not by inspection; equating them gives the useful by-product $\underline{R}_G = \underline{Q} \underline{R}_{\hat{G}} \underline{Q}^t$, which says that the spatial and referential shear rotations are the same rotation seen in the two frames, and explains why their matrices came out identical above.

3.5.4 Torsion

What this example is for. It is the last of the four and it is included mainly to be recognised: nothing new happens. Torsion differs from azimuthal shear only in *which* coordinate the azimuth is made to depend on, Z instead of R , and every formula transfers with $\hat{E}_R \mapsto \hat{E}_3$. What is genuinely new is the geometric reading, the generator becoming a helix, which Remarks 3.43 and 3.44 develop well past the book.

Consider

$$r = R, \quad \theta = \Theta + \tau Z, \quad z = Z, \quad (3.96)$$

where the constant τ is the *twist*, the rate of change of azimuth with axial coordinate. A cross-section $Z = \text{const}$ is rotated rigidly through the angle τZ , different sections by different amounts, which is what twisting a bar does.

The book compresses this into “the usual procedure leads again to (3.88) and (3.89)”. Written out, the usual procedure is this. Differentiating (3.96),

$$dr = dR, \quad d\theta = d\Theta + \tau dZ, \quad dz = dZ,$$

the middle relation because τ is a constant and θ depends on Z as well as on Θ . Substituting into the polar line element with $r = R$,

$$\begin{aligned} d\tilde{x} &= dR \underline{e}_r(\theta) + R[d\Theta + \tau dZ] \underline{e}_\theta(\theta) + dZ \underline{e}_3 \\ &= dR \underline{e}_r(\theta) + R d\Theta \underline{e}_\theta(\theta) + [R\tau \underline{e}_\theta(\theta) + \underline{e}_3] dZ, \end{aligned}$$

the terms in dZ having been collected. Now insert $dR = \hat{E}_R \cdot d\tilde{X}$, $R d\Theta = \hat{E}_\Theta \cdot d\tilde{X}$, $dZ = \hat{E}_3 \cdot d\tilde{X}$:

$$\begin{aligned} d\tilde{x} &= \underline{e}_r(\hat{E}_R \cdot d\tilde{X}) + \underline{e}_\theta(\hat{E}_\Theta \cdot d\tilde{X}) + (R\tau \underline{e}_\theta + \underline{e}_3)(\hat{E}_3 \cdot d\tilde{X}) \\ &= \left[\underline{e}_r \otimes \hat{E}_R + \underline{e}_\theta \otimes \hat{E}_\Theta + \underline{e}_3 \otimes \hat{E}_3 + R\tau \underline{e}_\theta \otimes \hat{E}_3 \right] d\tilde{X}, \end{aligned}$$

so that $\tilde{F} = \tilde{Q} + R\tau \underline{e}_\theta \otimes \hat{E}_3$ with the same $\tilde{Q}$ of (3.87). Compare this with (3.86): the extra dyad now has $\hat{E}_3$ in its second slot instead of $\hat{E}_R$, and that is the *only* difference between torsion and azimuthal shear. Factoring $\tilde{Q}$ out on either side exactly as before, using $\underline{e}_\theta = \tilde{Q} \hat{E}_\Theta$ on the left and $\hat{E}_3 = \tilde{Q}^t \underline{e}_3$ on the right, gives (3.88) and (3.89) again, now with

$$\hat{G} = \hat{I} + R\tau \hat{E}_\Theta(\Theta) \otimes \hat{E}_3, \quad \mathcal{G} = \mathcal{I} + R\tau \underline{e}_\theta(\theta) \otimes \underline{e}_3. \quad (3.97)$$

This is again a shearing deformation, of variable amount $\gamma(R) = R\tau$, now on the orthonormal pair $(\hat{E}_\Theta, \hat{E}_3)$, and it is isochoric by any of the four arguments given for azimuthal shear; the first of them reads $\det \hat{G} = 1 + R\tau \hat{E}_\Theta \cdot \hat{E}_3 = 1 + 0 = 1$.

REMARK 3.43 (what a *generator* of a cylinder is). A right circular cylinder is a *ruled* surface: it is swept out by translating a straight line along a circle, and the straight lines so produced, the ones lying in the surface and parallel to the axis, are its *generators*, or generatrices. In the referential coordinates of (3.73) a generator is the material line

$$R = R_0, \quad \Theta = \Theta_0, \quad Z \text{ varying}, \quad \underline{\tilde{X}}(Z) = R_0 \hat{\underline{\tilde{E}}}_R(\Theta_0) + Z \hat{\underline{\tilde{E}}}_3,$$

and its unit tangent is $\underline{\tilde{M}} = d\underline{\tilde{X}}/dZ = \hat{\underline{\tilde{E}}}_3$, since R_0 and Θ_0 are fixed and $\hat{\underline{\tilde{E}}}_R(\Theta_0)$ is therefore a constant vector. That is why the book takes $\underline{\tilde{M}} = \hat{\underline{\tilde{E}}}_3$: it is not an arbitrary choice of test direction but the tangent to the one family of *straight* material lines that the undeformed cylinder contains. A cylinder has two such rulings only if it is degenerate; the circular cylinder has exactly one, the axial family, and the circles $Z = \text{const}$ are of course not straight.

A generator therefore maps to

$$\mu \underline{\tilde{m}} = \underline{\tilde{F}} \hat{\underline{\tilde{E}}}_3 = \underline{\tilde{Q}} \hat{\underline{\tilde{E}}}_3 + R\tau \underline{\tilde{e}}_\theta = \underline{\tilde{e}}_3 + R\tau \underline{\tilde{e}}_\theta(\theta); \quad \mu = \sqrt{1 + (R\tau)^2}, \quad (3.98)$$

using $\underline{\tilde{Q}} \hat{\underline{\tilde{E}}}_3 = \underline{\tilde{e}}_3$ from Lemma 3.39 and $\hat{\underline{\tilde{E}}}_3 \cdot \hat{\underline{\tilde{E}}}_3 = 1$, with $|\underline{\tilde{e}}_3 + \gamma \underline{\tilde{e}}_\theta|^2 = 1 + \gamma^2$ by orthonormality.

REMARK 3.44 (the image is a circular helix, and everything about it). The book says the generator goes to “a helix” and stops. Here is the curve and its geometry, none of which is in the text.

Push the generator forward through (3.96). A point at axial station Z has $r = R_0$, $\theta = \Theta_0 + \tau Z$, $z = Z$, so

$$\underline{\tilde{x}}(Z) = R_0 \underline{\tilde{e}}_r(\Theta_0 + \tau Z) + Z \underline{\tilde{e}}_3.$$

The radius is constant, so the curve lies on the cylinder $r = R_0$, and the azimuth is an affine function of the axial coordinate. That is the definition of a *circular*, or cylindrical, helix. Differentiating, and using $d\underline{\tilde{e}}_r/d\theta = \underline{\tilde{e}}_\theta$ together with $d\theta/dZ = \tau$,

$$\frac{d\underline{\tilde{x}}}{dZ} = R_0\tau \underline{\tilde{e}}_\theta + \underline{\tilde{e}}_3, \quad \left| \frac{d\underline{\tilde{x}}}{dZ} \right| = \sqrt{1 + R_0^2\tau^2} = \mu,$$

which is (3.98) obtained a second way, by differentiating the curve rather than by applying $\underline{F}$ to its tangent. The agreement is the content of (2.67).

Pitch. One full turn needs $\Delta\theta = 2\pi$, hence $\Delta Z = 2\pi/\tau$. So the pitch is

$$P = \frac{2\pi}{\tau},$$

independent of R_0 : every generator of every coaxial cylinder becomes a helix of the same pitch, which is exactly what it means for the cross-sections to rotate rigidly. The helices differ only in radius.

Helix angle. The angle α_h between the deformed generator and the axis satisfies

$$\tan \alpha_h = \frac{R_0\tau |\underline{e}_\theta|}{|\underline{e}_3|} = R_0\tau = \gamma.$$

So γ is again the tangent of a tilt angle, as it was in (3.49) for the vertical line and in Remark 3.40 for the radius. The outer fibres are tilted most, the axis not at all, which is why a twisted bar fails at its surface.

Curvature and torsion. Writing $a = R_0$ and $b = 1/\tau$, the curve is the standard helix $(a \cos t, a \sin t, bt)$ with $t = \tau Z$, whose curvature and geometric torsion are

$$\kappa = \frac{a}{a^2 + b^2} = \frac{R_0\tau^2}{1 + R_0^2\tau^2}, \quad \varsigma = \frac{b}{a^2 + b^2} = \frac{\tau}{1 + R_0^2\tau^2}.$$

Both are constants along the curve, and the circular helix is the only curve in three-space for which both are constant and non-zero. Two checks: as $\tau \rightarrow 0$ both tend to 0 and the helix straightens back into the generator; and $\kappa/\varsigma = R_0\tau = \gamma$, so the ratio of curvature to torsion is the amount of shear.

REMARK 3.45 (three shears, one algebra). Sections 3.5.1, 3.5.3 and 3.5.4 are the same deformation three times. Each has $\hat{G} = \hat{I} + \gamma \underline{a} \otimes \underline{b}$ with $\underline{a}, \underline{b}$ orthonormal: $(\hat{E}_1, \hat{E}_2)$ for simple shear, $(\hat{E}_\Theta, \hat{E}_R)$ for azimuthal shear, $(\hat{E}_\Theta, \hat{E}_3)$ for torsion. Every consequence therefore follows once and for all from that single form. In particular $\det \hat{G} = 1$ by Problem 21(f), the principal stretches are $\frac{1}{2}(\gamma \pm \sqrt{4 + \gamma^2})$ and 1 by (3.60), and the stretch tensors are given by Problem 9 of §3.8 in closed form. The

only difference between the three is which pair of orthonormal directions carries the shear, and whether γ is constant. Recognising this saves solving three characteristic equations.

3.6 Homogeneous deformations

Where we are and why this section exists

Sections 3.1 to 3.4 built the machinery, $\underline{F} = \underline{R}\underline{U}$, and §3.5 exercised it on four deformations chosen because they can be written down. The obvious complaint is that real deformations cannot be written down. This section answers the complaint. It isolates the deformations for which $\underline{F}$ does not vary from point to point, proves three things about them, and then observes, at (3.102), that *every* deformation looks like one of these under a microscope. So the three properties proved here are the local structure of every deformation there is, and that is why the section is worth the trouble.

The definition and its integrated form

A deformation is *homogeneous* if $\underline{F}$ is the same at every material point, i.e. independent of $\underline{X}$. What does that buy? Start from the definition of $\underline{F}$, which is $d\underline{x} = \underline{F} d\underline{X}$, and ask whether the right-hand side is itself the differential of something. In general it is not, because $\underline{F}$ varies. If $\underline{F}$ is uniform it is, because a constant tensor passes through a differential untouched: $d(\underline{F}\underline{X}) = \underline{F} d\underline{X}$ exactly, with no term $(d\underline{F})\underline{X}$ to worry about. Hence

$$d\underline{x} = \underline{F} d\underline{X} = d(\underline{F}\underline{X}), \quad (3.99)$$

so $d(\underline{x} - \underline{F}\underline{X}) = \underline{0}$, and a field whose differential vanishes identically is constant in $\underline{X}$, though it may still depend on t , which the differential at fixed t cannot see. Calling that constant $\underline{c}(t)$,

$$\underline{x} = \underline{\chi}_\kappa(\underline{X}, t) = \underline{F}(t)\underline{X} + \underline{c}(t). \quad (3.100)$$

So homogeneous means *affine*: a fixed linear map plus a translation. The translation $\underline{c}$ is a rigid displacement of the whole body and carries no deformation, which is why it never appeared in §3.5, where only $\underline{F}$ was ever computed. Simple shear, §3.5.1, is homogeneous, as (3.49) shows directly; the three cylindrical deformations of §§3.5.2–3.5.4 are not, since in each the coefficients of $\underline{F}$ depend on R .

For homogeneous deformations, write (3.100) at two material points and subtract:

$$\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t) = [\underline{F}\underline{Y} + \underline{c}] - [\underline{F}\underline{X} + \underline{c}] = \underline{F}(\underline{Y} - \underline{X}),$$

the translations cancelling, which gives

$$\chi_\kappa(\underline{Y}, t) = \chi_\kappa(\underline{X}, t) + \underline{F}(t)(\underline{Y} - \underline{X}), \quad (3.101)$$

exactly, for *any* two points however far apart. For a general differentiable deformation (1.127) and (1.128) give only

$$\chi_\kappa(\underline{Y}, t) = \chi_\kappa(\underline{X}, t) + \underline{F}(\underline{X}, t)(\underline{Y} - \underline{X}) + o(\underline{Y} - \underline{X}). \quad (3.102)$$

So every smooth deformation is approximately homogeneous in a small enough neighbourhood of a material point. That is the reason for studying the class: it describes the local structure of everything else.

3.6.1 Some properties of homogeneous deformations

(a) Material planes deform to planes; parallel planes map to parallel planes

Consider a fixed plane in κ with uniform unit normal $\underline{N}$. Its differential equation is

$$0 = d\underline{X} \cdot \underline{N} = d(\underline{X} \cdot \underline{N}), \quad (3.103)$$

the second form because $\underline{N}$ is constant, and this integrates to

$$\underline{X} \cdot \underline{N} = D, \quad (3.104)$$

D being the perpendicular distance from the origin to the plane, and parallel planes being distinguished by the value of D alone.

Invert (3.100) to get $\underline{\tilde{X}} = \underline{\tilde{F}}^{-1}(\underline{\tilde{x}} - \underline{\tilde{c}})$ and substitute into (3.104). Moving $\underline{\tilde{F}}^{-1}$ across the dot product by the definition of the transpose, $\underline{\tilde{A}}\underline{\tilde{u}} \cdot \underline{\tilde{v}} = \underline{\tilde{u}} \cdot \underline{\tilde{A}}^t \underline{\tilde{v}}$ applied with $\underline{\tilde{A}} = \underline{\tilde{F}}^{-1}$ and $\underline{\tilde{A}}^t = \underline{\tilde{F}}^{-t}$ by Problem 18 of Chapter 1, and then distributing,

$$D = \underline{\tilde{F}}^{-1}(\underline{\tilde{x}} - \underline{\tilde{c}}) \cdot \underline{\tilde{N}} = (\underline{\tilde{x}} - \underline{\tilde{c}}) \cdot \underline{\tilde{F}}^{-t} \underline{\tilde{N}} = \underline{\tilde{x}} \cdot \underline{\tilde{F}}^{-t} \underline{\tilde{N}} - \underline{\tilde{c}} \cdot \underline{\tilde{F}}^{-t} \underline{\tilde{N}},$$

so, moving the last term to the other side,

$$\underline{\tilde{x}} \cdot \underline{\tilde{F}}^{-t} \underline{\tilde{N}} = d, \quad d = D + \underline{\tilde{c}} \cdot \underline{\tilde{F}}^{-t} \underline{\tilde{N}}. \quad (3.105)$$

The left-hand side is linear in $\underline{\tilde{x}}$ with a constant coefficient vector, so this is the equation of a plane in κ_t whose normal is parallel to $\underline{\tilde{F}}^{-t} \underline{\tilde{N}}$; the Piola–Nanson formulas (2.54) and (2.55) say the same thing, since $\underline{\tilde{n}} da = \underline{\tilde{F}}^* \underline{\tilde{N}} dA = J \underline{\tilde{F}}^{-t} \underline{\tilde{N}} dA$. Different values of D give different values of d with the same coefficient vector, so parallel planes map to parallel planes.

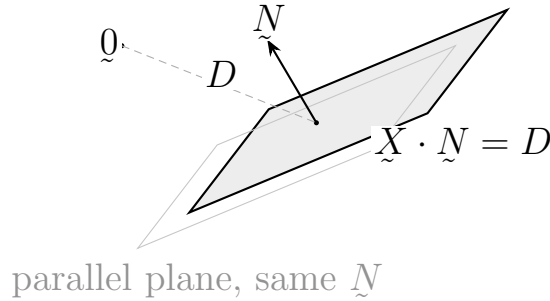


Figure 3.4: A material plane in κ . It is fixed by its unit normal $\underline{\tilde{N}}$ and the perpendicular distance D from the origin; parallel planes share $\underline{\tilde{N}}$ and differ only in D . Under a homogeneous deformation the whole family goes to a parallel family with common normal along $\underline{\tilde{F}}^{-t} \underline{\tilde{N}}$.

(b) Straight lines deform to straight lines; parallel lines map to parallel lines

The method is the same as in (a), one dimension lower: characterise the object by a differential equation, integrate it, push it through (3.100), and recognise what comes out.

A straight line is the curve whose unit tangent never changes. Writing S for arclength, that is $d\tilde{X}/dS = \tilde{M}$ with $\tilde{M}$ a fixed vector, or

$$d\tilde{X} = \tilde{M} dS = d(\tilde{M}S), \quad (3.106)$$

the second form because $\tilde{M}$ is constant and so passes through the differential. Hence $d(\tilde{X} - \tilde{M}S) = 0$, the bracket is constant, and naming that constant $\tilde{X}_0 = \tilde{X}(0)$,

$$\tilde{X}(S) = \tilde{M}S + \tilde{X}_0, \quad \tilde{X}_0 = \tilde{X}(0). \quad (3.107)$$

Now push it through (3.100), term by term, using linearity of $\tilde{F}$:

$$\tilde{x}(S) = \tilde{F}\tilde{X}(S) + \tilde{c} = \tilde{F}(\tilde{M}S + \tilde{X}_0) + \tilde{c} = S \tilde{F}\tilde{M} + \underbrace{\tilde{F}\tilde{X}_0 + \tilde{c}}_{=: \tilde{x}_0}.$$

The image is therefore of the form (constant vector) times (scalar parameter) plus (fixed point), which is again a straight line. To display it in the same normalised form as (3.107), write $\tilde{F}\tilde{M}$ as its length times its direction, $\tilde{F}\tilde{M} = \mu \tilde{m}$ with

$$\mu = |\tilde{F}\tilde{M}|, \quad \tilde{m} = \mu^{-1}\tilde{F}\tilde{M}, \quad |\tilde{m}| = 1,$$

μ being the stretch of the line by (2.67), and put $s = \mu S$. Then $S \tilde{F}\tilde{M} = S\mu \tilde{m} = s \tilde{m}$ and

$$\tilde{x}(s) = s \tilde{m} + \tilde{x}_0, \quad (3.108)$$

in which s is arclength on the deformed line, since $\tilde{m}$ is a unit vector and $|d\tilde{x}| = |\tilde{m}| ds = ds$.

Two consequences read straight off. The deformed tangent $\tilde{m}$ depends only on $\tilde{M}$, not on $\tilde{X}_0$; so *parallel* lines in κ , which by definition share $\tilde{M}$ and differ only in $\tilde{X}_0$, go to lines sharing $\tilde{m}$ and differing only in $\tilde{x}_0$: they stay parallel. And the stretch $\mu = |\tilde{F}\tilde{M}|$ likewise depends only on $\tilde{M}$: every line of a parallel family is stretched by the same factor.

REMARK 3.46 (the converse, which the book does not state). Property (b) is proved in one direction: homogeneous implies straight lines to straight lines. The converse is also true, and it is the more useful half,

because it turns a *geometric* observation into a statement about $\underline{F}$. Suppose only that the deformation is invertible and twice differentiable and that it carries every straight material line onto a straight line. Then it is homogeneous.

Proof. Let $\underline{X}(S) = \underline{X}_0 + S\underline{M}$ be any line. Its image is by hypothesis a straight line, so $\underline{x}(S) = \underline{x}_0 + g(S)\underline{m}$ for some scalar function g and some fixed unit vector $\underline{m}$ depending on $\underline{M}$ but not on S . Differentiating and using the chain rule,

$$\underline{F}(\underline{X}_0 + S\underline{M})\underline{M} = \underline{x}'(S) = g'(S)\underline{m}.$$

So for every $\underline{X}$ and every $\underline{M}$, the vector $\underline{F}(\underline{X})\underline{M}$ points along a direction $\underline{m}(\underline{M})$ that does not depend on $\underline{X}$: only its length does. Write $\underline{F}(\underline{X})\underline{M} = \mu(\underline{X}, \underline{M})\underline{m}(\underline{M})$. Fix two points $\underline{X}_1$ and $\underline{X}_2$ and put $\underline{A} = \underline{F}(\underline{X}_1)^{-1}\underline{F}(\underline{X}_2)$. Then, for every $\underline{M}$,

$$\underline{A}\underline{M} = \underline{F}(\underline{X}_1)^{-1}[\mu(\underline{X}_2, \underline{M})\underline{m}(\underline{M})] = \frac{\mu(\underline{X}_2, \underline{M})}{\mu(\underline{X}_1, \underline{M})}\underline{M},$$

so every vector is a principal vector of $\underline{A}$. A tensor with that property is a multiple of the identity: for independent $\underline{u}, \underline{v}$ with $\underline{A}\underline{u} = \alpha\underline{u}$, $\underline{A}\underline{v} = \beta\underline{v}$ and $\underline{A}(\underline{u} + \underline{v}) = \sigma(\underline{u} + \underline{v})$, subtracting gives $(\alpha - \sigma)\underline{u} + (\beta - \sigma)\underline{v} = \underline{0}$, hence $\alpha = \beta = \sigma$ by independence. Therefore

$$\underline{F}(\underline{X}) = c(\underline{X})\underline{F}_0, \quad \underline{F}_0 := \underline{F}(\underline{X}_1),$$

for a positive scalar field c with $c(\underline{X}_1) = 1$.

It remains to show c is constant, and here the fact that $\underline{F}$ is a *gradient* does the work. From $F_{iA} = \chi_{i,A}$ and the equality of mixed partial derivatives, $F_{iA,B} = \chi_{i,AB} = \chi_{i,BA} = F_{iB,A}$. Applying this to $F_{iA} = c(\underline{F}_0)_{iA}$, in which $\underline{F}_0$ is constant,

$$c_{,B}(\underline{F}_0)_{iA} = c_{,A}(\underline{F}_0)_{iB} \quad \text{for all } i, A, B.$$

Read with i fixed, this says $\underline{g} \otimes \underline{f}_i = \underline{f}_i \otimes \underline{g}$, where $\underline{g} = \text{Grad } c$ and $\underline{f}_i$ is the i th row of $\underline{F}_0$; by Problem 17(a) that forces $\underline{g}$ parallel to $\underline{f}_i$ or $\underline{g} = \underline{0}$. The three rows of the invertible $\underline{F}_0$ are linearly independent, so no single

non-zero $\underline{g}$ can be parallel to all three. Hence $\text{Grad } c = \underline{0}$, c is constant, $c \equiv 1$, and $\underline{F} \equiv \underline{F}_0$. $\square$

The moral is that “straight lines stay straight” is not one property of homogeneous deformations among many: it *characterises* them. The same is true of (a): a deformation carrying every material plane to a plane is homogeneous, by the same argument applied to $\underline{F}^{-t}$ and normals. Property (c) is not characterising, since every deformation carries small spheres to approximate ellipsoids by (3.102); what is special in the homogeneous case is that the statement is exact and the ellipsoid is the same at every point.

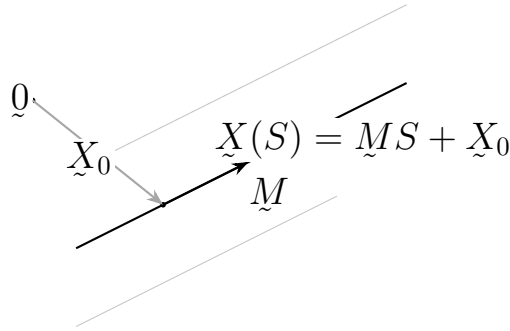


Figure 3.5: A material line in κ , fixed by a point $\underline{X}_0$ on it and its uniform unit tangent $\underline{M}$. Parallel lines share $\underline{M}$. A homogeneous deformation sends the family to a parallel family with common tangent $\underline{m} = \underline{F}\underline{M}/|\underline{F}\underline{M}|$ and common stretch $\mu = |\underline{F}\underline{M}|$.

(c) A spherical material surface is mapped to the surface of an ellipsoid

Consider a unit sphere in κ centred at $\underline{X}$, and let $\underline{Y}$ be a point on it, so that the radius vector is $\hat{\underline{r}} = \underline{Y} - \underline{X}$ with $|\hat{\underline{r}}| = 1$. After deformation the centre and the point go to $\underline{x} = \underline{\chi}_\kappa(\underline{X}, t)$ and $\underline{y} = \underline{\chi}_\kappa(\underline{Y}, t)$, and (3.101) gives

$$\underline{r} = \underline{F}\hat{\underline{r}}, \quad \underline{r} = \underline{y} - \underline{x}. \quad (3.109)$$

Hence $\hat{\underline{r}} = \underline{F}^{-1}\underline{r}$ and

$$1 = \hat{\underline{r}} \cdot \hat{\underline{r}} = \underline{F}^{-1}\underline{r} \cdot \underline{F}^{-1}\underline{r} = \underline{r} \cdot \underline{F}^{-t}\underline{F}^{-1}\underline{r} = \underline{r} \cdot \underline{B}^{-1}\underline{r}, \quad (3.110)$$

using $\tilde{B}^{-1} = (\tilde{F}\tilde{F}^t)^{-1} = \tilde{F}^{-t}\tilde{F}^{-1}$, which is Problem 19 of Chapter 1 combined with Problem 18. Substituting the spectral representation

$$\tilde{B}^{-1} = \sum_i \lambda_i^{-2} \underline{v}_i \otimes \underline{v}_i \quad (3.111)$$

from (3.42) and Remark 3.10, the substitution runs as follows. Apply $\tilde{B}^{-1}$ to $\underline{r}$ using $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$:

$$\tilde{B}^{-1}\underline{r} = \sum_i \lambda_i^{-2} (\underline{v}_i \cdot \underline{r}) \underline{v}_i,$$

then dot with $\underline{r}$ and move the scalars out:

$$\underline{r} \cdot \tilde{B}^{-1}\underline{r} = \sum_i \lambda_i^{-2} (\underline{v}_i \cdot \underline{r}) (\underline{r} \cdot \underline{v}_i) = \sum_i \lambda_i^{-2} (\underline{v}_i \cdot \underline{r})^2.$$

Writing $r_i = \underline{v}_i \cdot \underline{r}$ for the components of $\underline{r}$ on the spatial principal frame, and setting the whole thing equal to 1 by (3.110),

$$1 = (r_1/\lambda_1)^2 + (r_2/\lambda_2)^2 + (r_3/\lambda_3)^2, \quad (3.112)$$

which is the equation of an ellipsoid with axes along $\underline{v}_i$ and semi-axis lengths λ_i , centred at $\underline{x}$.

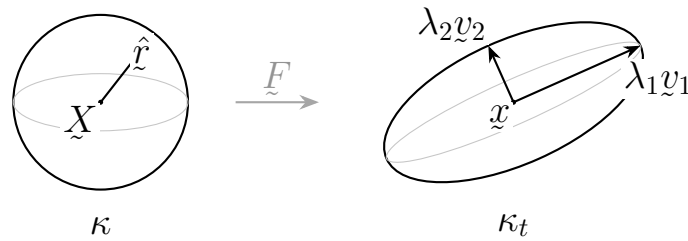


Figure 3.6: A unit sphere of material in κ is carried by a homogeneous deformation to an ellipsoid in κ_t whose axes lie along the spatial principal directions $\underline{v}_i$ and whose semi-axes have lengths equal to the principal stretches λ_i . Equation (3.112) is the algebraic statement, and $\tilde{B}^{-1}$ is the tensor that carries it.

REMARK 3.47 (what the numerical test in Figure D actually checks). Panels (c) and (d) are the substance; (a) and (b) only make the result

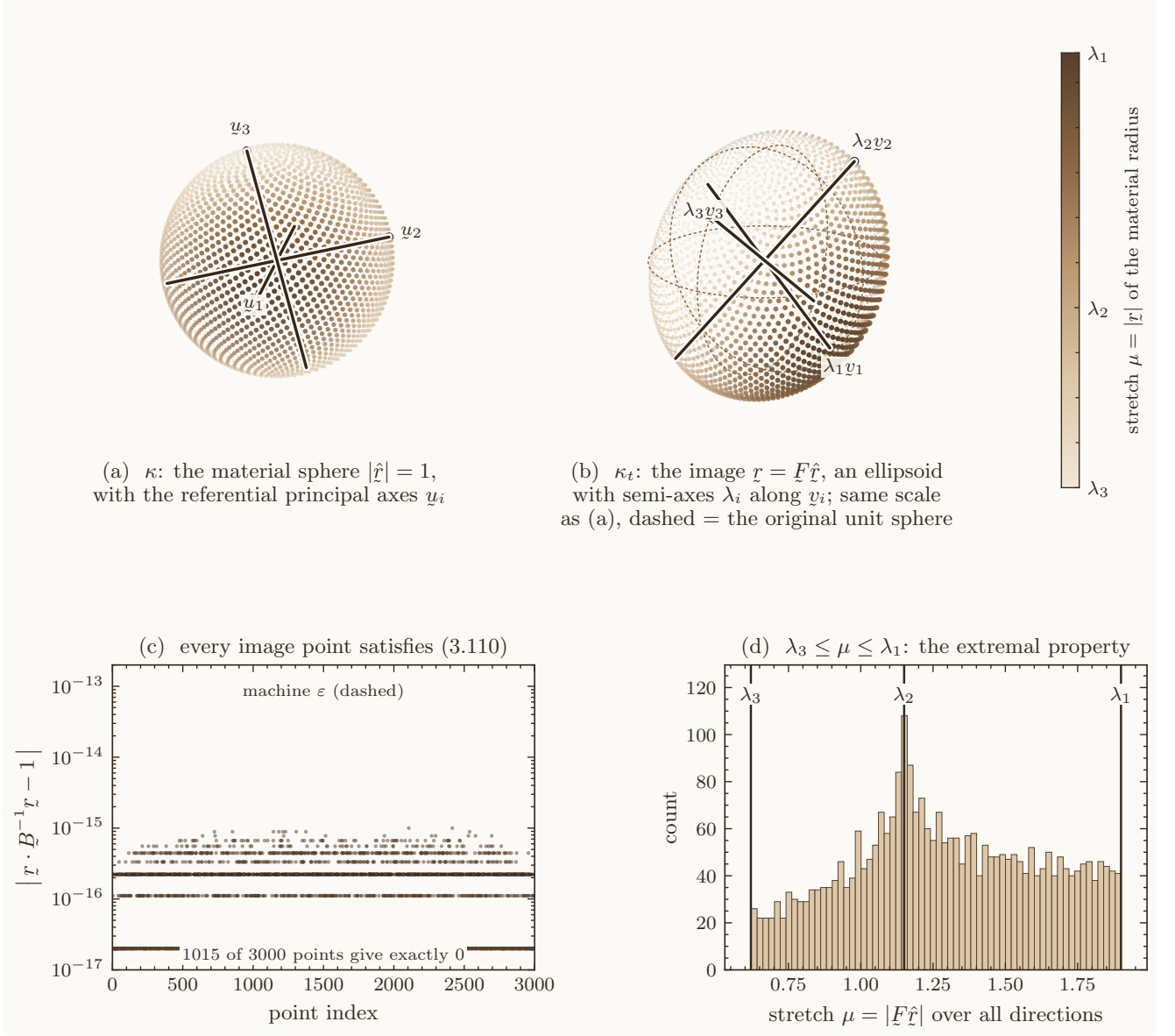


Figure D: Property (c) tested numerically, by the script `sphere_to_ellipsoid.py`. Three thousand points are spread over the unit material sphere by a golden-angle spiral and pushed forward by one fixed $\underline{\tilde{F}} = \underline{\tilde{R}}\underline{\tilde{U}}$ with principal stretches $\lambda = (1.90, 1.15, 0.62)$ and $J = \det \underline{\tilde{F}} = 1.3547$. Every marker in the figure is an individual vector point, not a raster image. (a) the sphere in κ , each point shaded by the stretch that material radius will undergo, with the referential principal axes u_i . (b) the image in κ_t , at the same scale, with the original sphere retained as dashed great circles for comparison; the point cloud lies on an ellipsoid with semi-axes λ_i along v_i , exactly as (3.112) asserts. (c) the residual $|\underline{\tilde{r}} \cdot \underline{\tilde{B}}^{-1}\underline{\tilde{r}} - 1|$ of (3.110) for every one of the three thousand image points, at or below machine precision throughout, with 1015 of them exactly zero. (d) the distribution of the stretch $\mu = |\underline{\tilde{F}}\hat{r}|$ over all directions, confined between λ_3 and λ_1 as the extremal property (3.26) requires, with λ_2 falling inside.

visible. Four separate predictions of the text are tested, and all four hold to about 10^{-15} :

- (3.110), $\underline{r} \cdot \underline{B}^{-1} \underline{r} = 1$, panel (c);
- (3.112), $\sum_i (r_i/\lambda_i)^2 = 1$ with $r_i = \underline{v}_i \cdot \underline{r}$, the same statement resolved on the principal frame;
- (3.17), $\lambda_1 \lambda_2 \lambda_3 = J$, so that the ellipsoid volume $\frac{4}{3}\pi \lambda_1 \lambda_2 \lambda_3$ is exactly J times the sphere volume;
- (3.44), $\underline{F} \underline{u}_i = \lambda_i \underline{v}_i$, so the axes really are the images of the referential principal directions.

One check is deliberately *not* at machine precision, and it should not be. The semi-axis lengths measured from the cloud, $\max_{\text{cloud}} |\underline{v}_i \cdot \underline{r}|$, come out about 4×10^{-4} below λ_i . That is not an error in (3.112) but a consequence of sampling: with finitely many points, none lands exactly on an axis tip, so the measured semi-axis always undershoots. The script confirms the diagnosis by refining the cloud, and the discrepancy falls from 3.3×10^{-4} at $N = 3000$ to 1.8×10^{-5} at $N = 30\,000$ and 1.5×10^{-6} at $N = 300\,000$, that is like $1/N$. Distinguishing a discretisation error from a genuine one is the whole skill of testing a formula numerically.

The script also runs the test backwards. It builds $\underline{F}$ from a chosen $\underline{R}$ and $\underline{U}$, then throws them away and recovers them from $\underline{F}$ alone by Theorem 3.23, that is by the closed form $\underline{U} = (-\underline{C}^2 + (i_1^2 - i_2)\underline{C} + i_1 i_3 \hat{\underline{I}})/(\underline{i}_1 \underline{i}_2 - \underline{i}_3)$ with $\underline{i}_1$ from the quartic $(*_2)$. The recovered $\underline{U}$ and $\underline{R}$ agree with the originals to 5×10^{-15} , $\underline{R}^t \underline{R} = \hat{\underline{I}}$ and $\det \underline{R} = 1$ hold to the same order, and both $\underline{F} = \underline{R} \underline{U}$ and $\underline{F} = \underline{V} \underline{R}$ are reproduced. So the closed form of §3.4 is not merely derived, it is executed.

REMARK 3.48 (the ellipsoid is the picture of the polar decomposition). Property (c) is $\underline{F} = \underline{V} \underline{R}$ drawn. The rotation $\underline{R}$ takes the unit sphere to the unit sphere, since it preserves lengths, and therefore contributes nothing to the shape; all of the shape comes from $\underline{V}$, which stretches the sphere by λ_i along $\underline{v}_i$. Conversely, if one measures the ellipsoid in κ_t one recovers $\underline{V}$ completely and $\underline{R}$ not at all. This is exactly the sense in which the ellipsoid, and not $\underline{F}$, is what the material experiences.

3.7 Rigid-body motions

Where we are and why this section closes the chapter

The chapter opened with a promise: separate the part of $\underline{F}$ that a material feels from the part it does not. §3.4 delivered the separation, $\underline{F} = \underline{R}\underline{U}$, and asserted that $\underline{R}$ is the part that costs nothing. That assertion has not yet been proved. This section proves it, in the sharpest possible form: the motions in which *nothing at all* happens to the material are exactly the motions with $\underline{U} = \hat{\underline{I}}$, that is exactly the motions in which $\underline{F}$ is a rotation. Once that is established, the case for writing constitutive laws in terms of $\underline{C}$ rather than $\underline{F}$ is complete, and the chapter is finished.

The logic runs the opposite way from §3.6. There we assumed something about $\underline{F}$ and deduced geometry; here we assume something purely geometric, that distances are preserved, and deduce what $\underline{F}$ must be.

A motion is *rigid* if it preserves the distance between every pair of material points, i.e. if $|\underline{\chi}_\kappa(\underline{Y}, t) - \underline{\chi}_\kappa(\underline{X}, t)| = |\underline{Y} - \underline{X}|$, or, squaring to remove the square root,

$$[\underline{\chi}_\kappa(\underline{Y}, t) - \underline{\chi}_\kappa(\underline{X}, t)] \cdot [\underline{\chi}_\kappa(\underline{Y}, t) - \underline{\chi}_\kappa(\underline{X}, t)] = (\underline{Y} - \underline{X}) \cdot (\underline{Y} - \underline{X}) \quad (3.113)$$

for all $\underline{X}, \underline{Y}$. Squaring is not cosmetic: the square root in the definition is awkward to differentiate at $\underline{Y} = \underline{X}$, whereas the quadratic form on each side of (3.113) is a polynomial and differentiates cleanly.

The plan. We have a functional equation, (3.113), holding for all pairs of points, and we want a statement about $\underline{F}$. The route is to differentiate twice, once with respect to $\underline{Y}$ and once with respect to $\underline{X}$, because each differentiation converts a difference of positions into an $\underline{F}$, by the definition $d\underline{\chi}_\kappa = \underline{F} d\underline{X}$. Two differentiations therefore produce a relation between two $\underline{F}$'s, and that relation will turn out to say both that $\underline{F}$ is orthogonal and that it does not vary.

First differentiation. Differentiate (3.113). Both sides have the form $\underline{a} \cdot \underline{a}$, whose differential is $d(\underline{a} \cdot \underline{a}) = d\underline{a} \cdot \underline{a} + \underline{a} \cdot d\underline{a} = 2\underline{a} \cdot d\underline{a}$, the two

terms being equal because the dot product is symmetric. The factor 2 appears on both sides and cancels, leaving

$$[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] \cdot d[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] = (\underline{Y} - \underline{X}) \cdot d(\underline{Y} - \underline{X}), \quad (3.114)$$

in which the derivative may be taken with respect to $\underline{X}$ or to $\underline{Y}$; we choose $\underline{Y}$ first.

So fix $\underline{X}$, which makes $\chi_\kappa(\underline{X}, t)$ a constant vector, and let $\underline{Y}$ vary. Then

$$d[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] = d\chi_\kappa(\underline{Y}, t) - 0 = \underline{F}(\underline{Y}, t) d\underline{Y}, \quad d(\underline{Y} - \underline{X}) = d\underline{Y},$$

the first by the definition (2.34) of the deformation gradient at the point $\underline{Y}$. Substituting into (3.114),

$$[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] \cdot \underline{F}(\underline{Y}, t) d\underline{Y} = (\underline{Y} - \underline{X}) \cdot d\underline{Y} \quad (3.115)$$

for all $d\underline{Y}$. Move $\underline{F}$ to the other factor by the definition of the transpose, $\underline{a} \cdot \underline{F}\underline{b} = \underline{F}^t \underline{a} \cdot \underline{b}$, and collect:

$$\left(\underline{F}^t(\underline{Y}, t) [\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] - (\underline{Y} - \underline{X}) \right) \cdot d\underline{Y} = 0 \quad \text{for all } d\underline{Y}.$$

A vector orthogonal to every vector is 0 , so the bracket vanishes:

$$\underline{F}^t(\underline{Y}, t) [\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] = \underline{Y} - \underline{X}. \quad (3.116)$$

Second differentiation. Now reverse the roles: fix $\underline{Y}$, so that $\underline{F}^t(\underline{Y}, t)$ and $\chi_\kappa(\underline{Y}, t)$ are both constant, and differentiate (3.116) with respect to $\underline{X}$. On the left, the only $\underline{X}$ -dependent piece is $-\chi_\kappa(\underline{X}, t)$, whose differential is $-\underline{F}(\underline{X}, t) d\underline{X}$, and the constant tensor $\underline{F}^t(\underline{Y}, t)$ multiplies it. On the right, the only $\underline{X}$ -dependent piece is $-\underline{X}$. Hence

$$-\underline{F}^t(\underline{Y}, t) \underline{F}(\underline{X}, t) d\underline{X} = -d\underline{X} \quad \text{for all } d\underline{X},$$

and cancelling the minus signs and using once more that two tensors agreeing on every vector are equal,

$$\underline{F}^t(\underline{Y}, t) \underline{F}(\underline{X}, t) = \hat{I} \quad \text{for all } \underline{X}, \underline{Y} \in \kappa. \quad (3.117)$$

Putting $\underline{Y} = \underline{X}$,

$$\underline{F}^t(\underline{X}, t) \underline{F}(\underline{X}, t) = \hat{\underline{I}} \quad \text{for all } \underline{X} \in \kappa, \quad (3.118)$$

so $\underline{F}(\underline{X}, t)$ is orthogonal at every $\underline{X}$ and $\underline{F}^t = \underline{F}^{-1}$. Substituting this back into (3.117) gives $\underline{F}^{-1}(\underline{Y}, t) \underline{F}(\underline{X}, t) = \hat{\underline{I}}$, i.e.

$$\underline{F}(\underline{Y}, t) = \underline{F}(\underline{X}, t) \quad \text{for all } \underline{X}, \underline{Y} \in \kappa, \quad (3.119)$$

so $\underline{F}$ does not depend on $\underline{X}$ at all:

$$\underline{F}(\underline{X}, t) = \underline{Q}(t), \quad (3.120)$$

a uniform orthogonal tensor, and a rotation if κ is occupiable, by (2.50). Rigid motions are therefore homogeneous, and (3.100) gives

$$\underline{x} = \underline{\chi}_\kappa(\underline{X}, t) = \underline{Q}(t) \underline{X} + \underline{c}(t). \quad (3.121)$$

Conversely, substituting (3.121) into the left side of (3.113) gives $\underline{Q}(\underline{Y} - \underline{X}) \cdot \underline{Q}(\underline{Y} - \underline{X}) = (\underline{Y} - \underline{X}) \cdot \underline{Q}^t \underline{Q}(\underline{Y} - \underline{X}) = (\underline{Y} - \underline{X}) \cdot (\underline{Y} - \underline{X})$, so (3.121) is necessary *and* sufficient for rigidity.

The polar decomposition is trivial in this case:

$$\underline{R} = \underline{Q}, \quad \underline{U} = \hat{\underline{I}}, \quad \underline{V} = \underline{I}, \quad (3.122)$$

and of course $\underline{C} = \hat{\underline{I}}$ and $\underline{B} = \underline{I}$. Uniqueness makes this immediate: $\underline{Q} = \underline{Q} \hat{\underline{I}}$ is a factorisation of the required type, hence *the* one.

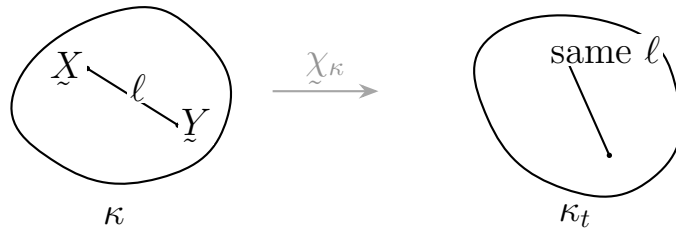


Figure 3.7: Rigid-body motion. Every material distance is preserved, so the body may translate and turn but cannot change shape. The computation (3.113) to (3.120) shows that this forces $\underline{F}$ to be a uniform rotation, hence $\underline{U} = \hat{\underline{I}}$ and $\underline{C} = \hat{\underline{I}}$: no stretching anywhere.

REMARK 3.49 (why this is the theorem the whole chapter was for). (3.122) says: rigid motion $\iff \underline{U} = \underline{\hat{I}} \iff \underline{C} = \underline{\hat{I}} \iff \underline{E} = \underline{\hat{O}}$. The strain tensors vanish exactly on motions in which nothing happens to the material. That is the property one demands of a strain measure, and it is why $\underline{E} = \frac{1}{2}(\underline{C} - \underline{\hat{I}})$ was defined with the $\underline{\hat{I}}$ subtracted. Note what is *not* true: $\underline{F} = \underline{I}$ is not the condition for rigidity, since $\underline{F} = \underline{Q}(t)$ with $\underline{Q} \neq \underline{I}$ is rigid too. A constitutive law written in terms of $\underline{F}$ would therefore assign energy to a body that is merely being carried around; a law written in terms of $\underline{C}$ cannot. This is the whole reason for §3.4.

3.8 Problems

All nine problems of the chapter are solved below in the book's order and notation. They fall into three groups, and it helps to know which is which before starting.

Problems 1, 4, 5, 6 are further worked deformations, in the style of §3.5: write down $\underline{x}$, differentiate, read off $\underline{F}$, then extract what is asked. The recipe of §3.5 applies verbatim, and rules (R1)–(R4) are used throughout. Problem 1 adds area change through Piola–Nanson, Problem 4 is spherically symmetric rather than cylindrical, Problem 5 is a fourth shear, and Problem 6 is bending.

Problems 2, 7, 8 are algebra, filling gaps that §§3.1–3.4 left open: that products of rotations are rotations, which was used at (3.95); that the invariants are the elementary symmetric functions of the principal values, which was used at (3.17) and in Remark 3.11; and the Cayley–Hamilton theorem, which is what makes Theorem 3.23 and Problem 9 possible.

Problems 3 and 9 are the pay-off. Problem 3 completes torsion, and Problem 9 produces the closed form for $\underline{U}$ that removes eigenvectors from the subject entirely, in two dimensions; Theorem 3.23 did the same in three.

PROBLEM 1 (Triaxial stretch: area change and normals). Consider the deformation $x_1 = \lambda_1 X_1$, $x_2 = \lambda_2 X_2$, $x_3 = \lambda_3 X_3$ with constant λ_i

and $\{\underline{e}_i\} = \{\underline{E}_A\}$. (a) Obtain $\underline{F}$ and state a restriction on the λ_i for the deformation to be physically possible. (b) With $\lambda_3 = \lambda_2$ and the deformation isochoric, consider a plane material surface in κ whose unit normal is inclined at angle ϕ to the 1-axis. Find, in terms of λ_1 and ϕ , the ratio of its deformed area to its undeformed area. (c) What is the unit normal to that surface after deformation?

Solution. **(a)** Differentiating the three relations at fixed t , $dx_i = \lambda_i dX_i$ (no sum), and $dX_A = \hat{E}_A \cdot d\underline{X}$, so

$$d\underline{x} = dx_i \underline{e}_i = \sum_{A=1}^3 \lambda_A \underline{e}_A (\hat{E}_A \cdot d\underline{X}) = \left(\sum_{A=1}^3 \lambda_A \underline{e}_A \otimes \hat{E}_A \right) d\underline{X},$$

whence

$$\underline{F} = \lambda_1 \underline{e}_1 \otimes \hat{E}_1 + \lambda_2 \underline{e}_2 \otimes \hat{E}_2 + \lambda_3 \underline{e}_3 \otimes \hat{E}_3.$$

This is already in the form (3.43) with orthonormal frames on both sides, so if the λ_A are positive they *are* the principal stretches, $\underline{U} = \sum \lambda_A \hat{E}_A \otimes \hat{E}_A$, $\underline{V} = \sum \lambda_A \underline{e}_A \otimes \underline{e}_A$ and $\underline{R} = \underline{1}$. The Jacobian is obtained as follows. First the images of the referential frame, each by (R1) applied to the three dyads:

$$\underline{F} \hat{E}_1 = \lambda_1 \underline{e}_1, \quad \underline{F} \hat{E}_2 = \lambda_2 \underline{e}_2, \quad \underline{F} \hat{E}_3 = \lambda_3 \underline{e}_3,$$

since for instance $\underline{F} \hat{E}_1 = \lambda_1 (\hat{E}_1 \cdot \hat{E}_1) \underline{e}_1 + \lambda_2 (\hat{E}_2 \cdot \hat{E}_1) \underline{e}_2 + \lambda_3 (\hat{E}_3 \cdot \hat{E}_1) \underline{e}_3 = \lambda_1 \underline{e}_1$. Then rule (R4), with the denominator $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = 1$ by Lemma 3.28 and the scalars taken out one slot at a time by (R3):

$$\begin{aligned} J = \det \underline{F} &= \frac{[\underline{F} \hat{E}_1, \underline{F} \hat{E}_2, \underline{F} \hat{E}_3]}{[\hat{E}_1, \hat{E}_2, \hat{E}_3]} = \frac{[\lambda_1 \underline{e}_1, \lambda_2 \underline{e}_2, \lambda_3 \underline{e}_3]}{1} \\ &= \lambda_1 \lambda_2 \lambda_3 [\underline{e}_1, \underline{e}_2, \underline{e}_3] = \lambda_1 \lambda_2 \lambda_3 \cdot 1 = \lambda_1 \lambda_2 \lambda_3. \end{aligned}$$

The restriction on the λ_A

Two requirements, excluding different things.

Invertibility. If $\lambda_1 = 0$ then $\hat{E}_1$ and $2\hat{E}_1$ both map to $\underline{0}$, contradicting (2.1). So $\lambda_A \neq 0$ and $J \neq 0$.

Sign. Along a motion, $\underline{F}(t_0) = \underline{1}$ by (2.79), so $J(t_0) = 1 > 0$. Now $J(t)$ is continuous and never zero, so by the intermediate value theorem it cannot change sign. Hence

$$J = \lambda_1 \lambda_2 \lambda_3 > 0$$

This is weaker than $\lambda_A > 0$. Assuming no signs,

$$\underline{C} = \sum_A \lambda_A^2 \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A, \quad \underline{U} = \sqrt{\underline{C}} = \sum_A |\lambda_A| \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A,$$

the positive root by Theorem 3.16, and with $s_A = \text{sgn } \lambda_A$,

$$\underline{R} = \underline{F} \underline{U}^{-1} = \sum_A \frac{\lambda_A}{|\lambda_A|} \underline{e}_A \otimes \hat{\underline{E}}_A = \sum_A s_A \underline{e}_A \otimes \hat{\underline{E}}_A,$$

$$\det \underline{R} = s_1 s_2 s_3 = \text{sgn } J.$$

So $\underline{R}$ is a rotation exactly when $J > 0$. The four sign patterns:

(s_A)	J	$\underline{R}$	
$(+, +, +)$	> 0	$\underline{1}$	pure stretch
$(-, +, +)$	< 0	$\text{diag}(-1, 1, 1)$	mirror, excluded
$(-, -, +)$	> 0	$\text{diag}(-1, -1, 1)$	half turn about $\underline{e}_3$, admissible
$(-, -, -)$	< 0	$-\underline{1}$	inversion, excluded

The stretches are $|\lambda_A|$, not λ_A . Parts (b) and (c) speak of stretches, so take $\lambda_A > 0$ below.

(b) Isochoric with $\lambda_3 = \lambda_2$ means $\lambda_1 \lambda_2^2 = 1$, so

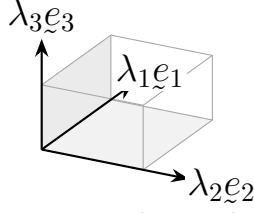
$$\lambda_2 = \lambda_3 = \lambda_1^{-1/2}.$$

Take the normal in the $(1, 2)$ plane without loss of generality, since the 2 and 3 directions are now equivalent:

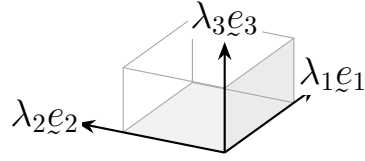
$$\underline{N} = \cos \phi \hat{\underline{E}}_1 + \sin \phi \hat{\underline{E}}_2.$$

The Piola–Nanson formula (2.54) with (2.55) gives $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$ and $\underline{F}^* = J \underline{F}^{-t} = \underline{F}^{-t}$ since $J = 1$. From the dyadic form of $\underline{F}$,

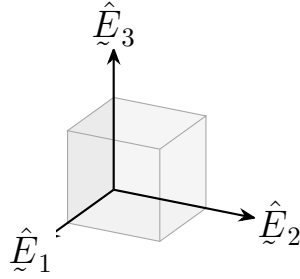
$$\underline{F}^{-1} = \sum_A \lambda_A^{-1} \hat{\underline{E}}_A \otimes \underline{e}_A, \quad \underline{F}^{-t} = \sum_A \lambda_A^{-1} \underline{e}_A \otimes \hat{\underline{E}}_A,$$



$\lambda = (-1.5, 1.1, 0.7)$: $[\tilde{F}\hat{E}_1, \tilde{F}\hat{E}_2, \tilde{F}\hat{E}_3] = -1.155$
 now left handed; $\tilde{R}$ is a mirror $\times$



$\lambda = (-1.5, -1.1, 0.7)$: $[\tilde{F}\hat{E}_1, \tilde{F}\hat{E}_2, \tilde{F}\hat{E}_3] = +1.155$
 still right handed; $\tilde{R}$ is the half turn about e_3 $\checkmark$



κ : the unit cube,
 $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = +1$

Figure E: Why the restriction is $\lambda_1\lambda_2\lambda_3 > 0$ and not $\lambda_A > 0$. Bottom, the reference cube with its right-handed frame. Middle, two of the λ 's negative: the frame is turned through π about e_3 but is still right handed, the box product is still positive, and the deformation is a genuine one, a pure stretch followed by a half turn. Top, one λ negative: the frame has become left handed, the box product is negative, and $\tilde{R}$ is a mirror. No continuous motion starting from $\tilde{F} = \underline{1}$ can reach it, because J would have to pass through zero and $\tilde{F}$ would cease to be invertible at that instant.

the first because

$$\left(\sum_A \lambda_A^{-1} \hat{E}_A \otimes \underline{e}_A \right) \left(\sum_B \lambda_B \underline{e}_B \otimes \hat{E}_B \right) = \sum_{A,B} \lambda_A^{-1} \lambda_B \delta_{AB} \hat{E}_A \otimes \hat{E}_B = \hat{I}.$$

Therefore

$$\underline{F}^* \underline{N} = \lambda_1^{-1} \cos \phi \underline{e}_1 + \lambda_2^{-1} \sin \phi \underline{e}_2,$$

and taking the length of both sides of $\underline{n} da = \underline{F}^* \underline{N} dA$, with $|\underline{n}| = 1$,

$$\frac{da}{dA} = |\underline{F}^* \underline{N}| = \sqrt{\lambda_1^{-2} \cos^2 \phi + \lambda_2^{-2} \sin^2 \phi} = \sqrt{\lambda_1^{-2} \cos^2 \phi + \lambda_1 \sin^2 \phi}$$

where $\lambda_2^{-2} = (\lambda_1^{-1/2})^{-2} = \lambda_1$ was used.

Checks. At $\lambda_1 = 1$ the deformation is the identity and the formula gives $\sqrt{\cos^2 \phi + \sin^2 \phi} = 1$. At $\phi = 0$ the surface is normal to the stretching axis and the ratio is λ_1^{-1} , which is right: that face has area $\lambda_2 \lambda_3 = \lambda_1^{-1}$ times its original. At $\phi = \pi/2$ the ratio is $\lambda_1^{1/2} = \lambda_1 \lambda_3 = \lambda_1 \lambda_1^{-1/2}$, also right.

(c) Dividing $\underline{F}^* \underline{N}$ by its length,

$$\underline{n} = \frac{\lambda_1^{-1} \cos \phi \underline{e}_1 + \lambda_1^{1/2} \sin \phi \underline{e}_2}{\sqrt{\lambda_1^{-2} \cos^2 \phi + \lambda_1 \sin^2 \phi}},$$

using $\lambda_2^{-1} = \lambda_1^{1/2}$. Equivalently, the deformed normal makes an angle ψ with the 1-axis given by

$$\tan \psi = \lambda_1^{3/2} \tan \phi.$$

Note that the normal does *not* rotate the way the material does: for $\lambda_1 > 1$ it turns *away* from the stretching axis, because $\underline{F}^{-t}$, not $\underline{F}$, transports normals. This is the standard trap that Piola–Nanson exists to avoid. $\square$

PROBLEM 2 (Products of orthogonal tensors). Prove that the product of two orthogonal tensors is orthogonal, and that the product of two rotations is a rotation.

Solution. Let $\underline{Q}_1, \underline{Q}_2$ be orthogonal, meaning $\underline{Q}_k^t \underline{Q}_k = \underline{I}$. Using $(\underline{AB})^t = \underline{B}^t \underline{A}^t$ from Problem 20 of Chapter 1 and associativity,

$$(\underline{Q}_1 \underline{Q}_2)^t (\underline{Q}_1 \underline{Q}_2) = \underline{Q}_2^t \underline{Q}_1^t \underline{Q}_1 \underline{Q}_2 = \underline{Q}_2^t \underline{I} \underline{Q}_2 = \underline{Q}_2^t \underline{Q}_2 = \underline{I},$$

so $\underline{Q}_1 \underline{Q}_2$ is orthogonal.

Two gaps to close. First, $\underline{Q}^t \underline{Q} = \underline{I}$ implies $\underline{Q} \underline{Q}^t = \underline{I}$: if $\underline{Q} \underline{a} = \underline{0}$ then $\underline{a} = \underline{Q}^t \underline{Q} \underline{a} = \underline{0}$, so $\underline{Q}$ is injective, hence bijective in finite dimensions, hence $\underline{Q}^{-1} = \underline{Q}^t$ and $\underline{Q} \underline{Q}^t = \underline{I}$. Second, $\det \underline{Q}^t \det \underline{Q} = \det \underline{I} = 1$ with $\det \underline{Q}^t = \det \underline{Q}$ gives $(\det \underline{Q})^2 = 1$, so $\det \underline{Q} = \pm 1$.

What orthogonality means. The three statements

$$\underline{Q}^t \underline{Q} = \underline{I}, \quad \underline{Q} \underline{a} \cdot \underline{Q} \underline{b} = \underline{a} \cdot \underline{b} \quad \forall \underline{a}, \underline{b}, \quad |\underline{Q} \underline{a}| = |\underline{a}| \quad \forall \underline{a}$$

are equivalent. The first gives the second by $\underline{Q} \underline{a} \cdot \underline{Q} \underline{b} = \underline{a} \cdot \underline{Q}^t \underline{Q} \underline{b} = \underline{a} \cdot \underline{b}$; the second gives the third at $\underline{b} = \underline{a}$; and the third gives the second by polarisation,

$$\underline{a} \cdot \underline{b} = \frac{1}{2}(|\underline{a} + \underline{b}|^2 - |\underline{a}|^2 - |\underline{b}|^2),$$

each term of which is preserved. The second then gives the first, since $\underline{a} \cdot (\underline{Q}^t \underline{Q} - \underline{I}) \underline{b} = 0$ for all $\underline{a}, \underline{b}$. So a product of orthogonal tensors is orthogonal for a reason needing no algebra: doing two length-preserving maps in succession preserves lengths.

A *rotation* is an orthogonal tensor with $\det = +1$. By Problem 21(b) of Chapter 1,

$$\det(\underline{Q}_1 \underline{Q}_2) = (\det \underline{Q}_1)(\det \underline{Q}_2) = (+1)(+1) = +1,$$

so a product of rotations is a rotation.

The version actually used at (3.95) is the two-point one: if $\underline{Q}_1 \in \text{Orth}(\hat{\mathbb{E}}^3, \mathbb{E}^3)$ and $\underline{Q}_2 \in \text{Orth}(\hat{\mathbb{E}}^3, \hat{\mathbb{E}}^3)$ then $\underline{Q}_1 \underline{Q}_2 \in \text{Orth}(\hat{\mathbb{E}}^3, \mathbb{E}^3)$. The computation is identical, the only change being that $\underline{Q}_2^t \underline{Q}_2 = \hat{\underline{I}}$ and $\underline{Q}_1^t \underline{Q}_1 = \underline{I}$, so the chain reads $(\underline{Q}_1 \underline{Q}_2)^t (\underline{Q}_1 \underline{Q}_2) = \underline{Q}_2^t \hat{\underline{I}} \underline{Q}_2 = \underline{I}$; and the determinant statement uses the extension recorded in Remark 3.20. $\square$

PROBLEM 3 (Torsion: Cauchy–Green tensors and principal stretches). Evaluate the left and right Cauchy–Green tensors for the torsion problem. Find the principal stretches in terms of the twist τ . Verify that azimuthal shear and torsion are isochoric.

Solution. From (3.88) and (3.97), $\underline{\tilde{F}} = \underline{\tilde{Q}}\hat{G} = \underline{\tilde{G}}\underline{\tilde{Q}}$ with

$$\hat{G} = \hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3, \quad \underline{\tilde{G}} = \underline{\tilde{I}} + \gamma \underline{\tilde{e}}_\theta \otimes \underline{\tilde{e}}_3, \quad \gamma = \gamma(R) = R\tau.$$

Since $\underline{\tilde{Q}}^t \underline{\tilde{Q}} = \underline{\tilde{I}}$ and $\underline{\tilde{Q}} \underline{\tilde{Q}}^t = \underline{\tilde{I}}$, exactly as in (3.91),

$$\underline{\tilde{C}} = \hat{G}^t \underline{\tilde{Q}}^t \underline{\tilde{Q}} \hat{G} = \hat{G}^t \hat{G}, \quad \underline{\tilde{B}} = \underline{\tilde{G}} \underline{\tilde{Q}} \underline{\tilde{Q}}^t \underline{\tilde{G}}^t = \underline{\tilde{G}} \underline{\tilde{G}}^t.$$

With $\hat{G}^t = \hat{I} + \gamma \hat{E}_3 \otimes \hat{E}_\Theta$,

$$\begin{aligned} \underline{\tilde{C}} &= (\hat{I} + \gamma \hat{E}_3 \otimes \hat{E}_\Theta)(\hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3) \\ &= \hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3 + \gamma \hat{E}_3 \otimes \hat{E}_\Theta + \gamma^2 (\hat{E}_\Theta \cdot \hat{E}_\Theta) \hat{E}_3 \otimes \hat{E}_3 \\ &= \hat{I} + \gamma [\hat{E}_\Theta(\Theta) \otimes \hat{E}_3 + \hat{E}_3 \otimes \hat{E}_\Theta(\Theta)] + \gamma^2 \hat{E}_3 \otimes \hat{E}_3, \end{aligned}$$

and in the same way

$$\underline{\tilde{B}} = \underline{\tilde{I}} + \gamma [\underline{\tilde{e}}_\theta(\theta) \otimes \underline{\tilde{e}}_3 + \underline{\tilde{e}}_3 \otimes \underline{\tilde{e}}_\theta(\theta)] + \gamma^2 \underline{\tilde{e}}_\theta(\theta) \otimes \underline{\tilde{e}}_\theta(\theta).$$

Note again the shift of the squared term from $\hat{E}_3$ in $\underline{\tilde{C}}$ to $\underline{\tilde{e}}_\theta$ in $\underline{\tilde{B}}$, exactly as in Remark 3.31.

Principal stretches. In the orthonormal basis $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ the matrix of $\underline{\tilde{C}}$ is

$$(C_{AB}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \gamma \\ 0 & \gamma & 1 + \gamma^2 \end{pmatrix},$$

which is (3.55) with the roles of the axes permuted: the shear acts on the pair $(\hat{E}_\Theta, \hat{E}_3)$ and $\hat{E}_R$ is inert. Hence one principal stretch is 1, with principal direction $\hat{E}_R$, and the other two solve the same quadratic as before. Reading (3.60) with $\gamma = R\tau$,

$$\lambda_1 = \frac{1}{2}(R\tau + \sqrt{4 + R^2\tau^2}), \quad \lambda_2 = \lambda_1^{-1}, \quad \lambda_3 = 1.$$

The stretches depend on R : fibres near the axis are barely deformed, fibres near the outer surface most. At $R = 0$ all three are 1, which is right, since the axis of a twisted bar is not deformed at all.

Isochoric. For torsion, by Problem 21(f) of Chapter 1,

$$\det \hat{G} = \det(\hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3) = 1 + \gamma \hat{E}_\Theta \cdot \hat{E}_3 = 1 + 0 = 1,$$

and $\det \underline{\underline{F}} = (\det \underline{\underline{Q}})(\det \underline{\underline{\hat{G}}}) = 1 \cdot 1 = 1$. For azimuthal shear the same computation with (3.88) gives $\det \underline{\underline{\hat{G}}} = 1 + \gamma \underline{\underline{\hat{E}}}_\Theta \cdot \underline{\underline{\hat{E}}}_R = 1$. Both deformations are isochoric.

An independent check: $\det \underline{\underline{C}} = (\det \underline{\underline{F}})^2$ should be 1. Expanding the matrix above along the first row, $\det(C_{AB}) = 1 \cdot [(1)(1 + \gamma^2) - \gamma^2] = 1$. Consistent.

Geometrically the reason is the one visible in Remark 3.45: every shear $\underline{\underline{\hat{I}}} + \gamma \underline{\underline{a}} \otimes \underline{\underline{b}}$ with $\underline{\underline{a}} \perp \underline{\underline{b}}$ slides planes over one another without moving any of them closer, so no volume can change. $\square$

PROBLEM 4 (Spherically symmetric deformation). Consider the static deformation $\underline{\underline{x}} = \underline{\underline{\chi}}_\kappa(\underline{\underline{X}}) = \lambda(R)\underline{\underline{1}}\underline{\underline{X}}$, where $R = |\underline{\underline{X}}|$ and $\lambda(R)$ is given. (a) Describe what it does to a spherical annulus $A \leq R \leq B$. (b) Derive $\underline{\underline{F}}$ and state the restrictions on λ . (c) Find $\lambda(R)$ if the deformation is isochoric with $\lambda(A) = a/A$.

Solution. **(a)** Since $|\underline{\underline{x}}| = \lambda(R)|\underline{\underline{1}}\underline{\underline{X}}| = \lambda(R)R$ and the direction of $\underline{\underline{x}}$ is that of $\underline{\underline{1}}\underline{\underline{X}}$, every material point moves along its own radius; the sphere $R = \text{const}$ goes to the sphere $r = \lambda(R)R$. The annulus $A \leq R \leq B$ therefore becomes the annulus $a \leq r \leq b$ with $a = \lambda(A)A$ and $b = \lambda(B)B$, and each concentric material sphere remains a concentric sphere. The deformation is a purely radial inflation or contraction, with no rotation and no distortion of the angular structure. It is the natural model for a pressurised spherical shell.

Not a similarity: λ depends on R , so $\Delta r/\Delta R$ varies. Every direction perpendicular to $\underline{\underline{N}}$ is equivalent, so there are only two distinct stretches, the second repeated: the repeated-root case of Remark 3.8.

(b) Write $\underline{\underline{N}} = \underline{\underline{X}}/R$, the referential radial unit vector, so $\underline{\underline{1}}\underline{\underline{X}} = R\underline{\underline{1}}\underline{\underline{N}}$. Differentiating $\underline{\underline{x}} = \lambda(R)\underline{\underline{1}}\underline{\underline{X}}$,

$$d\underline{\underline{x}} = \lambda'(R) dR \underline{\underline{1}}\underline{\underline{X}} + \lambda(R) \underline{\underline{1}} d\underline{\underline{X}}.$$

The increment dR follows from $R^2 = \underline{\underline{X}} \cdot \underline{\underline{X}}$: differentiating, $2R dR = 2\underline{\underline{X}} \cdot d\underline{\underline{X}}$, so $dR = (\underline{\underline{X}}/R) \cdot d\underline{\underline{X}} = \underline{\underline{N}} \cdot d\underline{\underline{X}}$. Hence

$$d\underline{\underline{x}} = \lambda'R(\underline{\underline{1}}\underline{\underline{N}})(\underline{\underline{N}} \cdot d\underline{\underline{X}}) + \lambda\underline{\underline{1}} d\underline{\underline{X}} = [\lambda\underline{\underline{1}} + R\lambda'(\underline{\underline{1}}\underline{\underline{N}}) \otimes \underline{\underline{N}}] d\underline{\underline{X}},$$

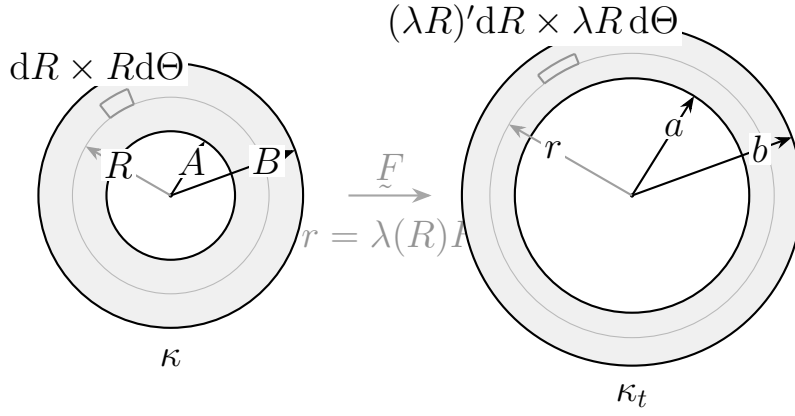


Figure F: Problem 4, on a diametral section. Every point moves along its own radius, so $A \leq R \leq B$ goes to $a \leq r \leq b$ with $a = \lambda(A)A$, $b = \lambda(B)B$. The element shows the two stretches, $dR \mapsto (\lambda R)'dR$ and $R d\Theta \mapsto \lambda R d\Theta$; with the second, equal, tangential stretch, $J = \lambda^2(\lambda R)'$.

so, writing $\underline{n}_0 = \underline{1}N$ for the shifted radial direction,

$$\underline{F} = \lambda(R) \underline{1} + R\lambda'(R) \underline{n}_0 \otimes N.$$

Its action splits cleanly. On the radial direction,

$$\underline{F}N = \lambda\underline{n}_0 + R\lambda'(N \cdot N)\underline{n}_0 = (\lambda + R\lambda')\underline{n}_0 = (\lambda R)'\underline{n}_0,$$

so the radial stretch is $(\lambda R)' = dr/dR$ with $r = \lambda R$; on any $\underline{M} \perp N$,

$$\underline{F}\underline{M} = \lambda\underline{1}\underline{M} + R\lambda'(N \cdot \underline{M})\underline{n}_0 = \lambda\underline{1}\underline{M},$$

so the hoop stretch is λ , twice. The deformation gradient is therefore already in the form (3.43), with

$$\underline{U} = (\lambda R)'\underline{N} \otimes \underline{N} + \lambda(\hat{\underline{I}} - \underline{N} \otimes \underline{N}), \quad \underline{R} = \underline{1},$$

using $\hat{\underline{I}} - \underline{N} \otimes \underline{N} = \mathbb{P}_{(N)}$, the projection of Problem 10 of Chapter 1 onto the tangent plane of the sphere. For the Jacobian, take as test triad $\{\underline{N}, \underline{M}_1, \underline{M}_2\}$, any right-handed orthonormal frame whose first member is the radial direction, so that $\underline{M}_1, \underline{M}_2$ are tangent to the sphere. The images were just computed: $\underline{F}\underline{N} = (\lambda R)'\underline{n}_0$, $\underline{F}\underline{M}_1 = \lambda\underline{1}\underline{M}_1$, $\underline{F}\underline{M}_2 = \lambda\underline{1}\underline{M}_2$. Since $\underline{1}$ is an isometry with $\det \underline{1} = 1$, the triad $\{\underline{n}_0, \underline{1}\underline{M}_1, \underline{1}\underline{M}_2\}$

is again right-handed orthonormal, so both box products in (R4) equal 1 by Lemma 3.28, and pulling out one scalar per slot,

$$\begin{aligned} J = \det \underline{\underline{F}} &= \frac{[(\lambda R)' \underline{\underline{n}}_0, \lambda \underline{\underline{1}} \underline{\underline{M}}_1, \lambda \underline{\underline{1}} \underline{\underline{M}}_2]}{[\underline{\underline{N}}, \underline{\underline{M}}_1, \underline{\underline{M}}_2]} = \frac{(\lambda R)' \lambda \lambda \cdot 1}{1} \\ &= \lambda^2 (\lambda R)' = \frac{r^2 r'}{R^2}, \quad r = \lambda R, \end{aligned}$$

the last form because $r^2/R^2 = \lambda^2$ and $r' = (\lambda R)'$. Physical possibility requires $J > 0$, hence

$$\lambda(R) > 0 \quad \text{and} \quad (\lambda R)' > 0,$$

i.e. $r(R)$ must be positive and strictly increasing: distinct material spheres must remain distinct and must not turn inside out.

(c) Isochoric means $J = 1$, i.e. $r^2 r' = R^2$, i.e. $\frac{1}{3}(r^3)' = \frac{1}{3}(R^3)'$. Integrating,

$$r^3 - R^3 = \text{const.}$$

The condition $\lambda(A) = a/A$ says $r(A) = \lambda(A)A = a$, so the constant is $a^3 - A^3$ and

$$r^3 = R^3 + a^3 - A^3, \quad \text{that is} \quad \lambda(R) = \frac{(R^3 + a^3 - A^3)^{1/3}}{R}.$$

Checks. $\lambda(A) = (A^3 + a^3 - A^3)^{1/3}/A = a/A$ as required. If $a = A$ then $\lambda \equiv 1$ and the deformation is the identity, as it must be, since an isochoric radial deformation that fixes the inner surface fixes everything. If $a > A$ the shell is inflated, and $\lambda(R) \rightarrow 1$ as $R \rightarrow \infty$: the disturbance dies away, which is the familiar statement that an incompressible medium transmits a cavity expansion as a $1/R^3$ perturbation. $\square$

PROBLEM 5 (Telescopic, or anti-plane, shear). Consider $\underline{\underline{x}} = \underline{\underline{\chi}}_\kappa(\underline{\underline{X}}, t) = \underline{\underline{1}}\underline{\underline{X}} + w(R, t)\underline{\underline{e}}_3$ with $\underline{\underline{X}} = R\hat{\underline{\underline{E}}}_R(\Theta) + Z\hat{\underline{\underline{E}}}_3$. (a) Show that $\underline{\underline{F}} = \underline{\underline{1}} + \gamma(R, t)\underline{\underline{e}}_3 \otimes \hat{\underline{\underline{E}}}_R(\Theta)$ with $\gamma = \partial w / \partial R$, derive $\underline{\underline{C}}$ and $\underline{\underline{B}}$, and justify the terminology. (b) Find the stretch μ and current tangent $\underline{\underline{m}}$ of the material curve with $\underline{\underline{M}} = \frac{\sqrt{2}}{2}[\hat{\underline{\underline{E}}}_\Theta + \hat{\underline{\underline{E}}}_3]$, and the same for $\underline{\underline{M}} = \frac{\sqrt{2}}{2}[\hat{\underline{\underline{E}}}_R + \hat{\underline{\underline{E}}}_\Theta]$.

Solution. (a) Differentiating at fixed t and using $dR = \hat{E}_R \cdot dX$,

$$d\tilde{x} = \underline{1} d\tilde{X} + \frac{\partial w}{\partial R} dR \underline{e}_3 = [\underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R(\Theta)] d\tilde{X}, \quad \gamma = \frac{\partial w}{\partial R},$$

so $\tilde{F} = \underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R(\Theta)$ as claimed. Choose the shifter so that $\underline{1}\hat{E}_3 = \underline{e}_3$ and $\underline{1}\hat{E}_R = \underline{e}_r$, which is the alignment of (2.79).

For $\tilde{C}$, transpose and multiply, using Problems 17(b),(c) and $\underline{1}^t \underline{e}_3 = \hat{E}_3$:

$$\begin{aligned} \tilde{C} = \tilde{F}^t \tilde{F} &= (\underline{1}^t + \gamma \hat{E}_R \otimes \underline{e}_3)(\underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R) \\ &= \hat{I} + \gamma (\underline{1}^t \underline{e}_3) \otimes \hat{E}_R + \gamma \hat{E}_R \otimes (\underline{1}^t \underline{e}_3) + \gamma^2 (\underline{e}_3 \cdot \underline{e}_3) \hat{E}_R \otimes \hat{E}_R \\ &= \hat{I} + \gamma [\hat{E}_3 \otimes \hat{E}_R(\Theta) + \hat{E}_R(\Theta) \otimes \hat{E}_3] + \gamma^2 \hat{E}_R(\Theta) \otimes \hat{E}_R(\Theta), \end{aligned}$$

and similarly, with $\underline{1}\hat{E}_R = \underline{e}_r$,

$$\begin{aligned} \tilde{B} = \tilde{F} \tilde{F}^t &= (\underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R)(\underline{1}^t + \gamma \hat{E}_R \otimes \underline{e}_3) \\ &= \hat{I} + \gamma (\underline{1}\hat{E}_R) \otimes \underline{e}_3 + \gamma \underline{e}_3 \otimes (\underline{1}\hat{E}_R) + \gamma^2 \underline{e}_3 \otimes \underline{e}_3 \\ &= \hat{I} + \gamma [\underline{e}_r(\theta) \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_r(\theta)] + \gamma^2 \underline{e}_3 \otimes \underline{e}_3. \end{aligned}$$

This is again a simple shear of amount $\gamma(R, t)$, now on the orthonormal pair $(\hat{E}_3, \hat{E}_R)$, so everything in Remark 3.45 applies; in particular $\det \tilde{F} = 1 + \gamma \underline{e}_3 \cdot \underline{1}\hat{E}_R = 1 + \gamma \underline{e}_3 \cdot \underline{e}_r = 1$ by Problem 21(g) of Chapter 1, so the deformation is isochoric.

Terminology. A material point at (R, Θ, Z) moves to $(R, \Theta, Z + w(R))$. Its radial and azimuthal coordinates do not change at all, and the displacement is purely axial, i.e. purely *out of* the (R, Θ) plane: hence *anti-plane* shear. Since w depends on R alone, each concentric material cylinder $R = \text{const}$ slides rigidly along the axis by a different amount, exactly as the sections of a telescope slide over one another: hence *telescopic* shear.

(b) Both computations use (2.67), $\mu \tilde{m} = \tilde{F} \tilde{M}$, with $\tilde{F} \hat{E}_R = \underline{e}_r + \gamma \underline{e}_3$, $\tilde{F} \hat{E}_\Theta = \underline{e}_\theta$, $\tilde{F} \hat{E}_3 = \underline{e}_3$, which follow from $\tilde{F} = \underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R$ and $\hat{E}_R \cdot \hat{E}_\Theta = \hat{E}_R \cdot \hat{E}_3 = 0$, $\hat{E}_R \cdot \hat{E}_R = 1$.

First curve, $\tilde{M} = \frac{\sqrt{2}}{2}(\hat{E}_\Theta + \hat{E}_3)$:

$$\tilde{F} \tilde{M} = \frac{\sqrt{2}}{2}(\tilde{F} \hat{E}_\Theta + \tilde{F} \hat{E}_3) = \frac{\sqrt{2}}{2}(\underline{e}_\theta + \underline{e}_3), \quad \mu = |\tilde{F} \tilde{M}| = \frac{\sqrt{2}}{2} \sqrt{1+1} = 1,$$

so $\mu = 1$ and $\underline{m} = \frac{\sqrt{2}}{2}(\underline{e}_\theta + \underline{e}_3)$: *this helix does not stretch at all*, whatever γ . The reason is visible in $\underline{C}$: the shear couples $\hat{\underline{E}}_R$ to $\hat{\underline{E}}_3$, and this $\underline{M}$ has no $\hat{\underline{E}}_R$ component. Confirming with (2.83),

$$\underline{C}\underline{M} = \frac{\sqrt{2}}{2}(\hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 + \gamma\hat{\underline{E}}_R), \quad \mu^2 = \underline{M} \cdot \underline{C}\underline{M} = \frac{1}{2}(1 + 1 + 0) = 1. \checkmark$$

Second curve, $\underline{M} = \frac{\sqrt{2}}{2}(\hat{\underline{E}}_R + \hat{\underline{E}}_\Theta)$:

$$\underline{F}\underline{M} = \frac{\sqrt{2}}{2}(\underline{e}_r + \gamma\underline{e}_3 + \underline{e}_\theta), \quad \mu = \frac{\sqrt{2}}{2}\sqrt{1 + \gamma^2 + 1} = \sqrt{1 + \frac{\gamma^2}{2}},$$

and

$$\underline{m} = \frac{\underline{e}_r(\theta) + \underline{e}_\theta(\theta) + \gamma\underline{e}_3}{\sqrt{2 + \gamma^2}}.$$

Confirming with $\underline{C}$: $\underline{C}\hat{\underline{E}}_R = (1 + \gamma^2)\hat{\underline{E}}_R + \gamma\hat{\underline{E}}_3$ and $\underline{C}\hat{\underline{E}}_\Theta = \hat{\underline{E}}_\Theta$, so

$$\begin{aligned} \mu^2 &= \underline{M} \cdot \underline{C}\underline{M} = \frac{1}{2}(\hat{\underline{E}}_R + \hat{\underline{E}}_\Theta) \cdot [(1 + \gamma^2)\hat{\underline{E}}_R + \gamma\hat{\underline{E}}_3 + \hat{\underline{E}}_\Theta] \\ &= \frac{1}{2}[(1 + \gamma^2) + 1] = 1 + \frac{\gamma^2}{2} \quad \checkmark \end{aligned}$$

So the curve inclined at 45° between the radial and azimuthal directions does stretch, and by a factor that grows with the local shear. $\square$

PROBLEM 6 (Flexure of a block). Consider the static deformation $\underline{X} \rightarrow \underline{x}$ with $\underline{X} = X_A \hat{\underline{E}}_A$ and $\underline{x} = r \underline{e}_r(\theta) + z \underline{e}_3$, where $r = f(X_1) > 0$, $\theta = g(X_2)$, $z = Z = X_3$. (a) Describe the images of the lines $X_1 = \text{const}$ and $X_2 = \text{const}$ and justify calling this a flexure. (b) Derive $\underline{F}$. (c) Show $\det \underline{F} > 0 \iff ff'g' > 0$. (d) Assuming $f > 0$, $g' > 0$, obtain $\underline{R}, \underline{U}, \underline{V}$ by inspection, and find f, g if the deformation is isochoric.

Solution. **(a)** A line $X_1 = c_1$, along which X_2 varies, maps to the set $r = f(c_1) = \text{const}$ with $\theta = g(X_2)$ sweeping an interval: a *circular arc* of radius $f(c_1)$ centred on the axis. A line $X_2 = c_2$, along which X_1 varies, maps to $\theta = g(c_2) = \text{const}$ with $r = f(X_1)$ varying: a *radial straight segment*. So the two families of coordinate lines of the block go to concentric arcs and radial rays respectively, and a rectangular block is carried into an annular wedge. That is precisely bending: parallel

planes become coaxial cylinders and normal cross-sections remain plane and normal. Hence *flexure*.

Explicitly $(X_1, X_2) \mapsto (r, \theta) = (f(X_1), g(X_2))$, so

lines $X_1 = \text{const} \mapsto \text{arcs } r = \text{const},$

lines $X_2 = \text{const} \mapsto \text{rays } \theta = \text{const}.$

A block $0 \leq X_1 \leq T, 0 \leq X_2 \leq L$ becomes the wedge $f(0) \leq r \leq f(T), g(0) \leq \theta \leq g(L)$. The bend angle is $g(L) - g(0)$; the ring closes when it equals 2π .

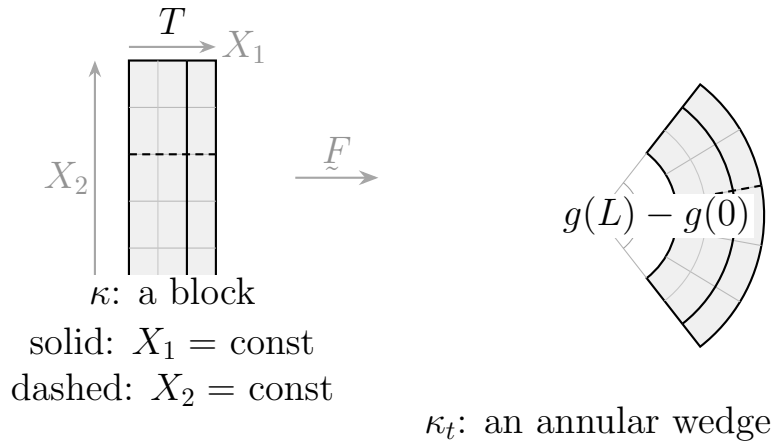


Figure G: Problem 6: flexure. The through-thickness coordinate X_1 becomes the radius, $r = f(X_1)$, so lines $X_1 = \text{const}$ (solid) become concentric arcs; the along-length coordinate X_2 becomes the azimuth, $\theta = g(X_2)$, so lines $X_2 = \text{const}$ (dashed) become radial rays. A block of thickness T and length L is wrapped into a wedge subtending $g(L) - g(0)$. Cross-sections stay plane and normal to the arcs, which is why $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$ and $\{\hat{\underline{E}}_A\}$ pair off in $\underline{F}$ with no shear between them.

(b) In polar coordinates $d\tilde{x} = dr \underline{e}_r + r d\theta \underline{e}_\theta + dz \underline{e}_3$, with $dr = f'(X_1)dX_1$, $d\theta = g'(X_2)dX_2$, $dz = dX_3$, and $dX_A = \hat{\underline{E}}_A \cdot d\tilde{X}$. Therefore

$$d\tilde{x} = f' \underline{e}_r (\hat{\underline{E}}_1 \cdot d\tilde{X}) + f g' \underline{e}_\theta (\hat{\underline{E}}_2 \cdot d\tilde{X}) + \underline{e}_3 (\hat{\underline{E}}_3 \cdot d\tilde{X}),$$

and since $d\tilde{X}$ is arbitrary,

$$\underline{F} = f'(X_1) \underline{e}_r \otimes \hat{\underline{E}}_1 + f(X_1) g'(X_2) \underline{e}_\theta \otimes \hat{\underline{E}}_2 + \underline{e}_3 \otimes \hat{\underline{E}}_3$$

with the primes denoting derivatives with respect to the indicated argument.

(c) Apply rule (R4) with the referential frame $\{\hat{E}_1, \hat{E}_2, \hat{E}_3\}$. First its images, each by (R1) on the three dyads of $\tilde{F}$:

$$\begin{aligned}\tilde{F}\hat{E}_1 &= f'(\hat{E}_1 \cdot \hat{E}_1)\underline{e}_r + fg'(\hat{E}_2 \cdot \hat{E}_1)\underline{e}_\theta + (\hat{E}_3 \cdot \hat{E}_1)\underline{e}_3 = f'\underline{e}_r, \\ \tilde{F}\hat{E}_2 &= f' \cdot 0 \cdot \underline{e}_r + fg' \cdot 1 \cdot \underline{e}_\theta + 0 \cdot \underline{e}_3 = fg'\underline{e}_\theta, \\ \tilde{F}\hat{E}_3 &= f' \cdot 0 \cdot \underline{e}_r + fg' \cdot 0 \cdot \underline{e}_\theta + 1 \cdot \underline{e}_3 = \underline{e}_3.\end{aligned}$$

The denominator is $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = 1$ by Lemma 3.28. In the numerator, pull out f' from the first slot, then fg' from the second, then evaluate the surviving box product of the right-handed orthonormal polar frame by Lemma 3.29:

$$\begin{aligned}\det \tilde{F} &= \frac{[f'\underline{e}_r, fg'\underline{e}_\theta, \underline{e}_3]}{[\hat{E}_1, \hat{E}_2, \hat{E}_3]} = \frac{f'[\underline{e}_r, fg'\underline{e}_\theta, \underline{e}_3]}{1} \\ &= f'(fg')[\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = f'(fg') \cdot 1 = ff'g' .\end{aligned}$$

Hence $\det \tilde{F} > 0 \iff ff'g' > 0$. The three factors are the three stretches: f' radially, fg' azimuthally, 1 axially, and their product is the volume ratio exactly as in (3.17).

(d) With $f > 0$ and $g' > 0$, part (c) forces $f' > 0$, so all three coefficients in $\tilde{F}$ are positive. But then $\tilde{F}$ is already displayed in the form (3.43), $\tilde{F} = \sum_i \lambda_i \underline{v}_i \otimes \underline{u}_i$, with orthonormal referential triad $\{\hat{E}_1, \hat{E}_2, \hat{E}_3\}$ and orthonormal spatial triad $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$. By the uniqueness of the polar decomposition, the factors can be read off:

$$\begin{aligned}\lambda_1 &= f', & \lambda_2 &= fg', & \lambda_3 &= 1, \\ \underline{U} &= f'\hat{E}_1 \otimes \hat{E}_1 + fg'\hat{E}_2 \otimes \hat{E}_2 + \hat{E}_3 \otimes \hat{E}_3, \\ \underline{V} &= f'\underline{e}_r \otimes \underline{e}_r + fg'\underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3, \\ \underline{R} &= \underline{e}_r \otimes \hat{E}_1 + \underline{e}_\theta \otimes \hat{E}_2 + \underline{e}_3 \otimes \hat{E}_3.\end{aligned}$$

That these do the job is checked at once: $\underline{R}\underline{U} = \sum \lambda_i \underline{v}_i \otimes \underline{u}_i = \tilde{F}$ by Problem 17(b), $\underline{U}$ is symmetric with positive principal values, and $\underline{R}$ pairs two right-handed orthonormal triads so $\underline{R}^t \underline{R} = \hat{I}$ and $\det \underline{R} = 1$. Uniqueness (Step 7 of §3.4) says there is nothing else to find.

Isochoric case. $\det \underline{F} = f f' g' = 1$ gives

$$f(X_1) f'(X_1) = \frac{1}{g'(X_2)}.$$

The left side depends only on X_1 , the right only on X_2 , so both equal a constant c , and $c > 0$ since $f, f', g' > 0$. Then

$$g' = \frac{1}{c} \implies g(X_2) = \frac{X_2}{c} + g_0,$$

a linear map of the height coordinate to the azimuth, and

$$f f' = c \implies \frac{1}{2}(f^2)' = c \implies f(X_1) = \sqrt{2cX_1 + d}$$

for a constant d making $f > 0$ on the block. So an isochoric flexure bends the block through an angle proportional to its height and maps the thickness coordinate by a square root, exactly the profile (3.77) found for isochoric radial deformation, as it must be, since both express conservation of area in the cross-section. $\square$

PROBLEM 7 (Invariants in terms of principal values; Cayley–Hamilton). Let $\underline{A}$ be symmetric. (a) Show that $I_1 = \lambda_1 + \lambda_2 + \lambda_3$, $I_2 = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3$, $I_3 = \lambda_1\lambda_2\lambda_3$. (b) Show that $\underline{A}^3 - I_1\underline{A}^2 + I_2\underline{A} - I_3\underline{I} = \underline{O}$.

Solution. **(a)** Use the spectral representation (3.12) throughout, together with $\text{tr}(\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{v}$ from Theorem 1.51 of Chapter 1.

For I_1 ,

$$I_1 = \text{tr } \underline{A} = \text{tr} \left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i \right) = \sum_i \lambda_i \underline{u}_i \cdot \underline{u}_i = \sum_i \lambda_i = \lambda_1 + \lambda_2 + \lambda_3,$$

the trace being linear and each $\underline{u}_i$ a unit vector.

For I_2 we need $\text{tr}(\underline{A}^2)$. By Remark 3.10, $\underline{A}^2 = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i$, so $\text{tr}(\underline{A}^2) = \sum_i \lambda_i^2$. Then, by (3.3)₂,

$$\begin{aligned} I_2 &= \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)] = \frac{1}{2} \left[\left(\sum_i \lambda_i \right)^2 - \sum_i \lambda_i^2 \right] \\ &= \frac{1}{2} \left[\sum_i \lambda_i^2 + 2(\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3) - \sum_i \lambda_i^2 \right] \\ &= \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3, \end{aligned}$$

the square having been expanded in full and the pure squares cancelling.

For I_3 , equation (3.17) already gives $I_3 = \det \underline{A} = \lambda_1 \lambda_2 \lambda_3$.

These three are the elementary symmetric functions of the principal values, which is what one expects: writing (3.2) as $-(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = 0$ and expanding gives $-\lambda^3 + (\sum \lambda_i)\lambda^2 - (\sum_{i < j} \lambda_i \lambda_j)\lambda + \lambda_1 \lambda_2 \lambda_3$, and comparing coefficients with (3.2) reproduces all three at once.

(b) Each principal value satisfies the characteristic equation (3.2), so

$$-\lambda_i^3 + I_1 \lambda_i^2 - I_2 \lambda_i + I_3 = 0, \quad \text{i.e.} \quad \lambda_i^3 - I_1 \lambda_i^2 + I_2 \lambda_i - I_3 = 0$$

for $i = 1, 2, 3$. Now evaluate the tensor expression using Remark 3.10 for the powers and (3.11) for the identity:

$$\begin{aligned} \underline{A}^3 - I_1 \underline{A}^2 + I_2 \underline{A} - I_3 \underline{I} &= \sum_i \lambda_i^3 \underline{u}_i \otimes \underline{u}_i - I_1 \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i \\ &\quad + I_2 \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i - I_3 \sum_i \underline{u}_i \otimes \underline{u}_i \\ &= \sum_i (\lambda_i^3 - I_1 \lambda_i^2 + I_2 \lambda_i - I_3) \underline{u}_i \otimes \underline{u}_i \\ &= \sum_i 0 \cdot \underline{u}_i \otimes \underline{u}_i = \underline{O}. \end{aligned}$$

This is the *Cayley–Hamilton* formula. It holds for non-symmetric tensors too, though the proof above does not cover that case, since it used a spectral representation. It is meaningful only for tensors mapping a space to itself: for a two-point $\underline{F}$ the term $\underline{F}^2$ does not exist, as $\underline{F}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$ and cannot be applied twice, so the formula cannot even be written. $\square$

PROBLEM 8 (Two-dimensional Cayley–Hamilton). Let $\underline{A}$ be symmetric on a two-dimensional space, with spectral decomposition $\underline{A} = \sum_{i=1}^2 \lambda_i \underline{u}_i \otimes \underline{u}_i$. Prove $\underline{A}^2 - (\text{tr } \underline{A})\underline{A} + (\det \underline{A})\underline{I} = \underline{O}$.

Solution. In two dimensions $\underline{I} = \underline{u}_1 \otimes \underline{u}_1 + \underline{u}_2 \otimes \underline{u}_2$, by the same verification as (3.11), and by the argument of Remark 3.10, $\underline{A}^2 = \sum_{i=1}^2 \lambda_i^2 \underline{u}_i \otimes \underline{u}_i$. The two invariants are

$$\text{tr } \underline{A} = \lambda_1 + \lambda_2, \quad \det \underline{A} = \frac{[\underline{A}\underline{u}_1, \underline{A}\underline{u}_2]}{[\underline{u}_1, \underline{u}_2]} = \frac{[\lambda_1 \underline{u}_1, \lambda_2 \underline{u}_2]}{[\underline{u}_1, \underline{u}_2]} = \lambda_1 \lambda_2,$$

where $[\cdot, \cdot]$ is the two-dimensional area form, linear in each slot and alternating, so the scalars come out and the common factor cancels. Substituting,

$$\begin{aligned} \underline{\underline{A}}^2 - (\text{tr } \underline{\underline{A}})\underline{\underline{A}} + (\det \underline{\underline{A}})\underline{\underline{I}} &= \sum_{i=1}^2 \left[\lambda_i^2 - (\lambda_1 + \lambda_2)\lambda_i + \lambda_1\lambda_2 \right] \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i \\ &= \sum_{i=1}^2 (\lambda_i - \lambda_1)(\lambda_i - \lambda_2) \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i. \end{aligned}$$

For $i = 1$ the first factor vanishes and for $i = 2$ the second does, so every term is zero and the sum is $\underline{\underline{O}}$. Explicitly, for $i = 1$: $\lambda_1^2 - \lambda_1^2 - \lambda_2\lambda_1 + \lambda_1\lambda_2 = 0$; for $i = 2$: $\lambda_2^2 - \lambda_1\lambda_2 - \lambda_2^2 + \lambda_1\lambda_2 = 0$. $\square$

REMARK 3.50 (three consequences, used at once). *(i) The characteristic equation.* Feeding $\underline{\underline{u}}_i$ to the identity gives $\lambda_i^2 - (\text{tr } \underline{\underline{A}})\lambda_i + \det \underline{\underline{A}} = 0$, so the principal values are

$$\lambda_{1,2} = \frac{1}{2} \left(\text{tr } \underline{\underline{A}} \pm \sqrt{(\text{tr } \underline{\underline{A}})^2 - 4 \det \underline{\underline{A}}} \right),$$

the discriminant being $(\lambda_1 - \lambda_2)^2 \geq 0$, hence real roots, as Theorem 3.4 promises.

(ii) The inverse without cofactors. If $\det \underline{\underline{A}} \neq 0$, multiply the identity by $\underline{\underline{A}}^{-1}$:

$$\underline{\underline{A}} - (\text{tr } \underline{\underline{A}})\underline{\underline{I}} + (\det \underline{\underline{A}})\underline{\underline{A}}^{-1} = \underline{\underline{O}} \implies \underline{\underline{A}}^{-1} = \frac{(\text{tr } \underline{\underline{A}})\underline{\underline{I}} - \underline{\underline{A}}}{\det \underline{\underline{A}}}.$$

This is what Problem 9 uses to invert $\underline{\underline{U}}$.

(iii) Every power reduces. $\underline{\underline{A}}^2 = (\text{tr } \underline{\underline{A}})\underline{\underline{A}} - (\det \underline{\underline{A}})\underline{\underline{I}}$, so by induction $\underline{\underline{A}}^n = p_n \underline{\underline{A}} + q_n \underline{\underline{I}}$ for scalars p_n, q_n : in two dimensions no tensor function ever needs more than $\underline{\underline{I}}$ and $\underline{\underline{A}}$. The three-dimensional analogue, Problem 7(b), reduces everything to $\underline{\underline{I}}$, $\underline{\underline{A}}$, $\underline{\underline{A}}^2$, and that is what Theorem 3.23 exploits.

Without the spectral form. The identity also follows from components alone, which shows it does not need symmetry. For $(A_{ij}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$\text{tr } \underline{\underline{A}} = a + d$ and $\det \underline{\underline{A}} = ad - bc$, so

$$\begin{aligned} \underline{\underline{A}}^2 &= \begin{pmatrix} a^2 + bc & b(a + d) \\ c(a + d) & d^2 + bc \end{pmatrix}, & (\text{tr } \underline{\underline{A}})\underline{\underline{A}} &= \begin{pmatrix} a(a + d) & b(a + d) \\ c(a + d) & d(a + d) \end{pmatrix}, \\ (\det \underline{\underline{A}})\underline{\underline{I}} &= \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}. \end{aligned}$$

The off-diagonal entries cancel between the first two. The diagonal ones give

$$\begin{aligned} (1, 1) : & \quad a^2 + bc - a^2 - ad + ad - bc = 0, \\ (2, 2) : & \quad d^2 + bc - ad - d^2 + ad - bc = 0, \end{aligned}$$

so the sum is $\underline{\underline{O}}$.

PROBLEM 9 (The stretch tensor without eigenvectors). Show that the two-dimensional right stretch tensor $\underline{\underline{U}}$ and right Cauchy–Green tensor $\underline{\underline{C}}$, both mapping $\hat{\mathbb{E}}^2$ to itself, satisfy $\underline{\underline{I}}\underline{\underline{U}} = \underline{\underline{C}} + J\hat{\underline{\underline{I}}}$ with $I = \text{tr } \underline{\underline{U}}$, $J = \det \underline{\underline{U}}$; show $J^2 = \det \underline{\underline{C}}$ and $I^2 = \text{tr } \underline{\underline{C}} + 2J$. Apply this to the simple shear $\underline{\underline{F}} = \underline{\underline{1}} + \gamma \underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2$, obtain $\underline{\underline{U}}$, build the three-dimensional stretch and rotation, recover the principal stretches, and derive $\underline{\underline{V}}$ directly. Do the same for azimuthal shear and torsion.

Solution. The identity. Apply Problem 8 to $\underline{\underline{U}}$, which is symmetric and positive definite:

$$\underline{\underline{U}}^2 - (\text{tr } \underline{\underline{U}})\underline{\underline{U}} + (\det \underline{\underline{U}})\hat{\underline{\underline{I}}} = \underline{\underline{O}}.$$

Since $\underline{\underline{U}}^2 = \underline{\underline{C}}$ by definition of the right stretch tensor, and writing $I = \text{tr } \underline{\underline{U}}$, $J = \det \underline{\underline{U}}$,

$$\underline{\underline{I}}\underline{\underline{U}} = \underline{\underline{C}} + J\hat{\underline{\underline{I}}}$$

which determines $\underline{\underline{U}}$ as soon as the two scalars I and J are known, and $I > 0$ because $\underline{\underline{U}}$ is positive definite so $\text{tr } \underline{\underline{U}} = \lambda_1 + \lambda_2 > 0$.

The two scalars are obtained from $\underline{\underline{C}}$ alone. By Problem 21(b) of Chapter 1,

$$\det \underline{\underline{C}} = \det(\underline{\underline{U}}\underline{\underline{U}}) = (\det \underline{\underline{U}})^2 = J^2, \quad \text{so} \quad J = +\sqrt{\det \underline{\underline{C}}}$$

with the positive root because $J = \lambda_1 \lambda_2 > 0$. And

$$\operatorname{tr} \mathcal{C} = \operatorname{tr}(U^2) = \lambda_1^2 + \lambda_2^2 = (\lambda_1 + \lambda_2)^2 - 2\lambda_1 \lambda_2 = I^2 - 2J,$$

so

$$I = +\sqrt{\operatorname{tr} \mathcal{C} + 2J}.$$

Hence $\mathcal{U} = (\mathcal{C} + J\hat{I})/I$ is computed from $\mathcal{C}$ by two square roots of scalars, with no characteristic equation and no eigenvectors. That is the point of the problem.

Verification, in the abstract. That this $\mathcal{U}$ really is $\sqrt{\mathcal{C}}$ can be checked without ever mentioning λ_i . Square it and use Cayley–Hamilton for $\mathcal{C}$, which by Problem 8 reads $\mathcal{C}^2 = (\operatorname{tr} \mathcal{C})\mathcal{C} - (\det \mathcal{C})\hat{I} = (I^2 - 2J)\mathcal{C} - J^2\hat{I}$:

$$\begin{aligned} I^2 \mathcal{U}^2 &= (\mathcal{C} + J\hat{I})^2 = \mathcal{C}^2 + 2J\mathcal{C} + J^2\hat{I} \\ &= [(I^2 - 2J)\mathcal{C} - J^2\hat{I}] + 2J\mathcal{C} + J^2\hat{I} = I^2 \mathcal{C}, \end{aligned}$$

so $\mathcal{U}^2 = \mathcal{C}$. It is symmetric, being a linear combination of the symmetric $\mathcal{C}$ and $\hat{I}$. Its principal values are $(\lambda_i^2 + J)/I$, and with $J = \lambda_1 \lambda_2$, $I = \lambda_1 + \lambda_2$,

$$\frac{\lambda_1^2 + \lambda_1 \lambda_2}{\lambda_1 + \lambda_2} = \frac{\lambda_1(\lambda_1 + \lambda_2)}{\lambda_1 + \lambda_2} = \lambda_1 > 0,$$

and likewise λ_2 ; so $\mathcal{U}$ is positive definite, and by Theorem 3.16 it is *the* square root. Nothing was assumed about $\mathcal{C}$ beyond symmetry and positive definiteness.

Simple shear. With $\mathcal{F} = \mathbf{1} + \gamma \mathbf{e}_1 \otimes \hat{\mathbf{E}}_2$ the two-dimensional part of (3.53) is

$$\mathcal{C} = \hat{I} + \gamma(\hat{\mathbf{E}}_1 \otimes \hat{\mathbf{E}}_2 + \hat{\mathbf{E}}_2 \otimes \hat{\mathbf{E}}_1) + \gamma^2 \hat{\mathbf{E}}_2 \otimes \hat{\mathbf{E}}_2, \quad (C_{AB}) = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix}.$$

Then

$$\begin{aligned} \det \mathcal{C} &= (1)(1 + \gamma^2) - \gamma^2 = 1 \Rightarrow J = 1, \\ \operatorname{tr} \mathcal{C} &= 2 + \gamma^2 \Rightarrow I = \sqrt{2 + \gamma^2 + 2} = \sqrt{4 + \gamma^2}, \end{aligned}$$

and therefore

$$\mathcal{U} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2\hat{I} + \gamma(\hat{\mathbf{E}}_1 \otimes \hat{\mathbf{E}}_2 + \hat{\mathbf{E}}_2 \otimes \hat{\mathbf{E}}_1) + \gamma^2 \hat{\mathbf{E}}_2 \otimes \hat{\mathbf{E}}_2 \right].$$

REMARK 3.51 (a misprint in the problem statement). The book prints the last term as $\gamma^2 \hat{E}_1 \otimes \hat{E}_1$. That cannot be right, and the check is immediate. In matrix form the printed version is $\underline{U} = \frac{1}{I} \begin{pmatrix} 2 + \gamma^2 & \gamma \\ \gamma & 2 \end{pmatrix}$, whose square is

$$\frac{1}{4 + \gamma^2} \begin{pmatrix} (2 + \gamma^2)^2 + \gamma^2 & * \\ * & \gamma^2 + 4 \end{pmatrix} = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix} = \underline{B}^{(2)} \neq \underline{C}.$$

With $\gamma^2 \hat{E}_2 \otimes \hat{E}_2$ instead, $\underline{U} = \frac{1}{I} \begin{pmatrix} 2 & \gamma \\ \gamma & 2 + \gamma^2 \end{pmatrix}$ and

$$\begin{aligned} \underline{U}^2 &= \frac{1}{4 + \gamma^2} \begin{pmatrix} 4 + \gamma^2 & 2\gamma + \gamma(2 + \gamma^2) \\ \gamma(2 + \gamma^2) + 2\gamma & \gamma^2 + (2 + \gamma^2)^2 \end{pmatrix} \\ &= \frac{1}{4 + \gamma^2} \begin{pmatrix} 4 + \gamma^2 & \gamma(4 + \gamma^2) \\ \gamma(4 + \gamma^2) & (4 + \gamma^2)(1 + \gamma^2) \end{pmatrix}, \end{aligned}$$

the last entry because $\gamma^2 + 4 + 4\gamma^2 + \gamma^4 = 4 + 5\gamma^2 + \gamma^4 = (4 + \gamma^2)(1 + \gamma^2)$. Cancelling, $\underline{U}^2 = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix} = \underline{C}$. So the printed expression is in fact $\underline{V}$ with hats attached, and the subscript on the squared term should be 2, not 1.

Principal stretches recovered. The principal values of $\underline{U}$ solve $\lambda^2 - I\lambda + J = 0$ by Problem 8, that is $\lambda^2 - \sqrt{4 + \gamma^2} \lambda + 1 = 0$, so

$$\lambda = \frac{\sqrt{4 + \gamma^2} \pm \sqrt{4 + \gamma^2 - 4}}{2} = \frac{1}{2}(\sqrt{4 + \gamma^2} \pm \gamma),$$

which is (3.60).

Three-dimensional stretch and rotation. The third direction is inert, so

$$\underline{U}^{(3)} = \frac{2\hat{I}^{(2)} + \gamma(\hat{E}_1 \otimes \hat{E}_2 + \hat{E}_2 \otimes \hat{E}_1) + \gamma^2 \hat{E}_2 \otimes \hat{E}_2}{\sqrt{4 + \gamma^2}} + \hat{E}_3 \otimes \hat{E}_3.$$

For the rotation, invert $\underline{U}$ using the same identity: from $\underline{U}^2 - I\underline{U} + J\hat{I} = \hat{Q}$, multiply by $\underline{U}^{-1}$ to get $\underline{U} - I\hat{I} + J\underline{U}^{-1} = \hat{Q}$, i.e.

$$\underline{U}^{-1} = \frac{I\hat{I} - \underline{U}}{J} = I\hat{I} - \underline{U} = \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 + \gamma^2 & -\gamma \\ -\gamma & 2 \end{pmatrix},$$

since $J = 1$ and $I^2 = 4 + \gamma^2$. Then, with $(F_{iA}) = \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix}$,

$$\begin{aligned} (R_{iA}) &= (F_{iB})(U_{BA}^{-1}) = \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 + \gamma^2 & -\gamma \\ -\gamma & 2 \end{pmatrix} \\ &= \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 & \gamma \\ -\gamma & 2 \end{pmatrix}, \end{aligned}$$

the $(1, 1)$ entry being $2 + \gamma^2 - \gamma^2 = 2$ and the $(1, 2)$ entry $-\gamma + 2\gamma = \gamma$. This is orthogonal: each column has length $\sqrt{4 + \gamma^2}/\sqrt{4 + \gamma^2} = 1$, the columns are orthogonal since $2\gamma - 2\gamma = 0$, and $\det = (4 + \gamma^2)/(4 + \gamma^2) = 1$. So $\underline{R}$ is the plane rotation through β with

$$\cos \beta = \frac{2}{\sqrt{4 + \gamma^2}}, \quad \sin \beta = \frac{-\gamma}{\sqrt{4 + \gamma^2}}, \quad \tan \beta = -\frac{\gamma}{2},$$

confirming exactly and to all orders the estimate $\beta \approx -\gamma/2$ of Remark 3.33. Comparing with (3.70) gives $\sin 2\alpha = 2/\sqrt{4 + \gamma^2}$ and $\cos 2\alpha = \gamma/\sqrt{4 + \gamma^2}$, which is consistent with (3.69): at $\gamma = 1$, $\tan \alpha = 2/(1 + \sqrt{5}) = 0.618$, so $\sin 2\alpha = 2(0.618)/(1 + 0.618^2) = 0.894 = 2/\sqrt{5}$ and $\cos 2\alpha = (1 - 0.382)/1.382 = 0.447 = 1/\sqrt{5}$.

The left stretch, directly. The same identity applies to $\underline{V}$, since $\underline{V}^2 = \underline{B}$ and $\underline{B}$ has the same invariants as $\underline{C}$ (indeed $\underline{B} = \underline{R}\underline{C}\underline{R}^t$). Hence $\underline{IV} = \underline{B} + \underline{JI}$ with the same I, J , and from (3.56),

$$\underline{V} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2\underline{I} + \gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1) + \gamma^2 \underline{e}_1 \otimes \underline{e}_1 \right],$$

the squared term now sitting on $\underline{e}_1$, as Remark 3.31 predicts and Remark 3.51 confirms.

Azimuthal shear. By (3.92), in the plane spanned by $\{\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta\}$ with $\gamma = R\phi'$,

$$(C_{AB}) = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix} \quad \text{in the ordered basis } (\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta),$$

so again $\det \underline{C} = 1$, $J = 1$, $\text{tr } \underline{C} = 2 + \gamma^2$, $I = \sqrt{4 + \gamma^2}$, and

$$\underline{U} = \frac{2\hat{\underline{I}} + \gamma(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_R) + \gamma^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R}{\sqrt{4 + \gamma^2}} + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,$$

the $\hat{E}_3$ term restoring the inert third direction. The rotation follows from $\underline{R}_{\hat{G}} = \hat{G}\underline{U}^{-1}$ with $\underline{U}^{-1} = I\hat{I} - \underline{U}$ in the plane, and then $\underline{R} = \underline{Q}\underline{R}_{\hat{G}}$ by (3.95). Note that here γ depends on R , so $\underline{U}$ and $\underline{R}$ do too: the deformation is not homogeneous, and the principal axes turn as one moves outward through the cylinder.

Torsion. By Problem 3, in the plane spanned by $\{\hat{E}_\Theta, \hat{E}_3\}$ with $\gamma = R\tau$,

$$(C_{AB}) = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix} \quad \text{in the ordered basis } (\hat{E}_\Theta, \hat{E}_3),$$

so $J = 1$, $I = \sqrt{4 + R^2\tau^2}$ and

$$\underline{U} = \frac{2\hat{I} + R\tau(\hat{E}_\Theta \otimes \hat{E}_3 + \hat{E}_3 \otimes \hat{E}_\Theta) + R^2\tau^2 \hat{E}_3 \otimes \hat{E}_3}{\sqrt{4 + R^2\tau^2}} + \hat{E}_R \otimes \hat{E}_R,$$

with $\underline{V}$ obtained by the same formula from $\underline{B}$, replacing $\hat{E}_\Theta \rightarrow \underline{e}_\theta$, $\hat{E}_3 \rightarrow \underline{e}_3$ and moving the squared term onto $\underline{e}_\theta$:

$$\underline{V} = \frac{2\underline{I} + R\tau(\underline{e}_\theta \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_\theta) + R^2\tau^2 \underline{e}_\theta \otimes \underline{e}_\theta}{\sqrt{4 + R^2\tau^2}} + \underline{e}_r \otimes \underline{e}_r.$$

The rotation is $\underline{R} = \underline{Q}\underline{R}_{\hat{G}}$ with $\underline{R}_{\hat{G}} = \hat{G}\underline{U}^{-1}$, computed exactly as above; in the $(\hat{E}_\Theta, \hat{E}_3)$ plane its matrix is $\frac{1}{\sqrt{4+\gamma^2}} \begin{pmatrix} 2 & \gamma \\ -\gamma & 2 \end{pmatrix}$ with $\gamma = R\tau$, so the local shear-induced rotation is through $\arctan(-R\tau/2)$, growing linearly with radius near the axis. $\square$

Closing summary

The logical spine of the chapter, in one paragraph. A symmetric tensor is three numbers on three orthogonal axes, (3.12); it is positive definite exactly when the three numbers are positive, Theorem 3.13; a positive definite symmetric tensor has exactly one positive definite symmetric square root, Theorem 3.16. Applying these to $\underline{C} = \underline{F}^t \underline{F}$, which Chapter

2 proved symmetric and positive definite, produces $\underline{U} = \sqrt{\underline{C}}$, and then $\underline{R} := \underline{F}\underline{U}^{-1}$ is forced to be a rotation, (3.37), and the factorisation is unique, Step 7. The numbers λ_i are the stretches of material fibres momentarily aligned with the principal axes, (3.44); the rotation is the part of $\underline{F}$ that does nothing to the material, and $\underline{U} = \hat{\underline{I}}$ characterises the motions in which nothing at all happens, (3.122).

Two conceptual points that the algebra does not announce. First, a *material* vector is one carrying no time dependence when written as a function of $\underline{X}$: it is glued to matter, and only its image $\underline{F}\underline{M}$ moves, Remark 3.26. Second, the principal directions $\underline{u}_i$ of $\underline{U}$ are *not* material, even though they live in the referential space; they sweep across the material as the deformation proceeds, at a rate computed explicitly in Remark 3.27. Confusing the two is the commonest error in reading (3.44).

Three shears, one algebra: simple shear, azimuthal shear and torsion are the same tensor $\hat{\underline{I}} + \gamma \underline{a} \otimes \underline{b}$ with $\underline{a} \perp \underline{b}$, on three different pairs of orthonormal directions, Remark 3.45; and Problem 9 gives all of them a closed form, $\underline{U} = (\underline{C} + J\hat{\underline{I}})/I$, with no eigenvector computation anywhere.

Chapter 4

Rate of Deformation

Chapter 3 answered “how much has the material been deformed”. This one answers “how fast is it being deformed now”. The two questions are not the same, and the difference is the whole content of the chapter: $\underline{F}$, $\underline{C}$, $\underline{U}$ compare the present with a reference chosen once and for all, whereas $\underline{D}$ and $\underline{W}$ below compare the present with the present, and need no reference configuration at all. That is why the rate quantities are *spatial* tensors while the strain quantities were referential.

4.1 Velocity gradient, stretching and spin

Let $\underline{v}$ be the material velocity field. By (2.19)–(2.21) and (2.37) its referential gradient is

$$\text{Grad } \underline{v} = \underline{L}\underline{F}, \quad \underline{L} = \text{grad } \underline{v}, \quad (4.1)$$

$\underline{L}$ being the *spatial* velocity gradient. In components, using $\hat{v}_i = \partial\chi_i/\partial t$ and $F_{iA} = \chi_{i,A}$,

$$\begin{aligned} \text{Grad } \underline{v} &= \hat{v}_{i,A} \underline{e}_i \otimes \hat{\underline{E}}_A = \frac{\partial}{\partial X_A} \left(\frac{\partial \chi_i}{\partial t} \right) \underline{e}_i \otimes \hat{\underline{E}}_A \\ &= \frac{\partial}{\partial t} \left(\frac{\partial \chi_i}{\partial X_A} \right) \underline{e}_i \otimes \hat{\underline{E}}_A = \dot{F}_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A, \end{aligned} \quad (4.2)$$

the interchange being permissible when $\chi_i(X_B, t)$ is twice continuously differentiable in X_B and t jointly. So $\text{Grad } \underline{v} = \dot{\underline{F}}$, and (4.1) gives

$$\dot{\underline{F}} = \underline{L}\underline{F}, \quad \text{equivalently} \quad \underline{L} = \dot{\underline{F}}\underline{F}^{-1}. \quad (4.3)$$

REMARK 4.1 (reading (4.2): what every mark in it means). Four different marks appear in these three lines. Each one is there for a reason, and the derivation stops making sense if any of them is misread.

The caret on $\hat{v}_i$. Remark 2.6 explained that the same velocity is three different functions, distinguished by what they take as input. The caret selects the one that takes a reference place, so $\hat{v}_i = \hat{v}_i(\underline{X}, t)$. It must be that one. The definition $\hat{v}_i = \partial \chi_i / \partial t$ is a partial derivative at fixed $\underline{X}$, and holding $\underline{X}$ fixed is an instruction that only makes sense in the referential description, where $\underline{X}$ is an independent variable. Written $\tilde{v}_i$ the same formula would say nothing: in the spatial description the independent variable is $\underline{x}$, holding it fixed follows no particle at all, and χ_i is not a function of $\underline{x}$ to begin with. The caret is therefore not an ornament. It records which of the three functions is meant, and only one of the three makes the formula meaningful.

The two bases. The vectors $\underline{e}_i$ span the current translation space $\mathbb{E}^3$, and $\hat{\underline{E}}_A$ spans the referential one $\hat{\mathbb{E}}^3$. These are different spaces, and $\underline{F}$ has one index in each: $\underline{F} = F_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A$. A tensor of this kind is called two-point, because it has one foot in the reference configuration and one in the current one. This is why the index alphabet changes, with Latin i, j, k for the present and capital A, B, C for the reference. The convention earns its place the first time a formula has to be checked. In $\underline{L} = \dot{\underline{F}} \underline{F}^{-1}$, for instance, $\dot{\underline{F}}$ carries $\underline{e}_i \otimes \hat{\underline{E}}_A$ and $\underline{F}^{-1}$ carries $\hat{\underline{E}}_A \otimes \underline{e}_j$. The two capital indices meet and are summed away, leaving $\underline{e}_i \otimes \underline{e}_j$. So $\underline{L}$ has both feet in the current configuration, which is exactly the claim (4.1) made when it called $\underline{L}$ spatial. Had the answer come out with a capital index left over, the formula would have been wrong, and the alphabet is what makes that visible at a glance.

The overdot. $\dot{F}_{iA}$ is the material derivative of Theorem 2.8, meaning the derivative taken while holding the material point fixed. When a quantity is already written as a function of $(\underline{X}, t)$, holding the material point fixed is the same as holding $\underline{X}$ fixed, so the material derivative is just $\partial / \partial t$. That is why (4.2) produces $\dot{F}_{iA}$ directly, with no extra term. The convective term $(\text{grad } \cdot) \underline{v}$ that appears in (2.20) is the cost of asking

the same question in the spatial description, and here the question was never asked in that description.

The interchange of derivatives. Swapping $\partial/\partial X_A$ with $\partial/\partial t$ is the only analytic step in the derivation. It is Clairaut’s theorem: mixed partial derivatives are equal where they are continuous. That is the entire content of the phrase “twice continuously differentiable in X_B and t jointly”. The hypothesis is not recited for form. Without it the two orders of differentiation can genuinely differ, and then $\text{Grad } \underline{v}$ and $\dot{\underline{F}}$ would be different tensors and (4.3) would fail.

REMARK 4.2 (why $\underline{L}$ and not $\dot{\underline{F}}$). $\dot{\underline{F}}$ is a two-point tensor and inherits the reference configuration: change κ and $\dot{\underline{F}}$ changes. $\underline{L} = \dot{\underline{F}}\underline{F}^{-1}$ maps $\mathbb{E}^3$ to $\mathbb{E}^3$ and does not: replacing κ by another reference replaces $\underline{F}$ by $\underline{F}\underline{K}$ with $\underline{K}$ fixed in time, so $(\underline{F}\underline{K})(\underline{F}\underline{K})^{-1} = \dot{\underline{F}}\underline{K}\underline{K}^{-1}\underline{F}^{-1} = \underline{L}$. The rate of deformation is a property of the motion at the present instant alone, and $\underline{L}$ is the object that says so. Figure A puts both statements as commuting diagrams.

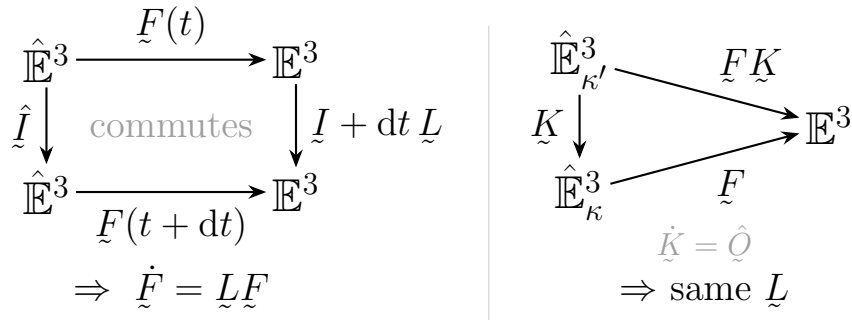


Figure A: Left: the reference space does not move, so the vertical arrow on that side is the identity, while the configuration at t is carried to the one at $t + dt$ by $\underline{x} \mapsto \underline{x} + dt \underline{v}$, whose gradient is $\underline{I} + dt \underline{L}$. Commutativity of the square is $\underline{F}(t + dt) = (\underline{I} + dt \underline{L})\underline{F}(t)$, which is (4.3). Right: changing the reference from κ to κ' replaces $\underline{F}$ by $\underline{F}\underline{K}$ with $\underline{K}$ fixed in time. The triangle commutes, and since $\dot{\underline{K}} = \dot{\underline{Q}}$ the factor cancels in $\dot{\underline{F}}\underline{F}^{-1}$: the two references disagree about $\underline{F}$ and about $\dot{\underline{F}}$, but not about $\underline{L}$.

The polar factors and their rates

Differentiate (3.32), $\underline{\underline{F}} = \underline{\underline{R}}\underline{\underline{U}} = \underline{\underline{V}}\underline{\underline{R}}$, with the product rule for tensors (Problem 1(a)):

$$\dot{\underline{\underline{F}}} = \dot{\underline{\underline{R}}}\underline{\underline{U}} + \underline{\underline{R}}\dot{\underline{\underline{U}}} = \dot{\underline{\underline{V}}}\underline{\underline{R}} + \underline{\underline{V}}\dot{\underline{\underline{R}}}. \quad (4.4)$$

We want $\underline{\underline{L}}$, and by (4.3) that means post-multiplying by $\underline{\underline{F}}^{-1}$. The two forms of $\underline{\underline{F}}$ give two forms of the inverse. From $\underline{\underline{F}} = \underline{\underline{R}}\underline{\underline{U}}$, reversing the order and inverting each factor, $\underline{\underline{F}}^{-1} = \underline{\underline{U}}^{-1}\underline{\underline{R}}^{-1} = \underline{\underline{U}}^{-1}\underline{\underline{R}}^t$, the last step because $\underline{\underline{R}}$ is orthogonal so $\underline{\underline{R}}^{-1} = \underline{\underline{R}}^t$. From $\underline{\underline{F}} = \underline{\underline{V}}\underline{\underline{R}}$ in the same way, $\underline{\underline{F}}^{-1} = \underline{\underline{R}}^{-1}\underline{\underline{V}}^{-1} = \underline{\underline{R}}^t\underline{\underline{V}}^{-1}$.

Take the first form and multiply (4.4) on the right by $\underline{\underline{U}}^{-1}\underline{\underline{R}}^t$, term by term:

$$\begin{aligned} \dot{\underline{\underline{R}}}\underline{\underline{U}} \cdot \underline{\underline{U}}^{-1}\underline{\underline{R}}^t &= \dot{\underline{\underline{R}}}(\underline{\underline{U}}\underline{\underline{U}}^{-1})\underline{\underline{R}}^t = \dot{\underline{\underline{R}}}\underline{\underline{R}}^t, \\ \underline{\underline{R}}\dot{\underline{\underline{U}}} \cdot \underline{\underline{U}}^{-1}\underline{\underline{R}}^t &= \underline{\underline{R}}\dot{\underline{\underline{U}}}\underline{\underline{U}}^{-1}\underline{\underline{R}}^t, \end{aligned}$$

where the first line used $\underline{\underline{U}}\underline{\underline{U}}^{-1} = \hat{I}$ and the second had nothing to cancel, since $\dot{\underline{\underline{U}}}$ sits between $\underline{\underline{R}}$ and $\underline{\underline{U}}^{-1}$ and tensors do not commute. Adding the two gives the first expression below. The second form goes the same way with $\underline{\underline{R}}^t\underline{\underline{V}}^{-1}$ on the right: $\dot{\underline{\underline{V}}}\underline{\underline{R}} \cdot \underline{\underline{R}}^t\underline{\underline{V}}^{-1} = \dot{\underline{\underline{V}}}\underline{\underline{V}}^{-1}$ using $\underline{\underline{R}}\underline{\underline{R}}^t = \underline{\underline{I}}$, and $\underline{\underline{V}}\dot{\underline{\underline{R}}} \cdot \underline{\underline{R}}^t\underline{\underline{V}}^{-1}$ left as it stands. So

$$\underline{\underline{L}} = \dot{\underline{\underline{R}}}\underline{\underline{R}}^t + \underline{\underline{R}}\dot{\underline{\underline{U}}}\underline{\underline{U}}^{-1}\underline{\underline{R}}^t = \dot{\underline{\underline{V}}}\underline{\underline{V}}^{-1} + \underline{\underline{V}}\dot{\underline{\underline{R}}}\underline{\underline{R}}^t\underline{\underline{V}}^{-1}. \quad (4.5)$$

The two expressions are equal because both are $\dot{\underline{\underline{F}}}\underline{\underline{F}}^{-1}$, computed by grouping the same product in two different ways.

REMARK 4.3 (why we may choose the reference, and why we choose this one). Equation (4.5) is correct but not yet usable. It expresses $\underline{\underline{L}}$ through $\underline{\underline{R}}$, $\underline{\underline{U}}$, $\underline{\underline{V}}$ together with their rates, which is six objects where we wanted a statement about rates alone. Getting rid of the first three is done by a manoeuvre that recurs all through the subject, so it is worth setting out in three parts.

First, why we are free to choose at all. Remark 4.2 established that $\underline{\underline{L}}$ does not depend on which reference configuration was used to define $\underline{\underline{F}}$. That fact is what licenses everything that follows. $\underline{\underline{L}}$ is a fixed object, determined by the motion at the present instant, while κ is a

bookkeeping device we introduced ourselves. So we may pick whichever κ is convenient, prove a formula for $\underline{L}$ with it, and the formula holds for every other choice as well, because $\underline{L}$ was the same object all along. No generality is lost.

Second, which one to pick. We want the terms containing $\underline{R}$, $\underline{U}$, $\underline{V}$ themselves to disappear, leaving only $\dot{\underline{R}}$, $\dot{\underline{U}}$, $\dot{\underline{V}}$. The deformation of a configuration relative to itself is the identity, so the way to make $\underline{F}$ trivial at one instant is to use the configuration the body occupies at that very instant as the reference. Setting $\kappa = \kappa_{t_0}$ gives $\underline{F} = \underline{1}$ at $t = t_0$ by (2.79). Uniqueness of the polar decomposition then forces $\underline{R} = \underline{1}$, $\underline{U} = \hat{\underline{I}}$ and $\underline{V} = \underline{I}$, since those values reproduce $\underline{F} = \underline{1}$ and no other values can. Every factor and every inverse in (4.5) becomes an identity, and the expression collapses to (4.6).

Third, the point that is easy to get wrong. The equalities $\underline{R} = \underline{1}$, $\underline{U} = \hat{\underline{I}}$, $\underline{V} = \underline{I}$ hold at t_0 and at no other time. An instant later the body has moved away from κ_{t_0} and the polar factors are no longer trivial. Their derivatives are therefore not required to vanish, and in general they do not. This is the asymmetry the argument runs on: the values are trivial, the rates are free. It is what allows (4.6) to isolate $\dot{\underline{U}}$ and $\dot{\underline{R}}$ while all the non-differentiated factors drop out.

To see how much damage the wrong reading does, suppose one took $\underline{U} = \hat{\underline{I}}$ to hold for all t . Then $\dot{\underline{U}} = \hat{\underline{Q}}$, and (4.6) would give $\underline{L} = \dot{\underline{R}}\underline{1}^t$, which is skew, so $\underline{D} = \underline{Q}$ and every motion would be rigid. The conclusion is absurd, and the error was a single confusion between a value at one instant and a value on an interval. Because the reference is chosen to sit at the present placement, this κ is often called the instantaneous reference, and $\underline{D}$ and $\underline{W}$ are said to be measured relative to the present configuration.

Now choose $\kappa = \kappa_{t_0}$. By (2.79) and the uniqueness of the polar factors, at $t = t_0$

$$\underline{R} = \underline{1}, \quad \underline{U} = \hat{\underline{I}}, \quad \underline{V} = \underline{I},$$

so $\underline{\underline{U}}^{-1} = \hat{\underline{\underline{I}}}$ and $\underline{\underline{V}}^{-1} = \underline{\underline{I}}$, and (4.5) collapses to

$$\underline{\underline{L}} = \dot{\underline{\underline{R}}} \underline{\underline{1}}^t + \underline{\underline{1}} \dot{\underline{\underline{U}}} \underline{\underline{1}}^t = \dot{\underline{\underline{V}}} + \dot{\underline{\underline{R}}} \underline{\underline{1}}^t, \quad (4.6)$$

all derivatives evaluated at t_0 . By Problem 1(c), $\dot{\underline{\underline{R}}} \underline{\underline{1}}^t$ is skew whenever $\underline{\underline{R}}$ is orthogonal, so $\dot{\underline{\underline{R}}} \underline{\underline{1}}^t$ is skew at t_0 ; and $\underline{\underline{1}} \dot{\underline{\underline{U}}} \underline{\underline{1}}^t$ is symmetric because $\underline{\underline{U}}$ is, $\underline{\underline{U}}$ being symmetric at every instant. Splitting

$$\underline{\underline{L}} = \underline{\underline{D}} + \underline{\underline{W}}; \quad \underline{\underline{D}} = \text{sym } \underline{\underline{L}} = \frac{1}{2}(\underline{\underline{L}} + \underline{\underline{L}}^t), \quad \underline{\underline{W}} = \text{skw } \underline{\underline{L}} = \frac{1}{2}(\underline{\underline{L}} - \underline{\underline{L}}^t), \quad (4.7)$$

and invoking the uniqueness of that splitting, Problem 14 of Chapter 1,

$$\underline{\underline{D}} = \underline{\underline{1}} \dot{\underline{\underline{U}}} \underline{\underline{1}}^t = \dot{\underline{\underline{V}}}, \quad \underline{\underline{W}} = \dot{\underline{\underline{R}}} \underline{\underline{1}}^t \quad \text{at } t_0. \quad (4.8)$$

So $\underline{\underline{D}}$ is the rate of stretch and $\underline{\underline{W}}$ the rate of rotation, both measured against the configuration occupied at that instant. $\underline{\underline{D}}$ is the *stretching* tensor, $\underline{\underline{W}}$ the *spin* tensor; Figure B shows what each half does. In components,

$$\begin{aligned} \underline{\underline{D}} &= D_{ij} \underline{\underline{e}}_i \otimes \underline{\underline{e}}_j, & D_{ij} &= \frac{1}{2}(v_{i,j} + v_{j,i}), \\ \underline{\underline{W}} &= W_{ij} \underline{\underline{e}}_i \otimes \underline{\underline{e}}_j, & W_{ij} &= \frac{1}{2}(v_{i,j} - v_{j,i}). \end{aligned} \quad (4.9)$$

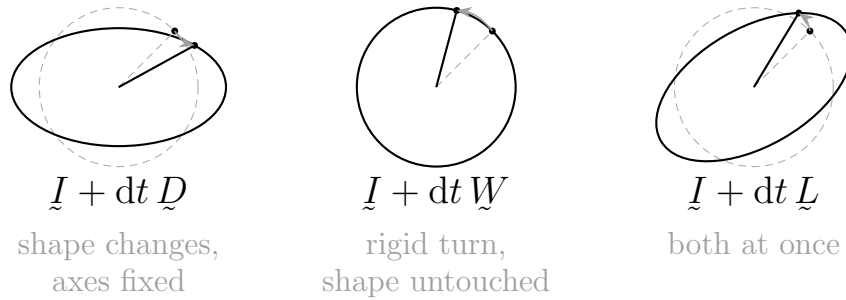


Figure B: What the two halves of (4.7) do to a disc of material directions at $\underline{\underline{x}}$, over a time dt , the dashed circle being the disc as it stands now. The symmetric part $\underline{\underline{D}}$ changes its shape without turning its axes; the skew part $\underline{\underline{W}}$ turns it without changing its shape; and $\underline{\underline{L}}$ does both, since to first order in dt , $\underline{\underline{I}} + dt \underline{\underline{L}} = (\underline{\underline{I}} + dt \underline{\underline{W}})(\underline{\underline{I}} + dt \underline{\underline{D}}) + O(dt^2)$. The marked material direction moves in all three panels, which is Remark 4.9: even the shape-only panel reorients it.

REMARK 4.4 (the two “rates of stretch” are not the same tensor). (4.8) holds *only* at the instant t_0 at which $\kappa = \kappa_t$. At any other instant $\underline{\underline{D}} \neq \underline{\underline{\dot{V}}}$ and $\underline{\underline{D}} \neq \underline{\underline{1}}\underline{\underline{\dot{U}}}\underline{\underline{1}}^t$; the general statements are (4.5) together with (4.7), and they contain $\underline{\underline{R}}$ and $\underline{\underline{V}}$ explicitly. This is the commonest trap in the chapter: $\underline{\underline{D}}$ is *not* the material derivative of any strain measure, and in particular $\underline{\underline{D}} \neq \underline{\underline{\dot{E}}}$. What is true at all times is (4.11) below, which carries $\underline{\underline{F}}$ on both sides.

Rates of the Cauchy–Green tensors

Differentiate $\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}}$, using the product rule and Problem 1(b), $(\underline{\underline{F}}^t)^\cdot = \underline{\underline{\dot{F}}}^t$, then (4.3):

$$\begin{aligned} \underline{\underline{\dot{C}}} &= (\underline{\underline{F}}^t \underline{\underline{F}})^\cdot = \underline{\underline{\dot{F}}}^t \underline{\underline{F}} + \underline{\underline{F}}^t \underline{\underline{\dot{F}}} = (\underline{\underline{L}} \underline{\underline{F}})^t \underline{\underline{F}} + \underline{\underline{F}}^t \underline{\underline{L}} \underline{\underline{F}} \\ &= \underline{\underline{F}}^t \underline{\underline{L}}^t \underline{\underline{F}} + \underline{\underline{F}}^t \underline{\underline{L}} \underline{\underline{F}} = \underline{\underline{F}}^t (\underline{\underline{L}} + \underline{\underline{L}}^t) \underline{\underline{F}} = 2 \underline{\underline{F}}^t \underline{\underline{D}} \underline{\underline{F}}. \end{aligned} \quad (4.10)$$

The Green strain of (2.97) is $\underline{\underline{E}} = \frac{1}{2}(\underline{\underline{C}} - \underline{\underline{\hat{I}}})$. Differentiating it costs nothing extra, because $\underline{\underline{\hat{I}}}$ is a fixed tensor and $\frac{1}{2}$ a fixed number, so $\underline{\underline{\dot{E}}} = \frac{1}{2}\underline{\underline{\dot{C}}}$. Therefore

$$\underline{\underline{F}}^t \underline{\underline{D}} \underline{\underline{F}} = \frac{1}{2} \underline{\underline{\dot{C}}} = \underline{\underline{\dot{E}}}. \quad (4.11)$$

REMARK 4.5 (why $\underline{\underline{C}}$ is the right thing to differentiate, and what (4.11) is saying). Two questions are worth asking here. Why differentiate $\underline{\underline{C}}$ rather than $\underline{\underline{F}}$ or $\underline{\underline{U}}$, and what kind of statement is (4.11)?

Why $\underline{\underline{C}}$. Differentiating $\underline{\underline{F}}$ gives $\underline{\underline{\dot{F}}} = \underline{\underline{L}} \underline{\underline{F}}$, which is (4.3) and tells us nothing new. Differentiating $\underline{\underline{U}}$ is what one would like, since $\underline{\underline{U}}$ is the pure stretch, but $\underline{\underline{U}}$ is defined through a square root, $\underline{\underline{U}} = \sqrt{\underline{\underline{C}}}$, and the derivative of a tensor square root has no simple closed form. $\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}}$ has neither problem. It is a polynomial in $\underline{\underline{F}}$, so the product rule applies directly, and it is already free of the rotation: substituting $\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}}$ gives $\underline{\underline{C}} = \underline{\underline{U}} \underline{\underline{R}}^t \underline{\underline{R}} \underline{\underline{U}} = \underline{\underline{U}}^2$, with $\underline{\underline{R}}$ cancelling. So $\underline{\underline{C}}$ carries exactly the stretch information, in a form that can be differentiated in one line. That combination is why every rate calculation in finite elasticity goes through $\underline{\underline{C}}$ rather than $\underline{\underline{U}}$.

What (4.11) says. On the right sits $\dot{\underline{\underline{E}}}$, a referential tensor: it lives on $\hat{\mathbb{E}}^3$ and is what an observer using the reference configuration would call the strain rate. On the left sits $\underline{\underline{D}}$, a spatial tensor, sandwiched between $\underline{\underline{F}}^t$ and $\underline{\underline{F}}$. That sandwich is the standard operation for moving a spatial tensor back to the reference configuration, and it is forced by index bookkeeping alone: $\underline{\underline{F}}^t$ carries $\hat{\underline{\underline{E}}}_A \otimes \underline{\underline{e}}_i$, $\underline{\underline{D}}$ carries $\underline{\underline{e}}_i \otimes \underline{\underline{e}}_j$, $\underline{\underline{F}}$ carries $\underline{\underline{e}}_j \otimes \hat{\underline{\underline{E}}}_B$, and the Latin indices contract in pairs to leave $\hat{\underline{\underline{E}}}_A \otimes \hat{\underline{\underline{E}}}_B$, which is what a referential tensor must carry.

So (4.11) is not a formula relating two different physical quantities. It says that $\underline{\underline{D}}$ and $\dot{\underline{\underline{E}}}$ are the same measurement expressed in two configurations, with $\underline{\underline{F}}$ doing the translating. This also settles the warning of Remark 4.4 in a sharper form. It is tempting to read (4.11) as $\underline{\underline{D}} = \dot{\underline{\underline{E}}}$ with the $\underline{\underline{F}}$'s ignored, but the $\underline{\underline{F}}$'s cannot be ignored: they are the only reason the two sides live in the same space. At the one instant when $\kappa = \kappa_{t_0}$ we have $\underline{\underline{F}} = \underline{\underline{1}}$, the sandwich disappears, and $\underline{\underline{D}} = \dot{\underline{\underline{E}}}$ holds. At every other instant it does not.

The same computation on $\underline{\underline{B}} = \underline{\underline{F}}\underline{\underline{F}}^t$ gives

$$\dot{\underline{\underline{B}}} = \dot{\underline{\underline{F}}}\underline{\underline{F}}^t + \underline{\underline{F}}\dot{\underline{\underline{F}}^t} = \underline{\underline{L}}\underline{\underline{F}}\underline{\underline{F}}^t + \underline{\underline{F}}\underline{\underline{F}}^t\underline{\underline{L}}^t = \underline{\underline{L}}\underline{\underline{B}} + \underline{\underline{B}}\underline{\underline{L}}^t. \quad (4.12)$$

Evaluating (4.11) at t_0 , where $\underline{\underline{F}} = \underline{\underline{1}}$,

$$\underline{\underline{D}} = \underline{\underline{1}}\dot{\underline{\underline{E}}}\underline{\underline{1}}^t, \quad (4.13)$$

so $\underline{\underline{D}}$ might equally be called the *straining* tensor.

REMARK 4.6 ($\underline{\underline{D}} = \underline{\underline{Q}}$ characterises rigidity). In a rigid motion $\underline{\underline{F}} = \underline{\underline{Q}}(t)$, so $\underline{\underline{C}} = \hat{\underline{\underline{I}}}$ and $\underline{\underline{B}} = \underline{\underline{I}}$, both constant; then $\dot{\underline{\underline{C}}} = \dot{\underline{\underline{Q}}}$ and (4.11) gives $\underline{\underline{F}}^t \underline{\underline{D}} \underline{\underline{F}} = \dot{\underline{\underline{Q}}}$, whence $\underline{\underline{D}} = \underline{\underline{Q}}$ since $\underline{\underline{F}}$ is invertible. The converse is also true but is not a one-line matter: $\underline{\underline{D}} = \underline{\underline{Q}}$ says only that $\underline{\underline{L}}$ is skew at each point, and one must use that $\underline{\underline{L}}$ is a *gradient* to conclude that it is also spatially uniform. Granting that, $\underline{\underline{D}} = \underline{\underline{Q}}$ is necessary and sufficient for rigidity, the rate analogue of (3.122).

4.2 Material curves

The object to differentiate is (2.67), $\mu \underline{m} = \underline{F} \underline{M}$. It is worth being clear about what each symbol in it is before touching it, because the whole derivation turns on which of them depend on time.

$\underline{M}$ is a unit vector in the reference configuration, pointing along a chosen material fibre as that fibre was laid out in κ . It is a label for the fibre, in the way $\underline{X}$ is a label for a particle, and labels do not change: $\dot{\underline{M}} = \underline{0}$ by Remark 3.26. The pair $(\mu, \underline{m})$ on the left is what that fibre has become. $\underline{m}$ is a unit vector in the current configuration giving the fibre's present direction, and $\mu > 0$ is its present stretch, the factor by which its length has grown. Splitting $\underline{F} \underline{M}$ into a length and a direction this way is a deliberate choice, and it is the choice that makes the section work: it separates the question “how much longer is the fibre” from the question “which way is it now pointing”, and the two answers turn out to be carried by $\underline{D}$ and by the whole of $\underline{L}$ respectively.

Differentiate, using the product rule on the left and $\dot{\underline{M}} = \underline{0}$ with $\dot{\underline{F}} = \underline{L} \underline{F}$ from (4.3) on the right:

$$\dot{\mu} \underline{m} + \mu \dot{\underline{m}} = \dot{\underline{F}} \underline{M} = \underline{L} \underline{F} \underline{M} = \mu \underline{L} \underline{m}, \quad (4.14)$$

the last step replacing $\underline{F} \underline{M}$ by $\mu \underline{m}$, which is the equation we started from.

Equation (4.14) contains both unknowns, $\dot{\mu}$ and $\dot{\underline{m}}$, and one equation cannot give both. The way to separate them is to project. Take the inner product of (4.14) with $\underline{m}$:

$$\dot{\mu} \underline{m} \cdot \underline{m} + \mu \underline{m} \cdot \dot{\underline{m}} = \mu \underline{m} \cdot \underline{L} \underline{m}.$$

Two of these pieces vanish, each for its own reason.

First, $\underline{m} \cdot \underline{m} = 1$ because $\underline{m}$ is a unit vector, so the coefficient of $\dot{\mu}$ is simply 1.

Second, $\underline{m} \cdot \dot{\underline{m}} = 0$. To see it, differentiate the identity $\underline{m} \cdot \underline{m} = 1$ with respect to time. The inner product is bilinear, so it obeys a product rule,

$$\frac{d}{dt} \underline{m} \cdot \underline{m} = \dot{\underline{m}} \cdot \underline{m} + \underline{m} \cdot \dot{\underline{m}} = 2 \underline{m} \cdot \dot{\underline{m}},$$

the two terms being equal by symmetry of the inner product. The left side is the derivative of the constant 1, hence zero, so $2\dot{\underline{m}} \cdot \underline{\dot{m}} = 0$ and therefore $\underline{m} \cdot \underline{\dot{m}} = 0$. Nothing about the motion was used; any unit vector, however it moves, has its derivative perpendicular to itself.

What survives is

$$\dot{\mu} = \mu \underline{m} \cdot \underline{Lm}.$$

The right side can be simplified once more. Split $\underline{L} = \underline{D} + \underline{W}$ by (4.7). Then $\underline{m} \cdot \underline{Lm} = \underline{m} \cdot \underline{Dm} + \underline{m} \cdot \underline{Wm}$, and the second term is zero because $\underline{W}$ is skew: by Problem 6(a) of Chapter 1, $\underline{u} \cdot \underline{Wu} = \underline{W}^t \underline{u} \cdot \underline{u} = -\underline{Wu} \cdot \underline{u} = -\underline{u} \cdot \underline{Wu}$ for every $\underline{u}$, and a number equal to its own negative is zero. So

$$\dot{\mu} = \mu \underline{m} \cdot \underline{Dm}.$$

This already says everything, but it is customarily written in a tidier form. Divide by μ , which is legitimate because a stretch is strictly positive, $\mu > 0$, so we are never dividing by zero:

$$\frac{\dot{\mu}}{\mu} = \underline{m} \cdot \underline{Dm}.$$

Now recognise the left side. By the chain rule, for any positive differentiable $\mu(t)$,

$$\frac{d}{dt} \ln \mu = \frac{1}{\mu} \frac{d\mu}{dt} = \frac{\dot{\mu}}{\mu},$$

so the left side is the material derivative of $\ln \mu$ and

$$(\ln \mu)^{\cdot} = \underline{m} \cdot \underline{Dm}. \quad (4.15)$$

REMARK 4.7 (why the logarithm is the natural variable here). The step from $\dot{\mu}/\mu$ to $(\ln \mu)^{\cdot}$ is only a rewriting, and one may ask why anyone bothers. Three reasons, and they are the same reasons the logarithmic strain is used throughout finite elasticity.

The first is that $\dot{\mu}/\mu$ is the quantity with physical meaning. A fibre growing from 1 to 1.01 and a fibre growing from 100 to 101 have the same $\dot{\mu}$ and are not stretching at the same rate; the second is barely changing. Rates of growth are relative, which is why interest, inflation

and population growth are all quoted as fractions of the current amount, and $\dot{\mu}/\mu$ is exactly that fraction per unit time.

The second is that logarithms turn the composition of stretches into addition. Stretch a fibre by μ_1 and then by μ_2 and the total stretch is the product $\mu_1\mu_2$, not the sum. Taking logarithms makes the total $\ln \mu_1 + \ln \mu_2$, so $\ln \mu$ accumulates the way one expects a strain to accumulate, and its rate is what (4.15) delivers.

The third is that it makes the statement integrable. Reading (4.15) backwards,

$$\ln \frac{\mu(t)}{\mu(t_0)} = \int_{t_0}^t \underline{m} \cdot \underline{D}\underline{m} \, d\tau,$$

so the total stretch of a fibre is recovered from the history of one diagonal component of $\underline{D}$ along it. Had we stopped at $\dot{\mu} = \mu \underline{m} \cdot \underline{D}\underline{m}$ we would have had a differential equation to solve rather than an integral to evaluate.

Now return to (4.14) for the other half of the answer. Solve it for $\dot{\underline{m}}$ by isolating that term,

$$\mu \dot{\underline{m}} = \mu \underline{L}\underline{m} - \dot{\mu} \underline{m},$$

divide through by $\mu > 0$,

$$\dot{\underline{m}} = \underline{L}\underline{m} - \frac{\dot{\mu}}{\mu} \underline{m},$$

and substitute $\dot{\mu}/\mu = \underline{m} \cdot \underline{D}\underline{m}$ from (4.15) together with $\underline{L} = \underline{D} + \underline{W}$:

$$\dot{\underline{m}} = \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m} + \underline{W}\underline{m}. \quad (4.16)$$

So one equation, (4.14), has been made to yield two, by taking its component along $\underline{m}$ and what is left over. That is the standard way to get two statements out of one vector equation, and it works here because $\underline{m}$ is the natural direction to resolve along.

REMARK 4.8 (what $\dot{\underline{m}}$ is: the overdot as a direction of change). The object $\dot{\underline{m}}$ deserves a closer look, because the overdot is saying something more specific here than a general rate of change.

Recall what $\underline{m}$ is. It is the unit vector along a material fibre as that fibre stands at the present instant. Since $\underline{m}$ always has length one, following it in time traces a curve on the unit sphere, and $\dot{\underline{m}}$ is the velocity along that curve. Two consequences follow at once.

It is perpendicular to $\underline{m}$. This was proved above in the course of deriving (4.15): differentiating $\underline{m} \cdot \underline{m} = 1$ gives $\underline{m} \cdot \dot{\underline{m}} = 0$. So $\dot{\underline{m}}$ carries no information about length. It cannot, since the length is pinned at one. What it points along is the direction in which the fibre is swinging, and its magnitude is the angular speed of that swing, measured in radians per unit time. The information about length was separated off into (4.15), which is why (4.15) and (4.16) together contain everything (4.14) contained, no more and no less.

The middle term is what enforces the perpendicularity. Look at the three terms of (4.16) in turn. The term $\underline{W}\underline{m}$ is perpendicular to $\underline{m}$ automatically, since $\underline{m} \cdot \underline{W}\underline{m} = 0$ for skew $\underline{W}$. The term $\underline{D}\underline{m}$ is not, in general; its component along $\underline{m}$ is exactly $\underline{m} \cdot \underline{D}\underline{m}$. The remaining term, $-(\underline{m} \cdot \underline{D}\underline{m})\underline{m}$, subtracts precisely that component and nothing else. So the formula says: apply $\underline{L}$ to the fibre, then remove the part that would change its length. In compact form,

$$\dot{\underline{m}} = (\underline{I} - \underline{m} \otimes \underline{m})\underline{L}\underline{m},$$

where $\underline{I} - \underline{m} \otimes \underline{m}$ is the projection onto the plane perpendicular to $\underline{m}$, the complement of the projection $\mathbb{P}_{(\underline{n})}$ of Chapter 1. To check that the compact form agrees with (4.16), expand it: $(\underline{I} - \underline{m} \otimes \underline{m})\underline{L}\underline{m} = \underline{L}\underline{m} - \underline{m}(\underline{m} \cdot \underline{L}\underline{m}) = \underline{D}\underline{m} + \underline{W}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m}$, using $\underline{m} \cdot \underline{L}\underline{m} = \underline{m} \cdot \underline{D}\underline{m}$ once more.

Which clock the dot keeps. As everywhere in this chapter, the overdot follows the material. So $\dot{\underline{m}}$ is the turning rate an observer riding the particle would measure, not the rate at which the direction field $\underline{m}(\underline{x}, t)$ changes at a fixed place in space. The two differ by the convective term of (2.20), and mistaking one for the other is the usual way to get a wrong sign in a rotating flow. Problem 10 is built on exactly this distinction. In the potential vortex the direction field at any fixed place is steady in time, while every material fibre passing through that place

is turning.

REMARK 4.9 (the trap: $\underline{W}$ is not the spin of material fibres). $\underline{m}$ is a unit vector, so $\dot{\underline{m}}$ is pure reorientation. One is tempted to read (4.16) as saying that the reorientation is $\underline{W}\underline{m}$. It is not: the first two terms are generally non-zero, and they are what makes Problem 6(d) come out with $\dot{\underline{m}} \neq \underline{0}$ even though $\underline{W} = \underline{Q}$. The stretching tensor turns material fibres as well, and it turns different fibres by different amounts. Only in the special case of (4.18) does $\underline{W}$ tell the whole story. Figure D shows the failure concretely, in the setting of Problem 6.

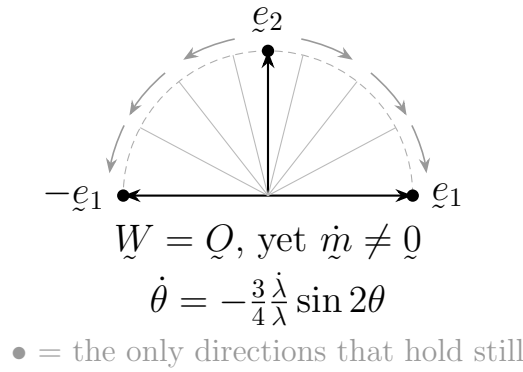


Figure D: The trap of Remark 4.9, in the $(\underline{e}_1, \underline{e}_2)$ section of Problem 6. The spin tensor vanishes identically, so nothing is rotating in the sense of $\underline{W}$; nevertheless every material fibre except four drifts, and it drifts towards $\pm \underline{e}_1$, the direction that is stretching fastest. The reason is (4.16): with $\underline{W} = \underline{Q}$ it reads $\dot{\underline{m}} = \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m}$, the component of $\underline{D}\underline{m}$ perpendicular to $\underline{m}$, which vanishes exactly when $\underline{m}$ is a principal direction of $\underline{D}$. Stretching alone reorients material, and that is why $\underline{W}$ cannot be called “the spin of the material”. Only directions in the upper half plane are drawn; the fourth fixed direction, $-\underline{e}_2$, is the mirror image of the one at the top.

Now specialise. Suppose that at some instant t_* the fibre direction $\underline{m}$ happens to coincide with a principal vector $\underline{\nu}$ of $\underline{D}$, so that

$$\underline{D}\underline{\nu} = \eta\underline{\nu}, \quad \underline{m}|_{t_*} = \underline{\nu},$$

with η the corresponding principal value. Feed this into (4.15). The right side becomes

$$\underline{m} \cdot \underline{D}\underline{m} = \underline{\nu} \cdot \underline{D}\underline{\nu} = \underline{\nu} \cdot \eta \underline{\nu} = \eta \underline{\nu} \cdot \underline{\nu} = \eta,$$

since $\underline{\nu}$ is a unit vector. So

$$\eta = (\ln \mu)' \big|_{t_*}, \quad (4.17)$$

which is the promised interpretation of the principal values: they are the logarithmic stretch rates of material curves momentarily lying along the principal directions of $\underline{D}$. The eigenvalues of $\underline{D}$ are not abstract numbers; each is a rate of stretching that could be measured by marking a fibre and timing its growth.

The same substitution does something to (4.16) as well. At t_* ,

$$\underline{D}\underline{m} = \eta \underline{\nu} \quad \text{and} \quad (\underline{m} \cdot \underline{D}\underline{m})\underline{m} = \eta \underline{\nu},$$

the second because the scalar $\underline{m} \cdot \underline{D}\underline{m}$ was just computed to be η and $\underline{m} = \underline{\nu}$. The two are equal, so the first two terms of (4.16) cancel exactly and only the spin survives:

$$\dot{\underline{m}} \big|_{t_*} = \underline{W}\underline{m} \big|_{t_*}. \quad (4.18)$$

REMARK 4.10 (why (4.18) is a statement about one instant only). It is tempting to read (4.18) as saying that principal directions turn with the spin, full stop. It does not, and the reason is worth being careful about.

The cancellation used $\underline{m} = \underline{\nu}$, which holds at t_* by assumption. But the two sides of that assumption evolve for different reasons and at different rates. The fibre direction $\underline{m}$ is convected by the motion, obeying (4.16). The principal direction $\underline{\nu}$ is whatever direction happens to diagonalise $\underline{D}$ at each instant, and $\underline{D}$ is itself changing in time, so $\underline{\nu}$ moves for reasons that have nothing to do with any particular fibre. There is no reason for the two to stay together, and in general they separate immediately after t_* . Once they have, $\underline{D}\underline{m}$ is no longer parallel to $\underline{m}$, the first two terms of (4.16) no longer cancel, and (4.18) fails.

The special motions in which a fibre stays principal for an interval rather than an instant are exactly those in which the principal axes are frozen in the material, and Problem 8 works out what that costs. Figure D shows the generic situation, in which $\underline{W} = \underline{Q}$ yet fibres still drift, precisely because they are not principal directions and (4.18) does not apply to them.

Rate of change of the angle between two fibres

Let $\underline{M}_1, \underline{M}_2$ be two material directions. By (4.16) with $\underline{D} + \underline{W} = \underline{L}$,

$$\dot{\underline{m}}_1 = \underline{L}\underline{m}_1 - (\underline{m}_1 \cdot \underline{D}\underline{m}_1)\underline{m}_1, \quad \dot{\underline{m}}_2 = \underline{L}\underline{m}_2 - (\underline{m}_2 \cdot \underline{D}\underline{m}_2)\underline{m}_2. \quad (4.19)$$

Both fibres are unit vectors, so the angle θ between them satisfies $\cos \theta = \underline{m}_1 \cdot \underline{m}_2$. This is the equation to differentiate, and it is worth noticing why we differentiate the cosine rather than θ itself: θ is not directly expressible in the fibres, whereas its cosine is a plain inner product, and inner products obey a product rule.

Differentiate the left side by the chain rule, $\frac{d}{dt} \cos \theta = -(\sin \theta)\dot{\theta}$, and the right side by the product rule for the inner product:

$$\begin{aligned} -(\sin \theta)\dot{\theta} &= \dot{\underline{m}}_1 \cdot \underline{m}_2 + \underline{m}_1 \cdot \dot{\underline{m}}_2 \\ &= \underline{m}_2 \cdot \underline{L}\underline{m}_1 - (\underline{m}_1 \cdot \underline{D}\underline{m}_1)(\underline{m}_1 \cdot \underline{m}_2) \\ &\quad + \underline{m}_1 \cdot \underline{L}\underline{m}_2 - (\underline{m}_2 \cdot \underline{D}\underline{m}_2)(\underline{m}_1 \cdot \underline{m}_2). \end{aligned} \quad (4.20)$$

The second equality substituted (4.19) for each dotted fibre and then expanded. In the first substitution, $\dot{\underline{m}}_1 \cdot \underline{m}_2 = \underline{L}\underline{m}_1 \cdot \underline{m}_2 - (\underline{m}_1 \cdot \underline{D}\underline{m}_1)\underline{m}_1 \cdot \underline{m}_2$, and the first term was rewritten as $\underline{m}_2 \cdot \underline{L}\underline{m}_1$ by symmetry of the inner product. The second substitution is the same with the labels exchanged.

Now specialise to the case of interest. Suppose the two fibres are momentarily perpendicular at an instant t_* , so that $\underline{m}_1 \cdot \underline{m}_2 = 0$ and $\theta = \pi/2$. Two simplifications follow at once. The factor $\underline{m}_1 \cdot \underline{m}_2$ multiplies both of the subtracted terms, so both vanish. And $\sin \theta = \sin(\pi/2) = 1$, so the left side is just $-\dot{\theta}$. What remains is

$$-\dot{\theta}|_{t_*} = \underline{m}_2 \cdot \underline{L}\underline{m}_1 + \underline{m}_1 \cdot \underline{L}\underline{m}_2.$$

Move $\underline{L}$ off the first fibre in the first term using the definition of the transpose, $\underline{m}_2 \cdot \underline{L}\underline{m}_1 = \underline{L}^t \underline{m}_2 \cdot \underline{m}_1 = \underline{m}_1 \cdot \underline{L}^t \underline{m}_2$, so that both terms act on $\underline{m}_2$:

$$-\dot{\theta}|_{t_*} = \underline{m}_1 \cdot \underline{L}^t \underline{m}_2 + \underline{m}_1 \cdot \underline{L} \underline{m}_2 = \underline{m}_1 \cdot (\underline{L} + \underline{L}^t) \underline{m}_2 = 2 \underline{m}_1 \cdot \underline{D} \underline{m}_2,$$

using linearity to combine the two and then $\underline{L} + \underline{L}^t = 2\underline{D}$ from (4.7). Therefore

$$\dot{\theta}|_{t_*} = -2 \underline{m}_1 \cdot \underline{D} \underline{m}_2. \quad (4.21)$$

The sign says that a positive off-diagonal component of $\underline{D}$ makes the angle decrease, which is why one speaks of the fibres closing on one another. So the *off-diagonal* components of $\underline{D}$ are half the rates at which initially perpendicular fibres close on one another, while by (4.17) the *diagonal* components are logarithmic stretch rates. That is the complete interpretation of $\underline{D}$, and Figure C is the whole of it in one picture.

REMARK 4.11 (why $\underline{W}$ vanished from (4.21), and what it would have done). Something happened between (4.20) and (4.21) that is easy to pass over: the spin tensor went into the calculation and did not come out. The algebra that removed it was the combination $\underline{L} + \underline{L}^t = 2\underline{D}$, in which the skew parts cancel. But the cancellation is not a lucky accident of the manipulation, and seeing why shows the difference between the two halves of $\underline{L}$ more clearly than anything else in the chapter.

Write out the two places $\underline{W}$ enters. Splitting $\underline{L} = \underline{D} + \underline{W}$ in (4.20), its contribution to $-(\sin \theta)\dot{\theta}$ is

$$\underline{m}_2 \cdot \underline{W} \underline{m}_1 + \underline{m}_1 \cdot \underline{W} \underline{m}_2.$$

By skewness the first equals $-\underline{W} \underline{m}_2 \cdot \underline{m}_1 = -\underline{m}_1 \cdot \underline{W} \underline{m}_2$, so the two are negatives of one another and their sum is zero. That is the algebra.

The reason behind the algebra is this. The angle θ is a relative quantity. It depends on the two fibres only through how they are placed with respect to each other, not on where either one sits in space. The spin tensor turns the pair bodily, moving both fibres at the same angular

rate about the same axis, so it changes where each of $\underline{m}_1$ and $\underline{m}_2$ points while leaving the relation between them untouched. Any quantity built out of the pair alone must therefore be unable to detect $\underline{W}$, and $\dot{\theta}$ is such a quantity. The skewness of $\underline{W}$ is the algebraic form of the sentence: a rigid turn does not deform anything.

This also explains the caption of Figure C, which says the picture is drawn with $\underline{W} = \underline{Q}$. If spin were present, the figure would show both arms swinging round together in addition to closing on one another, and $\dot{\theta}$ would come out the same. Setting $\underline{W} = \underline{Q}$ puts the pair in the frame that makes the closing look symmetric, so that each arm appears to contribute half of it. Equation (4.21) fixes only the sum $\dot{\theta}$ and says nothing about how that sum is divided between the two fibres, because the division depends on the frame in exactly the way $\underline{W}$ does. Remark 4.9 is the same warning stated for one fibre instead of two.

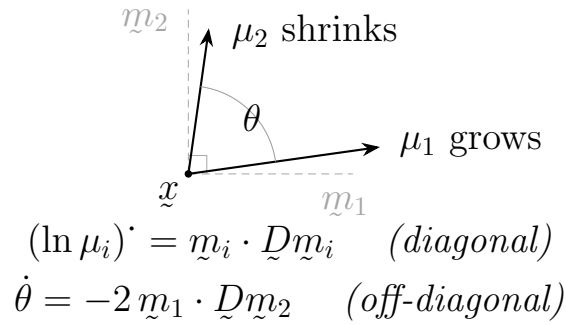


Figure C: The entire meaning of $\underline{D}$, read off two material fibres that are momentarily perpendicular at $\underline{x}$. Their lengths change at the logarithmic rates $\underline{m}_i \cdot \underline{D} \underline{m}_i$, which are the diagonal components of $\underline{D}$ in the basis $\{\underline{m}_1, \underline{m}_2\}$, by (4.15). The right angle between them closes at the rate $2 \underline{m}_1 \cdot \underline{D} \underline{m}_2$, twice the off-diagonal component, by (4.21). Drawn with $\underline{W} = \underline{Q}$, so that the closing is shared equally; in general $\underline{W}$ turns the pair bodily as well, and (4.21) fixes only the sum.

Vorticity

By Problem 16 of Chapter 1 every skew $\underline{W}$ has an axial vector $\underline{w}$ with $\underline{W} \underline{u} = \underline{w} \times \underline{u}$. For the spin tensor $\underline{w}$ is the *vorticity*.

REMARK 4.12 (where $w_i = \frac{1}{2}\varepsilon_{ijk}W_{kj}$ comes from). This formula is not a fresh definition. It is Proposition 1.44 solved for $\underline{w}$, and both the factor $\frac{1}{2}$ and the reversed index pair are forced rather than chosen. Here is the whole derivation.

Start with why an axial vector exists at all. A skew tensor satisfies $W_{ij} = -W_{ji}$, so its diagonal entries vanish and the entries below the diagonal are determined by those above it. In three dimensions that leaves three independent numbers, namely W_{12} , W_{13} , W_{23} . A vector in three dimensions also has three components. The correspondence between skew tensors and vectors rests on this coincidence and exists in no other dimension: in four dimensions a skew tensor has six independent entries and no vector to match them.

The correspondence is pinned down by requiring $\underline{W}\underline{u} = \underline{w} \times \underline{u}$ for every $\underline{u}$. Write both sides in components. The left side is $W_{ij}u_j$. For the right side, the cross product in components is $(\underline{w} \times \underline{u})_i = \varepsilon_{ijk}w_ju_k$, so

$$W_{ij}u_j = \varepsilon_{ijk}w_ju_k.$$

The free index is i and $\underline{u}$ is arbitrary, so the coefficients of u_j must agree on both sides. On the right, rename the summed index k to j and the summed index j to k , which is allowed because both are dummies:

$$\varepsilon_{ijk}w_ju_k = \varepsilon_{ikj}w_ku_j.$$

Matching coefficients of u_j now gives

$$W_{ij} = \varepsilon_{ikj}w_k = -\varepsilon_{ijk}w_k,$$

the last step using the antisymmetry $\varepsilon_{ikj} = -\varepsilon_{ijk}$. This is the statement recorded in Proposition 1.44, and it expresses $\underline{W}$ in terms of $\underline{w}$.

To go the other way, contract both sides with ε_{ijl} and sum over i and j :

$$\varepsilon_{ijl}W_{ij} = -\varepsilon_{ijl}\varepsilon_{ijk}w_k.$$

The double contraction is handled by Theorem 1.29, which is stated with the repeated pair in the last two slots. A cyclic shift puts each factor

into that form, $\varepsilon_{ijl} = \varepsilon_{lij}$ and $\varepsilon_{ijk} = \varepsilon_{kij}$, so

$$\varepsilon_{ijl}\varepsilon_{ijk} = \varepsilon_{lij}\varepsilon_{kij} = 2\delta_{lk},$$

and therefore

$$\varepsilon_{ijl}W_{ij} = -2\delta_{lk}w_k = -2w_l \quad \implies \quad w_l = -\frac{1}{2}\varepsilon_{ijl}W_{ij} = -\frac{1}{2}\varepsilon_{lij}W_{ij}.$$

Now the factor $\frac{1}{2}$ has an explanation. It is the 2 in $\varepsilon_{lij}\varepsilon_{kij} = 2\delta_{lk}$, and that 2 counts the two orderings of the summed pair of indices. Each independent entry of $\underline{W}$ is reached twice by the contraction, once as W_{ij} and once as W_{ji} , and since the two are negatives of one another and ε also changes sign, the two contributions add rather than cancel. Dividing by 2 undoes the double count. Nobody chose the factor for tidiness.

Finally, the form used below is $w_i = \frac{1}{2}\varepsilon_{ijk}W_{kj}$, with the indices of W in the order kj rather than jk . This is the same statement written differently. Relabel the dummies $j \leftrightarrow k$ in $\varepsilon_{ijk}W_{kj}$ to get $\varepsilon_{ikj}W_{jk}$, then use $\varepsilon_{ikj} = -\varepsilon_{ijk}$:

$$\frac{1}{2}\varepsilon_{ijk}W_{kj} = \frac{1}{2}\varepsilon_{ikj}W_{jk} = -\frac{1}{2}\varepsilon_{ijk}W_{jk},$$

which is what the boxed result above gives with l renamed to i . Writing W_{kj} instead of W_{jk} is a way of carrying the minus sign in the index order rather than in front of the expression, and it is why (4.22) runs through with no stray signs to keep track of.

In components, with $w_i = \frac{1}{2}\varepsilon_{ijk}W_{kj}$ and (4.9),

$$\begin{aligned} w_i &= \frac{1}{2}\varepsilon_{ijk}W_{kj} = \frac{1}{4}\varepsilon_{ijk}(v_{k,j} - v_{j,k}) \\ &= \frac{1}{4}(\varepsilon_{ijk}v_{k,j} + \varepsilon_{ikj}v_{j,k}) \\ &= \frac{1}{4}(\varepsilon_{ijk}v_{k,j} + \varepsilon_{ijk}v_{k,j}) = \frac{1}{2}\varepsilon_{jki}v_{k,j}, \end{aligned} \tag{4.22}$$

Each line deserves a word, since three different index manipulations are used in quick succession.

The first line substituted $W_{kj} = \frac{1}{2}(v_{k,j} - v_{j,k})$ from (4.9), which contributed the extra $\frac{1}{2}$ and made the total $\frac{1}{4}$.

The second line rewrote the subtraction as an addition. Look at the second piece, $-\frac{1}{4}\varepsilon_{ijk}v_{j,k}$. Using antisymmetry in the last two slots, $-\varepsilon_{ijk} = \varepsilon_{ikj}$, so the piece equals $+\frac{1}{4}\varepsilon_{ikj}v_{j,k}$. The minus sign has been absorbed into the order of the indices on ε , which is the same trick Remark 4.12 described.

The third line renamed the dummy indices in that second piece, $j \leftrightarrow k$, which is always permitted for summed indices and changes nothing. It turns $\varepsilon_{ikj}v_{j,k}$ into $\varepsilon_{ijk}v_{k,j}$, which is exactly the first piece. So the two pieces are identical, they add, and $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

The fourth line applied the cyclic symmetry $\varepsilon_{ijk} = \varepsilon_{jki}$ to put the free index i last, which is the form in which the curl was written in Chapter 1. Comparing with (1.165) there,

$$\underline{w} = \frac{1}{2} \text{curl } \underline{v}, \quad (4.23)$$

curl being the spatial curl.

REMARK 4.13 (the factor $\frac{1}{2}$, and what vorticity measures). The factor of $\frac{1}{2}$ in (4.23) is a standing source of confusion, because different books define vorticity as $\underline{w}$ and as $\text{curl } \underline{v}$ without always saying which. The way to keep it straight is to remember what each one counts, and the cleanest test is rigid rotation.

Let the body rotate rigidly at angular velocity $\underline{\Omega}$, so that $\underline{v} = \underline{\Omega} \times \underline{r}$ with $\underline{r}$ measured from a point on the axis. Then $\underline{L} = \text{grad } \underline{v}$ is the skew tensor whose axial vector is $\underline{\Omega}$, so $\underline{D} = \underline{O}$, $\underline{W} = \underline{L}$, and by definition $\underline{w} = \underline{\Omega}$. The vorticity is the angular velocity itself, with no factor.

Now compute the curl of the same field. Since $\underline{w} = \frac{1}{2} \text{curl } \underline{v}$ and $\underline{w} = \underline{\Omega}$, we get $\text{curl } \underline{v} = 2\underline{\Omega}$: the curl is *twice* the angular velocity. The factor is real and it is in the curl, not in the vorticity.

The reason is that $\text{curl } \underline{v}$ measures the circulation per unit area around a small loop, and a rigidly rotating disc of radius a has circulation $\oint \underline{v} \cdot d\underline{x} = (\Omega a)(2\pi a) = 2\pi\Omega a^2$ around its rim, while its area is πa^2 . The ratio is 2Ω , not Ω . Circulation counts the whole way round; angular velocity counts the turning rate. Defining $\underline{w}$ with the $\frac{1}{2}$ is what makes the vorticity equal to the angular velocity of the local rigid rota-

tion, which is the physically useful normalisation and the one this book uses.

The disc of the next section is the concrete version of this argument, and Problem 10 shows how badly intuition fails when the flow is not rigid: a potential vortex has curved streamlines and zero vorticity everywhere off the axis.

4.3 Example: a spinning disc

A disc spins about a fixed centre $\underline{x} \in \kappa_t$. With $\underline{y}$ a point of the rim and $\underline{r} = \underline{y} - \underline{x}$,

$$\dot{\underline{r}} = \omega \underline{n} \times \underline{r}, \quad (4.24)$$

ω the constant rotation rate and $\underline{n}$ a fixed unit normal to the disc (Figure 4.1). Taking $\underline{v}(\underline{x}, t) = \underline{0}$,

$$\underline{v}(\underline{y}, t) - \underline{v}(\underline{x}, t) = \omega \underline{n} \times (\underline{y} - \underline{x}). \quad (4.25)$$

The right side is linear in $\underline{y} - \underline{x}$, so there is a spatially uniform $\underline{L}(t)$ with $\underline{v}(\underline{y}, t) - \underline{v}(\underline{x}, t) = \underline{L}(t)(\underline{y} - \underline{x})$. It is skew, so $\underline{L} = \underline{W}$, $\underline{D} = \underline{O}$, and the axial vector is

$$\underline{w}(t) = \omega \underline{n}. \quad (4.26)$$

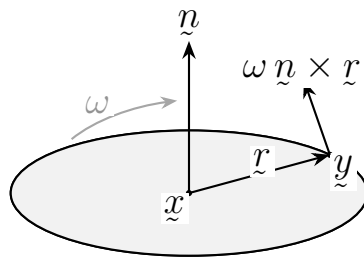


Figure 4.1: The spinning disc. Every point of the body moves with $\underline{v}(\underline{y}) - \underline{v}(\underline{x}) = \omega \underline{n} \times (\underline{y} - \underline{x})$, which is linear in $\underline{y} - \underline{x}$ with a skew coefficient. Hence $\underline{D} = \underline{O}$: the motion is rigid. The vorticity is $\underline{w} = \omega \underline{n}$, the angular velocity itself, because $\text{curl } \underline{v}$ of a rigid field is twice the angular velocity and (4.23) halves it.

In components (4.25) reads

$$v_i(\underline{y}, t) - v_i(\underline{x}, t) = \omega \varepsilon_{ijk} n_j (y_k - x_k), \quad (4.27)$$

so, matching against $L_{ik}(y_k - x_k)$,

$$L_{ik} = \omega \varepsilon_{ijk} n_j.$$

Then $L_{ki} = \omega \varepsilon_{kji} n_j = -\omega \varepsilon_{ijk} n_j = -L_{ik}$, so $D_{ik} = 0$ and $W_{ik} = \omega \varepsilon_{ijk} n_j$: the motion is rigid. The vorticity follows from (4.22),

$$\begin{aligned} \underline{w} &= w_j \underline{e}_j = \frac{1}{2} \varepsilon_{jki} v_{i,k} \underline{e}_j = \frac{1}{2} \varepsilon_{jki} L_{ik} \underline{e}_j \\ &= \frac{1}{2} \omega \varepsilon_{jki} \varepsilon_{ilk} n_l \underline{e}_j = \frac{1}{2} \omega (2\delta_{jl}) n_l \underline{e}_j = \omega n_j \underline{e}_j = \omega \underline{n}, \end{aligned} \quad (4.28)$$

the contraction being $\varepsilon_{jki} \varepsilon_{ilk} = \varepsilon_{ijk} \varepsilon_{ilk} = 2\delta_{jl}$ after cyclic shifts. This confirms (4.26).

REMARK 4.14 (two misprints in the printed (4.27)–(4.28)). The text writes “ $L_{ij} = \omega e_{ijk} n_j$ ” and, two lines later, “ $W_{ij} = \omega e_{ijk} n_j$ ”. Neither can be right as printed: the index j occurs three times, once free on the left and twice on the right, so the equation is not index balanced. Matching (4.27) against $v_i - v_i = L_{ik}(y_k - x_k)$ shows that the free indices are i and k and that j is the summed one, giving $L_{ik} = \omega \varepsilon_{ijk} n_j$ and $W_{ik} = \omega \varepsilon_{ijk} n_j$, as used above. Nothing downstream changes.

Second, the printed chain for (4.28) carries ω in the line $\frac{1}{2} \omega \varepsilon_{jki} L_{ik} \underline{e}_j$ and again in the following line. Since L already contains ω , the factor enters exactly once, at the substitution of L , as written above. Again the final answer $\underline{w} = \omega \underline{n}$ is unaffected.

4.4 Rigid-body motions

A rigid motion is (3.121), $\underline{x} = \underline{Q}(t)\underline{X} + \underline{c}(t)$, with $\underline{Q}$ orthogonal. Then $\underline{F} = \underline{Q}$ and, by (4.3) with $\underline{Q}^{-1} = \underline{Q}^t$,

$$\underline{L} = \dot{\underline{Q}} \underline{Q}^t. \quad (4.29)$$

By Problem 1(c) this is skew, so

$$\underline{D} = \underline{O}, \quad \underline{W} = \dot{\underline{Q}} \underline{Q}^t,$$

and $\underline{W}$ is spatially uniform, depending on t alone. To recover the velocity field, integrate at fixed t ; uniformity of $\underline{W}$ is what lets it pass inside the differential, exactly as at (3.99):

$$d\underline{v} = \underline{L} d\underline{x} = \underline{W}(t) d\underline{x} = d(\underline{W}\underline{x}), \quad (4.30)$$

so $d(\underline{v} - \underline{W}\underline{x}) = \underline{0}$ and the bracket is independent of $\underline{x}$:

$$\underline{v}(\underline{x}, t) = \underline{W}(t)\underline{x} + \underline{d}(t) = \underline{w}(t) \times \underline{x} + \underline{d}(t), \quad (4.31)$$

with $\underline{d}(t) \in \mathbb{E}^3$ spatially uniform. Subtracting the same relation at two points,

$$\underline{v}(\underline{y}, t) - \underline{v}(\underline{x}, t) = \underline{w}(t) \times (\underline{y} - \underline{x}), \quad (4.32)$$

of which the spinning disc is the special case $\underline{w} = \omega \underline{n}$.

Alternatively differentiate (3.121) directly and eliminate $\underline{X}$ by $\underline{X} = \underline{Q}^t(\underline{x} - \underline{c})$:

$$\underline{v} = \dot{\underline{Q}}\underline{X} + \dot{\underline{c}} = \dot{\underline{Q}}\underline{Q}^t(\underline{x} - \underline{c}) + \dot{\underline{c}} = \underline{W}\underline{x} + (\dot{\underline{c}} - \underline{W}\underline{c}), \quad (4.33)$$

which is (4.31) with $\underline{d} = \dot{\underline{c}} - \underline{W}\underline{c}$.

REMARK 4.15 (what survives from Chapter 3). Three statements now say the same thing about rigidity, at three levels:

$$\text{geometry: } |\underline{\chi}(\underline{Y}, t) - \underline{\chi}(\underline{X}, t)| = |\underline{Y} - \underline{X}|, \quad (3.113)$$

$$\text{deformation: } \underline{U} = \hat{\underline{I}}, \quad \underline{C} = \hat{\underline{I}}, \quad \underline{E} = \hat{\underline{Q}}, \quad (3.122)$$

$$\text{rate: } \underline{D} = \underline{Q}.$$

The third is the useful one in practice, because it is a condition on the present velocity field alone and needs no reference configuration. It is also the reason constitutive laws for fluids are written in $\underline{D}$: a fluid cannot feel how far it has come from a reference state, only how fast it is being distorted now.

4.5 Problems

PROBLEM 1 (Rules for time derivatives of tensors). (a) Product rule. (b) Differentiation commutes with transposition. (c) If $\underline{R}$ is orthogonal,

$\dot{\underline{R}}\underline{R}^t$ is skew. (d) Differentiation does *not* commute with inversion; find $(\underline{A}^{-1})^\cdot$.

Solution. Every part of this problem is settled the same way. A tensor equation is nine scalar equations in components, the components are ordinary functions of t , and the ordinary rules of calculus apply to them. The only care needed is that tensors do not commute, so the *order* of factors must be preserved when the component statement is read back as a tensor statement. All four parts are used repeatedly in this chapter and the next, and (a) and (c) in particular are what (4.4) and (4.6) rest on.

(a) In components $(\underline{A}\underline{B})_{ij} = A_{ik}B_{kj}$, a sum of products of scalar functions of t , so the scalar product rule applies to each term of the sum:

$$[(\underline{A}\underline{B})^\cdot]_{ij} = (A_{ik}B_{kj})^\cdot = \dot{A}_{ik}B_{kj} + A_{ik}\dot{B}_{kj}.$$

Read the two terms back as tensors. In the first, the free indices i and j sit on $\dot{A}$ and B in that order with k summed between them, so it is $\dot{\underline{A}}\underline{B}$; the second is $\underline{A}\dot{\underline{B}}$ by the same reading. Hence

$$(\underline{A}\underline{B})^\cdot = \dot{\underline{A}}\underline{B} + \underline{A}\dot{\underline{B}}.$$

The order matters: $\underline{B}\dot{\underline{A}}$ would be a different tensor, and writing the rule with the factors commuted is the commonest slip in the subject.

(b) $(\underline{A}^t)_{ij} = A_{ji}$, so $[(\underline{A}^t)^\cdot]_{ij} = \dot{A}_{ji} = (\dot{\underline{A}}^t)_{ij}$.

(c) Differentiate $\underline{R}\underline{R}^t = \underline{I}$ by (a) and (b):

$$\dot{\underline{R}}\underline{R}^t + \underline{R}\dot{\underline{R}}^t = \underline{O} \implies \dot{\underline{R}}\underline{R}^t = -\underline{R}\dot{\underline{R}}^t = -(\dot{\underline{R}}\underline{R}^t)^t,$$

which is skewness.

(d) Differentiate $\underline{A}\underline{A}^{-1} = \underline{I}$:

$$\dot{\underline{A}}\underline{A}^{-1} + \underline{A}(\underline{A}^{-1})^\cdot = \underline{O} \implies (\underline{A}^{-1})^\cdot = -\underline{A}^{-1}\dot{\underline{A}}\underline{A}^{-1}.$$

This is not $(\dot{\underline{A}})^{-1}$: the two differ already for $\underline{A} = t\underline{I}$, where $(\underline{A}^{-1})^\cdot = -t^{-2}\underline{I}$ while $(\dot{\underline{A}})^{-1} = \underline{I}$. The scalar case $d(1/a)/dt = -\dot{a}/a^2$ is the same statement, with the two copies of $\underline{A}^{-1}$ collapsing to a^{-2} only because scalars commute. $\square$

PROBLEM 2 ($\dot{J} = J \operatorname{tr} \underline{L}$, and $\partial J / \partial F_{iA} = F_{iA}^*$). (a) Differentiate $J[\underline{u}, \underline{v}, \underline{w}] = [\underline{F}\underline{u}, \underline{F}\underline{v}, \underline{F}\underline{w}]$ at fixed $\underline{u}, \underline{v}, \underline{w}$ and use the box-product definition of the trace to get $\dot{J}/J = \operatorname{tr} \underline{L}$. (b) Compare with $\dot{J} = (\partial J / \partial F_{iA}) \dot{F}_{iA}$ to conclude $\partial J / \partial F_{iA} = F_{iA}^*$.

Solution. **(a)** The box product is trilinear, so its derivative obeys a three-term product rule. With $\underline{u}, \underline{v}, \underline{w}$ fixed and $\dot{\underline{F}} = \underline{L}\underline{F}$,

$$\begin{aligned} \dot{J}[\underline{u}, \underline{v}, \underline{w}] &= [\dot{\underline{F}}\underline{u}, \underline{F}\underline{v}, \underline{F}\underline{w}] + [\underline{F}\underline{u}, \dot{\underline{F}}\underline{v}, \underline{F}\underline{w}] + [\underline{F}\underline{u}, \underline{F}\underline{v}, \dot{\underline{F}}\underline{w}] \\ &= [\underline{L}\underline{a}, \underline{b}, \underline{c}] + [\underline{a}, \underline{L}\underline{b}, \underline{c}] + [\underline{a}, \underline{b}, \underline{L}\underline{c}], \end{aligned}$$

where $\underline{a} = \underline{F}\underline{u}$, $\underline{b} = \underline{F}\underline{v}$, $\underline{c} = \underline{F}\underline{w}$. The last line is exactly Definition 1.49₁ of the trace, so it equals $(\operatorname{tr} \underline{L})[\underline{a}, \underline{b}, \underline{c}] = (\operatorname{tr} \underline{L})J[\underline{u}, \underline{v}, \underline{w}]$. Cancelling the non-zero box product,

$$\dot{J} = J \operatorname{tr} \underline{L} = J \operatorname{tr} \underline{D}$$

the second form because $\operatorname{tr} \underline{W} = 0$ for skew $\underline{W}$. So J is constant in time exactly when $\operatorname{div} \underline{v} = \operatorname{tr} \underline{L} = 0$: that is what *isochoric* means for a motion.

(b) By the chain rule, J being a function of the nine F_{iA} ,

$$\dot{J} = \frac{\partial J}{\partial F_{iA}} \dot{F}_{iA}.$$

On the other hand, by (a) and $L_{ii} = \dot{F}_{iA} F_{Ai}^{-1}$ from (4.3),

$$\dot{J} = J L_{ii} = J F_{Ai}^{-1} \dot{F}_{iA} = (J F_{iA}^{-t}) \dot{F}_{iA}.$$

The two hold for every $\dot{F}_{iA}$, which are arbitrary, so the coefficients agree:

$$\frac{\partial J}{\partial F_{iA}} = J F_{iA}^{-t} = F_{iA}^*,$$

the last step by (2.55). So the cofactor is the derivative of the determinant, which is the identity that produces the Piola stress in Chapter 6.

□

PROBLEM 3 (A material derivative from a spatial field). With $\theta(\underline{x}, t) = (A \cos t)/r$, $r^2 = x_1^2 + x_2^2$, and $v_1 = x_1^2 x_2 + x_3^2$, $v_2 = -x_1^3 - x_1 x_2^2$, $v_3 = 0$, find $\dot{\theta}$.

Solution. Use (2.17), $\dot{\theta} = \theta' + \text{grad } \theta \cdot \underline{v}$.

Spatial time derivative, at fixed $\underline{x}$:

$$\theta' = \frac{\partial}{\partial t} \left(\frac{A \cos t}{r} \right) = -\frac{A \sin t}{r}.$$

Spatial gradient. From $r^2 = x_1^2 + x_2^2$, $r_{,1} = x_1/r$, $r_{,2} = x_2/r$, $r_{,3} = 0$. Then $\theta = A \cos t r^{-1}$ gives

$$\theta_{,i} = -A \cos t r^{-2} r_{,i} \implies \text{grad } \theta = -\frac{A \cos t}{r^3} (x_1 \underline{e}_1 + x_2 \underline{e}_2).$$

Convective term. Only v_1, v_2 contribute:

$$\begin{aligned} x_1 v_1 + x_2 v_2 &= x_1 (x_1^2 x_2 + x_3^2) + x_2 (-x_1^3 - x_1 x_2^2) \\ &= x_1^3 x_2 + x_1 x_3^2 - x_1^3 x_2 - x_1 x_2^3 = x_1 (x_3^2 - x_2^3), \end{aligned}$$

the $x_1^3 x_2$ terms cancelling. Hence

$$\dot{\theta} = -\frac{A \sin t}{r} - \frac{A \cos t}{r^3} x_1 (x_3^2 - x_2^3), \quad r = \sqrt{x_1^2 + x_2^2}.$$

On the axis $x_1 = x_2 = 0$ both terms are singular, as θ itself is; off the axis the first term is the cooling seen by a fixed thermometer and the second is what the moving material point picks up by travelling through the spatial gradient. $\square$

PROBLEM 4 (The convective acceleration in Lamb form). Show $\dot{\underline{v}} = \underline{v}' + \frac{1}{2} \text{grad } (|\underline{v}|^2) + 2 \underline{w} \times \underline{v}$.

Solution. The strategy is to compute both sides in terms of $\underline{D}$ and $\underline{W}$ and see that they agree. The left side comes straight from Chapter 2.

The left side. Apply (2.20) to the vector field $\underline{v}$ itself, which is (2.21), and then split $\underline{L}$ into its symmetric and skew parts by (4.7):

$$\dot{\underline{v}} = \underline{v}' + (\text{grad } \underline{v})\underline{v} = \underline{v}' + \underline{L}\underline{v} = \underline{v}' + \underline{D}\underline{v} + \underline{W}\underline{v}.$$

The gradient term. Compute $\frac{1}{2} \text{grad } |\underline{v}|^2$ in components. Write $|\underline{v}|^2 = v_k v_k$ and differentiate with the product rule, noting that the two terms produced are equal because k is summed:

$$\left[\frac{1}{2} \text{grad } |\underline{v}|^2 \right]_i = \frac{1}{2} (v_k v_k)_{,i} = \frac{1}{2} (v_{k,i} v_k + v_k v_{k,i}) = v_k v_{k,i}.$$

Now recognise what $v_k v_{k,i}$ is. Since $L_{ki} = v_{k,i}$, the expression is $L_{ki} v_k = (L^t)_{ik} v_k = (\underline{L}^t \underline{v})_i$, the transpose appearing because the summed index sits in the *first* slot of L . So

$$\frac{1}{2} \text{grad } |\underline{v}|^2 = \underline{L}^t \underline{v} = (\underline{D} + \underline{W})^t \underline{v} = (\underline{D} - \underline{W}) \underline{v},$$

using $\underline{D}^t = \underline{D}$ and $\underline{W}^t = -\underline{W}$.

The cross-product term. By the definition of the axial vector, Proposition 1.44 with $\underline{W}$ in place of $\underline{A}$, we have $\underline{W} \underline{v} = \underline{w} \times \underline{v}$ for every $\underline{v}$.

Putting them together. Substitute both into the claimed right side:

$$\underline{v}' + \frac{1}{2} \text{grad } |\underline{v}|^2 + 2 \underline{w} \times \underline{v} = \underline{v}' + (\underline{D} - \underline{W}) \underline{v} + 2 \underline{W} \underline{v} = \underline{v}' + \underline{D} \underline{v} + \underline{W} \underline{v},$$

since $-\underline{W} \underline{v} + 2 \underline{W} \underline{v} = \underline{W} \underline{v}$. This is exactly the expression computed for $\dot{\underline{v}}$ above, so the two sides agree and the identity is proved.

Why anyone bothers. The convective term $(\text{grad } \underline{v}) \underline{v}$ is quadratic in the velocity and is the source of every difficulty in fluid mechanics. The Lamb form splits it into two pieces of quite different character. The first, $\frac{1}{2} \text{grad } |\underline{v}|^2$, is a gradient, so it integrates to a potential and contributes nothing to the circulation around a closed material curve, exactly as was seen in Problem 1 of Chapter 5. The second, $2 \underline{w} \times \underline{v}$, is perpendicular to $\underline{v}$ and therefore does no work on the flow; it is the term that vanishes when the motion is irrotational. In a steady irrotational flow both simplifications apply at once, $\underline{v}' = \underline{0}$ and $\underline{w} = \underline{0}$, so $\dot{\underline{v}}$ is a pure gradient and the equation of motion of Chapter 6 integrates immediately. That integral is Bernoulli's theorem, and this rearrangement is where it comes from. $\square$

PROBLEM 5 (A given velocity field). For $v_1 = ax_2x_3$, $v_2 = -ax_1x_3$, $v_3 = bx_3$ find (a) $\underline{L}$, (b) $\underline{D}, \underline{W}$, (c) $\underline{w}$, (d) the conditions on a, b for the motion to be (i) isochoric and (ii) irrotational.

Solution. **(a)** $L_{ij} = v_{i,j}$, differentiating each component with respect to each coordinate:

$$(L_{ij}) = \begin{pmatrix} 0 & ax_3 & ax_2 \\ -ax_3 & 0 & -ax_1 \\ 0 & 0 & b \end{pmatrix}.$$

(b) $\tilde{D} = \frac{1}{2}(\tilde{L} + \tilde{L}^t)$ and $\tilde{W} = \frac{1}{2}(\tilde{L} - \tilde{L}^t)$, entry by entry. The $(1, 2)$ entries cancel in $\tilde{D}$ and double in $\tilde{W}$:

$$(D_{ij}) = \begin{pmatrix} 0 & 0 & \frac{1}{2}ax_2 \\ 0 & 0 & -\frac{1}{2}ax_1 \\ \frac{1}{2}ax_2 & -\frac{1}{2}ax_1 & b \end{pmatrix}, \quad (W_{ij}) = \begin{pmatrix} 0 & ax_3 & \frac{1}{2}ax_2 \\ -ax_3 & 0 & -\frac{1}{2}ax_1 \\ -\frac{1}{2}ax_2 & \frac{1}{2}ax_1 & 0 \end{pmatrix}.$$

(c) From $\tilde{W}\underline{u} = \underline{w} \times \underline{u}$ one has $W_{ij} = -\varepsilon_{ijk}w_k$, so $w_1 = -W_{23}$, $w_2 = W_{13}$, $w_3 = -W_{12}$:

$$\underline{w} = \frac{a}{2}(x_1\underline{e}_1 + x_2\underline{e}_2) - ax_3\underline{e}_3.$$

Check against (4.23), $\underline{w} = \frac{1}{2} \text{curl } \underline{v}$:

$$\begin{aligned} (\text{curl } \underline{v})_1 &= v_{3,2} - v_{2,3} = 0 + ax_1 = ax_1, \\ (\text{curl } \underline{v})_2 &= v_{1,3} - v_{3,1} = ax_2 - 0 = ax_2, \\ (\text{curl } \underline{v})_3 &= v_{2,1} - v_{1,2} = -ax_3 - ax_3 = -2ax_3, \end{aligned}$$

whose half is the $\underline{w}$ just found.

(d) (i) Isochoric means $\tilde{J} = 0$, i.e. $\text{tr } \tilde{L} = 0$ by Problem 2. Here $\text{tr } \tilde{L} = 0 + 0 + b = b$, so the condition is

$$b = 0.$$

(ii) Irrotational means $\underline{w} = \underline{0}$ at every point; the component $w_1 = \frac{1}{2}ax_1$ must vanish for all x_1 , so

$$a = 0,$$

and then $\underline{w} = \underline{0}$ whatever b is. Note the two conditions are independent: b controls the volume change, a the rotation. $\square$

PROBLEM 6 (Spin of a fibre when $\underline{W} = \underline{Q}$). Let $\underline{F} = \lambda \underline{e}_1 \otimes \hat{\underline{E}}_1 + \lambda^{-1/2}(\underline{e}_2 \otimes \hat{\underline{E}}_2 + \underline{e}_3 \otimes \hat{\underline{E}}_3)$ with $\lambda = \lambda(t)$. (a) Show the motion is isochoric. (b) Verify the stated $\underline{F}^{-1}$. (c) Find $\underline{L}$ and show $\underline{W} = \underline{Q}$. (d) With $\underline{m} = \cos \phi \underline{e}_r(\theta) + \sin \phi \underline{e}_3$, find $\dot{\theta}$ and $\dot{\phi}$, and identify the directions with $\dot{\underline{m}} = \underline{0}$.

Solution. **(a)** $\underline{F}$ is already in the form (3.43) with orthonormal frames on both sides, so its principal stretches are the coefficients and

$$J = \lambda \cdot \lambda^{-1/2} \cdot \lambda^{-1/2} = \lambda^{1-\frac{1}{2}-\frac{1}{2}} = 1.$$

(b) Multiply, using $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c})\underline{a} \otimes \underline{d}$; only the three diagonal products survive:

$$\underline{F}\underline{F}^{-1} = \lambda\lambda^{-1}\underline{e}_1 \otimes \underline{e}_1 + \lambda^{-1/2}\lambda^{1/2}(\underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3) = \underline{I}.$$

(c) Differentiate $\underline{F}$ at fixed $\underline{X}$, the bases being constant:

$$\dot{\underline{F}} = \dot{\lambda} \underline{e}_1 \otimes \hat{\underline{E}}_1 - \frac{1}{2}\lambda^{-3/2}\dot{\lambda}(\underline{e}_2 \otimes \hat{\underline{E}}_2 + \underline{e}_3 \otimes \hat{\underline{E}}_3),$$

and multiply by $\underline{F}^{-1}$ of part (b):

$$\underline{L} = \dot{\underline{F}}\underline{F}^{-1} = \frac{\dot{\lambda}}{\lambda} \underline{e}_1 \otimes \underline{e}_1 - \frac{\dot{\lambda}}{2\lambda}(\underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3).$$

This is symmetric, so $\underline{W} = \text{skw } \underline{L} = \underline{Q}$ and $\underline{D} = \underline{L}$. Its trace is $\dot{\lambda}/\lambda - \dot{\lambda}/\lambda = 0$, consistent with (a) and Problem 2.

(d) Write $\alpha = \dot{\lambda}/\lambda$, so $\underline{D} = \alpha \underline{e}_1 \otimes \underline{e}_1 - \frac{\alpha}{2}(\underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3)$, and

$$\begin{aligned} \underline{m} &= (\cos \phi \cos \theta, \cos \phi \sin \theta, \sin \phi), \\ \underline{D}\underline{m} &= \left(\alpha \cos \phi \cos \theta, -\frac{\alpha}{2} \cos \phi \sin \theta, -\frac{\alpha}{2} \sin \phi\right). \end{aligned}$$

Differentiating the parametrisation, with $\underline{e}_\theta = \underline{e}'_r(\theta)$ and $\underline{e}_\phi = -\sin \phi \underline{e}_r + \cos \phi \underline{e}_3$,

$$\dot{\underline{m}} = \cos \phi \dot{\theta} \underline{e}_\theta + \dot{\phi} \underline{e}_\phi \implies \dot{\theta} = \frac{\underline{e}_\theta \cdot \dot{\underline{m}}}{\cos \phi}, \quad \dot{\phi} = \underline{e}_\phi \cdot \dot{\underline{m}},$$

since $\{\underline{m}, \underline{e}_\theta, \underline{e}_\phi\}$ is orthonormal. By (4.16) with $\underline{W} = \underline{Q}$, $\dot{\underline{m}} = \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m}$, and both $\underline{e}_\theta$ and $\underline{e}_\phi$ are perpendicular to $\underline{m}$, so the second

term drops out of both projections: $\underline{e}_\theta \cdot \dot{\underline{m}} = \underline{e}_\theta \cdot \underline{D}\underline{m}$ and $\underline{e}_\phi \cdot \dot{\underline{m}} = \underline{e}_\phi \cdot \underline{D}\underline{m}$.
Now

$$\begin{aligned}\underline{e}_\theta \cdot \underline{D}\underline{m} &= -\sin \theta \alpha \cos \phi \cos \theta + \cos \theta \left(-\frac{\alpha}{2} \cos \phi \sin \theta \right) \\ &= -\frac{3\alpha}{2} \cos \phi \sin \theta \cos \theta = -\frac{3\alpha}{4} \cos \phi \sin 2\theta,\end{aligned}$$

$$\begin{aligned}\underline{e}_\phi \cdot \underline{D}\underline{m} &= -\alpha \sin \phi \cos \phi \cos^2 \theta + \frac{\alpha}{2} \sin \phi \cos \phi \sin^2 \theta - \frac{\alpha}{2} \sin \phi \cos \phi \\ &= \alpha \sin \phi \cos \phi \left[-\cos^2 \theta + \frac{1}{2} \sin^2 \theta - \frac{1}{2} \right] = -\frac{3\alpha}{2} \sin \phi \cos \phi \cos^2 \theta,\end{aligned}$$

using $\frac{1}{2} \sin^2 \theta - \frac{1}{2} = -\frac{1}{2} \cos^2 \theta$. Hence

$$\dot{\theta} = -\frac{3\dot{\lambda}}{4\lambda} \sin 2\theta, \quad \dot{\phi} = -\frac{3\dot{\lambda}}{4\lambda} \sin 2\phi \cos^2 \theta.$$

Both are generally non-zero although $\underline{W} = \underline{Q}$: this is Remark 4.9 made quantitative, and it is the point of the problem.

When is $\dot{\underline{m}} = \underline{0}$? Since $\underline{W} = \underline{Q}$, (4.16) gives $\dot{\underline{m}} = \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m}$, which vanishes exactly when $\underline{D}\underline{m}$ is parallel to $\underline{m}$, i.e. when $\underline{m}$ is a principal direction of $\underline{D}$. Reading the boxed formulas, $\dot{\theta} = 0$ needs $\sin 2\theta = 0$ and $\dot{\phi} = 0$ needs $\sin 2\phi \cos^2 \theta = 0$, so

$$\begin{aligned}\theta &= \frac{\pi}{2} \text{ or } \frac{3\pi}{2} : \cos \theta = 0, \text{ so any } \phi \text{ works,} \\ \theta &= 0 \text{ or } \pi : \phi = 0 \text{ or } \frac{\pi}{2}.\end{aligned}$$

The first family is every direction in the $(\underline{e}_2, \underline{e}_3)$ plane, the second gives $\underline{m} = \pm \underline{e}_1$ and $\underline{m} = \pm \underline{e}_3$. That is exactly the principal set of $\underline{D}$: the value $-\alpha/2$ is repeated, so the whole $(\underline{e}_2, \underline{e}_3)$ plane is principal, and $\underline{e}_1$ carries the isolated value α . $\square$

PROBLEM 7 (Rate of area change and rotation of the normal). With $\alpha = da/dA$ the local area ratio of a material surface, (a) show $\alpha > 0$; (b) establish $\dot{\alpha}/\alpha = \text{tr } \underline{D} - \underline{n} \cdot \underline{D}\underline{n}$ and $\dot{\underline{n}} = (\underline{n} \cdot \underline{D}\underline{n})\underline{n} - \underline{L}^t \underline{n}$.

Solution. **(a)** By Piola–Nanson (2.54), $\alpha \underline{n} = \underline{F}^* \underline{N}$ with $|\underline{n}| = 1$, so $\alpha = |\underline{F}^* \underline{N}|$. Now $\underline{F}^* = J \underline{F}^{-t}$ is invertible, since $\det \underline{F}^* = J^2 \neq 0$, and $\underline{N} \neq \underline{0}$; hence $\underline{F}^* \underline{N} \neq \underline{0}$ and $\alpha > 0$.

(b) Differentiate $\alpha \underline{\underline{n}} = J \underline{\underline{F}}^{-t} \underline{\underline{N}}$ at fixed $\underline{\underline{N}}$, which is legitimate because $\underline{\underline{N}}$ is a material vector. Two ingredients are needed. First $\dot{J} = J \operatorname{tr} \underline{\underline{D}}$ from Problem 2. Second, by Problem 1(b),(d) and $\dot{\underline{\underline{F}}} = \underline{\underline{L}} \underline{\underline{F}}$,

$$(\underline{\underline{F}}^{-t})^\cdot = \left((\underline{\underline{F}}^{-1})^\cdot \right)^t = \left(-\underline{\underline{F}}^{-1} \underline{\underline{L}} \underline{\underline{F}} \underline{\underline{F}}^{-1} \right)^t = -(\underline{\underline{F}}^{-1} \underline{\underline{L}})^t = -\underline{\underline{L}}^t \underline{\underline{F}}^{-t}.$$

Therefore

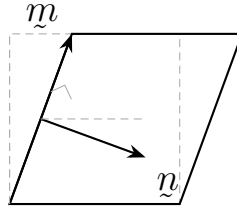
$$\dot{\alpha} \underline{\underline{n}} + \alpha \dot{\underline{\underline{n}}} = \dot{J} \underline{\underline{F}}^{-t} \underline{\underline{N}} + J (\underline{\underline{F}}^{-t})^\cdot \underline{\underline{N}} = (\operatorname{tr} \underline{\underline{D}}) \alpha \underline{\underline{n}} - \underline{\underline{L}}^t \alpha \underline{\underline{n}},$$

having recognised $J \underline{\underline{F}}^{-t} \underline{\underline{N}} = \alpha \underline{\underline{n}}$ twice. Dot with $\underline{\underline{n}}$, using $\underline{\underline{n}} \cdot \dot{\underline{\underline{n}}} = 0$ and $\underline{\underline{n}} \cdot \underline{\underline{L}}^t \underline{\underline{n}} = \underline{\underline{n}} \cdot \underline{\underline{L}} \underline{\underline{n}} = \underline{\underline{n}} \cdot \underline{\underline{D}} \underline{\underline{n}}$:

$$\frac{\dot{\alpha}}{\alpha} = \operatorname{tr} \underline{\underline{D}} - \underline{\underline{n}} \cdot \underline{\underline{D}} \underline{\underline{n}}$$

Substituting this back and dividing by α ,

$$\dot{\underline{\underline{n}}} = (\operatorname{tr} \underline{\underline{D}}) \underline{\underline{n}} - \underline{\underline{L}}^t \underline{\underline{n}} - (\operatorname{tr} \underline{\underline{D}} - \underline{\underline{n}} \cdot \underline{\underline{D}} \underline{\underline{n}}) \underline{\underline{n}} = (\underline{\underline{n}} \cdot \underline{\underline{D}} \underline{\underline{n}}) \underline{\underline{n}} - \underline{\underline{L}}^t \underline{\underline{n}}.$$



dashed horizontal: where convecting $\underline{\underline{n}}$ with $\underline{\underline{L}}$ would leave it

$$\dot{\underline{\underline{m}}} = \underline{\underline{L}} \underline{\underline{m}} - (\underline{\underline{m}} \cdot \underline{\underline{D}} \underline{\underline{m}}) \underline{\underline{m}} \quad \dot{\underline{\underline{n}}} = (\underline{\underline{n}} \cdot \underline{\underline{D}} \underline{\underline{n}}) \underline{\underline{n}} - \underline{\underline{L}}^t \underline{\underline{n}}$$

Figure E: Simple shear, with the square as it stands now dashed. The left edge is a material fibre and also a material surface; its unit tangent $\underline{\underline{m}}$ and unit normal $\underline{\underline{n}}$ must stay perpendicular, and they do. That is what the transpose in $-\underline{\underline{L}}^t \underline{\underline{n}}$ buys: here $\underline{\underline{L}} = \kappa \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_2$, so $\underline{\underline{L}} \underline{\underline{m}}$ tips the fibre to the right while $-\underline{\underline{L}}^t \underline{\underline{n}} = -\kappa \underline{\underline{e}}_2$ tips the normal down by the same angle. Convecting the normal with $\underline{\underline{L}}$ instead, the common slip, gives $\underline{\underline{L}} \underline{\underline{e}}_1 = \underline{\underline{0}}$: the normal would not move at all, and would stop being normal.

Compare with (4.16): a material *line* element is convected by $\underline{L}$, a material *normal* by $-\underline{L}^t$. That sign and transpose are the rate form of the difference between $\underline{F}$ and $\underline{F}^{-t}$ in Piola–Nanson, and forgetting them is the standard error. $\square$

PROBLEM 8 (Stretching when the principal axes are frozen in the body). If $\dot{\underline{u}}_i = \underline{0}$, show $\underline{D} = \sum_i (\ln \lambda_i)^\cdot \underline{v}_i \otimes \underline{v}_i$.

Solution. With $\underline{U} = \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i$ and $\dot{\underline{u}}_i = \underline{0}$, only the coefficients move:

$$\dot{\underline{U}} = \sum_i \dot{\lambda}_i \underline{u}_i \otimes \underline{u}_i, \quad \underline{U}^{-1} = \sum_i \lambda_i^{-1} \underline{u}_i \otimes \underline{u}_i,$$

the inverse by Remark 3.10. Now multiply them. The product of the two sums has nine terms, one for each pair (i, j) :

$$\dot{\underline{U}}\underline{U}^{-1} = \sum_i \sum_j \frac{\dot{\lambda}_i}{\lambda_j} (\underline{u}_i \otimes \underline{u}_i) (\underline{u}_j \otimes \underline{u}_j).$$

Each product of dyads collapses by the rule $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c}) \underline{a} \otimes \underline{d}$, which is the definition of the tensor product applied twice. Here $\underline{b} = \underline{u}_i$ and $\underline{c} = \underline{u}_j$, so the scalar produced is $\underline{u}_i \cdot \underline{u}_j = \delta_{ij}$ by orthonormality of the principal basis. That kills every term with $i \neq j$, six of the nine, and leaves the three diagonal ones with λ_j replaced by λ_i :

$$\dot{\underline{U}}\underline{U}^{-1} = \sum_i \frac{\dot{\lambda}_i}{\lambda_i} \underline{u}_i \otimes \underline{u}_i = \sum_i (\ln \lambda_i)^\cdot \underline{u}_i \otimes \underline{u}_i,$$

the last step being the same chain rule that produced (4.15), $\dot{\lambda}/\lambda = (\ln \lambda)^\cdot$. The result is *symmetric*, being a sum of terms $\underline{u} \otimes \underline{u}$ with real coefficients, and that observation is what the rest of the solution turns on. Now look at (4.5):

$$\underline{L} = \underbrace{\dot{\underline{R}}\underline{R}^t}_{\text{skew, Problem 1(c)}} + \underbrace{\underline{R}(\dot{\underline{U}}\underline{U}^{-1})\underline{R}^t}_{\text{symmetric}},$$

the second being symmetric because $\underline{R}\underline{S}\underline{R}^t$ is symmetric whenever $\underline{S}$ is. By the uniqueness of the symmetric–skew splitting, Problem 14 of

Chapter 1, these two *are* $\underline{\underline{D}}$ and $\underline{\underline{W}}$. Hence, using $(\underline{\underline{R}}a) \otimes (\underline{\underline{R}}b) = \underline{\underline{R}}(a \otimes b)\underline{\underline{R}}^t$ from (3.41) and $\underline{\underline{v}}_i = \underline{\underline{R}}\underline{\underline{u}}_i$,

$$\underline{\underline{D}} = \underline{\underline{R}} \left(\sum_i (\ln \lambda_i)^\cdot \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i \right) \underline{\underline{R}}^t = \sum_i (\ln \lambda_i)^\cdot \underline{\underline{v}}_i \otimes \underline{\underline{v}}_i$$

and $\underline{\underline{W}} = \dot{\underline{\underline{R}}}\underline{\underline{R}}^t$.

Two readings. The principal values of $\underline{\underline{D}}$ are the logarithmic stretch rates, in agreement with (4.17); and $\underline{\underline{D}}$ is here coaxial with $\underline{\underline{V}}$, which is exactly the case in which the Hencky strain of Remark 3.18 satisfies $\underline{\underline{D}} = (\ln \underline{\underline{V}})^\cdot$. In general, when $\dot{\underline{\underline{u}}}_i \neq 0$, that last identity is false. $\square$

PROBLEM 9 (An azimuthally graded axial flow). For $\underline{\underline{v}} = (\theta/\theta_0)W\underline{\underline{e}}_3$ with W, θ_0 constant, find (a) $\underline{\underline{L}}, \underline{\underline{D}}, \underline{\underline{W}}$; (b) the principal values and vectors of $\underline{\underline{D}}$, and $\underline{\underline{w}}$.

Solution. **(a)** $\underline{\underline{v}} = f\underline{\underline{e}}_3$ with $f = W\theta/\theta_0$ a scalar field and $\underline{\underline{e}}_3$ constant, so by Problem 24(f) of Chapter 1, $\text{grad}(f\underline{\underline{e}}_3) = \underline{\underline{e}}_3 \otimes \text{grad } f$. By Problem 25 of Chapter 1 in cylindrical form, $\text{grad } \theta = \underline{\underline{e}}_\theta/r$, whence

$$\underline{\underline{L}} = \frac{W}{\theta_0 r} \underline{\underline{e}}_3 \otimes \underline{\underline{e}}_\theta$$

and, splitting by (4.7) with $c := W/(2\theta_0 r)$,

$$\underline{\underline{D}} = c(\underline{\underline{e}}_3 \otimes \underline{\underline{e}}_\theta + \underline{\underline{e}}_\theta \otimes \underline{\underline{e}}_3), \quad \underline{\underline{W}} = c(\underline{\underline{e}}_3 \otimes \underline{\underline{e}}_\theta - \underline{\underline{e}}_\theta \otimes \underline{\underline{e}}_3).$$

Note $\text{tr } \underline{\underline{D}} = 2c \underline{\underline{e}}_3 \cdot \underline{\underline{e}}_\theta = 0$: the flow is isochoric.

(b) In the ordered orthonormal frame $(\underline{\underline{e}}_r, \underline{\underline{e}}_\theta, \underline{\underline{e}}_3)$,

$$(\underline{\underline{D}}) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & c \\ 0 & c & 0 \end{pmatrix},$$

a pure shear on the $(\underline{\underline{e}}_\theta, \underline{\underline{e}}_3)$ pair with $\underline{\underline{e}}_r$ inert. Its characteristic equation is $-\eta(\eta^2 - c^2) = 0$, so

$$\eta = 0, +c, -c; \quad \underline{\underline{\nu}}_0 = \underline{\underline{e}}_r, \quad \underline{\underline{\nu}}_\pm = \frac{\sqrt{2}}{2}(\underline{\underline{e}}_\theta \pm \underline{\underline{e}}_3).$$

For the vorticity, $\underline{W} = \underline{a} \otimes \underline{b} - \underline{b} \otimes \underline{a}$ has axial vector $\underline{b} \times \underline{a}$, since

$$(\underline{a} \otimes \underline{b} - \underline{b} \otimes \underline{a})\underline{u} = (\underline{b} \cdot \underline{u})\underline{a} - (\underline{a} \cdot \underline{u})\underline{b} = (\underline{b} \times \underline{a}) \times \underline{u}$$

by Problem 11(a). With $\underline{a} = c \underline{e}_3$, $\underline{b} = \underline{e}_\theta$ and $\underline{e}_\theta \times \underline{e}_3 = \underline{e}_r$,

$$\underline{w} = \frac{W}{2\theta_0 r} \underline{e}_r$$

which checks against (4.23): $\text{curl}(f \underline{e}_3) = \text{grad } f \times \underline{e}_3 = \frac{W}{\theta_0 r} \underline{e}_\theta \times \underline{e}_3 = \frac{W}{\theta_0 r} \underline{e}_r$, half of which is $\underline{w}$. $\square$

REMARK 4.16 (the field of Problem 9 is not single valued). θ jumps by 2π on crossing the half plane $\theta = 0$, so $\underline{v} = (W\theta/\theta_0)\underline{e}_3$ jumps by $2\pi W/\theta_0$ there: the velocity field is discontinuous across that surface, and the motion it describes tears the material. The *gradient* $\underline{L}$ is nevertheless perfectly single valued, because $\text{grad } \theta = \underline{e}_\theta/r$ is, which is why the computation above goes through without complaint. This is a useful warning: $\underline{L}$, $\underline{D}$, $\underline{W}$ can be smooth where $\underline{v}$ is not, and a velocity field must be checked for single-valuedness separately.

PROBLEM 10 (Circular flow). For $\underline{v} = rf(r)\underline{e}_\theta$ in cylindrical coordinates, show $\underline{L} = [\partial_r(rf)]\underline{e}_\theta \otimes \underline{e}_r - f \underline{e}_r \otimes \underline{e}_\theta$ and find $\underline{D}$ and $\underline{W}$.

Solution. Apply (1.145), the polar form of $\text{grad } \underline{u}$, to $u_r = 0$, $u_\theta = rf(r)$, $u_z = 0$, all independent of θ and z . The six coefficients are

$$\begin{aligned} u_{r,r} &= 0, & \frac{1}{r}(u_{r,\theta} - u_\theta) &= -f, & u_{\theta,r} &= (rf)', \\ \frac{1}{r}(u_{\theta,\theta} + u_r) &= 0, & u_{z,r} &= 0, & \frac{1}{r}u_{z,\theta} &= 0, \end{aligned}$$

so

$$\underline{L} = (rf)' \underline{e}_\theta \otimes \underline{e}_r - f \underline{e}_r \otimes \underline{e}_\theta$$

as stated. Splitting by (4.7), and using $(rf)' = f + rf'$ so that $(rf)' - f = rf'$ and $(rf)' + f = 2f + rf'$,

$$\underline{D} = \frac{rf'}{2}(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r), \quad \underline{W} = \frac{2f + rf'}{2}(\underline{e}_\theta \otimes \underline{e}_r - \underline{e}_r \otimes \underline{e}_\theta).$$

By the axial-vector rule of Problem 9, with $\underline{a} = \underline{e}_\theta$, $\underline{b} = \underline{e}_r$ and $\underline{e}_r \times \underline{e}_\theta = \underline{e}_3$,

$$\underline{w} = \frac{2f + rf'}{2} \underline{e}_3 = \frac{1}{2r} (r^2 f)' \underline{e}_3.$$

Two special cases show what the two tensors separate. If f is constant the flow is a rigid rotation: then $f' = 0$, so $\underline{D} = \underline{Q}$, as Remark 4.6 requires, and $\underline{w} = f \underline{e}_3$, the angular velocity. If instead $f = k/r^2$, then $r^2 f = k$ is constant and $\underline{w} = 0$: the flow is irrotational although every particle travels in a circle. This is the potential vortex, and it is the standard counterexample to the belief that curved streamlines imply vorticity. Its stretching is $\underline{D} = -\frac{k}{r^2} (\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r)$, which is not zero: the material is being sheared even though it is not spinning.

The two cases are worth seeing side by side, because they separate the two things $\underline{L}$ measures. Put a radial and an azimuthal material fibre at the same point and let Ω denote the rate at which each turns about $\underline{e}_3$. From $\dot{\underline{m}} = \underline{L}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m}$ with $\underline{e}_r \cdot \underline{D}\underline{e}_r = \underline{e}_\theta \cdot \underline{D}\underline{e}_\theta = 0$,

$$\dot{\underline{e}}_r = (rf)' \underline{e}_\theta, \quad \dot{\underline{e}}_\theta = -f \underline{e}_r, \quad \text{so} \quad \Omega_r = (rf)' = f + rf', \quad \Omega_\theta = f.$$

Their mean is the vorticity and their difference is the shear:

$$\frac{1}{2}(\Omega_r + \Omega_\theta) = \frac{2f + rf'}{2}, \quad \Omega_\theta - \Omega_r = -rf' = -2 \underline{e}_r \cdot \underline{D}\underline{e}_\theta,$$

in agreement with $\underline{w}$ above and with (4.21). Rigid rotation is $\Omega_r = \Omega_\theta = f$; the potential vortex $f = k/r^2$ has $rf' = -2f$, hence $\Omega_r = -f$ and $\Omega_\theta = +f$, equal and opposite. $\square$

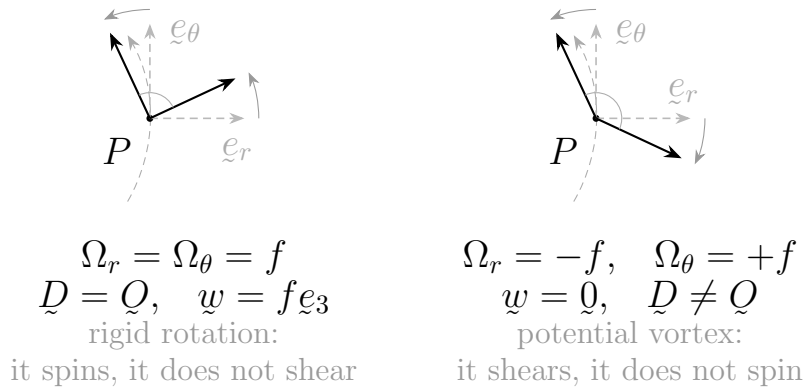


Figure F: Problem 10, at one point P of a circular streamline, with the two material fibres as they stand now shown dashed. In rigid rotation both turn at the same rate, so their mean rate, the vorticity, is f and their relative rate, the shearing, is zero. In the potential vortex $f = k/r^2$ the two rates are equal and opposite: the mean vanishes, so a paddle wheel placed at P would circle the origin without ever turning, yet the right angle between the fibres opens at rate $2f$, so the material is being sheared throughout. Curved streamlines therefore say nothing about vorticity.

Chapter 5

Transport Formulas

The balance laws of the next chapter are statements about *integrals*: how much mass, momentum or energy a lump of material carries, and how fast that amount changes. Every one of them therefore needs an answer to a single question:

$$\frac{d}{dt} \int_{P_t} (\cdot) dv = ?$$

The question is harder than it looks because *both* the integrand and the region of integration depend on time. This chapter answers it, in four settings of increasing generality: a material region (Reynolds' formula), a moving surface (the flux derivative), a region that is not material at all, and a region across which the integrand jumps. Nothing here is physics yet; it is the calculus the physics will be written in.

Chapter 1 already proved the divergence theorem and the two variants that follow from it, Theorem 1.100 and Corollary 1.101. We do not prove them again. Section 5.1 instead collects them in the form the balance laws will use and then makes a point Chapter 1 had no reason to make: two of the three are *symbolic* statements, and it matters to know exactly what they mean.

5.1 The divergence theorem and its variants

For a vector field $\underline{a}(\underline{x})$ on a volume $R \subset \mathbb{E}^3$ bounded by a piecewise smooth surface ∂R , with $\underline{n}$ the piecewise continuous exterior unit normal, Theorem 1.100 is

$$\int_R \operatorname{div} \underline{a} \, dv = \int_{\partial R} \underline{a} \cdot \underline{n} \, da, \quad (5.1)$$

or, in Cartesian coordinates and components,

$$\int_R a_{i,i} \, dv = \int_{\partial R} a_i n_i \, da. \quad (5.2)$$

Both of the variants we need come from (5.1) by the same device: feed it a field built from an *arbitrary constant vector* $\underline{c}$, and then let the arbitrariness of $\underline{c}$ do the work. It is worth isolating the device, because it is used four times in this chapter and a dozen more in the next.

LEMMA 5.1 (the constant-vector argument). *Let $\underline{u}$ be a fixed vector and suppose $\underline{c} \cdot \underline{u} = 0$ for every vector $\underline{c}$. Then $\underline{u} = \underline{0}$.*

Proof. The hypothesis holds for *every* $\underline{c}$, so in particular for $\underline{c} = \underline{u}$, giving $\underline{u} \cdot \underline{u} = |\underline{u}|^2 = 0$; positive definiteness of the inner product, Definition 1.5(d), then forces $\underline{u} = \underline{0}$. ■

REMARK 5.2 (why one may not simply “cancel $\underline{c}$ ”). The step is worth stating as a lemma because the tempting shortcut is wrong. From $\underline{c} \cdot \underline{u} = \underline{c} \cdot \underline{w}$ one may *not* conclude $\underline{u} = \underline{w}$ for a *particular* $\underline{c}$: any $\underline{u} - \underline{w}$ perpendicular to that $\underline{c}$ satisfies it. It is the quantifier “for all $\underline{c}$ ” that carries the argument, and the choice $\underline{c} = \underline{u}$ is what converts it into a statement about a norm. The same pattern will reappear in the localization theorem, where “for all subvolumes” plays the role that “for all $\underline{c}$ ” plays here.

The gradient of a scalar

Let $\varphi(\underline{x})$ be a smooth scalar field and put $\underline{a} = \varphi \underline{c}$ with $\underline{c}$ spatially uniform. The choice looks arbitrary; it is not. We want a statement

about $\text{grad } \varphi$, and (5.1) only knows how to handle a divergence, so we need a vector field whose divergence produces $\text{grad } \varphi$. Multiplying φ by a constant vector is the simplest thing that does it, and keeping the vector constant is what makes the product rule collapse to a single term.

In components $a_i = \varphi c_i$. Differentiate, remembering that the c_i are numbers and not functions of position, so they pass through $\partial/\partial x_i$ untouched:

$$\text{div } \underline{a} = a_{i,i} = (\varphi c_i)_{,i} = \varphi_{,i} c_i + \varphi c_{i,i} = \varphi_{,i} c_i,$$

the last term vanishing because $c_{i,i} = 0$ for a uniform field. On the boundary,

$$\underline{a} \cdot \underline{n} = a_i n_i = \varphi c_i n_i.$$

Put both into (5.2):

$$\int_R \varphi_{,i} c_i \, dv = \int_{\partial R} \varphi c_i n_i \, da.$$

The c_i are constants, so they may be taken outside the integral signs. Doing that on both sides and moving everything to one side,

$$c_i \left[\int_R \varphi_{,i} \, dv - \int_{\partial R} \varphi n_i \, da \right] = 0.$$

The bracket depends on i but not on $\underline{c}$, so the display says $\underline{c} \cdot \underline{u} = 0$ where $\underline{u} = u_i \underline{e}_i$ has components

$$u_i = \int_R \varphi_{,i} \, dv - \int_{\partial R} \varphi n_i \, da. \quad (5.3)$$

As $\underline{c}$ is arbitrary, Lemma 5.1 gives $\underline{u} = \underline{0}$, i.e.

$$\int_R \varphi_{,i} \, dv = \int_{\partial R} \varphi n_i \, da, \quad (5.4)$$

valid for every smooth scalar field φ .

The divergence of a tensor

Let $\underline{A}(\underline{x})$ be a smooth tensor field and put $\underline{a} = \underline{A}^t \underline{c}$, again with $\underline{c}$ uniform. Then $a_j = (\underline{A}^t \underline{c})_j = A_{ij} c_i$, so

$$\operatorname{div} \underline{a} = a_{j,j} = c_i A_{ij,j}, \quad \underline{a} \cdot \underline{n} = a_j n_j = c_i A_{ij} n_j,$$

and (5.2) yields $\underline{c} \cdot \underline{u} = 0$ with

$$u_i = \int_R A_{ij,j} \, dv - \int_{\partial R} A_{ij} n_j \, da. \quad (5.5)$$

The arbitrariness of $\underline{c}$ again gives $\underline{u} = \underline{0}$, hence

$$\int_R A_{ij,j} \, dv = \int_{\partial R} A_{ij} n_j \, da \quad (5.6)$$

for every smooth tensor field. We may write (5.4) and (5.6) in the symbolic forms

$$\int_R \operatorname{grad} \varphi \, dv = \int_{\partial R} \varphi \underline{n} \, da \quad \text{and} \quad \int_R \operatorname{div} \underline{A} \, dv = \int_{\partial R} \underline{A} \underline{n} \, da, \quad (5.7)$$

which are Corollary 1.101. There is an important caveat attached to them, and it is the subject of the next few paragraphs.

What “symbolic” means here

REMARK 5.3 (the integrals in (5.7) are not integrals of vectors). An integral is a limit of sums, and to add vectors that live at different points of R one must first agree on how to compare them. In $\mathbb{E}^3$ with a *fixed* orthonormal basis there is such an agreement, because parallel transport is trivial there, and that is exactly what (5.3) and (5.5) use when they integrate *components*. The symbolic forms (5.7) are shorthand for those component statements. They are not independent definitions, and they are not valid for an arbitrary basis field.

The point deserves the check the book gives it. What we must verify is that the object $\underline{u}$ built in (5.3) really is a vector: that its components in

two fixed orthonormal bases are related the way components of a vector are. Let $\{\underline{e}'_i\}$ be a second fixed orthonormal basis and $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, so that by Section 1.1.2 the rule to be confirmed is $u_i = a_{ij}u'_j$. With $\underline{x} = x_i \underline{e}_i = x'_j \underline{e}'_j$, hence $x'_j = a_{ij}x_i$, and with $\varphi'_{,j} = \partial\varphi/\partial x'_j$, the chain rule turns (5.3) into

$$u_i = \int_R a_{ij} \varphi'_{,j} dv - \int_{\partial R} a_{ij} n'_j da = a_{ij} \left(\int_R \varphi'_{,j} dv - \int_{\partial R} \varphi n'_j da \right) = a_{ij} u'_j, \quad (5.8)$$

as required. The whole step turns on being able to move a_{ij} across the integral sign, and that is legitimate precisely because the a_{ij} are *constants*. Consequently the statement $\underline{u} = \underline{0}$ is basis independent: $u'_i = 0$ if and only if $u_i = 0$.

REMARK 5.4 (where it fails). Replace the fixed basis by the polar frame of (1.122). Then the matrix relating the two sets of components depends on position, cannot be taken outside the integral, and (5.8) collapses. So an equation like $\int \text{grad } \varphi dv = \int \varphi \underline{n} da$ read with $\text{grad } \varphi$ and $\underline{n}$ resolved on $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$ is simply false. Equation (5.1), by contrast, needs no caveat at all: its integrands $\text{div } \underline{a}$ and $\underline{a} \cdot \underline{n}$ are scalars, and a scalar is the same number in every basis.

5.1.1 Example: The Piola identity

Here is the first use, and a good one: a purely kinematical identity falls out of (5.7) with the most trivial integrand imaginable.

REMARK 5.5 (why one feeds an identity its most boring input). The choice $\varphi = 1$ below looks like it must give nothing, and that impression is worth correcting, because the move is standard and will be used again.

A general identity holds for *every* admissible input. Its content is therefore a whole family of statements, and one is free to pick the member that says the most. Picking a complicated φ makes both sides complicated and relates two hard things; picking $\varphi = 1$ annihilates the left side, since $\text{grad } 1 = \underline{0}$, and leaves the right side standing alone with nothing to be balanced against. A statement of the form “this expression equals

zero” is far stronger than “this expression equals that one”, so the most trivial input yields the sharpest conclusion. This is the same logic by which $\underline{c} = \underline{u}$ was the useful choice in Lemma 5.1, and by which $\varphi = 1$ will give (5.16) in the next section.

The general principle: to extract something specific from something general, look for the input that makes one side collapse.

Let $S \subset B$ be an arbitrary part of the body, occupying $P_t \subset \kappa_t$ now and $P \subset \kappa$ in the reference configuration, so that $P_t = \chi_\kappa(P, t)$. Apply (5.7)₁ on P_t with $\varphi = 1$. Its gradient vanishes, so the left side is zero and

$$0 = \int_{\partial P_t} \underline{n} \, da = \int_{\partial P} \underline{F}^* \underline{N} \, dA = \int_P \text{Div } \underline{F}^* \, dV, \quad (5.9)$$

where the middle step is the Piola–Nanson formula (2.54), $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$, with $\underline{N}$ the exterior unit normal on ∂P and $\underline{F}^*$ the cofactor of the deformation gradient (Definition 1.54), and the last step is the referential form of (5.7)₂. Here Div differentiates with respect to $\underline{X} \in \kappa$; in components

$$\text{Div } \underline{F}^* = F_{iA,A}^* e_i. \quad (5.10)$$

What (5.9) has established is that

$$\int_P \text{Div } \underline{F}^* \, dV = 0$$

for one particular P . That alone says nothing pointwise: an integral can vanish while its integrand does not, as $\int_{-1}^1 x \, dx = 0$ shows. The extra ingredient is that S was an arbitrary part of the body, so P is an arbitrary subvolume of κ , and the integral vanishes for *every* one of them at once.

That is exactly the hypothesis of the localization theorem, Theorem 2.20, whose other hypothesis, continuity of the integrand, holds because $\underline{F}^*$ is built polynomially from $\underline{F}$ and $\underline{F}$ is continuously differentiable. The theorem’s argument is worth recalling in one line, since it is the same shape as Lemma 5.1: if the integrand were non-zero at some point, it would be non-zero with one sign throughout a small ball around

that point by continuity, and integrating over that ball alone would give a non-zero answer, contradicting the hypothesis. Freedom to choose the region plays here the part that freedom to choose $\mathcal{C}$ played there.

Applying it componentwise gives the *Piola identity*

$$\operatorname{Div} \underline{F}^* = \underline{0} \quad \text{at every } \underline{X} \in \kappa. \quad (5.11)$$

REMARK 5.6 (what has just happened). Read (5.9) from left to right and the identity looks like a fact about integrals; read it from right to left and it is a fact about the deformation alone, with no integral in sight. That is the pattern of the whole subject: an integral statement, valid for every part, plus localization, equals a pointwise statement. Chapter 6 will run the same argument on mass, momentum and energy. Here the “physics” is the empty statement that a closed surface has zero total normal. Read geometrically, $\int_{\partial P_t} \underline{n} \, da = \underline{0}$ says that a uniform pressure acting all round a closed surface cannot push the body anywhere. Even that empty-looking statement yields something: the nine derivatives of $\underline{F}^*$ are not independent.

5.2 Reynolds’ transport formula

Let φ be a scalar field on the body, the volumetric density of some quantity, so that $\varphi = \hat{\varphi}(\underline{X}, t) = \tilde{\varphi}(\underline{x}, t)$ with $\underline{x} = \underline{\chi}_\kappa(\underline{X}, t)$. We want

$$\frac{d}{dt} \int_{P_t} \tilde{\varphi}(\underline{x}, t) \, dv \quad \text{at fixed } S, \quad (5.12)$$

the derivative following a fixed set of *material points*. Both the integrand and the domain move, and there is no rule for differentiating under an integral sign whose limits are unknown functions of t . The remedy is the one used throughout Chapter 2: pull back to the reference configuration, where the domain does not move at all.

With $dv = J dV$ and P fixed in time,

$$\begin{aligned} \frac{d}{dt} \int_{P_t} \tilde{\varphi}(\underline{x}, t) dv &= \frac{d}{dt} \int_P \hat{\varphi}(\underline{X}, t) J dV \\ &= \int_P (\varphi J)^\cdot dV \\ &= \int_P (\dot{\varphi} J + \varphi \dot{J}) dV, \end{aligned} \tag{5.13}$$

the interchange of d/dt with $\int_P$ being permissible because P no longer depends on t , provided φJ and $(\varphi J)^\cdot$ are continuous in $\underline{X}$ and t jointly and dominated on P by an integrable function of $\underline{X}$.

REMARK 5.7 (the one move this chapter keeps making). Step (5.13)₁ is the whole method, and it will be made three more times, so it is worth naming: *when a domain moves, change variables until it does not, then differentiate.*

The obstacle was never the integrand, since differentiating under an integral sign is routine. The obstacle was the domain. There is no general rule for $\frac{d}{dt} \int_{D(t)}$, and the substitution $dv = J dV$ removes the difficulty entirely by trading a moving region P_t for the fixed region P and paying with an extra factor J inside. All the time dependence is then in the integrand, where the ordinary rules apply. The price is that we must know how J changes, and that is exactly what Problem 2 of Chapter 4 supplied.

This is the same manoeuvre as pulling back to the reference configuration in Chapter 2, and it is the reason the reference configuration is worth carrying around at all: not because it is physically distinguished, for it is not and Remark 4.2 of Chapter 4 insisted on the point, but because it holds still. Watch for it again at (5.20), where Piola–Nanson plays the role J plays here and converts a moving surface into a fixed one.

And watch for where it *fails*: in Section 5.3.3 the region is not material, so there is no fixed P to pull back to, the method has nothing to bite on, and we are thrown back on the definition of the derivative. That is the reason (5.23), alone in this chapter, is proved from first principles.

Now dispose of the $\dot{J}$ that appeared in (5.13). By Problem 2 of Chapter 4, $\dot{J} = J \operatorname{tr} \underline{L}$ with $\underline{L} = \operatorname{grad} \underline{v}$, and $\operatorname{tr} \underline{L} = \operatorname{tr}(\operatorname{grad} \underline{v}) = \operatorname{div} \underline{v}$ by Proposition 1.92. Substitute this into the integrand of (5.13) and factor out the J that both terms then contain:

$$\dot{\varphi}J + \varphi\dot{J} = \dot{\varphi}J + \varphi J \operatorname{div} \underline{v} = J(\dot{\varphi} + \varphi \operatorname{div} \underline{v}).$$

This factoring is the point of the whole calculation. The integral now reads

$$\int_P (\dot{\varphi} + \varphi \operatorname{div} \underline{v}) J \, dV,$$

and $J \, dV$ is dv , which is the substitution of (5.13)₁ run backwards. Undoing it carries the integral back to the moving region and leaves

$$\frac{d}{dt} \int_{P_t} \varphi \, dv = \int_{P_t} (\dot{\varphi} + \varphi \operatorname{div} \underline{v}) \, dv. \quad (5.14)$$

Notice what the round trip accomplished. We went to the reference configuration only to be allowed to differentiate, and came straight back. Nothing referential survives in (5.14): both sides are integrals over P_t and every symbol in it is spatial. The reference configuration was scaffolding, put up to make one step legal and taken down immediately afterwards. That is why the formula holds even for materials for which no natural reference configuration exists.

This is already a transport formula, and a useful one, but it is written in terms of the material derivative $\dot{\varphi}$. For a spatial description we want φ' instead. Substitute $\dot{\varphi} = \varphi' + \operatorname{grad} \varphi \cdot \underline{v}$ from (2.20) into the integrand of (5.14):

$$\dot{\varphi} + \varphi \operatorname{div} \underline{v} = \varphi' + \operatorname{grad} \varphi \cdot \underline{v} + \varphi \operatorname{div} \underline{v}.$$

The last two terms are exactly the two halves of a product rule. By Problem 24(c) of Chapter 1,

$$\operatorname{div}(\varphi \underline{v}) = \operatorname{grad} \varphi \cdot \underline{v} + \varphi \operatorname{div} \underline{v},$$

so those two terms collapse into a single divergence and the integrand becomes $\varphi' + \operatorname{div}(\varphi \underline{v})$. Splitting the integral in two,

$$\frac{d}{dt} \int_{P_t} \varphi \, dv = \int_{P_t} \varphi' \, dv + \int_{P_t} \operatorname{div}(\varphi \underline{v}) \, dv.$$

The second integral is a divergence over a volume, which is precisely what (5.1) converts into a surface integral, with $\underline{a} = \varphi \underline{v}$:

$$\int_{P_t} \operatorname{div}(\varphi \underline{v}) \, dv = \int_{\partial P_t} \varphi \underline{v} \cdot \underline{n} \, da = \int_{\partial P_t} \varphi \underline{v} \cdot \underline{n} \, da,$$

the scalar φ coming out of the inner product. Therefore

$$\frac{d}{dt} \int_{P_t} \varphi \, dv = \int_{P_t} \varphi' \, dv + \int_{\partial P_t} \varphi \underline{v} \cdot \underline{n} \, da \quad (5.15)$$

This is *Reynolds' transport formula*.

It is worth seeing why the product rule was the right thing to look for. We had a volume integral and wanted a surface integral, and (5.1) is the only tool that makes that conversion. But (5.1) accepts a divergence and nothing else, so the whole step amounted to asking whether the leftover terms could be assembled into one. They could, and not by luck: $\operatorname{grad} \varphi \cdot \underline{v} + \varphi \operatorname{div} \underline{v}$ is the full expansion of $\operatorname{div}(\varphi \underline{v})$, so the two terms that appeared were never anything but a divergence written out. Recognising a product rule run backwards is the recurring skill in this subject, and it is worth practising on the simpler scalar case $\frac{d}{dx}(fg) = f'g + fg'$ until the pattern is automatic.

REMARK 5.8 (the two clocks: $\dot{\varphi}$ against φ'). Three different time derivatives appear between (5.12) and (5.15), and the whole formula is really a statement about how they relate. It is worth setting them side by side, since every term of (5.15) is an answer to “which one?”.

$\frac{d}{dt} \int_{P_t} \varphi \, dv$	follows a fixed lump of <i>material</i> ; the domain moves,
$\dot{\varphi}$	follows a fixed material <i>point</i> ; hold $\underline{x}$ fixed,
φ'	follows a fixed <i>place</i> ; hold $\underline{x}$ fixed.

The first is what a balance law asks about, since it weighs the same piece of matter twice. The second is what a particle experiences. The third is what a fixed probe records. Chapter 2 already tied the last two

together in (2.20), $\dot{\varphi} = \varphi' + \text{grad } \varphi \cdot \underline{v}$: the difference is the convective term, the change you meet merely by being carried to a new place where φ happens to differ.

Equation (5.14) is the transport formula stated in the second clock, (5.15) the same content in the third. Neither is more correct; they differ in what has to be known to use them. If the data come as $\hat{\varphi}(\underline{X}, t)$, as a constitutive statement about the material would, then the overdot is the native derivative and (5.14) is the shorter road. If the data come as $\tilde{\varphi}(\underline{x}, t)$, read off a field in space, the prime is native and (5.15) is. The convective term is the toll paid at the border, and Remark 2.11 of Chapter 2 is the warning about forgetting to pay it.

REMARK 5.9 (how to read (5.15)). The amount of φ -stuff inside the moving lump changes for exactly two reasons, and the formula separates them. The volume integral is the change you would measure standing still at each point of space: it is local creation. The surface integral is the change that comes from the boundary sweeping over new material: $\underline{v} \cdot \underline{n}$ is the outward normal speed of the boundary, $\varphi \underline{v} \cdot \underline{n} da$ the rate at which φ -stuff is swept past da . Nothing is created there; the lump merely takes in territory. Equation (5.14) says the same thing in the other split, following the material rather than the place. Both are exact; which one is convenient depends on whether the data are given as $\hat{\varphi}$ or $\tilde{\varphi}$.

Taking $\varphi = 1$, so that $\varphi' = 0$, gives the rate of change of a material volume as the flux of material velocity through its boundary:

$$\frac{d}{dt} \int_{P_t} dv = \int_{\partial P_t} \underline{v} \cdot \underline{n} da. \quad (5.16)$$

Comparing with (5.14) at $\varphi = 1$ recovers $\dot{J}/J = \text{div } \underline{v}$ in integrated form, which is a useful consistency check and also the reason $\text{div } \underline{v} = 0$ is the pointwise statement of isochoric motion.

5.3 Extensions and generalizations

5.3.1 Reynolds' formula for vector fields

Apply (5.15) to $\varphi = u_i$, one Cartesian component of a vector field $\underline{u}$ resolved on the fixed basis $\{\underline{e}_i\}$:

$$\frac{d}{dt} \int_{P_t} u_i \, dv = \int_{P_t} u'_i \, dv + \int_{\partial P_t} u_i \, \underline{v} \cdot \underline{n} \, da; \quad i \in \{1, 2, 3\}. \quad (5.17)$$

Multiply by $\underline{e}_i$ and sum. The basis vectors are constants and pass through both integrals; on the right, $u_i(\underline{v} \cdot \underline{n})\underline{e}_i = (\underline{v} \cdot \underline{n})\underline{u} = (\underline{u} \otimes \underline{v})\underline{n}$ by the definition of the tensor product. Hence, with the caveat of Remark 5.3 still in force,

$$\frac{d}{dt} \int_{P_t} \underline{u} \, dv = \int_{P_t} \underline{u}' \, dv + \int_{\partial P_t} (\underline{u} \otimes \underline{v})\underline{n} \, da. \quad (5.18)$$

5.3.2 The flux derivative

Balance laws for electromagnetic fields need the rate of change not of a volume integral but of the *flux* of a vector field through a moving material surface. The claim is that

$$\frac{d}{dt} \int_{\partial P_t} \underline{u} \cdot \underline{n} \, da = \int_{\partial P_t} \underline{u}^* \cdot \underline{n} \, da \quad (5.19)$$

for a field $\underline{u}^*$, the *flux derivative* of $\underline{u}$, to be determined. As before the derivative follows a fixed set of material points, here the ones occupying the evolving surface ∂P_t , and as before we transform to a fixed domain before differentiating.

REMARK 5.10 (why the cofactor, and not $\underline{F}$). The pull-back of Remark 5.7 needs a conversion factor for whatever measure is being integrated, and the factor is different for each of the three. Chapter 2 produced all three, and this chapter uses all three:

$$\begin{aligned} \text{line:} \quad d\underline{x} &= \underline{F} \, d\underline{X} && (\text{the definition of } \underline{F}), \\ \text{area:} \quad \underline{n} \, da &= \underline{F}^* \underline{N} \, dA && (\text{Piola–Nanson, (2.54)}), \\ \text{volume:} \quad dv &= J \, dV && (J = \det \underline{F}). \end{aligned}$$

$\underline{\underline{F}}$ maps a line element because a line element *is* a vector attached to a material fibre. An area element is not that kind of object: $\underline{\underline{n}} \, da$ is built from a *pair* of fibres through their cross product, and the tensor that carries cross products is by definition the cofactor, $\underline{\underline{F}}^*(\underline{\underline{a}} \times \underline{\underline{b}}) = \underline{\underline{F}}\underline{\underline{a}} \times \underline{\underline{F}}\underline{\underline{b}}$ (Remark 1.56). Feeding $\underline{\underline{n}}$ to $\underline{\underline{F}}$ would deform the normal as if it were a fibre, and normals are not fibres. Under a simple shear a normal tilts the opposite way to a fibre lying in the shearing plane. That is the whole reason $\underline{\underline{F}}^*$ exists, and it is why every surface integral in this chapter goes through Piola–Nanson while every volume integral goes through J .

The identity $\underline{\underline{F}}^* = J\underline{\underline{F}}^{-t}$ of Proposition 1.55 then says these are not three unrelated facts: the area factor is the volume factor divided by the line factor, transposed, which is dimensionally exactly what one would guess from “area = volume per length”.

Piola–Nanson, $\underline{\underline{n}} \, da = \underline{\underline{F}}^* \underline{\underline{N}} \, dA$, moves the integral to ∂P ; then $\underline{\underline{F}}^* = J\underline{\underline{F}}^{-t}$ (Proposition 1.55) lets us shift $\underline{\underline{F}}$ across the inner product, $\underline{\underline{u}} \cdot J\underline{\underline{F}}^{-t} \underline{\underline{N}} = J\underline{\underline{F}}^{-1} \underline{\underline{u}} \cdot \underline{\underline{N}}$:

$$\begin{aligned} \frac{d}{dt} \int_{\partial P_t} \underline{\underline{u}} \cdot \underline{\underline{n}} \, da &= \frac{d}{dt} \int_{\partial P} \underline{\underline{u}} \cdot \underline{\underline{F}}^* \underline{\underline{N}} \, dA = \frac{d}{dt} \int_{\partial P} J\underline{\underline{F}}^{-1} \underline{\underline{u}} \cdot \underline{\underline{N}} \, dA \\ &= \int_{\partial P} (J\underline{\underline{F}}^{-1} \underline{\underline{u}})^\cdot \cdot \underline{\underline{N}} \, dA \\ &= \int_{\partial P} [J\underline{\underline{F}}^{-1} \dot{\underline{\underline{u}}} + \dot{J}\underline{\underline{F}}^{-1} \underline{\underline{u}} + J(\underline{\underline{F}}^{-1})^\cdot \underline{\underline{u}}] \cdot \underline{\underline{N}} \, dA. \end{aligned} \quad (5.20)$$

The middle factor is reduced by a trick already used in Problem 1(d) of Chapter 4. Differentiate $\underline{\underline{F}}\underline{\underline{F}}^{-1} = \underline{\underline{I}}$:

$$\dot{\underline{\underline{F}}}\underline{\underline{F}}^{-1} + \underline{\underline{F}}(\underline{\underline{F}}^{-1})^\cdot = \underline{\underline{O}} \quad \implies \quad \underline{\underline{F}}(\underline{\underline{F}}^{-1})^\cdot = -\dot{\underline{\underline{F}}}\underline{\underline{F}}^{-1} = -\underline{\underline{L}},$$

using $\underline{\underline{L}} = \dot{\underline{\underline{F}}}\underline{\underline{F}}^{-1}$ from (4.3). To free $(\underline{\underline{F}}^{-1})^\cdot$, multiply on the left by $\underline{\underline{F}}^{-1}$:

$$\underline{\underline{F}}^{-1}\underline{\underline{F}}(\underline{\underline{F}}^{-1})^\cdot = (\underline{\underline{F}}^{-1})^\cdot = -\underline{\underline{F}}^{-1}\underline{\underline{L}}.$$

Now substitute this and $\dot{J} = J \operatorname{tr} \underline{\underline{L}}$ into the bracket of (5.20), taking the

three terms in order:

$$\begin{aligned}
J\tilde{F}^{-1}\dot{\underline{u}} & \text{ stays as it is,} \\
J\tilde{F}^{-1}\underline{u} & = J(\operatorname{tr} \tilde{L})\tilde{F}^{-1}\underline{u} = J\tilde{F}^{-1}(\operatorname{tr} \tilde{L})\underline{u}, \\
J(\tilde{F}^{-1})^\cdot \underline{u} & = J(-\tilde{F}^{-1}\tilde{L})\underline{u} = -J\tilde{F}^{-1}\tilde{L}\underline{u}.
\end{aligned}$$

In the second line the scalar $\operatorname{tr} \tilde{L}$ was moved across $\tilde{F}^{-1}$, which is allowed because scalars commute with everything. Every term now carries the same left factor $J\tilde{F}^{-1}$, so it can be taken out:

$$J\tilde{F}^{-1}\dot{\underline{u}} + J\tilde{F}^{-1}(\operatorname{tr} \tilde{L})\underline{u} - J\tilde{F}^{-1}\tilde{L}\underline{u} = J\tilde{F}^{-1}[\dot{\underline{u}} + (\operatorname{tr} \tilde{L})\underline{u} - \tilde{L}\underline{u}].$$

That was the point of the whole manipulation. The bracket is a single spatial vector, and the factor in front is exactly the one that converts back to the current configuration. Reversing the step that produced it, $J\tilde{F}^{-1}\underline{s} \cdot \underline{N} = \underline{s} \cdot J\tilde{F}^{-t}\underline{N} = \underline{s} \cdot \tilde{F}^*\underline{N}$ for any $\underline{s}$, and then reading $\tilde{F}^*\underline{N} \, dA$ as $\underline{n} \, da$ once more,

$$\begin{aligned}
\frac{d}{dt} \int_{\partial P_t} \underline{u} \cdot \underline{n} \, da & = \int_{\partial P} [\dot{\underline{u}} + (\operatorname{tr} \tilde{L})\underline{u} - \tilde{L}\underline{u}] \cdot \tilde{F}^*\underline{N} \, dA \\
& = \int_{\partial P_t} [\dot{\underline{u}} + (\operatorname{tr} \tilde{L})\underline{u} - \tilde{L}\underline{u}] \cdot \underline{n} \, da.
\end{aligned} \tag{5.21}$$

Comparison with (5.19) identifies $\underline{u}^* = \dot{\underline{u}} + (\operatorname{div} \underline{v})\underline{u} - \tilde{L}\underline{u}$. Finally trade the material derivative for the spatial one with (2.20), $\dot{\underline{u}} = \underline{u}' + (\operatorname{grad} \underline{u})\underline{v}$, and note that $(\operatorname{grad} \underline{u})\underline{v} + (\operatorname{div} \underline{v})\underline{u} = \operatorname{div}(\underline{u} \otimes \underline{v})$ by Proposition 1.95:

$$\underline{u}^* = \underline{u}' + \operatorname{div}(\underline{u} \otimes \underline{v}) - (\operatorname{grad} \underline{v})\underline{u}. \tag{5.22}$$

REMARK 5.11 (why $\underline{u}^*$ is not $\dot{\underline{u}}$). Three things change when a material surface is carried along by the motion: the field on it, its area, and its orientation. The first term of $\underline{u}^*$ accounts for the field, the second for the stretching of the area element, and the third, $-\tilde{L}\underline{u}$, for the tilting of $\underline{n}$. Drop the motion, $\underline{v} = \underline{0}$, and $\underline{u}^*$ collapses to $\underline{u}'$ as it must. Problem 2 gives $\underline{u}^*$ a second form in which the same three effects are repackaged as a curl, which is how it is usually met in electrodynamics.

5.3.3 Non-material regions

Reynolds' formula assumed the region was material, meaning that it pulls back to a fixed $P \subset \kappa$. Control volumes in fluid mechanics do not, and neither will the shock surfaces of Section 5.4. Let $V(t) \subset \kappa_t$ be a region whose boundary moves with local normal speed $u(\underline{x}, t)$ in the direction of its exterior unit normal $\underline{n}$, with φ defined on V . The formula to be proved is

$$\frac{d}{dt} \int_V \varphi \, dv = \int_V \varphi' \, dv + \int_{\partial V} \varphi u \, da. \quad (5.23)$$

It reduces to (5.15) when V is material, for then the boundary is carried by the motion and $u = \underline{v} \cdot \underline{n}$.

The pull-back of Remark 5.7 is unavailable here, because there is no fixed domain to pull back to. We therefore go back to the definition of the derivative:

$$\frac{d}{dt} \int_{V(t)} \varphi \, dv = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left[\int_{V(t+\varepsilon)} \tilde{\varphi}(\underline{x}, t + \varepsilon) \, dv - \int_{V(t)} \tilde{\varphi}(\underline{x}, t) \, dv \right]. \quad (5.24)$$

Two things differ between the two integrals in (5.24): the integrand has moved from time t to time $t + \varepsilon$, and so has the domain. A difference in which two things changed at once is hard to estimate, so we separate them by the standard device of adding and subtracting an intermediate term. The intermediate we want is the one that agrees with the first integral in its integrand and with the second in its domain, namely

$$\int_{V(t)} \tilde{\varphi}(\underline{x}, t + \varepsilon) \, dv,$$

the integral over the *old* domain of the *new* integrand. Adding and subtracting it inside the bracket of (5.24) costs nothing, since the two copies cancel, and it splits the bracket into two differences in each of which only one thing has changed. The first difference has the same integrand on two different domains; the second has two different integrands on the same domain. Writing the limit of the sum as the sum of the limits,

which is legitimate once both limits are known to exist,

$$\begin{aligned}
\frac{d}{dt} \int_{V(t)} \varphi \, dv &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left[\int_{V(t+\varepsilon)} \tilde{\varphi}(\underline{x}, t + \varepsilon) \, dv - \int_{V(t)} \tilde{\varphi}(\underline{x}, t + \varepsilon) \, dv \right] \\
&\quad + \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left[\int_{V(t)} \tilde{\varphi}(\underline{x}, t + \varepsilon) \, dv - \int_{V(t)} \tilde{\varphi}(\underline{x}, t) \, dv \right] \\
&= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_{V(t+\varepsilon) \setminus V(t)} \tilde{\varphi}(\underline{x}, t + \varepsilon) \, dv + \int_{V(t)} \varphi' \, dv.
\end{aligned} \tag{5.25}$$

Both simplifications in the last line deserve a word.

The second limit is now an ordinary derivative. The domain $V(t)$ is held fixed while ε runs to zero, so the limit may be taken inside the integral, and what is left inside is

$$\lim_{\varepsilon \rightarrow 0} \frac{\tilde{\varphi}(\underline{x}, t + \varepsilon) - \tilde{\varphi}(\underline{x}, t)}{\varepsilon} = \frac{\partial \tilde{\varphi}}{\partial t} \Big|_{\underline{x}} = \varphi',$$

the derivative at fixed place, by definition. This is where the prime of Remark 5.8 enters the formula, and it enters for a plain reason: the domain was frozen at $V(t)$ for this piece, so the only thing still moving was the field, and a field watched at a frozen place is watched at fixed $\underline{x}$.

The first limit simplifies because the two domains overlap. The region $V(t)$ sits inside $V(t + \varepsilon)$ when the boundary is advancing, so the integral over $V(t)$ cancels the part of the integral over $V(t + \varepsilon)$ taken over the same points, and what survives is the integral over the thin shell $V(t + \varepsilon) \setminus V(t)$ between the two boundaries. Where the boundary is receding the shell counts with the opposite sign, which is handled automatically once u is allowed to be negative there. That shell integral is the whole content of the formula, and evaluating it is the business of the next paragraph.

The shell is thin, and that is what makes it computable. Take one element da of $\partial V(t)$ and ask how much volume the boundary sweeps over it in time ε . The boundary moves with normal speed u , so in time ε it advances a distance $u\varepsilon$ along $\underline{n}$. The piece of shell standing over da is therefore a slab of base da and height $u\varepsilon$, of volume $u\varepsilon \, da$, together with a correction.

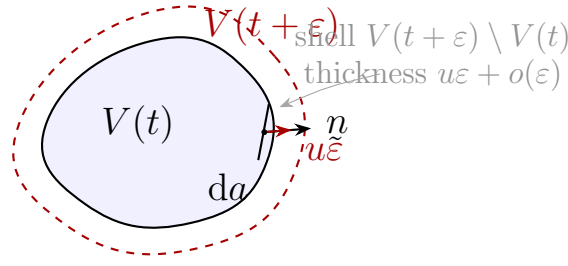


Figure 5.1: Evolution of a non-material region. In time ε the boundary advances a distance $u\varepsilon$ along its own normal, so the shell swept out has volume measure $dv = u\varepsilon da + o(\varepsilon)$ over each element da of $\partial V(t)$. Dividing by ε and letting $\varepsilon \rightarrow 0$ leaves $\int_{\partial V} \varphi u da$: only the normal motion of the boundary contributes, since sliding the boundary along itself sweeps no volume.

The correction is $o(\varepsilon)$, meaning it vanishes faster than ε does, and it has two sources. The boundary need not move exactly along $\underline{n}$; but by the argument after (5.27) the tangential part sweeps no volume to first order, since sliding a surface along itself moves no material across it. And the surface curves, so the slab is not exactly a cylinder; but the deviation is of order ε^2 , being a product of the advance $u\varepsilon$ with the change in $\underline{n}$ over that advance, itself of order ε . Both corrections therefore disappear on dividing by ε and letting ε tend to zero, which is the only thing we will do with them.

So, with reference to Figure 5.1, the volume measure on the shell is

$$dv = u\varepsilon da + o(\varepsilon),$$

and the integral over the shell becomes an integral over $\partial V(t)$. Therefore

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_{V(t+\varepsilon) \setminus V(t)} \tilde{\varphi}(\underline{x}, t + \varepsilon) dv &= \lim_{\varepsilon \rightarrow 0} \int_{\partial V(t)} \tilde{\varphi}(\underline{x}, t + \varepsilon) u da + \frac{o(\varepsilon)}{\varepsilon} \\ &= \int_{\partial V(t)} \tilde{\varphi}(\underline{x}, t) u da, \end{aligned} \tag{5.26}$$

which with (5.25) gives (5.23).

It is convenient to carry a vector field $\underline{u}(\underline{x}, t)$ on $\partial V(t)$ with $\underline{u} \cdot \underline{n} = u$. Only the normal component has any meaning: the tangential part

$(\underline{I} - \underline{n} \otimes \underline{n})\underline{u}$ slides the boundary along itself and sweeps no volume, so it cannot appear in (5.23). Then

$$\frac{d}{dt} \int_V \varphi \, dv = \int_V \varphi' \, dv + \int_{\partial V} \varphi \underline{v} \cdot \underline{n} \, da, \quad (5.27)$$

in exact analogy with (5.15). The analogy is only formal, however: $\underline{v}$ is defined throughout the body, $\underline{u}$ only on ∂V .

5.4 Discontinuous fields

5.4.1 Reynolds' formula for fields that suffer a discontinuity across a surface

Shock waves, phase boundaries and propagating cracks all present the same picture: a surface $s(t)$ sweeping through a material volume $P_t \subset \kappa_t$ at speed u along its unit normal $\underline{n}$, cutting P_t into two parts $V^\pm$ across which the fields jump. The two parts are *not* material regions, and s is not a material surface, whenever $u \neq \underline{v} \cdot \underline{n}$ (Figure 5.2).

Let φ be discontinuous across s , which we then call a *singular surface*. Its jump at $\underline{x} \in s$ is

$$[[\varphi]] = \varphi^+ - \varphi^-, \quad (5.28)$$

where $\varphi^\pm$ are the limits of φ as $\underline{x}$ is approached from the interiors of $V^\pm$. Throughout, $\underline{n}$ points from V^- into V^+ ; the sign of every jump depends on that convention and on nothing else. From the figure,

$$P_t = V^+ \cup V^-, \quad \partial P_t^\pm = \partial V^\pm \cap \partial P_t, \quad \partial P_t = \partial P_t^+ \cup \partial P_t^-. \quad (5.29)$$

Let φ be smooth in V^+ and in V^- separately. Each half is a non-material region, so (5.27) applies to each. The boundary of V^+ consists of ∂P_t^+ , which is material and therefore moves with normal speed $\underline{v} \cdot \underline{n}$, and of s , whose exterior normal *as seen from* V^+ is $-\underline{n}$, so that its normal speed in that direction is $-u$. For V^- the exterior normal on s

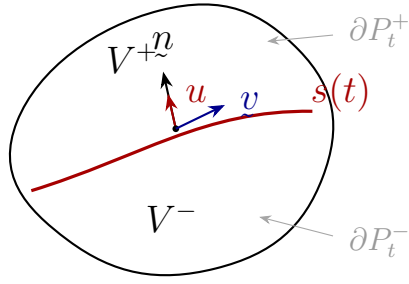


Figure 5.2: A material region $P_t \subset \kappa_t$ traversed by a propagating surface s . The normal $\underline{n}$ on s points from V^- into V^+ and s advances at speed u along it. Material crosses s whenever $u \neq \underline{v} \cdot \underline{n}$: the surface is then not made of the same particles from one instant to the next, and $V^\pm$ are non-material regions even though their union P_t is material. This is why (5.27), and not (5.15), is the formula to apply to each half.

is $+\underline{n}$ and the speed is $+u$. Hence

$$\begin{aligned} \frac{d}{dt} \int_{V^+} \varphi \, dv &= \int_{V^+} \varphi' \, dv + \int_{\partial P_t^+} \varphi \, \underline{v} \cdot \underline{n} \, da + \int_s \varphi^+ (-u) \, da \quad \text{and} \\ \frac{d}{dt} \int_{V^-} \varphi \, dv &= \int_{V^-} \varphi' \, dv + \int_{\partial P_t^-} \varphi \, \underline{v} \cdot \underline{n} \, da + \int_s \varphi^- u \, da. \end{aligned} \tag{5.30}$$

Before adding them, notice why the signs came out as they did, since this is the only delicate point in the section. Equation (5.27) was derived for a region whose boundary advances at speed u *along its own exterior normal*. The surface s is part of the boundary of both halves, but the two halves look at it from opposite sides. Seen from V^- , the outward direction is $+\underline{n}$ and the surface advances into V^+ , which is away from V^- : the region V^- is gaining, and the speed to use is $+u$. Seen from V^+ , the outward direction is $-\underline{n}$ and the surface is advancing into V^+ itself: the region is losing, and the speed to use is $-u$. One surface, one physical motion, two opposite bookkeeping signs, because the two regions are on opposite sides of it. The limiting value of φ likewise differs, being φ^+ from one side and φ^- from the other, which is exactly what makes the sum fail to cancel.

Now add the two lines of (5.30), one piece at a time. The volume

integrals combine, because $V^+ \cup V^- = P_t$ by (5.29)₁ and the two parts overlap only on s , a surface of zero volume:

$$\int_{V^+} \varphi' \, dv + \int_{V^-} \varphi' \, dv = \int_{P_t} \varphi' \, dv.$$

The material surface integrals combine for the same reason, using $\partial P_t = \partial P_t^+ \cup \partial P_t^-$ from (5.29)₃:

$$\int_{\partial P_t^+} \varphi \, \underline{v} \cdot \underline{n} \, da + \int_{\partial P_t^-} \varphi \, \underline{v} \cdot \underline{n} \, da = \int_{\partial P_t} \varphi \, \underline{v} \cdot \underline{n} \, da.$$

The two integrals over s are both over the same surface, so they combine into one integrand:

$$\int_s \varphi^+(-u) \, da + \int_s \varphi^- u \, da = - \int_s (\varphi^+ - \varphi^-) u \, da = - \int_s [\![\varphi]\!] u \, da,$$

the last step being the definition (5.28) of the jump. Collecting the three results,

$$\frac{d}{dt} \int_{P_t} \varphi \, dv = \int_{P_t} \varphi' \, dv + \int_{\partial P_t} \varphi \, \underline{v} \cdot \underline{n} \, da - \int_s [\![\varphi]\!] u \, da. \quad (5.31)$$

REMARK 5.12 (the sign, and what the extra term is). The minus sign is not a convention one may take or leave: it follows from $\underline{n}$ pointing into V^+ , which is also the convention that defines $[\![\varphi]\!]$ in (5.28). Reverse $\underline{n}$ and both $[\![\varphi]\!]$ and u change sign, so the product, and the formula, are unchanged, as they must be, since the left-hand side knows nothing about our choice. As for the meaning: if φ is smooth then $[\![\varphi]\!] = 0$ and (5.31) is (5.15). When it is not, the surface acts as a source. It is this term that will produce the Rankine–Hugoniot jump conditions in Chapter 6, in exactly the way (5.15) produces the smooth balance laws.

The passage to vector fields is the one already made from (5.15) to (5.18): apply (5.31) to each Cartesian component and reassemble.

5.4.2 The divergence theorem for discontinuous fields

The divergence theorem (5.1) needs the same repair. Recall that in $\underline{a} \cdot \underline{n}$ on its right-hand side, $\underline{a}$ means the limit taken as ∂R is approached *from inside* R . Let $\underline{a}(\underline{x}, t)$ be smooth in V^+ and V^- separately and suffer

$$[[a]] = a^+ - a^- \quad \text{at } \underline{x} \in s. \quad (5.32)$$

Apply (5.1) to each half separately. This is legitimate because $\underline{a}$ is smooth on each side taken by itself. The boundary of V^- is ∂P_t^- together with s , on which the exterior normal is $+\underline{n}$ and the relevant limit is $\underline{a}^-$; the boundary of V^+ is ∂P_t^+ together with s , on which the exterior normal is $-\underline{n}$ and the limit is $\underline{a}^+$:

$$\begin{aligned} \int_{\partial P_t^-} \underline{a} \cdot \underline{n} \, da + \int_s \underline{a}^- \cdot \underline{n} \, da &= \int_{V^-} \operatorname{div} \underline{a} \, dv \quad \text{and} \\ \int_{\partial P_t^+} \underline{a} \cdot \underline{n} \, da + \int_s \underline{a}^+ \cdot (-\underline{n}) \, da &= \int_{V^+} \operatorname{div} \underline{a} \, dv. \end{aligned} \quad (5.33)$$

The sign pattern is the same one that appeared in (5.30) and for the same reason: the two halves see the shared surface from opposite sides, so one of them reads its exterior normal as $+\underline{n}$ and the other as $-\underline{n}$. There is no motion in this argument at all, only geometry, which is why u never appears.

Add the two lines. The right sides combine into a single integral over P_t , since $V^+ \cup V^- = P_t$:

$$\int_{V^-} \operatorname{div} \underline{a} \, dv + \int_{V^+} \operatorname{div} \underline{a} \, dv = \int_{P_t} \operatorname{div} \underline{a} \, dv.$$

On the left, the two integrals over the outer boundary combine into one over ∂P_t by (5.29)₃, and the two over s give

$$\int_s \underline{a}^- \cdot \underline{n} \, da + \int_s \underline{a}^+ \cdot (-\underline{n}) \, da = - \int_s (\underline{a}^+ - \underline{a}^-) \cdot \underline{n} \, da = - \int_s [[a]] \cdot \underline{n} \, da,$$

using the linearity of the inner product in its first slot and then the definition (5.32). So the sum reads

$$\int_{\partial P_t} \underline{a} \cdot \underline{n} \, da - \int_s \llbracket \underline{a} \rrbracket \cdot \underline{n} \, da = \int_{P_t} \operatorname{div} \underline{a} \, dv,$$

and moving the jump term to the other side,

$$\int_{\partial P_t} \underline{a} \cdot \underline{n} \, da = \int_{P_t} \operatorname{div} \underline{a} \, dv + \int_s \llbracket \underline{a} \rrbracket \cdot \underline{n} \, da. \quad (5.34)$$

In every formula of this section the jump is taken with respect to a normal $\underline{n}$ on s directed from V^- to V^+ .

REMARK 5.13 (a way to remember both). Read (5.34) backwards: $\int_{P_t} \operatorname{div} \underline{a} \, dv$ *under-counts* the boundary flux by exactly the flux the jump carries across s . Equivalently, the distributional divergence of a piecewise smooth field is the classical divergence plus a surface layer of strength $\llbracket \underline{a} \rrbracket \cdot \underline{n}$ on s . The same reading applies to (5.31): the singular surface carries its own share of the balance, and localizing an integral statement over a region cut by s therefore produces two local statements: a field equation away from s , and a jump condition on s itself.

5.5 Problems

PROBLEM 1 (Transport of a line integral). Fill in the missing terms in the transport formula

$$\frac{d}{dt} \int_{C_t} \underline{a} \cdot d\underline{x} = \int_{C_t} (\underline{a}' + \dots) \cdot d\underline{x},$$

where C_t is a moving material curve and $\underline{a}'$ is the time derivative of the smooth spatial field $\underline{a}$ at fixed $\underline{x}$.

Solution. The pattern of the chapter applies unchanged: convert to a fixed domain, differentiate, convert back. The fixed domain here is the material curve $C \subset \kappa$ of which $C_t = \chi_\kappa(C, t)$ is the image.

A material line element obeys $d\tilde{x} = \tilde{F} d\tilde{X}$ by the definition of the deformation gradient, and $d\tilde{X}$ is fixed in time, so by (4.3)

$$(d\tilde{x})^\cdot = \dot{\tilde{F}} d\tilde{X} = \tilde{L}\tilde{F} d\tilde{X} = \tilde{L} d\tilde{x}.$$

This is the line-element counterpart of $\dot{J} = J \operatorname{tr} \tilde{L}$ for the volume element and of Piola–Nanson for the area element; the three together are the transport rules for the three measures.

Now differentiate the integral over the fixed C and use the product rule:

$$\begin{aligned} \frac{d}{dt} \int_{C_t} \underline{a} \cdot d\tilde{x} &= \int_C \left[\dot{\underline{a}} \cdot d\tilde{x} + \underline{a} \cdot (d\tilde{x})^\cdot \right] = \int_{C_t} \left[\dot{\underline{a}} \cdot d\tilde{x} + \underline{a} \cdot \tilde{L} d\tilde{x} \right] \\ &= \int_{C_t} (\dot{\underline{a}} + \tilde{L}^t \underline{a}) \cdot d\tilde{x}, \end{aligned}$$

The last step moved $\tilde{L}$ from the second slot of the inner product to the first. That is the definition of the transpose, $\underline{a} \cdot \tilde{L}\underline{s} = \tilde{L}^t \underline{a} \cdot \underline{s}$, applied with $\underline{s} = d\tilde{x}$, and then the two terms were collected under one inner product by linearity in the first slot.

Now trade the material derivative for the spatial one. By (2.20), $\dot{\underline{a}} = \underline{a}' + (\operatorname{grad} \underline{a})\underline{v}$, and $\tilde{L} = \operatorname{grad} \underline{v}$ so $\tilde{L}^t = (\operatorname{grad} \underline{v})^t$. Substituting both,

$$\frac{d}{dt} \int_{C_t} \underline{a} \cdot d\tilde{x} = \int_{C_t} \left(\underline{a}' + (\operatorname{grad} \underline{a})\underline{v} + (\operatorname{grad} \underline{v})^t \underline{a} \right) \cdot d\tilde{x}.$$

So the missing terms are $(\operatorname{grad} \underline{a})\underline{v} + (\operatorname{grad} \underline{v})^t \underline{a}$. The first is the convective change of $\underline{a}$, the change met by being carried to a new place. The second has a different origin: it is the change caused by the line element itself rotating and stretching, and it would be absent if one guessed the answer to be $\int \dot{\underline{a}} \cdot d\tilde{x}$. Guessing that is the natural mistake, and the reason it fails is that $d\tilde{x}$ is not a passive label but a material object that the motion deforms, exactly as $d\mathbf{v}$ and $d\mathbf{a}$ are.

A check worth doing. Take $\underline{a} = \underline{v}$, so that the integral is the circulation of the velocity around the curve. The integrand becomes

$$\underline{v}' \cdot d\tilde{x} + (\operatorname{grad} \underline{v})\underline{v} \cdot d\tilde{x} + (\operatorname{grad} \underline{v})^t \underline{v} \cdot d\tilde{x}.$$

The third term simplifies. In components, $[(\text{grad } \underline{v})^t \underline{v}]_i = v_{j,i} v_j = \frac{1}{2} (v_j v_j)_{,i} = \frac{1}{2} (|\underline{v}|^2)_{,i}$, using the product rule backwards, so

$$(\text{grad } \underline{v})^t \underline{v} = \frac{1}{2} \text{grad } (|\underline{v}|^2),$$

a gradient. The line integral of a gradient around a closed curve is the change of the potential around a closed loop, which is zero. So on a closed material curve the third term drops out entirely and

$$\frac{d}{dt} \oint_{C_t} \underline{v} \cdot d\underline{x} = \oint_{C_t} (\underline{v}' + (\text{grad } \underline{v}) \underline{v}) \cdot d\underline{x} = \oint_{C_t} \underline{\dot{v}} \cdot d\underline{x},$$

the last step recognising (2.21). This is Kelvin's circulation theorem in its purely kinematical form: the circulation around a closed material curve changes only through the acceleration. Once Chapter 6 supplies an equation of motion for $\underline{\dot{v}}$, this becomes the statement that circulation is conserved whenever the acceleration is itself a gradient, which is the usual form of the theorem in fluid mechanics. $\square$

PROBLEM 2 (A second form of the flux derivative). Show that the flux derivative of a vector field $\underline{u}$ may be written

$$\underline{u}^* = \underline{u}' + (\text{div } \underline{u}) \underline{v} - \text{curl}(\underline{v} \times \underline{u}).$$

Solution. Start from the identity for the curl of a cross product, Problem 24 of Chapter 1:

$$\text{curl}(\underline{a} \times \underline{b}) = (\text{div } \underline{b}) \underline{a} - (\text{div } \underline{a}) \underline{b} + (\text{grad } \underline{a}) \underline{b} - (\text{grad } \underline{b}) \underline{a}.$$

Put $\underline{a} = \underline{v}$, $\underline{b} = \underline{u}$:

$$\text{curl}(\underline{v} \times \underline{u}) = (\text{div } \underline{u}) \underline{v} - (\text{div } \underline{v}) \underline{u} + (\text{grad } \underline{v}) \underline{u} - (\text{grad } \underline{u}) \underline{v}.$$

Therefore

$$\begin{aligned} (\text{div } \underline{u}) \underline{v} - \text{curl}(\underline{v} \times \underline{u}) &= (\text{div } \underline{u}) \underline{v} - (\text{div } \underline{u}) \underline{v} + (\text{div } \underline{v}) \underline{u} \\ &\quad - (\text{grad } \underline{v}) \underline{u} + (\text{grad } \underline{u}) \underline{v} \\ &= \underbrace{(\text{grad } \underline{u}) \underline{v} + (\text{div } \underline{v}) \underline{u}}_{= \text{div}(\underline{u} \otimes \underline{v})} - (\text{grad } \underline{v}) \underline{u}, \end{aligned}$$

the brace by Proposition 1.95. Adding $\underline{u}'$ to both sides and comparing with (5.22),

$$\underline{u}^* = \underline{u}' + (\operatorname{div} \underline{u}) \underline{v} - \operatorname{curl}(\underline{v} \times \underline{u}).$$

The second form is the one that makes Faraday's law transparent. If $\operatorname{div} \underline{u} = 0$, as it is for the magnetic induction, the middle term drops and

$$\frac{d}{dt} \int_{\partial P_t} \underline{u} \cdot \underline{n} \, da = \int_{\partial P_t} [\underline{u}' - \operatorname{curl}(\underline{v} \times \underline{u})] \cdot \underline{n} \, da :$$

the flux through a material loop changes through the local rate $\underline{u}'$ and through the motional term $-\operatorname{curl}(\underline{v} \times \underline{u})$, whose line-integral form $\oint (\underline{v} \times \underline{u}) \cdot d\mathbf{x}$ is the motional electromotive force. The two forms (5.22) and the boxed one are of course the same field; which is preferable depends on whether $\operatorname{div} \underline{u}$ or $\operatorname{div}(\underline{u} \otimes \underline{v})$ is the quantity at hand. $\square$

PROBLEM 3 (The referential statements). Everything in this chapter is written in the spatial description. Derive the corresponding referential statements for fields suffering a discontinuity across a surface S propagating through a fixed region $P \subset \kappa$ at speed U along its unit normal $\underline{N}$. *Hint:* use $\dot{\underline{X}} = \underline{0}$.

Solution. The hint is the whole solution, once its consequence is seen. In the referential description the independent variable is $\underline{X}$, and material points do not move in κ : $\dot{\underline{X}} = \underline{0}$. So the “material velocity” of the referential picture vanishes, and every term in the spatial formulas built from $\underline{v}$ disappears.

Transport. Let $\hat{\Phi}(\underline{X}, t)$ be a referential density and let $S(t)$ propagate through the *fixed* region P at normal speed U along $\underline{N}$, cutting P into $P^\pm$ with $\underline{N}$ directed from P^- into P^+ . Apply the non-material formula (5.27) in κ to each half. Because ∂P is fixed, its normal speed is zero and the integral over ∂P that replaced $\int_{\partial P_t} \varphi \underline{v} \cdot \underline{n} \, da$ is absent; only the two integrals over S survive, with exterior normal $-\underline{N}$ and speed $-U$ for P^+ and $+\underline{N}$, $+U$ for P^- . Adding,

$$\frac{d}{dt} \int_P \hat{\Phi} \, dV = \int_P \frac{\partial \hat{\Phi}}{\partial t} \, dV - \int_S [[\hat{\Phi}]] U \, dA.$$

Compare (5.31): the boundary flux term is gone and the singular surface term is untouched. That is the referential picture in one line. Nothing is convected, because nothing moves; all that happens is that the discontinuity sweeps across the material.

Divergence. The argument of Section 5.4.2 used only the smoothness of the field on each side and the direction of the exterior normal, never the motion. It therefore transcribes verbatim, with Div for div and $\underline{N}$, dA , dV for $\underline{n}$, da , dv : for a referential field $\underline{A}(\underline{X}, t)$ smooth on each side of S ,

$$\int_{\partial P} \underline{A} \cdot \underline{N} \, dA = \int_P \text{Div } \underline{A} \, dV + \int_S [[\underline{A}]] \cdot \underline{N} \, dA.$$

Relation between U and u . The two speeds describe the same physical surface seen in two configurations and are not equal. A point of S with referential normal speed U maps to a point of s ; the displacement $U\varepsilon\underline{N}$ in κ carries over to $\underline{F}(U\varepsilon\underline{N})$ in κ_t , whose component along $\underline{n}$ is the normal advance there. Using Piola–Nanson to relate the normals, one finds

$$u - \underline{v} \cdot \underline{n} = \frac{J}{|\underline{F}^* \underline{N}|} U,$$

so the two vanish together: S is material in κ exactly when s is material in κ_t , as it must be. The combination $u - \underline{v} \cdot \underline{n}$, the speed of the surface *relative to the material*, is the one with physical content, and it is what Chapter 6 will meet in the mass jump condition. $\square$

Chapter 6

Balance Laws

Everything so far has been apparatus. Chapter 1 built the algebra and the calculus, Chapters 2 and 3 described what a deformation is, Chapter 4 differentiated it, and Chapter 5 answered the single question a balance law must ask, namely how fast the content of a moving lump of matter changes. None of it was physics. This chapter is where the physics arrives, in the form of four axioms: mass is conserved, linear and rotational momentum balance against force and torque, energy balances against power and heating, and entropy is produced but never destroyed.

The axioms are statements about *integrals* over an arbitrary part of the body. That is the only form in which they can be stated, since force and heating are things one applies to a region and measures on its boundary, not things one evaluates at a point. Turning them into differential equations is therefore the business of the chapter, and the machinery for doing it is already complete:

- Reynolds' transport formula (5.15), and its extension (5.31) to fields that jump across a moving surface, converts the left-hand side of every balance law into an integral of a pointwise expression plus a surface term.
- The divergence theorem (5.1), and its extension (5.34), converts every boundary flux into an integral of a pointwise expression plus a surface term.
- The localization theorem, Theorem 2.20, turns the resulting state-

ment, valid for *every* part, into two pointwise statements: a field equation away from the singular surface, and a jump condition on it.

That is the entire method, and it is used eight times in what follows without variation. Remark 5.13 of Chapter 5 predicted this shape, and Remark 5.12 promised that the surface terms would emerge as the Rankine–Hugoniot jump conditions. Section 6.1 delivers both promises at once, for a balance law of unspecified content, and every later section is an instance of it.

Two results in the chapter are not of that kind, and they are the deepest things in the book so far. Cauchy’s stress theorem, Section 6.5.3, proves that the force per unit area transmitted across a surface depends on the surface only through its unit normal and depends on that normal *linearly*. Nothing in Chapter 5 suggested such a thing; it comes from the balance of linear momentum alone, applied to a shrinking tetrahedron, and it is what brings the stress tensor into existence. Noll’s theorem, Section 6.5.2, proves the first half of that statement separately and shows it was never an assumption at all.

6.1 Generic forms of the balance laws

We will show that scalar balance laws are all of the form

$$\frac{d}{dt} \int_{P_t} \rho \varphi \, dv = \int_{P_t} \rho \sigma \, dv + \int_{\partial P_t} \underline{\pi} \cdot \underline{n} \, da, \quad (6.1)$$

where $P_t \subset \kappa_t$ is the region occupied by an arbitrary part S of the body B and $\underline{n}$ is the exterior unit normal to its boundary ∂P_t . Here $\varphi = \tilde{\varphi}(\underline{x}, t)$ is the *mass* density of the quantity considered, $\sigma = \tilde{\sigma}(\underline{x}, t)$ is the rate, per unit mass, at which that quantity is supplied, the vector field $\underline{\pi} = \tilde{\pi}(\underline{x}, t)$ is its *flux*, the rate of supply per unit area through the boundary, and ρ is the spatial mass density of (2.56). The time derivative on the left pertains to a fixed set of material points: it is the derivative

of Remark 5.8, which weighs the same piece of matter twice. The three balance laws of this kind are those of mass, energy and entropy.

Vector balance laws will be shown, with the caveat of Remark 5.3, to have the symbolic form

$$\frac{d}{dt} \int_{P_t} \rho \psi \, dv = \int_{P_t} \rho \tilde{r} \, dv + \int_{\partial P_t} \tilde{A} \tilde{n} \, da, \quad (6.2)$$

where $\psi = \tilde{\psi}(\underline{x}, t)$ is the mass density of the vector-valued quantity, $\tilde{r} = \tilde{r}(\underline{x}, t)$ is its supply rate per unit mass, and $\tilde{A} = \tilde{A}(\underline{x}, t)$ is its tensor-valued flux. The relevant examples are the balances of linear and rotational momentum, that is, the equations of motion.

REMARK 6.1 (why a balance law has exactly these three terms). The form of (6.1) is not a convention chosen for convenience. It is the statement that the quantity in question is *conserved except for what is added*, and there are only two ways of adding to the contents of a region: through its interior, and through its boundary.

The first term on the right is production in the bulk: radioactive heating, a chemical source, a body force doing work. It is written $\rho\sigma$ rather than as a volume density because supplies of this kind are almost always proportional to how much matter is present, so the per-unit-mass rate σ is the quantity a constitutive equation will later have to specify. The second term is transport across the boundary. Its integrand is a rate per unit area, and what makes it a *flux* rather than a mere boundary datum is that it is determined by the orientation of the surface it crosses.

Two features of the boundary term deserve to be flagged now, because each will be argued for later rather than assumed.

Linearity in $\tilde{n}$. Equation (6.1) writes the flux as $\tilde{\pi} \cdot \tilde{n}$, which is a linear function of the unit normal. For a quantity like heat this is a hypothesis, and it is the exact analogue of Cauchy's hypothesis for the traction. It will be *proved* for the traction in Section 6.5.3, and the proof will then be transplanted to the heat flux at (6.166).

Dependence on the surface only through $\tilde{n}$. Before one can ask whether the flux is linear in the normal one must know that it depends

on the surface only through the normal, and not on, say, the curvature. That is Noll's theorem, Section 6.5.2.

No term proportional to the surface area alone, and no term involving second derivatives of the boundary, appears in (6.1). Admitting such terms is possible and leads to theories of a different kind; the classical theory is the one that does not.

Before turning to particular balance laws we pause to derive the local forms of the generic balances (6.1) and (6.2), holding at points $\underline{x} \in \kappa_t$ and on any singular surface that may be present. Doing the work once, on an unspecified φ , saves doing it five times.

The scalar case

Start from the left-hand side of (6.1). It is the derivative of a volume integral over a material region, which is exactly what Chapter 5 computed, so substitute $\rho\varphi$ in place of φ in Reynolds' formula for a field that jumps across a singular surface, (5.31):

$$\frac{d}{dt} \int_{P_t} \rho\varphi \, dv = \int_{P_t} (\rho\varphi)' \, dv + \int_{\partial P_t} \rho\varphi \, \underline{v} \cdot \underline{n} \, da - \int_s [[\rho\varphi]] u \, da,$$

in which s is the singular surface, u its normal speed, and $\underline{n}$ on s is directed from V^- into V^+ as in (5.28).

The middle term is a boundary flux of the vector field $\rho\varphi \underline{v}$ and is therefore the business of the divergence theorem for discontinuous fields, (5.34), with $\underline{a} = \rho\varphi \underline{v}$:

$$\int_{\partial P_t} \rho\varphi \underline{v} \cdot \underline{n} \, da = \int_{P_t} \operatorname{div}(\rho\varphi \underline{v}) \, dv + \int_s [[\rho\varphi \underline{v}]] \cdot \underline{n} \, da.$$

Substituting, and collecting the two integrals over s ,

$$\begin{aligned} \frac{d}{dt} \int_{P_t} \rho\varphi \, dv &= \int_{P_t} (\rho\varphi)' \, dv + \int_{\partial P_t} \rho\varphi \underline{v} \cdot \underline{n} \, da - \int_s [[\rho\varphi]] u \, da \\ &= \int_{P_t} \{(\rho\varphi)' + \operatorname{div}(\rho\varphi \underline{v})\} \, dv + \int_s [[\rho\varphi(\underline{v} - u)]] \cdot \underline{n} \, da, \end{aligned} \tag{6.3}$$

where $\underline{u}$ is the vector defined on s by $\underline{u} \cdot \underline{n} = u$, the speed of the singular surface, exactly as in (5.27).

The last step is the one place in the section where something is moved across a jump bracket, so it is done in full. We have

$$[[\rho\varphi\underline{v}]] \cdot \underline{n} - [[\rho\varphi]]u.$$

Both u and $\underline{n}$ belong to the *surface*, not to the material: s has one normal and advances at one speed at each of its points, and neither of them takes two values there. A jump bracket subtracts the two limiting values of a field approached from the two sides, so a quantity with only one value passes through the bracket untouched: $[[\rho\varphi]]u = [[\rho\varphi u]]$, and $[[\underline{a}]] \cdot \underline{n}$ may be written $[[\underline{a} \cdot \underline{n}]]$. Writing $u = \underline{u} \cdot \underline{n}$ and using linearity of the inner product in its first slot,

$$[[\rho\varphi\underline{v}]] \cdot \underline{n} - [[\rho\varphi]]u \cdot \underline{n} = [[\rho\varphi\underline{v}]] \cdot \underline{n} - [[\rho\varphi\underline{u}]] \cdot \underline{n} = [[\rho\varphi(\underline{v} - \underline{u})]] \cdot \underline{n},$$

which is the integrand in (6.3)₂. The combination $\underline{v} - \underline{u}$ is the velocity of the material relative to the surface, and only its normal component survives the inner product:

$$(\underline{v} - \underline{u}) \cdot \underline{n} = \underline{v} \cdot \underline{n} - \underline{u} \cdot \underline{n} = \underline{v} \cdot \underline{n} - u,$$

which is the quantity Problem 3 of Chapter 5 identified as the one with physical content.

Now bring in the right-hand side of (6.1), treating its boundary term by (5.34) once more, this time with $\underline{a} = \underline{\pi}$:

$$\int_{\partial P_t} \underline{\pi} \cdot \underline{n} \, da = \int_{P_t} \operatorname{div} \underline{\pi} \, dv + \int_s [[\underline{\pi}]] \cdot \underline{n} \, da.$$

Equating (6.3)₂ to $\int \rho\sigma \, dv + \int \operatorname{div} \underline{\pi} \, dv + \int_s [[\underline{\pi}]] \cdot \underline{n} \, da$ and moving everything to one side,

$$\int_{P_t} \{(\rho\varphi)' + \operatorname{div}(\rho\varphi\underline{v} - \underline{\pi}) - \rho\sigma\} \, dv + \int_s [[\rho\varphi(\underline{v} - \underline{u}) - \underline{\pi}]] \cdot \underline{n} \, da = 0. \quad (6.4)$$

Two localizations, not one

Equation (6.4) holds for every part S of the body. It contains two integrals over two different sets, and the standard localization argument extracts one pointwise statement from each. The extraction is done in two stages, and the order matters.

Stage one: parts that miss the singular surface. Apply (6.4) to an arbitrary part P_t that contains no piece of a singular surface, so that $s = \emptyset$ and the second integral is absent (Figure 6.1). What remains is

$$\int_{P_t} \{(\rho\varphi)' + \operatorname{div}(\rho\varphi\mathbf{v} - \boldsymbol{\pi}) - \rho\sigma\} \, dv = 0$$

for every such P_t . Assuming, as always, that the integrand is continuous there, Theorem 2.20 applies and gives

$$(\rho\varphi)' + \operatorname{div}(\rho\varphi\mathbf{v}) = \operatorname{div} \boldsymbol{\pi} + \rho\sigma \quad (6.5)$$

at all points $\mathbf{x} \in \kappa_t$ that do not belong to a singular surface. The argument is the one recalled in Section 5.1.1: if the integrand were non-zero at a point it would keep its sign throughout a small ball around that point, and integrating over that ball alone would contradict the hypothesis. Freedom to choose the region is what carries it.

Stage two: parts that meet the singular surface. Now let P_t contain a piece s of a singular surface. It is cut by s into subvolumes $V^\pm$, and in the interior of each of them the fields are smooth, so (6.5) holds there by stage one. Therefore the volume integral in (6.4) vanishes, whatever P_t is, and (6.4) collapses to

$$\int_s [\rho\varphi(\mathbf{v} - \mathbf{u}) - \boldsymbol{\pi}] \cdot \mathbf{n} \, da = 0.$$

The part P_t is otherwise arbitrary, so its intersection s with the singular surface is an arbitrary piece of that surface, and a version of the localization theorem for surface integrals, together with continuity of the integrand on s , yields the *jump condition*

$$[\rho\varphi(\mathbf{v} - \mathbf{u}) - \boldsymbol{\pi}] \cdot \mathbf{n} = 0 \quad (6.6)$$

at every point of a singular surface.

REMARK 6.2 (why the two statements are independent). It is tempting to think that (6.6) is contained in (6.5), or follows from it by a limit. It does not, and the reason is the reason Chapter 5 needed Section 5.4.1 at all. Equation (6.5) is a statement about derivatives, and across s the fields have no derivatives: the classical divergence of a field that jumps is not defined there. What (6.4) says is that the balance, read distributionally, has a bulk part and a surface part, and that both must vanish separately because a part of the body may be chosen to contain the surface or to avoid it. That freedom of choice is exactly what Remark 5.13 anticipated.

The two-stage order is also not decorative. Had we tried to localize (6.4) in one step we would have concluded nothing, since a volume integral and a surface integral summing to zero does not by itself force either to vanish: they might cancel. What forces the split is the existence of parts with $s = \emptyset$, which kills the surface term while leaving the volume term arbitrary.

The jump condition (6.6) reduces to

$$[[\pi]] \cdot \underline{n} = 0 \tag{6.7}$$

if the singular surface is a *material* surface, that is, if it moves with the material, $u = \underline{v} \cdot \underline{n}$ with $[[\underline{v}]] = \underline{0}$. Then $\underline{v} - \underline{u}$ has vanishing normal component on each side and the convected term drops out. In fact (6.6) places no restriction whatever on the tangential discontinuity $(\underline{I} - \underline{n} \otimes \underline{n})[[\underline{v}]]$, since only the normal component of $\underline{v} - \underline{u}$ enters, so (6.7) survives under the weaker hypotheses $u = \underline{v} \cdot \underline{n}$ with $[[\underline{v}]] \cdot \underline{n} = 0$. A vortex sheet, across which the tangential velocity jumps while no material crosses the surface, is admitted.

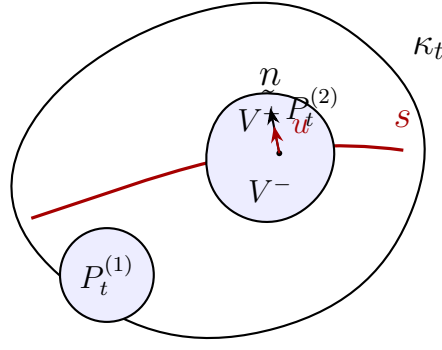


Figure 6.1: Parts of a body with and without a singular surface. Applying (6.4) to $P_t^{(1)}$, which misses s , kills the surface term and localizes to the field equation (6.5). Applying it to $P_t^{(2)}$, which is cut by s into $V^\pm$, then kills the volume term, because (6.5) already holds in each half, and what is left localizes on s to the jump condition (6.6). Both parts are arbitrary, which is why both conclusions follow. The normal $\underline{n}$ on s points from V^- into V^+ , and s advances at speed u along it, the convention of (5.28).

The vector case

Turning to (6.2), we proceed exactly as above, and as explained in Section 5.4.1, to cast (6.2) in the symbolic form

$$\int_{P_t} \{(\rho\psi)' + \operatorname{div}(\rho\psi \otimes \underline{v} - \underline{A}) - \rho\underline{r}\} \, dv + \int_s [\![\rho\psi \otimes (\underline{v} - \underline{u}) - \underline{A}]\!] \underline{n} \, da = 0, \quad (6.8)$$

use having been made of the relevant extensions of (5.31) and (5.34), namely

$$\begin{aligned} \frac{d}{dt} \int_{P_t} \rho\psi \, dv &= \int_{P_t} (\rho\psi)' \, dv + \int_{\partial P_t} (\rho\psi \otimes \underline{v}) \underline{n} \, da - \int_s [\![\rho\psi]\!] u \, da \quad \text{and} \\ \int_{\partial P_t} \underline{A} \underline{n} \, da &= \int_{P_t} \operatorname{div} \underline{A} \, dv + \int_s [\![\underline{A}]\!] \underline{n} \, da, \end{aligned} \quad (6.9)$$

respectively.

Both lines of (6.9) are obtained in the way described after (5.31) and again at (5.18): apply the scalar statement to each Cartesian component

on a fixed orthonormal basis, multiply by $\underline{e}_i$ and sum. The caveat of Remark 5.3 is in force throughout, so each display is shorthand for three component statements. In the first of them the surface integrand appeared as $\rho\psi_i \underline{v} \cdot \underline{n}$ in components, and $\rho\psi_i (\underline{v} \cdot \underline{n}) \underline{e}_i = (\underline{v} \cdot \underline{n}) \rho\psi = (\rho\psi \otimes \underline{v}) \underline{n}$ by the definition of the tensor product, Definition 1.32, which is the step already taken between (5.17) and (5.18).

The passage from $[[\rho\psi \otimes \underline{v}]] \underline{n} - [[\rho\psi]] u$ to $[[\rho\psi \otimes (\underline{v} - \underline{u})]] \underline{n}$ in (6.8) is the tensor-product version of the manipulation performed after (6.3): with $u = \underline{u} \cdot \underline{n}$ single-valued on s ,

$$[[\rho\psi]] \underline{u} \cdot \underline{n} = [[\rho\psi \underline{u} \cdot \underline{n}]] = [(\rho\psi \otimes \underline{u}) \underline{n}] = [[\rho\psi \otimes \underline{u}]] \underline{n},$$

and the two tensor products then subtract by bilinearity of $\otimes$.

The two localizations run as before. The local equation holding at points of κ_t off a singular surface is

$$(\rho\psi)' + \operatorname{div}(\rho\psi \otimes \underline{v}) = \operatorname{div} \underline{A} + \rho r, \quad (6.10)$$

and the jump condition holding at every point of a singular surface is

$$[[\rho\psi \otimes (\underline{v} - \underline{u}) - \underline{A}]] \underline{n} = \underline{0}, \quad (6.11)$$

reducing to

$$[[\underline{A}]] \underline{n} = \underline{0} \quad (6.12)$$

under the conditions stated after (6.7).

The foregoing statements are all expressed in the spatial description. Deriving their referential counterparts is a straightforward matter and is done in Problem 1, where the hint is the one Chapter 5 already supplied: in κ nothing moves, so $\dot{\underline{X}} = \underline{0}$ and every term built from the material velocity disappears.

6.2 Conservation of mass

Section 2.5 already introduced the idea that the mass of a fixed set $S \subset B$ of material points is conserved in the course of any motion the

body may undergo. As an axiom:

$$\frac{d}{dt} \int_{P_t} \rho(\underline{x}, t) dv = 0. \quad (6.13)$$

This is of the form (6.1) with

$$\varphi = 1, \quad \sigma = 0, \quad \underline{\pi} = \underline{0}.$$

Each of the three identifications says something. Taking $\varphi = 1$ says that the quantity being balanced is mass itself, whose mass density is one unit of mass per unit mass. Taking $\sigma = 0$ says that mass is not created in the interior. Taking $\underline{\pi} = \underline{0}$ says that mass does not enter through the boundary except by being carried there by the material, since the convective transport $\rho \underline{v} \cdot \underline{n}$ is already accounted for on the left-hand side by the fact that the region P_t moves with the material. A non-zero $\underline{\pi}$ would describe diffusion, and it is precisely the term one must restore in a mixture.

The two local statements now follow by substitution into (6.5) and (6.6). Since $\operatorname{div} \underline{\pi} = 0$ and $\rho \sigma = 0$, the field equation is

$$\rho' + \operatorname{div}(\rho \underline{v}) = 0, \quad (6.14)$$

holding wherever the fields are smooth, and the jump condition is

$$[[\rho(\underline{v} - \underline{u})]] \cdot \underline{n} = 0, \quad (6.15)$$

operative on singular surfaces.

REMARK 6.3 (the first Rankine–Hugoniot condition). Equation (6.15) is the promise of Remark 5.12 coming due. Write it out. With $\underline{n}$ and u single-valued on s it says

$$\rho^+(\underline{v}^+ \cdot \underline{n} - u) = \rho^-(\underline{v}^- \cdot \underline{n} - u),$$

so the scalar

$$m^* := \rho(\underline{v} \cdot \underline{n} - u)$$

takes the same value on both sides of the surface. Its meaning is direct: $\underline{v} \cdot \underline{n} - u$ is the normal speed of the material relative to the surface, so m^* is the mass crossing unit area of s per unit time. Conservation of mass says that whatever flows in on one side flows out on the other, which is the only thing it could say.

Two limiting cases are worth separating. If $m^* = 0$, no material crosses s : the surface is material, $u = \underline{v} \cdot \underline{n}$ on both sides, and ρ may jump freely. That is a contact discontinuity, or a phase boundary at rest relative to the material. If $m^* \neq 0$, material is being processed by the surface, which is then a shock: neither ρ nor $\underline{v}$ need be continuous, but their combination m^* is. Every further jump condition in this chapter will be written most naturally in terms of m^* , because m^* passes through jump brackets.

Equation (6.13) must be modified in the presence of diffusion. In that case σ is the supply of the diffusing species per unit mass of the mixture and π is the mass flux of that species through the boundary. Diffusion is not treated in this book.

Three equivalent forms

Equation (6.14) is stated in the spatial clock. Trading it for the material one is a matter of the product rule. By Problem 24(c) of Chapter 1,

$$\operatorname{div}(\rho \underline{v}) = \operatorname{grad} \rho \cdot \underline{v} + \rho \operatorname{div} \underline{v}, \quad (6.16)$$

and $\operatorname{div} \underline{v} = \operatorname{tr} \underline{L}$ with $\underline{L} = \operatorname{grad} \underline{v}$ the spatial velocity gradient of (4.1), by Proposition 1.92 and Definition 1.91. Substituting (6.16) into (6.14) and recognising the first two terms as a material derivative through (2.20), $\dot{\rho} = \rho' + \operatorname{grad} \rho \cdot \underline{v}$,

$$0 = \rho' + \operatorname{grad} \rho \cdot \underline{v} + \rho \operatorname{div} \underline{v} = \dot{\rho} + \rho \operatorname{div} \underline{v}.$$

By Problem 2 of Chapter 4, $\dot{J} = J \operatorname{tr} \underline{L}$, so $\operatorname{div} \underline{v} = \dot{J}/J$ and

$$0 = \dot{\rho} + \rho \operatorname{div} \underline{v} = \dot{\rho} + \rho (\dot{J}/J). \quad (6.17)$$

Multiplying (6.17) through by the positive number J assembles the two terms into a single product rule,

$$0 = J\dot{\rho} + \rho\dot{J} = (\rho J)^\cdot,$$

so ρJ is constant in time along each material point. With reference to (2.59), $\rho_\kappa = \rho J$, this is equivalent to the vanishing of the material derivative of the referential mass density:

$$\dot{\rho}_\kappa = 0. \tag{6.18}$$

Recalling that $\dot{\rho}_\kappa = \partial\rho_\kappa(\underline{X}, t)/\partial t$, since the material derivative in the referential description is simply the partial time derivative at fixed $\underline{X}$, it follows that the referential density, expressed as a function of $\underline{X}$ and t , is a function of $\underline{X}$ alone. We indicate this by writing $\rho_\kappa(\underline{X})$.

An alternative derivation of the same result is obtained by differentiating (2.56), assuming $P \subset \kappa$ to be a region in which ρ_κ and $\dot{\rho}_\kappa$ satisfy the continuity conditions stated after (5.13). Because P does not depend on time, the derivative passes under the integral sign at once, with no transport formula needed:

$$0 = \frac{d}{dt} \int_P \rho_\kappa dV = \int_P \dot{\rho}_\kappa dV. \tag{6.19}$$

Choosing P to be otherwise arbitrary, we localize as usual by Theorem 2.20 to arrive at (6.18) directly.

REMARK 6.4 (which form to use, and why the third is the best). The three statements

$$\rho' + \operatorname{div}(\rho \underline{v}) = 0, \quad \dot{\rho} + \rho \operatorname{div} \underline{v} = 0, \quad \dot{\rho}_\kappa = 0$$

are the same axiom in the three clocks of Remark 5.8, and (6.19) shows how cheap the third one is: pull back to a region that does not move, and conservation of mass becomes the observation that a time-independent integral has a vanishing derivative.

The third form is the one to remember, and the reason is stated at (6.136) below. In the referential description $\rho_\kappa(\underline{X})$ is *given data*, part

of the description of the body, and the mass balance is then not an equation to be solved at all: it has already been solved. In the spatial description $\rho(\underline{x}, t)$ is an unknown field obeying the partial differential equation (6.14), coupled to the velocity. One equation has been traded for none, and that is the whole advantage of the referential formulation of the equations of motion.

What mass conservation does to the generic balances

The results just derived effect a significant simplification of the generic local balances (6.5) and (6.10). Concerning the former, work on its left-hand side. Expand the divergence by Problem 24(c) of Chapter 1, applied to the scalar field $\rho\varphi$:

$$\begin{aligned} (\rho\varphi)' + \operatorname{div}(\rho\varphi\underline{v}) &= (\rho\varphi)' + \operatorname{grad}(\rho\varphi) \cdot \underline{v} + \rho\varphi \operatorname{div} \underline{v} \\ &= (\rho\varphi)^{\cdot} + \rho\varphi \operatorname{div} \underline{v} \\ &= (\dot{\rho} + \rho \operatorname{div} \underline{v})\varphi + \rho\dot{\varphi} \\ &= \rho\dot{\varphi}, \end{aligned} \tag{6.20}$$

where conservation of mass in the form (6.17)₁ has been applied in the last step. The second line used (2.20) on the field $\rho\varphi$; the third expanded $(\rho\varphi)^{\cdot} = \dot{\rho}\varphi + \rho\dot{\varphi}$ by the product rule and then grouped the two terms carrying φ . That grouping is the whole point of the calculation: it isolates exactly the combination that mass conservation annihilates. Accordingly, (6.5) simplifies to

$$\rho\dot{\varphi} = \operatorname{div} \underline{\pi} + \rho\sigma. \tag{6.21}$$

In the same way, for the vector case, we use Problem 24(g) of Chapter 1, which is Proposition 1.95 in the form

$$\operatorname{div}(\rho\underline{\psi} \otimes \underline{v}) = [\operatorname{grad}(\rho\underline{\psi})]\underline{v} + (\rho \operatorname{div} \underline{v})\underline{\psi}, \tag{6.22}$$

together with conservation of mass. The steps are the same three:

$$\begin{aligned} (\rho\underline{\psi})' + \operatorname{div}(\rho\underline{\psi} \otimes \underline{v}) &= (\rho\underline{\psi})' + [\operatorname{grad}(\rho\underline{\psi})]\underline{v} + \rho(\operatorname{div} \underline{v})\underline{\psi} \\ &= (\rho\underline{\psi})^{\cdot} + \rho(\operatorname{div} \underline{v})\underline{\psi} = (\dot{\rho} + \rho \operatorname{div} \underline{v})\underline{\psi} + \rho\dot{\underline{\psi}} = \rho\dot{\underline{\psi}}, \end{aligned}$$

the second equality being (2.20) applied componentwise to $\rho\psi$. So (6.10) simplifies to

$$\rho\dot{\psi} = \operatorname{div} \mathcal{A} + \rho\dot{r}. \quad (6.23)$$

Alternatively, these reduced equations may be derived directly by localizing the global statements (6.1) and (6.2), if we first replace their left-hand sides by

$$\begin{aligned} \frac{d}{dt} \int_{P_t} \rho\varphi \, dv &= \frac{d}{dt} \int_P \rho_\kappa \varphi \, dV = \int_P \rho_\kappa \dot{\varphi} \, dV = \int_{P_t} \rho \dot{\varphi} \, dv \quad \text{and} \\ \frac{d}{dt} \int_{P_t} \rho\psi \, dv &= \frac{d}{dt} \int_P \rho_\kappa \psi \, dV = \int_P \rho_\kappa \dot{\psi} \, dV = \int_{P_t} \rho \dot{\psi} \, dv, \end{aligned} \quad (6.24)$$

respectively, where use has been made of (2.50), $dv = J \, dV$, of (2.59), $\rho_\kappa = \rho J$, so that $\rho \, dv = \rho_\kappa \, dV$, and of mass conservation in the form (6.18), which is what allows ρ_κ to be treated as a constant when the derivative is taken inside the integral over the fixed region P .

REMARK 6.5 (the shortest route, and why both are kept). The route through (6.24) is three lines and the route through (6.20) is eight. The longer one was taken first for two reasons.

The first is that (6.24) assumes smoothness throughout P : the step $\frac{d}{dt} \int_P = \int_P \frac{d}{dt}$ is Remark 5.7 in action and has nothing to say when the integrand jumps. Every jump condition in this chapter comes from the long route, and none from the short one.

The second is that (6.20) shows *where* mass conservation enters: in exactly one place, as the coefficient of φ in the line

$$(\rho\varphi)' + \operatorname{div}(\rho\varphi\mathbf{v}) = \underbrace{(\dot{\rho} + \rho \operatorname{div} \mathbf{v})}_{=0 \text{ by (6.17)}} \varphi + \rho\dot{\varphi}.$$

The reduced equation (6.21) is therefore not a new axiom but the old one with mass conservation already spent. The compact form

$$(\rho\varphi)' + \operatorname{div}(\rho\varphi\mathbf{v}) = \rho\dot{\varphi}$$

is used at every balance law from here to the end of the book, each an instance of $\rho\dot{\varphi} = \operatorname{div} \pi + \rho\sigma$ with a particular reading of the three letters.

6.3 Linear and rotational momenta; forces and torques

The equations of motion for a continuum, due to Euler, are relations connecting the linear and rotational momenta of a set $S \subset B$ of material points to the forces and torques acting on it. This section defines the four objects; the next asserts the two relations.

6.3.1 Linear and rotational momenta

The linear momentum of an arbitrary sub-body is the vector

$$\underline{l}(S, t) = \int_{P_t} \rho \underline{v} \, dv = \int_P \rho_\kappa \underline{v} \, dV, \quad (6.25)$$

and the rotational momentum relative to an origin located at position ϱ , also called the moment of momentum, is

$$\underline{h}_\varrho(S, t) = \int_{P_t} \rho \underline{x} \times \underline{v} \, dv = \int_P \rho_\kappa \underline{x} \times \underline{v} \, dV, \quad (6.26)$$

where $\underline{x}$ is the position vector of a material point $p \in S$ relative to ϱ , in the sense of Proposition 1.63 and with the abuse of notation recorded in Remark 1.64.

The second equality in each of (6.25) and (6.26) is not a change of notation. It is the substitution $\rho \, dv = \rho_\kappa \, dV$, which combines the change of variables (2.50), $dv = J \, dV$, with the definition (2.59), $\rho_\kappa = \rho J$. The integrands are unchanged because $\underline{v}$ and $\underline{x}$ are the same objects read at the same material point; only the measure and the domain have been transferred to κ . This is the pull-back of Remark 5.7 once more, and it is available here precisely because mass conservation makes ρ_κ independent of t .

REMARK 6.6 (rotational momentum depends on the origin, and that is not a defect). The subscript on $\underline{h}_\varrho$ is doing work. Replace ϱ by $\varrho' = \varrho + \underline{c}$ with $\underline{c}$ fixed. Then the position of a material point relative to the new

origin is $\underline{x}' = \underline{x} - \underline{c}$, and

$$\underline{h}_{\underline{o}'} = \int_{P_t} \rho (\underline{x} - \underline{c}) \times \underline{v} \, dv = \underline{h}_{\underline{o}} - \underline{c} \times \int_{P_t} \rho \underline{v} \, dv = \underline{h}_{\underline{o}} - \underline{c} \times \underline{l}.$$

So rotational momentum is origin dependent unless the linear momentum vanishes. The same computation applied to (6.31) and (6.32) gives $\underline{m}_{\underline{o}'} = \underline{m}_{\underline{o}} - \underline{c} \times \underline{f}$, so the torque shifts by exactly the amount needed for the balance (6.35) to survive the change, as it must: a physical law cannot depend on where one chooses to put the origin. Section 6.4.1 makes the dependence explicit by choosing the one origin that simplifies it, the center of mass, and (6.47) is the result.

6.3.2 Forces

We assume the forces acting on S to be of two types. The first, called *body forces*, are expressible in the form

$$\underline{f}_b(S, t) = \int_{P_t} \rho \underline{b} \, dv = \int_P \rho_\kappa \underline{b} \, dV, \quad (6.27)$$

where $\underline{b} = \underline{\tilde{b}}(\underline{x}, t) = \underline{\hat{b}}(\underline{X}, t)$ is the body force per unit mass, the carets and tildes carrying their usual meaning from Remark 2.6. For example, near a point on the surface of the Earth the body force due to gravity is $\underline{b} = -g\underline{k}$, where $g \simeq 9.81 \text{ m/s}^2$ and $\underline{k}$ is a fixed unit vector directed vertically upward from the point in question.

The second type are the *contact forces*, arising from contact of S with its surroundings. These act on the surface ∂P_t and are expressed in the form

$$\underline{f}_c(S, t) = \int_{\partial P_t} \underline{t} \, da, \quad (6.28)$$

where $\underline{t} = \underline{\tilde{t}}(\underline{x}, t)$, the force per unit area, is the *Cauchy traction*, or simply the traction. It is often convenient to express this same force as an integral over ∂P . Thus

$$\underline{f}_c(S, t) = \int_{\partial P} \underline{p} \, dA, \quad (6.29)$$

in which $\underline{p} = \hat{\underline{p}}(\underline{X}, t)$, the *Piola traction*, furnishes the same force but measured per unit area of ∂P rather than of ∂P_t .

REMARK 6.7 (per unit mass, per unit area, and the fiction in $\underline{p}$). Three conventions are fixed in (6.27) to (6.29) and each is a choice about what the quantity is proportional to.

Why $\underline{b}$ is per unit mass. Gravity pulls on matter, not on volume, and the coefficient it pulls with is the mass. Writing the body force density as $\rho \underline{b}$ rather than as a single field therefore isolates in $\underline{b}$ the part that is a property of the external agency and not of the body. In a uniform gravitational field $\underline{b}$ is the same vector for lead and for cork.

Why $\underline{t}$ is per unit area. A contact force is exerted by what is on the other side of a surface, and how much of the other side is in contact is measured by area. This is not a definition one is free to make differently: (6.28) asserts that the contact force on a part is an *integral* over its boundary, that is, that it is absolutely continuous with respect to area. Contact forces concentrated on curves or points within the surface are thereby excluded.

What $\underline{p}$ is, and what it is not. While $\underline{p}$ is defined on ∂P , the actual contact forces act on ∂P_t . Nothing is pushing on the reference configuration; κ is a labelling device, as Remark 4.2 insisted. The Piola traction is the same force with the bookkeeping changed, so that the total is recovered by integrating over a surface that does not move. Equating (6.28) and (6.29) for every part will yield $\underline{p} dA = \underline{t} da$ at (6.142), and Piola–Nanson (2.54) will convert that into the relation $\underline{P} = \underline{T} \underline{F}^*$ between the two stresses. Confusing $\underline{p}$ with a traction that something exerts on κ is the standard way of getting the factors of $\underline{J}$ wrong.

The net force acting on S is

$$\underline{f}(S, t) = \underline{f}_b(S, t) + \underline{f}_c(S, t). \quad (6.30)$$

6.3.3 Torques

The moment of the body forces, relative to the origin located at position $\underline{q}$, is

$$\underline{m}_b(S, t) = \int_{P_t} \rho \underline{x} \times \underline{b} \, dv = \int_P \rho_\kappa \underline{x} \times \underline{b} \, dV, \quad (6.31)$$

and that of the contact forces is

$$\underline{m}_c(S, t) = \int_{\partial P_t} \underline{x} \times \underline{t} \, da = \int_{\partial P} \underline{x} \times \underline{p} \, dA. \quad (6.32)$$

In this book we follow the development of classical continuum mechanics as conceived by Cauchy, and thus assume that *intrinsic* body and contact torques are absent. Continua of this kind are called *non-polar* materials. The net torque acting on S , relative to $\underline{q}$, is then

$$\underline{m}_{\underline{q}}(S, t) = \underline{m}_b(S, t) + \underline{m}_c(S, t). \quad (6.33)$$

REMARK 6.8 (what “non-polar” assumes, and what it will cost). Read (6.31) and (6.32) carefully. Every torque appearing there is the moment of a *force* about $\underline{q}$: it is of the form $\underline{x} \times (\text{something})$, and it therefore vanishes at $\underline{x} = \underline{0}$. Nothing in the definitions allows for a torque that acts at a point without acting through a lever arm.

Such torques exist. Consider a material reinforced by a continuous distribution of embedded fibres. If the fibres resist local bending and torsion, then cutting the body across the fibres transmits not only a force per unit area but also a moment per unit area, a *couple traction*, wherever the fibres cross the cut. A medium with couple tractions is called *polar*, and for it (6.32) must be augmented by a term $\int_{\partial P_t} \underline{m} \, da$ and (6.26) by an intrinsic spin density, since the material points then carry angular momentum of their own rather than merely orbiting the origin.

The assumption is not free. What it buys is the symmetry of the Cauchy stress, (6.117): rotational momentum balance will turn out to add no differential equation at all, only the algebraic statement $\underline{T} = \underline{T}^t$, and that statement is exactly the statement that no couple tractions

are present. What it costs is that the theory has no internal length scale coming from the stress. Whether a real material is well described this way is decided by whether its response depends on the size of the specimen at fixed shape, and for ordinary engineering materials at ordinary scales it does not.

6.4 Euler's postulates

Euler's postulates apply to an arbitrary part S of a body. They entail the fundamental assumption that the motion of the part is governed by the same equations that govern the motion of a rigid body. Thus the rate of change of the linear momentum of a fixed set of material points is balanced by the net force acting on it,

$$\frac{d}{dt}l(S, t) = \tilde{f}(S, t), \quad (6.34)$$

and the rate of change of the rotational momentum is balanced by the net torque,

$$\frac{d}{dt}h_{\varrho}(S, t) = \tilde{m}_{\varrho}(S, t). \quad (6.35)$$

A frame of reference in which these apply, consisting of point spaces $\mathcal{E}^3$ and $\hat{\mathcal{E}}^3$, associated translation spaces $\mathbb{E}^3$ and $\hat{\mathbb{E}}^3$ respectively, and an origin located at a fixed position ϱ , is called an *inertial frame*. Euler's postulates are thus tantamount to the assumption that an inertial frame exists. Chapter 7 will show that there are infinitely many of them, and will obtain the equations holding relative to arbitrary, generally non-inertial, frames.

REMARK 6.9 (which word in (6.34) is doing the work). Neither (6.34) nor (6.35) is new as a statement about a whole body: they are Newton's and Euler's laws for a rigid body, written down. What is new, and what makes continuum mechanics possible, is the phrase *arbitrary part*.

A rigid body has six degrees of freedom, and (6.34) and (6.35) are six equations. Applied once, to the body as a whole, they would determine

a rigid motion and nothing else; for a deformable body they would be six equations for infinitely many unknowns and would determine nothing. Applied to *every* part, they become infinitely many equations, one for each subregion, and that is exactly the hypothesis the localization theorem needs. The step from (6.36) to (6.109) is nothing but the conversion of “for every part” into “at every point”. Remark 5.2 of Chapter 5 made the same observation about the quantifier “for all $\mathcal{C}$ ”; here the quantifier ranges over subregions, and Remark 5.6 predicted that this would be the pattern of the whole subject.

There is a genuine assumption hidden in the phrase as well. A part S is not isolated: it is glued to the rest of the body. Postulating (6.34) for S requires that everything the rest of the body does to it be representable as a body force plus a surface traction, with no further term. Section 6.5.1 shows that this hypothesis is strong enough to force action and reaction, which in particle mechanics is a separate axiom.

REMARK 6.10 (why “postulates”, and why Euler’s). The name records two decisions, one about logical status and one about attribution. Both are worth unpacking, because a reader arrives here with Newton in mind and expects (6.34) to be a law already proved.

Newton’s laws are laws for a mass point. A point has no extension. The moment $\underline{x} \times \underline{f}$ of a force acting on it is a statement about the choice of origin, not about the particle, and it carries nothing that $\underline{f} = m\underline{\ddot{x}}$ does not already carry: cross $m\ddot{\underline{x}} = \underline{f}$ with $\underline{x}$ on the left and (6.35) for a single particle drops out, since $\underline{x} \times m\ddot{\underline{x}} = (\underline{x} \times m\dot{\underline{x}})^\cdot$ by the cancellation displayed after (6.37). So nothing among Newton’s three laws plays the part of (6.35).

For a finite system of particles the first balance is a theorem. Let N particles carry masses m_a and positions $\underline{x}_a$, let $\underline{f}_a^e$ be the force applied to the a th from outside the system, let $\underline{f}_{ab}$ be the force applied to it by the b th, and let each particle obey Newton’s second law,

$$m_a \ddot{\underline{x}}_a = \underline{f}_a^e + \sum_{b \neq a} \underline{f}_{ab}.$$

Sum over a and collect the interior forces in pairs:

$$\dot{\underline{l}} = \sum_a m_a \ddot{\underline{x}}_a = \sum_a \underline{f}_a^e + \sum_{a < b} (\underline{f}_{ab} + \underline{f}_{ba}) = \sum_a \underline{f}_a^e,$$

the pair sum vanishing term by term because Newton's third law gives $\underline{f}_{ba} = -\underline{f}_{ab}$. That is (6.34) for the system, proved rather than assumed.

The second balance is not a theorem. Repeat the computation with the moment arm attached:

$$\dot{\underline{h}}_o = \sum_a m_a \underline{x}_a \times \ddot{\underline{x}}_a = \sum_a \underline{x}_a \times \underline{f}_a^e + \sum_{a < b} (\underline{x}_a \times \underline{f}_{ab} + \underline{x}_b \times \underline{f}_{ba}),$$

the left side again by $(\underline{x} \times \dot{\underline{x}})^\cdot = \underline{x} \times \ddot{\underline{x}}$. The third law turns each pair term into

$$\underline{x}_a \times \underline{f}_{ab} - \underline{x}_b \times \underline{f}_{ab} = (\underline{x}_a - \underline{x}_b) \times \underline{f}_{ab},$$

which vanishes for every pair if and only if $\underline{f}_{ab}$ is parallel to $\underline{x}_a - \underline{x}_b$ for every pair. Newton's third law as usually stated says only that the pair forces are equal and opposite. That their common line of action joins the two particles is a further assumption, called the strong form of the law, and it is exactly what the moment balance needs. Even in particle mechanics, then, (6.35) is not a consequence of Newton's three laws.

For a continuum there are no pairs to appeal to. The interior action is not given as a sum over particle pairs but as the traction field of (6.28), and Sections 6.5.1 and 6.5.2 will show that the two properties one might have hoped to use are themselves consequences of (6.34): Cauchy's lemma $\underline{t}_{21} = -\underline{t}_{12}$ at (6.83), which is the continuum form of the third law, and Cauchy's hypothesis (6.84) itself, by Noll's theorem. The logical order is the reverse of the particle case. Nothing independent of (6.34) is left from which (6.35) might be deduced, and Proposition 6.11 shows that the deduction is in fact impossible.

Why the word "postulate". Three things are assumed here and none of them can be checked inside the theory. Both statements are carried over from whole rigid bodies, where they are theorems of particle mechanics with central interior forces, to *every part* of *every* body, deformable or not; that extension is the subject of Remark 6.9. Both

presuppose a frame and an origin in which they hold, so together they assert that an inertial frame exists. And the second asserts something the first does not. Truesdell, whose account of this material shaped the modern presentation, called them Euler’s *laws* of mechanics; the two names carry the same content, and “postulate” merely keeps the assumed status in view.

Why Euler. The *Principia* of 1687 states three laws for what Newton calls bodies, but the mathematics carried out with them is the mathematics of mass points, and no moment of momentum principle appears among them. Euler, in a memoir of 1750 whose title announces the discovery of a new principle of mechanics, took $\underline{f} = m\underline{a}$ and applied it to every element of an extended body, which is the first appearance of the quantifier that Remark 6.9 identifies as the whole content of the phrase “arbitrary part”. In 1775 he stated the moment of momentum principle as a second and independent law, holding for every body and for every part of one. Both (6.34) and (6.35), in the form used here, are therefore his.

Newton’s law did not disappear in the process. Equation (6.44) is exactly $\underline{f} = m\underline{a}$, and Section 6.4.1 obtains it as a *theorem*: it is the statement Euler’s first postulate makes about the center of mass. Newton’s second law is thus a corollary of Euler’s first, recovered by applying the postulate to a whole part instead of to every part of it. Naming the pair after Newton would hide both that containment and the fact that the second of them is not his at all.

Proposition 6.11 makes the independence precise. It anticipates two results of Section 6.5: the local form (6.109) of the linear momentum balance, and the symmetry (6.117) of the Cauchy stress, which is everything the rotational balance contributes at a point.

PROPOSITION 6.11 (the second postulate is independent of the first). *The symmetry of the Cauchy stress is not a consequence of the local linear momentum balance. There exist a motion and fields $\underline{T}$ and $\underline{b}$ satisfying*

$$\rho \underline{\dot{v}} = \rho \underline{b} + \operatorname{div} \underline{T}$$

at every point of κ_t , with $\underline{T} \neq \underline{T}^t$ everywhere.

Proof. Take any motion whatever; it fixes ρ and $\dot{\underline{v}}$. For the stress take the constant field

$$\underline{T} = \underline{e}_1 \otimes \underline{e}_2,$$

which is not symmetric, since $\underline{T}^t = \underline{e}_2 \otimes \underline{e}_1$ and the two dyads differ, and which has $\operatorname{div} \underline{T} = \underline{0}$ because its Cartesian components are constants. Now *define* the body force by

$$\underline{b} := \dot{\underline{v}} - \rho^{-1} \operatorname{div} \underline{T} = \dot{\underline{v}}.$$

This is an admissible body force: (6.27) imposes nothing on $\underline{b}$ beyond enough regularity for the integral to exist, and $\dot{\underline{v}}$ is as regular as the motion. With these choices the displayed balance holds identically while $\underline{T}$ is nowhere symmetric. ■

The construction is deliberately trivial, and its triviality is the point. Equation (6.109) is three scalar equations, and the stress it constrains has nine components; the three components of $\operatorname{skw} \underline{T}$ never enter it. Those three are precisely what the rotational balance supplies, through (6.117). Counting the same fact from the other side: drop the non-polar assumption of Section 6.3.3 and admit couple tractions, and (6.109) survives unchanged while (6.117) fails, as Remark 6.8 records. A statement that can fail while (6.34) continues to hold is not a theorem of (6.34).

On invoking conservation of mass and making use of (6.24), Euler's postulates reduce to

$$\int_{P_t} \rho \dot{\underline{v}} \, dv = \int_{P_t} \rho \underline{b} \, dv + \int_{\partial P_t} \underline{t} \, da \quad (6.36)$$

and

$$\int_{P_t} \rho \underline{x} \times \dot{\underline{v}} \, dv = \int_{P_t} \rho \underline{x} \times \underline{b} \, dv + \int_{\partial P_t} \underline{x} \times \underline{t} \, da, \quad (6.37)$$

respectively. For (6.36) take $\underline{\psi} = \underline{v}$ in (6.24)₂; for (6.37) take $\underline{\psi} = \underline{x} \times \underline{v}$, and then use

$$(\underline{x} \times \underline{v})^\cdot = \dot{\underline{x}} \times \underline{v} + \underline{x} \times \dot{\underline{v}} = \underline{v} \times \underline{v} + \underline{x} \times \dot{\underline{v}} = \underline{x} \times \dot{\underline{v}},$$

since $\dot{\underline{x}} = \underline{v}$ by Definition 2.3 and $\underline{v} \times \underline{v} = \underline{0}$ because the cross product of a vector with itself vanishes, by the antisymmetry recorded in Definition 1.21. The moment of momentum therefore changes only through the acceleration, exactly as in particle mechanics, and the same cancellation will be met again in components at (6.116).

Equivalently, the linear momentum balance may be expressed in the form

$$\int_P \rho_\kappa \dot{\underline{v}} \, dV = \int_P \rho_\kappa \underline{b} \, dV + \int_{\partial P} \underline{p} \, dA, \quad (6.38)$$

and the rotational momentum balance in the form

$$\int_P \rho_\kappa \underline{x} \times \dot{\underline{v}} \, dV = \int_P \rho_\kappa \underline{x} \times \underline{b} \, dV + \int_{\partial P} \underline{x} \times \underline{p} \, dA, \quad (6.39)$$

with $\underline{x} = \underline{\chi}_\kappa(\underline{X}, t)$. These are (6.36) and (6.37) with every integral pulled back to κ , using $\rho \, dv = \rho_\kappa \, dV$ in the volume integrals and (6.29) in place of (6.28) on the boundary.

6.4.1 Center of mass

The *center of mass* of S , occupying position $\bar{\underline{x}}$ relative to $\underline{o}$, is located at the mass-weighted average of position,

$$m(S) \bar{\underline{x}}(t) = \int_{P_t} \rho \underline{x} \, dv = \int_P \rho_\kappa \underline{x} \, dV, \quad (6.40)$$

where $m(S)$ is the fixed mass of S ; that $m(S)$ is fixed is (6.13). The position of a material point $p \in S$ relative to the center of mass is

$$\underline{\pi} = \underline{x} - \bar{\underline{x}}. \quad (6.41)$$

The letter clashes with the flux vector of (6.1); the two never appear in the same computation, and from here to the end of Section 6.4.2 $\underline{\pi}$ means relative position.

Accordingly the mass-weighted average of the relative position vanishes identically:

$$\begin{aligned}\int_{P_t} \rho \underline{\pi} \, dv &= \int_{P_t} \rho (\underline{x} - \underline{\bar{x}}) \, dv \\ &= \left[m(S) - \int_{P_t} \rho \, dv \right] \underline{\bar{x}} = \underline{0},\end{aligned}\tag{6.42}$$

the first term having been evaluated by (6.40) and the second by taking the constant vector $\underline{\bar{x}}$ outside the integral, after which $\int_{P_t} \rho \, dv = m(S)$ makes the bracket vanish. The statement is a tautology: the average of the deviation from the average is zero. Its usefulness is that it holds at every instant, so it may be differentiated.

Differentiating, and invoking conservation of mass in the form (6.24)₂ with $\underline{\psi} = \underline{\pi}$,

$$\begin{aligned}\underline{0} &= \frac{d}{dt} \int_{P_t} \rho \underline{\pi} \, dv = \int_{P_t} \rho \dot{\underline{\pi}} \, dv \\ &= \int_{P_t} \rho \underline{v} \, dv - m(S) \dot{\underline{\bar{x}}},\end{aligned}\tag{6.43}$$

where the last line used $\dot{\underline{\pi}} = \dot{\underline{x}} - \dot{\underline{\bar{x}}} = \underline{v} - \dot{\underline{\bar{x}}}$, legitimate because $\underline{\bar{x}}$ is a function of t alone and is therefore the same vector at every material point, so that $\int_{P_t} \rho \dot{\underline{\bar{x}}} \, dv = \dot{\underline{\bar{x}}} \int_{P_t} \rho \, dv = m(S) \dot{\underline{\bar{x}}}$.

Comparing with (6.25), this says $\underline{l}(S, t) = m(S) \dot{\underline{\bar{x}}}$: the linear momentum of S is identical to that of a single particle of the same mass located at the center of mass. Euler's postulate (6.34) is thus equivalent, granted conservation of mass, to

$$m(S) \ddot{\underline{\bar{x}}} = \underline{f}(S, t),\tag{6.44}$$

where $\ddot{\underline{\bar{x}}}$ is the acceleration of the center of mass.

Using (6.41), we also have

$$\begin{aligned}\underline{h}_o(S, t) &= \int_{P_t} \rho \underline{x} \times \dot{\underline{x}} \, dv = \int_{P_t} \rho (\underline{\bar{x}} + \underline{\pi}) \times [\dot{\underline{\bar{x}}} + \dot{\underline{\pi}}] \, dv \\ &= m(S) \underline{\bar{x}} \times \dot{\underline{\bar{x}}} + \underline{\bar{x}} \times \int_{P_t} \rho \dot{\underline{\pi}} \, dv + \left[\int_{P_t} \rho \underline{\pi} \, dv \right] \times \dot{\underline{\bar{x}}} + \underline{h}(S, t),\end{aligned}\tag{6.45}$$

where

$$\underline{\underline{h}}(S, t) = \int_{P_t} \rho \underline{\underline{\pi}} \times \dot{\underline{\underline{x}}} \, dv \quad (6.46)$$

is the rotational momentum relative to the center of mass. The expansion in (6.45) used bilinearity of the cross product to produce four terms, and then moved the factors that do not depend on the material point outside their integrals: $\underline{\underline{x}}$ and $\dot{\underline{\underline{x}}}$ are functions of t only, so $\int \rho \underline{\underline{x}} \times \dot{\underline{\underline{x}}} \, dv = m(S) \underline{\underline{x}} \times \dot{\underline{\underline{x}}}$, and in the second and third terms one of the two slots is likewise constant over P_t .

Noting that the two integrals appearing in (6.45) vanish, the second by (6.42) and the first by (6.43)₂, and using $\underline{\underline{l}} = m(S) \dot{\underline{\underline{x}}}$, we conclude that

$$\underline{\underline{h}}_o(S, t) = \underline{\underline{x}}(t) \times \underline{\underline{l}}(S, t) + \underline{\underline{h}}(S, t). \quad (6.47)$$

We also have

$$\begin{aligned} \underline{\underline{m}}_o(S, t) &= \int_{P_t} \rho (\underline{\underline{x}} + \underline{\underline{\pi}}) \times \underline{\underline{b}} \, dv + \int_{\partial P_t} (\underline{\underline{x}} + \underline{\underline{\pi}}) \times \underline{\underline{t}} \, da \\ &= \underline{\underline{x}}(t) \times \underline{\underline{f}}(S, t) + \underline{\underline{m}}(S, t), \end{aligned} \quad (6.48)$$

where

$$\underline{\underline{m}}(S, t) = \int_{P_t} \rho \underline{\underline{\pi}} \times \underline{\underline{b}} \, dv + \int_{\partial P_t} \underline{\underline{\pi}} \times \underline{\underline{t}} \, da \quad (6.49)$$

is the net torque about the center of mass. Here the terms carrying $\underline{\underline{x}}$ were collected as $\underline{\underline{x}} \times [\int \rho \underline{\underline{b}} \, dv + \int \underline{\underline{t}} \, da] = \underline{\underline{x}} \times \underline{\underline{f}}$ by (6.27), (6.28) and (6.30).

REMARK 6.12 (a misprint in the printed (6.48) and (6.49)). The printed text carries a factor ρ in the surface integrals of both displays, writing $\int_{\partial P_t} \rho (\underline{\underline{x}} + \underline{\underline{\pi}}) \times \underline{\underline{t}} \, da$ and $\int_{\partial P_t} \rho \underline{\underline{\pi}} \times \underline{\underline{t}} \, da$. That factor should not be there. The contact torque is defined in (6.32) as $\int_{\partial P_t} \underline{\underline{x}} \times \underline{\underline{t}} \, da$ with no density: $\underline{\underline{t}}$ is already a force per unit area, whereas $\underline{\underline{b}}$ is a force per unit *mass* and needs the ρ to convert it. With the stray ρ the last equality of (6.48) would fail, since the bracket would no longer be $\underline{\underline{f}}$, and (6.51) would not follow. The displays above are corrected.

We thus recast Euler's postulate (6.35). Differentiate (6.47) and set the result equal to (6.48):

$$\bar{\underline{x}}(t) \times \underline{\underline{f}}(S, t) + \underline{\underline{m}}(S, t) = \frac{d}{dt} \underline{\underline{h}}(S, t) + \dot{\underline{\underline{x}}}(t) \times \underline{\underline{l}}(S, t) + \bar{\underline{x}}(t) \times \frac{d}{dt} \underline{\underline{l}}(S, t), \quad (6.50)$$

the product rule for the cross product being legitimate because the cross product is bilinear. Now dispose of two of the three terms on the right. By (6.34), $\frac{d}{dt} \underline{\underline{l}} = \underline{\underline{f}}$, so the last term is $\bar{\underline{x}} \times \underline{\underline{f}}$ and cancels the first term on the left. By (6.43) in the form $\underline{\underline{l}} = m(S) \dot{\underline{\underline{x}}}$,

$$\dot{\underline{\underline{x}}} \times \underline{\underline{l}} = \dot{\underline{\underline{x}}} \times m(S) \dot{\underline{\underline{x}}} = m(S) (\dot{\underline{\underline{x}}} \times \dot{\underline{\underline{x}}}) = \underline{\underline{0}},$$

a vector crossed with itself. What survives is

$$\underline{\underline{m}}(S, t) = \frac{d}{dt} \underline{\underline{h}}(S, t). \quad (6.51)$$

Thus Euler's postulate also holds relative to the center of mass, its motion in the inertial frame notwithstanding. This is a genuinely useful fact and not a triviality: the center of mass is in general accelerating, so the frame that follows it is not inertial, and one would expect the balance law to acquire correction terms. It does not, and (6.50) shows why: the two terms that could have produced corrections are the two that vanished, one because $\underline{\underline{l}}$ is parallel to $\dot{\underline{\underline{x}}}$ and the other because the force balance (6.34) already holds.

6.4.2 Rigid bodies

The foregoing development naturally subsumes the motions of rigid bodies. These are necessarily of the form (3.121), that is,

$$\underline{\underline{x}} = \underline{\underline{Q}}(t) \underline{\underline{X}} + \underline{\underline{c}}(t), \quad (6.52)$$

where $\underline{\underline{Q}}(t)$ is a spatially uniform two-point rotation tensor and $\underline{\underline{c}}(t)$ is a vector-valued function. Accordingly

$$\underline{\underline{\pi}} = \underline{\underline{Q}}(t) \underline{\underline{\Pi}}, \quad \text{where} \quad \underline{\underline{\Pi}} = \underline{\underline{X}} - \bar{\underline{\underline{X}}} \quad (6.53)$$

is the referential position of $p \in S$ relative to the center of mass of P , located at position $\bar{X}$. That $\bar{X}$ really is the center of mass of $P \subset \kappa$, and hence that (6.53) is consistent with (6.41), is Problem 2. Granting it, (6.53) follows from (6.52) by subtraction: $\bar{x} = Q\bar{X} + \underline{c}$, so $\pi = x - \bar{x} = Q(X - \bar{X}) = Q\Pi$, the constant $\underline{c}$ cancelling.

Thus

$$\dot{\pi} = \dot{Q}\Pi = \underline{\Omega}(t)\pi = \underline{\omega}(t) \times \pi, \quad (6.54)$$

where $\underline{\omega}$ is the axial vector, in the sense of Proposition 1.44, of the skew tensor

$$\underline{\Omega} = \dot{Q}Q^t, \quad (6.55)$$

this being simply a special case of (4.33). Two steps in (6.54) need saying. First, Π does not depend on t : both X and $\bar{X}$ are referential positions, and $\bar{X}$ is fixed because ρ_κ is a function of X alone by (6.18). So differentiating (6.53) differentiates only Q . Second, $\dot{Q}\Pi = \dot{Q}(Q^tQ)\Pi = (\dot{Q}Q^t)(Q\Pi) = \underline{\Omega}\pi$, using $Q^tQ = \hat{I}$ and then (6.53). That $\underline{\Omega}$ is skew follows by differentiating $QQ^t = I$:

$$\dot{Q}Q^t + Q\dot{Q}^t = O, \quad \text{that is} \quad \underline{\Omega} + \underline{\Omega}^t = O.$$

Consider a material curve with unit tangent M at X . From (2.67) and (3.120) we have that the unit tangent $\underline{m}$ to the image of this curve after deformation, and the associated stretch μ , satisfy

$$\mu\underline{m} = \underline{F}M = QM; \quad \text{hence} \quad \mu = 1 \quad \text{and} \quad \underline{m} = QM, \quad (6.56)$$

because $\underline{F} = Q$ for the motion (6.52) and because an orthogonal tensor preserves length, $|\underline{Q}M|^2 = \underline{Q}M \cdot \underline{Q}M = M \cdot Q^tQM = M \cdot \hat{I}M = 1$, by Proposition 1.48. Differentiating $\underline{m} = QM$, and using $\dot{M} = 0$ because M is a material vector in the sense of Definition 3.25 and Remark 3.26,

$$\dot{\underline{m}} = \underline{\Omega}\underline{m} = \underline{\omega} \times \underline{m}, \quad (6.57)$$

by the same insertion of $Q^tQ = \hat{I}$ that produced (6.54).

Let $\{\hat{E}_i\}$ be a fixed orthonormal basis for $\hat{\mathbb{E}}^3$ and let

$$e_i^*(t) = Q(t)\hat{E}_i \in \mathbb{E}^3. \quad (6.58)$$

Then $\{\underline{e}_i^*\}$ is fixed in the body and rotates with it: it is orthonormal at every instant, again by Proposition 1.48, and it is the image under the motion of a triad of material fibres. On identifying $\underline{m}$ in (6.57) with an element of this basis we have simply

$$\dot{\underline{e}}_i^* = \underline{\omega} \times \underline{e}_i^*. \quad (6.59)$$

The same kinematics, read from Chapter 4

Everything from (6.52) to (6.59) is Chapter 4 specialised to a rigid motion, and reading it that way does three things. It identifies the angular velocity of rigid-body dynamics with an object already constructed there. It supplies proofs of $\mu = 1$ in (6.56) and of $\rho = \rho_\kappa$ that never mention $\det Q$. And it locates (6.57) as the one case in which a warning issued in Chapter 4 does not apply.

The angular velocity tensor is the spin tensor. For the motion (6.52) the deformation gradient is $\underline{F} = \underline{Q}(t)$, spatially uniform because $\underline{Q}$ is. So by (4.3) the spatial velocity gradient is

$$\underline{L} = \dot{\underline{F}}\underline{F}^{-1} = \dot{\underline{Q}}\underline{Q}^{-1} = \dot{\underline{Q}}\underline{Q}^t,$$

the last step by $\underline{Q}^{-1} = \underline{Q}^t$. The right-hand side is (6.55). Hence

$$\underline{\Omega} = \underline{L},$$

which is (4.29), obtained in Section 4.4 as the velocity gradient of a rigid motion. Chapter 4 recorded there that it is skew, by Problem 1(c) of that chapter, which is the computation repeated below (6.55). Splitting $\underline{L}$ by (4.7) and invoking Remark 4.6, which states that $\underline{D} = \underline{Q}$ characterises rigidity,

$$\underline{D} = \underline{Q}, \quad \underline{W} = \underline{L} = \underline{\Omega}.$$

The angular velocity tensor of (6.55) and the spin tensor of (4.7) are one tensor under two names. Their axial vectors agree as well, so $\underline{\omega}$ is the vorticity $\underline{w}$ of (4.22), and (4.23) then gives a formula for the angular velocity that Section 6.4.2 does not otherwise produce:

$$\underline{\omega} = \frac{1}{2} \operatorname{curl} \underline{v}.$$

Section 4.3 carried out exactly this computation for a spinning disc and obtained $\underline{\omega} = \omega \underline{n}$ at (4.26).

(6.54) is (4.32). Chapter 4 integrated the velocity field of a rigid motion at (4.30) and reached (4.32),

$$\underline{v}(\underline{y}, t) - \underline{v}(\underline{x}, t) = \underline{w}(t) \times (\underline{y} - \underline{x}).$$

Take $\underline{y}$ to be the position of the material point $p \in S$ and $\underline{x}$ the position $\bar{\underline{x}}$ of the center of mass. The left side is $\dot{\underline{y}} - \dot{\bar{\underline{x}}} = \dot{\underline{\pi}}$ by (6.41), and the right side is $\underline{\omega} \times \underline{\pi}$. That is (6.54). The book cites (4.33) here; (4.32) is the same statement with the arbitrary spatially uniform vector $\underline{d}$ already removed by subtracting the velocity at one point from the velocity at another, and subtraction is what (6.54) needs, since $\underline{\pi}$ is a difference of positions.

$\mu = 1$, *by a rate argument*. The derivation of (6.56) above used $|\underline{Q}\underline{M}| = |\underline{M}|$. Chapter 4 reaches the same conclusion without naming $\underline{Q}$. Equation (4.15) reads

$$(\ln \mu)^{\cdot} = \underline{m} \cdot \underline{D}\underline{m},$$

and $\underline{D} = \underline{Q}$, so $(\ln \mu)^{\cdot} = 0$ and μ does not change in time. At the instant at which the body occupies κ we have $\underline{F} = \underline{1}$ and therefore $\mu = 1$, so $\mu \equiv 1$ for all time. The rate argument says more than the algebraic one: it says that no material curve in a rigid motion is stretching at any instant, which is the differential form of the distance-preserving property (3.113).

$\rho = \rho_{\kappa}$, *by a rate argument*. Equation (6.66) needed $J = 1$ and took it from $\det \underline{Q} = 1$. Chapter 4 gets it from the skewness of $\underline{\Omega}$ alone. Problem 2 of Chapter 4 established $\dot{J} = J \operatorname{tr} \underline{L}$, so here

$$\dot{J} = J \operatorname{tr} \underline{\Omega} = 0,$$

because the trace of a skew tensor vanishes: $\operatorname{tr} \underline{\Omega} = \operatorname{tr} \underline{\Omega}^t = \operatorname{tr}(-\underline{\Omega}) = -\operatorname{tr} \underline{\Omega}$, so $\operatorname{tr} \underline{\Omega} = 0$. Hence J is constant in time, and it equals 1 at the instant at which κ is occupied. By (2.59) this is $\rho = \rho_{\kappa}$, and by (6.17) it is also $\dot{\rho} = 0$: a rigid motion is isochoric, and carries its density along unchanged.

(6.57) is (4.16) with $\underline{D} = \underline{Q}$. The general law for the rate of a material direction is (4.16),

$$\dot{\underline{m}} = \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m} + \underline{W}\underline{m}.$$

Setting $\underline{D} = \underline{Q}$ removes the first two terms and leaves $\dot{\underline{m}} = \underline{W}\underline{m} = \underline{\omega} \times \underline{m}$, which is (6.57). The contrast with Chapter 4 is sharp. Equation (4.18) asserted $\dot{\underline{m}} = \underline{W}\underline{m}$ only at an instant t_* , and only for the special fibres momentarily aligned with a principal direction of $\underline{D}$; Remark 4.9 warned that for a general fibre in a general motion the identity is false, so that the spin tensor is not the spin of the material. In a rigid motion it is, for every fibre and at every instant, and (6.58) to (6.59) hang an entire orthonormal frame on that fact. The frame $\{\underline{e}_i^*\}$ exists because rigidity is exactly the hypothesis under which all three of its members obey the same rotation law.

Rigidity is not quite the right condition, and the sharp statement is the following.

PROPOSITION 6.13 (when the spin tensor is the spin of the material). *At a point and an instant, $\dot{\underline{m}} = \underline{W}\underline{m}$ for every material direction $\underline{m}$ if and only if $\underline{D}$ is spherical, that is $\underline{D} = \alpha \underline{I}$ for some scalar α . The motion is rigid there if and only if in addition $\alpha = 0$.*

Proof. By (4.16), $\dot{\underline{m}} - \underline{W}\underline{m} = \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m}$ for every unit $\underline{m}$, so the stated condition is

$$\underline{D}\underline{m} = (\underline{m} \cdot \underline{D}\underline{m})\underline{m} \quad \text{for every unit } \underline{m},$$

that is, every direction is a principal direction of $\underline{D}$.

Suppose this holds and let $\underline{D} = \sum_i d_i \underline{v}_i \otimes \underline{v}_i$ be the spectral representation supplied by Theorem 3.7, the $\underline{v}_i$ orthonormal. If two principal values differed, say $d_1 \neq d_2$, take $\underline{m} = (\underline{v}_1 + \underline{v}_2)/\sqrt{2}$, a unit vector. Then $\underline{D}\underline{m} = (d_1 \underline{v}_1 + d_2 \underline{v}_2)/\sqrt{2}$ and $\underline{m} \cdot \underline{D}\underline{m} = \frac{1}{2}(d_1 + d_2)$, so

$$\begin{aligned} \underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m} &= \frac{1}{\sqrt{2}} \left[\left(d_1 - \frac{d_1 + d_2}{2} \right) \underline{v}_1 + \left(d_2 - \frac{d_1 + d_2}{2} \right) \underline{v}_2 \right] \\ &= \frac{d_1 - d_2}{2\sqrt{2}} (\underline{v}_1 - \underline{v}_2), \end{aligned}$$

which is not $\mathbf{0}$, because $d_1 \neq d_2$ and $\underline{v}_1 \neq \underline{v}_2$ are independent. This contradicts the condition, so all three principal values coincide and $\underline{D} = \alpha \underline{I}$ with α their common value.

Conversely, if $\underline{D} = \alpha \underline{I}$ then $\underline{D}\underline{m} = \alpha \underline{m}$ and $\underline{m} \cdot \underline{D}\underline{m} = \alpha$ for every unit $\underline{m}$, so $\underline{D}\underline{m} - (\underline{m} \cdot \underline{D}\underline{m})\underline{m} = \mathbf{0}$ identically.

For the last statement, $\underline{D} = \underline{Q}$ characterises rigidity by Remark 4.6, and $\alpha \underline{I} = \underline{Q}$ if and only if $\alpha = 0$. ■

The intermediate case $\alpha \neq 0$ is not empty. By Problem 2 of Chapter 4, $\underline{D} = \alpha \underline{I}$ gives $\dot{J}/J = \text{tr } \underline{D} = 3\alpha$, so the motion is instantaneously a rigid rotation together with a uniform dilatation at rate 3α . Every fibre turns with $\underline{W}$ and every fibre stretches at the same logarithmic rate α , by (4.15). That is the widest class for which the sentence “ $\underline{W}$ is the spin of the material” is true without qualification, and rigid motions are the members of it that also preserve length.

One consequence carried forward. Since $\underline{L} = \underline{W}$ in a rigid motion and $\underline{T}$ is symmetric by (6.117), the stress power density of (6.153) is

$$\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{W} = 0$$

by Lemma 2.36, a symmetric tensor contracted with a skew one vanishing. Section 6.6 uses precisely this to decouple the thermal and the mechanical problem in a rigid body.

Let $\underline{u} \in \mathbb{E}^3$ be an arbitrary vector, possibly time dependent. We decompose it in the rotating basis, $\underline{u} = u_i \underline{e}_i^*$ with $u_i = \underline{u} \cdot \underline{e}_i^*$. Thus

$$\begin{aligned} \dot{\underline{u}} &= \dot{u}_i \underline{e}_i^* + u_i \dot{\underline{e}}_i^* \\ &= \overset{\circ}{\underline{u}} + \underline{\omega} \times \underline{u}, \end{aligned} \tag{6.60}$$

where

$$\overset{\circ}{\underline{u}} = \dot{u}_i \underline{e}_i^*, \tag{6.61}$$

the *co-rotational derivative*, is the derivative relative to the rotating basis. The second line of (6.60) used (6.59) and then the linearity of the cross product: $u_i(\underline{\omega} \times \underline{e}_i^*) = \underline{\omega} \times (u_i \underline{e}_i^*) = \underline{\omega} \times \underline{u}$. Thus $\overset{\circ}{\underline{u}}$ vanishes if and only if $\underline{u}$ is fixed in the rotating body, since it vanishes exactly when all

the components of $\underline{u}$ on a basis that rotates with the body are constant. From (6.54) it follows that $\underline{\pi}$ is such a vector.

REMARK 6.14 (two rates of change, and why both are needed). Equation (6.60) splits $\dot{\underline{u}}$ into the part due to the vector changing and the part due to the frame turning. Neither piece is frame-free by itself and the sum is: $\dot{\underline{u}}$ is the honest material derivative, computed with respect to the fixed basis of the inertial frame.

Whether one wants $\dot{\underline{u}}$ or $\dot{\underline{u}}$ depends on the question. A vector painted on the body has $\dot{\underline{u}} = \underline{0}$ and $\dot{\underline{u}} = \underline{\omega} \times \underline{u} \neq \underline{0}$: it is not constant, it is merely not changing *relative to the body*. The inertia tensor of (6.65) is an object of exactly this kind, and (6.68) is the tensor version of the same split. The same construction reappears in Problem 12(d), where the corotational rate of the left stretch tensor turns out to be the deformation measure conjugate to the Bell stress.

From (6.46) and (6.54) we have that

$$\underline{h}(S, t) = \int_{P_t} \rho \underline{\pi} \times (\underline{\omega} \times \underline{\pi}) \, dv \quad (6.62)$$

for rigid bodies. We would like to exploit the spatial uniformity of the spin $\underline{\omega}$ to factor it out of the integral, and we cannot do so directly because $\underline{\omega}$ sits inside a cross product with the integration variable. The obstacle is removed by the result of Problem 11 of Chapter 1, $\underline{a} \times (\underline{b} \times \underline{c}) = (\underline{a} \cdot \underline{c})\underline{b} - (\underline{a} \cdot \underline{b})\underline{c}$, applied with $\underline{a} = \underline{c} = \underline{\pi}$ and $\underline{b} = \underline{\omega}$:

$$\begin{aligned} \underline{\pi} \times (\underline{\omega} \times \underline{\pi}) &= (\underline{\pi} \cdot \underline{\pi})\underline{\omega} - (\underline{\pi} \cdot \underline{\omega})\underline{\pi} \\ &= [(\underline{\pi} \cdot \underline{\pi})\underline{I} - \underline{\pi} \otimes \underline{\pi}]\underline{\omega}, \end{aligned} \quad (6.63)$$

the second line by $\underline{I}\underline{\omega} = \underline{\omega}$ and $(\underline{\pi} \cdot \underline{\omega})\underline{\pi} = (\underline{\pi} \otimes \underline{\pi})\underline{\omega}$, which is the definition of the tensor product, Definition 1.32. The point of the manoeuvre is that $\underline{\omega}$ now stands to the right of a tensor that does not involve it, so it can be taken outside the integral. This yields

$$\underline{h}(S, t) = \underline{J}(S, t) \underline{\omega}(t), \quad (6.64)$$

where

$$\underline{\underline{J}}(S, t) = \int_{P_t} \rho [(\underline{\underline{\pi}} \cdot \underline{\underline{\pi}}) \underline{\underline{I}} - \underline{\underline{\pi}} \otimes \underline{\underline{\pi}}] \, dv \quad (6.65)$$

is the symmetric *inertia tensor* of the rigid body. It is symmetric because $\underline{\underline{I}}$ and $\underline{\underline{\pi}} \otimes \underline{\underline{\pi}}$ both are, the latter by $(\underline{\underline{a}} \otimes \underline{\underline{a}})^t = \underline{\underline{a}} \otimes \underline{\underline{a}}$. Problem 4 shows that it is in fact positive definite.

REMARK 6.15 (a collision of letters). The book writes J for the inertia tensor and J for $\det \underline{\underline{F}}$ in the same paragraph. Here the inertia tensor is $\underline{\underline{J}}$, carrying the under-tilde that marks every tensor in these notes, and $J = \det \underline{\underline{F}}$ keeps its meaning from (2.42). The next display uses both.

Equation (6.65) may be simplified on noting that $J = \det \underline{\underline{Q}} = 1$ for a rigid motion, and hence, by (2.59), that $\rho = \rho_\kappa$ and $\rho \, dv = \rho_\kappa \, dV$. Further, (6.53) implies

$$\underline{\underline{\pi}} \cdot \underline{\underline{\pi}} = \underline{\underline{Q}} \underline{\underline{\Pi}} \cdot \underline{\underline{Q}} \underline{\underline{\Pi}} = \underline{\underline{\Pi}} \cdot \underline{\underline{Q}}^t \underline{\underline{Q}} \underline{\underline{\Pi}} = \underline{\underline{\Pi}} \cdot \underline{\underline{\Pi}},$$

so that

$$\begin{aligned} \underline{\underline{J}}(S, t) &= \int_P \rho_\kappa [(\underline{\underline{\Pi}} \cdot \underline{\underline{\Pi}}) \underline{\underline{Q}} \underline{\underline{Q}}^t - \underline{\underline{Q}} \underline{\underline{\Pi}} \otimes \underline{\underline{Q}} \underline{\underline{\Pi}}] \, dV \\ &= \underline{\underline{Q}}(t) \underline{\underline{J}}_\kappa(S) \underline{\underline{Q}}(t)^t, \end{aligned} \quad (6.66)$$

where $\underline{\underline{J}}_\kappa(S)$ is the fixed referential tensor defined by

$$\underline{\underline{J}}_\kappa(S) = \int_P \rho_\kappa [(\underline{\underline{\Pi}} \cdot \underline{\underline{\Pi}}) \underline{\underline{I}} - \underline{\underline{\Pi}} \otimes \underline{\underline{\Pi}}] \, dV. \quad (6.67)$$

The first line of (6.66) replaced the spatial identity by $\underline{\underline{I}} = \underline{\underline{Q}} \underline{\underline{Q}}^t$, which is orthogonality read in the direction $\mathbb{E}^3 \rightarrow \mathbb{E}^3$, and replaced $\underline{\underline{\pi}}$ by $\underline{\underline{Q}} \underline{\underline{\Pi}}$. The second line used the dyad rule $\underline{\underline{Q}}(\underline{\underline{\Pi}} \otimes \underline{\underline{\Pi}}) \underline{\underline{Q}}^t = \underline{\underline{Q}} \underline{\underline{\Pi}} \otimes \underline{\underline{Q}} \underline{\underline{\Pi}}$ of Lemma 2.15 backwards, and then took the constant tensors $\underline{\underline{Q}}$ and $\underline{\underline{Q}}^t$ outside the integral, which is legitimate because they do not depend on $\underline{\underline{X}}$.

What (6.66) says is that the inertia tensor is fixed in the body: its referential version $\underline{\underline{J}}_\kappa$ does not depend on time at all, and all the time dependence of $\underline{\underline{J}}$ is the rotation carrying it along. Stated as a rate, this is the vanishing of the co-rotational derivative $\mathring{\underline{\underline{J}}}$, defined by

$$\mathring{\underline{\underline{J}}} = \dot{\underline{\underline{J}}} - \underline{\underline{\Omega}} \underline{\underline{J}} + \underline{\underline{J}} \underline{\underline{\Omega}}. \quad (6.68)$$

This follows easily from (6.55) and (6.66). Differentiating (6.66)₂ and using $\dot{\underline{Q}} = \underline{\Omega} \underline{Q}$ from (6.55), together with $\dot{\underline{Q}}^t = (\dot{\underline{Q}})^t = (\underline{\Omega} \underline{Q})^t = \underline{Q}^t \underline{\Omega}^t = -\underline{Q}^t \underline{\Omega}$,

$$\dot{\underline{J}} = \dot{\underline{Q}} \underline{J}_\kappa \underline{Q}^t + \underline{Q} \underline{J}_\kappa \dot{\underline{Q}}^t = \underline{\Omega} \underline{Q} \underline{J}_\kappa \underline{Q}^t - \underline{Q} \underline{J}_\kappa \underline{Q}^t \underline{\Omega} = \underline{\Omega} \underline{J} - \underline{J} \underline{\Omega},$$

whence $\dot{\underline{J}} = \underline{Q}$. Problem 3 shows that (6.68) is the tensor counterpart of (6.61): $\dot{\underline{J}}$ is the tensor whose components on the rotating basis are the derivatives of the components of $\underline{J}$.

The form of (6.68) is not a convention. Once the vector rate (6.61) is fixed, the tensor rate is determined by the demand that it annihilate tensors built from vectors fixed in the body, and both signs follow from that demand.

PROPOSITION 6.16 (the tensor rate is forced by the vector rate). *Write $\dot{\underline{u}} = \dot{\underline{u}} - \underline{\Omega} \underline{u}$, which is (6.60) rearranged. If $\underline{a}$ and $\underline{b}$ satisfy $\dot{\underline{a}} = \dot{\underline{b}} = \underline{0}$ and $\underline{A} = \underline{a} \otimes \underline{b}$, then*

$$\dot{\underline{A}} = \underline{\Omega} \underline{A} - \underline{A} \underline{\Omega},$$

so that $\dot{\underline{A}} = \underline{Q}$ with $\dot{\underline{A}}$ defined as in (6.68).

Proof. $\dot{\underline{a}} = \underline{0}$ means $\dot{\underline{a}} = \underline{\Omega} \underline{a}$, and likewise $\dot{\underline{b}} = \underline{\Omega} \underline{b}$. The product rule applied to the tensor product gives

$$\dot{\underline{A}} = \dot{\underline{a}} \otimes \underline{b} + \underline{a} \otimes \dot{\underline{b}} = (\underline{\Omega} \underline{a}) \otimes \underline{b} + \underline{a} \otimes (\underline{\Omega} \underline{b}).$$

Rewrite each term by testing it on an arbitrary $\underline{c} \in \mathbb{E}^3$. For the first,

$$[(\underline{\Omega} \underline{a}) \otimes \underline{b}] \underline{c} = (\underline{b} \cdot \underline{c}) \underline{\Omega} \underline{a} = \underline{\Omega} [(\underline{b} \cdot \underline{c}) \underline{a}] = \underline{\Omega} (\underline{a} \otimes \underline{b}) \underline{c} = \underline{\Omega} \underline{A} \underline{c},$$

where the second equality is linearity of $\underline{\Omega}$ and the third is Definition 1.32. Hence $(\underline{\Omega} \underline{a}) \otimes \underline{b} = \underline{\Omega} \underline{A}$. For the second,

$$[\underline{a} \otimes (\underline{\Omega} \underline{b})] \underline{c} = (\underline{\Omega} \underline{b} \cdot \underline{c}) \underline{a} = (\underline{b} \cdot \underline{\Omega}^t \underline{c}) \underline{a} = (\underline{a} \otimes \underline{b}) \underline{\Omega}^t \underline{c} = -\underline{A} \underline{\Omega} \underline{c},$$

the second equality being the definition of the transpose and the last using $\underline{\Omega}^t = -\underline{\Omega}$. Hence $\underline{a} \otimes (\underline{\Omega} \underline{b}) = -\underline{A} \underline{\Omega}$. Adding the two gives $\dot{\underline{A}} = \underline{\Omega} \underline{A} - \underline{A} \underline{\Omega}$, and substituting into (6.68) gives $\dot{\underline{A}} = \underline{Q}$. $\blacksquare$

The asymmetry between the two signs in (6.68) is therefore nothing but the transpose in the second computation meeting the skewness of $\underline{\Omega}$. The first leg of the dyad is acted on from the left and the second from the right, and $\underline{\Omega}^t = -\underline{\Omega}$ turns the second action into a subtraction.

This yields a second proof of $\dot{\underline{J}} = \underline{Q}$, one that never passes through (6.66). Read (6.65) ingredient by ingredient. The measure $\rho \, dv = \rho_\kappa \, dV$ does not depend on t , by (6.18). The scalar $\underline{\pi} \cdot \underline{\pi}$ does not either, since

$$(\underline{\pi} \cdot \underline{\pi})^\cdot = 2\underline{\pi} \cdot \dot{\underline{\pi}} = 2\underline{\pi} \cdot \underline{\omega} \times \underline{\pi} = 0$$

by (6.54), a cross product $\underline{\omega} \times \underline{\pi}$ being perpendicular to $\underline{\pi}$. The identity satisfies $\dot{\underline{I}} = \dot{\underline{I}} - \underline{\Omega}\underline{I} + \underline{I}\underline{\Omega} = \underline{Q}$, its material derivative being zero and the two correction terms cancelling. And $\underline{\pi}$ is fixed in the body, $\dot{\underline{\pi}} = \underline{0}$, so Proposition 6.16 applies to $\underline{\pi} \otimes \underline{\pi}$. Every ingredient of (6.65) thus has vanishing co-rotational rate, and so, term by term, does the integral.

Thus the rotational momentum balance (6.51) combines with (6.64) to give

$$\begin{aligned} \underline{m}(S, t) &= (\underline{J}\underline{\omega})^\cdot = \dot{\underline{J}}\underline{\omega} + \underline{J}\dot{\underline{\omega}} \\ &= (\underline{\Omega}\underline{J} - \underline{J}\underline{\Omega})\underline{\omega} + \underline{J}\dot{\underline{\omega}}. \end{aligned} \quad (6.69)$$

This simplifies further on noting that $\underline{\Omega}\underline{\omega} = \underline{\omega} \times \underline{\omega}$ vanishes, so that the term $\underline{J}\underline{\Omega}\underline{\omega}$ disappears, while $\underline{\Omega}\underline{J}\underline{\omega} = \underline{\omega} \times (\underline{J}\underline{\omega})$ by the definition of the axial vector. Thus

$$\underline{m}(S, t) = \underline{\omega} \times \underline{J}\underline{\omega} + \underline{J}\dot{\underline{\omega}}, \quad \text{with} \quad \dot{\underline{\omega}} = \underline{\dot{\omega}}, \quad (6.70)$$

the last identification being (6.60) applied to $\underline{u} = \underline{\omega}$, whose correction term $\underline{\omega} \times \underline{\omega}$ vanishes.

Equations (6.70) and (6.44) are the classical equations of rigid-body dynamics, determining the motion of the center of mass and the rotation in terms of the net force and torque about the mass center. Conversely, they yield the net force and torque required to effect a prescribed motion of the center of mass and a prescribed rotation.

REMARK 6.17 (where the gyroscopic term comes from). The term $\underline{\omega} \times \underline{J}\underline{\omega}$ in (6.70) is the one that makes rigid-body dynamics unlike particle

dynamics, and (6.69) shows exactly where it originates: in $\underline{\underline{J}}$, the fact that the inertia tensor changes in the inertial frame even though the body is rigid. Had $\underline{\underline{J}}$ been a constant tensor the equation would read $\underline{\underline{m}} = \underline{\underline{J}}\underline{\underline{\dot{\omega}}}$, the rotational analogue of $\underline{\underline{f}} = m\underline{\underline{\ddot{x}}}$, and a torque-free body would spin at constant $\underline{\underline{\omega}}$.

It does not, in general. Set $\underline{\underline{m}} = \underline{\underline{0}}$ in (6.70):

$$\underline{\underline{J}}\underline{\underline{\dot{\omega}}} = -\underline{\underline{\omega}} \times \underline{\underline{J}}\underline{\underline{\omega}},$$

which is non-zero whenever $\underline{\underline{\omega}}$ is not a principal direction of $\underline{\underline{J}}$. That is the tennis racket theorem and the precession of a free top, and it is a consequence of (6.66) alone. The co-rotational derivative was introduced for precisely this reason: in the body frame the inertia tensor is constant, so (6.70) written with $\underline{\underline{\dot{\omega}}}$ becomes Euler's equations with constant coefficients, at the price of the extra term.

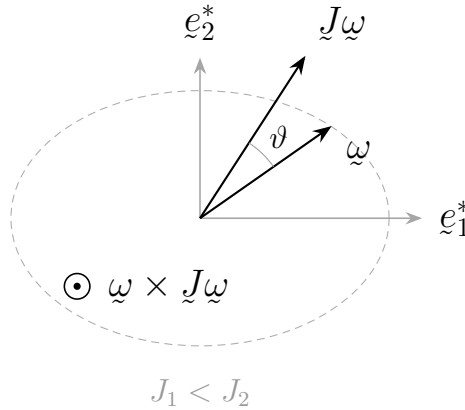


Figure A: Why (6.70) carries a term that $\underline{\underline{f}} = m\underline{\underline{\ddot{x}}}$ does not. The rotational momentum $\underline{\underline{J}}\underline{\underline{\omega}}$ is not parallel to $\underline{\underline{\omega}}$ unless $\underline{\underline{\omega}}$ lies along a principal axis: the inertia tensor weights the components of $\underline{\underline{\omega}}$ unequally, and tilts the result toward the axis of larger moment of inertia. The angle ϑ between them is what the gyroscopic term $\underline{\underline{\omega}} \times \underline{\underline{J}}\underline{\underline{\omega}}$ measures, and that term points out of the plane of the two vectors. The dashed curve is the section of $\underline{\underline{u}} \cdot \underline{\underline{J}}\underline{\underline{u}} = \text{const}$, elongated along the axis of *smaller* inertia because the quadratic form is smaller there.

Euler's equations in the principal basis

Equation (6.70) is a vector equation in the inertial frame whose coefficient tensor $\underline{J}$ depends on time. Resolving it on a basis fixed in the body removes that dependence completely, and the classical form of rigid-body dynamics appears.

By (6.67) the referential inertia tensor $\underline{J}_\kappa$ is symmetric, and Problem 4 shows it is positive definite. The spectral theorem, Theorem 3.7, therefore supplies an orthonormal basis $\{\hat{\underline{E}}_i\}$ of $\hat{\mathbb{E}}^3$ and positive numbers J_i with

$$\underline{J}_\kappa = \sum_i J_i \hat{\underline{E}}_i \otimes \hat{\underline{E}}_i.$$

The J_i are the *principal moments of inertia* and the $\hat{\underline{E}}_i$ the *principal axes of inertia*. This basis is available for use in (6.58), and the choice pays at once. Carrying it through (6.66) with the dyad rule of Lemma 2.15,

$$\underline{J} = \underline{Q} \underline{J}_\kappa \underline{Q}^t = \sum_i J_i (\underline{Q} \hat{\underline{E}}_i) \otimes (\underline{Q} \hat{\underline{E}}_i) = \sum_i J_i \underline{e}_i^* \otimes \underline{e}_i^*,$$

by (6.58). So the principal axes of the spatial inertia tensor are the rotating basis vectors themselves, and its principal values are the same constants J_i at every instant. All the time dependence that Remark 6.17 blamed for the gyroscopic term sits in the dyads, none of it in the numbers.

Write $\underline{\omega} = \omega_i \underline{e}_i^*$. Then

$$\underline{J} \underline{\omega} = \sum_i J_i \omega_i \underline{e}_i^*,$$

and, because $\dot{\underline{\omega}} = \dot{\underline{\omega}}$ by (6.70)₂ and $\dot{\underline{\omega}} = \dot{\omega}_i \underline{e}_i^*$ by (6.61),

$$\underline{J} \dot{\underline{\omega}} = \sum_i J_i \dot{\omega}_i \underline{e}_i^*.$$

The gyroscopic term is a cross product of two vectors whose components on the orthonormal triad $\{\underline{e}_i^*\}$ are now both known, so it may be written

out:

$$\omega \times \mathcal{J}\omega = [(J_3 - J_2)\omega_2\omega_3]\mathcal{e}_1^* + [(J_1 - J_3)\omega_3\omega_1]\mathcal{e}_2^* + [(J_2 - J_1)\omega_1\omega_2]\mathcal{e}_3^*,$$

the first component being $\omega_2(J_3\omega_3) - \omega_3(J_2\omega_2)$ and the other two following by cyclic permutation of 123. Equating components in (6.70) gives *Euler's equations*,

$$\begin{aligned} J_1\dot{\omega}_1 + (J_3 - J_2)\omega_2\omega_3 &= m_1, \\ J_2\dot{\omega}_2 + (J_1 - J_3)\omega_3\omega_1 &= m_2, \\ J_3\dot{\omega}_3 + (J_2 - J_1)\omega_1\omega_2 &= m_3, \end{aligned}$$

where $m_i = \mathfrak{m}(S, t) \cdot \mathcal{e}_i^*$ are the components of the torque about the center of mass on the same rotating triad. Every coefficient is a constant. That is what the body basis was for, and it is the reason the co-rotational derivative was introduced in (6.61) rather than being avoided.

The torque-free body

Set $\mathfrak{m}(S, t) = 0$. Two scalars are then constant in time, and between them they explain the assertion of Remark 6.17 that a free body does not spin steadily.

The first is immediate from (6.51): $\dot{\mathfrak{h}} = 0$, so the rotational momentum $\mathfrak{h} = \mathcal{J}\omega$ is a fixed vector of $\mathbb{E}^3$, and in particular $|\mathfrak{h}|$ is constant.

The second is the kinetic energy of the rotation, $T = \frac{1}{2}\omega \cdot \mathcal{J}\omega$. Differentiate, using the symmetry of $\mathcal{J}$ to combine the first and third terms:

$$\dot{T} = \frac{1}{2}\dot{\omega} \cdot \mathcal{J}\omega + \frac{1}{2}\omega \cdot \dot{\mathcal{J}}\omega + \frac{1}{2}\omega \cdot \mathcal{J}\dot{\omega} = \omega \cdot \mathcal{J}\dot{\omega} + \frac{1}{2}\omega \cdot \dot{\mathcal{J}}\omega.$$

The last term vanishes. From the derivation of (6.68), $\dot{\mathcal{J}} = \Omega\mathcal{J} - \mathcal{J}\Omega$, so

$$\omega \cdot (\Omega\mathcal{J} - \mathcal{J}\Omega)\omega = \omega \cdot \Omega\mathcal{J}\omega - \omega \cdot \mathcal{J}\Omega\omega = -\Omega\omega \cdot \mathcal{J}\omega - 0 = 0,$$

the first term because Ω is skew, so that moving it across the inner product changes the sign, and the second because $\Omega\omega = \omega \times \omega = 0$. Hence $\dot{T} = \omega \cdot \mathcal{J}\dot{\omega}$. Now take the inner product of (6.70) with ω :

$$\omega \cdot \mathfrak{m}(S, t) = \omega \cdot \omega \times \mathcal{J}\omega + \omega \cdot \mathcal{J}\dot{\omega} = \omega \cdot \mathcal{J}\dot{\omega} = \dot{T},$$

since $\underline{\omega} \times \underline{J}\underline{\omega}$ is perpendicular to $\underline{\omega}$. So in general

$$\dot{T} = \underline{\omega} \cdot \underline{m}(S, t),$$

the rotational power of the torque about the mass center, and T is constant when that torque vanishes. This is the rotational counterpart of the mechanical energy balance (6.150), and it is exact rather than approximate because a rigid body stores no strain energy: its stress power vanished at the end of Section 6.4.2.

On the body basis the two constants read

$$|\underline{h}|^2 = J_1^2 \omega_1^2 + J_2^2 \omega_2^2 + J_3^2 \omega_3^2, \quad 2T = J_1 \omega_1^2 + J_2 \omega_2^2 + J_3 \omega_3^2,$$

two ellipsoids in the space of the components ω_i . The motion of $\underline{\omega}$ relative to the body is confined to their intersection, which is a closed curve. Whether that curve stays near an axis or wanders far from it is the question of stability, and Euler's equations answer it in three lines.

Order the moments $J_1 < J_2 < J_3$ and set $\underline{m} = \underline{0}$. Steady rotation about the first axis is $\underline{\omega} = \Omega \underline{e}_1^*$, and it does solve the equations, since then $\omega_2 = \omega_3 = 0$ kills every product. Perturb it: put $\omega_1 = \Omega$ and treat ω_2, ω_3 as small, so that products of two small quantities are discarded. The second and third equations become

$$J_2 \dot{\omega}_2 = (J_3 - J_1) \Omega \omega_3, \quad J_3 \dot{\omega}_3 = (J_1 - J_2) \Omega \omega_2,$$

and differentiating the first and substituting the second,

$$\ddot{\omega}_2 = \frac{(J_3 - J_1)(J_1 - J_2)}{J_2 J_3} \Omega^2 \omega_2.$$

With J_1 the smallest, $J_3 - J_1 > 0$ and $J_1 - J_2 < 0$, so the coefficient is negative and ω_2 oscillates: the perturbation stays small. The same computation about the third axis gives $(J_1 - J_3)(J_3 - J_2) < 0$ again, so that axis is stable too. About the intermediate axis, however, $\underline{\omega} = \Omega \underline{e}_2^*$ gives

$$\ddot{\omega}_1 = \frac{(J_2 - J_3)(J_1 - J_2)}{J_1 J_3} \Omega^2 \omega_1,$$

in which both factors in the numerator are negative and the coefficient is positive. The perturbation grows exponentially. That is the tennis racket theorem, promised in Remark 6.17, and every step of it came from (6.66): the inertia tensor is fixed in the body, not in space.

6.5 Forces of mutual interaction in the body and the local equations of motion

So far the traction $\underline{t}$ has been an undetermined field on the boundary of whatever part is under consideration. This section determines its structure completely, in three steps. First, the traction that one part of a body exerts on an adjoining part is equal and opposite to the traction exerted back; that is a theorem, not an axiom. Second, the traction at a point depends on the cutting surface only through its unit normal there; that is Noll's theorem. Third, that dependence is linear, so there is a tensor field $\underline{T}$ with $\underline{t} = \underline{T}\underline{n}$; that is Cauchy's theorem. Only then can the linear momentum balance be localized.

6.5.1 Action and reaction

Consider an arbitrary part P_t of the region κ_t occupied by the body at time t . Let σ be a surface dividing it into two sub-parts P_{t1} and P_{t2} (Figure 6.4). The letter σ is the book's, and in this subsection it denotes that surface and not the supply rate of (6.1). Let

$$\partial P'_t = \partial P_{t1} \cap \partial P_t, \quad \partial P''_t = \partial P_{t2} \cap \partial P_t,$$

with $\partial P_{t1} = \partial P'_t \cup \sigma$ and $\partial P_{t2} = \partial P''_t \cup \sigma$. Then $P_t = P_{t1} \cup P_{t2}$ and $\partial P_t = \partial P'_t \cup \partial P''_t$. The last statement is the one that will matter: the dividing surface is part of the boundary of each half and of the boundary of neither the whole.

The forces acting on P_{t1} are the body force and the contact force.

We express the former as

$$\underline{f}_{b1} = \int_{P_{t1}} \rho(\bar{\underline{b}} + \underline{b}_{12}) \, dv, \quad (6.71)$$

where $\bar{\underline{b}}$ is due to sources external to P_t and $\underline{b}_{12}$ is the body force per unit mass exerted on P_{t1} by P_{t2} . The latter may be due to the gravitational attraction of the two parts for one another, for example. The contact force is

$$\underline{f}_{c1} = \int_{\partial P_{t1}} \underline{t} \, da = \int_{\partial P'_t} \underline{t} \, da + \int_{\sigma} \underline{t}_{12} \, da, \quad (6.72)$$

where $\underline{t}_{12}$ is the traction transmitted to P_{t1} by P_{t2} across σ . The net force applied to P_{t1} is

$$\underline{f}_1 = \underline{f}_{b1} + \underline{f}_{c1}. \quad (6.73)$$

Similarly, the net force acting on P_{t2} is

$$\underline{f}_2 = \underline{f}_{b2} + \underline{f}_{c2}, \quad (6.74)$$

where

$$\underline{f}_{b2} = \int_{P_{t2}} \rho(\bar{\underline{b}} + \underline{b}_{21}) \, dv \quad \text{and} \quad \underline{f}_{c2} = \int_{\partial P''_t} \underline{t} \, da + \int_{\sigma} \underline{t}_{21} \, da, \quad (6.75)$$

wherein $\underline{b}_{21}$ is due to P_{t1} and $\underline{t}_{21}$ is the traction exerted by P_{t1} on P_{t2} .

The linear momenta of the parts P_{t1} and P_{t2} are

$$\underline{l}_1 = \int_{P_{t1}} \rho \underline{v} \, dv \quad \text{and} \quad \underline{l}_2 = \int_{P_{t2}} \rho \underline{v} \, dv, \quad (6.76)$$

respectively, and the momentum of $P_t = P_{t1} \cup P_{t2}$ is $\underline{l} = \underline{l}_1 + \underline{l}_2$, the two pieces overlapping only on σ , a set of zero volume. Euler's postulate (6.34), applied to P_{t1} and P_{t2} separately, yields

$$\begin{aligned} \int_{P_{t1}} \rho(\bar{\underline{b}} + \underline{b}_{12}) \, dv + \int_{\partial P'_t} \underline{t} \, da + \int_{\sigma} \underline{t}_{12} \, da &= \frac{d}{dt} \underline{l}_1 \quad \text{and} \\ \int_{P_{t2}} \rho(\bar{\underline{b}} + \underline{b}_{21}) \, dv + \int_{\partial P''_t} \underline{t} \, da + \int_{\sigma} \underline{t}_{21} \, da &= \frac{d}{dt} \underline{l}_2. \end{aligned} \quad (6.77)$$

On adding these and recalling that $\partial P_t = \partial P'_t \cup \partial P''_t$, so that the two integrals of $\underline{t}$ over the outer boundary combine into one, we obtain

$$\int_{P_t} \rho \bar{\underline{b}} \, dv + \int_{P_{t1}} \rho \underline{b}_{12} \, dv + \int_{P_{t2}} \rho \underline{b}_{21} \, dv + \int_{\partial P_t} \underline{t} \, da + \int_{\sigma} (\underline{t}_{12} + \underline{t}_{21}) \, da = \frac{d}{dt} \underline{l}. \quad (6.78)$$

On the other hand, Euler's postulate applied to the whole of P_t reads

$$\int_{P_t} \rho \bar{\underline{b}} \, dv + \int_{\partial P_t} \underline{t} \, da = \frac{d}{dt} \underline{l}, \quad (6.79)$$

since the only body force acting on P_t from outside is $\bar{\underline{b}}$, the mutual attractions $\underline{b}_{12}$ and $\underline{b}_{21}$ being internal to it. Subtracting (6.79) from (6.78), the two terms they have in common cancel and

$$\int_{P_{t2}} \rho \underline{b}_{21} \, dv + \int_{\sigma} \underline{t}_{21} \, da = - \left[\int_{P_{t1}} \rho \underline{b}_{12} \, dv + \int_{\sigma} \underline{t}_{12} \, da \right]. \quad (6.80)$$

The force applied to P_{t2} by P_{t1} is thus opposite to the force applied to P_{t1} by P_{t2} .

REMARK 6.18 (action and reaction is a theorem here). In particle mechanics the equality of action and reaction is Newton's third law, an independent axiom. For continua it is a *consequence* of the balance of linear momentum, and (6.79) is where the consequence comes from.

The substance of the argument is that Euler's postulate is asserted for *every* part, and that the momentum, the external body force and the external traction of the whole are the sums of those of the halves. Given that, the internal interactions must cancel, because they appear in the two half-statements and are absent from the whole-statement. Any theory in which they did not cancel would be inconsistent: applying the same axiom at two different levels of subdivision would give two different answers.

The word *external* carries the weight. Equation (6.71) separates $\bar{\underline{b}}$ from $\underline{b}_{12}$ by fiat, and (6.79) then asserts that only $\bar{\underline{b}}$ acts on the whole. That is the modelling assumption: interactions come in pairs internal

to the body, plus whatever the outside world applies. It is the continuum version of the statement that a system cannot lift itself by its own bootstraps.

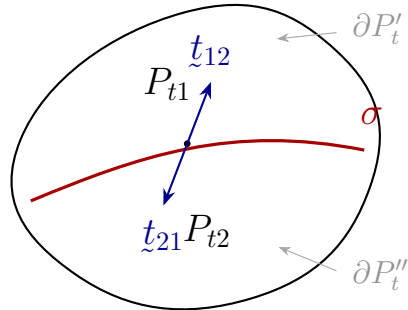


Figure 6.4: An arbitrary part of a body divided into two sub-parts by an interior surface σ . Euler's postulate holds for P_{t1} , for P_{t2} and for P_t ; the first two together must reproduce the third, and the difference is exactly (6.80). Shrinking P_t onto σ kills the volume terms and leaves Cauchy's lemma $\underline{t}_{21} = -\underline{t}_{12}$. Note that σ belongs to ∂P_{t1} and to ∂P_{t2} but not to ∂P_t : that is the whole of the bookkeeping.

Collapsing the parts onto the surface

Equation (6.80) is a statement about volumes and a surface together. To isolate the surface part, consider a sequence of parts P_t , all with the same interior surface σ and collapsing onto it in the limit. Each of these parts has associated sub-parts P_{t1} and P_{t2} whose volumes tend to zero, while σ is held fixed throughout. Assuming $|\rho \underline{b}_{21}|$ and $|\rho \underline{b}_{12}|$ to be bounded, say by a constant k , the volume integrals in (6.80) are bounded in magnitude by $k V(P_{t1})$ and $k V(P_{t2})$ and therefore vanish in the limit, yielding

$$\int_{\sigma} (\underline{t}_{21} + \underline{t}_{12}) da = 0. \quad (6.81)$$

Substituting this back into (6.80), the two surface integrals there cancel and we conclude that

$$\int_{P_{t2}} \rho \underline{b}_{21} dv = - \int_{P_{t1}} \rho \underline{b}_{12} dv. \quad (6.82)$$

So the two statements separate: the contact interactions balance among themselves, and so do the body-force interactions.

Because P_t is arbitrary, σ is an arbitrary surface in κ_t , and (6.81) holds for every one of them. Assuming the integrand to be continuous, we may invoke a version of the localization theorem for surface integrals, in the manner already used to pass from (6.4) to (6.6), to arrive at *Cauchy's lemma*:

$$\underline{t}_{21} = -\underline{t}_{12}. \quad (6.83)$$

REMARK 6.19 (why the body forces had to be assumed bounded). The hypothesis that $|\rho \underline{b}_{12}|$ is bounded is the only analytic input in the argument, and it is doing exactly one job: making the volume terms of (6.80) vanish faster than the surface terms as the parts collapse. If the mutual body force were allowed to become unbounded as the two parts approach one another, as happens for point masses in Newtonian gravitation at zero separation, then the volume integrals need not vanish and (6.81) fails.

The same competition between a volume and a surface decides Noll's theorem and Cauchy's theorem below. It is the structural fact of the whole section: *a surface integral over a shrinking region is of one order higher than a volume integral over it*, so in any limit that divides by the surface measure, the bulk terms disappear and only the surface terms survive. That is why all three theorems are proved by shrinking something.

6.5.2 Cauchy's hypothesis and Noll's theorem

Let $\underline{t}_s(\underline{x}, t)$ be the traction acting at $\underline{x} \in s$, a surface in κ_t . The subscript conveys the notion that the traction depends on the surface s and not merely on the point $\underline{x}$. That it must depend on something beyond $\underline{x}$ is clear from a simple thought experiment. Take a bar carrying an axial force F and cut it at a point $\underline{x}$, once by a plane s_1 perpendicular to the axis and once by a plane s_2 inclined to it. The same force F is transmitted in both cases, but it is spread over different areas, so the

force per unit area differs. The traction at $\underline{x}$ therefore cannot be a function of $\underline{x}$ alone.

Since the traction is defined locally at points of the surface, logically it should depend on a local measure of the surface orientation, namely a unit normal $\underline{n}$ to s at $\underline{x}$. This assumption is the content of the famous *Cauchy hypothesis*: there exists a function $\underline{t}(\underline{x}, t; \cdot)$ such that

$$\underline{t}_s(\underline{x}, t) = \underline{t}(\underline{x}, t; \underline{n}). \quad (6.84)$$

Cauchy's lemma (6.83) then assumes the form

$$\underline{t}(\underline{x}, t; -\underline{n}) = -\underline{t}(\underline{x}, t; \underline{n}), \quad (6.85)$$

since two parts adjoining along a surface see the same surface with opposite exterior normals.

REMARK 6.20 (what (6.84) rules out). Equation (6.84) looks weak and is not. What it forbids is the following. The traction might have depended on the curvature of s at $\underline{x}$, so that a cut along a sphere transmitted a different force from a cut along its tangent plane. It might have depended on the diameter of the part being cut off, or on the shape of s far from $\underline{x}$. Each of these is a coherent physical possibility, and each is excluded by (6.84).

That is why Noll's theorem matters. Cauchy's hypothesis has occupied a central place in continuum mechanics since it was proposed in the early nineteenth century, and in the mid-twentieth century Noll strengthened it to the status of a proved theorem: it is a consequence of Euler's linear momentum balance, given only continuity and boundedness of the fields involved. Nothing has to be assumed about how the traction responds to the shape of the cut, because the balance law already decides.

Sketch of Noll's theorem

With reference to Figure 6.7, let s_1 and s_2 be two surfaces in κ_t intersecting at a point with position $\underline{x}_0$, and suppose they share a common tangent plane T , with unit normal $\underline{n}$, at this point.

Let P_{t_1} be the part of the body bounded by $\partial P_{t_1} = d_1 \cup f_1 \cup e$, where d_1 is a subset of s_1 , f_1 is a piece of the lateral surface of a circular cylinder of radius R and axis $\underline{n}$, and e is the part of the surface of that cylinder common to both ∂P_{t_1} and ∂P_{t_2} . The part P_{t_2} , not shown separately, is bounded above by s_2 , with $\partial P_{t_2} = d_2 \cup f_2 \cup e$ and corresponding definitions of d_2 and f_2 .

The construction in coordinates

The estimates that follow are the whole content of the argument, and they depend on how deep the shared face e is placed, so the geometry is fixed precisely first.

Put the origin at $\underline{x}_0$, let T be the plane through it with normal $\underline{n}$, and let ξ denote position in T measured from $\underline{x}_0$. Near $\underline{x}_0$ each surface is a graph over T ,

$$s_\alpha : \quad z = g_\alpha(\xi), \quad g_\alpha(\underline{0}) = 0, \quad \nabla g_\alpha(\underline{0}) = \underline{0},$$

the first condition saying that s_α passes through $\underline{x}_0$ and the second that its tangent plane there is T . Differentiability of g_α at $\underline{0}$ then says precisely that

$$g_\alpha(\xi) = o(|\xi|) \quad \text{as } \xi \rightarrow \underline{0}, \quad (*)$$

and that single statement is the hypothesis the theorem runs on.

Now build the pillbox. Let the cylinder be $|\xi| \leq R$ about the axis through $\underline{x}_0$ along $\underline{n}$, and put the shared face at the depth

$$h(R) = \max_{\alpha \in \{1,2\}} \left(\max_{|\xi| \leq R} |g_\alpha(\xi)| \right), \quad (**)$$

in which the inner maximum is taken over the closed disc $D_R = \{\xi \in T : |\xi| \leq R\}$ and the outer one over the two-element index set $\{1, 2\}$. Both are attained and finite: D_R is closed and bounded, hence compact, and $|g_\alpha|$ is continuous on it, so the extreme value theorem applies; the outer maximum is that of two real numbers. Thus $h(R)$ is a well-defined non-negative number for each R , and by construction

$$-h(R) \leq g_\alpha(\xi) \leq h(R) \quad \text{for all } \xi \in D_R \text{ and } \alpha = 1, 2,$$

so the disc e of radius R placed at height $-h(R)$ lies below both caps, both parts P_{t1} and P_{t2} lie above it, and $\partial P_{t\alpha} = d_\alpha \cup f_\alpha \cup e$ closes up.

The one property of h that the proof uses is

$$h(R) = o(R) \quad \text{as } R \rightarrow 0^+, \quad (***)$$

that is, $h(R)/R \rightarrow 0$, in the sense of Definition 1.66. It is $(*)$ read on the disc. Let $\eta > 0$ be given. Tangency supplies $r_0 > 0$ with

$$|g_\alpha(\xi)| \leq \eta |\xi| \quad \text{whenever } |\xi| \leq r_0, \alpha = 1, 2.$$

For $0 < R \leq r_0$ every $\xi \in D_R$ has $|\xi| \leq R$, so the right-hand side is at most ηR ; taking the maximum over D_R and then over α preserves the bound, and

$$h(R) \leq \eta R, \quad \text{that is} \quad \frac{h(R)}{R} \leq \eta \quad \text{for } 0 < R \leq r_0.$$

As η was arbitrary, $h(R)/R \rightarrow 0$.

REMARK 6.21 (the depth of e is not free). The figure in the book marks the height of the cylinder as R , the same as its radius. That cannot be the intended construction, because it contradicts (6.86)₂.

With the face at a constant depth h , the wall of the pillbox reaches from $-h$ up to g_α , so its height nowhere exceeds $h + M_\alpha(R)$, where

$$M_\alpha(R) = \max_{|\xi|=R} |g_\alpha(\xi)|$$

is the largest departure of s_α from T on the rim of the disc, attained because the rim is compact and $|g_\alpha|$ continuous. By $(*)$, $M_\alpha(R) = o(R)$, so

$$A(f_\alpha) \leq 2\pi R[h + o(R)], \quad \frac{A(f_\alpha)}{\pi R^2} \leq \frac{2h}{R} + o(1).$$

Taking $h = R$ leaves the ratio bounded below by 2: the wall is *twice* the cap in the limit, not negligible beside it, and (6.90) does not follow. Taking $h = h(R)$ as above gives $2h(R)/R \rightarrow 0$ by $(***)$ and the wall disappears.

A numerical check on paraboloids $g_\alpha = \frac{1}{2}c_\alpha|\xi|^2$, which are tangent to T at $\underline{x}_0$, makes the contrast plain. With $h = h(R)$ the ratio $A(f_1)/A(d_1)$ runs 0.209, 0.105, 0.052 as R runs 0.1, 0.05, 0.025, halving with R ; with $h = R$ the same ratio runs 2.121, 2.063, 2.032 and settles on 2.

So the pillbox must be flat, and it must flatten faster than it narrows. That is the geometric content of tangency, and Figure 6.7 is drawn to scale in that respect.

Denote by $A(s)$ the area of a surface s and by $V(P)$ the volume of a region P . Then

$$A(d_\alpha) = \pi R^2 + o(R^2) \quad \text{and} \quad A(f_\alpha) = o(R^2); \quad \alpha = 1, 2, \quad (6.86)$$

and

$$V(P_{t\alpha}) = O(R^3) = o(R^2); \quad \alpha = 1, 2. \quad (6.87)$$

All three follow from $(*)$ and $(***)$, in the sense of the Landau symbols of Definition 1.66, and each is a one-line integral over the disc $|\xi| \leq R$.

The caps. A graph $z = g_\alpha$ has area element $\sqrt{1 + |\nabla g_\alpha|^2}$ times the area element of T , so

$$A(d_\alpha) = \int_{|\xi| \leq R} \sqrt{1 + |\nabla g_\alpha|^2} \, da_\xi.$$

The gradient vanishes at $\mathfrak{O}$ and is continuous, so $|\nabla g_\alpha| = o(1)$ on the disc and the integrand is $1 + o(1)$. Integrating the constant over a disc of area πR^2 ,

$$A(d_\alpha) = \pi R^2 [1 + o(1)] = \pi R^2 + o(R^2),$$

which is $(6.86)_1$. The cap is a disc to leading order, and the correction is the curvature, which is second order.

The lateral strips. The wall stands on the circle $|\xi| = R$, of circumference $2\pi R$, and reaches from height $-h(R)$ up to g_α , so its height nowhere exceeds $h(R) + |g_\alpha| \leq 2h(R)$. Therefore

$$A(f_\alpha) \leq 2\pi R \cdot 2h(R) = 4\pi R h(R) = o(R^2)$$

by $(***)$, which is (6.86)₂.

This is the estimate the whole proof turns on. As a ratio rather than an order,

$$\frac{A(f_\alpha)}{A(d_\alpha)} \leq \frac{4h(R)}{R} [1 + o(1)] \longrightarrow 0.$$

The wall is negligible *beside the cap*, and it is negligible only because the two surfaces are tangent. Were they to cross at a fixed angle, so that $g_\alpha \approx m_\alpha |\xi|$ with $m_\alpha \neq 0$, then $h(R)$ would be $O(R)$ rather than $o(R)$, the ratio above would tend to a non-zero constant depending on the m_α but not on R , and no amount of shrinking would remove the wall. Figure 6.7 shows the two cases side by side.

The volumes. The region $P_{t\alpha}$ lies between e , at height $-h(R)$, and the cap d_α , at height g_α , over the disc D_R , so its height above each ξ is $g_\alpha + h(R)$, which by $(**)$ lies between 0 and $2h(R)$. Hence

$$V(P_{t\alpha}) = \int_{D_R} [g_\alpha(\xi) + h(R)] da_\xi \leq 2h(R) \int_{D_R} da_\xi = 2\pi R^2 h(R). \quad (\S)$$

It remains to convert this into (6.87), and the conversion is the one place where the two Landau symbols must be told apart. Write $C = 2\pi$ and use $h(R) \leq R$, which holds for all small R by $(***)$ with $\eta = 1$:

$$V(P_{t\alpha}) \leq 2\pi R^2 h(R) \leq 2\pi R^3 = C R^3,$$

which is the definition of $V(P_{t\alpha}) = O(R^3)$: the ratio V/R^3 is bounded by the *single* constant C . Lemma 1.67(ii) with $p = 3$ and $q = 2$ then converts this into a little-oh statement one power down. Explicitly, given $\eta > 0$,

$$\frac{V(P_{t\alpha})}{R^2} \leq \frac{C R^3}{R^2} = C R < \eta \quad \text{whenever } R < \eta/C,$$

so the ratio V/R^2 falls below *every* constant, and

$$V(P_{t\alpha}) = O(R^3) = o(R^2),$$

which is (6.87). The two symbols differ in exactly one quantifier: big-oh asks for some constant bound, little-oh for every one, and one extra

power of R is what upgrades the first into the second. The same reading applies to $A(f_\alpha) = o(R^2)$ above, where the extra power was supplied by $h(R)$ rather than by R itself.

Relative to the cap, (§) and $A(d_\alpha) = \pi R^2[1 + o(1)]$ give

$$\frac{V(P_{t\alpha})}{A(d_\alpha)} \leq \frac{2\pi R^2 h(R)}{\pi R^2[1 + o(1)]} = 2h(R)[1 + o(1)] \longrightarrow 0.$$

The balances of linear momentum for P_{t1} and P_{t2} are given by (6.36):

$$\int_{P_{t1}} \rho(\dot{\underline{v}} - \underline{b}) \, dv = \int_{\partial P_{t1}} \underline{t}_{\partial P_{t1}} \, da \quad \text{and} \quad \int_{P_{t2}} \rho(\dot{\underline{v}} - \underline{b}) \, dv = \int_{\partial P_{t2}} \underline{t}_{\partial P_{t2}} \, da, \quad (6.88)$$

which, when subtracted and rearranged so that the two caps stand alone on the left, give

$$\begin{aligned} \int_{d_1} \underline{t}_{d_1} \, da - \int_{d_2} \underline{t}_{d_2} \, da &= \int_{P_{t1}} \rho(\dot{\underline{v}} - \underline{b}) \, dv - \int_{P_{t2}} \rho(\dot{\underline{v}} - \underline{b}) \, dv \\ &\quad + \int_{f_2} \underline{t}_{f_2} \, da - \int_{f_1} \underline{t}_{f_1} \, da. \end{aligned} \quad (6.89)$$

The integrals over e have cancelled, and the reason is the step the construction was designed around. Both parts lie *above* e , by the choice of the depth $h(R)$, so e enters ∂P_{t1} and ∂P_{t2} as the same surface with the same exterior normal, namely $-\underline{n}$. The traction on it is therefore literally the same field in the two balances, whatever that field may be, and subtracting one balance from the other removes it.

Nothing has been assumed about its value, and in particular Cauchy's hypothesis has not been used: at this stage the traction is still allowed to depend on the whole surface, and the argument needs only that e is one and the same surface seen twice. Had the two parts been placed on opposite sides of e , the exterior normals would have been opposite, the two tractions would have been opposite by Cauchy's lemma (6.83), and subtracting would have *doubled* the term instead of cancelling it. That is the reason e is pushed below both caps rather than laid in the tangent plane.

Assuming all the integrands to be continuous, and hence bounded on their respective compact domains of integration, the three terms on the right of (6.89) are bounded in magnitude by a constant times $V(P_{t_1})$, $V(P_{t_2})$, $A(f_2)$ and $A(f_1)$ respectively, all of which are $o(R^2)$ by (6.86)₂ and (6.87). Hence

$$\int_{d_1} \underline{t}_{d_1} da = \int_{d_2} \underline{t}_{d_2} da + \underline{o}(R^2), \quad (6.90)$$

where $\underline{o}(\varepsilon)$ denotes a vector whose magnitude is $o(\varepsilon)$. Dividing by πR^2 and invoking (6.86)₁, which says that $A(d_\alpha)$ and πR^2 may be interchanged at the cost of a further $o(1)$, we obtain

$$\frac{1}{A(d_1)} \int_{d_1} \underline{t}_{d_1} da = \frac{1}{A(d_2)} \int_{d_2} \underline{t}_{d_2} da + \frac{\underline{o}(R^2)}{R^2}. \quad (6.91)$$

Under our continuity hypothesis the mean-value theorem, essentially the same thing as the localization theorem, is applicable and implies that

$$\lim_{R \rightarrow 0} \frac{1}{A(d_\alpha)} \int_{d_\alpha} \underline{t}_{d_\alpha}(\underline{x}, t) da = \underline{t}_{d_\alpha}(\underline{x}_0, t); \quad \alpha = 1, 2. \quad (6.92)$$

The average of a continuous function over a set shrinking to a point tends to its value at that point. On passing to the limit in (6.91), the last term disappearing because $|\underline{o}(R^2)|/R^2 \rightarrow 0$ by the definition of the symbol, we finally reach

$$\underline{t}_{d_1}(\underline{x}_0, t) = \underline{t}_{d_2}(\underline{x}_0, t), \quad (6.93)$$

and thus conclude that the traction at a point takes the same value for all surfaces having the same unit normal at that point. Accordingly the traction is determined by $\underline{n}$, and (6.84) is proved. It follows that the traction acting on a surface at a particular point is identical to the traction acting on the tangent plane to the surface at the point in question. Here, for definiteness, $\underline{n}$ is taken to be the exterior unit normal to ∂P_{t_1} and ∂P_{t_2} at $\underline{x}_0$. Rigorous proofs may be found in the references; the argument above is a sketch in the sense that the error estimates have been carried informally.

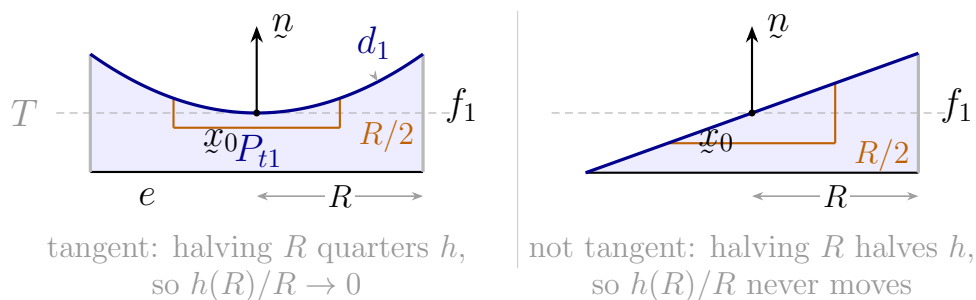


Figure 6.7: Noll's pillbox in axial cross-section, and the one thing that decides the argument: how the depth $h(R)$ of the shared face behaves as the radius R shrinks. The second surface s_2 is omitted for clarity; it caps the same box from the other side. In each panel the grey box has radius R and the orange one radius $R/2$, and the vertical direction is stretched by the same factor in both, so the comparison between them is faithful. Left, s_1 tangent to T at $\underline{x}_0$: the departure from T is quadratic, so h falls by four when R falls by two, the box flattens as it narrows, and the wall f_1 of area $2\pi R h$ dies beside the cap d_1 of area πR^2 . Right, s_1 crossing T at a fixed angle: the departure is linear, every box is a scaled copy of every other, and the ratio $A(f_1)/A(d_1)$ is the same at every radius. Tangency is not a convenience of the drawing; it is the hypothesis, and Remark 6.21 gives the numbers.

6.5.3 Cauchy's stress theorem

Sections 6.5.1 and 6.5.2 have reduced the contact force of (6.28) as far as the linear momentum balance alone can take it: by (6.84) the traction at $\underline{x}$ depends on the cut only through its unit normal, and by (6.85) it reverses when the normal does. The present section extracts the last consequence, and it is the one that makes the subject computable. A function on the unit sphere carries infinitely many values; (6.94) below says that this particular function is the restriction to the sphere of a linear map, hence is fixed by nine numbers. Linearity is nowhere assumed. It is produced by (6.36), and the mechanism is the elementary fact that a tetrahedron has volume of one order higher in its size than its faces have area.

THEOREM 6.22 (Cauchy's stress theorem). *Let $\underline{x}_0 \in \kappa_t$ and suppose that*

- (H1) *the traction obeys Cauchy's hypothesis (6.84) and Cauchy's lemma (6.85);*
- (H2) *for each fixed unit vector $\underline{n}$ the field $\underline{x} \mapsto \underline{t}(\underline{x}, t; \underline{n})$ is continuous on a neighbourhood of $\underline{x}_0$;*
- (H3) *the field $\underline{x} \mapsto \rho(\underline{\dot{v}} - \underline{b})$ is continuous on a neighbourhood of $\underline{x}_0$.*

Then there is a tensor $\underline{T}(\underline{x}_0, t) \in \text{Lin}(\mathbb{E}^3, \mathbb{E}^3)$, unique, such that

$$\underline{t}(\underline{x}, t; \underline{n}) = \underline{T}(\underline{x}, t)\underline{n} \quad (6.94)$$

for every unit vector $\underline{n}$. It is called the Cauchy stress tensor.

The proof occupies the rest of the section. Three lemmas are separated out first, because each is used twice and none of them is about mechanics: the first is the geometry of a tetrahedron, the second replaces the mean value theorem by a statement that is true for vector-valued integrands, and the third is the extension from one octant to all of them.

LEMMA 6.23 (geometry of the Cauchy tetrahedron). *Fix a unit vector $\underline{m}$ with components $m_i = \underline{e}_i \cdot \underline{m} > 0$ on the orthonormal basis $\{\underline{e}_i\}$*

of $\mathbb{E}^3$, fix $\delta > 0$, and let $\tau = \tau(\underline{m}, \delta)$ be the tetrahedron with vertex at $\underline{x}_0$ bounded by the three coordinate planes through $\underline{x}_0$ and by the plane $\underline{m} \cdot (\underline{x} - \underline{x}_0) = \delta$. Write s for the oblique face, s_i for the face lying in the plane normal to $\underline{e}_i$, and $A(\cdot)$, $V(\cdot)$ for area and volume. Then

$$\begin{aligned} h_i &= \frac{\delta}{m_i}, & A_i &:= A(s_i) = \frac{\delta^2}{2m_j m_k}, \\ A(s) &= \frac{\delta^2}{2m_1 m_2 m_3}, & V(\tau) &= \frac{\delta^3}{6m_1 m_2 m_3}, \end{aligned} \quad (*_1)$$

where h_i is the intercept of the oblique plane on the $\underline{e}_i$ axis and $\{i, j, k\}$ are distinct. Consequently

$$A(s) = c_1 \delta^2, \quad V(\tau) = c_2 \delta^3, \quad \frac{V(\tau)}{A(s)} = \frac{\delta}{3}, \quad (*_2)$$

with $c_1 = 1/(2m_1 m_2 m_3)$ and $c_2 = c_1/3$ positive and independent of δ .

Proof. Take $\underline{x}_0$ as origin. A point $h\underline{e}_i$ lies on the oblique plane when $\underline{m} \cdot h\underline{e}_i = hm_i = \delta$, so $h_i = \delta/m_i$, positive because $m_i > 0$.

The face s_i is the triangle with vertices $\mathbf{0}$, $h_j \underline{e}_j$ and $h_k \underline{e}_k$, a right triangle with legs h_j and h_k , whence $A_i = \frac{1}{2} h_j h_k = \delta^2/(2m_j m_k)$. The tetrahedron is the corner solid on the three mutually perpendicular edges $h_i \underline{e}_i$, so $V(\tau) = \frac{1}{6} h_1 h_2 h_3$, which is $(*_1)_4$.

For $A(s)$, the oblique face is the triangle with vertices $h_i \underline{e}_i$, whose area is half the modulus of the cross product of two of its edges, by Proposition 1.24. Writing $d = [\underline{e}_1, \underline{e}_2, \underline{e}_3] = \pm 1$ for the orientation of the triad, so that $\underline{e}_2 \times \underline{e}_3 = d \underline{e}_1$ and its two cyclic partners,

$$\begin{aligned} 2A(s) &= |(h_2 \underline{e}_2 - h_1 \underline{e}_1) \times (h_3 \underline{e}_3 - h_1 \underline{e}_1)| \\ &= |d(h_2 h_3 \underline{e}_1 + h_1 h_3 \underline{e}_2 + h_1 h_2 \underline{e}_3)| = \frac{\delta^2}{m_1 m_2 m_3} |m_1 \underline{e}_1 + m_2 \underline{e}_2 + m_3 \underline{e}_3|, \end{aligned}$$

the second line by expanding the product, the third by $h_j h_k = \delta^2/(m_j m_k) = (\delta^2/m_1 m_2 m_3) m_i$ together with $|d| = 1$. The last modulus is $|\underline{m}| = 1$,

which gives $(*_1)_3$. Only the orthonormality of $\{e_i\}$ has been used, never its handedness: the factor d leaves a modulus untouched, and $h_i > 0$ makes the volume positive whichever sign d carries. The lemma therefore holds verbatim for any orthonormal triad, a fact Lemma 6.25 will need.

Both c_1 and c_2 are built from $\underline{m}$ alone, and dividing the last two members of $(*_1)$ gives $V(\tau)/A(s) = \delta/3$. ■

Geometrically the last of $(*_2)$ is the statement that τ is a cone of apex $\underline{x}_0$ over the base s , whose apex stands at height δ ; and the reason c_1, c_2 do not move with δ is that $\underline{m}$ is held fixed throughout, so the tetrahedra $\tau(\underline{m}, \delta)$ are homothetic copies of one another about $\underline{x}_0$, scaled by δ . Area scales as the square of a length and volume as the cube, and the whole proof lives in the gap between those two powers.

LEMMA 6.24 (averages over a shrinking surface). *Let g be a $\mathbb{E}^3$ -valued field, continuous at $\underline{x}_0$, and let $\{\sigma_\delta\}_{\delta>0}$ be surfaces of positive area whose radius about $\underline{x}_0$,*

$$\varepsilon(\delta) := \sup_{\underline{x} \in \sigma_\delta} |\underline{x} - \underline{x}_0| = \text{the least } r \text{ with } \sigma_\delta \subseteq \{\underline{x} : |\underline{x} - \underline{x}_0| \leq r\},$$

satisfies $\varepsilon(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. The supremum is written rather than a maximum because σ_δ need not contain the point at which the distance is largest; it is finite as soon as σ_δ is bounded. Then

$$\lim_{\delta \rightarrow 0} \frac{1}{A(\sigma_\delta)} \int_{\sigma_\delta} \underline{g}(\underline{x}) \, da = \underline{g}(\underline{x}_0).$$

Proof. Since $A(\sigma_\delta)^{-1} \int_{\sigma_\delta} da = 1$, the constant $\underline{g}(\underline{x}_0)$ may be taken inside the integral, and the triangle inequality for integrals gives

$$\begin{aligned} \left| \frac{1}{A(\sigma_\delta)} \int_{\sigma_\delta} \underline{g} \, da - \underline{g}(\underline{x}_0) \right| &= \left| \frac{1}{A(\sigma_\delta)} \int_{\sigma_\delta} [\underline{g}(\underline{x}) - \underline{g}(\underline{x}_0)] \, da \right| \\ &\leq \frac{1}{A(\sigma_\delta)} \int_{\sigma_\delta} |\underline{g}(\underline{x}) - \underline{g}(\underline{x}_0)| \, da \leq \omega(\varepsilon(\delta)), \end{aligned}$$

where

$$\omega(r) = \sup_{|\underline{x} - \underline{x}_0| \leq r} |\underline{g}(\underline{x}) - \underline{g}(\underline{x}_0)|$$

denotes the largest departure of $\underline{g}$ from its value at $\underline{x}_0$ over the closed ball of radius r ; the last inequality holds because every $\underline{x} \in \sigma_\delta$ satisfies $|\underline{x} - \underline{x}_0| \leq \varepsilon(\delta)$, so the integrand is bounded by $\omega(\varepsilon(\delta))$, and the average of a constant is that constant.

It remains to show $\omega(r) \rightarrow 0$ as $r \rightarrow 0^+$, which is continuity of $\underline{g}$ at $\underline{x}_0$ restated. Let $\eta > 0$. Continuity supplies $r_0 > 0$ with $|\underline{g}(\underline{x}) - \underline{g}(\underline{x}_0)| < \eta$ whenever $|\underline{x} - \underline{x}_0| < r_0$; taking the supremum over the ball of any radius $r < r_0$ gives $\omega(r) \leq \eta$. Since $\varepsilon(\delta) \rightarrow 0$, we have $\varepsilon(\delta) < r_0$ for all small δ , and then $\omega(\varepsilon(\delta)) \leq \eta$. ■

This is the step the book performs with the mean value theorem, and the two are not interchangeable. For a scalar integrand the mean value theorem does produce a point $\underline{x}^* \in \sigma$ at which the integrand equals its own average; for a vector-valued integrand no such point need exist. The map $\underline{f}(\underline{x}) = (\cos 2\pi \underline{x}, \sin 2\pi \underline{x})$ on $[0, 1]$ has average $\underline{0}$ while $|\underline{f}(\underline{x})| = 1$ everywhere, so no $\underline{x}^*$ obliges. Applied componentwise the mean value theorem is of course legitimate, and then it yields three sample points, one per component, all of which tend to $\underline{x}_0$; the conclusion is the same. Lemma 6.24 obtains it in one line and for the vector at once.

LEMMA 6.25 (from one octant to all). *Assume (H1) and (H2), and suppose that*

$$\underline{t}(\underline{x}_0; \underline{n}) = \sum_i (\underline{f}_i \cdot \underline{n}) \underline{t}(\underline{x}_0; \underline{f}_i) \quad (\ddagger)$$

holds for every orthonormal triad $\{\underline{f}_i\}$ of $\mathbb{E}^3$ and every unit $\underline{n}$ with $\underline{f}_i \cdot \underline{n} > 0$ for $i = 1, 2, 3$. Then $\underline{t}(\underline{x}_0; \underline{n}) = \underline{T}\underline{n}$ for every unit $\underline{n}$, with $\underline{T}$ the single tensor $\underline{t}(\underline{x}_0; \underline{e}_i) \otimes \underline{e}_i$ built on any one fixed orthonormal triad $\{\underline{e}_i\}$.

Proof. Suppose first that no component of $\underline{n}$ on $\{\underline{e}_i\}$ vanishes, and put $\varepsilon_i = \text{sgn}(\underline{e}_i \cdot \underline{n}) \in \{\pm 1\}$. The triad $\underline{f}_i = \varepsilon_i \underline{e}_i$ is orthonormal, and $\underline{f}_i \cdot \underline{n} = \varepsilon_i \underline{e}_i \cdot \underline{n} = |\underline{e}_i \cdot \underline{n}| > 0$, so $(\ddagger)$ applies to it and yields

$$\begin{aligned} \underline{t}(\underline{x}_0; \underline{n}) &= \sum_i (\varepsilon_i \underline{e}_i \cdot \underline{n}) \underline{t}(\underline{x}_0; \varepsilon_i \underline{e}_i) = \sum_i \varepsilon_i (\underline{e}_i \cdot \underline{n}) \varepsilon_i \underline{t}(\underline{x}_0; \underline{e}_i) \\ &= \sum_i (\underline{e}_i \cdot \underline{n}) \underline{t}(\underline{x}_0; \underline{e}_i) = \underline{T}\underline{n}, \end{aligned}$$

where the middle equality used (6.85) on each coordinate face and the last $\varepsilon_i^2 = 1$.

The unit vectors with some vanishing component form the union of three great circles, whose complement is dense in the unit sphere. Both $\underline{n} \mapsto \underline{t}(\underline{x}_0; \underline{n})$ and $\underline{n} \mapsto \underline{T}\underline{n}$ are continuous there, the first by (H2) and the second because every linear map on a finite-dimensional space is continuous, and two continuous maps agreeing on a dense set agree everywhere. ■

Proof of Theorem 6.22

Fix $\underline{m}$ with $m_i = \underline{e}_i \cdot \underline{m} > 0$ and let $\tau = \tau(\underline{m}, \delta)$ be the tetrahedron of Lemma 6.23, so that the exterior unit normal on $\partial\tau$ is $\underline{m}$ on s and $-\underline{e}_i$ on s_i , the solid lying on the positive side of each coordinate plane. Euler's first postulate in the reduced form (6.36), applied to τ and with the two volume terms collected on the right, reads

$$\int_{\partial\tau} \underline{t} \, da = \int_{\tau} \rho(\dot{\underline{v}} - \underline{b}) \, dv. \quad (6.95)$$

By Remark 5.3 this is symbolic notation for the vector equation

$$\underline{e}_i \int_{\partial\tau} t_i \, da = \underline{e}_i \int_{\tau} \rho(\dot{v}_i - b_i) \, dv, \quad (6.96)$$

and since $\{\underline{e}_i\}$ is a basis, (6.96) is equivalent by Theorem 1.33 to the three scalar equations

$$\int_{\partial\tau} t_i \, da = \int_{\tau} \rho(\dot{v}_i - b_i) \, dv; \quad i = 1, 2, 3. \quad (6.97)$$

Every estimate below is carried out on (6.97), so that it is an estimate on numbers and the objection of Remark 5.3 does not arise.

Step 1: the volume term is negligible beside the faces. Fix once and for all a $\delta_0 > 0$ small enough that $\tau(\underline{m}, \delta_0)$ lies inside the neighbourhood of (H3). For $0 < \delta \leq \delta_0$ one has $\tau(\underline{m}, \delta) \subseteq \tau(\underline{m}, \delta_0)$, the tetrahedra being homothetic about $\underline{x}_0$, and $\tau(\underline{m}, \delta_0)$ is compact, so the

continuous functions $\rho(\dot{v}_i - b_i)$ attain a finite maximum modulus on it. Setting

$$k_i = \max_{x \in \tau(\underline{m}, \delta_0)} |\rho(\dot{v}_i - b_i)|, \quad (6.99)$$

where the maximum is taken over the whole of the largest tetrahedron and denotes the largest value the modulus attains there, we obtain three numbers k_1, k_2, k_3 that do not depend on δ . They exist because $\tau(\underline{m}, \delta_0)$ is compact and the integrand continuous, and they dominate the corresponding maxima over every smaller tetrahedron, since

$$\tau(\underline{m}, \delta) \subseteq \tau(\underline{m}, \delta_0) \implies \max_{\tau(\underline{m}, \delta)} |\rho(\dot{v}_i - b_i)| \leq k_i \quad (0 < \delta \leq \delta_0),$$

a maximum over a subset never exceeding one over the set. This is the point at which (6.99) becomes a genuine constant rather than a function of δ , and it is what makes the order estimate (6.100) meaningful. Now (6.97) and the triangle inequality for integrals give

$$\left| \int_{\partial\tau} t_i \, da \right| = \left| \int_{\tau} \rho(\dot{v}_i - b_i) \, dv \right| \leq \int_{\tau} |\rho(\dot{v}_i - b_i)| \, dv \leq k_i V(\tau). \quad (6.98)$$

Divide by $A(s)$, which is positive, and invoke $(*)_3$:

$$\left| \frac{1}{A(s)} \int_{\partial\tau} t_i \, da \right| \leq k_i \frac{V(\tau)}{A(s)} = \frac{k_i}{3} \delta, \quad \text{that is} \quad \frac{1}{A(s)} \int_{\partial\tau} t_i \, da = O(\delta) \quad (6.100)$$

in the sense of Definition 1.66: the ratio of the left-hand side to δ is bounded by the constant $k_i/3$, uniformly in δ . By Lemma 1.67(ii) with $p = 1$ and q any smaller positive number this is also $o(\delta^q)$, but nothing so strong is needed here: $O(\delta)$ already forces the limit, since $k_i\delta/3 \rightarrow 0$. Reverting to symbolic notation, the three components together give

$$\lim_{\delta \rightarrow 0} \frac{1}{A(s)} \int_{\partial\tau} \underline{t} \, da = \underline{0}. \quad (6.101)$$

Nothing has been assumed about the size of $\underline{v}$ or $\underline{b}$; they may be arbitrarily large, provided only that they are bounded near $\underline{x}_0$, which is what (H3) guarantees. The conclusion is therefore independent of the motion and of the material, and that is why $\underline{T}$ exists in every continuum.

Step 2: the surface term splits over the four faces. The boundary $\partial\tau = s \cup s_1 \cup s_2 \cup s_3$ is a union with pairwise intersections of zero area, so the integral splits. On each piece the normal is constant, and (6.84) permits the traction to be evaluated at the normal of the piece:

$$\begin{aligned} \int_{\partial\tau} \underline{t}(\underline{x}; \underline{n}) \, da &= \int_s \underline{t}(\underline{x}; \underline{m}) \, da + \sum_{i=1}^3 \int_{s_i} \underline{t}(\underline{x}; -\underline{e}_i) \, da \\ &= \int_s \underline{t}(\underline{x}; \underline{m}) \, da - \sum_{i=1}^3 \int_{s_i} \underline{t}(\underline{x}; \underline{e}_i) \, da, \end{aligned} \quad (6.102)$$

the second line by Cauchy's lemma (6.85). The passive argument t is suppressed throughout. Applying the mean value theorem to each scalar component, as in (6.97), gives points at which the average is attained,

$$\begin{aligned} \int_s \underline{t}(\underline{x}; \underline{m}) \, da &= A(s) \underline{t}(\underline{x}^*; \underline{m}) \quad \text{for some } \underline{x}^* \in s, \\ \int_{s_i} \underline{t}(\underline{x}; \underline{e}_i) \, da &= A_i \underline{t}(\underline{x}_i^*; \underline{e}_i); \quad i \in \{1, 2, 3\} \text{ fixed, } \underline{x}_i^* \in s_i, \end{aligned} \quad (6.103)$$

which is how the argument is usually written; strictly the sample points carry a component index as well, and Lemma 6.24 dispenses with them altogether. It is the lemma that will be used at Step 4.

Step 3: the coordinate faces are the shadows of the oblique face. Dividing $(*)_2$ by $(*)_3$,

$$\frac{A_i}{A(s)} = \frac{\delta^2 / (2m_j m_k)}{\delta^2 / (2m_1 m_2 m_3)} = m_i,$$

so that

$$A_i = A(s) \underline{e}_i \cdot \underline{m}. \quad (6.104)$$

The same identity follows with no computation from the first equality in (5.9), which states that the integral of the exterior unit normal over a closed surface vanishes. On $\partial\tau$ that normal is piecewise constant, so the integral is $A(s)\underline{m} - A_j \underline{e}_j$, and

$$A(s)\underline{m} - A_j \underline{e}_j = \underline{0}.$$

Taking the inner product with $\underline{e}_i$ and using orthonormality returns (6.104). In either reading, A_i is the area of the shadow that s casts on the i th coordinate plane, and the shadow of a plane region is its area times the cosine of the angle between the planes. This is the only place in the proof where the normal $\underline{m}$ enters linearly, and it is where the linearity of (6.94) comes from.

Step 4: the limit. Divide (6.102) by $A(s)$ and substitute (6.103) and (6.104):

$$\frac{1}{A(s)} \int_{\partial\tau} \underline{t}(\underline{x}; \underline{n}) \, da = \underline{t}(\underline{x}^*; \underline{m}) - \sum_{i=1}^3 (\underline{e}_i \cdot \underline{m}) \underline{t}(\underline{x}_i^*; \underline{e}_i). \quad (6.105)$$

The size of the tetrahedron has disappeared from the right-hand side except through the location of the sample points. Now let $\delta \rightarrow 0$ with $\underline{m}$ held fixed. Every point of τ , hence of each of its four faces, is a convex combination of $\underline{x}_0$ and the three intercepts $h_i \underline{e}_i$, so it lies within

$$\max_i h_i = \max_i \frac{\delta}{m_i} = \frac{\delta}{\min_i m_i}$$

of $\underline{x}_0$, where $\max_i$ and $\min_i$ denote the largest and the smallest of the three numbers indexed by $i = 1, 2, 3$, and the second equality holds because $t \mapsto \delta/t$ is decreasing on $t > 0$. Since $\underline{m}$ is fixed, $\min_i m_i$ is a positive constant and the bound tends to zero with δ . The radius $\varepsilon(\delta)$ of Lemma 6.24 therefore satisfies $\varepsilon(\delta) \leq \delta / \min_i m_i \rightarrow 0$ for each of the four faces, so that lemma applies to each of the four averages in (6.102) divided by its own area, with limits $\underline{t}(\underline{x}_0; \underline{m})$ and $\underline{t}(\underline{x}_0; \underline{e}_i)$ by (H2). The left-hand side of (6.105) tends to 0 by (6.101). Therefore

$$\underline{t}(\underline{x}_0; \underline{m}) = \underline{T}(\underline{x}_0) \underline{m}, \quad \text{where} \quad \underline{T}(\underline{x}_0) = \underline{t}(\underline{x}_0; \underline{e}_i) \otimes \underline{e}_i, \quad (6.106)$$

the identification of the sum $\sum_i (\underline{e}_i \cdot \underline{m}) \underline{t}(\underline{x}_0; \underline{e}_i)$ as a tensor applied to $\underline{m}$ being Definition 1.32 read backwards. Reinstating the argument t ,

$$\underline{T}(\underline{x}, t) = \underline{t}_i \otimes \underline{e}_i, \quad \text{where} \quad \underline{t}_i = \underline{t}(\underline{x}, t; \underline{e}_i) \quad (6.107)$$

is the traction on the i th coordinate plane at $\underline{x} \in \kappa_t$.

Step 5: all directions, and uniqueness. Steps 1 to 4 used nothing about $\{\underline{e}_i\}$ beyond its orthonormality, as the proof of Lemma 6.23 was careful to record, and nothing about $\underline{m}$ beyond $m_i > 0$, which is what puts the three intercepts on the positive axes. What they establish is therefore exactly the hypothesis (†) of Lemma 6.25: the identity

$$\underline{t}(\underline{x}_0; \underline{n}) = \sum_i (\underline{f}_i \cdot \underline{n}) \underline{t}(\underline{x}_0; \underline{f}_i)$$

for every orthonormal triad $\{\underline{f}_i\}$ and every unit $\underline{n}$ positive on it. That lemma removes the restriction and delivers (6.94) for every unit vector, with the one tensor (6.106). For uniqueness, suppose $\underline{T}_1$ and $\underline{T}_2$ both satisfy (6.94). Then $(\underline{T}_1 - \underline{T}_2)\underline{n} = \underline{0}$ for every unit $\underline{n}$, hence for every $\underline{n} \in \mathbb{E}^3$ by homogeneity, so $\underline{T}_1 = \underline{T}_2$ by Theorem 1.33. This completes the proof. $\blacksquare$

Three features of the argument are worth recording before it is used.

The object produced in (6.106) is a tensor by construction and not by assertion: it is a sum of tensor products of vectors of $\mathbb{E}^3$, hence a linear map $\mathbb{E}^3 \rightarrow \mathbb{E}^3$ in the sense of Definition 1.32. Its components on the fixed basis are

$$T_{ij} = \underline{e}_i \cdot \underline{T} \underline{e}_j = \underline{e}_i \cdot \underline{t}_j, \quad (\dagger)$$

by Theorem 1.33, so that the j th *column* of the matrix (T_{ij}) is the traction on the plane whose normal is $\underline{e}_j$, and conversely $\underline{t}_j = T_{ij}\underline{e}_i$. Although (6.107) is written on one basis, the tensor it defines does not depend on that choice, since by Step 5 it is the unique linear map agreeing with $\underline{t}(\underline{x}_0; \cdot)$ on the whole unit sphere.

Linearity in $\underline{n}$ was manufactured rather than assumed. The inputs were (6.84) and (6.85) from Sections 6.5.1 and 6.5.2, the boundedness supplied by (H3), and the geometric identity (6.104). Of these only (6.104) carries any dependence on $\underline{m}$, and it is linear in $\underline{m}$; the linear structure of (6.94) is that identity surviving the limit.

Finally, the theorem is a statement of structure and says nothing about *which* tensor $\underline{T}$ is. Fixing it is the business of a constitutive equation, and the balance laws will constrain it only twice more: the rotational momentum balance forces $\underline{T} = \underline{T}^t$ at (6.117), three scalar

conditions that Proposition 6.11 has already shown do not follow from (6.36), and the second law restricts it further in Chapter 8. What remains free is the material. Equation (6.94) holds for steel and for honey alike.

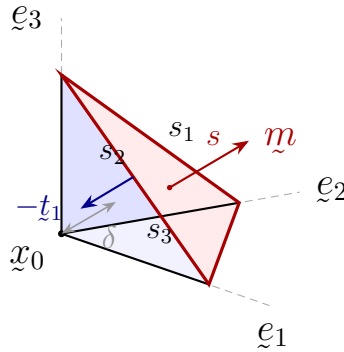


Figure 6.8: Cauchy tetrahedron with vertex at $\underline{x}_0$, the oblique face s at distance δ and unit normal $\underline{m}$ with $\underline{m} \cdot \underline{e}_i > 0$. The three coordinate faces have areas $A_i = A(s)\underline{e}_i \cdot \underline{m}$, since s_i is the shadow of s on the i th coordinate plane. Since $\underline{x}_0$ lies at distance δ from the base s , the volume is $\frac{1}{3}A(s)\delta$, so dividing the momentum balance by $A(s)$ leaves the inertia and the body force bounded by $k\delta/3$; they vanish in the limit while all four surface terms survive. What is left is $\underline{t}(\underline{x}_0; \underline{m}) = \sum_i (\underline{e}_i \cdot \underline{m})\underline{t}(\underline{x}_0; \underline{e}_i)$, and that sum is linear in $\underline{m}$.

The equations of motion

The foregoing construction may be repeated with a small tetrahedron-like region, having a curved oblique surface, adjoined to the boundary of an arbitrary part P_t of a body with piecewise smooth boundary ∂P_t . Such a region approaches a tetrahedron in the limit as it collapses onto a point, the curvature of the oblique face contributing only at higher order in δ exactly as the tangency estimates of (6.86) did, and we again recover (6.94) at the boundary. Every occurrence of $\underline{t}$ in a surface integral may therefore be replaced by $\underline{T}\underline{n}$, and Euler's linear momentum balance (6.36) becomes

$$\int_{P_t} \rho \underline{\dot{v}} dv = \int_{P_t} \rho \underline{b} dv + \int_{\partial P_t} \underline{T}\underline{n} da. \quad (6.108)$$

This is now of the generic vector form (6.2), with the identifications

$$\underline{\psi} = \underline{v}, \quad \underline{r} = \underline{b}, \quad \underline{A} = \underline{T}.$$

Comparison with (6.23), which is (6.2) localized and reduced by conservation of mass, immediately delivers the local form of the linear momentum balance:

$$\rho \dot{\underline{v}} = \rho \underline{b} + \operatorname{div} \underline{T}, \quad (6.109)$$

holding in a region of smoothness, or

$$\rho \dot{v}_i = \rho b_i + T_{ij,j} \quad (6.110)$$

in terms of Cartesian coordinates, where $T_{ij} = \underline{e}_i \cdot \underline{T} \underline{e}_j = \underline{e}_i \cdot \underline{t}(\underline{x}, t; \underline{e}_j) = \underline{e}_i \cdot \underline{t}_j$ by Proposition 1.94 and Theorem 1.33. Thus the traction on a surface with normal $\underline{e}_j$ is $\underline{t}_j = (\underline{e}_i \cdot \underline{t}_j) \underline{e}_i = T_{ij} \underline{e}_i$: reading down the j th column of the stress matrix gives the traction on the j th coordinate plane, its diagonal entry being the normal component and the two off-diagonal entries the shears.

The relevant jump condition at a singular surface is given by (6.11) with the same identifications:

$$[[\rho \underline{v} \otimes (\underline{v} - \underline{u}) - \underline{T}]] \underline{n} = \underline{0}. \quad (6.111)$$

REMARK 6.26 (the second Rankine–Hugoniot condition). Written with the mass flux $m^* = \rho(\underline{v} \cdot \underline{n} - \underline{u})$ of Remark 6.3, and using that m^* is continuous across s by (6.15), the condition (6.111) becomes

$$m^* [[\underline{v}]] = [[\underline{T}]] \underline{n} = [[\underline{t}]],$$

since $[[\rho \underline{v} \otimes (\underline{v} - \underline{u})]] \underline{n} = \rho((\underline{v} - \underline{u}) \cdot \underline{n}) \underline{v} = m^* \underline{v}$ and the scalar m^* passes through the jump bracket. So the traction on s jumps by exactly the momentum being carried across it per unit area per unit time. If no material crosses, $m^* = 0$ and the traction is continuous, which is the familiar statement that a surface of contact transmits the same force to both sides. Together with (6.15) and the energy condition of Problem 11, this is the Rankine–Hugoniot system for a shock, and Remark 5.12 is now discharged.

6.5.4 Rotational momentum balance

The combination of (6.94) with $\underline{x} \times \underline{\dot{v}} = (\underline{x} \times \underline{v})^\cdot$, established after (6.37), reduces the rotational momentum balance (6.37) to

$$\int_{P_t} \rho (\underline{x} \times \underline{v})^\cdot dv = \int_{P_t} \rho \underline{x} \times \underline{\dot{b}} dv + \int_{\partial P_t} \underline{x} \times \underline{T} \underline{n} da, \quad (6.112)$$

which is symbolic notation for the vector equation

$$\varepsilon_i \varepsilon_{ijk} \int_{P_t} \rho (x_j v_k)^\cdot dv = \varepsilon_i \varepsilon_{ijk} \int_{P_t} \rho x_j \dot{b}_k dv + \varepsilon_i \varepsilon_{ijk} \int_{\partial P_t} x_j T_{kl} n_l da, \quad (6.113)$$

where ε_{ijk} is the permutation symbol of Definition 1.23 and $(\underline{a} \times \underline{c})_i = \varepsilon_{ijk} a_j c_k$ by Proposition 1.24.

This is of the form (6.2). Taking (6.24)₂ into account, we have the identifications

$$\psi_i = \varepsilon_{ijk} x_j v_k, \quad r_i = \varepsilon_{ijk} x_j \dot{b}_k \quad \text{and} \quad A_{il} = \varepsilon_{ijk} x_j T_{kl}. \quad (6.114)$$

The third identification is read off the boundary integral: the flux tensor is whatever $\underline{A}$ makes $A_{il} n_l$ the surface integrand, and $\varepsilon_{ijk} x_j T_{kl} n_l = (\underline{x} \times \underline{T} \underline{n})_i$ is exactly that. Inserting (6.114) into the local balance law (6.23), that is into

$$\rho \dot{\psi}_i = \rho r_i + A_{il,l}, \quad (6.115)$$

and expanding both sides gives the result we are after. Take the two sides in turn.

The left side. By the product rule, and using $\dot{x}_j = v_j$ from Definition 2.3,

$$\dot{\psi}_i = \varepsilon_{ijk} (\dot{x}_j v_k + x_j \dot{v}_k) = \varepsilon_{ijk} v_j v_k + \varepsilon_{ijk} x_j \dot{v}_k = \varepsilon_{ijk} x_j \dot{v}_k,$$

since $\varepsilon_{ijk} v_j v_k = 0$: the array $v_j v_k$ is symmetric in j and k while ε_{ijk} is antisymmetric in them, and the contraction of a symmetric array with an antisymmetric one vanishes, by the argument given for Problem 5 of Chapter 1.

The right side. The divergence term is a product and must be differentiated as one:

$$A_{il,l} = \varepsilon_{ijk}(x_{j,l}T_{kl} + x_j T_{kl,l}) = \varepsilon_{ijk}(\delta_{jl}T_{kl} + x_j T_{kl,l}) = \varepsilon_{ijk}T_{kj} + \varepsilon_{ijk}x_j T_{kl,l},$$

using $x_{j,l} = \delta_{jl}$, which is Proposition 1.63: the gradient of the position field is the identity. Notice that the first of the two terms carries no factor of $\underline{x}$ at all. That is the term which will survive, and its appearance is entirely due to differentiating the lever arm rather than the stress.

Substituting both expansions into (6.115) and collecting the terms that carry x_j on the left,

$$\varepsilon_{ijk}x_j\{\rho(\dot{v}_k - b_k) - T_{kl,l}\} = \varepsilon_{ijk}T_{kj}. \quad (6.116)$$

The terms in the braces vanish by (6.110), which says precisely that $\rho\dot{v}_k - \rho b_k - T_{kl,l} = 0$, leaving the algebraic restriction $\varepsilon_{ijk}T_{kj} = 0$. Recalling Problem 5 of Chapter 1 and Remark 1.104, we conclude that $\underline{T}$ is a symmetric spatial tensor,

$$\underline{T} = \underline{T}^t. \quad (6.117)$$

REMARK 6.27 (a sign in the printed (6.116)). The printed text has $+T_{kl,l}$ inside the braces of (6.116). The sign should be negative, as the derivation above shows and as the sentence immediately following it in the book requires: the braces are said to vanish by (6.110), and (6.110) reads $\rho\dot{v}_k = \rho b_k + T_{kl,l}$, that is $\rho(\dot{v}_k - b_k) - T_{kl,l} = 0$. With a plus sign the braces would equal $2T_{kl,l}$ on solutions of (6.110) and nothing would follow.

REMARK 6.28 (what rotational momentum balance did and did not give). This is the one balance law of the chapter that produces no differential equation. Every term involving $\underline{x}$ cancelled, and what remained was an algebraic condition on $\underline{T}$ at each point. The reason is the following.

The moment balance is the force balance multiplied by a lever arm and integrated. Once the force balance holds pointwise, the moment of

the forces about any point is automatically balanced by the moment of the inertia, *except* for the contribution of the lever arm's own variation, which is the term $\varepsilon_{ijk}T_{kj}$ produced by $x_{j,l} = \delta_{jl}$. So the moment balance carries exactly one piece of information beyond the force balance, and that piece is $\varepsilon_{ijk}T_{kj} = 0$.

This is where the non-polar assumption of Remark 6.8 is paid for. Had couple tractions been admitted, (6.112) would carry an extra surface integral $\int_{\partial P_t} \underline{m} \, da$, its divergence would appear on the right of (6.116), and the conclusion would be not $\varepsilon_{ijk}T_{kj} = 0$ but $\varepsilon_{ijk}T_{kj} = -(\text{couple stress divergence})_i$: the skew part of $\underline{T}$ would be determined by the couple stresses instead of vanishing. Symmetry of the Cauchy stress is therefore not a theorem of continuum mechanics as such; it is a theorem of *non-polar* continuum mechanics, and it is the exact mathematical content of that hypothesis. Chapters 8 to 10 use it constantly, for instance in the step $\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{D}$ that opens every discussion of stress power.

The jump condition (6.11) at a singular surface is, with the identifications (6.114),

$$\begin{aligned} \underline{0} &= \underline{e}_i \llbracket \rho \psi_i (v_l - u_l) - A_{il} n_l \rrbracket \\ &= \underline{e}_i \varepsilon_{ijk} \llbracket \rho x_j v_k (v_l - u_l) - x_j T_{kl} n_l \rrbracket. \end{aligned} \quad (6.118)$$

If the position $\underline{x}$ of a material point is continuous across the singular surface, that is if the motion of the body introduces no gaps or fissures in the material, then $\llbracket x_j \rrbracket = 0$ and the x_j may be taken outside the jump brackets, reducing (6.118) to

$$\underline{x} \times \{ \llbracket \rho \underline{v} \otimes (\underline{v} - \underline{u}) - \underline{T} \rrbracket \underline{n} \} = \underline{0}, \quad (6.119)$$

in which the term in braces vanishes by (6.111). In this case the jump condition is identically satisfied and adds nothing. In the event that position is discontinuous, however, (6.118) yields a non-trivial condition, which is one way of seeing that a crack or a cavitating interface is not covered by the smooth theory.

6.5.5 Decomposition of the stress into normal and shear components

Consider the traction acting on the tangent plane to a surface $s \subset \kappa_t$ at the point $\underline{x}$ (Figure 6.11). By Noll's theorem this is the same as the traction on s itself, and by Cauchy's theorem it is $\underline{t} = \underline{T}\underline{n}$, where $\underline{n}$ is a unit normal to the tangent plane. Evidently

$$\underline{t} = \underline{I}\underline{t} = (\underline{n} \otimes \underline{n})\underline{t} + \mathbb{P}_{(\underline{n})}\underline{t}, \quad (6.120)$$

where $\underline{I}$ is the spatial identity and

$$\mathbb{P}_{(\underline{n})} = \underline{I} - \underline{n} \otimes \underline{n} \quad (6.121)$$

is the projection onto the tangent plane, Problem 10 of Chapter 1. All (6.120) does is add and subtract $(\underline{n} \otimes \underline{n})\underline{t}$; its content is that the two pieces so produced are respectively parallel and perpendicular to $\underline{n}$, which is what Problem 10 established.

We write this in the form

$$\underline{t} = \sigma \underline{n} + \tau \underline{s}, \quad (6.122)$$

where

$$\sigma = \underline{n} \cdot \underline{t} \quad \text{and} \quad \tau \underline{s} = \mathbb{P}_{(\underline{n})}\underline{t}, \quad (6.123)$$

with $|\underline{s}| = 1$ and

$$\tau = |\mathbb{P}_{(\underline{n})}\underline{t}|. \quad (6.124)$$

The first equality of (6.123) is $(\underline{n} \otimes \underline{n})\underline{t} = (\underline{n} \cdot \underline{t})\underline{n}$ read off Definition 1.32, and the second merely names the magnitude and direction of the tangential part; $\underline{s}$ is well defined whenever $\tau \neq 0$ and is arbitrary when $\tau = 0$, in which case the term it multiplies vanishes anyway.

We refer to σ and τ respectively as the *normal* and *shear* components of the traction. These are given in terms of the stress tensor by

$$\sigma = \underline{n} \cdot \underline{T}\underline{n} \quad \text{and} \quad \tau = \underline{s} \cdot \underline{T}\underline{n}. \quad (6.125)$$

The first is (6.123)₁ with $\underline{t} = \underline{T}\underline{n}$. For the second, take the inner product of (6.122) with $\underline{s}$ and use the two facts that define $\underline{s}$, namely $\underline{s} \cdot \underline{n} = 0$ and $\underline{s} \cdot \underline{s} = 1$:

$$\underline{s} \cdot \underline{t} = \underline{s} \cdot (\sigma \underline{n} + \tau \underline{s}) = \sigma \underbrace{\underline{s} \cdot \underline{n}}_{=0} + \tau \underbrace{\underline{s} \cdot \underline{s}}_{=1} = \tau,$$

while the left side is $\underline{s} \cdot \underline{T}\underline{n}$ by (6.94).

The normal part can be positive, negative or zero, according as the material on the far side of the plane is pulling, pushing, or doing neither. The shear part is non-negative, being a magnitude by (6.124); its sign has been absorbed into the direction $\underline{s}$.

We observe that

$$\underline{s} \cdot \underline{T}\underline{n} = \underline{n} \cdot \underline{T}^t \underline{s} = \underline{n} \cdot \underline{T}\underline{s},$$

the first equality by the definition of the transpose, Definition 1.41, and the second by the symmetry (6.117). This is called the *complementary property of shear*: the shear traction on the plane with normal $\underline{n}$, resolved along $\underline{s}$, equals the shear traction on the plane with normal $\underline{s}$, resolved along $\underline{n}$. It is the symmetry of the stress made visible, and it is what forces shear stresses on perpendicular faces of a small block to occur in balancing pairs.

It is also clear that the shear traction is determined entirely by $\underline{T}$ and $\underline{n}$, since eliminating σ between (6.122) and (6.125)₁ gives

$$\tau \underline{s} = \underline{T}\underline{n} - (\underline{n} \cdot \underline{T}\underline{n})\underline{n}. \quad (6.126)$$

If τ vanishes on the tangent plane at the point in question, then

$$\underline{T}\underline{n} = \sigma \underline{n}, \quad (6.127)$$

and $\underline{n}$ is a principal vector of $\underline{T}$, in the sense of Definition 3.1, associated with the principal stress σ . Because of the symmetry of $\underline{T}$ there are, by the spectral theorem, Theorem 3.7, three real-valued principal stresses and an associated orthonormal triad of principal vectors. Thus there exist three mutually orthogonal planes intersecting at each $\underline{x} \in \kappa_t$ on

which the shear tractions vanish. These are uniquely determined if all the σ 's are distinct, and Chapter 3's discussion of repeated principal values applies verbatim if they are not.

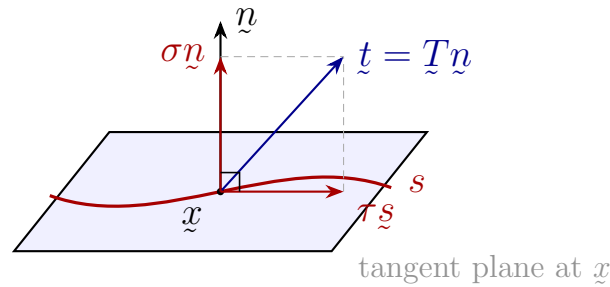


Figure 6.11: Resolution of the traction into normal and shear components. By Noll's theorem the traction on s at $\underline{x}$ is the traction on the tangent plane there, and by Cauchy's theorem it is $\underline{T}\underline{n}$. Splitting it with the projection (6.121) gives $\sigma\underline{n}$ along the normal and $\tau\underline{s}$ in the plane. The normal part $\sigma = \underline{n} \cdot \underline{T}\underline{n}$ may have either sign; the shear part $\tau = |\mathbb{P}_{(\underline{n})}\underline{T}\underline{n}|$ cannot be negative, its sign having been absorbed into $\underline{s}$. Planes on which $\tau = 0$ are exactly the principal planes of $\underline{T}$, and by Theorem 3.7 three mutually perpendicular ones exist at every point.

Examples of states of stress

1. Pure pressure

In a state of pure pressure the traction on every plane containing a point $\underline{x}$ is perpendicular to that plane. Thus $\underline{t}(\underline{x}, t; \underline{n}) = -p(\underline{x}, t)\underline{n}$ for all unit vectors $\underline{n}$, where p is the pressure. Suppressing the passive arguments $\underline{x}$ and t , we have $\underline{T}\underline{n} = -p\underline{n}$. On multiplying by an arbitrary scalar ν we conclude that

$$\underline{T}\underline{v} = -p\underline{v} = (-p\underline{I})\underline{v}, \quad \text{in which } \underline{v} = \nu\underline{n}$$

is an arbitrary vector: every non-zero vector is a positive multiple of a unit vector, and the case $\underline{v} = \underline{0}$ is covered because both sides are then zero by linearity. Two linear maps agreeing on every vector are the same map, by Definition 1.31, so the stress is

$$\underline{T} = -p\underline{I}, \tag{6.128}$$

which is clearly symmetric, as (6.117) requires.

Alternatively, in components, $T_{ij} = \underline{e}_i \cdot \underline{T} \underline{e}_j = \underline{e}_i \cdot \underline{t}(\underline{e}_j) = -p \underline{e}_i \cdot \underline{e}_j = -p \delta_{ij}$, which is the component form of the same result. We may also derive it from (6.107): $\underline{T} = \underline{t}(\underline{x}, t; \underline{e}_i) \otimes \underline{e}_i = -p \underline{e}_i \otimes \underline{e}_i = -p \underline{I}$, the last step by the resolution of the identity $\underline{e}_i \otimes \underline{e}_i = \underline{I}$ of Theorem 1.33.

Note the sign convention. A positive p makes $\underline{t}$ point *into* the region whose exterior normal is $\underline{n}$, that is, the surroundings push. This is why a fluid at rest carries $\underline{T} = -p \underline{I}$ with $p > 0$, and why the hydrostatic example of Chapter 10 finds $p = p_a + \rho g(h - x_3)$ rather than its negative.

2. Uniaxial tension and compression

In this example the traction is of the form $\underline{t}(\underline{x}, t; \underline{n}) = \alpha(\underline{x}, t; \underline{n}) \underline{e}$ for all unit vectors $\underline{n}$, where α is a scalar and $\underline{e}$ is a fixed unit vector. Whatever plane one cuts, the force transmitted is along the one direction $\underline{e}$; only its magnitude depends on the cut.

Thus $\underline{T}(\underline{x}, t) \underline{n} = \alpha(\underline{x}, t; \underline{n}) \underline{e}$. Let $\alpha^*(\underline{x}, t; \cdot)$ be an extension of $\alpha(\underline{x}, t; \cdot)$ from the set of unit vectors to the set of all vectors, so that $\alpha^*(\underline{x}, t; \underline{v}) = \alpha(\underline{x}, t; \underline{v})$ for all $\underline{v}$ with $|\underline{v}| = 1$. Then $\underline{T}(\underline{x}, t) \underline{v} = \alpha^*(\underline{x}, t; \underline{v}) \underline{e}$ for all $\underline{v}$. Clearly $\underline{T}(\underline{x}, t) \underline{v}$ is a linear function of $\underline{v}$, and $\underline{e}$ is a fixed non-zero vector, so $\alpha^*(\underline{x}, t; \underline{v})$ is a linear function of $\underline{v}$ as well. It follows from the Riesz representation theorem, Theorem 1.15, that there is a vector function $\underline{a}(\underline{x}, t)$ with

$$\alpha^*(\underline{x}, t; \underline{v}) = \underline{a}(\underline{x}, t) \cdot \underline{v}.$$

The extension is not a technicality to be skipped. Theorem 1.15 represents a linear functional on a vector space, and the set of unit vectors is not a vector space: an arbitrary linear combination of unit vectors is not a unit vector, and the zero vector is not a unit vector at all. Without extending α to a domain that is closed under addition and scalar multiplication, the word “linear” has no meaning and the theorem has nothing to act on. The extension is available because $\underline{T} \underline{v}$ is already defined for every $\underline{v}$ and $\underline{e} \neq \underline{0}$, so one may simply define α^* by $\alpha^*(\underline{v}) \underline{e} = \underline{T} \underline{v}$.

We thus have $\underline{T}\underline{v} = (\underline{a} \cdot \underline{v})\underline{e} = (\underline{e} \otimes \underline{a})\underline{v}$ for all $\underline{v}$, and hence $\underline{T} = \underline{e} \otimes \underline{a}$. The symmetry condition (6.117) requires $\underline{e} \otimes \underline{a} = \underline{a} \otimes \underline{e}$, and this in turn yields

$$\underline{a} = \underline{a}(\underline{e} \cdot \underline{e}) = (\underline{a} \otimes \underline{e})\underline{e} = (\underline{e} \otimes \underline{a})\underline{e} = \sigma \underline{e}, \quad \text{where } \sigma = \underline{a} \cdot \underline{e}. \quad (6.129)$$

The four equalities are, in order: the first inserts $\underline{e} \cdot \underline{e} = 1$, the second recognises $\underline{a}(\underline{e} \cdot \underline{e})$ as $(\underline{a} \otimes \underline{e})\underline{e}$ by Definition 1.32, the third applies the symmetry just imposed, and the fourth is $(\underline{e} \otimes \underline{a})\underline{e} = \underline{e}(\underline{a} \cdot \underline{e})$, again by Definition 1.32. So symmetry forces $\underline{a}$ to be parallel to $\underline{e}$. Finally,

$$\underline{T}(\underline{x}, t) = \sigma(\underline{x}, t) \underline{e} \otimes \underline{e}, \quad (6.130)$$

where σ is the uniaxial tension or compression according as it is positive or negative.

REMARK 6.29 (what symmetry cost the uniaxial state). Before symmetry was imposed, the state $\underline{T} = \underline{e} \otimes \underline{a}$ had three free parameters, the components of $\underline{a}$; after it, one, namely σ . Two constraints were spent, which is right: the skew part of a three-dimensional tensor has three components, and $\text{skw}(\underline{e} \otimes \underline{a}) = \frac{1}{2}(\underline{e} \otimes \underline{a} - \underline{a} \otimes \underline{e})$ vanishes exactly when $\underline{a} \parallel \underline{e}$, one condition of codimension two on the direction of $\underline{a}$.

The state (6.130) is exactly the reactive stress found for an inextensible fibre family in (10.28), with $\underline{e}$ the fibre direction, and its traction on a plane with normal $\underline{n}$ is $\sigma(\underline{e} \cdot \underline{n})\underline{e}$: proportional to the cosine of the angle between the cut and the fibre, and always directed along the fibre. Cut parallel to $\underline{e}$ and nothing at all is transmitted, which is what one would expect of a bundle of strings.

6.5.6 Referential equations of motion

The reader will have noticed that the local equations and their jump conditions have been obtained entirely in the spatial description. In particular we have not used the referential expressions (6.25)₂, (6.27)₂ or (6.29) in deducing local equations from the linear momentum balance

(6.34). We do so here for the local equations and leave the associated jump conditions to Problem 1.

Combining (6.38) with (6.29) gives the linear momentum balance for an arbitrary part $P \subset \kappa$:

$$\int_{\partial P} \underline{\underline{p}} \, dA = \int_P \rho_\kappa (\dot{\underline{\underline{v}}} - \underline{\underline{b}}) \, dV, \quad (6.131)$$

where $\underline{\underline{p}}$ is the Piola traction.

| READER'S NOTE (*assembling (6.131) from the axiom*). AXIOM

$$\frac{d}{dt} \underline{\underline{l}}(S, t) = \underline{\underline{f}}(S, t) = \underline{\underline{f}}_b(S, t) + \underline{\underline{f}}_c(S, t) \quad (6.34), (6.30)$$

No configuration is named: S is a set of material points.

PULL BACK, ONE LINE EACH

$$\rho \, dv = \rho(J \, dV) = (\rho J) \, dV = \rho_\kappa \, dV \quad (2.50) \times (2.59)$$

$$\underline{\underline{l}} = \int_{P_t} \rho \underline{\underline{v}} \, dv = \int_P \rho_\kappa \underline{\underline{v}} \, dV \quad (6.25)_2$$

$$\frac{d}{dt} \int_P \rho_\kappa \underline{\underline{v}} \, dV = \int_P \rho_\kappa \frac{\partial \hat{\underline{\underline{v}}}(X, t)}{\partial t} \, dV = \int_P \rho_\kappa \dot{\underline{\underline{v}}} \, dV \quad \dot{\rho}_\kappa = 0, \quad (6.18)$$

$$\underline{\underline{f}}_b = \int_{P_t} \rho \underline{\underline{b}} \, dv = \int_P \rho_\kappa \underline{\underline{b}} \, dV \quad (6.27)_2$$

$$\underline{\underline{f}}_c = \int_{\partial P} \underline{\underline{p}} \, dA \quad (6.29), \text{ definition of } \underline{\underline{p}}$$

ASSEMBLE

$$\int_P \rho_\kappa \dot{\underline{\underline{v}}} \, dV = \int_P \rho_\kappa \underline{\underline{b}} \, dV + \int_{\partial P} \underline{\underline{p}} \, dA \quad (6.38)$$

$$\therefore \int_{\partial P} \underline{\underline{p}} \, dA = \int_P \rho_\kappa (\dot{\underline{\underline{v}}} - \underline{\underline{b}}) \, dV \quad \text{linearity} = (6.131)$$

SHAPE $\int_{\partial P} (\text{unknown}) \, dA = \int_P (\text{bounded}) \, dV$, that is $O(\ell^2) = O(\ell^3)$ under shrinking. This is the sole input to Noll and to Cauchy, and it recurs at (6.79) and at (6.165).

CAUTION Mass conservation is spent once, in the third line; with $\dot{\rho}_\kappa \neq 0$ a term $\int \dot{\rho}_\kappa \underline{\underline{v}} \, dV$ survives and (6.131) is false. Nothing is yet known about $\underline{\underline{p}}$: (6.29) only names it.

We could use this in a version of Noll's theorem, run with regions in κ rather than in κ_t , to conclude that $\underline{\underline{p}}$ depends on ∂P through its local

unit normal $\underline{N}$, and then adapt Cauchy's theorem, with a tetrahedron in κ , to conclude that

$$\underline{p}(\underline{X}, t; \underline{N}) = \underline{P}(\underline{X}, t) \underline{N}, \quad (6.132)$$

where $\underline{P}$, the *Piola stress*, is a two-point tensor field, with Cartesian-component representation

$$\underline{P} = P_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A. \quad (6.133)$$

Neither argument needs any change: both proceed by shrinking a region and comparing a surface measure with a volume measure, and both are indifferent to which configuration the region lives in. Nothing in them used the motion.

| READER'S NOTE (*Noll and Cauchy read again in κ*). DICTIONARY

$$\kappa_t \rightarrow \kappa, \quad \underline{x} \rightarrow \underline{X}, \quad \underline{n} \rightarrow \underline{N}, \quad da \rightarrow dA, \quad dv \rightarrow dV, \quad \underline{t} \rightarrow \underline{p}.$$

NOLL The traction entered only through $|\int_\sigma \underline{t} da| \leq (\max_\sigma |\underline{t}|) A(\sigma)$ and the boundedness of the volume term. Under the dictionary the bound is the same statement, and the volume term is $\rho_\kappa(\dot{\underline{v}} - \underline{b})$, bounded whenever $\rho_\kappa, \dot{\underline{v}}, \underline{b}$ are continuous. Hence

$$\underline{p} = \underline{p}(\underline{X}, t; \underline{N}), \quad \underline{p}(\underline{X}, t; -\underline{N}) = -\underline{p}(\underline{X}, t; \underline{N}).$$

CAUCHY Tetrahedron in κ : vertex $\underline{X}_0$, coordinate faces on $\{\hat{\underline{E}}_A\}$, oblique normal $\underline{M}$, height δ .

$$A_A = A(s) \hat{\underline{E}}_A \cdot \underline{M} \quad (6.104) \text{ in } \kappa$$

$$\frac{1}{A(s)} \int_{\partial\tau} \underline{p} dA = \underline{p}(\underline{X}^*; \underline{M}) - \sum_A (\hat{\underline{E}}_A \cdot \underline{M}) \underline{p}(\underline{X}_A^*; \hat{\underline{E}}_A) \quad (6.105) \text{ in } \kappa$$

$$\left| \frac{1}{A(s)} \int_{\partial\tau} \underline{p} dA \right| \leq \max_\tau |\rho_\kappa(\dot{\underline{v}} - \underline{b})| \frac{V(\tau)}{A(s)} = O(\delta) \quad (6.131), \text{ Lem. 6.23}$$

$$\delta \rightarrow 0: \quad \underline{p}(\underline{X}_0, t; \underline{M}) = \underline{P} \underline{M}, \quad \underline{P} = \underline{p}(\underline{X}_0, t; \hat{\underline{E}}_A) \otimes \hat{\underline{E}}_A \quad (6.132), (6.133)$$

TYPE $\underline{N} \in \hat{\mathbb{E}}^3$ in, $\underline{p} \in \mathbb{E}^3$ out $\Rightarrow \underline{P}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$. The two-point character is forced by reckoning the surface in κ and the force in κ_t ; it is not a convention. Consequence: Remark 6.31.

We can combine this with the referential form of the divergence theorem, for smooth and hence continuous fields, to obtain

$$\int_P \rho_\kappa(\dot{\underline{v}} - \underline{b}) dV = \int_{\partial P} \underline{P} \underline{N} dA = \int_P \text{Div } \underline{P} dV, \quad (6.134)$$

where, in terms of Cartesians,

$$\text{Div } \underline{P} = P_{iA,A} \underline{e}_i, \quad (6.135)$$

the comma now denoting differentiation with respect to X_A , and then localize by Theorem 2.20 to conclude that

$$\rho_\kappa \dot{\underline{v}} = \rho_\kappa \underline{b} + \text{Div } \underline{P}, \quad (6.136)$$

that is

$$\rho_\kappa \dot{v}_i = \rho_\kappa b_i + P_{iA,A}, \quad (6.137)$$

at all points $\underline{X} \in \kappa$ that do not belong to a singular surface.

| READER'S NOTE (*Div $\underline{P}$ in components, and the localization*). WHAT $\underline{P}\underline{N}$ IS

$$\underline{P}\underline{N} = (P_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A) \underline{N} = P_{iA} (\hat{\underline{E}}_A \cdot \underline{N}) \underline{e}_i = P_{iA} N_A \underline{e}_i, \quad \underline{e}_i \cdot \underline{P}\underline{N} = P_{iA} N_A.$$

THREE ORDINARY VECTOR FIELDS Freeze i and put $\underline{p}^{(i)} = P_{iA} \hat{\underline{E}}_A \in \hat{\mathbb{E}}^3$, so that $\underline{e}_i \cdot \underline{P}\underline{N} = \underline{p}^{(i)} \cdot \underline{N}$. Then

$$\begin{aligned} \int_{\partial P} \underline{p}^{(i)} \cdot \underline{N} \, dA &= \int_P \text{Div } \underline{p}^{(i)} \, dV = \int_P P_{iA,A} \, dV \quad (5.1) \text{ in } \kappa \\ \times \underline{e}_i, \sum_i : \int_{\partial P} \underline{P}\underline{N} \, dA &= \int_P P_{iA,A} \underline{e}_i \, dV = \int_P \text{Div } \underline{P} \, dV \quad (6.134), (6.135) \end{aligned}$$

the sum over i being legitimate because $\{\underline{e}_i\}$ is fixed (Remark 5.3).

TYPE CHECK $\partial/\partial X_A$ makes a referential index; the only referential index on $\underline{P}$ is A ; so Div closes the referential leg and $\text{Div } \underline{P} \in \mathbb{E}^3$, the space of $\rho_\kappa \dot{\underline{v}}$. Writing $\text{div } \underline{P}$ would leave a free referential index and equate a $\mathbb{E}^3$ vector to an $\hat{\mathbb{E}}^3$ vector.

LOCALIZE

$$\int_P [\rho_\kappa (\dot{\underline{v}} - \underline{b}) - \text{Div } \underline{P}] \, dV = 0 \quad \forall P \xrightarrow{\text{Thm. 2.20}} \boxed{\rho_\kappa \dot{\underline{v}} = \rho_\kappa \underline{b} + \text{Div } \underline{P}}$$

componentwise (6.137). On a singular surface the integrand is not continuous and the step fails: hence the proviso, and Problem 1.

REMARK 6.30 (why anyone bothers with (6.136)). Equation (6.136) looks like (6.109) with hats on, and one may reasonably ask what has been gained by rewriting a correct equation. Two things, and both are practical.

The density is known. In (6.136) the coefficient $\rho_\kappa(\underline{X})$ is a specified function, part of the description of the body, by (6.18). In (6.109) the spatial density $\rho(\underline{x}, t)$ is an unknown field which must satisfy the partial differential equation (6.14), involving the velocity, and is not known in advance. So the referential formulation solves conservation of mass for free and reduces the number of unknown fields by one.

The acceleration is a partial derivative. In the referential description $\dot{\underline{v}} = \partial \hat{\underline{v}}(\underline{X}, t) / \partial t$ is simply the partial time derivative at fixed $\underline{X}$: no convective term, since material points do not move in κ . In the spatial description $\dot{\underline{v}} = \underline{v}' + (\text{grad } \underline{v})\underline{v}$ by (2.21), and the second term is quadratic in the unknown. The nonlinearity has not disappeared, it has moved into the constitutive relation between $\underline{P}$ and the deformation, which is where a solid mechanician wants it.

The price is that the domain κ must be known, which is why the referential form is the natural one for solids and the spatial form the natural one for fluids.

The relation between the two stresses

The connection (2.59) between the referential and spatial mass densities, combined with (6.109), suggests that if $\underline{P}$ is such that

$$J \operatorname{div} \underline{T} = \operatorname{Div} \underline{P}, \quad (6.138)$$

then the referential and spatial forms of the balance of linear momentum are equivalent. Multiplying (6.109) by $J > 0$ and substituting $J\rho = \rho_\kappa$ from (2.59) term by term,

$$\begin{aligned} \rho \dot{\underline{v}} &= \rho \underline{b} + \operatorname{div} \underline{T} & \Big| \times J \\ J\rho \dot{\underline{v}} &= J\rho \underline{b} + J \operatorname{div} \underline{T} \\ \rho_\kappa \dot{\underline{v}} &= \rho_\kappa \underline{b} + J \operatorname{div} \underline{T}, \end{aligned}$$

and comparing the last line with the referential balance (6.136),

$$\rho_\kappa \dot{\underline{v}} = \rho_\kappa \underline{b} + \operatorname{Div} \underline{P},$$

the two agree for every motion precisely when their third terms do, that is, precisely when $J \operatorname{div} \underline{T} = \operatorname{Div} \underline{P}$. So (6.138) is not an extra assumption about $\underline{P}$ but the exact condition under which the two descriptions carry the same content.

| **READER'S NOTE** (*why the factor is J , and what (6.138) is*). DENSITIES

$$(6.109) : \text{force per unit } dv, \quad (6.136) : \text{force per unit } dV, \quad \frac{dv}{dV} = J \quad (2.50).$$

Sanity check on the first term: $J\rho\dot{\underline{v}} = \rho_\kappa\dot{\underline{v}}$ by (2.59), and ρ_κ is mass per unit reference volume.

QUANTIFIER

$$\left. \begin{array}{l} \rho_\kappa\dot{\underline{v}} = \rho_\kappa\dot{\underline{b}} + J \operatorname{div} \underline{T} \\ \rho_\kappa\dot{\underline{v}} = \rho_\kappa\dot{\underline{b}} + \operatorname{Div} \underline{P} \end{array} \right\} \text{ for every motion } \iff J \operatorname{div} \underline{T} = \operatorname{Div} \underline{P}.$$

MORAL Both arrows hold, so (6.138) is an equivalence, not an assumption about $\underline{P}$. Nor is it an equation to be solved: (6.139) constructs the $\underline{P}$ that satisfies it and Problem 8 verifies it afterwards.

To determine the requisite expression for $\underline{P}$ we combine (6.94) with (2.50) and the Piola–Nanson formula (2.54), obtaining

$$\begin{aligned} \int_P J \operatorname{div} \underline{T} dV &= \int_{P_t} \operatorname{div} \underline{T} dv = \int_{\partial P_t} \underline{T} \underline{n} da = \int_{\partial P} \underline{T} \underline{F}^* \underline{N} dA \\ &= \int_P \operatorname{Div} (\underline{T} \underline{F}^*) dV. \end{aligned} \quad (6.139)$$

Read the chain from left to right. The first equality is the change of variables $dv = J dV$ of (2.50). The second is the divergence theorem (5.7)₂ in κ_t . The third substitutes $\underline{n} da = \underline{F}^* \underline{N} dA$, Piola–Nanson (2.54), and moves the constant-in-the-integration tensor $\underline{T}$ through: $\underline{T}(\underline{F}^* \underline{N}) = (\underline{T} \underline{F}^*) \underline{N}$ is associativity of composition. The fourth is the divergence theorem again, this time in κ . Note that $\underline{T} \underline{F}^*$ is a two-point tensor, $\underline{F}^*: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ followed by $\underline{T}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$, so Div is the right operator for it, in the sense of (6.135).

Collecting the first and last integrals of (6.139) and localizing, which is legitimate since P is arbitrary and the integrands are continuous, gives (6.138) together with

$$\underline{P} = \underline{T} \underline{F}^*, \quad (6.140)$$

or, in terms of components,

$$P_{iA} = T_{ij}F_{jA}^*. \quad (6.141)$$

Problem 8 establishes (6.138) directly, by differentiating (6.141) and invoking the Piola identity (5.11); that calculation is the one that shows where $\text{Div } \underline{\underline{F}}^* = \underline{\underline{0}}$ is needed.

| **READER'S NOTE** (*the chain (6.139) in components*).

$$\begin{aligned} \int_P J T_{ij,j} \, dV &= \int_{P_t} T_{ij,j} \, dv && dv = J \, dV, \quad (2.50) \\ &= \int_{\partial P_t} T_{ij} n_j \, da && \text{div. thm. in } \kappa_t, \quad (5.7)_2 \\ &= \int_{\partial P} T_{ij} F_{jA}^* N_A \, dA && n_j \, da = F_{jA}^* N_A \, dA, \quad (2.54) \\ &= \int_{\partial P} P_{iA} N_A \, dA && P_{iA} := T_{ij} F_{jA}^* \quad (a \text{ choice}) \\ &= \int_P P_{iA,A} \, dV && \text{div. thm. in } \kappa, \quad (5.1) \end{aligned}$$

THE ONE STEP THAT IS NOT BOOKKEEPING The fourth: a product is given a name. It works because the contracted j is spatial on both factors, leaving one spatial and one referential index — the structure (6.133) demands. In direct notation, $\underline{\underline{T}}(\underline{\underline{F}}^* \underline{\underline{N}}) = (\underline{\underline{T}} \underline{\underline{F}}^*) \underline{\underline{N}}$, and $\underline{\underline{F}}^*: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ followed by $\underline{\underline{T}}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$ admits only this composite.

LOCALIZE

$$\int_P [J T_{ij,j} - P_{iA,A}] \, dV = 0 \quad \forall P \implies \boxed{J \, \text{div } \underline{\underline{T}} = \text{Div } \underline{\underline{P}}, \quad \underline{\underline{P}} = \underline{\underline{T}} \underline{\underline{F}}^*}$$

one result and the definition that makes it true, not two results.

Indeed, the two expressions (6.28) and (6.29) for the contact force imply that

$$\int_{\partial P} \underline{\underline{p}} \, dA = \int_{\partial P_t} \underline{\underline{t}} \, da = \int_{\partial P_t} \underline{\underline{T}} \underline{\underline{n}} \, da = \int_{\partial P} \underline{\underline{T}} \underline{\underline{F}}^* \underline{\underline{N}} \, dA, \quad (6.142)$$

which is identically satisfied by (6.132) and (6.140). Since ∂P is arbitrary this also gives the pointwise relation $\underline{\underline{p}} \, dA = \underline{\underline{t}} \, da$ promised in Remark 6.7: the Piola traction is the Cauchy traction rescaled by the ratio of the two area elements, and it is larger or smaller than $\underline{\underline{t}}$ according as the surface has contracted or expanded.

| READER'S NOTE ($\underline{p} \, dA = \underline{t} \, da$, and one case done in the head). GET IT

$$\int_{\partial P} \underline{p} \, dA = \int_{\partial P_t} \underline{t} \, da \quad \forall \partial P \quad \Longrightarrow \quad \underline{p} \, dA = \underline{t} \, da$$

one force, two areas; localization on area rather than volume.

THREE CONSEQUENCES

$$\frac{|\underline{p}|}{|\underline{t}|} = \frac{da}{dA} = |\underline{F}^* \underline{N}| \quad (2.54)$$

$\underline{p} \parallel \underline{t}$ a positive scalar, never a rotation

$$\underline{P} \underline{N} = \underline{T} \underline{F}^* \underline{N} \quad \forall \underline{N} \quad \Longrightarrow \quad \underline{P} = \underline{T} \underline{F}^* \quad \text{third route to (6.140)}$$

CHECK: UNIFORM DILATATION $x_i = \lambda \delta_{iA} X_A$

$$\begin{aligned} F_{iA} &= \lambda \delta_{iA}, & J &= \lambda^3, & F_{iA}^* &= \lambda^3 \lambda^{-1} \delta_{iA} = \lambda^2 \delta_{iA}, \\ da &= \lambda^2 dA, & \underline{p} &= \lambda^2 \underline{t}, & P_{iA} &= T_{ij} F_{jA}^* = \lambda^2 T_{iA}. \quad \checkmark \end{aligned}$$

Stretch by two in every direction: the force on a material face is unchanged, its area is four times larger, $\underline{t}$ falls by four, $\underline{p}$ does not move.

The symmetry of the Cauchy stress imposes a restriction on the Piola stress and the deformation jointly. This follows on combining (6.140) with (2.55), $\underline{F}^* = J \underline{F}^{-t}$:

$$\underline{P} \underline{F}^t = \underline{T} \underline{F}^* \underline{F}^t = \underline{T} (J \underline{F}^{-t}) \underline{F}^t = J \underline{T} (\underline{F}^t)^{-1} \underline{F}^t = J \underline{T},$$

which is symmetric by (6.117). A tensor equal to its own transpose gives $\underline{P} \underline{F}^t = (\underline{P} \underline{F}^t)^t = \underline{F} \underline{P}^t$, so

$$\underline{P} \underline{F}^t = \underline{F} \underline{P}^t, \quad (6.143)$$

that is $P_{iA} F_{jA} = F_{iA} P_{jA}$.

| READER'S NOTE ((6.143) in components, and how many conditions). DERIVE

$$\underline{P} \underline{F}^t = \underline{T} \underline{F}^* \underline{F}^t = \underline{T} (J \underline{F}^{-t}) \underline{F}^t = J \underbrace{\underline{T} (\underline{F}^t)^{-1} \underline{F}^t}_{\underline{I}} = J \underline{T} \quad (6.140), \quad (2.55)$$

$$\underline{P} \underline{F}^t = (\underline{P} \underline{F}^t)^t = (\underline{F}^t)^t \underline{P}^t = \underline{F} \underline{P}^t \quad \underline{T} = \underline{T}^t, \quad (6.117); \quad J > 0$$

with $\underline{F}^{-t} := (\underline{F}^t)^{-1} = (\underline{F}^{-1})^t$, the two agreeing because transposing $\underline{F}^{-1} \underline{F} = \hat{\underline{I}}$ gives $\underline{F}^t (\underline{F}^{-1})^t = \hat{\underline{I}}$.

COMPONENTS

$$M_{ij} := (\underline{P} \underline{F}^t)_{ij} = P_{iA} (\underline{F}^t)_{Aj} = P_{iA} F_{jA}, \quad M_{ij} = M_{ji} \iff P_{iA} F_{jA} = F_{iA} P_{jA}.$$

Both free indices are spatial, so symmetry is askable. “ $P_{iA} = P_{Ai}$ ” is not a sentence: i and A run over different alphabets.

COUNT $M_{12} = M_{21}$, $M_{13} = M_{31}$, $M_{23} = M_{32}$: three conditions, $9 \rightarrow 6$, the same count as $\underline{T} = \underline{T}^t$. But the coefficients are the F_{iA} , so *which* six depends on $\underline{F}$; there is no fixed six-dimensional subspace of two-point tensors to work in. Hence (6.144), whose $\underline{S} = \underline{S}^t$ constrains $\underline{S}$ alone. Problem 6 runs this arithmetic on numbers.

REMARK 6.31 (why $\underline{P}$ is not symmetric, and what (6.143) means). Asking whether $\underline{P}$ is symmetric is asking a meaningless question, and Remark 2.33 of Chapter 2 explains why: $\underline{P}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$, so $\underline{P}^t$ maps $\mathbb{E}^3$ to $\hat{\mathbb{E}}^3$, and two maps between different pairs of spaces cannot be compared. Symmetry is defined only for a tensor mapping a space to itself.

What (6.143) says instead is that the particular *spatial* tensor $\underline{P}\underline{F}^t$, which does map $\mathbb{E}^3$ to $\mathbb{E}^3$, is symmetric. It is a restriction on $\underline{P}$ and $\underline{F}$ together, three scalar conditions rather than the three that $\underline{T} = \underline{T}^t$ imposes, and it is not a restriction on $\underline{P}$ alone: given any $\underline{F}$, the nine components of $\underline{P}$ are cut down to six by (6.143), but which six depends on $\underline{F}$. This is the reason that the purely referential stress of (6.144) is introduced: it restores an honest symmetry.

The purely referential *Piola–Kirchhoff stress* $\underline{S}$, defined by

$$\underline{P} = \underline{F}\underline{S}, \quad (6.144)$$

is often useful in the formulation of constitutive equations, especially for solids. Its relationship to the Cauchy stress follows from (6.140) and (2.55): substituting (6.144) into $\underline{P}\underline{F}^t = \underline{J}\underline{T}$, computed just above,

$$\underline{J}\underline{T} = \underline{F}\underline{S}\underline{F}^t. \quad (6.145)$$

Note that $\underline{S} = \underline{F}^{-1}\underline{P}$ maps $\hat{\mathbb{E}}^3$ to $\hat{\mathbb{E}}^3$, since $\underline{P}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ and $\underline{F}^{-1}: \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, so asking whether it is symmetric is a legitimate question. It is, and the proof is three lines. Transpose (6.145), using $(\underline{A}\underline{B}\underline{C})^t = \underline{C}^t\underline{B}^t\underline{A}^t$ and $\underline{J}^t = \underline{J}$ for the scalar:

$$\underline{J}\underline{T}^t = (\underline{F}\underline{S}\underline{F}^t)^t = (\underline{F}^t)^t \underline{S}^t \underline{F}^t = \underline{F}\underline{S}^t \underline{F}^t.$$

The Cauchy stress is symmetric by (6.117), so the left side is $J\tilde{T} = \tilde{F}\tilde{S}\tilde{F}^t$, again by (6.145). Equating,

$$\tilde{F}\tilde{S}^t\tilde{F}^t = \tilde{F}\tilde{S}\tilde{F}^t.$$

Both $\tilde{F}$ and $\tilde{F}^t$ are invertible, $J = \det \tilde{F} \neq 0$, so multiplying on the left by $\tilde{F}^{-1}$ and on the right by $\tilde{F}^{-t}$ removes them:

$$\tilde{F}^{-1}\tilde{F}\tilde{S}^t\tilde{F}^t\tilde{F}^{-t} = \tilde{F}^{-1}\tilde{F}\tilde{S}\tilde{F}^t\tilde{F}^{-t} \implies \tilde{S}^t = \tilde{S}.$$

So the symmetry of $\tilde{T}$, which came from Euler's second postulate, transfers to $\tilde{S}$ but not to $\tilde{P}$; the difference is that the congruence $\tilde{A} \mapsto \tilde{F}\tilde{A}\tilde{F}^t$ preserves symmetry while the one-sided product $\tilde{A} \mapsto \tilde{F}\tilde{A}$ does not. In terms of components,

$$P_{iA} = F_{iB}S_{BA} \quad \text{and} \quad JT_{ij} = F_{iA}F_{jB}S_{AB}. \quad (6.146)$$

| READER'S NOTE (*the components of (6.146), and the six conversions*). COMPONENTS

$$P_{iA} = F_{iB}S_{BA} \quad \tilde{P} = \tilde{F}\tilde{S}, \text{ (6.144); closes the referential } B$$

$$(\tilde{F}\tilde{S}\tilde{F}^t)_{ij} = F_{iA}S_{AB}(\tilde{F}^t)_{Bj} = F_{iA}F_{jB}S_{AB} = JT_{ij} \text{ (6.145); } S_{AB} = S_{BA} \Rightarrow i \leftrightarrow j$$

THE SIX CONVERSIONS

	from $\tilde{T}$	from $\tilde{P}$	from $\tilde{S}$
$\tilde{T}$	— — —	$J^{-1}\tilde{P}\tilde{F}^t$	$J^{-1}\tilde{F}\tilde{S}\tilde{F}^t$
$\tilde{P}$	$\tilde{T}\tilde{F}^* = J\tilde{T}\tilde{F}^{-t}$	— — —	$\tilde{F}\tilde{S}$
$\tilde{S}$	$J\tilde{F}^{-1}\tilde{T}\tilde{F}^{-t}$	$\tilde{F}^{-1}\tilde{P}$	— — —

Two are definitions, (6.140) and (6.144); the rest follow by multiplying by $\tilde{F}$, $\tilde{F}^{-1}$ or $J^{\pm 1}$ on the right side.

THE SAME TABLE AS A TYPE CHECK

	maps	indices	symmetric?	force per unit
$\tilde{T}$	$\mathbb{E}^3 \rightarrow \mathbb{E}^3$	T_{ij}	yes, (6.117)	da
$\tilde{P}$	$\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$	P_{iA}	meaningless	dA
$\tilde{S}$	$\hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^3$	S_{AB}	yes	dA, pulled back

An equation in this chapter is wrong the moment a spatial index is set equal to a referential one. Problem 7 computes all three for one deformation.

WHAT $\underline{S}$ COSTS $\underline{S}\underline{N}$ is the force on nothing: the pull-back $\underline{F}^{-1}$ has undone the geometry along with the bookkeeping. $\underline{T}$ is what a gauge reads, $\underline{P}$ is what a free-body diagram in κ balances, $\underline{S}$ is what a constitutive equation is written in.

REMARK 6.32 (three stresses, one state). The three tensors $\underline{T}$, $\underline{P}$ and $\underline{S}$ describe the same physical state and are related by invertible transformations, so no information is gained or lost in passing between them. What differs is what each is convenient for.

$\underline{T}$ is what is measured. Force per unit *current* area is what a gauge on the deformed body reads, and $\underline{T}$ is the only one of the three that is a spatial tensor and symmetric.

$\underline{P}$ is what balances. It appears in (6.136) with the known coefficient ρ_κ and a divergence taken over a fixed domain. It is a two-point tensor and satisfies no symmetry of its own.

$\underline{S}$ is what a constitutive equation gives. It is referential and symmetric, so it has six components and lives in the same space as $\underline{C}$ and $\underline{E}$. Chapter 8 will write constitutive equations as maps from $\underline{C}$ to $\underline{S}$ for exactly this reason, and (6.160) below shows that $\underline{S}$ is the stress power conjugate to $\underline{E}$.

A useful mnemonic for the factors: $\underline{P} = \underline{T}\underline{F}^*$ converts area elements, since $\underline{F}^*$ is the tensor that carries $\underline{n} \, da$ to $\underline{N} \, dA$ by Remark 5.10, and $\underline{S} = \underline{F}^{-1}\underline{P}$ then pulls the remaining spatial leg back to κ . Every factor of $\underline{F}$, $\underline{F}^*$ or $\underline{J}$ in this subsection is there to convert a measure, and nothing else.

6.5.7 Mechanical energy balance

In particle mechanics the power generated by the force $\underline{f}$ acting on a particle is balanced by the rate of change of the kinetic energy of the particle. This follows from Newton's law $\underline{f} = m\dot{\underline{v}}$, where m and $\underline{v}$ are the particle mass and velocity: the power is

$$\mathcal{P} = \underline{f} \cdot \underline{v} = m \underline{v} \cdot \dot{\underline{v}} = \frac{1}{2}m(\underline{v} \cdot \dot{\underline{v}} + \dot{\underline{v}} \cdot \underline{v}) = \frac{1}{2}m(\underline{v} \cdot \underline{v})',$$

the middle step by the symmetry of the inner product, the last by the product rule read backwards. Thus $\mathcal{P} = \dot{\mathcal{K}}$, where $\mathcal{K} = \frac{1}{2}m|\underline{v}|^2$ is the kinetic energy. We derive an analogous result for continua, and the interest lies entirely in the extra term that appears.

To achieve this we form the inner product of (6.109) with the material velocity $\underline{v}$, obtaining

$$\begin{aligned} \frac{1}{2}\rho(|\underline{v}|^2)^\cdot &= \rho \underline{b} \cdot \underline{v} + \operatorname{div} \underline{T} \cdot \underline{v} \\ &= \rho \underline{b} \cdot \underline{v} + \operatorname{div} (\underline{T}^t \underline{v}) - \underline{T} \cdot \underline{L}, \end{aligned} \quad (6.147)$$

where $\underline{L}$ is the spatial velocity gradient defined by (4.1), the tensor inner product is that of Definition 1.57, and use has been made of the identity of Proposition 1.95, $\underline{u} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{u}) - \operatorname{tr}[\underline{A}^t \operatorname{grad} \underline{u}]$, together with $\operatorname{tr}(\underline{T}^t \underline{L}) = T_{ij}L_{ij} = \underline{T} \cdot \underline{L}$ from Theorem 1.58. The left-hand side is $\rho \dot{\underline{v}} \cdot \underline{v} = \frac{1}{2}\rho(\underline{v} \cdot \underline{v})^\cdot$ by the same product rule as in the particle case.

A reduction employing Cartesian coordinates is arguably more enlightening:

$$\operatorname{div} \underline{T} \cdot \underline{v} = T_{ij,j}v_i = (T_{ij}v_i)_{,j} - T_{ij}v_{i,j} = \operatorname{div} (\underline{T}^t \underline{v}) - \underline{T} \cdot \underline{L}. \quad (6.148)$$

The middle equality is the product rule for a single ordinary derivative, run backwards; the identification of the two resulting terms uses $(\underline{T}^t \underline{v})_j = T_{ij}v_i$ and $v_{i,j} = L_{ij}$ from (4.1). So the whole content of (6.147) is that $T_{ij,j}v_i$ is not a divergence by itself and becomes one only at the cost of the term $T_{ij}v_{i,j}$.

| **READER'S NOTE** ((6.147) *assembled from nothing*).

$$\begin{aligned} \rho \dot{\underline{v}} &= \rho \underline{b} + \operatorname{div} \underline{T} && (6.109) \\ \cdot \cdot \underline{v} : \quad \rho \dot{\underline{v}} \cdot \underline{v} &= \rho \underline{b} \cdot \underline{v} + \operatorname{div} \underline{T} \cdot \underline{v} && \text{linearity of } \cdot \cdot \\ (\underline{v} \cdot \underline{v})^\cdot &= 2 \dot{\underline{v}} \cdot \underline{v} \implies \dot{\underline{v}} \cdot \underline{v} = \frac{1}{2}(|\underline{v}|^2)^\cdot && \text{product rule, } \cdot \cdot \cdot \text{ symmetric} \\ \underline{v} \cdot \operatorname{div} \underline{T} &= \operatorname{div} (\underline{T}^t \underline{v}) - \operatorname{tr} [\underline{T}^t \operatorname{grad} \underline{v}] && \text{Prop. 1.95} \\ \operatorname{tr} (\underline{T}^t \underline{L}) &= T_{ij}L_{ij} = \underline{T} \cdot \underline{L} && \operatorname{grad} \underline{v} = \underline{L}, (4.1); \text{ Thm. 1.58} \\ \therefore \quad \frac{1}{2}\rho(|\underline{v}|^2)^\cdot &= \rho \underline{b} \cdot \underline{v} + \operatorname{div} (\underline{T}^t \underline{v}) - \underline{T} \cdot \underline{L} && (6.147) \end{aligned}$$

THE MECHANISM, IN INDICES

$$\underbrace{T_{ij,j}v_i}_{\operatorname{div} \underline{T} \cdot \underline{v}} = \underbrace{(T_{ij}v_i)_{,j}}_{\operatorname{div}(\underline{T}^t \underline{v})} - \underbrace{T_{ij}v_{i,j}}_{\underline{T} \cdot \underline{L}}, \quad (\underline{T}^t \underline{v})_j = T_{ij}v_i, \quad v_{i,j} = L_{ij}.$$

Read right to left: $T_{ij}v_i$ is a vector, its divergence integrates to a boundary term, and the price of manufacturing it is $T_{ij}v_{i,j}$. A particle has no $v_{i,j}$; hence no stress power.

Integrating (6.147) over an arbitrary part $P_t = \chi(S, t) \subset \kappa_t$ and invoking conservation of mass in the form (6.24)₁ with $\varphi = \frac{1}{2}|v|^2$ gives

$$\frac{d}{dt} \int_{P_t} \frac{1}{2} \rho |v|^2 dv + \int_{P_t} \underline{T} \cdot \underline{L} dv = \int_{P_t} \rho \underline{b} \cdot \underline{v} dv + \int_{\partial P_t} \underline{t} \cdot \underline{v} da, \quad (6.149)$$

where the time derivative pertains to the fixed set S of material points, and we have used the divergence theorem for smooth fields to convert the integral of $\text{div}(\underline{T}^t \underline{v})$ over P_t into an integral over ∂P_t of

$$\underline{T}^t \underline{v} \cdot \underline{n} = \underline{v} \cdot \underline{T} \underline{n} = \underline{t} \cdot \underline{v},$$

the first equality by the definition of the transpose and the second by Cauchy's theorem (6.94).

| READER'S NOTE (*integrating (6.147): four terms*).

$$\begin{aligned} \int_{P_t} \frac{1}{2} \rho (|v|^2) \cdot dv &= \frac{d}{dt} \int_{P_t} \frac{1}{2} \rho |v|^2 dv && (6.24)_1, \varphi = \frac{1}{2}|v|^2 \\ \int_{P_t} \rho \underline{b} \cdot \underline{v} dv &= \int_{P_t} \rho \underline{b} \cdot \underline{v} dv && \text{already final} \\ \int_{P_t} \text{div}(\underline{T}^t \underline{v}) dv &= \int_{\partial P_t} \underline{T}^t \underline{v} \cdot \underline{n} da = \int_{\partial P_t} \underline{t} \cdot \underline{v} da && (5.1); \underline{T}^t \underline{v} \cdot \underline{n} = \underline{v} \cdot \underline{T} \underline{n}, (6.94) \\ \int_{P_t} \underline{T} \cdot \underline{L} dv &\text{ stays inside} && \text{not a boundary term} \end{aligned}$$

Moving the last to the left gives (6.149); naming the four objects gives (6.150)–(6.153).

CAUTION The derivative in the first line is taken at fixed *material* set S , not fixed region P_t . Differentiating under the integral over a moving region without (6.24) produces two extra terms.

UNITS $[\underline{T} \cdot \underline{L}] = \text{N m}^{-2} \text{s}^{-1} = \text{W/m}^3$ and $[\rho |v|^2] = \text{J/m}^3$: every term of (6.150) is a watt.

Whence the *mechanical energy balance*:

$$\frac{d}{dt} \mathcal{K}(S, t) + \mathcal{S}(S, t) = \mathcal{P}(S, t), \quad (6.150)$$

where

$$\mathcal{K}(S, t) = \frac{1}{2} \int_{P_t} \rho |\underline{v}|^2 dv \quad (6.151)$$

is the kinetic energy of S ,

$$\mathcal{P}(S, t) = \int_{P_t} \rho \underline{b} \cdot \underline{v} dv + \int_{\partial P_t} \underline{t} \cdot \underline{v} da \quad (6.152)$$

is the power of the forces acting on S , and

$$\mathcal{S}(S, t) = \int_{P_t} \underline{T} \cdot \underline{L} dv \quad (6.153)$$

is the *stress power* contained in S .

We observe that the stress power has no analogue in particle mechanics. This is due to the fact that ideal particles have no surface area, so the concepts of traction and stress have no meaning for them. For a continuum, the applied power (6.152) does not all go into kinetic energy: some of it is spent deforming the material, and (6.153) is the rate at which that happens.

REMARK 6.33 (reading the stress power). Two simplifications of $\underline{T} \cdot \underline{L}$ are used so often that they are worth recording here.

Only the stretching does work. Split $\underline{L} = \underline{D} + \underline{W}$ by (4.7). Since $\underline{T}$ is symmetric and $\underline{W}$ is skew, $\underline{T} \cdot \underline{W} = 0$ by Lemma 2.36, so

$$\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{D}.$$

Spin costs nothing. This is the algebraic expression of the fact that rigidly rotating a stressed body does no work on it, and it is the first thing Chapter 10 uses, at (10.15).

Two states in which it vanishes. For a pure pressure (6.128), $\underline{T} \cdot \underline{L} = -p \underline{I} \cdot \underline{L} = -p \operatorname{tr} \underline{L} = -p \operatorname{div} \underline{v} = -p \dot{J}/J$, which vanishes in an isochoric motion. For a rigid motion, $\underline{D} = \underline{Q}$ by Remark 4.6 and the stress power vanishes whatever $\underline{T}$ is. Both facts return in Section 6.6 as the two cases in which the thermal and mechanical problems decouple.

Referential forms, and power conjugacy

The referential forms of the kinetic energy and power are

$$\mathcal{K}(S, t) = \frac{1}{2} \int_P \rho_\kappa |\underline{\underline{v}}|^2 dV \quad \text{and} \quad \mathcal{P}(S, t) = \int_P \rho_\kappa \underline{\underline{b}} \cdot \underline{\underline{v}} dV + \int_{\partial P} \underline{\underline{p}} \cdot \underline{\underline{v}} dA, \quad (6.154)$$

respectively, obtained by the substitutions $\rho dv = \rho_\kappa dV$ and $\underline{\underline{t}} da = \underline{\underline{p}} dA$ of (6.142). As for the stress power, we use (6.140) in the form $\underline{\underline{T}} = J^{-1} \underline{\underline{P}} \underline{\underline{F}}^t$, which is (6.145) divided by J :

$$\begin{aligned} \int_{P_t} \underline{\underline{T}} \cdot \underline{\underline{L}} dv &= \int_{P_t} J^{-1} \underline{\underline{P}} \underline{\underline{F}}^t \cdot \underline{\underline{L}} dv \\ &= \int_P \underline{\underline{P}} \underline{\underline{F}}^t \cdot \underline{\underline{L}} dV = \int_P \text{tr}(\underline{\underline{P}} \underline{\underline{F}}^t \underline{\underline{L}}^t) dV. \end{aligned} \quad (6.155)$$

The second line used $dv = J dV$, whose factor J cancels the J^{-1} , and the third is Definition 1.57.

| **READER'S NOTE** (*why (6.155) carries no factor of J*).

$$\underline{\underline{T}} \cdot \underline{\underline{L}} dv = \underbrace{(J^{-1} \underline{\underline{P}} \underline{\underline{F}}^t)}_{\text{stress: } da \rightarrow dA} \cdot \underline{\underline{L}} \underbrace{(J dV)}_{\text{volume: } dv \rightarrow dV} = (\underline{\underline{P}} \underline{\underline{F}}^t) \cdot \underline{\underline{L}} dV = \text{tr}(\underline{\underline{P}} \underline{\underline{F}}^t \underline{\underline{L}}^t) dV$$

by (6.145)/ J , (2.50) and Definition 1.57, the scalar J^{-1} passing through the inner product.

WHY THEY CANCEL Both factors are the same Jacobian, once because area elements change and once because volume elements do, and the stress power is a force-density times a volume. Special to the stress power: the kinetic energy (6.154)₁ keeps its factor, absorbed into $\rho_\kappa = \rho J$.

Using $\underline{\underline{F}}^t \underline{\underline{L}}^t = (\underline{\underline{L}} \underline{\underline{F}})^t = \dot{\underline{\underline{F}}}^t$, which is (4.3), $\underline{\underline{L}} = \dot{\underline{\underline{F}}} \underline{\underline{F}}^{-1}$, rearranged as $\dot{\underline{\underline{F}}} = \underline{\underline{L}} \underline{\underline{F}}$ and transposed, we then have $\text{tr}(\underline{\underline{P}} \underline{\underline{F}}^t \underline{\underline{L}}^t) = \text{tr}(\underline{\underline{P}} \dot{\underline{\underline{F}}}^t) = \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}}$. In terms of components, $\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = P_{iA} \dot{F}_{iA}$. We note that whereas the trace of a two-point tensor is not defined, the product $\underline{\underline{P}} \dot{\underline{\underline{F}}}^t$ is a *spatial* tensor, $\dot{\underline{\underline{F}}}^t$ carrying $\mathbb{E}^3$ to $\hat{\mathbb{E}}^3$ and $\underline{\underline{P}}$ carrying $\hat{\mathbb{E}}^3$ back to $\mathbb{E}^3$, so its trace is meaningful and furnishes a well-defined inner product of two-point tensors. Accordingly we reach

$$\mathcal{S}(S, t) = \int_P \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} dV. \quad (6.156)$$

| READER'S NOTE ($\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}}$: how a two-point tensor gets an inner product). SUBSTITUTION

$$\underline{\underline{L}} = \dot{\underline{\underline{F}}} \underline{\underline{F}}^{-1} \implies \dot{\underline{\underline{F}}} = \underline{\underline{L}} \underline{\underline{F}} \implies \dot{\underline{\underline{F}}}^t = \underline{\underline{F}}^t \underline{\underline{L}}^t \quad (4.3), \quad (\underline{\underline{A}} \underline{\underline{B}})^t = \underline{\underline{B}}^t \underline{\underline{A}}^t$$

so $\text{tr}(\underline{\underline{P}} \underline{\underline{F}}^t \underline{\underline{L}}^t) = \text{tr}(\underline{\underline{P}} \dot{\underline{\underline{F}}}^t)$, with no rearrangement.

LEGS

$$\dot{\underline{\underline{F}}}^t: \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3, \quad \underline{\underline{P}}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3, \quad \implies \quad \underline{\underline{P}} \dot{\underline{\underline{F}}}^t: \mathbb{E}^3 \rightarrow \mathbb{E}^3 \quad \text{— trace defined.}$$

$$\text{tr}(\underline{\underline{P}} \dot{\underline{\underline{F}}}^t) = (\underline{\underline{P}} \dot{\underline{\underline{F}}}^t)_{ii} = P_{iA} (\dot{\underline{\underline{F}}}^t)_{Ai} = P_{iA} \dot{F}_{iA} =: \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}}, \quad \mathcal{S} = \int_P \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} \, dV.$$

CAUTION $\text{tr} \underline{\underline{P}}$ alone is undefined: P_{iA} has no diagonal, i and A being indices of different alphabets. And $\underline{\underline{P}} \cdot \underline{\underline{S}}$ would ask for $P_{iA} S_{iA}$, spatial against referential — not a sentence. Two two-point tensors pair only when they have the *same* legs.

Indeed the energy balance (6.150), with kinetic energy, power and stress power given by (6.154) and (6.156), may be derived directly by repeating the foregoing procedure with (6.109) replaced by (6.136).

A stress measure that is juxtaposed with the material derivative of a deformation measure in an expression for the stress power is said to be *power conjugate* to that deformation measure. Thus (6.156) says that the Piola stress is power conjugate to the deformation gradient. We show that the Piola–Kirchhoff stress is power conjugate to the Lagrange strain $\underline{\underline{E}}$ defined by (2.97). This is proved by combining (2.83)₂, $\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}}$, with (2.97), $\underline{\underline{E}} = \frac{1}{2}(\underline{\underline{C}} - \hat{\underline{\underline{I}}})$, to obtain

$$\begin{aligned} \dot{\underline{\underline{E}}} &= \frac{1}{2}(\underline{\underline{F}}^t \underline{\underline{F}} - \hat{\underline{\underline{I}}})^\cdot = \frac{1}{2}(\underline{\underline{F}}^t \dot{\underline{\underline{F}}} + \dot{\underline{\underline{F}}}^t \underline{\underline{F}}) \\ &= \frac{1}{2}[\underline{\underline{F}}^t \dot{\underline{\underline{F}}} + (\underline{\underline{F}}^t \dot{\underline{\underline{F}}})^t] = \text{sym}(\underline{\underline{F}}^t \dot{\underline{\underline{F}}}), \end{aligned} \quad (6.157)$$

the constant $\hat{\underline{\underline{I}}}$ dropping out on differentiation and $(\underline{\underline{F}}^t \dot{\underline{\underline{F}}})^t = \dot{\underline{\underline{F}}}^t \underline{\underline{F}}$ by the rule for the transpose of a product. Then

$$\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \underline{\underline{F}} \underline{\underline{S}} \cdot \dot{\underline{\underline{F}}} = \text{tr}(\underline{\underline{F}} \underline{\underline{S}} \dot{\underline{\underline{F}}}^t) = \text{tr}(\underline{\underline{S}} \dot{\underline{\underline{F}}}^t \underline{\underline{F}}), \quad (6.158)$$

in which the first trace pertains to a spatial tensor, yielding an inner product of two-point tensors, and the second pertains to a referential tensor. That the two are equal is the component identity

$$\text{tr}(\underline{\underline{F}} \underline{\underline{S}} \dot{\underline{\underline{F}}}^t) = F_{iA} S_{AB} \dot{F}_{iB} = S_{AB} (\dot{F}_{iB} F_{iA}) = \text{tr}(\underline{\underline{S}} \dot{\underline{\underline{F}}}^t \underline{\underline{F}}),$$

the same sum of products of the same numbers written in a different order; it is the manoeuvre already used at (10.17) of Chapter 10 and justified there in the same way. Thus $\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \underline{\underline{S}} \cdot \underline{\underline{F}}^t \dot{\underline{\underline{F}}}$, since $\text{tr}(\underline{\underline{S}}\underline{\underline{M}}) = \underline{\underline{S}} \cdot \underline{\underline{M}}^t$ with $\underline{\underline{M}} = \underline{\underline{F}}^t \underline{\underline{F}}$ and $\underline{\underline{M}}^t = \underline{\underline{F}}^t \underline{\underline{F}}$, and the symmetry of the Piola–Kirchhoff stress reduces this by Lemma 2.36 to

$$\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \underline{\underline{S}} \cdot \text{sym}(\underline{\underline{F}}^t \dot{\underline{\underline{F}}}), \quad (6.159)$$

yielding, by (6.157), the stress power in the form

$$\mathcal{S}(S, t) = \int_P \underline{\underline{S}} \cdot \dot{\underline{\underline{E}}} \, dV. \quad (6.160)$$

| READER'S NOTE ($\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \underline{\underline{S}} \cdot \dot{\underline{\underline{E}}}$ in components).

$$\begin{aligned} \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} &= P_{iA} \dot{F}_{iA} = F_{iB} S_{BA} \dot{F}_{iA} & (6.146)_1 \\ &= S_{AB} (\dot{F}_{iB} F_{iA}) & \text{numbers commute; } A \leftrightarrow B \\ \dot{F}_{iB} F_{iA} &= (\dot{F}^t)_{Bi} F_{iA} = (\dot{F}^t \underline{\underline{F}})_{BA} =: M_{BA} & \text{referential, both indices} \\ \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} &= S_{AB} M_{BA} = \text{tr}(\underline{\underline{S}}\underline{\underline{M}}) & (6.158) \\ &= S_{AB} (\text{sym } \underline{\underline{M}})_{BA} & \underline{\underline{S}} = \underline{\underline{S}}^t, \text{ Lem. 2.36} \\ \text{sym } \underline{\underline{M}} &= \frac{1}{2}(\dot{F}^t \underline{\underline{F}} + \underline{\underline{F}}^t \dot{\underline{\underline{F}}}) = \frac{1}{2}(\underline{\underline{F}}^t \underline{\underline{F}})^{\cdot} = \frac{1}{2} \dot{\underline{\underline{C}}} = \dot{\underline{\underline{E}}} & (2.83)_2, (2.97), \dot{\underline{\underline{I}}} = \dot{\underline{\underline{O}}} \\ \therefore \quad \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} &= \underline{\underline{S}} \cdot \dot{\underline{\underline{E}}} & \text{integrate } \Rightarrow (6.160) \end{aligned}$$

WHY $\underline{\underline{E}}$ AND NOT $\underline{\underline{C}}$ The same lines with $\dot{\underline{\underline{C}}}$ give $\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \frac{1}{2} \underline{\underline{S}} \cdot \dot{\underline{\underline{C}}}$. The $\frac{1}{2}$ in the definition (2.97) of $\underline{\underline{E}}$ is chosen exactly so that the pairing carries coefficient one, and so that $\underline{\underline{S}} = W_{\underline{\underline{E}}}$ has no stray factor.

REMARK 6.34 (what power conjugacy is good for). Three expressions for the same stress power have now been obtained:

$$\begin{aligned} \underline{\underline{T}} \cdot \underline{\underline{D}} & \quad \text{per unit current volume,} \\ \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} \quad \text{and} \quad \underline{\underline{S}} \cdot \dot{\underline{\underline{E}}} & \quad \text{per unit reference volume.} \end{aligned}$$

They are equal, so nothing physical distinguishes them. What distinguishes them is which pair of variables a constitutive equation is written in.

Suppose the material stores energy: there is a function $W(\underline{\underline{F}})$, per unit reference volume, whose rate along every motion is the stress power

density. The hypothesis and the chain rule give two expressions for the same scalar,

$$\dot{W} = W_{\underline{F}} \cdot \dot{\underline{F}} \quad \text{and} \quad \dot{W} = \underline{P} \cdot \dot{\underline{F}},$$

the second being (6.156). Subtracting,

$$(\underline{P} - W_{\underline{F}}) \cdot \dot{\underline{F}} = 0 \quad \text{for every } \dot{\underline{F}},$$

and since $\dot{\underline{F}}$ ranges over all two-point tensors as the motion varies, while the bracket does not depend on it, the bracket vanishes:

$$\underline{P} = W_{\underline{F}}.$$

The same argument run on (6.160), with W regarded as a function of $\underline{E}$ and $\dot{\underline{E}}$ ranging over all symmetric referential tensors, gives $\underline{S} = W_{\underline{E}}$. Neither statement would be available if the pairing were wrong, and this is why the list of conjugate pairs is extended. Problem 12 adds two more: the Biot stress conjugate to the right stretch $\underline{U}$ and the Bell stress conjugate to the co-rotational rate of the left stretch $\underline{V}$.

| READER'S NOTE *(the cancellation in Remark 6.34, with its hypotheses).* **NOTATION**

$$(W_{\underline{F}})_{iA} := \frac{\partial W}{\partial F_{iA}}, \quad \dot{W} = \frac{\partial W}{\partial F_{iA}} \dot{F}_{iA} = W_{\underline{F}} \cdot \dot{\underline{F}} \quad (\text{chain rule in nine variables})$$

so $W_{\underline{F}}$ carries the index structure of $\underline{P}$ and may be compared with it.

ARGUMENT

$$\begin{aligned} \dot{W} &= W_{\underline{F}} \cdot \dot{\underline{F}} = \underline{P} \cdot \dot{\underline{F}} \quad \text{hypothesis, (6.156)} \\ (\underline{P} - W_{\underline{F}}) \cdot \dot{\underline{F}} &= 0 \quad \forall \dot{\underline{F}} \quad \underline{\dot{F}} = \underline{L}\underline{F} \text{ with } \underline{L} \text{ free, (4.3)} \\ \underline{A} &:= \underline{P} - W_{\underline{F}}, \quad \underline{\dot{F}} = \underline{A} : \quad |\underline{A}|^2 = A_{iA}A_{iA} = 0 \quad \text{nine squares} \\ \therefore \quad \underline{A} &= \underline{Q}, \quad \underline{P} = W_{\underline{F}} \end{aligned}$$

This is Lemma 5.1 with a tensor in place of a vector, and the choice $\underline{\dot{F}} = \underline{A}$ is the choice $\underline{c} = \underline{u}$ made there.

SMALL PRINT ON $\underline{S} = W_{\underline{E}}$ $\underline{\dot{E}} = \text{sym}(\underline{F}^t \underline{\dot{F}})$ is symmetric always, so $\underline{\dot{E}}$ ranges over six dimensions, not nine. The conclusion weakens to $\text{sym}(\underline{S} - W_{\underline{E}}) = \hat{\underline{Q}}$ — enough, since $\underline{S} = \underline{S}^t$ and $W_{\underline{E}}$ may be symmetrized, but a caveat (6.156) does not carry.

6.6 Energy balance

The mechanical energy balance (6.150) does not account for the inter-convertibility of heat and mechanical power, a fundamental aspect of thermomechanical phenomena. It was not derived from an axiom at all: it is the linear momentum balance dotted with the velocity, so it contains no information that (6.109) did not already contain. To introduce heat we need a genuinely new postulate, the *balance of energy*. Thus we assume the existence of scalars $\mathcal{U}(S, t)$ and $\mathcal{H}(S, t)$, the *internal energy* and the *heating* of $S \subset B$ respectively, such that

$$\frac{d}{dt} [\mathcal{K}(S, t) + \mathcal{U}(S, t)] = \mathcal{P}(S, t) + \mathcal{H}(S, t). \quad (6.161)$$

This is the first law of thermodynamics for a part of a continuum: the energy of S , kinetic plus internal, changes at the rate at which work is done on it plus the rate at which it is heated.

We assume that $\mathcal{U}$ can be represented as an integral over P_t , that is, that there exists a function $\varepsilon(\underline{x}, t)$, the internal energy per unit mass, such that

$$\mathcal{U}(S, t) = \int_{P_t} \rho \varepsilon \, dv, \quad (6.162)$$

and that the heating is expressible in the form

$$\mathcal{H}(S, t) = \int_{P_t} \rho r \, dv + \int_{\partial P_t} h \, da, \quad (6.163)$$

where $r(\underline{x}, t)$ is the heating per unit mass, through radioactive decay for example, and $h(\underline{x}, t)$ is the heating influx supplied via conduction, for example by contact with material outside P_t . The two terms of (6.163) are the bulk supply and the boundary flux of the generic form (6.1), and (6.162) is the assertion that internal energy, like mass, is distributed and not concentrated.

| READER'S NOTE (*where ε came from, and what the first law asserts*). THE CALORIC PICTURE, AND WHAT KILLED IT Until the 1840s heat was a substance. Caloric was conserved, and an engine produced work by letting caloric fall

from a hot body to a cold one as water falls through a mill wheel. Two measurements destroyed the picture.

Rumford, 1798 : boring cannon: unlimited heat from a blunt tool
an inexhaustible supply is not a substance

Joule, 1843–1850 : stirring, electrical heating, compressing gas
all give 4.19 J cal^{-1}

Mayer (1842) and Helmholtz (1847) drew the consequence: work and heat are two ways of transferring one conserved quantity.

WHAT THE FIRST LAW ACTUALLY CLAIMS The content is *path independence*. Along a process from state A to state B ,

$$\begin{aligned} \int_A^B \mathcal{P} dt &\text{ depends on the path } \text{work is not a state function} \\ \int_A^B \mathcal{H} dt &\text{ depends on the path } \text{heat is not a state function} \\ \int_A^B (\mathcal{P} + \mathcal{H}) dt &= [\mathcal{K} + \mathcal{U}]_A^B \quad \text{the sum is, (6.161)} \end{aligned}$$

So (6.161) posits a function ε whose increment along *any* process is the work plus the heating, less the change in kinetic energy. That such an ε exists is the whole of the first law; (6.162) adds only that it is distributed rather than concentrated, in the sense of (6.1).

WHY IT IS NOT ENOUGH Nothing in (6.161) distinguishes a direction. Let a bar with a hot end and a cold end equalise its temperature: energy is conserved. Let the same bar spontaneously separate into a hot end and a cold end: energy is conserved again. Only the first occurs, and no sharpening of the constitutive equations can exclude the second, since both processes use the same material. What is missing is a one-sided statement, and Section 6.7 supplies it.

Combining the mechanical energy balance with (6.161) produces the *thermal energy balance*. The mechanical balance (6.150) reads $\dot{\mathcal{K}} + \mathcal{S} = \mathcal{P}$, so

$$\mathcal{P} = \dot{\mathcal{K}} + \mathcal{S}.$$

Substitute this for $\mathcal{P}$ on the right of (6.161) and expand the left:

$$\begin{aligned} \dot{\mathcal{K}} + \dot{\mathcal{U}} &= \mathcal{P} + \mathcal{H} = (\dot{\mathcal{K}} + \mathcal{S}) + \mathcal{H} \\ \dot{\mathcal{U}} &= \mathcal{S} + \mathcal{H}, \end{aligned}$$

the second line by cancelling $\dot{\mathcal{K}}$, which occurs once on each side. Rearranged,

$$\mathcal{H}(S, t) = \frac{d}{dt} \mathcal{U}(S, t) - \mathcal{S}(S, t). \quad (6.164)$$

| READER'S NOTE (*the cancellation of (6.164), with every integral*). FIRST LAW, WRITTEN OUT

$$\begin{aligned} \frac{d}{dt} \int_{P_t} \frac{1}{2} \rho |\underline{v}|^2 dv + \frac{d}{dt} \int_{P_t} \rho \varepsilon dv &= \int_{P_t} \rho \underline{b} \cdot \underline{v} dv + \int_{\partial P_t} \underline{t} \cdot \underline{v} da \\ &\quad + \int_{P_t} \rho r dv + \int_{\partial P_t} h da \end{aligned}$$

by (6.161) with (6.151), (6.162), (6.152), (6.163).

MECHANICAL BALANCE, REARRANGED

$$\underbrace{\int_{P_t} \rho \underline{b} \cdot \underline{v} dv + \int_{\partial P_t} \underline{t} \cdot \underline{v} da}_{\mathcal{P}} = \frac{d}{dt} \int_{P_t} \frac{1}{2} \rho |\underline{v}|^2 dv + \int_{P_t} \underline{T} \cdot \underline{L} dv \quad (6.149)$$

SUBSTITUTE AND CANCEL The kinetic-energy integral then stands once on each side with the same sign; subtract it:

$$\frac{d}{dt} \int_{P_t} \rho \varepsilon dv = \int_{P_t} \underline{T} \cdot \underline{L} dv + \int_{P_t} \rho r dv + \int_{\partial P_t} h da, \quad \text{i.e.} \quad \dot{\mathcal{U}} = \mathcal{S} + \mathcal{H}.$$

WHAT WAS USED Only that the same $\mathcal{P}$ appears in (6.150) and in (6.161). The mechanical balance is a theorem, not an axiom, so it may be spent to eliminate $\mathcal{P}$; what survives contains no force, and the stress enters solely through $\underline{T} \cdot \underline{L}$ — the whole of the thermomechanical coupling.

REMARK 6.35 (what the cancellation means). The kinetic energy has dropped out of (6.164) entirely, and that is the point of forming it. The mechanical energy balance is a *consequence* of the equations of motion and therefore adds nothing to them; subtracting it from (6.161) removes exactly the part of the first law that was already known, and leaves the part that is new. What is new is (6.164): internal energy changes through heating and through the stress power.

The stress power is the only term the two balances share, and it is therefore the sole channel of thermomechanical coupling. That observation is illustrated by two cases.

Inviscid incompressible fluids. If the stress is a pure pressure, then by Remark 6.33 $\underline{T} \cdot \underline{L} = -p \operatorname{tr} \underline{L} = -p \dot{J}/J$, and if the motion is isochoric,

$\dot{J} = 0$, the stress power vanishes. There is then no coupling at all, and the thermal and mechanical problems may be solved independently.

Rigid bodies. For a rigid motion $\underline{D} = \underline{Q}$ by Remark 4.6, so $\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{W}$, which vanishes by the symmetry (6.117) and Lemma 2.36. Again the two problems decouple. This is why traditional courses on heat transfer treat bodies as rigid, with no consideration given to mechanical response. It can only be valid approximately, since real bodies are not perfectly rigid, just as real fluids are not perfectly inviscid; but the approximation is a good one whenever the stress power is small compared with the heating. Chapter 10 returns to this from the other side: a material that stores no strain energy can also tell you nothing about its stress.

Proceeding, we write the thermal energy balance, granted conservation of mass, as

$$\int_{\partial P_t} h \, da = \int_{P_t} [\rho(\dot{\varepsilon} - r) - \underline{T} \cdot \underline{L}] \, dv. \quad (6.165)$$

The steps are: $\frac{d}{dt}\mathcal{U} = \frac{d}{dt} \int_{P_t} \rho \varepsilon \, dv = \int_{P_t} \rho \dot{\varepsilon} \, dv$ by (6.24)₁ with $\varphi = \varepsilon$; then (6.164) reads $\int_{P_t} \rho r \, dv + \int_{\partial P_t} h \, da = \int_{P_t} \rho \dot{\varepsilon} \, dv - \int_{P_t} \underline{T} \cdot \underline{L} \, dv$, and moving the bulk terms to the right gives (6.165).

| READER'S NOTE ((6.165): three steps, and the shape it is put into).

$$\begin{aligned} \frac{d}{dt}\mathcal{U} &= \frac{d}{dt} \int_{P_t} \rho \varepsilon \, dv = \int_{P_t} \rho \dot{\varepsilon} \, dv \quad (6.24)_1, \varphi = \varepsilon \\ \int_{P_t} \rho r \, dv + \int_{\partial P_t} h \, da &= \int_{P_t} \rho \dot{\varepsilon} \, dv - \int_{P_t} \underline{T} \cdot \underline{L} \, dv \quad (6.164) \\ \therefore \int_{\partial P_t} h \, da &= \int_{P_t} [\rho(\dot{\varepsilon} - r) - \underline{T} \cdot \underline{L}] \, dv \quad \text{fuse the volume terms} \end{aligned}$$

THE SHAPE

$$\underbrace{\int_{\partial P} p \, dA}_{(6.131)} = \int_P \rho_\kappa(\dot{v} - \underline{b}) \, dV, \quad \underbrace{\int_{\partial P_t} h \, da}_{(6.165)} = \int_{P_t} [\rho(\dot{\varepsilon} - r) - \underline{T} \cdot \underline{L}] \, dv.$$

Same statement, different letters: unknown boundary field on the left, bounded interior field on the right. That, and only that, is what Noll and Cauchy consume, so the next subsection transplants both proofs onto h with no new work. Writing (6.165) this way round is not cosmetic.

The heat flux vector

Equation (6.165) has exactly the structure that Sections 6.5.2 and 6.5.3 exploited: a surface integral of an unknown boundary field equal to a volume integral of bounded quantities. Assuming ρ , ε , r and $\underline{T} \cdot \underline{L}$ to be continuous, and hence bounded on compact subsets, we can therefore adapt Noll's theorem to conclude that h depends on ∂P_t through its local unit normal $\underline{n}$, and then adapt Cauchy's theorem to conclude that there is a vector field $\underline{q}(\underline{x}, t)$, the *heat flux vector*, such that

$$h(\underline{x}, t; \underline{n}) = -\underline{q}(\underline{x}, t) \cdot \underline{n}, \quad (6.166)$$

in which the minus sign is inserted to conform to the convention that $\underline{q} \cdot \underline{n} > 0$ corresponds to heat outflow, $h < 0$, and $\underline{q} \cdot \underline{n} < 0$ to heat inflow, $h > 0$.

REMARK 6.36 (the adaptation is word for word). Neither proof used the vectorial character of the traction, so both apply verbatim to a scalar. The three places where $\underline{t}$ entered are these.

Noll's argument, (6.86) to (6.93), used $\underline{t}$ only through the bound

$$\left| \int_{\sigma} \underline{t} \, da \right| \leq \left(\max_{\sigma} |\underline{t}| \right) A(\sigma),$$

and $|h|$ obeys the same bound with the same proof. The other input was boundedness of the volume term in (6.165), which is the hypothesis just made.

Cauchy's argument then runs on the tetrahedron with h in place of $\underline{t}$: the analogue of (6.105) is

$$\frac{1}{A(s)} \int_{\partial \tau} h \, da = h(\underline{x}^*; \underline{m}) - \sum_{i=1}^3 (\underline{e}_i \cdot \underline{m}) h(\underline{x}_i^*; \underline{e}_i),$$

the left side is $O(\delta)$ by the cone estimate of Lemma 6.23, and the limit gives $h(\underline{x}_0; \underline{m}) = \sum_i (\underline{e}_i \cdot \underline{m}) h(\underline{x}_0; \underline{e}_i) = \underline{a} \cdot \underline{m}$ with $\underline{a} = h(\underline{x}_0; \underline{e}_i) \underline{e}_i$. Setting $\underline{q} = -\underline{a}$ gives (6.166). So the heating influx is a linear function of the normal for the same reason the traction is, and the whole content

of Fourier's law, when it comes in Chapter 8, will be a constitutive statement about $\underline{q}$ and not about h .

The minus sign is pure convention and could have been absorbed into $\underline{q}$. Its purpose is to make $\underline{q}$ point in the direction heat *flows*, so that the second law will read $\underline{q} \cdot \text{grad } \theta \leq 0$, heat flowing down the temperature gradient, rather than the reverse.

Then

$$\int_{\partial P_t} h \, da = - \int_{\partial P_t} \underline{q} \cdot \underline{n} \, da = - \int_{P_t} \text{div } \underline{q} \, dv, \quad (6.167)$$

for smooth fields, by the divergence theorem (5.1). The printed text has ∂P_r in the middle integral here and again in (6.170); ∂P_t is meant in both places. Substituting (6.167) into (6.165), all terms become volume integrals over an arbitrary P_t , and we can localize by Theorem 2.20 to arrive at the local form

$$\rho \dot{\epsilon} = \underline{T} \cdot \underline{L} - \text{div } \underline{q} + \rho r, \quad (6.168)$$

holding at points in κ_t that do not belong to a singular surface.

| READER'S NOTE (*getting (6.168), and reading its four terms*). TWO LINES

$$- \int_{P_t} \text{div } \underline{q} \, dv = \int_{P_t} [\rho(\dot{\epsilon} - r) - \underline{T} \cdot \underline{L}] \, dv \quad (6.167) \text{ into } (6.165)$$

$$\int_{P_t} [\rho \dot{\epsilon} - \rho r - \underline{T} \cdot \underline{L} + \text{div } \underline{q}] \, dv = 0 \quad \forall P_t \quad \text{Thm. 2.20}$$

$$\rho \dot{\epsilon} = \underline{T} \cdot \underline{L} - \text{div } \underline{q} + \rho r \quad \text{in components} \quad \rho \dot{\epsilon} = T_{ij} v_{i,j} - q_{i,i} + \rho r.$$

BUDGET, PER UNIT dv PER UNIT TIME

term	what	positive when
$\rho \dot{\epsilon}$	storage	the parcel warms or strains
$\underline{T} \cdot \underline{L}$	stress power	stress does work on it
$-\text{div } \underline{q}$	net conduction in	more enters than leaves
ρr	bulk supply	a source is present

THE MINUS SIGN, ACCOUNTED FOR Two conventions, one surviving sign: the divergence theorem uses the *exterior* normal, so $\text{div } \underline{q} > 0$ is net outflow; and

(6.166) sets $h = -\underline{q} \cdot \underline{n}$ so that $\underline{q}$ points where heat goes. Their product makes $-\operatorname{div} \underline{q}$ a gain.

STATUS (6.168) is the first equation of the chapter that does *not* follow from (6.34). It needs the independent axiom (6.161), which is why ε , r , $\underline{q}$ are new letters.

Alternatively, this follows directly on making the appropriate identifications in (6.1) and (6.21), namely

$$\varphi = \varepsilon, \quad \underline{\pi} = -\underline{q}, \quad \rho\sigma = \rho r + \underline{T} \cdot \underline{L},$$

and (6.6) then delivers the associated jump condition

$$[\![\rho\varepsilon(\underline{v} - \underline{u}) + \underline{q}]\!] \cdot \underline{n} = 0, \quad (6.169)$$

holding at all points on a singular surface. Note that the stress power has been counted as part of the bulk supply $\rho\sigma$ and therefore contributes nothing to the jump condition, a volume term being invisible on a surface of zero volume. In terms of the mass flux m^* of Remark 6.3, (6.169) reads $m^*[\![\varepsilon]\!] + [\![\underline{q}]\!] \cdot \underline{n} = 0$: the internal energy convected across the surface is balanced by the jump in the conducted heat.

| READER'S NOTE (*the identifications, and why a supply never reaches a jump condition*). READ OFF (6.168)

$$\rho \underbrace{\dot{\varepsilon}}_{\varphi} = \operatorname{div} \underbrace{(-\underline{q})}_{\underline{\pi}} + \underbrace{\rho r + \underline{T} \cdot \underline{L}}_{\rho\sigma} \quad \text{against} \quad \rho\dot{\varphi} = \operatorname{div} \underline{\pi} + \rho\sigma \quad (6.21)$$

The flux of the generic law points *into* the region and $\underline{q}$ points out, hence $\underline{\pi} = -\underline{q}$.

INTO (6.6)

$$[\![\rho\varphi(\underline{v} - \underline{u}) - \underline{\pi}]\!] \cdot \underline{n} = 0 \longrightarrow [\![\rho\varepsilon(\underline{v} - \underline{u}) + \underline{q}]\!] \cdot \underline{n} = 0, \quad \sigma \text{ has vanished.}$$

WHY IT HAD TO VANISH Pillbox of thickness η , face area A : volume ηA , surface $2A + O(\eta)$. Divide by A , let $\eta \rightarrow 0$; every volume term carries η and dies, every surface term survives. So only φ and $\underline{\pi}$ can appear, never σ , however large. Hence the stress power — the entire bulk difference between the mechanical and thermal balances — makes no difference on a shock. Problem 11 uses exactly this.

MASS-FLUX FORM With $m^* = \rho(\underline{v} \cdot \underline{n} - u)$ and $[\![m^*]\!] = 0$ from (6.15),

$$m^*[\![\varepsilon]\!] + [\![\underline{q}]\!] \cdot \underline{n} = 0; \quad m^* = 0 \implies [\![\underline{q}]\!] \cdot \underline{n} = 0,$$

the interface condition of every heat-conduction problem.

Referential form

A referential form of the energy balance proceeds by rewriting the thermal energy balance as

$$\begin{aligned} \int_P [\rho_\kappa(\dot{\varepsilon} - r) - \underline{\underline{P}} \cdot \underline{\underline{\dot{F}}}] \, dV &= - \int_{\partial P_t} \underline{\underline{q}} \cdot \underline{\underline{n}} \, da = - \int_{\partial P} \underline{\underline{q}} \cdot \underline{\underline{F}}^* \underline{\underline{N}} \, dA \\ &= - \int_{\partial P} \underline{\underline{q}}_\kappa \cdot \underline{\underline{N}} \, dA, \end{aligned} \quad (6.170)$$

where

$$\underline{\underline{q}}_\kappa(\underline{\underline{X}}, t) = (\underline{\underline{F}}^*)^t \underline{\underline{q}} = J \underline{\underline{F}}^{-1} \underline{\underline{q}} \quad (6.171)$$

is the *referential heat flux vector*, with components

$$q_{(\kappa)A} = F_{iA}^* q_i, \quad (6.172)$$

in which the q_i are those of the spatial vector $\underline{\underline{q}}$. The left side of (6.170) is (6.165) pulled back: the volume term by $\rho \, dv = \rho_\kappa \, dV$, and the stress power by (6.155) and (6.156). On the right, the first equality is (6.167)₁, the second is Piola–Nanson (2.54), and the third moves $\underline{\underline{F}}^*$ from one slot of the inner product to the other by the definition of the transpose, which is what (6.171) records. The second form in (6.171) is (2.55): $(\underline{\underline{F}}^*)^t = (J \underline{\underline{F}}^{-t})^t = J \underline{\underline{F}}^{-1}$.

Then, for smooth fields,

$$\int_{\partial P} \underline{\underline{q}}_\kappa \cdot \underline{\underline{N}} \, dA = \int_P \text{Div } \underline{\underline{q}}_\kappa \, dV, \quad \text{where} \quad \text{Div } \underline{\underline{q}}_\kappa = q_{(\kappa)A,A}, \quad (6.173)$$

and we localize, P being arbitrary, to obtain

$$\rho_\kappa \dot{\varepsilon} = \underline{\underline{P}} \cdot \underline{\underline{\dot{F}}} - \text{Div } \underline{\underline{q}}_\kappa + \rho_\kappa r \quad (6.174)$$

at points $\underline{\underline{X}} \in \kappa$ removed from a singular surface. The derivation of the associated jump condition is Problem 1.

This result may also be derived directly if (6.168) is multiplied by J and use is made of the easily proved identity

$$J \, \text{div } \underline{\underline{q}} = \text{Div } (J \underline{\underline{F}}^{-1} \underline{\underline{q}}), \quad (6.175)$$

which is Problem 8. Multiplying (6.168) by J turns $\rho\dot{\varepsilon}$ into $\rho_\kappa\dot{\varepsilon}$ and ρr into $\rho_\kappa r$ by (2.59), turns $J\tilde{T} \cdot \tilde{L}$ into $\tilde{P} \cdot \dot{\tilde{F}}$ by the computation of (6.155), and turns $J \operatorname{div} \underline{q}$ into $\operatorname{Div} \underline{q}_\kappa$ by (6.175). Note the shape of (6.175): it is (6.138) for a vector instead of a tensor, and both are consequences of the Piola identity $\operatorname{Div} \tilde{F}^* = \underline{0}$ of (5.11). That identity is the reason every referential balance law in this chapter has exactly the same form as its spatial counterpart, with no leftover terms.

| READER'S NOTE (*the referential energy balance, both routes*). ROUTE 1: PULL THE INTEGRAL STATEMENT BACK

$$\begin{aligned}
\int_{\partial P_t} h \, da &= - \int_{\partial P_t} \underline{q} \cdot \underline{n} \, da & (6.166) \\
&= - \int_{\partial P} \underline{q} \cdot \tilde{F}^* \underline{N} \, dA & \underline{n} \, da = \tilde{F}^* \underline{N} \, dA, \quad (2.54) \\
&= - \int_{\partial P} (\tilde{F}^*)^t \underline{q} \cdot \underline{N} \, dA & \text{definition of the transpose} \\
&= - \int_{\partial P} \underline{q}_\kappa \cdot \underline{N} \, dA & \underline{q}_\kappa := (\tilde{F}^*)^t \underline{q} = J \tilde{F}^{-1} \underline{q}, \quad (6.171)
\end{aligned}$$

with the volume terms by $\rho \, dv = \rho_\kappa \, dV$ and the stress power by (6.155). Then (6.173) and localization give (6.174).

ROUTE 2: MULTIPLY (6.168) BY J

spatial	$\times J$	by
$\rho\dot{\varepsilon}$	$\rho_\kappa\dot{\varepsilon}$	$\rho J = \rho_\kappa$, (2.59)
$\tilde{T} \cdot \tilde{L}$	$\tilde{P} \cdot \dot{\tilde{F}}$	the computation of (6.155)
$-\operatorname{div} \underline{q}$	$-\operatorname{Div} \underline{q}_\kappa$	(6.175)
ρr	$\rho_\kappa r$	$\rho J = \rho_\kappa$

Four lines, and their sum is (6.174).

WHERE THE RESIDUE WOULD BE A priori $J \operatorname{div} \underline{q} = \operatorname{Div} (J \tilde{F}^{-1} \underline{q}) + q_i F_{iA,A}^*$, and the second term vanishes identically by the Piola identity (5.11). That, and nothing else, is why every referential balance in this chapter has the shape of its spatial counterpart. Geometrically $\operatorname{Div} \tilde{F}^* = \underline{0}$ is the differentiated form of $\int_{\partial P} \tilde{F}^* \underline{N} \, dA = \int_{\partial P_t} \underline{n} \, da = \underline{0}$: the vector area of a closed surface vanishes.

6.7 Entropy and the Clausius–Duhem inequality

To account for the thermodynamical concept of dissipation attending any real physical process, we introduce a further hypothesis, the *entropy balance*. This is assumed to take the form

$$\frac{d}{dt} \int_{P_t} \rho \eta \, dv = \int_{P_t} \theta^{-1} (\rho r) \, dv + \int_{\partial P_t} \theta^{-1} h \, da + \int_{P_t} \rho s \, dv, \quad (6.176)$$

where $\eta(\underline{x}, t)$ is the entropy per unit mass contained in S , $\theta(\underline{x}, t) > 0$ is the absolute temperature, and s is the entropy production rate per unit mass.

| READER'S NOTE (*why anyone needed entropy: Carnot, Clausius, Boltzmann*). Finding the one-sided statement took forty years, and the route explains the shape of (6.176).

CARNOT, 1824 Carnot asked for the greatest work obtainable from a given quantity of heat, and answered before the first law was known. His theorem: no engine working between two reservoirs beats a reversible one, and all reversible engines between the same pair have the same efficiency, whatever the working substance.

The proof needs no analysis. Were X more efficient than a reversible R , run X forward and R backward on X 's work; the pair returns to its initial state, delivers no net work, and moves heat from cold to hot as its sole effect. Hence

$$\varepsilon_{\text{eff}} = 1 - \frac{|Q_C|}{Q_H} = 1 - f(\theta_C, \theta_H) \quad \text{a universal function of the two temperatures}$$

$$f(\theta_C, \theta_H) = \frac{g(\theta_C)}{g(\theta_H)} \quad \text{two cycles in series}$$

$$\frac{Q_H}{\theta_H} + \frac{Q_C}{\theta_C} = 0 \quad \text{Kelvin, 1848 : } g(\theta) = \theta$$

around a reversible cycle, heat counted positive inwards. The combination Q/θ balances; Q does not.

CLAUSIUS, 1850–1865 Approximating an arbitrary cycle by Carnot cycles turns the last display into the *Clausius inequality*

$$\oint \frac{dQ}{\theta} \leq 0, \quad \text{with equality iff the cycle is reversible.}$$

Its consequence is the definition of entropy. If Γ_1, Γ_2 are reversible paths from

A to B , then Γ_1 forward and Γ_2 backward is a reversible cycle, so

$$\int_{\Gamma_1} \frac{dQ}{\theta} = \int_{\Gamma_2} \frac{dQ}{\theta} :$$

the integral depends on the end states alone and defines a function of state up to a constant. Clausius named it in 1865, coining *Entropie* from $\tau\rho\omicron\pi\acute{\eta}$, transformation, shaped to sound like *Energie*, and closed the paper with the two laws in the form still quoted: the energy of the world is constant, the entropy of the world tends to a maximum. For an irreversible process the inequality is strict,

$$S(B) - S(A) \geq \int_A^B \frac{dQ}{\theta},$$

which is (6.177) for a body at a single temperature. Passing to (6.177) replaces dQ by the supply and flux of (6.163) and lets θ vary from point to point, which is the whole of the modelling step.

BOLTZMANN, 1872–1877 Clausius left entropy defined but unexplained: a state function that increases, with no account of why. Boltzmann counted. If a macrostate is realised by W microstates,

$$S = k \log W, \quad k = 1.38 \times 10^{-23} \text{ J K}^{-1}.$$

The logarithm is forced rather than chosen: entropy is additive over independent subsystems while configuration counts multiply, $W = W_1 W_2$, and up to a constant the only continuous solution of $S(W_1 W_2) = S(W_1) + S(W_2)$ is a logarithm.

This explains the direction. A system moves to a macrostate of larger W not because a law forbids the reverse but because the ratios are of order e^N with $N \sim 10^{23}$. Two objections sharpened the claim. Loschmidt (1876) noted that reversing every velocity carries a solution to a solution, so no strictly decreasing functional of the mechanical state exists; Zermelo (1896), using Poincaré recurrence, noted that a bounded Hamiltonian system returns arbitrarily close to its initial state. Both are correct and both are compatible with the H -theorem, which is a statement about a distribution under an assumption of molecular chaos, not about one trajectory. Recurrence times for 10^{23} particles exceed the age of the universe by many orders.

WHAT THIS CHAPTER TAKES FROM IT Not the counting. Continuum mechanics takes the licence: entropy is a field, extensive, defined out of equilibrium, and its production is non-negative. Equation (6.176) postulates that and nothing more, and Remark 6.37 is right that the theory stands or falls by experiment rather than by $S = k \log W$.

The first integral on the right represents the entropy supplied by sources in the bulk and the second the flux of entropy through the surface.

Their forms are suggested by the classical thermostatics of reversible processes, in which an amount of heat dQ delivered at temperature θ carries entropy dQ/θ ; each heating term of (6.163) accordingly reappears in (6.176) divided by θ . The last integral, which vanishes in a reversible process, represents the net entropy production attending an irreversible one. These are, by definition, such that this integral is positive and hence that $s > 0$. All admissible processes are then subject to the *Clausius–Duhem inequality*

$$\frac{d}{dt} \int_{P_t} \rho \eta \, dv \geq \int_{P_t} \theta^{-1} (\rho r) \, dv + \int_{\partial P_t} \theta^{-1} h \, da, \quad (6.177)$$

obtained by deleting the non-negative production term from (6.176).

| READER’S NOTE ((6.176) read against (6.163), and where the inequality is born). THE POSTULATE IS THE HEATING WITH θ^{-1} PASTED ON

$$\mathcal{H} = \int_{P_t} \rho r \, dv + \int_{\partial P_t} h \, da \quad (6.163)$$

$$\text{supply + flux in (6.176)} = \int_{P_t} \theta^{-1} \rho r \, dv + \int_{\partial P_t} \theta^{-1} h \, da \quad dS = dQ/\theta, \text{ applied locally}$$

The whole modelling content is the word *locally*: each parcel of heat is divided by the temperature at the point where it is delivered.

BIRTH OF THE INEQUALITY

$$X = Y + \int_{P_t} \rho s \, dv \quad \text{and} \quad \int_{P_t} \rho s \, dv \geq 0 \quad \implies \quad X \geq Y.$$

Nothing more. The production is not computed or bounded but *discarded*, and the result is weaker than the start.

WHY DISCARD IT s is unobservable, so an equation containing it constrains nothing; (6.177) contains only η , θ , r , h . Same trade as at (6.184), where r goes.

UNITS $[\theta^{-1}h] = \text{J K}^{-1}\text{m}^{-2}\text{s}^{-1}$, $[\eta] = \text{J K}^{-1}\text{kg}^{-1}$: every term an entropy per unit time.

Truesdell’s motivation for (6.177)

The inequality may be motivated by an argument due to Truesdell. We assume that there is a function $\mathcal{B}(S, t)$, depending on material constitution, that bounds the heating from above, $\mathcal{B}(S, t) \geq \mathcal{H}(S, t)$. From the

energy balance (6.161) we then have that, in accordance with experience, there is a limit to the rate at which heat can be converted to energy in the absence of forces, $\mathcal{P} = 0$:

$$\frac{d}{dt} [\mathcal{K}(S, t) + \mathcal{U}(S, t)] \leq \mathcal{B}(S, t). \quad (6.178)$$

For uniformly distributed temperatures, $\text{grad } \theta = 0$, we define the entropy

$$\mathcal{C}(S, t) = \int_{P_t} \rho \eta \, dv \quad (6.179)$$

by

$$\mathcal{C}(S, t) = \int_{t_0}^t \theta(\tau)^{-1} \mathcal{B}(S, \tau) \, d\tau, \quad (6.180)$$

where t_0 is a fixed time. Then, differentiating (6.180) by the fundamental theorem of calculus and multiplying by $\theta(t)$,

$$\theta(t) \frac{d}{dt} \mathcal{C}(S, t) = \mathcal{B}(S, t) \geq \mathcal{H}(S, t), \quad (6.181)$$

which is just the inequality (6.177) with the uniform reciprocal temperature factored outside the integrals on the right-hand side: dividing by θ and writing $\mathcal{H}$ out by (6.163),

$$\frac{d}{dt} \int_{P_t} \rho \eta \, dv \geq \theta^{-1} \left[\int_{P_t} \rho r \, dv + \int_{\partial P_t} h \, da \right].$$

Placing θ^{-1} inside the integrals, which changes nothing while θ is uniform, and making the further assumption that the resulting inequality also applies to non-uniform temperature fields, then yields the Clausius–Duhem inequality.

REMARK 6.37 (how much of this is proof). Very little, and it is important to say so. The argument assumes the existence of $\mathcal{B}$, defines $\mathcal{C}$ by (6.180) so that (6.181) holds identically, asserts without further evidence that the $\mathcal{C}$ so defined is the integral (6.179) of a mass density η , and then extends an inequality established for uniform temperature fields to

non-uniform ones by fiat. The last step is the substantive one and it is not deducible from the preceding.

What the argument does is show that (6.177) is the natural continuum generalization of a statement one is willing to make about a body at uniform temperature, namely that there is a ceiling on the rate at which heat can be absorbed and that the ceiling is proportional to the temperature. The inequality is a postulate, on the same footing as (6.34) and (6.161), and its justification is that the theory built on it agrees with experiment.

The entropy supply and flux written in (6.176) are also not the most general that could be contemplated. One could allow an entropy flux not equal to $\theta^{-1}h$, and theories of that kind exist. The classical theory is the one that makes the simplest choice.

Local forms

Combining (6.177) with (6.166) and localizing as usual yields the local inequality. The identifications in the generic scalar balance (6.1), now read with $\geq$ in place of $=$, are

$$\varphi = \eta, \quad \sigma = \theta^{-1}r, \quad \pi = -\theta^{-1}\underline{q},$$

the last because the surface integrand is $\theta^{-1}h = -\theta^{-1}\underline{q} \cdot \underline{n} = (-\theta^{-1}\underline{q}) \cdot \underline{n}$. Then (6.21) gives

$$\rho\dot{\eta} \geq \theta^{-1}(\rho r) - \operatorname{div}(\theta^{-1}\underline{q}), \quad (6.182)$$

holding in regions of smoothness, whereas the jump condition

$$[\![\rho\eta(\underline{v} - \underline{u}) + \theta^{-1}\underline{q}]\!] \cdot \underline{n} \geq 0, \quad (6.183)$$

operative on singular surfaces, follows by making the same identifications in (6.6), with the equality replaced by an inequality.

Here we recall, with reference to Figure 5.2 of Chapter 5, that $\underline{n}$ is directed into the “+” region, and hence, with the definition (5.28) of the jump, that this inequality is independent of the orientation of $\underline{n}$. The

check is one line. Reversing $\underline{n}$ exchanges the labels $+$ and $-$, so every jump changes sign, $[[\cdot]] \mapsto -[[\cdot]]$; and $\underline{n}$ itself changes sign. The product of the two therefore does not change, so the left side of (6.183) is unaltered and the inequality means the same thing. Had the two sign changes not cancelled, (6.183) would have asserted that a shock dissipates when read from one side and creates entropy from the other, which is not a statement about the world but about the person reading it.

| READER'S NOTE (*the entropy identifications, and the orientation check*). PUT THE FLUX IN THE REQUIRED SHAPE

$$\theta^{-1}h \stackrel{(6.166)}{=} \theta^{-1}(-\underline{q} \cdot \underline{n}) = (-\theta^{-1}\underline{q}) \cdot \underline{n} \quad \implies \quad \underline{\pi} = -\theta^{-1}\underline{q}, \quad \sigma = \theta^{-1}r.$$

SUBSTITUTE

$$(6.21) : \quad \rho\dot{\eta} \geq \theta^{-1}(\rho r) - \operatorname{div}(\theta^{-1}\underline{q}) \quad (6.182)$$

$$(6.6) : \quad [[\rho\eta(\underline{v} - \underline{u}) + \theta^{-1}\underline{q}]] \cdot \underline{n} \geq 0 \quad (6.183)$$

the two minus signs cancelling as they did at (6.169).

ORIENTATION Reverse the normal: the labels $\pm$ swap, so $[[\cdot]] \mapsto -[[\cdot]]$ and $\underline{n} \mapsto -\underline{n}$; hence

$$(-[[\cdot]]) \cdot (-\underline{n}) = [[\cdot]] \cdot \underline{n},$$

unchanged. Every jump condition in this chapter is linear in $[[\cdot]]$ and in $\underline{n}$, exactly once each, which is what makes them statements about the world rather than about the reader.

The reduced inequality

A more useful form of the local inequality (6.182) is obtained by eliminating the radiative heating r , which is external data and which no constitutive equation should have to mention. Solve the thermal energy balance (6.168) for it,

$$\rho r = \rho\dot{\varepsilon} - \underline{T} \cdot \underline{L} + \operatorname{div} \underline{q},$$

and substitute into (6.182):

$$\rho\dot{\eta} \geq \theta^{-1}(\rho\dot{\varepsilon} - \underline{T} \cdot \underline{L} + \operatorname{div} \underline{q}) - \operatorname{div}(\theta^{-1}\underline{q}), \quad (6.184)$$

which, since $\theta > 0$, is equivalent to the inequality obtained on multiplying through by θ . Doing so, and expanding the last term by Problem

24(c) of Chapter 1,

$$\operatorname{div}(\theta^{-1}\underline{q}) = \operatorname{grad}(\theta^{-1}) \cdot \underline{q} + \theta^{-1} \operatorname{div} \underline{q} = -\theta^{-2} \operatorname{grad} \theta \cdot \underline{q} + \theta^{-1} \operatorname{div} \underline{q},$$

where $\operatorname{grad}(\theta^{-1}) = -\theta^{-2} \operatorname{grad} \theta$ by the chain rule, Theorem 1.78. Hence

$$\theta \operatorname{div}(\theta^{-1}\underline{q}) = -\theta^{-1} \operatorname{grad} \theta \cdot \underline{q} + \operatorname{div} \underline{q},$$

and multiplying (6.184) by θ gives

$$\rho\theta\dot{\eta} \geq \rho\dot{\varepsilon} - \underline{T} \cdot \underline{L} + \operatorname{div} \underline{q} + \theta^{-1}\underline{q} \cdot \operatorname{grad} \theta - \operatorname{div} \underline{q}.$$

The two divergences cancel, which is the whole purpose of the expansion, and rearranging so that everything stands on one side,

$$\rho(\theta\dot{\eta} - \dot{\varepsilon}) + \underline{T} \cdot \underline{L} - \theta^{-1}\underline{q} \cdot \operatorname{grad} \theta \geq 0. \quad (6.185)$$

On introducing the *Helmholtz free energy* per unit mass ψ , defined by

$$\psi = \varepsilon - \theta\eta, \quad (6.186)$$

we finally reduce the local Clausius–Duhem inequality to a form in which the entropy appears only through ψ and θ . Differentiating (6.186) along a material point,

$$\dot{\psi} = \dot{\varepsilon} - \dot{\theta}\eta - \theta\dot{\eta}, \quad \text{so} \quad \theta\dot{\eta} - \dot{\varepsilon} = -\dot{\psi} - \eta\dot{\theta},$$

and substituting into (6.185),

$$-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{T} \cdot \underline{L} - \theta^{-1}\underline{q} \cdot \operatorname{grad} \theta \geq 0. \quad (6.187)$$

This will play a central role in the discussion of constitutive equations in subsequent chapters.

| READER'S NOTE (*the reduction (6.182)→(6.187), and the referential form*). THE LEDGER

$$\rho\dot{\eta} \geq \theta^{-1}(\rho r) - \operatorname{div}(\theta^{-1}\underline{q}) \quad (6.182)$$

$$\rho\theta\dot{\eta} \geq \rho\dot{\varepsilon} - \underline{T} \cdot \underline{L} + \operatorname{div} \underline{q} - \theta \operatorname{div}(\theta^{-1}\underline{q}) \quad \text{drop } \rho r, (6.168); \times \theta > 0$$

$$\rho\theta\dot{\eta} \geq \rho\dot{\varepsilon} - \underline{T} \cdot \underline{L} + \theta^{-1}\underline{q} \cdot \operatorname{grad} \theta \quad \operatorname{div} \underline{q} \text{ cancels}$$

$$\rho(\theta\dot{\eta} - \dot{\varepsilon}) + \underline{T} \cdot \underline{L} - \theta^{-1}\underline{q} \cdot \operatorname{grad} \theta \geq 0 \quad \text{collect} = (6.185)$$

$$-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{T} \cdot \underline{L} - \theta^{-1}\underline{q} \cdot \operatorname{grad} \theta \geq 0 \quad (6.186) = (6.187)$$

$$\begin{aligned}
\operatorname{div}(\theta^{-1} \underline{q}) &= \operatorname{grad}(\theta^{-1}) \cdot \underline{q} + \theta^{-1} \operatorname{div} \underline{q} && \text{Prob. 24(c), Ch. 1} \\
\operatorname{grad}(\theta^{-1}) &= -\theta^{-2} \operatorname{grad} \theta && \text{Thm. 1.78} \\
\therefore \operatorname{div} \underline{q} - \theta \operatorname{div}(\theta^{-1} \underline{q}) &= \theta^{-1} \underline{q} \cdot \operatorname{grad} \theta
\end{aligned}$$

The point of expanding is that $\operatorname{div} \underline{q}$ — a *derivative* of the flux, against which no constitutive equation for $\underline{q}$ could be tested pointwise — drops out.

$$\psi = \varepsilon - \theta \eta \implies \dot{\psi} = \dot{\varepsilon} - \dot{\theta} \eta - \theta \dot{\eta} \implies \theta \dot{\eta} - \dot{\varepsilon} = -\dot{\psi} - \eta \dot{\theta}.$$

$$\begin{aligned}
-J\rho(\dot{\psi} + \eta \dot{\theta}) &= -\rho_{\kappa}(\dot{\psi} + \eta \dot{\theta}) && \rho J = \rho_{\kappa}, \text{ (2.59)} \\
J \underline{T} \cdot \underline{L} &= \underline{P} \cdot \underline{\dot{F}} && \text{the computation of (6.155)} \\
\underline{q}_{\kappa} \cdot \operatorname{Grad} \theta &= J \underline{F}^{-1} \underline{q} \cdot \underline{F}^t \operatorname{grad} \theta = J \underline{q} \cdot \operatorname{grad} \theta && \text{(6.171), Thm. 2.14}
\end{aligned}$$

$$-\rho_{\kappa}(\dot{\psi} + \eta \dot{\theta}) + \underline{P} \cdot \underline{\dot{F}} - \theta^{-1} \underline{q}_{\kappa} \cdot \operatorname{Grad} \theta \geq 0$$

at every $\underline{X} \in \kappa$ off a singular surface, with jump condition

$$[\![\rho_{\kappa} \eta]\!] U - [\![\theta^{-1} \underline{q}_{\kappa}]\!] \cdot \underline{N} \leq 0,$$

the reciprocal temperature staying *inside* the bracket since θ may itself jump, and the sense reversing for the reason set out in Problem 1. The arguments of the boxed inequality are $\underline{F}$, θ , $\operatorname{Grad} \theta$ and their rates — which is what Chapters 8 and 9 write constitutive equations in.

The referential form of the Clausius–Duhem inequality, and its associated local form and jump condition, are obtained by the same route as (6.174); the reader’s note just given writes them out. The only new ingredient is that $\operatorname{grad} \theta$ is replaced through $\operatorname{Grad} \theta = \underline{F}^t \operatorname{grad} \theta$, which is Theorem 2.14.

REMARK 6.38 (why (6.187) is the form that gets used). Three features of (6.187) explain why it, and not (6.177), is what Chapters 8 and 9 quote.

The supplies have gone. Neither r nor any boundary datum appears. Everything in (6.187) is either a response of the material (ψ , η , $\underline{T}$, $\underline{q}$) or a description of the process ($\underline{L}$, θ , $\operatorname{grad} \theta$). An inequality of that shape can be turned into a restriction on constitutive equations, which is exactly what Chapter 8 does with it.

The free energy replaced the internal energy. The change of variable (6.186) is a Legendre transform trading η for θ . It matters because θ is measurable and controllable while η is neither: a constitutive equation of the form $\psi = \hat{\psi}(\underline{F}, \theta)$ can be fitted to experiments, one of the form $\varepsilon = \hat{\varepsilon}(\underline{F}, \eta)$ cannot. The pairing η with $-\partial\psi/\partial\theta$ that Chapter 8 derives is the thermodynamic analogue of the power conjugacy of Remark 6.34.

The reaction stresses are invisible to it. The stress enters (6.187) only through $\underline{T} \cdot \underline{L}$, so any part of $\underline{T}$ that is workless in every admissible motion drops out identically. That is the observation on which Chapter 10 builds the whole theory of constraints on, and it is the reason the pressure in an incompressible fluid is thermodynamically indeterminate.

Finally, note that (6.187) is a restriction on *materials*, not on processes. Read as a restriction on processes it would say that certain motions are forbidden, which cannot be right, since one may impose whatever boundary data one likes. What it says is that a constitutive equation must be such that (6.187) holds for every process the material can undergo, and constitutive equations that fail that test are discarded.

| READER'S NOTE (*what (6.187) decides: the Coleman–Noll theorem*). Coleman and Noll (1963) showed what (6.187) is worth. Take a constitutive assumption of the widest reasonable form and the inequality returns three equalities and one inequality, none of them assumed.

HYPOTHESIS

$$\psi = \hat{\psi}(\underline{F}, \theta, \underline{G}), \quad \eta = \hat{\eta}(\underline{F}, \theta, \underline{G}), \quad \underline{P} = \hat{\underline{P}}(\underline{F}, \theta, \underline{G}), \quad \underline{q}_\kappa = \hat{\underline{q}}_\kappa(\underline{F}, \theta, \underline{G}),$$

with $\underline{G} = \text{Grad } \theta$, subject to the referential form of (6.187) in every process.

EXPAND AND COLLECT

$$\begin{aligned} \dot{\psi} &= \hat{\psi}_{\underline{F}} \cdot \dot{\underline{F}} + \hat{\psi}_{\theta} \dot{\theta} + \hat{\psi}_{\underline{G}} \cdot \dot{\underline{G}} && \text{chain rule} \\ (\underline{P} - \rho_\kappa \hat{\psi}_{\underline{F}}) \cdot \dot{\underline{F}} - \rho_\kappa (\hat{\psi}_{\theta} + \eta) \dot{\theta} - \rho_\kappa \hat{\psi}_{\underline{G}} \cdot \dot{\underline{G}} - \theta^{-1} \underline{q}_\kappa \cdot \underline{G} &\geq 0 && \text{substitute} \end{aligned}$$

THE ARGUMENT Fix a point and an instant. The left side is *affine* in the triple $(\dot{\underline{F}}, \dot{\theta}, \dot{\underline{G}})$, while its coefficients depend only on $(\underline{F}, \theta, \underline{G})$. The two may be assigned independently: any state and any rates are realised by some motion and temperature field, and the balances (6.136) and (6.174) are then met by choosing $\underline{b}$ and r , both of which are at one's disposal. An affine function with $\underline{a} \cdot \underline{x} + b \geq 0$ for every $\underline{x}$ has $\underline{a} = \underline{0}$, since otherwise $\underline{x} = -\lambda \underline{a}$ with λ large violates it. Applying this to each rate in turn,

$$\hat{\psi}_{\underline{G}} = \underline{0}, \quad \underline{P} = \rho_\kappa \hat{\psi}_{\underline{F}}, \quad \eta = -\hat{\psi}_{\theta}, \quad \underline{q}_\kappa \cdot \underline{G} \leq 0.$$

READING THE FOUR The free energy cannot depend on the temperature gradient, so one potential $\hat{\psi}(\underline{\tilde{F}}, \theta)$ delivers both stress and entropy by differentiation. The pairing $\eta = -\hat{\psi}_\theta$ is the thermodynamic analogue of the power conjugacy of Remark 6.34, and it is the statement Remark 6.38 anticipated. What survives is the *heat conduction inequality*: heat flows down the temperature gradient. Fourier's law $\underline{q}_\kappa = -\underline{K}\underline{G}$ satisfies it exactly when $\underline{K}$ is positive semi-definite, so a sign usually asserted is here proved. The minus sign put into (6.166) was chosen to make this inequality read in this direction.

DISSIPATION Naming the left side of (6.187) the entropy production,

$$\rho\theta s = \underbrace{-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{\tilde{T}} \cdot \underline{\tilde{L}}}_{\text{internal}} - \underbrace{\theta^{-1}\underline{q} \cdot \text{grad } \theta}_{\text{thermal}} \geq 0.$$

Under the boxed results the internal bracket vanishes identically, so conduction carries the whole production. A material of this kind is *thermoelastic*: it stores and returns energy without loss. Admitting $\underline{\tilde{L}}$ among the arguments restores viscosity and leaves a residual inequality in $\underline{\tilde{L}}$ as well.

TWO HYPOTHESES WORTH NAMING The rates must be assignable independently of the state. That uses the freedom in $\underline{\tilde{b}}$ and r , and it fails under an internal constraint, where the admissible $\dot{\underline{\tilde{F}}}$ span a proper subspace; the conclusion then holds only modulo the constraint, which is why the pressure in an incompressible fluid is left undetermined, as Remark 6.38 records.

The entropy flux is taken to be $\theta^{-1}h$. Müller (1967) and Liu (1972) drop that, admit an independent entropy flux, and treat the balance laws as constraints with Lagrange multipliers; the multiplier conjugate to the energy balance comes out as θ^{-1} , so the classical choice becomes a theorem rather than the postulate Remark 6.37 calls it.

6.8 Problems

PROBLEM 1 (The referential balance laws). Derive the referential forms of the local balance laws and jump conditions discussed in Section 6.1. *Hint*: recall Problem 3 of Chapter 5.

Solution. Every local statement of Section 6.1 was assembled from two theorems and a quantifier, and none of the three knows which configuration it is working in. What changes on passing to κ is one thing only: the material stands still. Problem 3 of Chapter 5 carried out the consequence, deleting from (5.31) and (5.34) every term built from the

material velocity, and what survives is

$$\begin{aligned} \frac{d}{dt} \int_P \hat{\Phi} dV &= \int_P \frac{\partial \hat{\Phi}}{\partial t} dV - \int_S [[\hat{\Phi}]] U dA, \\ \int_{\partial P} \underline{A} \cdot \underline{N} dA &= \int_P \text{Div } \underline{A} dV + \int_S [[\underline{A}]] \cdot \underline{N} dA. \end{aligned} \quad (*_3)$$

Nothing crosses ∂P , so there is no convective flux there; the only motion left in the picture belongs to the singular surface S , which sweeps through the material at speed U along $\underline{N}$. Everything below is these two formulas applied to a balance law, and the interest lies in exactly one of the three terms.

Transferring the three terms. Two of the three move without thought. For the left-hand side and the bulk supply of (6.1) the substitution is

$$\rho dv \stackrel{(2.50)}{=} \rho (J dV) = (\rho J) dV \stackrel{(2.59)}{=} \rho_\kappa dV,$$

so that

$$\int_{P_t} \rho \varphi dv = \int_P \rho_\kappa \varphi dV, \quad \int_{P_t} \rho \sigma dv = \int_P \rho_\kappa \sigma dV.$$

The fields φ and σ are untouched. They are reckoned per unit *mass*, and the mass of a material element does not depend on which configuration is used to measure its volume; had they been given per unit volume, each would have acquired a factor J and the final equations would not have had the shape they do.

The boundary flux is the term that costs something, and it costs two moves. First Piola–Nanson (2.54) converts the oriented area element, $\underline{n} da = \underline{F}^* \underline{N} dA$, which in components reads $n_i da = F_{iA}^* N_A dA$; and then the tensor so produced stands in the wrong slot of the inner product and has to be moved. In components,

$$\begin{aligned} \underline{\pi} \cdot \underline{F}^* \underline{N} &= \pi_i (\underline{F}^* \underline{N})_i = \pi_i F_{iA}^* N_A = (F_{iA}^* \pi_i) N_A \\ &= \left((\underline{F}^*)^t \underline{\pi} \right)_A N_A = (\underline{F}^*)^t \underline{\pi} \cdot \underline{N}, \end{aligned}$$

which is the definition of the transpose written out. Hence

$$\int_{\partial P_t} \underline{\pi} \cdot \underline{n} \, da = \int_{\partial P} \underline{\pi} \cdot \underline{F}^* \underline{N} \, dA = \int_{\partial P} (\underline{F}^*)^t \underline{\pi} \cdot \underline{N} \, dA,$$

and the referential flux is forced on us rather than chosen:

$$\underline{\pi}_\kappa = (\underline{F}^*)^t \underline{\pi} = J \underline{F}^{-1} \underline{\pi}, \quad \pi_{(\kappa)A} = F_{iA}^* \pi_i.$$

The second form follows from (2.55): transposing $\underline{F}^* = J \underline{F}^{-t}$ gives $(\underline{F}^*)^t = (J \underline{F}^{-t})^t = J((\underline{F}^{-1})^t)^t = J \underline{F}^{-1}$. That this is the same rule (6.171) met for the heat flux is no coincidence: the heat flux was a particular $\underline{\pi}$, and the calculation there was this one with a letter changed. The subscript κ is used rather than a capital Π because $\underline{\Pi}$ already denotes referential relative position in (6.53). Collecting the three terms, (6.1) reads

$$\frac{d}{dt} \int_P \rho_\kappa \varphi \, dV = \int_P \rho_\kappa \sigma \, dV + \int_{\partial P} \underline{\pi}_\kappa \cdot \underline{N} \, dA. \quad (*_4)$$

Localizing. Apply $(*_3)_1$ with $\hat{\Phi} = \rho_\kappa \varphi$ to the left of $(*_4)$:

$$\frac{d}{dt} \int_P \rho_\kappa \varphi \, dV = \int_P \frac{\partial(\rho_\kappa \varphi)}{\partial t} \, dV - \int_S \llbracket \rho_\kappa \varphi \rrbracket U \, dA,$$

and $(*_3)_2$ with $\underline{A} = \underline{\pi}_\kappa$ to the boundary term on the right:

$$\int_{\partial P} \underline{\pi}_\kappa \cdot \underline{N} \, dA = \int_P \text{Div } \underline{\pi}_\kappa \, dV + \int_S \llbracket \underline{\pi}_\kappa \rrbracket \cdot \underline{N} \, dA.$$

Substituting both into $(*_4)$ and moving everything to one side,

$$\int_P \left\{ \frac{\partial(\rho_\kappa \varphi)}{\partial t} - \text{Div } \underline{\pi}_\kappa - \rho_\kappa \sigma \right\} dV - \int_S \left\{ \llbracket \rho_\kappa \varphi \rrbracket U + \llbracket \underline{\pi}_\kappa \rrbracket \cdot \underline{N} \right\} dA = 0. \quad (*_5)$$

The two-stage argument of Section 6.1 now runs verbatim, and it is worth running rather than citing. Choose P disjoint from S . The surface integral is then absent, and

$$\int_P \left\{ \frac{\partial(\rho_\kappa \varphi)}{\partial t} - \text{Div } \underline{\pi}_\kappa - \rho_\kappa \sigma \right\} dV = 0$$

for every such P . Were the integrand non-zero at some $\underline{X}_0 \in \kappa$, continuity would keep it of one sign on a ball about $\underline{X}_0$, and integrating over that ball alone would give a non-zero answer, contradicting the hypothesis. Hence the brace vanishes pointwise. Now let P meet S . The volume integral in $(*_5)$ vanishes by what has just been proved, whatever P is, so

$$\int_S \left\{ \llbracket \rho_\kappa \varphi \rrbracket U + \llbracket \underline{\pi}_\kappa \rrbracket \cdot \underline{N} \right\} dA = 0,$$

and $S \cap P$ is an arbitrary piece of the singular surface, so the same argument on surfaces kills the brace there too. Finally, since ρ_κ is independent of t by (6.18),

$$\frac{\partial(\rho_\kappa \varphi)}{\partial t} = \rho_\kappa \frac{\partial \varphi}{\partial t} + \varphi \underbrace{\frac{\partial \rho_\kappa}{\partial t}}_{=0} = \rho_\kappa \dot{\varphi},$$

the material derivative in κ being the partial derivative at fixed $\underline{X}$. Therefore

$$\rho_\kappa \dot{\varphi} = \text{Div } \underline{\pi}_\kappa + \rho_\kappa \sigma, \quad \llbracket \rho_\kappa \varphi \rrbracket U + \llbracket \underline{\pi}_\kappa \rrbracket \cdot \underline{N} = 0,$$

the counterparts of (6.21) and (6.6).

| READER'S NOTE *(the sign comparison, one factor at a time).*

spatial :	$\llbracket \rho \varphi (\underline{v} - \underline{u}) - \underline{\pi} \rrbracket \cdot \underline{n} = 0$	(6.6)
(i)	$\underline{v} \rightarrow \underline{\mathfrak{v}}$	material points do not move in κ
(ii)	$\underline{u} \rightarrow U \underline{N}$	the surface does
	$\implies \underline{v} - \underline{u} \rightarrow -U \underline{N}$	the one sign change
(iii)	$(-\rho_\kappa \varphi U \underline{N}) \cdot \underline{N} = -\rho_\kappa \varphi U$	$ \underline{N} = 1$
(iv)	$-\left\{ \llbracket \rho_\kappa \varphi \rrbracket U + \llbracket \underline{\pi}_\kappa \rrbracket \cdot \underline{N} \right\} = 0$	$U, \underline{N}$ single-valued
	$\therefore \llbracket \rho_\kappa \varphi \rrbracket U + \llbracket \underline{\pi}_\kappa \rrbracket \cdot \underline{N} = 0$	$\times(-1)$

READING Not a discrepancy between two derivations: one factor of (-1) , introduced at (ii) and removed at the end. Spatially, $\underline{v} - \underline{u}$ is the velocity of matter relative to the surface; referentially the matter is at rest and the surface sweeps through it, so the same relative motion is described with the opposite sign.

The vector case. Nothing changes except the alphabet. Let $\underline{\psi}$ have components ψ_i and let the flux be the tensor $\underline{A}$. Piola–Nanson gives, componentwise,

$$(\underline{A}\underline{n})_i \, da = A_{ij}n_j \, da = A_{ij}F_{jA}^* N_A \, dA = (\underline{A}\underline{F}^*)_i N_A \, dA,$$

so that $\int_{\partial P_t} \underline{A}\underline{n} \, da = \int_{\partial P} \underline{A}\underline{F}^* \underline{N} \, dA$ and the referential flux is the two-point tensor

$$\underline{A}_\kappa = \underline{A}\underline{F}^*, \quad A_{(\kappa)iA} = A_{ij}F_{jA}^*.$$

Applying the scalar result to each component ψ_i with flux $\underline{\pi} = \underline{A}^t \underline{e}_i$, and reassembling on the fixed basis $\{\underline{e}_i\}$ — the caveat of Remark 5.3 being that the reassembly is componentwise and the basis must not move —

$$\rho_\kappa \dot{\underline{\psi}} = \text{Div } \underline{A}_\kappa + \rho_\kappa \underline{r}, \quad \llbracket \rho_\kappa \underline{\psi} \rrbracket U + \llbracket \underline{A}_\kappa \rrbracket \underline{N} = \underline{0}.$$

The three instances. What remains is bookkeeping, and it is worth doing in full because it shows how little is left to do: each balance law is a matter of naming three letters.

Mass. Here $\varphi = 1$, $\sigma = 0$, $\underline{\pi} = \underline{0}$, so $\underline{\pi}_\kappa = (\underline{F}^*)^t \underline{0} = \underline{0}$ and $\dot{\varphi} = 0$. The field equation reads $\rho_\kappa \cdot 0 = \text{Div } \underline{0} + 0$, which is vacuous; the content sits in the step before it, namely $\partial(\rho_\kappa \cdot 1)/\partial t = 0$, that is

$$\frac{\partial \rho_\kappa}{\partial t} = 0,$$

which is (6.18). The jump condition is $\llbracket \rho_\kappa \cdot 1 \rrbracket U + \llbracket \underline{0} \rrbracket \cdot \underline{N} = 0$, that is

$$\llbracket \rho_\kappa \rrbracket U = 0.$$

So either $U = 0$, the surface being stationary in κ and hence material, or $\llbracket \rho_\kappa \rrbracket = 0$ and the referential density is continuous across it.

Linear momentum. Here $\underline{\psi} = \underline{v}$, $\underline{r} = \underline{b}$, $\underline{A} = \underline{T}$, so that

$$\underline{A}_\kappa = \underline{T}\underline{F}^* \stackrel{(6.140)}{=} \underline{P},$$

and the two results read

$$\rho_\kappa \dot{\underline{v}} = \text{Div } \underline{P} + \rho_\kappa \underline{b}, \quad \llbracket \rho_\kappa \underline{v} \rrbracket U + \llbracket \underline{P} \rrbracket \underline{N} = \underline{0},$$

the first being (6.136).

Energy. Here $\varphi = \varepsilon$ and $\pi = -q$, so that

$$\pi_\kappa = (F^*)^t(-q) = -(F^*)^t q \stackrel{(6.171)}{=} -q_\kappa,$$

while the supply carries the stress power, $\rho_\kappa \sigma = \rho_\kappa r + P \cdot \dot{F}$. The field equation is

$$\rho_\kappa \dot{\varepsilon} = \text{Div}(-q_\kappa) + \rho_\kappa r + P \cdot \dot{F} = P \cdot \dot{F} - \text{Div} q_\kappa + \rho_\kappa r,$$

which is (6.174), and the jump condition is

$$[\![\rho_\kappa \varepsilon]\!]U + [\![-q_\kappa]\!] \cdot N = 0, \quad \text{that is} \quad [\![\rho_\kappa \varepsilon]\!]U - [\![q_\kappa]\!] \cdot N = 0,$$

which is the condition the book leaves to the reader after (6.174). The generic statement was proved once; these are three substitutions into it.

A consistency check. The spatial and referential jump conditions look different and must say the same thing. Problem 3 of Chapter 5 recorded the relation between the two normal speeds,

$$u - v \cdot n = \frac{J}{|F^* N|} U,$$

so the mass flux of Remark 6.3 is

$$m^* = \rho(v \cdot n - u) = -\rho(u - v \cdot n) = -\rho \frac{J}{|F^* N|} U = -\frac{\rho J}{|F^* N|} U = -\frac{\rho_\kappa}{|F^* N|} U,$$

using $\rho J = \rho_\kappa$ at the last step. Now U and N belong to the surface and are single-valued on it. So is $|F^* N|$, and the reason is worth stating: the motion χ_κ is continuous across S , so its derivatives *along* S agree when computed from either side, and $F^* N \, dA = n \, da$ is the cross product of two vectors tangent to S , hence built from precisely those tangential derivatives. Therefore the factor $-1/|F^* N|$ passes through the jump bracket, and

$$[\![m^*]\!] = -\frac{U}{|F^* N|} [\![\rho_\kappa]\!],$$

so that $[\![m^*]\!] = 0$, which is the spatial condition (6.15), holds if and only if $[\![\rho_\kappa]\!]U = 0$, which is the referential one. The other two convert by the same substitution. $\square$

PROBLEM 2 (The referential center of mass). Verify that $\bar{\tilde{X}}$ appearing in (6.53) is in fact the position of the center of mass of $P \subset \kappa$.

Solution. The center of mass in κ can only be defined one way, by copying (6.40) with referential quantities throughout:

$$m(S) \bar{\tilde{X}} = \int_P \rho_\kappa \tilde{X} \, dV, \quad m(S) \bar{\tilde{X}}_A = \int_P \rho_\kappa X_A \, dV. \quad (*_6)$$

What makes this worth writing down is that neither factor on the right depends on t : $\rho_\kappa = \rho_\kappa(\tilde{X})$ by (6.18), and P is a fixed region of κ . So $\bar{\tilde{X}}$ is a *constant* vector, once and for all, and

$$\dot{\bar{\tilde{X}}} = \underline{0}.$$

That is exactly what (6.53) needs of it, and it is why the definition is legitimate in a way its spatial counterpart is not: $\bar{x}(t)$ moves.

The mass $m(S)$ appearing in $(*_6)$ is the same number in both configurations. By (2.56) together with conservation of mass,

$$m(S) = \int_{P_t} \rho \, dv = \int_P \rho J \, dV = \int_P \rho_\kappa \, dV,$$

so that it may be used on either side without comment.

The two centers correspond. The question is whether $\bar{\tilde{X}}$ maps to $\bar{x}$ under the motion, and the answer turns on a single structural feature of (6.52). Transfer (6.40) to κ :

$$m(S) \bar{x}(t) = \int_{P_t} \rho \tilde{x} \, dv = \int_P \rho J \tilde{x} \, dV = \int_P \rho_\kappa \tilde{x} \, dV,$$

and substitute the rigid motion $\tilde{x} = Q(t)\tilde{X} + \underline{c}(t)$. Because Q and $\underline{c}$ depend on t alone, they are constant over the region of integration, and the integral splits. In components, with $x_i = Q_{iA}X_A + c_i$,

$$\begin{aligned} \int_P \rho_\kappa x_i \, dV &= \int_P \rho_\kappa (Q_{iA}X_A + c_i) \, dV \\ &= Q_{iA} \int_P \rho_\kappa X_A \, dV + c_i \int_P \rho_\kappa \, dV \\ &= Q_{iA} m(S) \bar{\tilde{X}}_A + c_i m(S), \end{aligned}$$

by $(*_6)$ and by the expression for $m(S)$ above. In direct notation this is

$$\int_P \rho_\kappa (\underline{Q} \underline{X} + \underline{c}) \, dV = \underline{Q} (m(S) \bar{\underline{X}}) + \underline{c} m(S),$$

$\underline{Q}$ passing outside as a constant linear map and $\underline{c}$ as a constant vector. Dividing by the positive number $m(S)$,

$$\bar{\underline{x}}(t) = \underline{Q}(t) \bar{\underline{X}} + \underline{c}(t).$$

The mass center is therefore carried by the very map that carries every other material point. Subtracting this from (6.52),

$$\underline{\pi} = \underline{x} - \bar{\underline{x}} = (\underline{Q} \underline{X} + \underline{c}) - (\underline{Q} \bar{\underline{X}} + \underline{c}) = \underline{Q} \underline{X} - \underline{Q} \bar{\underline{X}} = \underline{Q} (\underline{X} - \bar{\underline{X}}) = \underline{Q} \underline{\Pi},$$

which is (6.53), the third equality by linearity of $\underline{Q}$. The translation cancels, as it must: where the body happens to be cannot affect the position of one of its points relative to another.

The corollary. The same computation run on $\underline{\Pi}$ alone gives the referential counterpart of (6.42). Componentwise,

$$\begin{aligned} \int_P \rho_\kappa \Pi_A \, dV &= \int_P \rho_\kappa (X_A - \bar{X}_A) \, dV \\ &= \int_P \rho_\kappa X_A \, dV - \bar{X}_A \int_P \rho_\kappa \, dV \\ &= m(S) \bar{X}_A - \bar{X}_A m(S) = 0, \end{aligned}$$

so that

$$\int_P \rho_\kappa \underline{\Pi} \, dV = \underline{0}.$$

This is the identity (6.67) uses, and it is the reason the inertia tensor appearing in (6.64) is the one about the mass center rather than about some other point: about any other point the integral would not vanish and (6.45) would retain its cross terms.

As a check that the two statements agree, apply $\underline{Q}$ to the last display and use $\underline{\pi} = \underline{Q} \underline{\Pi}$ together with $\rho \, dv = \rho_\kappa \, dV$:

$$\int_{P_t} \rho \underline{\pi} \, dv = \int_P \rho_\kappa \underline{Q} \underline{\Pi} \, dV = \underline{Q} \int_P \rho_\kappa \underline{\Pi} \, dV = \underline{Q} \underline{0} = \underline{0},$$

which is (6.42).

Where rigidity entered. It is worth being explicit, since the calculation looks as though it might not have needed it. The step

$$\int_P \rho_\kappa \chi_\kappa(\underline{X}, t) dV = \underline{Q} \int_P \rho_\kappa \underline{X} dV + \underline{c} m(S)$$

used that χ_κ is *affine* in $\underline{X}$, and nothing weaker will let the motion through the integral sign: for a general χ_κ the integral $\int \rho_\kappa \chi_\kappa dV$ cannot be expressed through $\int \rho_\kappa \underline{X} dV$ at all. For a general motion the image of the referential mass center is therefore not the spatial mass center — bending a bar moves its centroid relative to the material — so (6.53) is a statement about rigid bodies and not a general identity. The present problem is precisely the check that it is available in the one place Section 6.4.2 uses it. $\square$

PROBLEM 3 (The co-rotational derivative in components). Show that the co-rotational derivative defined by (6.68) is expressible in the form $\dot{\underline{J}} = \dot{J}_{ij} \underline{e}_i^* \otimes \underline{e}_j^*$, where the J_{ij} are the components of $\underline{J}$ on the rotating basis (6.58).

Solution. The vector case (6.61) said that $\dot{\underline{u}}$ is what one gets by differentiating the components of $\underline{u}$ on a basis painted on the body and reassembling. A tensor differs from a vector in one respect only: it has two legs, and both of them turn. Everything below is that observation carried out, and the asymmetry between the two corrections in (6.68) is manufactured in one line of it.

Resolving on the rotating basis. By (6.58) the triad $\{\underline{e}_i^*\}$ is obtained from a fixed orthonormal triad by the rotation $\underline{Q}(t)$, so it is orthonormal at every instant by Proposition 1.48:

$$\underline{e}_i^* \cdot \underline{e}_j^* = \delta_{ij} \quad \text{for all } t.$$

Theorem 1.33 therefore resolves any spatial tensor on it,

$$\underline{J} = J_{ij} \underline{e}_i^* \otimes \underline{e}_j^*, \quad J_{ij} = \underline{e}_i^* \cdot \underline{J} \underline{e}_j^*, \quad (*7)$$

the second formula being the first tested on $\underline{e}_j^*$ and paired with $\underline{e}_i^*$:

$$\underline{e}_i^* \cdot \underline{J} \underline{e}_j^* = J_{kl} \underline{e}_i^* \cdot (\underline{e}_k^* \otimes \underline{e}_l^*) \underline{e}_j^* = J_{kl} \delta_{lj} \underline{e}_i^* \cdot \underline{e}_k^* = J_{kl} \delta_{lj} \delta_{ik} = J_{ij}.$$

Both J_{ij} and $\underline{e}_i^*$ depend on t , so differentiating $(*)_7$ is a product rule on three factors and produces one term for the components and one for each leg:

$$\dot{\underline{J}} = \dot{J}_{ij} \underline{e}_i^* \otimes \underline{e}_j^* + J_{ij} \dot{\underline{e}}_i^* \otimes \underline{e}_j^* + J_{ij} \underline{e}_i^* \otimes \dot{\underline{e}}_j^*. \quad (*_8)$$

The first term is the answer we want. The other two are the rotation of the frame, and (6.59), written with the tensor rather than its axial vector, gives $\dot{\underline{e}}_i^* = \underline{\Omega} \underline{e}_i^*$ for each i . What is instructive is that the two are disposed of by *different* identities.

The first leg. A tensor acting on a dyad from the left goes straight onto the first factor:

$$\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}.$$

To see it, test both sides on an arbitrary $\underline{w} \in \mathbb{E}^3$. The left gives $\underline{A}[(\underline{a} \otimes \underline{b})\underline{w}] = \underline{A}[(\underline{b} \cdot \underline{w})\underline{a}] = (\underline{b} \cdot \underline{w}) \underline{A}\underline{a}$, using linearity of $\underline{A}$ and Definition 1.32; the right gives $[(\underline{A}\underline{a}) \otimes \underline{b}]\underline{w} = (\underline{b} \cdot \underline{w}) \underline{A}\underline{a}$. They agree for every $\underline{w}$, so the tensors are equal. This is Lemma 2.15. Applying it with $\underline{A} = \underline{\Omega}$, $\underline{a} = \underline{e}_i^*$, $\underline{b} = \underline{e}_j^*$, and summing,

$$J_{ij} \dot{\underline{e}}_i^* \otimes \underline{e}_j^* = J_{ij} (\underline{\Omega} \underline{e}_i^*) \otimes \underline{e}_j^* = \underline{\Omega} (J_{ij} \underline{e}_i^* \otimes \underline{e}_j^*) = \underline{\Omega} \underline{J}.$$

The second leg. A tensor acting on a dyad from the right reaches the second factor only through its transpose:

$$(\underline{a} \otimes \underline{b}) \underline{B}^t = \underline{a} \otimes (\underline{B}\underline{b}).$$

Again test on $\underline{w}$. The left gives $(\underline{a} \otimes \underline{b})(\underline{B}^t \underline{w}) = \underline{b} \cdot \underline{B}^t \underline{w} \underline{a} = \underline{B} \underline{b} \cdot \underline{w} \underline{a}$, the last step by the definition of the transpose; the right gives $[\underline{a} \otimes (\underline{B}\underline{b})]\underline{w} = \underline{B} \underline{b} \cdot \underline{w} \underline{a}$. They agree. Now take $\underline{B} = \underline{\Omega}$. Since $\underline{\Omega}$ is skew, $\underline{\Omega}^t = -\underline{\Omega}$, and this is where the sign is made:

$$J_{ij} \underline{e}_i^* \otimes \dot{\underline{e}}_j^* = J_{ij} \underline{e}_i^* \otimes (\underline{\Omega} \underline{e}_j^*) = (J_{ij} \underline{e}_i^* \otimes \underline{e}_j^*) \underline{\Omega}^t = \underline{J} \underline{\Omega}^t = -\underline{J} \underline{\Omega}.$$

Assembling. Substituting both into $(*_8)$,

$$\dot{\underline{J}} = \dot{J}_{ij} \underline{e}_i^* \otimes \underline{e}_j^* + \underline{\Omega} \underline{J} - \underline{J} \underline{\Omega},$$

so that, by the definition (6.68),

$$\overset{\circ}{\underline{J}} = \dot{\underline{J}} - \underline{\Omega} \underline{J} + \underline{J} \underline{\Omega} = \dot{J}_{ij} \underline{e}_i^* \otimes \underline{e}_j^*.$$

Read backwards, this says that (6.68) was never a choice. The two correction terms are not selected for elegance or symmetry; they are exactly what must be removed to strip the rotation from the two legs, and the opposite signs are the arithmetic of the two paragraphs above and nothing else. One leg is acted on from the left, where $\underline{\Omega}$ arrives unchanged; the other from the right, where it arrives transposed, and $\underline{\Omega}^t = -\underline{\Omega}$ does the rest. A vector has one leg and needs one correction, (6.60); a tensor of order n would need n , alternating in sign by the same argument.

What vanishing means. By the boxed formula,

$$\overset{\circ}{\underline{J}} = \underline{Q} \iff \dot{J}_{ij} \underline{e}_i^* \otimes \underline{e}_j^* = \underline{Q} \iff \dot{J}_{ij} = 0 \text{ for all } i, j,$$

the last equivalence because $\{\underline{e}_i^* \otimes \underline{e}_j^*\}$ is a basis of the space of spatial tensors, being nine linearly independent dyads built from an orthonormal triad. So $\overset{\circ}{\underline{J}}$ vanishes precisely when $\underline{J}$ has constant components on a basis painted on the body. That is the exact content of (6.66), and it is what makes Euler's equations (6.70) a system with constant coefficients when written in the body frame — the one frame in which the inertia tensor stops moving. In the inertial frame $\underline{J}$ is emphatically not constant, and $\dot{\underline{J}} = \underline{\Omega} \underline{J} - \underline{J} \underline{\Omega}$ is precisely how much it moves; Remark 6.17 traces the gyroscopic term of (6.70) to this and nothing else. $\square$

PROBLEM 4 (The inertia tensor is positive definite). Recall the inertia tensor $\underline{J}(S, t) = \int_{P_t} \rho[(\underline{\pi} \cdot \underline{\pi}) \underline{I} - \underline{\pi} \otimes \underline{\pi}] dv$ associated with an arbitrary part S of a body, where $\underline{\pi}$ is the position of a point of P_t relative to the center of mass. (a) Show that the integrand is positive semi-definite, a symmetric tensor $\underline{A}$ being positive semi-definite if and only if $\underline{a} \cdot \underline{A} \underline{a} \geq 0$

for all $\underline{a} \in \mathbb{E}^3$. (b) Show that $\underline{J}$ is positive definite, that is $\underline{a} \cdot \underline{J}\underline{a} > 0$ for all $\underline{a} \neq \underline{0}$.

Solution. Both parts turn on Cauchy–Schwarz, and the whole interest of the problem is in its equality case: part (a) needs the inequality, part (b) needs to know exactly when it is tight.

(a) Write $\underline{A} = (\underline{\pi} \cdot \underline{\pi})\underline{I} - \underline{\pi} \otimes \underline{\pi}$ for the integrand and take any $\underline{a} \in \mathbb{E}^3$. Evaluating the two pieces on $\underline{a}$ by Definition 1.32,

$$[(\underline{\pi} \cdot \underline{\pi})\underline{I}]\underline{a} = (\underline{\pi} \cdot \underline{\pi})\underline{a}, \quad (\underline{\pi} \otimes \underline{\pi})\underline{a} = (\underline{\pi} \cdot \underline{a})\underline{\pi},$$

so that

$$\underline{A}\underline{a} = (\underline{\pi} \cdot \underline{\pi})\underline{a} - (\underline{\pi} \cdot \underline{a})\underline{\pi},$$

and pairing with $\underline{a}$,

$$\underline{a} \cdot \underline{A}\underline{a} = (\underline{\pi} \cdot \underline{\pi})\underline{a} \cdot \underline{a} - (\underline{\pi} \cdot \underline{a})\underline{a} \cdot \underline{\pi} = |\underline{\pi}|^2|\underline{a}|^2 - (\underline{\pi} \cdot \underline{a})^2,$$

using $\underline{a} \cdot \underline{a} = |\underline{a}|^2$ and the symmetry $\underline{a} \cdot \underline{\pi} = \underline{\pi} \cdot \underline{a}$ of the inner product. In components, with π_i and a_i the components on a fixed basis,

$$\underline{a} \cdot \underline{A}\underline{a} = (\pi_k \pi_k)(a_l a_l) - (\pi_k a_k)(\pi_l a_l).$$

By Cauchy–Schwarz, Theorem 1.6, $|\underline{\pi} \cdot \underline{a}| \leq |\underline{\pi}| |\underline{a}|$; both sides are non-negative, so squaring preserves the inequality and $(\underline{\pi} \cdot \underline{a})^2 \leq |\underline{\pi}|^2 |\underline{a}|^2$. Hence

$$\underline{a} \cdot \underline{A}\underline{a} \geq 0 \text{ for every } \underline{a}.$$

Symmetry of $\underline{A}$ is immediate: $\underline{I}^t = \underline{I}$, and $(\underline{\pi} \otimes \underline{\pi})^t = \underline{\pi} \otimes \underline{\pi}$ since $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$. So $\underline{A}$ is symmetric and positive semi-definite in the stated sense.

The geometry behind the algebra is worth naming, because it makes part (b) transparent. For $|\underline{a}| = 1$ the expression is $|\underline{\pi}|^2 - (\underline{\pi} \cdot \underline{a})^2$: the squared length of $\underline{\pi}$, less the square of its component along $\underline{a}$. Decomposing $\underline{\pi} = (\underline{\pi} \cdot \underline{a})\underline{a} + \underline{\pi}^\perp$ with $\underline{\pi}^\perp \cdot \underline{a} = 0$ and applying Pythagoras,

$$|\underline{\pi}|^2 = (\underline{\pi} \cdot \underline{a})^2 + |\underline{\pi}^\perp|^2, \quad \text{so} \quad \underline{a} \cdot \underline{A}\underline{a} = |\underline{\pi}^\perp|^2,$$

the squared distance from the point π to the axis through the mass center in the direction $\underline{a}$. Hence $\underline{a} \cdot \underline{J}\underline{a}$ is the moment of inertia about that axis, and (a) says that a moment of inertia is never negative — a fact one would want to be true before proving it.

The same decomposition locates the equality. Theorem 1.6 is tight exactly when π and $\underline{a}$ are linearly dependent, that is when $\pi^\perp = \underline{0}$, so for fixed $\underline{a} \neq \underline{0}$ the integrand vanishes at precisely those points where π is parallel to $\underline{a}$, the case $\pi = \underline{0}$ included. Those points form a single straight line through the mass center, namely $\{\pi = \lambda \underline{a} : \lambda \in \mathbb{R}\}$.

(b) Integrate. For any fixed $\underline{a}$,

$$\begin{aligned} \underline{a} \cdot \underline{J}\underline{a} &= \underline{a} \cdot \left(\int_{P_t} \rho \underline{A} \, dv \right) \underline{a} = \int_{P_t} \rho \underline{a} \cdot \underline{A}\underline{a} \, dv \\ &= \int_{P_t} \rho \left[|\pi|^2 |\underline{a}|^2 - (\pi \cdot \underline{a})^2 \right] dv, \end{aligned}$$

the interchange of $\underline{a}$ with the integral being legitimate because $\underline{a}$ is a fixed vector and the integral is of components on a fixed basis, in the sense of Remark 5.3: explicitly, $a_i J_{ij} a_j = a_i \left(\int_{P_t} \rho A_{ij} \, dv \right) a_j = \int_{P_t} \rho a_i A_{ij} a_j \, dv$, since the a_i are constants. The integrand is non-negative by (a) and by $\rho > 0$, so $\underline{a} \cdot \underline{J}\underline{a} \geq 0$ for every $\underline{a}$. It remains to exclude equality when $\underline{a} \neq \underline{0}$.

Suppose $\underline{a} \cdot \underline{J}\underline{a} = 0$ for some $\underline{a} \neq \underline{0}$. The integrand $f = \rho[|\pi|^2 |\underline{a}|^2 - (\pi \cdot \underline{a})^2]$ is continuous on P_t and non-negative, and its integral vanishes. A continuous non-negative function with vanishing integral over a region vanishes identically on it: were $f(\underline{x}_0) > 0$ at some point, continuity would give a ball $B \subset P_t$ about $\underline{x}_0$ on which $f > \frac{1}{2}f(\underline{x}_0) > 0$, whence

$$\int_{P_t} f \, dv \geq \int_B f \, dv \geq \frac{1}{2}f(\underline{x}_0) V(B) > 0,$$

the first inequality because $f \geq 0$ off B , contradicting the hypothesis. This is Theorem 2.20 run with an inequality in place of an equation.

So $f \equiv 0$ on P_t , and since $\rho > 0$ the bracket vanishes throughout. By the equality case recorded in (a), every point of P_t then has π parallel to $\underline{a}$: the whole of P_t lies on one straight line through the mass center.

A line has zero volume and P_t has positive volume, S being a genuine part of a body, so the supposition is untenable and

$$\underline{a} \cdot \underline{J} \underline{a} > 0 \text{ for every } \underline{a} \neq \underline{0}.$$

What positive definiteness buys. It is not an ornament. By Theorem 3.13 it gives $\underline{J}$ three strictly positive principal values J_1, J_2, J_3 , the principal moments of inertia, and by Theorem 3.7 an orthonormal triad of principal axes on which $\underline{J} = \sum_i J_i \underline{n}_i \otimes \underline{n}_i$. It also makes $\underline{J}$ invertible, since a tensor with $\underline{a} \cdot \underline{J} \underline{a} > 0$ for all $\underline{a} \neq \underline{0}$ has trivial kernel: $\underline{J} \underline{a} = \underline{0}$ would give $\underline{a} \cdot \underline{J} \underline{a} = 0$ and hence $\underline{a} = \underline{0}$. That is what allows (6.70) to be solved for the angular acceleration,

$$\dot{\underline{\omega}} = \underline{J}^{-1} [\underline{m}(S, t) - \underline{\omega} \times \underline{J} \underline{\omega}], \quad \underline{J}^{-1} = \sum_i J_i^{-1} \underline{n}_i \otimes \underline{n}_i.$$

Without it the rotational equation could not be put in normal form and the initial-value problem for a rigid body would not be well posed. The degenerate case excluded above, a body concentrated on a line, is exactly the case of a body with no resistance to rotation about that line — the moment of inertia about it would be zero — and it is excluded not by an extra hypothesis but by the continuum hypothesis itself. $\square$

PROBLEM 5 (Shear stress on an arbitrary plane; the octahedral shear). Let $\{\sigma_i\}$ and $\{\underline{n}_i\}$ be the principal Cauchy stresses and the associated orthonormal principal stress axes at a position $\underline{x}$ in a body. Use these axes to define the unit vector $\underline{n}(\theta, \varphi) = \cos \varphi \underline{e}_r(\theta) + \sin \varphi \underline{n}_3$, where $\underline{e}_r(\theta) = \cos \theta \underline{n}_1 + \sin \theta \underline{n}_2$, with $0 \leq \theta < 2\pi$ and $-\pi \leq \varphi \leq \pi$. (a) Compute the shear stress $\tau(\theta, \varphi)$ on the plane with unit normal $\underline{n}(\theta, \varphi)$. (b) Compute the octahedral shear stress, that is the shear stress on a plane whose normal is inclined at equal angles to each of the principal axes.

Solution. Setting up. By the spectral theorem, Theorem 3.7, applied to the symmetric $\underline{T}$ of (6.117),

$$\underline{T} = \sum_{i=1}^3 \sigma_i \underline{n}_i \otimes \underline{n}_i, \quad \underline{n}_i \cdot \underline{n}_j = \delta_{ij}.$$

Write $\underline{n} = \nu_i \underline{n}_i$. From the parametrisation given, $\underline{e}_r(\theta) = \cos \theta \underline{n}_1 + \sin \theta \underline{n}_2$ and $\underline{n} = \cos \varphi \underline{e}_r + \sin \varphi \underline{n}_3$, so

$$\nu_1 = \cos \varphi \cos \theta, \quad \nu_2 = \cos \varphi \sin \theta, \quad \nu_3 = \sin \varphi,$$

and

$$\nu_i \nu_i = \cos^2 \varphi \cos^2 \theta + \cos^2 \varphi \sin^2 \theta + \sin^2 \varphi = \cos^2 \varphi (\cos^2 \theta + \sin^2 \theta) + \sin^2 \varphi = 1,$$

confirming that $\underline{n}$ is a unit vector for every (θ, φ) . The traction is diagonal in the principal frame:

$$\underline{t} = \underline{T} \underline{n} = \sum_i \sigma_i (\underline{n}_i \otimes \underline{n}_i) \underline{n} = \sum_i \sigma_i \underline{n}_i \cdot \underline{n} \underline{n}_i = \sum_i \sigma_i \nu_i \underline{n}_i,$$

and orthonormality gives the two scalars everything will be built from,

$$\sigma = \underline{n} \cdot \underline{t} = \left(\sum_j \nu_j \underline{n}_j \right) \cdot \left(\sum_i \sigma_i \nu_i \underline{n}_i \right) = \sum_i \sigma_i \nu_i^2, \quad |\underline{t}|^2 = \underline{t} \cdot \underline{t} = \sum_i \sigma_i^2 \nu_i^2.$$

(a) The decomposition (6.122) writes $\underline{t} = \sigma \underline{n} + \tau \underline{s}$ with $\underline{n} \cdot \underline{s} = 0$ and $|\underline{s}| = |\underline{n}| = 1$, so the two pieces are orthogonal and

$$|\underline{t}|^2 = \sigma \underline{n} + \tau \underline{s} \cdot \sigma \underline{n} + \tau \underline{s} = \sigma^2 |\underline{n}|^2 + 2\sigma\tau \underbrace{\underline{n} \cdot \underline{s}}_{=0} + \tau^2 |\underline{s}|^2 = \sigma^2 + \tau^2,$$

whence

$$\tau^2 = |\underline{t}|^2 - \sigma^2 = \sum_i \sigma_i^2 \nu_i^2 - \left(\sum_i \sigma_i \nu_i^2 \right)^2.$$

This answers (a), and conceals what it means. Put $w_i = \nu_i^2$; then $w_i \geq 0$ and $\sum_i w_i = 1$, so w is a probability distribution on the three principal stresses, and the right-hand side is $\langle \sigma^2 \rangle - \langle \sigma \rangle^2$: the shear stress squared is the *variance* of the principal stresses under the weights w . A variance

is a sum over pairs, and writing it that way costs four lines:

$$\begin{aligned}
\sum_{i<j} w_i w_j (\sigma_i - \sigma_j)^2 &= \sum_{i<j} w_i w_j (\sigma_i^2 - 2\sigma_i \sigma_j + \sigma_j^2) \\
&= \sum_{i<j} w_i w_j (\sigma_i^2 + \sigma_j^2) - 2 \sum_{i<j} w_i w_j \sigma_i \sigma_j \\
&= \sum_i \sigma_i^2 w_i \left(\sum_{j \neq i} w_j \right) - 2 \sum_{i<j} w_i w_j \sigma_i \sigma_j \\
&= \sum_i \sigma_i^2 w_i (1 - w_i) - 2 \sum_{i<j} w_i w_j \sigma_i \sigma_j \\
&= \sum_i \sigma_i^2 w_i - \left(\sum_i \sigma_i^2 w_i^2 + 2 \sum_{i<j} w_i w_j \sigma_i \sigma_j \right) \\
&= \sum_i \sigma_i^2 w_i - \left(\sum_i \sigma_i w_i \right)^2.
\end{aligned}$$

The third line collected, for each fixed i , the two places where σ_i^2 occurs in the pair sum; the fourth used $\sum_{j \neq i} w_j = 1 - w_i$; and the last recognised $\left(\sum_i \sigma_i w_i \right)^2 = \sum_i \sigma_i^2 w_i^2 + 2 \sum_{i<j} w_i w_j \sigma_i \sigma_j$. Therefore

$$\tau^2(\theta, \varphi) = \sum_{i<j} (\sigma_i - \sigma_j)^2 \nu_i^2 \nu_j^2$$

with the ν_i above. Written out, the three pairs are

$$\begin{aligned}
\nu_1^2 \nu_2^2 &= \cos^4 \varphi \cos^2 \theta \sin^2 \theta = \frac{1}{4} \cos^4 \varphi \sin^2 2\theta, \\
\nu_1^2 \nu_3^2 &= \cos^2 \varphi \cos^2 \theta \sin^2 \varphi = \frac{1}{4} \sin^2 2\varphi \cos^2 \theta, \\
\nu_2^2 \nu_3^2 &= \cos^2 \varphi \sin^2 \theta \sin^2 \varphi = \frac{1}{4} \sin^2 2\varphi \sin^2 \theta,
\end{aligned}$$

using $2 \cos \theta \sin \theta = \sin 2\theta$ and $2 \cos \varphi \sin \varphi = \sin 2\varphi$, so that

$$\begin{aligned}
\tau^2 &= \frac{1}{4} \left[(\sigma_1 - \sigma_2)^2 \cos^4 \varphi \sin^2 2\theta \right. \\
&\quad \left. + \sin^2 2\varphi \left((\sigma_1 - \sigma_3)^2 \cos^2 \theta + (\sigma_2 - \sigma_3)^2 \sin^2 \theta \right) \right].
\end{aligned}$$

The pair form earns its keep immediately. Shear is driven by *differences* of principal stresses, so if all three coincide it vanishes on every

plane, which is the state of pure pressure (6.128); and if $\underline{n}$ is a principal direction, two of the ν_i vanish, killing every pair, which is (6.127) read backwards.

The largest value. This is worth deriving rather than quoting, and the derivation is an exercise in choosing coordinates. Order the principal stresses $\sigma_1 \geq \sigma_2 \geq \sigma_3$ and name the two gaps

$$a = \sigma_1 - \sigma_2 \geq 0, \quad b = \sigma_2 - \sigma_3 \geq 0, \quad \sigma_1 - \sigma_3 = a + b,$$

so that in the weights the boxed formula reads

$$\tau^2 = a^2 w_1 w_2 + (a + b)^2 w_1 w_3 + b^2 w_2 w_3, \quad w_i \geq 0, \quad \sum_i w_i = 1. \quad (\#)$$

Assume $a + b > 0$; the case $a + b = 0$ is the pure pressure already disposed of. Maximizing a quadratic form on a simplex invites Lagrange multipliers, but the constraint can be eliminated instead, and then the answer is visible. Hold w_2 fixed, put $s = 1 - w_2 = w_1 + w_3$, and split the two extreme weights symmetrically about their mean,

$$w_1 = \frac{s}{2} + u, \quad w_3 = \frac{s}{2} - u, \quad |u| \leq \frac{s}{2},$$

so that u measures how unevenly the extremes are loaded. Then $w_1 w_2 = w_2(\frac{s}{2} + u)$, $w_2 w_3 = w_2(\frac{s}{2} - u)$ and $w_1 w_3 = \frac{s^2}{4} - u^2$, so

$$\begin{aligned} \tau^2 &= w_2 \left[a^2 \left(\frac{s}{2} + u \right) + b^2 \left(\frac{s}{2} - u \right) \right] + (a + b)^2 \left(\frac{s^2}{4} - u^2 \right) \\ &= -(a + b)^2 u^2 + w_2 (a^2 - b^2) u + \frac{s}{2} w_2 (a^2 + b^2) + \frac{s^2}{4} (a + b)^2, \end{aligned}$$

a downward parabola in u , which one maximizes by completing the square. With $a^2 - b^2 = (a + b)(a - b)$ the vertex sits at

$$u^* = \frac{w_2(a^2 - b^2)}{2(a + b)^2} = \frac{w_2(a + b)(a - b)}{2(a + b)^2} = \frac{(a - b) w_2}{2(a + b)},$$

and completing the square gives

$$\tau^2 = \frac{(a + b)^2}{4} - ab w_2 - (a + b)^2 (u - u^*)^2. \quad (\#\#)$$

| READER'S NOTE (*the check that (##) is an identity*). ABBREVIATE

$$p = (a + b)^2, \quad q = (a - b)^2 \quad \implies \quad a^2 + b^2 = \frac{1}{2}(p + q), \quad q - p = -4ab.$$

COMPLETE THE SQUARE

$$\tau^2 = -p u^2 + w_2(a^2 - b^2)u + K, \quad K := \frac{s}{2}w_2(a^2 + b^2) + \frac{s^2}{4}p$$

$$u^* = \frac{w_2(a^2 - b^2)}{2p} = \frac{w_2(a + b)(a - b)}{2(a + b)^2} = \frac{(a - b)w_2}{2(a + b)} \quad a^2 - b^2 = (a + b)(a - b)$$

$$\tau^2 = -p(u - u^*)^2 + C, \quad C := p u^{*2} + K$$

EVALUATE C

$$4C = w_2^2 q + 2w_2(1 - w_2)(a^2 + b^2) + (1 - w_2)^2 p \quad p u^{*2} = \frac{1}{4}w_2^2 q, \quad s = 1 - w_2$$

$$= w_2^2 q + w_2(1 - w_2)(p + q) + (1 - w_2)^2 p \quad 2(a^2 + b^2) = p + q$$

$$= q w_2 \underbrace{[w_2 + 1 - w_2]}_1 + p(1 - w_2) \underbrace{[(1 - w_2) + w_2]}_1 \quad \text{collect on } p, q$$

$$= p + w_2(q - p) = p - 4ab w_2 \quad q - p = -4ab$$

$$\therefore C = \frac{(a + b)^2}{4} - ab w_2 \quad = (\#) \checkmark$$

An identity in a, b, u, w_2 : no inequality was used. Only afterwards are $a, b, w_2 \geq 0$ invoked, to read off the sign of each term.

WHY SUBSTITUTE AT ALL (#) is a quadratic form in three constrained variables; eliminating the constraint leaves a downward parabola in one free u , maximized by inspection, and w_2 then appears only in the manifestly non-positive $-ab w_2$. Both maximizations are read off (##) without differentiating.

Both corrections in (##) are non-positive, the first because $a, b, w_2 \geq 0$ and the second because it is a square times a positive coefficient, so

$$\tau^2 \leq \frac{1}{4}(\sigma_1 - \sigma_3)^2, \quad \tau_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3),$$

with equality only if both vanish, that is $ab w_2 = 0$ and $u = u^*$. For distinct principal stresses $a > 0$ and $b > 0$, so $ab > 0$ forces $w_2 = 0$; then $u^* = 0$ and the second condition gives $u = 0$, whence $s = 1$ and

$$w = \left(\frac{1}{2}, 0, \frac{1}{2}\right), \quad \nu_1 = \pm \frac{\sqrt{2}}{2}, \quad \nu_2 = 0, \quad \nu_3 = \pm \frac{\sqrt{2}}{2}, \quad \mathfrak{n} = \frac{\sqrt{2}}{2}(\mathfrak{n}_1 \pm \mathfrak{n}_3) :$$

the plane bisecting the directions of greatest and least principal stress, with the intermediate stress carrying no weight at all. Two coincident principal stresses make $ab = 0$ and leave w_2 free, so the maximizer stops

being unique — the degeneracy Chapter 3 met whenever a principal value repeated.

(b) A normal equally inclined to the three axes has $\nu_1^2 = \nu_2^2 = \nu_3^2$, and these sum to one, so each is $\frac{1}{3}$ and every pair in the boxed formula enters with weight $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$:

$$\tau_{\text{oct}}^2 = \frac{1}{9} \left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right],$$

that is

$$\tau_{\text{oct}} = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2},$$

with corresponding normal stress $\sigma_{\text{oct}} = \sum_i \sigma_i \cdot \frac{1}{3} = \frac{1}{3} \text{tr } \underline{T}$. In terms of the parametrisation, $\nu_3^2 = \sin^2 \varphi = \frac{1}{3}$ and $\nu_1^2 = \cos^2 \varphi \cos^2 \theta = \frac{1}{3}$; since $\cos^2 \varphi = 1 - \frac{1}{3} = \frac{2}{3}$, the second gives $\cos^2 \theta = \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2}$. So $\varphi = \pm \arcsin(1/\sqrt{3})$ and $\theta \in \{\pi/4, 3\pi/4, 5\pi/4, 7\pi/4\}$: eight normals in all, one into each octant, being the face normals of a regular octahedron centred at $\underline{x}$, which is where the name comes from.

Connection with the deviatoric stress. The octahedral shear turns out to be an invariant in disguise, and this is why it, rather than the shear on some particular plane, is what plasticity uses. Let $\underline{T}' = \underline{T} - \frac{1}{3}(\text{tr } \underline{T})\underline{I}$ be the deviatoric part, the splitting of Problem 15 of Chapter 1, with principal values $\sigma_i - \bar{\sigma}$ and $\bar{\sigma} = \frac{1}{3} \text{tr } \underline{T} = \frac{1}{3} \sum_i \sigma_i$. By Theorem 1.58 the squared norm is the sum of squares of the principal values, so

$$\begin{aligned} |\underline{T}'|^2 &= \sum_i (\sigma_i - \bar{\sigma})^2 = \sum_i \sigma_i^2 - 2\bar{\sigma} \sum_i \sigma_i + 3\bar{\sigma}^2 \\ &= \sum_i \sigma_i^2 - 2\bar{\sigma}(3\bar{\sigma}) + 3\bar{\sigma}^2 = \sum_i \sigma_i^2 - 3\bar{\sigma}^2 \\ &= \sum_i \sigma_i^2 - \frac{1}{3} \left(\sum_i \sigma_i \right)^2 = \frac{1}{3} \left[3 \sum_i \sigma_i^2 - \left(\sum_i \sigma_i \right)^2 \right] = \frac{1}{3} \sum_{i < j} (\sigma_i - \sigma_j)^2, \end{aligned}$$

the last equality being the identity $\sum_{i < j} (\sigma_i - \sigma_j)^2 = 3 \sum_i \sigma_i^2 - \left(\sum_i \sigma_i \right)^2$, which one checks by expanding the left side as $2 \sum_i \sigma_i^2 - 2 \sum_{i < j} \sigma_i \sigma_j$

and the right as $3 \sum_i \sigma_i^2 - \sum_i \sigma_i^2 - 2 \sum_{i < j} \sigma_i \sigma_j$. Comparing with the boxed result,

$$\tau_{\text{oct}}^2 = \frac{1}{9} \sum_{i < j} (\sigma_i - \sigma_j)^2 = \frac{1}{9} \cdot 3 |\underline{T}'|^2 = \frac{1}{3} |\underline{T}'|^2, \quad \text{so} \quad \tau_{\text{oct}} = \frac{|\underline{T}'|}{\sqrt{3}}.$$

A fixed multiple of $|\underline{T}'|$ depends on $\underline{T}$ alone and not on the frame, and is unchanged by superposed pressure: replacing $\underline{T}$ by $\underline{T} + p\underline{I}$ moves $\text{tr } \underline{T}$ by $3p$ and leaves $\underline{T}' = \underline{T} + p\underline{I} - \frac{1}{3}(\text{tr } \underline{T} + 3p)\underline{I}$ exactly where it was. Those are precisely the two properties one demands of a yield criterion, and they are why von Mises is stated in terms of this quantity. $\square$

PROBLEM 6 (Piola tractions and the symmetry restriction). A body undergoes a homogeneous deformation whose deformation gradient at a given time t has components

$$(F_{iA}) = \begin{pmatrix} 2 & 3 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with respect to the fixed basis $\{\underline{e}_i \otimes \hat{\underline{E}}_A\}$. The Piola tractions $\underline{p}(\underline{N})$ acting on the deformed surfaces whose unit normals in the reference configuration are $\hat{\underline{E}}_A$ are

$$\underline{p}(\hat{\underline{E}}_1) = a\underline{e}_1 + 10\underline{e}_2, \quad \underline{p}(\hat{\underline{E}}_2) = (b-6)\underline{e}_3 + 3\underline{e}_2, \quad \underline{p}(\hat{\underline{E}}_3) = c\underline{e}_2 + 3\underline{e}_3,$$

where a , b and c are constants. (a) Use $\underline{p}(\underline{N}) = \underline{P}\underline{N}$ together with $\hat{\underline{I}} = \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A$ to show that $\underline{P} = \underline{p}_A \otimes \hat{\underline{E}}_A$, where $\underline{p}_A = \underline{p}(\hat{\underline{E}}_A)$. (b) Find the components P_{iA} as functions of a , b , c . (c) Apply the rotational momentum balance $\underline{T} = \underline{T}^t$ to determine a , b , c . (d) Calculate the components T_{ij} of the Cauchy stress.

Solution. The problem gives three tractions and asks for a tensor, and the bridge between them is the observation that a tensor is determined by what it does to a basis.

(a) The referential identity resolves on the referential basis, $\hat{I} = \hat{E}_A \otimes \hat{E}_A$, which is Theorem 1.33 written in $\hat{\mathbb{E}}^3$; one checks it by testing on $\hat{E}_B$:

$$(\hat{E}_A \otimes \hat{E}_A) \hat{E}_B = \hat{E}_A \cdot \hat{E}_B \hat{E}_A = \delta_{AB} \hat{E}_A = \hat{E}_B.$$

Post-multiplying $\hat{P}$ by it changes nothing, and the dyad rule $\hat{A}(\hat{a} \otimes \hat{b}) = (\hat{A}\hat{a}) \otimes \hat{b}$ of Lemma 2.15 then moves $\hat{P}$ onto the first leg:

$$\hat{P} = \hat{P} \hat{I} = \hat{P}(\hat{E}_A \otimes \hat{E}_A) = (\hat{P} \hat{E}_A) \otimes \hat{E}_A = \hat{p}(\hat{E}_A) \otimes \hat{E}_A = \hat{p}_A \otimes \hat{E}_A,$$

the fourth equality by (6.132) with $N = \hat{E}_A$. Three tractions therefore determine $\hat{P}$ completely, and the identical argument run in $\mathbb{E}^3$ on T , with $I = e_i \otimes e_i$ and (6.94) in place of (6.132), gives (6.107).

| READER'S NOTE (*columns, not rows*).

$$\hat{P} \hat{E}_A = P_{iB} (\hat{E}_B \cdot \hat{E}_A) e_i = P_{iB} \delta_{BA} e_i = P_{iA} e_i \stackrel{(6.132)}{=} \hat{p}_A$$

and (P_{1A}, P_{2A}, P_{3A}) is the A th column, the first index being the row (Thm. 1.33). Hence

$$\text{Ath column of } (P_{iA}) = \text{the traction on } N = \hat{E}_A$$

MNEMONIC The free index of the traction is the row; the normal's index is the column. The tensor acts on the normal, and the normal contracts with the second slot.

READING THIS PROBLEM'S DATA

$$\hat{p}_1 = a e_1 + 10 e_2, \quad \hat{p}_2 = \underbrace{3 e_2 + (b-6) e_3}_{\text{stated as } (b-6) e_3 + 3 e_2}, \quad \hat{p}_3 = c e_2 + 3 e_3$$

$$\implies (a, 10, 0), \quad (0, 3, b-6), \quad (0, c, 3) \quad \text{as columns.}$$

Reading $\hat{p}_2$ in the order printed, or stacking as rows, gives the transpose — which then fails part (c) with different and wrong a, b, c .

(b) Extracting components from (a),

$$P_{iA} = e_i \cdot \hat{P} \hat{E}_A = e_i \cdot \hat{p}_A,$$

so the A th column of the matrix (P_{iA}) is the component triple of $\hat{p}_A$. Reading the three given vectors, and writing $\hat{p}_2$ with its terms in index

order — the problem states it as $(b-6)\underline{e}_3+3\underline{e}_2$, and taking the coefficients in the order printed is the commonest way of losing this problem —

$$\underline{p}_1 = a\underline{e}_1+10\underline{e}_2+0\underline{e}_3, \quad \underline{p}_2 = 0\underline{e}_1+3\underline{e}_2+(b-6)\underline{e}_3, \quad \underline{p}_3 = 0\underline{e}_1+c\underline{e}_2+3\underline{e}_3,$$

that is $\underline{p}_1 = (a, 10, 0)$, $\underline{p}_2 = (0, 3, b-6)$, $\underline{p}_3 = (0, c, 3)$. Stacking these as columns,

$$(P_{iA}) = \begin{pmatrix} a & 0 & 0 \\ 10 & 3 & c \\ 0 & b-6 & 3 \end{pmatrix}.$$

(c) The constants are fixed by a symmetry that $\underline{P}$ itself cannot possess. By (6.117) the Cauchy stress is symmetric, and by the computation preceding (6.143),

$$\underline{P}\underline{F}^t = \underline{T}\underline{F}^*\underline{F}^t = \underline{T}(\underline{J}\underline{F}^{-t})\underline{F}^t = \underline{J}\underline{T}(\underline{F}^t)^{-1}\underline{F}^t = \underline{J}\underline{T},$$

with $\underline{J} > 0$. So requiring $\underline{T} = \underline{T}^t$ is requiring $\underline{M} := \underline{P}\underline{F}^t$ to be symmetric, which is (6.143) in the form $P_{iA}F_{jA} = F_{iA}P_{jA}$. The virtue of this route is that it never needs $\underline{T}$: it works entirely with the data given.

Since $(\underline{F}^t)_{Aj} = F_{jA}$,

$$M_{ij} = P_{iA}(\underline{F}^t)_{Aj} = P_{iA}F_{jA},$$

which pairs the i th *row* of (P_{iA}) with the j th *row* of (F_{iA}) . The rows are

$$\begin{aligned} P_{1.} &= (a, 0, 0), & P_{2.} &= (10, 3, c), & P_{3.} &= (0, b-6, 3), \\ F_{1.} &= (2, 3, 0), & F_{2.} &= (1, 2, 0), & F_{3.} &= (0, 0, 1), \end{aligned}$$

and all nine entries are

$$\begin{aligned} M_{11} &= a \cdot 2 + 0 \cdot 3 + 0 \cdot 0 = 2a, & M_{12} &= a \cdot 1 + 0 \cdot 2 + 0 \cdot 0 = a, \\ M_{13} &= a \cdot 0 + 0 \cdot 0 + 0 \cdot 1 = 0, & M_{21} &= 10 \cdot 2 + 3 \cdot 3 + c \cdot 0 = 29, \\ M_{22} &= 10 \cdot 1 + 3 \cdot 2 + c \cdot 0 = 16, & M_{23} &= 10 \cdot 0 + 3 \cdot 0 + c \cdot 1 = c, \\ M_{31} &= 0 \cdot 2 + (b-6)3 = 3(b-6), & M_{32} &= 0 \cdot 1 + (b-6)2 = 2(b-6), \\ M_{33} &= (b-6) \cdot 0 + 3 \cdot 1 = 3. \end{aligned}$$

Symmetry is three conditions, one per off-diagonal pair, and they resolve in cascade rather than simultaneously:

$$\begin{aligned} M_{12} = M_{21} : \quad a &= 29, \\ M_{13} = M_{31} : \quad 0 &= 3(b - 6) \implies b = 6, \\ M_{23} = M_{32} : \quad c &= 2(b - 6) = 2 \cdot 0 = 0. \end{aligned}$$

The last two are consistent — both give $b = 6$ — which is a check that the data were not over-determined. Thus

$$a = 29, \quad b = 6, \quad c = 0.$$

(d) First the Jacobian. Expanding $\det \tilde{F}$ along the third row, whose only non-zero entry is $F_{33} = 1$,

$$J = \det \tilde{F} = 1 \cdot \det \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = 1 \cdot (2 \cdot 2 - 3 \cdot 1) = 4 - 3 = 1.$$

So $\tilde{T} = J^{-1} \tilde{P} \tilde{F}^t = \tilde{M}$, and with $a = 29$, $b = 6$, $c = 0$ the nine entries computed above become

$$\begin{aligned} M_{11} &= 2 \cdot 29 = 58, & M_{12} &= 29, & M_{13} &= 0, \\ M_{21} &= 29, & M_{22} &= 16, & M_{23} &= 0, \\ M_{31} &= 3 \cdot 0 = 0, & M_{32} &= 2 \cdot 0 = 0, & M_{33} &= 3, \end{aligned}$$

that is

$$(T_{ij}) = \begin{pmatrix} 58 & 29 & 0 \\ 29 & 16 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

It is symmetric, as it had to be, and its third row and column are uncoupled from the 1–2 plane, which is what one expects of a deformation leaving X_3 alone.

Check by inversion. Inverting the relation is worthwhile and makes one point that is easy to get wrong. With $J = 1$, (6.140) and (2.55) give $\tilde{P} = \tilde{T} \tilde{F}^* = J \tilde{T} \tilde{F}^{-t} = \tilde{T} \tilde{F}^{-t}$. Computing $\tilde{F}^{-1}$ from the adjugate, or by solving $\tilde{F} \tilde{F}^{-1} = \tilde{I}$ directly: the upper-left 2×2 block has determinant

1, so its inverse is $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$, and the third row and column are trivial. Hence

$$\tilde{F}^{-1} = \begin{pmatrix} 2 & -3 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tilde{F}^{-t} = \begin{pmatrix} 2 & -1 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which one verifies: $\tilde{F}\tilde{F}^{-1}$ has $(1, 1)$ entry $2 \cdot 2 + 3 \cdot (-1) = 1$, $(1, 2)$ entry $2 \cdot (-3) + 3 \cdot 2 = 0$, $(2, 1)$ entry $1 \cdot 2 + 2 \cdot (-1) = 0$ and $(2, 2)$ entry $1 \cdot (-3) + 2 \cdot 2 = 1$. Then, entry by entry,

$$\begin{aligned} \tilde{T}\tilde{F}^{-t} &= \begin{pmatrix} 58 & 29 & 0 \\ 29 & 16 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 58 \cdot 2 + 29 \cdot (-3) & 58 \cdot (-1) + 29 \cdot 2 & 0 \\ 29 \cdot 2 + 16 \cdot (-3) & 29 \cdot (-1) + 16 \cdot 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 29 & 0 & 0 \\ 10 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \end{aligned}$$

since $116 - 87 = 29$, $-58 + 58 = 0$, $58 - 48 = 10$ and $-29 + 32 = 3$. This is (P_{iA}) with the values found, as it should be.

Note that $\tilde{T} \neq \tilde{P}$ even though $J = 1$: the two differ by $\tilde{F}^{-t}$, which is not the identity, and a unit Jacobian says only that volumes are preserved, not areas. That is Remark 6.32 in miniature — the Piola stress carries a referential leg, and neither its symmetry nor its numerical values transfer to $\tilde{T}$ without the conversion. $\square$

PROBLEM 7 (Three stresses for one shear state). Consider the deformation

$$x_1 = a_1(X_1 + bX_2), \quad x_2 = a_2X_2, \quad x_3 = a_3X_3.$$

Suppose the Cauchy traction is $\underline{t} = \tau \underline{e}_1$, $\tau \underline{e}_2$ and $\underline{0}$ on the planes with unit normals $\underline{n} = \underline{e}_2$, $\underline{e}_1$ and $\underline{e}_3$ respectively; note the crossed correspondence in the first two. Find the stress tensors $\tilde{T}$, $\tilde{P}$ and $\tilde{S}$ in terms of τ , a_1 , a_2 , a_3 and b .

Solution. The Cauchy stress. The data prescribe $\underline{T}$ on a basis, which by (6.107) is $\underline{T}$ itself: the j th column of (T_{ij}) is $\underline{T}\underline{e}_j$, since $T_{ij} = \underline{e}_i \cdot \underline{T}\underline{e}_j$. Reading the three given values,

$$\underline{T}\underline{e}_1 = \tau\underline{e}_2 = (0, \tau, 0), \quad \underline{T}\underline{e}_2 = \tau\underline{e}_1 = (\tau, 0, 0), \quad \underline{T}\underline{e}_3 = \underline{0} = (0, 0, 0),$$

and stacking as columns,

$$\underline{T} = \tau(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1), \quad (T_{ij}) = \begin{pmatrix} 0 & \tau & 0 \\ \tau & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

That this comes out symmetric is a check on the data rather than a derivation, and the crossed correspondence the problem warns about is exactly what makes it work: had the problem prescribed $\underline{T}\underline{e}_2 = \tau\underline{e}_1$ together with $\underline{T}\underline{e}_1 = \underline{0}$, the matrix would have had $T_{12} = \tau$ and $T_{21} = 0$, violating (6.117), and no state of stress would exist.

The deformation gradient and its inverse. From $F_{iA} = \partial x_i / \partial X_A$ applied to $x_1 = a_1(X_1 + bX_2)$, $x_2 = a_2X_2$, $x_3 = a_3X_3$,

$$\begin{aligned} F_{11} &= a_1, & F_{12} &= a_1b, & F_{13} &= 0, \\ F_{21} &= 0, & F_{22} &= a_2, & F_{23} &= 0, \\ F_{31} &= 0, & F_{32} &= 0, & F_{33} &= a_3, \end{aligned} \quad (F_{iA}) = \begin{pmatrix} a_1 & a_1b & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix},$$

and $J = \det \underline{F} = a_1a_2a_3$, the determinant of an upper triangular matrix being the product of its diagonal.

Inverting by back substitution: solving $\underline{F}\underline{Y} = \underline{I}$ column by column, the third row gives $a_3Y_{3j} = \delta_{3j}$, so $Y_{31} = Y_{32} = 0$ and $Y_{33} = 1/a_3$; the second row gives $a_2Y_{2j} = \delta_{2j}$, so $Y_{21} = 0$, $Y_{22} = 1/a_2$, $Y_{23} = 0$; and the first row gives $a_1Y_{1j} + a_1bY_{2j} = \delta_{1j}$, whence $Y_{11} = 1/a_1$, $Y_{12} = -b/a_2$ and $Y_{13} = 0$. Thus

$$(F_{Ai}^{-1}) = \begin{pmatrix} 1/a_1 & -b/a_2 & 0 \\ 0 & 1/a_2 & 0 \\ 0 & 0 & 1/a_3 \end{pmatrix},$$

which one verifies directly: the $(1, 1)$ entry of $\tilde{F}\tilde{F}^{-1}$ is $a_1(1/a_1) + a_1b \cdot 0 = 1$, the $(1, 2)$ entry is $a_1(-b/a_2) + a_1b(1/a_2) = -a_1b/a_2 + a_1b/a_2 = 0$, and the remaining entries are immediate.

The cofactor. By (2.55), $\tilde{F}^* = J\tilde{F}^{-t}$, so transposing the inverse and multiplying every entry by $J = a_1a_2a_3$,

$$(F_{iA}^*) = a_1a_2a_3 \begin{pmatrix} 1/a_1 & 0 & 0 \\ -b/a_2 & 1/a_2 & 0 \\ 0 & 0 & 1/a_3 \end{pmatrix} = \begin{pmatrix} a_2a_3 & 0 & 0 \\ -a_1a_3b & a_1a_3 & 0 \\ 0 & 0 & a_1a_2 \end{pmatrix},$$

entry by entry: $a_1a_2a_3/a_1 = a_2a_3$; $a_1a_2a_3 \cdot (-b/a_2) = -a_1a_3b$; $a_1a_2a_3/a_2 = a_1a_3$; and $a_1a_2a_3/a_3 = a_1a_2$.

The Piola stress. By (6.140), $P_{iA} = T_{ij}F_{jA}^*$. Because $\tilde{T}$ has only two non-zero entries, $T_{12} = T_{21} = \tau$, the product merely selects rows of $\tilde{F}^*$:

$$\begin{aligned} P_{1A} &= T_{11}F_{1A}^* + T_{12}F_{2A}^* + T_{13}F_{3A}^* = \tau F_{2A}^*, \\ P_{2A} &= T_{21}F_{1A}^* + T_{22}F_{2A}^* + T_{23}F_{3A}^* = \tau F_{1A}^*, \\ P_{3A} &= 0. \end{aligned}$$

So the first row of $\tilde{P}$ is τ times the second row of $\tilde{F}^*$, the second row of $\tilde{P}$ is τ times the first row of $\tilde{F}^*$, and the third vanishes:

$$(P_{iA}) = \begin{pmatrix} -\tau a_1a_3b & \tau a_1a_3 & 0 \\ \tau a_2a_3 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This is not symmetric and must not be, $\tilde{P}$ having one referential and one spatial leg, so that its transpose is a different kind of object altogether; that is Remark 6.31. What holds instead is (6.143), and since it is the only available check it is worth carrying out in full. With $M_{ij} = P_{iA}F_{jA}$,

pairing rows of $\tilde{P}$ with rows of $\tilde{F}$,

$$\begin{aligned} M_{11} &= (-\tau a_1 a_3 b) a_1 + (\tau a_1 a_3)(a_1 b) + 0 = -\tau a_1^2 a_3 b + \tau a_1^2 a_3 b = 0, \\ M_{12} &= (-\tau a_1 a_3 b) \cdot 0 + (\tau a_1 a_3) a_2 + 0 = \tau a_1 a_2 a_3, \\ M_{13} &= (-\tau a_1 a_3 b) \cdot 0 + (\tau a_1 a_3) \cdot 0 + 0 \cdot a_3 = 0, \\ M_{21} &= (\tau a_2 a_3) a_1 + 0 \cdot (a_1 b) + 0 = \tau a_1 a_2 a_3, \\ M_{22} &= (\tau a_2 a_3) \cdot 0 + 0 \cdot a_2 + 0 = 0, \\ M_{23} &= 0, \quad M_{3j} = 0. \end{aligned}$$

Hence $\tilde{M} = \tau a_1 a_2 a_3 (\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1) = J\tilde{T}$, symmetric, as (6.140) and the symmetry of $\tilde{T}$ together require.

The Piola–Kirchhoff stress. By (6.144), $\tilde{S} = \tilde{F}^{-1}\tilde{P}$, that is $S_{AB} = F_{Ai}^{-1}P_{iB}$, pairing the A th row of $\tilde{F}^{-1}$ with the B th column of $\tilde{P}$:

$$(S_{AB}) = \begin{pmatrix} -2\tau a_3 b & \tau a_3 & 0 \\ \tau a_3 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and this one is symmetric, as (6.145) guarantees. That S_{12} and S_{21} arrive by entirely different routes and agree is the arithmetic check on the whole calculation.

| READER'S NOTE ($\tilde{S} = \tilde{F}^{-1}\tilde{P}$, all nine entries). $S_{AB} = F_{Ai}^{-1}P_{iB}$: row A of $\tilde{F}^{-1}$ against column B of $\tilde{P}$, with

$$(F_{Ai}^{-1}) = \begin{pmatrix} 1/a_1 & -b/a_2 & 0 \\ 0 & 1/a_2 & 0 \\ 0 & 0 & 1/a_3 \end{pmatrix}, \quad (P_{iA}) = \begin{pmatrix} -\tau a_1 a_3 b & \tau a_1 a_3 & 0 \\ \tau a_2 a_3 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$S_{11} = \frac{1}{a_1}(-\tau a_1 a_3 b) + \left(-\frac{b}{a_2}\right)(\tau a_2 a_3) + 0 = -\tau a_3 b - \tau a_3 b = -2\tau a_3 b,$$

$$S_{12} = \frac{1}{a_1}(\tau a_1 a_3) + 0 + 0 = \tau a_3,$$

$$S_{21} = 0 + \frac{1}{a_2}(\tau a_2 a_3) + 0 = \tau a_3,$$

$$S_{22} = \frac{1}{a_2} \cdot 0 = 0, \quad S_{13} = S_{23} = 0, \quad S_{3B} = \frac{1}{a_3} P_{3B} = 0.$$

THE CHECK S_{12} and S_{21} arrive by different routes — the $(1,1)$ entry of $\tilde{F}^{-1}$ on P_{12} , against the $(2,2)$ entry on P_{21} — and agree. That is (6.145) guaranteeing it in advance, and it is the strongest arithmetic check on the whole calculation: an error in $\tilde{F}$, $\tilde{F}^*$ or $\tilde{P}$ almost certainly breaks it.

S_{11}

$$T_{11} = 0 \quad \longrightarrow \quad P_{11} = -\tau a_1 a_3 b \quad \longrightarrow \quad S_{11} = -2\tau a_3 b.$$

Pure shear in κ_t ; $\underline{\underline{P}}$ picks up a normal entry from the tilt of the area element, and $\underline{\underline{S}}$ doubles it because $\underline{\underline{F}}^{-1}$ tilts the remaining leg the same way. The stress did not change — the surfaces on which it is reckoned did.

Reading the answer. In κ_t the state is a pure shear: no normal component on any coordinate plane, $T_{11} = T_{22} = T_{33} = 0$. Neither referential measure inherits that. Tracking the 11 entry through the three tensors,

$$T_{11} = 0 \quad \longrightarrow \quad P_{11} = -\tau a_1 a_3 b \quad \longrightarrow \quad S_{11} = -2\tau a_3 b,$$

so $\underline{\underline{P}}$ picks up a normal entry from the tilt of the area element, and $\underline{\underline{S}}$ carries twice what $\underline{\underline{P}}$ does, because $\underline{\underline{F}}^*$ tilts the area element once and $\underline{\underline{F}}^{-1}$ tilts the surviving leg the same way again. The stress did not change; the surfaces on which one reckons it did, and the shear b is what tilts them. Setting $b = 0$ makes $F_{21}^* = 0$ and returns both to pure shear form; setting $a_1 = a_2 = a_3 = 1$ with $b = 0$ makes $\underline{\underline{F}} = \underline{\underline{I}}$, hence $\underline{\underline{F}}^* = \underline{\underline{I}}$ and $J = 1$, and collapses all three tensors onto each other, which is Remark 6.32 at its trivial end. $\square$

PROBLEM 8 (The two Piola transforms, by direct calculation). Establish (6.138), $J \operatorname{div} \underline{\underline{T}} = \operatorname{Div} \underline{\underline{P}}$, and (6.175), $J \operatorname{div} \underline{\underline{q}} = \operatorname{Div}(\underline{\underline{JF}}^{-1} \underline{\underline{q}})$, by direct calculation with the aid of the Piola identity.

Solution. Both identities are the same statement, once for a tensor and once for a vector, and both rest on two facts about $\underline{\underline{F}}^*$.

The first is the Piola identity $\operatorname{Div} \underline{\underline{F}}^* = \underline{\underline{0}}$, that is $F_{iA,A}^* = 0$, proved as Problem 2(b) of Chapter 2 and revisited in Section 5.1.1. The second is the contraction

$$F_{kA} F_{jA}^* = J \delta_{kj}, \quad (*9)$$

which deserves a line of its own. Transposing (2.55), $\underline{\underline{F}}^* = J \underline{\underline{F}}^{-t}$, gives

$$(\underline{\underline{F}}^*)^t = (J \underline{\underline{F}}^{-t})^t = J \left((\underline{\underline{F}}^{-1})^t \right)^t = J \underline{\underline{F}}^{-1},$$

so that $\underline{F}(\underline{F}^*)^t = \underline{F}(J\underline{F}^{-1}) = J\underline{F}\underline{F}^{-1} = J\underline{I}$. In components, $(\underline{F}(\underline{F}^*)^t)_{kj} = F_{kA}((\underline{F}^*)^t)_{Aj} = F_{kA}F_{jA}^*$, which is $(*_9)$. The first fact kills a term; the second collapses the two-point machinery to a Kronecker delta. Nothing else is needed.

The stress. With $P_{iA} = T_{ij}F_{jA}^*$ from (6.141), the referential divergence differentiates a product of two $\underline{X}$ -dependent factors:

$$P_{iA,A} = \frac{\partial}{\partial X_A}(T_{ij}F_{jA}^*) = \frac{\partial T_{ij}}{\partial X_A}F_{jA}^* + T_{ij} \frac{\partial F_{jA}^*}{\partial X_A} = \frac{\partial T_{ij}}{\partial X_A}F_{jA}^* + T_{ij} \underbrace{F_{jA,A}^*}_{=0},$$

the second term vanishing by the Piola identity. In the first, $\underline{T}$ is a *spatial* field, $T_{ij} = T_{ij}(\underline{x}, t)$, and a referential coordinate reaches it only through the motion $\underline{x} = \underline{\chi}_\kappa(\underline{X}, t)$. The chain rule, exactly as at (2.37), therefore supplies a factor of $\underline{F}$:

$$\frac{\partial T_{ij}}{\partial X_A} = \frac{\partial T_{ij}}{\partial x_k} \frac{\partial x_k}{\partial X_A} = T_{ij,k} F_{kA},$$

summed on k . That factor is exactly what $(*_9)$ is waiting for:

$$P_{iA,A} = T_{ij,k} F_{kA} F_{jA}^* = T_{ij,k} J \delta_{kj} = J T_{ij,j},$$

the last step because the delta identifies k with j . In direct notation this is $\text{Div } \underline{P} = J \text{div } \underline{T}$, which is (6.138).

| READER'S NOTE (*the three moves in one line*).

$$\begin{aligned} (1) \quad & \frac{\partial T_{ij}}{\partial X_A} = \frac{\partial T_{ij}}{\partial x_k} \frac{\partial x_k}{\partial X_A} = T_{ij,k} F_{kA} \quad \text{chain rule, (2.37)} \\ (2) \quad & F_{kA} F_{jA}^* = (\underline{F} \underline{F}^{*t})_{kj} = J \delta_{kj} \quad \underline{F}^{*t} = J \underline{F}^{-1} \\ (3) \quad & T_{ij,k} J \delta_{kj} = J T_{ij,j} \quad \delta \text{ renames } k \rightarrow j \end{aligned}$$

Move (1) is why $\underline{F}$ appears at all: a spatial field differentiated by a referential coordinate must produce one deformation gradient. Move (3) turns a gradient into a divergence.

THE TERM THAT WAS DROPPED

$$P_{iA,A} = \frac{\partial T_{ij}}{\partial X_A} F_{jA}^* + T_{ij} \underbrace{F_{jA,A}^*}_{=0 \text{ by (5.11)}}.$$

Without the Piola identity, (6.138) would read $J \operatorname{div} \underline{T} = \operatorname{Div} \underline{P} - \underline{T} \operatorname{Div} \underline{F}^*$: the two descriptions would differ by a term in second derivatives of the motion, and would no longer be equivalent term by term. Same remark for $q_i F_{iA,A}^*$ in the heat flux.

The heat flux. The same argument runs with one index deleted. By (6.172) the referential flux has components $q_{(\kappa)A} = F_{iA}^* q_i$, so

$$\begin{aligned} q_{(\kappa)A,A} &= \frac{\partial}{\partial X_A} (F_{iA}^* q_i) = \underbrace{F_{iA,A}^*}_{=0} q_i + F_{iA}^* \frac{\partial q_i}{\partial X_A} \\ &= F_{iA}^* q_{i,k} F_{kA} = q_{i,k} F_{kA} F_{iA}^* \\ &= q_{i,k} J \delta_{ki} = J q_{i,i}, \end{aligned}$$

using the Piola identity, then the chain rule, then (*9) with j replaced by i . That is $\operatorname{Div} \underline{q}_\kappa = J \operatorname{div} \underline{q}$, which is (6.175) once $\underline{q}_\kappa = (\underline{F}^*)^t \underline{q} = J \underline{F}^{-1} \underline{q}$ is recognised from (6.171).

What has been proved. A single fact, applied twice: multiplying a spatial field by $\underline{F}^*$ and taking the referential divergence reproduces J times the spatial divergence, whatever the rank of the field. The Piola identity is what makes it work, and Remark 2.37 of Chapter 2 predicted this use.

Geometrically it is Piola–Nanson differentiated. The identity $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$ says $\underline{F}^*$ converts area elements, hence flux integrals; the divergence theorem converts those into volume integrals on either side; and $dv = J \, dV$ supplies the factor relating the two volume integrals. Chasing that around,

$$\int_P J \operatorname{div} \underline{q} \, dV = \int_{P_t} \operatorname{div} \underline{q} \, dv = \int_{\partial P_t} \underline{q} \cdot \underline{n} \, da = \int_{\partial P} \underline{q}_\kappa \cdot \underline{N} \, dA = \int_P \operatorname{Div} \underline{q}_\kappa \, dV,$$

and localizing on the arbitrary P gives (6.175) again, this time without differentiating anything.

It is worth naming what would go wrong without $\operatorname{Div} \underline{F}^* = 0$. The dropped terms were $T_{ij} F_{jA,A}^*$ and $q_i F_{iA,A}^*$, so in general one would have

$$\operatorname{Div} \underline{P} = J \operatorname{div} \underline{T} + \underline{T} \operatorname{Div} \underline{F}^*, \quad \operatorname{Div} \underline{q}_\kappa = J \operatorname{div} \underline{q} + \underline{q} \cdot \operatorname{Div} \underline{F}^*,$$

and since F_{iA}^* is built from first derivatives of the motion, $F_{iA,A}^*$ involves second derivatives. The spatial and referential balances would then differ by a term carrying second derivatives of $\underline{\chi}_\kappa$ — a quantity with no physical name and no place in either formulation. That the term vanishes identically, for every motion, is what makes the correspondence between the two descriptions term-by-term rather than merely overall. $\square$

PROBLEM 9 (The virtual power equation, or weak form). Suppose a body is in equilibrium in the configuration κ_t . Derive the virtual power equation

$$\int_{P_t} T_{ij} u_{i,j} dv = \int_{P_t} \rho b_i u_i dv + \int_{\partial P_t} t_i u_i da,$$

where $P_t \subset \kappa_t$ is any part and $\underline{u}$ is any differentiable vector field whatsoever.

Solution. Equilibrium means $\dot{\underline{v}} = \underline{0}$, so (6.110) reduces to

$$T_{ij,j} + \rho b_i = 0. \quad (*_{10})$$

The identity to be proved has a boundary integral on the right and none on the left, so the natural direction of attack is to start at the boundary and work inward, converting the one term that can be converted.

By (6.94), $t_i = T_{ij} n_j$, so for each fixed i the surface integrand $t_i u_i$ — summed on i — is

$$t_i u_i = T_{ij} n_j u_i = (T_{ij} u_i) n_j,$$

the normal component of the vector field with components $v_j := T_{ij} u_i$, summed on i . The divergence theorem, Theorem 1.100, applies to it — both $\underline{T}$ and $\underline{u}$ being differentiable, so v_j is — and gives

$$\int_{\partial P_t} t_i u_i da = \int_{\partial P_t} (T_{ij} u_i) n_j da = \int_{P_t} (T_{ij} u_i)_{,j} dv.$$

Expanding the derivative by the product rule produces exactly two terms, one of which (*₁₀) removes:

$$(T_{ij} u_i)_{,j} = T_{ij,j} u_i + T_{ij} u_{i,j} = (-\rho b_i) u_i + T_{ij} u_{i,j} = -\rho b_i u_i + T_{ij} u_{i,j}.$$

Integrating over P_t ,

$$\int_{\partial P_t} t_i u_i \, da = - \int_{P_t} \rho b_i u_i \, dv + \int_{P_t} T_{ij} u_{i,j} \, dv,$$

and carrying the body-force term across,

$$\int_{P_t} T_{ij} u_{i,j} \, dv = \int_{P_t} \rho b_i u_i \, dv + \int_{\partial P_t} t_i u_i \, da$$

for every differentiable $\underline{u}$ and every part P_t . The field $\underline{u}$ was never required to be a velocity, or to satisfy any equation at all, which is what makes the identity useful: one is free to choose it.

Choosing $\underline{u}$ constant. Take $\underline{u} = \underline{c}$ with $\underline{c}$ a fixed vector. Then $u_{i,j} = c_{i,j} = 0$, so the left side vanishes and

$$0 = \int_{P_t} \rho b_i c_i \, dv + \int_{\partial P_t} t_i c_i \, da = c_i \left[\int_{P_t} \rho b_i \, dv + \int_{\partial P_t} t_i \, da \right],$$

the constants c_i coming out of both integrals. In direct notation,

$$\underline{c} \cdot \left(\int_{P_t} \rho \underline{b} \, dv + \int_{\partial P_t} \underline{t} \, da \right) = 0 \quad \text{for every } \underline{c},$$

so the bracket vanishes by Lemma 5.1, and by (6.30) that bracket is $\underline{f}(S, t)$. This is (6.34) in equilibrium.

Choosing $\underline{u}$ a rigid rotation field. Take $\underline{u} = \underline{w} \times \underline{x}$ with $\underline{w}$ fixed. By Proposition 1.44, $\text{grad } \underline{u}$ is the skew tensor with axial vector $\underline{w}$; explicitly $u_i = \varepsilon_{ipq} w_p x_q$, so

$$u_{i,j} = \varepsilon_{ipq} w_p x_{q,j} = \varepsilon_{ipq} w_p \delta_{qj} = \varepsilon_{ipj} w_p,$$

which is antisymmetric in i and j because ε_{ipj} is. The left side vanishes again, but for a different reason than before: not because the gradient is zero but because

$$T_{ij} u_{i,j} = T_{ij} \varepsilon_{ipj} w_p = 0,$$

the contraction of the symmetric T_{ij} with the skew ε_{ipj} over the pair (i, j) , which vanishes by Lemma 2.36. What remains is

$$0 = \int_{P_t} \rho b_i u_i \, dv + \int_{\partial P_t} t_i u_i \, da = \int_{P_t} \rho \underline{b} \cdot \underline{w} \times \underline{x} \, dv + \int_{\partial P_t} \underline{t} \cdot \underline{w} \times \underline{x} \, da,$$

and the scalar triple product identity $\underline{b} \cdot \underline{w} \times \underline{x} = \underline{w} \cdot \underline{x} \times \underline{b}$ — both sides being the determinant $[\underline{w}, \underline{x}, \underline{b}]$ up to the same cyclic permutation — converts each term:

$$\underline{w} \cdot \left(\int_{P_t} \rho \underline{x} \times \underline{b} \, dv + \int_{\partial P_t} \underline{x} \times \underline{t} \, da \right) = 0 \quad \text{for every } \underline{w},$$

so the bracket vanishes by Lemma 5.1 again, and by (6.31)–(6.33) it is $m_o(S, t)$. This is (6.35) in equilibrium. One scalar equation, quantified over all $\underline{u}$, therefore carries both of Euler’s postulates, and the two are recovered by the two simplest choices of $\underline{u}$ available.

What the left side sees. The second computation generalises. In direct notation the left side is $\int_{P_t} \underline{T} \cdot \text{grad } \underline{u} \, dv$, and splitting $\text{grad } \underline{u}$ into symmetric and skew parts as in (4.7),

$$\underline{T} \cdot \text{grad } \underline{u} = \underline{T} \cdot \text{sym grad } \underline{u} + \underline{T} \cdot \text{skw grad } \underline{u} = \underline{T} \cdot \text{sym grad } \underline{u},$$

the second term vanishing by Lemma 2.36 since $\underline{T}$ is symmetric. So a virtual field with skew gradient contributes nothing, and by (4.31) those are exactly the rigid virtual velocities $\underline{u} = \underline{w} \times \underline{x} + \underline{d}$. The weak form does not see them, which is the statement that a body in equilibrium does no work in a rigid virtual motion.

Why it is called weak. The derivation used $T_{ij,j}$, but the result does not: the boxed identity contains only $\underline{T}$ itself and the first derivatives of $\underline{u}$. It therefore continues to make sense for stress fields that are merely integrable, whereas the local equation (6.110) demands that $\underline{T}$ be differentiable and is correspondingly called the strong form. The logical relation is that (6.110) implies the boxed identity, as just shown, and conversely the boxed identity for all $\underline{u}$ implies (6.110) wherever $\underline{T}$ is differentiable — run the argument backwards, integrate by parts the other

way, and localize. Every finite element computation in solid mechanics solves the boxed equation restricted to a finite-dimensional space of fields $\underline{u}$, and it does so precisely because the weaker demand on $\underline{T}$ admits approximations that are not differentiable across element boundaries. $\square$

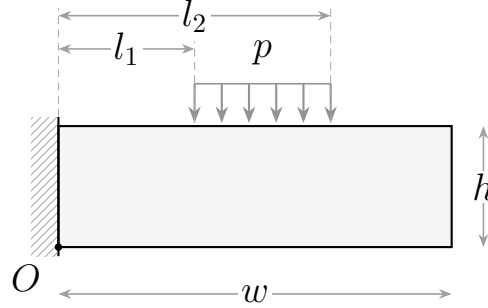


Figure 6.13: The block of Problem 10(d), of width w and height h , clamped on the face $x_1 = 0$ and carrying a uniform pressure p on the part $l_1 < x_1 < l_2$ of its top surface. The origin is the lower corner of the clamped face, x_1 runs along the length and x_2 upward, and the thickness in the x_3 direction is taken as unity. The reactions on the clamped face are unknown, which is why the calculation goes through $\bar{T}_{21}$ and uses the symmetry of the mean stress to recover $\bar{T}_{12}$.

PROBLEM 10 (The mean stress in a body at rest). Let $V(\kappa_t)$ be the volume of a configuration κ_t and define the mean Cauchy stress

$$\bar{T} = [V(\kappa_t)]^{-1} \int_{\kappa_t} T \, dv.$$

(a) If the body is in equilibrium, show that

$$V(\kappa_t) \bar{T}_{ij} = \int_{\partial\kappa_t} t_i x_j \, da + \int_{\kappa_t} \rho b_i x_j \, dv.$$

(b) Apply the global rotational momentum balance to show that $e_{kij} \bar{T}_{ij} = 0$ in equilibrium. (c) Suppose the body rests on a rigid plane under gravity $\underline{b} = -g\mathbf{e}_3$, with no applied tractions other than those due to contact. Show that $\bar{T}_{33} = -\bar{\rho}g\bar{x}_3$, where $\bar{\rho}$ is the mean density and $\bar{x}_3 = \bar{\mathbf{x}} \cdot \mathbf{e}_3$ with the origin on the plane. The result appears to depend on the choice of origin; show that it does not. (d) For the clamped block of Figure 6.13,

carrying a uniform pressure p on the interval $l_1 < x_1 < l_2$ of its top surface, evaluate $\bar{T}_{12}$.

Solution. The mean stress is a volume average and the data are boundary data, so the problem is to manufacture a volume integral of $\underline{T}$ out of surface integrals. The device is that the position field differentiates to a constant, $x_{j,k} = \delta_{jk}$ by Proposition 1.63, so inserting x_j under a divergence produces $\underline{T}$ itself as one of the two terms of the product rule.

(a) Fix j and apply the divergence theorem to the vector field with components $T_{ik}x_j$, k being the index summed against the normal:

$$\begin{aligned} \int_{\partial\kappa_t} t_i x_j \, da &= \int_{\partial\kappa_t} T_{ik} n_k x_j \, da = \int_{\partial\kappa_t} (T_{ik} x_j) n_k \, da = \int_{\kappa_t} (T_{ik} x_j)_{,k} \, dv \\ &= \int_{\kappa_t} (T_{ik,k} x_j + T_{ik} x_{j,k}) \, dv = \int_{\kappa_t} (T_{ik,k} x_j + T_{ik} \delta_{jk}) \, dv, \end{aligned}$$

the first equality by (6.94). Now $T_{ik} \delta_{jk} = T_{ij}$, and in equilibrium $T_{ik,k} = -\rho b_i$ by (*10), so

$$\int_{\partial\kappa_t} t_i x_j \, da = - \int_{\kappa_t} \rho b_i x_j \, dv + \int_{\kappa_t} T_{ij} \, dv.$$

The last integral is $V(\kappa_t) \bar{T}_{ij}$ by the definition of the mean, so moving the body-force term across,

$$V(\kappa_t) \bar{T}_{ij} = \int_{\partial\kappa_t} t_i x_j \, da + \int_{\kappa_t} \rho b_i x_j \, dv.$$

This is Problem 9 with the particular choice $u_i = c_i x_j$, one index j at a time: then $u_{i,k} = c_i \delta_{jk}$ and the left side of the virtual power equation is $c_i \int T_{ij} \, dv$. The weak form was general enough to contain this.

(b) In equilibrium the global rotational balance (6.37) reads

$$\int_{\kappa_t} \rho \underline{x} \times \underline{b} \, dv + \int_{\partial\kappa_t} \underline{x} \times \underline{t} \, da = \underline{0}.$$

Its k th component uses $(\underline{a} \times \underline{c})_k = e_{kji} a_j c_i$, so

$$e_{kji} \int_{\kappa_t} \rho x_j b_i \, dv + e_{kji} \int_{\partial\kappa_t} x_j t_i \, da = 0,$$

that is, the constants e_{kji} coming outside,

$$e_{kji} \left[\int_{\kappa_t} \rho b_i x_j dv + \int_{\partial \kappa_t} t_i x_j da \right] = 0.$$

The bracket is $V \bar{T}_{ij}$ by (a), and $V > 0$, so

$$e_{kji} \bar{T}_{ij} = 0, \quad \text{equivalently} \quad e_{kij} \bar{T}_{ij} = 0,$$

the two differing only by the sign $e_{kji} = -e_{kij}$. By Problem 5 of Chapter 1 a tensor annihilated in this way by the permutation symbol is symmetric — writing the three components out, $e_{1ji} \bar{T}_{ij} = \bar{T}_{32} - \bar{T}_{23}$ and cyclically — so $\bar{\underline{T}} = \bar{\underline{T}}^t$. This is of course the average of the pointwise symmetry (6.117), but the content of the exercise is that the global balance alone delivers it, without ever passing to a point.

(c) Put $i = j = 3$ in (a). The plane is $x_3 = 0$, and by hypothesis the only tractions act where the body touches it, so x_3 vanishes on the whole support of $\underline{t}$ and the surface integral vanishes term by term. With $b_3 = -g$ constant,

$$V \bar{T}_{33} = \int_{\kappa_t} \rho(-g)x_3 dv = -g \int_{\kappa_t} \rho x_3 dv = -g m(S) \bar{x}_3,$$

the last step by (6.40), whose third component reads $m(S) \bar{x}_3 = \int_{\kappa_t} \rho x_3 dv$. Writing $\bar{\rho} = m(S)/V$ for the mean density,

$$\bar{T}_{33} = -\bar{\rho} g \bar{x}_3.$$

The apparent dependence on the origin is not real. The statement fixes the origin on the plane but not where on it, so the admissible origins differ by translations *within* the plane, $\underline{q} \mapsto \underline{q} + \underline{c}$ with $\underline{c} \cdot \underline{e}_3 = 0$. Such a translation sends $x_3 = \underline{x} \cdot \underline{e}_3$ to $(\underline{x} - \underline{c}) \cdot \underline{e}_3 = x_3 - \underline{c} \cdot \underline{e}_3 = x_3$, so x_3 and hence $\bar{x}_3$ are untouched. Thus $\bar{x}_3$ is the height of the mass center above the plane, a geometric quantity, and the formula is unambiguous. Moving the origin off the plane would change $\bar{x}_3$ and would also invalidate the derivation, since the contact tractions would no longer sit at $x_3 = 0$ and the surface integral would not vanish.

(d) The block occupies $0 \leq x_1 \leq w$, $0 \leq x_2 \leq h$ with unit thickness in x_3 , so $V = wh$, and $\underline{b} = \underline{0}$.

Computing $\bar{T}_{12}$ from (a) would need $\int t_1 x_2 \, da$ over the clamped face $x_1 = 0$, where the reactions t_1 are unknown; that route is blocked. But $\bar{T}$ is symmetric by (b), so compute $\bar{T}_{21}$ instead:

$$V\bar{T}_{21} = \int_{\partial\kappa_t} t_2 x_1 \, da.$$

Now every contribution is known. Going round the boundary: on the clamped face $x_1 = 0$, so the integrand carries the factor $x_1 = 0$ and the unknown reactions drop out — the whole point of choosing this component; on the bottom $x_2 = 0$, on the free end $x_1 = w$, and on the unloaded part of the top the surface is traction-free, so $\underline{t} = \underline{0}$ there and each contributes nothing; and on the loaded part of the top the outward normal is $\underline{e}_2$ and the pressure acts inward, so $\underline{t} = -p\underline{e}_2$, giving $t_2 = -p$, with x_1 running from l_1 to l_2 and $da = dx_1$ for unit thickness. Hence

$$V\bar{T}_{21} = \int_{l_1}^{l_2} (-p) x_1 \, dx_1 = -p \left[\frac{x_1^2}{2} \right]_{l_1}^{l_2} = -\frac{p}{2}(l_2^2 - l_1^2),$$

and dividing by $V = wh$,

$$\bar{T}_{12} = \bar{T}_{21} = -\frac{p(l_2^2 - l_1^2)}{2wh}.$$

The negative sign says the mean shear opposes the applied load, and the factor $l_2^2 - l_1^2$, growing with the distance from the clamp, says that pressure applied further out counts for more — both as one expects of a cantilever. The computation is a small demonstration of what (6.117) is worth: it converted an integral over an unknown boundary into one over a known boundary. $\square$

PROBLEM 11 (The energy flux and the jump conditions it carries). All the scalar balance laws considered have the generic form (6.1). Take φ to be the total energy per unit mass, $\varphi = \varepsilon + \frac{1}{2}|\underline{v}|^2$. What is the energy flux π ? Obtain the relevant jump condition and compare it with (6.169). Thus derive the jump condition for (6.150).

Solution. The flux. The balance of total energy is (6.161), and to read a flux off it one has to write every term as an integral and then force the surface integrand into the shape $\pi \cdot \underline{n}$ that (6.1) demands. Using (6.151) and (6.162) on the left and (6.152) and (6.163) on the right,

$$\frac{d}{dt} \int_{P_t} \rho(\varepsilon + \tfrac{1}{2}|\underline{v}|^2) dv = \int_{P_t} \rho(\underline{b} \cdot \underline{v} + r) dv + \int_{\partial P_t} (\underline{t} \cdot \underline{v} + h) da,$$

the bulk terms having been collected over the common factor ρ and the surface terms over the common measure.

Of the two boundary contributions one is already in the required form: (6.166) reads $h = -\underline{q} \cdot \underline{n} = (-\underline{q}) \cdot \underline{n}$. The other is not, and making it so is the only step in the problem that uses anything. Inserting $\underline{t} = \underline{T}\underline{n}$ and moving $\underline{T}$ across the inner product by the definition of the transpose, then using the symmetry (6.117) to put it back onto the other factor,

$$\underline{t} \cdot \underline{v} = \underline{T}\underline{n} \cdot \underline{v} = \underline{n} \cdot \underline{T}^t \underline{v} = \underline{T}^t \underline{v} \cdot \underline{n} = \underline{T}\underline{v} \cdot \underline{n},$$

the third equality by symmetry of the inner product and the fourth by $\underline{T}^t = \underline{T}$. Hence the whole surface integrand is

$$\underline{t} \cdot \underline{v} + h = \underline{T}\underline{v} \cdot \underline{n} + (-\underline{q}) \cdot \underline{n} = (\underline{T}\underline{v} - \underline{q}) \cdot \underline{n},$$

and comparing with (6.1) term by term,

$$\varphi = \varepsilon + \tfrac{1}{2}|\underline{v}|^2, \quad \sigma = \underline{b} \cdot \underline{v} + r, \quad \pi = \underline{T}\underline{v} - \underline{q}$$

The flux of total energy is the working of the traction less the heat conducted out, which is what one would guess. What the derivation adds is that the guess needs $\underline{T} = \underline{T}^t$: without symmetry the working is $\underline{n} \cdot \underline{T}^t \underline{v}$, the normal component of $\underline{T}^t \underline{v}$ rather than of $\underline{T}\underline{v}$, and while that is still a vector, the identification of the flux would carry the transpose along with it into every later formula.

The jump condition. Substituting the three letters into (6.6), $\llbracket \rho\varphi(\underline{v} - \underline{u}) - \pi \rrbracket \cdot \underline{n} = 0$, gives

$$\llbracket \rho(\varepsilon + \tfrac{1}{2}|\underline{v}|^2)(\underline{v} - \underline{u}) - \underline{T}\underline{v} + \underline{q} \rrbracket \cdot \underline{n} = 0$$

at every point of a singular surface, the two minus signs in $-\pi = -\underline{T}\underline{v} + \underline{q}$ being those of the definition just made.

Set this beside (6.169), which is what the thermal balance alone carries:

$$\llbracket \rho \varepsilon(\underline{v} - \underline{u}) + \underline{q} \rrbracket \cdot \underline{n} = 0.$$

The total-energy condition has two terms the thermal one lacks, namely the convected kinetic energy $\rho \frac{1}{2} |\underline{v}|^2 (\underline{v} - \underline{u})$ and the stress working $-\underline{T}\underline{v}$, and no others: the terms $\rho \varepsilon(\underline{v} - \underline{u})$ and $+\underline{q}$ occur in both, with the same signs.

The mechanical jump condition. Subtracting is therefore legitimate — both are equalities at the same point of the same surface — and everything common cancels:

$$\begin{aligned} & \llbracket \rho \left(\varepsilon + \frac{1}{2} |\underline{v}|^2 \right) (\underline{v} - \underline{u}) - \underline{T}\underline{v} + \underline{q} \rrbracket \cdot \underline{n} - \llbracket \rho \varepsilon(\underline{v} - \underline{u}) + \underline{q} \rrbracket \cdot \underline{n} \\ &= \llbracket \rho \varepsilon(\underline{v} - \underline{u}) + \rho \frac{1}{2} |\underline{v}|^2 (\underline{v} - \underline{u}) - \underline{T}\underline{v} + \underline{q} - \rho \varepsilon(\underline{v} - \underline{u}) - \underline{q} \rrbracket \cdot \underline{n} \\ &= \llbracket \rho \frac{1}{2} |\underline{v}|^2 (\underline{v} - \underline{u}) - \underline{T}\underline{v} \rrbracket \cdot \underline{n}, \end{aligned}$$

the jump bracket being linear, so that

$$\llbracket \rho \frac{1}{2} |\underline{v}|^2 (\underline{v} - \underline{u}) - \underline{T}\underline{v} \rrbracket \cdot \underline{n} = 0,$$

which is the condition belonging to the mechanical energy balance (6.150).

The result may be checked head-on rather than by subtraction. Equation (6.150) is of the form (6.1) with

$$\varphi = \frac{1}{2} |\underline{v}|^2, \quad \pi = \underline{T}\underline{v}, \quad \sigma = \underline{b} \cdot \underline{v} - \rho^{-1} \underline{T} \cdot \underline{L},$$

the last because the stress power appears on the left of (6.150) and must be moved to the right as a supply with a sign change and a factor ρ^{-1} , the generic σ being per unit mass. Feeding these into (6.6) reproduces the display.

One observation. That last check makes the real point of the problem visible. The stress power $\underline{T} \cdot \underline{L}$, which is the entire difference between the mechanical and thermal balances in the bulk, appears in none of the three jump conditions. It cannot: it entered as part of σ , and σ

is a volume term. Collapsing a part onto a surface kills volume terms and spares surface terms — a pillbox of thickness η and face area A has volume ηA and surface $2A + O(\eta)$, so dividing by A and letting $\eta \rightarrow 0$ removes everything carrying the factor η — so only φ and π survive into (6.6), however large σ may be. This is the shrinking argument used at (6.81) and quantified by Lemma 6.23, and it is why the three balances — total, thermal, mechanical — differ in the bulk by the stress power but differ on a shock only by convected and worked terms. $\square$

PROBLEM 12 (Cofactors, and two more power-conjugate pairs). Let $\underline{A}$ and $\underline{B}$ be invertible tensors. (a) Show that $(\underline{A}\underline{B})^* = \underline{A}^*\underline{B}^*$. (b) For invertible $\underline{Q}$, show that $\underline{Q}^* = \underline{Q}$ if and only if $\underline{Q}$ is a rotation. (c) The referential *Biot stress* is $\underline{\sigma} = \text{sym}(\underline{P}^t \underline{R})$, with $\underline{R}$ the rotation of the polar decomposition of $\underline{F}$. Show that $\underline{\sigma}$ is power conjugate to the right stretch $\underline{U}$, that is, that the stress power per unit reference volume is $\underline{\sigma} \cdot \dot{\underline{U}}$. (d) The spatial *Bell stress* is $\underline{\Sigma} = \text{sym}(\underline{P} \underline{R}^t)$. Derive an expression for the tensor $\underline{A}$ such that $\underline{\Sigma} \cdot \underline{A}$ is the stress power per unit reference volume.

Solution. **(a)** The cofactor is characterised by what it does to cross products,

$$\underline{A}^*(\underline{u} \times \underline{v}) = \underline{A}\underline{u} \times \underline{A}\underline{v} \quad \text{for all } \underline{u}, \underline{v},$$

which is Definition 1.54. Multiplicativity follows by using that characterisation three times in succession:

$$\begin{aligned} (\underline{A}\underline{B})^*(\underline{u} \times \underline{v}) &= (\underline{A}\underline{B}\underline{u}) \times (\underline{A}\underline{B}\underline{v}) = \underline{A}(\underline{B}\underline{u}) \times \underline{A}(\underline{B}\underline{v}) \\ &= \underline{A}^*[(\underline{B}\underline{u}) \times (\underline{B}\underline{v})] = \underline{A}^*\underline{B}^*(\underline{u} \times \underline{v}), \end{aligned}$$

the third equality reading the defining property for $\underline{A}$ with the vectors $\underline{B}\underline{u}$ and $\underline{B}\underline{v}$, and the fourth reading it for $\underline{B}$. Since every vector of $\mathbb{E}^3$ is a cross product of two others — given $\underline{w} \neq \underline{0}$, pick any $\underline{p}$ with $\underline{p} \cdot \underline{w} = 0$ and $|\underline{p}| = 1$ and set $\underline{u} = \underline{p}$, $\underline{v} = \underline{w} \times \underline{p}/|\underline{w}|$, whence $\underline{u} \times \underline{v} = \underline{w}$ — the two tensors $(\underline{A}\underline{B})^*$ and $\underline{A}^*\underline{B}^*$ agree on all of $\mathbb{E}^3$ and are equal.

(b) One direction is a computation. If $\underline{Q}$ is a rotation then $\det \underline{Q} = 1$ and $\underline{Q}^t = \underline{Q}^{-1}$, so $\underline{Q}^{-t} = (\underline{Q}^{-1})^t = (\underline{Q}^t)^t = \underline{Q}$, and

$$\underline{Q}^* = (\det \underline{Q})\underline{Q}^{-t} = 1 \cdot \underline{Q} = \underline{Q}.$$

The other is a trick with determinants. Suppose $\underline{\underline{Q}}^* = \underline{\underline{Q}}$. Since the cofactor of an invertible tensor is $(\det \underline{\underline{Q}})\underline{\underline{Q}}^{-t}$, this reads $(\det \underline{\underline{Q}})\underline{\underline{Q}}^{-t} = \underline{\underline{Q}}$, and post-multiplying by $\underline{\underline{Q}}^t$,

$$(\det \underline{\underline{Q}}) \underline{\underline{Q}}^{-t} \underline{\underline{Q}}^t = \underline{\underline{Q}} \underline{\underline{Q}}^t, \quad \text{that is} \quad \underline{\underline{Q}} \underline{\underline{Q}}^t = (\det \underline{\underline{Q}}) \underline{\underline{I}},$$

since $\underline{\underline{Q}}^{-t} \underline{\underline{Q}}^t = \underline{\underline{I}}$. Take determinants of both sides. On the left, $\det(\underline{\underline{Q}} \underline{\underline{Q}}^t) = \det \underline{\underline{Q}} \det \underline{\underline{Q}}^t = (\det \underline{\underline{Q}})^2$; on the right, $\det((\det \underline{\underline{Q}}) \underline{\underline{I}}) = (\det \underline{\underline{Q}})^3$, since $\det(\alpha \underline{\underline{I}}) = \alpha^3$ in three dimensions. So

$$(\det \underline{\underline{Q}})^2 = (\det \underline{\underline{Q}})^3.$$

Invertibility gives $\det \underline{\underline{Q}} \neq 0$, so dividing by $(\det \underline{\underline{Q}})^2$ leaves $\det \underline{\underline{Q}} = 1$, and substituting back into $\underline{\underline{Q}} \underline{\underline{Q}}^t = (\det \underline{\underline{Q}}) \underline{\underline{I}}$ gives $\underline{\underline{Q}} \underline{\underline{Q}}^t = \underline{\underline{I}}$. Thus $\underline{\underline{Q}}$ is orthogonal with determinant +1: a rotation. The dimension enters only in that exponent, which is worth noticing — the statement is three-dimensional, and in n dimensions the same argument gives $(\det \underline{\underline{Q}})^2 = (\det \underline{\underline{Q}})^n$.

(c) Parts (c) and (d) follow one pattern: differentiate a polar decomposition, and dispose of the term carrying $\dot{\underline{\underline{R}}}$ by a symmetry argument. By (6.156) the stress power per unit reference volume is $\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}}$, and differentiating $\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}}$ from (3.32),

$$\dot{\underline{\underline{F}}} = \dot{\underline{\underline{R}}} \underline{\underline{U}} + \underline{\underline{R}} \dot{\underline{\underline{U}}}, \quad \text{so} \quad \underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \underline{\underline{P}} \cdot (\dot{\underline{\underline{R}}} \underline{\underline{U}}) + \underline{\underline{P}} \cdot (\underline{\underline{R}} \dot{\underline{\underline{U}}}).$$

The second term is the one that survives. Using $\underline{\underline{A}} \cdot \underline{\underline{B}} = \text{tr}(\underline{\underline{A}} \underline{\underline{B}}^t)$ from Definition 1.57, then $(\underline{\underline{R}} \dot{\underline{\underline{U}}})^t = \dot{\underline{\underline{U}}}^t \underline{\underline{R}}^t$, then cyclicity of the trace,

$$\underline{\underline{P}} \cdot (\underline{\underline{R}} \dot{\underline{\underline{U}}}) = \text{tr}(\underline{\underline{P}}(\underline{\underline{R}} \dot{\underline{\underline{U}}})^t) = \text{tr}(\underline{\underline{P}} \dot{\underline{\underline{U}}}^t \underline{\underline{R}}^t) = \text{tr}(\underline{\underline{R}}^t \underline{\underline{P}} \dot{\underline{\underline{U}}}^t) = (\underline{\underline{R}}^t \underline{\underline{P}}) \cdot \dot{\underline{\underline{U}}},$$

in which $\underline{\underline{R}}^t \underline{\underline{P}}$ maps $\hat{\mathbb{E}}^3$ to $\hat{\mathbb{E}}^3$, both legs now referential: $\underline{\underline{P}}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ and $\underline{\underline{R}}^t: \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$. So the inner product is one of referential tensors. Since $\underline{\underline{U}}$ is symmetric at every instant, so is $\dot{\underline{\underline{U}}}$, and Lemma 2.36 discards the skew part of $\underline{\underline{R}}^t \underline{\underline{P}}$:

$$\begin{aligned} (\underline{\underline{R}}^t \underline{\underline{P}}) \cdot \dot{\underline{\underline{U}}} &= \left[\text{sym}(\underline{\underline{R}}^t \underline{\underline{P}}) + \text{skw}(\underline{\underline{R}}^t \underline{\underline{P}}) \right] \cdot \dot{\underline{\underline{U}}} = \text{sym}(\underline{\underline{R}}^t \underline{\underline{P}}) \cdot \dot{\underline{\underline{U}}} \\ &= \text{sym}(\underline{\underline{P}}^t \underline{\underline{R}}) \cdot \dot{\underline{\underline{U}}} = \underline{\underline{\sigma}} \cdot \dot{\underline{\underline{U}}}, \end{aligned}$$

the penultimate equality because $\text{sym } \underline{X} = \text{sym } \underline{X}^t$ and $(\underline{R}^t \underline{P})^t = \underline{P}^t (\underline{R}^t)^t = \underline{P}^t \underline{R}$.

The first term must vanish, and the reason is the symmetry restriction (6.143) in disguise. Put $\underline{G} = \underline{R}^t \underline{P}$, so that $\underline{P} = \underline{R} \underline{G}$; then $\underline{F}^t = (\underline{R} \underline{U})^t = \underline{U}^t \underline{R}^t$ and $\underline{P}^t = \underline{G}^t \underline{R}^t$, so $\underline{P} \underline{F}^t = \underline{F} \underline{P}^t$ becomes

$$\underline{R} \underline{G} \underline{U} \underline{R}^t = \underline{R} \underline{U} \underline{G}^t \underline{R}^t.$$

Cancelling the invertible $\underline{R}$ on the left and $\underline{R}^t$ on the right — multiply by $\underline{R}^t$ in front and $\underline{R}$ behind, using $\underline{R}^t \underline{R} = \hat{\underline{I}}$ — gives

$$\underline{G} \underline{U} = \underline{U} \underline{G}^t,$$

and the right-hand form says $\underline{G} \underline{U}$ is symmetric, since $(\underline{G} \underline{U})^t = \underline{U}^t \underline{G}^t = \underline{U} \underline{G}^t = \underline{G} \underline{U}$. Meanwhile

$$\begin{aligned} \underline{P} \cdot (\dot{\underline{R}} \underline{U}) &= \text{tr} (\underline{P} (\dot{\underline{R}} \underline{U})^t) = \text{tr} (\underline{P} \underline{U}^t \dot{\underline{R}}^t) = \text{tr} (\underline{P} \underline{U} \dot{\underline{R}}^t) \\ &= \text{tr} (\underline{R} \underline{G} \underline{U} \dot{\underline{R}}^t) = \text{tr} (\underline{G} \underline{U} \dot{\underline{R}}^t \underline{R}), \end{aligned}$$

using $\underline{U}^t = \underline{U}$ and then cyclicity. Now $\dot{\underline{R}}^t \underline{R}$ is skew: differentiating $\underline{R}^t \underline{R} = \hat{\underline{I}}$ gives

$$\dot{\underline{R}}^t \underline{R} + \underline{R}^t \dot{\underline{R}} = \hat{\underline{O}},$$

and $\underline{R}^t \dot{\underline{R}} = (\dot{\underline{R}}^t \underline{R})^t$, so $\dot{\underline{R}}^t \underline{R} + (\dot{\underline{R}}^t \underline{R})^t = \hat{\underline{O}}$, which is skewness. A symmetric tensor contracted with a skew one vanishes, by Lemma 2.36 once more, so the whole term is zero and

$$\underline{P} \cdot \dot{\underline{F}} = \underline{\sigma} \cdot \dot{\underline{U}}$$

The Biot stress is power conjugate to $\underline{U}$, and no correction was needed.

(d) The same pattern on the other factorisation $\underline{F} = \underline{V} \underline{R}$, for which $\dot{\underline{F}} = \dot{\underline{V}} \underline{R} + \underline{V} \dot{\underline{R}}$. The first term gives, by the trace manipulation above with $\underline{R}$ now on the right,

$$\underline{P} \cdot (\dot{\underline{V}} \underline{R}) = \text{tr} (\underline{P} (\dot{\underline{V}} \underline{R})^t) = \text{tr} (\underline{P} \underline{R}^t \dot{\underline{V}}^t) = (\underline{P} \underline{R}^t) \cdot \dot{\underline{V}} = \underline{\Sigma} \cdot \dot{\underline{V}},$$

$\underline{P} \underline{R}^t$ being spatial — $\underline{R}^t: \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$ followed by $\underline{P}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ — and $\dot{\underline{V}}$ symmetric, so that as in (c) only $\text{sym}(\underline{P} \underline{R}^t) = \underline{\Sigma}$ survives.

This time the remaining term does *not* vanish. Put $\underline{H} = \underline{P}\underline{R}^t$, so $\underline{P} = \underline{H}\underline{R}$, and let $\underline{\Omega}_R = \underline{\dot{R}}\underline{R}^t$, spatial and skew by the argument of (6.55). Differentiating $\underline{R}\underline{R}^t = \underline{I}$ gives $\underline{\dot{R}}\underline{R}^t + \underline{R}\underline{\dot{R}}^t = \underline{O}$, so $\underline{R}\underline{\dot{R}}^t = -\underline{\Omega}_R$, and

$$\begin{aligned} \underline{P} \cdot (\underline{V}\underline{\dot{R}}) &= \text{tr} (\underline{P}(\underline{V}\underline{\dot{R}})^t) = \text{tr} (\underline{P}\underline{\dot{R}}^t\underline{V}^t) = \text{tr} (\underline{P}\underline{\dot{R}}^t\underline{V}) \\ &= \text{tr} (\underline{H}\underline{R}\underline{\dot{R}}^t\underline{V}) = -\text{tr} (\underline{H}\underline{\Omega}_R\underline{V}) = -\text{tr} (\underline{V}\underline{H}\underline{\Omega}_R), \end{aligned}$$

using $\underline{V}^t = \underline{V}$ and cyclicity at the last step: a leftover proportional to the spin of the polar rotation.

The claim is that restoring it is exactly what converts $\underline{\dot{V}}$ into its co-rotational rate

$$\underline{\overset{\circ}{V}} = \underline{\dot{V}} - \underline{\Omega}_R\underline{V} + \underline{V}\underline{\Omega}_R,$$

formed as in (6.68) but about $\underline{R}$ rather than the rigid $\underline{Q}$. To see it, compute the correction against $\underline{\Sigma}$. Since $\underline{\Omega}_R^t = -\underline{\Omega}_R$ and $\underline{\tilde{V}}^t = \underline{V}$,

$$(\underline{\Omega}_R\underline{V} - \underline{V}\underline{\Omega}_R)^t = \underline{\tilde{V}}^t\underline{\Omega}_R^t - \underline{\Omega}_R^t\underline{\tilde{V}}^t = -\underline{V}\underline{\Omega}_R + \underline{\Omega}_R\underline{V} = \underline{\Omega}_R\underline{V} - \underline{V}\underline{\Omega}_R,$$

so the commutator is symmetric and

$$\begin{aligned} \underline{\Sigma} \cdot (\underline{\Omega}_R\underline{V} - \underline{V}\underline{\Omega}_R) &= \text{tr} \left(\underline{\Sigma}(\underline{\Omega}_R\underline{V} - \underline{V}\underline{\Omega}_R)^t \right) = \text{tr} (\underline{\Sigma}\underline{\Omega}_R\underline{V}) - \text{tr} (\underline{\Sigma}\underline{V}\underline{\Omega}_R) \\ &= \text{tr} (\underline{V}\underline{\Sigma}\underline{\Omega}_R) - \text{tr} (\underline{\Sigma}\underline{V}\underline{\Omega}_R) = \text{tr} [(\underline{V}\underline{\Sigma} - \underline{\Sigma}\underline{V})\underline{\Omega}_R], \end{aligned}$$

the third equality by cyclicity applied to the first trace alone. Write $\underline{H} = \underline{\Sigma} + \underline{W}_H$ with $\underline{W}_H = \text{skw } \underline{H}$. The symmetry restriction in the form $\underline{H}\underline{V} = \underline{V}\underline{H}^t$, obtained as in (c) but with $\underline{F} = \underline{V}\underline{R}$, reads

$$(\underline{\Sigma} + \underline{W}_H)\underline{V} = \underline{V}(\underline{\Sigma} - \underline{W}_H), \quad \text{that is} \quad \underline{\Sigma}\underline{V} - \underline{V}\underline{\Sigma} + \underline{V}\underline{W}_H + \underline{W}_H\underline{V} = \underline{O},$$

using $\underline{H}^t = \underline{\Sigma} - \underline{W}_H$. This says precisely that $\underline{\Sigma}\underline{V} + \underline{V}\underline{W}_H$ is symmetric: its transpose is $\underline{V}\underline{\Sigma} + \underline{W}_H^t\underline{V} = \underline{V}\underline{\Sigma} - \underline{W}_H\underline{V}$, and the displayed identity is exactly the statement that the two agree. Its contraction with the skew $\underline{\Omega}_R$ therefore vanishes, leaving

$$\text{tr} [(\underline{V}\underline{\Sigma} - \underline{\Sigma}\underline{V})\underline{\Omega}_R] = \text{tr} (\underline{V}\underline{H}\underline{\Omega}_R),$$

which is precisely the leftover with its sign reversed. Assembling,

$$\begin{aligned}\underline{\Sigma} \cdot \dot{\underline{V}} &= \underline{\Sigma} \cdot \dot{\underline{V}} - \underline{\Sigma} \cdot (\underline{\Omega}_R \underline{V} - \underline{V} \underline{\Omega}_R) = \underline{\Sigma} \cdot \dot{\underline{V}} - \text{tr} (\underline{V} \underline{H} \underline{\Omega}_R) \\ &= \underline{P} \cdot (\dot{\underline{V}} \underline{R}) + \underline{P} \cdot (\underline{V} \dot{\underline{R}}) = \underline{P} \cdot \dot{\underline{F}},\end{aligned}$$

so that

$$\underline{A} = \dot{\underline{V}} = \dot{\underline{V}} - \underline{\Omega}_R \underline{V} + \underline{V} \underline{\Omega}_R, \quad \underline{\Omega}_R = \dot{\underline{R}} \underline{R}^t.$$

Reading the pair. The two halves differ in one structural respect, and it accounts for everything. Part (c) pairs a referential stress with the rate of a referential stretch, and both live in a space that does not move; no correction arises. Part (d) pairs a spatial stress with a spatial stretch, and $\underline{V}$ sits in a frame the material carries around with it, so its bare rate $\dot{\underline{V}}$ measures the turning of that frame along with the stretching. The correction is the co-rotational derivative of (6.68), met first for the inertia tensor and promised in Remark 6.14, and the rotation is $\underline{R}$ rather than $\underline{Q}$ because in a general deformation $\underline{R}$ is what plays the part the rigid rotation played there. Using $\dot{\underline{V}}$ in place of $\dot{\underline{V}}$ gives the wrong answer whenever $\underline{\Omega}_R \underline{V} \neq \underline{V} \underline{\Omega}_R$, which is the generic case; the two commute only when $\underline{V}$ and the axis of $\underline{\Omega}_R$ are compatibly aligned. $\square$

Chapter 7

Observers and Frames of Reference

Every formula written so far belongs to one observer. Chapter 2 gave that observer a point space and let him record where each material point is; Chapters 3 and 4 let him measure how the body has been distorted and how fast; Chapters 5 and 6 let him balance mass, momentum and energy. At no stage was it asked whether a second observer, watching the same body from a platform of his own, would write the same things down.

He would not. Two observers disagree about the velocity of a particle, about its acceleration, about the spin of the material, and about the components of every vector and tensor in sight. They nevertheless agree about the distance between two material points, about the elapsed time between two events, about the density, and about the stress power. The business of this chapter is to sort the one list from the other, and to do it by calculation rather than by intuition, because intuition gets it wrong: the stretching tensor $\underline{D}$ survives a change of observer and the spin tensor $\underline{W}$ does not, although the two were built side by side out of the same $\underline{L}$ in (4.7).

The sorting matters for a reason that will only become visible in Chapter 8. A constitutive equation is a claim about a material, and a material cannot know who is looking at it. So a constitutive equation may be built only out of quantities on which all observers agree, in the

precise sense set out in Section 7.3 below. Every result of this chapter is therefore a licence or a prohibition for the chapters that follow: $\underline{D}$ is admissible in a constitutive law and $\underline{W}$ is not, $\underline{C}$ is admissible and $\underline{F}$ is not, and the reason in each case is a transformation law derived here.

One object generates the entire chapter, the relation

$$\underline{x}^+ = Q(t)\underline{x} + \underline{c}(t)$$

between the place $\underline{x}$ that one observer assigns to a material point and the place $\underline{x}^+$ that the other assigns to the same point at the same instant. Section 7.1 derives it from two physical assumptions. Everything afterwards is obtained by differentiating it, with respect to position to get $\underline{F}$, $\underline{H}$ and the polar factors, and with respect to time to get $\underline{v}$, $\underline{a}$, $\underline{L}$, $\underline{D}$ and $\underline{W}$. The single term $\dot{Q}$, which appears whenever the second differentiation is performed, is the whole difficulty and the whole content.

7.1 Observers

An observer O , perceiving a configuration κ_t of a body, is equipped for the purpose with a point space $\mathbb{E}^3$ and its associated translation space, also written $\mathbb{E}^3$. These constitute the *frame of reference* adopted by O . We are concerned with the relationship between this observer's perception and that of an alternative observer O^+ , who perceives the same configuration as $\kappa_{t^+}^+$ relative to a frame of reference consisting of the point space $\mathbb{E}^{3+}$ and translation space $\mathbb{E}^{3+}$.

REMARK 7.1 (a frame of reference is not a coordinate system). This is the distinction that has to be fixed before anything else, and it is the one most often lost.

A *basis* is chosen inside a single frame. Changing it, as in Section 1.1.2, leaves every geometric object exactly where it is and merely re-expresses it: the vector $\underline{u}$ is one vector, and $u_i = \underline{u} \cdot \underline{e}_i$ and $u'_i = \underline{u} \cdot \underline{e}'_i$ are two lists of numbers naming it. The change-of-basis matrix a_{ij} of Theorem 1.19 is a fixed array of numbers; nothing in that discussion had

any time dependence, and nothing could, because the question “which basis?” is not a question about when.

A *frame* is a whole space, carried by an observer. Changing it replaces the point $\underline{x} \in \mathbb{E}^3$ by a different point $\underline{x}^+$ of a different space $\mathbb{E}^{3+}$. The tensor $\underline{Q}(t)$ that relates them is a two-point tensor with one foot in each space, in exactly the sense of Remark 4.1, and it depends on t because the two observers may be in relative motion.

The practical test: if $\dot{\underline{Q}}$ can be non-zero, you are changing observers; if it cannot, you are changing bases. Every difficulty in this chapter, and every term in (7.9) and (7.63), comes from $\dot{\underline{Q}} \neq \underline{Q}$. A change of basis produces no such term and no such difficulty, which is why Chapters 1 to 6 could ignore the whole subject.

In classical non-relativistic physics, including classical continuum mechanics, all observers are presumed to agree on two things: the time intervals between successive events, and the distance between arbitrary pairs of material points. Thus, if t and t^+ are the times on the watches worn by O and O^+ respectively, then

$$t^+ = t + a, \tag{7.1}$$

where a is a constant; and if $\underline{x}, \underline{y} \in \mathbb{E}^3$ and $\underline{x}^+, \underline{y}^+ \in \mathbb{E}^{3+}$ are the positions of material points p and q as perceived by O and O^+ (Figure 7.1), then

$$|\underline{x}^+ - \underline{y}^+| = |\underline{x} - \underline{y}|. \tag{7.2}$$

REMARK 7.2 (what the two assumptions cost, and what they buy). Neither statement is a theorem. Both are the classical idealisation, and both fail in relativity: (7.1) fails because simultaneity is not absolute, and (7.2) fails because lengths contract. Writing them down is therefore the point at which this book commits to Newtonian physics, and it is worth knowing exactly how much is being committed.

Very little, and very much. Very little, because the two statements concern only clocks and rulers and say nothing about matter. Very much, because (7.2) alone will be shown to force the form of the change

of observer completely, leaving only the arbitrary $\tilde{Q}(t)$ and $\tilde{c}(t)$; and the constant a in (7.1) will be shown to be irrelevant, since only $dt^+/dt = 1$ is ever used. From two sentences about measurement comes the entire kinematics of the chapter.

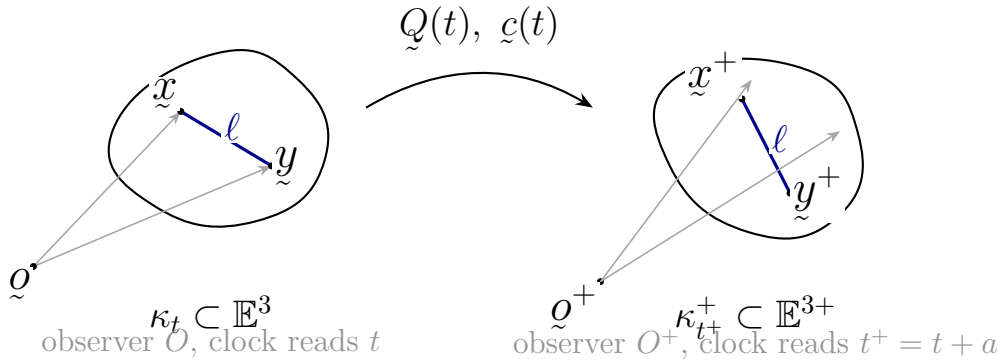


Figure 7.1: A body as perceived by two observers. The same two material points are seen at $\underline{x}, \underline{y}$ by O and at $\underline{x}^+, \underline{y}^+$ by O^+ , and the segment joining them has the same length ℓ in both pictures: that is (7.2), and it is the only geometric agreement assumed. The two observers need not agree about which direction the segment points, about where the origin is, or about anything else. Section 7.1 shows that this single agreement forces the relation between the pictures to be (7.3), a uniform orthogonal $\tilde{Q}(t)$ followed by a translation $\tilde{c}(t)$, and that both may depend on time in any way whatever.

Assume the two point spaces are related by a differentiable mapping. Fix t , and let ϕ denote the map that sends the place $\underline{x} \in \mathbb{E}^3$ assigned by O to a material point to the place $\underline{x}^+ = \phi(\underline{x}) \in \mathbb{E}^{3+}$ assigned by O^+ to the same material point at the same instant. Condition (7.2) reads

$$|\phi(\underline{y}) - \phi(\underline{x})| = |\underline{y} - \underline{x}| \quad \text{for all } \underline{x}, \underline{y},$$

which is word for word the hypothesis (3.113) of the analysis of rigid-body motion in Section 3.7, with ϕ in place of $\chi_\kappa(\cdot, t)$. The argument given there therefore applies verbatim. It is short enough to be worth repeating in the present notation, because the conclusion is the pivot of the chapter.

Square both sides to remove the square root, so that each side is a polynomial and differentiates cleanly:

$$[\phi(\underline{y}) - \phi(\underline{x})] \cdot [\phi(\underline{y}) - \phi(\underline{x})] = (\underline{y} - \underline{x}) \cdot (\underline{y} - \underline{x}).$$

Fix $\underline{x}$ and differentiate with respect to $\underline{y}$. Both sides have the form $\underline{a} \cdot \underline{a}$, whose differential is $d\underline{a} \cdot \underline{a} + \underline{a} \cdot d\underline{a} = 2\underline{a} \cdot d\underline{a}$, the two terms being equal by the symmetry of the inner product; the factor 2 appears on both sides and cancels. Writing $\underline{G}(\underline{y})$ for the gradient of ϕ at $\underline{y}$, a two-point tensor $\mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$, so that $d\phi = \underline{G} d\underline{y}$,

$$[\phi(\underline{y}) - \phi(\underline{x})] \cdot \underline{G}(\underline{y}) d\underline{y} = (\underline{y} - \underline{x}) \cdot d\underline{y}.$$

Move $\underline{G}(\underline{y})$ to the other slot by the definition of the transpose, $\underline{a} \cdot \underline{G}\underline{b} = \underline{G}^t \underline{a} \cdot \underline{b}$, and collect:

$$\left(\underline{G}^t(\underline{y}) [\phi(\underline{y}) - \phi(\underline{x})] - (\underline{y} - \underline{x}) \right) \cdot d\underline{y} = 0 \quad \text{for all } d\underline{y}.$$

A vector orthogonal to every vector is $\underline{0}$ by Lemma 5.1, so

$$\underline{G}^t(\underline{y}) [\phi(\underline{y}) - \phi(\underline{x})] = \underline{y} - \underline{x}.$$

Now reverse the roles: fix $\underline{y}$, so that $\underline{G}^t(\underline{y})$ and $\phi(\underline{y})$ are both constant, and differentiate with respect to $\underline{x}$. The only $\underline{x}$ -dependent piece on the left is $-\phi(\underline{x})$, whose differential is $-\underline{G}(\underline{x}) d\underline{x}$; on the right it is $-\underline{x}$. Hence

$$-\underline{G}^t(\underline{y}) \underline{G}(\underline{x}) d\underline{x} = -d\underline{x} \quad \text{for all } d\underline{x},$$

and since two tensors agreeing on every vector are equal,

$$\underline{G}^t(\underline{y}) \underline{G}(\underline{x}) = \underline{I} \quad \text{for all } \underline{x}, \underline{y} \in \mathbb{E}^3,$$

$\underline{I}$ being the identity of $\mathbb{E}^3$. Setting $\underline{y} = \underline{x}$ gives $\underline{G}^t(\underline{x}) \underline{G}(\underline{x}) = \underline{I}$, so $\underline{G}(\underline{x})$ is orthogonal at every $\underline{x}$ and $\underline{G}^t = \underline{G}^{-1}$ by Proposition 1.48; substituting that back gives $\underline{G}^{-1}(\underline{y}) \underline{G}(\underline{x}) = \underline{I}$, i.e. $\underline{G}(\underline{y}) = \underline{G}(\underline{x})$ for every pair of places. So $\underline{G}$ does not depend on position at all. Call its common value $\underline{Q}(t)$, the dependence on t surviving because t was held fixed throughout. Finally $d\underline{x}^+ = \underline{Q}(t) d\underline{x} = d(\underline{Q}(t)\underline{x})$, $\underline{Q}$ being uniform, so $d(\underline{x}^+ - \underline{Q}\underline{x}) = \underline{0}$

and the bracket is independent of $\underline{x}$; this is the integration step already used at (3.99) and (4.30). Therefore

$$\underline{x}^+ = \underline{Q}(t)\underline{x} + \underline{c}(t), \quad (7.3)$$

where $\underline{c}(t) \in \mathbb{E}^{3+}$ is arbitrary and $\underline{Q}(t)$ is an arbitrary spatially uniform two-point orthogonal tensor mapping $\mathbb{E}^3$ to $\mathbb{E}^{3+}$.

The converse is immediate and is what makes the condition sharp: if (7.3) holds then

$$|\underline{x}^+ - \underline{y}^+|^2 = \underline{Q}(\underline{x} - \underline{y}) \cdot \underline{Q}(\underline{x} - \underline{y}) = (\underline{x} - \underline{y}) \cdot \underline{Q}^t \underline{Q}(\underline{x} - \underline{y}) = |\underline{x} - \underline{y}|^2,$$

so (7.2) is recovered. Thus (7.2) holds *if and only if* (7.3) does.

REMARK 7.3 (the shape of that proof, and what each differentiation rules out). The derivation is four lines long but it is not obvious why it should work at all, and the reason is worth extracting, because the same three moves are used at (3.113) and in every rigidity argument in the book.

The hypothesis (7.2) is a statement about *pairs* of points; the conclusion (7.3) is a statement about *one* point at a time. The entire difficulty is the passage from the first kind of statement to the second, and each step performs one part of it.

Squaring is not cosmetic. The function $|\cdot|$ is not differentiable where its argument vanishes, and $|\cdot|^2$ is a polynomial in the components. Both sides being non-negative, nothing is lost.

The first differentiation, in $\underline{y}$, makes the statement local. It replaces the map ϕ by its gradient and yields $\underline{G}^t(\underline{y})[\phi(\underline{y}) - \phi(\underline{x})] = \underline{y} - \underline{x}$. This is still a statement about a pair of points, but it is now linear in the increment at $\underline{y}$; and it says nothing yet about whether $\underline{G}$ is the same at different places, which is exactly the thing to be proved.

The second differentiation, in $\underline{x}$, is what makes $\underline{G}$ uniform. Both $\phi(\underline{y})$ and $\underline{y}$ are constants of that differentiation and disappear, leaving $\underline{G}^t(\underline{y})\underline{G}(\underline{x}) = \underline{I}$ for *every pair* of places. That single relation carries two conclusions: putting $\underline{y} = \underline{x}$ gives orthogonality at each place, and feeding that back gives $\underline{G}(\underline{y}) = \underline{G}(\underline{x})$ everywhere. The second is the one

that matters. A map that preserves all distances cannot bend, because bending is precisely a $\underline{G}$ that varies from place to place, and two places whose $\underline{G}$ differed would betray themselves in the distance between a third pair.

The integration turns a uniform gradient into an affine map, as after (3.99) and (4.30). It is here, and only here, that the arbitrary $\underline{c}(t)$ enters: an affine map is determined by its linear part only up to a translation, which is the analytic form of the statement of Remark 7.4 that $\underline{c}$ is arbitrary because the two origins are.

Two footnotes on the hypotheses, since both are easy to overlook. Differentiability of ϕ was assumed before the argument began; it is a real assumption here, though a removable one, because a distance-preserving map of one Euclidean space onto another is affine with no smoothness assumed at all. Nothing in this book needs the sharper statement. And what comes out is orthogonality, not rotation: $\underline{Q}^t \underline{Q} = \underline{I}$ gives $\det \underline{Q} = \pm 1$ by Proposition 1.48, and no step of the argument prefers one sign to the other. Remark 7.6 takes up what the other sign would mean.

Because tensors operate on vectors rather than on positions, (7.3) should be interpreted to mean

$$\underline{x}^+ - \underline{y}^+ = \underline{Q}(t)(\underline{x} - \underline{y}). \quad (7.4)$$

In fact, as suggested by Figure 7.1, (7.3) also holds if the positions therein are replaced by position vectors.

REMARK 7.4 (why (7.4) is the honest statement and (7.3) the usable one). A point is not a vector. The difference of two points is a vector, by Definition 1.61, and it is on differences that $\underline{Q}$ acts. That is all (7.4) says, and it is the only form of the statement that is correct before any origin has been named.

Once each observer names an origin, $\underline{o} \in \mathbb{E}^3$ for O and $\underline{o}^+ \in \mathbb{E}^{3+}$ for O^+ , the deliberate abuse of Remark 1.64 lets the same letter denote a point and its position vector $\underline{x} - \underline{o}$. Apply (7.4) to the pair $(\underline{x}, \underline{o})$, which gives $\phi(\underline{x}) - \phi(\underline{o}) = \underline{Q}(t)(\underline{x} - \underline{o})$, and insert the place $\phi(\underline{o})$ between

$\underline{x}^+ = \phi(\underline{x})$ and $\underline{o}^+$ by Chasles' relation, Definition 1.61(P2):

$$\begin{aligned}\underline{x}^+ - \underline{o}^+ &= (\phi(\underline{x}) - \phi(\underline{o})) + (\phi(\underline{o}) - \underline{o}^+) \\ &= \underline{Q}(t)(\underline{x} - \underline{o}) + \underline{c}(t), \quad \underline{c}(t) = \phi(\underline{o}) - \underline{o}^+, \end{aligned}$$

which is (7.3) verbatim for position vectors, with $\underline{c}(t)$ now identified as the position, in O^+ 's frame, of the place O^+ assigns to O 's origin. So nothing is lost by reading (7.3) as an equation between position vectors, and that is how it will be read from here on. What is lost if the distinction is forgotten altogether is the ability to see why $\underline{c}$ is arbitrary: it is arbitrary precisely because the choice of the two origins is.

REMARK 7.5 (the two freedoms in (7.3), and which one is dangerous). The change of observer carries two free functions of time, and they are not on the same footing.

$\underline{c}(t)$ is a relative translation. It disappears from every statement about *differences* of positions, and (7.4) is the proof: no $\underline{c}$ appears there. Consequently $\underline{c}$ can never affect a deformation gradient, a strain, or a stress, all of which are built from differences. It survives only in the velocity, through $\dot{\underline{c}}$, and in the acceleration, through $\ddot{\underline{c}}$.

$\underline{Q}(t)$ is a relative rotation, and it does not disappear from differences. It is the source of every non-trivial transformation law in this chapter. The truly dangerous part of it is $\dot{\underline{Q}}$: if $\underline{Q}$ were constant the whole chapter would collapse into the change-of-basis formulas of Chapter 1, because a constant $\underline{Q}$ commutes with d/dt and every rate would transform exactly as the quantity itself does. It is because $\underline{Q}$ turns as time passes, and hence fails to commute with differentiation, that $\underline{D}$ and $\underline{W}$ part company at (7.34), that $\underline{a}$ acquires the extra terms of (7.9), and that the equation of motion acquires the inertial force of (7.63).

REMARK 7.6 (may one observer be the mirror image of another?). Nothing said so far forbids it, and it is worth seeing that this is not an oversight.

He is a legitimate observer. A reflection preserves every distance, so it satisfies (7.2) exactly, and the derivation of (7.3) returned only

$\tilde{Q}^t \tilde{Q} = \underline{I}$, hence $\det \tilde{Q} = \pm 1$ by Proposition 1.48. Had the book wanted to exclude mirror observers it would have had to say so, and it would have had to justify saying so by something other than clocks and rulers, because clocks and rulers cannot tell left from right.

What is unaffected. Every transformation law derived in this chapter. Each was obtained from $\tilde{Q}^t \tilde{Q} = \underline{I}$ alone, and none of them used $\det \tilde{Q} = 1$: $\tilde{U}^+ = \tilde{K} \tilde{U} \tilde{K}^t$, $\tilde{T}^+ = \tilde{Q} \tilde{T} \tilde{Q}^t$ and the rest hold verbatim for a mirror observer. So does everything metric, lengths and angles being exactly what (7.2) protects.

What he does not agree about. Handedness. A right-handed helix is left-handed in his picture (Figure A); a right-handed basis of $\mathbb{E}^3$ has a left-handed image in $\mathbb{E}^{3+}$; and every cross product changes sign relative to what a rotation would have given, because the box product carries the factor $\det \tilde{Q}$ by Theorem 1.52. The consequence worth recording is that an axial vector is not a spatial vector in the sense of Definition 7.7. If $\tilde{A}$ is skew with axial vector $\underline{a}$ in the sense of Proposition 1.44, and if $\tilde{A}$ is objective, $\tilde{A}^+ = \tilde{Q} \tilde{A} \tilde{Q}^t$, then for every $\underline{s} \in \mathbb{E}^{3+}$

$$\begin{aligned} \tilde{A}^+ \underline{s} &= \tilde{Q} \tilde{A} \tilde{Q}^t \underline{s} = \tilde{Q} (\underline{a} \times \tilde{Q}^t \underline{s}) \\ &= (\det \tilde{Q}) (\tilde{Q} \underline{a}) \times (\tilde{Q} \tilde{Q}^t \underline{s}) = (\det \tilde{Q}) (\tilde{Q} \underline{a}) \times \underline{s}, \end{aligned}$$

using $\tilde{Q}(\underline{u} \times \underline{w}) = (\det \tilde{Q})(\tilde{Q} \underline{u}) \times (\tilde{Q} \underline{w})$, which is Theorem 1.52 written for an orthogonal $\tilde{Q}$. Hence

$$\underline{a}^+ = (\det \tilde{Q}) \tilde{Q} \underline{a},$$

which is $\tilde{Q} \underline{a}$ for a rotation and $-\tilde{Q} \underline{a}$ for a reflection. This is why the whole of this chapter is written in the skew tensor $\tilde{\Omega}$ and never in its axial vector $\tilde{\omega}$: the tensor transforms by one rule for both signs and the vector does not. The same caution applies to $\tilde{W}$, and it is the reason (7.34)₂ is stated as a tensor equation.

What (7.19) costs him. Having chosen a mirrored frame, O^+ is no longer free in his choice of reference configuration. The requirement that his reference configuration be occupiable is $J^+ > 0$, and that forced $\det \tilde{Q} \det \tilde{K} = 1$; so $\det \tilde{Q} = -1$ obliges $\det \tilde{K} = -1$ as well. A mirror

observer must mirror his reference copy of the body too. This is not a restriction on physics but a consistency condition on bookkeeping, and it is the only place in the chapter where the sign of a determinant does any work.

Why the question is not idle. Whether a material can tell itself from its mirror image is a genuine physical question, and for some materials the answer is yes: a quartz crystal, a cholesteric liquid crystal, a composite whose fibres wind one way. Chapter 8 will require constitutive equations to be indifferent to a change of observer, and by this remark that requirement may be posed over the full orthogonal group Orth or over the proper orthogonal group alone. The two demands are different, and the difference is exactly whether the material is permitted to be chiral. Nothing in the present chapter decides between them, because (7.2) does not.

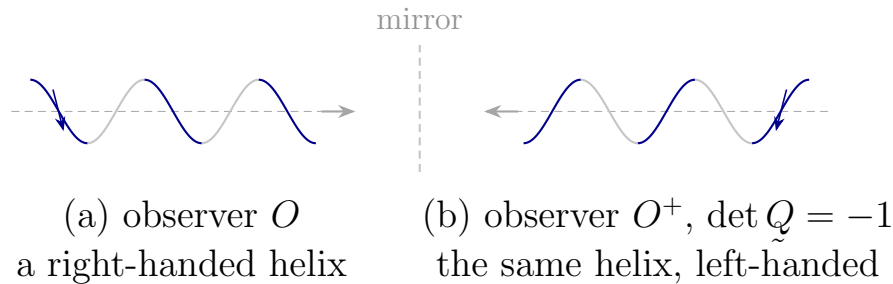


Figure A: A mirror observer. The solid strands are in front and the pale ones behind, and the grey arrow is the direction of advance along the axis, so the two panels are genuinely different curves in space and not the same curve redrawn: no rotation carries one onto the other. Yet every distance between two points of the coil is the same in both, which is all (7.2) demands, so $\tilde{Q}$ with $\det \tilde{Q} = -1$ is an admissible change of observer. What the two observers disagree about is handedness, and with it the sign of every cross product and of every axial vector, by the rule $\underline{a}^+ = (\det \tilde{Q})\tilde{Q}\underline{a}$ established in Remark 7.6. The disagreement is invisible in $\underline{U}$, $\underline{\tilde{C}}$ and $\underline{\tilde{D}}$, which are built from lengths and angles alone; it becomes visible the moment a formula contains $\underline{\omega}$ rather than $\underline{\Omega}$, or asks whether a material is chiral.

We can now name the two properties that the rest of the chapter is occupied with sorting.

DEFINITION 7.7 (objective and invariant). A field carried by the body is *objective*, or *frame indifferent*, if the values recorded by the two observers are related as the type of the field demands, that is, if

$$\begin{aligned} \text{a scalar } \varphi : & \quad \varphi^+ = \varphi, \\ \text{a spatial vector } \underline{u} : & \quad \underline{u}^+ = Q\underline{u}, \\ \text{a spatial tensor } \underline{A} : & \quad \underline{A}^+ = Q\underline{A}Q^t. \end{aligned}$$

A quantity is *invariant* under the change of observer if both observers assign it the same value, $(\cdot)^+ = (\cdot)$. An *equation* is invariant, or form invariant, if it has the same form in the two frames, that is, if starring every symbol in it produces a true statement.

REMARK 7.8 (the distinction, and a counterexample on each side). Objectivity is a property of a *quantity*; invariance is a property of a *quantity's value* or of an *equation*. Confusing the two is the standard error in this subject, so it is worth having a counterexample in each direction ready before the calculations start. All four are proved in the sections that follow; they are collected here as signposts.

Objective but not invariant. The Cauchy stress. By (7.40), $\underline{T}^+ = Q\underline{T}Q^t$, which is exactly what Definition 7.7 requires of a spatial tensor, so $\underline{T}$ is objective. But $Q\underline{T}Q^t \neq \underline{T}$ in general: the two observers write down different tensors, with different components. Objectivity is not sameness; it is agreement about *how* to differ.

Invariant but not objective, in the sense that the test does not apply. The right Cauchy–Green tensor. By (7.30)₁, $\underline{C}^+ = \underline{K}\underline{C}\underline{K}^t$ with $\underline{K}$ a *fixed* tensor that has nothing to do with the relative motion of the observers; if the two observers happen to use a common reference configuration, $\underline{K} = \underline{\hat{I}}$ and $\underline{C}^+ = \underline{C}$ literally. So $\underline{C}$ carries no trace of Q at all. It is not objective in the sense of Definition 7.7 because it is not a spatial tensor and the third line of that definition is not the right test for it. Set beside $\underline{B}^+ = Q\underline{B}Q^t$ of (7.30)₂, this is instructive: $\underline{C}$

and $\tilde{B}$ encode the same deformation and are as close to being the same tensor as two tensors can be, and yet one of them ignores the observer completely while the other turns with him. The difference is only which space they live in.

An invariant scalar built from non-objective ingredients. The velocity is not objective, by (7.8): $\tilde{v}^+ \neq Q\tilde{v}$. Yet $\operatorname{div}^+ \tilde{v}^+ = \operatorname{div} \tilde{v}$ by (7.56). A non-objective field can have an invariant scalar hidden inside it, and the extra terms can cancel.

An invariant scalar built from one objective and one non-objective ingredient. $\tilde{T}$ is objective and $\tilde{L}$ is not, by (7.33); nevertheless the stress power density is invariant, $\tilde{T}^+ \cdot \tilde{L}^+ = \tilde{T} \cdot \tilde{L}$, by (7.58). The offending part of $\tilde{L}$ is annihilated by the symmetry of $\tilde{T}$.

The moral is that invariance must be checked, never assumed from the ingredients, and that is what Section 7.5 does law by law. The one law that fails the check, linear momentum, fails it in a way that is itself instructive.

7.2 Transformation of the motion

Following the development of Section 2.1, the motions of $p \in B$, as perceived by O and O^+ , are expressible as

$$\tilde{x} = \chi(p, t) \quad \text{and} \quad \tilde{x}^+ = \chi^+(p, t^+), \quad (7.5)$$

respectively. Accordingly, substituting both into (7.3),

$$\chi^+(p, t^+) = Q(t)\chi(p, t) + \mathfrak{c}(t). \quad (7.6)$$

REMARK 7.9 (why the argument is p and not $\tilde{X}$). Equations (7.5) take the material point itself as the independent variable, which is the material description of Definition 2.2 and Remark 2.6, and they have to. The body B is the one thing the two observers share. It is not a subset of $\mathbb{E}^3$, nor of $\mathbb{E}^{3+}$; it is a set of material points, and each observer supplies his own map of it, as Remark 2.1 insisted. Had we written $\chi_\kappa(\tilde{X}, t)$ here, we would have had to say which observer chose κ , and

the statement (7.6) would have been circular. Section 7.3 introduces reference configurations only after (7.6) has been secured without them, and pays for the convenience with the extra tensor $\underline{K}$.

The velocity of p in the frame $\mathbb{E}^{3+}$ is $\underline{v}^+ = \partial \underline{\chi}^+ / \partial t^+$, by Definition 2.3 written in that frame. The two time derivatives are the same operation: $t^+ = t + a$ with a constant by (7.1), so $dt^+/dt = 1$ and the chain rule gives $\partial/\partial t^+ = (dt/dt^+) \partial/\partial t = \partial/\partial t$, both taken at fixed p . Differentiating (7.6) at fixed p , with the product rule applied to $\underline{Q}(t)\underline{\chi}$ and remembering that $\underline{Q}$ and $\underline{c}$ depend on t alone while $\underline{\chi}$ depends on p and t ,

$$\underline{v}^+ = \frac{\partial}{\partial t^+} \underline{\chi}^+ = \frac{\partial}{\partial t} \underline{\chi}^+ = \dot{\underline{Q}} \underline{x} + \underline{Q} \underline{v} + \dot{\underline{c}}, \quad (7.7)$$

where $\underline{v} = \partial \underline{\chi} / \partial t$ is the velocity of p in the frame $\mathbb{E}^3$ and $\underline{x} = \underline{\chi}(p, t)$.

This is the first transformation law of the chapter that is not simply a conjugation, and the reason is visible in the algebra: of the three terms on the right, only the middle one came from differentiating the motion. The other two came from differentiating the relation between the observers. If the body stands still for O , so that $\underline{v} = \underline{0}$, then $\underline{v}^+ = \dot{\underline{Q}} \underline{x} + \dot{\underline{c}}$, which vanishes only if the observers are at relative rest. *The velocity is not objective.*

It is convenient to eliminate $\underline{x}$, which belongs to O 's frame, in favour of $\underline{x}^+$. From (7.3), $\underline{Q} \underline{x} = \underline{x}^+ - \underline{c}$, so

$$\dot{\underline{Q}} \underline{x} = \dot{\underline{Q}} (\underline{Q}^t \underline{Q}) \underline{x} = (\dot{\underline{Q}} \underline{Q}^t) (\underline{Q} \underline{x}) = \underline{\Omega} (\underline{x}^+ - \underline{c}),$$

where the identity $\underline{Q}^t \underline{Q} = \underline{I}$ of Proposition 1.48 was inserted and the factors regrouped. Thus the velocities in the two frames are related by

$$\underline{v}^+ - \underline{Q} \underline{v} = \underline{\Omega} (\underline{x}^+ - \underline{c}) + \dot{\underline{c}}, \quad \text{where} \quad \underline{\Omega}(t) = \dot{\underline{Q}} \underline{Q}^t \quad (7.8)$$

is the spin of $\mathbb{E}^{3+}$.

Two properties of $\underline{\Omega}$ are needed constantly and are established once here.

$\underline{\Omega}$ is skew. Differentiate the orthogonality relation $\underline{Q} \underline{Q}^t = \underline{I}^+$, in which $\underline{I}^+$ is the identity of $\mathbb{E}^{3+}$ and is therefore constant. By the product

rule and the commutation of differentiation with transposition, Problem 1(a) and Problem 1(b) of Chapter 4,

$$\dot{\underline{Q}}\underline{Q}^t + \underline{Q}\dot{\underline{Q}}^t = \dot{\underline{Q}}\underline{Q}^t + (\dot{\underline{Q}}\underline{Q}^t)^t = \underline{Q}^+,$$

the middle step because $\underline{Q}\dot{\underline{Q}}^t = \underline{Q}(\dot{\underline{Q}})^t = (\dot{\underline{Q}}\underline{Q}^t)^t$ by the rule $(\underline{A}\underline{B})^t = \underline{B}^t\underline{A}^t$. Hence $\underline{\Omega} + \underline{\Omega}^t = \underline{Q}^+$, i.e. $\underline{\Omega}^t = -\underline{\Omega}$.

$\underline{\Omega}$ is a spatial tensor of the frame of O^+ . The derivative of a two-point tensor is a two-point tensor of the same type, so $\dot{\underline{Q}}: \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$, while $\underline{Q}^t: \mathbb{E}^{3+} \rightarrow \mathbb{E}^3$. The composition $\underline{\Omega} = \dot{\underline{Q}}\underline{Q}^t$ therefore maps $\mathbb{E}^{3+}$ into itself. This is why $\underline{\Omega}$ may be added to $\underline{Q}\underline{L}\underline{Q}^t$ in (7.33) without an index mismatch, and it is worth checking by the alphabet test of Remark 4.1: $\dot{\underline{Q}}$ carries $\underline{e}_i^+ \otimes \underline{e}_j$, $\underline{Q}^t$ carries $\underline{e}_j \otimes \underline{e}_k^+$, the unstarred indices contract in pairs and what is left is $\underline{e}_i^+ \otimes \underline{e}_k^+$.

REMARK 7.10 (what $\dot{\underline{Q}}\underline{Q}^t$ is). The combination has been met before. Equation (4.29) says that in a rigid motion $\underline{x} = \underline{Q}(t)\underline{X} + \underline{c}(t)$ the spatial velocity gradient is $\underline{L} = \dot{\underline{Q}}\underline{Q}^t$, that the stretching vanishes, $\underline{D} = \underline{Q}$, and that the whole of $\underline{L}$ is spin, $\underline{W} = \dot{\underline{Q}}\underline{Q}^t$. Compare that motion with (7.3) and they are the same formula. A change of observer is, formally, a rigid motion: not a motion of the body, but of one frame relative to the other.

So $\underline{\Omega}$ is the velocity gradient of the rigid displacement that carries O 's frame onto O^+ 's, and it is a *pure spin* for exactly the reason (4.29) gives, namely that a rigid displacement stretches nothing. By Proposition 1.44 the skew tensor $\underline{\Omega}$ has a unique axial vector $\underline{\omega}$ with $\underline{\Omega}\underline{s} = \underline{\omega} \times \underline{s}$, and $\underline{\omega}$ is the angular velocity of O 's frame as recorded by O^+ . If the platform of O^+ turns with angular velocity $\underline{\omega}_p$ relative to O , then $\underline{\omega} = -\underline{\omega}_p$: what O^+ sees is the rest of the world going round the other way. Problem 3 computes both for an observer standing on a spinning planet.

Everything else in this chapter follows from this one identification. The extra term in (7.33) is a pure spin, so it lands entirely in the skew part of $\underline{L}^+$ and leaves the symmetric part untouched: that is (7.34), and it is the whole reason $\underline{D}$ is objective and $\underline{W}$ is not. The extra terms in (7.9) are what a pure spin does to an acceleration, and they are the Coriolis and centrifugal terms.

REMARK 7.11 ((7.8) read as a decomposition, and a one-line route to (7.33)). Equation (7.8) is usually met in elementary mechanics in words, and the words are worth attaching to the symbols, because they make the rest of Section 7.3 predictable.

What the second and third terms are. Put $\underline{v} = \underline{0}$ in (7.8). A particle at rest in the frame of O , sitting at the place O^+ calls $\underline{x}^+$, is recorded by O^+ as having the velocity

$$\underline{v}_{\text{tr}}^+(\underline{x}^+, t) = \underline{\Omega}(\underline{x}^+ - \underline{c}) + \dot{\underline{c}}.$$

This is the *transport* velocity: the velocity O^+ attributes to the point of O 's frame that the particle happens to occupy. So (7.8) says

$$\underline{v}^+ = \underbrace{\underline{Q}\underline{v}}_{\text{what } O \text{ records, rotated}} + \underbrace{\underline{v}_{\text{tr}}^+}_{\text{what the frame contributes}},$$

which is the familiar statement that a velocity relative to a moving frame differs from a velocity relative to a fixed one by the velocity of the frame at that place. Figure B(a) is the triangle.

The transport field is rigid. As a field of $\underline{x}^+$ it is a constant $\dot{\underline{c}} - \underline{\Omega}\underline{c}$ plus a skew tensor applied to $\underline{x}^+$, which is exactly the form (4.29) of a rigid velocity field, and Remark 7.10 said as much. Its gradient is $\text{grad}^+ \underline{v}_{\text{tr}}^+ = \underline{\Omega}$, a constant tensor and a pure spin; it stretches nothing, anywhere.

Hence (7.33) in one line. Take grad^+ of (7.8). The first term gives $\underline{Q}(\text{grad } \underline{v})\underline{Q}^t = \underline{Q}\underline{L}\underline{Q}^t$, by the chain rule with $\partial \underline{x}/\partial \underline{x}^+ = \underline{Q}^t$ from (7.3), which is the computation performed in general at (7.48); the second gives $\underline{\Omega}$; and the left-hand side is $\underline{L}^+$. So

$$\underline{L}^+ = \underline{Q}\underline{L}\underline{Q}^t + \underline{\Omega},$$

which is (7.33) without passing through $\dot{\underline{F}}$ at all. The route through (7.31) is the book's and is worth having, because it exhibits $\underline{\Omega}$ as arising from $\dot{\underline{Q}}$; but this route exhibits it as the gradient of a rigid field, and that is the fact (7.34) needs. The splitting into $\underline{D}$ and $\underline{W}$ can now be predicted before it is done: the observer adds the gradient of a rigid field, a rigid field has no stretching, so the observer cannot touch $\underline{D}$.

Why the velocity itself is beyond repair. Notice what the decomposition does *not* say. There is no way to subtract $\underline{v}_{\text{tr}}^+$ and be left with something both observers can compute, because $\underline{v}_{\text{tr}}^+$ is not a property of the body: it depends on $\underline{\Omega}$, $\underline{c}$ and $\underline{x}^+$, and O has no access to any of them. This is the difference between the velocity, which is irreparably non-objective, and the velocity *gradient*, whose offending part is a single constant skew tensor that the splitting (4.7) isolates and discards.

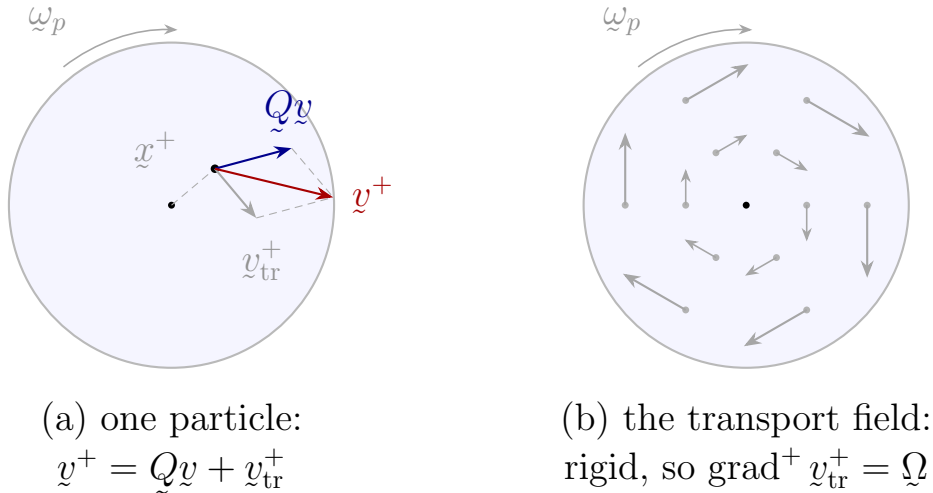


Figure B: Equation (7.8) as a decomposition, drawn in the frame of O^+ with $\underline{c} = \underline{0}$ for clarity. In (a) the velocity O^+ records for one particle is the sum of the velocity O records, carried over by $\underline{Q}$, and the transport velocity $\underline{v}_{\text{tr}}^+$, which is what O^+ would record if the particle were nailed to O 's frame. Since the platform of O^+ turns at $\underline{\omega}_p$, the transport velocity runs backwards, $\underline{\Omega}\underline{s} = -\underline{\omega}_p \times \underline{s}$, as Remark 7.10 requires. In (b) the same field is drawn at twelve places: its arrows grow linearly with the distance from the axis and are everywhere tangential, which is to say that it is the velocity field of a rigid rotation, (4.29). That is the whole reason $\underline{D}$ survives a change of observer and $\underline{W}$ does not: the observer adds the gradient of the field in (b), and the field in (b) stretches nothing.

Using this result it is straightforward, though not short, to show that the accelerations $\underline{a}^+ \in \mathbb{E}^{3+}$ and $\underline{a} \in \mathbb{E}^3$ of p are related by

$$\underline{a}^+ - \underline{Q}\underline{a} = \underline{\ddot{c}} + 2\underline{\Omega}(\underline{v}^+ - \underline{\dot{c}}) + (\underline{\dot{\Omega}} - \underline{\Omega}^2)(\underline{x}^+ - \underline{c}). \quad (7.9)$$

This is Problem 1, where it is derived in full from (7.7) and (7.8) with no step omitted. The reader who wants the algebra now should turn to it; what is needed here is only the reading of the four terms, and the fact that the left-hand side is not zero.

REMARK 7.12 (the four terms of (7.9)). Each term answers to one way in which the frame of O^+ can fail to be at rest relative to that of O , and each is familiar from elementary mechanics under a different name.

$\ddot{\underline{c}}$ is the acceleration of the origin of O^+ 's frame. It is the only term that survives when the frames do not rotate, and it is what throws a passenger forward when a bus brakes.

$2\Omega(\underline{v}^+ - \dot{\underline{c}})$ is the *Coriolis* term. It is proportional to the velocity of the particle relative to the moving frame, and it vanishes for a particle at rest in that frame; the factor 2 is not a convention but comes out of the calculation, as Problem 1 shows, because $\dot{\underline{Q}}$ is differentiated once in $\underline{a}^+$ and once again through $\underline{v}$.

$-\Omega^2(\underline{x}^+ - \underline{c})$ is the *centrifugal* term. With $\Omega \underline{s} = \underline{\omega} \times \underline{s}$, it is $-\underline{\omega} \times (\underline{\omega} \times \underline{s})$, which points away from the axis of $\underline{\omega}$ with magnitude $|\underline{\omega}|^2$ times the distance from that axis.

$\dot{\Omega}(\underline{x}^+ - \underline{c})$ is the *Euler* or azimuthal term, present only when the relative rotation is speeding up or slowing down.

Note that all four are proportional to nothing that the body is doing. They depend on the relative motion of the observers and on the particle's position and velocity, and on nothing else. That is exactly the property that will let them be moved to the other side of the equation of motion in (7.62) and called forces.

We observe that if $\ddot{\underline{c}} = \underline{0}$, i.e. if $\underline{c} = \underline{V}t + \underline{c}_0$ for constant $\underline{V}$ and $\underline{c}_0$, and if $\underline{Q}$ is independent of t , then $\dot{\underline{Q}} = \underline{0}$, so $\Omega = \underline{0}$ and $\dot{\Omega} = \underline{0}$, and (7.9) reduces term by term to $\underline{a}^+ - \underline{Q}\underline{a} = \underline{0}$. Thus

$$\underline{x}^+ = \underline{Q}\underline{x} + \underline{V}t + \underline{c}_0 \quad \text{and} \quad \underline{a}^+ = \underline{Q}\underline{a}. \quad (7.10)$$

This is called a *Galilean transformation*. Its significance will be discussed later, in connection with the equations of motion. Note that $\underline{V}$

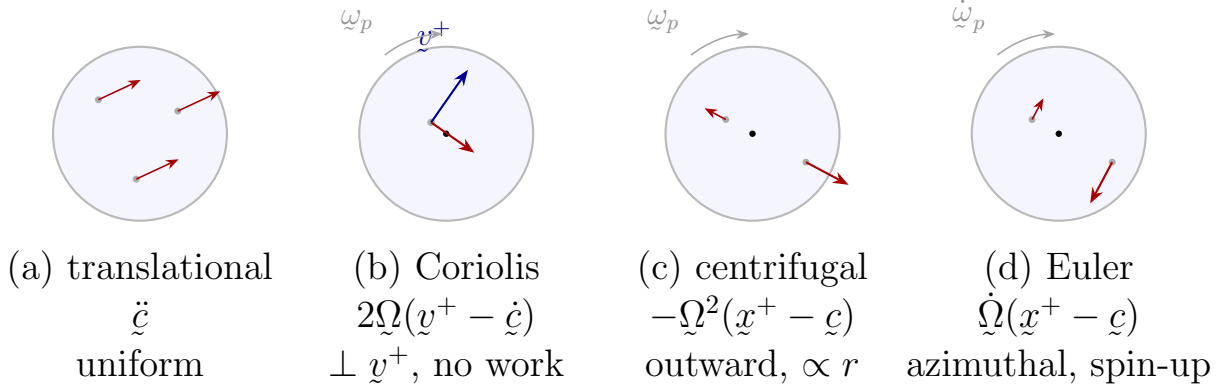


Figure C: The four terms of (7.9), drawn in the frame of O^+ as the four contributions they make to the inertial force (7.63). Each answers to one way in which the frame can move, and each has a signature that identifies it. (a) depends on neither the place nor the velocity of the particle, so it is indistinguishable from a uniform gravitational field, which is the content of Remark 7.30. (b) is perpendicular to $\underline{v}^+$, because Ω is skew, and therefore does no work: $\Omega \underline{v}^+ \cdot \underline{v}^+ = 0$ by Lemma 2.36. It vanishes for a particle at rest in the frame. (c) points away from the axis with magnitude proportional to the distance from it, and is the only one of the four that a particle at rest in a steadily spinning frame feels. (d) appears only while the spin rate is changing and is azimuthal, in the sense opposite to the change. A frame may exhibit any combination of the four; the spinning planet of Problem 3 has (b) and (c) and nothing else.

here is a constant velocity and has nothing to do with the left stretch tensor $\underline{V}$ of (7.27); the book uses the same letter for both and no formula contains the two together.

REMARK 7.13 (what a Galilean transformation does and does not change). Under (7.10) the acceleration is objective while the velocity is still not: (7.8) with $\underline{\Omega} = \underline{Q}$ and $\underline{\dot{c}} = \underline{V}$ gives $\underline{v}^+ = \underline{Q}\underline{v} + \underline{V}$, so the two observers disagree about velocity by a constant vector no matter what the body does. This is the familiar statement that velocity is relative and acceleration is not, and (7.9) shows precisely how much of it survives when the rotation is allowed to depend on time: none of it.

The class of Galilean transformations is exactly the class that preserves the form of the equation of motion, which is proved at (7.64). It is worth noticing already that the class is large: $\underline{Q}$ is any fixed orthogonal tensor, $\underline{V}$ any constant vector, $\underline{c}_0$ any fixed vector, and a in (7.1) any constant. That is the reason for the remark after (7.64) that if one inertial frame exists then infinitely many do.

REMARK 7.14 (where the name comes from, and what the transformation is for). Equation (7.10) takes three lines to write and is named after a man who never wrote it. Both facts repay a paragraph, because the transformation carries more of this book than its brevity suggests.

What it is, counted. A Galilean transformation is the change of observer (7.3) restricted by $\underline{\dot{Q}} = \underline{Q}$ and $\underline{\ddot{c}} = \underline{0}$, together with the clock shift (7.1):

$$\underline{x}^+ = \underline{Q}\underline{x} + \underline{V}t + \underline{c}_0, \quad t^+ = t + a.$$

It carries ten free constants: three in the fixed rotation $\underline{Q}$, three in the boost $\underline{V}$, three in the translation $\underline{c}_0$, and one in a . These are the ten ways in which two observers may be set up differently without either being able to find it out: pointing their axes differently, drifting apart at constant speed, placing their origins differently, and starting their clocks at different moments. (Strictly $\underline{Q}$ is orthogonal rather than a rotation, so the family has a second piece consisting of the mirrored observers of Remark 7.6; the ten constants count the piece containing the identity.)

Why they form a group, and why that matters. Compose two of them. By the calculation before (7.65), $\tilde{Q}^{++} = \tilde{Q}^+ \tilde{Q}$ and $\tilde{c}^{++} = \tilde{Q}^+ \tilde{c} + \tilde{c}^+$; a product of constant tensors is constant, and $\tilde{Q}^+(Vt + \tilde{c}_0) + V^+t + \tilde{c}_0^+$ is again linear in t . The identity and the inverses are immediate. So the Galilean transformations form a group, and that is precisely what licenses the sentence after (7.64). Being inertial is not a property a frame has by itself but a relation between two frames, and such a relation is usable only if it is transitive. Were the composition of two Galilean transformations not Galilean, a frame inertial relative to an inertial frame need not have been inertial, and the whole notion would have dissolved. Remark 7.31 is the same closure property stated for the spins.

What it is for. One thing, and (7.64) is it. The Galilean transformations are exactly the changes of observer under which (7.53)₃ keeps its bare form with no inertial force; by Remark 7.17 the condition $\tilde{i}^+ = \tilde{0}$, imposed as an identity in $\tilde{x}^+$ and $\tilde{v}^+$, is equivalent to (7.10) and to nothing weaker. So (7.10) is not a computational convenience but a definition in disguise: it names the class of frames in which the balance laws of Chapters 5 and 6 may be read exactly as they were written.

How Galileo came to it. Not as a formula. The *Dialogo* of 1632 is a defence of the claim that the Earth moves, and the standing objection to that claim was mechanical rather than astronomical: if the Earth turned, a stone dropped from a tower would be left behind by the tower, and a cannon fired westward would outrange one fired eastward. Galileo's answer, put into the mouth of Salviati, is the ship. Shut yourself below decks with some flies, a bowl of water holding a fish, a bottle dripping into a basin, and a friend to throw things to; watch everything carefully while the ship stands still; then have the ship move as fast as you please, provided only that the motion is uniform and does not fluctuate this way and that. You will not find the least change in any of those effects, nor will any of them tell you whether the ship is moving or standing still. That is Figure D, and it is (7.10)₂ stated two centuries before anyone wrote $\tilde{a}^+ = \tilde{Q}\tilde{a}$.

Two qualifications keep the history honest. Galileo argued the *prin-*

ciple, that a uniform motion shared by everything in a closed room is undetectable inside it; the *transformation*, and with it the group, is a much later codification, and the name was attached to it only when there was a Lorentz transformation to set beside it. And Galileo's own notion of uniform motion still admitted uniform motion in a circle, since what he needed was for a stone dropped from a mast to keep pace with a turning Earth. The sharpening to uniform motion in a straight line is Descartes' and Newton's, and (7.10), with its *constant* $\tilde{V}$ and *constant* $\tilde{Q}$, is the sharpened version. Newton states the conclusion as Corollary V to the laws of motion in the *Principia* of 1687: the motions of bodies among themselves are the same whether the space that contains them is at rest or moves uniformly forwards in a straight line without circular motion. That last clause is the requirement in (7.10) that $\tilde{Q}$ be constant, and Problem 3 is what happens when it is dropped.

Where it stops. The Galilean group is the symmetry of (7.1) and (7.2) and of nothing else. Maxwell's equations do not share it, and the attempt to make them share it is where the two assumptions of Remark 7.2 were given up in favour of the Lorentz transformation. The relation between the two groups is not rivalry but limit: the Galilean group is what the Poincaré group becomes as $c \rightarrow \infty$. Inside a theory that has already committed itself to (7.1) and (7.2), as this one has, (7.10) is the last word.

7.3 Transformation of kinematic variables

To analyze the motion, observers O and O^+ select reference configurations κ and κ^+ respectively, in which the positions of p are $\tilde{X} = \kappa(p)$ and $\tilde{X}^+ = \kappa^+(p)$. The motions of p in the two frames are then given by

$$\tilde{x} = \chi_{\kappa}(\tilde{X}, t) \quad \text{and} \quad \tilde{x}^+ = \chi_{\kappa^+}^+(\tilde{X}^+, t^+), \quad (7.11)$$

where $\chi_{\kappa}(\tilde{X}, t) = \chi(\kappa^{-1}(\tilde{X}), t)$ and $\chi_{\kappa^+}^+(\tilde{X}^+, t^+) = \chi^+((\kappa^+)^{-1}(\tilde{X}^+), t^+)$, exactly as in Definition 2.4. Thus (7.6) becomes

$$\chi_{\kappa^+}^+(\tilde{X}^+, t^+) = \tilde{Q}(t) \chi_{\kappa}(\tilde{X}, t) + \tilde{c}(t). \quad (7.12)$$

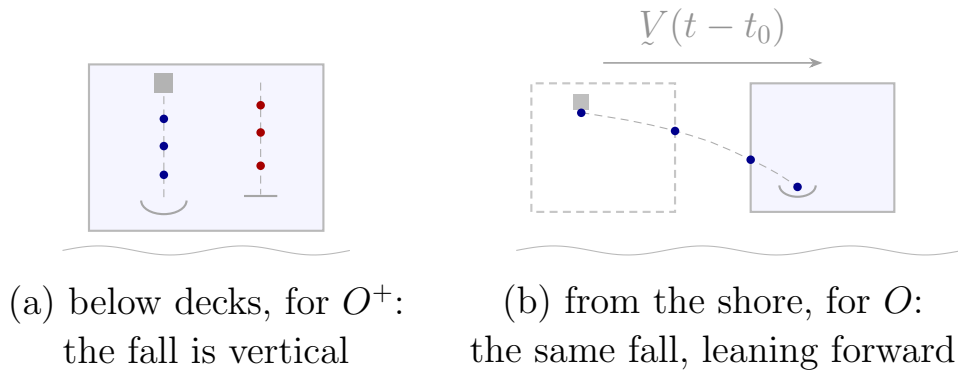


Figure D: Salviati's ship, the argument of the *Dialogo* of 1632 and the physical content of (7.10). The cabin sails at the constant velocity $\tilde{V}$. In (a) the observer O^+ shut inside it sees the drops leave the bottle and fall vertically into the basin, and so does everything else he tries; nothing he can do below decks tells him whether the ship is moving. In (b) the observer O on the shore sees the same drop follow a track that leans forward, because to its fall he must add the ship's displacement $\tilde{V}(t - t_0)$. The two records differ, and by exactly the amount (7.10)₁ predicts; but they differ by a term linear in t , whose second derivative vanishes, so the two observers agree about every acceleration, which is (7.10)₂. That agreement is what lets O^+ use the equations of Chapter 6 unaltered, and it is what fails, and is repaired by (7.62), the moment the ship begins to turn or to change speed.

The two sides are functions of different arguments, $\underline{\underline{X}}^+$ on the left and $\underline{\underline{X}}$ on the right, and the equation is to be read as holding when both label the same material point p . Finding the relation between $\underline{\underline{X}}$ and $\underline{\underline{X}}^+$ is therefore the first order of business.

Suppose the observers adopt κ_{t_0} and $\kappa_{t_0^+}^+$ as their respective reference configurations, the two being the placement of the body at one and the same instant, since $t_0^+ = t_0 + a$ by (7.1). Following the development of Section 2.6, we interpret this to mean

$$\chi_\kappa(\underline{\underline{X}}, t_0) = \underline{\underline{1}}\underline{\underline{X}} \quad \text{and} \quad \chi_{\kappa^+}^+(\underline{\underline{X}}^+, t_0^+) = \underline{\underline{1}}^+\underline{\underline{X}}^+, \quad (7.13)$$

with $\underline{\underline{X}}$ and $\underline{\underline{X}}^+$ regarded as position vectors in $\hat{\mathbb{E}}^3$ and $\hat{\mathbb{E}}^{3+}$ respectively, and where $\underline{\underline{1}}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ and $\underline{\underline{1}}^+: \hat{\mathbb{E}}^{3+} \rightarrow \mathbb{E}^{3+}$ are the uniform, orthogonal shifters of Definition 2.23 in the two frames. Equation (7.13) is (2.76) written twice, once for each observer.

Now set $t = t_0$ in (7.12) and use (7.13) on both sides:

$$\underline{\underline{1}}^+\underline{\underline{X}}^+ = \underline{\underline{Q}}(t_0)\underline{\underline{1}}\underline{\underline{X}} + \underline{\underline{c}}(t_0).$$

Multiply on the left by $(\underline{\underline{1}}^+)^t$ and use $(\underline{\underline{1}}^+)^t\underline{\underline{1}}^+ = \hat{I}^+$, which is (2.72) read in the frame of O^+ , $\hat{I}^+$ being the identity of $\hat{\mathbb{E}}^{3+}$:

$$\underline{\underline{X}}^+ = \underline{\underline{K}}\underline{\underline{X}} + \underline{\underline{c}}^+, \quad (7.14)$$

where

$$\underline{\underline{K}} = (\underline{\underline{1}}^+)^t\underline{\underline{Q}}(t_0)\underline{\underline{1}} \quad \text{and} \quad \underline{\underline{c}}^+ = (\underline{\underline{1}}^+)^t\underline{\underline{c}}(t_0), \quad (7.15)$$

with the shifters evaluated at the times t_0 and t_0^+ . Here $\underline{\underline{K}}: \hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^{3+}$ is a fixed orthogonal tensor and $\underline{\underline{c}}^+ \in \hat{\mathbb{E}}^{3+}$ is a fixed vector.

Both claims about $\underline{\underline{K}}$ need checking, and one of them is the most important single fact in the chapter.

$\underline{\underline{K}}$ is orthogonal. It is a product of three orthogonal maps, $\underline{\underline{1}}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, then $\underline{\underline{Q}}(t_0): \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$, then $(\underline{\underline{1}}^+)^t: \mathbb{E}^{3+} \rightarrow \hat{\mathbb{E}}^{3+}$, so the composition maps $\hat{\mathbb{E}}^3$ to $\hat{\mathbb{E}}^{3+}$ as claimed, and

$$\underline{\underline{K}}^t\underline{\underline{K}} = \underline{\underline{1}}^t\underline{\underline{Q}}^t(t_0)\underline{\underline{1}}^+(\underline{\underline{1}}^+)^t\underline{\underline{Q}}(t_0)\underline{\underline{1}} = \underline{\underline{1}}^t\underline{\underline{Q}}^t(t_0)\underline{\underline{I}}^+\underline{\underline{Q}}(t_0)\underline{\underline{1}} = \underline{\underline{1}}^t\underline{\underline{I}}\underline{\underline{1}} = \underline{\underline{1}}^t\underline{\underline{1}} = \hat{I},$$

using $\underline{1}^+(\underline{1}^+)^t = \underline{I}^+$ from (2.73), $\underline{Q}^t \underline{Q} = \underline{I}$ from Proposition 1.48, and $\underline{1}^t \underline{1} = \underline{\hat{I}}$ from (2.72).

$\underline{K}$ is fixed. Every one of the three factors in (7.15)₁ is evaluated at the single instant t_0 . The tensor $\underline{Q}(t)$ moves; the tensor $\underline{Q}(t_0)$ is one particular tensor, chosen once, and does not.

REMARK 7.15 (why the whole chapter splits in two at (7.15)). The fixity of $\underline{K}$ is the reason the transformation laws that follow come in exactly two families, and it is worth stating before any of them is derived, because it makes all of them predictable.

Referential objects, those living on $\hat{\mathbb{E}}^3$, are related to their counterparts on $\hat{\mathbb{E}}^{3+}$ by $\underline{K}$ alone: $\underline{U}$, $\underline{C}$, $\underline{E}$, $\underline{S}$, $\underline{q}_\kappa$, $\underline{N}$. Spatial objects, those living on $\mathbb{E}^3$, are related by $\underline{Q}(t)$ alone: $\underline{B}$, $\underline{V}$, $\underline{T}$, $\underline{q}$, $\underline{n}$, $\underline{D}$. Two-point objects get one of each: $\underline{F}$, $\underline{R}$, $\underline{P}$, $\underline{H}$, $\underline{1}$. The rule is not a coincidence but the alphabet test of Remark 4.1 applied to the transformation law itself, and Remark 2.39 of Chapter 2 already made the general point that the transformation law identifies the tensor's type.

The consequence is decisive. Since $\underline{K}$ does not depend on time, $\dot{\underline{K}} = \underline{\hat{Q}}$, and differentiating a referential quantity produces no extra term at all: whatever survives a change of observer before differentiation survives it afterwards. Since $\underline{Q}$ does depend on time, differentiating a spatial quantity produces $\dot{\underline{Q}}$, and that is where every anomaly of this chapter comes from. Compare (7.30)₁, in which $\underline{C}^+ = \underline{K} \underline{C} \underline{K}^t$ is as clean after differentiation as before, with (7.33), in which $\underline{L}$ picks up $\underline{\Omega}$. Nothing else has changed between the two; only which tensor does the conjugating.

This is also why constitutive theory is written in terms of referential strain measures whenever possible. A referential measure cannot be polluted by the observer's motion, because the observer's motion has been evaluated at t_0 and frozen.

Fixing t , and hence t^+ , and combining (2.34) with (7.3) and $d\underline{X}^+ = \underline{K} d\underline{X}$, which follows from (7.14) because $\underline{c}^+$ is constant, we find that

the deformation gradients in the two frames are related by

$$\underline{\underline{F}}^+ d\underline{\underline{X}}^+ = d\underline{\underline{x}}^+ = \underline{\underline{Q}}(t) d\underline{\underline{x}} = \underline{\underline{Q}}(t) \underline{\underline{F}} d\underline{\underline{X}} = \underline{\underline{Q}}(t) \underline{\underline{F}} \underline{\underline{K}}^t d\underline{\underline{X}}^+, \quad (7.16)$$

i.e., since $d\underline{\underline{X}}^+$ is arbitrary, that

$$\underline{\underline{F}}^+ = \underline{\underline{Q}}(t) \underline{\underline{F}} \underline{\underline{K}}^t. \quad (7.17)$$

Every step of (7.16) deserves its justification. The first equality is the definition (2.34) of the deformation gradient in the frame of O^+ . The second is (7.4) applied to a pair of neighbouring places, which is legitimate at fixed t because $\underline{\underline{Q}}$ is then a constant tensor and a differential is taken at fixed t . The third is (2.34) in the frame of O . The fourth substitutes $d\underline{\underline{X}} = \underline{\underline{K}}^t d\underline{\underline{X}}^+$, obtained from $d\underline{\underline{X}}^+ = \underline{\underline{K}} d\underline{\underline{X}}$ by multiplying on the left by $\underline{\underline{K}}^t$ and using $\underline{\underline{K}}^t \underline{\underline{K}} = \underline{\underline{I}}$.

This is a linear map from $\hat{\mathbb{E}}^{3+}$ to $\mathbb{E}^{3+}$, as the alphabet test confirms: $\underline{\underline{K}}^t$ carries $\hat{\underline{\underline{E}}}_A \otimes \hat{\underline{\underline{E}}}_B^+$, $\underline{\underline{F}}$ carries $\underline{\underline{e}}_i \otimes \hat{\underline{\underline{E}}}_A$, $\underline{\underline{Q}}$ carries $\underline{\underline{e}}_j^+ \otimes \underline{\underline{e}}_i$, and the contractions leave $\underline{\underline{e}}_j^+ \otimes \hat{\underline{\underline{E}}}_B^+$. Because the shifter $\underline{\underline{1}}^+$ is also such a map, it is natural to assume that

$$\underline{\underline{1}}^+ = \underline{\underline{Q}}(t) \underline{\underline{1}} \underline{\underline{K}}^t. \quad (7.18)$$

This recovers (7.15)₁ at time t_0 : substituting (7.18) into $(\underline{\underline{1}}^+)^t \underline{\underline{Q}}(t_0) \underline{\underline{1}}$ and using $\underline{\underline{Q}}^t(t_0) \underline{\underline{Q}}(t_0) = \underline{\underline{I}}$ and $\underline{\underline{1}}^t \underline{\underline{1}} = \underline{\underline{I}}$,

$$(\underline{\underline{1}}^+)^t \underline{\underline{Q}}(t_0) \underline{\underline{1}} = (\underline{\underline{Q}}(t_0) \underline{\underline{1}} \underline{\underline{K}}^t)^t \underline{\underline{Q}}(t_0) \underline{\underline{1}} = \underline{\underline{K}} \underline{\underline{1}}^t \underline{\underline{Q}}^t(t_0) \underline{\underline{Q}}(t_0) \underline{\underline{1}} = \underline{\underline{K}} \underline{\underline{1}}^t \underline{\underline{1}} = \underline{\underline{K}} \underline{\underline{I}} = \underline{\underline{K}}.$$

It also ensures that $\mathbb{E}^{3+}$ and O^+ evolve together as time passes, and it yields $\underline{\underline{F}}^+(\underline{\underline{X}}^+, t_0^+) = \underline{\underline{1}}^+$ in accordance with (2.79) and the chosen reference configurations: by (7.17) and (2.79), $\underline{\underline{F}}^+(\underline{\underline{X}}^+, t_0^+) = \underline{\underline{Q}}(t_0) \underline{\underline{F}}(\underline{\underline{X}}, t_0) \underline{\underline{K}}^t = \underline{\underline{Q}}(t_0) \underline{\underline{1}} \underline{\underline{K}}^t = \underline{\underline{1}}^+$.

REMARK 7.16 (why (7.18) is assumed rather than derived). The shifter is not determined by anything said so far. Its construction in Definition 2.23 required a choice of fixed orthonormal vectors $\underline{\underline{E}}_A$ in the current translation space, and O^+ must likewise choose fixed orthonormal $\underline{\underline{E}}_A^+ \in \mathbb{E}^{3+}$. Nothing in (7.1) to (7.17) says which. Equation (7.18) is

that choice, and the two properties just verified are exactly the reasons for making it: it agrees with (7.15) at the instant the reference configurations were fixed, and it keeps $\underline{1}^+$ playing for O^+ the role $\underline{1}$ plays for O at every later instant as well.

The alternative would be to let O^+ pick his $\underline{E}_A^+$ once and leave them alone in his own frame. That is a legitimate choice too, but then $\underline{1}^+$ would not satisfy (7.18), the relation $\underline{H} = \underline{F} - \underline{1}$ would fail to transform correctly, and the two observers would disagree about the displacement gradient. The discussion after (7.26) is that failure worked out in detail for the sharper case in which the shifter is dropped altogether.

An *inertial frame* is a frame that neither accelerates nor spins. In this book we always regard the frame $\mathbb{E}^3$ of O as being inertial. Accordingly the shifter $\underline{1}$ is fixed, independent of time. It then follows from (7.18), with $\underline{\dot{Q}} = \underline{Q}$ under a Galilean transformation, that $(\underline{1}^+)' = \underline{\dot{Q}}\underline{1}\underline{K}^t = \underline{Q}$: the shifter is fixed in any frame related to O by a Galilean transformation, i.e. in all inertial frames. In a frame that does spin, $\underline{1}^+$ is not fixed, and Problem 2 computes exactly how fast it turns. What an inertial frame is, what the two ways of failing to be one are, and what it means for a shifter not to be fixed, are taken apart in the remark that follows.

REMARK 7.17 (what *inertial* means, and which half of it the shifter feels). The paragraph just ended packs a definition, a standing assumption and a computation into four sentences. Each repays unpacking, the more so because the standing assumption is what allowed Chapter 6 to be written without mentioning observers at all, and because the last sentence is the only place in the chapter where an object is allowed to depend on time for a reason having nothing to do with the body.

Exactly two things can move. A change of observer is (7.3), $\underline{x}^+ = \underline{Q}(t)\underline{x} + \underline{c}(t)$, and it has exactly two ingredients. Each may depend on t , and each carries one of the two words in the definition.

The translation $\underline{c}$ is not an abstract offset. Setting $\underline{x} = \underline{0}$ in (7.3) gives $\underline{x}^+ = \underline{c}(t)$: the vector $\underline{c}(t)$ is the place at which O^+ sees the origin of O 's frame. So $\underline{\dot{c}}$ is the velocity of that origin and $\underline{\ddot{c}}$ its acceleration,

both as recorded by O^+ , and $\ddot{\mathfrak{c}} \neq \mathfrak{0}$ is what *accelerates* means. The reading is worth keeping in mind at (7.63), where it explains a fact that surprises everyone who meets it: in Problem 3 the observer stands on a spinning planet, his station is certainly being accelerated, and yet $\ddot{\mathfrak{c}} = \mathfrak{0}$ there. It is O 's origin, the centre of the planet, that $\mathfrak{c}$ tracks, and the observer sees it sitting still a distance R straight down. The acceleration of his own station is not lost; it reappears in the centrifugal term.

The rotation $\tilde{Q}(t)$ is the identification of directions. By (7.4) a vector $\underline{u}$ of $\mathbb{E}^3$ is seen by O^+ as $\tilde{Q}(t)\underline{u}$, so $\tilde{Q}(t_2)\tilde{Q}^t(t_1)$ is the rotation carrying the direction O^+ assigned to $\underline{u}$ at t_1 onto the one he assigns at t_2 . If $\tilde{Q}$ is constant, that rotation is the identity for every pair of instants and the two observers agree permanently about which way is which. If it is not, the instantaneous rate is $\tilde{\Omega} = \dot{\tilde{Q}}\tilde{Q}^t$, with axial vector $\underline{\omega}$ by Proposition 1.44, and Remark 7.10 has already identified it: $\underline{\omega}$ is the angular velocity of O 's frame as recorded by O^+ , the negative of the rate $\underline{\omega}_p$ at which O^+ 's own platform turns. That is what *spins* means.

There is no third possibility. A frame is a Euclidean space and has no shape to distort; (7.2) forbids it a change of scale; and (7.3), which was derived and not assumed, exhausts what is left. Position and orientation, two ingredients, two defects. Figure E is the whole of the classification.

Why acceleration and not motion. The definition says *accelerates*, not *moves*, and the asymmetry is not a convenience. A frame drifting at constant velocity is inertial, because by Remark 7.13 the two observers then disagree about velocity by the fixed vector $\underline{V}$ and about nothing else, and no measurement returns $\underline{V}$ without first nominating a frame in which to make it. A frame whose drift is speeding up is a different matter, and the passenger thrown forward when the bus brakes is the measurement.

The definition is the vanishing of the inertial force. The two conditions are not an extra postulate laid on top of the chapter. They are (7.63) read as an identity. Suppose $\ddot{\mathfrak{z}}^+ = \mathfrak{0}$ for every particle, that is, for every $\underline{x}^+$ and every $\underline{v}^+$. The only term of (7.63) containing $\underline{v}^+$ is $2\tilde{\Omega}\underline{v}^+$, so subtracting the statement for two particles at the same place

with different velocities leaves $\underline{\Omega}(v_1^+ - v_2^+) = \underline{0}$ for arbitrary velocities, whence $\underline{\Omega} = \underline{Q}$. If that holds at every instant then $\dot{\underline{\Omega}} = \underline{Q}$ and $\underline{\Omega}^2 = \underline{Q}$ as well, and what remains of (7.63) is $\underline{\ddot{c}} = \underline{0}$. Conversely all four terms vanish when $\underline{\Omega} = \underline{Q}$ and $\underline{\ddot{c}} = \underline{0}$; and $\underline{\Omega} = \underline{Q}$ returns $\dot{\underline{Q}} = \underline{\Omega}\underline{Q} = \underline{Q}$. Hence

$$\begin{aligned} i^+ = \underline{0} \text{ for every particle} &\iff \dot{\underline{Q}} = \underline{Q} \text{ and } \underline{\ddot{c}} = \underline{0} \\ &\iff \underline{x}^+ = \underline{Q}\underline{x} + \underline{V}t + \underline{c}_0, \text{ which is (7.10).} \end{aligned}$$

The first line says that a body free of tractions and of body force satisfies $\underline{a}^+ = \underline{0}$, i.e. that free particles move uniformly; the second is the definition of an inertial frame; the third is a Galilean transformation. The three are one statement wearing three costumes, and (7.64) is the fourth.

Inertial relative to what. Neither $\underline{Q}$ nor $\underline{c}$ exists until two observers have been named, so “neither accelerates nor spins” is, as it stands, a statement about a *pair* of frames. The theory does not pretend otherwise. What is postulated is that one inertial frame exists, which is the content of Euler’s postulates (6.34) and (6.35); the sentence after (7.64) then manufactures infinitely many others, one for each Galilean transformation; and the standing assumption that $\mathbb{E}^3$ is inertial merely names one member of that class. Every statement made afterwards is a statement about the class, not about the member.

The postulate is nevertheless more than a definition, because a single frame can be tested from inside it. Isolate a particle from all contact and all body force and watch it: by (7.62) it moves uniformly if and only if $i^+ = \underline{0}$. Two instruments split the test in two. An accelerometer reports $\underline{\ddot{c}}$, subject to the caveat of Remark 7.30 that no local measurement separates a uniform $\underline{\ddot{c}}$ from a uniform $\underline{b}^+$. A gyroscope, or a Foucault pendulum, reports $\underline{\Omega}$, and that measurement carries no such caveat, because the caveat rests on both quantities being uniform and the two terms $\underline{\Omega}$ contributes to (7.63) are not: one grows with the particle’s distance from the axis and the other with its velocity. No body force does that. Uniform motion is undetectable from inside a closed room; rotation is not.

Which half of it the shifter feels. Return now to $\underline{1}^+ = Q(t)\underline{1}K^t$. Of the three factors $\underline{1}$ is fixed, being built in Definition 2.23 from vectors chosen once, and K is fixed by (7.15), every factor in it having been evaluated at the single instant t_0 . Only Q can move, so

$$(\underline{1}^+)^\cdot = \dot{Q}\underline{1}K^t = (\dot{Q}Q^t)(Q\underline{1}K^t) = \Omega\underline{1}^+,$$

which is Problem 2. The formula deserves reading twice, once for what is in it and once for what is not.

What is not in it is $\ddot{c}$. A frame that accelerates in a straight line, however violently, has a fixed shifter. The reason is that the shifter is a tensor, tensors act on vectors, and vectors are differences of positions in which the translation cancels exactly as it does in (7.4). The whole translational half of non-inertiality is invisible to it.

What is in it is Ω , alone and undiluted. The shifter turns if and only if the frame spins, and then at precisely the frame's own rate. So of the two ways a frame can fail to be inertial the shifter feels one and ignores the other, and knowing which is which saves a good deal of worrying: whatever else an observer who merely accelerates disturbs, it is not the shifter.

What a moving shifter actually looks like. The statement can be made completely concrete. Let O^+ label his reference space by $\hat{E}_A^+ = K\hat{E}_A$, which is legitimate because K is orthogonal and therefore carries an orthonormal basis of $\hat{\mathbb{E}}^3$ onto one of $\hat{\mathbb{E}}^{3+}$, and which is the choice that gives $\underline{X}$ and $\underline{X}^+$ the same components in (7.14). Then $K = K\hat{I} = (K\hat{E}_A) \otimes \hat{E}_A = \hat{E}_A^+ \otimes \hat{E}_A$, so $K^t = \hat{E}_A \otimes \hat{E}_A^+$, and (7.18) with $\underline{1} = \underline{E}_B \otimes \underline{E}_B$ and Lemma 2.15 gives

$$\begin{aligned} \underline{1}^+ &= Q(\underline{E}_B \otimes \underline{E}_B)(\hat{E}_A \otimes \hat{E}_A^+) = Q(\underline{E}_A \otimes \hat{E}_A^+) = \underline{E}_A^+ \otimes \hat{E}_A^+, \\ \underline{E}_A^+(t) &= Q(t)\underline{E}_A, \end{aligned}$$

and therefore $\dot{\underline{E}}_A^+ = \dot{Q}\underline{E}_A = \Omega\underline{E}_A^+$.

Read as an instruction this says something quite definite. The three directions O^+ must use to build his shifter are the images of O 's. They are *not* the directions he calls north, east and up, which are fixed in

his own frame and which he has every reason to regard as fixed; they are a triad that keeps pointing where O 's triad points. For an observer on a turntable those are the directions of the fixed stars, and his shifter triad is a gyroscope. He is quite entitled to say that it turns, because in his frame it does, backwards at the rate of his own platform. For an observer standing on the Earth it comes round once per sidereal day, at $\omega \simeq 7.29 \times 10^{-5} \text{ s}^{-1}$: slow, but not zero. Figure F is this paragraph in two pictures, and Problem 3 is this observer.

What is given up, and what is not. Definition 2.23 asked for *fixed* orthonormal $\underline{E}_A$, and $\underline{E}_A^+(t) = \underline{Q}(t)\underline{E}_A$ is not fixed, so (7.18) does relax that definition for every observer but the inertial one. It is worth being precise about how much is relaxed, because the answer is: the word *fixed*, and nothing else. At each instant $\{\underline{E}_A^+(t)\}$ is still orthonormal, $\underline{Q}$ being orthogonal, so $\underline{1}^+$ is still a uniform orthogonal two-point tensor and every identity of Proposition 2.24 holds of it at every instant; in particular $(\underline{1}^+)^t \underline{1}^+ = \underline{\hat{I}}^+$ and $\underline{1}^+(\underline{1}^+)^t = \underline{I}^+$, which were used at (7.14) and again at (7.21), remain available. Uniformity in $\underline{X}$, which is what makes $\underline{1}^+ \underline{X}^+$ differentiate to $\underline{1}^+$ and hence makes (7.25) work, is untouched. Constancy in t is the single property surrendered, and it is surrendered only where the frame spins.

Why the surrender is the right way round. The alternative offered in the second paragraph of Remark 7.16 can now be priced. Suppose O^+ nails his $\underline{E}_A^+$ to the walls of his laboratory. His shifter is then constant, $(\underline{1}^+)^{\cdot} = \underline{Q}$, but $\underline{F}^+$ is not: for a body sitting motionless in O 's frame, $\underline{\dot{F}}^+ = \underline{\dot{Q}}\underline{F}\underline{K}^t = \underline{\Omega}\underline{F}^+$. The difference $\underline{H}^+ = \underline{F}^+ - \underline{1}^+$ would then change in time while absolutely nothing happened to the body, and O^+ would read his own rotation as a growing displacement gradient. With (7.18) instead, both terms obey the same evolution law and $\underline{\dot{H}}^+ = \underline{\Omega}\underline{H}^+$: the displacement gradient rides round with the observer and reports the body alone. Problem 2 works this out; it is the dynamic form of the objection raised after (7.26). The shifter of a spinning observer must be gyroscopic, and if it is not, his displacement gradient measures him.

Nothing physical is being smuggled in. It is worth closing on why

none of this should be troubling. The shifter is not a field on the body and carries no mechanical information whatever; by Remark 2.25 it exists only so that $\chi_\kappa(\underline{X}, t) - \underline{1}\underline{X}$ has both of its terms in the same space. Requiring it to turn therefore costs nothing, because there was nothing in it to conserve. What the requirement buys is that $\underline{H}$ means the same thing to every observer, which is precisely the property Remark 7.19 shows cannot be arranged by any choice of bases.

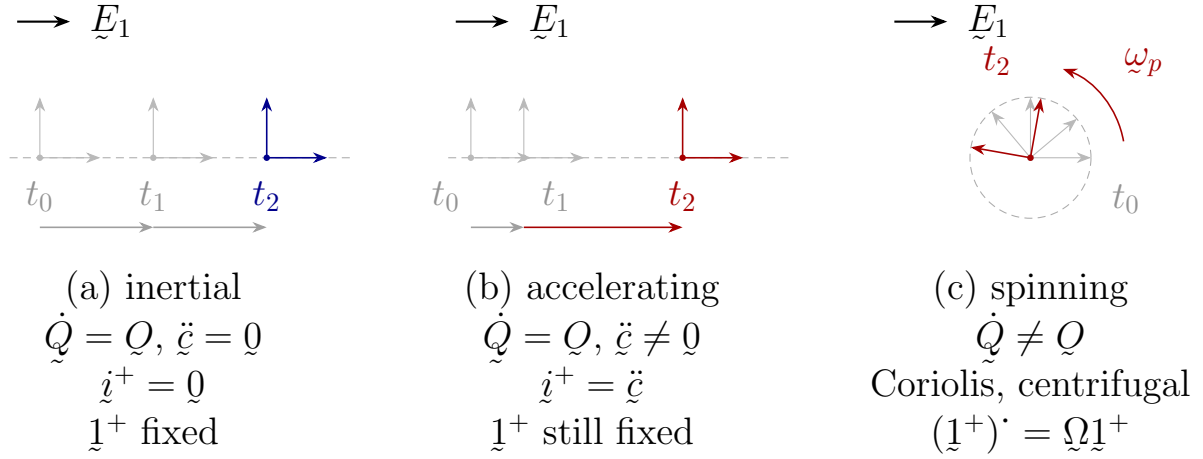
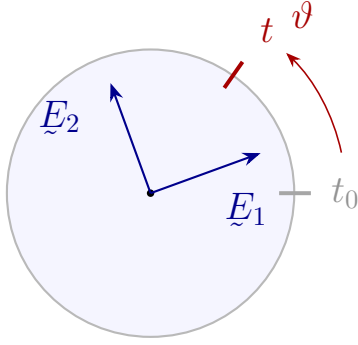
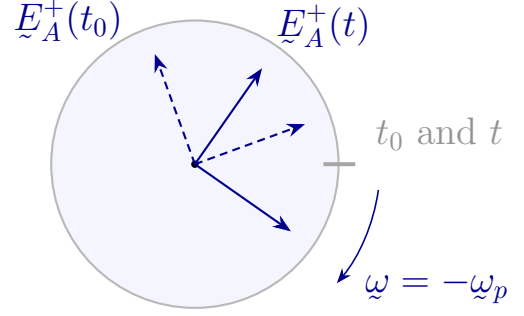


Figure E: The two ways a frame can fail to be inertial, and which of them the shifter feels. Each panel is drawn in the frame of $\underline{O}$ and shows the frame of $\underline{O}^+$ — a pair of axes, that is, the directions $\underline{O}^+$ calls fixed — at three successive instants, beside one direction $\underline{E}_1$ that really is fixed in $\mathbb{E}^3$. In (a) the frame drifts at constant velocity, so that the steps taken in equal times are equal: $\underline{Q}$ is constant, $\underline{c}$ is linear in t , the transformation is the Galilean one (7.10), and $\underline{O}^+$ is inertial. In (b) the steps grow, so $\ddot{\underline{c}} \neq \underline{0}$, while the axes nevertheless stay parallel to $\underline{E}_1$ for ever. In (c) the origin does not move at all and the axes turn, $\dot{\underline{Q}} \neq \underline{Q}$, and the velocity- and position-dependent terms of (7.63) appear. The last line of each panel is the point of the figure: by (7.18) the shifter is carried by $\underline{Q}$ alone, so it is blind to (b) and sensitive only to (c). The shifter records the orientation of the frame and nothing else.

Because both observers have selected occupiable reference configurations, we have that $0 < J^+ = \det \underline{F}^+ = J \det \underline{Q} \det \underline{K}$, in which $J = \det \underline{F} > 0$ by (2.50). The determinant of a two-point tensor is com-



(a) as O records it:
the platform turns, $\underline{E}_A$ does not



(b) as O^+ records it:
the platform is at rest, $\underline{E}_A^+(t)$ turns

Figure F: What it means for the shifter not to be fixed. An observer O^+ stands on a turntable spinning at ω_p , and the mark on the rim records how far it has gone round, through the angle $\vartheta = |\omega_p|(t - t_0)$. The two panels are the same physical situation, and the blue triad, dashed at t_0 and solid at t , is the same pair of directions in both. In (a) the vectors $\underline{E}_A$ of Definition 2.23 do not move; the platform does. In (b) the platform is at rest, by definition, and it is the triad $\underline{E}_A^+(t) = Q(t)\underline{E}_A$ carrying the shifter $\underline{1}^+ = \underline{E}_A^+ \otimes \hat{\underline{E}}_A^+$ that turns, backwards at the observer's own rate: $\dot{\underline{E}}_A^+ = \underline{\Omega}\underline{E}_A^+$, which is $(\underline{1}^+)^{\cdot} = \underline{\Omega}\underline{1}^+$ of Problem 2. The practical content is in panel (b). Observer O^+ cannot build his shifter out of the walls of his own laboratory; he must sight three fixed stars, or mount a gyroscope. Had he used the walls, his shifter would be constant, (7.18) would fail, and $\underline{H}^+ = \underline{F}^+ - \underline{1}^+$ would change in time for a body on which nothing whatever was happening.

puted from its component matrix relative to a right-handed orthonormal basis in each of the two spaces, and $\det(\underline{\underline{A}}\underline{\underline{B}}) = \det \underline{\underline{A}} \det \underline{\underline{B}}$ holds for the corresponding matrices by Theorem 1.52; that is all that is being used, together with $\det \underline{\underline{K}}^t = \det \underline{\underline{K}}$. Then, because $\underline{\underline{Q}}$ and $\underline{\underline{K}}$ are orthogonal, each of $\det \underline{\underline{Q}}$ and $\det \underline{\underline{K}}$ is ± 1 by Proposition 1.48, so their product is ± 1 ; and since $J > 0$ and $J^+ > 0$, that product cannot be -1 . Hence $\det \underline{\underline{Q}} \det \underline{\underline{K}} = 1$, and

$$J^+ = J. \quad (7.19)$$

It is then a simple matter to show that the cofactor of $\underline{\underline{F}}$ transforms to

$$(\underline{\underline{F}}^+)^* = \underline{\underline{Q}} \underline{\underline{F}}^* \underline{\underline{K}}^t. \quad (7.20)$$

Here is the computation. By Proposition 1.55, $\underline{\underline{A}}^* = (\det \underline{\underline{A}}) \underline{\underline{A}}^{-t}$, so $(\underline{\underline{F}}^+)^* = J^+ (\underline{\underline{F}}^+)^{-t} = J (\underline{\underline{F}}^+)^{-t}$ by (7.19). Invert (7.17), reversing the order of the factors as in Proposition 1.46 and using $\underline{\underline{Q}}^{-1} = \underline{\underline{Q}}^t$, $(\underline{\underline{K}}^t)^{-1} = \underline{\underline{K}}$:

$$(\underline{\underline{F}}^+)^{-1} = (\underline{\underline{Q}} \underline{\underline{F}} \underline{\underline{K}}^t)^{-1} = (\underline{\underline{K}}^t)^{-1} \underline{\underline{F}}^{-1} \underline{\underline{Q}}^{-1} = \underline{\underline{K}} \underline{\underline{F}}^{-1} \underline{\underline{Q}}^t,$$

and transposing, $(\underline{\underline{F}}^+)^{-t} = (\underline{\underline{K}} \underline{\underline{F}}^{-1} \underline{\underline{Q}}^t)^t = \underline{\underline{Q}} \underline{\underline{F}}^{-t} \underline{\underline{K}}^t$. Therefore

$$(\underline{\underline{F}}^+)^* = J \underline{\underline{Q}} \underline{\underline{F}}^{-t} \underline{\underline{K}}^t = \underline{\underline{Q}} (J \underline{\underline{F}}^{-t}) \underline{\underline{K}}^t = \underline{\underline{Q}} \underline{\underline{F}}^* \underline{\underline{K}}^t,$$

the scalar J passing through $\underline{\underline{Q}}$ because scalars commute with everything. So the cofactor transforms exactly as $\underline{\underline{F}}$ does, which had to be the case: by Remark 1.56 the cofactor carries area elements, and an area element is built from a pair of line elements, each of which transforms with $\underline{\underline{Q}}$ and $\underline{\underline{K}}$.

With reference to (2.77), the displacement of p in the frame of O^+ is

$$\begin{aligned} \underline{\underline{u}}^+(\underline{\underline{X}}^+, t^+) &= \underline{\underline{\chi}}_{\kappa^+}^+(\underline{\underline{X}}^+, t^+) - \underline{\underline{1}}^+ \underline{\underline{X}}^+ \\ &= \underline{\underline{Q}}(t) [\underline{\underline{\chi}}_{\kappa}(\underline{\underline{X}}, t) - \underline{\underline{1}} \underline{\underline{K}}^t (\underline{\underline{K}} \underline{\underline{X}} + \underline{\underline{c}}^+)] + \underline{\underline{c}}(t) \\ &= \underline{\underline{Q}}(t) [\underline{\underline{\chi}}_{\kappa}(\underline{\underline{X}}, t) - \underline{\underline{1}} \underline{\underline{X}}] + \underline{\underline{c}}(t) - \underline{\underline{Q}}(t) \underline{\underline{1}} \underline{\underline{K}}^t \underline{\underline{c}}^+ \\ &= \underline{\underline{Q}}(t) \underline{\underline{u}}(\underline{\underline{X}}, t) + \underline{\underline{d}}(t), \end{aligned} \quad (7.21)$$

where $\underline{\underline{u}}$ is the displacement in the frame of O and $\underline{\underline{d}}(t) = \underline{\underline{c}}(t) - \underline{\underline{Q}}(t) \underline{\underline{1}} \underline{\underline{K}}^t \underline{\underline{c}}^+$ vanishes at time t_0 . Here we have used the fact that $\underline{\underline{K}}^t \underline{\underline{K}} =$

$\hat{I}$, the identity for $\hat{\mathbb{E}}^3$: it is what turns $\underline{1}\underline{K}^t\underline{K}\underline{X}$ into $\underline{1}\underline{X}$ between the second and third lines. The second line substituted (7.12) for the first term and (7.18) together with (7.14) for the second.

That $\underline{d}(t_0) = \underline{0}$ is worth verifying, since it is the statement that the two displacements are both zero at the instant from which they are measured. At $t = t_0$, (7.18) gives $\underline{Q}(t_0)\underline{1}\underline{K}^t = \underline{1}^+\underline{K}\underline{K}^t = \underline{1}^+\hat{I}^+ = \underline{1}^+$, so

$$\underline{d}(t_0) = \underline{c}(t_0) - \underline{1}^+\underline{c}^+ = \underline{c}(t_0) - \underline{1}^+(\underline{1}^+)^t\underline{c}(t_0) = \underline{c}(t_0) - \underline{I}^+\underline{c}(t_0) = \underline{0},$$

using (7.15)₂ and then $\underline{1}^+(\underline{1}^+)^t = \underline{I}^+$.

REMARK 7.18 (the displacement is not objective, but its gradient is). Equation (7.21) is $\underline{u}^+ = \underline{Q}\underline{u} + \underline{d}$, not $\underline{u}^+ = \underline{Q}\underline{u}$, so the displacement fails the test of Definition 7.7. It could not pass it. A displacement is the difference between a position now and a position at t_0 , and the two observers were in different relative positions at those two instants; the mismatch is exactly $\underline{d}(t)$.

The gradient does pass, because $\underline{d}$ is uniform in $\underline{X}$ and a uniform field has zero gradient. This is the same phenomenon as in Remark 7.5: translations disappear on taking differences. It is also the reason strain measures may be built from $\underline{H}$ but not from $\underline{u}$.

Let $\underline{H}$ be the gradient of $\underline{u}$ with respect to $\underline{X}$, i.e. $\underline{H} \, d\underline{X} = d\underline{u}$ at fixed t . It follows from (2.77) by taking the referential gradient of $\underline{u} = \underline{\chi}_\kappa - \underline{1}\underline{X}$, the shifter being uniform so that its gradient is itself, that

$$\underline{H} = \underline{F} - \underline{1}. \quad (7.22)$$

In terms of Cartesian components, cf. (2.78) and (2.70),

$$\underline{H} = H_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A, \quad \text{where} \quad H_{iA} = \hat{u}_{i,A} = F_{iA} - \delta_{iA}. \quad (7.23)$$

This is a two-point tensor, mapping $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$.

Let $\underline{H}^+$ be the gradient of $\underline{u}^+$ with respect to $\underline{X}^+$. Then, from (7.21), differentiating at fixed t so that $\underline{d}(t)$ contributes nothing,

$$\underline{H}^+ d\underline{X}^+ = d\underline{u}^+ = \underline{Q}(t) d\underline{u} = \underline{Q}(t) \underline{H} d\underline{X} = \underline{Q}(t) \underline{H} \underline{K}^t d\underline{X}^+, \quad (7.24)$$

yielding, since $d\tilde{X}^+$ is arbitrary,

$$\tilde{H}^+ = Q(t)\tilde{H}\tilde{K}^t = \tilde{F}^+ - \tilde{1}^+. \quad (7.25)$$

The second equality is (7.22) transformed: subtracting (7.18) from (7.17), $\tilde{F}^+ - \tilde{1}^+ = Q\tilde{F}\tilde{K}^t - Q\tilde{1}\tilde{K}^t = Q(\tilde{F} - \tilde{1})\tilde{K}^t = Q\tilde{H}\tilde{K}^t$, the middle step by the distributivity of tensor multiplication. This, of course, is also a two-point tensor, mapping $\hat{\mathbb{E}}^{3+}$ to $\mathbb{E}^{3+}$.

It is at this stage that the significance of the shifter becomes apparent. In particular, it ensures that every tensor in (7.22) is of the same type, and hence that the relation remains invariant under basis transformations in a given frame. It also ensures that the relationship between the deformation and displacement gradients is the same in all frames: (7.22) holds for O and (7.25)₂, which is the same sentence with every symbol starred, holds for O^+ .

In contrast, in the extant literature the shifter in (7.22) is typically replaced by the referential identity $\hat{I}$. That tensor transforms differently. From (7.14),

$$\hat{I}^+ d\tilde{X}^+ = d\tilde{X}^+ = \tilde{K} d\tilde{X} = \tilde{K}\tilde{I} d\tilde{X} = \tilde{K}\tilde{I}\tilde{K}^t d\tilde{X}^+,$$

whence, $d\tilde{X}^+$ being arbitrary,

$$\hat{I}^+ = \tilde{K}\tilde{I}\tilde{K}^t. \quad (7.26)$$

Now, if (7.22) is replaced by $\tilde{H} = \tilde{F} - \tilde{I}$, then (7.25)₁ yields

$$\begin{aligned} \tilde{H}^+ &= Q(t)\tilde{H}\tilde{K}^t = Q(t)(\tilde{F} - \tilde{I})\tilde{K}^t \\ &= Q(t)\tilde{F}\tilde{K}^t - Q(t)\tilde{I}\tilde{K}^t = \tilde{F}^+ - Q(t)\tilde{I}\tilde{K}^t \neq \tilde{F}^+ - \hat{I}^+, \end{aligned}$$

since $Q(t)\tilde{I}\tilde{K}^t$ and $\tilde{K}\tilde{I}\tilde{K}^t$ differ whenever $Q(t)$ differs from $\tilde{K}$, which it does at every instant other than t_0 and, in general, even there. So the proposed relationship, if deemed valid by O , must be regarded by O^+ as being invalid. This would imply that O enjoys a privileged status, not shared by other observers. Such a state of affairs runs counter to the philosophy underlying modern physics. We will consider the displacement gradient again later, in the context of linear elasticity theory.

REMARK 7.19 (the same objection, stated so that it cannot be evaded). The temptation is to answer that $\hat{\underline{I}}$ and $\underline{1}$ are the same thing when the two reference bases are aligned with the spatial one, so that δ_{iA} really are Kronecker deltas, and that the distinction is therefore pedantry. Remark 2.25 already answered that for one observer: $\underline{F} - \hat{\underline{I}}$ subtracts a map $\hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^3$ from a map $\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ and is meaningless whatever the numerical values of the components. The present objection is stronger and does not depend on type checking at all.

Grant, for the sake of argument, that O has aligned his bases so that the components of $\underline{1}$ and of $\hat{\underline{I}}$ agree, and that he therefore sees no difference between the two formulas. The tensor $\underline{Q}(t)$ still turns. At any instant $t \neq t_0$ the observer O^+ , aligning *his* bases in the same way, computes $\hat{\underline{I}}^+$ by (7.26) and $\underline{1}^+$ by (7.18), and these are not the same tensor, because one is built from the frozen $\underline{K}$ and the other from the moving $\underline{Q}(t)$. So a convention that costs O nothing costs O^+ the correctness of his formula. There is no way to align bases once and for all that repairs this, because $\underline{Q}(t)$ is a function of time and a choice of bases is not.

The general lesson is worth extracting, because it recurs in Chapter 8. An identity that holds in one frame and is written with objects that transform by different rules is not an identity at all; it is a statement about that frame. Testing a formula against a change of observer is the cheapest available check on whether it is a law or a convention.

Continuing, we have the polar decompositions (3.32) of $\underline{F}$ and the polar decompositions

$$\underline{F}^+ = \underline{R}^+ \underline{U}^+ = \underline{V}^+ \underline{R}^+ \quad (7.27)$$

of $\underline{F}^+$ in terms of the symmetric, positive-definite stretch tensors $\underline{U}^+$ and $\underline{V}^+$ and the two-point rotation $\underline{R}^+$, all guaranteed to exist and to be unique by Theorem 3.19 applied in the frame of O^+ . Then, substituting (7.17) and inserting an identity in the one place where it does the required

work,

$$\begin{aligned} R^+ U^+ &= \underline{\underline{Q}} \underline{\underline{R}} \underline{\underline{U}} \underline{\underline{K}}^t = \underline{\underline{Q}} \underline{\underline{R}} \hat{\underline{\underline{I}}} \underline{\underline{U}} \underline{\underline{K}}^t = (\underline{\underline{Q}} \underline{\underline{R}} \underline{\underline{K}}^t) (\underline{\underline{U}} \underline{\underline{K}}^t) \quad \text{and} \\ V^+ R^+ &= \underline{\underline{Q}} \underline{\underline{V}} \underline{\underline{R}} \underline{\underline{K}}^t = \underline{\underline{Q}} \underline{\underline{V}} \underline{\underline{I}} \underline{\underline{R}} \underline{\underline{K}}^t = (\underline{\underline{Q}} \underline{\underline{V}} \underline{\underline{Q}}^t) (\underline{\underline{Q}} \underline{\underline{R}} \underline{\underline{K}}^t), \end{aligned} \quad (7.28)$$

the inserted identities being $\hat{\underline{\underline{I}}} = \underline{\underline{K}}^t \underline{\underline{K}}$ in the first line and $\underline{\underline{I}} = \underline{\underline{Q}}^t \underline{\underline{Q}}$ in the second. The device is the one used throughout Chapter 3: to convert a product into a product of factors of prescribed type, insert an identity written as a product of the tensor that has the right domain and its transpose.

The uniqueness of the polar decompositions then furnishes

$$\underline{\underline{U}}^+ = \underline{\underline{K}} \underline{\underline{U}} \underline{\underline{K}}^t, \quad \underline{\underline{V}}^+ = \underline{\underline{Q}} \underline{\underline{V}} \underline{\underline{Q}}^t \quad \text{and} \quad \underline{\underline{R}}^+ = \underline{\underline{Q}} \underline{\underline{R}} \underline{\underline{K}}^t, \quad (7.29)$$

but only once it has been checked that the two brackets in each line of (7.28) really are of the kind that Theorem 3.19 declares unique. Three checks are needed.

$\underline{\underline{Q}} \underline{\underline{R}} \underline{\underline{K}}^t$ is a two-point rotation $\hat{\mathbb{E}}^{3+} \rightarrow \mathbb{E}^{3+}$. It is a product of orthogonal maps, hence orthogonal, by the computation already performed for $\underline{\underline{K}}$ after (7.15); and its determinant is $(\det \underline{\underline{Q}})(\det \underline{\underline{R}})(\det \underline{\underline{K}}) = (\det \underline{\underline{Q}})(\det \underline{\underline{K}}) = 1$ by (7.19) and $\det \underline{\underline{R}} = 1$.

$\underline{\underline{K}} \underline{\underline{U}} \underline{\underline{K}}^t$ is symmetric and positive definite on $\hat{\mathbb{E}}^{3+}$. Symmetry: $(\underline{\underline{K}} \underline{\underline{U}} \underline{\underline{K}}^t)^t = \underline{\underline{K}} \underline{\underline{U}}^t \underline{\underline{K}}^t = \underline{\underline{K}} \underline{\underline{U}} \underline{\underline{K}}^t$, using $(\underline{\underline{A}} \underline{\underline{B}} \underline{\underline{C}})^t = \underline{\underline{C}}^t \underline{\underline{B}}^t \underline{\underline{A}}^t$ and the symmetry of $\underline{\underline{U}}$. Positive definiteness: for $\underline{\underline{a}} \neq \underline{\underline{0}}$ in $\hat{\mathbb{E}}^{3+}$,

$$\underline{\underline{a}} \cdot \underline{\underline{K}} \underline{\underline{U}} \underline{\underline{K}}^t \underline{\underline{a}} = \underline{\underline{K}}^t \underline{\underline{a}} \cdot \underline{\underline{U}} \underline{\underline{K}}^t \underline{\underline{a}} > 0,$$

by Definition 3.12 applied to $\underline{\underline{U}}$, since $\underline{\underline{K}}^t \underline{\underline{a}} \neq \underline{\underline{0}}$: $\underline{\underline{K}}^t$ is invertible.

$\underline{\underline{Q}} \underline{\underline{V}} \underline{\underline{Q}}^t$ is symmetric and positive definite on $\mathbb{E}^{3+}$. The same two computations with $\underline{\underline{Q}}$ for $\underline{\underline{K}}$ and $\underline{\underline{V}}$ for $\underline{\underline{U}}$.

So each line of (7.28) exhibits $\underline{\underline{F}}^+$ as a rotation times a symmetric positive definite tensor, in the two possible orders, and uniqueness identifies the factors term by term, giving (7.29).

The right- and left-Cauchy–Green tensors in the frame of O^+ are thus

given by

$$\begin{aligned} \underline{\underline{C}}^+ &= (\underline{\underline{U}}^+)^2 = \underline{\underline{K}}\underline{\underline{U}}\underline{\underline{K}}^t\underline{\underline{K}}\underline{\underline{U}}\underline{\underline{K}}^t = \underline{\underline{K}}\underline{\underline{U}}\hat{\underline{\underline{I}}}\underline{\underline{U}}\underline{\underline{K}}^t = \underline{\underline{K}}\underline{\underline{U}}^2\underline{\underline{K}}^t = \underline{\underline{K}}\underline{\underline{C}}\underline{\underline{K}}^t \quad \text{and} \\ \underline{\underline{B}}^+ &= (\underline{\underline{V}}^+)^2 = \underline{\underline{Q}}\underline{\underline{V}}\underline{\underline{Q}}^t\underline{\underline{Q}}\underline{\underline{V}}\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{V}}\underline{\underline{I}}\underline{\underline{V}}\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{V}}^2\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{B}}\underline{\underline{Q}}^t, \end{aligned} \quad (7.30)$$

respectively. These also follow directly from (7.17) with $\underline{\underline{C}}^+ = (\underline{\underline{F}}^+)^t\underline{\underline{F}}^+$ and $\underline{\underline{B}}^+ = \underline{\underline{F}}^+(\underline{\underline{F}}^+)^t$:

$$\begin{aligned} \underline{\underline{C}}^+ &= (\underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t)^t\underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t = \underline{\underline{K}}\underline{\underline{F}}^t\underline{\underline{Q}}^t\underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t = \underline{\underline{K}}\underline{\underline{F}}^t\underline{\underline{I}}\underline{\underline{F}}\underline{\underline{K}}^t = \underline{\underline{K}}\underline{\underline{C}}\underline{\underline{K}}^t, \\ \underline{\underline{B}}^+ &= \underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t(\underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t)^t = \underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t\underline{\underline{K}}\underline{\underline{F}}^t\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{F}}\hat{\underline{\underline{I}}}\underline{\underline{F}}^t\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{B}}\underline{\underline{Q}}^t. \end{aligned}$$

REMARK 7.20 (which measures of deformation see the observer, and why). Equations (7.29) and (7.30) are the complete answer to the question the chapter was written for, at least as far as deformation is concerned, and it is worth reading them as a list.

$\underline{\underline{U}}$ and $\underline{\underline{C}}$ do not see $\underline{\underline{Q}}$ at all. They are referential, and by Remark 7.15 they can only be touched by the fixed $\underline{\underline{K}}$. If both observers use the same reference configuration, so that $\underline{\underline{K}} = \hat{\underline{\underline{I}}}$, they write down literally the same tensors. This is the algebraic form of the statement, made intuitively in Remark 2.29, that $\underline{\underline{C}}$ measures lengths and angles of material fibres: lengths and angles are what (7.2) guarantees the two observers agree about.

$\underline{\underline{V}}$ and $\underline{\underline{B}}$ see only $\underline{\underline{Q}}$, and see it in the objective way of Definition 7.7. They measure the same lengths and angles, but record them in the current space, which the two observers hold differently.

$\underline{\underline{R}}$ and $\underline{\underline{F}}$ see both, one on each side. They are two-point tensors and cannot be otherwise.

The practical consequence, which Chapter 8 will use and Chapter 10 will use again, is that a scalar function of $\underline{\underline{C}}$ is automatically agreed on by all observers, whereas a scalar function of $\underline{\underline{F}}$ is not. That is why constitutive equations reduce to functions of $\underline{\underline{C}}$, and why Chapter 10 will be able to say that a kinematic constraint may not see $\underline{\underline{R}}$: by (7.29)₃ the rotation is exactly the part of $\underline{\underline{F}}$ that a change of observer alters. The reduction is not a modelling preference; it is forced by (7.29)₃.

Equation (7.17) also furnishes, on differentiating with respect to t and remembering that $\underline{\underline{K}}$ is fixed so that $\dot{\underline{\underline{K}}} = \hat{\underline{\underline{Q}}}$,

$$\dot{\underline{\underline{F}}}^+ = \underline{\underline{Q}}\dot{\underline{\underline{F}}}\underline{\underline{K}}^t + \dot{\underline{\underline{Q}}}\underline{\underline{F}}\underline{\underline{K}}^t. \quad (7.31)$$

Using $\dot{\underline{\underline{F}}}^+ = \underline{\underline{L}}^+\underline{\underline{F}}^+$ and $\dot{\underline{\underline{F}}} = \underline{\underline{L}}\underline{\underline{F}}$, both instances of (4.3), where $\underline{\underline{L}}^+$ and $\underline{\underline{L}}$ respectively are the spatial velocity gradients as perceived by O^+ and O , the left side of (7.31) becomes $\underline{\underline{L}}^+\underline{\underline{F}}^+$ and the first term on the right becomes $\underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{F}}\underline{\underline{K}}^t$. Both terms on the right now contain the factor $\underline{\underline{F}}\underline{\underline{K}}^t$, and (7.17) identifies it: $\underline{\underline{F}}\underline{\underline{K}}^t = \underline{\underline{Q}}^t\underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{K}}^t = \underline{\underline{Q}}^t\underline{\underline{F}}^+$. Substituting,

$$\underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{F}}\underline{\underline{K}}^t = \underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{Q}}^t\underline{\underline{F}}^+, \quad \dot{\underline{\underline{Q}}}\underline{\underline{F}}\underline{\underline{K}}^t = \dot{\underline{\underline{Q}}}\underline{\underline{Q}}^t\underline{\underline{F}}^+ = \underline{\underline{\Omega}}\underline{\underline{F}}^+,$$

the second using the definition (7.8)₂ of $\underline{\underline{\Omega}}$. We conclude that

$$\underline{\underline{L}}^+\underline{\underline{F}}^+ = \underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{Q}}^t\underline{\underline{F}}^+ + \dot{\underline{\underline{Q}}}\underline{\underline{Q}}^t\underline{\underline{F}}^+, \quad (7.32)$$

and right-multiplication by $(\underline{\underline{F}}^+)^{-1}$, which exists because $\det \underline{\underline{F}}^+ = J > 0$ by (7.19), gives

$$\underline{\underline{L}}^+ = \underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{Q}}^t + \underline{\underline{\Omega}}, \quad (7.33)$$

where $\underline{\underline{\Omega}}$ is the skew tensor defined in (7.8)₂.

The stretching and spin tensors thus transform as

$$\underline{\underline{D}}^+ = \underline{\underline{Q}}\underline{\underline{D}}\underline{\underline{Q}}^t \quad \text{and} \quad \underline{\underline{W}}^+ = \underline{\underline{Q}}\underline{\underline{W}}\underline{\underline{Q}}^t + \underline{\underline{\Omega}}. \quad (7.34)$$

The passage from (7.33) to (7.34) is the splitting (4.7) applied in the frame of O^+ , and it goes through in three steps. First, conjugation by $\underline{\underline{Q}}$ commutes with transposition:

$$(\underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{Q}}^t)^t = \underline{\underline{Q}}\underline{\underline{L}}^t\underline{\underline{Q}}^t,$$

so that

$$\text{sym}(\underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{Q}}^t) = \frac{1}{2}\underline{\underline{Q}}(\underline{\underline{L}} + \underline{\underline{L}}^t)\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{D}}\underline{\underline{Q}}^t, \quad \text{skw}(\underline{\underline{Q}}\underline{\underline{L}}\underline{\underline{Q}}^t) = \underline{\underline{Q}}\underline{\underline{W}}\underline{\underline{Q}}^t,$$

using the linearity of $\underline{\underline{A}} \mapsto \underline{\underline{Q}}\underline{\underline{A}}\underline{\underline{Q}}^t$ to take the factor $\frac{1}{2}$ and the sum through. Second, $\underline{\underline{\Omega}}$ is skew, so $\text{sym} \underline{\underline{\Omega}} = \underline{\underline{Q}}^+$ and $\text{skw} \underline{\underline{\Omega}} = \underline{\underline{\Omega}}$: the extra

term contributes nothing at all to the symmetric part and all of itself to the skew part. Third, the splitting of a tensor into symmetric and skew parts is unique, Problem 14 of Chapter 1 and Definition 1.59, so the symmetric parts of the two sides of (7.33) may be equated and likewise the skew parts. That is (7.34).

REMARK 7.21 (why $\underline{D}$ survives and $\underline{W}$ does not). This is the central result of the chapter and it should not be left as two lines of algebra.

The mechanism. By Remark 7.10 the contamination $\underline{\Omega}$ is the velocity gradient of a rigid displacement, and (4.29) established that a rigid displacement has zero stretching and pure spin. So the pollution introduced by the observer is, by its very nature, spin and nothing but spin. The splitting (4.7) sorts $\underline{L}$ into a stretching part and a spinning part; the observer can only add to the second bin. That is the whole reason, and it explains why the result could have been predicted without any calculation.

Why it is not an accident of the algebra. $\underline{D}$ measures rates of change of lengths and angles. Equation (4.15) makes this exact: $\underline{m} \cdot \underline{D}\underline{m} = (\ln \mu)^\cdot$, the rate of logarithmic stretch of the material fibre in the direction $\underline{m}$. Both observers agree about μ , because (7.2) says they agree about lengths, so they must agree about $\dot{\mu}/\mu$, and hence about $\underline{D}$ up to the rotation that relates their pictures. $\underline{W}$ measures no length and no angle; by Remark 4.9 it is not even the spin of a material fibre. There is therefore nothing in (7.2) to protect it, and nothing does.

The counterexample, which is the point. Let the body be at rest for O : $\underline{v} = \underline{0}$, hence $\underline{L} = \underline{O}$, $\underline{D} = \underline{O}$, $\underline{W} = \underline{O}$. Let O^+ spin steadily. Then (7.33) gives $\underline{L}^+ = \underline{\Omega}$ and (7.34) gives

$$\underline{D}^+ = \underline{O}^+, \quad \underline{W}^+ = \underline{\Omega} \neq \underline{O}^+.$$

The two observers agree that nothing is being deformed, and they disagree completely about the spin. O^+ records a non-zero $\underline{W}^+$ although no material fibre changes its length or its angle to any other. The spin was manufactured by the act of looking. Figure G is that example drawn.

What it forbids. Suppose someone proposed the constitutive equation $\underline{T} = 2\mu\underline{W}$ for some material. In the example above O would

compute $\underline{T} = \underline{Q}$ and O^+ would compute $\underline{T}^+ = 2\mu\Omega \neq \underline{Q}^+$. But by (7.40) the two are obliged to satisfy $\underline{T}^+ = \underline{Q}\underline{T}\underline{Q}^t = \underline{Q}^+$. The proposed law is therefore not merely inaccurate but self-contradictory: it asserts that a body sitting still on a table develops stress if the person looking at it starts to turn round. The same argument disqualifies $\underline{T} = f(\underline{L})$ for any f that actually depends on the skew part of $\underline{L}$, and it is why the viscous fluid of Chapter 9 is written with $\underline{D}$ and never with $\underline{L}$. That prohibition is worth the whole chapter.

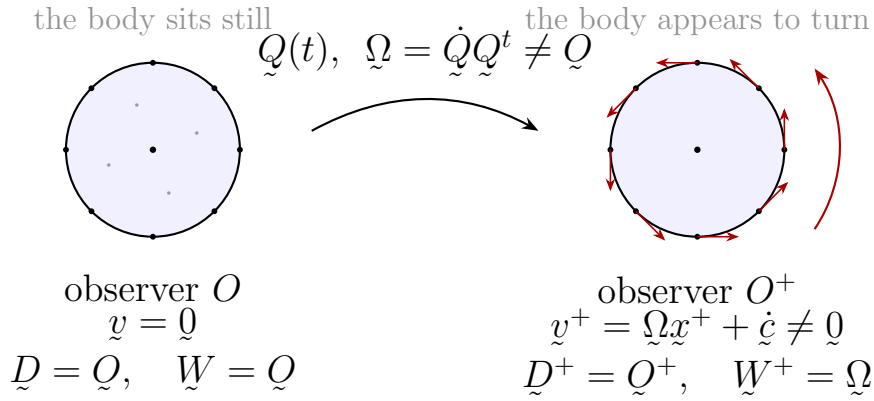


Figure G: One motion, two records: the counterexample of Remark 7.21. The body is at rest for O and the observer O^+ spins, so O^+ sees every material point moving on a circle. The circle of material drawn in each panel is a circle in both, and every marked fibre keeps its length and its angle to every other; that is what (7.2) guarantees, and it is why both observers record $\underline{D} = \underline{Q}$. The velocity fields differ, and so do the spins: O records $\underline{W} = \underline{Q}$ and O^+ records $\underline{W}^+ = \underline{\Omega}$, the entire spin having been manufactured by the observer's own rotation. A constitutive law written in $\underline{W}$ would give these two panels different stresses, which is impossible by (7.40).

To any unit vector $\underline{n} \in \mathbb{E}^3$ we can associate two positions $\underline{x}, \underline{y} \in \mathbb{E}^3$ by $\underline{n} = (\underline{y} - \underline{x})/|\underline{y} - \underline{x}|$ (Figure 7.2). Its counterpart in $\mathbb{E}^{3+}$ is

$$\underline{n}^+ = \frac{\underline{y}^+ - \underline{x}^+}{|\underline{y}^+ - \underline{x}^+|} = \frac{\underline{y}^+ - \underline{x}^+}{|\underline{y} - \underline{x}|} = \frac{\underline{Q}(\underline{y} - \underline{x})}{|\underline{y} - \underline{x}|} = \underline{Q}\underline{n}, \quad (7.35)$$

the second equality by (7.2) and the third by (7.4); the last step takes the scalar $|\underline{y} - \underline{x}|^{-1}$ inside the linear map $\underline{Q}$. Similarly, for a unit vector

$\tilde{N} \in \hat{\mathbb{E}}^3$ we have its counterpart in $\hat{\mathbb{E}}^{3+}$:

$$\tilde{N}^+ = \frac{\tilde{Y}^+ - \tilde{X}^+}{|\tilde{Y}^+ - \tilde{X}^+|} = \frac{\tilde{Y}^+ - \tilde{X}^+}{|\tilde{Y} - \tilde{X}|} = \frac{\tilde{K}(\tilde{Y} - \tilde{X})}{|\tilde{Y} - \tilde{X}|} = \tilde{K}\tilde{N}, \quad (7.36)$$

now using (7.14), from which $\tilde{Y}^+ - \tilde{X}^+ = \tilde{K}(\tilde{Y} - \tilde{X})$ because the constant $\tilde{c}^+$ cancels in the difference, and $|\tilde{K}(\tilde{Y} - \tilde{X})| = |\tilde{Y} - \tilde{X}|$ because $\tilde{K}$ is orthogonal.

REMARK 7.22 (why (7.35) is a theorem and (7.38) will be an assumption). It would have been possible to declare that unit vectors transform by $\tilde{n}^+ = \tilde{Q}\tilde{n}$ and be done with it. The book does not, and the reason is instructive.

A unit vector attached to a configuration is not a primitive object. It is built from two positions, and positions are the primitives about which (7.2) and (7.3) already say everything. So its transformation law is a consequence, and (7.35) derives it in three lines with nothing assumed that was not assumed in Section 7.1.

Contrast (7.38) below. A traction is not built out of positions; it is a primitive of the mechanical theory, introduced in Chapter 6 by a postulate of its own. Nothing in the geometry of Section 7.1 can determine how it transforms, and the statement $\tilde{t}^+ = \tilde{Q}\tilde{t}$ has to be adopted as a further physical hypothesis. Keeping the two kinds of statement apart is what makes it possible to say later exactly which physical assumptions the theory rests on.

7.4 Further transformations

We assume the density, energy, heating supply, temperature and entropy to be absolute scalars, and hence the same for all observers. Thus

$$\rho^+ = \rho, \quad \varepsilon^+ = \varepsilon, \quad r^+ = r, \quad h^+ = h, \quad \theta^+ = \theta \quad \text{and} \quad \eta^+ = \eta. \quad (7.37)$$

The first of these is a natural consequence of the fact that neither the mass of a body, nor its volume, are affected by a change from one observer

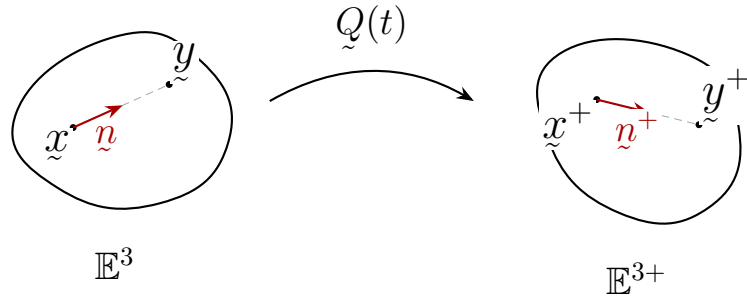


Figure 7.2: A unit vector as perceived by two observers. The vector is not posited but built: $\underline{n}$ is the direction from $\underline{x}$ to $\underline{y}$, normalised by the distance between them. Both the numerator and the denominator are already governed by Section 7.1, the first by (7.4) and the second by (7.2), so the transformation law $\underline{n}^+ = \underline{Q}\underline{n}$ of (7.35) is derived rather than assumed. The same construction in the reference configuration, with (7.14) in place of (7.4), gives $\underline{N}^+ = \underline{K}\underline{N}$, and the change from $\underline{Q}$ to $\underline{K}$ is the whole difference between a spatial and a referential vector.

to another. Mass is assigned to the body itself in Section 2.5, before any observer appears; and volume is unaffected because, by (7.4),

$$\begin{aligned} dv^+ &= |[\underline{d}\underline{x}^+, \underline{d}\underline{y}^+, \underline{d}\underline{z}^+]| \\ &= |[\underline{Q}\underline{d}\underline{x}, \underline{Q}\underline{d}\underline{y}, \underline{Q}\underline{d}\underline{z}]| = |\det \underline{Q}| |[\underline{d}\underline{x}, \underline{d}\underline{y}, \underline{d}\underline{z}]| = dv, \end{aligned}$$

by Theorem 1.52 and $|\det \underline{Q}| = 1$. Equation (7.37)₁ combines with (7.19) to yield the equality of the referential mass densities: since $\rho_\kappa = \rho J$ by the mass conservation of Section 2.5, and both factors are unchanged, $\rho_{\kappa^+}^+ = \rho^+ J^+ = \rho J = \rho_\kappa$.

REMARK 7.23 (why (7.37) is an assumption, and how one would notice if it were wrong). Nothing derived so far bears on any of the six quantities in (7.37). Sections 7.1 to 7.3 concerned only clocks, rulers and the geometry that follows from them; mass, energy and temperature were never mentioned. So (7.37) is new physical input, and it is worth being explicit about that rather than letting it pass as obvious.

It is also worth seeing that it has content. Suppose temperature were not absolute, and that a moving observer read a different value. Then the Clausius–Duhem inequality (7.53)₅, which carries θ in two places,

would hold for one observer and could fail for another, and the second law of thermodynamics would become a statement about who is looking. The assumption (7.37) is precisely what forbids that, and (7.59)₃ is the pay-off.

The one entry that is not simply a postulate is $\rho^+ = \rho$, since it follows from mass and volume separately, and the volume computation above is the proof.

Further, we assume that the Cauchy and Piola tractions $\underline{t}, \underline{p} \in \mathbb{E}^3$ and the body force density $\underline{b} \in \mathbb{E}^3$ transform as spatial vectors, i.e.

$$\underline{t}^+ = \underline{Q}\underline{t}, \quad \underline{p}^+ = \underline{Q}\underline{p} \quad \text{and} \quad \underline{b}^+ = \underline{Q}\underline{b}. \quad (7.38)$$

The Cauchy stress $\underline{T}^+$ in the frame of O^+ is then such that

$$\underline{T}^+ \underline{n}^+ = \underline{t}^+ = \underline{Q}\underline{t} = \underline{Q}\underline{T}\underline{n} = \underline{Q}\underline{T}\underline{Q}^t \underline{n}^+ \quad (7.39)$$

for all unit vectors $\underline{n}^+$; thus

$$\underline{T}^+ = \underline{Q}\underline{T}\underline{Q}^t. \quad (7.40)$$

The chain (7.39) uses Cauchy's relation $\underline{t} = \underline{T}\underline{n}$ of Chapter 6 in each frame, (7.38)₁ in the middle, and finally $\underline{n} = \underline{Q}^t \underline{n}^+$, which is (7.35) solved for $\underline{n}$ by multiplying on the left by $\underline{Q}^t$ and using $\underline{Q}^t \underline{Q} = \underline{I}$.

Two small points make the passage from (7.39) to (7.40) legal. First, every unit vector $\underline{n}^+ \in \mathbb{E}^{3+}$ arises as $\underline{Q}\underline{n}$ for a unit $\underline{n} \in \mathbb{E}^3$, namely $\underline{n} = \underline{Q}^t \underline{n}^+$, which is a unit vector because $\underline{Q}^t$ is orthogonal; so (7.39) really does hold for every unit vector of the starred frame and not merely for those that happen to be images. Second, two tensors that agree on every *unit* vector agree on every vector: given $\underline{s} \neq \underline{0}$, write $\underline{s} = |\underline{s}| \underline{n}$ with $\underline{n} = \underline{s}/|\underline{s}|$ a unit vector, and linearity gives $\underline{A}\underline{s} = |\underline{s}| \underline{A}\underline{n} = |\underline{s}| \underline{B}\underline{n} = \underline{B}\underline{s}$, while $\underline{A}\underline{0} = \underline{0} = \underline{B}\underline{0}$. Hence $\underline{T}^+$ and $\underline{Q}\underline{T}\underline{Q}^t$ are the same tensor.

REMARK 7.24 (what kind of statement (7.38) is, and how much of it is independent). Three statements were written down in (7.38), and it matters that they are not three.

Two are assumptions and the third is a theorem. Equation (7.42) will derive $\underline{p}^+ = \underline{Q}\underline{p}$ from (7.38)₁ and the kinematics of Section 7.3 alone, so (7.38)₂ adds nothing. What is genuinely assumed is that the Cauchy traction and the body force density are objective spatial vectors. Nothing proves either, and Remark 7.22 said why: a traction is a primitive of Chapter 6 rather than a construction out of positions, and the geometry of Section 7.1 has no hold on it. This is the point at which the mechanics of the chapter, as opposed to its geometry, enters.

What the assumption says physically. That the force transmitted across a surface is a property of the two pieces of material on either side of it, and not of the person watching. Cut the body and put a spring balance in the cut: both observers read the same needle, and they disagree only about which direction in their own space the reading points along, by exactly the $\underline{Q}$ that relates their spaces. Stated that way it is nearly a tautology about what the word *force* means, and that is why it usually passes without comment. It is nevertheless an assumption, and it is a strong one: it is what forbids a force whose magnitude depends on the observer's motion.

Where it fails, and why the failure is the whole story of Section 7.5. The one force in this chapter that is *not* objective is the inertial force $\underline{i}^+$ of (7.63), and it could not be, since by construction it is built out of $\underline{Q}$ and $\underline{c}$ themselves. That is the precise content of the word *fictitious*. A centrifugal force has the units of a body force density, enters the equations of motion exactly like one, and is distinguished from a real one by a single property: it fails the test of Definition 7.7. Chapter 6's postulates concerned real body forces, (7.38)₃ is the statement that they are objective, and (7.62) is what happens when the two kinds are added together.

What rests on it. Because (7.38)₁ is an assumption, so is (7.40), and with it the whole of Chapter 8. Had the traction carried an extra term, in the manner of (7.8), Cauchy's relation $\underline{t} = \underline{T}\underline{n}$ would have produced a $\underline{T}^+$ that is not a conjugate of $\underline{T}$; the stress would not have been objective; the stress power would not have been invariant, so (7.59)₂ and (7.59)₃

would have failed as well; and there would have been nothing left on the stress side of a constitutive equation for both observers to agree about. One assumed line, (7.38)₁, carries all of that.

The Piola stress $\underline{\underline{P}}^+$ follows from $\underline{\underline{P}} = \underline{\underline{T}}\underline{\underline{F}}^*$, the definition of Chapter 6, written in each frame and combined with (7.40) and (7.20):

$$\underline{\underline{P}}^+ = \underline{\underline{T}}^+(\underline{\underline{F}}^+)^* = \underline{\underline{Q}}\underline{\underline{T}}\underline{\underline{Q}}^t\underline{\underline{Q}}\underline{\underline{F}}^*\underline{\underline{K}}^t = \underline{\underline{Q}}\underline{\underline{T}}\underline{\underline{I}}\underline{\underline{F}}^*\underline{\underline{K}}^t = \underline{\underline{Q}}\underline{\underline{P}}\underline{\underline{K}}^t. \quad (7.41)$$

This furnishes

$$\underline{\underline{p}}^+ = \underline{\underline{P}}^+\underline{\underline{N}}^+ = \underline{\underline{Q}}\underline{\underline{P}}\underline{\underline{K}}^t\underline{\underline{K}}\underline{\underline{N}} = \underline{\underline{Q}}\underline{\underline{P}}\underline{\underline{\hat{I}}}\underline{\underline{N}} = \underline{\underline{Q}}\underline{\underline{P}}\underline{\underline{N}} = \underline{\underline{Q}}\underline{\underline{p}}, \quad (7.42)$$

in accordance with (7.38)₂, using (7.36) for $\underline{\underline{N}}^+$ and $\underline{\underline{K}}^t\underline{\underline{K}} = \underline{\underline{\hat{I}}}$. So (7.38)₂ was not an independent assumption: it is forced by (7.38)₁ together with the kinematics of Section 7.3. Lastly, the Piola–Kirchhoff stress $\underline{\underline{S}}^+$ follows from $\underline{\underline{P}}^+ = \underline{\underline{F}}^+\underline{\underline{S}}^+$. Solving,

$$\underline{\underline{S}}^+ = (\underline{\underline{F}}^+)^{-1}\underline{\underline{P}}^+ = (\underline{\underline{K}}\underline{\underline{F}}^{-1}\underline{\underline{Q}}^t)(\underline{\underline{Q}}\underline{\underline{P}}\underline{\underline{K}}^t) = \underline{\underline{K}}\underline{\underline{F}}^{-1}\underline{\underline{I}}\underline{\underline{P}}\underline{\underline{K}}^t = \underline{\underline{K}}(\underline{\underline{F}}^{-1}\underline{\underline{P}})\underline{\underline{K}}^t,$$

using the inverse of (7.17) computed after (7.20); and $\underline{\underline{F}}^{-1}\underline{\underline{P}} = \underline{\underline{S}}$ by the same relation $\underline{\underline{P}} = \underline{\underline{F}}\underline{\underline{S}}$ in the frame of O . We obtain

$$\underline{\underline{S}}^+ = \underline{\underline{K}}\underline{\underline{S}}\underline{\underline{K}}^t. \quad (7.43)$$

REMARK 7.25 (three stresses, three transformation laws). Put (7.40), (7.41) and (7.43) side by side and they say what Remark 2.39 of Chapter 2 promised: the transformation law announces the type.

$$\begin{aligned} \underline{\underline{T}}^+ &= \underline{\underline{Q}}\underline{\underline{T}}\underline{\underline{Q}}^t && \text{spatial: } \mathbb{E}^3 \rightarrow \mathbb{E}^3, \\ \underline{\underline{P}}^+ &= \underline{\underline{Q}}\underline{\underline{P}}\underline{\underline{K}}^t && \text{two-point: } \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3, \\ \underline{\underline{S}}^+ &= \underline{\underline{K}}\underline{\underline{S}}\underline{\underline{K}}^t && \text{referential: } \hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^3. \end{aligned}$$

The pattern is exactly that of $\underline{\underline{B}}$, $\underline{\underline{F}}$, $\underline{\underline{C}}$ in (7.30) and (7.17), and for the same reason: each stress measures force per unit area, and the three differ only in which configuration supplies the force and which supplies the area. $\underline{\underline{T}}$ takes both from the current configuration, $\underline{\underline{S}}$ takes both from the reference one, and $\underline{\underline{P}}$ takes one from each.

The practically important entry is the last. Because $\underline{K}$ is fixed, a constitutive equation written for $\underline{S}$ in terms of $\underline{C}$ relates two objects neither of which sees $\underline{Q}$, and observer agreement is then automatic. Written for $\underline{T}$ in terms of $\underline{B}$ it is equally automatic, since both sides carry one $\underline{Q}$ on each side. Written for $\underline{T}$ in terms of $\underline{C}$ it is not, and that mismatch is what Chapter 8 has to repair.

The relationship between the heat flux h through a surface with unit normal $\underline{n}$ and the heat flux vector $\underline{q}$ is $h = -\underline{q} \cdot \underline{n}$, cf. (6.166). The corresponding relationship in the frame of O^+ is $h^+ = -\underline{q}^+ \cdot \underline{n}^+$. We thus have, from (7.35) and (7.37)₄, that

$$\underline{q}^+ \cdot \underline{n}^+ = \underline{q} \cdot \underline{n} = \underline{q} \cdot \underline{Q}^t \underline{n}^+ = \underline{Q} \underline{q} \cdot \underline{n}^+$$

for all unit vectors $\underline{n}^+$, the last step being the definition of the transpose. Hence $(\underline{q}^+ - \underline{Q} \underline{q}) \cdot \underline{n}^+ = 0$ for every unit $\underline{n}^+$, and therefore for every vector of $\mathbb{E}^{3+}$ by the scaling argument used after (7.39); Lemma 5.1 then gives

$$\underline{q}^+ = \underline{Q} \underline{q}, \quad (7.44)$$

and the referential heat flux vectors in the two frames, defined by $\underline{q}_\kappa = (\underline{F}^*)^t \underline{q}$ in Chapter 6, are related by

$$\begin{aligned} \underline{q}_{\kappa^+}^+ &= [(\underline{F}^+)^*]^t \underline{q}^+ = (\underline{Q} \underline{F}^* \underline{K}^t)^t \underline{Q} \underline{q} \\ &= \underline{K} (\underline{F}^*)^t \underline{Q}^t \underline{Q} \underline{q} = \underline{K} (\underline{F}^*)^t \underline{I} \underline{q} = \underline{K} \underline{q}_\kappa. \end{aligned} \quad (7.45)$$

REMARK 7.26 (the whole of it is one commutative square). Everything derived in Sections 7.3 and 7.4 is a statement about the diagram of Figure H, and once that diagram is in front of one, the dictionary below can be read off instead of memorised.

Four spaces, two vertical maps. There are four spaces in play: the reference spaces $\hat{\mathbb{E}}^3$ and $\hat{\mathbb{E}}^{3+}$ and the current spaces $\mathbb{E}^3$ and $\mathbb{E}^{3+}$. Changing observer means going down, by $\underline{K}$ on the reference side and by $\underline{Q}(t)$ on the current side, and the whole content of Remark 7.15 is the difference between those two verticals: the left one was frozen at t_0 by (7.15) and the right one moves.

Every two-point law says that the square commutes. Equation (7.17) is $\tilde{F}^+ = \tilde{Q}\tilde{F}\tilde{K}^t$, and multiplying on the right by $\tilde{K}$ and using $\tilde{K}^t\tilde{K} = \hat{I}$ turns it into

$$\tilde{F}^+\tilde{K} = \tilde{Q}\tilde{F},$$

that is, into the statement that the two routes from $\hat{\mathbb{E}}^3$ to $\mathbb{E}^{3+}$ agree: deform and then change observer, or change observer and then deform. Read that way, (7.20), (7.25), (7.29)₃ and (7.41) are the same sentence for $\tilde{F}^*$, $\tilde{H}$, $\tilde{R}$ and $\tilde{P}$. And (7.18) is the demand that the shifter be a horizontal arrow of the same square. Seen from here it stops looking like a convention: it is what puts $\underline{1}$ into the diagram at all, and Remark 7.16 is the observation that nothing else would.

Every one-space law is a conjugation. A tensor mapping a space to itself is a loop at one corner, and carrying it down the diagram means going down, round the loop, and back up: $\tilde{A}^+ = \tilde{Q}\tilde{A}\tilde{Q}^t$ on the right, $\tilde{A}^+ = \tilde{K}\tilde{A}\tilde{K}^t$ on the left. That is the entire difference between the second and third groups of the dictionary, and the polar decomposition displays it: by (7.29) and Figure H(b), $\tilde{U}$ is a loop on the reference side, $\tilde{V}$ a loop on the current side, and $\tilde{R}$ the horizontal arrow they share. The pairs $(\tilde{C}, \tilde{B})$ and $(\tilde{S}, \tilde{T})$ are loops in the same two places. A vector is a third case: it lives *at* a corner rather than on an arrow, and it is carried down by a single application of the vertical, which is $\tilde{N}^+ = \tilde{K}\tilde{N}$ and $\tilde{q}_{\kappa^+}^+ = \tilde{K}\tilde{q}_{\kappa}$ on the left and $\tilde{n}^+ = \tilde{Q}\tilde{n}$, $\tilde{t}^+ = \tilde{Q}\tilde{t}$, $\tilde{q}^+ = \tilde{Q}\tilde{q}$ on the right.

What the diagram cannot show. One thing only, and it is the subject of the rest of the chapter: the right-hand vertical depends on t . Every law free of time derivatives is a statement about the square as drawn. Every law containing one must differentiate that vertical too, and $\dot{\tilde{Q}}$ is what comes out. That is why $\tilde{v}$, $\tilde{a}$, $\tilde{L}$ and $\tilde{W}$ are absent from the figure and present in the last group of the dictionary: they are statements about how the square changes, not about the square.

Using it as a check. The alphabet test of Remark 4.1 is precisely the instruction to read a formula's corners off this square, and it catches nearly every error available in this chapter. A term $\tilde{K}\tilde{T}\tilde{K}^t$ is wrong

because $\tilde{T}$ is a loop on the right while $\tilde{K}$ runs down the left; $\tilde{Q}\tilde{C}\tilde{Q}^t$ is wrong for the mirror-image reason; the sum $\tilde{F} - \underline{1}$ is legitimate and $\tilde{F} - \hat{I}$ is not, because the first subtracts one horizontal arrow from another and the second subtracts a loop from a horizontal arrow. That last is the objection of Remark 7.19, drawn.

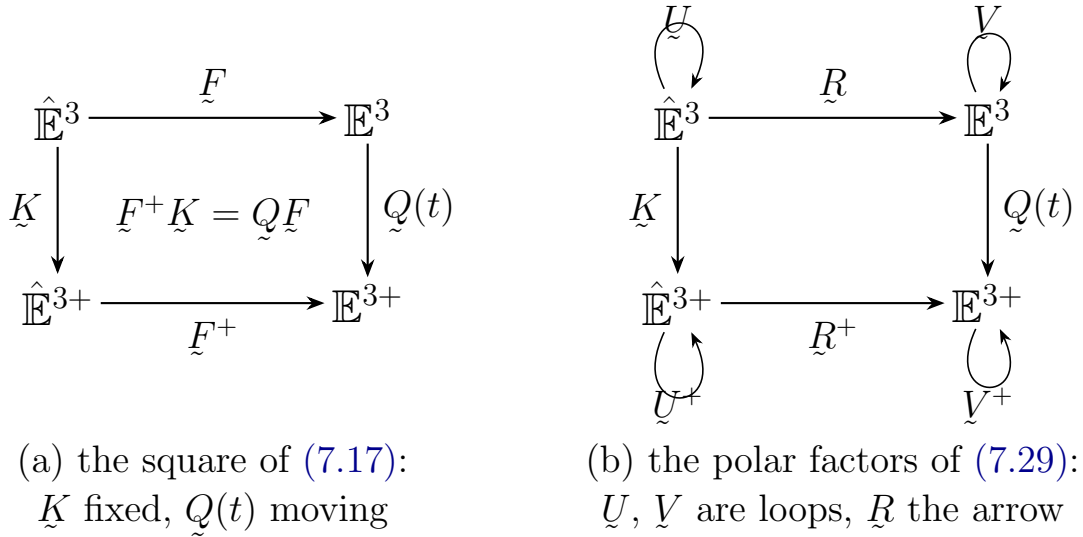


Figure H: The diagram the chapter is about. Going right is deforming; going down is changing observer. (a) Equation (7.17) says that the square commutes, and the same statement with $\tilde{F}$ replaced by $\tilde{F}^*$, $\tilde{R}$, $\tilde{H}$, $\tilde{P}$ or $\underline{1}$ is (7.20), (7.29)₃, (7.25), (7.41) and (7.18) respectively. The two verticals are not alike: $\tilde{K}$ is one fixed tensor, every factor of it evaluated at t_0 by (7.15), while $\tilde{Q}(t)$ turns, and every anomaly of this chapter is $\dot{\tilde{Q}}$. (b) A tensor that maps a space to itself is a loop at a corner, and a loop is carried down the diagram by conjugation: $\tilde{U}^+ = \tilde{K}\tilde{U}\tilde{K}^t$ on the left and $\tilde{V}^+ = \tilde{Q}\tilde{V}\tilde{Q}^t$ on the right, which is (7.29). The pairs $(\tilde{C}, \tilde{B})$ and $(\tilde{S}, \tilde{T})$ are loops in the same two places, and $\tilde{D}$ is a loop on the right that happens also to be objective; a vector such as $\tilde{N}$ or $\tilde{n}$ sits at a corner instead and is carried down by one application of the vertical. The dictionary below is this figure sorted into a list.

The dictionary

It is convenient to have the results of Sections 7.3 and 7.4 in one place, since the remaining sections and the whole of Chapters 8 to 10 consult them constantly. Sorted by what each quantity sees:

$$\text{nothing: } \rho, \varepsilon, r, h, \theta, \eta, J, \rho_\kappa \quad (7.19), (7.37)$$

$$\underline{K} \text{ only: } \underline{N}^+ = \underline{K}\underline{N}, \quad \underline{U}^+ = \underline{K}\underline{U}\underline{K}^t, \quad (7.29)$$

$$\underline{C}^+ = \underline{K}\underline{C}\underline{K}^t, \quad (7.30)$$

$$\underline{S}^+ = \underline{K}\underline{S}\underline{K}^t, \quad \underline{q}_{\kappa^+}^+ = \underline{K}\underline{q}_\kappa, \quad (7.43), (7.45)$$

$$\underline{\hat{I}}^+ = \underline{K}\underline{\hat{I}}\underline{K}^t \quad (7.26)$$

$$\underline{Q} \text{ only: } \underline{n}^+ = \underline{Q}\underline{n}, \quad \underline{t}^+ = \underline{Q}\underline{t}, \quad (7.35), (7.38)$$

$$\underline{b}^+ = \underline{Q}\underline{b}, \quad \underline{q}^+ = \underline{Q}\underline{q}, \quad (7.38), (7.44)$$

$$\underline{V}^+ = \underline{Q}\underline{V}\underline{Q}^t, \quad \underline{B}^+ = \underline{Q}\underline{B}\underline{Q}^t, \quad (7.29), (7.30)$$

$$\underline{T}^+ = \underline{Q}\underline{T}\underline{Q}^t, \quad (7.40)$$

$$\underline{D}^+ = \underline{Q}\underline{D}\underline{Q}^t \quad (7.34)$$

$$\text{both: } \underline{F}^+ = \underline{Q}\underline{F}\underline{K}^t, \quad \underline{R}^+ = \underline{Q}\underline{R}\underline{K}^t, \quad (7.17), (7.29)$$

$$\underline{H}^+ = \underline{Q}\underline{H}\underline{K}^t, \quad (7.25)$$

$$\underline{P}^+ = \underline{Q}\underline{P}\underline{K}^t, \quad (\underline{F}^+)^* = \underline{Q}\underline{F}^*\underline{K}^t, \quad (7.41), (7.20)$$

$$\underline{1}^+ = \underline{Q}\underline{1}\underline{K}^t \quad (7.18)$$

$\dot{\underline{Q}}$ as well:

$$\underline{v}^+ = \underline{Q}\underline{v} + \underline{\Omega}(\underline{x}^+ - \underline{c}) + \underline{\dot{c}}, \quad (7.8)$$

$$\underline{L}^+ = \underline{Q}\underline{L}\underline{Q}^t + \underline{\Omega}, \quad (7.33)$$

$$\underline{W}^+ = \underline{Q}\underline{W}\underline{Q}^t + \underline{\Omega}, \quad (7.34)$$

$$\underline{a}^+ = \underline{Q}\underline{a} + \underline{\dot{i}}^+ \quad (7.9)$$

The first four groups are the objective and the observer-blind quantities, in the sense of Definition 7.7; they are the raw material of constitutive theory. The last group is the list of everything that goes wrong, and every entry in it contains $\dot{\underline{Q}}$, whether written as $\dot{\underline{Q}}$, as $\underline{\Omega}$, or as $\underline{\dot{i}}^+$.

7.5 Transformation of the balance laws

Let $\underline{w}(\underline{x}, t)$ be an arbitrary spatial vector in the frame of O , and suppose $\underline{w}^+(\underline{x}^+, t^+)$ is its counterpart in the frame of O^+ , i.e. that $\underline{w}$ is objective in the sense of Definition 7.7:

$$\underline{w}^+(\underline{x}^+, t^+) = \underline{Q}(t)\underline{w}(\underline{x}, t). \quad (7.46)$$

Then, differentiating at fixed t , so that $\underline{Q}$ is a constant tensor throughout and passes through the differential,

$$\underline{Q}(\text{grad } \underline{w})d\underline{x} = \underline{Q} d\underline{w} = d\underline{w}^+ = (\text{grad}^+ \underline{w}^+)d\underline{x}^+ = (\text{grad}^+ \underline{w}^+)\underline{Q} d\underline{x}, \quad (7.47)$$

using the definition of the gradient of a vector field in each frame, Definition 1.85, and $d\underline{x}^+ = \underline{Q} d\underline{x}$ from (7.4). As $d\underline{x}$ is arbitrary the two ends are equal as tensors, $\underline{Q}(\text{grad } \underline{w}) = (\text{grad}^+ \underline{w}^+)\underline{Q}$, and multiplying on the right by $\underline{Q}^t$ therefore gives

$$\text{grad}^+ \underline{w}^+ = \underline{Q}(\text{grad } \underline{w})\underline{Q}^t. \quad (7.48)$$

The gradient of an objective vector field is an objective tensor field.

This implies that the divergence is the same in both frames:

$$\begin{aligned} \text{div}^+ \underline{w}^+ &= \text{tr}(\text{grad}^+ \underline{w}^+) = \text{tr} [\underline{Q}(\text{grad } \underline{w})\underline{Q}^t] \\ &= \text{tr} [\underline{Q}^t \underline{Q}(\text{grad } \underline{w})] = \text{tr}(\text{grad } \underline{w}) = \text{div } \underline{w}, \end{aligned} \quad (7.49)$$

by Proposition 1.92 at the two ends and the cyclic property $\text{tr}(\underline{A}\underline{B}) = \text{tr}(\underline{B}\underline{A})$ in the middle.

REMARK 7.27 (two traces, on two different spaces). The middle step of (7.49) needs the care that Remark 1.106 taught. The first trace is taken on $\mathbb{E}^{3+}$, since $\underline{Q}(\text{grad } \underline{w})\underline{Q}^t$ maps that space to itself; the second is taken on $\mathbb{E}^3$, since $\underline{Q}^t \underline{Q}(\text{grad } \underline{w})$ maps $\mathbb{E}^3$ to itself. They are different operations on different spaces, and the assertion is that they produce the same number.

They do, and the reason is the component identity behind Remark 1.106: with $\underline{Q} = Q_{\alpha i} \underline{e}_\alpha^+ \otimes \underline{e}_i$ and $\text{grad } \underline{w} = w_{i,j} \underline{e}_i \otimes \underline{e}_j$,

$$\text{tr} [\underline{Q}(\text{grad } \underline{w})\underline{Q}^t] = Q_{\alpha i} w_{i,j} Q_{\alpha j}, \quad \text{tr} [\underline{Q}^t \underline{Q}(\text{grad } \underline{w})] = Q_{\alpha j} Q_{\alpha i} w_{i,j},$$

the same sum of products of the same numbers, written in a different order. Both traces are legitimate, each on its own space, and they are equal. Every use of the cyclic property in this chapter is of that kind and none of them is an abuse.

From (7.40),

$$\underline{\underline{T}}^+ \underline{\underline{w}}^+ = \underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t \underline{\underline{Q}} \underline{\underline{w}} = \underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{I}} \underline{\underline{w}} = \underline{\underline{Q}} (\underline{\underline{T}} \underline{\underline{w}}). \quad (7.50)$$

If $\underline{\underline{w}}$ is independent of $\underline{\underline{x}}$, i.e. $\text{grad } \underline{\underline{w}} = \underline{\underline{O}}$, then $\underline{\underline{w}}^+$ is independent of $\underline{\underline{x}}^+$, since $\underline{\underline{Q}}$ is spatially uniform and (7.48) gives $\text{grad}^+ \underline{\underline{w}}^+ = \underline{\underline{O}}^+$; and the definition (1.148) of the divergence of a tensor, Definition 1.93, yields

$$\begin{aligned} \underline{\underline{Q}}^t (\text{div}^+ \underline{\underline{T}}^+) \cdot \underline{\underline{w}} &= \text{div}^+ \underline{\underline{T}}^+ \cdot \underline{\underline{w}}^+ = \text{div}^+ [(\underline{\underline{T}}^+)^t \underline{\underline{w}}^+] \\ &= \text{div}(\underline{\underline{T}}^t \underline{\underline{w}}) = \text{div } \underline{\underline{T}} \cdot \underline{\underline{w}}, \end{aligned} \quad (7.51)$$

leading, as $\underline{\underline{w}}$ is arbitrary, to the conclusion that

$$\text{div}^+ \underline{\underline{T}}^+ = \underline{\underline{Q}} (\text{div } \underline{\underline{T}}). \quad (7.52)$$

Each equality of (7.51) carries its own justification. The first moves $\underline{\underline{Q}}^t$ across the inner product by the definition of the transpose and then uses (7.46). The second is Definition 1.93 in the frame of O^+ , applicable because $\underline{\underline{w}}^+$ is uniform there. The third is (7.49) applied to the vector field $\underline{\underline{s}} = \underline{\underline{T}}^t \underline{\underline{w}}$, which is objective:

$$(\underline{\underline{T}}^+)^t \underline{\underline{w}}^+ = (\underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t)^t \underline{\underline{Q}} \underline{\underline{w}} = \underline{\underline{Q}} \underline{\underline{T}}^t \underline{\underline{Q}}^t \underline{\underline{Q}} \underline{\underline{w}} = \underline{\underline{Q}} \underline{\underline{T}}^t \underline{\underline{w}} = \underline{\underline{Q}} \underline{\underline{s}} = \underline{\underline{s}}^+.$$

The fourth is Definition 1.93 in the frame of O . The passage from (7.51) to (7.52) is Lemma 5.1: the vector $\underline{\underline{Q}}^t \text{div}^+ \underline{\underline{T}}^+ - \text{div } \underline{\underline{T}}$ has zero inner product with every uniform $\underline{\underline{w}}$, hence vanishes, and multiplying by $\underline{\underline{Q}}$ gives (7.52).

Suppose now that O is an inertial observer. The laws holding in the frame of this observer are the conservation of mass, the balance of energy, the balances of linear and rotational momenta, and the Clausius–Duhem

inequality:

$$\begin{aligned}
\dot{\rho} + \rho \operatorname{div} \underline{v} &= 0, \\
\rho \dot{\varepsilon} + \operatorname{div} \underline{q} - \underline{T} \cdot \underline{L} &= \rho r, \\
\rho \underline{a} - \operatorname{div} \underline{T} &= \rho \underline{b}, \quad \underline{T}^t = \underline{T}, \\
\text{and} \quad -\rho(\dot{\psi} + \eta \dot{\theta}) + \underline{T} \cdot \underline{L} - \theta^{-1} \underline{q} \cdot \operatorname{grad} \theta &\geq 0,
\end{aligned} \tag{7.53}$$

where $\underline{a}$ is the material acceleration. We seek the corresponding laws pertaining to an arbitrary observer O^+ . To this end we use (7.44), (7.46) and (7.49) to infer that

$$\operatorname{div}^+ \underline{q}^+ = \operatorname{div} \underline{q}, \tag{7.54}$$

the heat flux vector being objective by (7.44) and hence an admissible $\underline{w}$ in (7.49).

Also, from $\operatorname{grad}^+ \theta^+ \cdot d\underline{x}^+ = d\theta^+ = d\theta = \operatorname{grad} \theta \cdot d\underline{x} = \operatorname{grad} \theta \cdot \underline{Q}^t d\underline{x}^+ = \underline{Q} \operatorname{grad} \theta \cdot d\underline{x}^+$, valid for every $d\underline{x}^+$ because θ is an absolute scalar by (7.37)₅ and hence has the same increment for both observers, Lemma 5.1 gives $\operatorname{grad}^+ \theta^+ = \underline{Q}(\operatorname{grad} \theta)$, and therefore

$$\begin{aligned}
\operatorname{grad}^+ \theta^+ &= \underline{Q}(\operatorname{grad} \theta), \quad \underline{q}^+ \cdot \operatorname{grad}^+ \theta^+ \\
&= \underline{Q} \underline{q} \cdot \underline{Q} \operatorname{grad} \theta = \underline{q} \cdot \underline{Q}^t \underline{Q} \operatorname{grad} \theta = \underline{q} \cdot \operatorname{grad} \theta.
\end{aligned} \tag{7.55}$$

The transformation formula (7.8) for the material velocity is not of the form (7.46). Nevertheless

$$\operatorname{div}^+ \underline{v}^+ = \operatorname{div} \underline{v}. \tag{7.56}$$

This follows from (7.33), i.e.

$$\operatorname{div}^+ \underline{v}^+ = \operatorname{tr} \underline{L}^+ = \operatorname{tr} (\underline{Q} \underline{L} \underline{Q}^t + \underline{\Omega}) = \operatorname{tr} (\underline{Q} \underline{L} \underline{Q}^t) = \operatorname{tr} (\underline{Q}^t \underline{Q} \underline{L}) = \operatorname{tr} \underline{L} = \operatorname{div} \underline{v}, \tag{7.57}$$

the third equality because $\operatorname{tr} \underline{\Omega} = 0$: a skew tensor has $\Omega_{\alpha\alpha} = -\Omega_{\alpha\alpha}$ for each α , with no sum, so every diagonal component vanishes.

Further, the symmetry of $\underline{T}$, and hence that of $\underline{Q} \underline{T} \underline{Q}^t$, yields

$$\begin{aligned}
\underline{T}^+ \cdot \underline{L}^+ &= \underline{Q} \underline{T} \underline{Q}^t \cdot (\underline{Q} \underline{L} \underline{Q}^t + \underline{\Omega}) = \underline{Q} \underline{T} \underline{Q}^t \cdot \underline{Q} \underline{L} \underline{Q}^t \\
&= \operatorname{tr} [\underline{Q} \underline{T} \underline{Q}^t (\underline{Q} \underline{L} \underline{Q}^t)] = \operatorname{tr} (\underline{Q} \underline{T} \underline{L} \underline{Q}^t) \\
&= \operatorname{tr} (\underline{Q}^t \underline{Q} \underline{T} \underline{L}) = \operatorname{tr} (\underline{T} \underline{L}) = \underline{T} \cdot \underline{L}.
\end{aligned} \tag{7.58}$$

The second equality dropped $\underline{Q}\underline{T}\underline{Q}^t \cdot \underline{\Omega}$, which vanishes by Lemma 2.36 because $\underline{Q}\underline{T}\underline{Q}^t$ is symmetric, $(\underline{Q}\underline{T}\underline{Q}^t)^t = \underline{Q}\underline{T}^t\underline{Q}^t = \underline{Q}\underline{T}\underline{Q}^t$, and $\underline{\Omega}$ is skew. The third used Definition 1.57, $\underline{A} \cdot \underline{B} = \text{tr}(\underline{A}\underline{B}^t)$, with $(\underline{Q}\underline{L}\underline{Q}^t)^t = \underline{Q}\underline{L}^t\underline{Q}^t$; the fourth cancelled $\underline{Q}^t\underline{Q} = \underline{I}$ in the middle; the fifth is the cyclic property once more.

REMARK 7.28 (the two invariant scalars, and what they teach about Definition 7.7). Equations (7.56) and (7.58) are the counterexamples promised in Remark 7.8, and they are worth pausing over because they are the sharpest available warning against reasoning from ingredients.

In (7.56) both sides are built from a field, $\underline{v}$, that is not objective: (7.8) carries two extra terms. The extra terms survive differentiation, since $\underline{L}^+$ carries $\underline{\Omega}$; they are annihilated only at the very last step, by the trace. So a non-objective field can have a perfectly invariant scalar buried inside it, and no inspection of the field alone would reveal it.

In (7.58) the ingredients are mixed: $\underline{T}$ is objective, $\underline{L}$ is not. What kills the extra term is not a property of $\underline{L}$ at all but the symmetry of $\underline{T}$, which is itself a balance law, (7.53)₄. Had the material admitted couple stresses, so that $\underline{T}$ were not symmetric, the stress power would not have been invariant and the energy balance (7.59)₂ would have failed to have the same form for both observers. The invariance of the stress power is therefore not a kinematic fact but a consequence of the balance of rotational momentum. That is worth remembering, because Chapters 8 and 9 use $\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{D}$ constantly, and the deletion of $\underline{W}$ there is the same deletion as here.

With these results and (7.37), together with the invariance of the Helmholtz free energy, cf. (6.186), and of the material derivative, we conclude that the balances of mass and energy, and the Clausius–Duhem inequality, are the same in all frames:

$$\begin{aligned} \dot{\rho}^+ + \rho^+ \text{div}^+ \underline{v}^+ &= 0, \\ \rho^+ \dot{\varepsilon}^+ + \text{div}^+ \underline{q}^+ - \underline{T}^+ \cdot \underline{L}^+ &= \rho^+ r^+, \\ \text{and} \quad -\rho^+ (\dot{\psi}^+ + \eta^+ \dot{\theta}^+) + \underline{T}^+ \cdot \underline{L}^+ - (\theta^+)^{-1} \underline{q}^+ \cdot \text{grad}^+ \theta^+ &\geq 0, \end{aligned} \tag{7.59}$$

and, of course, (7.40) and (7.53)₄ ensure that

$$(\tilde{T}^+)^t = \tilde{T}^+. \quad (7.60)$$

The verification is term by term, and it is entirely a matter of substituting the results already obtained. In (7.59)₁, $\rho^+ = \rho$ by (7.37)₁ and $\operatorname{div}^+ \underline{v}^+ = \operatorname{div} \underline{v}$ by (7.56); the material derivative is unchanged because it follows the material point p and, by (7.1), $\partial/\partial t^+ = \partial/\partial t$, so $\dot{\rho}^+ = \dot{\rho}$. Every term of (7.59)₁ therefore equals the corresponding term of (7.53)₁, and the equation holds because that one does. In (7.59)₂, use (7.37)_{1,2,3}, (7.54) and (7.58). In (7.59)₃, use (7.37) again, (7.58) and (7.55)₂. And (7.60) is $(\tilde{Q}\tilde{T}\tilde{Q}^t)^t = \tilde{Q}\tilde{T}^t\tilde{Q}^t = \tilde{Q}\tilde{T}\tilde{Q}^t$.

REMARK 7.29 (why the symmetry of $\tilde{T}$ survives a rotating observer). Equation (7.60) was obtained by conjugation in one line, and the ease of it hides a question worth asking. The symmetry of $\tilde{T}$ is not an algebraic identity. It is the local form of the balance of moment of momentum, derived in Chapter 6 for an inertial observer. In the frame of O^+ that balance picks up the moment of the inertial force, which does not vanish and which varies from place to place. Why is the conclusion unchanged?

Because a body force never enters the symmetry argument. Taking moments about a point of $\mathbb{E}^{3+}$ adds the term $\int \rho^+ \underline{s}^+ \times \underline{i}^+ dv$, a volume integral of exactly the same form as the moment $\int \rho^+ \underline{s}^+ \times \underline{b}^+ dv$ of the true body force. In the localisation of Chapter 6 the volume terms are one order higher in the diameter of the shrinking region than the surface terms, and it is the surface terms alone that survive to give $\varepsilon_{ijk}T_{jk} = 0$. Adding $\underline{i}^+$ to $\underline{b}^+$ therefore changes the balance of moment of momentum in precisely the way that changes nothing in its local consequence.

The observation generalises, and explains why (7.62) is the only casualty. By (7.63) a change of observer contributes to the mechanics through a term of body-force type and through nothing else. Any law whose local content is insensitive to the body force is therefore untouched: the balance of moment of momentum for the reason just given, and the mass balance and the Clausius–Duhem inequality because no force appears in them at all. Only the balance of linear momentum reads the body force pointwise, and only it changes.

The energy balance is the interesting case. No body force appears in (7.53)₂, but one does appear in the statement of Chapter 6 from which it came, through the working $\int \rho \underline{b} \cdot \underline{v} \, dv$; it cancels against the same term in the kinetic energy theorem, and (7.59)₂ is what is left after that cancellation. Adding $\underline{i}^+$ to $\underline{b}^+$ adds $\int \rho^+ \underline{i}^+ \cdot \underline{v}^+ \, dv$ to both sides of the cancellation, so it survives in neither. An inertial force does work on a body in a rotating frame, and none of that work reaches the internal energy.

Exactly one of the four terms of (7.63) does no work at all, and it is worth saying which. The Coriolis term is orthogonal to $\underline{v}^+$ whenever $\underline{\dot{c}} = \underline{0}$, since $\underline{\Omega} \underline{v}^+ \cdot \underline{v}^+ = 0$ for a skew $\underline{\Omega}$ by Lemma 2.36: a Coriolis force deflects a particle and never speeds it up or slows it down. The Euler term is orthogonal to $\underline{s}^+$, by the same lemma applied to the skew $\underline{\dot{\Omega}}$, which is what makes it azimuthal; but it is not orthogonal to $\underline{v}^+$, and it does work. The other two do work as well, and they are the two that possess potentials:

$$-\underline{\Omega}^2 \underline{s}^+ = \text{grad}^+ \psi, \quad \psi = \frac{1}{2} |\underline{\Omega} \underline{s}^+|^2, \quad \underline{\ddot{c}} = \text{grad}^+ \underline{\ddot{c}} \cdot \underline{x}^+,$$

the first because $\underline{\Omega}^2 \underline{s} \cdot \underline{s} = -|\underline{\Omega} \underline{s}|^2$ and $\underline{\Omega}^2$ is symmetric. With $\underline{\Omega} \underline{s} = \underline{\omega} \times \underline{s}$ the potential ψ is the familiar $\frac{1}{2} |\underline{\omega}|^2 r^2$, r being the distance from the axis. That is why a steadily rotating frame admits an effective potential energy, and why the Coriolis force cannot be absorbed into one.

The situation for the linear momentum balance is different. To investigate this we combine (7.38)₃ and (7.53)₃, obtaining

$$\begin{aligned} \rho^+ \underline{b}^+ &= \rho^+ \underline{Q} \underline{b} = \rho \underline{Q} \underline{b} = \underline{Q} (\rho \underline{a} - \text{div} \underline{T}) \\ &= \rho \underline{Q} \underline{a} - \underline{Q} (\text{div} \underline{T}) = \rho^+ \underline{Q} \underline{a} - \text{div}^+ \underline{T}^+, \end{aligned} \tag{7.61}$$

the last step by (7.52) and (7.37)₁ read backwards.

Every term of (7.53)₃ has now been carried into the frame of O^+ intact except one. The divergence of the stress went over by (7.52), the body force by (7.38)₃, and the density is invariant by (7.37)₁; each of the three appears in the starred frame as the image under $\underline{Q}$ of what O wrote, which is what objectivity means. The acceleration is the single

exception, and (7.9) says by exactly how much it fails. Give that failure a name:

$$\dot{\underline{i}}^+ := \underline{a}^+ - Q\underline{a}, \quad \text{equivalently} \quad Q\underline{a} = \underline{a}^+ - \dot{\underline{i}}^+.$$

This introduces nothing. The quantity $\dot{\underline{i}}^+$ is the right-hand side of (7.9), already computed in Problem 1 out of $\underline{Q}(t)$ and $\underline{c}(t)$ alone; the definition merely gives it a symbol so that it can be moved about. Substituting it into (7.61) we arrive at

$$\rho^+ \underline{b}^+ = \rho^+ (\underline{a}^+ - \dot{\underline{i}}^+) - \operatorname{div}^+ \underline{T}^+,$$

i.e., moving $\operatorname{div}^+ \underline{T}^+$ and $\rho^+ \dot{\underline{i}}^+$ to the left, the linear momentum balance

$$\operatorname{div}^+ \underline{T}^+ + \rho^+ (\underline{b}^+ + \dot{\underline{i}}^+) = \rho^+ \underline{a}^+, \quad (7.62)$$

where

$$\dot{\underline{i}}^+ = \ddot{\underline{c}} + 2\Omega(\underline{v}^+ - \dot{\underline{c}}) + (\dot{\Omega} - \Omega^2)(\underline{x}^+ - \underline{c}) \quad (7.63)$$

is the contribution to the net body force arising from the spin and translation of the frame of O^+ . This is typically referred to as the *inertial force*.

Three features of the symbol are worth fixing before it is used. It is an *acceleration*, of the same dimensions as $\underline{a}^+$, which is why it may stand beside $\underline{b}^+$ inside the bracket of (7.62) and why that bracket must be multiplied by ρ^+ before it becomes a force per unit volume. Its sign is not a convention but is forced by the substitution: $Q\underline{a}$ was replaced by $\underline{a}^+ - \dot{\underline{i}}^+$, and carrying the resulting $-\rho^+ \dot{\underline{i}}^+$ from the right-hand side to the left turned it into $+\rho^+ \dot{\underline{i}}^+$. And it has no star-free counterpart. For O himself the change of observer is the identity, $\underline{Q} = \underline{I}$ and $\underline{c} = \underline{0}$, so that $\underline{\Omega} = \underline{0}$ and (7.63) returns $\dot{\underline{i}} = \underline{0}$; (7.62) then collapses to (7.53)₃. The symbol exists only because a second observer was introduced, and it measures that observer and nothing else.

Notice, finally, what (7.63) does *not* contain. There is no $\underline{T}$, no $\underline{b}$, no material constant, nothing about what is being done to the body: only $\underline{c}$ and $\underline{\Omega}$, which describe the relation between the two frames, and $\underline{x}^+$ and $\underline{v}^+$, which say where the particle is and how fast it is moving

as O^+ records it. That is precisely the property that licenses moving $\dot{\underline{z}}^+$ to the force side of the equation and treating it as though it were a body force, in the manner of Remark 7.30: O^+ can evaluate it once, as a field over his whole frame, before he knows anything whatever about the body that is going to occupy it. Figure C is that field, term by term.

However, if the frame of O^+ is related to that of O by a Galilean transformation, then $\underline{\Omega} = \underline{Q}$, $\dot{\underline{\Omega}} = \underline{Q}$ and $\ddot{\underline{c}} = \underline{Q}$, so every term of (7.63) vanishes, $\dot{\underline{z}}^+ = \underline{Q}$, and the balance of linear momentum is identical in both frames, i.e.

$$\operatorname{div}^+ T^+ + \rho^+ b^+ = \rho^+ a^+. \quad (7.64)$$

Thus, if an inertial frame exists, cf. Euler's postulates (6.34) and (6.35), then there are infinitely many, simply because there are infinitely many Galilean transformations.

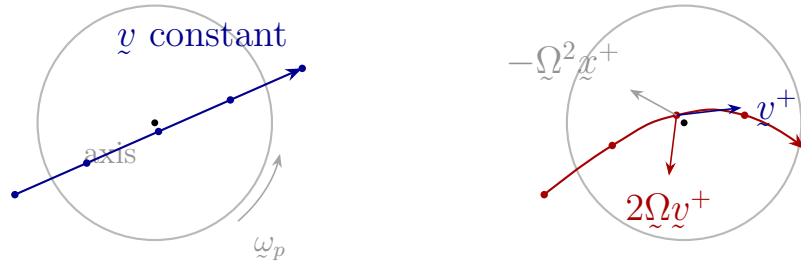
REMARK 7.30 (what an inertial force is, and what it is not). Three things about (7.62) repay attention.

It is not a repair. Nothing was added to the physics. Equation (7.62) is (7.53)₃ rewritten, and every symbol in $\dot{\underline{z}}^+$ was computed in Section 7.2 from the relative motion of the two observers. The non-inertial observer is not doing worse mechanics; he is doing the same mechanics with a longer right-hand side, and he can compute the extra term himself once he knows $\underline{Q}(t)$ and $\underline{c}(t)$.

Why it looks like a body force. In (7.62) the inertial force enters multiplied by ρ^+ , exactly as b^+ does, and that is not a notational accident: $\dot{\underline{z}}^+$ is an acceleration, and multiplying an acceleration by a density produces a force per unit volume. The consequence is that no measurement made inside a small region can distinguish a uniform $\ddot{\underline{c}}$ from a uniform gravitational field, since both act on every particle in proportion to its mass. That is the observation from which general relativity starts. Within the present theory it means only that the split of the right-hand side of (7.62) into b^+ and $\dot{\underline{z}}^+$ is a matter of bookkeeping unless the frames are specified.

Why only this law fails. Look back at (7.59). The mass balance contains no acceleration; the energy balance contains $\underline{a}$ nowhere, only $\dot{\underline{e}}$

and the stress power; the Clausius–Duhem inequality likewise. Linear momentum is the only one of the five in which the material acceleration appears, and $\underline{a}$ is the only quantity in the whole dictionary of Section 7.4 whose transformation law carries $\underline{\ddot{c}}$ and $\underline{\dot{\Omega}}$. One non-objective ingredient, one non-invariant law. The correspondence is exact, and it is the clearest illustration in the book of Remark 7.8.



(a) observer O , inertial (b) observer O^+ , spinning with the disc
 $\underline{b} = \underline{0}$ and $\underline{a} = \underline{0}$: no force, no bending $\underline{\dot{\Omega}} = \underline{0}$ but $\underline{a}^+ \neq \underline{0}$: the deficit is $\underline{i}^+$

Figure I: Why a rotating observer manufactures forces. A free particle crosses a spinning disc. In (a) the inertial observer records a straight path at constant speed, consistent with (7.64) and no force at all. In (b) the observer turning with the disc records the same journey as a curved track, so $\underline{a}^+ \neq \underline{0}$; since the tractions and the true body force are unchanged, (7.64) cannot hold in his frame. Equation (7.62) restores the balance by crediting him with the inertial force $\underline{i}^+$ of (7.63), whose Coriolis part $2\Omega\underline{v}^+$ bends the track and whose centrifugal part $-\Omega^2\underline{x}^+$ pushes outward. Nothing touches the particle in either panel; the difference is entirely in $\underline{\dot{Q}}$.

It is possible to show that

$$\underline{a}^{++} - \underline{i}^{++} = \underline{Q}^+(t^+)(\underline{a}^+ - \underline{i}^+) \quad (7.65)$$

under a further transformation

$$\underline{x}^{++} = \underline{Q}^+(t^+)\underline{x}^+ + \underline{c}^+(t^+), \quad t^{++} = t^+ + a^+ \quad (7.66)$$

from the frame of O^+ to that of another observer O^{++} , where $\underline{i}^{++}$ is the inertial force in the latter frame, and hence that

$$\operatorname{div}^{++} \underline{T}^{++} + \rho^{++}(\underline{b}^{++} + \underline{i}^{++}) = \rho^{++}\underline{a}^{++}. \quad (7.67)$$

Thus the linear momentum balance, with the inertial forces included, is valid in all frames. Moreover, the jump conditions derived in Chapter 6 are identical in all frames.

The proof of (7.65) is two observations and one line, and it is worth giving, because it is what makes the theory consistent rather than merely serviceable.

First, changes of observer compose. Substituting (7.3) into (7.66),

$$\tilde{x}^{++} = \tilde{Q}^+(Q\tilde{x} + \tilde{c}) + \tilde{c}^+ = (\tilde{Q}^+Q)\tilde{x} + (\tilde{Q}^+\tilde{c} + \tilde{c}^+) =: \tilde{Q}^{++}\tilde{x} + \tilde{c}^{++},$$

and $\tilde{Q}^{++} = \tilde{Q}^+Q$ is orthogonal as a product of orthogonal tensors, mapping $\mathbb{E}^3$ to $\mathbb{E}^{\tilde{3}^{++}}$. Likewise $t^{++} = t + (a + a^+)$, again of the form (7.1). So the passage from O directly to O^{++} is itself a change of observer of exactly the kind studied in Section 7.1, and everything proved for a single change applies to it.

Second, (7.9) is the definition of the inertial force. Comparing (7.9) with (7.63), the content of the two together is

$$\tilde{i}^+ = \tilde{a}^+ - Q\tilde{a},$$

the acceleration deficit of the frame of O^+ measured against the inertial frame of O . Applying the same statement to the composite change, with $\tilde{Q}^{++}$ in place of $\tilde{Q}$, $\tilde{i}^{++} = \tilde{a}^{++} - \tilde{Q}^{++}\tilde{a}$. Therefore

$$\tilde{a}^{++} - \tilde{i}^{++} = \tilde{Q}^{++}\tilde{a} = \tilde{Q}^+Q\tilde{a} = \tilde{Q}^+(\tilde{a}^+ - \tilde{i}^+),$$

which is (7.65). And (7.67) is (7.62) written for the composite change of observer, which is legitimate precisely because the first observation showed the composite to be a change of observer.

REMARK 7.31 (spins add). The composition rule has a corollary worth recording, since it is the statement that the Ω of (7.8) behaves like an angular velocity. Differentiating $\tilde{Q}^{++} = \tilde{Q}^+Q$, all time derivatives being with respect to the common time by (7.1) and (7.66)₂,

$$\tilde{\Omega}^{++} = \dot{\tilde{Q}}^{++}(\tilde{Q}^{++})^t = (\dot{\tilde{Q}}^+Q + \tilde{Q}^+\dot{Q})\tilde{Q}^t(Q^+)^t = \dot{\tilde{Q}}^+(Q^+)^t + \tilde{Q}^+(\dot{Q}Q^t)(Q^+)^t,$$

that is, $\underline{\Omega}^{++} = \underline{\Omega}_{\text{rel}}^+ + \underline{Q}^+ \underline{\Omega}(\underline{Q}^+)^t$, where $\underline{\Omega}_{\text{rel}}^+ = \dot{\underline{Q}}^+(\underline{Q}^+)^t$ is the spin of the second change taken by itself. The spin of the composite is the spin of the second plus the spin of the first, the latter carried over by $\underline{Q}^+$ into the final frame. Written with axial vectors, by Proposition 1.44, this reads $\underline{\omega}^{++} = \underline{\omega}_{\text{rel}}^+ + \underline{Q}^+ \underline{\omega}$: angular velocities add. Nothing in this chapter would work if they did not, since (7.65) would then depend on the route taken from O to O^{++} .

REMARK 7.32 (what the chapter licenses, and what it forbids). It is worth setting down in one place the principle that the whole chapter exists to make precise, because Chapter 8 will invoke it on every page and will not repeat the derivation.

The principle. A constitutive equation is a claim about a material, and a material has no access to the identity of the observer looking at it. A constitutive equation must therefore survive the substitution of (7.3) for arbitrary $\underline{Q}(t)$ and $\underline{c}(t)$: if it holds between the quantities recorded by O , then the same equation with every symbol starred must hold between the quantities recorded by O^+ . This is the *principle of material frame indifference*. It is a restriction on the constitutive equation and never on the motion: the motion is whatever it is, and it is the form of the law that is being constrained.

Why it is not vacuous. The dictionary of Section 7.4 is the list of what each variable does under that substitution, and its entries are not alike. A law written in variables drawn from the first four groups satisfies the principle automatically, because the $\underline{Q}$'s cancel in pairs or never appear at all. A law containing any entry from the last group does not, because $\dot{\underline{Q}}$ enters with nothing to cancel against and $\dot{\underline{Q}}$ is arbitrary. The principle is therefore a genuine sieve, and the dictionary is that sieve written out.

The three prohibitions established here. Each is a mistake that has been made.

- No constitutive law may contain $\underline{W}$, nor $\underline{L}$ otherwise than through $\underline{D}$. Remark 7.21: the observer manufactures spin out of nothing, so such a law would predict stress in a body lying still on a table.

- No constitutive law may contain $\underline{v}$ or $\underline{a}$. By (7.8) and (7.9) the two observers disagree about both even when the body is doing nothing whatever.
- No constitutive law may contain $\underline{F}$ except through $\underline{C}$, or equivalently none may see $\underline{R}$. By (7.29)₃ the rotation is exactly the part of $\underline{F}$ that a change of observer alters.

The corresponding licences are the first four groups of the dictionary, and the safest of them is the second: a law relating $\underline{S}$ to $\underline{C}$ involves nothing that sees $\underline{Q}$ at all, which is why constitutive theory is written there.

One thing the principle does not govern. It says nothing about the balance laws, which are not constitutive statements. Section 7.5 showed that four of the five are form invariant on their own and that the fifth is repaired by (7.62). A frequent confusion is to read the failure of (7.64) in a rotating frame as a failure of frame indifference. It is not. Frame indifference is a demand made on constitutive equations; (7.62) is a theorem about a balance law; and the theorem tells the engineer in the rotating frame which extra terms to add, not that he must stop.

How strong the principle is. What has been established in this chapter is that the theory is consistent with the principle and what the principle sieves out. Whether the principle is exactly true is a separate question, and it has been argued over since Müller and Truesdell: for a gas whose molecular mean free path is not negligible against the scale of the flow, kinetic theory does predict a weak dependence of the constitutive response on $\underline{\Omega}$, and the principle is then an excellent approximation rather than an axiom. Nothing in this book is at that scale, and the principle is used at full strength throughout; but it is an assumption about materials, not a theorem about geometry, and it is worth knowing which of the two one is holding. Compare Remark 7.24, where the same point was made about (7.38)₁: the geometry of Section 7.1 is proved, and everything in this chapter that concerns matter rather than space is assumed.

7.6 Problems

PROBLEM 1 (The transformation of the acceleration). Derive (7.9).

Solution. The acceleration in the frame of O^+ is $\underline{a}^+ = \partial \underline{v}^+ / \partial t^+$ at fixed p , by Definition 2.3, and $\partial / \partial t^+ = \partial / \partial t$ by the argument given before (7.7). So the whole problem is to differentiate (7.7) once more and then to eliminate the unstarred quantities $\underline{x}$ and $\underline{v}$ in favour of the starred ones. Neither operation is difficult; what makes the result look complicated is that both $\underline{x}$ and $\underline{v}$ have to be eliminated, and the second elimination feeds on the first. The elimination is therefore split into two steps below, giving four in all.

Step 1: differentiate. From (7.7), $\underline{v}^+ = \dot{\underline{Q}}\underline{x} + \underline{Q}\underline{v} + \dot{\underline{c}}$. Differentiate at fixed p , using the product rule of Problem 1(a) of Chapter 4 on the first two terms and remembering that $\dot{\underline{x}} = \underline{v}$ and $\dot{\underline{v}} = \underline{a}$ because $\underline{x}$ and $\underline{v}$ are being followed along the motion of the material point:

$$\underline{a}^+ = \ddot{\underline{Q}}\underline{x} + \dot{\underline{Q}}\underline{v} + \dot{\underline{Q}}\underline{v} + \underline{Q}\underline{a} + \ddot{\underline{c}} = \ddot{\underline{Q}}\underline{x} + 2\dot{\underline{Q}}\underline{v} + \underline{Q}\underline{a} + \ddot{\underline{c}}.$$

The two middle terms are equal and are the origin of the factor 2 in the Coriolis term: one copy comes from differentiating $\dot{\underline{Q}}\underline{x}$ and the other from differentiating $\underline{Q}\underline{v}$. Rearranged,

$$\underline{a}^+ - \underline{Q}\underline{a} = \ddot{\underline{c}} + 2\dot{\underline{Q}}\underline{v} + \ddot{\underline{Q}}\underline{x}.$$

Everything now depends on rewriting the last two terms.

Step 2: eliminate $\underline{x}$. From (7.3), $\underline{Q}\underline{x} = \underline{x}^+ - \underline{c}$. Also, from $\underline{\Omega} = \dot{\underline{Q}}\underline{Q}^t$ and $\underline{Q}^t\underline{Q} = \underline{I}$,

$$\dot{\underline{Q}} = \dot{\underline{Q}}\underline{I} = \dot{\underline{Q}}\underline{Q}^t\underline{Q} = \underline{\Omega}\underline{Q},$$

which is the identity that will be used three times. Differentiating it,

$$\ddot{\underline{Q}} = (\underline{\Omega}\underline{Q})^\cdot = \dot{\underline{\Omega}}\underline{Q} + \underline{\Omega}\dot{\underline{Q}} = \dot{\underline{\Omega}}\underline{Q} + \underline{\Omega}(\underline{\Omega}\underline{Q}) = (\dot{\underline{\Omega}} + \underline{\Omega}^2)\underline{Q}.$$

Hence

$$\ddot{\underline{Q}}\underline{x} = (\dot{\underline{\Omega}} + \underline{\Omega}^2)\underline{Q}\underline{x} = (\dot{\underline{\Omega}} + \underline{\Omega}^2)(\underline{x}^+ - \underline{c}).$$

Step 3: eliminate $\underline{v}$. Solve (7.7) for $\underline{Q}\underline{v}$:

$$\underline{Q}\underline{v} = \underline{v}^+ - \dot{\underline{Q}}\underline{x} - \dot{\underline{c}} = \underline{v}^+ - \dot{\underline{c}} - \underline{\Omega}(\underline{x}^+ - \underline{c}),$$

using the result of Step 2 for $\dot{\underline{Q}}\underline{x} = \underline{\Omega}\underline{Q}\underline{x} = \underline{\Omega}(\underline{x}^+ - \underline{c})$. This is (7.8) rearranged, as it must be. Therefore

$$\begin{aligned} 2\dot{\underline{Q}}\underline{v} &= 2\underline{\Omega}\underline{Q}\underline{v} = 2\underline{\Omega}\left[\underline{v}^+ - \dot{\underline{c}} - \underline{\Omega}(\underline{x}^+ - \underline{c})\right] \\ &= 2\underline{\Omega}(\underline{v}^+ - \dot{\underline{c}}) - 2\underline{\Omega}^2(\underline{x}^+ - \underline{c}). \end{aligned}$$

Step 4: collect. Substituting the results of Steps 2 and 3 into the display at the end of Step 1,

$$\begin{aligned} \underline{a}^+ - \underline{Q}\underline{a} &= \ddot{\underline{c}} + 2\underline{\Omega}(\underline{v}^+ - \dot{\underline{c}}) - 2\underline{\Omega}^2(\underline{x}^+ - \underline{c}) + (\dot{\underline{\Omega}} + \underline{\Omega}^2)(\underline{x}^+ - \underline{c}) \\ &= \ddot{\underline{c}} + 2\underline{\Omega}(\underline{v}^+ - \dot{\underline{c}}) + (-2\underline{\Omega}^2 + \dot{\underline{\Omega}} + \underline{\Omega}^2)(\underline{x}^+ - \underline{c}), \end{aligned}$$

the two brackets in $\underline{x}^+ - \underline{c}$ having been collected by linearity. Since $-2\underline{\Omega}^2 + \underline{\Omega}^2 = -\underline{\Omega}^2$,

$$\underline{a}^+ - \underline{Q}\underline{a} = \ddot{\underline{c}} + 2\underline{\Omega}(\underline{v}^+ - \dot{\underline{c}}) + (\dot{\underline{\Omega}} - \underline{\Omega}^2)(\underline{x}^+ - \underline{c}),$$

which is (7.9).

Two checks. First, put $\dot{\underline{Q}} = \underline{Q}$ and $\ddot{\underline{c}} = \underline{0}$. Then $\underline{\Omega} = \underline{Q}$, $\dot{\underline{\Omega}} = \underline{Q}$, and every term on the right vanishes, leaving $\underline{a}^+ = \underline{Q}\underline{a}$: this is (7.10)₂, so the Galilean case is recovered. Second, put $\underline{v}^+ = \dot{\underline{c}}$ and $\underline{x}^+ = \underline{c}$, meaning a particle sitting at the origin of O^+ 's frame and moving with it. Then only $\ddot{\underline{c}}$ survives, which is right: such a particle has, for O^+ , no acceleration at all, and the whole discrepancy is the acceleration of the frame itself.

Where the minus sign comes from. The centrifugal term appears as $-\underline{\Omega}^2$ and not $+\underline{\Omega}^2$ because the two contributions to $\underline{\Omega}^2$ have opposite signs and different origins: $+\underline{\Omega}^2$ comes from $\dot{\underline{Q}}$, the turning of the frame's turning, while $-2\underline{\Omega}^2$ comes from the Coriolis term acting on the part of $\underline{v}$ that is itself due to the frame's rotation. Their sum is $-\underline{\Omega}^2$. With $\underline{\Omega}\underline{s} = \underline{\omega} \times \underline{s}$ this is $-\underline{\omega} \times (\underline{\omega} \times \underline{s})$, which by the expansion of the triple

cross product is $|\underline{\omega}|^2 \underline{s} - \underline{\omega} \cdot \underline{s} \underline{\omega}$, the component of $|\underline{\omega}|^2 \underline{s}$ perpendicular to the axis. It points away from the axis, as centrifugal effects do.

The result is what (7.62) and (7.63) are built on: the entire difference between the two observers' equations of motion is the boxed expression multiplied by ρ . $\square$

PROBLEM 2 (The rate of the shifter in a spinning frame). Show that $(\underline{1}^+)^{\cdot} = \underline{\Omega} \underline{1}^+$, where $\underline{\Omega} = \dot{\underline{Q}} \underline{Q}^t$.

Solution. Everything is in (7.18), $\underline{1}^+ = \underline{Q}(t) \underline{1} \underline{K}^t$, once it is noticed which of the three factors can move.

The shifter $\underline{1}$ of the observer O is fixed, because the frame of O is taken to be inertial, as stated after (7.18); so $\dot{\underline{1}} = \underline{0}$ in the sense that $\underline{1}$ is a constant two-point tensor. The tensor $\underline{K}$ is fixed by (7.15), every factor in it having been evaluated at the single instant t_0 ; so $\dot{\underline{K}} = \underline{0}$ and hence $\dot{\underline{K}}^t = (\dot{\underline{K}})^t = \underline{0}$ by Problem 1(b) of Chapter 4. Only $\underline{Q}(t)$ moves. Differentiating (7.18) with the product rule, two of the three terms therefore vanish:

$$(\underline{1}^+)^{\cdot} = \dot{\underline{Q}} \underline{1} \underline{K}^t + \underline{Q} \dot{\underline{1}} \underline{K}^t + \underline{Q} \underline{1} \dot{\underline{K}}^t = \dot{\underline{Q}} \underline{1} \underline{K}^t.$$

Now insert $\underline{I} = \underline{Q}^t \underline{Q}$, which is legitimate by Proposition 1.48, and re-group so that (7.18) reappears on the right:

$$\dot{\underline{Q}} \underline{1} \underline{K}^t = \dot{\underline{Q}} \underline{I} \underline{1} \underline{K}^t = (\dot{\underline{Q}} \underline{Q}^t) (\underline{Q} \underline{1} \underline{K}^t) = \underline{\Omega} \underline{1}^+,$$

using (7.8)₂ for the first bracket and (7.18) for the second. Hence

$$(\underline{1}^+)^{\cdot} = \underline{\Omega} \underline{1}^+, \quad \underline{\Omega} = \dot{\underline{Q}} \underline{Q}^t.$$

What the formula says. Compare it with (4.3), $\dot{\underline{F}} = \underline{L} \underline{F}$. The two are the same statement with $\underline{1}^+$ in place of $\underline{F}$ and $\underline{\Omega}$ in place of $\underline{L}$. That is exactly right, and it is Remark 7.10 in another guise: the frame of O^+ undergoes a rigid displacement relative to that of O , whose velocity gradient is $\underline{\Omega}$ by (4.29), and $\underline{1}^+$ is the two-point tensor that gets carried

along by it. The shifter of a spinning observer turns with him, at the rate $\underline{\Omega}$.

The consistency check that matters. The relation $\underline{H}^+ = \underline{F}^+ - \underline{1}^+$ of (7.25)₂ would be worthless if its two terms evolved by different rules. They do not: $\dot{\underline{F}}^+ = \underline{L}^+ \underline{F}^+$ by (4.3), and for a frame at rest relative to the material, so that $\underline{L}^+$ reduces to the frame's own contribution $\underline{\Omega}$ by (7.33), the two rates coincide. In particular, if the body is at rest for O then $\underline{F}$ is constant, and

$$\dot{\underline{F}}^+ = (\underline{Q} \underline{F} \underline{K}^t)^\cdot = \dot{\underline{Q}} \underline{F} \underline{K}^t = \underline{\Omega} \underline{F}^+,$$

by the same insertion of $\underline{I} = \underline{Q}^t \underline{Q}$, so $\dot{\underline{H}}^+ = \underline{\Omega}(\underline{F}^+ - \underline{1}^+) = \underline{\Omega} \underline{H}^+$ and the displacement gradient simply rides round with the observer. Had the shifter been replaced by $\hat{\underline{I}}$, its counterpart $\hat{\underline{I}}^+ = \underline{K} \hat{\underline{I}} \underline{K}^t$ of (7.26) would have satisfied $(\hat{\underline{I}}^+)^\cdot = \hat{\underline{Q}}^+$ instead: it would have stood still while $\underline{F}^+$ turned, and the difference $\underline{F}^+ - \hat{\underline{I}}^+$ would have changed in time even though nothing whatever was happening to the body. That is the dynamic version of the objection raised after (7.26), and it is the sharper of the two.

When it vanishes. If O^+ is related to O by a Galilean transformation then $\dot{\underline{Q}} = \underline{Q}$, so $\underline{\Omega} = \underline{Q}$ and $(\underline{1}^+)^\cdot = \underline{Q}$: the shifter is fixed in every inertial frame, which is the claim made in Remark 7.16. $\square$

PROBLEM 3 (An observer standing on a spinning planet). Observer O^+ adopts a frame of reference attached to the surface of a planet of radius R , with origin located at azimuth θ (degrees longitude) and elevation φ (degrees latitude). The planet spins about its axis, the unit vector $\underline{e}_3$, at the constant rate ω , i.e. $\theta = \omega t + \theta_0$ with $\theta_0 = \theta(0)$. What are the equations of motion for this observer?

Solution. Required: (7.62) with (7.63), i.e. $\underline{c}, \underline{\Omega}, \dot{\underline{\Omega}}$ from $\underline{Q}(t), \underline{c}(t)$ of (7.3). Geometry: Figures J and K. Data:

$$\begin{aligned} \underline{e}_i \cdot \underline{e}_j &= \delta_{ij}, & \underline{e}_1 \times \underline{e}_2 &= \underline{e}_3, & \underline{e}_2 \times \underline{e}_3 &= \underline{e}_1, & \underline{e}_3 \times \underline{e}_1 &= \underline{e}_2, \\ \theta &= \omega t + \theta_0, & \dot{\theta} &= \omega, & \dot{\varphi} &= 0, & \dot{\underline{e}}_i &= \underline{0}. \end{aligned}$$

Equatorial plane (Figure K): unit vector at angle θ from $\underline{e}_1$, and the same at $\theta + \pi/2$:

$$\begin{aligned}
\underline{f}_\rho &= \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2, \\
\underline{f}_E &= \cos(\theta + \frac{\pi}{2}) \underline{e}_1 + \sin(\theta + \frac{\pi}{2}) \underline{e}_2 = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2, \\
\underline{e}_3 \times \underline{f}_\rho &= \cos \theta \underline{e}_3 \times \underline{e}_1 + \sin \theta \underline{e}_3 \times \underline{e}_2 = \cos \theta \underline{e}_2 - \sin \theta \underline{e}_1 = \underline{f}_E, \\
|\underline{f}_\rho|^2 &= \cos^2 \theta + \sin^2 \theta = 1, \quad |\underline{f}_E|^2 = \sin^2 \theta + \cos^2 \theta = 1, \\
\underline{f}_\rho \cdot \underline{f}_E &= -\cos \theta \sin \theta + \sin \theta \cos \theta = 0, \quad \underline{f}_\rho \cdot \underline{e}_3 = \underline{f}_E \cdot \underline{e}_3 = 0, \\
\underline{f}_\rho \times \underline{f}_E &= \cos^2 \theta \underline{e}_1 \times \underline{e}_2 - \sin^2 \theta \underline{e}_2 \times \underline{e}_1 = (\cos^2 \theta + \sin^2 \theta) \underline{e}_3 = \underline{e}_3, \\
\underline{f}_E \times \underline{e}_3 &= -\sin \theta \underline{e}_1 \times \underline{e}_3 + \cos \theta \underline{e}_2 \times \underline{e}_3 = \sin \theta \underline{e}_2 + \cos \theta \underline{e}_1 = \underline{f}_\rho.
\end{aligned}$$

Meridian plane (Figure J(a)), spanned by $\underline{f}_\rho, \underline{e}_3$: unit vector at angle φ from $\underline{f}_\rho$, and the same at $\varphi + \pi/2$:

$$\begin{aligned}
\underline{f}_U &= \cos \varphi \underline{f}_\rho + \sin \varphi \underline{e}_3, \quad \underline{o}^+ = R \underline{f}_U, \\
\underline{f}_N &= \cos(\varphi + \frac{\pi}{2}) \underline{f}_\rho + \sin(\varphi + \frac{\pi}{2}) \underline{e}_3 = -\sin \varphi \underline{f}_\rho + \cos \varphi \underline{e}_3, \\
|\underline{f}_U|^2 &= \cos^2 \varphi |\underline{f}_\rho|^2 + 2 \cos \varphi \sin \varphi \underline{f}_\rho \cdot \underline{e}_3 + \sin^2 \varphi |\underline{e}_3|^2 \\
&= \cos^2 \varphi + 0 + \sin^2 \varphi = 1, \\
|\underline{f}_N|^2 &= \sin^2 \varphi |\underline{f}_\rho|^2 - 2 \sin \varphi \cos \varphi \underline{f}_\rho \cdot \underline{e}_3 + \cos^2 \varphi |\underline{e}_3|^2 \\
&= \sin^2 \varphi + 0 + \cos^2 \varphi = 1, \\
\underline{f}_N \cdot \underline{f}_U &= -\sin \varphi \cos \varphi |\underline{f}_\rho|^2 + (\cos^2 \varphi - \sin^2 \varphi) \underline{f}_\rho \cdot \underline{e}_3 + \cos \varphi \sin \varphi |\underline{e}_3|^2 \\
&= -\sin \varphi \cos \varphi + 0 + \cos \varphi \sin \varphi = 0, \\
\underline{f}_E \cdot \underline{f}_U &= \cos \varphi \underline{f}_E \cdot \underline{f}_\rho + \sin \varphi \underline{f}_E \cdot \underline{e}_3 = 0, \\
\underline{f}_E \cdot \underline{f}_N &= -\sin \varphi \underline{f}_E \cdot \underline{f}_\rho + \cos \varphi \underline{f}_E \cdot \underline{e}_3 = 0, \\
\underline{f}_E \times \underline{f}_N &= -\sin \varphi \underline{f}_E \times \underline{f}_\rho + \cos \varphi \underline{f}_E \times \underline{e}_3 = \sin \varphi \underline{e}_3 + \cos \varphi \underline{f}_\rho = \underline{f}_U, \\
[\underline{f}_E, \underline{f}_N, \underline{f}_U] &= \underline{f}_E \times \underline{f}_N \cdot \underline{f}_U = \underline{f}_U \cdot \underline{f}_U = 1.
\end{aligned}$$

Hence, for $\alpha, \beta \in \{E, N, U\}$, $\underline{f}_\alpha \cdot \underline{f}_\beta = \delta_{\alpha\beta}$ and $\underline{f}_E \times \underline{f}_N = \underline{f}_U$, $\underline{f}_N \times \underline{f}_U = \underline{f}_E$, $\underline{f}_U \times \underline{f}_E = \underline{f}_N$. Inverse of the last two definitions: $\cos \varphi \times \text{first} - \sin \varphi \times \text{second}$, and $\sin \varphi \times \text{first} + \cos \varphi \times \text{second}$,

$$\begin{aligned}
\cos \varphi \underline{f}_U - \sin \varphi \underline{f}_N &= (\cos^2 \varphi + \sin^2 \varphi) \underline{f}_\rho + (\cos \varphi \sin \varphi - \sin \varphi \cos \varphi) \underline{e}_3 = \underline{f}_\rho, \\
\sin \varphi \underline{f}_U + \cos \varphi \underline{f}_N &= (\sin \varphi \cos \varphi - \cos \varphi \sin \varphi) \underline{f}_\rho + (\sin^2 \varphi + \cos^2 \varphi) \underline{e}_3 = \underline{e}_3,
\end{aligned}$$

and in the equatorial plane in the same way

$$\begin{aligned}\cos \theta \, \underline{f}_\rho - \sin \theta \, \underline{f}_E &= (\cos^2 \theta + \sin^2 \theta) \, \underline{e}_1 + (\cos \theta \sin \theta - \sin \theta \cos \theta) \, \underline{e}_2 = \underline{e}_1, \\ \sin \theta \, \underline{f}_\rho + \cos \theta \, \underline{f}_E &= (\sin \theta \cos \theta - \cos \theta \sin \theta) \, \underline{e}_1 + (\sin^2 \theta + \cos^2 \theta) \, \underline{e}_2 = \underline{e}_2.\end{aligned}$$

O^+ : fixed basis $\{\underline{e}_E^+, \underline{e}_N^+, \underline{e}_U^+\}$ of $\mathbb{E}^{3+}$, $\underline{e}_\alpha^+ \cdot \underline{e}_\beta^+ = \delta_{\alpha\beta}$, $\underline{e}_E^+ \times \underline{e}_N^+ = \underline{e}_U^+$ cyclically. Q of (7.3) with $Q\underline{f}_\alpha = \underline{e}_\alpha^+$, by Lemma 2.15:

$$\begin{aligned}\underline{Q} &= \underline{e}_\alpha^+ \otimes \underline{f}_\alpha, & \underline{Q}^t &= \underline{f}_\alpha \otimes \underline{e}_\alpha^+, & \underline{Q}\underline{f}_\beta &= \underline{e}_\alpha^+ \underline{f}_\alpha \cdot \underline{f}_\beta = \underline{e}_\alpha^+ \delta_{\alpha\beta} = \underline{e}_\beta^+, \\ \underline{Q}\underline{Q}^t &= (\underline{e}_\alpha^+ \otimes \underline{f}_\alpha)(\underline{f}_\beta \otimes \underline{e}_\beta^+) = \underline{f}_\alpha \cdot \underline{f}_\beta \, \underline{e}_\alpha^+ \otimes \underline{e}_\beta^+ \\ &= \delta_{\alpha\beta} \, \underline{e}_\alpha^+ \otimes \underline{e}_\beta^+ = \underline{e}_\alpha^+ \otimes \underline{e}_\alpha^+ = \underline{I}^+, \\ \underline{Q}^t \underline{Q} &= (\underline{f}_\alpha \otimes \underline{e}_\alpha^+)(\underline{e}_\beta^+ \otimes \underline{f}_\beta) = \underline{e}_\alpha^+ \cdot \underline{e}_\beta^+ \, \underline{f}_\alpha \otimes \underline{f}_\beta \\ &= \delta_{\alpha\beta} \, \underline{f}_\alpha \otimes \underline{f}_\beta = \underline{f}_\alpha \otimes \underline{f}_\alpha = \underline{I}.\end{aligned}$$

Station at the origin of $\mathbb{E}^{3+}$:

$$\underline{0} = \underline{Q}\underline{o}^+ + \underline{c} = R \underline{Q}\underline{f}_U + \underline{c} = R \underline{e}_U^+ + \underline{c} \implies \underline{c} = -R \underline{e}_U^+, \quad \dot{\underline{c}} = \ddot{\underline{c}} = \underline{0}.$$

Rates, with $\dot{\theta} = \omega$, $\dot{\varphi} = 0$, $\dot{\underline{e}}_i = \underline{0}$, and the inverse relation for $\underline{f}_\rho$:

$$\begin{aligned}\dot{\underline{f}}_\rho &= \dot{\theta}(-\sin \theta \, \underline{e}_1 + \cos \theta \, \underline{e}_2) = \omega \, \underline{f}_E, \\ \dot{\underline{f}}_E &= \dot{\theta}(-\cos \theta \, \underline{e}_1 - \sin \theta \, \underline{e}_2) = -\omega \, \underline{f}_\rho = -\omega \cos \varphi \, \underline{f}_U + \omega \sin \varphi \, \underline{f}_N, \\ \dot{\underline{f}}_N &= -\sin \varphi \, \dot{\underline{f}}_\rho + \cos \varphi \, \dot{\underline{e}}_3 = -\omega \sin \varphi \, \underline{f}_E, \\ \dot{\underline{f}}_U &= \cos \varphi \, \dot{\underline{f}}_\rho + \sin \varphi \, \dot{\underline{e}}_3 = \omega \cos \varphi \, \underline{f}_E.\end{aligned}$$

Spin, (7.8)₂ and Lemma 2.15:

$$\begin{aligned}\dot{\underline{Q}} &= \dot{\underline{e}}_\alpha^+ \otimes \dot{\underline{f}}_\alpha, & \underline{\Omega} &= \dot{\underline{Q}}\underline{Q}^t = (\dot{\underline{e}}_\alpha^+ \otimes \dot{\underline{f}}_\alpha)(\underline{f}_\beta \otimes \underline{e}_\beta^+) \\ &= \dot{\underline{f}}_\alpha \cdot \underline{f}_\beta \, \underline{e}_\alpha^+ \otimes \underline{e}_\beta^+ = \Omega_{\alpha\beta} \, \underline{e}_\alpha^+ \otimes \underline{e}_\beta^+, \\ \Omega_{\alpha\beta} &= \dot{\underline{f}}_\alpha \cdot \underline{f}_\beta : & \Omega_{EE} &= \dot{\underline{f}}_E \cdot \underline{f}_E = 0, & \Omega_{EN} &= \dot{\underline{f}}_E \cdot \underline{f}_N = \omega \sin \varphi, \\ \Omega_{EU} &= \dot{\underline{f}}_E \cdot \underline{f}_U = -\omega \cos \varphi, & \Omega_{NE} &= \dot{\underline{f}}_N \cdot \underline{f}_E = -\omega \sin \varphi, \\ \Omega_{UE} &= \dot{\underline{f}}_U \cdot \underline{f}_E = \omega \cos \varphi, & \Omega_{NN} &= \Omega_{NU} = \Omega_{UN} = \Omega_{UU} = 0, \\ [\Omega_{\alpha\beta}] &= \omega \begin{pmatrix} 0 & \sin \varphi & -\cos \varphi \\ -\sin \varphi & 0 & 0 \\ \cos \varphi & 0 & 0 \end{pmatrix} = -[\Omega_{\alpha\beta}]^t, \\ \Omega_{\alpha\beta} + \Omega_{\beta\alpha} &= \dot{\underline{f}}_\alpha \cdot \underline{f}_\beta + \underline{f}_\alpha \cdot \dot{\underline{f}}_\beta = \frac{d}{dt} \underline{f}_\alpha \cdot \underline{f}_\beta = \frac{d}{dt} \delta_{\alpha\beta} = 0.\end{aligned}$$

Axial vector, Proposition 1.44 with (E, N, U) as $(1, 2, 3)$:

$$\begin{aligned}
w_\gamma &= -\frac{1}{2}\varepsilon_{\gamma\alpha\beta}\Omega_{\alpha\beta}, & w_E &= -\frac{1}{2}(\Omega_{NU} - \Omega_{UN}) = 0, \\
w_N &= -\frac{1}{2}(\Omega_{UE} - \Omega_{EU}) = -\frac{1}{2}(\omega \cos \varphi + \omega \cos \varphi) = -\omega \cos \varphi, \\
w_U &= -\frac{1}{2}(\Omega_{EN} - \Omega_{NE}) = -\frac{1}{2}(\omega \sin \varphi + \omega \sin \varphi) = -\omega \sin \varphi, \\
\underline{w} &= -\omega(\cos \varphi \underline{e}_N^+ + \sin \varphi \underline{e}_U^+) =: -\underline{\omega}, \\
\Omega \underline{s} &= \underline{w} \times \underline{s} = -\underline{\omega} \times \underline{s} \quad \forall \underline{s} \in \mathbb{E}^{3+}, \\
Q \underline{e}_3 &= Q(\sin \varphi \underline{f}_U + \cos \varphi \underline{f}_N) = \sin \varphi \underline{e}_U^+ + \cos \varphi \underline{e}_N^+ = \underline{\omega}/\omega, \\
\dot{\underline{\omega}} &= \underline{0} \implies \dot{\underline{\Omega}} = \underline{Q}^+.
\end{aligned}$$

Sign: (7.8) with $\underline{v} = \underline{0}$, $\dot{\underline{c}} = \underline{0}$ gives $\underline{v}^+ = \underline{\Omega}(\underline{x}^+ - \underline{c}) = -\underline{\omega} \times (\underline{x}^+ - \underline{c})$, a point at rest for O circles the axis backwards for O^+ . θ_0 enters $\underline{Q}$ only, not $\underline{\Omega}$.

(7.63) with $\ddot{\underline{c}} = \dot{\underline{c}} = \underline{0}$, $\dot{\underline{\Omega}} = \underline{Q}^+$, $\underline{s}^+ := \underline{x}^+ - \underline{c} = \underline{x}^+ + R \underline{e}_U^+$:

$$\begin{aligned}
\underline{i}^+ &= 2\underline{\Omega}\underline{v}^+ - \underline{\Omega}^2 \underline{s}^+, & \underline{\Omega}\underline{v}^+ &= -\underline{\omega} \times \underline{v}^+, \\
\underline{\Omega}^2 \underline{s}^+ &= \underline{\Omega}(\underline{\Omega}\underline{s}^+) = \underline{\Omega}(-\underline{\omega} \times \underline{s}^+) = -\underline{\omega} \times (-\underline{\omega} \times \underline{s}^+) = \underline{\omega} \times (\underline{\omega} \times \underline{s}^+),
\end{aligned}$$

$$\begin{aligned}
\operatorname{div}^+ \underline{T}^+ + \rho^+ (\underline{b}^+ + \underline{i}^+) &= \rho^+ \underline{a}^+, \\
\underline{i}^+ &= -2\underline{\omega} \times \underline{v}^+ - \underline{\omega} \times (\underline{\omega} \times \underline{s}^+), & \underline{\omega} &= \omega(\cos \varphi \underline{e}_N^+ + \sin \varphi \underline{e}_U^+).
\end{aligned}$$

(7.59), (7.60) and the jump conditions are unchanged.

Centrifugal term, by $\underline{a} \times (\underline{b} \times \underline{c}) = \underline{a} \cdot \underline{c} \underline{b} - \underline{a} \cdot \underline{b} \underline{c}$ and $|\underline{\omega}|^2 = \omega^2(\cos^2 \varphi + \sin^2 \varphi) = \omega^2$:

$$-\underline{\omega} \times (\underline{\omega} \times \underline{s}^+) = -\underline{\omega} \cdot \underline{s}^+ \underline{\omega} + \underline{\omega} \cdot \underline{\omega} \underline{s}^+ = \omega^2 \underline{s}^+ - \underline{\omega} \cdot \underline{s}^+ \underline{\omega},$$

ω^2 times the part of $\underline{s}^+$ perpendicular to $\underline{\omega}$. At the station $\underline{x}^+ = \underline{0}$, $\underline{s}^+ = R \underline{e}_U^+$, $\underline{\omega} \cdot \underline{e}_U^+ = \omega \sin \varphi$:

$$\begin{aligned}
-\underline{\omega} \times (\underline{\omega} \times R \underline{e}_U^+) &= \omega^2 R \underline{e}_U^+ - R \omega \sin \varphi \cdot \omega (\cos \varphi \underline{e}_N^+ + \sin \varphi \underline{e}_U^+) \\
&= \omega^2 R (1 - \sin^2 \varphi) \underline{e}_U^+ - \omega^2 R \sin \varphi \cos \varphi \underline{e}_N^+ \\
&= \omega^2 R \cos \varphi (\cos \varphi \underline{e}_U^+ - \sin \varphi \underline{e}_N^+),
\end{aligned}$$

$$|\dots| = \omega^2 R \cos \varphi \sqrt{\cos^2 \varphi + \sin^2 \varphi} = \omega^2 R \cos \varphi,$$

$$\cos \varphi \underline{e}_U^+ - \sin \varphi \underline{e}_N^+ = \underline{Q}(\cos \varphi \underline{f}_U - \sin \varphi \underline{f}_N) = \underline{Q} \underline{f}_\rho,$$

$\omega^2 \times$ the distance $R \cos \varphi$ from the axis, along the outward horizontal of the equatorial plane (Figure J(a)). With gravity $-g \underline{e}_U^+$,

$$\begin{aligned} \underline{g}_{\text{eff}} &= -g \underline{e}_U^+ + \omega^2 R \cos \varphi (\cos \varphi \underline{e}_U^+ - \sin \varphi \underline{e}_N^+) \\ &= -(g - \omega^2 R \cos^2 \varphi) \underline{e}_U^+ - \omega^2 R \sin \varphi \cos \varphi \underline{e}_N^+, \\ \tan \delta &= \frac{|\text{horizontal part}|}{|\text{vertical part}|} = \frac{\omega^2 R \sin \varphi \cos \varphi}{g - \omega^2 R \cos^2 \varphi}, \end{aligned}$$

Figure J(b),(c), Remark 7.30: the term is uniform and $\propto \rho^+$ like gravity, hence indistinguishable from it locally.

Coriolis term, $\underline{v}^+ = v_E \underline{e}_E^+ + v_N \underline{e}_N^+ + v_U \underline{e}_U^+$:

$$\begin{aligned} \underline{e}_N^+ \times \underline{v}^+ &= v_E \underline{e}_N^+ \times \underline{e}_E^+ + v_U \underline{e}_N^+ \times \underline{e}_U^+ = -v_E \underline{e}_U^+ + v_U \underline{e}_E^+, \\ \underline{e}_U^+ \times \underline{v}^+ &= v_E \underline{e}_U^+ \times \underline{e}_E^+ + v_N \underline{e}_U^+ \times \underline{e}_N^+ = v_E \underline{e}_N^+ - v_N \underline{e}_E^+, \\ -2\omega \times \underline{v}^+ &= -2\omega \cos \varphi \underline{e}_N^+ \times \underline{v}^+ - 2\omega \sin \varphi \underline{e}_U^+ \times \underline{v}^+ \\ &= -2\omega \cos \varphi (v_U \underline{e}_E^+ - v_E \underline{e}_U^+) - 2\omega \sin \varphi (-v_N \underline{e}_E^+ + v_E \underline{e}_N^+) \\ &= 2\omega \left[(v_N \sin \varphi - v_U \cos \varphi) \underline{e}_E^+ - v_E \sin \varphi \underline{e}_N^+ + v_E \cos \varphi \underline{e}_U^+ \right]. \end{aligned}$$

$\sin \varphi > 0$: $v_N > 0 \Rightarrow$ push along $+\underline{e}_E^+$, $v_E > 0 \Rightarrow$ push along $-\underline{e}_N^+$; both to the right of the motion, reversed for $\sin \varphi < 0$.

Earth: $\omega = 2\pi/86164 \text{ s} = 7.29 \times 10^{-5} \text{ s}^{-1}$, $R = 6.37 \times 10^6 \text{ m}$, $g = 9.81 \text{ m s}^{-2}$:

$$\begin{aligned} \omega^2 R &= (7.29 \times 10^{-5})^2 \times 6.37 \times 10^6 = 5.32 \times 10^{-9} \times 6.37 \times 10^6 \\ &= 3.39 \times 10^{-2} \text{ m s}^{-2} = 0.0035 g, \\ 2\omega |\underline{v}^+| &= 1.46 \times 10^{-4} \text{ m s}^{-2} \quad \text{at } |\underline{v}^+| = 1 \text{ m s}^{-1}, \\ \tan \delta &= \frac{\varepsilon \sin \varphi \cos \varphi}{1 - \varepsilon \cos^2 \varphi} = \frac{\frac{1}{2}\varepsilon \sin 2\varphi}{1 - \varepsilon \cos^2 \varphi}, \quad \varepsilon = \frac{\omega^2 R}{g} = 0.00345, \\ \frac{d}{d\varphi} \tan \delta &= 0 \Leftrightarrow \varepsilon \cos 2\varphi + O(\varepsilon^2) = 0 \Leftrightarrow \varphi = 45^\circ + O(\varepsilon), \\ \tan \delta_{\text{max}} &= \frac{\varepsilon/2}{1 - \varepsilon/2} = 1.73 \times 10^{-3}, \quad \delta_{\text{max}} = 0.099^\circ = 5.9'. \end{aligned}$$

Negligible against g in a laboratory; decisive in the ocean and the atmosphere, where the comparison is with the residual accelerations left after gravity is balanced by pressure. $\square$

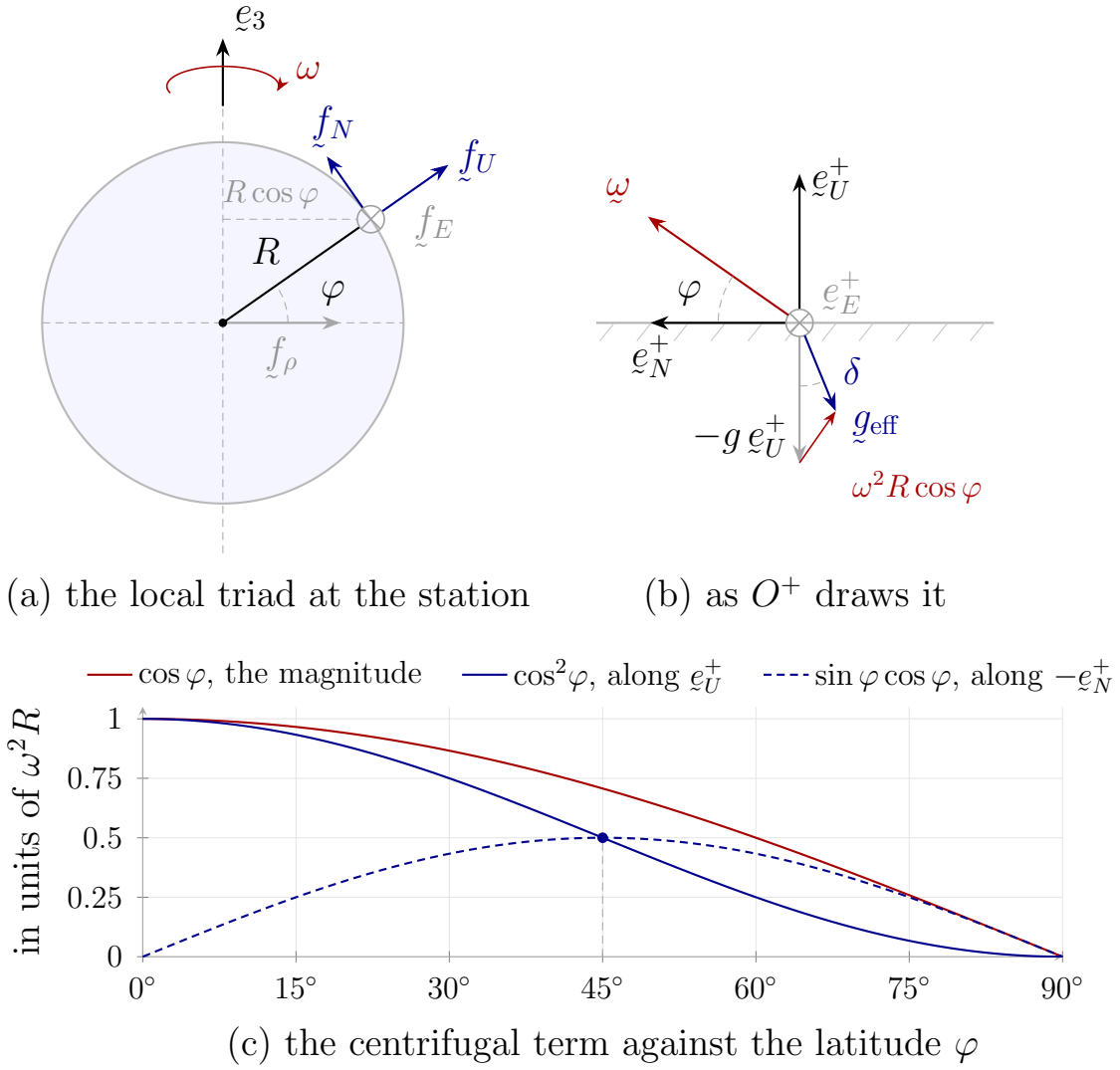


Figure J: The geometry of Problem 3 at latitude $\varphi = 35^\circ$. (a) The meridional section through the station: the planet turns at ω about e_3 , f_ρ is the equatorial radial direction at the station's longitude, and $f_U = \cos \varphi f_\rho + \sin \varphi e_3$, $f_N = -\sin \varphi f_\rho + \cos \varphi e_3$, with f_E into the page, is the local triad; the station is R from the centre and $R \cos \varphi$ from the axis, and that second distance is the whole of the centrifugal term. (b) The same picture as O^+ draws it, e_U^+ vertical, north to the left, east into the page: the angular velocity ω points at the celestial pole, at elevation φ above the northern horizon, and the shifter triad of Remark 7.17 turns backwards about it once a sidereal day; the centrifugal term, perpendicular to the axis, tilts the effective gravity g_{eff} from $-ge_U^+$ towards the equator by δ , so a plumb line does not point at the centre of the planet ($\omega^2 R/g$ is drawn as 0.55; for the Earth it is 0.0035). (c) Magnitude and components of the centrifugal term against latitude.

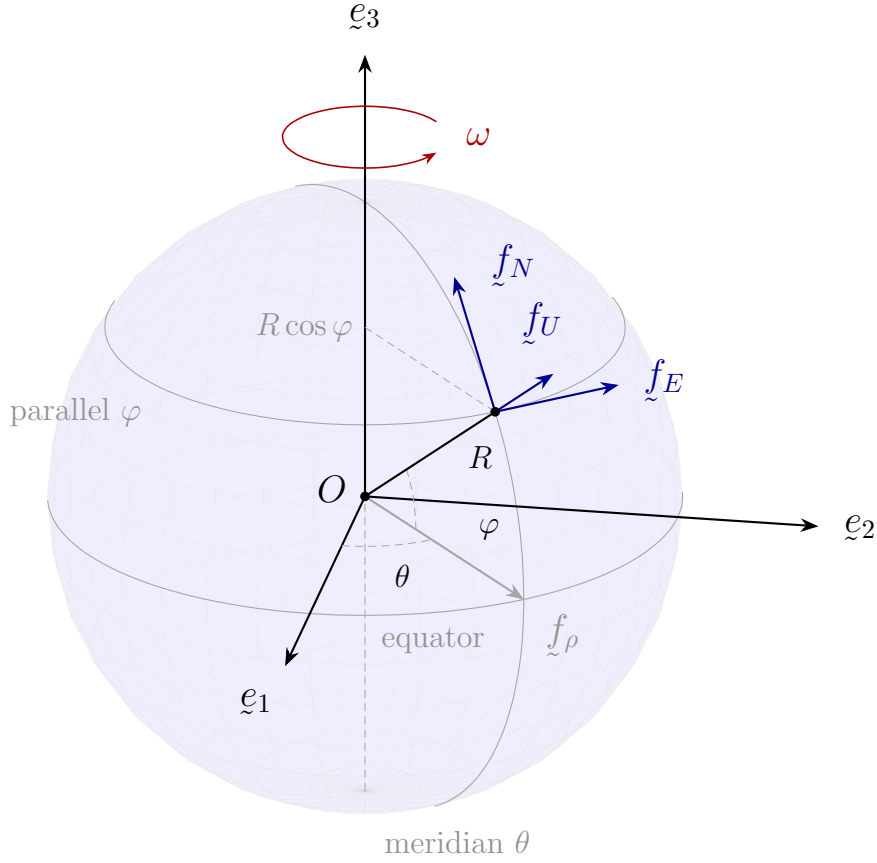


Figure K: Figure J(a) in three dimensions: what is where, for a station at longitude $\theta = 40^\circ$ and latitude $\varphi = 35^\circ$. O is the inertial observer at the centre with the fixed triad $\{\underline{e}_i\}$, $\underline{e}_3$ the axis of the spin ω . The station lies on the meridian of longitude $\theta = \omega t + \theta_0$, measured in the equatorial plane from $\underline{e}_1$, and on the parallel of latitude φ , measured from the equator along that meridian; $\underline{f}_\rho$ is the radial direction in the equatorial plane at that longitude. The station is R from O and $R \cos \varphi$ from the axis. At the station, $\underline{f}_U$ points straight up along O –station, $\underline{f}_N$ points north along the meridian and $\underline{f}_E$ points east along the parallel; only the near side of the three circles is drawn.

Chapter 8

Constitutive Equations

The balance laws of Chapter 6 are true of every material. Mass is conserved in steel and in honey; linear momentum balances in glass and in air. That universality is what made them worth proving once and for all, and it is also precisely why they cannot, by themselves, predict anything at all. A law that does not distinguish steel from honey cannot tell you what either of them will do.

The deficit can be counted exactly, and the count is the reason this chapter exists. The unknown fields at a point of the body are the mass density ρ , the motion $\underline{\chi}$ or equivalently the velocity $\underline{v}$ (three components), the temperature θ , the Cauchy stress $\underline{T}$ (six components, the symmetry $\underline{T} = \underline{T}^t$ having been extracted in Chapter 6 from the balance of moment of momentum), the heat flux $\underline{q}$ (three), the internal energy density ε and the entropy density η . That is

$$1 + 3 + 1 + 6 + 3 + 1 + 1 = 16$$

scalar fields. Against them Chapter 6 supplies conservation of mass (one equation), the balance of linear momentum (three) and the balance of energy (one): five. The Clausius–Duhem inequality (6.187) adds nothing to the tally, since an inequality selects among candidates but determines none of them.

Eleven relations are therefore missing, and it is no accident that the set $C' = \{\underline{T}, \underline{q}, \varepsilon, \eta\}$ of (8.2) below has exactly eleven components. Supply those eleven as functions of the rest and the system closes: five

equations left, five unknowns left, namely ρ , χ and θ . The eleven relations are the *constitutive equations*, and they are where the particular material enters the theory. Everything before this chapter was true of all matter; everything in it is true of some matter and false of the rest.

Such relations are idealizations. They are not deduced from anything more fundamental within continuum mechanics, and they are judged solely by whether the predictions they deliver agree with what is measured. For any real substance that agreement is to be expected only within a limited range of conditions, so a constitutive equation encodes both the material and the circumstances: the same steel is elastic under a light load, plastic under a heavy one, and molten at 1700 degrees, and no single relation covers all three. This book treats only the classical cases, viscous and inviscid fluids and elastic solids, which are the most fully developed and most widely used, and which are the prerequisite for everything more elaborate.

What this chapter does *not* do is write down a particular constitutive equation. It asks the prior question: what may a constitutive equation look like at all? Four answers are given to it, and only the first is a free modelling choice.

- *Locality*, Section 8.1: the response at a material point is settled by what happens in an arbitrarily small neighbourhood of that point, and, in the class treated here, by the values of the deformation, the velocity and the temperature there to first order. This is an assumption, and a restrictive one.
- *Material symmetry*, Section 8.2: certain rearrangements of the reference neighbourhood are undetectable by any experiment, and the set of them is a group. Whether that group consists of rotations or of all volume-preserving maps is the difference between a solid and a fluid.
- *Noll's rule*, Section 8.3: the symmetry group belongs jointly to the material and to the reference configuration, and changing the reference conjugates the group. This is a theorem, not a choice.

- *Frame invariance*, Section 8.4: two observers must agree about the material, and the demand that they use the same response function restricts that function. This too is forced.

A fifth restriction, compatibility with the Clausius–Duhem inequality of Section 6.7, is stated at the end of the chapter and exploited in the two that follow.

8.1 Elastic materials with heat conduction and viscosity

The quantities for which constitutive equations are sought are the present values of the stress $\underline{T}$, the heat flux $\underline{q}$, the energy density ε and the entropy density η , all at a material point $p \in B$ whose position in the reference configuration κ is $\underline{X}$. We construct these relations from the standpoint of a single observer O , and only in Section 8.4 bring in a second observer O^+ and ask what the two must agree about.

The order matters, and it is worth saying why we do not impose observer agreement from the start. Material frame indifference is the requirement that constitutive relations written by O^+ describe the same material as those written by O . If they did not, then what a substance *is* would depend on who was watching it, and every laboratory would have to publish its own elasticity of copper. Experience says otherwise: material behaviour is intrinsic. Agreement between the two observers is therefore not a further physical hypothesis but a consistency condition, and consistency conditions are cleanest to apply to a structure that already exists. So we build the structure first.

Locality, and the expansions it licenses

The first restriction to suggest itself is *locality*: the response of the material at p should depend on the values of the variables constituting a process in a neighbourhood of p , that is, in a neighbourhood of $\underline{X}$, and

on nothing further afield. This rules out action at a distance: what the material does here is not directly affected by what is happening a metre away, except through the intervening material.

We narrow this further. The processes to which the material is deemed sensitive are taken to consist of the current deformation $\chi_\kappa(\underline{X}', t)$, the current material velocity $\hat{v}(\underline{X}', t)$ and the current temperature $\hat{\theta}(\underline{X}', t)$ for all $\underline{X}' \in \kappa$ in an arbitrarily small neighbourhood of $\underline{X}$. The carets are the ones introduced in Remark 2.6: they select the referential description, the function whose input is a place in κ , which is the only description in which “a neighbourhood of the material point” is a neighbourhood of a fixed set.

Assuming these to be smooth functions of position in κ , each may be expanded about $\underline{X}$. The expansions are ordinary first-order Taylor expansions, and the coefficient of $\underline{X}' - \underline{X}$ in each is the referential gradient of the quantity being expanded, evaluated at $\underline{X}$:

$$\begin{aligned}\chi_\kappa(\underline{X}', t) &= \chi_\kappa(\underline{X}, t) + \underline{F}(\underline{X}, t)(\underline{X}' - \underline{X}) + o(|\underline{X}' - \underline{X}|), \\ \hat{v}(\underline{X}', t) &= \hat{v}(\underline{X}, t) + \dot{\underline{F}}(\underline{X}, t)(\underline{X}' - \underline{X}) + o(|\underline{X}' - \underline{X}|), \\ \hat{\theta}(\underline{X}', t) &= \hat{\theta}(\underline{X}, t) + (\text{Grad } \hat{\theta}(\underline{X}, t)) \cdot (\underline{X}' - \underline{X}) + o(|\underline{X}' - \underline{X}|).\end{aligned}\tag{8.1}$$

Each of the three coefficients has already been computed in an earlier chapter, and it is worth saying which is which, because the middle one is the only one that is not a definition.

The first line is the statement that $\underline{F} = \text{Grad } \chi_\kappa$, written as an expansion; it is (3.102), the local approximation of an arbitrary deformation by a homogeneous one.

The second line has coefficient $\text{Grad } \hat{v}$, and (4.2) showed that $\text{Grad } \hat{v} = \dot{\underline{F}}$: differentiating χ_i with respect to X_A and then with respect to t gives the same thing in either order, so the referential gradient of the velocity is the material time derivative of the deformation gradient. That interchange, and hence the appearance of $\dot{\underline{F}}$ rather than some new tensor, is the whole content of the second line.

The third line is the definition of the gradient of a scalar field, Theorem 1.70, transplanted to κ .

The symbol $o(|\underline{\tilde{X}}' - \underline{\tilde{X}}|)$ is the Landau symbol of Definition 1.66: a quantity which, divided by $|\underline{\tilde{X}}' - \underline{\tilde{X}}|$, tends to zero as $\underline{\tilde{X}}' \rightarrow \underline{\tilde{X}}$. It is what makes “for $|\underline{\tilde{X}}' - \underline{\tilde{X}}|$ sufficiently small these are dominated by the terms shown” a precise statement rather than a hope.

REMARK 8.1 (what the truncation is, and what it costs). Nothing so far forces a stopping point. One could keep the next term in each expansion, which for the first line would be the second gradient $\text{Grad } \underline{\tilde{F}}$, and obtain a wider class of models. Truncating at first order is a modelling decision and should be recognised as one.

The decision says: whatever the material at p is sensitive to, it is captured to sufficient accuracy by the value and the first derivative of the deformation, the velocity and the temperature at p . If two processes agree in those data at p , the material at p cannot tell them apart.

What it costs is every phenomenon carried by the higher gradients. A material whose response involves $\text{Grad } \underline{\tilde{F}}$ has an intrinsic length scale, since $\text{Grad } \underline{\tilde{F}}$ carries dimensions of one over length while $\underline{\tilde{F}}$ is dimensionless; such theories are used for surface tension, for the width of a phase boundary and for the size effects seen in very small specimens. None of that is available here. In exchange, the theory becomes a theory of *fields at a point*, and the whole apparatus of Sections 8.2 to 8.4 becomes statements about tensors rather than about functionals.

A material whose response depends on the deformation only through $\underline{\tilde{F}}$ is called *simple*, following Noll. The present class is that of simple materials with heat conduction and viscosity.

The two lists

Accordingly, if we mean to account for effects at this order, we assume the constitutive response of the material, consisting of the set

$$C' = \{\underline{\tilde{T}}, \underline{\tilde{q}}, \varepsilon, \eta\}, \quad (8.2)$$

evaluated at $\underline{\tilde{X}}$ and t , to be determined by the list

$$L' = \{\underline{\tilde{x}}, \underline{\tilde{v}}, \theta, \underline{\tilde{F}}, \underline{\tilde{F}}, \text{Grad } \theta\}, \quad (8.3)$$

also evaluated at $\underline{X}$ and t . Here $\underline{x} = \underline{\chi}_\kappa(\underline{X}, t)$, $\underline{v} = \underline{\hat{v}}(\underline{X}, t)$ and $\theta = \hat{\theta}(\underline{X}, t)$: the list contains the values at $\underline{X}$ from (8.1) together with the three gradients that accompany them.

It will prove more convenient to work not with C' but with the equivalent set

$$C = \{\underline{T}, q, \psi, \eta\}, \quad (8.4)$$

where ψ is the Helmholtz free energy per unit mass, defined in Chapter 6 by (6.186),

$$\psi = \varepsilon - \theta\eta.$$

The two sets carry the same information. Given θ , which is in the list L' and therefore always available, the relation $\psi = \varepsilon - \theta\eta$ determines ψ from (ε, η) and $\varepsilon = \psi + \theta\eta$ recovers ε from (ψ, η) ; the map is a bijection with η and θ held fixed. Nothing is gained or lost in the exchange, which is why (8.4) may replace (8.2) without comment.

What is gained is convenience later. The local Clausius–Duhem inequality was reduced in Chapter 6 to (6.187),

$$-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{T} \cdot \underline{L} - \theta^{-1}q \cdot \text{grad } \theta \geq 0,$$

in which ψ , not ε , is the energy that appears. Since that inequality is the instrument by which the chapters that follow will extract restrictions on the constitutive functions, it pays to have the constitutive list already written in the variable the inequality uses. The exchange $\varepsilon \mapsto \psi$ is a Legendre transformation trading the entropy for the temperature as the independent thermal variable, and θ is what appears in the list (8.3), not η .

Two gradients of the temperature

The list L' contains $\text{Grad } \theta$, a referential vector. The list we shall actually use contains $\text{grad } \theta$, a spatial one. They determine each other, and the relation between them is worth deriving rather than quoting, since the same two-line argument recurs three more times in this chapter.

Let $d\tilde{X}$ be a material line element at $\tilde{X}$ and $d\tilde{x} = \tilde{F}d\tilde{X}$ its image, (2.34). The increment of temperature between the two ends of that element is one number, and each description computes it its own way:

$$\text{Grad } \hat{\theta} \cdot d\tilde{X} = d\theta = \text{grad } \tilde{\theta} \cdot d\tilde{x} = \text{grad } \tilde{\theta} \cdot \tilde{F} d\tilde{X}, \quad (8.5)$$

the first and third equalities being the definition of the gradient in κ and in κ_t respectively, and the last the substitution $d\tilde{x} = \tilde{F}d\tilde{X}$. Now move $\tilde{F}$ across the inner product using the definition of the transpose, Definition 1.41, $a \cdot \tilde{F}b = \tilde{F}^t a \cdot b$:

$$\text{grad } \tilde{\theta} \cdot \tilde{F} d\tilde{X} = \tilde{F}^t \text{grad } \tilde{\theta} \cdot d\tilde{X}.$$

The two ends of (8.5) are therefore inner products of $d\tilde{X}$ with two fixed vectors, and they agree:

$$(\text{Grad } \hat{\theta} - \tilde{F}^t \text{grad } \tilde{\theta}) \cdot d\tilde{X} = 0.$$

The element $d\tilde{X}$ was arbitrary, so Lemma 5.1, the constant-vector argument of Chapter 5, applies and the bracket vanishes:

$$\text{Grad } \hat{\theta} = \tilde{F}^t \text{grad } \tilde{\theta}. \quad (8.6)$$

This is Theorem 2.14 of Chapter 2 specialised to a scalar field. Note that it is the arbitrariness of $d\tilde{X}$ that does the work, exactly as Remark 5.2 warned: from a single $d\tilde{X}$ one could conclude nothing, since any vector perpendicular to it would satisfy the display.

Since $\tilde{F}$ is invertible, (8.6) may be read either way, $\text{grad } \tilde{\theta} = \tilde{F}^{-t} \text{Grad } \hat{\theta}$, so the two gradients determine each other once $\tilde{F}$ is known. And $\tilde{F}$ is known: it is already in the list. The same is true of the other exchange we are about to make. By (4.3), $\tilde{L} = \dot{\tilde{F}} \tilde{F}^{-1}$, so $\dot{\tilde{F}} = \tilde{L} \tilde{F}$ and $\tilde{L} = \dot{\tilde{F}} \tilde{F}^{-1}$ pass between $\dot{\tilde{F}}$ and $\tilde{L}$ in either direction, again using only $\tilde{F}$. Hence L' is equivalent to the list

$$L = \{\tilde{x}, \tilde{v}, \theta, \tilde{F}_\kappa, \tilde{L}, \text{grad } \theta\}, \quad (8.7)$$

where $\tilde{L}$ is the spatial velocity gradient of (4.3) and the subscript κ has been attached to the deformation gradient to record the reference

configuration relative to which it is computed. None of the other entries of (8.7) depends on a reference configuration: $\underline{x}$, $\underline{v}$, θ , $\underline{L}$ and $\text{grad } \theta$ are all spatial, and $\underline{L}$ in particular was shown in Remark 4.2 to be indifferent to the choice of κ . The deformation gradient is the sole entry that is not, and the whole of Sections 8.2 and 8.3 is the consequence of that single fact.

REMARK 8.2 (three letters that look alike). Equation (8.7) contains both L and $\underline{L}$, and they are unrelated. The plain italic L is the *list*, a set of six entries; the under-tilde $\underline{L}$ is the spatial velocity gradient, one of them. The book distinguishes them by boldface, we by the under-tilde, and the distinction is not decorative: the sentence “ $\underline{L} \in L$ ” is true and “ $L \in L$ ” is nonsense.

The same warning will be needed twice more. From (8.10) onward, F without an under-tilde denotes a generic member of the response set C , so that F may stand for the stress, the heat flux, the free energy or the entropy, while $\underline{F}$ remains the deformation gradient and F_κ is a *function*, not a value. And κ denotes the reference configuration while $\underline{\kappa}$ denotes the map $p \mapsto \underline{X}$ that produces it. In each pair the two objects sit in the same formula, so reading the decoration is not optional.

Why this particular list

The constitutive assumption adopted here may be motivated by an analogy with a spring and a damper. The force in a spring is fixed by its present elongation; the force in a damper is fixed by the present rate of elongation. A device made of the two in parallel responds to the pair (elongation, rate of elongation) and to nothing else, in particular not to how it arrived at that state. The list (8.7) is the continuum version of that pair: $\underline{F}_\kappa$ is the local elongation and $\underline{L}$ its rate. So the class of materials studied here is the class that exhibits both elasticity and viscosity, and we should expect the constitutive functions to split, eventually, into a part depending on $\underline{F}$ that stores energy and a part depending on $\underline{L}$ that dissipates it. The chapters that follow carry that split out, Chapter 9 doing it for viscous fluids first.

REMARK 8.3 (what this class cannot do). The analogy also marks the boundary. A real polymer stretched and held does not settle instantly to the force its present elongation would dictate; the force decays over minutes or hours. Its stress at time t depends on the whole history $\tilde{F}(s)$ for $s \leq t$, not on $\tilde{F}(t)$ and $\dot{\tilde{F}}(t)$ alone. The response is then a *functional* on a space of histories rather than a function on a finite list, and the entire apparatus of this chapter, which manipulates the arguments of a function, has to be rebuilt.

That is the general theory of materials with memory, and it lies outside the scope of this book. The class treated here is not general viscoelasticity, and it should not be represented as such. It is, however, enough for the two applications that follow, and it is the case in which every idea of the subject can be seen without the analytic machinery that histories require.

8.2 Change of reference configuration and material symmetry

The list L of variables presumed to constitute the relevant process depends on the choice of reference configuration, and depends on it through exactly one entry, the deformation gradient. To see what that dependence does to the theory, change the reference configuration from κ to μ (Figure 8.1).

Because the body and any reference configuration are in one-to-one correspondence, there is an invertible map $\mu(p)$ taking $p \in B$ to $\tilde{Y} \in \mu$. And B and κ are in one-to-one correspondence as well, so the same is true of κ and μ : composing the inverse of one placement with the other gives a map from κ straight to μ that never mentions B . We assume it smooth, and call it λ :

$$\tilde{Y} = \lambda(\tilde{X}) \quad \text{and} \quad d\tilde{Y} = \tilde{R}(\tilde{X})d\tilde{X}, \quad \text{where } \tilde{R}(\tilde{X}) = \text{Grad } \lambda. \quad (8.8)$$

The second statement is the first differentiated: λ is a differentiable vector field on κ , so by Definition 1.85 its increment is $\lambda(\tilde{X} + d\tilde{X}) -$

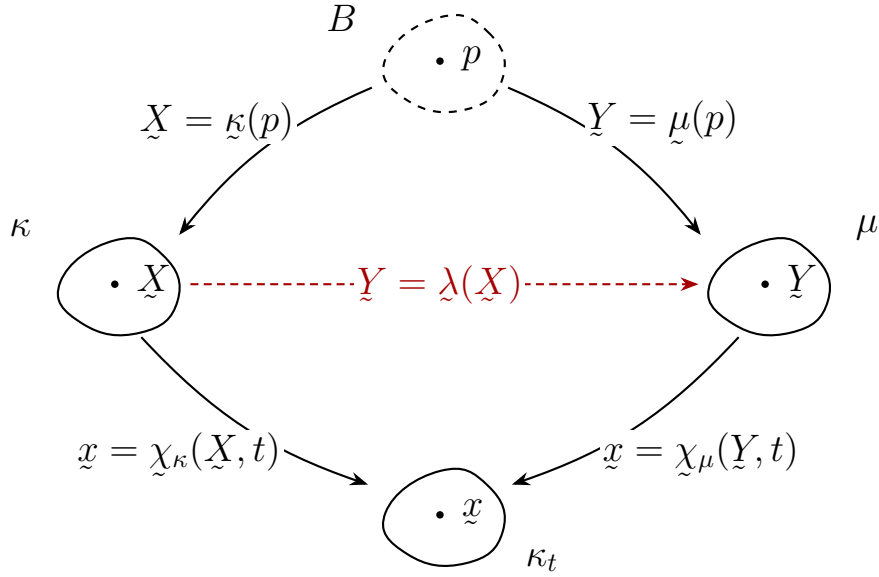


Figure 8.1: Change of reference configuration. The body B is a set of material points and is not a region of space; κ and μ are two ways of laying it out in $\hat{\mathbb{E}}^3$, and χ_κ , χ_μ are the two descriptions of one and the same motion. Because both placements are one to one, the diagram can be closed along the top: $\lambda = \mu \circ \kappa^{-1}$ takes κ to μ without passing through B , and it is the gradient of this map, and nothing about B itself, that enters (8.8) and (8.9). The current configuration κ_t is reached by two routes and does not care which was taken; that is the content of $F_\kappa = F_\mu R$.

$\lambda(\underline{X}) = (\text{Grad } \lambda) d\underline{X} + o(|d\underline{X}|)$, and the tensor $\text{Grad } \lambda$ carrying $d\underline{X}$ to $d\underline{Y}$ is what (8.8) calls $\underline{R}$.

The deformation gradients relative to κ and to μ are now related. Take one material line element and write its image in κ_t in the two ways the diagram permits:

$$\underline{F}_\kappa d\underline{X} = d\underline{x} = \underline{F}_\mu d\underline{Y} = \underline{F}_\mu \underline{R} d\underline{X}; \quad \text{hence} \quad \underline{F}_\kappa = \underline{F}_\mu \underline{R}. \quad (8.9)$$

The word “hence” is doing a small job here and it should be watched. What the first chain gives is $(\underline{F}_\kappa - \underline{F}_\mu \underline{R}) d\underline{X} = 0$ for the particular $d\underline{X}$ chosen; but $d\underline{X}$ was any material element at $\underline{X}$, so the tensor in the bracket annihilates every vector of $\hat{\mathbb{E}}_\kappa^3$, and a tensor that sends every vector to 0 is the zero tensor by Definition 1.31. That is what licenses cancelling $d\underline{X}$, and it is the same quantifier argument as in (8.6).

Two consequences of (8.9) are worth extracting at once. Taking determinants and using Theorem 1.52,

$$\det \underline{F}_\kappa = (\det \underline{F}_\mu)(\det \underline{R}),$$

and both deformation gradients have positive determinant because both κ and μ are occupiable configurations; hence $\det \underline{R} > 0$ and $\underline{R}$ is invertible. Second, $\underline{R}$ does not depend on t : it is built from two reference configurations, neither of which moves.

REMARK 8.4 (this $\underline{R}$ is not the rotation of the polar decomposition). The letter is unfortunate and the book flags it explicitly, so we do too. The $\underline{R}$ of (3.32), $\underline{F} = \underline{R}\underline{U} = \underline{V}\underline{R}$, is orthogonal and maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$, from the reference translation space to the current one. The $\underline{R}$ of (8.8) maps $\hat{\mathbb{E}}_\kappa^3$ to $\hat{\mathbb{E}}_\mu^3$, from one *reference* translation space to another, and it is not orthogonal: nothing has been assumed about it beyond invertibility and $\det \underline{R} > 0$.

The index test of Remark 4.1 separates them at a glance. The polar rotation carries $\underline{e}_i \otimes \hat{\underline{E}}_A$, one Latin index and one capital. The present $\underline{R}$ carries $\hat{\underline{E}}_A \otimes \hat{\underline{E}}_B$, two capitals, and that is exactly why it can stand on the right of $\underline{F}_\mu$ in (8.9) and produce a two-point tensor again. Confusing the two would make $\underline{R}^t \underline{R} = \hat{\underline{I}}$ look like a triviality instead of what it becomes in (8.19), namely the defining property of a solid.

The response function, and why it must change with κ

Let $F \in C$ be an entry in the list of variables characterizing the response of the material: F may be $\underline{T}$, or $\underline{q}$, or ψ , or η : whichever of the four is under discussion. Its value is determined by the entries of L , which we have just seen depends on the choice of reference. Choosing κ ,

$$F(p, t) = F_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}), \quad (8.10)$$

where the right-hand side is the constitutive function delivering the values of F . The semicolon is not a typographical flourish: the entries before it are the process variables, which change from instant to instant, while $\underline{X}$ after it identifies which material point is being asked about and is a parameter, fixed once the point is chosen. The dependence on $\underline{X}$ is what allows a body to be inhomogeneous.

REMARK 8.5 (the value does not depend on κ ; the function must). This is the pivot of the whole section, and it is one sentence: $F(p, t)$ is a number, or a vector, or a tensor, that the material actually has, whereas F_κ is a rule we wrote down for computing it.

The rule takes $\underline{F}_\kappa$ among its arguments, and $\underline{F}_\kappa$ changes when κ changes. The output does not change: the stress at p at time t is whatever it is, and no bookkeeping we perform can alter it. So if the input changes and the output does not, the rule must change to compensate. That is the entire argument, and (8.12) is its statement.

This is the same distinction as Remark 2.6, where one velocity was carried by three different functions $\underline{v}$, $\hat{\underline{v}}$, $\tilde{\underline{v}}$ according to what was fed in. Here one response is carried by a family of functions F_κ , one for each reference configuration. Miss the distinction and (8.16), the definition of a symmetry transformation, reads as the false claim that $\underline{F}$ and $\underline{F}\underline{R}$ are the same tensor. They are not. The claim is that a particular function does not distinguish them.

Using (8.8) and (8.9) we may rewrite (8.10) in terms of μ . The recipe is mechanical: wherever $\underline{X}$ appears, substitute $\underline{X} = \underline{\lambda}^{-1}(\underline{Y})$, and

wherever $\underline{F}_\kappa$ appears, substitute $\underline{F}_\kappa = \underline{F}_\mu \underline{R}$ with $\underline{R}$ evaluated at that same $\underline{X}$. Then

$$F(p, t) = F_\mu(\underline{x}, \underline{v}, \theta, \underline{F}_\mu, \underline{L}, \text{grad } \theta; \underline{Y}), \quad (8.11)$$

where

$$\begin{aligned} & F_\mu(\underline{x}, \underline{v}, \theta, \underline{F}_\mu, \underline{L}, \text{grad } \theta; \underline{Y}) \\ &= F_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\mu \underline{R}(\underline{\lambda}^{-1}(\underline{Y})), \underline{L}, \text{grad } \theta; \underline{\lambda}^{-1}(\underline{Y})), \end{aligned} \quad (8.12)$$

which produces the constitutive function pertaining to any choice of reference configuration once the function pertaining to one is given.

Read (8.12) carefully, because it is a definition dressed as an equation. The left-hand side is not known in advance; the right-hand side is a formula built from the known F_κ , from $\underline{\lambda}$ and from $\underline{R}$, and (8.12) defines F_μ to be that formula. What is being asserted, then, is not the equality itself but two things that follow from it.

First, F_μ exists: a change of reference never destroys the possibility of a constitutive description, it merely relabels it. Second, and this is the substantive point, the *list of arguments* is unchanged. On the right-hand side of (8.12) the fourth slot is occupied by a deformation gradient, the sixth by a spatial temperature gradient, and so on, exactly as on the left. No new kind of variable had to be introduced to make the transfer work. Hence the list of variables deemed relevant to material response is the same for all reference configurations, even though the function of them is not. Had this failed, the constitutive assumption of Section 8.1 would have been an assumption about κ rather than about the material, and it would have been worthless.

Localizing the change of reference

Constitutive response is defined pointwise, so nothing is lost by confining attention to changes of reference in a neighbourhood of one material point p_0 , rather than over the whole body (Figure 8.2). Let $\underline{\lambda}(\underline{X})$ be smooth and invertible on an open neighbourhood $N_\kappa(p_0) \subset \kappa$, mapping it onto

another open neighbourhood $N_\mu(p_0)$, and let $\underline{X}_0 = \underline{\kappa}(p_0)$ be a *pivot point* of the map:

$$\underline{Y}_0 = \underline{\lambda}(\underline{X}_0) = \underline{X}_0.$$

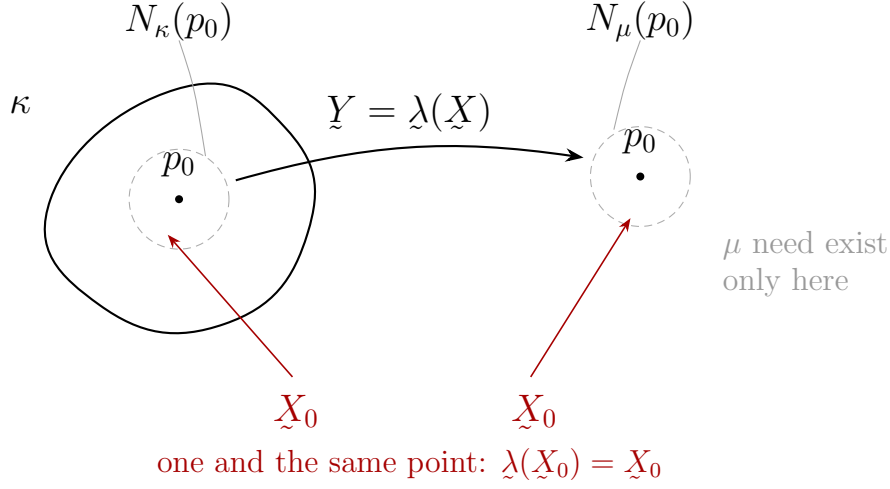


Figure 8.2: Local change of reference. Only a neighbourhood of the one material point p_0 is relabelled, so $\underline{\lambda}$ need not, and in general does not, extend to a map of the whole body; μ is a local reference configuration and nothing more. The two arrows on the right carry the same label because $\underline{X}_0$ is a pivot: p_0 is assigned the same position vector by both placements, which is what allows the two response functions in (8.13) to be compared at a single point. The pivot costs no generality, since translating μ leaves $\underline{R} = \text{Grad } \underline{\lambda}$ untouched.

The pivot condition looks like an extra hypothesis and is not one. The placement $\underline{\mu}$ is ours to choose, and replacing $\underline{\mu}$ by $\underline{\mu} + \underline{c}$ for a constant vector $\underline{c}$ shifts every $\underline{Y}$ by $\underline{c}$ and leaves $\underline{R} = \text{Grad } \underline{\lambda}$ completely unchanged, a constant having zero gradient. So one may always translate $\underline{\mu}$ until $\underline{\lambda}(\underline{X}_0) = \underline{X}_0$, and since $\underline{R}$ is the only feature of $\underline{\lambda}$ that will be used, nothing has been given up. What has been bought is that both response functions may be evaluated at the same label $\underline{X}_0$, so that (8.13) compares like with like:

$$\begin{aligned} F(p_0, t) &= F_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}_0) \\ &= F_\mu(\underline{x}, \underline{v}, \theta, \underline{F}_\mu, \underline{L}, \text{grad } \theta; \underline{X}_0), \quad \text{with } \underline{F}_\kappa = \underline{F}_\mu \underline{R}(\underline{X}_0), \end{aligned} \tag{8.13}$$

all arguments being evaluated at $\underline{X}_0$.

One experiment, two neighbourhoods

Now the definition of material symmetry. Subject both $N_\kappa(p_0)$ and $N_\mu(p_0)$ to the *same* experiment, consisting of a prescribed deformation $\chi(\cdot, t)$ and temperature distribution $\theta(\cdot, t)$. Acting on $N_\kappa(p_0)$ the experiment produces the fields

$$\begin{aligned}\underline{\underline{\chi}}(X, t) &= \underline{\underline{\chi}}(X_0, t) + \underline{\underline{F}}(X_0, t)(X - X_0) + o(|X - X_0|), \\ \underline{\underline{\hat{v}}}(X, t) &= \underline{\underline{\hat{v}}}(X_0, t) + \underline{\underline{\dot{F}}}(X_0, t)(X - X_0) + o(|X - X_0|), \\ \underline{\underline{\theta}}(X, t) &= \underline{\underline{\theta}}(X_0, t) + \text{Grad } \underline{\underline{\theta}}(X_0, t) \cdot (X - X_0) + o(|X - X_0|),\end{aligned}\tag{8.14}$$

for $X \in N_\kappa(p_0)$, with $\underline{\underline{\hat{v}}}(X_0, t) = \underline{\underline{\dot{\chi}}}(X_0, t)$ and $\underline{\underline{F}}(X_0, t) = \text{Grad } \underline{\underline{\chi}}|_{X_0}$; and acting on $N_\mu(p_0)$ it produces

$$\begin{aligned}\underline{\underline{\chi}}(Y, t) &= \underline{\underline{\chi}}(X_0, t) + \underline{\underline{F}}(X_0, t)(Y - X_0) + o(|Y - X_0|), \\ \underline{\underline{\hat{v}}}(Y, t) &= \underline{\underline{\hat{v}}}(X_0, t) + \underline{\underline{\dot{F}}}(X_0, t)(Y - X_0) + o(|Y - X_0|), \\ \underline{\underline{\theta}}(Y, t) &= \underline{\underline{\theta}}(X_0, t) + \text{Grad } \underline{\underline{\theta}}(X_0, t) \cdot (Y - X_0) + o(|Y - X_0|),\end{aligned}\tag{8.15}$$

for $Y \in N_\mu(p_0)$. The point of writing both out is that the coefficients are literally the same symbols: the values of the fields and of their gradients agree at X_0 , which is possible only because X_0 is a pivot and therefore belongs to both neighbourhoods.

Now compute the response twice. Using $N_\kappa(p_0)$ as reference, it is $F_\kappa(\underline{\underline{x}}, \underline{\underline{v}}, \underline{\underline{\theta}}, \underline{\underline{F}}, \underline{\underline{L}}, \text{grad } \underline{\underline{\theta}}; X_0)$ with

$$\begin{aligned}\underline{\underline{x}} &= \underline{\underline{\chi}}(X_0, t), \quad \underline{\underline{v}} = \underline{\underline{\dot{\chi}}}(X_0, t), \quad \underline{\underline{\theta}} = \underline{\underline{\theta}}(X_0, t), \quad \underline{\underline{F}} = \underline{\underline{F}}(X_0, t), \\ \underline{\underline{L}} &= \underline{\underline{\dot{F}}}(X_0, t)\underline{\underline{F}}(X_0, t)^{-1}, \quad \text{grad } \underline{\underline{\theta}} = \underline{\underline{F}}(X_0, t)^{-t} \text{Grad } \underline{\underline{\theta}}(X_0, t),\end{aligned}$$

the last two being (4.3) and (8.6) read backwards. Using $N_\mu(p_0)$ as reference it is $F_\mu(\underline{\underline{x}}, \underline{\underline{v}}, \underline{\underline{\theta}}, \underline{\underline{F}}, \underline{\underline{L}}, \text{grad } \underline{\underline{\theta}}; X_0)$, and by (8.15) the six arguments are the very same six numbers, vectors and tensors. So we have two functions evaluated at identical arguments.

Having subjected two different local neighbourhoods of p_0 to the same experiment, we should expect the two response functions, evaluated at the same arguments, to return *different* values, and (8.13) says exactly

why: the two are related by an insertion of $\underline{R}$ in the fourth slot, and a function need not be blind to that. However, there may be distinguished maps $\underline{\lambda}$ for which the two responses at p_0 are nevertheless identical. For such a map no experiment whatever can tell the two neighbourhoods apart. These are the *material symmetry transformations*.

Suppose then that $N_\kappa(p_0)$ and $N_\mu(p_0)$ are related by a symmetry transformation, so that

$$F_\kappa(\underline{x}, \underline{v}, \theta, \underline{G}, \underline{L}, \text{grad } \theta; \underline{X}_0) = F_\mu(\underline{x}, \underline{v}, \theta, \underline{G}, \underline{L}, \text{grad } \theta; \underline{X}_0)$$

for every admissible value $\underline{G}$ of the fourth argument. Equation (8.13) read as an identity between functions says

$$F_\mu(\underline{x}, \underline{v}, \theta, \underline{G}, \underline{L}, \text{grad } \theta; \underline{X}_0) = F_\kappa(\underline{x}, \underline{v}, \theta, \underline{G}\underline{R}, \underline{L}, \text{grad } \theta; \underline{X}_0),$$

again for every $\underline{G}$, since (8.13) holds whatever deformation the material happens to be undergoing. Eliminating F_μ between the two displays and renaming $\underline{G}$ as $\underline{F}$,

$$F_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}, \underline{L}, \text{grad } \theta; \underline{X}_0) = F_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}\underline{R}, \underline{L}, \text{grad } \theta; \underline{X}_0), \quad (8.16)$$

where $\underline{R} = \text{Grad } \underline{\lambda}(\underline{X})|_{\underline{X}_0}$.

Two features of (8.16) deserve emphasis before it is used.

It is a statement about *one* function. The second reference configuration has vanished from it entirely, which is what makes it usable: (8.16) is a functional equation constraining F_κ , and one may test a candidate constitutive function against it without ever constructing μ .

And it is required to hold *for all admissible* $\underline{F}$, with $\underline{R}$ fixed. This quantifier is not decoration. Every argument in the rest of the section, and the whole of Section 8.3, proceeds by substituting some new tensor for $\underline{F}$ in (8.16), and each such substitution is legal only because the equation was demanded of every $\underline{F}$ to begin with.

REMARK 8.6 (symmetry is a local notion). The map $\underline{\lambda}$ in this discussion is defined on a neighbourhood of the material point in question, and on nothing more. In particular $\underline{R}$ need not be the gradient of a single

function defined on the whole body: there is no requirement that the local relabellings at different points fit together into a global change of reference configuration.

Two things follow. Different points of one body may have different symmetries, so the theory accommodates a composite, in which the type of material varies from place to place, or a polycrystal, in which the lattice orientation does. And, less obviously, the freedom to choose λ point by point is exactly what Section 8.3 will exploit and what makes Noll's rule matter in plasticity, where the local relaxed configurations at neighbouring points genuinely fail to fit together.

The symmetry group

It follows from (8.16) that $\underline{F}\underline{R}$ is the value of an admissible deformation gradient at $\underline{X}_0$: the right-hand side evaluates the constitutive function with $\underline{F}\underline{R}$ in the deformation-gradient slot, and a constitutive function is only defined for arguments the material can actually present. If κ is an occupiable reference configuration this requires

$$0 < \det(\underline{F}\underline{R}) = (\det \underline{F})(\det \underline{R}), \quad \text{with } \det \underline{F} > 0,$$

whence $\det \underline{R} > 0$. Denote by $\mathcal{G}_{\kappa(p_0)}$ the set of those $\underline{R}$ for which (8.16) holds for all admissible values of $\underline{F}$:

$$\mathcal{G}_{\kappa(p_0)} = \{\underline{R} : \det \underline{R} > 0 \text{ and (8.16) holds}\}. \quad (8.17)$$

This set is a *group*. Three properties have to be checked, and each is a two-line calculation in which every step uses the “for all $\underline{F}$ ” clause of (8.16). Throughout, all arguments not involving $\underline{R}$ are suppressed, so that $F_\kappa(\underline{F})$ abbreviates $F_\kappa(x, y, \theta, \underline{F}, L, \text{grad } \theta; \underline{X}_0)$.

(1) $\hat{\underline{I}} \in \mathcal{G}_{\kappa(p_0)}$. With $\underline{R} = \hat{\underline{I}}$, (8.16) reads $F_\kappa(\underline{F}) = F_\kappa(\underline{F}\hat{\underline{I}}) = F_\kappa(\underline{F})$, which is true of any function whatever, and $\det \hat{\underline{I}} = 1 > 0$. The identity is a symmetry of every material: doing nothing to the reference neighbourhood is certainly undetectable.

(2) If $\underline{R}_1, \underline{R}_2 \in \mathcal{G}_{\kappa(p_0)}$ then $\underline{R}_1 \underline{R}_2 \in \mathcal{G}_{\kappa(p_0)}$. Compute, for arbitrary admissible $\underline{F}$,

$$F_{\kappa}(\underline{F}(\underline{R}_1 \underline{R}_2)) = F_{\kappa}((\underline{F} \underline{R}_1) \underline{R}_2) = F_{\kappa}(\underline{F} \underline{R}_1) = F_{\kappa}(\underline{F}).$$

The first equality is associativity of the tensor product, nothing more. The second applies (8.16) for $\underline{R}_2$ with $\underline{F}$ replaced by $\underline{F} \underline{R}_1$, which is permitted because $\underline{F} \underline{R}_1$ is itself an admissible deformation gradient and (8.16) was demanded for all such. The third applies (8.16) for $\underline{R}_1$. Finally $\det(\underline{R}_1 \underline{R}_2) = (\det \underline{R}_1)(\det \underline{R}_2) > 0$, so the product qualifies on both counts.

(3) If $\underline{R} \in \mathcal{G}_{\kappa(p_0)}$ then $\underline{R}^{-1} \in \mathcal{G}_{\kappa(p_0)}$. The inverse exists because $\det \underline{R} > 0$. Apply (8.16) with $\underline{F}$ replaced by $\underline{F} \underline{R}^{-1}$:

$$F_{\kappa}(\underline{F} \underline{R}^{-1}) = F_{\kappa}((\underline{F} \underline{R}^{-1}) \underline{R}) = F_{\kappa}(\underline{F} \hat{I}) = F_{\kappa}(\underline{F}),$$

which is (8.16) for $\underline{R}^{-1}$. And $\det(\underline{R}^{-1}) = 1/\det \underline{R} > 0$.

Associativity needs no proof, being inherited from the composition of linear maps, so $\mathcal{G}_{\kappa(p_0)}$ is a subgroup of the group of invertible tensors of positive determinant. The physical reading of the three properties is worth pausing on: (1) says that doing nothing is undetectable, (2) that two undetectable rearrangements performed in succession are undetectable, and (3) that undoing an undetectable rearrangement is undetectable. Stated that way the group structure is obvious, which is the point; the calculations above are only the translation of the obvious into the language of (8.16).

The unimodular restriction

Property (2) applied repeatedly gives $\underline{R}^n \in \mathcal{G}_{\kappa(p_0)}$ for every positive integer n : if $\underline{R}^{n-1}$ is in the group then so is $\underline{R}^{n-1} \underline{R} = \underline{R}^n$, and the induction starts at $\underline{R}^1 = \underline{R}$. Hence $F_{\kappa}(\underline{F}) = F_{\kappa}(\underline{F} \underline{R}^n)$ for every n , and

$$\det(\underline{F} \underline{R}^n) = (\det \underline{F})(\det \underline{R}^n) = (\det \underline{F})(\det \underline{R})^n,$$

using Theorem 1.52 twice. Let $n \rightarrow \infty$ and read off the two cases.

If $\det \underline{R} > 1$ then $(\det \underline{R})^n \rightarrow \infty$, so $\det(\underline{F} \underline{R}^n) \rightarrow \infty$. If $\det \underline{R} < 1$ then $(\det \underline{R})^n \rightarrow 0$ and $\det(\underline{F} \underline{R}^n) \rightarrow 0$. In the first case the material at p_0 is dilated without bound and in the second compacted without bound, and in neither case does its response change at all: the left-hand side $F_\kappa(\underline{F})$ never moved.

That is untenable, and it is worth being explicit about why rather than resting on the word. By (2.50) and (2.59) the current density is $\rho = \rho_\kappa(\underline{X})/\det \underline{F}$, so a deformation gradient with $\det \underline{F} \rightarrow 0$ is one in which a finite mass is squeezed into vanishing volume and $\rho \rightarrow \infty$, while $\det \underline{F} \rightarrow \infty$ is unlimited rarefaction. A material insensitive to either is a material one could compress to a point at no cost in stress and no change in energy. No substance behaves so. To avoid the conclusion we must restrict $\underline{R}$ so that neither limit can be reached, that is, so that $(\det \underline{R})^n$ does not run away, and for a positive number that means $\det \underline{R} = 1$. Therefore

$$\mathcal{G}_{\kappa(p_0)} \subset U = \{\underline{R} : \det \underline{R} = 1\}, \quad (8.18)$$

which is called the *unimodular group*. It is a group by the same determinant identities used above: $\det \hat{\underline{I}} = 1$, $\det(\underline{R}_1 \underline{R}_2) = 1 \cdot 1 = 1$ and $\det(\underline{R}^{-1}) = 1/1 = 1$.

REMARK 8.7 (what (8.18) does and does not say). It does not say that symmetry transformations preserve shape. It says they preserve *volume*, by Remark 1.53, since the determinant is the ratio of oriented volumes. A unimodular $\underline{R}$ may stretch the reference neighbourhood by a factor of ten in one direction provided it compresses it correspondingly in others, and (8.18) permits that; whether the material notices is a separate question, answered by (8.19).

Nor is U the largest group one might have written. Some authors define the unimodular group by $|\det \underline{R}| = 1$, which admits reflections. Here $\det \underline{R} > 0$ was forced before (8.18) was reached, by the requirement that $\underline{F} \underline{R}$ be an admissible deformation gradient, so reflections are excluded from the outset. They are excluded for a physical reason and not a conventional one: a reflection of the reference neighbourhood would have to be realised by an actual motion of material, and no motion

can pass continuously from $\det \underline{F} > 0$ to $\det \underline{F} < 0$ without passing through $\det \underline{F} = 0$, at which the deformation ceases to be invertible (Remark 2.17).

Finally, (8.18) is an upper bound and nothing else. Every material of the class considered satisfies it, in every reference configuration. The interesting question is how far below the bound a given material sits, and that is the question the next two definitions answer.

Solid and fluid, defined by the symmetry group

It was Noll's idea to classify materials as solid or fluid on the basis of the symmetry group. The classification is not a description of how a substance looks or pours; it is a statement about which rearrangements of a material neighbourhood the substance can detect.

Fluids. Take the simplest constitutive equation there is: the stress in a compressible inviscid fluid under isothermal conditions,

$$\underline{T} = -p(\rho)\underline{I},$$

where $p(\rho)$ is the constitutive function for the density-dependent pressure. Ask which $\underline{R}$ satisfy (8.16) for it. From (2.50) and (2.59),

$$\rho = \frac{\rho_\kappa(\underline{X})}{\det \underline{F}} = \frac{\rho_\kappa(\underline{X})}{\det(\underline{F}\underline{R})} \quad \text{provided } \underline{R} \in U,$$

the second equality holding because $\det(\underline{F}\underline{R}) = (\det \underline{F})(\det \underline{R}) = \det \underline{F}$ when $\det \underline{R} = 1$. So replacing $\underline{F}$ by $\underline{F}\underline{R}$ leaves ρ unchanged, hence leaves $p(\rho)$ unchanged, hence leaves $\underline{T}$ unchanged: (8.16) holds for every $\underline{R} \in U$. Therefore $U \subset \mathcal{G}_{\kappa(p_0)}$, and with (8.18) in the opposite direction,

$$\mathcal{G}_{\kappa(p_0)} = U.$$

This motivates the definition: a *fluid* is a material of the class considered here for which there exists a local reference configuration $\kappa^*(p_0)$ such that $\mathcal{G}_{\kappa^*(p_0)} = U$.

The stress depended on $\underline{F}$ only through $\det \underline{F}$, and $\det$ is exactly the feature of $\underline{F}$ that a unimodular map cannot change. A fluid has

the largest symmetry group available: it is a material that cannot detect any volume-preserving rearrangement of its own reference neighbourhood whatever. That is what it means to say a fluid has no preferred shape, expressed constitutively rather than anecdotally.

Solids. For these the situation is different, and the difference is one sentence of physics: experience indicates that a strain induced by the map λ is experimentally detectable. A solid remembers a shape, and deforming it away from that shape produces stress that can be measured.

If λ is to be a symmetry transformation, and hence undetectable, it must therefore be strain free. Strain was defined in Chapter 2 through the right Cauchy–Green tensor, (2.96) and (2.97): a deformation with gradient $\underline{R}$ is strain free exactly when the associated $\underline{C} = \underline{R}^t \underline{R}$ equals $\hat{\underline{I}}$, for then the Lagrange strain $\underline{E} = \frac{1}{2}(\underline{C} - \hat{\underline{I}})$ vanishes and, by Proposition 2.26, every material line element keeps its length and every pair of them its angle. So a symmetry transformation of a solid satisfies $\underline{R}^t \underline{R} = \hat{\underline{I}}$. Combined with (8.18) this makes $\underline{R}$ a rotation: $\underline{R}^t \underline{R} = \hat{\underline{I}}$ makes $\underline{R}$ orthogonal, hence $\det \underline{R} = \pm 1$ by Proposition 1.48, and $\det \underline{R} = 1$ selects the proper orthogonal case.

Accordingly a *solid* is defined to be a material of the class considered for which there exists a local reference $\kappa^*(p_0)$ such that $\mathcal{G}_{\kappa^*(p_0)}$ is contained in Orth^+ , the set of proper-orthogonal tensors, or simply the rotations:

$$\mathcal{G}_{\kappa^*(p_0)} \subset \text{Orth}^+ = \{\underline{R} : \underline{R} \in U \text{ and } \underline{R}^t \underline{R} = \hat{\underline{I}}\}. \quad (8.19)$$

We note that $\kappa^*(p_0)$ need bear no relationship to any particular reference configuration one may use for the purposes of analysis. In a solid it would correspond to a unit cell of an undistorted lattice of a crystalline material. Such cells are mapped to themselves by certain discrete rotations, and those rotations are what characterize the crystalline symmetry at hand: four of them about a four-fold axis for a square cell, twenty-four in all for a cubic crystal, and only the identity for a triclinic one. For this reason $\kappa^*(p_0)$ is often called an *undistorted* local configuration of the solid.

REMARK 8.8 (isotropy, and the whole ladder in one line). Definitions (8.18) and (8.19) between them organise every material of the class considered on a single scale, and the scale is worth writing out because the later chapters use its rungs by name. Every symmetry group satisfies

$$\{\hat{I}\} \subseteq \mathcal{G}_{\kappa(p_0)} \subseteq U,$$

with the lower end forced by property (1) above and the upper by (8.18). Between the ends:

- $\mathcal{G} = \{\hat{I}\}$: no symmetry at all. A triclinic crystal. Every rearrangement, however slight, is detectable.
- $\mathcal{G}$ a finite subgroup of Orth^+ : a crystalline solid, the finite group being its point group.
- $\mathcal{G} \subseteq \text{Orth}^+$, possibly infinite: a solid. A transversely isotropic solid, for instance, has the continuous group of rotations about one axis.
- $\mathcal{G} = \text{Orth}^+$: an *isotropic* solid. It is the largest a solid's group can be, since (8.19) caps it, so an isotropic solid is a solid with as much symmetry as the definition of a solid allows: no direction in κ^* is distinguished from any other.
- $\mathcal{G} = U$: a fluid.

Two consequences follow immediately from $\text{Orth}^+ \subset U$, and both are the kind of statement that is obvious once the group language is in place and obscure without it.

Every fluid is isotropic. A fluid's group is U , which contains Orth^+ ; so a fluid has at least the symmetry an isotropic solid has. The converse fails: an isotropic solid has Orth^+ and not U , and the elements of U outside Orth^+ are exactly the volume-preserving distortions it can feel and a fluid cannot. Steel and water are both indifferent to direction; only one of them is indifferent to shape.

The classification is exhaustive but not complete. A group strictly between Orth^+ and U describes a material that is neither a solid nor

a fluid in the above sense. Such groups exist and such materials exist, liquid crystals among them. The two definitions are the two extremes of a range, and the range in between is where much of modern constitutive theory lives.

A SOLID

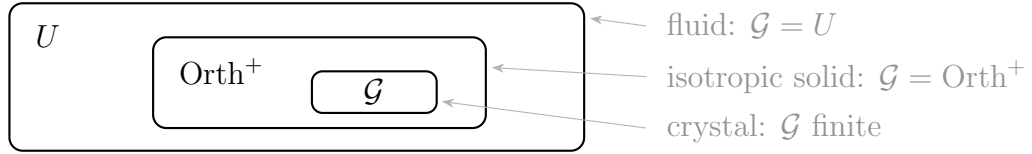
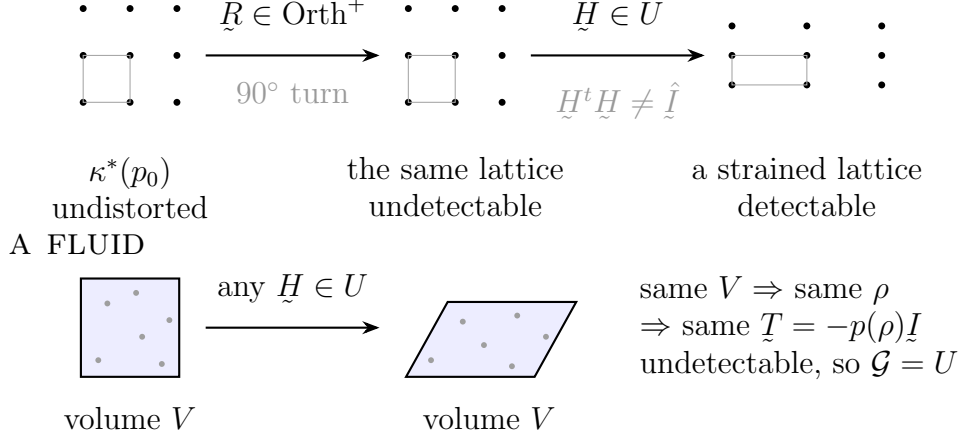


Figure A: Solid and fluid, told apart by the symmetry group. Top: in an undistorted configuration a crystalline solid is carried onto itself by certain rotations, which no experiment can detect, but a unimodular map that is not a rotation strains the lattice, and by (8.19) that is detectable; the group is confined to Orth^+ . Middle: the response of an inviscid fluid depends on $\tilde{F}$ only through $\det \tilde{F}$, and no member of U changes that, so every volume-preserving rearrangement is undetectable and the group is the whole of U . Bottom: the two definitions as nested sets, with $\text{Orth}^+ \subset U$ showing at a glance that every fluid is isotropic while an isotropic solid is not a fluid.

8.3 Noll's rule

Material symmetry, as (8.16) defines it, is a characteristic of two things at once: of the material, and of the reference configuration adopted.

The definitions of solid and fluid both had to say “there exists a local reference configuration such that...”, which is an admission that the group could look different in another one. It is therefore of interest to determine exactly how much the choice of reference matters. The answer is complete and is due to Noll.

Consider two reference configurations κ_1 and κ_2 in which the positions of $p_0 \in B$ are $\underline{X}_1$ and $\underline{X}_2$ respectively, and assume a smooth invertible map $\underline{\pi}$, defined on a neighbourhood of $\underline{X}_1$, with $\underline{X}_2 = \underline{\pi}(\underline{X}_1)$. Write $\underline{K} = \text{Grad } \underline{\pi}|_{\underline{X}_1}$. This is the situation of (8.8) with new names, so (8.12) applies verbatim; suppressing the variables that do not involve a reference configuration, it reads

$$\begin{aligned} F_{\kappa_2}(\underline{F}; \underline{X}_2) &= F_{\kappa_1}(\underline{F}\underline{K}; \underline{X}_1), & \text{where } \underline{K} \\ &= \text{Grad } \underline{\pi}|_{\underline{X}_1}. \end{aligned} \quad (8.20)$$

Here $\underline{F}$ in the argument slot is the deformation gradient computed from κ_2 , and $\underline{F}\underline{K}$ is the corresponding one computed from κ_1 , which is (8.9) again.

Everything that follows is obtained by putting different tensors into the one slot of (8.20), so it is worth repeating that this is legal: (8.20) holds for every admissible $\underline{F}$, and $\det \underline{K} > 0$ because both configurations are occupiable, so $\underline{F}\underline{K}$, $\underline{F}\underline{K}^{-1}$ and $\underline{F}\underline{R}\underline{K}^{-1}$ are all admissible whenever $\underline{F}$ is.

Now replace $\underline{F}$ by $\underline{F}\underline{K}^{-1}$ in (8.20). The right-hand side becomes $F_{\kappa_1}(\underline{F}\underline{K}^{-1}\underline{K}; \underline{X}_1) = F_{\kappa_1}(\underline{F}\hat{I}; \underline{X}_1) = F_{\kappa_1}(\underline{F}; \underline{X}_1)$, so

$$F_{\kappa_2}(\underline{F}\underline{K}^{-1}; \underline{X}_2) = F_{\kappa_1}(\underline{F}; \underline{X}_1). \quad (8.21)$$

This is (8.20) solved the other way round, and it is the form we shall use, because it has κ_1 , whose symmetry group is known, standing alone on the right.

Now suppose $\underline{R} \in \mathcal{G}_{\kappa_1(p_0)}$, so that by (8.16)

$$F_{\kappa_1}(\underline{F}; \underline{X}_1) = F_{\kappa_1}(\underline{F}\underline{R}; \underline{X}_1) \quad (8.22)$$

for every admissible $\underline{F}$. Replacing $\underline{F}$ by $\underline{F}\underline{R}$ in (8.21),

$$F_{\kappa_1}(\underline{F}\underline{R}; \underline{X}_1) = F_{\kappa_2}(\underline{F}\underline{R}\underline{K}^{-1}; \underline{X}_2). \quad (8.23)$$

Chaining (8.23), (8.22) and (8.21) in that order gives

$$F_{\kappa_2}(\underline{\tilde{F}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1}; \underline{\tilde{X}}_2) = F_{\kappa_1}(\underline{\tilde{F}}; \underline{\tilde{X}}_1) = F_{\kappa_2}(\underline{\tilde{F}} \underline{\tilde{K}}^{-1}; \underline{\tilde{X}}_2). \quad (8.24)$$

The two ends of (8.24) are values of the same function F_{κ_2} , so the statement is now entirely about κ_2 : κ_1 has served its purpose and may be discarded. All that remains is to recognise (8.16) in it.

Set $\hat{\underline{\tilde{F}}} = \underline{\tilde{F}} \underline{\tilde{K}}^{-1}$, so that $\underline{\tilde{F}} = \hat{\underline{\tilde{F}}} \underline{\tilde{K}}$ and

$$\underline{\tilde{F}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1} = \hat{\underline{\tilde{F}}} \underline{\tilde{K}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1}.$$

Then (8.24) becomes

$$F_{\kappa_2}(\hat{\underline{\tilde{F}}}; \underline{\tilde{X}}_2) = F_{\kappa_2}(\hat{\underline{\tilde{F}}} \underline{\tilde{K}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1}; \underline{\tilde{X}}_2). \quad (8.25)$$

As $\underline{\tilde{F}}$ runs over all admissible tensors so does $\hat{\underline{\tilde{F}}} = \underline{\tilde{F}} \underline{\tilde{K}}^{-1}$, because $\underline{\tilde{F}} \mapsto \underline{\tilde{F}} \underline{\tilde{K}}^{-1}$ is a bijection of the admissible set onto itself; so (8.25) holds for every admissible $\hat{\underline{\tilde{F}}}$. Comparing it with (8.16), and noting

$$\det(\underline{\tilde{K}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1}) = (\det \underline{\tilde{K}})(\det \underline{\tilde{R}})(\det \underline{\tilde{K}})^{-1} = \det \underline{\tilde{R}} > 0,$$

we conclude that $\underline{\tilde{K}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1} \in \mathcal{G}_{\kappa_2(p_0)}$. We thus arrive at *Noll's rule*:

$$\mathcal{G}_{\kappa_2(p_0)} = \{\underline{\tilde{K}} \underline{\tilde{R}} \underline{\tilde{K}}^{-1} : \underline{\tilde{R}} \in \mathcal{G}_{\kappa_1(p_0)}\}. \quad (8.26)$$

REMARK 8.9 (the missing half of (8.26)). What the calculation established is one inclusion, $\underline{\tilde{K}} \mathcal{G}_{\kappa_1(p_0)} \underline{\tilde{K}}^{-1} \subseteq \mathcal{G}_{\kappa_2(p_0)}$: every conjugate of a κ_1 symmetry is a κ_2 symmetry. Equality needs the other direction, and it costs one line.

The situation is symmetric in the two configurations. Running the whole argument again with κ_1 and κ_2 interchanged uses $\underline{\tilde{\pi}}^{-1}$ in place of $\underline{\tilde{\pi}}$, whose gradient at $\underline{\tilde{X}}_2$ is $\underline{\tilde{K}}^{-1}$, and yields $\underline{\tilde{K}}^{-1} \mathcal{G}_{\kappa_2(p_0)} \underline{\tilde{K}} \subseteq \mathcal{G}_{\kappa_1(p_0)}$. So if $\underline{\tilde{S}} \in \mathcal{G}_{\kappa_2(p_0)}$ then $\underline{\tilde{K}}^{-1} \underline{\tilde{S}} \underline{\tilde{K}} \in \mathcal{G}_{\kappa_1(p_0)}$, and

$$\underline{\tilde{S}} = \underline{\tilde{K}} (\underline{\tilde{K}}^{-1} \underline{\tilde{S}} \underline{\tilde{K}}) \underline{\tilde{K}}^{-1}$$

exhibits $\underline{\tilde{S}}$ as a conjugate of a member of $\mathcal{G}_{\kappa_1(p_0)}$. Both inclusions hold, so (8.26) is an equality. The gap is worth closing because the two statements are not interchangeable: the inclusion alone would leave open the possibility that κ_2 has *more* symmetry than κ_1 , and the whole force of the section is that it has exactly as much.

Why conjugation, and why nothing else would do

Equation (8.26) is stated compactly and can be read as a formula that fell out of an algebraic manipulation. It is not. Conjugation is the only thing that could have appeared, and there are three ways to see that, each of which is worth having.

First: a symmetry is an object, and $\tilde{K}$ is a change of description. Recall what $\tilde{R}$ is. By (8.16) and the pivot arrangement of Figure 8.2, $\tilde{R} = \text{Grad } \tilde{\lambda}$ where $\tilde{\lambda}$ carries a neighbourhood of $\tilde{X}_0$ to another neighbourhood of the same point. So $\tilde{R}$ is a linear map of the local reference translation space to *itself*: it takes a material fibre of the reference neighbourhood and returns another material fibre of the same neighbourhood, and it is a symmetry when the material cannot tell the two apart.

Now change the reference. The tensor $\tilde{K}$ maps $\hat{\mathbb{E}}_{\kappa_1}^3$ to $\hat{\mathbb{E}}_{\kappa_2}^3$: it is the dictionary between the two descriptions of the same set of material fibres. A fibre described in κ_2 by $d\tilde{X}_2$ is described in κ_1 by $d\tilde{X}_1 = \tilde{K}^{-1}d\tilde{X}_2$. To apply the rearrangement, translate into κ_1 , apply $\tilde{R}$ there, and translate back:

$$d\tilde{X}_2 \xrightarrow{\tilde{K}^{-1}} \tilde{K}^{-1}d\tilde{X}_2 \xrightarrow{\tilde{R}} \tilde{R}\tilde{K}^{-1}d\tilde{X}_2 \xrightarrow{\tilde{K}} \tilde{K}\tilde{R}\tilde{K}^{-1}d\tilde{X}_2,$$

which is Figure B. That is the whole of (8.26), and it is the same statement as Theorem 1.39, the transformation law for the components of a tensor under a change of basis, with one difference: there the change of basis was orthogonal and the conjugation read $\tilde{Q}\tilde{A}\tilde{Q}^t$; here $\tilde{K}$ is merely invertible and the conjugation reads $\tilde{K}\tilde{A}\tilde{K}^{-1}$. Remark 2.39 made the point that the transformation law identifies the type of an object. Equation (8.26) is that law, and it identifies $\tilde{R}$ as a linear map of a translation space to itself.

Second: any other rule would break under composition.

Change the reference twice, from κ_1 to κ_2 with gradient $\tilde{K}_{12}$, then from κ_2 to κ_3 with gradient $\tilde{K}_{23}$. The composite map has gradient $\tilde{K}_{13} = \tilde{K}_{23}\tilde{K}_{12}$, by the chain rule. Applying (8.26) twice,

$$\tilde{K}_{23}(\tilde{K}_{12}\mathcal{G}_{\kappa_1(p_0)}\tilde{K}_{12}^{-1})\tilde{K}_{23}^{-1} = (\tilde{K}_{23}\tilde{K}_{12})\mathcal{G}_{\kappa_1(p_0)}(\tilde{K}_{23}\tilde{K}_{12})^{-1} = \tilde{K}_{13}\mathcal{G}_{\kappa_1(p_0)}\tilde{K}_{13}^{-1},$$

using $(\underline{K}_{23}\underline{K}_{12})^{-1} = \underline{K}_{12}^{-1}\underline{K}_{23}^{-1}$, which is the same as applying (8.26) once with $\underline{K}_{13}$. Doing it in two steps agrees with doing it in one. A rule such as $\underline{R} \mapsto \underline{K}\underline{R}\underline{K}^t$ or $\underline{R} \mapsto \underline{K}^t\underline{R}\underline{K}$ fails this test at once, since $\underline{K}^t$ does not compose in the required way, and it would then matter through which intermediate configurations one had passed. Nothing physical can depend on that.

Third: conjugation is an isomorphism, and that is the real content. The map $\underline{R} \mapsto \underline{K}\underline{R}\underline{K}^{-1}$ is one to one, is onto $\mathcal{G}_{\kappa_2(p_0)}$ by (8.26), and respects products:

$$(\underline{K}\underline{R}_1\underline{K}^{-1})(\underline{K}\underline{R}_2\underline{K}^{-1}) = \underline{K}\underline{R}_1(\underline{K}^{-1}\underline{K})\underline{R}_2\underline{K}^{-1} = \underline{K}(\underline{R}_1\underline{R}_2)\underline{K}^{-1}.$$

So the two symmetry groups are isomorphic as groups. Whatever is intrinsic to the group survives a change of reference: how many elements it has, whether it is finite, the order of each element, whether it is abelian. What does *not* survive is how the group sits inside the set of all tensors, which is to say whether its elements happen to be rotations. The distinction between the abstract group and its representation is exactly the distinction between what a material is and how a particular reference configuration displays it.

REMARK 8.10 ($\underline{K}$ is local, and what that buys). As with $\underline{R}$ in Remark 8.6, $\underline{K}$ is the gradient at $\underline{X}_1$ of a map defined on a neighbourhood of $\underline{X}_1$. It need not be the gradient of any map defined on the whole body, and in the applications that matter most it is not.

Plasticity is the standard example. There one imagines the deformation gradient factored as $\underline{F} = \underline{F}_e\underline{F}_p$, where $\underline{F}_p$ carries the reference neighbourhood to a locally relaxed state, from which $\underline{F}_e$ produces the actual current state elastically. The relaxed states are precisely local reference configurations in the sense of this section, one for each material point, chosen so that the lattice is undistorted there. In general no single configuration of the body relaxes every neighbourhood at once: the relaxed neighbourhoods do not fit together, $\underline{F}_p$ is not a gradient, and the residual stresses of a plastically worked body are the visible consequence. Noll's rule is what governs how the symmetry group is carried along by

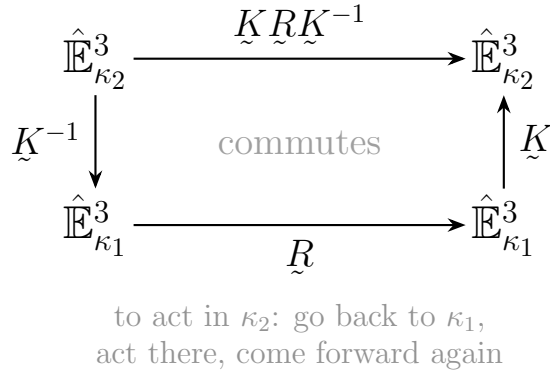


Figure B: Noll's rule as a change of description. A symmetry transformation is a linear map of the local reference translation space to itself, and $\tilde{K}$ is the dictionary between the two descriptions of that space. Going round the square the long way and the short way must give the same answer, and that single requirement is (8.26). Since $\det(\tilde{K}\tilde{R}\tilde{K}^{-1}) = \det \tilde{R}$, the top arrow is unimodular exactly when the bottom one is; but $\tilde{K}\tilde{R}\tilde{K}^{-1}$ is orthogonal only under special circumstances, which is why U is preserved and Orth^+ is not.

$\tilde{F}_p$ as the material yields, and it is the reason the theory can speak of a lattice orientation that evolves while the crystal class does not.

A worked example: a four-fold axis in a distorted configuration

Take the smallest interesting symmetry group and watch what (8.26) does to it. Let $\{\hat{E}_A\}$ be a fixed orthonormal basis of $\hat{\mathbb{E}}^3$ and let κ_1 be a configuration in which the material is a lattice generated by $\hat{E}_1$ and $\hat{E}_2$, with a four-fold axis along $\hat{E}_3$. Its symmetry group is cyclic of order four,

$$\mathcal{G}_{\kappa_1(p_0)} = \{\hat{I}, \tilde{R}, \tilde{R}^2, \tilde{R}^3\}, \quad \tilde{R}\hat{E}_1 = \hat{E}_2, \quad \tilde{R}\hat{E}_2 = -\hat{E}_1, \quad \tilde{R}\hat{E}_3 = \hat{E}_3,$$

so that $\tilde{R}$ is the rotation through 90° about $\hat{E}_3$, with components

$$[\tilde{R}] = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Since $\tilde{R}^t \tilde{R} = \hat{I}$ and $\det \tilde{R} = 1$, this group lies in Orth^+ and κ_1 is an undistorted configuration in the sense of (8.19): the material is a solid and κ_1 displays it as one.

Now change the reference by a pure stretch that preserves volume,

$$\tilde{K} = \alpha \hat{\tilde{E}}_1 \otimes \hat{\tilde{E}}_1 + \alpha^{-1} \hat{\tilde{E}}_2 \otimes \hat{\tilde{E}}_2 + \hat{\tilde{E}}_3 \otimes \hat{\tilde{E}}_3, \quad [\tilde{K}] = \begin{pmatrix} \alpha & 0 & 0 \\ 0 & \alpha^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

with $\alpha > 0$ and $\alpha \neq 1$; note $\det \tilde{K} = 1$, so κ_2 holds the same volume of material as κ_1 . Its inverse is $[\tilde{K}^{-1}] = \text{diag}(\alpha^{-1}, \alpha, 1)$. Conjugate, multiplying the matrices in order:

$$[\tilde{K}][\tilde{R}] = \begin{pmatrix} \alpha & 0 & 0 \\ 0 & \alpha^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\alpha & 0 \\ \alpha^{-1} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and then

$$[\tilde{H}] := [\tilde{K}][\tilde{R}][\tilde{K}^{-1}] = \begin{pmatrix} 0 & -\alpha & 0 \\ \alpha^{-1} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha^{-1} & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\alpha^2 & 0 \\ \alpha^{-2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Now read off what has and has not changed.

The determinant is unchanged. $\det \tilde{H} = -(-\alpha^2)(\alpha^{-2}) \cdot 1$ computed along the third row gives $\det \tilde{H} = 1$, so $\tilde{H} \in U$, as (8.26) guarantees in general.

Orthogonality is destroyed.

$$[\tilde{H}^t][\tilde{H}] = \begin{pmatrix} 0 & \alpha^{-2} & 0 \\ -\alpha^2 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -\alpha^2 & 0 \\ \alpha^{-2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \alpha^{-4} & 0 & 0 \\ 0 & \alpha^4 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is $\hat{I}$ only when $\alpha = 1$. So $\mathcal{G}_{\kappa_2(p_0)} \not\subset \text{Orth}^+$: in κ_2 the symmetry is no longer a rotation, and κ_2 is a distorted configuration.

The group structure is untouched. Squaring,

$$[\tilde{H}]^2 = \begin{pmatrix} 0 & -\alpha^2 & 0 \\ \alpha^{-2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = [\tilde{R}^2],$$

the two factors of α cancelling, and $\underline{H}^4 = (\underline{H}^2)^2 = \underline{\hat{I}}$. So $\underline{H}$ still has order four and $\mathcal{G}_{\kappa_2(p_0)} = \{\underline{\hat{I}}, \underline{H}, \underline{H}^2, \underline{H}^3\}$ is still cyclic of order four, as Remark 8.9 and the isomorphism argument require. The characteristic polynomial is likewise preserved by conjugation, so $\underline{H}$ has the same eigenvalues 1 and $\pm i$ as the rotation it came from, even though it is not a rotation.

And the geometry is exactly what one would draw. In κ_2 the lattice is generated by $\underline{K}\underline{\hat{E}}_1 = \alpha\underline{\hat{E}}_1$ and $\underline{K}\underline{\hat{E}}_2 = \alpha^{-1}\underline{\hat{E}}_2$, a rectangle rather than a square. Apply $\underline{H}$ to those generators:

$$\underline{H}(\alpha\underline{\hat{E}}_1) = \alpha \cdot \alpha^{-2}\underline{\hat{E}}_2 = \alpha^{-1}\underline{\hat{E}}_2, \quad \underline{H}(\alpha^{-1}\underline{\hat{E}}_2) = \alpha^{-1} \cdot (-\alpha^2)\underline{\hat{E}}_1 = -\alpha\underline{\hat{E}}_1.$$

Each generator is carried onto the other, so $\underline{H}$ maps the rectangular lattice onto itself, exactly as $\underline{R}$ mapped the square one onto itself. The symmetry is still there and is still detectable by no experiment; what has changed is that expressing it now requires a map that stretches one lattice direction by α^{-2} while compressing the other by α^2 . A rotation could not do it, because the two lattice directions no longer have equal spacing.

Consequences: fluidity is intrinsic, undistortedness is not

Two conclusions follow from (8.26), and they are the reason the definitions of Section 8.2 were phrased as they were.

Orth⁺ is not preserved. The worked example is a counterexample: $\mathcal{G}_{\kappa_1(p_0)} \subset \text{Orth}^+$ while $\mathcal{G}_{\kappa_2(p_0)}$ is not. So if $\kappa_1(p_0)$ is an undistorted state of a crystalline lattice, the lattice is distorted in $\kappa_2(p_0)$, and (8.19) fails there. This is why the existence of *some* local configuration in which (8.19) holds is an essential part of the definition of a solid: solidity is the statement that an undistorted configuration exists, not that the one in front of us is undistorted.

It is worth recording exactly when nothing goes wrong. If $\underline{K} = c\underline{Q}$ with $c > 0$ and $\underline{Q}$ orthogonal, then $\underline{K}^{-1} = c^{-1}\underline{Q}^t$ and

$$\underline{K}\underline{R}\underline{K}^{-1} = c\underline{Q}\underline{R}c^{-1}\underline{Q}^t = \underline{Q}\underline{R}\underline{Q}^t,$$

the scalars cancelling, and $\underline{Q}\underline{R}\underline{Q}^t$ is orthogonal whenever $\underline{R}$ is, since $(\underline{Q}\underline{R}\underline{Q}^t)^t(\underline{Q}\underline{R}\underline{Q}^t) = \underline{Q}\underline{R}^t\underline{Q}^t\underline{Q}\underline{R}\underline{Q}^t = \underline{Q}\underline{R}^t\underline{R}\underline{Q}^t = \underline{Q}\underline{Q}^t = \hat{I}$. So rotating an undistorted configuration, or dilating it uniformly, leaves it undistorted, and the undistorted configurations of a solid form a family closed under those operations and no others. Anything that strains the neighbourhood spoils it, which is the same statement as (8.19) viewed from the other side.

U is preserved. Suppose $\mathcal{G}_{\kappa_1(p_0)} = U$. Every $\underline{R} \in U$ has $\det(\underline{K}\underline{R}\underline{K}^{-1}) = \det \underline{R} = 1$, so $\underline{K}U\underline{K}^{-1} \subseteq U$; conversely every $\underline{S} \in U$ is $\underline{K}(\underline{K}^{-1}\underline{S}\underline{K})\underline{K}^{-1}$ with $\det(\underline{K}^{-1}\underline{S}\underline{K}) = \det \underline{S} = 1$, so $U \subseteq \underline{K}U\underline{K}^{-1}$. Hence $\underline{K}U\underline{K}^{-1} = U$ and, by (8.26), $\mathcal{G}_{\kappa_2(p_0)} = U$ as well.

So (8.18) is satisfied in all configurations, and more: the *identification* of the symmetry group with the unimodular group is independent of the reference configuration. Being a fluid is a property of the substance and of nothing else, and no change of reference, however violent, can make a fluid look like a solid or a solid like a fluid. That is in accord with the intuitive notion of fluidity, and it is the reason the definition of a fluid could have been stated without the words “there exists a local reference configuration”, whereas the definition of a solid could not.

8.4 Frame invariance

In building the class of constitutive equations of Section 8.1 we adopted the perspective of one particular observer O . Chapter 7 was written to make the second observer available, and this section spends it.

Recall the setting. An observer is equipped with a point space and its translation space, and two observers agree about time intervals and about distances between material points. Section 7.1 showed that these two agreements force the relation between their perceptions to be (7.3),

$$\underline{x}^+ = \underline{Q}(t)\underline{x} + \underline{c}(t),$$

with $\underline{c}(t)$ arbitrary and $\underline{Q}(t)$ a spatially uniform, time-dependent two-point orthogonal tensor, together with $t^+ = t + a$ from (7.1). Sections 7.3

and 7.4 then computed what every kinematic and mechanical variable does under such a change, and those computations are the dictionary this section uses. The entries needed here are

$$\theta^+ = \theta, \quad \varepsilon^+ = \varepsilon, \quad \eta^+ = \eta, \quad (7.37): \text{ absolute scalars,}$$

$$\underline{\underline{F}}_{\kappa}^+ = \underline{\underline{Q}} \underline{\underline{F}}_{\kappa} \underline{\underline{K}}^t, \quad (7.17),$$

$$\underline{\underline{L}}^+ = \underline{\underline{Q}} \underline{\underline{L}} \underline{\underline{Q}}^t + \underline{\underline{\Omega}}, \quad \underline{\underline{\Omega}} = \dot{\underline{\underline{Q}}} \underline{\underline{Q}}^t, \quad (7.33),$$

$$\underline{\underline{T}}^+ = \underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t, \quad (7.40),$$

$$\underline{\underline{q}}^+ = \underline{\underline{Q}} \underline{\underline{q}}, \quad (7.44),$$

where $\underline{\underline{K}}$ is the fixed orthogonal two-point tensor of (7.14) relating the two observers' reference placements, $\underline{\underline{X}}^+ = \underline{\underline{K}} \underline{\underline{X}} + \underline{\underline{c}}^+$. One entry of the list (8.7) was not computed in Chapter 7 and takes one line here. Temperature is an absolute scalar, so the increment $d\theta$ across a given pair of neighbouring places is the same for both observers, while $d\underline{\underline{x}}^+ = \underline{\underline{Q}} d\underline{\underline{x}}$ by (7.3); therefore

$$\begin{aligned} \text{grad}^+ \theta^+ \cdot \underline{\underline{Q}} d\underline{\underline{x}} &= d\theta^+ = d\theta \\ &= \text{grad } \theta \cdot d\underline{\underline{x}} \implies (\underline{\underline{Q}}^t \text{grad}^+ \theta^+ - \text{grad } \theta) \cdot d\underline{\underline{x}} = 0 \end{aligned}$$

for every $d\underline{\underline{x}}$, whence $\underline{\underline{Q}}^t \text{grad}^+ \theta^+ = \text{grad } \theta$ and, multiplying by $\underline{\underline{Q}}$ and using $\underline{\underline{Q}} \underline{\underline{Q}}^t = \underline{\underline{I}}$,

$$\text{grad}^+ \theta^+ = \underline{\underline{Q}} \text{grad } \theta.$$

The move is (8.5)–(8.6) once more, with $\underline{\underline{Q}}$ in place of $\underline{\underline{F}}$; it is the third time in this chapter that the arbitrariness of a line element has converted a statement about increments into a statement about gradients.

What both observers must be given

Material frame indifference is the requirement that all observers agree on the nature of the material at hand, and hence on the class of processes to which the material responds. With reference to (8.4) and (8.7), then, the response functions in the frame of O^+ should deliver the values of

the elements of the set

$$C^+ = \{\tilde{T}^+, \tilde{q}^+, \psi^+, \eta^+\} \quad (8.27)$$

in terms of the elements of the list

$$L^+ = \{\tilde{x}^+, \tilde{v}^+, \theta^+, \tilde{F}_{\kappa^+}^+, \tilde{L}^+, \text{grad}^+ \theta^+\}, \quad (8.28)$$

all evaluated at the material point $p \in B$ and the time t^+ . So if O adopts the constitutive function $F_\kappa(\tilde{x}, \tilde{v}, \theta, \tilde{F}_\kappa, \tilde{L}, \text{grad } \theta; \tilde{X})$ to determine the values of $F(p, t) \in C$, then O^+ should adopt a function $F_{\kappa^+}^+$ of the starred arguments to evaluate its counterpart $F^+(p, t^+) \in C^+$.

Note what (8.27) and (8.28) are asserting, since at first sight they assert nothing. They say that the *form* of the constitutive hypothesis is the same for both observers: O^+ has a set of four responses determined by a list of six process variables, in one-to-one correspondence with O 's. If this failed, the two observers would not be describing the same material at all, and there would be nothing left to compare. It is the counterpart, for a change of observer, of the statement made after (8.12) for a change of reference.

REMARK 8.11 (the star on κ). The reference configuration carries a star too, and it is worth saying why this changes little. A reference configuration is a placement of the body chosen for bookkeeping, not a thing either observer watches, and by (7.14) the two placements differ by $\tilde{X}^+ = \tilde{K}\tilde{X} + \tilde{c}^+$ with $\tilde{K}$ a *fixed orthogonal* two-point tensor and $\tilde{c}^+$ a fixed vector. Fixed, because it is built in (7.15) from $\tilde{Q}(t_0)$ and the shifters at the initial instant, and none of those depends on t .

Two things follow. First, $\tilde{K}$ is a change of reference configuration of exactly the kind treated in Section 8.3, so its effect on the symmetry group is Noll's rule, and since $\tilde{K}$ is orthogonal the conjugation is $\tilde{K}\mathcal{G}\tilde{K}^t$, which by the calculation after (8.26) carries Orth^+ into itself: a change of observer never distorts an undistorted configuration. Second, if the two observers agree to use the same reference placement, as one always may by taking $\tilde{Q}(t_0) = \tilde{I}$ and equal shifters, then $\tilde{K} = \tilde{I}$, $\tilde{c}^+ = \mathbf{0}$, $\tilde{X}^+ = \tilde{X}$, $\kappa^+ = \kappa$ and (7.17) reduces to $\tilde{F}^+ = \tilde{Q}\tilde{F}$. That is the convention used

in every calculation below. It costs nothing, and it removes a star that would otherwise clutter every formula without changing any conclusion.

The four transformation statements

Take the four members of C in turn. If the free energy and entropy are determined by O to be

$$\begin{aligned} \psi &= \hat{\psi}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}) \\ \text{and } \eta &= \hat{\eta}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}), \end{aligned} \quad (8.29)$$

in terms of constitutive functions $\hat{\psi}_\kappa$ and $\hat{\eta}_\kappa$, then the corresponding constitutive equations in the frame of O^+ are

$$\begin{aligned} \hat{\psi}_{\kappa^+}(\underline{x}^+, \underline{v}^+, \theta^+, \underline{F}_{\kappa^+}^+, \underline{L}^+, \text{grad}^+ \theta^+; \underline{X}^+) &= \hat{\psi}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}) \\ \text{and } \hat{\eta}_{\kappa^+}(\underline{x}^+, \underline{v}^+, \theta^+, \underline{F}_{\kappa^+}^+, \underline{L}^+, \text{grad}^+ \theta^+; \underline{X}^+) &= \hat{\eta}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}), \end{aligned} \quad (8.30)$$

respectively. There is no $\underline{Q}$ anywhere in (8.30), and the reason is (6.186) together with (7.37): the energy ε , the temperature θ and the entropy η are absolute scalars, hence $\psi^+ = \varepsilon^+ - \theta^+ \eta^+ = \varepsilon - \theta \eta = \psi$, and a scalar has no components to rotate.

Similarly, if the heat flux is determined by O to be

$$\underline{q} = \hat{\underline{q}}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}) \quad (8.31)$$

in terms of a constitutive function $\hat{\underline{q}}_\kappa$, then, with reference to (7.44), the corresponding constitutive equation in the frame of O^+ is

$$\hat{\underline{q}}_{\kappa^+}^+(\underline{x}^+, \underline{v}^+, \theta^+, \underline{F}_{\kappa^+}^+, \underline{L}^+, \text{grad}^+ \theta^+; \underline{X}^+) = \underline{Q} \hat{\underline{q}}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta; \underline{X}), \quad (8.32)$$

where $\underline{Q}$ is the orthogonal transformation of (7.3) that characterizes the change of frame. The single factor $\underline{Q}$ is the mark of a spatial vector: one index, one rotation.

REMARK 8.12 ($\hat{\underline{q}}_\kappa$ is not $\underline{q}_\kappa$). The subscript κ on $\hat{\underline{q}}_\kappa$ records that the function takes $\underline{F}_\kappa$ among its arguments and therefore depends on the

reference configuration, exactly as in (8.10). It does *not* mean that the output is referential. The output is the spatial heat flux $\underline{q}$, a vector in $\mathbb{E}^3$, and it transforms with one factor of $\underline{Q}$.

The referential heat flux of (6.171), $\underline{q}_\kappa = (\underline{F}^*)^t \underline{q} = J \underline{F}^{-1} \underline{q}$, is a different object living in $\hat{\mathbb{E}}^3$, and by (7.45) it transforms with $\underline{K}$ rather than $\underline{Q}$. The two are distinguished here by the caret, which marks a constitutive *function* rather than a value, and by nothing else, so the caret has to be read. Getting them confused would put a $\underline{Q}$ where a $\underline{K}$ belongs and would make a frame-indifferent theory look frame dependent.

Lastly, if O finds the stress to be determined constitutively by

$$\underline{T} = \hat{\underline{T}}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}; \underline{X}), \quad (8.33)$$

then the stress is determined by O^+ to be

$$\underline{T}^+ = \hat{\underline{T}}_{\kappa^+}(\underline{x}^+, \underline{v}^+, \theta^+, \underline{F}_{\kappa^+}, \underline{L}^+; \underline{X}^+), \quad (8.34)$$

where, by (7.40),

$$\hat{\underline{T}}_{\kappa^+}(\underline{x}^+, \underline{v}^+, \theta^+, \underline{F}_{\kappa^+}, \underline{L}^+; \underline{X}^+) = \underline{Q} \hat{\underline{T}}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}; \underline{X}) \underline{Q}^t. \quad (8.35)$$

Two factors of $\underline{Q}$, one for each index, which is what Theorem 1.39 says a second-order spatial tensor must do.

The dependence of the stress on the temperature gradient has been suppressed in (8.33) to (8.35). This is in accord with classical treatments of the present class of constitutive equations, and it is a restriction on the model rather than a consequence of anything proved so far; the entropy inequality is what eventually justifies it, and the matter is taken up again in the chapters that follow.

Where the restriction actually comes from

Equations (8.30), (8.32) and (8.35) look like restrictions and, taken by themselves, are not. Read literally, each of them *defines* the starred function on the left in terms of the unstarred one on the right: given $\hat{\underline{T}}_\kappa$, and given the dictionary of Chapter 7 that expresses the starred

arguments through the unstarred, the right-hand side of (8.35) is a computable expression in the starred arguments, and (8.35) names it $\hat{T}_{\kappa^+}^+$. Nothing has been forbidden.

The physical principle is the further demand that the two observers use *the same function*:

$$\hat{\psi}_{\kappa^+}^+ = \hat{\psi}_{\kappa}, \quad \hat{\eta}_{\kappa^+}^+ = \hat{\eta}_{\kappa}, \quad \hat{q}_{\kappa^+}^+ = \hat{q}_{\kappa}, \quad \hat{T}_{\kappa^+}^+ = \hat{T}_{\kappa},$$

under the convention $\kappa^+ = \kappa$ of Remark 8.11. This is the content of material frame indifference: the elasticity of copper is a property of copper, so the function that gives the stress of copper cannot depend on which laboratory bench it was measured from. Impose it on (8.35) and the definition turns into a functional equation,

$$\hat{T}_{\kappa}(x^+, v^+, \theta, QF, QLQ^t + \Omega; X) = Q \hat{T}_{\kappa}(x, v, \theta, F, L; X) Q^t,$$

required to hold for every change of frame, that is, for every choice of the functions $Q(t)$ and $c(t)$. Now something is forbidden, and what is forbidden is found by choosing Q and c cleverly. The chapters that follow carry this out in full for the two sub-models, viscous fluids and elastic solids. Two consequences are worth extracting here, the first because Section 8.1 promised it.

REMARK 8.13 (the constitutive functions cannot depend on x or v). Fix an instant t later than the instant t_0 at which the reference placements were fixed, and consider the change of frame with

$$Q(\tau) = I \text{ for all } \tau, \quad c(t_0) = 0, \quad c(t) = a, \quad \dot{c}(t) = b,$$

where a and b are arbitrary vectors. Such a c exists: three conditions at two distinct instants are met by, for example, a cubic polynomial in τ , and no relation between a and b is implied because $t \neq t_0$.

Work out what this frame change does to each entry of the list, at the instant t .

- $Q(t_0) = I$ and equal shifters give $K = \hat{I}$ and $c^+ = 0$ in (7.14), so $\hat{X}^+ = \hat{X}$ and $\kappa^+ = \kappa$.

- $\underline{x}^+ = \underline{Q}\underline{x} + \underline{c} = \underline{x} + \underline{a}$ by (7.3).
- $\underline{\Omega} = \dot{\underline{Q}}\underline{Q}^t = \underline{O}$ since $\underline{Q}$ is constant, so (7.8) gives $\underline{v}^+ = \underline{v} + \underline{\dot{c}} = \underline{v} + \underline{b}$.
- $\underline{F}^+ = \underline{Q}\underline{F}\underline{K}^t = \underline{F}$ and, by (7.33), $\underline{L}^+ = \underline{Q}\underline{L}\underline{Q}^t + \underline{\Omega} = \underline{L}$.
- $\theta^+ = \theta$ and $\text{grad}^+ \theta^+ = \underline{Q} \text{grad } \theta = \text{grad } \theta$.

So this change of frame moves $\underline{x}$ and $\underline{v}$ and moves nothing else at all. Since $\underline{Q} = \underline{I}$, (8.35) reads $\hat{\underline{T}}_\kappa^+ = \hat{\underline{T}}_\kappa$, and frame indifference then gives

$$\hat{\underline{T}}_\kappa(\underline{x} + \underline{a}, \underline{v} + \underline{b}, \theta, \underline{F}, \underline{L}; \underline{X}) = \hat{\underline{T}}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}, \underline{L}; \underline{X})$$

for every $\underline{a}$ and every $\underline{b}$. A function whose value is unchanged when its first two arguments are shifted by arbitrary amounts does not depend on those arguments. The same argument applied to (8.30) and (8.32), in which $\underline{Q} = \underline{I}$ likewise removes every factor of $\underline{Q}$, disposes of $\underline{x}$ and $\underline{v}$ in $\hat{\underline{\psi}}_\kappa$, $\hat{\underline{\eta}}_\kappa$ and $\hat{\underline{q}}_\kappa$ as well. This is the claim announced without proof in Section 8.1, and the reason it holds is worth stating in words: $\underline{x}$ is where the particle happens to be and $\underline{v}$ is how fast it happens to be going, and both are artefacts of the observer's choice of origin and of state of motion. A material cannot respond to either, because neither is a property of the material.

The reduced lists are therefore

$$C = \{\underline{T}, \underline{q}, \psi, \eta\} \quad \text{determined by} \quad \{\theta, \underline{F}_\kappa, \underline{L}, \text{grad } \theta\},$$

and it is these four arguments, not six, that the following chapters work with.

REMARK 8.14 (a preview: why viscous stress is written in $\underline{D}$). One more choice of $\underline{Q}$ is instructive, and it explains a feature of every viscous constitutive equation the reader has ever seen.

Choose $\underline{Q}(\tau)$ with $\underline{Q}(t_0) = \underline{I}$ and $\underline{Q}(t) = \underline{I}$ but $\dot{\underline{Q}}(t) = \underline{\Omega}$ an arbitrary skew tensor; a rotation that starts at the identity, wanders, and returns to the identity at time t with non-zero angular velocity will do. At the instant t we then have $\underline{Q} = \underline{I}$, so

$$\underline{F}^+ = \underline{F}, \quad \theta^+ = \theta, \quad \text{grad}^+ \theta^+ = \text{grad } \theta, \quad \underline{T}^+ = \underline{T},$$

while (7.33) gives $\underline{L}^+ = \underline{L} + \underline{\Omega}$. Frame indifference applied to (8.35), with $\underline{x}$ and $\underline{v}$ already gone by Remark 8.13, therefore requires

$$\hat{\underline{T}}_\kappa(\theta, \underline{F}, \underline{L} + \underline{\Omega}; \underline{X}) = \hat{\underline{T}}_\kappa(\theta, \underline{F}, \underline{L}; \underline{X}) \quad \text{for every skew } \underline{\Omega}.$$

Choose $\underline{\Omega} = -\underline{W}$, the spin tensor of (4.7), which is admissible because $\underline{W}$ is skew. Then $\underline{L} + \underline{\Omega} = \underline{D} + \underline{W} - \underline{W} = \underline{D}$ and

$$\hat{\underline{T}}_\kappa(\theta, \underline{F}, \underline{L}; \underline{X}) = \hat{\underline{T}}_\kappa(\theta, \underline{F}, \underline{D}; \underline{X}) :$$

the stress can depend on the velocity gradient only through its symmetric part. The same holds for $\hat{\underline{\psi}}_\kappa$, $\hat{\underline{\eta}}_\kappa$ and $\hat{\underline{q}}_\kappa$.

This is the constitutive counterpart of Remark 4.9. The spin $\underline{W}$ measures rotation relative to the observer's frame, and an observer who spins along with the material sees a different $\underline{W}$; (7.34) says so explicitly, $\underline{W}^+ = \underline{Q}\underline{W}\underline{Q}^t + \underline{\Omega}$, with the inhomogeneous term $\underline{\Omega}$ that $\underline{D}^+ = \underline{Q}\underline{D}\underline{Q}^t$ does not have. A material cannot respond to a quantity that an observer can change by turning around, so $\underline{W}$ must drop out and $\underline{D}$ must remain. Every constitutive law for a viscous fluid is written in $\underline{D}$ for this reason and for no other.

What frame invariance leaves undone

Frame invariance alone does not suffice to restrict the class of admissible constitutive functions. Nothing said so far prevents a material from generating energy out of nothing, or from conducting heat from cold to hot; those are thermodynamic prohibitions and the requirement that two observers agree cannot see them. We must also ensure that the constitutive functions are compatible with the balances of mass, energy and momentum and with the Clausius–Duhem inequality of Section 6.7.

Of those four, three are quickly disposed of. Conservation of mass imposes nothing on the list, since it merely determines ρ from ρ_κ and $\det \underline{F}$. The balance of energy imposes nothing either, since the heat supply r is at our disposal and may be chosen to satisfy it whatever the other fields do; this is exactly the freedom that the Coleman–Noll argument below exploits. The balance of linear momentum imposes

nothing beyond what (8.35) already contains, but the balance of moment of momentum does: it requires $\underline{T} = \underline{T}^t$, hence $\hat{T}_\kappa = \hat{T}_\kappa^t$ identically in its arguments, which is six scalar conditions rather than nine.

The Clausius–Duhem inequality is the substantial one, and the following remark sketches how it is used, both because it is the organising principle of every constitutive chapter that follows and because it shows why Section 8.1 took the trouble to trade ε for ψ .

REMARK 8.15 (how the entropy inequality bites). The local inequality (6.187) of Chapter 6 is

$$-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{T} \cdot \underline{\dot{L}} - \theta^{-1}\underline{q} \cdot \text{grad } \dot{\theta} \geq 0,$$

and it is required to hold in *every* admissible process, not in some. Substitute the constitutive equations in their reduced form,

$$\psi = \hat{\psi}_\kappa(\theta, \underline{F}, \underline{L}, \text{grad } \theta; X)$$

and so on, and expand $\dot{\psi}$ by the chain rule:

$$\dot{\psi} = \frac{\partial \hat{\psi}_\kappa}{\partial \theta} \dot{\theta} + \frac{\partial \hat{\psi}_\kappa}{\partial \underline{F}} \cdot \underline{\dot{F}} + \frac{\partial \hat{\psi}_\kappa}{\partial \underline{L}} \cdot \underline{\dot{L}} + \frac{\partial \hat{\psi}_\kappa}{\partial (\text{grad } \theta)} \cdot (\text{grad } \dot{\theta}).$$

Collect terms:

$$\begin{aligned} & -\rho \left(\frac{\partial \hat{\psi}_\kappa}{\partial \theta} + \eta \right) \dot{\theta} - \rho \frac{\partial \hat{\psi}_\kappa}{\partial \underline{L}} \cdot \underline{\dot{L}} - \rho \frac{\partial \hat{\psi}_\kappa}{\partial (\text{grad } \theta)} \cdot (\text{grad } \dot{\theta}) \\ & - \rho \frac{\partial \hat{\psi}_\kappa}{\partial \underline{F}} \cdot \underline{\dot{F}} + \underline{T} \cdot \underline{\dot{L}} - \theta^{-1} \underline{q} \cdot \text{grad } \dot{\theta} \geq 0. \end{aligned}$$

Now the decisive observation. The three quantities $\dot{\theta}$, $\underline{\dot{L}}$ and $(\text{grad } \dot{\theta}) \cdot$ appear *linearly*, and their coefficients do not contain them: the coefficients are functions of $(\theta, \underline{F}, \underline{L}, \text{grad } \theta)$ alone. But at a given material point and instant one may prescribe a process realising any values of $\dot{\theta}$, $\underline{\dot{L}}$ and $(\text{grad } \dot{\theta}) \cdot$ one pleases, with $(\theta, \underline{F}, \underline{L}, \text{grad } \theta)$ held at whatever values are under consideration. A quantity of the form $\alpha z + \beta$, with α and β independent of z and z free to take any value including large

negative ones, is bounded below only if $\alpha = 0$. Hence each of the three coefficients vanishes:

$$\eta = -\frac{\partial \hat{\psi}_\kappa}{\partial \theta}, \quad \frac{\partial \hat{\psi}_\kappa}{\partial \underline{L}} = \underline{0}, \quad \frac{\partial \hat{\psi}_\kappa}{\partial (\text{grad } \theta)} = \underline{0}.$$

Three results at once, and none of them was assumed. The free energy depends on θ and $\underline{F}$ only, so it is purely elastic; the entropy is not an independent constitutive function at all but the temperature derivative of the free energy; and the viscosity, which lives in the dependence on $\underline{L}$, has been banished from the energy and confined to the stress.

What remains, after $\dot{\underline{F}} = \underline{L}\underline{F}$ from (4.3) is used to write $(\partial \hat{\psi}_\kappa / \partial \underline{F}) \cdot \dot{\underline{F}} = [(\partial \hat{\psi}_\kappa / \partial \underline{F}) \underline{F}^t] \cdot \underline{L}$, is

$$\left(\underline{T} - \rho \frac{\partial \hat{\psi}_\kappa}{\partial \underline{F}} \underline{F}^t \right) \cdot \underline{L} - \theta^{-1} \underline{q} \cdot \text{grad } \theta \geq 0,$$

the *residual dissipation inequality*. It does not determine anything, because $\underline{L}$ and $\text{grad } \theta$ are among the constitutive arguments and so cannot be varied freely against $\underline{T}$ and $\underline{q}$. What it does is split the stress into an equilibrium part $\rho(\partial \hat{\psi}_\kappa / \partial \underline{F}) \underline{F}^t$, delivered by the free energy, and a dissipative remainder that must do non-negative work, and it forces heat to flow down the temperature gradient. Those two statements are the whole thermodynamic content of the theory, and the following chapters develop them in the contexts of distinct sub-models for viscous fluids and elastic solids.

The sketch also settles a question left open at (8.33): why the temperature gradient could be dropped from the stress. It is not dropped by fiat. Once $\hat{\psi}_\kappa$ is known to be independent of $\text{grad } \theta$, the residual inequality above must hold for $\text{grad } \theta = \underline{0}$ as well, and the argument that then bounds the $\text{grad } \theta$ dependence of $\underline{T}$ is the one carried out in the chapters that follow.

With that the general theory stops. Nothing more can be said about constitutive equations without committing to a particular kind of material, since everything assembled here holds equally of steel and of honey:

the locality assumption of Section 8.1, the functional equation (8.16) and the group it defines, Noll's rule (8.26), and the frame-invariance requirements (8.30), (8.32) and (8.35). The commitment is made along the fault line that Section 8.2 opened. Take $\mathcal{G} = U$ and the material is a fluid, which is the subject of the next chapter; take $\mathcal{G} \subseteq \text{Orth}^+$ in some undistorted configuration and it is a solid. The constraints of this chapter are what both of them are built inside.

Chapter 9

Viscous Fluids

Chapters 1 to 5 built the algebra and the calculus, Chapter 6 the balance laws, Chapter 7 the comparison of observers and Chapter 8 the shape a constitutive equation may take. None of that, taken alone, predicts anything. In this chapter the machinery is applied to a class of real materials, and what comes out is the Navier–Stokes equations. They are usually written down as a plausible postulate; here they appear as what is left once fluidity, frame indifference and the entropy inequality have each been imposed.

The route is worth seeing whole before it is walked, because each step removes something from the constitutive function and the removals compound. Fluidity comes first. The symmetry group of a fluid is the whole unimodular group, a unimodular map cannot change $\det \underline{F}$, and so of the deformation gradient only its determinant survives; by conservation of mass that determinant is a function of the density. Frame indifference comes next. It removes the position and the velocity outright, and it removes the spin $\underline{W}$ from the velocity gradient, so that of $\underline{L}$ only $\underline{D}$ remains. The device that achieves this, a comparison of two relative motions of the observers carried out through a third observer, is the subject of Section 9.1, and it is there that the positions $\underline{x}_1$, $\underline{x}_2$ and $\underline{x}^+$ appear. The same requirement then makes the stress an isotropic function of $\underline{D}$, which is a consequence and not an extra hypothesis, and the representation theorem of Section 9.1.1 converts this into a polynomial of degree two in $\underline{D}$ whose three coefficients are functions of three scalars.

Linearity in $\underline{D}$, the one modelling assumption made rather than derived, reduces those coefficients to two numbers, the viscosities. Finally the Clausius–Duhem inequality fixes the signs of the two viscosities and of the conductivity, and it identifies the pressure at rest with a derivative of the free energy.

Chapter 8 ended with the general constitutive statement (8.33) for the stress,

$$\underline{T} = \hat{\underline{T}}_{\kappa}(\underline{x}, \underline{v}, \theta, \underline{F}_{\kappa}, \underline{L}; \underline{X}),$$

together with (8.31) for the heat flux and (8.29) for the free energy and the entropy. We begin by imposing on (8.33) the symmetry condition that defines a fluid.

Fluidity, and the disappearance of the reference configuration

For fluids the constitutive equation (8.33) for the stress, in the frame of O , satisfies the material symmetry condition (8.16) with $\mathcal{G}_{\kappa(p_0)} = U$, that is

$$\hat{\underline{T}}_{\kappa}(\underline{x}, \underline{v}, \theta, \underline{F}, \underline{L}; \underline{X}) = \hat{\underline{T}}_{\kappa}(\underline{x}, \underline{v}, \theta, \underline{F}\underline{R}, \underline{L}; \underline{X}) \quad (9.1)$$

for all fixed $\underline{R} \in U$, the unimodular group of (8.18).

To derive a necessary condition for (9.1) we consider a fixed instant t_* , say, and select

$$\underline{R} = J^{1/3}(\underline{1}^t \underline{F})^{-1},$$

in which $\underline{1}$ is the fixed shifter of Definition 2.23 in the frame of O and $\underline{F}$ is fixed at the value $\underline{F}(\underline{X}, t_*)$, with determinant J . In terms of components,

$$\underline{1}^t \underline{F} = \delta_{iA} F_{iB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B.$$

Three properties of this choice have to be checked before it may be used, and each takes a line.

The first is that $\underline{R}$ is a referential tensor. A symmetry transformation is required by (8.8) to be the gradient of a map from one reference configuration to another, hence a linear map of $\hat{\mathbb{E}}^3$ to itself. Now $\underline{F}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$, so $\underline{F}^{-1}$ maps $\mathbb{E}^3$ to $\hat{\mathbb{E}}^3$, and $\underline{F}^{-1}$ alone will not do:

something is needed that first returns to $\hat{\mathbb{E}}^3$. The shifter supplies it, since $\underline{1}^t: \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, so that $\underline{1}^t \underline{F}: \hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^3$ and its inverse are referential. This is Remark 2.25 at work; without the shifter the expression would not even be well typed.

The second is that $\underline{R}$ is unimodular. By Problem 21(a) of Chapter 1, $\det(\alpha A) = \alpha^3 \det A$, and by Proposition 1.46, $\det(A^{-1}) = (\det A)^{-1}$. The matrix of $\underline{1}^t \underline{F}$ displayed above is $\delta_{iA} F_{iB}$, which is the matrix $[F_{AB}]$ with its first index relabelled, so $\det(\underline{1}^t \underline{F}) = \det \underline{F} = J$. Hence

$$\det \underline{R} = (J^{1/3})^3 \det [(\underline{1}^t \underline{F})^{-1}] = J \cdot J^{-1} = 1,$$

so $\underline{R} \in U$. It is fixed, because $\underline{F}$ was frozen at its value at t_* .

The third, and the reason for the choice, is that $\underline{R}$ removes $\underline{F}$. Since $\underline{1}$ is orthogonal, $(\underline{1}^t)^{-1} = \underline{1}$, and by Proposition 1.46, $(\underline{1}^t \underline{F})^{-1} = \underline{F}^{-1}(\underline{1}^t)^{-1} = \underline{F}^{-1} \underline{1}$. Therefore

$$\underline{F} \underline{R} = J^{1/3} \underline{F} \underline{F}^{-1} \underline{1} = J^{1/3} \underline{1}.$$

The arbitrary $\underline{F}$ has been replaced by a multiple of the shifter, and the only trace of $\underline{F}$ left is the scalar J . Substituting into (9.1),

$$\hat{T}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \underline{L}; \underline{X}) = \hat{T}_\kappa(\underline{x}, \underline{v}, \theta, J^{1/3} \underline{1}, \underline{L}; \underline{X}). \quad (9.2)$$

Conversely, (9.2) satisfies (9.1) for all unimodular $\underline{R}$, and is thus also sufficient for fluidity at the instant t_* : replacing $\underline{F}$ by $\underline{F} \underline{R}$ with $\det \underline{R} = 1$ leaves $\det(\underline{F} \underline{R}) = (\det \underline{F})(\det \underline{R}) = J$ unchanged, so the right-hand side of (9.2) does not move, and neither therefore does the left. The instant is arbitrary, however, and so (9.2) holds at all instants.

Further, from (2.59) and the conservation of mass (6.18) we have that

$$J = \frac{\rho_\kappa(\underline{X})}{\rho}.$$

Thus the stress at $\underline{X}$ is determined by the list $\{\underline{x}, \underline{v}, \theta, \rho, \underline{L}\}$: the deformation gradient has been traded for a single scalar, and that scalar is the current density, a spatial field that owes nothing to κ . Then, using

$\underline{X} = \underline{\kappa}(p)$, which is (2.3), we may eliminate the reference configuration from the constitutive function altogether and write

$$\underline{T} = \mathcal{G}(\underline{x}, \underline{v}, \theta, \rho, \underline{L}; p). \quad (9.3)$$

The calligraphic letter is the book's. Chapter 8 used it for the symmetry group $\mathcal{G}_{\kappa(p_0)}$, which always carried a subscript and has now done its work; from (9.3) on, $\mathcal{G}$ without a subscript is the constitutive function of the stress.

REMARK 9.1 (what has just been bought). Compare (9.3) with (8.10). Two changes have occurred, and the second is the important one.

The obvious change is arithmetic: a tensor argument with nine components has become a scalar argument with one. The subtle change is that the parameter after the semicolon is now the material point p itself and not its label $\underline{X}$ in some chosen configuration. Every other chapter of this book has had to carry a reference configuration because $\underline{F}$ has no meaning without one, and Sections 8.2 and 8.3 were entirely about the consequences of that dependence: how the response function changes when κ changes, and how the symmetry group is conjugated. For a fluid all of that machinery becomes vacuous, because κ has vanished from (9.3).

That is what “a fluid has no preferred shape” means when it is written out. A solid remembers a configuration and its constitutive function must say which; a fluid remembers nothing but how much of itself is packed into a given volume. Everything else in this chapter is a consequence of the shortness of the list in (9.3), and the list is short because $\mathcal{G}_{\kappa(p_0)} = U$ is as large as a symmetry group can be.

9.1 Material frame indifference

Of course any other observer, O^+ say, can follow the same line of reasoning to conclude that the stress is expressible in the form

$$\underline{T}^+ = \mathcal{G}^+(\underline{x}^+, \underline{v}^+, \theta^+, \rho^+, \underline{L}^+; p), \quad (9.4)$$

for some suitable constitutive function $\mathcal{G}^+$. This function is related to $\mathcal{G}$ by

$$\mathcal{G}^+(\underline{x}^+, \underline{v}^+, \theta^+, \rho^+, \underline{L}^+; p) = \underline{Q} \mathcal{G}(\underline{x}, \underline{v}, \theta, \rho, \underline{L}; p) \underline{Q}^t, \quad (9.5)$$

where $\underline{Q}(t): \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$ is the orthogonal tensor associated with the change of frame. Equation (9.5) is (8.35) with the reduced list of (9.3), the two factors of $\underline{Q}$ coming from the transformation rule (7.40) of the Cauchy stress. We wish to use this statement to derive restrictions on the constitutive function $\mathcal{G}$ pertaining to O . That pertaining to O^+ , if desired, can then be obtained a posteriori; Problem 2 does exactly that.

What (9.5) asks of $\mathcal{G}$

Taken by itself, for one change of frame, (9.5) forbids nothing. At a fixed instant the dictionary of Chapter 7 can be inverted. From (7.3), (7.8), (7.33) and (7.37), with $\underline{\Omega} = \dot{\underline{Q}}\underline{Q}^t$ and using $\underline{Q}^t\underline{Q} = \underline{I}$,

$$\begin{aligned} \underline{x} &= \underline{Q}^t(\underline{x}^+ - \underline{c}), & \underline{v} &= \underline{Q}^t[\underline{v}^+ - \underline{\Omega}(\underline{x}^+ - \underline{c}) - \dot{\underline{c}}], \\ \underline{L} &= \underline{Q}^t(\underline{L}^+ - \underline{\Omega})\underline{Q}, & \theta &= \theta^+, & \rho &= \rho^+. \end{aligned}$$

Every argument on the right of (9.5) is therefore a known expression in the plus arguments, and (9.5) may be read simply as the definition of $\mathcal{G}^+$: substitute these five expressions into $\underline{Q} \mathcal{G}(\dots) \underline{Q}^t$ and call the result $\mathcal{G}^+$. Whatever $\mathcal{G}$ is, that produces some function of the plus arguments, and nothing has been restricted.

The content of frame indifference lies in the words “a function of the plus arguments”. Section 8.4 required the response functions of O^+ to deliver the set C^+ in terms of the list L^+ of (8.28), and that list contains what O^+ perceives and nothing else. In particular it does not contain $\underline{Q}(t)$ or $\underline{c}(t)$, which record how O^+ happens to be moving relative to O and are not a property of the material; the same holds for $\mathcal{G}$ and the list of O . So (9.5) is to hold for every change of frame, with one and the same pair of functions $\mathcal{G}$ and $\mathcal{G}^+$. Read in this way the definition above becomes a demand: the function of the plus arguments computed from $\mathcal{G}$ must come out the same whichever relative motion $(\underline{Q}, \underline{c})$ was used

to compute it. That is a genuine restriction on $\mathcal{G}$, and the rest of this section extracts it.

The extraction compares two members of the family. Let $\tilde{Q}_1(t), \underline{c}_1(t)$ and $\tilde{Q}_2(t), \underline{c}_2(t)$ characterize two relative motions of the observers. Hold fixed the particle p , the instant t , and everything that O^+ perceives at that particle and instant; then ask what O perceives in each case.

The meaning of $\underline{x}_1, \underline{x}_2$ and $\underline{x}^+$

Figure A shows the situation for positions. The vector $\underline{x}^+$ is the position at which O^+ sees the particle p at time t , and it is held fixed throughout. If the frame of O is related to that of O^+ by the first relative motion, then by (7.3) the position that O assigns to p is the vector $\underline{x}_1$ satisfying $\tilde{Q}_1 \underline{x}_1 + \underline{c}_1 = \underline{x}^+$; if by the second, it is the vector $\underline{x}_2$ satisfying $\tilde{Q}_2 \underline{x}_2 + \underline{c}_2 = \underline{x}^+$. So the subscripts do not label two particles, and they do not label two instants. They label the two answers that O would give, for the same particle at the same instant, under the two relative motions; $\underline{x}_1$ and $\underline{x}_2$ are two elements of the one space $\mathbb{E}^3$ of O , and $\underline{x}^+$ is an element of $\mathbb{E}^{3+}$.

Geometrically, $\underline{c}_k$ locates the origin of O as it appears in the frame of O^+ , and $\tilde{Q}_k$ describes how the basis of O appears turned there, $\underline{e}_i \mapsto \tilde{Q}_k \underline{e}_i$. The vector $\tilde{Q}_k \underline{x}_k = \underline{x}^+ - \underline{c}_k$ is the arrow from that origin to p , drawn in the frame of O^+ , and its components on the turned basis are

$$\tilde{Q}_k \underline{x}_k \cdot \tilde{Q}_k \underline{e}_i = \underline{x}_k \cdot \tilde{Q}_k^t \tilde{Q}_k \underline{e}_i = \underline{x}_k \cdot \underline{e}_i,$$

by the definition of the transpose and $\tilde{Q}_k^t \tilde{Q}_k = I$. These are the three numbers that O reads off, the components of $\underline{x}_k$ on its own basis. They differ between the two cases for the plain reason that the frame of O was laid down differently in the two cases, while the particle stayed where it was.

From the perspective of O^+ , then, the positions $\underline{x}_1$ and $\underline{x}_2$ of p , as perceived by O , are related by (compare (7.3))

$$\tilde{Q}_1(t) \underline{x}_1 + \underline{c}_1(t) = \underline{x}^+ = \tilde{Q}_2(t) \underline{x}_2 + \underline{c}_2(t), \quad (9.6)$$

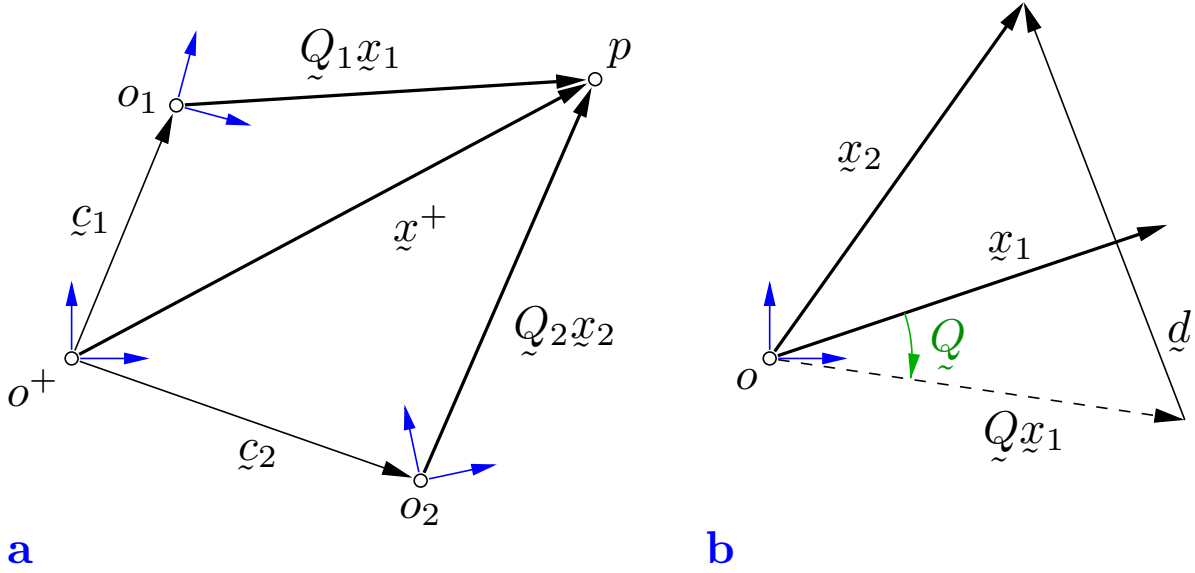


Figure A: What the subscripts in (9.6) mean. Panel a is drawn in the space of O^+ : the particle p is one point, $\underline{x}^+$ is its position vector from the origin o^+ , and the short blue arrows at o^+ , o_1 and o_2 are bases. The frame of O can be related to that of O^+ in more than one way, and two ways are drawn. In the first the origin of O appears at o_1 , displaced by $\underline{c}_1$, and its basis appears turned into $\underline{Q}_1 \underline{e}_1, \underline{Q}_1 \underline{e}_2$; in the second it appears at o_2 with basis $\underline{Q}_2 \underline{e}_1, \underline{Q}_2 \underline{e}_2$. The one point p is reached along three routes, $\underline{x}^+ = \underline{c}_1 + \underline{Q}_1 \underline{x}_1 = \underline{c}_2 + \underline{Q}_2 \underline{x}_2$, which is (9.6). Panel b is drawn in the space of O itself. What O records in the two cases are two different vectors $\underline{x}_1$ and $\underline{x}_2$ of the one space $\mathbb{E}^3$, related by the rigid map (9.12): turn $\underline{x}_1$ by $\underline{Q} = \underline{Q}_2^t \underline{Q}_1$ (green arc), then shift by $\underline{d} = \underline{Q}_2^t (\underline{c}_1 - \underline{c}_2)$. Both panels are to scale for turning angles of -15° and 12° , so that $\underline{Q}$ is a turn through -27° and $|\underline{d}| = |\underline{c}_1 - \underline{c}_2|$.

and from (7.8) and (7.33), the associated velocities and velocity gradients are related by

$$\mathcal{Q}_1 \mathfrak{v}_1 + \dot{\mathcal{Q}}_1 \mathcal{Q}_1^t (\mathfrak{x}^+ - \mathfrak{c}_1) + \dot{\mathfrak{c}}_1 = \mathfrak{v}^+ = \mathcal{Q}_2 \mathfrak{v}_2 + \dot{\mathcal{Q}}_2 \mathcal{Q}_2^t (\mathfrak{x}^+ - \mathfrak{c}_2) + \dot{\mathfrak{c}}_2 \quad (9.7)$$

and

$$\mathcal{Q}_1 \mathcal{L}_1 \mathcal{Q}_1^t + \dot{\mathcal{Q}}_1 \mathcal{Q}_1^t = \mathcal{L}^+ = \mathcal{Q}_2 \mathcal{L}_2 \mathcal{Q}_2^t + \dot{\mathcal{Q}}_2 \mathcal{Q}_2^t, \quad (9.8)$$

respectively, whereas (7.37) implies that $\rho_1 = \rho^+ = \rho_2$ and $\theta_1 = \theta^+ = \theta_2$. Each of (9.6) to (9.8) computes one plus quantity in two ways. The density and the temperature are absolute scalars and need no conversion at all. The velocity picks up the velocity of the moving frame and the velocity gradient its angular rate $\dot{\mathcal{Q}}_k \mathcal{Q}_k^t$, which is why the two sides of (9.7) and (9.8) carry terms that (9.6) does not. Denoting the common scalar readings simply by ρ and θ , we then have, from (9.5) applied to each of the two relative motions in turn,

$$\begin{aligned} \mathcal{Q}_1 \mathcal{G}(\mathfrak{x}_1, \mathfrak{v}_1, \theta, \rho, \mathcal{L}_1; p) \mathcal{Q}_1^t &= \mathcal{G}^+(\mathfrak{x}^+, \mathfrak{v}^+, \theta^+, \rho^+, \mathcal{L}^+; p) \\ &= \mathcal{Q}_2 \mathcal{G}(\mathfrak{x}_2, \mathfrak{v}_2, \theta, \rho, \mathcal{L}_2; p) \mathcal{Q}_2^t, \end{aligned} \quad (9.9)$$

and hence the restriction

$$\mathcal{G}(\mathfrak{x}_2, \mathfrak{v}_2, \theta, \rho, \mathcal{L}_2; p) = \mathcal{Q} \mathcal{G}(\mathfrak{x}_1, \mathfrak{v}_1, \theta, \rho, \mathcal{L}_1; p) \mathcal{Q}^t \quad (9.10)$$

on the constitutive equation used by O , where

$$\mathcal{Q} = \mathcal{Q}_2^t \mathcal{Q}_1: \mathbb{E}^3 \rightarrow \mathbb{E}^3 \quad (9.11)$$

is an arbitrary orthogonal tensor mapping $\mathbb{E}^3$ to itself.

The middle member of (9.9) is the same in both lines because it is one function evaluated at one list of arguments, and this is exactly where the independence of $\mathcal{G}^+$ from the relative motion is used. The passage to (9.10) is then algebra. Abbreviate $\mathcal{G}_k = \mathcal{G}(\mathfrak{x}_k, \mathfrak{v}_k, \theta, \rho, \mathcal{L}_k; p)$, so that the two ends of (9.9) give $\mathcal{Q}_1 \mathcal{G}_1 \mathcal{Q}_1^t = \mathcal{Q}_2 \mathcal{G}_2 \mathcal{Q}_2^t$. Multiply on the left by $\mathcal{Q}_2^t$ and on the right by $\mathcal{Q}_2$. The right-hand side becomes $\mathcal{Q}_2^t \mathcal{Q}_2 \mathcal{G}_2 \mathcal{Q}_2^t \mathcal{Q}_2 = \mathcal{G}_2$, because $\mathcal{Q}_2^t \mathcal{Q}_2 = \mathcal{I}$ is the identity on $\mathbb{E}^3$ by Proposition 1.48. The left-hand side becomes $(\mathcal{Q}_2^t \mathcal{Q}_1) \mathcal{G}_1 (\mathcal{Q}_1^t \mathcal{Q}_2)$, and $\mathcal{Q}_1^t \mathcal{Q}_2 = (\mathcal{Q}_2^t \mathcal{Q}_1)^t = \mathcal{Q}^t$ by Proposition 1.42. That is (9.10).

Two features of $\tilde{Q}$ deserve notice. It is orthogonal, being a product of two orthogonal tensors, and it maps $\mathbb{E}^3$ to $\mathbb{E}^3$, not to $\mathbb{E}^{3+}$: the excursion into the space of O^+ and back has cancelled. Figure B records this as a diagram. The restriction (9.10) is therefore a statement made entirely within the frame of O , and it is usable for exactly that reason: both sides are values of the one function $\mathcal{G}$, at two lists of arguments in the same space, so the equation constrains $\mathcal{G}$. The observer O^+ has served as a hinge. It is held fixed, it is itself never constrained, and it cannot be dispensed with, since without a common report to compare against nothing would connect the two descriptions by O .

The tensor $\tilde{Q}$ is moreover arbitrary, and so is the translation that goes with it. Given any orthogonal $Q(t): \mathbb{E}^3 \rightarrow \mathbb{E}^3$ and any vector function $\underline{d}(t)$ in $\mathbb{E}^3$, pick any orthogonal $\tilde{Q}_2(t): \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$ and any $\underline{c}_2(t)$, and put

$$\tilde{Q}_1 = \tilde{Q}_2 \tilde{Q}, \quad \underline{c}_1 = \underline{c}_2 + \tilde{Q}_2 \underline{d}.$$

Then $\tilde{Q}_2^t \tilde{Q}_1 = \tilde{Q}_2^t \tilde{Q}_2 \tilde{Q} = \tilde{Q}$ and $\tilde{Q}_2^t (\underline{c}_1 - \underline{c}_2) = \tilde{Q}_2^t \tilde{Q}_2 \underline{d} = \underline{d}$. So every time-dependent rigid map of $\mathbb{E}^3$ onto itself, reflections included, arises from some pair of relative motions, and that is what licenses the particular choices made below. No condition at the reference instant t_0 is needed here, as it will be in Chapters 10 and 11, because the reference configuration, and with it the tensor $\underline{K}$ of (7.14) that depends on $\tilde{Q}(t_0)$, disappeared at (9.3).

REMARK 9.2 (how this relates to “the same function” of Chapter 8). Section 8.4 stated frame indifference as the requirement that the two observers use the same function, after agreeing to use the same reference placement. Taken literally that formulation identifies the space $\mathbb{E}^{3+}$ of O^+ with the space $\mathbb{E}^3$ of O ; without the identification, $\mathcal{G}^+ = \mathcal{G}$ would not even make sense, since the two functions take their arguments in different spaces and return tensors on different spaces.

The device of this section never makes the identification. It does not compare $\mathcal{G}^+$ with $\mathcal{G}$ at all; it uses O^+ only as a common reference against which two descriptions by O are compared, and every tensor in the final restriction (9.10) lives on $\mathbb{E}^3$. If one does identify the spaces,

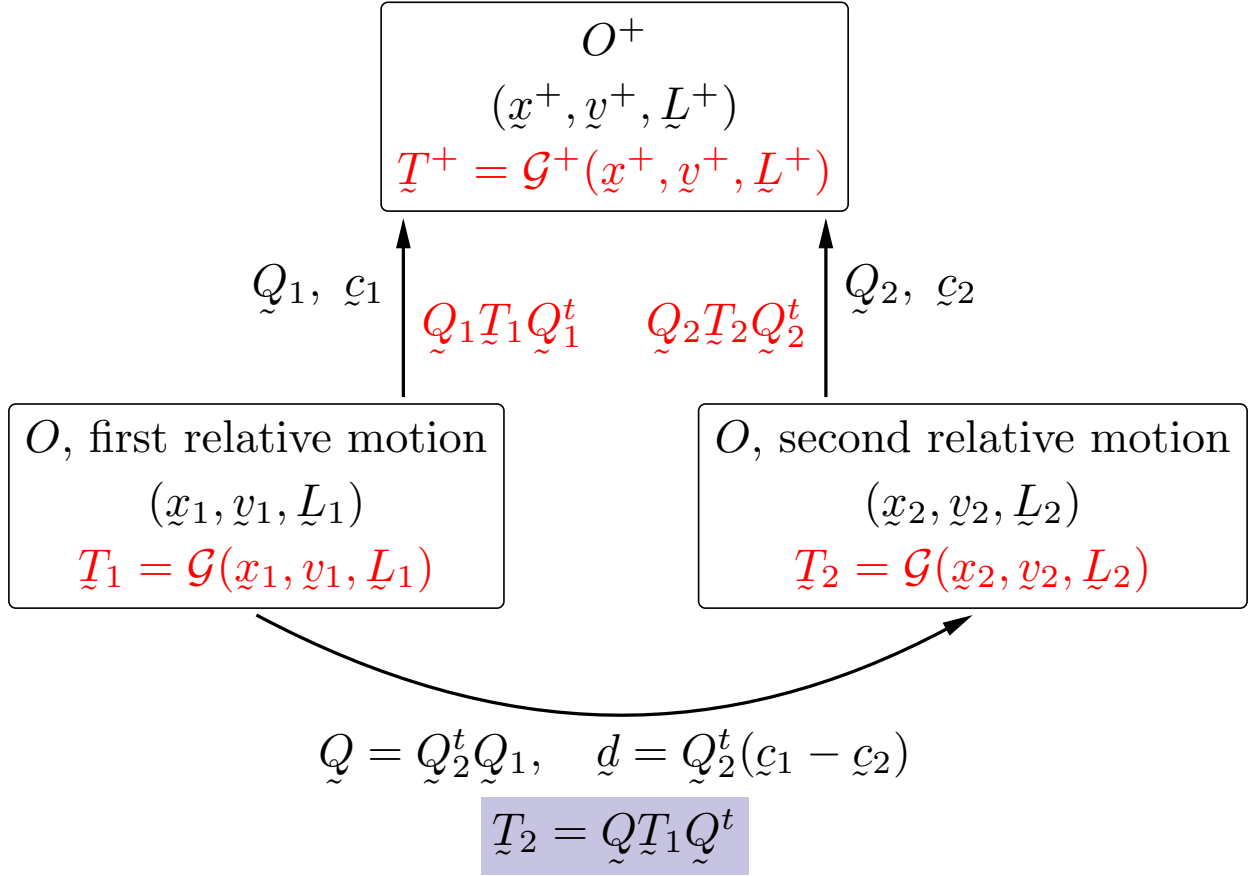


Figure B: The logic of (9.9) and (9.10) as a diagram; θ , ρ and p are common to all three boxes and are not written. The top box is the one report of O^+ , held fixed. Each upward arrow is a change of frame: along it the data change by (9.6) to (9.8), and the stress is carried over by conjugation, as in (9.5). The function $\mathcal{G}^+$ is a function of the plus data alone, so both upward routes must land on the same $\underline{T}^+$. Going up one side and down the other gives the bottom arrow, which never leaves $\mathbb{E}^3$: it is the rigid map $\underline{x}_2 = \underline{Q}\underline{x}_1 + \underline{d}$ of (9.12), and along it the single function $\mathcal{G}$ must satisfy the boxed relation $\underline{T}_2 = \underline{Q}\underline{T}_1\underline{Q}^t$. That is (9.10), a statement about $\mathcal{G}$ alone.

by taking $\mathbb{E}^{3+} = \mathbb{E}^3$ and letting the second relative motion be the trivial one, $\underline{Q}_2 = \underline{I}$ and $\underline{c}_2 = \underline{0}$, then $\underline{x}_2 = \underline{x}^+$, $\underline{v}_2 = \underline{v}^+$ and $\underline{L}_2 = \underline{L}^+$, the second line of (9.9) says $\underline{\mathcal{G}}^+ = \underline{\mathcal{G}}$, and (9.10) becomes the functional equation displayed in Section 8.4. The two formulations agree on every conclusion; the present one assumes less.

How the two descriptions are related

Straightforward manipulations making use of (9.6) to (9.8) yield

$$\begin{aligned}\underline{x}_2 &= \underline{Q}\underline{x}_1 + \underline{d}(t), & \text{where } \underline{d}(t) &= \underline{Q}_2^t(\underline{c}_1 - \underline{c}_2), \\ \underline{v}_2 &= \underline{Q}\underline{v}_1 + \underline{e},\end{aligned}\tag{9.12}$$

where $\underline{e}$ is a function of $\underline{x}^+$ and t , and

$$\begin{aligned}\underline{L}_2 &= \underline{Q}\underline{L}_1\underline{Q}^t + \dot{\underline{Q}}\underline{Q}^t; & \text{hence} \\ \underline{D}_2 &= \underline{Q}\underline{D}_1\underline{Q}^t & \text{and} & \quad \underline{W}_2 = \underline{Q}\underline{W}_1\underline{Q}^t + \dot{\underline{Q}}\underline{Q}^t,\end{aligned}\tag{9.13}$$

in which $\underline{Q}$ is given by (9.11). The manipulations are worth doing in full, because everything below rests on them; Problem 1 collects them again.

Positions come first. From (9.6), $\underline{Q}_2\underline{x}_2 = \underline{Q}_1\underline{x}_1 + \underline{c}_1 - \underline{c}_2$; multiplying on the left by $\underline{Q}_2^t$ and using $\underline{Q}_2^t\underline{Q}_2 = \underline{I}$ and (9.11) gives the first line of (9.12). The vector $\underline{d}$ is the displacement $\underline{c}_1 - \underline{c}_2$ from the second placement of the origin of O to the first, which is measured in the frame of O^+ , read back into the frame of O by $\underline{Q}_2^t$. In Figure A it is the arrow from o_2 to o_1 seen from the second placement, which is why $|\underline{d}| = |\underline{c}_1 - \underline{c}_2|$. It depends on t alone.

For the velocity there are two routes, and both are instructive. The first is algebra on (9.7): moving the terms of the first relative motion to one side and multiplying on the left by $\underline{Q}_2^t$ gives $\underline{v}_2 = \underline{Q}\underline{v}_1 + \underline{e}$ with

$$\underline{e} = \underline{Q}_2^t \left[\dot{\underline{Q}}_1 \underline{Q}_1^t (\underline{x}^+ - \underline{c}_1) - \dot{\underline{Q}}_2 \underline{Q}_2^t (\underline{x}^+ - \underline{c}_2) + \dot{\underline{c}}_1 - \dot{\underline{c}}_2 \right].$$

Every symbol here is a function of t except $\underline{x}^+$, so $\underline{e}$ is a function of $\underline{x}^+$ and t , as (9.12) states. The second route is kinematics. The vectors $\underline{x}_1(t)$ and

$\underline{x}_2(t)$ are the positions of the one particle p in the two descriptions, the first line of (9.12) holds at every instant, and differentiating it following p , with $\dot{\underline{x}}_k = \underline{v}_k$, gives

$$\underline{v}_2 = \dot{\underline{Q}}\underline{x}_1 + \underline{Q}\underline{v}_1 + \dot{\underline{d}}, \quad \text{so that} \quad \underline{e} = \dot{\underline{Q}}\underline{x}_1 + \dot{\underline{d}}.$$

The two expressions for $\underline{e}$ agree. To see it, expand $\dot{\underline{Q}} = \dot{\underline{Q}}_2^t \underline{Q}_1 + \underline{Q}_2^t \dot{\underline{Q}}_1$ and $\dot{\underline{d}} = \dot{\underline{Q}}_2^t(\underline{c}_1 - \underline{c}_2) + \underline{Q}_2^t(\dot{\underline{c}}_1 - \dot{\underline{c}}_2)$ by the product rule, and use $\underline{Q}_1 \underline{x}_1 = \underline{x}^+ - \underline{c}_1$ from (9.6) and $\underline{x}_1 = \underline{Q}_1^t(\underline{x}^+ - \underline{c}_1)$:

$$\begin{aligned} \dot{\underline{Q}}\underline{x}_1 + \dot{\underline{d}} &= \dot{\underline{Q}}_2^t \underline{Q}_1 \underline{x}_1 + \underline{Q}_2^t \dot{\underline{Q}}_1 \underline{x}_1 + \dot{\underline{Q}}_2^t(\underline{c}_1 - \underline{c}_2) + \underline{Q}_2^t(\dot{\underline{c}}_1 - \dot{\underline{c}}_2) \\ &= \dot{\underline{Q}}_2^t(\underline{x}^+ - \underline{c}_2) + \underline{Q}_2^t[\dot{\underline{Q}}_1 \underline{Q}_1^t(\underline{x}^+ - \underline{c}_1) + \dot{\underline{c}}_1 - \dot{\underline{c}}_2]. \end{aligned}$$

Differentiating $\underline{Q}_2^t \underline{Q}_2 = \underline{I}$ gives $\dot{\underline{Q}}_2^t \underline{Q}_2 = -\underline{Q}_2^t \dot{\underline{Q}}_2$, and multiplying on the right by $\underline{Q}_2^t$, with $\underline{Q}_2 \underline{Q}_2^t = \underline{I}^+$,

$$\dot{\underline{Q}}_2^t = -\underline{Q}_2^t \dot{\underline{Q}}_2 \underline{Q}_2^t.$$

With this the first term becomes $-\underline{Q}_2^t \dot{\underline{Q}}_2 \underline{Q}_2^t(\underline{x}^+ - \underline{c}_2)$, and the whole is the algebraic expression for $\underline{e}$.

The velocity gradient takes one more step. By definition $\underline{L}_2 = \text{grad}_2 \underline{v}_2$, the gradient of the velocity field of the second description with respect to its own position variable $\underline{x}_2$. Regard $\underline{v}_2 = \dot{\underline{Q}}\underline{x}_1 + \underline{Q}\underline{v}_1 + \dot{\underline{d}}$ as such a field. At fixed t the first line of (9.12) inverts to $\underline{x}_1 = \underline{Q}^t(\underline{x}_2 - \underline{d})$, so that $\partial \underline{x}_1 / \partial \underline{x}_2 = \underline{Q}^t$, while $\dot{\underline{Q}}$ and $\dot{\underline{d}}$ do not depend on position. By the chain rule, with $\text{grad}_1 \underline{v}_1 = \underline{L}_1$,

$$\underline{L}_2 = (\dot{\underline{Q}} + \underline{Q}\underline{L}_1)\underline{Q}^t = \underline{Q}\underline{L}_1\underline{Q}^t + \dot{\underline{Q}}\underline{Q}^t,$$

which is the first statement of (9.13). The same result follows from (9.8) by the manipulation that produced (9.10); Problem 1 carries it out.

Now split into symmetric and skew parts. Conjugation by an orthogonal tensor preserves symmetry and skewness, since $(\underline{Q}\underline{A}\underline{Q}^t)^t = \underline{Q}\underline{A}^t\underline{Q}^t$ by Proposition 1.42; so $\underline{Q}\underline{D}_1\underline{Q}^t$ is symmetric and $\underline{Q}\underline{W}_1\underline{Q}^t$ is skew. The term $\dot{\underline{Q}}\underline{Q}^t$ is skew outright: differentiating $\underline{Q}\underline{Q}^t = \underline{I}$ gives $\dot{\underline{Q}}\underline{Q}^t + \underline{Q}\dot{\underline{Q}}^t = \underline{O}$, that is, $\dot{\underline{Q}}\underline{Q}^t = -(\dot{\underline{Q}}\underline{Q}^t)^t$. Since the symmetric and skew parts of

a tensor are unique (Problem 14 of Chapter 1), the second and third statements of (9.13) can be read off.

That asymmetry is the whole point of the section. The stretching $\underline{D}$ transforms by conjugation alone, with nothing left over, so it is objective in the sense of Chapter 7. The spin $\underline{W}$ picks up the extra $\dot{\underline{Q}}\underline{Q}^t$, which records how the two descriptions turn relative to one another and has nothing to do with the material. A constitutive function allowed to see $\underline{W}$ would let the stress depend on that turning, and the second choice below shows that frame indifference forbids it.

REMARK 9.3 (two letters, and the book's grouping). The vector $\underline{e}$ of (9.12) is the book's, and it is not a basis vector: in this section $\underline{e}$ without a subscript is the velocity offset, while $\underline{e}_1, \underline{e}_2, \underline{e}_3$ keep their meaning as a fixed orthonormal basis of $\mathbb{E}^3$. Similarly $\underline{d}$ here is a translation, and Section 9.2 will reuse the same letter for the temperature gradient. The book numbers the position and velocity relations together as (9.12) and the three velocity-gradient relations together as (9.13), and these notes follow it; Chapter 11 cites (9.12) for $\underline{x}_2$ and $\underline{v}_2$ in exactly this sense.

REMARK 9.4 (this is (7.34) in a mirror). The pair $\underline{D}_2 = \underline{Q}\underline{D}_1\underline{Q}^t$, $\underline{W}_2 = \underline{Q}\underline{W}_1\underline{Q}^t + \dot{\underline{Q}}\underline{Q}^t$ is formally (7.34), and the resemblance is not an accident, but the two statements are about different things and the difference matters.

Equation (7.34) compares what O sees with what O^+ sees: $\underline{D}$ lives on $\mathbb{E}^3$ and $\underline{D}^+$ on $\mathbb{E}^{3+}$, and $\underline{Q}$ is the two-point tensor between them. The present statement compares two descriptions both belonging to O , of the same motion under two relative motions of the pair of observers; every tensor in it lives on $\mathbb{E}^3$ and $\underline{Q}$ maps $\mathbb{E}^3$ to itself. That is precisely why (9.10) is a restriction on $\mathcal{G}$ and (9.5) is not: a functional equation whose two sides are evaluated in the same space can be read as a constraint on the function, and one whose sides live in different spaces cannot.

The mechanism that produces $\underline{W}^+ \neq \underline{Q}\underline{W}\underline{Q}^t$ is the same in both: an observer who spins sees an extra rotation and no extra stretching. Remark 4.9 of Chapter 4 warned that $\underline{W}$ is not the spin of material

fibres; the present formula says something stronger, that $\underline{W}$ is not even a property of the material, since it can be changed by turning around.

REMARK 9.5 (a superposed rigid motion, and why the book does not say so). Since $\underline{e} = \underline{\dot{Q}}\underline{x}_1 + \underline{\dot{d}}$, the pair (9.12), (9.13) says that $\underline{x}_2$, $\underline{v}_2$ and $\underline{L}_2$ are exactly the position, velocity and velocity gradient of the motion $\underline{Q}(t)\underline{\chi}(p, t) + \underline{d}(t)$, where $\underline{\chi}$ is the motion recorded in the first description. So (9.10) can be read as follows: if a rigid motion is superposed on the motion of the body, the stress delivered by $\underline{\mathcal{G}}$ is merely turned by $\underline{Q}$. Many texts take that sentence as their starting point, under the name of the principle of invariance under superposed rigid body motions.

The book reaches the same equation by a different road. It follows Murdoch, its reference [13], whose paper sets out to discard that principle in favour of purely objective considerations, and the difference is one of what has to be believed. The principle compares two different motions of the body and asserts how the material responds to a motion it is not actually undergoing; that is a physical claim about the material, and the kinetic theory of rarefied gases is the standard example in which it is in doubt. The argument of this section compares no motions at all. There is one motion, the one O^+ sees, and two descriptions of it by O , which differ only because O is imagined in two different states of motion relative to O^+ . What is assumed is only that the constitutive function of each observer is a function of that observer's own perceptions. The formal consequence, (9.10) with (9.12) and (9.13), is the same, and so is every conclusion drawn from it in this chapter.

First choice: $\underline{Q} = \underline{I}$, and the loss of $\underline{x}$ and $\underline{v}$

To derive a necessary condition for (9.10) we exploit the arbitrariness of $\underline{Q}$ and select $\underline{Q}(t) = \underline{I}$, the identity on $\mathbb{E}^3$. Inserting this into (9.11) and left-multiplying by $\underline{Q}_2$ gives $\underline{Q}_2\underline{I} = \underline{I}^+\underline{Q}_1$, where $\underline{I}^+$ is the identity on $\mathbb{E}^{3+}$. The left- and right-hand sides of this are simply $\underline{Q}_2$ and $\underline{Q}_1$, respectively, and our choice thus yields $\underline{Q}_2 = \underline{Q}_1$: the two relative motions turn in the same way and differ only by a translation, which in Figure A

would make the bases at o_1 and o_2 parallel. We find that $\underline{e}$ then reduces to a function of t alone, given by

$$\underline{e}(t) = \underline{Q}_1^t(\dot{\underline{c}}_1 - \dot{\underline{c}}_2) + \dot{\underline{Q}}_1^t(\underline{c}_1 - \underline{c}_2), \quad (9.14)$$

and that (9.10) reduces, on dropping the subscript 1, to

$$\mathcal{G}(\underline{x} + \underline{d}, \underline{v} + \underline{e}, \theta, \rho, \underline{L}; p) = \mathcal{G}(\underline{x}, \underline{v}, \theta, \rho, \underline{L}; p), \quad (9.15)$$

for arbitrary $\underline{d}$ and $\underline{e}$.

Both statements follow at once from the kinematic form of $\underline{e}$. With $\underline{Q} = \underline{I}$ constant, $\dot{\underline{Q}} = \underline{Q}$, so $\underline{e} = \dot{\underline{Q}}\underline{x}_1 + \underline{d} = \dot{\underline{d}}$, and by the product rule applied to $\underline{d} = \underline{Q}_1^t(\underline{c}_1 - \underline{c}_2)$,

$$\underline{e} = \dot{\underline{d}} = \dot{\underline{Q}}_1^t(\underline{c}_1 - \underline{c}_2) + \underline{Q}_1^t(\dot{\underline{c}}_1 - \dot{\underline{c}}_2),$$

which is (9.14). The same comes out of the algebraic form: setting $\underline{Q}_2 = \underline{Q}_1$ there, the two terms containing $\underline{x}^+$ cancel,

$$\begin{aligned} \underline{e} &= \underline{Q}_1^t \left[\dot{\underline{Q}}_1 \underline{Q}_1^t(\underline{x}^+ - \underline{c}_1) - \dot{\underline{Q}}_1 \underline{Q}_1^t(\underline{x}^+ - \underline{c}_2) + \dot{\underline{c}}_1 - \dot{\underline{c}}_2 \right] \\ &= -(\underline{Q}_1^t \dot{\underline{Q}}_1 \underline{Q}_1^t)(\underline{c}_1 - \underline{c}_2) + \underline{Q}_1^t(\dot{\underline{c}}_1 - \dot{\underline{c}}_2), \end{aligned}$$

and $\underline{Q}_1^t \dot{\underline{Q}}_1 \underline{Q}_1^t = -\dot{\underline{Q}}_1^t$ by the identity just proved for $\underline{Q}_2$. Further, $\dot{\underline{Q}} = \underline{Q}$ makes $\underline{L}_2 = \underline{L}_1$ in (9.13), and $\underline{Q} = \underline{I}$ removes the conjugation from (9.10), which is what (9.15) records.

REMARK 9.6 (a misprint in the printed (9.14)). The book prints the second term of (9.14) as $\dot{\underline{Q}}_1 \underline{Q}_1^t(\underline{c}_1 - \underline{c}_2)$. That expression cannot be right, and the reason is visible without any calculation. The vectors $\underline{c}_k$ lie in $\mathbb{E}^{3+}$, since (9.6) adds them to $\underline{Q}_k \underline{x}_k$ there. Now $\underline{Q}_1^t$ carries $\mathbb{E}^{3+}$ to $\mathbb{E}^3$ and $\dot{\underline{Q}}_1$ carries $\mathbb{E}^3$ back to $\mathbb{E}^{3+}$, so $\dot{\underline{Q}}_1 \underline{Q}_1^t(\underline{c}_1 - \underline{c}_2)$ lands in $\mathbb{E}^{3+}$; but $\underline{e}$ must lie in $\mathbb{E}^3$, because (9.12) adds it to $\underline{Q}\underline{v}_1$ there. The calculation above supplies the correct term, $\dot{\underline{Q}}_1^t(\underline{c}_1 - \underline{c}_2)$, which is also $-\underline{Q}_1^t \dot{\underline{Q}}_1 \underline{Q}_1^t(\underline{c}_1 - \underline{c}_2)$: compared with it the printed term has lost a leading factor $\underline{Q}_1^t$ and its sign. The check $\underline{e} = \dot{\underline{d}}$ settles the matter, as it must when the two descriptions differ by a pure translation.

It follows from (9.15) that the constitutive function is insensitive to variations of its first two arguments, and hence that it is independent of those arguments. The quantifier carries the conclusion, so it is worth checking that we really have it. The vectors $\underline{c}_1$ and $\underline{c}_2$ are arbitrary functions of time, so at the instant under consideration $\underline{c}_1 - \underline{c}_2$ and its derivative $\dot{\underline{c}}_1 - \dot{\underline{c}}_2$ may be assigned independently; by (9.12) and (9.14) the same is then true of $\underline{d}$ and $\underline{e}$. A simple instance is $\underline{Q}_1 = \underline{Q}_2$ constant, $\underline{c}_1 = \underline{0}$ and $\underline{c}_2 = \underline{Q}_1 \underline{a}(t)$: then $\underline{d} = -\underline{a}(t)$ and $\underline{e} = -\dot{\underline{a}}(t)$, and $\underline{a}(t_*)$ and $\dot{\underline{a}}(t_*)$ may be prescribed independently at any instant t_* . A function whose value is unchanged when its first two arguments are shifted by arbitrary amounts cannot depend on them: fix $\underline{x}_0$ and $\underline{v}_0$, and for any $\underline{x}, \underline{v}$ choose $\underline{d} = \underline{x}_0 - \underline{x}$ and $\underline{e} = \underline{v}_0 - \underline{v}$, whereupon (9.15) gives $\mathcal{G}(\underline{x}, \underline{v}, \dots) = \mathcal{G}(\underline{x}_0, \underline{v}_0, \dots)$, a value independent of $\underline{x}$ and $\underline{v}$. We thus arrive at the major simplification

$$\mathcal{G}(\underline{x}, \underline{v}, \theta, \rho, \underline{L}; p) = \underline{\mathcal{G}}(\theta, \rho, \underline{L}; p) \quad (9.16)$$

for some function $\underline{\mathcal{G}}$. Substituting back into (9.10), and using the first statement of (9.13) in the form $\underline{L}_2 = \underline{Q} \underline{L}_1 \underline{Q}^t + \underline{\Omega}$ with $\underline{\Omega} = \dot{\underline{Q}} \underline{Q}^t$, we find that this function must be such that

$$\underline{\mathcal{G}}(\theta, \rho, \underline{Q} \underline{L} \underline{Q}^t + \underline{\Omega}; p) = \underline{Q} \underline{\mathcal{G}}(\theta, \rho, \underline{L}; p) \underline{Q}^t \quad (9.17)$$

for arbitrary orthogonal $\underline{Q}(t): \mathbb{E}^3 \rightarrow \mathbb{E}^3$, with $\underline{\Omega} = \dot{\underline{Q}} \underline{Q}^t$.

REMARK 9.7 (three functions, two letters). The book keeps the functions apart typographically, and it helps to follow it. The calligraphic $\mathcal{G}$ of (9.3) to (9.15) is a function of five arguments; the $\underline{\mathcal{G}}$ introduced in (9.16), printed bold in the book and with an under-tilde here, is a different function, of three. Equation (9.16) says that the first is the second composed with the map that forgets $\underline{x}$ and $\underline{v}$.

The one genuine overloading comes at (9.21) below, where $\underline{\mathcal{G}}$ is written with $\underline{L}$ on one side and $\underline{D}$ on the other. That equation means $\underline{\mathcal{G}}(\theta, \rho, \underline{L}; p) = \underline{\mathcal{G}}(\theta, \rho, \text{sym } \underline{L}; p)$ for every $\underline{L}$, a statement about the one function $\underline{\mathcal{G}}$, and from then on only the values of $\underline{\mathcal{G}}$ at symmetric third arguments are ever used. Remark 2.6 made the same point about $\underline{v}$, $\hat{v}$ and $\tilde{v}$: the argument list is part of the name.

Second choice: an arbitrary spin, and the loss of $\underline{W}$

At any fixed instant t_* , say, we may choose $\underline{Q}$ and $\underline{\Omega}$ independently. This is the crucial observation, and it is not obvious: $\underline{\Omega} = \dot{\underline{Q}}\underline{Q}^t$ is built from $\underline{Q}$, and one might expect the value of $\underline{Q}$ at an instant to constrain the value of $\underline{\Omega}$ there. It does not, because $\underline{\Omega}$ involves the derivative of $\underline{Q}$, and the value of a function at a point and the value of its derivative there are independent data. An explicit family shows it. With reference to (C.24) of Appendix C, consider the orthogonal tensor

$$\underline{Q}(t) = \exp[(t - t_*)\hat{\underline{\Omega}}] \hat{\underline{Q}}, \quad (9.18)$$

in which $\hat{\underline{\Omega}}$ and $\hat{\underline{Q}}$, respectively, are arbitrary fixed skew and orthogonal tensors. Then $\underline{Q}(t_*) = \hat{\underline{Q}}$ and $\dot{\underline{Q}}\underline{Q}^t|_{t_*} = \hat{\underline{\Omega}}$. The hat marks a chosen constant, the subscript star a value at the instant t_* .

These claims take a few lines. Write $\underline{A}(s) = \exp(s\hat{\underline{\Omega}})$, so that $\underline{Q}(t) = \underline{A}(t - t_*)\hat{\underline{Q}}$. The defining series begins $\underline{A}(s) = \underline{I} + s\hat{\underline{\Omega}} + \frac{1}{2}s^2\hat{\underline{\Omega}}^2 + \dots$, so $\underline{A}(0) = \underline{I}$ and $\underline{Q}(t_*) = \hat{\underline{Q}}$. Differentiating the series term by term, as in (C.24), gives $\underline{A}'(s) = \hat{\underline{\Omega}}\underline{A}(s)$. The tensor $\underline{A}(s)$ is orthogonal for every s , because

$$(\underline{A}^t \underline{A})' = (\underline{A}')^t \underline{A} + \underline{A}^t \underline{A}' = \underline{A}^t \hat{\underline{\Omega}}^t \underline{A} + \underline{A}^t \hat{\underline{\Omega}} \underline{A} = \underline{A}^t (-\hat{\underline{\Omega}} + \hat{\underline{\Omega}}) \underline{A} = \underline{O},$$

using $\hat{\underline{\Omega}}^t = -\hat{\underline{\Omega}}$, so that $\underline{A}^t \underline{A}$ is constant and equal to its value $\underline{I}$ at $s = 0$. Hence $\underline{Q}(t)$ is orthogonal, as a product of orthogonal tensors, and at every t

$$\dot{\underline{Q}}\underline{Q}^t = \hat{\underline{\Omega}}\underline{A}\hat{\underline{Q}}\hat{\underline{Q}}^t\underline{A}^t = \hat{\underline{\Omega}}\underline{A}\underline{A}^t = \hat{\underline{\Omega}}.$$

The fixed tensors $\hat{\underline{Q}}$ and $\hat{\underline{\Omega}}$ were chosen independently, so at the instant t_* the orientation and the angular rate of the relative motion can be prescribed separately, as claimed.

REMARK 9.8 (where the bracket in (9.18) belongs). The book prints (9.18) as $\underline{Q}(t) = [\exp(t - t_*)\hat{\underline{\Omega}}]\hat{\underline{Q}}$. Read literally, $\exp(t - t_*)$ is the number e^{t-t_*} , and then $\underline{Q}(t) = e^{t-t_*}\hat{\underline{\Omega}}\hat{\underline{Q}}$ is a skew tensor times an orthogonal one:

it is not orthogonal, and at $t = t_*$ it equals $\hat{\underline{\underline{\Omega}}}\hat{\underline{\underline{Q}}}$ rather than $\hat{\underline{\underline{Q}}}$. What is meant is the tensor exponential of Appendix C applied to $(t - t_*)\hat{\underline{\underline{\Omega}}}$, whose argument (C.22) to (C.24) always write inside the parentheses; the bracket should enclose $(t - t_*)\hat{\underline{\underline{\Omega}}}$, as in (9.18) above.

From (9.17), evaluated at $t = t_*$,

$$\underline{\underline{G}}(\theta, \rho, \hat{\underline{\underline{Q}}}(\underline{\underline{D}}_* + \underline{\underline{W}}_*)\hat{\underline{\underline{Q}}}^t + \hat{\underline{\underline{\Omega}}}; p) = \hat{\underline{\underline{Q}}} \underline{\underline{G}}(\theta, \rho, \underline{\underline{L}}_*; p) \hat{\underline{\underline{Q}}}^t, \quad (9.19)$$

where $\underline{\underline{L}}_*$ is the velocity gradient at the instant t_* , and $\underline{\underline{D}}_* = \text{sym } \underline{\underline{L}}_*$ and $\underline{\underline{W}}_* = \text{skw } \underline{\underline{L}}_*$ are the associated stretching and spin tensors of (4.7).

Now, since the material point p is fixed in this discussion, all arguments of the constitutive function depend on time alone; in particular $\underline{\underline{W}}_*$ is one fixed skew tensor, not a field, and a spatially uniform relative motion can be tuned to it. We are thus free to choose $\hat{\underline{\underline{\Omega}}} = -\underline{\underline{W}}_*$, which is admissible precisely because $\underline{\underline{W}}_*$ is skew. Setting $\hat{\underline{\underline{Q}}} = \underline{\underline{I}}$ as well, the argument on the left of (9.19) becomes

$$\underline{\underline{I}}(\underline{\underline{D}}_* + \underline{\underline{W}}_*)\underline{\underline{I}} - \underline{\underline{W}}_* = \underline{\underline{D}}_*,$$

and the right-hand side becomes $\underline{\underline{I}} \underline{\underline{G}}(\theta, \rho, \underline{\underline{L}}_*; p) \underline{\underline{I}} = \underline{\underline{G}}(\theta, \rho, \underline{\underline{L}}_*; p)$. We arrive at

$$\underline{\underline{G}}(\theta, \rho, \underline{\underline{L}}_*; p) = \underline{\underline{G}}(\theta, \rho, \underline{\underline{D}}_*; p). \quad (9.20)$$

Figure C shows what the choice does. The two descriptions coincide at t_* , since $\hat{\underline{\underline{Q}}} = \underline{\underline{I}}$, but the second is turning relative to the first at exactly the rate that cancels the spin of the element at p ; the stretching is untouched. This is a local choice at the selected p and t_* , and a pointwise constitutive equation requires nothing more.

Since t_* is arbitrary, we conclude that, for (9.17) to hold, it is necessary that

$$\underline{\underline{G}}(\theta, \rho, \underline{\underline{L}}; p) = \underline{\underline{G}}(\theta, \rho, \underline{\underline{D}}; p) \quad (9.21)$$

at all t , and hence that the stress can depend on the velocity gradient only through the stretching tensor.

On substituting back into (9.17) and recalling (9.13), we find that this dependence must be such that

$$\underline{\underline{G}}(\theta, \rho, \hat{\underline{\underline{Q}}}\underline{\underline{D}}\hat{\underline{\underline{Q}}}^t; p) = \hat{\underline{\underline{Q}}} \underline{\underline{G}}(\theta, \rho, \underline{\underline{D}}; p) \hat{\underline{\underline{Q}}}^t \quad (9.22)$$

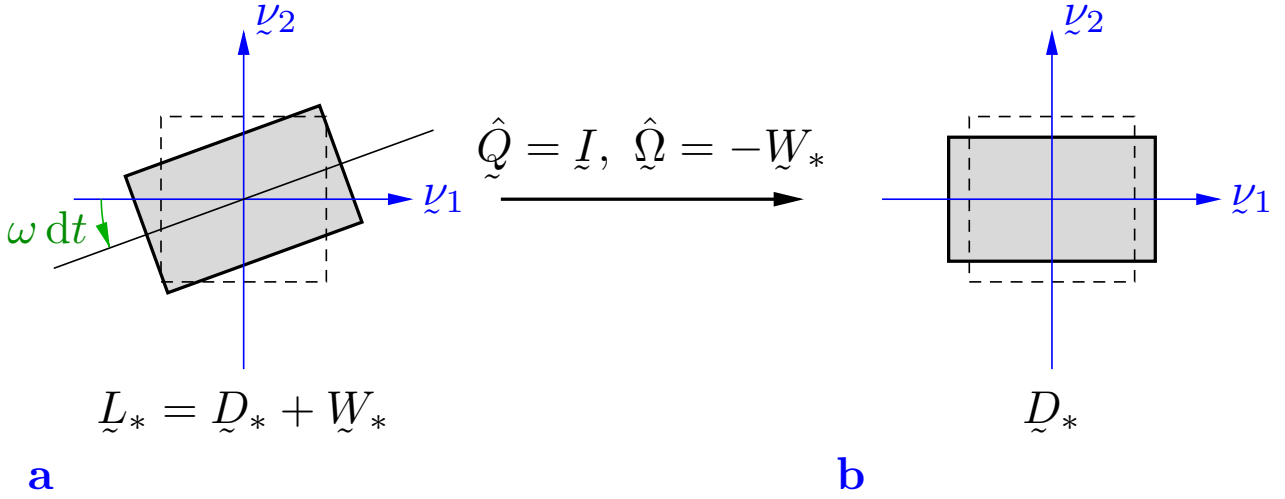


Figure C: The second choice, (9.18) to (9.20), drawn for a square element (dashed) whose sides lie along the principal axes ν_1, ν_2 of $\underline{D}_*$, and its image a short time dt after t_* (grey, exaggerated). a: in the first description the element is stretched along ν_1 , shortened along ν_2 , and turned through the angle ωdt carried by the spin $\underline{W}_*$; the thin line is the material fibre that lay along ν_1 at t_* . b: the second description coincides with the first at t_* , since $\hat{\underline{Q}} = \underline{I}$, but its frame turns relative to the first at the rate $\hat{\underline{\Omega}} = -\underline{W}_*$, so that its velocity gradient is $\hat{\underline{Q}}(\underline{D}_* + \underline{W}_*)\hat{\underline{Q}}^t + \hat{\underline{\Omega}} = \underline{D}_*$. In it the turn has disappeared and the stretch has not. Both descriptions are served by the one function $\underline{G}$, and at t_* they must report the same stress, so $\underline{G}$ cannot tell $\underline{L}_*$ from $\underline{D}_*$.

for all orthogonal $\underline{Q}(t): \mathbb{E}^3 \rightarrow \mathbb{E}^3$. The substitution is one line. By (9.21) the function sees only the symmetric part of its third argument, and the symmetric part of $\underline{Q}\underline{L}\underline{Q}^t + \underline{\Omega}$ is $\underline{Q}\underline{D}\underline{Q}^t$, since $\underline{\Omega}$ is skew and $\text{sym}(\underline{Q}\underline{L}\underline{Q}^t) = \underline{Q}(\text{sym } \underline{L})\underline{Q}^t$ as shown after (9.13). So the left side of (9.17) is the left side of (9.22), and (9.21) applied on the right of (9.17) turns $\underline{L}$ into $\underline{D}$ there too. The word “all” includes the improper orthogonal tensors, since $\underline{Q}_1$ and $\underline{Q}_2$ in (7.3) are only required to be orthogonal; Section 9.2 will need exactly that.

Since this function delivers the values of the Cauchy stress, it must also be symmetric: $\underline{T} = \underline{T}^t$ was extracted in Chapter 6 from the balance of moment of momentum, and it must therefore hold identically in the arguments of $\underline{G}$.

REMARK 9.9 (what (9.21) really rules out). It is tempting to read (9.21) as a technical simplification. It is not; it is a prohibition with observable content, and the cleanest way to see it is to ask what a fluid whose stress depended on $\underline{W}$ would do.

Consider a body of such a fluid at rest on a turntable, rotating rigidly with the table. By Remark 4.6, rigid motion has $\underline{D} = \underline{Q}$ but $\underline{W} \neq \underline{Q}$. A stress depending on $\underline{W}$ would therefore be non-zero, and it would depend on how fast the table turned. But an observer riding the table sees a fluid at rest, with $\underline{W} = \underline{Q}$ in that observer’s frame, and would compute a different stress. Both are looking at the same material and the stress is what it is, so one of them is wrong, and there is no principle to say which. The only escape is that the stress does not depend on $\underline{W}$ at all.

This is the constitutive twin of Remark 4.2 of Chapter 4, which asked why one works with $\underline{L}$ rather than $\dot{\underline{F}}$. Here the question is pushed one stage further: of the two halves of $\underline{L}$, only the half that survives a change of observer can carry material information, and (7.34) identifies that half as $\underline{D}$.

9.1.1 Isotropic symmetric tensor-valued functions of a symmetric tensor

Dropping the passive variables θ , ρ and p for convenience, we seek a general representation of a symmetric tensor-valued function $\underline{G}$, of an arbitrary symmetric tensor $\underline{D}$, satisfying

$$\underline{G}(\underline{Q}\underline{D}\underline{Q}^t) = \underline{Q} \underline{G}(\underline{D}) \underline{Q}^t \quad (9.23)$$

for all orthogonal $\underline{Q}(t): \mathbb{E}^3 \rightarrow \mathbb{E}^3$. Such a function is said to be *isotropic*, despite the fact that (9.22) pertains to frame invariance rather than to the notion of isotropic material symmetry, the latter to be discussed in Chapter 11.

REMARK 9.10 (two things called isotropy). The clash of names is unfortunate and it is worth separating the two statements before the proof, since they have different sources and different scopes.

Isotropic material symmetry is a statement about the symmetry group, $\mathcal{G}_{\kappa^*(p_0)} = \text{Orth}^+$ for an isotropic solid, $\mathcal{G}_{\kappa(p_0)} = U$ for a fluid; it says that no direction in a particular reference configuration is distinguished. It is a property some materials have and others do not, and Section 8.2 placed it on a scale running from the trivial group up to U .

Isotropy of a function, in the sense of (9.23), is a property of a map, and here it was derived: it came from (9.22), which came from frame indifference. Every fluid of this class has it, including a hypothetical fluid with no material symmetry at all, because the two observers of Section 9.1 would still have to agree.

The two coincide for the class treated here only because a fluid happens to be isotropic as a material as well, its group U containing Orth^+ ; for a solid the two come apart completely, and Chapter 10 and beyond have to keep them apart.

We will prove that the most general function of this type is of the form

$$\underline{G}(\underline{D}) = \varphi_0(I_1, I_2, I_3)\underline{I} + \varphi_1(I_1, I_2, I_3)\underline{D} + \varphi_2(I_1, I_2, I_3)\underline{D}^2, \quad (9.24)$$

where $\varphi_{0,1,2}$ are scalar-valued functions of the principal invariants of $\underline{D}$ (see (3.3)):

$$I_1(\underline{D}) = \text{tr } \underline{D}, \quad I_2(\underline{D}) = \frac{1}{2}[I_1^2 - \text{tr}(\underline{D}^2)] \quad \text{and} \quad I_3(\underline{D}) = \det \underline{D}. \quad (9.25)$$

REMARK 9.11 (why the series stops at $\underline{D}^2$). Before the proof, the answer to the question the statement provokes. Nothing in (9.23) forbids $\underline{D}^3$, $\underline{D}^4$ and so on from appearing: each of them satisfies $\underline{Q}\underline{D}^n\underline{Q}^t = (\underline{Q}\underline{D}\underline{Q}^t)^n$ and so is an isotropic function of $\underline{D}$ in its own right. The series stops because the higher powers are not new.

The Cayley–Hamilton theorem, Problem 7(b) of Chapter 3, states that every symmetric $\underline{D}$ satisfies its own characteristic equation,

$$\underline{D}^3 - I_1\underline{D}^2 + I_2\underline{D} - I_3\underline{I} = \underline{O},$$

so that

$$\underline{D}^3 = I_1\underline{D}^2 - I_2\underline{D} + I_3\underline{I}.$$

Multiplying by $\underline{D}$ and substituting for $\underline{D}^3$ again expresses $\underline{D}^4$ in terms of $\underline{I}$, $\underline{D}$, $\underline{D}^2$ with coefficients that are again polynomials in the invariants, and by induction every power does the same. So $\{\underline{I}, \underline{D}, \underline{D}^2\}$ spans everything that could be built from $\underline{D}$ by multiplication, and (9.24) is not a truncation of an infinite series but a complete statement.

The proof given below never invokes Cayley–Hamilton explicitly. It reaches the same place by counting dimensions: Part 3 shows that $\{\underline{I}, \underline{D}, \underline{D}^2\}$ is linearly independent and that the space it must span is three-dimensional, so three elements are exactly enough and a fourth would be redundant. The two arguments are the same fact seen from opposite ends, and it is worth having both, because the dimension count explains why Cayley–Hamilton has degree three: the space of tensors diagonal in a fixed basis of $\mathbb{E}^3$ has dimension three, and a polynomial identity is forced as soon as one has four elements of it.

The proof of this important result is in four parts.

Part 1: an isotropic scalar function is a function of the invariants

Let the scalar-valued function $\varphi(\underline{\underline{D}})$ be isotropic, which means that $\varphi(\underline{\underline{Q}}\underline{\underline{D}}\underline{\underline{Q}}^t) = \varphi(\underline{\underline{D}})$ for every orthogonal $\underline{\underline{Q}}$. We show that it is necessary and sufficient that

$$\varphi(\underline{\underline{D}}) = \bar{\varphi}(I_1, I_2, I_3)$$

for some function $\bar{\varphi}$ of three variables.

To prove necessity, it suffices to show that φ is determined by the invariants, i.e., that $\varphi(\underline{\underline{A}}) = \varphi(\underline{\underline{B}})$ whenever $I_k(\underline{\underline{A}}) = I_k(\underline{\underline{B}})$, $k = 1, 2, 3$. Here $\underline{\underline{A}}$ and $\underline{\underline{B}}$ are arbitrary symmetric tensors. The step deserves a word: to say that a function of $\underline{\underline{D}}$ “is” a function of the three invariants is to say that its value never distinguishes two tensors that the invariants do not distinguish. If that holds, one may define $\bar{\varphi}$ on triples of numbers by choosing any $\underline{\underline{D}}$ with those invariants and setting $\bar{\varphi}(I_1, I_2, I_3) = \varphi(\underline{\underline{D}})$, and the definition is unambiguous exactly because of the property just stated.

Thus, suppose that $I_k(\underline{\underline{A}}) = I_k(\underline{\underline{B}})$. From (3.2) the principal values are the roots of $\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0$, a polynomial built from the invariants alone, so $\underline{\underline{A}}$ and $\underline{\underline{B}}$ share the same principal values. Their spectral representations, Theorem 3.7, are

$$\underline{\underline{A}} = \sum_i \lambda_i \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i \quad \text{and} \quad \underline{\underline{B}} = \sum_i \lambda_i \underline{\underline{v}}_i \otimes \underline{\underline{v}}_i, \quad (9.26)$$

where $\{\underline{\underline{u}}_i\}$ and $\{\underline{\underline{v}}_i\}$ are orthonormal sets of principal vectors. We use them to construct the orthogonal tensor $\tilde{\underline{\underline{Q}}} = \underline{\underline{u}}_i \otimes \underline{\underline{v}}_i$, summed on i .

That $\tilde{\underline{\underline{Q}}}$ is orthogonal, and that $\underline{\underline{A}} = \tilde{\underline{\underline{Q}}} \underline{\underline{B}} \tilde{\underline{\underline{Q}}}^t$, are both computations with the dyad product rule $(\underline{\underline{a}} \otimes \underline{\underline{b}})(\underline{\underline{c}} \otimes \underline{\underline{d}}) = (\underline{\underline{b}} \cdot \underline{\underline{c}}) \underline{\underline{a}} \otimes \underline{\underline{d}}$ of Lemma 2.15. For orthogonality, using $\tilde{\underline{\underline{Q}}}^t = \underline{\underline{v}}_i \otimes \underline{\underline{u}}_i$ and $\underline{\underline{u}}_i \cdot \underline{\underline{u}}_j = \delta_{ij}$,

$$\tilde{\underline{\underline{Q}}}^t \tilde{\underline{\underline{Q}}} = (\underline{\underline{v}}_i \otimes \underline{\underline{u}}_i)(\underline{\underline{u}}_j \otimes \underline{\underline{v}}_j) = \delta_{ij} \underline{\underline{v}}_i \otimes \underline{\underline{v}}_j = \underline{\underline{v}}_i \otimes \underline{\underline{v}}_i = \underline{\underline{I}},$$

the last step being the resolution of the identity on the orthonormal basis

$\{v_i\}$. For the conjugation, in two stages,

$$\tilde{Q}B = (\underline{u}_i \otimes \underline{v}_i) \left(\sum_j \lambda_j \underline{v}_j \otimes \underline{v}_j \right) = \sum_{i,j} \lambda_j \delta_{ij} \underline{u}_i \otimes \underline{v}_j = \sum_i \lambda_i \underline{u}_i \otimes \underline{v}_i,$$

and then

$$\begin{aligned} (\tilde{Q}B)\tilde{Q}^t &= \left(\sum_i \lambda_i \underline{u}_i \otimes \underline{v}_i \right) (\underline{v}_k \otimes \underline{u}_k) \\ &= \sum_{i,k} \lambda_i \delta_{ik} \underline{u}_i \otimes \underline{u}_k = \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i = \underline{A}. \end{aligned}$$

Recalling that $\varphi(\tilde{Q}B\tilde{Q}^t) = \varphi(B)$ by hypothesis, we arrive at $\varphi(\underline{A}) = \varphi(\underline{B})$, and so φ is determined by the invariants, as claimed.

To prove sufficiency, suppose $\varphi(\underline{A}) = \bar{\varphi}(I_1(\underline{A}), I_2(\underline{A}), I_3(\underline{A}))$. Then for any orthogonal Q ,

$$\begin{aligned} I_1(Q\underline{A}Q^t) &= \text{tr}(Q\underline{A}Q^t) = \text{tr}(Q^tQ\underline{A}) = \text{tr } \underline{A} = I_1(\underline{A}), \\ I_2(Q\underline{A}Q^t) &= \frac{1}{2} [I_1^2 - \text{tr}(\underline{Q}\underline{A}Q^t\underline{Q}\underline{A}Q^t)] \\ &= \frac{1}{2} [I_1^2 - \text{tr}(\underline{Q}\underline{A}^2Q^t)] = I_2(\underline{A}), \\ I_3(Q\underline{A}Q^t) &= \det(\underline{Q}\underline{A}Q^t) = (\det Q)^2 \det \underline{A} = I_3(\underline{A}), \end{aligned}$$

so that $\varphi(\underline{A}) = \varphi(Q\underline{A}Q^t)$. Three properties were used and each is from Chapter 1: the cyclic property of the trace, Theorem 1.51; the collapse $Q^tQ = \underline{I}$ in the middle of $\underline{Q}\underline{A}Q^t\underline{Q}\underline{A}Q^t$, giving $\underline{Q}\underline{A}^2Q^t$; and the product rule for determinants, Theorem 1.52, together with $\det Q = \pm 1$ from Proposition 1.48, so that $(\det Q)^2 = 1$ whether Q is a rotation or a reflection.

Part 2: $G(\underline{D})$ is coaxial with $\underline{D}$

Let η_i be principal values of $\underline{D}$, and let $\underline{v}_i$ be the associated principal vectors, so that $\underline{D} = \sum_i \eta_i \underline{v}_i \otimes \underline{v}_i$. We show that $\underline{v}_i$ are also principal vectors of $G(\underline{D})$.

To prove this claim consider the orthogonal tensor $\bar{Q} = 2\underline{v}_1 \otimes \underline{v}_1 - \underline{I}$, representing a 180° rotation about the axis $\underline{v}_1$, i.e., $\bar{Q}\underline{v}_1 = \underline{v}_1$, $\bar{Q}\underline{v}_2 = -\underline{v}_2$

and $\bar{Q}\underline{\nu}_3 = -\underline{\nu}_3$. That $\bar{Q}$ is what it is claimed to be takes two lines. It is symmetric, being a combination of $\underline{I}$ and a symmetric dyad, and

$$\bar{Q}^2 = (2\underline{\nu}_1 \otimes \underline{\nu}_1 - \underline{I})(2\underline{\nu}_1 \otimes \underline{\nu}_1 - \underline{I}) = 4\underline{\nu}_1 \otimes \underline{\nu}_1 - 2\underline{\nu}_1 \otimes \underline{\nu}_1 - 2\underline{\nu}_1 \otimes \underline{\nu}_1 + \underline{I} = \underline{I},$$

using $(\underline{\nu}_1 \otimes \underline{\nu}_1)(\underline{\nu}_1 \otimes \underline{\nu}_1) = |\underline{\nu}_1|^2 \underline{\nu}_1 \otimes \underline{\nu}_1 = \underline{\nu}_1 \otimes \underline{\nu}_1$; so $\bar{Q}^t \bar{Q} = \bar{Q}^2 = \underline{I}$ and $\bar{Q}$ is orthogonal. Its action on the principal basis is read off directly: $\bar{Q}\underline{\nu}_1 = 2\underline{\nu}_1 - \underline{\nu}_1 = \underline{\nu}_1$ and, for $i = 2, 3$, $\bar{Q}\underline{\nu}_i = 2(\underline{\nu}_1 \cdot \underline{\nu}_i)\underline{\nu}_1 - \underline{\nu}_i = -\underline{\nu}_i$. Its principal values are therefore 1, -1, -1 and $\det \bar{Q} = 1$: it is a rotation, through 180° , about $\underline{\nu}_1$.

From (9.23), and using $Q(\underline{a} \otimes \underline{b})Q^t = (Q\underline{a}) \otimes (Q\underline{b})$,

$$\bar{Q} G(\underline{D}) \bar{Q}^t = G\left(\sum_i \eta_i \bar{Q}\underline{\nu}_i \otimes \bar{Q}\underline{\nu}_i\right) = G\left(\sum_i \eta_i \underline{\nu}_i \otimes \underline{\nu}_i\right) = G(\underline{D}), \quad (9.27)$$

the middle equality holding because each sign change appears twice in a tensor product and $(-\underline{\nu}_i) \otimes (-\underline{\nu}_i) = \underline{\nu}_i \otimes \underline{\nu}_i$. This is the whole trick of Part 2: the rotation $\bar{Q}$ moves the principal vectors but leaves $\underline{D}$ itself exactly where it was, so (9.23) says that it must leave $G(\underline{D})$ where it was too.

Thus $\bar{Q} G(\underline{D}) = G(\underline{D}) \bar{Q}$, on multiplying (9.27) on the right by $\bar{Q}$ and using $\bar{Q}^t \bar{Q} = \underline{I}$, and

$$\bar{Q} G(\underline{D})\underline{\nu}_1 = G(\underline{D}) \bar{Q}\underline{\nu}_1 = G(\underline{D})\underline{\nu}_1. \quad (9.28)$$

Therefore $G(\underline{D})\underline{\nu}_1$ is also an axis of $\bar{Q}$, meaning a vector that $\bar{Q}$ leaves fixed. But this axis is unique, up to scaling: for a rotation through an angle other than zero the fixed vectors form a one-dimensional subspace, which is the statement proved in Appendix C and visible here directly, since $\bar{Q}\underline{w} = \underline{w}$ with $\underline{w} = w_i \underline{\nu}_i$ forces $w_1 \underline{\nu}_1 - w_2 \underline{\nu}_2 - w_3 \underline{\nu}_3 = w_1 \underline{\nu}_1 + w_2 \underline{\nu}_2 + w_3 \underline{\nu}_3$, hence $w_2 = w_3 = 0$. Therefore $G(\underline{D})\underline{\nu}_1 = \beta_1(\underline{D})\underline{\nu}_1$ for some scalar function β_1 , and $\underline{\nu}_1$ is a principal vector of $G(\underline{D})$. Repeating this argument with $\underline{\nu}_1$ replaced by $\underline{\nu}_2$ or $\underline{\nu}_3$, we conclude that $\{\underline{\nu}_i\}$ is a principal basis for $G(\underline{D})$, as claimed, and hence that

$$G(\underline{D}) = \sum_i \beta_i(\underline{D}) \underline{\nu}_i \otimes \underline{\nu}_i, \quad (9.29)$$

for some scalar-valued functions β_i . Figure D draws the conclusion: stretching and stress share their principal axes, so the planes on which the fluid is stretching fastest carry no shear traction at all.

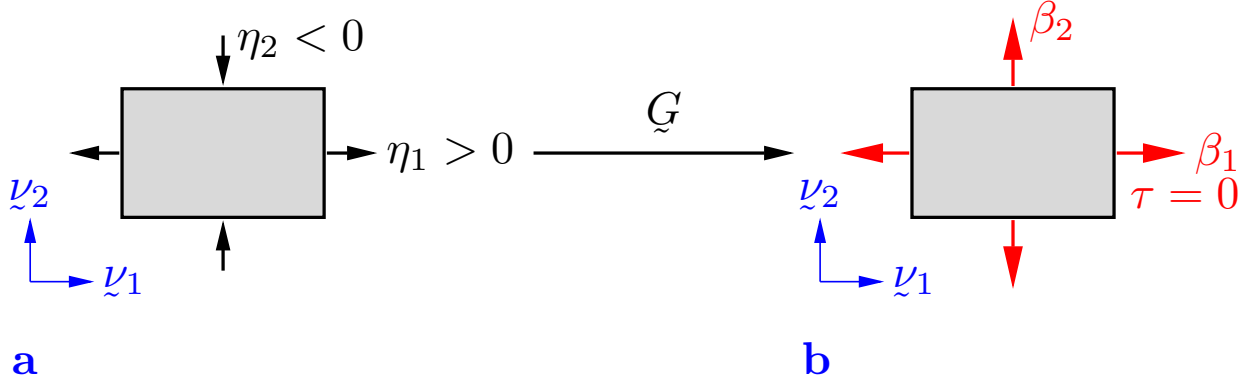


Figure D: What Part 2 proves. a: the stretching $\underline{D}$ has principal axes $\{\underline{\nu}_i\}$, shown by the blue triad, and principal values η_i , the rates at which material fibres along those axes lengthen or shorten; drawn for $\eta_1 > 0 > \eta_2$. b: by (9.29) the stress $\underline{G}(\underline{D})$ has the same axes, with new principal values β_i , so the traction on a face normal to a principal direction of stretching is purely normal and no shear traction acts there. The argument is (9.27): a 180° turn about $\underline{\nu}_1$ leaves $\underline{D}$ untouched, so by (9.23) it must leave $\underline{G}(\underline{D})$ untouched, and the only directions a rotation fixes are those of its own axis. Nothing about the material has been used, only the requirement that two observers agree.

Part 3: $\{\underline{I}, \underline{D}, \underline{D}^2\}$ is a basis for the diagonal tensors

We show that the set $\{\underline{I}, \underline{D}, \underline{D}^2\}$ is linearly independent. To prove this, we first note that since the constitutive function $\underline{G}$ is defined for all symmetric $\underline{D}$, it is defined for $\underline{D}$ having distinct principal values η_i . For such $\underline{D}$, we will assume that $\underline{I}$, $\underline{D}$ and $\underline{D}^2$ are linearly dependent and reach a contradiction. Thus, we assume that

$$\underline{Q} = a\underline{I} + b\underline{D} + c\underline{D}^2 = \sum_i (a + b\eta_i + c\eta_i^2) \underline{\nu}_i \otimes \underline{\nu}_i, \quad (9.30)$$

with at least one element of $\{a, b, c\}$ unequal to zero. The middle step used the spectral representation three times: $\underline{I} = \sum_i \underline{\nu}_i \otimes \underline{\nu}_i$, $\underline{D} = \sum_i \eta_i \underline{\nu}_i \otimes \underline{\nu}_i$ and, by Remark 3.10, $\underline{D}^2 = \sum_i \eta_i^2 \underline{\nu}_i \otimes \underline{\nu}_i$. Since the dyads $\underline{\nu}_i \otimes \underline{\nu}_i$ are themselves linearly independent, which follows on applying the right-hand side of (9.30) to $\underline{\nu}_k$ and using orthonormality, (9.30) implies that $a + b\eta_i + c\eta_i^2 = 0$ for $i = 1, 2, 3$, and hence that the η_i are all the roots of the same quadratic equation. But such equations have at most two distinct roots, contrary to our choice of $\underline{D}$. This contradiction implies that the premise is false, and hence that the set $\{\underline{I}, \underline{D}, \underline{D}^2\}$ is linearly independent, as claimed.

Note that

$$\underline{I}, \underline{D}, \underline{D}^2 \in S = \text{Span}\{\underline{\nu}_1 \otimes \underline{\nu}_1, \underline{\nu}_2 \otimes \underline{\nu}_2, \underline{\nu}_3 \otimes \underline{\nu}_3\}, \quad (9.31)$$

a three-dimensional vector space. Since n linearly independent elements of an n -dimensional vector space constitute a basis for that space, by Definition 1.3 and Theorem 1.4, we have

$$S = \text{Span}\{\underline{I}, \underline{D}, \underline{D}^2\}. \quad (9.32)$$

Part 4: assembling the representation

It follows from (9.29) that $\underline{G}(\underline{D}) \in S$. Thus, (9.32) implies that there are unique scalars $\psi_0(\underline{D})$, $\psi_1(\underline{D})$ and $\psi_2(\underline{D})$ such that

$$\underline{G}(\underline{D}) = \psi_0(\underline{D})\underline{I} + \psi_1(\underline{D})\underline{D} + \psi_2(\underline{D})\underline{D}^2. \quad (9.33)$$

To see this explicitly, we combine it with (9.29) to get

$$\beta_i(\underline{D}) = \psi_0 + \psi_1\eta_i + \psi_2\eta_i^2; \quad i = 1, 2, 3, \quad (9.34)$$

i.e.,

$$\begin{pmatrix} 1 & \eta_1 & \eta_1^2 \\ 1 & \eta_2 & \eta_2^2 \\ 1 & \eta_3 & \eta_3^2 \end{pmatrix} \begin{Bmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{Bmatrix} = \begin{Bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{Bmatrix}, \quad (9.35)$$

in which the coefficient matrix has determinant $(\eta_1 - \eta_2)(\eta_2 - \eta_3)(\eta_3 - \eta_1)$. This is non-zero for the family of $\underline{D}$'s considered. Then, since the η_i are determined by the $I_k(\underline{D})$, it follows that the ψ 's are determined by $\underline{D}$.

The determinant is the Vandermonde determinant and it is worth evaluating rather than quoting, since the whole of Part 4 turns on its not vanishing. Subtract the first row from the second and from the third, which changes no determinant:

$$\begin{aligned} \det \begin{pmatrix} 1 & \eta_1 & \eta_1^2 \\ 1 & \eta_2 & \eta_2^2 \\ 1 & \eta_3 & \eta_3^2 \end{pmatrix} &= \det \begin{pmatrix} 1 & \eta_1 & \eta_1^2 \\ 0 & \eta_2 - \eta_1 & \eta_2^2 - \eta_1^2 \\ 0 & \eta_3 - \eta_1 & \eta_3^2 - \eta_1^2 \end{pmatrix} \\ &= (\eta_2 - \eta_1)(\eta_3 - \eta_1) \det \begin{pmatrix} 1 & \eta_2 + \eta_1 \\ 1 & \eta_3 + \eta_1 \end{pmatrix}, \end{aligned}$$

where the common factors have been taken out of the second and third rows using $\eta^2 - \eta_1^2 = (\eta - \eta_1)(\eta + \eta_1)$, and the expansion along the first column has been performed. The remaining two-by-two determinant is $\eta_3 - \eta_2$, so

$$\det = (\eta_2 - \eta_1)(\eta_3 - \eta_1)(\eta_3 - \eta_2) = (\eta_1 - \eta_2)(\eta_2 - \eta_3)(\eta_3 - \eta_1),$$

the second form following from the first by changing the sign of two of the three factors. It vanishes if and only if two of the η_i coincide, which the choice of $\tilde{D}$ has excluded.

REMARK 9.12 (a misprint in the printed (9.35)). The book prints the determinant as $(\eta_1 - \eta_1)(\eta_2 - \eta_3)(\eta_3 - \eta_1)$, whose first factor is identically zero and whose vanishing would destroy the argument. The intended factor is $\eta_1 - \eta_2$, as the evaluation above confirms.

Now, from (9.33) we have

$$\begin{aligned} G(\tilde{Q}\tilde{D}\tilde{Q}^t) - \tilde{Q}G(\tilde{D})\tilde{Q}^t \\ = \tilde{Q} \left\{ [\psi_0(\tilde{Q}\tilde{D}\tilde{Q}^t) - \psi_0(\tilde{D})]\tilde{I} + [\psi_1(\tilde{Q}\tilde{D}\tilde{Q}^t) - \psi_1(\tilde{D})]\tilde{D} \right. \\ \left. + [\psi_2(\tilde{Q}\tilde{D}\tilde{Q}^t) - \psi_2(\tilde{D})]\tilde{D}^2 \right\} \tilde{Q}^t, \end{aligned} \quad (9.36)$$

where we have used $\tilde{Q}\tilde{Q}^t = \tilde{I}$ and $(\tilde{Q}\tilde{D}\tilde{Q}^t)^2 = \tilde{Q}\tilde{D}^2\tilde{Q}^t$. Writing out the step: (9.33) applied at the argument $\tilde{Q}\tilde{D}\tilde{Q}^t$ gives

$$G(\tilde{Q}\tilde{D}\tilde{Q}^t) = \psi_0(\tilde{Q}\tilde{D}\tilde{Q}^t)\tilde{I} + \psi_1(\tilde{Q}\tilde{D}\tilde{Q}^t)\tilde{Q}\tilde{D}\tilde{Q}^t + \psi_2(\tilde{Q}\tilde{D}\tilde{Q}^t)\tilde{Q}\tilde{D}^2\tilde{Q}^t,$$

and the first term may be written $\underline{Q}\underline{I}\underline{Q}^t$, so every term on the right carries $\underline{Q} \cdots \underline{Q}^t$ and the common factors may be taken outside; subtracting $\underline{Q}\underline{G}(\underline{D})\underline{Q}^t$, itself expanded by (9.33), leaves (9.36).

We impose (9.23), which makes the left-hand side of (9.36) vanish, and left- and right-multiply the resulting equation by $\underline{Q}^t$ and $\underline{Q}$ respectively, which removes the outer factors. Using the result of Part 3, that $\{\underline{I}, \underline{D}, \underline{D}^2\}$ is linearly independent and so the only vanishing combination is the one with zero coefficients, we obtain $\psi_i(\underline{Q}\underline{D}\underline{Q}^t) = \psi_i(\underline{D})$. Finally, we apply the result of Part 1 to conclude that $\psi_i(\underline{D}) = \bar{\psi}_i(I_1, I_2, I_3)$ and substitute into (9.33) to arrive at (9.24). Conversely, it is a simple matter to confirm that (9.24) ensures the isotropy of $\underline{G}(\underline{D})$: the calculation is the one just performed, run in reverse, and Problem 3 carries out its exact counterpart for the heat flux.

This result subsumes the case in which $\underline{D}$ has only two distinct principal values, or just one. The first of these is covered by taking $\varphi_2 = 0$, and the second by taking $\varphi_1 = \varphi_2 = 0$. The book says “the third” in the last clause, but only two cases have been named, and the clause refers to the second of them, a $\underline{D}$ with a single principal value.

REMARK 9.13 (the degenerate cases, and what continuity costs). Parts 3 and 4 assumed the η_i distinct, so the argument as given proves (9.24) only on that set of $\underline{D}$ ’s, and the sentence above disposes of the rest rather briskly. Two separate things need saying.

Existence is not in doubt. Suppose exactly two principal values coincide, $\eta_2 = \eta_3 \neq \eta_1$. Every unit vector in the plane of $\underline{\nu}_2$ and $\underline{\nu}_3$ is then a principal vector of $\underline{D}$, so by Part 2 it is a principal vector of $\underline{G}(\underline{D})$ as well; a symmetric tensor for which every vector of a plane is principal has a single principal value on that plane, so $\beta_2 = \beta_3$ and

$$\underline{G}(\underline{D}) = \beta_1 \underline{\nu}_1 \otimes \underline{\nu}_1 + \beta_2 (\underline{\nu}_2 \otimes \underline{\nu}_2 + \underline{\nu}_3 \otimes \underline{\nu}_3).$$

The two tensors on the right span the same plane of tensors as $\underline{I}$ and $\underline{D}$, which are independent here, so $\underline{G}(\underline{D}) = \psi_0 \underline{I} + \psi_1 \underline{D}$ with unique coefficients, and $\varphi_2 = 0$ is one admissible choice. If all three principal values coincide, $\underline{D} = \eta \underline{I}$, every vector is principal, and $\underline{G}(\underline{D})$ is a multiple of $\underline{I}$.

Uniqueness is what is lost, and with it, possibly, continuity. On the degenerate set $\underline{D}^2$ is itself a combination of $\underline{I}$ and $\underline{D}$, so the three coefficients are no longer fixed by $\underline{G}$. The book's recipe, $\varphi_2 = 0$ there, is a valid representation, but it need not be the continuous one. The simplest example is $\underline{G}(\underline{D}) = \underline{D}^2$, for which $\varphi_0 = \varphi_1 = 0$ and $\varphi_2 = 1$ wherever the η_i are distinct. At a $\underline{D}$ with principal values $\eta_1 \neq \eta_2 = \eta_3$ the spectral forms give

$$\underline{D}^2 = (\eta_1 + \eta_2)\underline{D} - \eta_1\eta_2\underline{I},$$

as one checks on $\underline{\nu}_1$, where both sides give η_1^2 , and on $\underline{\nu}_2$ and $\underline{\nu}_3$, where both give η_2^2 . So the recipe assigns $\varphi_1 = \eta_1 + \eta_2$ and $\varphi_0 = -\eta_1\eta_2$ at such a point, while at every nearby $\underline{D}$ with distinct principal values $\varphi_1 = \varphi_0 = 0$: the coefficient functions jump. Since (9.38) goes on to assume the φ 's continuous, the right reading is that on the degenerate set the coefficients are defined by continuity, where the limits exist, rather than by the rule $\varphi_2 = 0$. At $\underline{D} = \underline{Q}$, the only degenerate point that (9.38) actually uses, no difficulty arises, since $\underline{D}$ and $\underline{D}^2$ both vanish there and only φ_0 survives, whatever the others are.

Reinstating the passive variables, we conclude that the stress at a material point p in a viscous fluid is of the form

$$\begin{aligned} \underline{T} = & \varphi_0(I_1, I_2, I_3, \theta, \rho; p)\underline{I} + \varphi_1(I_1, I_2, I_3, \theta, \rho; p)\underline{D} \\ & + \varphi_2(I_1, I_2, I_3, \theta, \rho; p)\underline{D}^2, \end{aligned} \quad (9.37)$$

where the I_k are the principal invariants of $\underline{D}$. It is a matter of experiment to determine the three functions φ_0 , φ_1 and φ_2 in terms of the five variables I_1 , I_2 , I_3 , θ and ρ . While this task presents a formidable challenge, it is far more tractable than that presented by the original constitutive hypothesis embodied in (9.3).

REMARK 9.14 (the size of the reduction). The gain is worth counting, because it is the reason theorems of this kind are proved at all.

Equation (9.3) asks for six functions, the independent components of a symmetric $\underline{T}$, of eight scalar arguments: three components of $\underline{x}$, three

of $\underline{v}$, plus θ and ρ , and then nine more from $\underline{L}$, seventeen in all. To measure such a thing is out of the question.

Equation (9.37) asks for three functions of five arguments, and the five are all scalars with direct physical meaning. Better still, the tensorial structure is now completely fixed: whatever the material, the stress must be a combination of $\underline{I}$, $\underline{D}$ and $\underline{D}^2$ and of nothing else. An experiment no longer has to discover the form of the answer, only three numbers at each value of five parameters.

Every step of the reduction came from a principle rather than from data: fluidity removed $\underline{F}$, frame indifference removed $\underline{x}$, $\underline{v}$ and $\underline{W}$, and isotropy fixed the tensorial form. That is the return on Chapters 7 and 8.

Assuming the φ 's to be continuous functions of the invariants, we observe that the stress reduces to

$$\underline{T}|_{\underline{D}=\underline{0}} = -\tilde{p}(\theta, \rho; p)\underline{I} \quad (9.38)$$

in the absence of stretching, where $\tilde{p} = -\varphi_0(0, 0, 0, \theta, \rho; p)$ is the pressure. The step is immediate once one notices that all three invariants vanish with $\underline{D}$: $I_1(\underline{Q}) = \text{tr } \underline{Q} = 0$, $I_2(\underline{Q}) = 0$ and $I_3(\underline{Q}) = \det \underline{Q} = 0$. So the arguments of the φ 's go to $(0, 0, 0, \theta, \rho)$ and, $\underline{D}$ and $\underline{D}^2$ having vanished, only the term in $\underline{I}$ survives.

If (9.38) holds in the presence of stretching, then the fluid is *inviscid*. The latter model is descriptive in regions exterior to “boundary layers” in which the effects of viscosity are important. The inviscid fluid model therefore plays a major role in the theories of aero- and hydro-dynamics.

REMARK 9.15 (what (9.38) says about a fluid at rest). Equation (9.38) is the constitutive statement that a fluid at rest cannot support a shear stress, and it has been derived rather than assumed. It is worth seeing that it is genuinely a theorem here.

The traction on a surface with unit normal $\underline{n}$ is $\underline{T}\underline{n}$, and with $\underline{T} = -\tilde{p}\underline{I}$ this is $-\tilde{p}\underline{n}$: whatever the orientation of the surface, the traction is along its own normal and of the same magnitude. A fluid at rest therefore

pushes equally hard in every direction, which is Pascal's principle, and it is a consequence of (9.37) together with the continuity of the φ 's.

What made it work is that the only tensor available at $\underline{D} = \underline{Q}$ is $\underline{I}$, and $\underline{I}$ has no preferred direction. A solid at rest can carry shear because its constitutive function still has $\underline{F}$ to build a tensor from, and $\underline{F}$ does have preferred directions. The whole difference between a fluid and a solid, in the end, is which tensors survive into the list.

9.2 The heat flux

Beginning with (8.31), we may proceed, as in the passage from (9.1) to (9.22), to conclude that the heat flux is given constitutively in the form

$$\underline{q} = \underline{g}(\theta, \rho, \underline{D}, \text{grad } \theta; p), \quad (9.39)$$

in the frame of O , where the function $\underline{g}$ is such that

$$\underline{g}(\theta, \rho, \underline{Q}\underline{D}\underline{Q}^t, \underline{Q} \text{grad } \theta; p) = \underline{Q} \underline{g}(\theta, \rho, \underline{D}, \text{grad } \theta; p) \quad (9.40)$$

for all orthogonal $\underline{Q}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$.

The repetition is genuine and only two things change. The heat flux is a vector, so (8.32) carries a single factor $\underline{Q}$ rather than the two of (8.35), and (9.40) inherits that. And the list retains $\text{grad } \theta$, which (8.33) had dropped from the stress; under the change of frame it transforms as a spatial vector, $\text{grad}^+ \theta^+ = \underline{Q} \text{grad } \theta$, as was shown in Section 8.4, so it appears inside $\underline{g}$ with a $\underline{Q}$ attached to it. Everything else is word for word: fluidity replaces $\underline{F}$ by ρ and removes κ ; the choice $\underline{Q} = \underline{I}$ removes $\underline{x}$ and $\underline{v}$; the choice $\hat{\underline{\Omega}} = -\underline{W}_*$ removes $\underline{W}$.

We will show that the general form of such a function is

$$\underline{g}(\theta, \rho, \underline{D}, \text{grad } \theta; p) = \underline{K} \text{grad } \theta, \quad (9.41)$$

where

$$\begin{aligned} \underline{K}(\theta, \rho, \underline{D}, \text{grad } \theta; p) = & \phi_0(I_1, \dots, I_6, \theta, \rho; p) \underline{I} + \phi_1(I_1, \dots, I_6, \theta, \rho; p) \underline{D} \\ & + \phi_2(I_1, \dots, I_6, \theta, \rho; p) \underline{D}^2, \end{aligned} \quad (9.42)$$

with

$$\begin{aligned} I_4(\underline{\underline{D}}, \text{grad } \theta) &= \text{grad } \theta \cdot \underline{\underline{D}}(\text{grad } \theta), \\ I_5(\underline{\underline{D}}, \text{grad } \theta) &= \text{grad } \theta \cdot \underline{\underline{D}}^2(\text{grad } \theta), \\ I_6(\text{grad } \theta) &= |\text{grad } \theta|. \end{aligned} \quad (9.43)$$

REMARK 9.16 ($\underline{\underline{K}}$ here is not the $\underline{\underline{K}}$ of Chapter 7). The letter is overloaded and the two objects are unrelated. The $\underline{\underline{K}}$ of (7.14) is the fixed orthogonal two-point tensor relating the reference placements adopted by two observers, $\underline{\underline{X}}^+ = \underline{\underline{K}}\underline{\underline{X}} + \underline{\underline{c}}^+$; it maps $\hat{\mathbb{E}}^3$ to $\hat{\mathbb{E}}^{3+}$ and it played its part in Section 8.4. The $\underline{\underline{K}}$ of (9.42) is the *thermal conductivity tensor*: it maps $\mathbb{E}^3$ to itself, it is not orthogonal, it need not even be invertible, and it depends on the state of the fluid. The two never meet in the same formula in this chapter, since Chapter 7's $\underline{\underline{K}}$ disappeared along with κ at (9.3).

The scalar K of (9.59) and (9.61) is a third object, the conductivity of an isotropic conductor, and it is what $\underline{\underline{K}}$ becomes when $\underline{\underline{D}} = \underline{\underline{Q}}$.

To simplify the notation we again suppress the passive variables and write (9.40) as

$$\underline{\underline{g}}(\underline{\underline{Q}}\underline{\underline{D}}\underline{\underline{Q}}^t, \underline{\underline{Q}}\underline{\underline{d}}) = \underline{\underline{Q}} \underline{\underline{g}}(\underline{\underline{D}}, \underline{\underline{d}}), \quad (9.44)$$

for all symmetric $\underline{\underline{D}}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$ and all $\underline{\underline{d}} = \text{grad } \theta \in \mathbb{E}^3$, where $\underline{\underline{Q}}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$ is an arbitrary orthogonal tensor. The letter $\underline{\underline{d}}$ is the book's and it is not the translation vector of (9.12), which has finished its work; from here to the end of the section $\underline{\underline{d}}$ means the temperature gradient.

For $\underline{\underline{Q}} = -\underline{\underline{I}}$ this yields

$$\underline{\underline{g}}(\underline{\underline{D}}, -\underline{\underline{d}}) = -\underline{\underline{g}}(\underline{\underline{D}}, \underline{\underline{d}}). \quad (9.45)$$

The choice is legitimate and instructive. The tensor $-\underline{\underline{I}}$ is orthogonal, since $(-\underline{\underline{I}})^t(-\underline{\underline{I}}) = \underline{\underline{I}}$, though improperly so: $\det(-\underline{\underline{I}}) = (-1)^3 \det \underline{\underline{I}} = -1$ by Problem 21(a) of Chapter 1. It is available because (9.44) was demanded for all orthogonal $\underline{\underline{Q}}$, reflections included, and not merely for rotations. Its effect on the two arguments is different: it leaves the tensor alone, since $(-\underline{\underline{I}})\underline{\underline{D}}(-\underline{\underline{I}})^t = \underline{\underline{D}}$, the two sign changes cancelling, while it reverses the vector. So (9.44) becomes (9.45), which says that reversing

the temperature gradient reverses the heat flux and changes nothing else. Physically that is the statement that the fluid has no built-in sense of direction which could make heat run more easily one way than the other.

Our proof is again in four parts.

Part 1: even dependence on $\underline{d}$ is dependence on $\underline{d} \otimes \underline{d}$

Suppose $\varphi(\underline{D}, \underline{d}) = \varphi(\underline{D}, -\underline{d})$. We show that φ then depends on $\underline{d}$ via $\underline{d} \otimes \underline{d}$. Dropping the passive variable $\underline{D}$, our task is to show that if $\psi(\underline{d}) = \psi(-\underline{d})$, then $\psi(\underline{d}) = \bar{\psi}(\underline{d} \otimes \underline{d})$, i.e., that $\psi(\underline{a}) = \psi(\underline{b})$ whenever $\underline{a} \otimes \underline{a} = \underline{b} \otimes \underline{b}$.

Suppose, then, that the latter is true. Apply the tensor equation $\underline{a} \otimes \underline{a} = \underline{b} \otimes \underline{b}$ first to $\underline{a}$ and then to $\underline{b}$, using $(\underline{u} \otimes \underline{v})\underline{w} = (\underline{v} \cdot \underline{w})\underline{u}$ from Definition 1.32. It follows that

$$|\underline{a}|^2 \underline{a} = (\underline{a} \cdot \underline{b})\underline{b} \quad \text{and} \quad |\underline{b}|^2 \underline{b} = (\underline{a} \cdot \underline{b})\underline{a}, \quad (9.46)$$

which imply that $|\underline{a}| = |\underline{b}|$ and $(\underline{a} \cdot \underline{b})^2 = |\underline{a}|^2 |\underline{b}|^2$. Both implications come from taking inner products. Dotting (9.46)₁ with $\underline{a}$ gives $|\underline{a}|^4 = (\underline{a} \cdot \underline{b})^2$, and dotting (9.46)₂ with $\underline{b}$ gives $|\underline{b}|^4 = (\underline{a} \cdot \underline{b})^2$; so $|\underline{a}|^4 = |\underline{b}|^4$, and since norms are non-negative, $|\underline{a}| = |\underline{b}|$, whence $(\underline{a} \cdot \underline{b})^2 = |\underline{a}|^4 = |\underline{a}|^2 |\underline{b}|^2$.

But by Definition 1.8 there is $\alpha \in \mathbb{R}$ such that $\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \alpha$. Then $\cos \alpha = \pm 1$ and either of (9.46) gives $\underline{b} = \pm \underline{a}$: from (9.46)₁, $\underline{a} \cdot \underline{b} = \pm |\underline{a}| |\underline{b}| = \pm |\underline{a}|^2$, so $|\underline{a}|^2 \underline{a} = \pm |\underline{a}|^2 \underline{b}$ and, if $\underline{a} \neq \underline{0}$, $\underline{b} = \pm \underline{a}$; while if $\underline{a} = \underline{0}$ then $\underline{a} \otimes \underline{a} = \underline{0}$ forces $\underline{b} = \underline{0}$ as well. The solution $\underline{b} = \underline{a}$ trivially yields $\psi(\underline{b}) = \psi(\underline{a})$, and the solution $\underline{b} = -\underline{a}$ gives $\psi(\underline{b}) = \psi(-\underline{a})$. But $\psi(\underline{a}) = \psi(-\underline{a})$ by hypothesis, and therefore $\psi(\underline{a}) = \psi(\underline{b})$ whenever $\underline{a} \otimes \underline{a} = \underline{b} \otimes \underline{b}$. Thus, $\psi(\underline{d}) = \bar{\psi}(\underline{d} \otimes \underline{d})$ for some function $\bar{\psi}$, as claimed. Conversely, if $\psi(\underline{d}) = \bar{\psi}(\underline{d} \otimes \underline{d})$, then it follows immediately that $\psi(\underline{d}) = \psi(-\underline{d})$, since $(-\underline{d}) \otimes (-\underline{d}) = \underline{d} \otimes \underline{d}$.

REMARK 9.17 (the point of trading $\underline{d}$ for $\underline{d} \otimes \underline{d}$). The manoeuvre looks like a change of notation and is not. A function of $\underline{d}$ that happens to be even is an awkward object: the evenness is a constraint one must

remember to impose, and it is invisible in the formula. A function of $\underline{d} \otimes \underline{d}$ has the evenness built into its argument, so no constraint has to be carried.

The exchange is exactly Remark 5.5's principle applied in reverse. There one fed an identity its most trivial input to make one side collapse. Here one changes variables so that a symmetry becomes invisible, which is the standard way of disposing of a symmetry: not by enforcing it at every step, but by working in variables that cannot violate it. The same idea produced $\underline{C} = \underline{F}^t \underline{F}$ in Chapter 2, whose whole merit is that it cannot see the rotation in $\underline{F}$.

Part 2: $\{\underline{d}, \underline{D}\underline{d}, \underline{D}^2\underline{d}\}$ is a basis

We show that if $\underline{d}$ is arbitrary and $\underline{D}$ is an arbitrary symmetric tensor, then $\{\underline{d}, \underline{D}\underline{d}, \underline{D}^2\underline{d}\}$ is a basis for $\mathbb{E}^3$. We again discuss only the case in which the principal values η_i of $\underline{D}$ are distinct. Thus, we assume these vectors to be linearly dependent, i.e., that

$$a\underline{d} + b\underline{D}\underline{d} + c\underline{D}^2\underline{d} = \underline{0}, \quad (9.47)$$

with at least one element of $\{a, b, c\}$ unequal to zero. Using the spectral decomposition of $\underline{D}$, we write this as

$$\sum_i (a + b\eta_i + c\eta_i^2) d_i \underline{\nu}_i = \underline{0}, \quad \text{where } d_i = \underline{\nu}_i \cdot \underline{d}. \quad (9.48)$$

The rewriting is the calculation of Part 3 of the previous section with a vector attached. With $\underline{D} = \sum_i \eta_i \underline{\nu}_i \otimes \underline{\nu}_i$ and $\underline{D}^2 = \sum_i \eta_i^2 \underline{\nu}_i \otimes \underline{\nu}_i$, and with $\underline{d} = \sum_i d_i \underline{\nu}_i$ resolved on the principal basis,

$$\underline{D}\underline{d} = \sum_i \eta_i d_i \underline{\nu}_i, \quad \underline{D}^2\underline{d} = \sum_i \eta_i^2 d_i \underline{\nu}_i,$$

and collecting the three terms of (9.47) on each $\underline{\nu}_i$ gives (9.48). Since $\{\underline{\nu}_i\}$ is a basis, each coefficient must vanish separately: $(a + b\eta_i + c\eta_i^2)d_i = 0$ for each i . When every component d_i is non-zero it may be cancelled, and this requires $a + b\eta_i + c\eta_i^2 = 0$ for $i = 1, 2, 3$. This is the contradiction

encountered previously: three distinct numbers cannot all be roots of one quadratic unless that quadratic is identically zero. It follows that a, b, c all vanish and hence that the set $\{\underline{d}, \underline{D}\underline{d}, \underline{D}^2\underline{d}\}$ is linearly independent, for every $\underline{D}$ with distinct principal values and every $\underline{d}$ with no zero component on the principal basis. Remark 9.18 explains why that proviso cannot be dropped, and why it costs nothing. Three independent vectors in the three-dimensional $\mathbb{E}^3$ form a basis, by Theorem 1.4, and every vector, in particular the value $\underline{g}(\underline{D}, \underline{d})$, is a combination of them. Thus,

$$\underline{g}(\underline{D}, \underline{d}) = \psi_0(\underline{D}, \underline{d})\underline{d} + \psi_1(\underline{D}, \underline{d})\underline{D}\underline{d} + \psi_2(\underline{D}, \underline{d})\underline{D}^2\underline{d}. \quad (9.49)$$

REMARK 9.18 (the claim that opens Part 2 is too strong). The book opens Part 2 with the sentence “if $\underline{d}$ is arbitrary and $\underline{D}$ is an arbitrary symmetric tensor, then $\{\underline{d}, \underline{D}\underline{d}, \underline{D}^2\underline{d}\}$ is a basis for $\mathbb{E}^3$ ”, and it passes from (9.48) to the vanishing of $a + b\eta_i + c\eta_i^2$ with the words “as $\underline{d}$ is arbitrary”. As stated, the claim is false. Take $\underline{D}$ with distinct principal values and $\underline{d} = \underline{\nu}_1$: then $\underline{D}\underline{d} = \eta_1\underline{\nu}_1$ and $\underline{D}^2\underline{d} = \eta_1^2\underline{\nu}_1$, three parallel vectors. More generally, if one component d_k vanishes, all three vectors lie in the plane of the other two principal vectors, and if two principal values coincide they lie in a plane as well. The appeal to the arbitrariness of $\underline{d}$ does not help, because $\underline{d}$ is the fixed argument at which $\underline{g}$ is being represented and a, b, c may depend on it; one may not vary $\underline{d}$ inside a relation that is asserted at one $\underline{d}$ only.

What is true is what was proved above: $\{\underline{d}, \underline{D}\underline{d}, \underline{D}^2\underline{d}\}$ is a basis if and only if the η_i are distinct and every $d_i = \underline{\nu}_i \cdot \underline{d}$ is non-zero. Call such pairs $(\underline{D}, \underline{d})$ generic. Every pair is a limit of generic pairs, obtained by perturbing $\underline{D}$ and $\underline{d}$ slightly, so (9.49) and everything built on it in Parts 3 and 4 hold on a dense set, and for a continuous $\underline{g}$ with continuous coefficient functions the final representation (9.41), (9.42) extends to all pairs by continuity.

The exceptional pairs can also be treated directly, and doing so shows that the form of (9.41) survives without any appeal to continuity. If $\underline{d} = d_1\underline{\nu}_1$ lies along a principal direction, the half turn $\bar{\underline{Q}} = 2\underline{\nu}_1 \otimes \underline{\nu}_1 - \underline{I}$ of Part 2 of the previous section fixes both $\underline{D}$ and $\underline{d}$, so (9.44) gives $\bar{\underline{Q}} \underline{g}(\underline{D}, \underline{d}) = \underline{g}(\underline{D}, \underline{d})$: the vector $\underline{g}$ lies along the axis $\underline{\nu}_1$ of $\bar{\underline{Q}}$, that is,

along $\underline{d}$, and $\underline{g} = \psi_0 \underline{d}$. If instead $\underline{d}$ lies in the plane of $\underline{\nu}_1$ and $\underline{\nu}_2$, the reflection $\underline{I} - 2\underline{\nu}_3 \otimes \underline{\nu}_3$ fixes $\underline{D}$ and $\underline{d}$, so $\underline{g}$ has no component along $\underline{\nu}_3$ and lies in the plane spanned by $\underline{d}$ and $\underline{D}\underline{d}$, which is that plane when $\eta_1 \neq \eta_2$ and $d_1 d_2 \neq 0$. This is a place where the full orthogonal group, and not only the rotations, is needed. Repeated principal values are dealt with in the same way, using reflections in planes that contain $\underline{d}$ and fix $\underline{D}$. So in every case $\underline{g}(\underline{D}, \underline{d}) = \underline{K}\underline{d}$ with $\underline{K}$ a combination of $\underline{I}$, $\underline{D}$ and $\underline{D}^2$; only the uniqueness of the coefficients is lost on the exceptional set, exactly as for the stress in Remark 9.13.

REMARK 9.19 (why the heat flux is forced to be a linear map of $\text{grad } \theta$). Equation (9.49) already contains (9.41), and it is worth seeing why, since the linearity of $\underline{g}$ in $\text{grad } \theta$ is usually presented as a modelling assumption named after Fourier and is here a consequence.

Collect the three terms of (9.49): each is a tensor applied to $\underline{d}$, so

$$\underline{g}(\underline{D}, \underline{d}) = [\psi_0 \underline{I} + \psi_1 \underline{D} + \psi_2 \underline{D}^2] \underline{d} = \underline{K}\underline{d}.$$

The dependence of $\underline{g}$ on the temperature gradient is therefore through a tensor acting on it, and that much is exact, however violent the temperature gradient. What is not linear, at this stage, is the tensor $\underline{K}$ itself: its coefficients ψ_i still depend on $\underline{d}$ and will be shown in Part 4 to depend on it through the three scalars I_4, I_5, I_6 . Fourier's law is the further statement that this residual dependence may be neglected, which is the content of (9.61).

So the structure “flux = tensor times gradient” is forced by frame indifference alone, and only the constancy of the tensor is a constitutive assumption. This is a good example of what representation theorems buy: not the answer, but the shape of the answer.

Part 3: the coefficients are even in $\underline{d}$

From the result of Part 2, applied at $-\underline{d}$ and using $\underline{D}(-\underline{d}) = -\underline{D}\underline{d}$ and $\underline{D}^2(-\underline{d}) = -\underline{D}^2\underline{d}$, we have

$$\underline{g}(\underline{D}, -\underline{d}) = -\psi_0(\underline{D}, -\underline{d})\underline{d} - \psi_1(\underline{D}, -\underline{d})\underline{D}\underline{d} - \psi_2(\underline{D}, -\underline{d})\underline{D}^2\underline{d}. \quad (9.50)$$

Imposing (9.45), that is, adding (9.50) to (9.49) and requiring the sum to vanish, yields

$$\begin{aligned} & [\psi_0(\underline{D}, \underline{d}) - \psi_0(\underline{D}, -\underline{d})]\underline{d} + [\psi_1(\underline{D}, \underline{d}) - \psi_1(\underline{D}, -\underline{d})]\underline{D}\underline{d} \\ & + [\psi_2(\underline{D}, \underline{d}) - \psi_2(\underline{D}, -\underline{d})]\underline{D}^2\underline{d} = \underline{0}, \end{aligned} \quad (9.51)$$

and therefore $\psi_i(\underline{D}, \underline{d}) = \psi_i(\underline{D}, -\underline{d})$, $i = 0, 1, 2$, by the linear independence established in Part 2. It then follows from Part 1 that

$$\psi_i(\underline{D}, \underline{d}) = \bar{\psi}_i(\underline{D}, \underline{d} \otimes \underline{d}); \quad i = 0, 1, 2. \quad (9.52)$$

Thus,

$$\begin{aligned} & g(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}) - \underline{Q} g(\underline{D}, \underline{d}) \\ & = [\bar{\psi}_0(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d} \otimes \underline{Q}\underline{d}) - \bar{\psi}_0(\underline{D}, \underline{d} \otimes \underline{d})]\underline{Q}\underline{d} \\ & + [\bar{\psi}_1(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d} \otimes \underline{Q}\underline{d}) - \bar{\psi}_1(\underline{D}, \underline{d} \otimes \underline{d})]\underline{Q}\underline{D}\underline{d} \\ & + [\bar{\psi}_2(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d} \otimes \underline{Q}\underline{d}) - \bar{\psi}_2(\underline{D}, \underline{d} \otimes \underline{d})]\underline{Q}\underline{D}^2\underline{d}. \end{aligned} \quad (9.53)$$

Two identities were used in assembling the right-hand side, and both are worth naming. Evaluating (9.49) at the arguments $(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d})$ produces terms $(\underline{Q}\underline{D}\underline{Q}^t)^n \underline{Q}\underline{d} = \underline{Q}\underline{D}^n \underline{Q}^t \underline{Q}\underline{d} = \underline{Q}\underline{D}^n \underline{d}$ for $n = 0, 1, 2$, which are the three vectors displayed; and (9.52) evaluated there involves $(\underline{Q}\underline{d}) \otimes (\underline{Q}\underline{d}) = \underline{Q}(\underline{d} \otimes \underline{d})\underline{Q}^t$, written in the book's shorthand as $\underline{Q}\underline{d} \otimes \underline{Q}\underline{d}$.

Imposing (9.44), which makes the left side of (9.53) vanish, left-multiplying by $\underline{Q}^t$, which converts the three vectors $\underline{Q}\underline{d}, \underline{Q}\underline{D}\underline{d}, \underline{Q}\underline{D}^2\underline{d}$ back into $\underline{d}, \underline{D}\underline{d}, \underline{D}^2\underline{d}$, and invoking the result of Part 2, we conclude that

$$\bar{\psi}_i(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d} \otimes \underline{Q}\underline{d}) = \bar{\psi}_i(\underline{D}, \underline{d} \otimes \underline{d}); \quad i = 0, 1, 2, \quad (9.54)$$

for all orthogonal $\underline{Q}$.

Part 4: the six invariants determine an isotropic scalar

Suppose $\varphi(\underline{D}, \underline{d} \otimes \underline{d}) = \varphi(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d} \otimes \underline{Q}\underline{d})$ for all orthogonal $\underline{Q}$. We demonstrate that φ is then determined by I_1 to I_6 , i.e., that $\varphi(\underline{A}, \underline{a} \otimes \underline{a}) =$

$\varphi(\underline{B}, \underline{b} \otimes \underline{b})$ whenever

$$\begin{aligned} & \{I_1(\underline{A}), I_2(\underline{A}), I_3(\underline{A}), I_4(\underline{A}, \underline{a}), I_5(\underline{A}, \underline{a}), I_6(\underline{a})\} \\ &= \{I_1(\underline{B}), I_2(\underline{B}), I_3(\underline{B}), I_4(\underline{B}, \underline{b}), I_5(\underline{B}, \underline{b}), I_6(\underline{b})\}, \end{aligned} \quad (9.55)$$

where I_1 to I_6 are defined in (9.25) and (9.43).

Thus, suppose these two sets are equal. Then the invariants I_1, I_2, I_3 of $\underline{A}$ and $\underline{B}$ coincide and (9.26) holds in particular, implying that $\underline{A} = \tilde{\underline{Q}} \underline{B} \tilde{\underline{Q}}^t$ and $\underline{A}^2 = \tilde{\underline{Q}} \underline{B}^2 \tilde{\underline{Q}}^t$, where $\tilde{\underline{Q}} = \underline{u}_i \otimes \underline{v}_i$ is orthogonal. The second follows from the first by squaring: $\tilde{\underline{Q}} \underline{B} \tilde{\underline{Q}}^t \tilde{\underline{Q}} \underline{B} \tilde{\underline{Q}}^t = \tilde{\underline{Q}} \underline{B}^2 \tilde{\underline{Q}}^t$.

Let

$$c_i = (\tilde{\underline{Q}}^t \underline{a} \cdot \underline{v}_i)^2 - (\underline{b} \cdot \underline{v}_i)^2. \quad (9.56)$$

Then the three equations expressing the equality of I_4, I_5 and I_6 in (9.55) may be summarized in the matrix form

$$\begin{pmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{pmatrix} \begin{Bmatrix} c_1 \\ c_2 \\ c_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}, \quad (9.57)$$

where λ_i are the principal values of $\underline{A}$ and $\underline{B}$.

The three rows deserve to be written out, since each is a small computation and the pattern of powers is what makes the system invertible. Throughout, $\underline{c} = \tilde{\underline{Q}}^t \underline{a}$, and $\{\underline{v}_i\}$ is an orthonormal basis, so that $|\underline{w}|^2 = \sum_i (\underline{w} \cdot \underline{v}_i)^2$ for every $\underline{w}$.

The first row comes from I_6 . Equality of $|\underline{a}|$ and $|\underline{b}|$ gives $|\underline{a}|^2 = |\underline{b}|^2$, and $|\underline{c}| = |\tilde{\underline{Q}}^t \underline{a}| = |\underline{a}|$ because $\tilde{\underline{Q}}$ is orthogonal and orthogonal tensors preserve norms (Proposition 1.48). Hence

$$\sum_i (\underline{c} \cdot \underline{v}_i)^2 = |\underline{c}|^2 = |\underline{a}|^2 = |\underline{b}|^2 = \sum_i (\underline{b} \cdot \underline{v}_i)^2, \quad \text{i.e.} \quad \sum_i c_i = 0.$$

The second row comes from I_4 . Using $\underline{A} = \tilde{\underline{Q}} \underline{B} \tilde{\underline{Q}}^t$ and the definition of the transpose,

$$I_4(\underline{A}, \underline{a}) = \underline{a} \cdot \underline{A} \underline{a} = \underline{a} \cdot \tilde{\underline{Q}} \underline{B} \tilde{\underline{Q}}^t \underline{a} = \tilde{\underline{Q}}^t \underline{a} \cdot \underline{B} \tilde{\underline{Q}}^t \underline{a} = \underline{c} \cdot \underline{B} \underline{c} = \sum_i \lambda_i (\underline{c} \cdot \underline{v}_i)^2,$$

the last step by the spectral form of $\underline{B}$ in (9.26). Likewise $I_4(\underline{B}, \underline{b}) = \sum_i \lambda_i (\underline{b} \cdot \underline{v}_i)^2$, so equality of the two gives $\sum_i \lambda_i c_i = 0$.

The third row comes from I_5 in the same way, with $\underline{A}^2 = \tilde{Q} \underline{B}^2 \tilde{Q}^t$ in place of $\underline{A}$ and $\underline{B}^2 = \sum_i \lambda_i^2 \underline{v}_i \otimes \underline{v}_i$ in place of $\underline{B}$: $\sum_i \lambda_i^2 c_i = 0$.

As we are concerned with tensors having distinct principal values, the determinant of the coefficient matrix, $(\lambda_1 - \lambda_2)(\lambda_2 - \lambda_3)(\lambda_3 - \lambda_1)$, is nonzero, and therefore all the c_i vanish, i.e.,

$$\tilde{Q}^t \underline{a} \cdot \underline{v}_i = \pm \underline{b} \cdot \underline{v}_i; \quad i = 1, 2, 3.$$

The matrix of (9.57) is the transpose of the one in (9.35), and transposition does not change a determinant, so the evaluation performed there serves here unchanged.

REMARK 9.20 (a misprint in the printed (9.57)). The book prints the last row of the matrix in (9.57) as $(\lambda_1^2, \lambda_2^2, \lambda_2^2)$. The third entry must be λ_3^2 : the row comes from I_5 , in which each principal value enters squared, as the computation of the third row above shows. The determinant the book quotes next, $(\lambda_1 - \lambda_2)(\lambda_2 - \lambda_3)(\lambda_3 - \lambda_1)$, belongs to the corrected matrix. With the printed row the determinant would instead be

$$\lambda_2^2(\lambda_2 - \lambda_3) + \lambda_1^2(\lambda_3 - \lambda_2) = -(\lambda_1 - \lambda_2)(\lambda_2 - \lambda_3)(\lambda_1 + \lambda_2),$$

expanding along the first row, and it vanishes whenever $\lambda_1 + \lambda_2 = 0$, for instance for the distinct values $(1, -1, 2)$; the conclusion that every c_i vanishes would then fail.

REMARK 9.21 (a step in the book's proof that needs one more line). At this point the book concludes " $\underline{a} = \pm \tilde{Q} \underline{b}$ " and finishes. The conclusion is true but does not follow from what has been proved, and the gap is worth repairing rather than stepping over, because the repair is short and because the naive reading is genuinely false.

What the vanishing of the c_i gives is three sign ambiguities, one for each i , and nothing says they are the same sign. They need not be. Take $\underline{A} = \underline{B}$ diagonal on $\{\underline{v}_i\} = \{\underline{e}_i\}$ with distinct λ_i , so that $\tilde{Q} = \underline{I}$ is an admissible choice, and take $\underline{a} = \underline{e}_1 + \underline{e}_2$, $\underline{b} = \underline{e}_1 - \underline{e}_2$. All six invariants

agree, since I_6 sees $|a| = |b| = \sqrt{2}$ and I_4, I_5 see $\lambda_1 + \lambda_2$ and $\lambda_1^2 + \lambda_2^2$ in both cases. But $\underline{b} \neq \pm \underline{a}$, and $\underline{a} \otimes \underline{a} \neq \underline{b} \otimes \underline{b}$.

The repair is to absorb the three signs into an orthogonal tensor. Fix $s_i \in \{+1, -1\}$ with $\tilde{Q}^t \underline{a} \cdot \underline{v}_i = s_i \underline{b} \cdot \underline{v}_i$, taking $s_i = +1$ when both sides vanish, and set

$$\underline{S} = \sum_i s_i \underline{v}_i \otimes \underline{v}_i.$$

Then $\underline{S}$ is orthogonal, since $\underline{S}^t \underline{S} = \sum_i s_i^2 \underline{v}_i \otimes \underline{v}_i = \sum_i \underline{v}_i \otimes \underline{v}_i = \underline{I}$; and $\underline{S}$ commutes with $\underline{B}$, indeed $\underline{S} \underline{B} \underline{S}^t = \sum_i s_i^2 \lambda_i \underline{v}_i \otimes \underline{v}_i = \underline{B}$, because both are diagonal on the same basis. Moreover

$$\tilde{Q}^t \underline{a} = \sum_i (\tilde{Q}^t \underline{a} \cdot \underline{v}_i) \underline{v}_i = \sum_i s_i (\underline{b} \cdot \underline{v}_i) \underline{v}_i = \underline{S} \underline{b}, \quad \text{so} \quad \underline{a} = \tilde{Q} \underline{S} \underline{b}.$$

Put $\hat{Q} = \tilde{Q} \underline{S}$, orthogonal as a product of orthogonal tensors. Then $\hat{Q} \underline{b} = \underline{a}$ and

$$\hat{Q} \underline{B} \hat{Q}^t = \tilde{Q} \underline{S} \underline{B} \underline{S}^t \tilde{Q}^t = \tilde{Q} \underline{B} \tilde{Q}^t = \underline{A}.$$

So the pair $(\underline{B}, \underline{b})$ is carried onto the pair $(\underline{A}, \underline{a})$ by a single orthogonal tensor, which is all the argument needs. In the counterexample above, $\hat{Q} = \text{diag}(1, -1, 1)$ does the job: it fixes $\underline{A}$ and sends $\underline{b}$ to $\underline{a}$.

The book's version is recoverable in the special case where all three signs happen to agree, and the general case differs only in that the sign pattern is itself a symmetry of $\underline{B}$ and may be absorbed. Nothing in the conclusion changes.

With $\hat{Q}$ so constructed, $\underline{a} \otimes \underline{a} = \hat{Q} \underline{b} \otimes \hat{Q} \underline{b}$, and the equality of the two sets of scalars I_1 to I_6 has been shown to imply that the two pairs are related by an orthogonal transformation. Finally, as $\varphi(\underline{B}, \underline{b} \otimes \underline{b}) = \varphi(\hat{Q} \underline{B} \hat{Q}^t, \hat{Q} \underline{b} \otimes \hat{Q} \underline{b})$ by hypothesis, we have $\varphi(\underline{A}, \underline{a} \otimes \underline{a}) = \varphi(\underline{B}, \underline{b} \otimes \underline{b})$, as claimed.

Collecting these results and reinstating the passive variables, we conclude that the heat flux is given by (9.41) and (9.42). This subsumes the cases in which the stretching tensor has only one or two distinct principal values. Conversely, it is a straightforward matter to show that

(9.41) and (9.42) yield (9.40) for all orthogonal $\underline{Q}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$, and hence that the two statements are equivalent; Problem 3 does it.

Further, assuming the ϕ 's in (9.42) to be continuous functions of the I_k , $k = 1, \dots, 6$, it follows from the continuity of this representation with respect to $\text{grad } \theta$ that

$$\underline{g}(\theta, \rho, \underline{D}, \underline{0}; p) = \underline{0}, \quad (9.58)$$

so that there is no heat conduction in the absence of a temperature gradient; and, from its continuity with respect to $\underline{D}$, that

$$\underline{g}(\theta, \rho, \underline{Q}, \text{grad } \theta; p) = K(I_6, \theta, \rho; p) \text{grad } \theta, \quad (9.59)$$

for some scalar function K .

Both statements are read off (9.41) and (9.42). For (9.58), setting $\text{grad } \theta = \underline{0}$ makes the explicit factor $\text{grad } \theta$ in (9.41) vanish, and the tensor $\underline{K}$ multiplying it stays bounded because the ϕ 's are continuous and I_4, I_5, I_6 all tend to zero with $\text{grad } \theta$; a bounded tensor applied to the zero vector is the zero vector. For (9.59), setting $\underline{D} = \underline{Q}$ kills the second and third terms of (9.42) outright and sends I_1 through I_5 to zero, leaving $\underline{K} = \phi_0(0, \dots, 0, I_6, \theta, \rho; p) \underline{I}$, a multiple of the identity; calling that multiple K gives (9.59). Notice that $I_6 = |\text{grad } \theta|^2$ survives, so the conductivity of a fluid at rest may still depend on the magnitude, though not on the direction, of the temperature gradient.

The Newtonian fluid and the Fourier law

Empirical facts indicate that fluids of the present class, such as air and water, are well described by constitutive equations for the stress and heat flux that are linear in $\underline{D}$ and $\text{grad } \theta$, respectively, and that the heat flux is insensitive to $\underline{D}$. Recalling that I_2 and I_3 are nonlinear functions of $\underline{D}$, whereas $I_1 = \text{tr } \underline{D}$ is linear, the relevant expression for the stress is given by the *Newtonian fluid*

$$\underline{T} = \left[-\tilde{p}(\theta, \rho; p) + \lambda(\theta, \rho; p) \text{tr } \underline{D} \right] \underline{I} + 2\mu(\theta, \rho; p) \underline{D}, \quad (9.60)$$

where $\tilde{p}$ is the pressure in the absence of flow, and λ , μ are viscosity coefficients; and that for the heat flux, by the *Fourier law*

$$\underline{q} = K(\theta, \rho; p) \text{grad } \theta, \quad (9.61)$$

where K is the thermal conductivity.

The passage from (9.37) to (9.60) is a linearisation and it is worth carrying out, because it is where the two viscosities are born and because the reader should see exactly which terms were discarded.

The three invariants scale differently with $\underline{D}$. Replacing $\underline{D}$ by $\varepsilon \underline{D}$ for a small number ε and using (9.25),

$$I_1(\varepsilon \underline{D}) = \varepsilon I_1(\underline{D}), \quad I_2(\varepsilon \underline{D}) = \varepsilon^2 I_2(\underline{D}), \quad I_3(\varepsilon \underline{D}) = \varepsilon^3 I_3(\underline{D}),$$

the second because both I_1^2 and $\text{tr}(\underline{D}^2)$ are quadratic, the third by Problem 21(a) of Chapter 1 again. So to first order in ε only I_1 is active, and the Taylor expansions of the three coefficient functions about $\underline{D} = \underline{Q}$ read

$$\begin{aligned} \varphi_0(I_1, I_2, I_3, \theta, \rho; p) &= \varphi_0(0, 0, 0, \theta, \rho; p) + \left. \frac{\partial \varphi_0}{\partial I_1} \right|_{(0,0,0)} I_1 + o(\varepsilon) \\ &= -\tilde{p}(\theta, \rho; p) + \lambda(\theta, \rho; p) \text{tr } \underline{D} + o(\varepsilon), \\ \varphi_1(I_1, I_2, I_3, \theta, \rho; p) &= \varphi_1(0, 0, 0, \theta, \rho; p) + O(\varepsilon) = 2\mu(\theta, \rho; p) + O(\varepsilon), \end{aligned}$$

in which the two viscosities are defined by

$$\lambda(\theta, \rho; p) = \left. \frac{\partial \varphi_0}{\partial I_1} \right|_{\underline{D}=\underline{Q}}, \quad 2\mu(\theta, \rho; p) = \varphi_1|_{\underline{D}=\underline{Q}},$$

and $\tilde{p}$ is the pressure already named at (9.38). The third term of (9.37) contributes $\varphi_2 \underline{D}^2$, which is $O(\varepsilon^2)$ whatever φ_2 does, and the correction to φ_1 multiplies $\underline{D}$ and so is $O(\varepsilon^2)$ as well. Dropping everything of second order and higher leaves (9.60) exactly.

So the two viscosities are not independent postulates. They are the two numbers left when the general response (9.37) is differentiated once with respect to the stretching at the state of rest: λ measures how the isotropic part of the stress responds to a rate of volume change, and

2μ is the value of φ_1 at rest, measuring how the stress responds to the stretching itself. Everything the material might do at second order in $\underline{D}$ has been discarded, and a fluid for which that is a bad approximation is called non-Newtonian.

The reduction of (9.41) to (9.61) is the same operation applied to $\underline{K}$: set $\phi_1 = \phi_2 = 0$, which is the statement that the heat flux is insensitive to $\underline{D}$, and evaluate ϕ_0 at $I_6 = 0$, which discards the dependence on the magnitude of the temperature gradient allowed by (9.59). What remains is a scalar conductivity $K(\theta, \rho; p)$ and the linear law (9.61).

REMARK 9.22 (the sign convention in (9.61)). Fourier's law is more often written $\underline{q} = -k \text{grad } \theta$ with $k \geq 0$, and the reader should be warned that (9.61) has no minus sign in front, so that $K = -k$ and K will turn out in (9.80) to be non-positive. The convention is forced by the definition of $\underline{q}$ in Chapter 6 and not chosen for convenience.

Recall (6.166): the heating influx through a surface with unit normal $\underline{n}$ is $h = -\underline{q} \cdot \underline{n}$, the minus sign being inserted so that $\underline{q} \cdot \underline{n} > 0$ means heat flowing out through that surface. Heat runs from hot to cold, that is, opposite to $\text{grad } \theta$, which points uphill in temperature by Remark 1.71. So $\underline{q}$ and $\text{grad } \theta$ point in opposite directions and the coefficient relating them is negative. Both conventions describe the same physics; the only danger is in mixing them, and the inequality (9.80) below is where the sign is settled by thermodynamics rather than by taste.

The Navier–Stokes equations

The Newtonian fluid is the point at which constitutive theory meets the balance laws, and the meeting produces the equations that carry the whole of classical fluid mechanics. Substitute (9.60) into the local balance of linear momentum (6.109), $\rho \underline{\dot{v}} = \rho \underline{b} + \text{div } \underline{T}$, taking λ and μ uniform in space so that they pass through the divergence.

The stress splits into three pieces and each is differentiated separately. For a scalar field φ ,

$$[\text{div}(\varphi \underline{I})]_i = (\varphi \delta_{ij})_{,j} = \varphi_{,i}, \quad \text{so} \quad \text{div}(\varphi \underline{I}) = \text{grad } \varphi,$$

which handles the first two pieces with $\varphi = -\tilde{p}$ and $\varphi = \lambda \operatorname{div} \underline{v}$, the latter using $\operatorname{tr} \underline{D} = \operatorname{tr} \underline{L} = \operatorname{div} \underline{v}$ from Proposition 1.92. For the third, with $D_{ij} = \frac{1}{2}(v_{i,j} + v_{j,i})$ from (4.7),

$$[\operatorname{div}(2\mu \underline{D})]_i = 2\mu D_{ij,j} = \mu(v_{i,jj} + v_{j,ij}) = \mu(\Delta v_i + (\operatorname{div} \underline{v})_{,i}),$$

where $\Delta = \partial_j \partial_j$ is the Laplacian and the second term was recognised by exchanging the order of the two derivatives, $v_{j,ij} = v_{j,ji} = (\operatorname{div} \underline{v})_{,i}$, which is legitimate for a twice continuously differentiable velocity field. Collecting,

$$\operatorname{div} \underline{T} = -\operatorname{grad} \tilde{p} + (\lambda + \mu) \operatorname{grad}(\operatorname{div} \underline{v}) + \mu \Delta \underline{v},$$

and the balance of linear momentum becomes

$$\rho[\underline{v}' + (\operatorname{grad} \underline{v})\underline{v}] = -\operatorname{grad} \tilde{p} + (\lambda + \mu) \operatorname{grad}(\operatorname{div} \underline{v}) + \mu \Delta \underline{v} + \rho \underline{b},$$

in which $\underline{\dot{v}} = \underline{v}' + (\operatorname{grad} \underline{v})\underline{v}$ is (2.21). These are the *Navier–Stokes equations* for a compressible Newtonian fluid. Together with conservation of mass (6.14), an equation of state giving $\tilde{p}$ in terms of θ and ρ , and the energy balance (6.168) with Fourier’s law in it, they close the system.

When the fluid is treated as incompressible, $\operatorname{div} \underline{v} = 0$ deletes the middle term and $\tilde{p}$ is replaced by the reaction pressure p of (9.83) below:

$$\rho[\underline{v}' + (\operatorname{grad} \underline{v})\underline{v}] = -\operatorname{grad} p + \mu \Delta \underline{v} + \rho \underline{b}, \quad \operatorname{div} \underline{v} = 0.$$

This is the form in which the equations are almost always met, and every one of Problems 5 to 12 is an exercise in solving it.

REMARK 9.23 (what was postulated and what was derived). It is worth being explicit about the status of each ingredient, because the Navier–Stokes equations are usually introduced as a physical hypothesis and they are not one.

Derived, and not assumed, are these: that the stress depends on the motion only through $\underline{D}$; that it is isotropic in $\underline{D}$; that it is therefore of the form (9.37); that the heat flux is a tensor times $\operatorname{grad} \theta$; and, below, that $\mu \geq 0$, $\kappa \geq 0$ and $K \leq 0$. All of it follows from fluidity, from the

requirement that two observers describe the same material, and from the second law.

Assumed are two things only: that the response is local and of first order, which was the constitutive hypothesis of Chapter 8; and that the dependence on $\underline{D}$ and on $\text{grad } \theta$ may be truncated at first order, which is the step from (9.37) to (9.60).

The second assumption is the only one with a free parameter in it, and it is the one that fails first in practice. Polymer solutions, blood and molten plastics are all fluids by the definition of Chapter 8, and all of them satisfy (9.37); what they do not satisfy is linearity, and the models used for them keep φ_2 , or allow the φ 's to depend on the invariants, or abandon the locality assumption altogether in favour of a memory functional, as the discussion of viscoelasticity in Section 8.1 described.

Regarding the constitutive functions for the Helmholtz energy and the entropy, the only consequences of frame invariance and symmetry under the unimodular group that are required for our purposes are that these functions are determined by the list $\{\theta, \rho, \underline{D}, \text{grad } \theta\}$. We write these simply as $\psi(\theta, \rho, \underline{D}, \text{grad } \theta; p)$ and $\eta(\theta, \rho, \underline{D}, \text{grad } \theta; p)$, respectively.

REMARK 9.24 (ψ has been used for two things). The coefficient functions in the proofs of Section 9.1.1 and of this section were called ψ_0 , ψ_1 and ψ_2 , following the book, and the free energy is also called ψ . The two have nothing to do with each other. The ψ_i appeared only inside the proofs, where the passive variables were suppressed, and they were renamed φ_i and ϕ_i the moment the proofs ended. From (9.62) onward ψ always means the Helmholtz free energy per unit mass of (6.186), and it never carries a subscript.

9.3 Thermodynamical considerations

The stress, energy and entropy must be related if the balance laws and the Clausius–Duhem inequality are to be satisfied. To see this we invoke

the latter inequality, (6.187),

$$-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{T} \cdot \underline{L} - \theta^{-1} \underline{q} \cdot \text{grad } \theta \geq 0,$$

at the material point $p \in B$ for a process consisting of arbitrary values of θ , ρ , $\underline{D}$ and $\text{grad } \theta$. In principle the balance laws for energy and momentum, together with the constitutive equations, furnish the energy supply r and body force $\underline{b}$ required to effect this process. Here, we note that θ and $\text{grad } \theta$ can be specified independently at any one point.

REMARK 9.25 (the licence that makes the whole argument work). Everything in this section rests on the claim that the fields entering the inequality may be assigned at will, and that claim deserves scrutiny because it looks like cheating: surely a fluid cannot be made to do anything one likes.

It can, and the reason is in the sentence above. The Clausius–Duhem inequality is required to hold in every admissible process, and a process is admissible if it satisfies the balance laws. Now the balance of energy (6.168) contains the heat supply r , which is at our disposal: whatever ϑ and θ we prescribe, we may solve (6.168) for the r that makes the energy balance hold. Likewise the balance of linear momentum (6.109) contains the body force $\underline{b}$, and we may solve it for the $\underline{b}$ that makes the momentum balance hold. So no choice of fields can be ruled inadmissible on the grounds of violating a balance law; the balance laws merely say what supply and what force would be needed.

This is the Coleman–Noll argument, sketched in general at the end of Chapter 8 and carried out in full here. It is worth noticing what it would cost to reject it. If one insisted that r and $\underline{b}$ be given in advance, the set of admissible processes would shrink, the inequality would be demanded of fewer of them, and fewer restrictions would follow. The conclusions of this section are therefore the strongest ones available, and they are strongest precisely because the supplies are free.

The last sentence before this remark makes the point that carries the most weight below: θ and $\text{grad } \theta$ are independent at a point. A field and its gradient at one place constrain each other not at all, since one

may add to θ any function vanishing at the point with arbitrary gradient there.

Proceeding, we use the chain rule to obtain

$$\dot{\psi} = \psi_{\theta}\dot{\theta} + \psi_{\rho}\dot{\rho} + \psi_{\underline{D}} \cdot \underline{\dot{D}} + \psi_{\text{grad } \theta} \cdot (\text{grad } \theta)^{\cdot}, \quad (9.62)$$

where subscripts are used to denote derivatives with respect to scalar, tensor or vector variables. So ψ_{θ} and ψ_{ρ} are ordinary partial derivatives, $\psi_{\underline{D}}$ is the symmetric tensor representing the derivative of ψ with respect to $\underline{D}$ in the tensor inner product of Definition 1.57, and $\psi_{\text{grad } \theta}$ is the vector representing the derivative with respect to $\text{grad } \theta$ in the inner product of $\mathbb{E}^3$. Both representations exist and are unique by Theorem 1.15, applied in $\mathbb{E}^3$ and in the nine-dimensional space Lin respectively; this is the same double construction that produced the gradient in Chapter 1.

Invoking (6.17)₁ in the form $\dot{\rho} = -\rho \text{tr } \underline{D} = -\rho \underline{I} \cdot \underline{D}$ and recalling that $\underline{T}$ is symmetric, we thus reduce the Clausius–Duhem inequality to the form

$$\begin{aligned} & -\rho(\psi_{\theta} + \eta)\dot{\theta} - \rho \psi_{\underline{D}} \cdot \underline{\dot{D}} - \rho \psi_{\text{grad } \theta} \cdot (\text{grad } \theta)^{\cdot} \\ & + (\underline{T} + \rho^2 \psi_{\rho} \underline{I}) \cdot \underline{D} - \theta^{-1} \underline{q} \cdot \text{grad } \theta \geq 0. \end{aligned} \quad (9.63)$$

Three substitutions were made and each is worth isolating.

The first is the density rate. Conservation of mass in the form (6.17) is $\dot{\rho} + \rho \text{div } \underline{v} = 0$, and $\text{div } \underline{v} = \text{tr } \underline{L} = \text{tr } \underline{D}$ because the trace of a skew tensor vanishes. So $\dot{\rho} = -\rho \text{tr } \underline{D}$, and $\text{tr } \underline{D}$ is $\underline{I} \cdot \underline{D}$ by Theorem 1.58, since $\underline{I} \cdot \underline{D} = \delta_{ij} D_{ij} = D_{ii}$. The term $-\rho \psi_{\rho} \dot{\rho}$ of (9.62), carried into the inequality with its factor $-\rho$, therefore becomes

$$-\rho \psi_{\rho} \dot{\rho} = -\rho \psi_{\rho} (-\rho \underline{I} \cdot \underline{D}) = (\rho^2 \psi_{\rho} \underline{I}) \cdot \underline{D},$$

which is where the second half of the bracket $\underline{T} + \rho^2 \psi_{\rho} \underline{I}$ comes from.

The second is the stress power, $\underline{T} \cdot \underline{L} = \underline{T} \cdot (\underline{D} + \underline{W}) = \underline{T} \cdot \underline{D} + \underline{T} \cdot \underline{W} = \underline{T} \cdot \underline{D}$, the last term vanishing because $\underline{T}$ is symmetric and $\underline{W}$ skew (Lemma 2.36). So the stress does no work on the spin, which is the same fact that made a rigid body dissipate nothing.

The third is only bookkeeping. The remaining terms of (9.62) are carried across unchanged, each multiplied by $-\rho$, and the term $-\rho\eta\dot{\theta}$ of (6.187) is collected with $-\rho\psi_\theta\dot{\theta}$ to give the first bracket.

At any fixed instant, t_* say, the values of θ , $\underline{D}$ and $\text{grad } \theta$ and those of their material derivatives $\dot{\theta}$, $\dot{\underline{D}}$ and $(\text{grad } \theta)^\cdot$ can be assigned independently. Accordingly this inequality is of the form

$$\underline{a}(\underline{z}) \cdot \underline{w} + b(\underline{z}) \geq 0 \quad (9.64)$$

for arbitrary, independent values of the arrays

$$\underline{z} = (\theta, \rho, \underline{D}, \text{grad } \theta) \quad \text{and} \quad \underline{w} = (\dot{\theta}, \dot{\underline{D}}, (\text{grad } \theta)^\cdot), \quad (9.65)$$

where

$$\underline{a}(\underline{z}) = (-\rho(\psi_\theta + \eta), -\rho\psi_{\underline{D}}, -\rho\psi_{\text{grad } \theta}) \quad (9.66)$$

and

$$b(\underline{z}) = (\underline{T} + \rho^2\psi_\rho \underline{I}) \cdot \underline{D} - \theta^{-1}q \cdot \text{grad } \theta. \quad (9.67)$$

The arrays $\underline{a}$ and $\underline{w}$ each carry $1 + 6 + 3 = 10$ independent numbers, a scalar together with a symmetric tensor and a vector, and $\underline{a} \cdot \underline{w}$ means the sum of the three separate inner products, one in $\mathbb{R}$, one in the space of symmetric tensors and one in $\mathbb{E}^3$. That is a genuine inner product on the ten-dimensional space of such arrays, so $|\underline{a}|^2 = \underline{a} \cdot \underline{a}$ vanishes only when all three entries do.

To derive a necessary condition for (9.64), we choose $\underline{w} = \delta \underline{a}(\underline{z})$ with $\delta \in \mathbb{R}$ arbitrary. Then $\delta |\underline{a}|^2 + b \geq 0$. If $|\underline{a}| \neq 0$, then this is violated for $\delta < -b/|\underline{a}|^2$, contrary to the requirement that it be satisfied for all δ . Hence the necessary conditions

$$\underline{a}(\underline{z}) = (0, \underline{Q}, 0) \quad \text{and} \quad b(\underline{z}) \geq 0 \quad (9.68)$$

for (9.64), which are obviously also sufficient. The choice of $\underline{w}$ is legitimate exactly because $\underline{w}$ and $\underline{z}$ are independent: $\underline{a}$ is a function of $\underline{z}$ alone, so once $\underline{z}$ is fixed the vector $\underline{a}(\underline{z})$ is a fixed array and $\delta \underline{a}(\underline{z})$ is an admissible value of $\underline{w}$ for every real δ . Sufficiency is immediate: if $\underline{a} = 0$ then (9.64) reads $b \geq 0$ whatever $\underline{w}$ is. Panel a of Figure E draws

the argument: along the ray the whole inequality becomes a statement about one straight line, and a straight line with non-zero slope cannot stay on one side of zero.

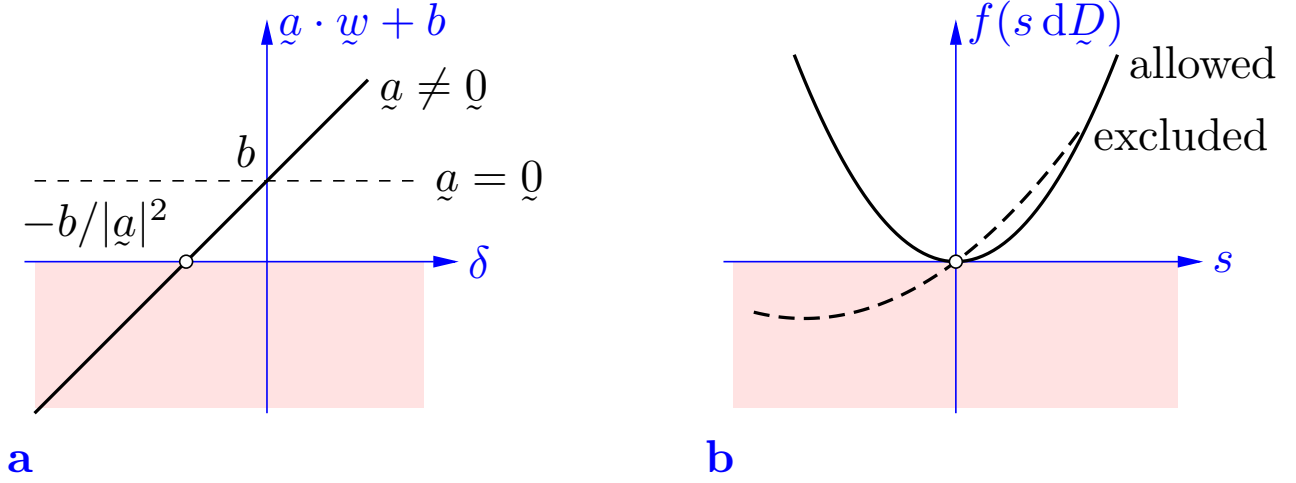


Figure E: The two uses of “for all” in Section 9.3. a: along the ray $w = \delta \underline{a}(\underline{z})$ the left side of (9.64) is an affine function of the single number δ with slope $|\underline{a}|^2$. If $\underline{a} \neq \underline{0}$ the line crosses zero at $\delta = -b/|\underline{a}|^2$ and beyond it enters the shaded half-plane that (9.64) forbids; only the level line of $\underline{a} = \underline{0}$ stays out of it for every δ , which is (9.68). b: along the ray $\underline{D} = s \, d\underline{D}$ the function f of (9.71) vanishes at $s = 0$ and may not go negative. A non-zero slope at $s = 0$ would carry it into the shaded half-plane on one side, as the dashed curve shows, so the slope, which is the left side of (9.72), must vanish.

We conclude that $\psi_{\underline{D}} = \underline{Q}$ and $\psi_{\text{grad } \theta} = \underline{0}$, and hence that the free energy is a function $\psi(\theta, \rho; p)$ of θ and ρ , and that

$$\eta = -\psi_{\theta}, \quad (9.69)$$

so that the entropy is also a function of θ and ρ . This leaves the residual dissipation inequality

$$(\underline{T} + \rho^2 \psi_{\rho} \underline{I}) \cdot \underline{D} - \theta^{-1} \underline{q} \cdot \text{grad } \theta \geq 0, \quad (9.70)$$

restricting the stress and heat flux jointly, for all values of θ , ρ , $\underline{D}$ and $\text{grad } \theta$. Moreover, as the material point p and the instant t_* are arbitrary, these conclusions hold at all material points and at all instants.

REMARK 9.26 (three results, and none of them assumed). Equation (9.68) has just delivered three statements of very different character from a single inequality, and they are worth separating.

The first is that the free energy is not viscous: $\psi_{\underline{D}} = \underline{Q}$ says that the stretching does not enter the stored energy at all. A fluid stores energy by being compressed, not by being sheared, and no matter how violently it is being deformed its free energy is the same as that of the same fluid at rest at the same θ and ρ . Viscosity is confined to the stress, where it dissipates rather than stores.

The second is that the entropy is not an independent constitutive function: equation (9.69) determines η once ψ is known. Of the four members of the response set $C = \{\underline{T}, \underline{q}, \psi, \eta\}$ of (8.4), one has just been eliminated, and (9.74) below will show that part of a second is determined by ψ as well.

The third is that heat conduction cannot be stored either: $\psi_{\text{grad } \theta} = 0$ removes the temperature gradient from the energy, which is the statement that a body does not become able to do work merely by being unevenly heated.

The pattern is the one flagged in Chapter 8: a term linear in a quantity that may be given any value, with a coefficient that cannot depend on it, must have vanishing coefficient. Here the free quantities are the rates $\dot{\theta}$, $\underline{\dot{D}}$ and $(\text{grad } \theta)^\cdot$, and the reason they are free is that ψ was allowed to depend on $\theta, \rho, \underline{D}, \text{grad } \theta$ but on none of their rates.

The thermodynamic pressure

Consider a process for which $\text{grad } \theta$ vanishes at a particular point and instant, and let

$$f(\underline{D}) = (\underline{T} + \rho^2 \psi_{\rho} \underline{I}) \cdot \underline{D} \quad (9.71)$$

in which θ and ρ are fixed. Then $f(\underline{Q}) = 0$ and $f(\underline{D}) \geq 0$ for all symmetric $\underline{D}$, so that f is minimized, and hence stationary, at $\underline{D} = \underline{Q}$, i.e.,

$$(\underline{T}|_{\underline{D}=\underline{Q}} + \rho^2 \psi_{\rho} \underline{I}) \cdot d\underline{D} = 0 \quad \text{for all symmetric } d\underline{D}. \quad (9.72)$$

This implies that

$$\underline{T}|_{\underline{D}=\underline{Q}} = -\rho^2 \psi_\rho \underline{I} \quad (9.73)$$

and hence, from (9.38), that

$$\tilde{p}(\theta, \rho; p) = \rho^2 \psi_\rho. \quad (9.74)$$

This is called the *thermodynamic pressure*.

The three steps each deserve a line. That $f(\underline{Q}) = 0$ is immediate from (9.71), since every term carries a factor $\underline{D}$. That $f(\underline{D}) \geq 0$ is (9.70) with $\text{grad } \theta = \underline{0}$. A differentiable function that is non-negative everywhere and zero at a point attains a minimum there, so its derivative vanishes there, and the derivative of (9.71) in the direction of an arbitrary symmetric increment $d\underline{D}$ is what (9.72) writes down: $\underline{T}$ and ψ_ρ are held at their values at $\underline{D} = \underline{Q}$ because differentiating the product would produce a second term carrying the factor $\underline{D}$, which vanishes there. Differentiability is in fact more than is needed. Put $\underline{D} = s d\underline{D}$ with s a real number; then

$$f(s d\underline{D}) = s (\underline{T}|_{\underline{D}=s d\underline{D}} + \rho^2 \psi_\rho \underline{I}) \cdot d\underline{D},$$

and $f \geq 0$ says that the inner product on the right is non-negative for $s > 0$ and non-positive for $s < 0$. Letting $s \rightarrow 0$ from either side, continuity of the stress in $\underline{D}$ alone makes the limit both, hence zero, which is (9.72); panel b of Figure E shows the picture. Finally, (9.72) says that a fixed symmetric tensor is orthogonal to every symmetric tensor, hence orthogonal to itself, hence zero by the positive definiteness of the tensor inner product, Theorem 1.58. That the arbitrary increment had to be symmetric costs nothing here, because the tensor in the bracket is itself symmetric and could not have been detected by a skew increment anyway.

REMARK 9.27 (what (9.74) accomplishes). Before (9.74) the fluid had two independent scalar functions of (θ, ρ) describing its equilibrium: the free energy ψ , which is a thermodynamic object, and the pressure $\tilde{p}$, which was introduced in (9.38) as a purely mechanical one, the value of $-\varphi_0$ at zero stretching. Nothing had connected them.

Equation (9.74) connects them, and the connection is not an identity but a consequence of the second law. It says that the pressure a fluid at rest exerts on its container is ρ^2 times the rate at which its stored energy rises with density, which is the familiar statement that pressure is the resistance to compression measured energetically. Written per unit volume rather than per unit mass, with $v = 1/\rho$ the specific volume, the chain rule gives $\partial\psi/\partial v = \psi_\rho(d\rho/dv) = -\rho^2\psi_\rho$, so (9.74) reads $\tilde{p} = -\partial\psi/\partial v$: the pressure is minus the derivative of the free energy with respect to volume at fixed temperature. That is the definition of pressure in classical thermodynamics, recovered here from continuum mechanics.

Together with (9.69), $\eta = -\psi_\theta$, the whole equilibrium response of the fluid is now packed into the single function $\psi(\theta, \rho; p)$. Two derivatives of it give the entropy and the pressure, and nothing else about equilibrium remains to be measured.

The signs of the viscosities

Invoking (9.70) for the Newtonian fluid model (9.60), again with $\text{grad } \theta = 0$, yields

$$\lambda(\text{tr } \underline{\underline{D}})^2 + 2\mu \underline{\underline{D}} \cdot \underline{\underline{D}} \geq 0, \quad (9.75)$$

for all symmetric $\underline{\underline{D}}$. The reduction uses (9.74) to cancel two terms. With $\tilde{p} = \rho^2\psi_\rho$,

$$\underline{\underline{T}} + \rho^2\psi_\rho\underline{\underline{I}} = [-\tilde{p} + \lambda \text{tr } \underline{\underline{D}}]\underline{\underline{I}} + 2\mu\underline{\underline{D}} + \tilde{p}\underline{\underline{I}} = \lambda(\text{tr } \underline{\underline{D}})\underline{\underline{I}} + 2\mu\underline{\underline{D}},$$

so that the pressure disappears entirely from the dissipation, and

$$(\lambda(\text{tr } \underline{\underline{D}})\underline{\underline{I}} + 2\mu\underline{\underline{D}}) \cdot \underline{\underline{D}} = \lambda(\text{tr } \underline{\underline{D}})\underline{\underline{I}} \cdot \underline{\underline{D}} + 2\mu\underline{\underline{D}} \cdot \underline{\underline{D}} = \lambda(\text{tr } \underline{\underline{D}})^2 + 2\mu \underline{\underline{D}} \cdot \underline{\underline{D}},$$

using $\underline{\underline{I}} \cdot \underline{\underline{D}} = \text{tr } \underline{\underline{D}}$ once more.

To derive necessary conditions for (9.75) we invoke the orthogonal decomposition

$$\underline{\underline{D}} = \text{Dev } \underline{\underline{D}} + \frac{1}{3}(\text{tr } \underline{\underline{D}})\underline{\underline{I}}, \quad (9.76)$$

where $\text{Dev } \underline{\underline{D}}$ is the deviatoric part of $\underline{\underline{D}}$ (see Problem 15 of Chapter 1 and Definition 1.59). Our inequality becomes

$$\kappa(\text{tr } \underline{\underline{D}})^2 + 2\mu |\text{Dev } \underline{\underline{D}}|^2 \geq 0, \quad (9.77)$$

for all $\underline{\underline{D}}$, where

$$\kappa = \lambda + \frac{2}{3}\mu. \quad (9.78)$$

The passage from (9.75) to (9.77) is the whole point of introducing (9.76) and is worth doing slowly. The decomposition is orthogonal in the sense of Proposition 1.60: the spherical and deviatoric parts satisfy $\text{Sph } \underline{\underline{A}} \cdot \text{Dev } \underline{\underline{B}} = 0$, so Pythagoras applies and

$$\begin{aligned} |\underline{\underline{D}}|^2 &= |\text{Dev } \underline{\underline{D}}|^2 + \left| \frac{1}{3}(\text{tr } \underline{\underline{D}})\underline{\underline{I}} \right|^2 \\ &= |\text{Dev } \underline{\underline{D}}|^2 + \frac{1}{9}(\text{tr } \underline{\underline{D}})^2 |\underline{\underline{I}}|^2 = |\text{Dev } \underline{\underline{D}}|^2 + \frac{1}{3}(\text{tr } \underline{\underline{D}})^2, \end{aligned}$$

where $|\underline{\underline{I}}|^2 = \underline{\underline{I}} \cdot \underline{\underline{I}} = \delta_{ij}\delta_{ij} = 3$. Substituting into (9.75),

$$\lambda(\text{tr } \underline{\underline{D}})^2 + 2\mu \left[|\text{Dev } \underline{\underline{D}}|^2 + \frac{1}{3}(\text{tr } \underline{\underline{D}})^2 \right] = \left(\lambda + \frac{2}{3}\mu \right) (\text{tr } \underline{\underline{D}})^2 + 2\mu |\text{Dev } \underline{\underline{D}}|^2,$$

which is (9.77) with κ as defined. The gain is that the two surviving terms involve independent pieces of $\underline{\underline{D}}$: one may change $\text{tr } \underline{\underline{D}}$ without touching $\text{Dev } \underline{\underline{D}}$ and conversely, which is exactly what (9.75) did not permit, since $\underline{\underline{D}} \cdot \underline{\underline{D}}$ mixes them.

Consider two flows, the first with $\underline{\underline{D}} = \underline{\underline{I}}$ (so that $\text{Dev } \underline{\underline{D}} = \underline{\underline{Q}}$) and the second with $\underline{\underline{D}} = \frac{1}{2}(\underline{\underline{a}} \otimes \underline{\underline{b}} + \underline{\underline{b}} \otimes \underline{\underline{a}})$, where $\underline{\underline{a}} \cdot \underline{\underline{b}} = 0$, the latter yielding $\text{tr } \underline{\underline{D}} = 0$ and hence $\underline{\underline{D}} = \text{Dev } \underline{\underline{D}}$. These yield the necessary and sufficient conditions

$$\kappa \geq 0 \quad \text{and} \quad \mu \geq 0. \quad (9.79)$$

The two evaluations are short. For the first, $\text{tr } \underline{\underline{I}} = 3$ and $\text{Dev } \underline{\underline{I}} = \underline{\underline{I}} - \frac{1}{3} \cdot 3 \cdot \underline{\underline{I}} = \underline{\underline{Q}}$, so (9.77) reads $9\kappa \geq 0$, giving $\kappa \geq 0$. For the second, using $\text{tr}(\underline{\underline{a}} \otimes \underline{\underline{b}}) = \underline{\underline{a}} \cdot \underline{\underline{b}}$ from Theorem 1.51,

$$\text{tr } \underline{\underline{D}} = \frac{1}{2}(\underline{\underline{a}} \cdot \underline{\underline{b}} + \underline{\underline{b}} \cdot \underline{\underline{a}}) = 0,$$

so the first term of (9.77) drops out and $\text{Dev } \underline{\underline{D}} = \underline{\underline{D}}$. Its magnitude is computed with the dyad rule of Lemma 2.15 and $\underline{\underline{a}} \cdot \underline{\underline{b}} = 0$:

$$\begin{aligned} \underline{\underline{D}}^2 &= \frac{1}{4}(\underline{\underline{a}} \otimes \underline{\underline{b}} + \underline{\underline{b}} \otimes \underline{\underline{a}})(\underline{\underline{a}} \otimes \underline{\underline{b}} + \underline{\underline{b}} \otimes \underline{\underline{a}}) \\ &= \frac{1}{4}\left[(\underline{\underline{b}} \cdot \underline{\underline{a}}) \underline{\underline{a}} \otimes \underline{\underline{b}} + (\underline{\underline{b}} \cdot \underline{\underline{b}}) \underline{\underline{a}} \otimes \underline{\underline{a}} + (\underline{\underline{a}} \cdot \underline{\underline{a}}) \underline{\underline{b}} \otimes \underline{\underline{b}} + (\underline{\underline{a}} \cdot \underline{\underline{b}}) \underline{\underline{b}} \otimes \underline{\underline{a}}\right] \\ &= \frac{1}{4}\left[|\underline{\underline{b}}|^2 \underline{\underline{a}} \otimes \underline{\underline{a}} + |\underline{\underline{a}}|^2 \underline{\underline{b}} \otimes \underline{\underline{b}}\right], \end{aligned}$$

so that, taking the trace and using $\text{tr}(\underline{\underline{a}} \otimes \underline{\underline{a}}) = |\underline{\underline{a}}|^2$,

$$|\underline{\underline{D}}|^2 = \underline{\underline{D}} \cdot \underline{\underline{D}} = \text{tr}(\underline{\underline{D}} \underline{\underline{D}}^t) = \text{tr}(\underline{\underline{D}}^2) = \frac{1}{2}|\underline{\underline{a}}|^2|\underline{\underline{b}}|^2 > 0$$

for $\underline{\underline{a}}, \underline{\underline{b}} \neq \underline{\underline{0}}$. Hence (9.77) reads $\mu|\underline{\underline{a}}|^2|\underline{\underline{b}}|^2 \geq 0$ and $\mu \geq 0$ follows. Sufficiency is immediate: if $\kappa \geq 0$ and $\mu \geq 0$ then both terms of (9.77) are non-negative for every $\underline{\underline{D}}$, since $(\text{tr } \underline{\underline{D}})^2$ and $|\text{Dev } \underline{\underline{D}}|^2$ are squares.

Because the second form of $\underline{\underline{D}}$ represents a shear flow, we refer to μ as the *shear viscosity*, whereas κ is the *bulk viscosity*. Figure F shows the two test motions and the split of the stress that goes with them.

REMARK 9.28 (Stokes' condition). Written with (9.76) and (9.78), the Newtonian law (9.60) becomes

$$\underline{\underline{T}} = \left[-\tilde{p} + \kappa \text{tr } \underline{\underline{D}}\right]\underline{\underline{I}} + 2\mu \text{Dev } \underline{\underline{D}},$$

since $2\mu \underline{\underline{D}} = 2\mu \text{Dev } \underline{\underline{D}} + \frac{2}{3}\mu(\text{tr } \underline{\underline{D}})\underline{\underline{I}}$ and the second piece joins the spherical term. Taking the trace and dividing by three,

$$\frac{1}{3} \text{tr } \underline{\underline{T}} = -\tilde{p} + \kappa \text{tr } \underline{\underline{D}},$$

because $\text{tr}(\text{Dev } \underline{\underline{D}}) = 0$. The quantity $-\frac{1}{3} \text{tr } \underline{\underline{T}}$ is the *mean normal pressure*, the average over all orientations of the normal traction, and it is what a pressure gauge in a moving fluid reports. The display says that it agrees with the thermodynamic pressure $\tilde{p}$ of (9.74) only when $\kappa \text{tr } \underline{\underline{D}} = 0$.

Stokes' condition is the assumption $\kappa = 0$, equivalently $\lambda = -\frac{2}{3}\mu$, which makes the two pressures agree in every flow whatever. Three things should be said about it.

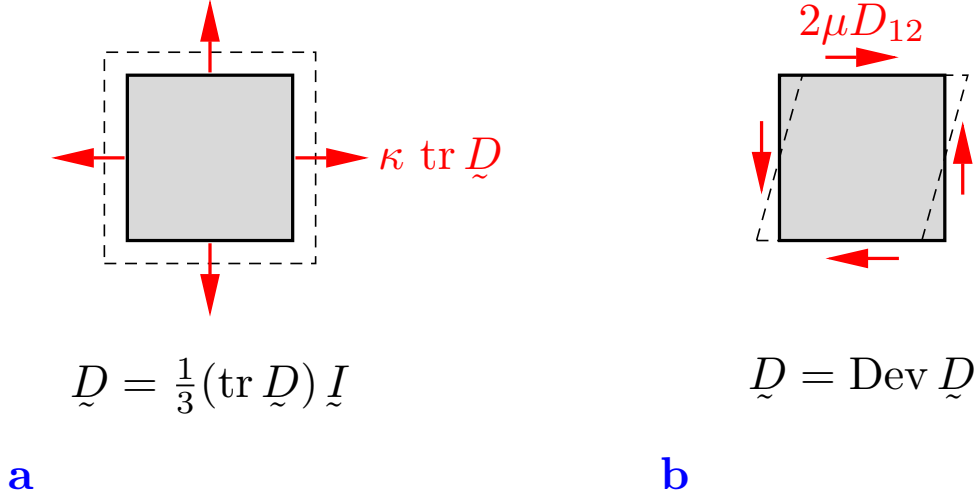


Figure F: The two viscosities, and the two motions that isolate them, drawn for a square element and, dashed, its shape a moment later (exaggerated). a: a pure dilatation, $\underline{\underline{D}} = \frac{1}{3}(\operatorname{tr} \underline{\underline{D}})\underline{\underline{I}}$, changes the volume and keeps the shape; on every face acts the viscous traction $\kappa(\operatorname{tr} \underline{\underline{D}})\underline{\underline{n}}$, normal and outward, pointing the way the face moves. b: a pure shear, $\underline{\underline{D}} = \operatorname{Dev} \underline{\underline{D}}$, keeps the volume and changes the shape; on the faces act the shear tractions $2\mu D_{12}$. The tractions drawn are those acting on the element, and in both panels they do positive work on its deformation; that work is the dissipation (9.77), $\kappa(\operatorname{tr} \underline{\underline{D}})^2$ in a and $2\mu|\operatorname{Dev} \underline{\underline{D}}|^2$ in b, so feeding (9.77) the first motion forces $\kappa \geq 0$ and the second $\mu \geq 0$. Written with (9.76) and (9.78), the Newtonian law (9.60) is $\underline{\underline{T}} = [-\tilde{p} + \kappa \operatorname{tr} \underline{\underline{D}}]\underline{\underline{I}} + 2\mu \operatorname{Dev} \underline{\underline{D}}$: an equilibrium pressure, a bulk viscous pressure proportional to the rate of volume change, and a shear stress proportional to the rate of change of shape.

It is admissible: (9.79)₁ requires $\kappa \geq 0$ and $\kappa = 0$ sits at the edge of that range, so nothing in the second law forbids it.

It is not forced: the same inequality permits any $\kappa > 0$, and measured bulk viscosities of polyatomic gases and of water are not zero. The bulk viscosity is the resistance a fluid offers to being compressed quickly, which arises when the energy fed into compression takes time to redistribute among the internal degrees of freedom of the molecules. A monatomic gas has no such degrees of freedom, and for it kinetic theory gives $\kappa = 0$ exactly.

It rarely matters: in an incompressible flow $\text{tr } \underline{D} = 0$ and the term $\kappa \text{tr } \underline{D}$ vanishes identically, so κ never appears. That is why Stokes' condition can be carried for a century in textbooks without doing much harm, and why it shows itself mainly in the absorption of sound, where compression is both rapid and unavoidable.

The sign of the conductivity

Invoking (9.70) for the classical Fourier law (9.61), with $\underline{D} = \underline{Q}$, we conclude that $K|\text{grad } \theta|^2 \leq 0$ and hence that the thermal conductivity is non-positive:

$$K \leq 0. \quad (9.80)$$

Setting $\underline{D} = \underline{Q}$ kills the first term of (9.70) entirely, leaving

$$-\theta^{-1} \underline{q} \cdot \text{grad } \theta = -\theta^{-1} K |\text{grad } \theta|^2 \geq 0.$$

The absolute temperature is positive, so $\theta^{-1} > 0$ and the inequality is equivalent to $K|\text{grad } \theta|^2 \leq 0$; since $|\text{grad } \theta|^2 > 0$ for any non-vanishing temperature gradient, and $\text{grad } \theta$ may be assigned freely, $K \leq 0$ follows.

In turn, these inequalities ensure that a model consisting of the Newtonian fluid and Fourier's law satisfy the Clausius–Duhem inequality without qualification, i.e., for all processes. The check is one line now that the pieces are assembled: substituting (9.60) and (9.61) into (9.70) and using (9.74), (9.77) and the computation above,

$$\underbrace{\kappa(\text{tr } \underline{D})^2}_{\geq 0} + \underbrace{2\mu|\text{Dev } \underline{D}|^2}_{\geq 0} + \underbrace{(-\theta^{-1}K)|\text{grad } \theta|^2}_{\geq 0} \geq 0,$$

a sum of three separately non-negative terms, for every $\underline{D}$ and every $\text{grad } \theta$. Sufficiency has therefore been established alongside necessity, and the three inequalities $\kappa \geq 0$, $\mu \geq 0$, $K \leq 0$ are exactly what compatibility with the second law amounts to for this model.

REMARK 9.29 (what the second law has and has not said). The three inequalities are the entire thermodynamic content of the Newtonian–Fourier model, and it is worth being precise about their force.

They are inequalities, not equations: thermodynamics tells us the sign of μ but not its size, and no argument of this chapter could determine that a particular fluid has $\mu = 10^{-3}$ rather than 10^{-5} in SI units. Measuring the magnitudes is what experiment is for; forbidding the wrong sign is what the second law is for.

Each has a plain physical reading. $\mu \geq 0$ says that shearing a fluid warms it and never cools it: a fluid with $\mu < 0$ would accelerate spontaneously, converting heat from its surroundings into ordered motion. $\kappa \geq 0$ says the same of compression. $K \leq 0$, which is $k = -K \geq 0$ in the usual convention, says heat runs from hot to cold.

And each was derived from one inequality, (6.187), by feeding it suitably chosen processes: a purely dilatational flow, a purely shearing flow, and a temperature gradient at rest. That is the same technique as everywhere else in this chapter and in Chapter 8. A statement required to hold for all inputs is tested against the inputs that make everything but one term vanish, and each such test yields one restriction. Remark 5.5 of Chapter 5 named the principle.

At this stage a remark is in order concerning our a priori omission of the temperature gradient from the list of independent variables in the constitutive function for the stress, which was made without justification at (8.33). If one does not omit the temperature gradient, then it is possible to show that the leading-order model emerging for small values of $|\underline{D}|$ and $|\text{grad } \theta|$ is precisely the Newtonian–Fourier model; and that, to satisfy the Clausius–Duhem inequality, the associated stress must be independent of the temperature gradient. The details of the fairly lengthy argument may be found in the reference cited by the book. So

the omission costs nothing: what was assumed for convenience turns out to be forced.

The incompressible Newtonian fluid

Many fluids are approximately incompressible under feasible conditions, including small deviations in temperature. Their motions are therefore approximately isochoric, satisfying $J = 1$ very nearly. It is thus appropriate to impose the constraint $\dot{J} = 0$ at a material point. This in turn is equivalent to $0 = \text{tr } \underline{\underline{D}} = \underline{\underline{I}} \cdot \underline{\underline{D}}$, by $\dot{J} = J \text{tr } \underline{\underline{L}}$ from Problem 2 of Chapter 4 together with $J > 0$ and $\text{tr } \underline{\underline{L}} = \text{tr } \underline{\underline{D}}$, reducing the dissipation inequality (9.70) to

$$\text{Dev } \underline{\underline{T}} \cdot \text{Dev } \underline{\underline{D}} - \theta^{-1} \underline{\underline{q}} \cdot \text{grad } \theta \geq 0, \quad (9.81)$$

where, for the incompressible Newtonian fluid,

$$\text{Dev } \underline{\underline{T}} = 2\mu \text{Dev } \underline{\underline{D}} = 2\mu \underline{\underline{D}}. \quad (9.82)$$

The reduction of the first term of (9.70) uses the constraint twice. First, $(\rho^2 \psi_\rho \underline{\underline{I}}) \cdot \underline{\underline{D}} = \rho^2 \psi_\rho \text{tr } \underline{\underline{D}} = 0$, so the pressure term drops out on its own. Second, splitting $\underline{\underline{T}}$ by (9.76),

$$\begin{aligned} \underline{\underline{T}} \cdot \underline{\underline{D}} &= \left(\text{Dev } \underline{\underline{T}} + \frac{1}{3}(\text{tr } \underline{\underline{T}}) \underline{\underline{I}} \right) \cdot \underline{\underline{D}} \\ &= \text{Dev } \underline{\underline{T}} \cdot \underline{\underline{D}} + \frac{1}{3}(\text{tr } \underline{\underline{T}}) \text{tr } \underline{\underline{D}} = \text{Dev } \underline{\underline{T}} \cdot \text{Dev } \underline{\underline{D}}, \end{aligned}$$

since $\text{tr } \underline{\underline{D}} = 0$ makes both the second term vanish and $\underline{\underline{D}} = \text{Dev } \underline{\underline{D}}$. Equation (9.82) is then (9.60) with $\text{tr } \underline{\underline{D}} = 0$: the term in λ is gone and $\text{Dev}(-\tilde{p} \underline{\underline{I}}) = \underline{\underline{Q}}$, leaving only $2\mu \underline{\underline{D}}$, which is already deviatoric.

The foregoing argument yields only (9.79)₂. That is not an oversight but a necessity: the flow that produced $\kappa \geq 0$ was $\underline{\underline{D}} = \underline{\underline{I}}$, which has $\text{tr } \underline{\underline{D}} = 3 \neq 0$ and is no longer an admissible motion. A constraint removes processes from the admissible set, and a smaller set yields fewer restrictions. The bulk viscosity of an incompressible fluid is not merely unrestricted, it is meaningless, since the motion that would reveal it cannot occur.

Moreover, conservation of mass, i.e., $(\rho J)^\cdot = 0$ from (6.17), then implies that $\dot{\rho} = 0$, and hence that ρ is fixed at a material point. It therefore enters the functions ψ , η and μ as a fixed parameter.

Thermodynamic considerations do not restrict the spherical part of the stress, which is thus constitutively indeterminate and hence outside the purview of our earlier discussion about observer consensus concerning material response. The stress is thus of the form

$$\underline{T} = -p(\underline{x}, t)\underline{I} + 2\mu\underline{D}, \quad (9.83)$$

where $p(\underline{x}, t)$ is entirely arbitrary. With $\underline{D} = \text{sym}(\text{grad } \underline{v})$, the linear momentum balance is a vector equation for the determination of $\underline{v}$, ρ and p , assuming the body force to be assigned. Conservation of mass provides the additional equation

$$\rho' + \text{grad } \rho \cdot \underline{v} = 0, \quad (9.84)$$

where $(\cdot)'$ is the time derivative in the spatial description, and the constraint furnishes the final equation

$$\text{div } \underline{v} = 0. \quad (9.85)$$

Thus, we have five equations for the five variables consisting of ρ , p and the components of $\underline{v}$. Clearly, this system would be overdetermined, and hence generally insoluble, if not for the presence of the additional unknown function p . Accordingly the latter takes whatever values that may be necessary to furnish a solution to the problem at hand.

REMARK 9.30 (why the pressure is not a constitutive quantity). This is the first appearance of a reactive stress and it is the theme of Chapter 10, so it is worth stating carefully what has happened.

Every stress met before (9.83) was delivered by a constitutive function: give the state, and the function returns the stress. The p of (9.83) is not of that kind. It is not returned by any function of $(\theta, \rho, \underline{D})$, and it cannot be, because the argument that would have determined it has been withdrawn. Look back at (9.72): the spherical part of $\underline{T}$ was pinned

by testing the dissipation against an increment $d\tilde{D}$ with $\text{tr } d\tilde{D} \neq 0$, and under the constraint no such increment exists. What determines p instead is the equations of motion together with the boundary data, which is to say the problem rather than the material.

The counting in the last paragraph is the practical form of the same statement. Before the constraint, five unknowns $(\rho, \underline{v})$ and five equations. Imposing $\text{div } \underline{v} = 0$ adds a sixth equation without adding an unknown, and the system would be overdetermined; the reactive pressure is the sixth unknown that restores the balance. A constraint and its reaction always arrive in pairs, and (9.85) and the p of (9.83) are one such pair.

Two consequences worth carrying forward. The pressure in an incompressible fluid can be changed by an arbitrary constant without changing the motion, since only $\text{grad } p$ enters the equation of motion, so p is determined only up to that constant by the field equations and is pinned by a boundary condition. And, despite being constitutively indeterminate, p is not observer dependent: Problem 4 shows that all observers agree on its value, which is what makes it a legitimate physical field rather than a bookkeeping device.

9.4 Problems

PROBLEM 1 (The two relative motions, verified). Verify (9.12) to (9.14).

Solution. All three statements were established in the text. The point of collecting them is to see that they come from one operation performed on (9.6), (9.7) and (9.8) in turn, and to record where the corrected (9.14) comes from. Each of (9.6) to (9.8) has the shape “something built from $\underline{Q}_1, \underline{c}_1$ equals the same thing built from $\underline{Q}_2, \underline{c}_2$ ”. In each case we solve for the object carrying the subscript 2, moving everything else to the other side and multiplying by $\underline{Q}_2^t$, which is legitimate because $\underline{Q}_2^t \underline{Q}_2 = \underline{I}$ on $\mathbb{E}^3$ (Proposition 1.48); the combination $\underline{Q}_2^t \underline{Q}_1$ that appears is the $\underline{Q}$ of (9.11).

(a) Position, the first line of (9.12). From (9.6), $\underline{Q}_2 \underline{x}_2 = \underline{Q}_1 \underline{x}_1 + \underline{c}_1 - \underline{c}_2$.

Multiplying on the left by $\underline{Q}_2^t$,

$$\underline{x}_2 = (\underline{Q}_2^t \underline{Q}_1) \underline{x}_1 + \underline{Q}_2^t (\underline{c}_1 - \underline{c}_2) = \underline{Q} \underline{x}_1 + \underline{d}(t), \quad \underline{d}(t) = \underline{Q}_2^t (\underline{c}_1 - \underline{c}_2),$$

and $\underline{d}$ depends on t alone, since $\underline{Q}_2$, $\underline{c}_1$ and $\underline{c}_2$ do.

(b) Velocity, the second line of (9.12). From (9.7),

$$\underline{Q}_2 \underline{v}_2 = \underline{Q}_1 \underline{v}_1 + \dot{\underline{Q}}_1 \underline{Q}_1^t (\underline{x}^+ - \underline{c}_1) - \dot{\underline{Q}}_2 \underline{Q}_2^t (\underline{x}^+ - \underline{c}_2) + \dot{\underline{c}}_1 - \dot{\underline{c}}_2,$$

and multiplying on the left by $\underline{Q}_2^t$ gives $\underline{v}_2 = \underline{Q} \underline{v}_1 + \underline{e}$ with

$$\underline{e} = \underline{Q}_2^t \left[\dot{\underline{Q}}_1 \underline{Q}_1^t (\underline{x}^+ - \underline{c}_1) - \dot{\underline{Q}}_2 \underline{Q}_2^t (\underline{x}^+ - \underline{c}_2) + \dot{\underline{c}}_1 - \dot{\underline{c}}_2 \right].$$

Every symbol on the right is a function of t except $\underline{x}^+$, so $\underline{e}$ is a function of $\underline{x}^+$ and t . The position $\underline{x}^+$ appears rather than $\underline{x}_1$ or $\underline{x}_2$ because both changes of frame were written from the standpoint of O^+ in (9.6), so that $\underline{x}^+$ is the one position the two descriptions share. The text showed that the same vector is $\underline{e} = \dot{\underline{Q}} \underline{x}_1 + \underline{d}$.

(c) Velocity gradient, (9.13). From (9.8), $\underline{Q}_2 \underline{L}_2 \underline{Q}_2^t = \underline{Q}_1 \underline{L}_1 \underline{Q}_1^t + \dot{\underline{Q}}_1 \underline{Q}_1^t - \dot{\underline{Q}}_2 \underline{Q}_2^t$. Multiplying on the left by $\underline{Q}_2^t$ and on the right by $\underline{Q}_2$, and cancelling $\underline{Q}_2^t \underline{Q}_2 = \underline{I}$ where it appears,

$$\underline{L}_2 = \underline{Q} \underline{L}_1 \underline{Q}^t + \underline{Q}_2^t \dot{\underline{Q}}_1 \underline{Q}_1^t \underline{Q}_2 - \underline{Q}_2^t \dot{\underline{Q}}_2.$$

The last two terms are $\dot{\underline{Q}} \underline{Q}^t$. By the product rule $\dot{\underline{Q}} = \dot{\underline{Q}}_2^t \underline{Q}_1 + \underline{Q}_2^t \dot{\underline{Q}}_1$, and $\underline{Q}^t = \underline{Q}_1^t \underline{Q}_2$, so

$$\dot{\underline{Q}} \underline{Q}^t = \dot{\underline{Q}}_2^t \underline{Q}_1 \underline{Q}_1^t \underline{Q}_2 + \underline{Q}_2^t \dot{\underline{Q}}_1 \underline{Q}_1^t \underline{Q}_2 = \dot{\underline{Q}}_2^t \underline{Q}_2 + \underline{Q}_2^t \dot{\underline{Q}}_1 \underline{Q}_1^t \underline{Q}_2,$$

using $\underline{Q}_1 \underline{Q}_1^t = \underline{I}^+$; and differentiating $\underline{Q}_2^t \underline{Q}_2 = \underline{I}$ gives $\dot{\underline{Q}}_2^t \underline{Q}_2 = -\underline{Q}_2^t \dot{\underline{Q}}_2$. Hence

$$\underline{L}_2 = \underline{Q} \underline{L}_1 \underline{Q}^t + \dot{\underline{Q}} \underline{Q}^t,$$

the chain-rule result of the text obtained by algebra alone; the book's own solution in Appendix E runs this way. The split into $\underline{D}_2 = \underline{Q} \underline{D}_1 \underline{Q}^t$ and $\underline{W}_2 = \underline{Q} \underline{W}_1 \underline{Q}^t + \dot{\underline{Q}} \underline{Q}^t$ follows from the skewness of $\dot{\underline{Q}} \underline{Q}^t$, as in the text.

(d) Equation (9.14). With the choice $\underline{Q} = \underline{I}$, which forces $\underline{Q}_2 = \underline{Q}_1$, the terms in $\underline{x}^+$ of part (b) cancel and, as computed in the text,

$$\underline{e}(t) = \underline{Q}_1^t(\dot{\underline{c}}_1 - \dot{\underline{c}}_2) + \dot{\underline{Q}}_1^t(\underline{c}_1 - \underline{c}_2) = \dot{\underline{d}}(t).$$

Remark 9.6 records that the printed (9.14) carries $\dot{\underline{Q}}_1 \underline{Q}_1^t$ where $\dot{\underline{Q}}_1^t$ belongs, and gives both the space-counting reason and the check $\underline{e} = \dot{\underline{d}}$.

The verification is not decoration. The whole of Section 9.1 rests on the claim that $\underline{d}$ and $\underline{e}$ may be prescribed arbitrarily and independently, and that claim can be read only from the explicit formulas: $\underline{d}$ and $\underline{e}$ are built from $\underline{c}_1 - \underline{c}_2$ and its derivative, and those two may be assigned at will at any one instant because $\underline{c}_1$ and $\underline{c}_2$ are arbitrary functions of time. Had $\underline{e}$ kept a dependence on $\underline{x}^+$ when $\underline{Q} = \underline{I}$, the step from (9.15) to (9.16) would have failed, since the velocity argument of $\underline{\mathcal{G}}$ could not then be shifted by an amount independent of position without also disturbing $\underline{L}$. $\square$

PROBLEM 2 (The same laws as the second observer writes them). What are the counterparts of (9.37) and (9.41) in the frame of O^+ ?

Solution. The short answer is that they are the same equations with stars on every symbol and no stars on the response functions, and the content of the problem is to see that the second statement is a theorem rather than a notational convenience.

From Chapter 7 and Section 8.4 we need the dictionary

$$\begin{aligned} \underline{T}^+ &= \underline{Q} \underline{T} \underline{Q}^t \text{ (7.40),} & \underline{q}^+ &= \underline{Q} \underline{q} \text{ (7.44),} & \underline{D}^+ &= \underline{Q} \underline{D} \underline{Q}^t \text{ (7.34),} \\ \theta^+ &= \theta, & \rho^+ &= \rho \text{ (7.37),} & \text{grad}^+ \theta^+ &= \underline{Q} \text{ grad } \theta. \end{aligned}$$

The invariants are absolute scalars. Part 1 of Section 9.1.1 already showed $I_k(\underline{Q} \underline{D} \underline{Q}^t) = I_k(\underline{D})$ for $k = 1, 2, 3$. For the three of (9.43), with $\underline{d} = \text{grad } \theta$,

$$\begin{aligned} I_4^+ &= \underline{Q} \underline{d} \cdot \underline{Q} \underline{D} \underline{Q}^t \underline{Q} \underline{d} = \underline{Q} \underline{d} \cdot \underline{Q} \underline{D} \underline{d} = \underline{d} \cdot \underline{Q}^t \underline{Q} \underline{D} \underline{d} = \underline{d} \cdot \underline{D} \underline{d} = I_4, \\ I_5^+ &= \underline{Q} \underline{d} \cdot (\underline{Q} \underline{D} \underline{Q}^t)^2 \underline{Q} \underline{d} = \underline{Q} \underline{d} \cdot \underline{Q} \underline{D}^2 \underline{d} = \underline{d} \cdot \underline{D}^2 \underline{d} = I_5, \\ I_6^+ &= |\underline{Q} \underline{d}| = |\underline{d}| = I_6, \end{aligned}$$

using $\underline{\underline{Q}}^t \underline{\underline{Q}} = \underline{\underline{I}}$ throughout and, in the last line, that orthogonal tensors preserve norms (Proposition 1.48). So O and O^+ compute the same six numbers.

For the stress, conjugate (9.37) by $\underline{\underline{Q}}$, term by term. Using $\underline{\underline{Q}} \underline{\underline{I}} \underline{\underline{Q}}^t = \underline{\underline{Q}} \underline{\underline{Q}}^t = \underline{\underline{I}}^+$ and $\underline{\underline{Q}} \underline{\underline{D}}^2 \underline{\underline{Q}}^t = (\underline{\underline{Q}} \underline{\underline{D}} \underline{\underline{Q}}^t)^2 = (\underline{\underline{D}}^+)^2$,

$$\begin{aligned} \underline{\underline{T}}^+ &= \underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t = \varphi_0(I_1, I_2, I_3, \theta, \rho; p) \underline{\underline{Q}} \underline{\underline{I}} \underline{\underline{Q}}^t \\ &\quad + \varphi_1(\cdots) \underline{\underline{Q}} \underline{\underline{D}} \underline{\underline{Q}}^t + \varphi_2(\cdots) \underline{\underline{Q}} \underline{\underline{D}}^2 \underline{\underline{Q}}^t \\ &= \varphi_0(I_1, I_2, I_3, \theta, \rho; p) \underline{\underline{I}}^+ \\ &\quad + \varphi_1(\cdots) \underline{\underline{D}}^+ + \varphi_2(\cdots) (\underline{\underline{D}}^+)^2, \end{aligned}$$

the scalars passing through the conjugation untouched. Now replace each argument by its starred equal, which changes nothing numerically:

$$\underline{\underline{T}}^+ = \varphi_0(I_1^+, I_2^+, I_3^+, \theta^+, \rho^+; p) \underline{\underline{I}}^+ + \varphi_1(\cdots) \underline{\underline{D}}^+ + \varphi_2(\cdots) (\underline{\underline{D}}^+)^2.$$

For the heat flux, from (9.41),

$$\underline{\underline{q}}^+ = \underline{\underline{Q}} \underline{\underline{q}} = \underline{\underline{Q}} \underline{\underline{K}} \text{grad } \theta = (\underline{\underline{Q}} \underline{\underline{K}} \underline{\underline{Q}}^t) \underline{\underline{Q}} \text{grad } \theta = \underline{\underline{K}}^+ \text{grad}^+ \theta^+,$$

where the middle step inserted $\underline{\underline{Q}}^t \underline{\underline{Q}} = \underline{\underline{I}}$, and where, by the same conjugation as for the stress,

$$\begin{aligned} \underline{\underline{K}}^+ &= \underline{\underline{Q}} \underline{\underline{K}} \underline{\underline{Q}}^t \\ &= \phi_0(I_1^+, \dots, I_6^+, \theta^+, \rho^+; p) \underline{\underline{I}}^+ \\ &\quad + \phi_1(\cdots) \underline{\underline{D}}^+ + \phi_2(\cdots) (\underline{\underline{D}}^+)^2. \end{aligned}$$

What has been shown is this. The functions φ_k and ϕ_k appearing in the starred equations are the same functions that appear in the unstarred ones, evaluated at the starred arguments, and not merely functions of the same form. That is material frame indifference in the concrete case, in the form “the same function” that Section 8.4 uses and Remark 9.2 discusses. It is also what makes a table of measured viscosities meaningful, since a laboratory on a rotating planet and a laboratory in free fall will fill in the same table.

The one quantity that behaves differently is $\underline{W}$, and it is instructive that it has already been eliminated. Had the stress been allowed to depend on $\underline{W}$, the conjugation above would have produced $\underline{Q}\underline{W}\underline{Q}^t$ where (7.34) demands $\underline{W}^+ = \underline{Q}\underline{W}\underline{Q}^t + \underline{\Omega}$, and the two observers would have needed different functions. $\square$

PROBLEM 3 (Sufficiency of the heat-flux representation). Show that (9.41) and (9.42) satisfy (9.40) for arbitrary orthogonal $\underline{Q}: \mathbb{E}^3 \rightarrow \mathbb{E}^3$.

Solution. Section 9.2 proved that (9.40) implies (9.41) and (9.42). This is the converse, and with it the two statements are equivalent, which is what entitles one to use the representation in place of the frame-indifference requirement.

Write $\underline{d} = \text{grad } \theta$ and suppress the passive variables, so that what must be shown is

$$\underline{g}(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}) = \underline{Q} \underline{g}(\underline{D}, \underline{d})$$

for $\underline{g}(\underline{D}, \underline{d}) = \underline{K}(\underline{D}, \underline{d})\underline{d}$ with $\underline{K} = \phi_0\underline{I} + \phi_1\underline{D} + \phi_2\underline{D}^2$ and the ϕ_k functions of $I_1, \dots, I_6$.

First, the arguments of the ϕ 's do not move. This was computed in Problem 2: $\underline{I}_k(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}) = \underline{I}_k(\underline{D}, \underline{d})$ for $k = 1, \dots, 6$. Hence

$$\phi_k(\underline{I}_1(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}), \dots) = \phi_k(\underline{I}_1(\underline{D}, \underline{d}), \dots),$$

and we may write ϕ_k for both without ambiguity. This is the step that uses the specific list (9.43): any other scalar built from $\underline{D}$ and $\underline{d}$ that failed to be invariant would break the argument here.

Second, the tensor conjugates. Evaluating (9.42) at the transformed arguments and using $\underline{I} = \underline{Q}\underline{I}\underline{Q}^t$ and $(\underline{Q}\underline{D}\underline{Q}^t)^2 = \underline{Q}\underline{D}^2\underline{Q}^t$,

$$\begin{aligned} \underline{K}(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}) &= \phi_0\underline{I} + \phi_1\underline{Q}\underline{D}\underline{Q}^t + \phi_2\underline{Q}\underline{D}^2\underline{Q}^t \\ &= \underline{Q}[\phi_0\underline{I} + \phi_1\underline{D} + \phi_2\underline{D}^2]\underline{Q}^t = \underline{Q} \underline{K}(\underline{D}, \underline{d}) \underline{Q}^t. \end{aligned}$$

Third, apply it to $\underline{Q}\underline{d}$. Therefore

$$\begin{aligned} \underline{g}(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}) &= \underline{K}(\underline{Q}\underline{D}\underline{Q}^t, \underline{Q}\underline{d}) \underline{Q}\underline{d} \\ &= \underline{Q}\underline{K}\underline{Q}^t\underline{Q}\underline{d} = \underline{Q}\underline{K}\underline{d} = \underline{Q} \underline{g}(\underline{D}, \underline{d}), \end{aligned}$$

using $\underline{\underline{Q}}^t \underline{\underline{Q}} = \underline{\underline{I}}$ in the third equality. Hence

$$\underline{\underline{g}}(\theta, \rho, \underline{\underline{Q}} \underline{\underline{D}} \underline{\underline{Q}}^t, \underline{\underline{Q}} \operatorname{grad} \theta; p) = \underline{\underline{Q}} \underline{\underline{g}}(\theta, \rho, \underline{\underline{D}}, \operatorname{grad} \theta; p)$$

for every orthogonal $\underline{\underline{Q}}$, which is (9.40).

A check on the odd part: setting $\underline{\underline{Q}} = -\underline{\underline{I}}$ recovers (9.45): $\underline{\underline{K}}$ is unchanged, since $(-\underline{\underline{I}}) \underline{\underline{D}} (-\underline{\underline{I}})^t = \underline{\underline{D}}$ and the invariants are even in $\underline{\underline{d}}$, while $\underline{\underline{d}}$ reverses, so $\underline{\underline{g}}$ reverses. That the representation automatically satisfies the reflection condition, without any further assumption, is the payoff of Part 1 of Section 9.2: the coefficients were forced to depend on $\underline{\underline{d}}$ only through $\underline{\underline{d}} \otimes \underline{\underline{d}}$, which cannot see the sign.

The same three steps, with the vector $\underline{\underline{d}}$ deleted and the outer factor $\underline{\underline{Q}}^t$ retained, prove the converse half of the stress representation promised after (9.36). $\square$

PROBLEM 4 (Observers agree on the reaction pressure). Show that all observers agree on the value of p in (9.83).

Solution. The point of the problem is that p is not delivered by a constitutive function, as Remark 9.30 stressed, so the agreement of observers cannot be inferred from (9.5) in the way Problem 2 inferred it for $\underline{\underline{T}}$. It has to be shown separately, and it follows from the constraint.

The first route takes the trace. Apply tr to (9.83), using linearity of the trace and $\operatorname{tr} \underline{\underline{I}} = 3$:

$$\operatorname{tr} \underline{\underline{T}} = -p \operatorname{tr} \underline{\underline{I}} + 2\mu \operatorname{tr} \underline{\underline{D}} = -3p + 2\mu \operatorname{div} \underline{\underline{v}} = -3p,$$

the last step by the constraint (9.85). Hence

$$p = -\frac{1}{3} \operatorname{tr} \underline{\underline{T}}.$$

So p , though not constitutively determined, is a definite function of the stress: it is the mean normal pressure. Now

$$\operatorname{tr} \underline{\underline{T}}^+ = \operatorname{tr} (\underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t) = \operatorname{tr} (\underline{\underline{Q}}^t \underline{\underline{Q}} \underline{\underline{T}}) = \operatorname{tr} \underline{\underline{T}}$$

by (7.40) and the cyclic property of the trace, Theorem 1.51. The observer O^+ writes (9.83) in the form $\underline{\underline{T}}^+ = -p^+ \underline{\underline{I}}^+ + 2\mu \underline{\underline{D}}^+$ with

$\text{tr } \underline{\underline{D}}^+ = \text{div}^+ \underline{\underline{v}}^+ = \text{div } \underline{\underline{v}} = 0$ by (7.56), so the same computation gives $p^+ = -\frac{1}{3} \text{tr } \underline{\underline{T}}^+$. Therefore

$$p^+ = -\frac{1}{3} \text{tr } \underline{\underline{T}}^+ = -\frac{1}{3} \text{tr } \underline{\underline{T}} = p.$$

The second route conjugates the constitutive equation. The same conclusion, obtained without dividing by three, shows where it could have failed. Conjugating (9.83),

$$\underline{\underline{T}}^+ = \underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t = \underline{\underline{Q}} (-p \underline{\underline{I}} + 2\mu \underline{\underline{D}}) \underline{\underline{Q}}^t = -p \underline{\underline{I}}^+ + 2\mu \underline{\underline{D}}^+,$$

using $\underline{\underline{Q}} \underline{\underline{I}} \underline{\underline{Q}}^t = \underline{\underline{I}}^+$ and $\underline{\underline{D}}^+ = \underline{\underline{Q}} \underline{\underline{D}} \underline{\underline{Q}}^t$, and noting that $\mu = \hat{\mu}(\theta, \rho; p)$ is unchanged because θ and ρ are absolute scalars by (7.37). But O^+ writes $\underline{\underline{T}}^+ = -p^+ \underline{\underline{I}}^+ + 2\mu \underline{\underline{D}}^+$. Subtracting, $(p^+ - p) \underline{\underline{I}}^+ = \underline{\underline{Q}}^+$, and since $\underline{\underline{I}}^+ \neq \underline{\underline{Q}}^+$ the scalar factor must vanish. The book's solution in Appendix E takes this second route and records the result as a statement about the two fields, $p^+(\underline{\underline{x}}^+, t^+) = p(\underline{\underline{Q}}(t)^t(\underline{\underline{x}}^+ - \underline{\underline{c}}(t)), t)$ with $t = t^+ - a$: the same number, attached by the two observers to the same event. It cites (7.1) and (7.2) for this, but the inversion $\underline{\underline{x}} = \underline{\underline{Q}}^t(\underline{\underline{x}}^+ - \underline{\underline{c}})$ it uses is that of (7.3); (7.2), the equality of distances, plays no part.

This matters for the following reason. A quantity that is determined by the problem rather than by the material might have been expected to be observer dependent, and if it were, (9.83) would be an equation two observers could not both write. The reason it is not is visible in the first route: the constraint $\text{tr } \underline{\underline{D}} = 0$ makes the viscous stress traceless, so the trace of $\underline{\underline{T}}$ isolates p exactly, and the trace is an invariant. Chapter 10 generalises this: the reactive stress associated with any constraint is determined up to a multiplier, and the multiplier is always an absolute scalar for the same reason. $\square$

PROBLEM 5 (Axial flow along a pipe, and the elliptical cross section). Consider an incompressible viscous liquid described by (9.83), and take μ to be constant. Consider the steady flow (that is, $\underline{\underline{v}}' = \underline{\underline{0}}$) defined by $\underline{\underline{v}} = f(x_1, x_2) \underline{\underline{e}}_3$. (a) Show that such a motion is isochoric, as required for incompressible fluids. (b) Assuming zero body force, show that the

pressure depends on x_3 only and that the pressure gradient $dp/dx_3 = P$, a constant. Thus reduce the problem to the equation $\Delta f = P/\mu$, where Δ is the two-dimensional Laplacian. (c) Solve this problem for flow through a pipe with an elliptical cross section in the x_1, x_2 plane with major and minor axes aligned with the x_1, x_2 axes, respectively. The semi-axis lengths are a and $b < a$. Assume the no-slip condition, $\underline{v} = \underline{0}$ on the boundary. *Hint:* if $\varphi(x_1, x_2) = 0$ is the equation of the boundary, then $c\varphi$, for an appropriate constant c , also solves the governing differential equation.

Solution. (a) The velocity has components $v_1 = v_2 = 0$ and $v_3 = f(x_1, x_2)$. By Proposition 1.92,

$$\operatorname{div} \underline{v} = v_{i,i} = v_{1,1} + v_{2,2} + v_{3,3} = 0 + 0 + \frac{\partial f(x_1, x_2)}{\partial x_3} = 0,$$

since f does not depend on x_3 . So $\operatorname{div} \underline{v} = 0$, which is (9.85), and by (6.17) together with $\dot{J} = J \operatorname{div} \underline{v}$ from Problem 2 of Chapter 4 this is $\dot{J} = 0$: the motion is isochoric.

Two features of the field conspire here and it is worth noticing which. The velocity points along $\underline{e}_3$, so only $v_{3,3}$ could contribute to the divergence; and v_3 is independent of x_3 , so that derivative vanishes. A flow along a fixed direction whose speed varies only across that direction is automatically isochoric.

(b) We need $\operatorname{div} \underline{T}$ and $\dot{\underline{v}}$.

Begin with the velocity gradient and the acceleration. By Theorem 1.86, $L_{ij} = v_{i,j}$, and the only non-zero entries are $L_{31} = f_{,1}$ and $L_{32} = f_{,2}$, so

$$\underline{L} = f_{,1} \underline{e}_3 \otimes \underline{e}_1 + f_{,2} \underline{e}_3 \otimes \underline{e}_2.$$

Then, using $(\underline{u} \otimes \underline{v})\underline{w} = (\underline{v} \cdot \underline{w})\underline{u}$ and $\underline{e}_1 \cdot \underline{e}_3 = \underline{e}_2 \cdot \underline{e}_3 = 0$,

$$\underline{L}\underline{v} = f[f_{,1}(\underline{e}_1 \cdot \underline{e}_3) + f_{,2}(\underline{e}_2 \cdot \underline{e}_3)]\underline{e}_3 = \underline{0},$$

so by (2.21) the material acceleration is $\dot{\underline{v}} = \underline{v}' + \underline{L}\underline{v} = \underline{0}$. The fluid is not accelerating at all: each particle moves in a straight line at constant speed, since it never leaves the line on which f takes its value.

Next, the divergence of the stress. Since μ is constant and $\operatorname{div} \underline{v} = 0$, the calculation performed after (9.61) gives

$$\operatorname{div} \underline{T} = -\operatorname{grad} p + \mu \Delta_3 \underline{v},$$

where $\Delta_3 = \partial_j \partial_j$ is the three-dimensional Laplacian. Applied to $\underline{v} = f \underline{e}_3$ with f independent of x_3 ,

$$\Delta_3 \underline{v} = (f_{,11} + f_{,22} + f_{,33}) \underline{e}_3 = (f_{,11} + f_{,22}) \underline{e}_3 = (\Delta f) \underline{e}_3,$$

Δ now denoting the two-dimensional Laplacian in x_1, x_2 , as the problem asks.

The equation of motion follows. With $\underline{b} = \underline{0}$ and $\underline{\dot{v}} = \underline{0}$, (6.109) reduces to $\operatorname{div} \underline{T} = \underline{0}$, that is

$$-\operatorname{grad} p + \mu(\Delta f) \underline{e}_3 = \underline{0}.$$

Resolving on the fixed basis,

$$p_{,1} = 0, \quad p_{,2} = 0, \quad p_{,3} = \mu \Delta f.$$

The first two say that p does not vary across the pipe: $p = p(x_3, t)$. But then $p_{,3}$ is a function of x_3 and t alone, while the right-hand side $\mu \Delta f$ is a function of x_1, x_2 alone. A function of one set of variables can equal a function of a disjoint set only if both are constant, so

$$\frac{dp}{dx_3} = P = \text{const}, \quad \Delta f = \frac{P}{\mu}.$$

In particular p is steady as well as being a function of x_3 alone, and $p = Px_3 + p_0$.

The argument is worth pausing on: nothing was assumed about p beyond what the equations of motion say, and the constancy of the pressure gradient, which is normally quoted as an assumption in textbook treatments of pipe flow, has been derived. The physical content is that a unidirectional flow can be driven only by a uniform push along the pipe.

(c) Let

$$\varphi(x_1, x_2) = \frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} - 1,$$

so that $\varphi = 0$ is the boundary ellipse and $\varphi < 0$ inside. Following the hint, try $f = c\varphi$. Then

$$\Delta\varphi = \varphi_{,11} + \varphi_{,22} = \frac{2}{a^2} + \frac{2}{b^2} = 2 \frac{a^2 + b^2}{a^2 b^2},$$

a constant, which is the “lucky coincidence” the hint refers to: the Laplacian of a quadratic is constant, and the right-hand side P/μ is also constant, so the two can be matched by one choice of c . Requiring $c\Delta\varphi = P/\mu$,

$$c = \frac{P}{\mu} \cdot \frac{a^2 b^2}{2(a^2 + b^2)}.$$

The no-slip condition is satisfied automatically, since $f = c\varphi = 0$ wherever $\varphi = 0$. Hence

$$\underline{v} = \frac{P a^2 b^2}{2\mu(a^2 + b^2)} \left(\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} - 1 \right) \underline{e}_3.$$

To read the answer, note that inside the ellipse the bracket is negative, so the flow is along $+\underline{e}_3$ exactly when $P < 0$: the fluid runs from high pressure to low, as it must. The maximum speed is on the axis,

$$|v|_{\max} = \frac{|P| a^2 b^2}{2\mu(a^2 + b^2)},$$

and the profile is a paraboloid over the elliptical cross section.

Two checks confirm it. First, the volume flux. With $x_1 = a\rho \cos \vartheta$, $x_2 = b\rho \sin \vartheta$, so that $\varphi = \rho^2 - 1$ and $da = ab \rho d\rho d\vartheta$,

$$\begin{aligned} Q &= \int f da = c ab \int_0^{2\pi} \int_0^1 (\rho^2 - 1) \rho d\rho d\vartheta \\ &= 2\pi c ab \left[\frac{1}{4} - \frac{1}{2} \right] = -\frac{\pi c ab}{2} = -\frac{\pi P a^3 b^3}{4\mu(a^2 + b^2)}. \end{aligned}$$

Setting $a = b = R$ gives $Q = -\pi P R^4/8\mu$, which is the Hagen–Poiseuille law for a circular pipe and agrees with Problem 6(b) below. Second, the stress: with $f = c\varphi$,

$$\underline{\underline{D}} = \text{sym } \underline{\underline{L}} = \frac{1}{2} \left[f_{,1} (\underline{e}_3 \otimes \underline{e}_1 + \underline{e}_1 \otimes \underline{e}_3) + f_{,2} (\underline{e}_3 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_3) \right],$$

which has zero trace, as (9.85) requires, and gives a wall shear traction of magnitude $\mu |\text{grad } f|$ directed along the pipe. The pressure drop over a length ℓ pays for the total wall friction: that is the balance (6.109) integrated over a slug of fluid, and it is a useful way to remember why P had to be constant. $\square$

PROBLEM 6 (Two classical unidirectional flows). Use the same model to solve the following problems, in which the body force is zero. (a) Steady flow of the form $\underline{v} = v_1(x_2)\underline{e}_1$, with $v_1(0) = 0$ and $v_1(d) = V$. Obtain a restriction on p and use it to find the function $v_1(x_2)$. (b) Steady pipe flow $\underline{v} = w(r)\underline{e}_3$ in polar coordinates, inside a pipe of radius a , with $w(a) = 0$. Again obtain a restriction on p and use it to find $w(r)$.

Solution. (a) This is plane Couette–Poiseuille flow. The velocity has $v_1 = v_1(x_2)$, $v_2 = v_3 = 0$, so

$$\text{div } \underline{v} = v_{1,1} = 0$$

because v_1 is independent of x_1 : the flow is isochoric, and (9.85) is satisfied. The velocity gradient has the single non-zero entry $L_{12} = v'_1$, so

$$\underline{L} = v'_1(x_2) \underline{e}_1 \otimes \underline{e}_2, \quad \underline{L}\underline{v} = v'_1 v_1 (\underline{e}_2 \cdot \underline{e}_1) \underline{e}_1 = \underline{0},$$

and again $\dot{\underline{v}} = \underline{v}' + \underline{L}\underline{v} = \underline{0}$.

With μ constant, $\text{div } \underline{v} = 0$ and $\underline{b} = \underline{0}$, the equation of motion $\text{div } \underline{T} = \underline{0}$ reads

$$-\text{grad } p + \mu \Delta_3 \underline{v} = \underline{0}, \quad \Delta_3 \underline{v} = v''_1(x_2) \underline{e}_1,$$

whose three components are

$$p_{,1} = \mu v''_1(x_2), \quad p_{,2} = 0, \quad p_{,3} = 0.$$

The last two give $p = p(x_1, t)$; then $p_{,1}$ is a function of x_1 and t while $\mu v''_1$ is a function of x_2 , so both are the same constant:

$$p = Px_1 + p_0, \quad \mu v''_1 = P = \text{const.}$$

This is the restriction on p : it is linear in x_1 and independent of x_2 and x_3 .

Integrating $v_1'' = P/\mu$ twice,

$$v_1(x_2) = \frac{P}{2\mu}x_2^2 + Ax_2 + B.$$

The condition $v_1(0) = 0$ gives $B = 0$; the condition $v_1(d) = V$ gives $\frac{P}{2\mu}d^2 + Ad = V$, hence $A = V/d - Pd/2\mu$. Therefore

$$v_1(x_2) = \frac{P}{2\mu}x_2(x_2 - d) + \frac{Vx_2}{d}.$$

The answer is the sum of two classical flows and it is worth separating them. With $P = 0$ it is $v_1 = Vx_2/d$, plane Couette flow: a linear profile driven purely by the moving wall at $x_2 = d$, with no pressure gradient. With $V = 0$ it is the parabola $v_1 = Px_2(x_2 - d)/2\mu$, plane Poiseuille flow, driven purely by the pressure gradient and vanishing at both walls. That the general solution is their sum is a consequence of the linearity of the problem, which itself came from the vanishing of the acceleration: the Navier–Stokes equations are nonlinear only through $(\text{grad } \underline{v})\underline{v}$, and that term is absent here. Figure G shows the whole family.

The wall traction is worth recording, since it is what one measures. On the plane $x_2 = 0$ with normal $\underline{e}_2$, (9.83) gives

$$\underline{T}\underline{e}_2 = -p\underline{e}_2 + 2\mu\underline{D}\underline{e}_2 = -p\underline{e}_2 + \mu v_1'(0)\underline{e}_1,$$

since $\underline{D} = \frac{1}{2}v_1'(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1)$. The shear stress at the wall is therefore $\mu v_1'(0) = \mu V/d - Pd/2$, which for pure Couette flow is the familiar $\mu V/d$.

(b) Hagen–Poiseuille flow is the case $f = w(r)$ of Problem 5 with $r^2 = x_1^2 + x_2^2$, so parts (a) and (b) of that problem apply verbatim: the motion is isochoric, the pressure satisfies $p = p(x_3)$ with

$$\frac{dp}{dx_3} = P = \text{const}$$

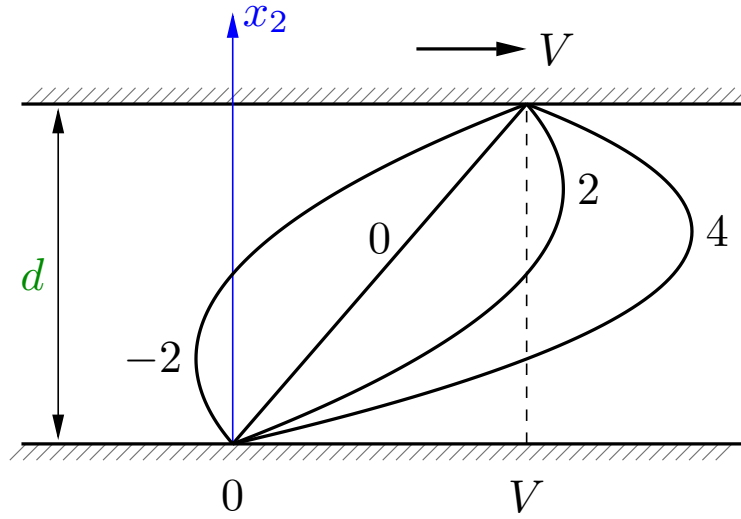


Figure G: The profiles of part (a), drawn in the dimensionless form $v_1/V = \xi + \Pi \xi(1 - \xi)$ with $\xi = x_2/d$ and $\Pi = -Pd^2/(2\mu V)$, and labelled with the value of Π ; the lower wall is fixed and the upper one moves with speed V . The line $\Pi = 0$ is Couette flow, driven by the moving wall alone. A favourable pressure gradient, $P < 0$ and so $\Pi > 0$, adds a parabola that vanishes at both walls. An adverse one, $\Pi < 0$, subtracts it, and once $\Pi < -1$ the slope $v'_1(0) = (V/d)(1 + \Pi)$ at the fixed wall changes sign and the fluid next to that wall runs backwards. Every curve is the line plus a multiple of one parabola, because the acceleration vanishes and the problem is linear.

and the governing equation is $\Delta w = P/\mu$ with Δ the two-dimensional Laplacian.

For a function of r alone the two-dimensional Laplacian is

$$\Delta w = \operatorname{div}(\operatorname{grad} w) = \frac{1}{r}(r w')',$$

which follows from (1.122) and the polar divergence formula quoted in Example 1.99: $\operatorname{grad} w = w'(r)\underline{e}_r$ by (1.122), since w depends on neither θ nor z , and then $\operatorname{div}(w_r \underline{e}_r) = w_{r,r} + w_r/r$ with $w_r = w'$, which is $w'' + w'/r = \frac{1}{r}(r w')'$.

Hence $(r w')' = P r / \mu$, and integrating,

$$r w' = \frac{P}{2\mu} r^2 + C_1, \quad w' = \frac{P}{2\mu} r + \frac{C_1}{r}, \quad w = \frac{P}{4\mu} r^2 + C_1 \ln r + C_2.$$

Boundedness of the velocity on the axis forces $C_1 = 0$, since $\ln r$ diverges as $r \rightarrow 0$; and $w(a) = 0$ gives $C_2 = -P a^2 / 4\mu$. Therefore

$$w(r) = \frac{P}{4\mu}(r^2 - a^2), \quad \underline{v} = w(r)\underline{e}_3.$$

Again $P < 0$ for flow along $+\underline{e}_3$, and the profile is a paraboloid with maximum speed $|P|a^2/4\mu$ on the axis.

The flux is found by integrating over the cross section,

$$\begin{aligned} Q &= \int_0^a w(r) 2\pi r \, dr = \frac{\pi P}{2\mu} \int_0^a (r^3 - a^2 r) \, dr \\ &= \frac{\pi P}{2\mu} \left[\frac{a^4}{4} - \frac{a^4}{2} \right] = -\frac{\pi P a^4}{8\mu}, \end{aligned}$$

the Hagen–Poiseuille law, in agreement with the elliptical result of Problem 5(c) at $a = b$. The fourth power of the radius is the reason a small change in the calibre of a pipe, or of an artery, changes the flow so much: halving a at fixed pressure gradient divides the flux by sixteen.

The wall shear is found as in part (a). On $r = a$ the exterior normal of the fluid is $\underline{e}_r$, and with $\underline{D} = \frac{1}{2}w'(\underline{e}_r \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_r)$,

$$\underline{T} \underline{e}_r = -p \underline{e}_r + \mu w'(a) \underline{e}_3 = -p \underline{e}_r + \frac{Pa}{2} \underline{e}_3,$$

so the fluid drags the wall along the flow direction. Over a length ℓ of pipe the total wall force is $2\pi a\ell \cdot |P|a/2 = \pi a^2\ell|P|$, which is exactly the pressure difference $|P|\ell$ acting on the cross-sectional area πa^2 : the slug of fluid is in equilibrium, as $\dot{v} = 0$ requires. $\square$

PROBLEM 7 (Rigid rotation, and the shape of the free surface). The same liquid experiences a steady velocity field of the form $\underline{v} = r\omega \underline{e}_\theta$ in a polar coordinate system, where ω is a constant. Assume the density ρ of the fluid to be uniform and independent of time in the spatial description. (a) Describe this motion in words. Is mass conservation satisfied? (b) Evaluate the tensors $\underline{L}$, $\underline{D}$, $\underline{W}$. (c) Suppose the fluid is subjected to the uniform gravitational body force $\underline{b} = -g\underline{e}_3$, with g constant. Derive the equation $z(r)$ of the surface of the liquid if the surface is exposed to the air and the air exerts the uniform atmospheric pressure p_a , a constant, on the entire surface. Ignore the viscosity of the air.

Solution. Throughout, $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$ is the cylindrical polar frame of (1.116), with $\underline{e}_3$ the axis and $z = x_3$.

(a) The azimuthal speed is $v_\theta = r\omega$, so every particle travels on a circle about the $\underline{e}_3$ axis, and the angular rate $v_\theta/r = \omega$ is the same on every circle. The whole body of fluid therefore turns as a rigid body at angular velocity ω about $\underline{e}_3$: nothing is being sheared, stretched or compressed. This is the case $f = \omega$ of the family $\underline{v} = rf(r)\underline{e}_\theta$ treated in Problem 10 of Chapter 4.

Mass conservation is (6.14), $\rho' + \text{div}(\rho\underline{v}) = 0$. By Problem 24(c) of Chapter 1, $\text{div}(\rho\underline{v}) = \text{grad } \rho \cdot \underline{v} + \rho \text{div } \underline{v}$, and the hypotheses give $\rho' = 0$ and $\text{grad } \rho = 0$, since ρ is uniform and time independent in the spatial description. So the requirement reduces to $\rho \text{div } \underline{v} = 0$, that is $\text{div } \underline{v} = 0$. From part (b) below, $\text{tr } \underline{L} = \omega(\underline{e}_\theta \cdot \underline{e}_r - \underline{e}_r \cdot \underline{e}_\theta) = 0$, so

$$\text{div } \underline{v} = 0 \quad \text{and mass conservation is satisfied.}$$

It could hardly be otherwise: a rigid motion moves no material relative to any other, so it cannot change any density.

(b) Problem 10 of Chapter 4 gives, for $\underline{v} = rf(r)\underline{e}_\theta$,

$$\begin{aligned}\underline{L} &= (rf)'\underline{e}_\theta \otimes \underline{e}_r - f\underline{e}_r \otimes \underline{e}_\theta, \quad \underline{D} \\ &= \frac{rf'}{2}(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r), \quad \underline{W} = \frac{2f + rf'}{2}(\underline{e}_\theta \otimes \underline{e}_r - \underline{e}_r \otimes \underline{e}_\theta).\end{aligned}$$

With $f = \omega$ constant, $f' = 0$ and $(rf)' = \omega$, so

$$\underline{L} = \omega(\underline{e}_\theta \otimes \underline{e}_r - \underline{e}_r \otimes \underline{e}_\theta), \quad \underline{D} = \underline{O}, \quad \underline{W} = \underline{L}.$$

The axial vector of $\underline{W}$ is $\underline{w} = \omega\underline{e}_3$, so the vorticity is $\text{curl } \underline{v} = 2\omega\underline{e}_3$. That $\underline{D} = \underline{O}$ is the characterisation of rigid motion recorded in Remark 4.6, and it is the fact that makes part (c) easy: by (9.83) the viscous stress $2\mu\underline{D}$ vanishes identically, so

$$\underline{T} = -p\underline{I}$$

throughout the fluid, exactly as if the liquid were inviscid. A rigid rotation costs no viscous dissipation, which is why a bucket of water spun up long enough eventually turns with no internal friction at all.

(c) The material acceleration is $\underline{\dot{v}} = \underline{v}' + \underline{L}\underline{v}$ by (2.21), with $\underline{v}' = \underline{0}$ because the field is steady. Using the dyad rule and $\underline{e}_r \cdot \underline{e}_\theta = 0$, $\underline{e}_\theta \cdot \underline{e}_\theta = 1$,

$$\begin{aligned}\underline{L}\underline{v} &= \omega(\underline{e}_\theta \otimes \underline{e}_r - \underline{e}_r \otimes \underline{e}_\theta)(r\omega\underline{e}_\theta) \\ &= r\omega^2[(\underline{e}_r \cdot \underline{e}_\theta)\underline{e}_\theta - (\underline{e}_\theta \cdot \underline{e}_\theta)\underline{e}_r] = -r\omega^2\underline{e}_r,\end{aligned}$$

the centripetal acceleration, directed inward and of magnitude $r\omega^2$ as it must be.

The equation of motion (6.109) with $\underline{T} = -p\underline{I}$ and $\text{div}(-p\underline{I}) = -\text{grad } p$ reads

$$-\rho r\omega^2\underline{e}_r = -\text{grad } p - \rho g\underline{e}_3, \quad \text{i.e.} \quad \text{grad } p = \rho r\omega^2\underline{e}_r - \rho g\underline{e}_3.$$

Resolving on the polar frame with (1.122), $\text{grad } p = p_{,r}\underline{e}_r + \frac{1}{r}p_{,\theta}\underline{e}_\theta + p_{,z}\underline{e}_3$, gives

$$p_{,r} = \rho\omega^2 r, \quad p_{,\theta} = 0, \quad p_{,z} = -\rho g.$$

The second says p is axisymmetric. The first and third integrate independently, since each right-hand side involves only its own variable, to

$$p = \frac{1}{2}\rho\omega^2 r^2 - \rho g z + C,$$

with C a constant. That the two integrations are consistent may be checked by cross-differentiating: $\partial_z(\rho\omega^2 r^2) = 0 = \partial_r(-\rho g)$, so the field $\text{grad } p$ is indeed a gradient.

On the free surface $z = z(r)$ the air exerts the uniform pressure p_a , and since the air is inviscid by hypothesis its traction is $-p_a \underline{n}$; the liquid's traction there is $-p \underline{n}$ by $\underline{T} = -p \underline{I}$, and the two must balance, so $p = p_a$ on the surface. Hence

$$p_a = \frac{1}{2}\rho\omega^2 r^2 - \rho g z(r) + C,$$

and solving for z ,

$$z(r) = z_0 + \frac{\omega^2 r^2}{2g}, \quad z_0 = z(0) = \frac{C - p_a}{\rho g}.$$

The free surface is a paraboloid of revolution, rising with the square of the distance from the axis, and its shape depends on neither ρ nor p_a : the constant z_0 , which fixes the height of the vertex, is set by the amount of liquid in the vessel. The result is the principle of the liquid-mirror telescope, whose parabolic reflector is a dish of spinning mercury, and of the rotating furnace used to cast telescope blanks. The focal length is $g/2\omega^2$, read off by comparing $z = r^2/4F$ with the boxed formula.

Notice what did the work. The viscosity never appeared, because $\underline{D} = \underline{Q}$ removed it; the whole problem is the balance between the pressure gradient and the centripetal acceleration, and $\omega^2 r$ in the acceleration is what produces r^2 in the surface. $\square$

PROBLEM 8 (Circular Couette flow in an annulus). Suppose this liquid experiences a steady velocity field of the kind described in Problem 10 of Chapter 4, $\underline{v} = r f(r) \underline{e}_\theta$, and suppose the body force vanishes. (a) Find the function f for the problem of flow in an annulus bounded by cylinders

of radii a and $b > a$. The inner and outer cylinders spin at the constant rates ω_a and ω_b , respectively, about the $\underline{e}_3$ axis. Assume the no-slip condition, i.e., the fluid velocity matches the cylindrical wall velocities at $r = a, b$. (b) Calculate the torque per unit length of the $\underline{e}_3$ axis needed to drive each cylinder. (c) Find the ratio ω_b/ω_a corresponding to irrotational flow, i.e., the ratio needed to nullify the vorticity everywhere in the fluid.

Solution. (a) From Problem 10 of Chapter 4, with $v_\theta = rf(r)$,

$$\underline{\underline{D}} = \frac{rf'}{2}(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r), \quad \text{tr } \underline{\underline{D}} = rf' \underline{e}_r \cdot \underline{e}_\theta = 0,$$

so the flow is isochoric for every f and (9.85) imposes nothing. By (9.83) the stress is

$$\underline{\underline{T}} = -p\underline{\underline{I}} + 2\mu\underline{\underline{D}} = -p\underline{\underline{I}} + \mu rf'(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r),$$

whose polar components are $T_{rr} = T_{\theta\theta} = T_{zz} = -p$ and $T_{r\theta} = T_{\theta r} = \mu rf'$, all other components zero.

For the divergence of the stress, split it as $\text{div } \underline{\underline{T}} = -\text{grad } p + \text{div}(2\mu\underline{\underline{D}})$, the first piece by $\text{div}(\varphi\underline{\underline{I}}) = \text{grad } \varphi$ and (1.122), the second by the polar formula (1.162) of Example 1.99 applied to $\underline{\underline{A}} = 2\mu\underline{\underline{D}}$, whose only non-zero components $A_{r\theta} = A_{\theta r} = \mu rf'$ depend on r alone. Reading (1.162) line by line,

$$\begin{aligned} \underline{e}_r \cdot \text{div}(2\mu\underline{\underline{D}}) &= A_{rr,r} + \frac{1}{r}(A_{rr} - A_{\theta\theta} + A_{r\theta,\theta}) = 0, \\ \underline{e}_\theta \cdot \text{div}(2\mu\underline{\underline{D}}) &= A_{\theta r,r} + \frac{1}{r}(A_{\theta r} + A_{r\theta} + A_{\theta\theta,\theta}) \\ &= \mu(rf')' + \frac{2\mu rf'}{r} = \mu(rf'' + 3f'), \\ \underline{e}_3 \cdot \text{div}(2\mu\underline{\underline{D}}) &= A_{zr,r} + \frac{1}{r}(A_{zr} + A_{z\theta,\theta}) = 0, \end{aligned}$$

where $(rf')' = f' + rf''$ was expanded in the second line.

For the acceleration, as in Problem 7, $\underline{v}' = \underline{0}$ and

$$\underline{\underline{L}}\underline{v} = \left[(rf')'\underline{e}_\theta \otimes \underline{e}_r - f \underline{e}_r \otimes \underline{e}_\theta \right] (rf \underline{e}_\theta) = -rf^2 \underline{e}_r = -\frac{v_\theta^2}{r} \underline{e}_r.$$

With $b = 0$, the equation of motion (6.109) resolves into

$$-\rho r f^2 = -p_{,r}, \quad 0 = -\frac{p_{,\theta}}{r} + \mu(r f'' + 3f'), \quad 0 = -p_{,z}.$$

The e_θ component gives $p_{,\theta} = \mu r(r f'' + 3f')$, whose right-hand side is a function of r alone. Integrating around a circuit of constant r , and requiring p to be single valued, forces

$$0 = \oint p_{,\theta} d\theta = 2\pi\mu r(r f'' + 3f'),$$

hence $p_{,\theta} = 0$ and

$$r f'' + 3f' = 0.$$

This is the governing equation, and it integrates in one step once one notices that

$$(r^3 f')' = 3r^2 f' + r^3 f'' = r^2(3f' + r f'') = 0.$$

Therefore $r^3 f' = -2B$ for a constant $-2B$, so $f' = -2B/r^3$ and

$$f(r) = A + \frac{B}{r^2}$$

with A a second constant.

It remains to impose the boundary conditions. No slip at $r = a$ means the fluid's azimuthal speed equals the wall's, $af(a) = a\omega_a$, i.e. $f(a) = \omega_a$; similarly $f(b) = \omega_b$. Thus

$$A + \frac{B}{a^2} = \omega_a, \quad A + \frac{B}{b^2} = \omega_b.$$

Subtracting, $B(a^{-2} - b^{-2}) = \omega_a - \omega_b$, and since $a^{-2} - b^{-2} = (b^2 - a^2)/a^2b^2$,

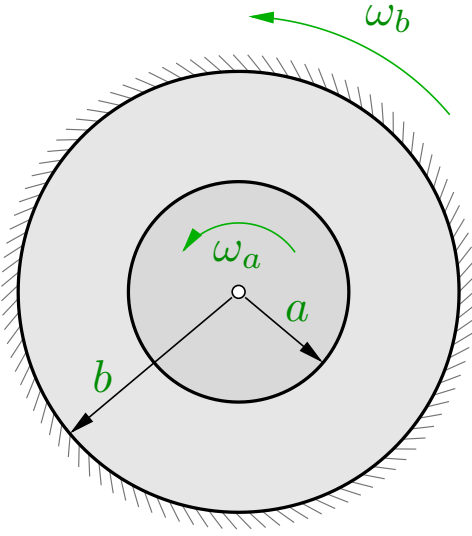
$$\begin{aligned} B &= \frac{a^2b^2(\omega_a - \omega_b)}{b^2 - a^2}, & A &= \omega_a - \frac{B}{a^2} \\ &= \frac{\omega_a(b^2 - a^2) - b^2(\omega_a - \omega_b)}{b^2 - a^2} = \frac{b^2\omega_b - a^2\omega_a}{b^2 - a^2}. \end{aligned}$$

Hence

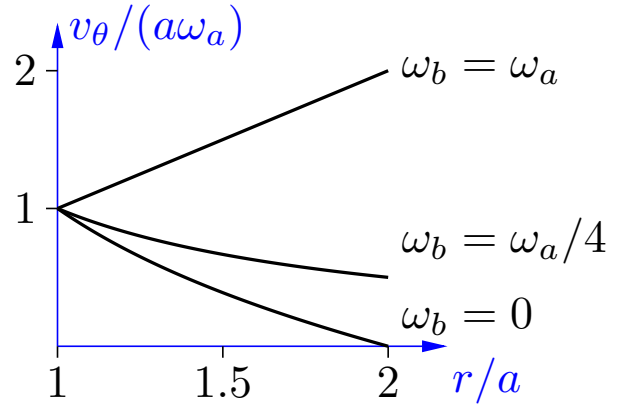
$$f(r) = \frac{b^2\omega_b - a^2\omega_a}{b^2 - a^2} + \frac{a^2b^2(\omega_a - \omega_b)}{(b^2 - a^2)r^2}, \quad \underline{v} = rf(r) \underline{e}_\theta.$$

The pressure follows by quadrature from $p_{,r} = \rho rf^2$ and is not needed below. This is circular Couette flow, the flow between rotating cylinders used since Couette's own measurements to determine μ .

Two limits check the answer. If $\omega_a = \omega_b = \omega$ then $B = 0$ and $A = \omega$, so $f = \omega$ and the fluid turns rigidly with the apparatus, which is Problem 7. If $b \rightarrow \infty$ with $\omega_b = 0$ then $A \rightarrow 0$ and $B \rightarrow a^2\omega_a$, giving $f = a^2\omega_a/r^2$, the potential vortex of Problem 10 of Chapter 4. Figure H compares these cases for $b = 2a$.



a



b

Figure H: Circular Couette flow. a: the annulus between the inner cylinder, of radius a and turning at ω_a , and the outer cylinder, of radius b and turning at ω_b . b: the speed $v_\theta = rf(r) = Ar + B/r$ for $b = 2a$ and three rates of the outer cylinder. The line $B = 0$ is rigid rotation, sheared nowhere and needing no torque. The hyperbola $A = 0$, reached when $\omega_b/\omega_a = a^2/b^2$, is the potential vortex of part (c): its vorticity $2A\underline{e}_3$ vanishes, yet it is sheared everywhere and needs the torque $4\pi\mu a^2\omega_a$ per unit length. With the outer cylinder at rest the profile lies below both.

(b) The traction exerted by the fluid on a cylindrical surface whose normal into the fluid is $\underline{e}_r$ is $\underline{T}\underline{e}_r$. From the components above, and using

$$(\underline{e}_r \otimes \underline{e}_\theta)\underline{e}_r = (\underline{e}_\theta \cdot \underline{e}_r)\underline{e}_r = 0 \text{ and } (\underline{e}_\theta \otimes \underline{e}_r)\underline{e}_r = \underline{e}_\theta,$$

$$\underline{T}\underline{e}_r = -p\underline{e}_r + \mu r f' \underline{e}_\theta, \quad \text{with} \quad \mu r f' = -\frac{2\mu B}{r^2}.$$

The pressure part is radial and exerts no moment about the axis, since $\underline{e}_r \times \underline{e}_r = 0$; only the shear part contributes, and $\underline{e}_r \times \underline{e}_\theta = \underline{e}_3$.

Consider the inner cylinder first. The surface $r = a$ has area $2\pi a$ per unit length, and the force per unit area exerted on the cylinder by the fluid is $\underline{T}\underline{e}_r$ evaluated at $r = a$, because the exterior normal of the fluid there is $-\underline{e}_r$ and Newton's third law reverses the sign twice. The moment per unit length about the axis is

$$\underline{m}_a = \int_0^{2\pi} (a\underline{e}_r) \times (\underline{T}\underline{e}_r) a \, d\theta = 2\pi a^2 \left(-\frac{2\mu B}{a^2} \right) \underline{e}_3 = -4\pi\mu B \underline{e}_3.$$

The torque that must be applied externally to keep the inner cylinder turning steadily is minus this, since the cylinder's angular momentum is constant:

$$\underline{m}_a^{\text{ext}} = +4\pi\mu B \underline{e}_3 = \frac{4\pi\mu a^2 b^2 (\omega_a - \omega_b)}{b^2 - a^2} \underline{e}_3 \quad \text{per unit length.}$$

At the outer cylinder the exterior normal of the fluid is $+\underline{e}_r$, so the traction on the cylinder from the fluid is $-\underline{T}\underline{e}_r$ evaluated at $r = b$, and

$$\underline{m}_b = \int_0^{2\pi} (b\underline{e}_r) \times (-\underline{T}\underline{e}_r) b \, d\theta = -2\pi b^2 \left(-\frac{2\mu B}{b^2} \right) \underline{e}_3 = +4\pi\mu B \underline{e}_3,$$

so the external torque required is

$$\underline{m}_b^{\text{ext}} = -4\pi\mu B \underline{e}_3 = -\underline{m}_a^{\text{ext}}.$$

The two external torques are equal and opposite, as they must be: the fluid between the cylinders has constant angular momentum, so the total moment exerted on it vanishes, and every unit of angular momentum fed in at one wall leaves at the other. Notice also that the torque is independent of r : the moment transmitted across any cylindrical surface

$a < r < b$ is $2\pi r^2 \cdot (-2\mu B/r^2) = -4\pi\mu B$, the same number, which is exactly the statement that angular momentum is being conducted radially at a uniform rate. That is what makes the device a viscometer: measure the torque, know B , and hence μ .

(c) The vorticity is twice the axial vector of $\underline{W}$, and from Problem 10 of Chapter 4,

$$\underline{W} = \frac{2f + rf'}{2}(\underline{e}_\theta \otimes \underline{e}_r - \underline{e}_r \otimes \underline{e}_\theta), \quad \underline{w} = \frac{2f + rf'}{2}\underline{e}_3 = \frac{1}{2r}(r^2f)'\underline{e}_3.$$

Irrotational flow requires $(r^2f)' = 0$, i.e. r^2f constant, i.e. $A = 0$ in $f = A + B/r^2$. From the expression for A above,

$$b^2\omega_b - a^2\omega_a = 0, \quad \text{i.e.} \quad \frac{\omega_b}{\omega_a} = \frac{a^2}{b^2}.$$

The flow is then the potential vortex $f = B/r^2$ of Chapter 4, in which every particle circles the axis while no particle turns about its own centre: a paddle wheel set in the fluid would be carried around without rotating. Yet $\underline{D} \neq \underline{Q}$ throughout, so the fluid is being sheared and the torque of part (b) is not zero; with $A = 0$ we have $B = a^2\omega_a$ and the required torque is $4\pi\mu a^2\omega_a$ per unit length. Vanishing vorticity and vanishing dissipation are entirely different conditions, and this problem separates them cleanly. $\square$

PROBLEM 9 (Archimedes). Suppose a configuration κ_t of a body is completely immersed in an incompressible fluid of constant density. The body and fluid are in equilibrium in a uniform gravitational field. (a) Show that the force on the body, due to the pressure distribution at its boundary, is equal to the weight of the fluid that is displaced by the body. (b) Show that the moment of this pressure distribution, relative to the centroid of the body, is zero. *Hint:* the centroid is located at $\underline{x}_c = V^{-1} \int_{\kappa_t} \underline{x} \, dv$, where V is the volume of κ_t . (c) Revisit this problem for the case in which the body is only partly immersed in the fluid.

Solution. First the pressure field. In equilibrium the fluid is at rest, so $\underline{v} = \underline{0}$, hence $\underline{D} = \underline{Q}$ and by (9.83) $\underline{T} = -p\underline{I}$ throughout the fluid. The

equation of motion (6.109) with $\dot{\underline{v}} = \underline{0}$ and $\underline{b} = -g\underline{e}_3$ reduces to

$$\underline{0} = \operatorname{div} \underline{T} + \rho \underline{b} = -\operatorname{grad} p - \rho g \underline{e}_3, \quad \text{so} \quad \operatorname{grad} p \\ = -\rho g \underline{e}_3, \quad p = p_0 - \rho g x_3,$$

with ρ and g constant. This is the hydrostatic pressure, and the whole problem is the geometry of integrating it over a surface.

(a) The force on the body is the resultant of the tractions its boundary receives from the fluid,

$$\underline{f} = \int_{\partial\kappa_t} \underline{T} \underline{n} \, da = - \int_{\partial\kappa_t} p \underline{n} \, da,$$

$\underline{n}$ being the exterior unit normal of the body. Now p is defined only in the fluid, but the formula $p = p_0 - \rho g x_3$ makes sense everywhere, so we may use it to extend p smoothly into the region κ_t occupied by the body. The values assigned there are fictitious and no physical claim is made about them; they exist only to let the divergence theorem be applied. With that extension, (5.7)₁ gives

$$\int_{\partial\kappa_t} p \underline{n} \, da = \int_{\kappa_t} \operatorname{grad} p \, dv = \int_{\kappa_t} (-\rho g \underline{e}_3) \, dv = -\rho g V \underline{e}_3,$$

V being the volume of κ_t . Therefore

$$\underline{f} = \rho g V \underline{e}_3.$$

The force is vertical, upward, and of magnitude $\rho g V$, which is precisely the weight of the fluid that would occupy the volume V : the weight of the displaced fluid. This is Archimedes' law of buoyancy.

The argument is worth reading once more for what it does not use. It does not use the shape of the body, nor its material, nor its density, nor whether it is in equilibrium under gravity alone. All it uses is the hydrostatic pressure field on the surface and the divergence theorem, which is why the same $\rho g V$ applies to a stone, a submarine and a balloon.

(b) With $\underline{y} = \underline{x} - \underline{x}_c$, the moment about the centroid is

$$\underline{m}_c = \int_{\partial\kappa_t} \underline{y} \times (-p \underline{n}) \, da, \quad m_i = - \int_{\partial\kappa_t} \varepsilon_{ijk} y_j p n_k \, da.$$

Apply (5.4) with the scalar field $y_j p$, one j at a time, and then use the product rule together with $y_{j,k} = \delta_{jk}$:

$$\begin{aligned} m_i &= -\varepsilon_{ijk} \int_{\kappa_t} (y_j p)_{,k} dv = -\varepsilon_{ijk} \int_{\kappa_t} (\delta_{jk} p + y_j p_{,k}) dv \\ &= -\varepsilon_{ijk} \int_{\kappa_t} y_j p_{,k} dv, \end{aligned}$$

the term in δ_{jk} dropping out because $\varepsilon_{ijk}\delta_{jk} = 0$: the permutation symbol is antisymmetric in j and k while δ_{jk} is symmetric, and Lemma 2.36 disposes of the contraction. With $p_{,k} = -\rho g \delta_{k3}$,

$$m_i = \rho g \varepsilon_{ij3} \int_{\kappa_t} y_j dv.$$

But by the definition of the centroid,

$$\int_{\kappa_t} y_j dv = \int_{\kappa_t} (x_j - x_{cj}) dv = V x_{cj} - V x_{cj} = 0,$$

so

$$\underline{m}_c = \underline{0}.$$

The buoyant force therefore acts through the centroid of the body, which in this fully immersed case is also the centroid of the displaced fluid. That point is called the *centre of buoyancy*.

(c) Now let the body be partly immersed, with the free surface of the liquid at $x_3 = 0$, the submerged part of the body occupying the region R , and the wetted part of its boundary called ∂_w , the dry part ∂_d . Take

$$p = p_a - \rho g x_3 \quad \text{for } x_3 < 0, \quad p = p_a \quad \text{for } x_3 > 0,$$

the second because the air is treated as weightless and exerts the uniform atmospheric pressure p_a , as in Problem 7(c).

For the force, split the surface integral and add and subtract the constant part:

$$\begin{aligned} \underline{f} &= - \int_{\partial_d} p_a \underline{n} da - \int_{\partial_w} (p_a - \rho g x_3) \underline{n} da \\ &= -p_a \int_{\partial \kappa_t} \underline{n} da + \rho g \int_{\partial_w} x_3 \underline{n} da. \end{aligned}$$

The first integral vanishes by (5.9): a closed surface has zero total normal, which is Remark 5.6's observation that uniform pressure all round pushes a body nowhere. For the second, close ∂_w with the plane waterline section A_0 , the part of the plane $x_3 = 0$ cut off inside the body, on which the exterior normal of R is $+\underline{e}_3$ and on which $x_3 = 0$. That piece contributes nothing, so, by (5.7)₁ and $\text{grad } x_3 = \underline{e}_3$,

$$\int_{\partial_w} x_3 \underline{n} \, da = \int_{\partial R} x_3 \underline{n} \, da - \int_{A_0} x_3 \underline{e}_3 \, da = \int_R \text{grad } x_3 \, dv - \underline{0} = V_{\text{sub}} \underline{e}_3,$$

V_{sub} being the submerged volume. Hence

$$\underline{f} = \rho g V_{\text{sub}} \underline{e}_3 :$$

the weight of the displaced fluid again, with V_{sub} in place of V .

For the moment, repeat part (b) with the same splitting. The atmospheric part contributes

$$-p_a \varepsilon_{ijk} \int_{\partial \kappa_t} y_j n_k \, da = -p_a \varepsilon_{ijk} \int_{\kappa_t} \delta_{jk} \, dv = 0,$$

and the hydrostatic part, closed with A_0 as before,

$$\rho g \varepsilon_{ijk} \int_{\partial_w} x_3 y_j n_k \, da = \rho g \varepsilon_{ijk} \int_R (\delta_{3k} y_j + x_3 \delta_{jk}) \, dv = \rho g \varepsilon_{ij3} \int_R y_j \, dv,$$

the second term again killed by $\varepsilon_{ijk} \delta_{jk} = 0$. Writing $\underline{x}_b = V_{\text{sub}}^{-1} \int_R \underline{x} \, dv$ for the centroid of the submerged volume, $\int_R y_j \, dv = V_{\text{sub}} (x_{bj} - x_{cj})$, and using $\varepsilon_{ij3} u_j = (\underline{u} \times \underline{e}_3)_i$,

$$\underline{m}_c = \rho g V_{\text{sub}} (\underline{x}_b - \underline{x}_c) \times \underline{e}_3 = (\underline{x}_b - \underline{x}_c) \times \underline{f}.$$

So the moment about the body's centroid no longer vanishes; it is the moment of the buoyant force applied at the centre of buoyancy $\underline{x}_b$, which for a partly immersed body is the centroid of the submerged volume and not of the body. Part (b) is the special case $R = \kappa_t$, $\underline{x}_b = \underline{x}_c$, in which the moment vanishes.

This is the whole of elementary ship stability. The weight acts down through $\underline{x}_c$ and the buoyancy up through $\underline{x}_b$; if the two are not on the same vertical line there is a couple, which rights the vessel or capsizes it according to which way it turns. Since $\underline{x}_b$ moves as the hull heels while $\underline{x}_c$ does not, whether the couple restores or overturns depends on the shape of the waterline section, and that is why A_0 appeared in the calculation at all. $\square$

PROBLEM 10 (The thermal energy balance for an incompressible fluid). Consider an incompressible Newtonian fluid in which the absolute temperature field is $\theta(\underline{x}, t)$. Suppose the internal energy per unit mass ε is some function $\hat{\varepsilon}(\theta)$ of temperature, and suppose the internal heat supply vanishes. Derive the thermal energy balance

$$\underline{T} \cdot \underline{L} = \operatorname{div} \underline{q},$$

where $\underline{q}$ is the spatial heat flux, assuming that the temperature does not vary with time at a material point, so that the material derivative $\dot{\theta}$ is zero. Under what restrictions does this remain valid for temperature fields that are steady in the sense of the spatial description, $\theta' = 0$?

Solution. The local balance of energy is (6.168),

$$\rho \dot{\varepsilon} = \underline{T} \cdot \underline{L} - \operatorname{div} \underline{q} + \rho r.$$

The internal heat supply vanishes, $r = 0$. The internal energy is $\varepsilon = \hat{\varepsilon}(\theta)$, a function of the temperature alone, so by the chain rule for material derivatives

$$\dot{\varepsilon} = \hat{\varepsilon}'(\theta) \dot{\theta},$$

and the hypothesis $\dot{\theta} = 0$ makes this vanish. Substituting both into (6.168),

$$0 = \underline{T} \cdot \underline{L} - \operatorname{div} \underline{q}, \quad \text{i.e.} \quad \underline{T} \cdot \underline{L} = \operatorname{div} \underline{q}.$$

It is worth seeing what the balance says. For the incompressible Newtonian fluid (9.83) the left-hand side simplifies. Using $\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{D}$,

which holds because $\underline{T}$ is symmetric and $\underline{W}$ skew (Lemma 2.36),

$$\underline{T} \cdot \underline{D} = (-p\underline{I} + 2\mu\underline{D}) \cdot \underline{D} = -p \operatorname{tr} \underline{D} + 2\mu \underline{D} \cdot \underline{D} = 2\mu |\underline{D}|^2,$$

since $\operatorname{tr} \underline{D} = \operatorname{div} \underline{v} = 0$ by (9.85). So the balance reads

$$2\mu |\underline{D}|^2 = \operatorname{div} \underline{q}.$$

The left-hand side is the rate at which mechanical work is being degraded into heat by viscosity, per unit volume, and by $\mu \geq 0$ it is never negative. The right-hand side is the rate at which heat leaves a unit volume by conduction. The equation says that in a steady state the two must match exactly at every point: the fluid is a furnace whose output is set by how hard it is being sheared, and the temperature field arranges itself so that conduction carries exactly that much heat away.

With Fourier's law (9.61) and constant K , $\operatorname{div} \underline{q} = K \Delta \theta$ and the balance becomes $2\mu |\underline{D}|^2 = K \Delta \theta$. Since $K \leq 0$ by (9.80), the temperature is superharmonic wherever the fluid is being sheared, $\Delta \theta \leq 0$, which is the analytic form of the statement that the interior is hotter than a purely conducting profile would make it.

Now the restriction asked for. If instead the field is steady in the spatial description, $\theta' = 0$, then by (2.20)

$$\dot{\theta} = \theta' + \operatorname{grad} \theta \cdot \underline{v} = \operatorname{grad} \theta \cdot \underline{v},$$

which need not vanish. The energy balance is then

$$\rho \hat{\varepsilon}'(\theta) \operatorname{grad} \theta \cdot \underline{v} = \underline{T} \cdot \underline{L} - \operatorname{div} \underline{q},$$

and the stated form survives if and only if the extra term is zero. Excluding the degenerate possibilities $\hat{\varepsilon}' = 0$, which would be a fluid with no heat capacity, and $\underline{v} = \underline{0}$, the condition is

$$\operatorname{grad} \theta \cdot \underline{v} = 0 :$$

the velocity must be everywhere tangent to the isothermal surfaces, so that no particle is convected from one isotherm to another. Streamlines must lie in surfaces of constant temperature.

This is exactly the distinction of Remark 5.8 of Chapter 5 between the two clocks. A field steady at a fixed place is not steady for a fixed particle unless the particle stays on a level set of the field, and the convective term $\text{grad } \theta \cdot \underline{v}$ is the toll for crossing. The condition is not artificial: it holds exactly in the circular Couette flow of Problem 8 with $\theta = \theta(r)$, since there $\text{grad } \theta = \theta'(r)\underline{e}_r$ by (1.122) while $\underline{v} = rf\underline{e}_\theta$, and $\underline{e}_r \cdot \underline{e}_\theta = 0$. That is what makes Problem 11 solvable in the form stated. $\square$

PROBLEM 11 (The temperature field between rotating cylinders). For the Newtonian fluid and the Fourier law, solve the thermal energy balance equation for the flow of Problem 8, assuming the temperature θ to be a function of r . Assume the viscosity μ and conductivity K to be constant; this is realistic if the temperature does not vary significantly at a material point. The inner and outer cylinders are held at the fixed temperatures θ_a and θ_b , respectively.

Solution. The balance applies in the form derived. With $\theta = \theta(r)$ and $\underline{v} = rf(r)\underline{e}_\theta$, Problem 10 has just shown that $\text{grad } \theta \cdot \underline{v} = 0$, so the balance $2\mu|\underline{D}|^2 = \text{div } \underline{q}$ applies as it stands.

The dissipation comes first. From Problem 8, $\underline{D} = \gamma(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r)$ with $\gamma = \frac{1}{2}rf'$, and $f = A + B/r^2$ gives $f' = -2B/r^3$, so

$$rf' = -\frac{2B}{r^2}, \quad \gamma = -\frac{B}{r^2}.$$

Its magnitude follows from Theorem 1.58, $(\underline{a} \otimes \underline{b}) \cdot (\underline{c} \otimes \underline{d}) = (\underline{a} \cdot \underline{c})(\underline{b} \cdot \underline{d})$, applied to the four cross terms with $\{\underline{e}_r, \underline{e}_\theta\}$ orthonormal:

$$|\underline{D}|^2 = \gamma^2[1 + 0 + 0 + 1] = 2\gamma^2 = \frac{2B^2}{r^4},$$

so that the heat generated per unit volume is

$$2\mu|\underline{D}|^2 = \frac{4\mu B^2}{r^4}.$$

Note that it is largest at the inner cylinder and falls off as r^{-4} : the shearing is concentrated where the gap curves most.

Next the conduction. With $\underline{q} = K \text{grad } \theta$, K constant, and θ a function of r alone, (1.122) gives $\text{grad } \theta = \theta'(r)\underline{e}_r$ and the polar divergence of Example 1.99 gives

$$\text{div } \underline{q} = K \left(\theta'' + \frac{\theta'}{r} \right) = \frac{K}{r} (r\theta')'.$$

The balance is therefore

$$\frac{K}{r} (r\theta')' = \frac{4\mu B^2}{r^4}, \quad \text{i.e.} \quad (r\theta')' = \frac{4\mu B^2}{K} r^{-3}.$$

Integrating once,

$$r\theta' = \frac{4\mu B^2}{K} \cdot \frac{r^{-2}}{-2} + C_1 = -\frac{2\mu B^2}{Kr^2} + C_1, \quad \theta' = -\frac{2\mu B^2}{Kr^3} + \frac{C_1}{r},$$

and again,

$$\theta(r) = \frac{\mu B^2}{Kr^2} + C_1 \ln r + C_2,$$

since $\int -2\mu B^2 K^{-1} r^{-3} dr = \mu B^2 K^{-1} r^{-2}$.

It is convenient to name the particular integral. Put

$$\Theta(r) = \theta(r) - \frac{\mu B^2}{Kr^2},$$

so that $\Theta = C_1 \ln r + C_2$ is the general radial solution of $\Delta\Theta = 0$. The boundary conditions $\theta(a) = \theta_a$ and $\theta(b) = \theta_b$ become

$$\Theta(a) = \theta_a - \frac{\mu B^2}{Ka^2}, \quad \Theta(b) = \theta_b - \frac{\mu B^2}{Kb^2},$$

and the harmonic function taking prescribed values at $r = a$ and $r = b$ is

$$\Theta(r) = \Theta(a) + [\Theta(b) - \Theta(a)] \frac{\ln(r/a)}{\ln(b/a)}.$$

Hence

$$\begin{aligned} \theta(r) = & \frac{\mu B^2}{Kr^2} + \left(\theta_a - \frac{\mu B^2}{Ka^2} \right) \\ & + \left[\left(\theta_b - \frac{\mu B^2}{Kb^2} \right) - \left(\theta_a - \frac{\mu B^2}{Ka^2} \right) \right] \frac{\ln(r/a)}{\ln(b/a)}, \end{aligned}$$

with

$$B = \frac{a^2 b^2 (\omega_a - \omega_b)}{b^2 - a^2}$$

from Problem 8.

The solution is the sum of two pieces and they have different origins.

The logarithmic piece is what one would get with no flow at all: it is the steady conduction profile between two cylinders held at different temperatures, and it is the two-dimensional analogue of the linear profile in a slab. It carries all the dependence on θ_a and θ_b .

The piece proportional to $\mu B^2/K$ is viscous heating, and it vanishes when $B = 0$, that is, when the two cylinders turn at the same rate and the fluid rotates rigidly. That is the expected answer, since by Problem 7 a rigid rotation has $\underline{D} = \underline{O}$ and dissipates nothing.

The sign of the heating term is worth checking, since K is negative by (9.80). Take $\theta_a = \theta_b = \theta_0$, so that the walls are at a common temperature and any departure must be due to dissipation. Then

$$\theta(r) - \theta_0 = \frac{\mu B^2}{K} \left[\left(\frac{1}{r^2} - \frac{1}{a^2} \right) + \left(\frac{1}{a^2} - \frac{1}{b^2} \right) \frac{\ln(r/a)}{\ln(b/a)} \right].$$

Write $s = \ln r$. The bracket is a convex function of s , namely e^{-2s} , plus a linear function of s , and it vanishes at both ends $r = a$ and $r = b$; a convex function vanishing at the ends of an interval is non-positive inside it. So the bracket is ≤ 0 , and since $K \leq 0$ and $\mu \geq 0$ the product is ≥ 0 :

$$\theta(r) \geq \theta_0 \quad \text{for } a \leq r \leq b.$$

The interior is warmer than the walls, and heat flows outward to both of them. That is what the second law demanded, and it is reassuring to see the two sign conditions $\mu \geq 0$ and $K \leq 0$ cooperating to produce it: had either sign been reversed the fluid would cool itself by being stirred. $\square$

PROBLEM 12 (A spherically symmetric flow). An incompressible Newtonian fluid with a constant shear viscosity experiences a spherically symmetric velocity field $\underline{v} = v(r)\underline{e}_\rho$ in the spatial description, where

$r = |\underline{x}|$ and $\underline{e}_\rho = r^{-1}\underline{x}$. There are no body forces acting, and the problem is steady in the sense that $\underline{v}' = \underline{0}$ and $\rho' = 0$. (a) Obtain an expression for the spatial velocity gradient $\underline{L}$. Assuming $v(a) = v_a$ to be a known constant, find the function $v(r)$ consistent with the incompressibility constraint, that is, consistent with the motion being isochoric. (b) What is the material acceleration? Consider a spherical material surface of radius a at time $t = 0$. Assuming $v_a > 0$, what is the time at which the radius of the material surface doubles? *Hint:* switch to the referential description. (c) Solve the equation of motion to obtain the pressure field $p(\underline{x})$, assuming that $\underline{T} \rightarrow -p_\infty \underline{I}$ as $r \rightarrow \infty$, where p_∞ is a known constant. *Hint:* use the Cartesian-coordinate formula for the divergence of a tensor, together with $r_{,i} = r^{-1}x_i$.

Solution. (a) Write the field in Cartesian components. With $\underline{e}_\rho = \underline{x}/r$,

$$v_i = \phi(r) x_i, \quad \text{where} \quad \phi(r) = \frac{v(r)}{r}.$$

By Theorem 1.86, $L_{ij} = v_{i,j}$, and using the product rule with $x_{i,j} = \delta_{ij}$ and $r_{,j} = x_j/r$ from (1.124),

$$L_{ij} = \phi'(r) r_{,j} x_i + \phi \delta_{ij} = \phi \delta_{ij} + \frac{\phi'}{r} x_i x_j,$$

that is,

$$\underline{L} = \phi \underline{I} + r\phi' \underline{e}_\rho \otimes \underline{e}_\rho = \frac{v}{r} \underline{I} + \left(v' - \frac{v}{r}\right) \underline{e}_\rho \otimes \underline{e}_\rho,$$

where $\underline{x} \otimes \underline{x} = r^2 \underline{e}_\rho \otimes \underline{e}_\rho$ was used, and where $r\phi' = r(v/r)' = r \cdot \frac{v'r-v}{r^2} = v' - v/r$.

Two things are visible at once. The tensor is symmetric, so

$$\underline{D} = \underline{L}, \quad \underline{W} = \underline{Q} :$$

the flow is irrotational, which is what one expects of a purely radial field. And its trace is

$$\text{tr } \underline{L} = 3\phi + r\phi' |\underline{e}_\rho|^2 = 3\frac{v}{r} + v' - \frac{v}{r} = v' + \frac{2v}{r},$$

using $\text{tr}(\underline{e}_\rho \otimes \underline{e}_\rho) = |\underline{e}_\rho|^2 = 1$ from Theorem 1.51.

The constraint (9.85) is $\text{div } \underline{v} = \text{tr } \underline{L} = 0$, so

$$v' + \frac{2v}{r} = 0 \quad \Longleftrightarrow \quad (r^2 v)' = r^2 v' + 2rv = r(rv' + 2v) = 0.$$

Hence $r^2 v$ is constant, and evaluating it at $r = a$ gives $r^2 v = a^2 v_a$:

$$v(r) = \frac{a^2 v_a}{r^2}.$$

This is the flow from a point source: the volume crossing any sphere of radius r per unit time is $4\pi r^2 v(r) = 4\pi a^2 v_a$, the same for every r , which is what incompressibility means for a radial flow. Substituting $v' = -2v/r$ back into $\underline{L}$,

$$\underline{L} = \underline{D} = \frac{a^2 v_a}{r^3} (\underline{I} - 3 \underline{e}_\rho \otimes \underline{e}_\rho),$$

whose trace is $\frac{a^2 v_a}{r^3} (3 - 3) = 0$, as it must be.

(b) By (2.21) and $v' = 0$,

$$\dot{\underline{v}} = \underline{L} \underline{v} = \underline{L} (v \underline{e}_\rho) = v \left[\frac{v}{r} \underline{e}_\rho + \left(v' - \frac{v}{r} \right) \underline{e}_\rho \right] = v v' \underline{e}_\rho,$$

using $(\underline{e}_\rho \otimes \underline{e}_\rho) \underline{e}_\rho = |\underline{e}_\rho|^2 \underline{e}_\rho = \underline{e}_\rho$. With $v = a^2 v_a / r^2$ and $v' = -2a^2 v_a / r^3$,

$$\dot{\underline{v}} = -\frac{2a^4 v_a^2}{r^5} \underline{e}_\rho.$$

The acceleration points inward although the flow is outward: a particle moving away from the origin is decelerating, because the same volume flux is spread over an ever larger sphere.

For the doubling time, switch to the referential description as the hint suggests. A material point that starts at radius R_0 stays on the radius through its starting point, by symmetry, and its distance $R(t)$ from the origin obeys

$$\dot{R} = v(R) = \frac{a^2 v_a}{R^2},$$

which is dR/dt evaluated following the particle, that is, the material derivative. Separating variables,

$$R^2 dR = a^2 v_a dt, \quad \frac{R^3}{3} = a^2 v_a t + \text{const.}$$

The material surface in question is the sphere $R = a$ at $t = 0$, so the constant is $a^3/3$ and

$$R(t)^3 = a^3 + 3a^2 v_a t.$$

Every point of the surface obeys the same equation, so the surface remains a sphere, which is why a single scalar suffices. Setting $R = 2a$,

$$8a^3 = a^3 + 3a^2 v_a t \quad \implies \quad t = \frac{7a}{3v_a}.$$

(c) Follow the hint and compute $\text{div } \underline{D}$ in Cartesian components. From part (a),

$$D_{ij} = a^2 v_a \left[\frac{\delta_{ij}}{r^3} - \frac{3x_i x_j}{r^5} \right],$$

and, using $r_{,j} = x_j/r$ throughout,

$$\begin{aligned} \left(\frac{1}{r^3} \right)_{,j} &= -\frac{3}{r^4} r_{,j} = -\frac{3x_j}{r^5}, \\ \left(\frac{x_i x_j}{r^5} \right)_{,j} &= \frac{x_i + 3x_i}{r^5} - \frac{5x_i x_j x_j}{r^6} \cdot \frac{1}{r} \\ &= \frac{4x_i}{r^5} - \frac{5x_i}{r^5} = -\frac{x_i}{r^5}, \end{aligned}$$

the middle expression using $\delta_{ij} x_j = x_i$, $x_i \delta_{jj} = 3x_i$ and $x_j x_j = r^2$. Hence

$$D_{ij,j} = a^2 v_a \left[-\frac{3x_i}{r^5} - 3 \left(-\frac{x_i}{r^5} \right) \right] = 0, \quad \text{that is} \quad \text{div } \underline{D} = 0.$$

The viscous force per unit volume vanishes identically. Equivalently, $\Delta \underline{y} = \underline{0}$, which is visible directly from $v_i = a^2 v_a x_i / r^3 = -a^2 v_a (1/r)_{,i}$ together with the harmonicity of $1/r$ away from the origin.

Therefore $\operatorname{div} \underline{T} = -\operatorname{grad} p$, and the equation of motion (6.109) with $\underline{b} = \underline{0}$ gives

$$\operatorname{grad} p = -\rho \dot{\underline{v}} = \frac{2\rho a^4 v_a^2}{r^5} \underline{e}_\rho.$$

Since $\underline{e}_\rho = \operatorname{grad} r$ by (1.124), this is $p_{,r} = 2\rho a^4 v_a^2 r^{-5}$ with no angular dependence, and integrating,

$$p = -\frac{\rho a^4 v_a^2}{2r^4} + C,$$

which may be verified by differentiating: $\frac{d}{dr}(-\frac{1}{2}\rho a^4 v_a^2 r^{-4}) = 2\rho a^4 v_a^2 r^{-5}$.

The constant is fixed by the condition at infinity. As $r \rightarrow \infty$ the stretching $\underline{D}$ decays like r^{-3} , so $\underline{T} \rightarrow -p\underline{I}$, and matching with $\underline{T} \rightarrow -p_\infty \underline{I}$ requires $p \rightarrow p_\infty$. Hence $C = p_\infty$ and

$$p(\underline{x}) = p_\infty - \frac{\rho a^4 v_a^2}{2r^4}, \quad \underline{T} = -p \underline{I} + \frac{2\mu a^2 v_a}{r^3} (\underline{I} - 3\underline{e}_\rho \otimes \underline{e}_\rho).$$

There is a check, and a surprise. Because $\underline{W} = \underline{0}$, the tensor $\underline{L}$ is symmetric and

$$\dot{\underline{v}} = \underline{L}\underline{v} = \underline{L}^t \underline{v} = (\operatorname{grad} \underline{v})^t \underline{v} = \frac{1}{2} \operatorname{grad} (|\underline{v}|^2),$$

the last step being the identity computed in Problem 1 of Chapter 5, $[(\operatorname{grad} \underline{v})^t \underline{v}]_i = v_{j,i} v_j = \frac{1}{2} (v_j v_j)_{,i}$. So the equation of motion integrates at once to

$$p + \frac{1}{2} \rho |\underline{v}|^2 = \text{const},$$

which is Bernoulli's theorem; with $|\underline{v}| = a^2 v_a / r^2$ it reproduces the boxed pressure exactly.

The surprise is that Bernoulli's theorem, an inviscid result, is exact here for a viscous fluid. The reason is that the viscous force $\mu \Delta \underline{v}$ vanishes identically for this particular field, even though the viscous stress $2\mu \underline{D}$ does not. Viscosity is present in the stress, and a sphere of fluid does feel a viscous traction, but the tractions on opposite faces of every material element balance, so the net viscous force is zero and the motion is governed by the Euler equations. Such flows are rare and valuable: they are the cases in which the nonlinear and the viscous terms of the Navier–Stokes equations can both be handled exactly. $\square$

Chapter 10

Kinematic Constraints

Not every motion is available to every material. Water resists a change of volume so much more stiffly than it resists a change of shape that, over the range of pressures met in ordinary problems, it is both more accurate and far more convenient to forbid the volume change outright than to model it. A stiff fibre embedded in a soft matrix tells the same story along a line rather than in bulk: the fibre will stretch, but by so little that setting its stretch to one costs less error than any elastic law for it would. Idealising a very stiff response as *no* response is what this chapter is about.

The idealisation is not free, and the price is the whole content of the chapter. If the material cannot perform a certain motion, then something must stop it, and that something is a stress. It is a stress of a new kind: no constitutive equation produces it, because a constitutive equation answers “what stress does this deformation cause”, and here the deformation in question never happens. What determines it instead is the equations of motion together with the boundary and initial data. The pressure in an incompressible fluid is the case everyone meets first, and it is worth saying at the outset what will be proved: that pressure is not a material property at all.

Guided by the incompressible fluid, we consider *local* constraints of the form

$$g_{\kappa}(\underline{F}) = 0, \tag{10.1}$$

from the perspective of an observer O , where g_{κ} is a scalar-valued func-

tion, $\tilde{F}$ is the deformation gradient of Section 2.4, and the subscript κ records the reference configuration relative to which $\tilde{F}$, and hence g_κ , is defined. For incompressibility, $g_\kappa(\tilde{F}) = \det \tilde{F} - 1$.

REMARK 10.1 (what a constraint is, and what it is not). Three distinctions are worth fixing before anything is computed, because each of them is a way of misreading (10.1).

A constraint is not a boundary condition. A boundary condition restricts the motion of particular material points, those on $\partial\kappa$, and it is data supplied with the problem. Equation (10.1) is imposed at *every* $\tilde{X} \in \kappa$ and at every t , and it is part of the description of the material, not of the problem. Two different problems for the same incompressible fluid have different boundary conditions and the same constraint.

A constraint is not a constitutive equation. A constitutive equation such as (8.33) is a map from the deformation to the stress; it answers a question. A constraint answers nothing; it deletes part of the domain on which the question may be asked. This is why the two must be stated separately, and why (10.14) below will split the stress in two rather than modifying $\hat{T}_\kappa$.

The constraint is on $\tilde{F}$, not on χ_κ . The deformation gradient is the local, first-order content of the motion at a point, so (10.1) restricts what the material may do in an arbitrarily small neighbourhood of $\tilde{X}$, which is what a statement about material response has to do. A restriction on χ_κ itself would say where the particle is allowed to be, which is a statement about the problem and not about the material. The same locality argument opened Chapter 8.

10.1 Observer consensus

Chapter 7 required all observers to agree on material response. Because a constraint characterises the nature of that response, we require that all observers agree that a constraint of the form (10.1) is in force. Thus the constraint, as perceived by an observer O^+ , is

$$g_{\kappa^+}^+(\tilde{F}^+) = 0, \quad (10.2)$$

where

$$g_{\kappa^+}^+(F^+) = g_{\kappa}(F). \quad (10.3)$$

Equation (10.3) is the substantive assumption and (10.2) is its consequence. The two observers are not merely required to reach the same verdict, zero or non-zero; they are required to compute the same *number*. That is what “agreeing on material response” means when the response is described by a scalar function.

From (7.17) the two deformation gradients are related by $\tilde{F}^+ = Q(t)\tilde{F}K^t$, in which $Q(t): \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$ is the orthogonal tensor of the change of observer and $K: \hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^{3+}$ is the fixed orthogonal tensor of (7.15) relating the two reference configurations. Solving for $\tilde{F}$,

$$Q^t\tilde{F}^+K = Q^tQ\tilde{F}K^tK = \underline{I}\tilde{F}\hat{I} = \tilde{F},$$

where $Q^tQ = \underline{I}$ is the identity on $\mathbb{E}^3$ and $K^tK = \hat{I}$ the identity on $\hat{\mathbb{E}}^3$, both by orthogonality (Proposition 1.48). Substituting into (10.3),

$$g_{\kappa^+}^+(F^+) = g_{\kappa}(Q^t\tilde{F}^+K). \quad (10.4)$$

As in Chapter 9, we now impose this requirement for *two* relative motions of the observers, both viewed from O^+ . This is the device that converts a statement relating two observers into a restriction on the single function g_{κ} used by one of them. Let the two relative motions be characterised by orthogonal tensors $Q_1(t)$ and $Q_2(t)$ in the sense of (9.6). Observer O^+ perceives one and the same $\tilde{F}^+$, so (10.4) applies twice with the same left side and gives

$$g_{\kappa}(Q_2^t\tilde{F}^+K) = g_{\kappa^+}^+(F^+) = g_{\kappa}(Q_1^t\tilde{F}^+K). \quad (10.5)$$

Here we assume $Q_2(t_0) = Q_1(t_0)$, so that K is the same for both relative motions; by (7.15)₁, K is built from Q at the single instant t_0 and from the two shifters, so it is this equality at t_0 , and nothing else, that is needed. Let $\tilde{Q} = Q_2^tQ_1: \mathbb{E}^3 \rightarrow \mathbb{E}^3$, exactly as in (9.11), and let

$$\tilde{F} = Q_1^t\tilde{F}^+K. \quad (10.6)$$

Then, using $\underline{\underline{Q}}_1 \underline{\underline{Q}}_1^t = \underline{\underline{I}}^+$,

$$\underline{\underline{Q}} \underline{\underline{F}} = \underline{\underline{Q}}_2^t \underline{\underline{Q}}_1 \underline{\underline{Q}}_1^t \underline{\underline{F}}^+ \underline{\underline{K}} = \underline{\underline{Q}}_2^t \underline{\underline{I}}^+ \underline{\underline{F}}^+ \underline{\underline{K}} = \underline{\underline{Q}}_2^t \underline{\underline{F}}^+ \underline{\underline{K}},$$

and taking determinants of the two ends, with $\det(\underline{\underline{A}}\underline{\underline{B}}) = \det \underline{\underline{A}} \det \underline{\underline{B}}$ and $\det \underline{\underline{Q}}_2^t = \det \underline{\underline{Q}}_2$,

$$\underline{\underline{Q}}_2^t \underline{\underline{F}}^+ \underline{\underline{K}} = \underline{\underline{Q}} \underline{\underline{F}}; \quad \text{hence} \quad (\det \underline{\underline{Q}}) J = (\det \underline{\underline{Q}}_2)(\det \underline{\underline{K}}) J^+, \quad (10.7)$$

in which $J = \det \underline{\underline{F}}$ and $J^+ = \det \underline{\underline{F}}^+$.

Recall the discussion leading to (7.19), on the choice of occupiable reference configurations. Both $\underline{\underline{Q}}_2$ and $\underline{\underline{K}}$ are orthogonal, so each determinant is ± 1 and their product is ± 1 ; occupiability forces $J > 0$ and $J^+ > 0$, and $J^+ = J(\det \underline{\underline{Q}}_2)(\det \underline{\underline{K}})$ then leaves only the choice $(\det \underline{\underline{Q}}_2)(\det \underline{\underline{K}}) = 1$, whence $J^+ = J$. Putting both into (10.7)₂ gives $(\det \underline{\underline{Q}}) J = J$, and since $J > 0$ we may divide by it:

$$\det \underline{\underline{Q}} = 1.$$

So $\underline{\underline{Q}}$ is a rotation. With (10.6) and (10.7)₁, equation (10.5) now reads

$$g_\kappa(\underline{\underline{Q}} \underline{\underline{F}}) = g_\kappa(\underline{\underline{F}}) \quad (10.8)$$

for arbitrary rotations $\underline{\underline{Q}}$. The two observers have been eliminated entirely: (10.8) is a statement about the one function g_κ that $\underline{\underline{O}}$ uses.

Reduction of (10.8)

To obtain a *necessary* condition we exploit the arbitrariness of $\underline{\underline{Q}}$ by choosing it as awkwardly as possible for the left side of (10.8). Take

$$\underline{\underline{Q}} = \underline{\underline{1}} \underline{\underline{R}}^t,$$

where $\underline{\underline{1}}$ is the shifter of Definition 2.23 and $\underline{\underline{R}}$ is the two-point rotation in the polar factorisation (3.32), $\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}}$, of Theorem 3.19. This is a legitimate choice. Since $\underline{\underline{R}}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ we have $\underline{\underline{R}}^t: \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, and $\underline{\underline{1}}: \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, so the product maps $\mathbb{E}^3$ to $\mathbb{E}^3$ as $\underline{\underline{Q}}$ must; it is orthogonal as

a product of orthogonal tensors, and its determinant is $(\det \underline{1})(\det \underline{R}) = 1$ because both factors are proper. With it,

$$\underline{Q}\underline{F} = \underline{1}\underline{R}^t\underline{R}\underline{U} = \underline{1}\hat{\underline{I}}\underline{U} = \underline{1}\underline{U},$$

the middle step by $\underline{R}^t\underline{R} = \hat{\underline{I}}$. So (10.8) furnishes

$$g_\kappa(\underline{F}) = g_\kappa(\underline{1}\hat{\underline{I}}\underline{U}) = g_\kappa(\underline{1}\underline{U}).$$

Because $\underline{1}$ is fixed in the frame of O , the right side is a function of $\underline{U}$ alone. The right stretch tensor is in turn uniquely determined by the right Cauchy–Green tensor $\underline{C} = \underline{F}^t\underline{F}$ through (3.38), $\underline{U} = \sqrt{\underline{C}}$, the square root being unique by Theorem 3.16. We therefore conclude that

$$g_\kappa(\underline{F}) = \bar{g}_\kappa(\underline{C}) \quad (10.9)$$

for some function $\bar{g}_\kappa$.

Conversely, suppose (10.9) holds. Then for every rotation $\underline{Q}$,

$$g_\kappa(\underline{Q}\underline{F}) = \bar{g}_\kappa((\underline{Q}\underline{F})^t\underline{Q}\underline{F}) = \bar{g}_\kappa(\underline{F}^t\underline{Q}^t\underline{Q}\underline{F}) = \bar{g}_\kappa(\underline{F}^t\underline{I}\underline{F}) = \bar{g}_\kappa(\underline{F}^t\underline{F}) = g_\kappa(\underline{F}),$$

so (10.9) is both necessary and sufficient for (10.8).

REMARK 10.2 (what (10.9) forbids). The content of (10.9) is negative and it is worth stating that way: a kinematic constraint may not see the rotation $\underline{R}$. Suppose someone proposed the constraint $R_{11} = 1$, that a certain material line never turn out of a certain plane. Chapter 3 showed that $\underline{R}$ is the part of $\underline{F}$ that carries no stretch at all, and Chapter 7 showed, in (7.29)₃, that $\underline{R}^+ = \underline{Q}\underline{R}\underline{K}^t$: the rotation is precisely the part of $\underline{F}$ that a change of observer alters. So the proposed constraint would be satisfied for one observer and violated for another, watching the same body. It would make “this material is inextensible” a statement about who is looking. Equation (10.9) is the exact statement of what survives that objection: only $\underline{C}$, and by Proposition 2.27 and Remark 2.29 $\underline{C}$ is the part of the deformation that measures lengths and angles of material fibres, which is what two observers can compare.

The constraint from the perspective of O^+ now follows from (10.4). Here $\tilde{Q}: \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$ is once more the orthogonal tensor of the change of observer, and $\tilde{K}$ is a fixed parameter:

$$\begin{aligned} g_{\kappa^+}^+(\tilde{F}^+) &= g_{\kappa}(\tilde{Q}^t \tilde{F}^+ \tilde{K}) = \bar{g}_{\kappa}((\tilde{Q}^t \tilde{F}^+ \tilde{K})^t \tilde{Q}^t \tilde{F}^+ \tilde{K}) \\ &= \bar{g}_{\kappa}(\tilde{K}^t (\tilde{F}^+)^t \tilde{Q} \tilde{Q}^t \tilde{F}^+ \tilde{K}) = \bar{g}_{\kappa}(\tilde{K}^t (\tilde{F}^+)^t \tilde{I}^+ \tilde{F}^+ \tilde{K}) \quad (10.10) \\ &= \bar{g}_{\kappa}(\tilde{K}^t (\tilde{F}^+)^t \tilde{F}^+ \tilde{K}) = \bar{g}_{\kappa}(\tilde{K}^t \tilde{C}^+ \tilde{K}) = \bar{g}_{\kappa^+}^+(\tilde{C}^+). \end{aligned}$$

The transpose of a product reverses the order, which produced the second line; $\tilde{Q}\tilde{Q}^t = \tilde{I}^+$ is orthogonality read in the other direction from the one used above, and it is the step at which the observer disappears. What remains, $\tilde{K}^t \tilde{C}^+ \tilde{K}$, involves only the two reference configurations, and $\tilde{K}$ does not depend on time. Comparing with (7.30)₁, $\tilde{C}^+ = \tilde{K} \tilde{C} \tilde{K}^t$, the last line simply says $\bar{g}_{\kappa^+}^+(\tilde{C}^+) = \bar{g}_{\kappa}(\tilde{C})$, which is (10.3) again with $\tilde{C}$ in place of $\tilde{F}$.

Examples

1. Incompressibility

Because $J > 0$, the constraint $J = 1$ associated with incompressibility is equivalent to $1 = J^2$, and

$$J^2 = (\det \tilde{F})^2 = (\det \tilde{F}^t)(\det \tilde{F}) = \det(\tilde{F}^t \tilde{F}) = \det \tilde{C},$$

using $\det \tilde{A}^t = \det \tilde{A}$ and the product rule for determinants. Hence

$$\bar{g}_{\kappa}(\tilde{C}) = \det \tilde{C} - 1. \quad (10.11)$$

The step $J = 1 \iff J^2 = 1$ needs $J > 0$ and would be false without it; $J > 0$ is guaranteed for a motion by Remark 1.53 and the requirement that material not turn itself inside out. Note also that (10.11) is manifestly of the form (10.9), as it had to be: $\det \tilde{F}$ by itself is *not* an admissible constraint function in the sense of (10.9), but on rotations $\det \tilde{Q} = 1$ so $\det(\tilde{Q}\tilde{F}) = \det \tilde{F}$ and (10.8) holds anyway. The two statements agree, as the equivalence just proved says they must.

2. Inextensibility

In the engineering theory of fibre-reinforced composites the fibres are idealised as continuously distributed material curves in κ with a unit tangent field $\underline{M}(\underline{X})$. If the fibres are very stiff compared with the matrix in which they are embedded, their stretches μ stay close to unity in any deformation the composite can survive, and it is then appropriate to impose $\mu = 1$ exactly. By (2.83), $\mu^2 = \underline{M} \cdot \underline{C} \underline{M}$, so with $\mu > 0$ the constraint $\mu = 1$ is $\underline{M} \cdot \underline{C} \underline{M} = 1$ and

$$\bar{g}_\kappa(\underline{C}) = \underline{M} \cdot \underline{C} \underline{M} - 1. \quad (10.12)$$

Here $\underline{M}$ is a material vector in the sense of Definition 3.25: it labels a fibre, and by Remark 3.26 it satisfies $\dot{\underline{M}} = \underline{0}$. That fact will be used when (10.12) is differentiated.

3. Rigidity

If the body is rigid then $\underline{F}$ is a spatially uniform rotation (Section 3.7), so $\underline{C} = \underline{F}^t \underline{F} = \underline{\hat{I}}$. Because $\underline{C}$ is symmetric, this single tensor equation is equivalent to six scalar constraints,

$$\bar{g}_{(\kappa)AB} = C_{AB} - \delta_{AB}, \quad (10.13)$$

one for each of the six independent entries of the symmetric matrix (C_{AB}) .

REMARK 10.3 (six is the maximum). There can be no more than six independent constraints, because there are at most six independent entries in (C_{AB}) . Rigidity is therefore the *maximal* constraint: it fixes every component of that matrix and leaves the material no freedom whatever beyond a rigid motion. The number six is the dimension of sym , computed in Chapter 1 by counting $C_{11}, C_{22}, C_{33}, C_{12}, C_{13}, C_{23}$, and the same six will reappear in (10.19) as the maximum number of multipliers. That is not a coincidence and Section 10.2 explains why: each independent constraint deletes one dimension from the space of admissible stretchings and pays for it with one dimension of reactive stress, and both spaces sit inside the same six-dimensional sym .

10.2 Modification of the stress to accommodate constraints

We have seen, in connection with the incompressible Newtonian fluid of Chapter 9, that the stress decomposes into the sum of a reactive part $-p\mathbb{I}$ and a constitutive part determined by the material. The salient feature of the reactive part is that it makes no contribution to the stress power density whenever the constraint is operative: by Definition 1.57, $-p\mathbb{I} \cdot \underline{\underline{D}} = -p \operatorname{tr} \underline{\underline{D}}$, which vanishes whenever $\operatorname{tr} \underline{\underline{D}} = 0$. We extend this idea to general constraints by assuming that the net Cauchy stress is

$$\underline{\underline{T}} = \underline{\underline{N}} + \underline{\underline{G}}, \quad (10.14)$$

where $\underline{\underline{G}}$ is a symmetric-tensor-valued constitutive function appropriate to the material and the conditions at hand, and $\underline{\underline{N}}$ is a symmetric tensor such that $\underline{\underline{N}} \cdot \underline{\underline{D}}$ vanishes in any motion compatible with the constraint. Once the structure of $\underline{\underline{N}}$ has been determined, any indeterminacy remaining in it must be such as to secure a solution, if one exists, to the equations of motion together with the boundary and initial conditions.

REMARK 10.4 (why worklessness, and why it is an assumption). Two questions press here. Why should the reaction do no work, and is that a theorem or a hypothesis?

Why worklessness is the right requirement. The stress power of a part S is $\mathcal{S}(S, t) = \int_{P_t} \underline{\underline{T}} \cdot \underline{\underline{L}} \, dv$ by (6.153). Since $\underline{\underline{T}}$ is symmetric and $\underline{\underline{W}}$ is skew, $\underline{\underline{T}} \cdot \underline{\underline{W}} = 0$ by Lemma 2.36, so with $\underline{\underline{L}} = \underline{\underline{D}} + \underline{\underline{W}}$ from (4.7) the stress power density is $\underline{\underline{T}} \cdot \underline{\underline{L}} = \underline{\underline{T}} \cdot \underline{\underline{D}}$. Under (10.14) this is $\underline{\underline{N}} \cdot \underline{\underline{D}} + \underline{\underline{G}} \cdot \underline{\underline{D}}$. If $\underline{\underline{N}} \cdot \underline{\underline{D}}$ vanished only sometimes, the reaction would be feeding energy into or draining it from the body, and by the energy balance (6.168) that energy would have to show up as heat or internal energy. A constraint is an idealisation of a very stiff *elastic* response, which stores energy and gives it back and dissipates none; in the limit the stored energy goes to zero along with the deformation that stores it. Requiring $\underline{\underline{N}} \cdot \underline{\underline{D}} = 0$ is the exact expression of that limit.

The same conclusion arrives from the thermodynamic side. The Clausius–Duhem inequality carries the stress only through $\underline{T} \cdot \underline{L}$, so $\underline{N}$ drops out of it identically. Thermodynamics therefore places no restriction whatever on $\underline{N}$, which is precisely the sentence in Chapter 9 that the spherical part of the stress in an incompressible fluid “is constitutively indeterminate and hence outside the purview of our earlier discussion about observer consensus concerning material response”. Worklessness and constitutive indeterminacy are two readings of one fact.

It is an assumption. Nothing proved so far forces (10.14) on us; a theory in which the constraint reaction did work could be written down and would not be self-contradictory. It would, however, require extra constitutive information to say how much work, and the whole economy of the constraint idea is that it requires none. This is the continuum version of the principle of d’Alembert, that the forces of constraint in a mechanical system do no virtual work, and it is adopted for the same reason: it is the assumption that makes constraints cost nothing to describe.

Suppose, then, that the constraint is operative at a particular material point throughout an interval I of time. We require that

$$\underline{N} \cdot \underline{D} = 0 \quad \text{for all } \underline{D} \text{ such that } \bar{g}_\kappa(\underline{C}(t)) = 0; \quad t \in I. \quad (10.15)$$

Because $\bar{g}_\kappa$ vanishes identically on I , the function $t \mapsto \bar{g}_\kappa(\underline{C}(t))$ is the zero function there, and the derivative of the zero function is zero. Writing g in place of $\bar{g}_\kappa$ to keep the notation tractable, and using the chain rule for a scalar-valued function of a tensor,

$$0 = \dot{g} = g_{\underline{C}} \cdot \dot{\underline{C}} = \text{tr} \left((g_{\underline{C}}) \dot{\underline{C}} \right), \quad (10.16)$$

where $g_{\underline{C}} = (g_{\underline{C}})^t$ is the symmetric referential tensor-valued gradient of g (Sections B.2 and B.3 of Appendix B), defined by $\text{d}g = g_{\underline{C}} \cdot \text{d}\underline{C}$ and existing uniquely by the Riesz argument of Theorem 1.15 applied on the inner-product space Lin .

Three points in (10.16) need spelling out.

Why the interval, and not the instant. The constraint must hold throughout I for the differentiation to be legal. Had (10.15) required only $\bar{g}_\kappa(\underline{C}(t_0)) = 0$ at one instant, the function $t \mapsto \bar{g}_\kappa(\underline{C}(t))$ would be zero at a single point and its derivative there would be unrestricted. A constraint is by its nature a standing condition, so this costs nothing, but it is the hypothesis the argument actually uses.

Why $g_{\underline{C}}$ may be taken symmetric. By Proposition 2.27, $\underline{C}(t)$ is symmetric at every t , so each difference quotient $(\underline{C}(t+h) - \underline{C}(t))/h$ is symmetric, and sym is a subspace and therefore closed: the limit $\dot{\underline{C}}$ is symmetric. By the orthogonality of the splitting $\text{Lin} = \text{sym} \oplus \text{skw}$ of Definition 1.59 and Lemma 2.36, $\text{skw}(g_{\underline{C}}) \cdot \dot{\underline{C}} = 0$, so the inner product in (10.16) picks out only $\text{sym}(g_{\underline{C}})$. The skew part of $g_{\underline{C}}$ is invisible to (10.16), which is what effectively defines $g_{\underline{C}}$; no generality is lost by taking it symmetric, and doing so makes it unique.

The order of operations. The function $g(\underline{C})$ is well defined for *all* symmetric positive definite $\underline{C}$, not only for those that satisfy the constraint, and $g_{\underline{C}}$ is therefore likewise well defined for all such $\underline{C}$. The derivative appearing in (10.16) is obtained by differentiating first and imposing the constraint afterwards.

REMARK 10.5 (what breaks if the constraint is imposed too early). Reverse the order and the entire theory collapses in one line. Impose the constraint first: then g is the zero function on the set where it is defined, its gradient is $\hat{Q}$, and (10.18) below gives $\underline{N} = \underline{Q}$. There would be no reaction, the incompressible fluid would carry no pressure, and a rigid body would be free of stress.

The error is the same one that Remark 4.3 of Chapter 4 warned against in another guise: confusing the value of a function on a set with the function itself. What the constraint says is that the motion never leaves the surface $g = 0$ in the space of symmetric positive definite tensors. The gradient $g_{\underline{C}}$ is the *normal* to that surface, and a normal is computed from how g behaves as one steps *off* the surface, which is exactly the information deleted by imposing the constraint first. That is why (10.21) is computed for all $\underline{C}$ and only then evaluated, in (10.22),

on the constraint set.

Now bring in Chapter 4. Recalling (4.11), $\tilde{F}^t \tilde{D} \tilde{F} = \frac{1}{2} \dot{\tilde{C}}$, that is $\dot{\tilde{C}} = 2 \tilde{F}^t \tilde{D} \tilde{F}$, equation (10.16) becomes $0 = 2 \operatorname{tr} (g_{\tilde{C}}(\tilde{F}^t \tilde{D} \tilde{F}))$, and dividing by the constant 2,

$$0 = \operatorname{tr} (g_{\tilde{C}}(\tilde{F}^t \tilde{D} \tilde{F})) = \operatorname{tr} ((\tilde{F}(g_{\tilde{C}})\tilde{F}^t)\tilde{D}) = \tilde{F}(g_{\tilde{C}})\tilde{F}^t \cdot \tilde{D}, \quad (10.17)$$

in which the first factor on the right-hand side of the third equality is a symmetric *spatial* tensor.

The middle equality of (10.17) is the one step here that is not purely formal, since the two products live on different spaces and the trace of a two-point tensor is not defined. In components, with $g_{\tilde{C}} = G_{AB} \hat{E}_A \otimes \hat{E}_B$, $\tilde{F} = F_{iA} \underline{e}_i \otimes \hat{E}_A$ and $\tilde{D} = D_{ij} \underline{e}_i \otimes \underline{e}_j$, the tensor $g_{\tilde{C}} \tilde{F}^t \tilde{D} \tilde{F}$ is referential with components $G_{AB} F_{iB} D_{ij} F_{jA}$, so its trace is

$$\operatorname{tr} (g_{\tilde{C}} \tilde{F}^t \tilde{D} \tilde{F}) = G_{AB} F_{iB} D_{ij} F_{jA},$$

whereas $\tilde{F}(g_{\tilde{C}})\tilde{F}^t$ is spatial with components $F_{iA} G_{AB} F_{jB}$, so

$$\tilde{F}(g_{\tilde{C}})\tilde{F}^t \cdot \tilde{D} = F_{iA} G_{AB} F_{jB} D_{ij}.$$

The two are the same sum of products of the same numbers, written in a different order. Both traces are legitimate, each on its own space, and they are equal. Note also that $(\tilde{F}(g_{\tilde{C}})\tilde{F}^t)^t = \tilde{F}(g_{\tilde{C}})^t \tilde{F}^t = \tilde{F}(g_{\tilde{C}})\tilde{F}^t$, so the tensor is symmetric, and that the index bookkeeping of Remark 4.1 confirms it is spatial: $\tilde{F}$ carries $\underline{e}_i \otimes \hat{E}_A$, $g_{\tilde{C}}$ carries $\hat{E}_A \otimes \hat{E}_B$, $\tilde{F}^t$ carries $\hat{E}_B \otimes \underline{e}_j$, and the capital indices contract in pairs leaving $\underline{e}_i \otimes \underline{e}_j$.

Equation (10.17) says exactly this: the stretchings $\tilde{D}$ that the constraint permits are those, and only those, perpendicular in sym to the fixed symmetric spatial tensor $\tilde{F}(g_{\tilde{C}})\tilde{F}^t$. A constraint on $\tilde{C}$, which is a statement about the deformation as a whole, has become a single linear equation for $\tilde{D}$, which is a statement about the present rate. The bridge is (4.11), and this is the sense in which the constraint restricts the stretching tensor of Chapter 4.

REMARK 10.6 (from a surface to a plane). The set of $\underline{C}$ satisfying $g(\underline{C}) = 0$ is a curved five-dimensional surface in the six-dimensional sym; the set of $\underline{D}$ satisfying (10.17) is a flat five-dimensional subspace. Nothing was lost in passing from one to the other, because a velocity is by definition tangent to the surface it moves on, and the tangent space to a level surface at a point is a subspace. Linearity is what makes the next step possible at all: one may speak of the orthogonal complement of a subspace, but not of the orthogonal complement of a curved surface.

The reaction is pinned down to one multiplier

By (10.15) we require $\underline{N}$ to be perpendicular, in sym, to every $\underline{D}$ that is perpendicular to $\underline{F}(g_{\underline{C}})\underline{F}^t$ (Figure 10.1). Accordingly $\underline{N}$ must be parallel to $\underline{F}(g_{\underline{C}})\underline{F}^t$. The following lemma is that inference, made precise; it is the same manoeuvre as Lemma 5.1 of Chapter 5, namely, feed the hypothesis the one special element that turns it into a statement about a norm.

LEMMA 10.7 (the double orthogonal complement of a line). *Let $\underline{A} \in \text{sym}$ with $\underline{A} \neq \underline{Q}$, and let $\underline{N} \in \text{sym}$ satisfy $\underline{N} \cdot \underline{D} = 0$ for every $\underline{D} \in \text{sym}$ with $\underline{A} \cdot \underline{D} = 0$. Then $\underline{N} = \lambda \underline{A}$ for exactly one scalar λ .*

Proof. By Theorem 1.58 the tensor inner product is positive definite, so $\underline{A} \cdot \underline{A} = |\underline{A}|^2 > 0$ and we may set

$$\lambda = \frac{\underline{N} \cdot \underline{A}}{\underline{A} \cdot \underline{A}}, \quad \underline{N}' = \underline{N} - \lambda \underline{A}.$$

Then $\underline{N}' \in \text{sym}$, being a difference of symmetric tensors, and

$$\underline{A} \cdot \underline{N}' = \underline{A} \cdot \underline{N} - \lambda \underline{A} \cdot \underline{A} = \underline{A} \cdot \underline{N} - \frac{\underline{N} \cdot \underline{A}}{\underline{A} \cdot \underline{A}} \underline{A} \cdot \underline{A} = 0.$$

So $\underline{N}'$ is one of the $\underline{D}$'s covered by the hypothesis, and taking $\underline{D} = \underline{N}'$ gives $\underline{N} \cdot \underline{N}' = 0$. On the other hand $\underline{N} = \underline{N}' + \lambda \underline{A}$, so

$$0 = \underline{N} \cdot \underline{N}' = (\underline{N}' + \lambda \underline{A}) \cdot \underline{N}' = |\underline{N}'|^2 + \lambda (\underline{A} \cdot \underline{N}') = |\underline{N}'|^2,$$

whence $\underline{N}' = \underline{Q}$ by positive definiteness again, and $\underline{N} = \lambda \underline{A}$. Uniqueness: if $\lambda_1 \underline{A} = \lambda_2 \underline{A}$ then $(\lambda_1 - \lambda_2) \underline{A} = \underline{Q}$, and $\underline{A} \neq \underline{Q}$ forces $\lambda_1 = \lambda_2$. ■

Applying the lemma with $\underline{A} = \underline{F}(g_{\underline{C}})\underline{F}^t$,

$$\underline{N} = \lambda(\underline{x}, t) \underline{F}(g_{\underline{C}})\underline{F}^t, \quad (10.18)$$

in which the multiplier $\lambda(\underline{x}, t)$ is determined solely by the equations of motion, with the stress given by (10.14), together with the constraint equation and the initial and boundary conditions.

REMARK 10.8 (the multiplier is a field, and it is a Lagrange multiplier). Read (10.18) beside the elementary rule for constrained stationarity and it is the same statement. To make f stationary on the surface $g = 0$ one sets $\nabla f = \lambda \nabla g$: the reaction is directed along the normal to the constraint surface, and λ is whatever number the constraint then requires. Here $g_{\underline{C}}$ is the normal to the constraint surface in the referential space of symmetric tensors, the sandwich $\underline{F}(\cdot)\underline{F}^t$ carries it to the spatial one, and (10.18) says the reaction stress lies along it.

Two features of λ are easy to misread. It is a *field*, $\lambda(\underline{x}, t)$, not a constant and not a material parameter: it is allowed to vary from point to point and instant to instant, and it must, since it is doing the work of enforcing the constraint everywhere at once. And it is *not* a function of $\underline{C}$, $\underline{F}$ or $\underline{D}$. Writing $\lambda = \hat{\lambda}(\underline{C})$ would smuggle a constitutive equation back in and would generally make the resulting system unsolvable, for the reason Chapter 9 gives after (9.85): without a free field the equations for an incompressible fluid outnumber the unknowns.

If n constraints $g^{(i)}(\underline{C}) = 0$, $i = 1, \dots, n$, are in force during an interval of time, then $\underline{N} \cdot \underline{D}$ must vanish for all $\underline{D}$ such that $\underline{F}(g_{\underline{C}}^{(i)})\underline{F}^t \cdot \underline{D}$ vanishes for each i . The n constraints are said to be independent provided that the elements of $\{\underline{F}(g_{\underline{C}}^{(i)})\underline{F}^t\}$ are linearly independent, the $g_{\underline{C}}^{(i)}$ being computed for all positive definite symmetric $\underline{C}$. In this case $n \leq 6$, because sym is six-dimensional and $\text{Span}\{\underline{F}(g_{\underline{C}}^{(i)})\underline{F}^t\}$ is then a subspace of sym. Thus

$$\underline{N} = \sum_{i=1}^{n \leq 6} \lambda_i(\underline{x}, t) \underline{F}(g_{\underline{C}}^{(i)})\underline{F}^t = \underline{F}\left(\sum_{i=1}^n \lambda_i g_{\underline{C}}^{(i)}\right)\underline{F}^t, \quad (10.19)$$

where the multipliers $\lambda_i(\underline{x}, t)$ are again such as to secure a solution to the equations of motion together with the n constraint equations and the initial and boundary conditions, and the derivatives $g_C^{(i)}$ are evaluated, after the fact, at those $\underline{C}$'s that satisfy all n constraints simultaneously. The second form of (10.19) follows from the first because $\underline{\underline{F}}(\cdot)\underline{\underline{F}}^t$ is linear in its argument.

The proof of (10.19) is Lemma 10.7 with a line replaced by a subspace. Let $\mathcal{W} = \text{Span}\{\underline{\underline{F}}(g_C^{(i)})\underline{\underline{F}}^t\} \subset \text{sym}$. The admissible $\underline{\underline{D}}$ are those in $\mathcal{W}^\perp$, and the hypothesis is $\underline{\underline{N}} \in (\mathcal{W}^\perp)^\perp$. Decompose $\underline{\underline{N}} = \underline{\underline{N}}_{\mathcal{W}} + \underline{\underline{N}}'$ with $\underline{\underline{N}}_{\mathcal{W}} \in \mathcal{W}$ and $\underline{\underline{N}}' \in \mathcal{W}^\perp$, which is possible because $\mathcal{W}$ is a finite-dimensional subspace of an inner-product space. Then $\underline{\underline{N}}'$ is an admissible $\underline{\underline{D}}$, so $0 = \underline{\underline{N}} \cdot \underline{\underline{N}}' = |\underline{\underline{N}}'|^2$ and $\underline{\underline{N}}' = \underline{\underline{0}}$; therefore $\underline{\underline{N}} \in \mathcal{W}$, which is (10.19).

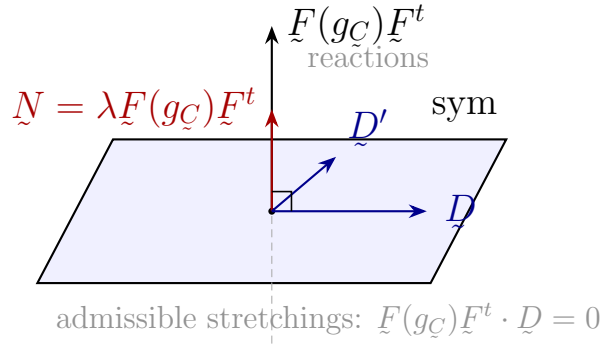


Figure 10.1: Orthogonality of $\underline{\underline{F}}(g_C)\underline{\underline{F}}^t$ and $\underline{\underline{D}}$ in the space of symmetric tensors. One constraint deletes one dimension from sym: the admissible stretchings fill the five-dimensional plane through the origin perpendicular to $\underline{\underline{F}}(g_C)\underline{\underline{F}}^t$, drawn here as a parallelogram. Requiring $\underline{\underline{N}}$ to be perpendicular to all of them leaves exactly the one-dimensional line along $\underline{\underline{F}}(g_C)\underline{\underline{F}}^t$, which is (10.18). The picture also shows why the count in Remark 10.3 comes out as it does: dimensions lost by $\underline{\underline{D}}$ are dimensions gained by $\underline{\underline{N}}$, and $5 + 1 = 6 = \dim \text{sym}$.

Examples

We determine the forms of the reactive stresses for the constraints of incompressibility, inextensibility and rigidity.

(1) Incompressibility

In the case of incompressibility we use (10.11), with (B.26) and (B.27) of Appendix B, to obtain

$$g_{\mathcal{C}} \cdot \dot{\mathcal{C}} = (\det \mathcal{C})^\cdot = \mathcal{C}^* \cdot \dot{\mathcal{C}} = (\det \mathcal{C}) \mathcal{C}^{-1} \cdot \dot{\mathcal{C}}, \quad (10.20)$$

for arbitrary symmetric $\dot{\mathcal{C}}$.

Both of the equalities after the first deserve a derivation, since they are the whole computation. Start from the box-product characterisation of the determinant, Theorem 1.52: for any invertible $\mathcal{A}(t)$ and fixed $\underline{u}, \underline{v}, \underline{w}$,

$$(\det \mathcal{A})[\underline{u}, \underline{v}, \underline{w}] = [\mathcal{A}\underline{u}, \mathcal{A}\underline{v}, \mathcal{A}\underline{w}].$$

Differentiating, the box product being trilinear so that its derivative obeys a three-term product rule, and writing $\underline{a} = \mathcal{A}\underline{u}$, $\underline{b} = \mathcal{A}\underline{v}$, $\underline{c} = \mathcal{A}\underline{w}$ together with $\dot{\mathcal{A}}\underline{u} = \dot{\mathcal{A}}\mathcal{A}^{-1}\underline{a}$ and likewise for the others,

$$\begin{aligned} (\det \mathcal{A})^\cdot[\underline{u}, \underline{v}, \underline{w}] &= [\dot{\mathcal{A}}\underline{u}, \mathcal{A}\underline{v}, \mathcal{A}\underline{w}] + [\mathcal{A}\underline{u}, \dot{\mathcal{A}}\underline{v}, \mathcal{A}\underline{w}] + [\mathcal{A}\underline{u}, \mathcal{A}\underline{v}, \dot{\mathcal{A}}\underline{w}] \\ &= [\dot{\mathcal{A}}\mathcal{A}^{-1}\underline{a}, \underline{b}, \underline{c}] + [\underline{a}, \dot{\mathcal{A}}\mathcal{A}^{-1}\underline{b}, \underline{c}] + [\underline{a}, \underline{b}, \dot{\mathcal{A}}\mathcal{A}^{-1}\underline{c}] \\ &= \text{tr}(\dot{\mathcal{A}}\mathcal{A}^{-1})[\underline{a}, \underline{b}, \underline{c}] = \text{tr}(\dot{\mathcal{A}}\mathcal{A}^{-1})(\det \mathcal{A})[\underline{u}, \underline{v}, \underline{w}], \end{aligned}$$

the third line by Definition 1.49₁, which is precisely that sum of three box products. Cancelling the non-zero $[\underline{u}, \underline{v}, \underline{w}]$,

$$(\det \mathcal{A})^\cdot = (\det \mathcal{A}) \text{tr}(\dot{\mathcal{A}}\mathcal{A}^{-1}).$$

This is Problem 2(a) of Chapter 4 with $\mathcal{A} = \mathcal{F}$; here we want it with $\mathcal{A} = \mathcal{C}$. Using $\text{tr}(\mathcal{A}\mathcal{B}) = \text{tr}(\mathcal{B}\mathcal{A})$ and then the symmetry of $\mathcal{C}^{-1}$ and $\dot{\mathcal{C}}$,

$$(\det \mathcal{C})^\cdot = (\det \mathcal{C}) \text{tr}(\dot{\mathcal{C}}\mathcal{C}^{-1}) = (\det \mathcal{C}) \text{tr}(\mathcal{C}^{-1}\dot{\mathcal{C}}) = (\det \mathcal{C}) \mathcal{C}^{-1} \cdot \dot{\mathcal{C}},$$

which is the last equality of (10.20). The middle one is then Proposition 1.55, $\mathcal{A}^* = (\det \mathcal{A})\mathcal{A}^{-t}$, read for the symmetric $\mathcal{C}$: $\mathcal{C}^* = (\det \mathcal{C})\mathcal{C}^{-1}$. So the cofactor of Definition 1.54 is the derivative of the determinant, which is the same fact that produced $\partial J / \partial F_{iA} = F_{iA}^*$ in Chapter 4.

Since (10.20) holds for arbitrary symmetric $\underline{\underline{C}}$ and both $g_{\underline{\underline{C}}}$ and $(\det \underline{\underline{C}})\underline{\underline{C}}^{-1}$ are symmetric, their difference is a symmetric tensor orthogonal to every symmetric tensor, hence orthogonal to itself, hence zero. Thus

$$g_{\underline{\underline{C}}} = (\det \underline{\underline{C}})\underline{\underline{C}}^{-1}, \quad (10.21)$$

which reduces to

$$g_{\underline{\underline{C}}} = \underline{\underline{C}}^{-1} \quad (10.22)$$

on the set of $\underline{\underline{C}}$'s that satisfy the constraint, where $\det \underline{\underline{C}} = 1$. Equations (10.21) and (10.22) are Remark 10.5 in action: the first is computed off the constraint set, the second is the first evaluated on it, and doing those two steps in the other order would have produced $g_{\underline{\underline{C}}} = \hat{Q}$.

Writing $\lambda = -\bar{p}$, the associated reactive stress follows from (10.18):

$$\underline{\underline{N}} = -\bar{p} \underline{\underline{F}} \underline{\underline{C}}^{-1} \underline{\underline{F}}^t = -\bar{p} \underline{\underline{F}} (\underline{\underline{F}}^t \underline{\underline{F}})^{-1} \underline{\underline{F}}^t = -\bar{p} \underline{\underline{I}}, \quad (10.23)$$

and is therefore mechanically equivalent to a pressure. The middle step substituted $\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}}$; the last used $(\underline{\underline{F}}^t \underline{\underline{F}})^{-1} = \underline{\underline{F}}^{-1} \underline{\underline{F}}^{-t}$, from Proposition 1.46, so that

$$\underline{\underline{F}} (\underline{\underline{F}}^t \underline{\underline{F}})^{-1} \underline{\underline{F}}^t = \underline{\underline{F}} \underline{\underline{F}}^{-1} \underline{\underline{F}}^{-t} \underline{\underline{F}}^t = \underline{\underline{I}} \underline{\underline{I}} = \underline{\underline{I}}.$$

The worklessness may now be verified directly rather than assumed. By Definition 1.57 and Problem 2(a) of Chapter 4,

$$\underline{\underline{N}} \cdot \underline{\underline{D}} = -\bar{p} \underline{\underline{I}} \cdot \underline{\underline{D}} = -\bar{p} \operatorname{tr} \underline{\underline{D}} = -\bar{p} \operatorname{div} \underline{\underline{v}} = -\bar{p} \frac{\dot{J}}{J},$$

which vanishes because the constraint $J \equiv 1$ gives $\dot{J} = 0$. The reaction does no work for the plainest of reasons: a pressure works only through a change of volume, and the constraint is that there is none.

For example, in the case of the Newtonian viscous fluid model (9.60), whose stress is the sum of $-\tilde{p}\underline{\underline{I}}$, a term proportional to $(\operatorname{tr} \underline{\underline{D}})\underline{\underline{I}}$, and $2\mu\underline{\underline{D}}$, the constraint annihilates the middle term and leaves the constitutive part $\underline{\underline{G}} = -\tilde{p}\underline{\underline{I}} + 2\mu\underline{\underline{D}}$. Adding the reaction (10.23) and writing $p = \tilde{p} + \bar{p}$,

$$\underline{\underline{T}} = -p \underline{\underline{I}} + 2\mu \underline{\underline{D}} \quad (10.24)$$

if the constraint is operative. The constitutive indeterminacy of $\bar{p}$ implies that p is likewise indeterminate, and we arrive at (9.83). Note what has become of the thermodynamic pressure $\tilde{p} = \rho^2 \psi_\rho$ of (9.74): it has not been contradicted, it has been absorbed. Since $\bar{p}$ is arbitrary so is the sum, and there is no way to recover $\tilde{p}$ from a measurement of $\underline{T}$.

EXAMPLE 10.9 (the pressure in a fluid at rest, and why it is not a material property). This is the example that makes the whole chapter concrete, so it is worked out completely.

The problem. Let the fluid occupy $0 \leq x_3 \leq h$, be at rest, and be acted on by gravity $\underline{b} = -g\underline{e}_3$. At rest $\underline{v} = \underline{0}$, so $\underline{L} = \underline{Q}$ and $\underline{D} = \underline{Q}$; the constraint $\text{tr } \underline{D} = 0$ holds, and (10.24) gives $\underline{T} = -p\underline{I}$.

The equation of motion. Substituting into $\text{div } \underline{T} + \rho \underline{b} = \rho \underline{\dot{v}}$, with $\underline{\dot{v}} = \underline{0}$ because the fluid is at rest. In components,

$$(\text{div } \underline{T})_i = T_{ij,j} = (-p\delta_{ij})_{,j} = -p_{,i}, \quad \text{that is} \quad \text{div}(-p\underline{I}) = -\text{grad } p,$$

so

$$\text{grad } p = \rho \underline{b} = -\rho g \underline{e}_3, \quad \text{i.e.} \quad p_{,1} = 0, \quad p_{,2} = 0, \quad p_{,3} = -\rho g.$$

The first two say p depends on x_3 alone; integrating the third, $p = c - \rho g x_3$ with c a constant of integration.

The boundary condition. On the free surface $x_3 = h$ the exterior unit normal is $\underline{e}_3$ and the air exerts the traction $-p_a \underline{e}_3$. Cauchy's relation gives $\underline{t} = \underline{T} \underline{e}_3 = -p \underline{e}_3$, so $p(h) = p_a$ and $c = p_a + \rho g h$. Therefore

$$p = p_a + \rho g(h - x_3).$$

The point. Not one step of that calculation consulted the material. The viscosity μ never appeared, because $\underline{D} = \underline{Q}$; the density entered only through the body force, and $\tilde{p}$ was absorbed into p at the start. What fixed p was the equation of motion, which is a balance law of Chapter 6 and holds for every material, together with one boundary condition.

To see that this is not an accident of the example, seal the container with a piston and press on it, so that the boundary condition at $x_3 = h$

becomes $p = p_a + \Delta$ (Figure A). The material is the same, the motion is the same, $\underline{D}$ is still $\underline{Q}$, and $\underline{G}$ is unchanged; yet now

$$p = p_a + \Delta + \rho g(h - x_3).$$

Two different stress fields in the same material undergoing the same motion. No constitutive equation can be responsible for the difference, because a constitutive equation is a function of the motion and the motion did not change. The pressure in an incompressible fluid is a reaction, and Δ is the price of the constraint.

A dynamic case, so that the split is visible. Take instead the shear flow $\underline{v} = \gamma x_2 \underline{e}_1$ with γ constant. Then $\underline{L} = \gamma \underline{e}_1 \otimes \underline{e}_2$, so $\text{tr } \underline{L} = 0$ and the flow is isochoric as the constraint demands, and by (4.7) $\underline{D} = \frac{1}{2}\gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1)$, which is trace free. Equation (10.24) gives

$$\begin{aligned} \underline{T} &= -p\underline{I} + \mu\gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1), & \text{so} & & T_{12} \\ &= \mu\gamma, & T_{11} &= T_{22} = T_{33} = -p. \end{aligned}$$

The shear component is constitutive: measure it and you have measured μ . The three normal components are the reaction: measure them and you have measured the boundary conditions.

Counting. With $\underline{D} = \text{sym}(\text{grad } \underline{v})$, the linear momentum balance is three equations, conservation of mass (9.84) is a fourth, and the constraint $\text{div } \underline{v} = 0$ of (9.85) is a fifth, for the five unknowns ρ , p and the three components of $\underline{v}$. Remove p and there are five equations for four unknowns: an overdetermined, and generally insoluble, system. The constraint added an equation, and the multiplier it brought with it added exactly one unknown to pay for it. That balance is not a coincidence of this example; it is what Lemma 10.7 guarantees in general, one multiplier per independent constraint.

(2) Inextensibility

For inextensibility of embedded fibres with unit tangent $\underline{M}$ in κ we use (10.12). Since $\underline{M}$ is a material vector, $\dot{\underline{M}} = \underline{0}$ by Remark 3.26, so differentiating $\underline{M} \cdot \underline{C}\underline{M} = 1$ leaves only the term in which $\underline{C}$ is differentiated,

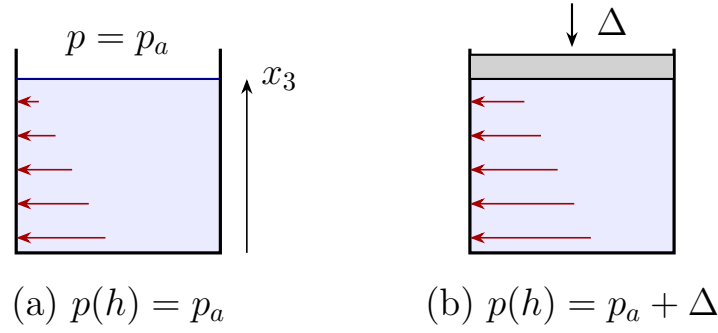


Figure A: The reaction pressure of Example 10.9. In both panels the fluid is the same, the motion is the same, namely rest, and the constitutive part $\underline{G}$ of the stress is the same, since $\underline{D} = \underline{Q}$. The arrows show the pressure the fluid exerts on the left wall, growing linearly with depth in both cases. Only the boundary condition at the top differs, and the whole pressure field shifts by the constant Δ throughout the body. A constitutive equation is a function of the motion and could not tell the two panels apart; the pressure is therefore not constitutive, but a reaction fixed by the equations of motion and the data.

and

$$g_{\underline{C}} \cdot \dot{\underline{C}} = \underline{M} \cdot \dot{\underline{C}} \underline{M} = \underline{M} \otimes \underline{M} \cdot \dot{\underline{C}} \quad (10.25)$$

for all symmetric $\dot{\underline{C}}$. The second equality is the component identity

$$(\underline{M} \otimes \underline{M}) \cdot \underline{A} = M_A M_B A_{AB} = M_A (\underline{A} \underline{M})_A = \underline{M} \cdot \underline{A} \underline{M},$$

using the component rule for the tensor product, Definition 1.32, and the component form of the tensor inner product. Thus, by the same arbitrariness argument used for (10.21), and noting that $\underline{M} \otimes \underline{M}$ is symmetric,

$$g_{\underline{C}} = \underline{M} \otimes \underline{M}. \quad (10.26)$$

Writing $\lambda = \sigma$, we use (10.18) to obtain

$$\underline{N} = \sigma \underline{F}(\underline{M} \otimes \underline{M}) \underline{F}^t = \sigma \underline{F} \underline{M} \otimes \underline{F} \underline{M}, \quad (10.27)$$

the second equality being the dyad rule $\underline{A}(\underline{a} \otimes \underline{b}) \underline{B}^t = \underline{A} \underline{a} \otimes \underline{B} \underline{b}$, which follows from Lemma 2.15, or directly in components: $A_{iC}(M_C M_D) B_{jD} = (A_{iC} M_C)(B_{jD} M_D)$. This reduces to

$$\underline{N} = \sigma \underline{m} \otimes \underline{m} \quad (10.28)$$

when the constraint is in effect, where $\underline{m}$ is the unit tangent to a fibre in κ_t . The step is (2.67), $\underline{F}\underline{M} = \mu\underline{m}$, together with the constraint $\mu = 1$: with $\mu \neq 1$ one would have $\underline{F}\underline{M} \otimes \underline{F}\underline{M} = \mu^2\underline{m} \otimes \underline{m}$, and the factor μ^2 is absorbable into σ in any case, which is why the constraint may be imposed here without loss. Thus the reactive stress is mechanically equivalent to a uniaxial stress along the fibres, exactly the state of stress of (6.130), with σ the fibre tension.

Worklessness is again immediate, and this time it is Chapter 4 that supplies it. From (10.28) and the identity used in (10.25),

$$\underline{N} \cdot \underline{D} = \sigma (\underline{m} \otimes \underline{m}) \cdot \underline{D} = \sigma \underline{m} \cdot \underline{D}\underline{m} = \sigma (\ln \mu)^\cdot = 0,$$

the third equality being (4.15) and the last the constraint $\mu \equiv 1$, whose logarithm is constant. A tension works only through a stretching of the line it acts along, and the constraint is that there is none. Compare the incompressible case: there the pair was pressure against rate of volume change, here it is tension against rate of fibre stretch, and in both the constraint annihilates the second member of the pair.

EXAMPLE 10.10 (what an inextensible fibre still permits). It is worth seeing that the constraint is far from paralysing. Take the fibres along $\underline{M} = \hat{\underline{E}}_1$ and let the body undergo the simple shear of Section 3.5.1 *parallel* to them,

$$x_1 = X_1 + \gamma X_2, \quad x_2 = X_2, \quad x_3 = X_3, \quad \text{so} \quad \underline{F} = \underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2.$$

Then $\underline{F}\hat{\underline{E}}_1 = \underline{1}\hat{\underline{E}}_1 + \gamma \underline{e}_1(\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_1) = \underline{e}_1$, since $\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_1 = 0$ and $\underline{1}\hat{\underline{E}}_1 = \underline{e}_1$ by (2.68). Comparing with $\underline{F}\underline{M} = \mu\underline{m}$ gives $\mu = 1$ and $\underline{m} = \underline{e}_1$: the constraint holds for *every* γ , however large. Equivalently $C_{11} = \underline{F}\hat{\underline{E}}_1 \cdot \underline{F}\hat{\underline{E}}_1 = 1$.

Shear the same body across the fibres instead, $x_2 = X_2 + \gamma X_1$, and $\underline{F}\hat{\underline{E}}_1 = \underline{e}_1 + \gamma \underline{e}_2$, so $\mu = \sqrt{1 + \gamma^2} \neq 1$ for $\gamma \neq 0$: forbidden. One constraint, one direction lost; five of the six dimensions of sym remain, which is Figure 10.1 in a concrete case.

The reaction in the permitted motion is $\underline{N} = \sigma \underline{e}_1 \otimes \underline{e}_1$, with $\sigma(\underline{x}, t)$ a free field. Its worklessness can be checked without any general theory.

Let the shear proceed in time, $\gamma = \gamma(t)$. Then $\underline{v} = \dot{\gamma} X_2 \underline{e}_1 = \dot{\gamma} x_2 \underline{e}_1$, since $x_2 = X_2$, so $\underline{L} = \text{grad } \underline{v} = \dot{\gamma} \underline{e}_1 \otimes \underline{e}_2$ and by (4.7)

$$\underline{D} = \frac{1}{2} \dot{\gamma} (\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1), \quad \text{so} \quad D_{11} = \underline{e}_1 \cdot \underline{D} \underline{e}_1 = 0,$$

and $\underline{N} \cdot \underline{D} = \sigma D_{11} = 0$. What σ actually is at each point is settled, as always, by $\text{div } \underline{T} + \rho \underline{b} = \rho \underline{\dot{v}}$ and the tractions applied to the ends of the fibres, not by any property of the fibre.

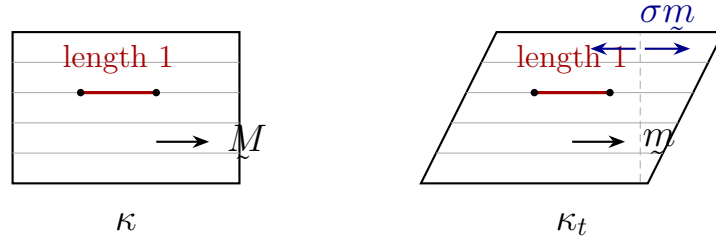


Figure B: An inextensible fibre family, $\underline{M}$ in κ and $\underline{m}$ in κ_t , under a shear parallel to the fibres. Every fibre keeps its length, so the constraint $\mu = 1$ holds for every amount of shear, while the matrix between the fibres shears freely: the constraint removes one dimension from sym and leaves five. The reaction (10.28) is the uniaxial stress $\sigma \underline{m} \otimes \underline{m}$, drawn as the pair of opposed tractions on a cut normal to $\underline{m}$. It does no work because $\underline{m} \cdot \underline{D} \underline{m} = (\ln \mu)^\cdot = 0$, and its magnitude σ is not a property of the fibre but whatever the equations of motion and the applied end tractions require.

(3) Rigidity

In the case of rigidity it follows from (10.13) that, for each ordered pair (A, B) ,

$$\begin{aligned} (g_{AB})_{\underline{C}} \cdot \dot{\underline{C}} &= \dot{C}_{AB} = (\hat{\underline{E}}_A \cdot \underline{C} \hat{\underline{E}}_B)^\cdot = \hat{\underline{E}}_A \cdot \dot{\underline{C}} \hat{\underline{E}}_B \\ &= (\hat{\underline{E}}_A \otimes \hat{\underline{E}}_B) \cdot \dot{\underline{C}} = \text{sym}(\hat{\underline{E}}_A \otimes \hat{\underline{E}}_B) \cdot \dot{\underline{C}}, \end{aligned} \quad (10.29)$$

on account of the symmetry of $\dot{\underline{C}}$, where

$$\text{sym}(\hat{\underline{E}}_A \otimes \hat{\underline{E}}_B) = \frac{1}{2} (\hat{\underline{E}}_A \otimes \hat{\underline{E}}_B + \hat{\underline{E}}_B \otimes \hat{\underline{E}}_A). \quad (10.30)$$

The first line of (10.29) used $\dot{\delta}_{AB} = 0$ and the fixity of the referential basis, $\dot{\hat{E}}_A = 0$; the second used the identity established after (10.25); the third dropped $\text{skw}(\hat{E}_A \otimes \hat{E}_B)$ by Lemma 2.36, since $\hat{C}$ is symmetric. Thus

$$(g_{AB})_{\hat{C}} = \text{sym}(\hat{E}_A \otimes \hat{E}_B) \quad (10.31)$$

and it follows from (10.19) that

$$\begin{aligned} \underline{N} &= \lambda_{AB} \underline{Q} [\text{sym}(\hat{E}_A \otimes \hat{E}_B)] \underline{Q}^t = \underline{Q} [\lambda_{(AB)} \hat{E}_A \otimes \hat{E}_B] \underline{Q}^t, \\ \text{with } \lambda_{(AB)} &= \frac{1}{2}(\lambda_{AB} + \lambda_{BA}), \end{aligned} \quad (10.32)$$

where the usual summation convention is in effect, the constraint has been imposed in the form $\underline{F} = \underline{Q}$, a rotation, and the λ_{AB} are arbitrary scalars.

The passage from the first form of (10.32) to the second is a relabelling worth doing once. Using (10.30),

$$\lambda_{AB} \text{sym}(\hat{E}_A \otimes \hat{E}_B) = \frac{1}{2} \lambda_{AB} \hat{E}_A \otimes \hat{E}_B + \frac{1}{2} \lambda_{AB} \hat{E}_B \otimes \hat{E}_A.$$

In the second sum both A and B are summed over, so they may be interchanged as names, turning it into $\frac{1}{2} \lambda_{BA} \hat{E}_A \otimes \hat{E}_B$. Adding,

$$\lambda_{AB} \text{sym}(\hat{E}_A \otimes \hat{E}_B) = \frac{1}{2}(\lambda_{AB} + \lambda_{BA}) \hat{E}_A \otimes \hat{E}_B = \lambda_{(AB)} \hat{E}_A \otimes \hat{E}_B.$$

Only the symmetric part of the array (λ_{AB}) survives, which is the same phenomenon as the symmetry of $g_{\hat{C}}$: the skew part is invisible because it is contracted against symmetric objects.

The term in square brackets to the right of the second equality in (10.32) is an arbitrary symmetric referential tensor: it has six free entries $\lambda_{(AB)}$, and every symmetric referential tensor is of that form. Conjugation $\underline{S} \mapsto \underline{Q} \underline{S} \underline{Q}^t$ is linear, carries symmetric tensors to symmetric tensors, since $(\underline{Q} \underline{S} \underline{Q}^t)^t = \underline{Q} \underline{S}^t \underline{Q}^t$, and is invertible with inverse $\underline{T} \mapsto \underline{Q}^t \underline{T} \underline{Q}$. It therefore maps the symmetric referential tensors *onto* the symmetric spatial tensors. Thus $\underline{N}$ is an arbitrary symmetric spatial tensor. It follows that the net stress $\underline{T}$ of (10.14) is constitutively indeterminate: the stress in a rigid body is indeterminate. Nevertheless, the equations of rigid-body dynamics of Section 6.4.2 suffice to determine the motion in terms of the applied forces and torques, and vice versa.

REMARK 10.11 (why rigidity determines nothing, in one line). The orthogonality condition (10.15) is the only thing that ever restricted $\underline{N}$, and in a rigid motion it says nothing at all. By Remark 4.6, $\underline{D} = \underline{Q}$ in a rigid motion, so $\underline{N} \cdot \underline{D} = 0$ holds for *every* symmetric $\underline{N}$ and the requirement is vacuous. Equation (10.32) is that observation written out. This also settles the count: six independent constraints remove all six dimensions of admissible stretching and return all six dimensions of sym as reaction, which is the extreme case of the arithmetic in Figure 10.1.

The same fact appears in the energy balance. Chapter 6 observes that for a rigid body the stress power is $\underline{T} \cdot \underline{L} = \underline{T} \cdot \underline{W}$, which vanishes by the symmetry of $\underline{T}$ and Lemma 2.36, so the thermal and mechanical problems decouple. A material that can store no strain energy can also tell you nothing about its stress.

10.3 Problems

PROBLEM 1 (The three constraints as O^+ writes them). Write the constraints of incompressibility, inextensibility and rigidity in the frame of observer O^+ .

Solution. The general rule is (10.10): whatever $\bar{g}_\kappa$ is, the function used by O^+ is

$$\bar{g}_{\kappa^+}^+(\underline{C}^+) = \bar{g}_\kappa(\underline{K}^t \underline{C}^+ \underline{K}),$$

and $\underline{C}^+ = \underline{K} \underline{C} \underline{K}^t$ by (7.30)₁, with $\underline{K}$ the fixed orthogonal tensor of (7.15). So each case is a substitution followed by a cancellation of $\underline{K}$'s.

Incompressibility. From (10.11),

$$\begin{aligned} \bar{g}_{\kappa^+}^+(\underline{C}^+) &= \det(\underline{K}^t \underline{C}^+ \underline{K}) - 1 \\ &= (\det \underline{K}^t)(\det \underline{C}^+)(\det \underline{K}) - 1 = (\det \underline{K})^2 \det \underline{C}^+ - 1, \end{aligned}$$

and $(\det \underline{K})^2 = 1$ because $\underline{K}$ is orthogonal, so

$$\bar{g}_{\kappa^+}^+(\underline{C}^+) = \det \underline{C}^+ - 1.$$

The function has exactly the same form. Its value is also the same number, since $\det \mathcal{C}^+ = (\det \mathcal{K})^2 \det \mathcal{C} = \det \mathcal{C}$. Equivalently, $J^+ = 1$, which is (7.19) with $J = 1$.

Inextensibility. From (10.12), moving $\mathcal{K}$ across the inner product by the definition of the transpose,

$$\bar{g}_{\kappa^+}^+(\mathcal{C}^+) = \mathcal{M} \cdot (\mathcal{K}^t \mathcal{C}^+ \mathcal{K}) \mathcal{M} - 1 = \mathcal{K} \mathcal{M} \cdot \mathcal{C}^+ \mathcal{K} \mathcal{M} - 1.$$

By (7.36) the counterpart in $\hat{\mathbb{E}}^{3+}$ of the referential unit vector $\mathcal{M}$ is $\mathcal{M}^+ = \mathcal{K} \mathcal{M}$, so

$$\bar{g}_{\kappa^+}^+(\mathcal{C}^+) = \mathcal{M}^+ \cdot \mathcal{C}^+ \mathcal{M}^+ - 1, \quad \mathcal{M}^+ = \mathcal{K} \mathcal{M}.$$

Again the same form, with the fibre direction carried to the new reference configuration by $\mathcal{K}$. That $\mathcal{M}^+$ is still a unit vector is orthogonality of $\mathcal{K}$: $|\mathcal{K} \mathcal{M}|^2 = \mathcal{K} \mathcal{M} \cdot \mathcal{K} \mathcal{M} = \mathcal{M} \cdot \mathcal{K}^t \mathcal{K} \mathcal{M} = \mathcal{M} \cdot \hat{I} \mathcal{M} = 1$.

Rigidity. From (10.13) the constraint is $\mathcal{C} = \hat{I}$, so

$$\mathcal{C}^+ = \mathcal{K} \hat{I} \mathcal{K}^t = \mathcal{K} \mathcal{K}^t = \hat{I}^+,$$

which is (7.26), the transformation law for the referential identity. Componentwise, resolve on the basis $\hat{\mathcal{E}}_A^+ = \mathcal{K} \hat{\mathcal{E}}_A$, which is orthonormal for the same reason $\mathcal{M}^+$ was a unit vector. Then

$$\begin{aligned} C_{AB}^+ &= \hat{\mathcal{E}}_A^+ \cdot \mathcal{C}^+ \hat{\mathcal{E}}_B^+ = \mathcal{K} \hat{\mathcal{E}}_A \cdot \mathcal{K} \mathcal{C} \mathcal{K}^t \mathcal{K} \hat{\mathcal{E}}_B \\ &= \hat{\mathcal{E}}_A \cdot \mathcal{K}^t \mathcal{K} \mathcal{C} \hat{\mathcal{E}}_B = \hat{\mathcal{E}}_A \cdot \mathcal{C} \hat{\mathcal{E}}_B = C_{AB}, \end{aligned}$$

so the components are literally the same numbers and

$$\bar{g}_{(\kappa^+)}^+{}_{AB} = C_{AB}^+ - \delta_{AB}.$$

What the three answers say together. In each case the constraint keeps its form and its numerical value; only the referential objects it is written with, $\mathcal{M}$ and $\hat{\mathcal{E}}_A$, are carried over by $\mathcal{K}$. That is precisely the requirement (10.3) from which the whole of Section 10.1 was deduced, now verified in the three cases that matter. Had any of the three come out differently, that constraint would have failed the test of (10.9) and would have had to be discarded, in the manner of Remark 10.2. $\square$

PROBLEM 2 (The reactive stress seen by another observer). Find an expression for the reactive stress $\underline{N}^+$ as perceived by O^+ , associated with a generic constraint. Show that all observers agree on the values of the various constitutively indeterminate multipliers.

Solution. Everything follows from (10.18) once the three ingredients $g_{\underline{C}^+}^+$, $\underline{F}^+$ and the transformation of $\underline{N}$ itself are in hand.

Step 1: the gradient transforms by $\underline{K}$. We claim

$$g_{\underline{C}^+}^+ = \underline{K} g_{\underline{C}} \underline{K}^t.$$

By definition, $g_{\underline{C}^+}^+$ is the tensor for which $dg^+ = g_{\underline{C}^+}^+ \cdot d\underline{C}^+$ for every symmetric increment $d\underline{C}^+$. From (10.10), $g^+(\underline{C}^+) = g(\underline{K}^t \underline{C}^+ \underline{K})$, so the increments are related by $d\underline{C} = \underline{K}^t d\underline{C}^+ \underline{K}$, $\underline{K}$ being fixed. Hence, for any symmetric $\underline{A}$,

$$\begin{aligned} dg^+ &= g_{\underline{C}} \cdot (\underline{K}^t \underline{A} \underline{K}) = \text{tr} (g_{\underline{C}} (\underline{K}^t \underline{A} \underline{K})^t) \\ &= \text{tr} (g_{\underline{C}} \underline{K}^t \underline{A}^t \underline{K}) = \text{tr} (\underline{K} g_{\underline{C}} \underline{K}^t \underline{A}^t) = (\underline{K} g_{\underline{C}} \underline{K}^t) \cdot \underline{A}, \end{aligned}$$

where the fourth equality moved the trailing $\underline{K}$ to the front. Both traces are defined, the first on $\hat{\mathbb{E}}^3$ and the second on $\hat{\mathbb{E}}^{3+}$, and they are equal for the same component reason checked after (10.17): the two products are the same sum of products of the same numbers. Comparing the two expressions for dg^+ and using the uniqueness of the gradient gives the claim. Note that $\underline{K} g_{\underline{C}} \underline{K}^t$ is symmetric whenever $g_{\underline{C}}$ is, as it must be.

Step 2: the sandwich is unchanged up to $\underline{Q}$. With $\underline{F}^+ = \underline{Q} \underline{F} \underline{K}^t$ from (7.17),

$$\begin{aligned} \underline{F}^+ g_{\underline{C}^+}^+ (\underline{F}^+)^t &= (\underline{Q} \underline{F} \underline{K}^t) (\underline{K} g_{\underline{C}} \underline{K}^t) (\underline{Q} \underline{F} \underline{K}^t)^t = \underline{Q} \underline{F} \underline{K}^t \underline{K} g_{\underline{C}} \underline{K}^t \underline{K} \underline{F}^t \underline{Q}^t \\ &= \underline{Q} \underline{F} \hat{\underline{I}} g_{\underline{C}} \hat{\underline{I}} \underline{F}^t \underline{Q}^t = \underline{Q} (\underline{F} g_{\underline{C}} \underline{F}^t) \underline{Q}^t. \end{aligned}$$

Every $\underline{K}$ has cancelled, which had to happen: the choice of reference configuration is bookkeeping and cannot survive in a spatial tensor. So by (10.18) as written by O^+ ,

$$\underline{N}^+ = \lambda^+ \underline{F}^+ g_{\underline{C}^+}^+ (\underline{F}^+)^t = \lambda^+ \underline{Q} (\underline{F} g_{\underline{C}} \underline{F}^t) \underline{Q}^t.$$

Step 3: $\underline{N}$ transforms as a spatial tensor. By (7.40) the net stress obeys $\underline{T}^+ = \underline{Q}\underline{T}\underline{Q}^t$, and by (8.35) the constitutive part obeys $\underline{G}^+ = \underline{Q}\underline{G}\underline{Q}^t$, which is exactly the statement that the two observers agree about material response. Since $\underline{N} = \underline{T} - \underline{G}$ and $\underline{N}^+ = \underline{T}^+ - \underline{G}^+$,

$$\underline{N}^+ = \underline{Q}\underline{T}\underline{Q}^t - \underline{Q}\underline{G}\underline{Q}^t = \underline{Q}(\underline{T} - \underline{G})\underline{Q}^t = \underline{Q}\underline{N}\underline{Q}^t = \lambda \underline{Q}(\underline{F}g_{\underline{C}}\underline{F}^t)\underline{Q}^t.$$

Step 4: the multipliers agree. Subtracting the two expressions for $\underline{N}^+$,

$$(\lambda^+ - \lambda) \underline{Q}(\underline{F}g_{\underline{C}}\underline{F}^t)\underline{Q}^t = \underline{Q}.$$

Now $\underline{F}g_{\underline{C}}\underline{F}^t \neq \underline{Q}$, since $\underline{F}$ is invertible and $g_{\underline{C}} \neq \hat{Q}$ for a non-degenerate constraint, and $\underline{Q}$ is invertible, so the conjugated tensor is non-zero. A scalar multiple of a non-zero tensor vanishes only if the scalar does. Therefore

$$\lambda^+ = \lambda.$$

Several constraints. With n constraints, Steps 1 and 2 apply to each $g^{(i)}$ separately, so (10.19) as written by O^+ reads $\underline{N}^+ = \sum_i \lambda_i^+ \underline{Q}(\underline{F}g_{\underline{C}}^{(i)}\underline{F}^t)\underline{Q}^t$, while Step 3 gives $\underline{N}^+ = \sum_i \lambda_i \underline{Q}(\underline{F}g_{\underline{C}}^{(i)}\underline{F}^t)\underline{Q}^t$. Subtracting,

$$\sum_{i=1}^n (\lambda_i^+ - \lambda_i) \underline{Q}(\underline{F}g_{\underline{C}}^{(i)}\underline{F}^t)\underline{Q}^t = \underline{Q},$$

and since conjugation by $\underline{Q}$ is an invertible linear map it preserves linear independence, so the tensors $\underline{Q}(\underline{F}g_{\underline{C}}^{(i)}\underline{F}^t)\underline{Q}^t$ are linearly independent whenever the constraints are independent. Every coefficient must therefore vanish: $\lambda_i^+ = \lambda_i$ for each i .

The three cases. For incompressibility $\lambda = -\bar{p}$, so $\bar{p}^+ = \bar{p}$. The thermodynamic pressure is also observer independent, since $\tilde{p} = \rho^2\psi_\rho$ by (9.74) and both ρ and ψ are absolute scalars by (7.37); hence $p^+ = \tilde{p}^+ + \bar{p}^+ = p$, which is Problem 4 of Chapter 9. For inextensibility $\sigma^+ = \sigma$: the fibre tension is a number both observers read off the same. For rigidity $\lambda_{AB}^+ = \lambda_{AB}$, although in that case the statement is empty of content, since by Remark 10.11 the multipliers are arbitrary to begin with.

Why this had to come out this way. A multiplier is a scalar, and Chapter 7 assumed absolute scalars to be the same for all observers. The calculation above is not an independent discovery of that principle but a consistency check on (10.18): had λ^+ differed from λ , the constraint reaction would have been observer dependent while the constraint itself was not, and (10.14) would have split the stress differently for different observers. $\square$

PROBLEM 3 (A laminate of inextensible sheets). Consider a laminated material formed from thin sheets of stiff paper interspersed in a soft matrix. Take the sheets to be continuously distributed parallel planes spanned by $\hat{E}_1$ and $\hat{E}_2$ in the reference configuration, so that the constraints are $\hat{E}_1 \cdot \hat{C} \hat{E}_2 = 0$ and $\hat{E}_1 \cdot \hat{C} \hat{E}_1 = \hat{E}_2 \cdot \hat{C} \hat{E}_2 = 1$. Show that there can be no extensional or shear strain in the plane of the sheets, but that transverse normal and shear strains are permitted. What is the form of the reactive stress?

Solution. The three constraints are statements about the upper left two by two block of the matrix (C_{AB}):

$$C_{11} = C_{22} = 1, \quad C_{12} = C_{21} = 0,$$

where $C_{AB} = \hat{E}_A \cdot \hat{C} \hat{E}_B$ as in Chapter 2. Nothing is said about C_{13} , C_{23} or C_{33} .

No in-plane strain, in any direction. Take any material fibre lying in a sheet, $\underline{M} = M_1 \hat{E}_1 + M_2 \hat{E}_2$ with $M_1^2 + M_2^2 = 1$. Its stretch follows from (2.83),

$$\begin{aligned} \mu^2 &= \underline{M} \cdot \hat{C} \underline{M} = M_A M_B C_{AB} \\ &= M_1^2 C_{11} + 2M_1 M_2 C_{12} + M_2^2 C_{22} = M_1^2 + M_2^2 = 1, \end{aligned}$$

so $\mu = 1$ for every in-plane direction: not just the two coordinate directions but every fibre in the sheet is inextensible. For the angles, (2.105) gives the cosine of the angle between the images of two reference fibres $\underline{M}$, $\underline{M}'$ as $\underline{M} \cdot \hat{C} \underline{M}' / (\mu \mu')$. With $\underline{M} = \hat{E}_1$ and $\underline{M}' = \hat{E}_2$ this is $C_{12} / (1 \cdot 1) = 0$: the right angle between the two coordinate fibres is

preserved, so there is no in-plane shear either. In terms of the Lagrange strain $\underline{\underline{E}} = \frac{1}{2}(\underline{\underline{C}} - \hat{\underline{\underline{I}}})$ of (2.97) the three statements are simply

$$E_{11} = E_{22} = E_{12} = 0.$$

Since $\underline{\underline{E}}$ restricted to the sheet plane vanishes identically, the sheet deforms rigidly in its own plane. Its local area is preserved as well: with $\underline{\underline{a}}_\alpha = \underline{\underline{F}} \hat{\underline{\underline{E}}}_\alpha$ for $\alpha = 1, 2$ we have $\underline{\underline{a}}_1 \cdot \underline{\underline{a}}_1 = C_{11} = 1$, $\underline{\underline{a}}_2 \cdot \underline{\underline{a}}_2 = C_{22} = 1$ and $\underline{\underline{a}}_1 \cdot \underline{\underline{a}}_2 = C_{12} = 0$, so $\{\underline{\underline{a}}_1, \underline{\underline{a}}_2\}$ is an orthonormal pair and $|\underline{\underline{a}}_1 \times \underline{\underline{a}}_2| = 1$.

What is still permitted. The entries C_{13}, C_{23}, C_{33} are unconstrained, so E_{13}, E_{23}, E_{33} are free: the sheets may slide over one another, tilt relative to their common normal, and move apart or together. Three of the six components of $\underline{\underline{E}}$ are frozen and three are free, which is the count of Remark 10.3 with $n = 3$.

The reactive stress. Write the three constraint functions as

$$g^{(1)} = \hat{\underline{\underline{E}}}_1 \cdot \underline{\underline{C}} \hat{\underline{\underline{E}}}_1 - 1, \quad g^{(2)} = \hat{\underline{\underline{E}}}_2 \cdot \underline{\underline{C}} \hat{\underline{\underline{E}}}_2 - 1, \quad g^{(3)} = \hat{\underline{\underline{E}}}_1 \cdot \underline{\underline{C}} \hat{\underline{\underline{E}}}_2.$$

Their gradients follow from the computation that produced (10.26) and (10.31), since $g^{(1)}$ and $g^{(2)}$ are instances of (10.12) and $g^{(3)}$ is an instance of (10.13) with $A \neq B$:

$$g_C^{(1)} = \hat{\underline{\underline{E}}}_1 \otimes \hat{\underline{\underline{E}}}_1, \quad g_C^{(2)} = \hat{\underline{\underline{E}}}_2 \otimes \hat{\underline{\underline{E}}}_2, \quad g_C^{(3)} = \text{sym}(\hat{\underline{\underline{E}}}_1 \otimes \hat{\underline{\underline{E}}}_2).$$

Substituting in (10.19) and using the dyad rule $\underline{\underline{F}}(\underline{\underline{a}} \otimes \underline{\underline{b}})\underline{\underline{F}}^t = \underline{\underline{F}}\underline{\underline{a}} \otimes \underline{\underline{F}}\underline{\underline{b}}$ from (10.27),

$$\underline{\underline{N}} = \lambda_1 \underline{\underline{a}}_1 \otimes \underline{\underline{a}}_1 + \lambda_2 \underline{\underline{a}}_2 \otimes \underline{\underline{a}}_2 + \frac{1}{2} \lambda_3 (\underline{\underline{a}}_1 \otimes \underline{\underline{a}}_2 + \underline{\underline{a}}_2 \otimes \underline{\underline{a}}_1).$$

Because $\{\underline{\underline{a}}_1, \underline{\underline{a}}_2\}$ was shown above to be orthonormal, write $\underline{\underline{m}}_1 = \underline{\underline{a}}_1$, $\underline{\underline{m}}_2 = \underline{\underline{a}}_2$ and let $\underline{\underline{m}}_3 = \underline{\underline{m}}_1 \times \underline{\underline{m}}_2$ be the unit normal to the deformed sheet, so that $\{\underline{\underline{m}}_1, \underline{\underline{m}}_2, \underline{\underline{m}}_3\}$ is an orthonormal basis of $\mathbb{E}^3$. Then

$$\underline{\underline{N}} = \lambda_1 \underline{\underline{m}}_1 \otimes \underline{\underline{m}}_1 + \lambda_2 \underline{\underline{m}}_2 \otimes \underline{\underline{m}}_2 + \frac{1}{2} \lambda_3 (\underline{\underline{m}}_1 \otimes \underline{\underline{m}}_2 + \underline{\underline{m}}_2 \otimes \underline{\underline{m}}_1).$$

Equivalently: $\underline{\underline{N}}$ is an arbitrary symmetric tensor with $\underline{\underline{N}}\underline{\underline{m}}_3 = \underline{\underline{0}}$, that is, an arbitrary state of *plane stress in the plane of the deformed sheets*. It has three free components, one per constraint.

Check that it does no work. By (10.17) the admissible stretchings are those with $\underline{m}_1 \cdot \underline{D}\underline{m}_1 = \underline{m}_2 \cdot \underline{D}\underline{m}_2 = \underline{m}_1 \cdot \underline{D}\underline{m}_2 = 0$, that is, those whose in-plane block vanishes. The reaction has only an in-plane block. Resolving both on $\{\underline{m}_i\}$, $\underline{N} \cdot \underline{D} = N_{ij} D_{ij}$ and every term has at least one factor from a vanishing block, so $\underline{N} \cdot \underline{D} = 0$. The reaction lives exactly where the stretching is forbidden, and the stretching survives exactly where the reaction is absent. That complementarity is Figure 10.1 again, now with a three-dimensional space of reactions and a three-dimensional space of admissible stretchings inside the six-dimensional sym.

Reading the answer. The reaction is the membrane stress carried by the stiff paper: two normal components in the sheet and one in-plane shear. The matrix is free to shear transversely and to change thickness, and the reaction supplies no stress for those modes, so the transverse response of the laminate is entirely constitutive, which is why a stack of paper is stiff in its plane and floppy across it. $\square$

PROBLEM 4 (A laminate whose sheets preserve area). The body is laminated as in the previous problem, but now constrained in such a way that the planes experience no change in local surface area as they deform. Find the form of the reactive stress.

Solution. The single constraint has to be expressed through $\underline{C}$ before anything else can be done, and Chapter 2 has the tool.

Step 1: the constraint. A material surface element of the sheet has reference unit normal $\hat{\underline{E}}_3$, since the sheets are spanned by $\hat{\underline{E}}_1$ and $\hat{\underline{E}}_2$. By the Piola–Nanson formula (2.54), $\underline{n} \, da = \underline{F}^* \hat{\underline{E}}_3 \, dA$, so taking magnitudes of both sides and using $|\underline{n}| = 1$,

$$\frac{da}{dA} = |\underline{F}^* \hat{\underline{E}}_3| = \sqrt{\hat{\underline{E}}_3 \cdot (\underline{F}^*)^t \underline{F}^* \hat{\underline{E}}_3}.$$

The tensor in the middle is the cofactor of $\underline{C}$. Indeed, by Proposition 1.55, $\underline{F}^* = J \underline{F}^{-t}$, so

$$(\underline{F}^*)^t \underline{F}^* = (J \underline{F}^{-t})^t (J \underline{F}^{-t}) = J^2 \underline{F}^{-1} \underline{F}^{-t} = J^2 (\underline{F}^t \underline{F})^{-1} = (\det \underline{C}) \underline{C}^{-1} = \underline{C}^*,$$

using $\tilde{F}^{-1}\tilde{F}^{-t} = (\tilde{F}^t\tilde{F})^{-1}$ from Proposition 1.46, $J^2 = \det \tilde{C}$ from (10.11), and $\tilde{C}^* = (\det \tilde{C})\tilde{C}^{-1}$ from the derivation of (10.21). This is Remark 1.56 in tensor form: the cofactor is the tensor that carries area elements. Hence $da = dA$ is $\hat{E}_3 \cdot \tilde{C}^* \hat{E}_3 = 1$, and

$$g(\tilde{C}) = \hat{E}_3 \cdot \tilde{C}^* \hat{E}_3 - 1 = C_{33}^* - 1 = C_{11}C_{22} - C_{12}^2 - 1,$$

the last step because the $(3,3)$ entry of the cofactor matrix is the determinant of the complementary two by two block, and $C_{21} = C_{12}$. So the constraint is $\det \tilde{C}_{\parallel} = 1$, where $\tilde{C}_{\parallel}$ denotes the in-plane block. Note in passing that the three constraints of Problem 3 imply this one, since they give $\det \tilde{C}_{\parallel} = 1 \cdot 1 - 0 = 1$; the present constraint is the weaker requirement that the in-plane block have unit determinant rather than be the identity.

Step 2: the gradient. Let $g_{\tilde{C}} = G_{AB} \hat{E}_A \otimes \hat{E}_B$ with G symmetric. For a symmetric increment $d\tilde{C}$,

$$g_{\tilde{C}} \cdot d\tilde{C} = G_{11}dC_{11} + G_{22}dC_{22} + G_{33}dC_{33} + 2G_{12}dC_{12} + 2G_{13}dC_{13} + 2G_{23}dC_{23},$$

the factors of two arising because each off-diagonal pair contributes twice to the sum $G_{AB}dC_{AB}$. Differentiating $g = C_{11}C_{22} - C_{12}^2 - 1$ directly,

$$dg = C_{22} dC_{11} + C_{11} dC_{22} - 2C_{12} dC_{12}.$$

Matching the coefficient of each independent increment,

$$G_{11} = C_{22}, \quad G_{22} = C_{11}, \quad G_{12} = -C_{12}, \quad G_{33} = G_{13} = G_{23} = 0,$$

that is,

$$g_{\tilde{C}} = C_{22} \hat{E}_1 \otimes \hat{E}_1 + C_{11} \hat{E}_2 \otimes \hat{E}_2 - C_{12} (\hat{E}_1 \otimes \hat{E}_2 + \hat{E}_2 \otimes \hat{E}_1).$$

The array (G_{AB}) is the adjugate of the in-plane block, extended by zeros, so

$$g_{\tilde{C}} = (\det \tilde{C}_{\parallel}) \tilde{C}_{\parallel}^{-1} \quad \text{extended by zero,}$$

which is (10.21) one dimension down, as one would expect from a constraint that is a determinant in two dimensions rather than three. On the

constraint set $\det \underline{C}_{\parallel} = 1$ this reduces to $\underline{C}_{\parallel}^{-1}$, the analogue of (10.22). As always the reduction is performed only after the differentiation, per Remark 10.5.

Step 3: the reaction. With $\underline{a}_{\alpha} = \underline{F} \hat{\underline{E}}_{\alpha}$, $\alpha = 1, 2$, and the dyad rule of (10.27),

$$\underline{F} g_{\underline{C}} \underline{F}^t = (\underline{C}_{\parallel}^{-1})_{\alpha\beta} \underline{a}_{\alpha} \otimes \underline{a}_{\beta},$$

the Greek indices running over 1, 2 only. Now observe that $\underline{C}_{\parallel}$ is precisely the Gram matrix of the pair $\{\underline{a}_1, \underline{a}_2\}$, since $\underline{a}_{\alpha} \cdot \underline{a}_{\beta} = \underline{F} \hat{\underline{E}}_{\alpha} \cdot \underline{F} \hat{\underline{E}}_{\beta} = \hat{\underline{E}}_{\alpha} \cdot \underline{C} \hat{\underline{E}}_{\beta} = C_{\alpha\beta}$. The tensor $\underline{P} = (\underline{C}_{\parallel}^{-1})_{\alpha\beta} \underline{a}_{\alpha} \otimes \underline{a}_{\beta}$ is then the orthogonal projection onto the plane they span. To verify this, apply it to $\underline{a}_{\gamma}$:

$$\underline{P} \underline{a}_{\gamma} = (\underline{C}_{\parallel}^{-1})_{\alpha\beta} \underline{a}_{\alpha} \underline{a}_{\beta} \cdot \underline{a}_{\gamma} = (\underline{C}_{\parallel}^{-1})_{\alpha\beta} C_{\beta\gamma} \underline{a}_{\alpha} = \delta_{\alpha\gamma} \underline{a}_{\alpha} = \underline{a}_{\gamma},$$

and to any $\underline{w}$ perpendicular to both: $\underline{P} \underline{w} = (\underline{C}_{\parallel}^{-1})_{\alpha\beta} \underline{a}_{\alpha} \underline{a}_{\beta} \cdot \underline{w} = \underline{0}$. A symmetric tensor that fixes a plane and annihilates its orthogonal complement is the orthogonal projection onto that plane, by Definition 1.59 and the uniqueness of the splitting. Writing $\underline{m}_3$ for the unit normal to the deformed sheet, $\underline{P} = \underline{I} - \underline{m}_3 \otimes \underline{m}_3$, and (10.18) gives

$$\underline{N} = \lambda(\underline{I} - \underline{m}_3 \otimes \underline{m}_3).$$

Reading the answer. Compare (10.23). Incompressibility is the statement that a volume element does not change, and its reaction is an isotropic pressure in three dimensions, $-\bar{p}\underline{I}$. The present constraint is the statement that an area element does not change, and its reaction is an isotropic pressure in the two dimensions of the sheet, $\lambda(\underline{I} - \underline{m}_3 \otimes \underline{m}_3)$. One constraint, one multiplier, and the reaction is the identity of the space in which the measure is frozen.

Consistency with Problem 3. The reaction found there had three free components in the sheet plane; the reaction found here has one, and it is the isotropic member of that three-parameter family, obtained by setting $\lambda_1 = \lambda_2 = \lambda$ and $\lambda_3 = 0$. That is the right relationship: Problem 3 imposes three constraints and Problem 4 imposes one of the consequences of those three, so it must buy a one-dimensional subspace

of the earlier three-dimensional space of reactions. Weaker constraint, smaller reaction, and the arithmetic of Remark 10.3 is respected in both.

Worklessness, checked. Admissible stretchings satisfy $\underline{\underline{P}} \cdot \underline{\underline{D}} = 0$ by (10.17), and $\underline{\underline{P}} \cdot \underline{\underline{D}} = \text{tr}(\underline{\underline{P}}\underline{\underline{D}}) = D_{11} + D_{22}$ resolved on $\{m_i\}$. So the constraint permits any $\underline{\underline{D}}$ whose in-plane trace vanishes, which is the statement that the sheet may stretch in one in-plane direction provided it contracts in the other, keeping the area fixed. And $\underline{\underline{N}} \cdot \underline{\underline{D}} = \lambda \underline{\underline{P}} \cdot \underline{\underline{D}} = 0$ for exactly those $\underline{\underline{D}}$. A two-dimensional pressure works only through a change of area, and the constraint is that there is none. $\square$

Chapter 11

Elastic Solids

An elastic material is what is left of the general model of Chapter 8 when the constitutive functions are made blind to the velocity gradient. Chapter 9 removed the reference configuration instead and arrived at fluids; here the reference configuration stays and the rate of deformation goes. The two deletions are independent, and the chapter shows how much survives each one.

The list of response functions is that of Chapter 8,

$$\mathcal{C} = \{\underline{T}, \underline{q}, \psi, \eta\}, \quad (11.1)$$

and the list of variables on which they are allowed to depend is (8.4) with $\underline{L}$ struck out,

$$\mathcal{L} = \{\underline{x}, \underline{v}, \theta, \underline{F}, \text{grad } \theta\}, \quad (11.2)$$

both evaluated at the same material point p . For the Helmholtz free energy of (6.186) this reads

$$\psi = \hat{\psi}_{\kappa}(\underline{x}, \underline{v}, \theta, \underline{F}_{\kappa}, \text{grad } \theta; p), \quad (11.3)$$

the subscript κ recording that the observer O computes $\underline{F}_{\kappa}$ from a reference configuration of its own choosing.

REMARK 11.1 (what has been deleted, and what that costs). Deleting $\underline{L}$ is not a small change. In Chapter 9 the stress of a viscous fluid was built out of $\underline{D}$, and (9.38) showed that when $\underline{D}$ is switched off nothing remains but a pressure. Here $\underline{D}$ is switched off from the start, and yet

Section 11.2 will produce a stress that depends on the deformation in full. The difference is the reference configuration: a fluid has none to depend on, so removing the rate leaves it with only J , while a solid remembers κ and can measure how far it has been carried from it.

The deletion also has a thermodynamical consequence that is not obvious. For fluids, Chapter 9 had to *assume* that the stress is insensitive to $\text{grad } \theta$. Here that insensitivity will be *proved*, at (11.16)₂, and the proof is the reason (11.13) is worth writing out in full.

Frame invariance, in the form (7.37) carried through the definition (6.186) of ψ , requires that the constitutive function used by a second observer O^+ deliver the same value:

$$\hat{\psi}_\kappa(\underline{x}, \underline{v}, \theta, \underline{F}_\kappa, \text{grad } \theta; p) = \hat{\psi}_{\kappa^+}(\underline{x}^+, \underline{v}^+, \theta^+, \underline{F}_{\kappa^+}, \text{grad}^+ \theta^+; p). \quad (11.4)$$

Now run the argument of Sections 9.1 and 10.1: compare two relative motions of the observers, as at (9.6), chosen so that $\underline{Q}_2(t_0) = \underline{Q}_1(t_0)$ and $\underline{c}_2(t_0) = \underline{c}_1(t_0)$, so that the two agree on the configuration at t_0 and differ only in how they are moving. The conclusion is

$$\hat{\psi}_\kappa(\underline{x}_2, \underline{v}_2, \theta, \underline{F}_2, (\text{grad } \theta)_2; p) = \hat{\psi}_\kappa(\underline{x}_1, \underline{v}_1, \theta, \underline{F}_1, (\text{grad } \theta)_1; p), \quad (11.5)$$

where $\underline{x}_2$ and $\underline{v}_2$ follow from $\underline{x}_1$ and $\underline{v}_1$ by (9.12) with $\underline{Q} = \underline{Q}_2^t \underline{Q}_1$ a rotation, and where

$$\underline{F}_2 = \underline{Q} \underline{F}_1 \quad \text{and} \quad (\text{grad } \theta)_2 = \underline{Q} (\text{grad } \theta)_1. \quad (11.6)$$

Choosing $\underline{Q} = \underline{I}$ leaves $\underline{F}$ and $\text{grad } \theta$ alone while moving $\underline{x}$ and $\underline{v}$, so $\hat{\psi}_\kappa$ can depend on neither of the latter two:

$$\psi = \Psi(\theta, \underline{F}, \text{grad } \theta; p) \quad (11.7)$$

for some function Ψ , the subscript κ now suppressed. Restoring general $\underline{Q}$ in (11.5) then leaves

$$\Psi(\theta, \underline{Q} \underline{F}, \underline{Q} \text{grad } \theta; p) = \Psi(\theta, \underline{F}, \text{grad } \theta; p) \quad \text{for all rotations } \underline{Q}. \quad (11.8)$$

The identical reasoning applied to the stress and the heat flux removes $\underline{x}$ and $\underline{v}$ from them too. For the heat flux,

$$\underline{q} = \hat{q}(\theta, \underline{F}, \text{grad } \theta; p), \quad (11.9)$$

where, $\underline{q}$ being a vector rather than a scalar, the transformation carries a $\underline{Q}$ on the left as well:

$$\hat{q}(\theta, \underline{Q}\underline{F}, \underline{Q} \text{grad } \theta; p) = \underline{Q} \hat{q}(\theta, \underline{F}, \text{grad } \theta; p) \quad \text{for all rotations } \underline{Q}. \quad (11.10)$$

11.1 Clausius–Duhem inequality

Before exploiting (11.8) and (11.10), which are statements about symmetry, we spend the second law. It is (6.187),

$$-\rho(\dot{\psi} + \eta\dot{\theta}) + \underline{T} \cdot \underline{L} - \theta^{-1} \underline{q} \cdot \text{grad } \theta \geq 0,$$

and the plan is to convert every term into a referential one and then read off what the inequality forbids.

Two ingredients are needed. The first is the chain rule applied to (11.7), the free energy now being a function of three arguments:

$$\dot{\psi} = \Psi_{\theta} \dot{\theta} + \Psi_{\underline{F}} \cdot \dot{\underline{F}} + \Psi_{\text{grad } \theta} \cdot (\text{grad } \theta)^{\cdot}, \quad (11.11)$$

in which Ψ_{θ} is a scalar, $\Psi_{\underline{F}}$ a two-point tensor and $\Psi_{\text{grad } \theta}$ a spatial vector, each paired with its argument by the appropriate inner product. The second converts the stress power density into one written on $\dot{\underline{F}}$:

$$\underline{T} \cdot \underline{L} = \underline{T} \cdot \dot{\underline{F}} \underline{F}^{-1} = \text{tr}(\underline{T} \underline{F}^{-t} \dot{\underline{F}}^t) = \underline{T} \underline{F}^{-t} \cdot \dot{\underline{F}}. \quad (11.12)$$

The middle step is $\underline{A} \cdot \underline{B} = \text{tr}(\underline{A} \underline{B}^t)$ from Definition 1.57 with $\underline{B} = \dot{\underline{F}} \underline{F}^{-1}$, whose transpose is $\underline{F}^{-t} \dot{\underline{F}}^t$; the last step reads the same trace backwards with the factors regrouped as $(\underline{T} \underline{F}^{-t})(\dot{\underline{F}}^t)$.

Now multiply the inequality by J , which is positive and so does not turn it around. Three conversions occur at once. By (2.59), $J\rho = \rho_{\kappa}$. By (2.55) and (6.140), $J\underline{T} \underline{F}^{-t} = \underline{T} \underline{F}^* = \underline{P}$, so the stress power becomes

$\underline{P} \cdot \dot{\underline{F}}$, which is (6.156) again. The dissipation term simply acquires the factor J . Substituting (11.11) and collecting the coefficient of each rate,

$$\begin{aligned} -\rho_\kappa(\Psi_\theta + \eta)\dot{\theta} + (\underline{P} - \rho_\kappa\Psi_{\underline{F}}) \cdot \dot{\underline{F}} - \rho_\kappa\Psi_{\text{grad } \theta} \cdot (\text{grad } \theta) \\ - J\theta^{-1}\underline{q} \cdot \text{grad } \theta \geq 0. \end{aligned} \quad (11.13)$$

This has the shape met at (9.64). Write

$$\begin{aligned} \underline{z} &= (\theta, \underline{F}, \text{grad } \theta), & \underline{w} &= (\dot{\theta}, \dot{\underline{F}}, (\text{grad } \theta)^\cdot), \\ \underline{a}(\underline{z}) &= (-\rho_\kappa(\Psi_\theta + \eta), \underline{P} - \rho_\kappa\Psi_{\underline{F}}, -\rho_\kappa\Psi_{\text{grad } \theta}) \end{aligned} \quad (11.14)$$

and

$$b(\underline{z}) = -J\theta^{-1}\underline{q} \cdot \text{grad } \theta, \quad (11.15)$$

so that (11.13) reads $\underline{a}(\underline{z}) \cdot \underline{w} + b(\underline{z}) \geq 0$. The Coleman–Noll argument of (9.68) applies verbatim: at a fixed state $\underline{z}$ the rates $\underline{w}$ may be assigned arbitrarily and independently, so a non-zero $\underline{a}(\underline{z})$ could be defeated by choosing $\underline{w} = -\lambda \underline{a}$ with λ large. Hence $\underline{a}(\underline{z}) = \underline{0}$ and $b(\underline{z}) \geq 0$. Written out, and using $\Psi_{\text{grad } \theta} = \underline{0}$ to drop $\text{grad } \theta$ from the list,

$$\eta = -\Psi_\theta(\theta, \underline{F}; p), \quad \underline{P} = W_{\underline{F}}(\theta, \underline{F}; \underline{X}), \quad \underline{q} \cdot \text{grad } \theta \leq 0, \quad (11.16)$$

where

$$W(\theta, \underline{F}; \underline{X}) = \rho_\kappa(\underline{X}) \Psi(\theta, \underline{F}; \kappa^{-1}(\underline{X})) \quad (11.17)$$

is the free energy per unit volume of κ . Passing from $\rho_\kappa\Psi_{\underline{F}}$ to $W_{\underline{F}}$ is legitimate because ρ_κ depends on $\underline{X}$ alone, by (6.18), and so rides through the differentiation with respect to $\underline{F}$ as a constant. The third of (11.16) follows from $b(\underline{z}) \geq 0$ after dividing by the positive numbers J and θ^{-1} and reversing the sign.

REMARK 11.2 (three results, and the one that is a surprise). The first of (11.16) is familiar: entropy is minus the temperature derivative of the free energy, exactly as in equilibrium thermostatics. The third is the statement that heat does not flow up a temperature gradient, which is the local form of the second law one expects.

The second is the one worth pausing on, and it says two things at once. That $\underline{P} = W_{\underline{F}}$ is the definition of hyperelasticity: the stress is the

gradient of a potential, so no closed cycle of deformation at fixed temperature can extract work. Chapter 9 had no such statement, because the viscous stress there is not a gradient and dissipates.

The other thing it says is hidden in the argument list. Nothing was assumed about the dependence of $\underline{P}$ on $\text{grad } \theta$; what was assumed is that Ψ may depend on it, and the inequality then forced $\Psi_{\text{grad } \theta} = 0$. The stress inherits that, and so the Piola stress, and with it the Cauchy stress by (6.140), cannot depend on the temperature gradient at all. Remark 11.1 promised this. For viscous fluids the same insensitivity had to be put in by hand, as Chapter 9 records.

11.2 Stress response

With $\text{grad } \theta$ gone from Ψ , the invariance requirement (11.8) becomes a statement about W alone. Multiplying (11.8) by ρ_κ and using (11.17),

$$W(\theta, Q\underline{F}; \underline{X}) = W(\theta, \underline{F}; \underline{X}) \quad \text{for all rotations } Q. \quad (11.18)$$

PROPOSITION 11.3 (the content of (11.18)). *A function W satisfies (11.18) if and only if it depends on $\underline{F}$ only through $\underline{C} = \underline{F}^t \underline{F}$, that is, if and only if*

$$W(\theta, \underline{F}; \underline{X}) = \bar{W}(\theta, \underline{C}; \underline{X}) \quad (11.19)$$

for some function $\bar{W}$.

Proof. If (11.19) holds then (11.18) follows, because $Q\underline{F}$ produces the same $\underline{C}$:

$$(Q\underline{F})^t(Q\underline{F}) = \underline{F}^t Q^t Q \underline{F} = \underline{F}^t I \underline{F} = \underline{C},$$

Q being orthogonal.

Conversely, suppose (11.18). Let $\underline{F} = \underline{R}_F \underline{U}$ be the polar decomposition (3.32), and take

$$Q = \underline{1} \underline{R}_F^t.$$

This is a rotation of $\mathbb{E}^3$ onto itself: $\underline{R}_F^t$ carries $\mathbb{E}^3$ to $\hat{\mathbb{E}}^3$ and the shifter $\underline{1}$ carries $\hat{\mathbb{E}}^3$ back to $\mathbb{E}^3$, both preserve inner products, and both have

determinant +1. Substituting,

$$\underline{\underline{Q}}\underline{\underline{F}} = \underline{\underline{1}}\underline{\underline{R}}_F^t \underline{\underline{R}}_F \underline{\underline{U}} = \underline{\underline{1}}\underline{\underline{I}}\underline{\underline{U}} = \underline{\underline{1}}\underline{\underline{U}},$$

so (11.18) gives $W(\theta, \underline{\underline{F}}; \underline{\underline{X}}) = W(\theta, \underline{\underline{1}}\underline{\underline{U}}; \underline{\underline{X}})$. The right-hand side involves $\underline{\underline{F}}$ only through $\underline{\underline{U}}$, and $\underline{\underline{U}}$ is determined by $\underline{\underline{C}}$, being its unique positive-definite square root by Theorem 3.16. Defining $\bar{W}(\theta, \underline{\underline{C}}; \underline{\underline{X}})$ to be $W(\theta, \underline{\underline{1}}\underline{\underline{C}}^{1/2}; \underline{\underline{X}})$ gives (11.19). ■

REMARK 11.4 (why $\underline{\underline{C}}$ and not $\underline{\underline{U}}$). The proof produced $\underline{\underline{U}}$ and then traded it for $\underline{\underline{C}}$. Both are legal arguments, and $\underline{\underline{C}}$ is chosen because it is a polynomial in $\underline{\underline{F}}$ while $\underline{\underline{U}}$ is a square root: differentiating $\bar{W}$ with respect to $\underline{\underline{C}}$ is elementary, while differentiating with respect to $\underline{\underline{U}}$ drags in the derivative of the square root, which is awkward: Theorem 3.16 delivers $\underline{\underline{U}}$ but not a formula for it, and the closed form of Theorem 3.23 degenerates when two principal stretches coincide. The same reason made $\underline{\underline{C}}$ rather than $\underline{\underline{U}}$ the working strain measure at (2.97).

To convert (11.19) into a formula for the stress, differentiate it along a motion. By the chain rule on both sides,

$$\underline{\underline{W}}_F \cdot \dot{\underline{\underline{F}}} = \bar{\underline{\underline{W}}}_C \cdot \dot{\underline{\underline{C}}} = \bar{\underline{\underline{W}}}_C \cdot (\dot{\underline{\underline{F}}}^t \underline{\underline{F}} + \underline{\underline{F}}^t \dot{\underline{\underline{F}}}), \quad (11.20)$$

where $\bar{\underline{\underline{W}}}_C = (\bar{\underline{\underline{W}}}_C)^t$ is symmetric, being the gradient of a scalar with respect to a symmetric tensor. The two terms are now moved onto $\dot{\underline{\underline{F}}}$ using the rules

$$\underline{\underline{A}}_1 \cdot \underline{\underline{A}}_2 \underline{\underline{A}}_3 = \underline{\underline{A}}_2^t \underline{\underline{A}}_1 \cdot \underline{\underline{A}}_3 = \underline{\underline{A}}_3 \underline{\underline{A}}_1^t \cdot \underline{\underline{A}}_2, \quad (11.21)$$

which are Problem 1 and follow from the cyclic property of the trace. Applying the first of them to $\bar{\underline{\underline{W}}}_C \cdot \underline{\underline{F}}^t \dot{\underline{\underline{F}}}$, with $\underline{\underline{A}}_2 = \underline{\underline{F}}^t$, gives $\underline{\underline{F}} \bar{\underline{\underline{W}}}_C \cdot \dot{\underline{\underline{F}}}$. Applying the second to $\bar{\underline{\underline{W}}}_C \cdot \dot{\underline{\underline{F}}}^t \underline{\underline{F}}$, with $\underline{\underline{A}}_2 = \dot{\underline{\underline{F}}}^t$ and $\underline{\underline{A}}_3 = \underline{\underline{F}}$, gives $\underline{\underline{F}} \bar{\underline{\underline{W}}}_C^t \cdot \dot{\underline{\underline{F}}}$, and the symmetry of $\bar{\underline{\underline{W}}}_C$ makes this the same thing. The two terms are therefore equal, and

$$[\underline{\underline{W}}_F - 2\underline{\underline{F}}(\bar{\underline{\underline{W}}}_C)] \cdot \dot{\underline{\underline{F}}} = 0. \quad (11.22)$$

The bracket is a two-point tensor and $\dot{\underline{\underline{F}}}$ is an arbitrary two-point tensor, the motion being at our disposal. Two-point tensors form a linear space,

so a fixed element of it whose inner product with every element vanishes is itself zero. Hence

$$W_{\tilde{F}} = 2\tilde{F}(\bar{W}_{\tilde{C}}). \quad (11.23)$$

Combining this with $\tilde{P} = W_{\tilde{F}}$ from (11.16)₂ and with $\tilde{P} = \tilde{F}\tilde{S}$ from (6.144), and cancelling the invertible $\tilde{F}$,

$$\tilde{S} = 2\bar{W}_{\tilde{C}}. \quad (11.24)$$

This is automatically symmetric, and (6.145) then makes the Cauchy stress symmetric as well.

REMARK 11.5 (rotational momentum balance comes for free). Chapter 6 obtained $\tilde{T} = \tilde{T}^t$ at (6.117) by localizing Euler's second postulate, and Proposition 6.11 showed that it is genuinely independent information. Here it costs nothing: (11.24) delivers a symmetric $\tilde{S}$ because $\bar{W}_{\tilde{C}}$ is the gradient of a scalar with respect to a symmetric argument, and (6.145) transports the symmetry to $\tilde{T}$.

Nothing has been proved twice. What happened is that (11.19), which came from frame invariance, already contains the symmetry: a constitutive function of $\tilde{C}$ cannot produce an unsymmetric stress. Constitutive theory and the balance laws have to be consistent, and this is the consistency being checked. Had (11.24) come out unsymmetric, the model would have been discarded.

11.2.1 Material symmetry

Frame invariance restricts how a constitutive function responds to rotating the *observer*. Material symmetry restricts how it responds to rotating the *reference configuration*, and the two are independent. In the form (8.16) and (8.17) it reads

$$W(\theta, \tilde{F}; \tilde{X}) = W(\theta, \tilde{F}\tilde{R}; \tilde{X}) \quad \text{for all } \tilde{R} \in \mathcal{G}_{\kappa}(p), \quad (11.25)$$

$\mathcal{G}_{\kappa}(p)$ being the symmetry group at p in the configuration κ . Note where $\tilde{R}$ sits: on the right of $\tilde{F}$, because it acts before the deformation, whereas

the $\underline{Q}$ of (11.18) sits on the left and acts after it. Passing to $\bar{W}$ through (11.19), and using $(\underline{F}\underline{R})^t(\underline{F}\underline{R}) = \underline{R}^t\underline{C}\underline{R}$,

$$\bar{W}(\theta, \underline{C}; \underline{X}) = \bar{W}(\theta, \underline{R}^t\underline{C}\underline{R}; \underline{X}). \quad (11.26)$$

REMARK 11.6 (two rotations that must not be confused). The $\underline{R}$ of (11.26) is a referential tensor, mapping $\hat{\mathbb{E}}^3$ to itself, and it is a member of the symmetry group. The $\underline{R}_F$ of the polar decomposition is a two-point tensor, mapping $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$, and it is determined by the deformation. They are different objects wearing the same letter, and the book warns about it at this point. In these notes the polar factor carries the subscript F wherever both appear in one argument, as in the proof of Proposition 11.3 and in (11.44) below.

The unimodular case: fluids again

Suppose $\mathcal{G}_\kappa(p) = U$, the unimodular group of (8.18). The argument of Chapter 9 then applies unchanged and forces W to depend on $\underline{F}$ only through $J = \det \underline{F}$, say $W(\theta, \underline{F}; \underline{X}) = w(\theta, J; \underline{X})$. Hold θ fixed and differentiate along a motion, using (11.16)₂ on the left and Problem 2(b) of Chapter 4, $\partial J / \partial F_{iA} = F_{iA}^*$, on the right:

$$\underline{P} \cdot \dot{\underline{F}} = \dot{W} = \dot{w} = w_J \dot{J} = w_J \underline{F}^* \cdot \dot{\underline{F}}. \quad (11.27)$$

As $\dot{\underline{F}}$ is arbitrary this gives $\underline{P} = w_J \underline{F}^*$, and (6.140) in the form $\underline{P} = \underline{T} \underline{F}^*$ then yields, after cancelling the invertible $\underline{F}^*$,

$$\underline{T} = w_J \underline{I}.$$

Combined with conservation of mass, which ties J to ρ through (2.59), this is exactly the constitutive equation (9.38) of an inviscid fluid, with the pressure $\tilde{p} = -w_J$. So the fluid is not a separate theory: it is the elastic solid whose symmetry group is as large as it can be.

Isotropy

For a solid there is a local undistorted configuration in which the symmetry group is contained in Orth^+ . If it is the whole of Orth^+ the material

is called *hemitropic* at the point: no rotation applied before the deformation can be detected by any experiment. Taking $\kappa(p)$ to be such a configuration,

$$W(\theta, \underline{\underline{F}}; \underline{\underline{X}}) = W(\theta, \underline{\underline{F}} \underline{\underline{R}}; \underline{\underline{X}}) \quad \text{for all } \underline{\underline{R}} \in \text{Orth}^+, \quad (11.28)$$

and therefore, by (11.26),

$$\bar{W}(\theta, \underline{\underline{C}}; \underline{\underline{X}}) = \bar{W}(\theta, \underline{\underline{R}}^t \underline{\underline{C}} \underline{\underline{R}}; \underline{\underline{X}}) \quad \text{for all } \underline{\underline{R}} \in \text{Orth}, \quad (11.29)$$

now over the *full* orthogonal group. The upgrade is free because $\text{Orth} = \text{Orth}^+ \cup \{-\hat{I}\} \text{Orth}^+$: every improper orthogonal tensor is $-\underline{\underline{R}}$ for some rotation $\underline{\underline{R}}$, and

$$(-\underline{\underline{R}})^t \underline{\underline{C}} (-\underline{\underline{R}}) = (-1)^2 \underline{\underline{R}}^t \underline{\underline{C}} \underline{\underline{R}} = \underline{\underline{R}}^t \underline{\underline{C}} \underline{\underline{R}},$$

so the two signs give the same restriction. A function obeying (11.29) is called *isotropic*.

The representation theorem proved in Part 1 of Section 9.1.1 now applies, a scalar isotropic function of a symmetric tensor being a function of its principal invariants:

$$\bar{W}(\theta, \underline{\underline{C}}; \underline{\underline{X}}) = \hat{W}(\theta, I_1(\underline{\underline{C}}), I_2(\underline{\underline{C}}), I_3(\underline{\underline{C}}); \underline{\underline{X}}), \quad (11.30)$$

where

$$I_1(\underline{\underline{C}}) = \text{tr } \underline{\underline{C}}, \quad I_2(\underline{\underline{C}}) = \frac{1}{2}[I_1^2 - \text{tr}(\underline{\underline{C}}^2)], \quad I_3(\underline{\underline{C}}) = \det \underline{\underline{C}} \quad (11.31)$$

are the principal invariants of (3.2). Substituting into (11.24) and writing $\hat{W}_k = \partial \hat{W} / \partial I_k$,

$$\left[\frac{1}{2} \underline{\underline{S}} - \sum_{k=1}^3 \hat{W}_k(I_k) \underline{\underline{C}} \right] \cdot \dot{\underline{\underline{C}}} = 0 \quad \text{for all symmetric } \dot{\underline{\underline{C}}}, \quad (11.32)$$

and the gradients of the invariants, computed in Section B.3 of Appendix B and reproduced here for use,

$$(I_1)_{\underline{\underline{C}}} = \hat{I}, \quad (I_2)_{\underline{\underline{C}}} = I_1 \hat{I} - \underline{\underline{C}}, \quad (I_3)_{\underline{\underline{C}}} = I_3 \underline{\underline{C}}^{-1},$$

deliver the Piola–Kirchhoff stress

$$\mathcal{S} = 2[(\hat{W}_1 + I_1 \hat{W}_2) \hat{I} - \hat{W}_2 \mathcal{C} + I_3 \hat{W}_3 \mathcal{C}^{-1}]. \quad (11.33)$$

Pushing this forward with (6.145), $J\mathcal{T} = \mathcal{F} \mathcal{S} \mathcal{F}^t$, and using

$$\begin{aligned} \mathcal{F} \hat{I} \mathcal{F}^t &= \mathcal{B}, & \mathcal{F} \mathcal{C} \mathcal{F}^t &= \mathcal{F} \mathcal{F}^t \mathcal{F} \mathcal{F}^t = \mathcal{B}^2, \\ \mathcal{F} \mathcal{C}^{-1} \mathcal{F}^t &= \mathcal{F} \mathcal{F}^{-1} \mathcal{F}^{-t} \mathcal{F}^t = \mathcal{I}, \end{aligned}$$

the middle one by $\mathcal{C} = \mathcal{F}^t \mathcal{F}$ and the last by $\mathcal{C}^{-1} = \mathcal{F}^{-1} \mathcal{F}^{-t}$, gives the Cauchy stress

$$J\mathcal{T} = 2[(\hat{W}_1 + I_1 \hat{W}_2) \mathcal{B} - \hat{W}_2 \mathcal{B}^2 + I_3 \hat{W}_3 \mathcal{I}], \quad (11.34)$$

$\mathcal{B}$ being the left Cauchy–Green tensor.

The invariants appearing here may be read off either tensor, because

$$\begin{aligned} I_1(\mathcal{C}) &= \lambda_1^2 + \lambda_2^2 + \lambda_3^2 = I_1(\mathcal{B}), \\ I_2(\mathcal{C}) &= \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_1^2 \lambda_3^2 = I_2(\mathcal{B}), \\ I_3(\mathcal{C}) &= \lambda_1^2 \lambda_2^2 \lambda_3^2 = I_3(\mathcal{B}), \end{aligned} \quad (11.35)$$

the λ_k being the principal stretches. This is Problem 3, and it follows from the spectral representations (3.38): $\mathcal{C}$ and $\mathcal{B}$ carry the same eigenvalues λ_k^2 on different eigenvectors, and the invariants see only the eigenvalues.

If the material is incompressible then $J = 1$, so $I_3 = 1$ throughout the motion and $\hat{W}_3$ is never sampled. The term it multiplied is replaced by the reaction of Chapter 10:

$$\begin{aligned} \mathcal{T} &= -p\mathcal{I} + 2[(W_1^* + I_1 W_2^*) \mathcal{B} - W_2^* \mathcal{B}^2], \\ \text{where } W^*(I_1, I_2; \mathcal{X}) &= \hat{W}(I_1, I_2, 1; \mathcal{X}), \end{aligned} \quad (11.36)$$

with $W_k^* = \partial W^* / \partial I_k$ and p a constitutively indeterminate pressure, the multiplier of Remark 10.8.

REMARK 11.7 (which symmetries a constraint tolerates). The constraint (10.11) of incompressibility, $\det \mathcal{C} = 1$, is unchanged when $\mathcal{C}$ is replaced

by $\underline{R}^t \underline{C} \underline{R}$ for any $\underline{R} \in \text{Orth}$, since $\det(\underline{R}^t \underline{C} \underline{R}) = \det \underline{C}$. So isotropy and incompressibility can hold together. The same computation works for any $\underline{R} \in U$, the determinant being all a unimodular map preserves, so incompressibility is compatible with every symmetry group, fluidity included.

Inextensibility is different. The constraint (10.12) singles out a direction $\underline{M}$, and there are infinitely many $\underline{R} \in \text{Orth}$ that move it. A material reinforced by a family of inextensible fibres therefore cannot be isotropic, which is Problem 5 and which Example 10.10 already suggested: the fibre is visible, so some rotations of the reference configuration are detectable.

Example: simple shear

Take the simple shear of Section 3.5.1 and let it be suffered by a uniform isotropic material, so that $\hat{W}$, or W^* in the incompressible case, carries no explicit $\underline{X}$.

The deformation is homogeneous, so by (3.54) the left Cauchy–Green tensor is a constant field. Hence the stress delivered by (11.34) with $J = 1$, or by (11.36), is also constant, its divergence vanishes, and (6.109) is satisfied with zero body force: simple shear is an equilibrium deformation for every uniform isotropic material. For the incompressible case one condition is added, that $\text{div}(p\underline{I}) = \text{grad } p$ vanish, so the reactive pressure must be uniform.

From (3.54), in the basis $\{\underline{e}_i\}$,

$$(B_{ij}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and squaring the matrix, entry by entry,

$$\begin{aligned} \underline{B}^2 = & [(1 + \gamma^2)^2 + \gamma^2] \underline{e}_1 \otimes \underline{e}_1 + (1 + \gamma^2) \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3 \\ & + \gamma(2 + \gamma^2) (\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1), \end{aligned} \quad (11.37)$$

the 11 entry being $(1 + \gamma^2)^2 + \gamma \cdot \gamma$, the 12 entry $(1 + \gamma^2)\gamma + \gamma \cdot 1 = \gamma(2 + \gamma^2)$, and the 22 entry $\gamma \cdot \gamma + 1 \cdot 1 = 1 + \gamma^2$.

REMARK 11.8 (a misprint in the printed (11.37) and (11.39)). The text prints the 11 component of $\tilde{B}^2$ as $(1 + \gamma^2) + \gamma^2$, both in (11.37) and in the third matrix of (11.39). It should be $(1 + \gamma^2)^2 + \gamma^2$. Squaring the matrix above, the 11 entry is

$$B_{11}B_{11} + B_{12}B_{21} + B_{13}B_{31} = (1 + \gamma^2)^2 + \gamma^2,$$

and at $\gamma = 1$ this is 5, whereas the printed expression gives 3; direct squaring of $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ confirms 5.

The error is not confined to that entry. Section 11.2.1 below derives the universal relation (11.42) from these components, and with the printed value it comes out as $T_{11} - T_{22} = \gamma^2(f_1 + f_2)$, which is not γT_{12} unless $f_2 = 0$. With the corrected value it comes out exactly. So (11.42), which the text calls remarkable and which is the experimentally interesting statement of the section, is inconsistent with (11.37) as printed and consistent with it as corrected.

The invariants follow. With γ the uniform shear,

$$I_1(\tilde{B}) = I_2(\tilde{B}) = 3 + \gamma^2, \quad (11.38)$$

and $I_3(\tilde{B}) = 1$. The first is a trace, $(1 + \gamma^2) + 1 + 1$. The second uses (11.31)₂ with the trace of (11.37),

$$\text{tr } \tilde{B}^2 = (1 + \gamma^2)^2 + \gamma^2 + (1 + \gamma^2) + 1 = 3 + 4\gamma^2 + \gamma^4,$$

so that

$$I_2 = \frac{1}{2}[(3 + \gamma^2)^2 - (3 + 4\gamma^2 + \gamma^4)] = \frac{1}{2}[9 + 6\gamma^2 + \gamma^4 - 3 - 4\gamma^2 - \gamma^4] = 3 + \gamma^2,$$

and $I_3 = \det \tilde{B} = (\det \tilde{F})^2 = 1$. This is Problem 6. Every coefficient in (11.34) and (11.36) is therefore a function of γ^2 alone.

Resolving those equations in the basis $\{\underline{e}_i \otimes \underline{e}_j\}$ and collecting com-

ponents,

$$\begin{aligned}
 (T_{ij}) = f_0 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + f_1(\gamma^2) \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 + f_2(\gamma^2) \begin{pmatrix} (1 + \gamma^2)^2 + \gamma^2 & \gamma(2 + \gamma^2) & 0 \\ \gamma(2 + \gamma^2) & 1 + \gamma^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{11.39}
 \end{aligned}$$

where f_1 and f_2 are constitutive functions and f_0 is another one in the compressible case, or minus the uniform constraint pressure in the incompressible case. Comparing with (11.34), the three matrices are those of $\underline{\underline{I}}$, $\underline{\underline{B}}$ and $\underline{\underline{B}}^2$.

The normal stress effect

Reading (11.39) off the third row and column, $T_{13} = T_{23} = 0$: the shear does not drag the $\underline{e}_3$ direction. The shear component is

$$T_{12} = \gamma\mu(\gamma^2), \quad \text{where} \quad \mu(\gamma^2) = f_1(\gamma^2) + (2 + \gamma^2)f_2(\gamma^2), \tag{11.40}$$

obtained by adding the 12 entries, $f_1\gamma + f_2\gamma(2 + \gamma^2)$, and taking out the common factor γ .

REMARK 11.9 (a second misprint, in (11.40)). The text prints $\mu(\gamma^2) = f_1(\gamma^2) + \gamma(2 + \gamma^2)f_2(\gamma^2)$, carrying a factor γ that does not belong. Two independent checks reject it. First, the 12 entry of (11.39) is $f_1\gamma + f_2\gamma(2 + \gamma^2)$, and dividing by γ leaves $f_1 + (2 + \gamma^2)f_2$. Second, the notation $\mu(\gamma^2)$ asserts that μ depends on γ only through γ^2 , and the printed expression contains a term odd in γ , so it contradicts its own argument list. The same factor also breaks (11.42).

If μ is differentiable then, since it is a function of γ^2 , its expansion about $\gamma = 0$ has no linear term:

$$T_{12} = G\gamma + O(\gamma^3), \tag{11.41}$$

where $G = \mu(0)$ is the classical shear modulus of small deformations. The correction is cubic, not quadratic, and that is a consequence of isotropy: reversing the shear must reverse the shear stress.

The diagonal components T_{11} , T_{22} and T_{33} are in general non-zero, and they satisfy

$$T_{11} - T_{22} = \gamma T_{12}. \quad (11.42)$$

To see it, subtract the 22 entries of (11.39) from the 11 entries. The f_0 terms cancel, being equal, and

$$\begin{aligned} T_{11} - T_{22} &= f_1[(1 + \gamma^2) - 1] + f_2[(1 + \gamma^2)^2 + \gamma^2 - (1 + \gamma^2)] \\ &= f_1\gamma^2 + f_2[(1 + \gamma^2)\gamma^2 + \gamma^2] = \gamma^2[f_1 + (2 + \gamma^2)f_2] = \gamma^2\mu, \end{aligned}$$

where the middle step factored $(1 + \gamma^2)^2 - (1 + \gamma^2) = (1 + \gamma^2)\gamma^2$. Comparing with (11.40), $\gamma T_{12} = \gamma^2\mu$ as well, which is (11.42). This is Problem 7.

REMARK 11.10 (what makes (11.42) worth having). No constitutive function survives in (11.42). Whatever $\hat{W}$ or W^* may be, and whether the material is compressible or not, a uniform isotropic body in simple shear satisfies it. A relation of this kind is called *universal*, and it is the rarest and most valuable thing a nonlinear theory can produce, because it can be checked in the laboratory without first identifying the material.

What it predicts is that shearing a block requires normal tractions as well as shear tractions, and that the difference between two of them is fixed by the shear stress and the shear. Hold the block between two plates and slide one of them: the block tries to change its thickness, and $T_{11} - T_{22}$ is what must be applied to stop it. The right-hand side is $\gamma T_{12} = \gamma^2\mu$, of order γ^2 , so the linear theory of Chapter 12, which discards everything beyond first order in the displacement gradient, predicts none of it. The normal stress effect is inherently nonlinear.

Pure simple shear is hard to realise in a laboratory. Torsion is not, and Section 3.5.4 showed that torsion is locally a simple shear, which is how the effect has been confirmed: a twisted bar tries to lengthen. Problem 12 works that case out.

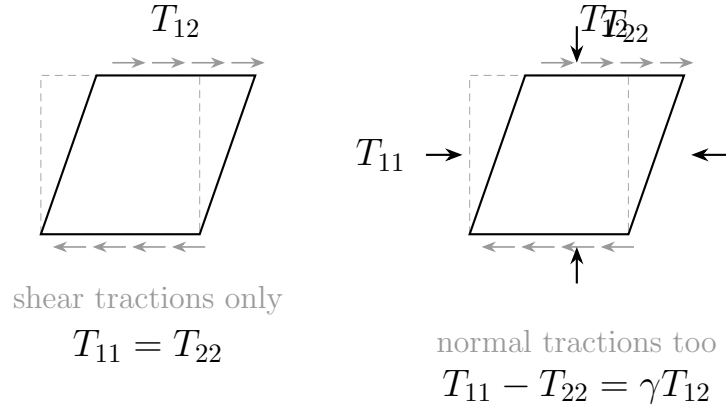


Figure A: The normal stress effect of (11.42), with the undeformed block dashed. Sliding the top face of a block over the bottom one is not achieved by shear tractions alone. A uniform isotropic solid held in simple shear also pushes on the faces normal to $\underline{e}_1$ and $\underline{e}_2$, and the difference of those two normal stresses is γT_{12} , whatever the material. Left is what the linear theory of Chapter 12 predicts, right what (11.34) and (11.36) require; the difference is of order γ^2 and vanishes with the approximation.

11.3 Heat flux

The stress has been reduced by (11.19) and then by isotropy. The same two steps are now applied to the heat flux, and the first of them has a different flavour, because $\hat{q}$ carries a free vector index and cannot be handled by the polar trick alone.

Let the material be hemitropic at p , so that the symmetry condition on the constitutive function for the heat flux reads

$$\hat{q}(\theta, \underline{F}, \text{grad } \theta; p) = \hat{q}(\theta, \underline{F} \underline{R}, \text{grad } \theta; p) \quad \text{for all } \underline{R} \in \text{Orth}^+. \quad (11.43)$$

For a necessary condition, choose the particular member $\underline{R} = \underline{R}_F^t \underline{1}$, where $\underline{1}$ is the shifter and $\underline{R}_F$ the two-point rotation of the polar factorization of $\underline{F}$. It is a referential rotation, being a composition of two orientation-preserving isometries $\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$. Using $\underline{F} = \underline{V} \underline{R}_F$ from (3.32),

$$\hat{q}(\theta, \underline{F}, \text{grad } \theta; p) = \hat{q}(\theta, \underline{V} \underline{R}_F \underline{R}_F^t \underline{1}, \text{grad } \theta; p) = \hat{q}(\theta, \underline{V} \underline{1}, \text{grad } \theta; p), \quad (11.44)$$

since $\underline{R}_F \underline{R}_F^t = \underline{I}$. The shifter is fixed in the frame of O and carries no information about the deformation, and $\underline{V}$ is determined by the left Cauchy–Green tensor $\underline{B} = \underline{F} \underline{F}^t = \underline{V}^2$, uniquely by Theorem 3.16. So the right-hand side is a function of $\underline{B}$:

$$\hat{q}(\theta, \underline{F}, \text{grad } \theta; p) = q^*(\theta, \underline{B}, \text{grad } \theta; \underline{X}) \quad (11.45)$$

for some function q^* . The condition is also sufficient. Substituting (11.45) and using $\underline{R} \underline{R}^t = \underline{I}$,

$$\begin{aligned} \hat{q}(\theta, \underline{F} \underline{R}, \text{grad } \theta; p) &= q^*(\theta, \underline{F} \underline{R} \underline{R}^t \underline{F}^t, \text{grad } \theta; \underline{X}) \\ &= q^*(\theta, \underline{F} \underline{I} \underline{F}^t, \text{grad } \theta; \underline{X}) \\ &= q^*(\theta, \underline{B}, \text{grad } \theta; \underline{X}) = \hat{q}(\theta, \underline{F}, \text{grad } \theta; p) \end{aligned} \quad (11.46)$$

for every $\underline{R} \in \text{Orth}^+$, which is (11.43). So (11.45) is necessary and sufficient for hemitropy.

Frame invariance (11.10) may now be rewritten on q^* . Under $\underline{F} \mapsto \underline{Q} \underline{F}$ the left Cauchy–Green tensor becomes $\underline{Q} \underline{F} \underline{F}^t \underline{Q}^t = \underline{Q} \underline{B} \underline{Q}^t$, so

$$\begin{aligned} q^*(\theta, \underline{Q} \underline{B} \underline{Q}^t, \underline{Q} \text{grad } \theta; \underline{X}) &= \underline{Q} q^*(\theta, \underline{B}, \text{grad } \theta; \underline{X}) \\ &\text{for all rotations } \underline{Q}: \mathbb{E}^3 \rightarrow \mathbb{E}^3. \end{aligned} \quad (11.47)$$

REMARK 11.11 (why $\underline{Q}$ is a rotation here and orthogonal in Chapter 9). The restriction (9.40) on the heat flux in a fluid looked the same but ranged over all orthogonal $\underline{Q}$. The difference is the reference configuration. For a fluid the symmetry group is unimodular, so Chapter 9 was able to absorb κ entirely and the argument lost track of orientation. For a solid κ survives explicitly, and it was chosen occupiable, so only orientation-preserving maps are admissible; the discussion leading to (10.8) made the same point.

The restriction to rotations can be lifted at the cost of one further physical assumption. Suppose the heat flux is odd in the temperature gradient, as it is in the Fourier law (9.61) and as it is in every material for which measurements exist:

$$\underline{q}^*(\theta, \underline{B}, -\text{grad } \theta; \underline{X}) = -\underline{q}^*(\theta, \underline{B}, \text{grad } \theta; \underline{X}). \quad (11.48)$$

Then (11.47) holds for every orthogonal Q . For if Q is improper, write $Q = -Q'$ with $Q' = -Q$ a rotation, and note $Q\tilde{B}Q^t = Q'BQ'^t$, the two signs cancelling, while $Q \text{ grad } \theta = -Q' \text{ grad } \theta$. Hence

$$\begin{aligned} q^*(\theta, Q\tilde{B}Q^t, Q \text{ grad } \theta; X) &= q^*(\theta, Q'BQ'^t, -Q' \text{ grad } \theta; X) \\ &= -q^*(\theta, Q'BQ'^t, Q' \text{ grad } \theta; X) \\ &= -Q'q^*(\theta, \tilde{B}, \text{ grad } \theta; X) \\ &= Qq^*(\theta, \tilde{B}, \text{ grad } \theta; X), \end{aligned}$$

the second equality by (11.48) and the third by (11.47) for the rotation Q' .

With Q now ranging over the full orthogonal group, the representation (9.41) with (9.42) applies and gives

$$\underline{q} = (\phi_0 \tilde{I} + \phi_1 \tilde{B} + \phi_2 \tilde{B}^2) \text{ grad } \theta, \quad (11.49)$$

where ϕ_0, ϕ_1, ϕ_2 are functions of θ , of the invariants $I_1(\tilde{B}), I_2(\tilde{B}), I_3(\tilde{B})$, and of

$$\begin{aligned} I_4(\tilde{B}, \text{ grad } \theta) &= \text{ grad } \theta \cdot \tilde{B} \text{ grad } \theta, & I_5(\tilde{B}, \text{ grad } \theta) &= \text{ grad } \theta \cdot \tilde{B}^2 \text{ grad } \theta, \\ I_6(\text{ grad } \theta) &= |\text{ grad } \theta|, \end{aligned} \quad (11.50)$$

and, for a non-uniform material, of X as well.

11.3.1 General form of the heat flux

Equation (11.49) is isotropic, and it exhibits the heat flux as a linear map applied to $\text{ grad } \theta$. That linearity is not a consequence of isotropy. It holds for every symmetry group, and the reason is the dissipation inequality (11.16)₃ alone.

REMARK 11.12 (a change of letter). The book calls the temperature-gradient variable $\underline{p}$ in what follows. That letter already carries two meanings in this chapter, the material point p of (11.3) and the reaction pressure p of (11.36), and the derivative $\nabla \underline{h}(\underline{p})$ would then be a gradient with respect to something that also indexes the body. These notes write $\underline{g}$ for the variable and keep $\underline{q}$ for the flux.

The claim to be established is

$$\underline{q} = \underline{K}(\underline{\theta}, \underline{F}, \text{grad } \underline{\theta}; \underline{X}) \text{ grad } \underline{\theta} \quad (11.51)$$

for a tensor-valued function $\underline{K}$, a result of [22] holding for every symmetry group.

Fix $\underline{\theta}$, $\underline{F}$ and $\underline{X}$, and hence p , and define

$$\underline{h}(\underline{g}) = \hat{q}(\underline{\theta}, \underline{F}, \underline{g}; p), \quad (11.52)$$

a vector-valued function of the single vector variable $\underline{g}$. The inequality (11.16)₃ says

$$\underline{g} \cdot \underline{h}(\underline{g}) \leq 0 \quad \text{for all } \underline{g}. \quad (11.53)$$

PROPOSITION 11.13 (no gradient, no flux). *If $\underline{h}$ is differentiable then $\underline{h}(\underline{0}) = \underline{0}$; that is, $\hat{q}(\underline{\theta}, \underline{F}, \underline{0}; p) = \underline{0}$.*

Proof. The scalar function $\underline{g} \mapsto \underline{g} \cdot \underline{h}(\underline{g})$ is bounded above by zero, by (11.53), and it attains the value zero at $\underline{g} = \underline{0}$. So it is maximized there, and being differentiable it is stationary there. Its differential at $\underline{g}$, in the direction $d\underline{g}$, is

$$d\underline{g} \cdot \underline{h}(\underline{g}) = d\underline{g} \cdot \underline{h}(\underline{g}) + \underline{g} \cdot [\nabla \underline{h}(\underline{g})] d\underline{g}, \quad (11.54)$$

$\nabla \underline{h}$ being the tensor-valued gradient of $\underline{h}$. Evaluating at $\underline{g} = \underline{0}$ kills the second term, whatever $\nabla \underline{h}(\underline{0})$ may be, and stationarity leaves

$$d\underline{g} \cdot \underline{h}(\underline{0}) = 0 \quad \text{for every } d\underline{g},$$

whence $\underline{h}(\underline{0}) = \underline{0}$ by Lemma 5.1. ■

$$\hat{q}(\underline{\theta}, \underline{F}, \underline{0}; p) = \underline{0}. \quad (11.55)$$

So there is no heat flux in the absence of a temperature gradient. The statement is worth isolating, because it is often taken as a modelling assumption and it is in fact a theorem of the second law.

Now fix $\underline{g}$ and travel out to it along a straight line. Put

$$\underline{y}(x) = \underline{h}(x\underline{g}), \quad x \in [0, 1], \quad (11.56)$$

a curve in $\mathbb{E}^3$ joining $\underline{y}(0) = \underline{h}(0) = 0$ to $\underline{y}(1) = \underline{h}(\underline{g})$. By the chain rule, the argument $\underline{xg}$ having derivative $\underline{g}$ with respect to $\underline{x}$,

$$\underline{y}'(\underline{x}) = [\nabla \underline{h}(\underline{xg})]\underline{g}. \quad (11.57)$$

Integrating from 0 to 1 and using $\underline{y}(0) = 0$ from Proposition 11.13,

$$\underline{h}(\underline{g}) = \int_0^1 \underline{y}'(\underline{x}) \, d\underline{x} = [\underline{K}(\underline{g})]\underline{g}, \quad \text{where} \quad \underline{K}(\underline{g}) = \int_0^1 \nabla \underline{h}(\underline{xg}) \, d\underline{x}, \quad (11.58)$$

the constant vector $\underline{g}$ coming outside the integral because the integration is over $\underline{x}$. Reinstating the passive variables gives (11.51).

REMARK 11.14 (what (11.51) is and is not). It is not a linear constitutive law. The tensor $\underline{K}$ depends on $\text{grad } \theta$, so (11.51) is a rewriting rather than a restriction: it says only that the flux vanishes with the gradient and that the dependence can always be factored through a tensor. Fourier's law is the special case in which $\underline{K}$ does not depend on $\text{grad } \theta$, and (11.49) is the special case in which $\underline{K}$ is the isotropic combination in parentheses.

Nor is $\underline{K}$ unique or symmetric. The construction integrates $\nabla \underline{h}$ along one particular path, the straight line, and a different path would give a different $\underline{K}$ with the same product $\underline{K} \text{grad } \theta$. In (11.49) the $\underline{K}$ that appears is symmetric, being a polynomial in the symmetric $\underline{B}$; in general it is not.

That last point costs nothing, because the dissipation inequality only ever sees the symmetric part. Writing (11.53) with (11.51),

$$\underline{K} \cdot \underline{g} \otimes \underline{g} \leq 0, \quad (11.59)$$

since $\underline{K} \cdot (\underline{g} \otimes \underline{g}) = K_{ij}g_i g_j = \underline{g} \cdot \underline{K} \underline{g}$, and a quadratic form is blind to the skew part of its matrix: $\underline{g} \cdot \underline{K} \underline{g} = \underline{g} \cdot (\text{sym } \underline{K}) \underline{g}$ by Lemma 2.36, $\underline{g} \otimes \underline{g}$ being symmetric.

11.4 Problems

PROBLEM 1 (The three rearrangement rules). Prove (11.21).

Solution. All three expressions are the same trace, read with the factors grouped differently. Start from Definition 1.57, $\underline{A} \cdot \underline{B} = \text{tr}(\underline{A}\underline{B}^t)$, and expand the left member:

$$\underline{A}_1 \cdot \underline{A}_2 \underline{A}_3 = \text{tr}(\underline{A}_1(\underline{A}_2 \underline{A}_3)^t) = \text{tr}(\underline{A}_1 \underline{A}_3^t \underline{A}_2^t),$$

using $(\underline{A}\underline{B})^t = \underline{B}^t \underline{A}^t$.

For the first equality, move $\underline{A}_2^t$ to the front by cyclic permutation and regroup:

$$\text{tr}(\underline{A}_1 \underline{A}_3^t \underline{A}_2^t) = \text{tr}(\underline{A}_2^t \underline{A}_1 \underline{A}_3^t) = \text{tr}((\underline{A}_2^t \underline{A}_1) \underline{A}_3^t) = \underline{A}_2^t \underline{A}_1 \cdot \underline{A}_3.$$

For the second, use $\text{tr} \underline{X} = \text{tr} \underline{X}^t$ on the same expression:

$$\begin{aligned} \text{tr}(\underline{A}_1 \underline{A}_3^t \underline{A}_2^t) &= \text{tr}(\underline{A}_2 \underline{A}_3 \underline{A}_1^t) = \text{tr}(\underline{A}_3 \underline{A}_1^t \underline{A}_2) \\ &= \text{tr}((\underline{A}_3 \underline{A}_1^t)(\underline{A}_2^t)^t) = \underline{A}_3 \underline{A}_1^t \cdot \underline{A}_2^t, \end{aligned}$$

the second step cyclic again. Hence

$$\begin{aligned} \underline{A}_1 \cdot \underline{A}_2 \underline{A}_3 &= \underline{A}_2^t \underline{A}_1 \cdot \underline{A}_3 \\ &= \underline{A}_3 \underline{A}_1^t \cdot \underline{A}_2^t \end{aligned}$$

The rules say that a factor may be moved from one side of an inner product to the other at the cost of a transpose, exactly as $\underline{A}\underline{u} \cdot \underline{v} = \underline{u} \cdot \underline{A}^t \underline{v}$ does for vectors. They are used at (11.22) and again in Problem 12 of Chapter 6. $\square$

PROBLEM 2 (The isotropic Cauchy stress). Derive (11.34).

Solution. Start from (11.33) and push it forward with (6.145), $\underline{J}\underline{T} = \underline{F}\underline{S}\underline{F}^t$:

$$\underline{J}\underline{T} = 2\underline{F}[(\hat{W}_1 + I_1 \hat{W}_2)\hat{\underline{I}} - \hat{W}_2 \underline{C} + I_3 \hat{W}_3 \underline{C}^{-1}]\underline{F}^t,$$

the scalar coefficients passing through $\underline{F}$ untouched. Three products are needed, and each is a one-line computation.

The identity term. $\underline{F}\hat{\underline{I}}\underline{F}^t = \underline{F}\underline{F}^t = \underline{B}$, by the definition of the left Cauchy–Green tensor at (2.83).

The $\underline{C}$ term. Substitute $\underline{C} = \underline{F}^t \underline{F}$ and regroup:

$$\underline{F} \underline{C} \underline{F}^t = \underline{F} (\underline{F}^t \underline{F}) \underline{F}^t = (\underline{F} \underline{F}^t) (\underline{F} \underline{F}^t) = \underline{B}^2.$$

The $\underline{C}^{-1}$ term. Inverting $\underline{C} = \underline{F}^t \underline{F}$ gives $\underline{C}^{-1} = \underline{F}^{-1} \underline{F}^{-t}$, so

$$\underline{F} \underline{C}^{-1} \underline{F}^t = \underline{F} \underline{F}^{-1} \underline{F}^{-t} \underline{F}^t = \underline{I} \underline{I} = \underline{I}.$$

Assembling,

$$J \underline{T} = 2 [(\hat{W}_1 + I_1 \hat{W}_2) \underline{B} - \hat{W}_2 \underline{B}^2 + I_3 \hat{W}_3 \underline{I}]$$

The pattern is worth noticing. Each referential tensor in (11.33) is raised by one power of $\underline{B}$: $\hat{\underline{I}} \mapsto \underline{B}$, $\underline{C} \mapsto \underline{B}^2$, $\underline{C}^{-1} \mapsto \underline{I}$. That is the general rule $\underline{F} \underline{C}^n \underline{F}^t = \underline{B}^{n+1}$, valid for every integer n , positive or negative, and it is why the isotropic stress is a quadratic polynomial in $\underline{B}$ exactly when the energy gradient is a linear one in $\underline{C}$ and $\underline{C}^{-1}$. $\square$

PROBLEM 3 ($\underline{C}$ and $\underline{B}$ have the same invariants). Prove (11.35).

Solution. **By the spectral representations.** The right Cauchy–Green tensor has the representation (3.38)₁,

$$\underline{C} = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i, \quad \text{and} \quad \underline{B} = \sum_i \lambda_i^2 \underline{v}_i \otimes \underline{v}_i,$$

the $\underline{u}_i$ being the referential principal axes, the $\underline{v}_i = \underline{R} \underline{u}_i$ the spatial ones, and the λ_i the principal stretches, common to both by Theorem 3.19. The principal invariants (11.31) are symmetric functions of the eigenvalues,

$$\begin{aligned} I_1 &= \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \\ I_2 &= \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_1^2 \lambda_3^2, \\ I_3 &= \lambda_1^2 \lambda_2^2 \lambda_3^2, \end{aligned}$$

and these expressions mention no eigenvector. Since $\underline{C}$ and $\underline{B}$ carry the same λ_i^2 , they carry the same invariants:

$$I_k(\underline{C}) = I_k(\underline{B}), \quad k = 1, 2, 3.$$

By similarity, in one line. Substituting the polar decomposition $\underline{\underline{F}} = \underline{\underline{R}}\underline{\underline{U}}$ into $\underline{\underline{B}} = \underline{\underline{F}}\underline{\underline{F}}^t$,

$$\underline{\underline{B}} = \underline{\underline{R}}\underline{\underline{U}}\underline{\underline{U}}^t\underline{\underline{R}}^t = \underline{\underline{R}}\underline{\underline{U}}^2\underline{\underline{R}}^t = \underline{\underline{R}}\underline{\underline{C}}\underline{\underline{R}}^t,$$

using $\underline{\underline{U}}^t = \underline{\underline{U}}$ and $\underline{\underline{C}} = \underline{\underline{U}}^2$ from (2.83). So $\underline{\underline{B}}$ and $\underline{\underline{C}}$ are related by a similarity transformation, and similar tensors have the same characteristic polynomial:

$$\begin{aligned} \det(\underline{\underline{B}} - \omega \underline{\underline{I}}) &= \det(\underline{\underline{R}}(\underline{\underline{C}} - \omega \underline{\underline{I}})\underline{\underline{R}}^t) \\ &= \det \underline{\underline{R}} \det(\underline{\underline{C}} - \omega \underline{\underline{I}}) \det \underline{\underline{R}}^t = \det(\underline{\underline{C}} - \omega \underline{\underline{I}}), \end{aligned}$$

since $\det \underline{\underline{R}} \det \underline{\underline{R}}^t = \det(\underline{\underline{R}}\underline{\underline{R}}^t) = 1$. The invariants are the coefficients of that polynomial, by (3.2), so they agree. This version does not need the spectral theorem, and it makes clear what is really going on: $\underline{\underline{B}}$ is $\underline{\underline{C}}$ seen from the deformed configuration, and a change of orthonormal frame cannot alter an invariant. $\square$

PROBLEM 4 (The isotropic response as O^+ writes it). Use (11.34), or (11.36), together with (11.49) to find expressions for the stress and the heat flux in the frame of an observer O^+ .

Solution. Chapter 7 supplies the transformations. Under a change of observer the deformation gradient becomes $\underline{\underline{F}}^+ = \underline{\underline{Q}}\underline{\underline{F}}$, with $\underline{\underline{Q}}(t)$ the relative rotation, and the spatial temperature gradient becomes $\text{grad}^+ \theta^+ = \underline{\underline{Q}} \text{grad} \theta$, while θ itself and the reference configuration are untouched.

What is invariant. First,

$$\underline{\underline{B}}^+ = \underline{\underline{F}}^+(\underline{\underline{F}}^+)^t = \underline{\underline{Q}}\underline{\underline{F}}\underline{\underline{F}}^t\underline{\underline{Q}}^t = \underline{\underline{Q}}\underline{\underline{B}}\underline{\underline{Q}}^t, \quad \underline{\underline{C}}^+ = (\underline{\underline{Q}}\underline{\underline{F}})^t(\underline{\underline{Q}}\underline{\underline{F}}) = \underline{\underline{F}}^t\underline{\underline{F}} = \underline{\underline{C}},$$

so $\underline{\underline{C}}$ does not move at all and $\underline{\underline{B}}$ moves by conjugation. Second, $J^+ = \det \underline{\underline{F}}^+ = \det \underline{\underline{Q}} \det \underline{\underline{F}} = J$. Third, the invariants are unchanged: $I_k(\underline{\underline{B}}^+) = I_k(\underline{\underline{Q}}\underline{\underline{B}}\underline{\underline{Q}}^t) = I_k(\underline{\underline{B}})$ by the similarity argument of Problem 3, and for the three of (11.50),

$$I_4(\underline{\underline{B}}^+, \text{grad}^+ \theta^+) = \underline{\underline{Q}} \text{grad} \theta \cdot \underline{\underline{Q}}\underline{\underline{B}}\underline{\underline{Q}}^t \underline{\underline{Q}} \text{grad} \theta = \text{grad} \theta \cdot \underline{\underline{B}} \text{grad} \theta = I_4,$$

using $\underline{\underline{Q}}^t \underline{\underline{Q}} = \underline{\underline{I}}$ twice and $\underline{\underline{Q}} \underline{\underline{a}} \cdot \underline{\underline{Q}} \underline{\underline{b}} = \underline{\underline{a}} \cdot \underline{\underline{b}}$; identically I_5 is unchanged, and $I_6 = |\text{grad } \theta|$ is unchanged because $\underline{\underline{Q}}$ preserves length. So every one of the response functions $\hat{W}_k$ and ϕ_k takes the same value for both observers.

The stress. Writing (11.34) in the frame of O^+ and substituting,

$$\begin{aligned} J \underline{\underline{T}}^+ &= 2[(\hat{W}_1 + I_1 \hat{W}_2) \underline{\underline{B}}^+ - \hat{W}_2 (\underline{\underline{B}}^+)^2 + I_3 \hat{W}_3 \underline{\underline{I}}] \\ &= 2[(\hat{W}_1 + I_1 \hat{W}_2) \underline{\underline{Q}} \underline{\underline{B}} \underline{\underline{Q}}^t - \hat{W}_2 \underline{\underline{Q}} \underline{\underline{B}}^2 \underline{\underline{Q}}^t + I_3 \hat{W}_3 \underline{\underline{Q}} \underline{\underline{Q}}^t] = \underline{\underline{Q}} (J \underline{\underline{T}}) \underline{\underline{Q}}^t, \end{aligned}$$

where $(\underline{\underline{B}}^+)^2 = \underline{\underline{Q}} \underline{\underline{B}} \underline{\underline{Q}}^t \underline{\underline{Q}} \underline{\underline{B}} \underline{\underline{Q}}^t = \underline{\underline{Q}} \underline{\underline{B}}^2 \underline{\underline{Q}}^t$ and $\underline{\underline{I}} = \underline{\underline{Q}} \underline{\underline{Q}}^t$. Cancelling J ,

$$\underline{\underline{T}}^+ = \underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t$$

The incompressible form (11.36) gives the same conclusion, the reaction transforming as $p^+ = p$ because $\underline{\underline{Q}}(-p \underline{\underline{I}}) \underline{\underline{Q}}^t = -p \underline{\underline{I}}$.

The heat flux. By (11.49),

$$\begin{aligned} \underline{\underline{q}}^+ &= (\phi_0 \underline{\underline{I}} + \phi_1 \underline{\underline{B}}^+ + \phi_2 (\underline{\underline{B}}^+)^2) \text{grad}^+ \theta^+ \\ &= \underline{\underline{Q}} (\phi_0 \underline{\underline{I}} + \phi_1 \underline{\underline{B}} + \phi_2 \underline{\underline{B}}^2) \underline{\underline{Q}}^t \underline{\underline{Q}} \text{grad } \theta = \underline{\underline{Q}} \underline{\underline{q}}. \end{aligned}$$

So

$$\underline{\underline{q}}^+ = \underline{\underline{Q}} \underline{\underline{q}}$$

Reading the answer. Nothing new has been derived. These are precisely the transformation rules that Chapter 7 established for the Cauchy stress and the heat flux from the balance laws, and the point of the exercise is that the constitutive equations reproduce them automatically. That is what frame invariance was imposed for: had (11.34) delivered anything other than $\underline{\underline{Q}} \underline{\underline{T}} \underline{\underline{Q}}^t$, the two observers would have disagreed about the material, and the model would be inadmissible. $\square$

PROBLEM 5 (Are the laminate constraints compatible with isotropy?). Are the constraints considered in Problems 3 and 4 of Chapter 10 compatible with isotropy?

Solution. The test is the one applied to incompressibility in Remark 11.7: a constraint is compatible with isotropy when it is unchanged by $\underline{C} \mapsto \underline{R}^t \underline{C} \underline{R}$ for every $\underline{R} \in \text{Orth}$. The answer in both cases is no, and for the same reason.

The inextensible laminate. The constraints of Problem 3 of Chapter 10 are

$$\hat{\underline{E}}_1 \cdot \underline{C} \hat{\underline{E}}_1 = \hat{\underline{E}}_2 \cdot \underline{C} \hat{\underline{E}}_2 = 1, \quad \hat{\underline{E}}_1 \cdot \underline{C} \hat{\underline{E}}_2 = 0.$$

Replace $\underline{C}$ by $\underline{R}^t \underline{C} \underline{R}$ and move $\underline{R}$ across the inner product:

$$\hat{\underline{E}}_A \cdot \underline{R}^t \underline{C} \underline{R} \hat{\underline{E}}_B = \underline{R} \hat{\underline{E}}_A \cdot \underline{C} \underline{R} \hat{\underline{E}}_B.$$

The transformed constraints are therefore the original ones written about the rotated directions $\underline{R} \hat{\underline{E}}_1$ and $\underline{R} \hat{\underline{E}}_2$. They agree with the originals for every $\underline{C}$ only if $\underline{R}$ maps the plane spanned by $\hat{\underline{E}}_1, \hat{\underline{E}}_2$ onto itself, and most rotations do not. Take $\underline{R}$ to be the quarter turn about $\hat{\underline{E}}_1$, which sends $\hat{\underline{E}}_2 \mapsto \hat{\underline{E}}_3$: the transformed constraint $\hat{\underline{E}}_3 \cdot \underline{C} \hat{\underline{E}}_3 = 1$ forbids transverse stretch, which the original explicitly permits. So the constraint set is not invariant, and the material is not isotropic.

The area-preserving laminate. The constraint of Problem 4 of Chapter 10 says that the sheets, whose reference normal is $\hat{\underline{E}}_3$, do not change area. By the Piola–Nanson formula (2.54) the area ratio is $|\underline{F}^* \hat{\underline{E}}_3|$, and

$$|\underline{F}^* \hat{\underline{E}}_3|^2 = \hat{\underline{E}}_3 \cdot \underline{F}^{*t} \underline{F}^* \hat{\underline{E}}_3 = J^2 \hat{\underline{E}}_3 \cdot \underline{F}^{-1} \underline{F}^{-t} \hat{\underline{E}}_3 = I_3 \hat{\underline{E}}_3 \cdot \underline{C}^{-1} \hat{\underline{E}}_3,$$

using $\underline{F}^* = J \underline{F}^{-t}$ and $J^2 = \det \underline{C} = I_3$. The constraint is therefore

$$I_3 \hat{\underline{E}}_3 \cdot \underline{C}^{-1} \hat{\underline{E}}_3 = 1.$$

Under $\underline{C} \mapsto \underline{R}^t \underline{C} \underline{R}$ the factor I_3 is invariant, but the second factor becomes $\underline{R} \hat{\underline{E}}_3 \cdot \underline{C}^{-1} \underline{R} \hat{\underline{E}}_3$, so again the constraint is carried to the same statement about a different direction. Only rotations fixing the axis $\hat{\underline{E}}_3$ leave it alone. Not isotropic.

Why incompressibility is the exception. Both of these constraints name a direction, $\hat{\underline{E}}_3$ or the plane normal to it, and a named

direction is exactly what a rotation of the reference configuration can move. Incompressibility names none: $\det \mathcal{C} = 1$ mentions only an invariant. That is the whole content of Remark 11.7, and Example 10.10 said the same thing physically. A fibre or a laminate can be seen, so the undeformed body is not the same in every direction. $\square$

PROBLEM 6 (The invariants in simple shear). Verify (11.38).

Solution. From (3.54) the left Cauchy–Green tensor of simple shear has the matrix

$$(B_{ij}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The first invariant is the trace, by (11.31)₁:

$$I_1 = B_{11} + B_{22} + B_{33} = (1 + \gamma^2) + 1 + 1 = 3 + \gamma^2.$$

The second needs $\text{tr}(\mathcal{B}^2)$. From (11.37), whose correction is the subject of Remark 11.8,

$$\text{tr}(\mathcal{B}^2) = [(1 + \gamma^2)^2 + \gamma^2] + (1 + \gamma^2) + 1.$$

Expanding $(1 + \gamma^2)^2 = 1 + 2\gamma^2 + \gamma^4$ and adding,

$$\text{tr}(\mathcal{B}^2) = 1 + 2\gamma^2 + \gamma^4 + \gamma^2 + 1 + \gamma^2 + 1 = 3 + 4\gamma^2 + \gamma^4.$$

Then (11.31)₂ gives

$$I_2 = \frac{1}{2}[(3 + \gamma^2)^2 - (3 + 4\gamma^2 + \gamma^4)] = \frac{1}{2}[9 + 6\gamma^2 + \gamma^4 - 3 - 4\gamma^2 - \gamma^4] = 3 + \gamma^2.$$

The third is a determinant, and simple shear is isochoric:

$$I_3 = \det \mathcal{B} = \det(\mathcal{F}\mathcal{F}^t) = (\det \mathcal{F})^2 = 1.$$

Hence

$$I_1(\mathcal{B}) = I_2(\mathcal{B}) = 3 + \gamma^2, \quad I_3(\mathcal{B}) = 1.$$

The coincidence $I_1 = I_2$ is not an accident of the algebra. For an isochoric deformation $I_3 = 1$, so the principal stretches satisfy $\lambda_1 \lambda_2 \lambda_3 = 1$, and then

$$I_2 = \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_3^2 \lambda_1^2 = \lambda_3^{-2} + \lambda_1^{-2} + \lambda_2^{-2},$$

which is I_1 computed for the inverse deformation. Simple shear happens to have the same stretches as its own inverse, up to a relabelling, which is why the two invariants agree here. $\square$

PROBLEM 7 (The universal relation). Derive (11.42).

Solution. Take the 11 and 22 components of (11.39). The first matrix contributes f_0 to each, so it cancels in the difference. The other two contribute

$$\begin{aligned} T_{11} &= f_0 + f_1(1 + \gamma^2) + f_2[(1 + \gamma^2)^2 + \gamma^2], \\ T_{22} &= f_0 + f_1 \cdot 1 + f_2(1 + \gamma^2). \end{aligned}$$

Subtracting,

$$\begin{aligned} T_{11} - T_{22} &= f_1[(1 + \gamma^2) - 1] + f_2[(1 + \gamma^2)^2 + \gamma^2 - (1 + \gamma^2)] \\ &= f_1\gamma^2 + f_2[(1 + \gamma^2)((1 + \gamma^2) - 1) + \gamma^2] \\ &= f_1\gamma^2 + f_2[(1 + \gamma^2)\gamma^2 + \gamma^2] = \gamma^2[f_1 + (2 + \gamma^2)f_2], \end{aligned}$$

the last step taking γ^2 out and simplifying $(1 + \gamma^2) + 1 = 2 + \gamma^2$. The bracket is $\mu(\gamma^2)$ by (11.40), so $T_{11} - T_{22} = \gamma^2\mu$.

On the other hand $T_{12} = \gamma\mu$, again by (11.40), so $\gamma T_{12} = \gamma^2\mu$ as well. Hence

$$T_{11} - T_{22} = \gamma T_{12}$$

Two remarks on the mechanism. The constitutive functions entered only through the combination μ , and they entered the two sides in the same way, which is why they cancel; that cancellation is the whole content of the word *universal*. And the computation used the corrected 11 entry of $\underline{B}^2$ throughout: with the printed value the f_2 bracket collapses to γ^2 instead of $\gamma^2(2 + \gamma^2)$, and the relation fails, which is the evidence given in Remark 11.8. $\square$

PROBLEM 8 (The neo-Hookean material, and a bar in tension). Consider the strain energy $W^* = \frac{1}{2}G(I_1 - 3)$ with G a positive constant. This model, the *neo-Hookean* material, describes incompressible rubber well while the principal stretches stay in the approximate range $\frac{1}{2} < \lambda_i < 2$. (a) Show that $\underline{\underline{T}} = -p\underline{\underline{I}} + G\underline{\underline{B}}$, and that equilibrium without body force requires $\text{grad } p = G \text{div } \underline{\underline{B}}$. (b) Use the power conjugacy of $\underline{\underline{P}}$ and $\underline{\underline{F}}$ to deduce that G is the ratio of the shear component of the Piola traction to the amount of shear in a simple shear. (c) For the deformation of Problem 6 of Chapter 4, find $\underline{\underline{C}}$ and $\underline{\underline{B}}$. (d) Impose $\underline{\underline{T}}\underline{\underline{n}} = \underline{\underline{0}}$ for every unit $\underline{\underline{n}}$ with $\underline{\underline{n}} \cdot \underline{\underline{e}}_1 = 0$, simulating the traction-free lateral surface of a prismatic bar with axis $\underline{\underline{e}}_1$, and deduce $p = G/\lambda$. (e) Show that $\underline{\underline{T}} = T(\lambda)\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$ with $T(\lambda) = G(\lambda^2 - \lambda^{-1})$. (f) Young's modulus is $E = T'(1)$. Find E in terms of G . (g) Show that $\underline{\underline{P}} = P(\lambda)\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_1$ and $\underline{\underline{S}} = S(\lambda)\hat{\underline{\underline{E}}}_1 \otimes \hat{\underline{\underline{E}}}_1$, determine P and S , and relate $P'(1)$ and $S'(1)$ to E .

Solution. (a) With $W^* = \frac{1}{2}G(I_1 - 3)$ the derivatives are $W_1^* = \frac{1}{2}G$ and $W_2^* = 0$, so (11.36) gives

$$\underline{\underline{T}} = -p\underline{\underline{I}} + 2\left[\left(\frac{1}{2}G + I_1 \cdot 0\right)\underline{\underline{B}} - 0 \cdot \underline{\underline{B}}^2\right] = -p\underline{\underline{I}} + G\underline{\underline{B}}$$

Equilibrium without body force is $\text{div } \underline{\underline{T}} = \underline{\underline{0}}$ by (6.109) with $\underline{\underline{v}} = \underline{\underline{0}}$. Since $\text{div}(p\underline{\underline{I}}) = \text{grad } p$, by Problem 24 of Chapter 1, and G is constant,

$$-\text{grad } p + G \text{div } \underline{\underline{B}} = \underline{\underline{0}}, \quad \text{that is} \quad \text{grad } p = G \text{div } \underline{\underline{B}}$$

The equation determines p up to a constant once the deformation is known, and it is solvable only if $\text{div } \underline{\underline{B}}$ is a gradient. That is a genuine restriction: not every isochoric deformation of a neo-Hookean body can be maintained without body force.

(b) By (6.156) the stress power per unit reference volume is $\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}}$, and for a hyperelastic material at fixed temperature that equals $\dot{W}^*$. In simple shear $\underline{\underline{F}} = \underline{\underline{1}} + \gamma\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2$ with $I_1 = 3 + \gamma^2$ by (11.38), so

$$W^* = \frac{1}{2}G(3 + \gamma^2 - 3) = \frac{1}{2}G\gamma^2, \quad \dot{W}^* = G\gamma\dot{\gamma}.$$

On the other side, $\dot{\underline{\underline{F}}} = \dot{\gamma}\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2$, and by Definition 1.57,

$$\underline{\underline{P}} \cdot \dot{\underline{\underline{F}}} = \dot{\gamma}\underline{\underline{P}} \cdot \underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2 = \dot{\gamma}\underline{\underline{e}}_1 \cdot \underline{\underline{P}}\hat{\underline{\underline{E}}}_2 = \dot{\gamma}P_{12}.$$

The reactive part contributes nothing, since its share of the power is $-p\tilde{F}^* \cdot \dot{\tilde{F}} = -p\dot{J} = 0$ by Problem 2 of Chapter 4 and $J \equiv 1$. Equating and cancelling the arbitrary $\dot{\gamma}$,

$$P_{12} = G\gamma, \quad G = P_{12}/\gamma$$

and $P_{12} = \underline{e}_1 \cdot P \hat{E}_2$ is the $\underline{e}_1$ component of the Piola traction on the material plane whose reference normal is $\hat{E}_2$, which is the shear component. So G is the shear stress per unit shear, the shear modulus, and here the identification is exact rather than a small-strain approximation.

(c) Problem 6 of Chapter 4 has

$$\tilde{F} = \lambda \underline{e}_1 \otimes \hat{E}_1 + \lambda^{-1/2} (\underline{e}_2 \otimes \hat{E}_2 + \underline{e}_3 \otimes \hat{E}_3),$$

an axial stretch λ with equal lateral contractions, isochoric because $\lambda \cdot \lambda^{-1/2} \cdot \lambda^{-1/2} = 1$. Multiplying out with $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c})\underline{a} \otimes \underline{d}$, only the diagonal terms surviving,

$$\begin{aligned} \underline{C} &= \tilde{F}^t \tilde{F} = \lambda^2 \hat{E}_1 \otimes \hat{E}_1 + \lambda^{-1} (\hat{E}_2 \otimes \hat{E}_2 + \hat{E}_3 \otimes \hat{E}_3), \\ \underline{B} &= \tilde{F} \tilde{F}^t = \lambda^2 \underline{e}_1 \otimes \underline{e}_1 + \lambda^{-1} (\underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3). \end{aligned}$$

They carry the same numbers on different bases, as Problem 3 requires.

(d) Let $\underline{n}$ be a unit vector with $\underline{n} \cdot \underline{e}_1 = 0$, so that $\underline{n}$ lies in the plane spanned by $\underline{e}_2$ and $\underline{e}_3$. On that plane $\underline{B}$ acts as λ^{-1} times the identity, by (c), so $\underline{B}\underline{n} = \lambda^{-1}\underline{n}$ and

$$\underline{T}\underline{n} = -p\underline{n} + G\lambda^{-1}\underline{n} = (-p + G\lambda^{-1})\underline{n}.$$

Setting this to zero, and $\underline{n} \neq 0$,

$$p = G/\lambda$$

The pressure is uniform, so $\text{grad } p = 0$; and $\underline{B}$ is uniform, so $\text{div } \underline{B} = 0$; hence (a) is satisfied and the deformation is equilibrated without body force.

(e) Substituting $p = G/\lambda$ into (a) and using (c),

$$\begin{aligned}\tilde{T} &= -\frac{G}{\lambda}\tilde{I} + G\left[\lambda^2\mathbf{e}_1 \otimes \mathbf{e}_1 + \lambda^{-1}(\mathbf{e}_2 \otimes \mathbf{e}_2 + \mathbf{e}_3 \otimes \mathbf{e}_3)\right] \\ &= G\left(\lambda^2 - \frac{1}{\lambda}\right)\mathbf{e}_1 \otimes \mathbf{e}_1 + G\left(\frac{1}{\lambda} - \frac{1}{\lambda}\right)(\mathbf{e}_2 \otimes \mathbf{e}_2 + \mathbf{e}_3 \otimes \mathbf{e}_3),\end{aligned}$$

the lateral coefficients cancelling exactly, which is what (d) arranged. Hence

$$\tilde{T} = T(\lambda) \mathbf{e}_1 \otimes \mathbf{e}_1, \quad T(\lambda) = G(\lambda^2 - \lambda^{-1}).$$

The bar carries a uniaxial stress, and $T(1) = 0$: the undeformed state is unstressed, as it must be for a model with no residual stress.

(f) Differentiating, $T'(\lambda) = G(2\lambda + \lambda^{-2})$, so

$$E = T'(1) = G(2 + 1) = 3G$$

This is the incompressible limit of the classical relation $E = 2G(1 + \nu)$ at $\nu = \frac{1}{2}$, and Chapter 12 will recover it from the linear theory.

(g) By (6.140), $\tilde{P} = \tilde{T}\tilde{F}^* = J\tilde{T}\tilde{F}^{-t}$ with $J = 1$. Problem 6(b) of Chapter 4 gives $\tilde{F}^{-1} = \lambda^{-1}\hat{\tilde{E}}_1 \otimes \mathbf{e}_1 + \lambda^{1/2}(\hat{\tilde{E}}_2 \otimes \mathbf{e}_2 + \hat{\tilde{E}}_3 \otimes \mathbf{e}_3)$, so

$$\tilde{F}^{-t} = \lambda^{-1}\mathbf{e}_1 \otimes \hat{\tilde{E}}_1 + \lambda^{1/2}(\mathbf{e}_2 \otimes \hat{\tilde{E}}_2 + \mathbf{e}_3 \otimes \hat{\tilde{E}}_3).$$

Multiplying, only the $\mathbf{e}_1$ term of $\tilde{T}$ surviving,

$$\tilde{P} = T(\lambda)(\mathbf{e}_1 \otimes \mathbf{e}_1)\tilde{F}^{-t} = T(\lambda)\lambda^{-1}\mathbf{e}_1 \otimes \hat{\tilde{E}}_1,$$

and by (6.144), $\tilde{S} = \tilde{F}^{-1}\tilde{P} = P(\lambda)\lambda^{-1}\hat{\tilde{E}}_1 \otimes \hat{\tilde{E}}_1$. So

$$P(\lambda) = \frac{T(\lambda)}{\lambda} = G(\lambda - \lambda^{-2}), \quad S(\lambda) = \frac{P(\lambda)}{\lambda} = G(1 - \lambda^{-3}).$$

Differentiating, $P'(\lambda) = G(1 + 2\lambda^{-3})$ and $S'(\lambda) = 3G\lambda^{-4}$, so

$$P'(1) = 3G = E, \quad S'(1) = 3G = E.$$

All three stress measures have the same slope at the undeformed state. That is the general reason why the linear theory of Chapter 12 need not say which stress it means: the three differ at second order in the strain, and the linear theory keeps only the first. $\square$

PROBLEM 9 (Anti-plane shear of a neo-Hookean tube). A circular cylindrical bar of neo-Hookean material, with axis $\hat{E}_3$ in κ and inner and outer radii A and B , undergoes the static anti-plane shear of Problem 5 of Chapter 3. It is in equilibrium under negligible body force, and $w(A) = W$, $w(B) = 0$ with W a specified constant. Obtain $w(R)$.

Solution. **The kinematics.** The deformation is $r = R$, $\theta = \Theta$, $z = Z + w(R)$: cylinders slide along the axis without changing radius. Identifying $\hat{E}_R$ with $\underline{e}_r$ and $\hat{E}_\Theta$ with $\underline{e}_\theta$ through the shifter, since $r = R$ and $\theta = \Theta$,

$$\underline{F} = \underline{1} + w'(r) \underline{e}_3 \otimes \hat{E}_R, \quad \det \underline{F} = 1,$$

the determinant being that of a matrix with unit diagonal and a single entry below it. Then

$$\underline{B} = \underline{F} \underline{F}^t = \underline{I} + w'(\underline{e}_r \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_r) + w'^2 \underline{e}_3 \otimes \underline{e}_3,$$

so the non-zero components are $B_{rr} = B_{\theta\theta} = 1$, $B_{33} = 1 + w'^2$ and $B_{r3} = B_{3r} = w'$.

The stress. By Problem 8(a),

$$T_{rr} = T_{\theta\theta} = -p + G, \quad T_{33} = -p + G(1 + w'^2), \quad T_{r3} = G w'(r),$$

all functions of r alone, the remaining components zero.

Equilibrium. With nothing depending on θ or z , the cylindrical components of $\operatorname{div} \underline{T}$ reduce to

$$(\operatorname{div} \underline{T})_r = \frac{\partial T_{rr}}{\partial r} + \frac{T_{rr} - T_{\theta\theta}}{r}, \quad (\operatorname{div} \underline{T})_3 = \frac{1}{r} \frac{\partial}{\partial r} (r T_{3r}),$$

and $(\operatorname{div} \underline{T})_\theta = 0$ identically. Since $T_{rr} = T_{\theta\theta}$, the radial equation is $-p'(r) = 0$, so the reactive pressure is uniform. The axial equation is the one that determines the motion:

$$\frac{1}{r} (G r w')' = 0 \quad \implies \quad r w'(r) = c$$

for a constant c , whence $w' = c/r$ and

$$w(r) = c \ln r + d.$$

The boundary conditions. Imposing $w(A) = W$ and $w(B) = 0$,

$$c \ln A + d = W, \quad c \ln B + d = 0,$$

so $d = -c \ln B$ and $c \ln(A/B) = W$. Therefore

$$w(R) = W \frac{\ln(R/B)}{\ln(A/B)}$$

which returns W at $R = A$ and 0 at $R = B$, as required.

Reading the answer. The profile is logarithmic, and it contains no material constant: G cancelled when the axial equation was divided through. So every neo-Hookean tube, stiff or soft, shears into the same shape, and G affects only the force needed. The axial traction on the inner surface is $T_{3r}(A) = Gc/A = GW/[A \ln(A/B)]$, and the total axial force there, over a length l , is $2\pi Al T_{3r}(A) = 2\pi lGW/\ln(A/B)$, independent of A ; the same magnitude with opposite sign acts on the outer surface, as it must for equilibrium of the tube. $\square$

PROBLEM 10 (Residual stress in an isotropic solid). An elastic solid is isotropic at all material points relative to the adopted reference configuration, and is stressed in its undeformed state, so that $\underline{T} \neq \underline{Q}$ when $\underline{F} = \underline{1}$. Suppose this residually stressed undeformed configuration is in equilibrium without body force, and that the traction vanishes on a portion of the boundary. Show that the residual stress vanishes identically.

Solution. **Step 1: isotropy forces the residual stress to be a pressure.** At $\underline{F} = \underline{1}$ the left Cauchy–Green tensor is $\underline{B} = \underline{1}\underline{1}^t = \underline{I}$, so

$$I_1 = \text{tr } \underline{I} = 3, \quad I_2 = \frac{1}{2}[9 - \text{tr } \underline{I}] = 3, \quad I_3 = \det \underline{I} = 1,$$

and $\underline{B}^2 = \underline{I}$. Substituting into (11.34), with $J = 1$,

$$\underline{T} = 2[(\hat{W}_1 + 3\hat{W}_2)\underline{I} - \hat{W}_2\underline{I} + \hat{W}_3\underline{I}] = 2[\hat{W}_1 + 2\hat{W}_2 + \hat{W}_3]\underline{I},$$

every derivative evaluated at $(\theta, 3, 3, 1)$. So

$$\underline{T} = -p_0(\underline{X})\underline{I}, \quad p_0 = -2[\hat{W}_1 + 2\hat{W}_2 + \hat{W}_3]|_{(3,3,1)}.$$

This is the whole force of isotropy: whatever the material, a residual stress compatible with isotropy can only be hydrostatic. It may still vary from point to point, since $\hat{W}$ may depend on $\underline{X}$.

Step 2: equilibrium forces it to be uniform. With no body force and no acceleration, (6.109) reads $\operatorname{div} \underline{T} = \underline{0}$, and by Problem 24 of Chapter 1,

$$\operatorname{div}(-p_0 \underline{I}) = -\operatorname{grad} p_0 = \underline{0},$$

so p_0 is constant throughout the body, the body being connected.

Step 3: the free boundary forces it to be zero. Let $\underline{n}$ be the unit normal at a point of the traction-free portion. There

$$\underline{0} = \underline{T} \underline{n} = -p_0 \underline{n},$$

and $|\underline{n}| = 1$, so $p_0 = 0$. Being constant, p_0 vanishes everywhere, and therefore

$$\underline{T} = \underline{Q} \text{ throughout the body.}$$

What each hypothesis did. Drop isotropy and Step 1 fails: an anisotropic solid can carry a residual stress with a non-zero deviator, which is what a shrink-fit ring or a cold-drawn wire does, and such stresses are the reason residual stress is a subject at all. Drop equilibrium and Step 2 fails. Drop the traction-free patch and Step 3 fails, leaving an arbitrary uniform pressure, which is exactly the indeterminacy met in Problem 13(e) below. So the result is sharp: an isotropic body cannot hold residual stress unless something holds it from outside. $\square$

PROBLEM 11 (An azimuthal anti-plane shear, and what it is). Consider the deformation $\underline{x} = \underline{1}\underline{X} + w(\Theta)\underline{e}_3$, with Θ the azimuthal angle of a cylindrical polar system $\{R, \Theta, Z\}$ in the reference configuration. (a) Using $\underline{x} = r\underline{e}_r(\theta) + z\underline{e}_3$ with a suitable choice of basis vectors, show that one may take $R = r$, $\Theta = \theta$ and $Z + w = z$ without loss of generality. (b) Show that $\underline{F} = \underline{1} + R^{-1}w'(\Theta)\underline{e}_3 \otimes \hat{\underline{E}}_\Theta$, obtain $\underline{B}$, and show $\det \underline{F} = 1$. (c) If a neo-Hookean material occupies an annular tube of radii A and B , is in equilibrium with zero body force, and has traction-free inner and outer cylindrical surfaces, show that $w(\Theta) = C\Theta + D$. Interpret the deformation.

Solution. (a) The shifter carries $\hat{E}_R(\Theta) \mapsto \underline{e}_r(\Theta)$, $\hat{E}_\Theta \mapsto \underline{e}_\theta$ and $\hat{E}_3 \mapsto \underline{e}_3$, so with $\underline{X} = R\hat{E}_R(\Theta) + Z\hat{E}_3$,

$$\underline{x} = \underline{1}\underline{X} + w(\Theta)\underline{e}_3 = R\underline{e}_r(\Theta) + (Z + w(\Theta))\underline{e}_3.$$

Comparing with $\underline{x} = r\underline{e}_r(\theta) + z\underline{e}_3$, and using that $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$ is a basis at each point, the radial components give $r = R$ and the polar angle of $\underline{e}_r$ gives $\theta = \Theta$, while the axial components give $z = Z + w$.

(b) Write $\underline{x} = \underline{1}\underline{X} + w(\Theta)\underline{e}_3$ and take the referential gradient term by term. The first term contributes $\underline{1}$, since $\text{Grad}(\underline{1}\underline{X}) = \underline{1}$. For the second, $\underline{e}_3$ is constant and w is a scalar field, so $\text{Grad}(w\underline{e}_3) = \underline{e}_3 \otimes \text{Grad } w$ by Problem 24(f) of Chapter 1, and the cylindrical gradient of a function of Θ alone is $\text{Grad } w = R^{-1}w'(\Theta)\hat{E}_\Theta$. Hence

$$\underline{F} = \underline{1} + \alpha \underline{e}_3 \otimes \hat{E}_\Theta, \quad \alpha = \frac{w'(\Theta)}{R}.$$

On the ordered bases $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ and $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$ its matrix is $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \alpha & 1 \end{pmatrix}$, lower triangular with unit diagonal, so $\det \underline{F} = 1$ and the deformation is isochoric. For the left Cauchy–Green tensor,

$$\begin{aligned} \underline{B} &= \underline{F}\underline{F}^t = (\underline{1} + \alpha \underline{e}_3 \otimes \hat{E}_\Theta)(\underline{1}^t + \alpha \hat{E}_\Theta \otimes \underline{e}_3) \\ &= \underline{1}\underline{1}^t + \alpha(\underline{1}\hat{E}_\Theta) \otimes \underline{e}_3 + \alpha \underline{e}_3 \otimes (\underline{1}\hat{E}_\Theta) + \alpha^2(\hat{E}_\Theta \cdot \hat{E}_\Theta)\underline{e}_3 \otimes \underline{e}_3 \\ &= \underline{I} + \alpha(\underline{e}_\theta \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_\theta) + \alpha^2 \underline{e}_3 \otimes \underline{e}_3, \end{aligned}$$

using $\underline{1}\underline{1}^t = \underline{I}$ and $\underline{1}\hat{E}_\Theta = \underline{e}_\theta$.

(c) By Problem 8(a) the stress is $\underline{T} = -p\underline{I} + G\underline{B}$, so

$$T_{rr} = T_{\theta\theta} = -p + G, \quad T_{33} = -p + G(1 + \alpha^2), \quad T_{\theta 3} = G\alpha,$$

the remaining components zero.

The lateral surfaces. On $r = A$ and $r = B$ the normal is $\pm \underline{e}_r$, and $\underline{T}\underline{e}_r = T_{rr}\underline{e}_r$ since $T_{r\theta} = T_{r3} = 0$. Traction-free therefore means $p = G$ on both.

The radial equation. With $T_{r\theta} = 0$ and $T_{rr} = T_{\theta\theta}$,

$$(\text{div } \underline{T})_r = \frac{\partial T_{rr}}{\partial r} + \frac{T_{rr} - T_{\theta\theta}}{r} = -\frac{\partial p}{\partial r} = 0,$$

so p does not depend on r . Combined with $p = G$ at $r = A$, this gives $p \equiv G$ throughout, and then $T_{rr} = T_{\theta\theta} = 0$ everywhere.

The axial equation. With $T_{3r} = 0$ and T_{33} independent of z ,

$$(\operatorname{div} \underline{\underline{T}})_3 = \frac{1}{r} \frac{\partial T_{3\theta}}{\partial \theta} = \frac{G}{r} \frac{\partial \alpha}{\partial \theta} = 0,$$

so $\alpha = w'(\theta)/r$ must not depend on θ . Hence $w'(\Theta)$ is a constant C and

$$w(\Theta) = C\Theta + D$$

The azimuthal equation is satisfied identically, every relevant component being independent of θ .

Interpretation. The deformation is $z = Z + C\Theta$: material at azimuth Θ is lifted along the axis in proportion to Θ . Going once around the tube, $\Theta \mapsto \Theta + 2\pi$ returns to the same material line but at an axial offset $2\pi C$, so w is not single valued on the tube unless $C = 0$. The deformation cannot be produced from the intact tube. It is produced by cutting the tube along a half plane $\Theta = \text{const}$, sliding one face axially by $2\pi C$, and welding the faces together again. That is a *screw dislocation* of Burgers vector $2\pi C \underline{\underline{e}}_3$ along the axis, and (11.42) has nothing to do with it: the stress computed above has $T_{rr} = T_{\theta\theta} = 0$ and the only non-zero component is the shear $T_{\theta 3} = GC/r$, which is singular on the axis and is why real dislocation cores need a separate model. Remark 4.16 of Chapter 4 met the same multivaluedness in a velocity field. $\square$

PROBLEM 12 (Torsion of a neo-Hookean bar, and the Poynting effect). A solid circular cylindrical bar of neo-Hookean material is subjected to the torsion of Section 3.5.4. It is in equilibrium without body force and the traction on its cylindrical surface vanishes. Obtain the stress field and use it to compute the net force and torque on a cross-section. Show that the torque and twist are related as classical strength of materials predicts, that the axial force is non-zero contrary to that theory, and that it is proportional to the square of the twist.

Solution. **The kinematics.** Torsion carries $\underline{\underline{X}} = R\hat{\underline{\underline{E}}}_R(\Theta) + Z\hat{\underline{\underline{E}}}_3$ to $\underline{\underline{x}} = R\underline{\underline{e}}_r(\Theta + \tau Z) + Z\underline{\underline{e}}_3$, with τ the twist per unit length. Differentiating

along the three referential directions,

$$\frac{\partial \underline{x}}{\partial R} = \underline{e}_r, \quad \frac{1}{R} \frac{\partial \underline{x}}{\partial \Theta} = \underline{e}_\theta, \quad \frac{\partial \underline{x}}{\partial Z} = \tau R \underline{e}_\theta + \underline{e}_3,$$

so

$$\underline{\underline{F}} = \underline{\underline{1}} + \gamma \underline{e}_\theta \otimes \hat{\underline{E}}_3, \quad \gamma = \tau R = \tau r,$$

and $\det \underline{\underline{F}} = 1$. The local shear is γ , growing linearly with radius, which is the sense in which torsion is locally a simple shear. Squaring as in Problem 11(b),

$$\underline{\underline{B}} = \underline{\underline{I}} + \gamma(\underline{e}_\theta \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_\theta) + \gamma^2 \underline{e}_\theta \otimes \underline{e}_\theta.$$

The stress. By Problem 8(a), $\underline{\underline{T}} = -p\underline{\underline{I}} + G\underline{\underline{B}}$, so

$$T_{rr} = -p + G, \quad T_{\theta\theta} = -p + G(1 + \gamma^2), \quad T_{zz} = -p + G, \quad T_{\theta z} = G\gamma,$$

the rest zero. Note $T_{rr} = T_{zz}$ and $T_{rr} - T_{\theta\theta} = -G\gamma^2 = -G\tau^2 r^2$.

Equilibrium. The azimuthal and axial components of $\operatorname{div} \underline{\underline{T}}$ vanish identically, nothing depending on θ or z and $T_{zr} = 0$. The radial component gives

$$\frac{\partial T_{rr}}{\partial r} + \frac{T_{rr} - T_{\theta\theta}}{r} = 0 \quad \implies \quad -p'(r) - G\tau^2 r = 0,$$

so $p'(r) = -G\tau^2 r$ and $p(r) = -\frac{1}{2}G\tau^2 r^2 + c$.

The free lateral surface. At $r = a$ the normal is $\underline{e}_r$ and $\underline{\underline{T}}\underline{e}_r = T_{rr}\underline{e}_r$, so $T_{rr}(a) = 0$, that is $p(a) = G$. Hence $c = G + \frac{1}{2}G\tau^2 a^2$ and

$$p(r) = G + \frac{1}{2}G\tau^2(a^2 - r^2)$$

Substituting back, the stress field is

$$\begin{aligned} T_{rr} &= T_{zz} = -\frac{1}{2}G\tau^2(a^2 - r^2), \\ T_{\theta\theta} &= G\tau^2\left(\frac{3}{2}r^2 - \frac{1}{2}a^2\right), \quad T_{\theta z} = G\tau r. \end{aligned}$$

The torque. The traction on a cross-section, whose normal is $\underline{e}_3$, is $\underline{\underline{T}}\underline{e}_3 = T_{\theta z}\underline{e}_\theta + T_{zz}\underline{e}_3$. Its moment about the axis is

$$M = \int_{\text{section}} [(r\underline{e}_r) \times \underline{\underline{T}}\underline{e}_3] \cdot \underline{e}_3 \, da = \int_0^a (r \cdot G\tau r) 2\pi r \, dr = 2\pi G\tau \frac{a^4}{4},$$

using $\underline{e}_r \times \underline{e}_\theta = \underline{e}_3$ and $da = 2\pi r dr$. Hence

$$M = \frac{1}{2}\pi G \tau a^4 = G J_p \tau, \quad J_p = \frac{1}{2}\pi a^4,$$

J_p being the polar second moment of the section. This is exactly the classical formula, with no nonlinear correction whatever: the torque is linear in the twist however large the twist becomes. The reason is visible in the computation: $T_{\theta z} = G\gamma$ exactly, the neo-Hookean shear response being linear in the amount of shear.

The axial force. On the same section,

$$N = \int_0^a T_{zz} 2\pi r dr = -\pi G \tau^2 \int_0^a (a^2 - r^2) r dr = -\pi G \tau^2 \left[\frac{a^4}{2} - \frac{a^4}{4} \right],$$

so

$$N = -\frac{1}{4}\pi G \tau^2 a^4$$

It is not zero, it is compressive, and it is quadratic in τ . Classical strength of materials, being linear, predicts $N = 0$.

What it means. To hold the bar at fixed length while twisting it, one must push on the ends with the force $|N|$. Release the ends and the bar lengthens instead. This is the *Poynting effect*, and it is the same phenomenon as the normal stress effect of Remark 11.10: torsion is locally a simple shear of amount $\gamma = \tau r$, and (11.42) demands normal stresses of order γ^2 to accompany it. Integrating those over the section is exactly the calculation just done. The ratio

$$\frac{|N|}{M/a} = \frac{\frac{1}{4}\pi G \tau^2 a^4}{\frac{1}{2}\pi G \tau a^3} = \frac{1}{2}\tau a$$

is half the surface shear, so the effect becomes measurable when the twist per unit length times the radius is not small. $\square$

PROBLEM 13 (Closing a wedge: the cylinder with a sector removed). A circular cylinder of radius A has a wedge removed, so that the angular extent of the material in κ is γ (Figure 11.1). The cross-section is deformed so that the gap closes, by

$$\begin{aligned} \underline{\tilde{x}} &= R \underline{\hat{E}}_R(\Theta) + Z \underline{\hat{E}}_3, & \underline{\tilde{x}} &= r \underline{e}_r(\theta) + z \underline{e}_3, \\ z &= Z, & r &= f(R), & \theta &= g(\Theta). \end{aligned}$$

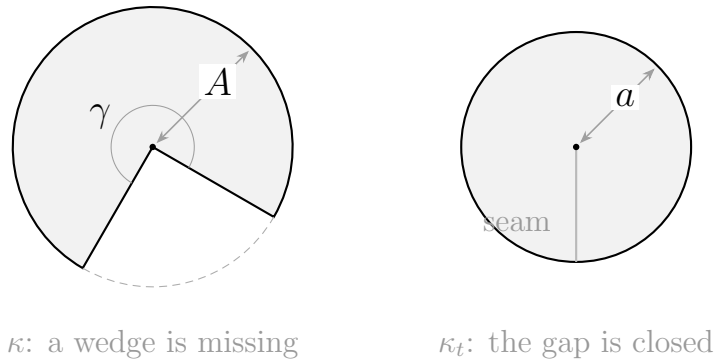


Figure 11.1: Problem 13. In the reference configuration the material of a circular cylinder occupies an angular extent $\gamma < 2\pi$, a wedge having been removed; the missing wedge is dashed. The cross-section is then deformed so that the two cut faces meet, the material spanning the full 2π . The map is $r = f(R)$, $\theta = g(\Theta)$, $z = Z$, and isochoricity together with regularity on the axis fixes both functions. The deformed radius comes out proportional to $\sqrt{\gamma}$, so closing a larger gap makes a thinner bar.

(a) Find $\underline{\underline{F}}$ for known f and g . (b) Assume the deformation isochoric and find f and g from the boundary conditions; show the deformed radius a is proportional to $\sqrt{\gamma}$. (c) Find $\underline{\underline{R}}$, $\underline{\underline{U}}$, $\underline{\underline{V}}$. (d) Obtain the principal stretches in terms of γ . (e) Solve the equilibrium problem without body force for a neo-Hookean material. Find the reactive pressure and show it is singular on the centreline. Evaluate the traction at the outer boundary required to maintain the deformation and show it is an arbitrary uniform pressure. What contact traction acts on the planar surfaces where the two faces of the wedge meet, if there is no traction at the outer boundary?

Solution. (a) Differentiating $\underline{\underline{x}}$ along the three referential directions, and remembering that $\underline{\underline{e}}_r$ depends on $\theta = g(\Theta)$ so that $\partial \underline{\underline{e}}_r / \partial \Theta = g'(\Theta) \underline{\underline{e}}_\theta$,

$$\frac{\partial \underline{\underline{x}}}{\partial R} = f'(R) \underline{\underline{e}}_r, \quad \frac{1}{R} \frac{\partial \underline{\underline{x}}}{\partial \Theta} = \frac{f(R)g'(\Theta)}{R} \underline{\underline{e}}_\theta, \quad \frac{\partial \underline{\underline{x}}}{\partial Z} = \underline{\underline{e}}_3,$$

so

$$\underline{\underline{F}} = f'(R) \underline{\underline{e}}_r \otimes \hat{\underline{\underline{E}}}_R + \frac{f(R)g'(\Theta)}{R} \underline{\underline{e}}_\theta \otimes \hat{\underline{\underline{E}}}_\Theta + \underline{\underline{e}}_3 \otimes \hat{\underline{\underline{E}}}_3.$$

(b) Both frames are orthonormal, so $\det \tilde{F}$ is the product of the three coefficients:

$$\det \tilde{F} = f' \frac{f g'}{R} \cdot 1 = 1.$$

The variables separate. Rearranged, $R/(f f') = g'(\Theta)$, and the left side depends on R alone while the right depends on Θ alone, so both equal a constant k :

$$g'(\Theta) = k, \quad f(R) f'(R) = \frac{R}{k}.$$

The gap closes when the material of angular extent γ covers 2π , so $g(0) = 0$ and $g(\gamma) = 2\pi$, giving

$$k = \frac{2\pi}{\gamma}, \quad g(\Theta) = \frac{2\pi}{\gamma} \Theta.$$

Integrating the radial equation, $\frac{1}{2} f^2 = R^2/(2k) + \text{const}$, and the axis must map to the axis, $f(0) = 0$, which kills the constant:

$$f(R) = \frac{R}{\sqrt{k}} = R \sqrt{\frac{\gamma}{2\pi}}, \quad g(\Theta) = \frac{2\pi}{\gamma} \Theta.$$

Hence the deformed radius is

$$a = f(A) = A \sqrt{\frac{\gamma}{2\pi}} \propto \sqrt{\gamma},$$

so closing a larger gap, that is starting from a smaller γ , gives a thinner bar. The area is preserved: $\pi a^2 = \pi A^2 \gamma / (2\pi) = \frac{1}{2} A^2 \gamma$, which is exactly the area of the original sector.

(c) With $f' = 1/\sqrt{k}$ and $f g'/R = (R/\sqrt{k})k/R = \sqrt{k}$,

$$\tilde{F} = \frac{1}{\sqrt{k}} \underline{e}_r \otimes \hat{E}_R + \sqrt{k} \underline{e}_\theta \otimes \hat{E}_\Theta + \underline{e}_3 \otimes \hat{E}_3,$$

which is already the canonical form $\tilde{F} = \sum_i \lambda_i \underline{v}_i \otimes \underline{u}_i$ of (3.43), both

triads being orthonormal. Reading off the polar factors,

$$\begin{aligned}\underline{\tilde{R}} &= \underline{\tilde{e}}_r \otimes \hat{\underline{\tilde{E}}}_R + \underline{\tilde{e}}_\theta \otimes \hat{\underline{\tilde{E}}}_\Theta + \underline{\tilde{e}}_3 \otimes \hat{\underline{\tilde{E}}}_3, \\ \underline{\tilde{U}} &= \frac{1}{\sqrt{k}} \hat{\underline{\tilde{E}}}_R \otimes \hat{\underline{\tilde{E}}}_R + \sqrt{k} \hat{\underline{\tilde{E}}}_\Theta \otimes \hat{\underline{\tilde{E}}}_\Theta + \hat{\underline{\tilde{E}}}_3 \otimes \hat{\underline{\tilde{E}}}_3, \\ \underline{\tilde{V}} &= \frac{1}{\sqrt{k}} \underline{\tilde{e}}_r \otimes \underline{\tilde{e}}_r + \sqrt{k} \underline{\tilde{e}}_\theta \otimes \underline{\tilde{e}}_\theta + \underline{\tilde{e}}_3 \otimes \underline{\tilde{e}}_3.\end{aligned}$$

Note that $\underline{\tilde{R}}$ is not the shifter: it carries $\hat{\underline{\tilde{E}}}_R(\Theta)$ to $\underline{\tilde{e}}_r(g(\Theta))$, and those are different directions unless $k = 1$.

(d) The principal stretches are the coefficients:

$$\lambda_1 = \frac{1}{\sqrt{k}} = \sqrt{\frac{\gamma}{2\pi}}, \quad \lambda_2 = \sqrt{k} = \sqrt{\frac{2\pi}{\gamma}}, \quad \lambda_3 = 1.$$

Their product is 1, as isochoricity requires. For $\gamma < 2\pi$ the material is stretched azimuthally and compressed radially by reciprocal factors, which is a state of pure shear in the cross-sectional plane.

(e) With $\underline{\tilde{B}} = \underline{\tilde{V}}^2$ from (c),

$$T_{rr} = -p + \frac{G}{k}, \quad T_{\theta\theta} = -p + Gk, \quad T_{zz} = -p + G,$$

off-diagonal components zero, and all quantities functions of r alone. The radial equilibrium equation is

$$\frac{\partial T_{rr}}{\partial r} + \frac{T_{rr} - T_{\theta\theta}}{r} = 0, \quad T_{rr} - T_{\theta\theta} = G\left(\frac{1}{k} - k\right),$$

so

$$p'(r) = \frac{G}{r}\left(\frac{1}{k} - k\right), \quad p(r) = G\left(\frac{1}{k} - k\right) \ln r + \text{const.}$$

Since $\gamma < 2\pi$ gives $k > 1$, the coefficient $1/k - k$ is negative and $p \rightarrow +\infty$ as $r \rightarrow 0$:

p is logarithmically singular on the centreline.

The outer traction. At $r = a$ the normal is $\underline{\tilde{e}}_r$ and $\underline{\tilde{T}}\underline{\tilde{e}}_r = T_{rr}(a)\underline{\tilde{e}}_r$, a normal traction, uniform because it depends only on r . The constant

of integration is not fixed by the field equations, so $T_{rr}(a)$ may be given any value: the deformation is maintained by an *arbitrary uniform pressure* on the outer boundary. This is the indeterminacy that Problem 10 removed by assuming a traction-free patch.

The seam. Fix the constant by requiring no traction at the outer boundary, $T_{rr}(a) = 0$, that is $p(a) = G/k$. Then

$$p(r) = \frac{G}{k} + G\left(\frac{1}{k} - k\right) \ln \frac{r}{a}.$$

The two faces of the wedge meet on a half plane whose normal is $\underline{e}_\theta$, and the traction there is $\underline{T}\underline{e}_\theta = T_{\theta\theta}\underline{e}_\theta$, purely normal since $T_{r\theta} = T_{z\theta} = 0$. Substituting,

$$T_{\theta\theta}(r) = G\left(k - \frac{1}{k}\right) \left[1 + \ln \frac{r}{a}\right], \quad k = \frac{2\pi}{\gamma}.$$

So the glue transmits a pure normal traction, with no shear. It is tensile at the rim, where $\ln(r/a) = 0$ and $T_{\theta\theta} = G(k - 1/k) > 0$, changes sign at $r = a/e$, and is compressive and logarithmically unbounded towards the axis. The bar is trying to spring open at its surface and is crushed together at its core, which is what one would expect of a wedge forced shut. $\square$

PROBLEM 14 (The referential energy balance for an elastic solid). Derive the energy balance equation

$$\rho_\kappa \dot{r} = \text{Div } \underline{q}_\kappa - \theta(W_\theta)^\cdot, \quad \text{where} \quad (W_\theta)^\cdot = W_{\theta\theta}\dot{\theta} + W_{\theta F} \cdot \dot{\underline{F}},$$

with $\underline{q}_\kappa$ the referential heat flux of (6.174).

Solution.

REMARK 11.15 (the overdot in the statement). The book writes the second factor as $\dot{W}$, but the expression given for it, $W_{\theta\theta}\dot{\theta} + W_{\theta F} \cdot \dot{\underline{F}}$, is the material derivative of W_θ , not of W . The chain rule for W itself would be $\dot{W} = W_\theta\dot{\theta} + W_F \cdot \dot{\underline{F}}$, with first derivatives. These notes write $(W_\theta)^\cdot$, and the derivation below shows it is the right object.

Step 1: the internal energy in terms of W . By (6.186), $\psi = \varepsilon - \theta\eta$, so $\varepsilon = \psi + \theta\eta$. Multiply by ρ_κ and use (11.17) on the first term:

$$\rho_\kappa \varepsilon = \rho_\kappa \psi + \theta \rho_\kappa \eta = W + \theta \rho_\kappa \eta.$$

The entropy is given by (11.16)₁, $\eta = -\Psi_\theta$, and ρ_κ passes through the θ derivative as a constant, so $\rho_\kappa \eta = -\rho_\kappa \Psi_\theta = -W_\theta$. Hence

$$\rho_\kappa \varepsilon = W - \theta W_\theta.$$

Step 2: differentiate. Taking the material derivative, and using the chain rule $\dot{W} = W_\theta \dot{\theta} + W_{\underline{F}} \cdot \dot{\underline{F}}$ on the first term and the product rule on the second,

$$\begin{aligned} \rho_\kappa \dot{\varepsilon} &= \dot{W} - \dot{\theta} W_\theta - \theta (W_\theta)^\cdot \\ &= (W_\theta \dot{\theta} + W_{\underline{F}} \cdot \dot{\underline{F}}) - W_\theta \dot{\theta} - \theta (W_\theta)^\cdot = W_{\underline{F}} \cdot \dot{\underline{F}} - \theta (W_\theta)^\cdot, \end{aligned}$$

the two $W_\theta \dot{\theta}$ terms cancelling. Since $\underline{P} = W_{\underline{F}}$ by (11.16)₂,

$$\rho_\kappa \dot{\varepsilon} = \underline{P} \cdot \dot{\underline{F}} - \theta (W_\theta)^\cdot.$$

Step 3: compare with the balance. The referential energy balance is (6.174),

$$\rho_\kappa \dot{\varepsilon} = \underline{P} \cdot \dot{\underline{F}} - \text{Div } \underline{q}_\kappa + \rho_\kappa r.$$

Subtracting the two displays, the stress power cancels and

$$-\theta (W_\theta)^\cdot = -\text{Div } \underline{q}_\kappa + \rho_\kappa r,$$

that is

$$\rho_\kappa r = \text{Div } \underline{q}_\kappa - \theta (W_\theta)^\cdot$$

Finally, W_θ is a function of θ and $\underline{F}$, so the chain rule gives $(W_\theta)^\cdot = W_{\theta\theta} \dot{\theta} + W_{\theta\underline{F}} \cdot \dot{\underline{F}}$, as stated.

Reading the result. The stress power has disappeared entirely. In a hyperelastic solid the mechanical working is stored, not dissipated, so it cannot appear as a source of heat; what remains is the conduction

term and a coupling term $-\theta(W_\theta)^\cdot$, which is the rate of latent heat associated with changes of temperature and deformation. Its second piece, $-\theta W_{\theta F} \cdot \dot{\tilde{F}}$, is the thermoelastic coupling: deforming the body at fixed temperature absorbs or releases heat unless $W_{\theta F} = \hat{Q}$, in which case the thermal and mechanical problems decouple as they did for rigid bodies at Section 6.6. $\square$

PROBLEM 15 (The referential heat flux). Show that if the spatial heat flux is (11.49), then the referential heat flux is

$$\underline{q}_\kappa = (\psi_0 \hat{I} + \psi_1 \underline{C} + \psi_2 \underline{C}^2) \text{Grad } \theta,$$

with ψ_0, ψ_1, ψ_2 functions of θ and of the invariants

$$I_1(\underline{C}), \quad I_2(\underline{C}), \quad I_3(\underline{C}), \quad I_4(\underline{C}, \text{Grad } \theta), \quad I_5(\underline{C}, \text{Grad } \theta), \quad I_6(\text{Grad } \theta).$$

Solution. **Two conversions.** By (6.171), $\underline{q}_\kappa = J \tilde{F}^{-1} \underline{q}$. By the chain rule applied to the temperature field, $\text{Grad } \theta = \tilde{F}^t \text{grad } \theta$, whence $\text{grad } \theta = \tilde{F}^{-t} \text{Grad } \theta$. Substituting (11.49),

$$\underline{q}_\kappa = J \tilde{F}^{-1} (\phi_0 \underline{I} + \phi_1 \underline{B} + \phi_2 \underline{B}^2) \tilde{F}^{-t} \text{Grad } \theta.$$

Pulling the three tensors back. Each is a one-line computation, the reverse of the ones in Problem 2:

$$\tilde{F}^{-1} \underline{I} \tilde{F}^{-t} = \tilde{F}^{-1} \tilde{F}^{-t} = (\tilde{F}^t \tilde{F})^{-1} = \underline{C}^{-1},$$

$$\tilde{F}^{-1} \underline{B} \tilde{F}^{-t} = \tilde{F}^{-1} (\tilde{F} \tilde{F}^t) \tilde{F}^{-t} = \hat{I},$$

$$\tilde{F}^{-1} \underline{B}^2 \tilde{F}^{-t} = \tilde{F}^{-1} \tilde{F} \tilde{F}^t \tilde{F} \tilde{F}^t \tilde{F}^{-t} = \tilde{F}^t \tilde{F} = \underline{C}.$$

Therefore

$$\underline{q}_\kappa = J (\phi_0 \underline{C}^{-1} + \phi_1 \hat{I} + \phi_2 \underline{C}) \text{Grad } \theta.$$

Removing the inverse. The result is not yet in the stated form, because of $\underline{C}^{-1}$. The Cayley–Hamilton theorem of Section 9.1.1, applied to $\underline{C}$, gives $\underline{C}^3 - I_1 \underline{C}^2 + I_2 \underline{C} - I_3 \hat{I} = \hat{Q}$, and multiplying by $\underline{C}^{-1}$ and rearranging,

$$\underline{C}^{-1} = \frac{1}{I_3} (\underline{C}^2 - I_1 \underline{C} + I_2 \hat{I}),$$

legitimate because $I_3 = \det \underline{C} = J^2 > 0$. Substituting and collecting powers of $\underline{C}$,

$$\underline{q}_\kappa = (\psi_0 \hat{I} + \psi_1 \underline{C} + \psi_2 \underline{C}^2) \text{Grad } \theta$$

with

$$\psi_0 = J \left(\phi_1 + \frac{I_2}{I_3} \phi_0 \right), \quad \psi_1 = J \left(\phi_2 - \frac{I_1}{I_3} \phi_0 \right), \quad \psi_2 = \frac{J}{I_3} \phi_0.$$

The argument lists match. The ϕ_k depend on θ , on the invariants of $\underline{B}$ and on the three scalars (11.50). The invariants of $\underline{B}$ are those of $\underline{C}$, by (11.35). For the others, substitute $\text{grad } \theta = \underline{F}^{-t} \text{Grad } \theta$ and pull back as above:

$$I_4(\underline{B}, \text{grad } \theta) = \text{Grad } \theta \cdot \underline{F}^{-1} \underline{B} \underline{F}^{-t} \text{Grad } \theta = \text{Grad } \theta \cdot \text{Grad } \theta = I_6(\text{Grad } \theta)^2,$$

$$I_5(\underline{B}, \text{grad } \theta) = \text{Grad } \theta \cdot \underline{C} \text{Grad } \theta = I_4(\underline{C}, \text{Grad } \theta),$$

$$\begin{aligned} I_6(\text{grad } \theta)^2 &= \text{Grad } \theta \cdot \underline{C}^{-1} \text{Grad } \theta \\ &= \frac{1}{I_3} [I_5(\underline{C}, \text{Grad } \theta) - I_1 I_4(\underline{C}, \text{Grad } \theta) + I_2 I_6(\text{Grad } \theta)^2], \end{aligned}$$

the last line using the same Cayley–Hamilton substitution, and with the referential scalars defined by $I_4(\underline{C}, \text{Grad } \theta) = \text{Grad } \theta \cdot \underline{C} \text{Grad } \theta$, $I_5(\underline{C}, \text{Grad } \theta) = \text{Grad } \theta \cdot \underline{C}^2 \text{Grad } \theta$ and $I_6(\text{Grad } \theta) = |\text{Grad } \theta|$. Every argument of the ϕ_k is therefore a function of the referential list, and so is every ψ_k .

The moral is the one of Remark 6.32 transposed to fluxes: the referential and spatial descriptions carry the same physical content, and converting between them replaces $\underline{B}$ by $\underline{C}$ and turns grad into Grad , at the price of factors of $\underline{F}$ that Cayley–Hamilton then tidies away. $\square$

PROBLEM 16 (A torsion with an axial temperature gradient). A solid circular cylindrical bar undergoes the static torsion of Section 3.5.4. The ϕ 's in (11.49) are independent of temperature, though they may depend on I_1 to I_6 , and the material is uniform. The temperature is independent of time and depends only on z , with the ends $z = 0$ and $z = l$ held at θ_0 and θ_l . Assuming this function linear, show that the energy balance

with zero internal heating supply is automatically satisfied, and that the lateral surface of the bar is insulated.

Solution. **The data.** From Problem 12,

$$\underline{B} = \underline{I} + \gamma(\underline{e}_\theta \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_\theta) + \gamma^2 \underline{e}_\theta \otimes \underline{e}_\theta, \quad \gamma = \tau r,$$

and the temperature is

$$\theta(z) = \theta_0 + \beta z, \quad \beta = \frac{\theta_l - \theta_0}{l}, \quad \text{grad } \theta = \beta \underline{e}_3,$$

a constant vector field.

The heat flux. Apply $\underline{B}$ and $\underline{B}^2$ to $\underline{e}_3$:

$$\underline{B}\underline{e}_3 = \underline{e}_3 + \gamma \underline{e}_\theta, \quad \underline{B}\underline{e}_\theta = (1 + \gamma^2)\underline{e}_\theta + \gamma \underline{e}_3,$$

so

$$\begin{aligned} \underline{B}^2 \underline{e}_3 &= \underline{B}(\underline{e}_3 + \gamma \underline{e}_\theta) = (\underline{e}_3 + \gamma \underline{e}_\theta) + \gamma[(1 + \gamma^2)\underline{e}_\theta + \gamma \underline{e}_3] \\ &= (1 + \gamma^2)\underline{e}_3 + \gamma(2 + \gamma^2)\underline{e}_\theta. \end{aligned}$$

Substituting into (11.49),

$$\underline{q} = \beta \left\{ [\phi_0 + \phi_1 + \phi_2(1 + \gamma^2)]\underline{e}_3 + \gamma[\phi_1 + \phi_2(2 + \gamma^2)]\underline{e}_\theta \right\}$$

The lateral surface is insulated. Its outward normal is $\underline{e}_r$, and the flux has no $\underline{e}_r$ component:

$$\underline{q} \cdot \underline{e}_r = 0,$$

so no heat crosses the cylindrical surface. The flux runs along the axis, as the boundary data demand, and around it, which is the thermal signature of the torsion.

The energy balance. The local spatial form is (6.168),

$$\rho \dot{\varepsilon} = \underline{T} \cdot \underline{L} - \text{div } \underline{q} + \rho r,$$

with $r = 0$ by hypothesis. The deformation is static, so $\underline{v} = \underline{0}$, $\underline{L} = \underline{0}$ and the stress power vanishes; and ε depends on θ and $\underline{F}$, neither of

which depends on time, so $\dot{\varepsilon} = 0$. What is left is $\text{div } \underline{q} = 0$, and this holds identically. To see it, note first what the coefficients depend on. The ϕ_k are independent of θ by hypothesis and of $\underline{X}$ by uniformity, so they depend only on the six invariants, and

$$\begin{aligned} I_1 = I_2 &= 3 + \gamma^2, & I_3 &= 1, \\ I_4 &= \text{grad } \theta \cdot \underline{B} \text{ grad } \theta = \beta^2 \underline{e}_3 \cdot \underline{e}_3 + \gamma \underline{e}_\theta = \beta^2, \\ I_5 &= \text{grad } \theta \cdot \underline{B}^2 \text{ grad } \theta = \beta^2(1 + \gamma^2), & I_6 &= |\beta|, \end{aligned}$$

the first line by (11.38), torsion being locally a simple shear of amount γ . Every one of these is a function of r alone, through $\gamma = \tau r$. So the boxed flux has the form

$$\underline{q} = q_\theta(r) \underline{e}_\theta + q_z(r) \underline{e}_3, \quad q_r = 0,$$

and the cylindrical divergence of such a field is

$$\text{div } \underline{q} = \frac{1}{r} \frac{\partial}{\partial r} (r q_r) + \frac{1}{r} \frac{\partial q_\theta}{\partial \theta} + \frac{\partial q_z}{\partial z} = 0 + 0 + 0 = 0,$$

the first term because $q_r = 0$, the second because q_θ does not depend on θ , the third because q_z does not depend on z . Hence

$$\text{div } \underline{q} = 0 \text{ identically, and the balance is satisfied with } r = 0.$$

Why it worked. Three separate facts conspired. The temperature gradient is uniform, so it contributes nothing to any derivative. The deformation is a function of r only, so the constitutive coefficients are too. And the flux has no radial component, so the one term of the divergence that would have differentiated an r -dependent quantity is absent. Remove any one of them, for instance by letting the ϕ 's depend on temperature, and $\text{div } \underline{q}$ would acquire a term in ∂_z through $\theta(z)$, and an internal supply r would be needed to maintain the assumed field. $\square$

Chapter 12

Linear Elasticity

Linear elasticity is the theory that carries almost all of engineering practice, and it is obtained from Chapter 11 by one approximation: the displacement gradient is small. The chapter works out what that approximation does to each object built in the preceding eleven chapters, and it takes some care over one point that most treatments pass over, the shifter.

For simplicity the theory is developed in its purely mechanical form, the free energy being independent of temperature and the temperature itself uniform and constant throughout the body. By (11.51) and Problem 14 of Chapter 11, the thermal energy balance is then satisfied identically provided the internal heating supply vanishes: with $\text{grad } \theta = 0$ the flux vanishes by (11.55), and with $\dot{\theta} = 0$ and W independent of θ the coupling term $\theta(W_\theta)^\cdot$ vanishes too.

12.1 Kinematics

The premise of the theory is that the displacement gradient $\underline{H}$ of (7.22) is small everywhere in the body. Taking $\underline{H}$ continuous in $\underline{X}$, this is

$$\varepsilon = \max_{\underline{X} \in \kappa} |\underline{H}(\underline{X}, t)| \ll 1, \quad (12.1)$$

where $|\underline{H}|^2 = \text{tr}(\underline{H}\underline{H}^t) = \text{tr}(\underline{H}^t\underline{H})$, so that $|\underline{H}| = \sqrt{H_{iA}H_{iA}}$ in Cartesian components. Since a sum of squares dominates each of its terms, $|H_{iA}| \ll 1$ for every pair iA .

REMARK 12.1 (the letter ε in this chapter). Three near-identical symbols are about to appear, and the book uses all three. The scalar ε of (12.1) is the small parameter. The referential infinitesimal strain of (12.8) is $\underline{\underline{\varepsilon}}$, the closed epsilon. The spatial infinitesimal strain of (12.17) is $\underline{\underline{\varepsilon}}$, the open one. Both strains carry the under-tilde that marks a tensor, so the scalar is distinguishable at sight; the two tensors are told apart by their shape, and the shifter is what converts one into the other. The internal energy, which also wore the letter ε in Chapter 6, does not appear here at all, the theory being purely mechanical.

Strain

By (7.22), $F = \underline{\underline{1}} + \underline{\underline{H}}$, so the right Cauchy–Green tensor is

$$\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}} = \underline{\underline{\hat{I}}} + \underline{\underline{1}}^t \underline{\underline{H}} + \underline{\underline{H}}^t \underline{\underline{1}} + \underline{\underline{H}}^t \underline{\underline{H}}, \quad (12.2)$$

the first term because $\underline{\underline{1}}^t \underline{\underline{1}} = \underline{\underline{\hat{I}}}$, the shifter being an isometry. The three remaining terms are all referential, but the middle two are built from a two-point tensor, and it is worth introducing the object that removes that awkwardness. Set

$$\underline{\underline{G}} = \underline{\underline{\hat{I}}} + \underline{\underline{G}} + \underline{\underline{G}}^t + \underline{\underline{G}}^t \underline{\underline{G}}, \quad \text{where} \quad \underline{\underline{G}} = \underline{\underline{1}}^t \underline{\underline{H}} \quad (12.3)$$

is a referential tensor with components $G_{AB} = \delta_{iA} H_{iB}$. The identification of the last term uses $\underline{\underline{G}}^t \underline{\underline{G}} = \underline{\underline{H}}^t \underline{\underline{1}} \underline{\underline{1}}^t \underline{\underline{H}} = \underline{\underline{H}}^t \underline{\underline{I}} \underline{\underline{H}} = \underline{\underline{H}}^t \underline{\underline{H}}$, and of the second $\underline{\underline{G}}^t = (\underline{\underline{1}}^t \underline{\underline{H}})^t = \underline{\underline{H}}^t \underline{\underline{1}}$.

What $\underline{\underline{G}}$ is becomes clear on applying it to a material line element:

$$\underline{\underline{G}} d\underline{\underline{X}} = \underline{\underline{1}}^t \underline{\underline{H}} d\underline{\underline{X}} = \underline{\underline{1}}^t d\underline{\underline{u}} = d\underline{\underline{w}}, \quad \text{where} \quad \underline{\underline{w}} = \underline{\underline{1}}^t \underline{\underline{u}} \quad (12.4)$$

is the *referential displacement*, $\underline{\underline{u}}$ being the spatial displacement of (2.77), and where the shifter has been taken through the differential because it is uniform. Hence

$$\underline{\underline{G}} = \text{Grad } \underline{\underline{w}}, \quad (12.5)$$

and $\underline{u} = \underline{1}\underline{w}$. In components $w_A = \delta_{iA}u_i$, $u_i = \delta_{iA}w_A$ and $G_{AB} = w_{A,B}$. Substituting $\underline{H} = \underline{1}\underline{G}$ into (7.22),

$$\underline{F} = \underline{1}(\hat{\underline{I}} + \underline{G}). \quad (12.6)$$

The norms agree,

$$|\underline{G}|^2 = \text{tr}(\underline{G}\underline{G}^t) = \text{tr}(\underline{1}^t \underline{H} \underline{H}^t \underline{1}) = \text{tr}(\underline{H} \underline{H}^t \underline{1} \underline{1}^t) = \text{tr}(\underline{H} \underline{H}^t) = |\underline{H}|^2,$$

by the cyclic property of the trace, so (12.1) is equivalent to $|\underline{G}| \ll 1$, and again $|G_{AB}| \ll 1$ componentwise.

Since $\underline{C}$ is close to $\hat{\underline{I}}$, it is more convenient to work with the Lagrange strain (2.97), which measures departure from $\hat{\underline{I}}$ directly. From (12.3),

$$\underline{E} = \frac{1}{2}(\underline{C} - \hat{\underline{I}}) = \frac{1}{2}(\underline{G} + \underline{G}^t + \underline{G}^t \underline{G}). \quad (12.7)$$

Discarding the quadratic term,

$$\underline{E} \simeq \underline{\epsilon}, \quad \text{where} \quad \underline{\epsilon} = \frac{1}{2}(\underline{G} + \underline{G}^t) \quad (12.8)$$

is the *referential infinitesimal strain tensor*, the symbol $\simeq$ marking a statement valid to leading order in $|\underline{G}|$.

REMARK 12.2 (small strain is weaker than small displacement gradient). Equation (12.8) runs one way only. If $|\underline{G}| \ll 1$ then $|\underline{E}| \ll 1$, since $|\underline{E}| \leq |\underline{G}| + \frac{1}{2}|\underline{G}|^2$. The converse fails, and the standard counterexample is a rigid rotation: take $\underline{F} = \underline{Q}$ with $\underline{Q}$ a finite rotation. Then $\underline{C} = \hat{\underline{I}}$ and $\underline{E} = \hat{\underline{Q}}$ exactly, so the strain is not merely small but zero, while $\underline{G} = \underline{1}^t \underline{Q} - \hat{\underline{I}}$ has norm of order the rotation angle and is not small at all.

The whole theory therefore assumes more than small strain. It assumes small *rotation* as well, and (12.20)₃ makes that explicit by producing a rotation tensor differing from the shifter by a term of order $|\underline{G}|$. A structure that strains by a thousandth while rotating through a radian is outside the theory, which is why the analysis of slender beams and thin shells needs more than what follows.

Two observers, and why the shifter cannot be dropped

These statements belong to the frame of one observer O . From (12.3)₂ together with (7.18), $\underline{1}^+ = Q\underline{1}K^t$, and (7.25), $H^+ = QHK^t$,

$$G^+ = KGK^t, \quad (12.9)$$

$K: \hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^{3+}$ being the fixed orthogonal tensor relating the two reference configurations. The computation is

$$G^+ = (\underline{1}^+)^t H^+ = (Q\underline{1}K^t)^t QHK^t = K\underline{1}^t Q^t QHK^t = K\underline{1}^t HK^t = KGK^t,$$

$Q^t Q = I$ removing the observer rotation entirely. Symmetrizing (12.7) and (12.8) under conjugation, which commutes with transposition because $(KAK^t)^t = K\tilde{A}^t K^t$,

$$E^+ = KEK^t, \quad (12.10)$$

a result that also follows from (7.20) and (7.26) through $E^+ = \frac{1}{2}[(F^+)^t F^+ - \hat{I}^+]$. Further (12.9) gives $|G^+| = |G|$, by the same cyclic-trace computation as above with K in place of $\underline{1}$, and

$$\epsilon^+ = \frac{1}{2}[G^+ + (G^+)^t] = K\epsilon K^t, \quad (12.11)$$

so that

$$E^+ \simeq \epsilon^+. \quad (12.12)$$

If one observer finds the strain small, so does every other, and (12.12) says the approximation itself survives the change of observer. This is Problem 1.

REMARK 12.3 (the equation that is not a tensor equation). The literature almost always writes the strain–displacement relation by dropping the shifter from (12.2), obtaining

$$2\tilde{E} \simeq \tilde{H} + \tilde{H}^t$$

in the frame of O . The left side is a referential tensor, mapping $\hat{\mathbb{E}}^3$ to $\hat{\mathbb{E}}^3$. On the right, $\tilde{H}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$ and $\tilde{H}^t$ maps $\mathbb{E}^3$ to $\hat{\mathbb{E}}^3$, so their

sum is not defined: the two terms do not even have the same domain, let alone the same codomain (Figure A). The expression is not a tensor equation and cannot be one.

The damage is visible under a change of observer. By (7.25),

$$\tilde{H}^+ + (\tilde{H}^+)^t = \tilde{Q}\tilde{H}\tilde{K}^t + \tilde{K}\tilde{H}^t\tilde{Q}^t,$$

which carries the observer rotation $\tilde{Q}$ on the outside, whereas (12.10) requires $2\tilde{E}^+ = \tilde{K}(2\tilde{E})\tilde{K}^t$, with $\tilde{K}$ on both sides and no $\tilde{Q}$ anywhere. So $\tilde{H}^+ + (\tilde{H}^+)^t$ cannot approximate $2\tilde{E}^+$ for the second observer even if $\tilde{H} + \tilde{H}^t$ did for the first. What makes (12.8) work is exactly that $\tilde{G} = \mathbb{1}^t\tilde{H}$ is referential on both legs, and (12.9) then carries no $\tilde{Q}$.

None of this is an obstacle in practice, as the last paragraph of the chapter records: an observer may choose the shifted basis $\tilde{E}_A = \mathbb{1}\tilde{E}_A$ of (2.68) to coincide with $\tilde{e}_i$, after which δ_{iA} are numerically the Kronecker components and the distinction is invisible. The point is that the invisibility is a consequence of a choice, not a licence to write an equation that is false in general.

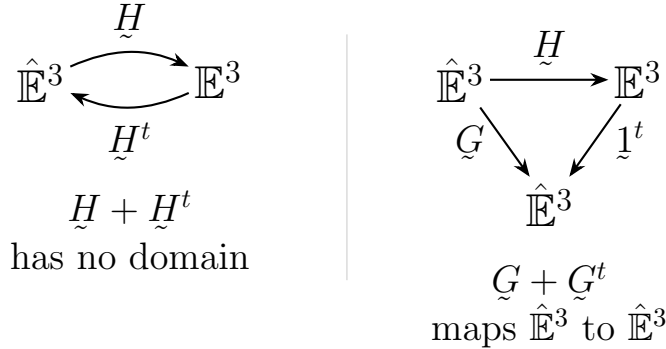


Figure A: Why Remark 12.3 insists on the shifter. The displacement gradient $\tilde{H}$ of (7.22) is a two-point tensor, so its transpose runs the other way and the sum $\tilde{H} + \tilde{H}^t$ is not defined on any space. Composing with $\mathbb{1}^t$ brings the second leg home, and the resulting $\tilde{G} = \mathbb{1}^t\tilde{H}$ is referential, so $\tilde{G} + \tilde{G}^t$ is a legitimate referential tensor and is what (12.8) uses. The triangle on the right commutes by the definition (12.3)₂ of $\tilde{G}$.

The spatial description

The displacement $\underline{u}$ is a spatial vector, by (2.77) and (2.78), so it has a spatial gradient, defined by $d\underline{u} = (\text{grad } \underline{u})d\underline{x}$ at fixed t . Since also $Hd\underline{X} = d\underline{u}$ and $d\underline{x} = \underline{F}d\underline{X}$,

$$\underline{H} = (\text{grad } \underline{u})\underline{F} \simeq (\text{grad } \underline{u})\underline{1}, \quad (12.13)$$

the estimate by (12.6) and $|\underline{G}| \ll 1$. In components, $\hat{u}_{i,A} = \tilde{u}_{i,j}F_{jA} = \tilde{u}_{i,j}\delta_{jB}(\delta_{BA} + G_{BA}) \simeq \tilde{u}_{i,j}\delta_{jA}$. This is an instance of the general estimate

$$(\cdot)_{,A} \simeq (\cdot)_{,j}\delta_{jA}, \quad (12.14)$$

valid to leading order: differentiating with respect to $\underline{X}$ and with respect to $\underline{x}$ differ only by $\underline{F}$, which is $\underline{1}$ to leading order. Combining (12.13) with (12.3)₂,

$$\underline{G} \simeq \underline{1}^t(\text{grad } \underline{u})\underline{1} \quad \text{and} \quad \text{grad } \underline{u} \simeq \underline{1}\underline{G}\underline{1}^t, \quad (12.15)$$

the second obtained from the first by multiplying by $\underline{1}$ on the left and $\underline{1}^t$ on the right and using $\underline{1}\underline{1}^t = \underline{I}$. Hence

$$|\underline{H}| = |\underline{G}| \simeq |\text{grad } \underline{u}|, \quad (12.16)$$

so $|\tilde{u}_{i,j}| \ll 1$ as well. Symmetrizing (12.15),

$$\underline{\epsilon} \simeq \underline{1}^t\underline{\varepsilon}\underline{1}, \quad \text{where} \quad \underline{\varepsilon} = \frac{1}{2}[\text{grad } \underline{u} + (\text{grad } \underline{u})^t] \quad (12.17)$$

is the *spatial infinitesimal strain tensor*, with components satisfying $\epsilon_{AB} \simeq \delta_{iA}\delta_{jB}\varepsilon_{ij}$. From the exact relation (12.13)₁, or from either estimate,

$$\text{grad}^+ \underline{u}^+ = \underline{Q}(\text{grad } \underline{u})\underline{Q}^t \quad \text{and} \quad \underline{\varepsilon}^+ = \underline{Q}\underline{\varepsilon}\underline{Q}^t, \quad (12.18)$$

$\underline{Q}: \mathbb{E}^3 \rightarrow \mathbb{E}^{3+}$ being the observer rotation. These also follow from (7.3) and (7.21). Note the contrast with (12.9) and (12.11): the spatial objects are conjugated by $\underline{Q}$, the referential ones by $\underline{K}$, and the shifter is what carries one rule into the other.

Stretch and rotation

Take the polar decomposition $\tilde{F} = \tilde{R}\tilde{U}$ of (3.32). From (12.3) and (12.8),

$$\tilde{U}^2 - \hat{\tilde{I}} = \tilde{C} - \hat{\tilde{I}} = 2\tilde{E} \simeq 2\tilde{\epsilon}. \quad (12.19)$$

Write $\tilde{U} = \hat{\tilde{I}} + \tilde{\beta}$, which defines $\tilde{\beta}$; then $\tilde{U}^2 = \hat{\tilde{I}} + 2\tilde{\beta} + \tilde{\beta}^2$, and discarding the square, $2\tilde{\beta} \simeq 2\tilde{\epsilon}$. Hence, with $\tilde{U}^{-1} = (\hat{\tilde{I}} + \tilde{\beta})^{-1} \simeq \hat{\tilde{I}} - \tilde{\beta}$,

$$\tilde{U} - \hat{\tilde{I}} \simeq \tilde{\epsilon}, \quad \tilde{U}^{-1} - \hat{\tilde{I}} \simeq -\tilde{\epsilon}, \quad \tilde{R} - \mathbb{1} \simeq \mathbb{1}\hat{\tilde{\omega}}, \quad (12.20)$$

where

$$\hat{\tilde{\omega}} = \frac{1}{2}(\tilde{G} - \tilde{G}^t) \quad (12.21)$$

is the *referential infinitesimal rotation tensor*. The third of (12.20) is the one that needs a computation. From $\tilde{R} = \tilde{F}\tilde{U}^{-1}$ with (12.6) and $\tilde{U}^{-1} \simeq \hat{\tilde{I}} - \tilde{\epsilon}$,

$$\tilde{R} \simeq \mathbb{1}(\hat{\tilde{I}} + \tilde{G})(\hat{\tilde{I}} - \tilde{\epsilon}) \simeq \mathbb{1}(\hat{\tilde{I}} + \tilde{G} - \tilde{\epsilon}) = \mathbb{1}(\hat{\tilde{I}} + \hat{\tilde{\omega}}),$$

the product $\tilde{G}\tilde{\epsilon}$ being of second order, and $\tilde{G} - \tilde{\epsilon} = \tilde{G} - \frac{1}{2}(\tilde{G} + \tilde{G}^t) = \frac{1}{2}(\tilde{G} - \tilde{G}^t) = \hat{\tilde{\omega}}$. Equivalently,

$$\tilde{R} - \mathbb{1} \simeq \omega \mathbb{1}, \quad \text{where} \quad \omega = \frac{1}{2}[\text{grad } \underline{u} - (\text{grad } \underline{u})^t] = \mathbb{1}\hat{\tilde{\omega}}\mathbb{1}^t \quad (12.22)$$

is the *spatial infinitesimal rotation tensor*, the two forms agreeing because $\omega \mathbb{1} = \mathbb{1}\hat{\tilde{\omega}}\mathbb{1}^t \mathbb{1} = \mathbb{1}\hat{\tilde{\omega}}$.

REMARK 12.4 (the linearized polar decomposition). Collecting (12.8), (12.21) and (12.20),

$$\tilde{G} = \tilde{\epsilon} + \hat{\tilde{\omega}}, \quad \tilde{F} = \mathbb{1}(\hat{\tilde{I}} + \tilde{\epsilon} + \hat{\tilde{\omega}}) \simeq \mathbb{1}(\hat{\tilde{I}} + \hat{\tilde{\omega}})(\hat{\tilde{I}} + \tilde{\epsilon}),$$

the last step discarding $\hat{\tilde{\omega}}\tilde{\epsilon}$. So at leading order the polar decomposition degenerates into the additive split of $\tilde{G}$ into its symmetric and skew parts, and the multiplicative structure that Chapter 3 worked so hard for collapses into a sum. That collapse is the whole simplification of linear elasticity, and Remark 12.2 is the price: the additive split is faithful only while the rotation is small, since $\hat{\tilde{I}} + \hat{\tilde{\omega}}$ is a rotation only to first order.

The additive split is also the linear analogue of (4.7). There $\text{grad } \underline{v}$ splits into stretching and spin; here $\text{grad } \underline{u}$ splits into strain and infinitesimal rotation. Chapter 4's split is exact and Chapter 12's is not, because velocity is a rate and displacement is not.

Volume and density

The local volume change is $J - 1$, and $J^2 = \det \underline{C}$. Writing $\underline{C} = \hat{\underline{I}} + \underline{A}$ with $\underline{A} = \underline{G} + \underline{G}^t + \underline{G}^t \underline{G}$ from (12.3), and using the result of Problem 23 of Chapter 1 for the expansion of a determinant,

$$J^2 = \det (\hat{\underline{I}} + \underline{A}) = \det (\underline{A} - \lambda \hat{\underline{I}}) \big|_{\lambda=-1}, \quad (12.23)$$

whose expansion is $1 + I_1(\underline{A}) + I_2(\underline{A}) + I_3(\underline{A})$. The last two invariants are quadratic and cubic in $\underline{A}$ and so are discarded, and $\text{tr}(\underline{G}^t \underline{G})$ is quadratic in $\underline{G}$, leaving

$$J^2 - 1 \simeq \text{tr } \underline{A} \simeq 2 \text{tr } \underline{G} \simeq 2 \text{tr}(\text{grad } \underline{u}), \quad (12.24)$$

the last estimate by (12.15) and $\text{tr}[\underline{1}^t(\text{grad } \underline{u})\underline{1}] = \text{tr}[(\text{grad } \underline{u})\underline{1}\underline{1}^t] = \text{tr}[(\text{grad } \underline{u})\underline{I}] = \text{tr}(\text{grad } \underline{u})$. Since J is close to 1, $J^2 - 1 = (J - 1)(J + 1) \simeq 2(J - 1)$, and therefore

$$J - 1 \simeq \text{Div } \underline{w} \simeq \text{div } \underline{u}. \quad (12.25)$$

The change of volume of an arbitrary part S follows by integrating, P being the region occupied by S in κ :

$$\Delta V = \int_P (J - 1) \, dV \simeq \int_P \text{Div } \underline{w} \, dV = \int_{\partial P} \underline{w} \cdot \underline{N} \, dA, \quad (12.26)$$

by (2.50) and the divergence theorem. Alternatively, from $J^{-1} - 1 \simeq -\text{div } \underline{u}$,

$$\Delta V \simeq \int_P \text{div } \underline{u} \, dV = \int_{P_t} J^{-1} \text{div } \underline{u} \, dv \simeq \int_{P_t} \text{div } \underline{u} \, dv = \int_{\partial P_t} \underline{u} \cdot \underline{n} \, da. \quad (12.27)$$

To leading order the volume change is the flux of the displacement through the boundary, computed with either description. That the two right-hand sides agree is Problem 4.

Finally the density. By (2.59), $\rho_\kappa = \rho J$, so

$$\Delta\rho = \rho - \rho_\kappa, \quad (12.28)$$

and $\rho = \rho_\kappa J^{-1} \simeq \rho_\kappa(1 - \operatorname{Div} \underline{w})$ gives

$$\Delta\rho \simeq -\rho_\kappa \operatorname{Div} \underline{w} \simeq -\rho_\kappa \operatorname{div} \underline{u}, \quad \text{yielding} \quad \rho \simeq \rho_\kappa \quad (12.29)$$

at leading order. The density change is of first order in $|\underline{G}|$ and the density itself is ρ_κ to zeroth order, which is why the linear equation of motion may carry either symbol.

12.2 Stress–deformation relation

We now want small-displacement-gradient estimates of the three stresses. Introduce the *strain-energy function*

$$U(\underline{E}; \underline{X}) = \bar{W}(\theta_0, \hat{\underline{I}} + 2\underline{E}; \underline{X}), \quad (12.30)$$

$\bar{W}$ being the free energy per unit reference volume of (11.19) and θ_0 the fixed uniform temperature. The substitution $\underline{C} = \hat{\underline{I}} + 2\underline{E}$ is (2.97) read backwards. Proceeding as at (11.20), and using $\dot{\underline{C}} = 2\dot{\underline{E}}$,

$$U_{\underline{E}} \cdot \dot{\underline{E}} = \bar{W}_{\underline{C}} \cdot \dot{\underline{C}} = 2\bar{W}_{\underline{C}} \cdot \dot{\underline{E}} = \underline{S} \cdot \dot{\underline{E}} \quad (12.31)$$

for every symmetric $\dot{\underline{E}}$, the last step by (11.24). Since $U_{\underline{E}}$ is symmetric, being a gradient with respect to a symmetric argument, and $\dot{\underline{E}}$ ranges over all symmetric tensors,

$$\underline{S} = U_{\underline{E}}. \quad (12.32)$$

Expanding the energy

A small displacement gradient implies a small strain, by Remark 12.2, so we want U near $\underline{E} = \hat{\underline{O}}$. Taylor's theorem gives

$$U(\underline{E}; \underline{X}) = U(\hat{\underline{O}}; \underline{X}) + \underline{S}_0(\underline{X}) \cdot \underline{E} + \frac{1}{2} \underline{E} \cdot \mathbb{D}(\underline{X})[\underline{E}] + o(|\underline{E}|^2), \quad (12.33)$$

where

$$\mathcal{S}_0(\mathcal{X}) = U_{\mathcal{E}}|_{\mathcal{E}=\hat{\mathcal{O}}} \quad \text{and} \quad \mathbb{D}(\mathcal{X}) = U_{\mathcal{E}\mathcal{E}}|_{\mathcal{E}=\hat{\mathcal{O}}} \quad (12.34)$$

is a referential fourth-order tensor, $\mathbb{D}(\mathcal{X})[\mathcal{E}]$ being a linear second-order-tensor-valued function of $\mathcal{E}$.

REMARK 12.5 ($\mathbb{C}$ here is not the complex numbers). The spatial counterpart of $\mathbb{D}$, introduced at (12.56), is written $\mathbb{C}$, following the book. Chapter 1 used the same blackboard letter for the complex numbers in the proof that the principal values of a symmetric tensor are real. The two never meet: this chapter has no complex numbers in it. Nor should $\mathbb{C}$ be confused with $\mathcal{C}$, the right Cauchy–Green tensor, which carries an under-tilde and is of second order.

Resolving on a basis, $\mathcal{E} = E_{CD}\hat{\mathcal{E}}_C \otimes \hat{\mathcal{E}}_D$, linearity gives

$$\mathbb{D}(\mathcal{X})[\mathcal{E}] = E_{CD} \mathbb{D}(\mathcal{X})[\hat{\mathcal{E}}_C \otimes \hat{\mathcal{E}}_D], \quad (12.35)$$

and therefore

$$\begin{aligned} \mathcal{E} \cdot \mathbb{D}(\mathcal{X})[\mathcal{E}] &= E_{AB}E_{CD} \mathbb{D}_{ABCD}(\mathcal{X}), \\ \text{where } \mathbb{D}_{ABCD} &= \hat{\mathcal{E}}_A \otimes \hat{\mathcal{E}}_B \cdot \mathbb{D}[\hat{\mathcal{E}}_C \otimes \hat{\mathcal{E}}_D] = \hat{\mathcal{E}}_A \cdot (\mathbb{D}[\hat{\mathcal{E}}_C \otimes \hat{\mathcal{E}}_D])\hat{\mathcal{E}}_B. \end{aligned} \quad (12.36)$$

Three symmetries may now be assumed without loss of generality, because only the combination $E_{AB}E_{CD}\mathbb{D}_{ABCD}$ ever appears. Relabelling the summation indices, and using that ordinary multiplication commutes,

$$E_{AB}E_{CD}\mathbb{D}_{ABCD} = E_{CD}E_{AB}\mathbb{D}_{CDAB} = E_{AB}E_{CD}\mathbb{D}_{CDAB}, \quad (12.37)$$

so nothing is lost by taking

$$\mathbb{D}_{ABCD} = \mathbb{D}_{CDAB}. \quad (12.38)$$

Relabelling again and using $E_{AB} = E_{BA}$,

$$E_{BA}E_{CD}\mathbb{D}_{BACD} = E_{AB}E_{CD}\mathbb{D}_{ABCD} = E_{BA}E_{CD}\mathbb{D}_{ABCD}, \quad (12.39)$$

so we may take

$$\mathbb{D}_{BACD} = \mathbb{D}_{ABCD} \quad (12.40)$$

and hence, combining with (12.38),

$$\mathbb{D}_{ABCD} = \mathbb{D}_{ABDC}. \quad (12.41)$$

In tensor form, let A_{AB} and B_{CD} be the components of arbitrary referential tensors. Then (12.36)₂, (12.38) and linearity give the *major symmetry*

$$\underline{A} \cdot \mathbb{D}[\underline{B}] = A_{AB} B_{CD} \mathbb{D}_{ABCD} = B_{CD} A_{AB} \mathbb{D}_{CDAB} = \underline{B} \cdot \mathbb{D}[\underline{A}] \quad (12.42)$$

for all $\underline{A}$ and $\underline{B}$, while (12.40) and (12.41) give the two *minor symmetries*

$$\underline{A}^t \cdot \mathbb{D}[\underline{B}] = \underline{A} \cdot \mathbb{D}[\underline{B}] \quad \text{and} \quad \underline{A} \cdot \mathbb{D}[\underline{B}^t] = \underline{A} \cdot \mathbb{D}[\underline{B}]. \quad (12.43)$$

Now differentiate (12.33), keeping terms through quadratic order. The rate of the linear map is

$$(\mathbb{D}(\underline{X})[\underline{E}])^\cdot = \dot{E}_{CD} \mathbb{D}(\underline{X})[\hat{\underline{E}}_C \otimes \hat{\underline{E}}_D] = \mathbb{D}(\underline{X})[\dot{\underline{E}}], \quad (12.44)$$

by (12.35), the basis dyads being fixed. Hence, using major symmetry to merge the two quadratic contributions,

$$\begin{aligned} U_{\underline{E}} \cdot \dot{\underline{E}} &= \underline{S}_0(\underline{X}) \cdot \dot{\underline{E}} + \frac{1}{2} \{ \underline{E} \cdot \mathbb{D}(\underline{X})[\dot{\underline{E}}] + \dot{\underline{E}} \cdot \mathbb{D}(\underline{X})[\underline{E}] \} \\ &= \{ \underline{S}_0(\underline{X}) + \mathbb{D}(\underline{X})[\underline{E}] \} \cdot \dot{\underline{E}}, \end{aligned} \quad (12.45)$$

for all symmetric $\dot{\underline{E}}$, and (12.32) with (12.40) delivers the Piola–Kirchhoff stress

$$\underline{S} = \underline{S}_0(\underline{X}) + \mathbb{D}(\underline{X})[\underline{E}]. \quad (12.46)$$

Here $\underline{S}_0$ is the *residual stress*, the stress carried at zero strain, and $\mathbb{D}$ is the *stiffness tensor*, also called the elastic modulus tensor.

Isotropy

For an isotropic material, (11.29) gives

$$\begin{aligned} U(\underline{E}; \underline{X}) &= \bar{W}(\theta_0, \hat{\underline{I}} + 2\underline{E}; \underline{X}) = \bar{W}(\theta_0, \underline{R}^t(\hat{\underline{I}} + 2\underline{E})\underline{R}; \underline{X}) \\ &= \bar{W}(\theta_0, \hat{\underline{I}} + 2\underline{R}^t \underline{E} \underline{R}; \underline{X}) = U(\underline{R}^t \underline{E} \underline{R}; \underline{X}) \end{aligned} \quad (12.47)$$

for all $\underline{R} \in \text{Orth}$, the third step by $\underline{R}^t \hat{\underline{I}} \underline{R} = \underline{R}^t \underline{R} = \hat{\underline{I}}$. So U is an isotropic scalar function of $\underline{E}$ and depends on it through the principal invariants $I_1(\underline{E}) = \text{tr } \underline{E} = \hat{\underline{I}} \cdot \underline{E}$, $I_2(\underline{E}) = \frac{1}{2}[I_1^2 - \text{tr}(\underline{E}^2)]$ and $I_3(\underline{E}) = \det \underline{E}$, or equivalently through $\hat{\underline{I}} \cdot \underline{E}$, $\underline{E} \cdot \underline{E}$ and $\det \underline{E}$. Keeping terms through quadratic order, and noting that $\det \underline{E}$ is cubic while I_2 is quadratic,

$$U(\underline{E}; \underline{X}) = U(\hat{\underline{O}}; \underline{X}) + \alpha(\underline{X}) \hat{\underline{I}} \cdot \underline{E} + \frac{1}{2} \lambda(\underline{X}) (\hat{\underline{I}} \cdot \underline{E})^2 + \mu(\underline{X}) \underline{E} \cdot \underline{E}, \quad (12.48)$$

the functions α, λ, μ being properties of the material. Differentiating term by term,

$$U_{\underline{E}} \cdot \dot{\underline{E}} = \{ [\alpha + \lambda(\text{tr } \underline{E})] \hat{\underline{I}} + 2\mu \underline{E} \} \cdot \dot{\underline{E}}, \quad (12.49)$$

since $(\hat{\underline{I}} \cdot \underline{E})^\cdot = \hat{\underline{I}} \cdot \dot{\underline{E}}$ and $(\underline{E} \cdot \underline{E})^\cdot = 2\underline{E} \cdot \dot{\underline{E}}$, and comparison with (12.46) gives

$$\underline{S}_0(\underline{X}) = \alpha(\underline{X}) \hat{\underline{I}} \quad \text{and} \quad \mathbb{D}(\underline{X})[\underline{E}] = \lambda(\underline{X})(\text{tr } \underline{E}) \hat{\underline{I}} + 2\mu(\underline{X}) \underline{E}. \quad (12.50)$$

REMARK 12.6 (what isotropy costs the residual stress). Equation (12.50)₁ says that an isotropic material can carry only a hydrostatic residual stress. That is the linearized form of the statement proved at Problem 10 of Chapter 11, and Problem 3 below turns it into the same conclusion: with equilibrium and one traction-free patch, an isotropic body cannot be residually stressed at all. The two constants λ and μ are the Lamé moduli, and μ is the shear modulus G of (11.41).

Linearizing, and the other two stresses

Equation (12.46) is accurate whenever $|\underline{E}|$ is small, and it is a nonlinear relation between $\underline{S}$ and $\underline{G}$, since $\underline{E}$ is quadratic in $\underline{G}$ by (12.7). If $|\underline{G}|$ is

also small, (12.8) linearizes it:

$$\mathcal{S} - \mathcal{S}_0(\mathcal{X}) \simeq \mathbb{D}(\mathcal{X})[\underline{\epsilon}] = \mathbb{D}(\mathcal{X})[\underline{G}], \quad (12.51)$$

the second equality by the minor symmetry (12.43)₂, which lets the skew part of $\underline{G}$ be discarded; in components $S_{AB} - S_{(0)AB} \simeq \mathbb{D}_{ABCD} w_{C,D}$. Classical linear elasticity assumes the residual stress vanishes, so

$$\mathcal{S} \simeq \mathbb{D}(\mathcal{X})[\underline{G}], \quad (12.52)$$

and for an isotropic material, by (12.50)₂ with $\underline{\epsilon} = \frac{1}{2}(\underline{G} + \underline{G}^t)$,

$$\mathcal{S} \simeq \lambda(\mathcal{X})(\text{tr } \underline{G})\hat{\underline{I}} + \mu(\mathcal{X})(\underline{G} + \underline{G}^t). \quad (12.53)$$

For the Piola stress, combine (6.144) with (12.6), which gives $\underline{F} \simeq \underline{1}$:

$$\underline{P} = \underline{F}\mathcal{S} \simeq \underline{1}\mathcal{S}. \quad (12.54)$$

To the same order $\underline{F}^* \simeq \underline{1}$, which is Problem 4, and either (6.140) or (6.145) then gives

$$\underline{T} \simeq \underline{1}\mathcal{S}\underline{1}^t \quad (12.55)$$

for the Cauchy stress. In components $P_{iA} \simeq \delta_{iB}S_{BA}$ and $T_{ij} \simeq \delta_{iA}\delta_{jB}S_{AB}$: all three stresses carry the same numbers, and differ only in which legs are referential.

With vanishing residual stress, (12.15) and (12.52) combine into

$$\underline{T} \simeq \mathbb{C}[\text{grad } \underline{u}], \quad \text{where} \quad \mathbb{C}[\text{grad } \underline{u}] = \underline{1}\{\mathbb{D}[\underline{1}^t(\text{grad } \underline{u})\underline{1}]\}\underline{1}^t. \quad (12.56)$$

The definition is easier to read in components. Using (12.36)₂ and linearity at each step,

$$\begin{aligned} T_{ij} &= e_i \cdot \underline{T} e_j = \underline{1}^t e_i \cdot (\mathbb{D}[u_{k,l} \delta_{kC} \delta_{lD} \hat{\underline{E}}_C \otimes \hat{\underline{E}}_D]) \underline{1}^t e_j \\ &= \delta_{iA} \delta_{jB} \hat{\underline{E}}_A \cdot (\mathbb{D}[\delta_{kC} \delta_{lD} u_{k,l} \hat{\underline{E}}_C \otimes \hat{\underline{E}}_D]) \hat{\underline{E}}_B \\ &= \delta_{iA} \delta_{jB} \delta_{kC} \delta_{lD} u_{k,l} \hat{\underline{E}}_A \cdot (\mathbb{D}[\hat{\underline{E}}_C \otimes \hat{\underline{E}}_D]) \hat{\underline{E}}_B = \mathbb{C}_{ijkl} u_{k,l}, \end{aligned} \quad (12.57)$$

where

$$\mathbb{C}_{ijkl} = \delta_{iA} \delta_{jB} \delta_{kC} \delta_{lD} \mathbb{D}_{ABCD}. \quad (12.58)$$

The spatial stiffness inherits every symmetry of the referential one, which is Problem 5: the major symmetry

$$\mathbb{C}_{ijkl} = \mathbb{C}_{klij} \quad (12.59)$$

and the two minor ones

$$\mathbb{C}_{ijkl} = \mathbb{C}_{jikl} \quad \text{and} \quad \mathbb{C}_{ijkl} = \mathbb{C}_{ijlk}. \quad (12.60)$$

The last of these lets $u_{k,l}$ in (12.57) be replaced by its symmetric part, so $T_{ij} = \mathbb{C}_{ijkl}\varepsilon_{kl}$ with ε_{ij} the components of (12.17). For an isotropic material, (12.15) and (12.53) give directly

$$\underline{T} \simeq \lambda(\operatorname{div} \underline{u})\underline{I} + \mu[\operatorname{grad} \underline{u} + (\operatorname{grad} \underline{u})^t]. \quad (12.61)$$

Tractions

The Piola traction is $\underline{p} = \underline{P}\underline{N}$ and the Cauchy traction $\underline{t} = \underline{T}\underline{n}$, by (6.132) and (6.94). The Piola–Nanson formula (2.54) with $\underline{F}^* \simeq \underline{1}$ gives

$$\underline{n} \, da \simeq \underline{1}\underline{N} \, dA, \quad (12.62)$$

so $da \simeq dA$, the shifter preserving length, and $\underline{n} \simeq \underline{1}\underline{N}$. Therefore

$$\underline{p} = \underline{P}\underline{N} = \underline{T}\underline{F}^*\underline{N} \simeq \underline{T}\underline{1}\underline{N} \simeq \underline{T}\underline{n} = \underline{t} : \quad (12.63)$$

the two tractions coincide at leading order, which is why elementary treatments need not distinguish them. By (12.54) this is the same as

$$\underline{1}^t \underline{p} \simeq \underline{S}\underline{N}. \quad (12.64)$$

In the frame of O^+ the counterparts follow from Chapter 7:

$$\begin{aligned} \underline{S}^+ &\simeq \mathbb{D}^+[G^+], & \underline{T}^+ &\simeq \underline{1}^+ \underline{S}^+ (\underline{1}^+)^t \simeq \mathbb{C}^+[\operatorname{grad}^+ \underline{u}^+], \\ (\underline{1}^+)^t \underline{p}^+ &\simeq \underline{S}^+ \underline{N}^+, \end{aligned} \quad (12.65)$$

where

$$\begin{aligned} \mathbb{D}^+[G^+] &= \underline{K}(\mathbb{D}[\underline{K}^t G^+ \underline{K}])\underline{K}^t, \\ \mathbb{C}^+[\operatorname{grad}^+ \underline{u}^+] &= \underline{Q}\{\mathbb{C}[\underline{Q}^t(\operatorname{grad}^+ \underline{u}^+)\underline{Q}]\}\underline{Q}^t. \end{aligned} \quad (12.66)$$

For an isotropic material these reduce to

$$\begin{aligned}\mathbb{D}^+[G^+] &= \lambda(\text{tr } G^+) \hat{I}^+ + \mu[G^+ + (G^+)^t], \\ \mathbb{C}^+[\text{grad}^+ \underline{u}^+] &= \lambda(\text{div}^+ \underline{u}^+) I^+ + \mu[\text{grad}^+ \underline{u}^+ + (\text{grad}^+ \underline{u}^+)^t],\end{aligned}\tag{12.67}$$

with the same two constants: the moduli are numbers and no observer can change them.

12.3 Equation of motion

Take the frame of O to be inertial. The referential equation of motion is (6.136) with $\underline{P} = \underline{F}\underline{S}$ and, the shifter being fixed,

$$\dot{\underline{v}} = \ddot{\underline{\chi}}_\kappa = (\underline{u} + \underline{1}X)^{\cdot\cdot} = \ddot{\underline{u}}.\tag{12.68}$$

The divergence of the Piola stress is $\text{Div } \underline{P} = P_{iA,A} \underline{e}_i$, and by the product rule together with $F_{iB} = \delta_{iB} + \delta_{iC} G_{CB}$ from (12.6),

$$\begin{aligned}P_{iA,A} &= (F_{iB} S_{BA})_{,A} = F_{iB} S_{BA,A} + \delta_{iC} G_{CB,A} S_{BA} \\ &\simeq \delta_{iB} S_{BA,A} + \delta_{iC} G_{CB,A} S_{BA}.\end{aligned}\tag{12.69}$$

With no residual stress, (12.52) makes the non-dimensionalized S_{BA} of order ε , the small number of (12.1). If we assume in addition that $G_{CB,A}$, non-dimensionalized by a length scale of the problem, is also of order ε , the second term is of order ε^2 and drops:

$$\text{Div } \underline{P} \simeq \underline{1}(\text{Div } \underline{S}).\tag{12.70}$$

REMARK 12.7 (the assumption hidden in (12.70)). Smallness of $\underline{G}$ says nothing about smallness of $\text{Grad } \underline{G}$. A field may be uniformly small and vary arbitrarily rapidly, and then the discarded term in (12.69) is not negligible. The restriction on $G_{CB,A}$ is therefore a second and independent hypothesis, tacitly made throughout the literature. It is what confines the theory to displacement fields whose variation is slow on the scale of the body, and it excludes, for instance, the immediate neighbourhood of a crack tip, where the strain is small but its gradient is not.

The same caution applies to the solution rather than the data. Once a boundary-value problem has been solved, both hypotheses should be checked against the answer, since nothing in the linear theory enforces them.

With these estimates, and applying $\underline{1}^t$ to $\underline{1} \operatorname{Div} \underline{S} + \rho_\kappa \underline{b} = \rho_\kappa \underline{\ddot{u}}$,

$$\operatorname{Div} \underline{S} + \rho_\kappa \underline{f} = \rho_\kappa \underline{\ddot{w}}, \quad (12.71)$$

where $\underline{f} = \underline{1}^t \underline{b}$ is the referential body force, $\underline{\ddot{w}} = (\underline{1}^t \underline{u})^{\cdot\cdot} = \underline{1}^t \underline{\ddot{u}}$ the referential acceleration, and

$$\underline{S} = \mathbb{D}[\operatorname{Grad} \underline{w}]. \quad (12.72)$$

This is a linear second-order partial differential system for $\underline{w}(X, t)$. A typical initial-boundary-value problem adds initial data $\underline{w}|_{t=0}$ and $\underline{\dot{w}}|_{t=0}$, together with $\underline{1}^t \underline{p}$ prescribed on a part $\partial\kappa_p$ of the boundary and $\underline{w}$ prescribed on the complement $\partial\kappa_w = \partial\kappa \setminus \partial\kappa_p$.

In the frame of O^+ ,

$$\operatorname{Div}^+ \underline{S}^+ = \underline{K}(\operatorname{Div} \underline{S}), \quad (12.73)$$

which is Problem 9, and hence, with (7.38)₃,

$$\operatorname{Div}^+ \underline{S}^+ + \rho_{\kappa^+}^+ \underline{f}^+ = \rho_{\kappa^+}^+ \underline{K} \underline{\ddot{w}}, \quad (12.74)$$

with

$$\underline{f}^+ = (\underline{1}^+)^t \underline{b}^+ \quad \text{and} \quad \underline{S}^+ = \mathbb{D}^+[\operatorname{Grad}^+ \underline{w}^+], \quad \underline{w}^+ = (\underline{1}^+)^t \underline{u}^+. \quad (12.75)$$

If the frame of O^+ is also inertial then the two are related by a Galilean transformation (7.10), with $\underline{Q}$ a fixed orthogonal tensor and $\underline{V}$ a fixed vector, and then $\underline{K} \underline{\ddot{w}} = \underline{\ddot{w}}^+$, so that

$$\operatorname{Div}^+ \underline{S}^+ + \rho_{\kappa^+}^+ \underline{f}^+ = \rho_{\kappa^+}^+ \underline{\ddot{w}}^+. \quad (12.76)$$

Linear elasticity therefore holds in every inertial frame, in the same form. The non-inertial case is Problem 10.

REMARK 12.8 (how to make the shifter disappear). The shifter has done real work: without it, (12.8) is not a tensor equation and (12.9) does not follow, as Remark 12.3 showed. In any single problem, however, it can be made invisible. The observer O is free to choose the shifted basis $\underline{E}_A = \underline{1}\hat{\underline{E}}_A$ of (2.68) to coincide with the spatial basis $\underline{e}_i$. Then δ_{iA} are numerically the components of the Kronecker delta, the components of a referential tensor and of its spatial shift are the same numbers, and one may write $2\underline{E} \simeq \underline{H} + \underline{H}^t$ without harm.

That is why the omission survives in the literature. It is a legitimate computational convenience for one observer, and an illegitimate statement about tensors. The distinction matters exactly when a second observer appears, which is the situation (12.65) and (12.66) describe.

12.4 Problems

PROBLEM 1 (The kinematic transformations). Derive (12.9) to (12.11). Show that $|\underline{G}^+| = |\underline{G}|$. What is the relationship between $\underline{w}^+$ and $\underline{w}$?

Solution. **(12.9).** By definition $\underline{G}^+ = (\underline{1}^+)^t \underline{H}^+$. Chapter 7 supplies both factors: $\underline{1}^+ = \underline{Q}\underline{1}\underline{K}^t$ by (7.18) and $\underline{H}^+ = \underline{Q}\underline{H}\underline{K}^t$ by (7.25). Hence

$$\underline{G}^+ = (\underline{Q}\underline{1}\underline{K}^t)^t \underline{Q}\underline{H}\underline{K}^t = \underline{K}\underline{1}^t \underline{Q}^t \underline{Q}\underline{H}\underline{K}^t = \underline{K}\underline{1}^t \underline{H}\underline{K}^t = \underline{K}\underline{G}\underline{K}^t$$

using $(\underline{A}\underline{B}\underline{C})^t = \underline{C}^t \underline{B}^t \underline{A}^t$ and $\underline{Q}^t \underline{Q} = \underline{I}$. The observer rotation cancels completely, which is the point: $\underline{G}$ is a referential tensor and only $\underline{K}$ can act on it.

(12.10). Substituting into (12.7) and using $(\underline{K}\underline{G}\underline{K}^t)^t = \underline{K}\underline{G}^t \underline{K}^t$,

$$\begin{aligned} \underline{E}^+ &= \frac{1}{2} [\underline{G}^+ + (\underline{G}^+)^t + (\underline{G}^+)^t \underline{G}^+] \\ &= \frac{1}{2} [\underline{K}\underline{G}\underline{K}^t + \underline{K}\underline{G}^t \underline{K}^t + \underline{K}\underline{G}^t \underline{K}^t \underline{K}\underline{G}\underline{K}^t] = \underline{K}\underline{E}\underline{K}^t, \end{aligned}$$

the last step by $\underline{K}^t \underline{K} = \hat{\underline{I}}$ in the quadratic term and by factoring $\underline{K}$ out on the left and $\underline{K}^t$ on the right. So (12.10) is exact, not an approximation.

(12.11). The same computation without the quadratic term gives $\underline{\epsilon}^+ = \frac{1}{2} [\underline{G}^+ + (\underline{G}^+)^t] = \underline{K}\underline{\epsilon}\underline{K}^t$. Comparing with (12.10), the two agree

to leading order exactly when $\underline{E} \simeq \underline{\epsilon}$ does, which is (12.12): the approximation is preserved under change of observer.

The norm. Using the cyclic property of the trace and $\underline{K}^t \underline{K} = \hat{I}$ twice,

$$\begin{aligned} |\underline{G}^+|^2 &= \text{tr} (\underline{G}^+ (\underline{G}^+)^t) = \text{tr} (\underline{K} \underline{G} \underline{K}^t \underline{K} \underline{G}^t \underline{K}^t) \\ &= \text{tr} (\underline{G} \underline{G}^t \underline{K}^t \underline{K}) = \text{tr} (\underline{G} \underline{G}^t) = |\underline{G}|^2, \end{aligned}$$

so $|\underline{G}^+| = |\underline{G}|$. The two observers agree on how small the displacement gradient is, and therefore on whether the linear theory applies at all.

The displacements. By (7.21), $\underline{u}^+ = \underline{Q} \underline{u} + \underline{d}(t)$ with $\underline{d}$ spatially uniform. Hence

$$\underline{w}^+ = (\underline{1}^+)^t \underline{u}^+ = \underline{K} \underline{1}^t \underline{Q}^t (\underline{Q} \underline{u} + \underline{d}) = \underline{K} \underline{w} + \underline{d}_\kappa(t), \quad \underline{d}_\kappa = \underline{K} \underline{1}^t \underline{Q}^t \underline{d}$$

so the referential displacements differ by a rotation and a spatially uniform translation. Taking Grad^+ kills the translation and returns (12.9), as it must: a uniform displacement is a rigid translation and strains nothing. $\square$

PROBLEM 2 (Energy balance in the purely mechanical theory). Show that in the purely mechanical theory $\frac{d}{dt} E(S, t) = P(S, t)$, where P is the power of the forces acting on $S \subset B$ and

$$E = \int_P U \, dV + K(S, t),$$

K being the kinetic energy and $P = \kappa(S)$. The letter P carries two meanings in this statement, the power on the left and the reference region under the integral sign; both are the book's, and they are told apart by position.

Solution. The mechanical energy balance (6.150) reads

$$\frac{d}{dt} K(S, t) + \mathcal{S}(S, t) = P(S, t),$$

with the stress power $\mathcal{S}$ given in referential form by (6.156), $\mathcal{S} = \int_P \underline{P} \cdot \underline{\dot{F}} \, dV$. Everything turns on identifying that integrand.

By (6.159), $\underline{P} \cdot \dot{\underline{F}} = \underline{S} \cdot \dot{\underline{E}}$, the Piola–Kirchhoff stress being power conjugate to the Lagrange strain. By (12.32), $\underline{S} = U_{\underline{E}}$, and the temperature being fixed, the chain rule for $U(\underline{E}; \underline{X})$ gives

$$\dot{U} = U_{\underline{E}} \cdot \dot{\underline{E}} = \underline{S} \cdot \dot{\underline{E}} = \underline{P} \cdot \dot{\underline{F}}.$$

Therefore

$$\mathcal{S}(S, t) = \int_P \dot{U} \, dV = \frac{d}{dt} \int_P U \, dV,$$

the differentiation passing through the integral because P is a fixed region of κ : it is the reference placement of the fixed set S of material points, and it does not move as the body deforms. This is the one place where the referential form of the stress power is essential; the spatial form would have a moving domain and the interchange would need Reynolds' formula.

Substituting,

$$\frac{d}{dt}K + \frac{d}{dt} \int_P U \, dV = P(S, t), \quad \text{that is} \quad \frac{d}{dt}E(S, t) = P(S, t)$$

with $E = \int_P U \, dV + K$.

What is being said. In a hyperelastic body at constant temperature the working of the external forces goes entirely into kinetic energy and stored strain energy, and none of it is lost. The stress power, which Remark 6.33 of Chapter 6 identified as the term with no counterpart in particle mechanics, turns out here to be the rate of a potential. Chapter 9 had no such statement, the viscous stress power being a dissipation; and Chapter 11 needed the full Clausius–Duhem inequality to see that $\underline{P} = W_{\underline{F}}$, which is what makes the potential exist. $\square$

PROBLEM 3 (Residual stress in the linear theory). Suppose the residually stressed undeformed configuration of an isotropic material is in equilibrium without body force, and that the associated traction vanishes on a portion of the boundary. Prove that the residual stress vanishes everywhere in the body.

Solution. This is the linearized twin of Problem 10 of Chapter 11, and it runs in three steps of the same shape.

Isotropy makes it hydrostatic. By (12.50)₁ the residual stress of an isotropic material is

$$\mathcal{S}_0(\underline{X}) = \alpha(\underline{X})\hat{\underline{I}},$$

a scalar multiple of the referential identity at each point. Isotropy permits no deviatoric part, because U can depend on $\underline{E}$ only through its invariants, and at $\underline{E} = \hat{\underline{Q}}$ the only tensor available from $I_1 = \hat{\underline{I}} \cdot \underline{E}$ is $\hat{\underline{I}}$ itself.

Equilibrium makes it uniform. With no body force and no acceleration, (12.71) reduces to $\text{Div } \mathcal{S}_0 = \underline{0}$. For a spherical field this is a gradient,

$$\text{Div } (\alpha \hat{\underline{I}}) = \text{Grad } \alpha = \underline{0},$$

by the referential form of Problem 24 of Chapter 1, so α is constant throughout the body, which is connected.

The free patch makes it zero. On the traction-free portion, (12.64) gives $\mathcal{S}_0 \underline{N} = \underline{0}$ for the unit normal $\underline{N}$ there, that is $\alpha \underline{N} = \underline{0}$; and $|\underline{N}| = 1$, so $\alpha = 0$. Being constant, α vanishes identically and

$$\begin{aligned} \mathcal{S}_0 &= \hat{\underline{Q}} \text{ throughout the body,} \\ &\text{and by (12.54) and (12.55) so do } \underline{P} \text{ and } \underline{T}. \end{aligned}$$

Why the linear theory may assume (12.52). The result is the justification for dropping $\mathcal{S}_0$ from (12.51). It is not a theorem about all materials: an anisotropic body can hold residual stress with a non-zero deviator, and the shrink-fitted ring and the toughened glass plate are engineered to do exactly that. What the problem shows is that isotropy plus equilibrium plus one free patch leaves no room for it. $\square$

PROBLEM 4 (The cofactor at leading order, and the two volume formulas). Show that

$$\underline{F}^* - \underline{1} \simeq \underline{1} [(\text{tr } \underline{G})\hat{\underline{I}} - \underline{G}^t],$$

and hence that $\underline{\underline{F}}^* \simeq \underline{\underline{1}}$ as claimed in the text. Use this to confirm $\underline{\underline{u}} \cdot \underline{\underline{n}} \, da \simeq \underline{\underline{w}} \cdot \underline{\underline{N}} \, dA$, and hence that the right-hand sides of (12.26) and (12.27) agree at leading order.

Solution. The cofactor. By (2.55), $\underline{\underline{F}}^* = J \underline{\underline{F}}^{-t}$, so both factors are needed to first order in $\underline{\underline{G}}$.

From (12.6), $\underline{\underline{F}} = \underline{\underline{1}}(\hat{\underline{\underline{I}}} + \underline{\underline{G}})$, and inverting a product,

$$\underline{\underline{F}}^{-1} = (\hat{\underline{\underline{I}}} + \underline{\underline{G}})^{-1} \underline{\underline{1}}^{-1} \simeq (\hat{\underline{\underline{I}}} - \underline{\underline{G}}) \underline{\underline{1}}^t,$$

the expansion $(\hat{\underline{\underline{I}}} + \underline{\underline{G}})^{-1} \simeq \hat{\underline{\underline{I}}} - \underline{\underline{G}}$ being the geometric series truncated after one term, legitimate because $|\underline{\underline{G}}| \ll 1$, and $\underline{\underline{1}}^{-1} = \underline{\underline{1}}^t$. Transposing,

$$\underline{\underline{F}}^{-t} \simeq \underline{\underline{1}}(\hat{\underline{\underline{I}}} - \underline{\underline{G}}^t).$$

From (12.25), $J \simeq 1 + \text{tr } \underline{\underline{G}}$. Multiplying and discarding the product of two first-order terms,

$$\underline{\underline{F}}^* = J \underline{\underline{F}}^{-t} \simeq (1 + \text{tr } \underline{\underline{G}}) \underline{\underline{1}}(\hat{\underline{\underline{I}}} - \underline{\underline{G}}^t) \simeq \underline{\underline{1}}[\hat{\underline{\underline{I}}} + (\text{tr } \underline{\underline{G}})\hat{\underline{\underline{I}}} - \underline{\underline{G}}^t],$$

that is

$$\underline{\underline{F}}^* - \underline{\underline{1}} \simeq \underline{\underline{1}}[(\text{tr } \underline{\underline{G}})\hat{\underline{\underline{I}}} - \underline{\underline{G}}^t].$$

The right-hand side has norm of order $|\underline{\underline{G}}|$, so $\underline{\underline{F}}^* \simeq \underline{\underline{1}}$ to leading order, as (12.55) and (12.62) assumed.

The two volume formulas. By Piola–Nanson (2.54) and the estimate just proved,

$$\underline{\underline{n}} \, da = \underline{\underline{F}}^* \underline{\underline{N}} \, dA \simeq \underline{\underline{1}} \underline{\underline{N}} \, dA.$$

Taking the inner product with the spatial displacement $\underline{\underline{u}}$, and moving the shifter across by the definition of the transpose,

$$\underline{\underline{u}} \cdot \underline{\underline{n}} \, da \simeq \underline{\underline{u}} \cdot \underline{\underline{1}} \underline{\underline{N}} \, dA = \underline{\underline{1}}^t \underline{\underline{u}} \cdot \underline{\underline{N}} \, dA = \underline{\underline{w}} \cdot \underline{\underline{N}} \, dA$$

by (12.4). Integrating over the boundary,

$$\int_{\partial P_t} \underline{\underline{u}} \cdot \underline{\underline{n}} \, da \simeq \int_{\partial P} \underline{\underline{w}} \cdot \underline{\underline{N}} \, dA,$$

which is exactly the equality of the right-hand sides of (12.27) and (12.26). Both compute the same ΔV , one over the deformed boundary and one over the reference boundary, and at leading order it does not matter which.

A check on the order of the errors. Every discarded term above was of order $|\underline{G}|^2$, while ΔV itself is of order $|\underline{G}|$ by (12.25). So the relative error in ΔV is of order $|\underline{G}|$, which is the accuracy the whole chapter works to. $\square$

PROBLEM 5 (The symmetries of the spatial stiffness). Verify (12.59) and (12.60).

Solution. Everything follows from (12.58),

$$\mathbb{C}_{ijkl} = \delta_{iA}\delta_{jB}\delta_{kC}\delta_{lD} \mathbb{D}_{ABCD},$$

in which the four shifter components are attached one to each index and are the same for every term. Relabelling the referential summation indices is therefore free, and the symmetries pass straight through.

Major symmetry. Interchange the index pairs (ij) and (kl) on the left, and correspondingly relabel $A \leftrightarrow C$, $B \leftrightarrow D$ on the right:

$$\mathbb{C}_{klij} = \delta_{kA}\delta_{lB}\delta_{iC}\delta_{jD} \mathbb{D}_{ABCD} = \delta_{iC}\delta_{jD}\delta_{kA}\delta_{lB} \mathbb{D}_{CDAB} = \mathbb{C}_{ijkl},$$

the middle step renaming the dummies $A \leftrightarrow C$, $B \leftrightarrow D$ and the last using (12.38), $\mathbb{D}_{CDAB} = \mathbb{D}_{ABCD}$. Hence $\mathbb{C}_{ijkl} = \mathbb{C}_{klij}$.

First minor symmetry. Interchanging i and j interchanges A and B after relabelling:

$$\mathbb{C}_{jikl} = \delta_{jA}\delta_{iB}\delta_{kC}\delta_{lD} \mathbb{D}_{ABCD} = \delta_{iA}\delta_{jB}\delta_{kC}\delta_{lD} \mathbb{D}_{BACD} = \mathbb{C}_{ijkl},$$

by (12.40). Hence $\mathbb{C}_{ijkl} = \mathbb{C}_{jikl}$.

Second minor symmetry. The same argument on the pair (kl) , using (12.41), gives $\mathbb{C}_{ijkl} = \mathbb{C}_{ijlk}$.

What each one is for. The first minor symmetry says the stress comes out symmetric, $T_{ij} = T_{ji}$, which is (6.117) and hence Euler's

second postulate. The second says only the symmetric part of $\text{grad } u$ enters, so $T_{ij} = \mathbb{C}_{ijkl}\varepsilon_{kl}$ and rigid infinitesimal rotations produce no stress. The major symmetry is the one that has no counterpart in the balance laws: it says the stiffness derives from a potential, and it is what makes Problem 2 work. Counting independent constants, a general fourth-order tensor in three dimensions has 81 components; the two minor symmetries reduce this to 36, and the major symmetry to 21. $\square$

PROBLEM 6 (The strain energy as a quadratic form). Establish that in the absence of residual stress the strain energy U is approximated at leading order by $\frac{1}{2}\underline{\underline{\varepsilon}} \cdot \mathbb{D}[\underline{\underline{\varepsilon}}]$, or equivalently by $\frac{1}{2}\underline{\underline{\varepsilon}} \cdot \mathbb{C}[\underline{\underline{\varepsilon}}]$. Hence show that $\underline{\underline{S}}$ and $\underline{\underline{T}}$ are approximated by the gradients $U_{\underline{\underline{\varepsilon}}}$ and $U_{\underline{\underline{\varepsilon}}}$.

Solution. **The quadratic form.** Take the energy datum at the undeformed state, $U(\hat{\underline{\underline{Q}}}; \underline{\underline{X}}) = 0$, which costs nothing since only differences of energy have meaning. With $\underline{\underline{S}}_0 = \hat{\underline{\underline{Q}}}$, (12.33) reduces to

$$U(\underline{\underline{E}}; \underline{\underline{X}}) = \frac{1}{2}\underline{\underline{E}} \cdot \mathbb{D}(\underline{\underline{X}})[\underline{\underline{E}}] + o(|\underline{\underline{E}}|^2),$$

and substituting $\underline{\underline{E}} \simeq \underline{\underline{\varepsilon}}$ from (12.8), whose error is of order $|\underline{\underline{G}}|^2$ and therefore contributes at order $|\underline{\underline{G}}|^3$ inside a quadratic form,

$$U \simeq \frac{1}{2}\underline{\underline{\varepsilon}} \cdot \mathbb{D}[\underline{\underline{\varepsilon}}]$$

The spatial form. Write both sides in components, using (12.17)₂ in the form $\epsilon_{AB} \simeq \delta_{iA}\delta_{jB}\varepsilon_{ij}$ and (12.58):

$$\begin{aligned} \underline{\underline{\varepsilon}} \cdot \mathbb{D}[\underline{\underline{\varepsilon}}] &= \epsilon_{AB}\epsilon_{CD}\mathbb{D}_{ABCD} \\ &\simeq \delta_{iA}\delta_{jB}\varepsilon_{ij}\delta_{kC}\delta_{lD}\varepsilon_{kl}\mathbb{D}_{ABCD} = \varepsilon_{ij}\varepsilon_{kl}\mathbb{C}_{ijkl} = \underline{\underline{\varepsilon}} \cdot \mathbb{C}[\underline{\underline{\varepsilon}}], \end{aligned}$$

so

$$U \simeq \frac{1}{2}\underline{\underline{\varepsilon}} \cdot \mathbb{C}[\underline{\underline{\varepsilon}}]$$

The strain energy is the same number computed on either description, as it must be, being a scalar.

The stresses as gradients. Differentiate the first boxed expression with respect to $\underline{\underline{\varepsilon}}$. For a quadratic form with major symmetry the

gradient is twice the linear map, halved by the factor $\frac{1}{2}$: taking the rate,

$$\dot{U} \simeq \frac{1}{2} \{ \dot{\epsilon} \cdot \mathbb{D}[\epsilon] + \epsilon \cdot \mathbb{D}[\dot{\epsilon}] \} = \mathbb{D}[\epsilon] \cdot \dot{\epsilon},$$

the two terms merging by (12.42). Since $\dot{\epsilon}$ is an arbitrary symmetric tensor,

$$U_{\epsilon} = \mathbb{D}[\epsilon] \simeq \mathcal{S}$$

by (12.52) and (12.51). The identical computation on the spatial form gives $U_{\epsilon} = \mathbb{C}[\epsilon] \simeq \mathcal{T}$ by (12.56). So

$$\mathcal{S} \simeq U_{\epsilon}, \quad \mathcal{T} \simeq U_{\epsilon}.$$

Both stresses are gradients of the same potential, taken with respect to the strain measure that lives in the same space. This is the linearized form of $\underline{P} = W_{\underline{F}}$ from (11.16)₂, and it is why the major symmetry (12.42) could not have been dispensed with. $\square$

PROBLEM 7 (Isotropic moduli, and when the energy is positive definite). Find expressions for the components $\mathbb{C}_{ijkl}$ and $\mathbb{D}_{ABCD}$ for isotropic solids. Linear elasticity always assumes the strain-energy function positive definite, that is $U \geq 0$ with equality only at zero strain. Show that for isotropic solids this holds if and only if $\mu > 0$ and $\kappa > 0$, where $\kappa = \lambda + \frac{2}{3}\mu$.

Solution. The components. By (12.50)₂, $\mathbb{D}[E] = \lambda(\text{tr } E)\hat{I} + 2\mu E$. In components this is $\mathbb{D}_{ABCD}E_{CD}$, so we need a $\mathbb{D}_{ABCD}$ reproducing $\lambda\delta_{AB}E_{CC} + 2\mu E_{AB}$. Take

$$\mathbb{D}_{ABCD} = \lambda\delta_{AB}\delta_{CD} + \mu(\delta_{AC}\delta_{BD} + \delta_{AD}\delta_{BC})$$

and check by contracting with E_{CD} :

$$\mathbb{D}_{ABCD}E_{CD} = \lambda\delta_{AB}E_{CC} + \mu(E_{AB} + E_{BA}) = \lambda(\text{tr } E)\delta_{AB} + 2\mu E_{AB},$$

using $E_{AB} = E_{BA}$. The expression has both minor symmetries by inspection and the major symmetry because it is unchanged under $(AB) \leftrightarrow (CD)$. By (12.58) the spatial components are the same numbers,

$$\mathbb{C}_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$$

the shifter components turning each referential Kronecker delta into a spatial one.

The quadratic form. By Problem 6, and with $\mathcal{S}_0 = \hat{Q}$,

$$2U = \underline{\epsilon} \cdot \mathbb{D}[\underline{\epsilon}] = \lambda (\text{tr } \underline{\epsilon})^2 + 2\mu \underline{\epsilon} \cdot \underline{\epsilon}.$$

The two terms are not independent as they stand, because $\underline{\epsilon} \cdot \underline{\epsilon}$ contains the trace. Split off the spherical part: write

$$\underline{\epsilon} = \underline{\epsilon}' + \frac{1}{3} (\text{tr } \underline{\epsilon}) \hat{I}, \quad \text{tr } \underline{\epsilon}' = 0,$$

the deviator of Chapter 1. The two pieces are orthogonal, since $\underline{\epsilon}' \cdot \hat{I} = \text{tr } \underline{\epsilon}' = 0$, so

$$\underline{\epsilon} \cdot \underline{\epsilon} = \underline{\epsilon}' \cdot \underline{\epsilon}' + \frac{1}{9} (\text{tr } \underline{\epsilon})^2 \hat{I} \cdot \hat{I} = |\underline{\epsilon}'|^2 + \frac{1}{3} (\text{tr } \underline{\epsilon})^2,$$

using $\hat{I} \cdot \hat{I} = \text{tr } \hat{I} = 3$. Substituting,

$$2U = \left(\lambda + \frac{2}{3}\mu \right) (\text{tr } \underline{\epsilon})^2 + 2\mu |\underline{\epsilon}'|^2 = \kappa (\text{tr } \underline{\epsilon})^2 + 2\mu |\underline{\epsilon}'|^2$$

with $\kappa = \lambda + \frac{2}{3}\mu$.

The criterion. The two terms are now genuinely independent: the scalar $\text{tr } \underline{\epsilon}$ and the traceless tensor $\underline{\epsilon}'$ may be chosen separately, and together they determine $\underline{\epsilon}$.

If $\mu > 0$ and $\kappa > 0$ then $2U$ is a sum of two non-negative terms, so $U \geq 0$; and $U = 0$ forces both $\text{tr } \underline{\epsilon} = 0$ and $\underline{\epsilon}' = \hat{Q}$, hence $\underline{\epsilon} = \hat{Q}$.

Conversely, suppose U is positive definite. Taking $\underline{\epsilon} = c\hat{I}$ with $c \neq 0$ gives $\underline{\epsilon}' = \hat{Q}$ and $\text{tr } \underline{\epsilon} = 3c$, so $2U = 9\kappa c^2 > 0$ and hence $\kappa > 0$. Taking $\underline{\epsilon}$ any non-zero traceless tensor, for instance $\hat{E}_1 \otimes \hat{E}_2 + \hat{E}_2 \otimes \hat{E}_1$, gives $\text{tr } \underline{\epsilon} = 0$ and $2U = 2\mu |\underline{\epsilon}'|^2 > 0$, hence $\mu > 0$. So

$$U \text{ positive definite} \iff \mu > 0 \text{ and } \kappa > 0.$$

Reading the two constants. The decomposition separates the two ways a body can be strained. A pure dilatation has $\underline{\epsilon}' = \hat{Q}$ and costs $\frac{1}{2}\kappa(\text{tr } \underline{\epsilon})^2$, so κ is the bulk modulus. A pure shear has $\text{tr } \underline{\epsilon} = 0$ and costs $\mu|\underline{\epsilon}'|^2$, so μ is the shear modulus, the G of (11.41). Positive definiteness

says both cost energy. Note that λ itself need not be positive: $\lambda = -\frac{2}{3}\mu$ gives $\kappa = 0$, which is excluded, but any $\lambda > -\frac{2}{3}\mu$ is admissible, and negative λ corresponds to materials that thicken laterally when compressed. $\square$

PROBLEM 8 (The spatial equation of motion). Show that Cauchy's equation of motion in an inertial frame is approximated at leading order in the small displacement gradient by

$$\operatorname{div} \underline{T} + \rho_\kappa \underline{b} = \rho_\kappa \underline{\ddot{u}}.$$

Solution. Start from the referential form (12.71) and push every term forward with the shifter.

The divergence. By (12.55), $T_{ij} \simeq \delta_{iA} \delta_{jB} S_{AB}$. Differentiating with respect to x_j and converting the derivative by (12.14) in the form $(\cdot)_{,j} \simeq (\cdot)_{,C} \delta_{jC}$,

$$T_{ij,j} \simeq \delta_{iA} \delta_{jB} S_{AB,C} \delta_{jC} = \delta_{iA} \delta_{BC} S_{AB,C} = \delta_{iA} S_{AB,B},$$

using $\delta_{jB} \delta_{jC} = \delta_{BC}$, which is $\underline{1}^t \underline{1} = \underline{\hat{I}}$ in components. Hence

$$\operatorname{div} \underline{T} \simeq \underline{1} (\operatorname{Div} \underline{S}).$$

The other terms. The body force satisfies $\underline{1} \underline{f} = \underline{1} \underline{1}^t \underline{b} = \underline{b}$, and the acceleration $\underline{1} \underline{\ddot{u}} = \underline{1} \underline{1}^t \underline{\ddot{u}} = \underline{\ddot{u}}$, both by $\underline{1} \underline{1}^t = \underline{I}$.

Assembling. Apply $\underline{1}$ to (12.71):

$$\underline{1} (\operatorname{Div} \underline{S}) + \rho_\kappa \underline{1} \underline{f} = \rho_\kappa \underline{1} \underline{\ddot{u}},$$

that is

$$\operatorname{div} \underline{T} + \rho_\kappa \underline{b} = \rho_\kappa \underline{\ddot{u}}$$

at leading order.

Which density. By (12.29), $\rho \simeq \rho_\kappa$, so the same equation may be written with ρ in place of ρ_κ : the difference is of first order in $|\underline{G}|$ and multiplies terms that are already of first order, so it is of second order and lies below the accuracy of the theory. That is why the literature is

free to write either, and it is worth remembering that the freedom is an approximation, not an identity. In the exact theory (6.109) carries ρ and (6.136) carries ρ_κ , and the two equations are genuinely different.

For an isotropic material with constant moduli, substituting (12.61) and using $\operatorname{div}[(\operatorname{grad} \underline{u})^t] = \operatorname{grad}(\operatorname{div} \underline{u})$ gives the Navier equation

$$\mu \operatorname{div} \operatorname{grad} \underline{u} + (\lambda + \mu) \operatorname{grad}(\operatorname{div} \underline{u}) + \rho_\kappa \underline{b} = \rho_\kappa \ddot{\underline{u}},$$

which is the equation almost every elasticity calculation begins from. $\square$

PROBLEM 9 (The equation of motion for a second inertial observer). Verify (12.73). Show that $\ddot{\underline{w}} \mapsto \ddot{\underline{w}}^+ = \underline{K} \ddot{\underline{w}}$ under a Galilean transformation, and hence that (12.76) holds if the frame of O^+ is inertial.

Solution. **(12.73).** By Problem 1 and (12.72), $\underline{S}$ transforms like $\underline{E}$, so $\underline{S}^+ = \underline{K} \underline{S} \underline{K}^t$, that is $S_{AB}^+ = K_{AC} K_{BD} S_{CD}$. The reference positions are related by $\underline{X}^+ = \underline{K} \underline{X} + \underline{c}^+$ with $\underline{c}^+$ constant, so $X_B^+ = K_{BE} X_E + c_B^+$ and, $\underline{K}$ being orthogonal, $\partial X_E / \partial X_B^+ = K_{BE}$. The chain rule then gives

$$\frac{\partial S_{AB}^+}{\partial X_B^+} = K_{AC} K_{BD} \frac{\partial S_{CD}}{\partial X_E} K_{BE} = K_{AC} \delta_{DE} S_{CD,E} = K_{AC} S_{CD,D},$$

using $K_{BD} K_{BE} = \delta_{DE}$, which is $\underline{K}^t \underline{K} = \hat{I}$. Hence

$$\operatorname{Div}^+ \underline{S}^+ = \underline{K} (\operatorname{Div} \underline{S})$$

The acceleration. By Problem 1, $\underline{w}^+ = \underline{K} \underline{w} + \underline{d}_\kappa$ with $\underline{d}_\kappa = \underline{K} \underline{1}^t \underline{Q}^t \underline{d}$. Differentiating twice, and noting that $\underline{K}$ is constant in time by hypothesis,

$$\ddot{\underline{w}}^+ = \underline{K} \ddot{\underline{w}} + \ddot{\underline{d}}_\kappa.$$

Under a Galilean transformation (7.10) the rotation $\underline{Q}$ is constant and $\underline{c}(t) = \underline{V}t + \underline{c}_0$ is linear in t . By (7.21) the uniform part of the displacement is $\underline{d}(t) = \underline{c}(t) - \underline{Q} \underline{1}^t \underline{K}^t \underline{c}^+$, which is therefore also linear in t , so $\ddot{\underline{d}} = \underline{0}$; and $\underline{d}_\kappa = \underline{K} \underline{1}^t \underline{Q}^t \underline{d}$ with all three tensors constant, so $\ddot{\underline{d}}_\kappa = \underline{0}$ too. Hence

$$\ddot{\underline{w}}^+ = \underline{K} \ddot{\underline{w}}$$

Assembling. Substituting into (12.74),

$$\text{Div}^+ \mathcal{S}^+ + \rho_{\kappa^+}^+ f^+ = \rho_{\kappa^+}^+ K \ddot{w} = \rho_{\kappa^+}^+ \ddot{w}^+,$$

which is (12.76). So the second observer writes the same equation, with the same moduli by (12.67): linear elasticity is Galilean invariant, and no experiment within it can distinguish one inertial frame from another. $\square$

PROBLEM 10 (A non-inertial second observer). How is (12.76) modified if the frame of O^+ is not inertial?

Solution. Only the acceleration changes; (12.73) and the constitutive transformations (12.66) used nothing about the motion of the frames.

The extra terms. For general $Q(t)$ and $\mathcal{C}(t)$ the accelerations are related by (7.9),

$$\mathcal{A}^+ - Q\mathcal{A} = \mathcal{A}_f, \quad \mathcal{A}_f = \ddot{\mathcal{C}} + 2\Omega(\mathcal{V}^+ - \dot{\mathcal{C}}) + (\dot{\Omega} - \Omega^2)(\mathcal{X}^+ - \mathcal{C}),$$

$\Omega = \dot{Q}Q^t$ being the spin of the relative motion. In the linear theory $\mathcal{A} = \ddot{u}$ and $\mathcal{A}^+ = \ddot{u}^+$, by (12.68) applied in each frame, so

$$\ddot{u}^+ = (\mathbb{1}^+)^t \mathcal{A}^+ = K \mathbb{1}^t Q^t (Q\mathcal{A} + \mathcal{A}_f) = K \mathbb{1}^t \mathcal{A} + K \mathbb{1}^t Q^t \mathcal{A}_f = K \ddot{w} + (\mathbb{1}^+)^t \mathcal{A}_f,$$

using $(\mathbb{1}^+)^t = K \mathbb{1}^t Q^t$ from (7.18) and $\mathbb{1}^t \mathcal{A} = \ddot{w}$. Hence $K \ddot{w} = \ddot{u}^+ - (\mathbb{1}^+)^t \mathcal{A}_f$, and substituting into (12.74),

$$\begin{aligned} \text{Div}^+ \mathcal{S}^+ + \rho_{\kappa^+}^+ [f^+ + f_f^+] &= \rho_{\kappa^+}^+ \ddot{w}^+, \\ \text{where } f_f^+ &= (\mathbb{1}^+)^t \mathcal{A}_f. \end{aligned}$$

The non-inertial observer writes the same equation with an additional body force per unit mass, and Chapter 7 already named its three parts: $\ddot{\mathcal{C}}$ is the translational term, $2\Omega(\mathcal{V}^+ - \dot{\mathcal{C}})$ the Coriolis term, and $(\dot{\Omega} - \Omega^2)(\mathcal{X}^+ - \mathcal{C})$ the Euler and centrifugal terms.

REMARK 12.9 (what the Coriolis term does to the equation). The apparent force is not merely an inconvenience of bookkeeping. Two of its parts spoil the structure of (12.76).

The Coriolis term contains $\underline{v}^+$, hence $\underline{\dot{u}}^+$, so the equation acquires a first derivative in time. It ceases to be the wave equation of (12.76) and becomes one with a gyroscopic term, which is why the vibration of a rotating shaft cannot be analysed with the non-rotating modes.

The centrifugal term contains $\underline{x}^+$, hence $\underline{u}^+$, so it adds a term proportional to the displacement itself. It behaves like a negative stiffness and can destabilize a structure that is stable at rest, which is the mechanism of spin softening in turbine blades. Both effects are linear in $\underline{u}^+$ and belong properly to the linear theory; nothing here required an additional approximation.

□

Appendix A

Dictionary to the book: Chapter 1 notation

Book	Here	Book	Here
(1.1)–(1.3)	Def. 1.2, Thm. 1.4	(1.79)–(1.84)	Thm. 1.51
ip axioms	Def. 1.5	(1.85), (1.86)	Def. 1.54, Prop. 1.55
(1.4)–(1.6)	Thm. 1.6–Cor. 1.7	(1.87)–(1.93)	Def. 1.57, Thm. 1.58
(1.7)–(1.16)	Def. 1.9–Prop. 1.11	(1.95)	Prop. 1.63
(1.17)–(1.19)	Thm. 1.15	(1.96), (1.97)	Def. 1.69, Def. 1.66
(1.20)–(1.22)	Thm. 1.19	(1.98), (1.99)	Thm. 1.70
(1.23)–(1.30)	Def. 1.21–Prop. 1.24	(1.100)–(1.108)	Thm. 1.74–Thm. 1.76
(1.31)–(1.39)	Def. 1.27–Cor. 1.30	(1.109)–(1.115)	Thm. 1.78–Ex. 1.81
(1.40)–(1.50)	Def. 1.31–Ex. 1.34	(1.116)–(1.125)	Ex. 1.82, Ex. 1.83
(1.55)–(1.58)	Thm. 1.39	(1.126)–(1.134)	Def. 1.85–Cor. 1.87
(1.59)–(1.62)	Prop. 1.42, Prop. 1.44	(1.135)–(1.145)	Ex. 1.88–Ex. 1.90
(1.63)–(1.72)	Prop. 1.46	(1.146)–(1.152)	Def. 1.91–Prop. 1.95
(1.73)–(1.77)	Prop. 1.48	(1.153)–(1.162)	Ex. 1.99
(1.78)	Def. 1.49	(1.163)–(1.166)	Def. 1.96–Rem. 1.98
sym/skw, sph/dev	Def. 1.59	(1.167), (1.168)	Thm. 1.100, Thm. 1.102

Lemma 1.25 (one dimensionality of alternating 3 forms) replaces the book's Appendix A computations in Propositions 1.28, 1.50 and Theorem 1.52. All twenty eight chapter problems are solved in §1.3.

Appendix B

Dictionary to the book: Chapter 2 notation

Book	Here	Book	Here
(2.1) motion	Def. 2.2	(2.34)–(2.37)	§2.4, Thm. 2.14
(2.2) velocity, accel.	Def. 2.3	(2.38)–(2.41)	eqs. here in §2.4
(2.3), (2.4) reference	Def. 2.4	(2.42)–(2.44)	Lem. 2.15
(2.5)–(2.7) χ_κ	eqs. in §2.1	(2.45), (2.46)	material curves, §2.4
(2.8)–(2.10) velocity forms	§2.1	(2.47)–(2.50) volume	eqs. here, Rem. 2.18
(2.11)–(2.13) descriptions	§2.2, Rem. 2.6	(2.51)–(2.54) Piola–Nanson	§2.4
(2.14)–(2.16) differentials	§2.2	(2.55) $\underline{F}^* = J\underline{F}^{-t}$	§2.4
(2.17)–(2.21) material deriv.	Thm. 2.8	(2.56)–(2.59) mass	§2.5
(2.22)–(2.24) coordinates	§2.3, Rem. 2.10	(2.60)–(2.65) localization	Thm. 2.20
(2.25)–(2.29) components	§2.3	Example (2.30)–(2.33)	Example 2.12
(2.66), (2.67) stretch	§2.6, Rem. 2.22	(2.89)–(2.95) $\underline{B}$	Prop. 2.28
(2.68)–(2.74) shifter	Def. 2.23, Prop. 2.24	(2.96)–(2.99) strains	Def. 2.31, Rem. 2.32
(2.75)–(2.78) displacement	§2.6, Rem. 2.25	(2.100)–(2.103) arclength	§2.7, Rem. 2.34
(2.79)–(2.81) undeformed	Prop. 2.26	(2.104)–(2.108) angles	§2.7, Rem. 2.35
(2.82)–(2.88) $\underline{C}$	Prop. 2.27	§2.8 Problems 1–3	§2.8

Figures 2.1–2.12 are described in Remarks 2.1, 2.5, 2.7, 2.10, 2.16, 2.18, 2.19, 2.22, 2.34 and 2.35. All algebra uses only Chapter 1: the tensor product and its component rule, the dyad product $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c})\underline{a} \otimes \underline{d}$, the transpose rules of Problems 17–20, the box product and determinant $[\underline{F}\underline{a}, \underline{F}\underline{b}, \underline{F}\underline{c}] = \det \underline{F} [\underline{a}, \underline{b}, \underline{c}]$, the cofactor identity $\underline{A}^t \underline{A}^* = (\det \underline{A})\underline{I}$ of Problem 22(c), the contracted ε – δ identities, and the coordinate free chain rule $d\underline{f} = \text{grad } \underline{f} \cdot d\underline{x}$.